PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 4

Cost and Management Accounting

Full Lessons

Every lesson as the app's Exam view (all sections except Under the hood), worked examples with their working shown, and the self-check with its answer key at the end of each lesson.

66 lessons in 15 chapters · phone edition · made 2026-10-20 from the PassBook lessons

The labs (step-through simulators) are not printed: open the lesson in the PassBook app for the interactive version.

Contents

1 Introduction to Cost and Management Accounting 7

1.1 Objectives, scope and users of cost and management accounting 7

1.2 Cost accounting department and installing a costing system 39

1.3 Cost, financial and management accounting, and financial management 69

1.4 Cost units, cost centres, cost objects and responsibility centres 94

1.5 Elements and classification of cost 127

1.6 Cost behaviour: fixed, variable, semi-variable and step costs 170

1.7 Cost control and cost reduction 202

1.8 Methods and techniques of costing, and digital costing 229

2 Cost Sheet 264

2.1 Cost sheet for a manufacturer 264

2.2 Estimated cost sheets, tenders and quotations 308

2.3 Cost sheet for the service sector 339

3 Material Cost 367

3.1 Procurement procedure, GeM, ERP and stock verification 367

3.2 Valuing material receipts 401

3.3 Pricing material issues: FIFO, LIFO and average methods 431

3.4 Stock levels: re-order, minimum, maximum, average and danger levels 466

3.5 Economic Order Quantity and quantity discounts 495

3.6 ABC, FSN, HML, VED, JIT and perpetual inventory 527

3.7 Inventory turnover ratios and stock discrepancies 566

3.8 Waste, scrap, spoilage, defectives and obsolescence 597

4 Employee Cost and Direct Expenses 631

4.1 Attendance, payroll and the elements of wages 631

4.2 Idle time and overtime 670

4.3 Employee turnover: measurement, causes, effects and cost 700

4.4 Time rate, piece rate and the Halsey and Rowan premium plans 729

4.5 Direct expenses 758

5 Overheads: Absorption Costing 782

5.1 Classifying overheads, allocation and apportionment 782

5.2 Primary distribution of factory overheads 813

5.3 Secondary distribution: direct, step, repeated and simultaneous methods 842

5.4 Absorption rates and the machine hour rate 876

5.5 Under- and over-absorption of overheads 919

5.6 Administration, selling, distribution and R&D overheads 954

6 Activity Based Costing 995

6.1 Activities, cost pools, cost drivers and the cost hierarchy 995

6.2 Product costs under ABC versus traditional absorption 1034

6.3 Activity based management and budgeting; benefits and limits 1075

7 Cost Accounting Systems 1110

7.1 Non-integrated and integrated accounting systems 1110

7.2 Reconciling cost and financial profit 1154

8 Unit and Batch Costing 1189

8.1 Unit costing 1189

8.2 Batch costing and the economic batch quantity 1215

9 Job Costing 1243

9.1 Job costing 1243

10 Process and Operation Costing 1280

10.1 Process accounts, normal loss, abnormal loss and abnormal gain 1280

10.2 Equivalent units: FIFO and weighted average 1318

10.3 Inter-process profit and operation costing 1365

11 Joint Products and By-Products 1395

11.1 Apportioning joint costs: physical units, sales value and net realisable value 1395

11.2 Sell at split-off or process further 1434

11.3 Treatment of by-products 1463

12 Service Costing 1499

12.1 Service costing and composite cost units 1499

12.2 Transport costing: passenger-km and tonne-km 1533

12.3 Hotels, hospitals, power, IT and other services 1566

13 Standard Costing 1606

13.1 Standards, their types and standard costing 1606

13.2 Material variances 1641

13.3 Labour variances, including idle time 1672

13.4 Variable overhead variances 1704

13.5 Fixed overhead variances 1729

13.6 Reconciling budgeted and actual profit, and accounting for variances 1759

14 Marginal Costing 1800

14.1 Marginal cost, contribution and the P/V ratio 1800

14.2 Break-even point, margin of safety and angle of incidence 1832

14.3 Break-even and profit-volume charts 1865

14.4 Marginal versus absorption costing and profit reconciliation 1901

14.5 Limiting factor and product mix 1934

14.6 Make or buy, shut down and other short-term decisions 1968

14.7 Multi-product break-even analysis 2005

15 Budgets and Budgetary Control 2032

15.1 Budgets, budgetary control, the budget manual and key factor 2032

15.2 Functional budgets and the master budget 2075

15.3 Flexible budgets 2117

15.4 Cash budget 2155

15.5 Zero based budgeting and performance budgeting 2190

15.6 Control ratios, budget variances, motivation, feedback and feed-forward 2224

CHAPTER 1

Introduction to Cost and Management Accounting

1.1 Objectives, scope and users of cost and management accounting

What cost and management accounting is for (five objectives), the seven functions that make up its scope, and who uses its information inside and outside the entity.

In plain words

One profit figure, forty cities, zero answers

Tapri Tea Co. Ltd has just published its annual report. Profit for the year: one neat number. Kesar reads it twice and is still worried.

> **Kesar:** We made a profit. Good. But which of my forty cities made it? What does one samosa pack cost us? What do I quote the railways for premix?
>
> **Ritu:** The annual report cannot tell you any of that. It was never meant to.
>
> **Kesar:** Then put it all under miscellaneous and let us move on.
>
> **Ritu:** Per unit, please, Kesar-ji. That is a different set of books.

That "different set of books" is **cost and management accounting**. Financial accounts tell outsiders how the whole company did over a year. Cost accounting finds out what each thing costs: a pack, a café, a delivery run. **Management accounting** takes that cost information, adds other information, and helps managers plan, control and decide.

It has five jobs, its **objectives**: know the cost; use it to set prices and see profit; keep spending in line with the plan (**cost control**); push the cost itself down for good (**cost reduction**); and help managers decide. Its **scope** is the list of things the system actually does, from working out a cost all the way to sending reports to managers and keeping the records the law asks for.

And it has many readers. Inside Tapri: Kesar and the board, managers, supervisors, workers. Outside: regulators, auditors, shareholders, and the bank that lent against Tapri's stock.

Why bother? Because BrewBucks is coming to every Tapri city. One way to beat a rival is **cost leadership**: making the product at the lowest cost while keeping the quality, so the price can compete. Nobody leads on cost without knowing it first.

Think of it like

The navigation app

Cost and management accounting works like the navigation app on Kesar's phone. It tells you exactly where you are (the cost), what the fare should be (the price), whether you are still on the planned route (cost control), whether a shorter road exists (cost reduction), and which turn to take next (decisions). The annual report is the photo of the destination: useful, but it cannot steer.

Key idea

Cost and management accounting exists to ascertain the cost of each cost object and to use it for pricing, cost control, cost reduction and decisions. It serves users inside the entity (who manage it) and outside it (who regulate, audit, own or lend to it).

How it works

First, ICAI's six definitions

The Study Material opens the chapter with Michael E. Porter's "Cost Leadership", one of three strategic dimensions for competitive advantage (the others are "Product differentiation" and "Focus or Niche"): producing goods or services at the lowest cost while maintaining quality, which can be achieved if an entity

has a robust Cost and Management Accounting system in place (SM 1.2). Then it defines the words.

Definition

The six terms, in ICAI's words (SM 1.2–1.3)

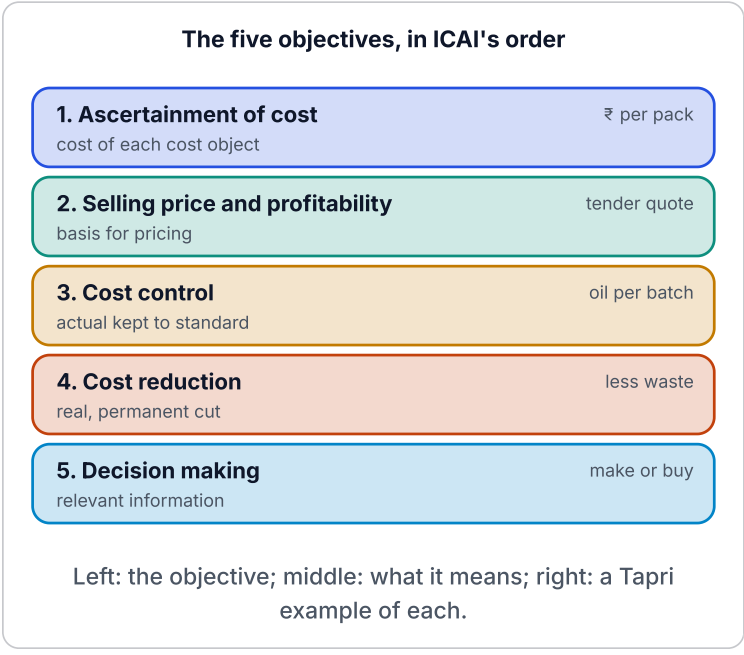
- **Cost**: the amount of resource given up in exchange of some goods or services. As a noun, "the amount of expenditure (actual or notional) incurred on or attributable to a specified article, product or activity"; as a verb, an action to ascertain the cost of a specified thing or activity.
- **Costing**: "the technique and process of ascertaining costs".
- **Cost Accounting**: "the process of accounting for cost which begins with the recording of income and expenditure or the bases on which they are calculated and ends with the preparation of periodical statements and reports for ascertaining and controlling costs."
- **Cost Accountancy**: "the application of costing and cost accounting principles, methods and techniques to the science, art and practice of cost control and the ascertainment of profitability. It includes the presentation of information derived there from for the purpose of managerial decision making."
- **Management Accounting** (CIMA Official Terminology): "the application of the principles of accounting and financial management to create, protect, preserve and increase value for the stakeholders of for-profit and not-for-profit

enterprises in the public and private sectors." It is an integral part of the management function and assists management with relevant information for planning, organising, controlling and decision making.

- **Cost Management**: "an application of management accounting concepts, methods of collections, analysis and presentation of data to provide the information needed to plan, monitor and control costs."

CIMA adds why it matters: an organisation's costing system is "the foundation of the internal financial information system for managers. It provides the information that management needs to plan and control the organisation's activities and to make decisions about the future" (SM 1.3).

The five objectives (SM 1.4–1.6)



Left: the objective; middle: what it means; right: a Tapri example of each.

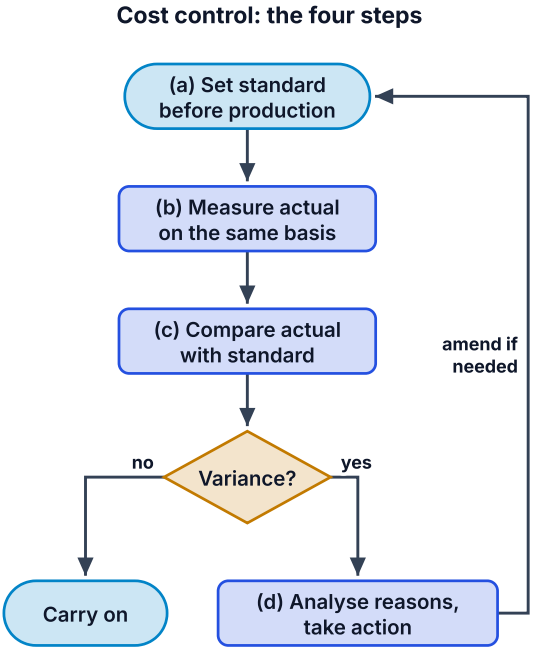
1. **Ascertainment of cost.** The main objective: costs are accumulated, assigned and ascertained for each **cost object** (a unit, job, operation, process, department or service). Cost objects are explained in cost units, cost centres and cost objects.
2. **Determination of selling price and profitability.** In a competitive market, selling prices are set by external factors, but the cost accounting system still provides a basis for price fixation and rate negotiation, and shows the profitability of each cost object.
3. **Cost control.** "Maintaining discipline in expenditure": expenditure is kept in line with predetermined

standards, and any variation is noted and reported continuously. Four steps, below.

4. **Cost reduction.** "The achievement of real and permanent reduction in the unit cost of goods manufactured or services rendered without impairing their suitability for the use intended or diminution in the quality of the product." No cost is treated as the lowest. Two actions: (a) segment each activity into value added and non-value added, and eliminate the non-value added ones without affecting the essential characteristics of the product or process (Porter's Value Chain Analysis is one method of value analysis); (b) continuous research and study to find the most optimal way to make the product or render the service.
5. **Assisting management in decision making.** By providing relevant information, internal and external, it helps management plan, implement, measure, control and evaluate activities.

The definition of cost reduction rests on a **three-fold assumption** (SM 1.5–1.6): (a) there is a saving in unit cost; (b) the saving is of a permanent nature; (c) the utility and quality of the goods and services remain unaffected, if not improved. How cost control and cost reduction differ, point by point, is in cost control and cost reduction.

Cost control in four steps (SM 1.4–1.5)



The variance is reported to the appropriate person or authority at step (c); at step (d) the standard itself is amended if developments make it necessary.

- (a) **Determination of pre-determined standard or results**: standard costs or performance targets for a cost object or cost centre are set before production or service begins.
- (b) **Measurement of actual performance**, in the same manner as the targets were set: if targets are set

operation-wise, actual costs are collected operation-wise too.

- (c) **Comparison of actual performance with set standard or target**: any deviation (variance) is noted and reported to the appropriate person or authority.
- (d) **Analysis of variance and action**: reasons are found and action taken to ensure compliance in future; if necessary, the standards are amended.

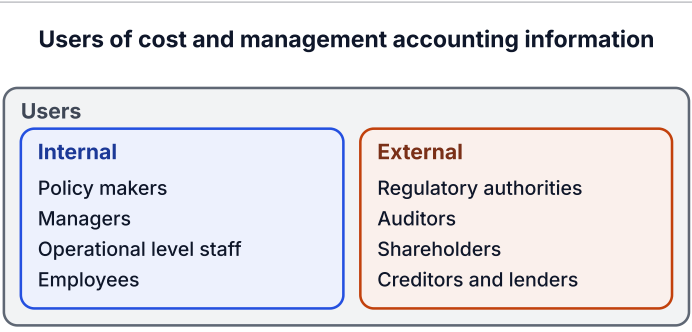
Standard costing turns these steps into variances: standards and standard costing.

Scope: seven functions (SM 1.7–1.8)

Function	What it covers
(i) Costing	The technique and process of ascertaining the costs of products or services, under cost accounting principles and rules; using historical costs, standard costs, process cost, operation cost etc.
(ii) Cost Accounting	Accounting for cost from the recording of expenditure to periodical statements and reports for ascertaining and controlling cost: a formal mechanism of cost ascertainment
(iii) Cost Analysis	Finding the factors responsible for the variance of actual costs from budgeted costs, and fixing responsibility for cost differences; it also helps in better cost management and strategic decisions

Function	What it covers
(iv) Cost Comparisons	Comparing the costs of alternative courses of action: different technology, different products and activities, the same product or service over time
(v) Cost Control	A detailed examination of each cost in the light of the advantage received from incurring it: is it within budget, and is further reduction possible?
(vi) Cost Reports	The ultimate function of cost accounting: reports for management at different levels, for planning and control, performance appraisal and decision making
(vii) Statutory Compliances	Maintaining cost accounting records as per the rules prescribed by the statute, on the use of materials, labour and other items of cost

Who reads all this: the users (SM 1.12–1.14)



ICAI's two groups and eight users, in the Study Material's order.

Internal users

User	Uses the information to
Policy makers	Formulate strategies: to achieve the goals (short and long term) that fulfil the organisation's objectives; to position the organisation in the competitive market; to design the organisational structure that gets policies and strategies implemented
Managers	Know the cost of a cost object and cost centre; know the price for the product or service; measure and evaluate the performance of responsibility centres; know profitability product-wise, department-wise, customer-wise; evaluate strategic options and decide

User	Uses the information to
Operational level staff (supervisors, foremen, team leaders)	Know their objectives and performance goals; know product and service specifications (volume, quality, process); know the parameters against which their performance is measured; know divisional (responsibility centre) profitability
Employees	Know time and attendance, incentives for work, performance standards

External users

User	Concerned with
Regulatory authorities	Cost data for tariff determination, providing subsidies, rate fixation; they require it on the basis of set standards and formats
Auditors	Costing information and reports reviewed by management, for the audit of financial accounts or a special purpose audit such as cost audit
Shareholders	Information that affects their investment: management's periodic communiqués and annual reports on new orders, product expansion, market share
Creditors and lenders	Information on the entity's ability to serve them: a lender against book debts and inventories wants regular reports on the net debt position and stock balances

Worked examples

Worked example **Ritu's week: name the function**

Ritu Sharma's diary for one week at Tapri Tea Co. Ltd reads:

(a) Worked out the cost of one 200 g chai premix pouch from its material, labour and overheads.

(b) Recorded every expense of the samosa factory, from purchase invoices to the monthly cost statement.

(c) Found out why the Pune café's milk cost was ₹38,000 above budget in August, and who was responsible.

(d) Compared the cost of firing kulhads on Tapri's own electric kiln with buying them from potters in Khurja.

(e) Examined each item of the cold-chain fleet's spending against the benefit it brought, to see that none exceeded its budget and whether any could be cut further.

(f) Sent the board and the city managers the monthly café-wise performance report.

(g) Updated the premix plant's cost records in the form the cost records rules under the law require.

Required: Identify the function within the scope of cost accounting to which each entry belongs.

MODEL ANSWER

Entry	Function within the scope of cost accounting
(a) Cost of one premix pouch	Costing: the technique and process of ascertaining costs of products
(b) Expenses recorded up to the monthly statement	Cost Accounting: recording of expenditure to periodical statements and reports
(c) Reasons and responsibility for the milk overrun	Cost Analysis: factors responsible for the variance of actual from budgeted cost, and fixation of responsibility
(d) Own kiln against buying from Khurja	Cost Comparisons: cost of alternative courses of action
(e) Each fleet cost against its benefit and budget	Cost Control: detailed examination of each cost in the light of the advantage received
(f) Monthly report to the board and managers	Cost Reports: reports for management at different levels for planning, control, performance appraisal and decision making

Entry	Function within the scope of cost accounting
(g) Cost records in the prescribed form	Statutory Compliances: cost accounting records maintained as per the rules prescribed by the statute

Worked example **Seven requests on Ritu's desk**

In one week, seven requests for cost and management accounting information reach Ritu:

1. The board, meeting on BrewBucks' entry, wants café profitability by city to decide where to open the next fifty cafés.
2. The head of the cafés division wants each café's cost per cup and its performance against budget.
3. Kitchen supervisors want this month's targets for cups per hour and wastage.
4. Café staff ask how the new incentive for fast service will be worked out.
5. A state food-processing department will grant a subsidy only on cost data in its prescribed format.
6. The bank that lent Tapri working capital against its inventories and book debts asks for the monthly stock statement and net debt position.
7. Shareholders at the annual general meeting ask about new orders from the railways and Tapri's

market share.

Required: Classify each request as coming from an internal or an external user, name the user as ICAI does, and explain how that user is concerned with the information.

MODEL ANSWER

Cost and Management Accounting information is used by different stakeholders, broadly categorised into users internal and external to the entity (SM 1.12).

Request
1

User
Policy makers (the board)

Internal or external
Internal

How the user is concerned
Formulate strategies to achieve the goals and position the organisation in the competitive market

Request
2

User
Managers

Internal or external
Internal

How the user is concerned
Know the cost of a cost object and measure the performance of responsibility centres

Request
3

User
Operational level staff

Internal or external
Internal

How the user is concerned
Know their objectives, performance goals and the parameters on which they are evaluated

Request
4

User
Employees

Internal or external
Internal

How the user is concerned
Information on time and attendance, incentives for work and performance standards

Request
5

User
Regulatory authority

Internal or external
External

How the user is concerned
Cost data for providing subsidies, required on the basis of set standards and formats

Request
6

User
Creditors and lenders

Internal or external
External

How the user is concerned
Ability to serve the loan: regular reporting on the net debt position and stock balances

Request
7

User
Shareholders

Internal or external
External

How the user is concerned
Information affecting their investment: new orders received, product expansion, market share

Worked example

Kesar's samosa factory gets a costing system

Tapri's new frozen-samosa factory in Pithampur sells to Tapri's own cafés and to supermarkets. Kesar has asked Ritu to introduce a cost and management accounting system there. He wants to know what a pack really costs, what to quote in a supermarket chain's tender, whether frying oil is being wasted, how to bring the cost per pack down without changing the

taste, and whether to keep making pastry sheets or buy them in.

The standard set for frying oil is 18 litres per 1,000 packs at ₹140 a litre. In August the factory made 60,000 packs and used 1,170 litres, all bought at ₹140.

Required:

(i) Describe the main objectives of introducing a cost and management accounting system in this factory, relating each to the facts.

(ii) Show the cost control steps for frying oil in August, with the figures.

MODEL ANSWER

(i) Objectives (SM 1.4–1.6)

1. **Ascertainment of cost.** Costs are accumulated, assigned and ascertained for each cost object: here the cost per pack of frozen samosas.
2. **Determination of selling price and profitability.** Supermarket prices are set by competition, but the cost system provides the basis for price fixation and rate negotiation: the floor for the tender quote, and the profit on each channel.
3. **Cost control.** Expenditure is kept in consonance with predetermined standards, and variations are noted and reported continuously: frying oil against its standard of 18 litres per 1,000 packs.

4. **Cost reduction.** A real and permanent reduction in the unit cost without impairing suitability or quality: activities are segmented into value added and non-value added, the non-value added ones eliminated, and continuous research finds the optimal process; the taste (quality) must remain unaffected.
5. **Assisting management in decision making.** Relevant information for decisions such as making or buying pastry sheets.

(ii) Cost control steps for frying oil, August

Step	Frying oil
(a) Pre-determined standard	18 litres per 1,000 packs at ₹140 = ₹2,520 per 1,000 packs
(b) Actual performance, measured on the same basis	1,170 litres × ₹140 = ₹1,63,800 for 60,000 packs
(c) Comparison	Standard for 60,000 packs: 60 × 18 = 1,080 litres = ₹1,51,200; excess 90 litres (8.33%), variance ₹12,600 (Adverse), reported to the factory manager
(d) Analysis and action	The reasons for the excess 90 litres are analysed (for example, how often the oil is changed, fryer temperature, leakage) and action is taken to ensure compliance in future; the standard is amended only if developments require it

The price was the standard ₹140, so the whole ₹12,600 comes from using 90 litres more than the standard.

Exam angle

In the exam

How ICAI has examined this lesson

In the ICAI documents read for this lesson (new-scheme papers, RTPs and MTPs from May 2024 to September 2026, and some old-scheme ones), the topic was asked in three of them, and the chapter asks it once more:

- **Users, 5 marks**: "Give two examples of internal users and three examples of external users and explain how they are concerned" (MTP Series I for January 2025 Q6(a)). The suggested answer reproduces the Study Material's list of all eight users with their uses, word for word.
- **Scope, 4 marks**: four descriptions to match with "the most relevant scope of Cost Accounting" (May 2025 Q6(c), an OR option); the answers were cost control, cost reports, cost comparison and cost analysis, matched on the Study Material's own phrases ("detailed examination of each cost", "alternative courses of action").
- **Objectives**: "Narrate the objectives of cost accounting" (RTP November 2023, old scheme, Q14(b)), answered in five points; and the chapter's own question, "Describe the main objectives of introduction of a Cost and Management Accounting System in a manufacturing organization" (SM 1.45).
- No question in the papers read asked for the six definitions alone, but they are the vocabulary of every theory answer in the paper.

In the exam

The wording traps

- "Explain how they are concerned": a list of names earns little; each user needs its use (tariff determination, subsidies and rate fixation for a regulator; net debt position and stock balances for a lender).
- A scope question matches on the Study Material's phrases. "Factors responsible for variance ... fixation of responsibility" is cost analysis, not cost control; "detailed examination of each cost in the light of advantage received" is cost control.
- The old-scheme answer to "objectives" (RTP November 2023) combines cost control with cost reduction and adds "ascertainment of profit of each activity". For January 2027, write the current Study Material's five, in its order.

Trick

Seven verbs for the scope

Ritu's week runs in ICAI's order: she **costs** it, **accounts** for it, **analyses** it, **compares** it, **controls** it, **reports** it and **complies** with the statute: Costing, Cost Accounting, Cost Analysis, Cost Comparisons, Cost Control, Cost Reports, Statutory Compliances (SM 1.7). It lists ICAI's seven functions only; it does not mean they happen one after another.

Answer frame

"Describe the main objectives of a cost and management accounting system"

1. **Ascertainment of cost**: accumulated, assigned and ascertained for each cost object (unit, job, operation, process, department, service).
2. **Determination of selling price and profitability**: a basis for price fixation and rate negotiation, even where prices are set by external factors.
3. **Cost control**: discipline in expenditure, in four steps: pre-determined standard; measurement of actual performance; comparison; analysis of variance and action.
4. **Cost reduction**: the definition ("real and permanent reduction in the unit cost … without impairing their suitability for the use intended or diminution in the quality"), value added and non-value added activities, continuous research.
5. **Assisting management in decision making**: relevant information for planning, implementing, measuring, controlling and evaluating.

In a case ("in a manufacturing organisation"), add one line of the case's facts under each heading.

"Give examples of internal and external users and explain how they are concerned" (5 marks)

Part of the answer	What earns the mark
Opening line	The information is used by different stakeholders, broadly categorised into internal and external users
Each internal user	Its name as ICAI gives it (policy makers, managers, operational level staff, employees) and its uses
Each external user	Its name (regulatory authorities, auditors, shareholders, creditors and lenders) and what it is concerned with
Number asked	Exactly as many of each as the question asks, the strongest ones first

Three answers to "Explain two external users"

Loses marks

"Auditors and banks."

Names only, and "banks" is not ICAI's term.

Passes

"Auditors use cost data for audit. Creditors and lenders use it to give loans."

Right users, thin uses.

Full marks

"Auditors need costing information and reports reviewed by management, for a financial or cost audit. Creditors and lenders need regular reports on the net debt position and stock balances."

ICAI's names, each with its concern.

Common mistakes

⚠ **Common mistake**

Hall of Shame: cost reduction that only kept costs on budget

"Cost reduction means keeping costs within the standards." That is cost control. **Fix:** cost reduction is a *real and permanent* reduction in the *unit* cost without impairing suitability or quality; it challenges the standards themselves. Test it against the three-fold assumption: a saving in unit cost, a permanent saving, quality unaffected if not improved.

⚠ **Common mistake**

The scope that lost two functions

Writing five functions and forgetting cost comparisons and statutory compliances, or calling cost control "the ultimate function". **Fix:** seven functions, and the Study Material calls **cost reports** the ultimate function of cost accounting (SM 1.8). Use the seven verbs.

⚠ **Common mistake**

Auditors inside, employees outside

Placing auditors among the internal users ("they come into the office every day") or forgetting employees altogether. **Fix:** ICAI's internal users are policy makers, managers, operational level staff and employees; the external users are regulatory authorities, auditors, shareholders, and creditors and lenders (SM 1.12).

Practice

Check yourself

1 Case scenario. Tapri's lender, a bank that financed its working capital against inventories and book debts, writes to Ritu every month. Which of the following correctly describes this user and its concern, as the Study Material gives it?

- **A** Internal user; it needs the cost per unit to fix Tapri's selling prices
- **B** External user (creditors and lenders); it needs regular reports on the net debt position and stock balances
- **C** External user (regulatory authority); it needs cost data for tariff determination
- **D** External user (shareholders); it needs information on new orders and market share

2 The standard for frying oil is 20 litres per 1,000 packs at ₹150 a litre. The factory made 45,000 packs and used 960 litres, bought at ₹150. By how many rupees did the actual oil cost exceed the standard cost for the output? (Enter the amount in rupees.)

Answer: _____

3 Which of the following is NOT part of the three-fold assumption in ICAI's definition of cost reduction?

- **A** There is a saving in unit cost
- **B** The saving is of a permanent nature
- **C** The utility and quality of the goods and services remain unaffected, if not improved
- **D** Costs are maintained in accordance with the established standards

4 Kesar says: "BrewBucks sets the market price of a cup, so our cost accounts are useless for pricing." According to the Study Material, which reply is correct?

- **A** He is right: in a competitive market, costing has no role in pricing
- **B** Even when prices are set by external factors, the cost accounting system provides a basis for price fixation and rate negotiation
- **C** Prices must always equal cost plus a fixed margin, so the market price is irrelevant
- **D** Pricing is a function of financial accounting, not cost accounting

5 Which of these does the Study Material list among the internal users of cost and management accounting information? Select all that apply.

- **A** Policy makers
- **B** Auditors
- **C** Operational level staff
- **D** Employees
- **E** Regulatory authorities

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Five objectives, in ICAI's order: ascertainment of cost, determination of selling price and profitability, cost control, cost reduction, and assisting management in decision making.
- Scope, seven functions: costing, cost accounting, cost analysis, cost comparisons, cost control, cost reports (the ultimate function) and statutory compliances.
- Internal users: policy makers, managers, operational level staff, employees. External users: regulatory authorities, auditors, shareholders, creditors and lenders.

- Cost control: pre-determined standard → measure actual (same basis) → compare and report → analyse variance and act (amend standards if needed).
- Cost reduction: real and permanent reduction in unit cost without impairing suitability or quality; three-fold assumption: saving in unit cost, permanent, quality unaffected if not improved.
- Costing: "the technique and process of ascertaining costs". Cost, as a noun: expenditure (actual or notional) incurred on or attributable to a specified article, product or activity.
- Cost leadership needs a robust cost and management accounting system. (No navigation app, no shortcut.)

Answer key · Check yourself

1 **B External user (creditors and lenders); it needs regular reports on the net debt position and stock balances**

A lender is an external user concerned with the entity's ability to serve it; ICAI's own example is a lender against book debts and inventories that wants the net debt position and stock balances. Tariffs are a regulator's concern, market share a shareholder's.

2 **9000**

Standard quantity = 45 × 20 = 900 litres = ₹1,35,000. Actual = 960 × ₹150 = ₹1,44,000. Excess = 60 litres × ₹150 = ₹9,000 (adverse). This is step (c) of cost control, comparison of actual with standard.

3 **D Costs are maintained in accordance with the established standards**

Keeping costs to established standards is cost control. Cost reduction assumes a saving in unit cost, a permanent saving, and unaffected (if not improved) utility and quality.

4 **B Even when prices are set by external factors, the cost accounting system provides a basis for price fixation and rate negotiation**

The second objective: "though in a competitive business environment selling prices are determined by external factors but cost accounting system provides a basis for price fixation and rate negotiation" (SM 1.4).

5 **A Policy makers; C Operational level staff; D Employees**

Internal: policy makers, managers, operational level staff, employees. Auditors and regulatory authorities are external, with shareholders and creditors and lenders.

1.2 Cost accounting department and installing a costing system

What the cost and management accounting department does, the nine factors to study before installing a costing system, the six features of a good one, its limitations, and the practical difficulties of installation.

In plain words

"Buy the software, sab adjust ho jayega"

Kesar has just bought his cousin's namkeen business, renamed it Tapri Namkeen, and wants a costing system there by Monday.

> **Kesar:** Ritu, buy the best costing software. Install it on Monday. Sab adjust ho jayega.
>
> **Ritu:** Software is the easy part. First I need to know why we want it, what we make, how we make it, and who will read the reports.
>
> **Kesar:** Everyone will read the reports!
>
> **Ritu:** Then they must be short enough to read. Per unit, please, not per kilogram of paper.

Ritu is right in the way ICAI is right. A costing system costs money, so the first question is whether it will pay for itself. Then, before installing anything, the business must be studied: the **objective** of the system (fixing prices? controlling costs?), the industry, the levels of

management and what each needs, the product, the production process, what other departments need, how the cost books will tie up with the financial books, the law, and the qualities the information must have.

A system built that way has six qualities: it is **informative and simple**, **accurate and authentic**, **uniform and consistent**, **integrated** with the other systems, **flexible**, and **trusted** by management.

Once it runs, the **cost and management accounting department** collects costs by element, assigns them to products and services, compares them with budgets and standards, gives managers the information they need to decide, and measures how each department performs.

And installing it is a people problem as much as a software problem: busy managers call it interference, the accounts staff fear for their jobs, supervisors hate the extra forms, and trained cost staff are hard to find.

Think of it like

The tailor-made suit

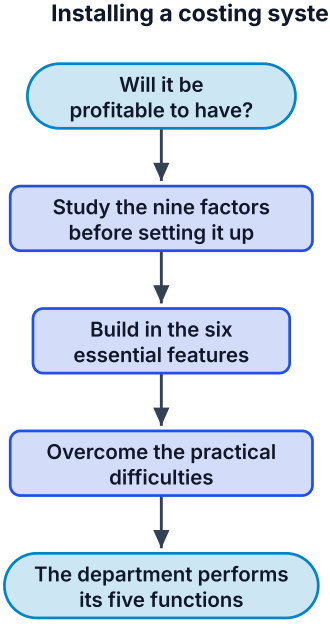
A costing system is a suit, not a T-shirt off the rack. The tailor first measures you: what the suit is for, your build, how you move. A good suit is comfortable (simple), fits exactly (accurate), uses one size chart (uniform), goes with your shoes (integrated), has seams that can be let out (flexible), and, most important, you actually wear it (trust). ICAI's own word for the system is "tailor-made".

Key idea

Study the business before installing the system, build the system so that people can and do use it, and remember that it must be worth its cost. A costing system that management does not trust produces reports that nobody reads.

How it works

From "is it worth it?" to a working department



The first three boxes and the last are from the Study Material (SM 1.11–1.17); the practical difficulties are from ICAI's RTP and MTP answers.

"As in the case of every other form of activity, it should be considered whether it would be profitable to have a Cost Accounting System." Management needs complete and accurate information to decide, and a well-established costing system should provide all relevant information as and when required by management and other stakeholders (SM 1.15).

What the department is for: role and functions (SM 1.11–1.12)

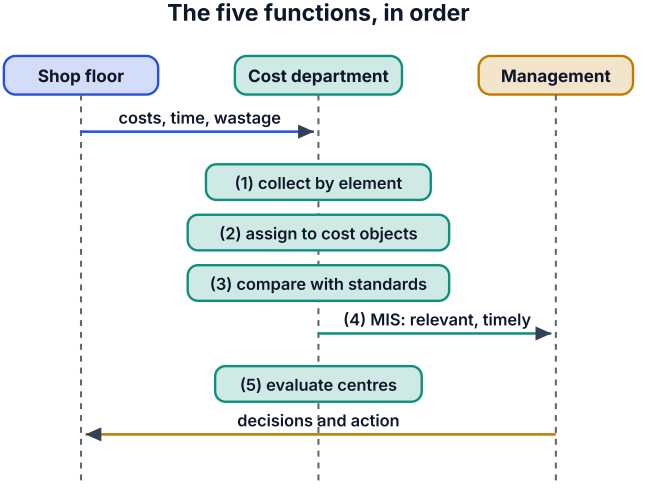
The role of a cost and management accounting system is to:

- provide relevant information to management for decision making;
- assist management in planning, measurement, evaluation and controlling of business activities;
- help in allocation of cost to products and inventories for both external and internal users.

Its **functions** are:

1. **Collection and accumulation of cost** for each element of cost.
2. **Assigning costs to cost objects** to ascertain cost.
3. **Budgets and standards**: the Cost and Management Accounting Department (whatever it is called) sets budgets and standards for a period or activity beforehand; they are compared with the assigned and ascertained cost, and deviations are analysed and reported, to control costs.
4. **Information for decisions**, the main function: an information system environment, popularly known as the **Management Information System (MIS)**, gives management at all levels relevant and timely information, internal and external, for decisions on cost optimisation, price fixation and plans for products, processes and marketing.

5. **Performance evaluation**: the performance of each responsibility centre is measured against the set standards; the department gathers data such as time taken, wastages and process idleness, analyses it, prepares reports and takes the necessary action.



Data comes in from the shop floor, information goes out to management, and management's decisions go back to the shop floor.

Responsibility centres, whose performance function 5 measures, are explained in cost units, cost centres and responsibility centres.

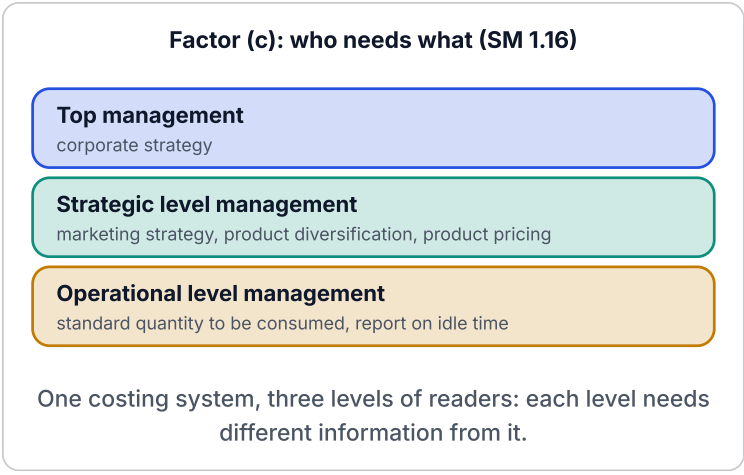
Before installing: the nine factors (SM 1.15–1.17)

"Before setting up a system of cost accounting the factors mentioned below should be studied":

- (a) **Objective**: the objective of setting up the costing system, for example whether it is being introduced for fixing prices or for establishing a system of cost control.
- (b) **Nature of business or industry**: every business or industry has its own uniqueness and objectives, and cost accounting methods follow its cost information requirement; an oil refinery, for example, maintains process-wise cost accounts to find the cost of a process such as crude refinement.
- (c) **Organisational hierarchy**: the system should fulfil the information requirements of the different levels of management (the diagram below).
- (d) **Knowing the product**: the nature of the product determines the type of costing system. A product with by-products needs a system that accounts for by-products; for perishable or short shelf-life products, **marginal costing** is appropriate, to know the contribution and the minimum price at which the products could be sold.
- (e) **Knowing the production process**: a good costing system can never be established without complete knowledge of the production process; costs can be apportioned on the most appropriate and scientific basis only if the cost accountant can identify the effort or resources each process consumes. This includes

basic technical know-how and the peculiarities of the process.

- (f) **Information synchronisation**: a department or system takes substantial organisational resources, so the information needs of other departments must be taken into account; for example, the accounts department submits a monthly stock statement to the lender bank and quantity-wise stock details when filing returns with the tax authorities.
- (g) **Method of maintenance of cost records**: decide beforehand how the cost and financial accounts could be inter-locked into a single integral accounting system, and how the results of separate cost and financial accounts could be reconciled by means of control accounts.
- (h) **Statutory compliances and audit**: records are maintained to comply with statutory requirements, and the applicable cost accounting standards are followed.
- (i) **Information attributes**: the information should have all the attributes of useful information, i.e. complete, accurate, timely and relevant, for an effective MIS.



One costing system, three levels of readers: each level needs different information from it.

How cost and financial accounts are kept separately or integrated (factor g) is in non-integrated and integrated accounting systems, and the reconciliation in reconciling cost and financial profit.

The six essential features of a good system (SM 1.14–1.15)

1. **Informative and simple**: tailor-made, practical, simple and capable of meeting the requirements of the business; it should not sacrifice utility by introducing inaccurate and unnecessary details.
2. **Accurate and authentic**: the data should be accurate and authenticated; otherwise it may distort the output and a wrong decision may be taken.
3. **Uniformity and consistency** in the classification, treatment and reporting of cost data, for benchmarking and comparability of results in both horizontal and vertical analysis.

4. **Integrated and inclusive**: integrated with other systems such as financial accounting, taxation, statistics and operational research, for a complete overview and clarity in results.
5. **Flexible and adaptive**: flexible enough to be amended for changes in technological, reporting, regulatory and other requirements.
6. **Trust on the system**: management should trust the system and its output; this needs an active role of management in developing a system that reflects a strong conviction in using information for decision making.

The price of a costing system: its limitations (SM 1.23–1.24)

1. **Expensive**: analysis, allocation and absorption of overheads require considerable additional work, and hence additional money.
2. **Requirement of reconciliation**: the results shown by the cost accounts differ from those shown by the financial accounts, so reconciliation statements are necessary to verify their accuracy.
3. **Duplication of work**: the organisation has to maintain two sets of accounts, financial accounts and cost accounts.

These three are why the first question is whether the system will be profitable to have, and why factor (g) asks how the two sets of accounts will be inter-locked or reconciled.

The practical difficulties, and how to overcome them

The current Study Material chapter does not list these, but ICAI's suggested answers do (RTP September 2026 Q16(a); MTP Series II for September 2024 Q6(a)). The difficulties:

1. **Lack of top management support**: the installation does not receive the support of top management. They consider it interference in their work, believe it will involve additional paperwork, and wrongly think it is meant for keeping a check on their activities.
2. **Resistance from the cost accounting department's staff**, who fear losing their jobs and importance after the new system is implemented.
3. **Non-co-operation from user departments**: foremen, supervisors and other staff may not co-operate in providing the data, as it adds to their responsibilities and to the paperwork of the whole team.
4. **Shortage of trained staff**: installation is specialised work, and trained staff may be short.

To overcome them:

- sell the idea to top management and convince them of the utility of the system;
- overcome resistance and non-co-operation by a behavioural approach, dealing with the staff concerned effectively;
- give proper training to the staff at each level;

- hold regular meetings with the cost accounting staff, user departments and top management to clarify their doubts and misgivings.

> **Note**
>
> **Where this list comes from**
>
> The July 2024 edition of the Study Material (reprint August 2025), the one for January 2027, does not list these points. ICAI still asks the question in its RTPs and MTPs and answers it with the points above, so learn them with the nine factors.

Worked examples

Worked example **Tapri Namkeen: what to study first**

Tapri Namkeen Pvt Ltd makes bhujia and sev in four stages (mixing, extruding, frying, packing), and fresh sweets with a shelf life of three days. The frying stage leaves spent oil and fried crumbs, sold to an animal-feed maker. The company has a cash-credit limit from its bank against stock and book debts. Kesar wants to price the sweets for the Diwali rush; the plant supervisors complain that nobody tells them their material and idle-time targets. The financial accounts are kept on accounting software that the plant does not use.

Ritu, sent from Tapri to set up the costing there, must decide what to study before installing a cost accounting

system.

Required: Advise Ritu on the factors to be studied before installing a cost accounting system, applying each to Tapri Namkeen.

MODEL ANSWER

Before setting up a system of cost accounting, the following factors should be studied (SM 1.15–1.17):

1. **Objective**: whether the system is introduced for fixing prices or for establishing cost control. Here both: Diwali pricing of sweets and material and idle-time control in the plant.
2. **Nature of business or industry**: cost accounting methods follow the industry's cost information requirement. Bhujia and sev pass through four stages, so process-wise cost accounts are needed (mixing, extruding, frying, packing).
3. **Organisational hierarchy**: top management needs corporate strategy information; strategic level management, marketing strategy and product pricing (Kesar's Diwali prices); operational level management, standard quantities to be consumed and idle-time reports (the supervisors' complaint).
4. **Knowing the product**: spent oil and crumbs are by-products, so the system must account for by-products; the sweets are perishable, so marginal costing is appropriate to know the contribution

and the minimum price at which they could be sold.

5. **Knowing the production process**: the cost accountant must know the effort and resources each of the four stages consumes, to apportion costs on the most appropriate and scientific basis.
6. **Information synchronisation**: the needs of other departments must be met, such as the monthly stock statement for the bank under the cash-credit limit, and quantity-wise stock details for tax returns.
7. **Method of maintenance of cost records**: decide beforehand whether cost and financial accounts will be inter-locked into a single integral system or kept separately and reconciled through control accounts, since the financial accounts are on software the plant does not use.
8. **Statutory compliances and audit**: cost records must comply with statutory requirements and applicable cost accounting standards.
9. **Information attributes**: the information must be complete, accurate, timely and relevant, for an effective MIS.

Worked example

The cold-chain system nobody liked

Tapri Cold Chain Pvt Ltd installed a costing system three years ago. Ritu's review finds:

1. The monthly cost report runs to 46 pages, with tyre costs analysed by tread pattern; depot managers do not read it.
2. Diesel issued at depots is entered from memory at the end of the week.
3. The Indore depot treats tyre replacement as maintenance; the Nagpur depot treats it as running cost, so the two depots' cost per km cannot be compared.
4. Cost data is re-typed into the financial accounting software every month, and the two never agree.
5. When a new reporting requirement for e-way bills arrived, the system could not add the field for five months.
6. The managing director decides routes by instinct and ignores the reports: "My gut is cheaper than your software."

Required: Identify, for each finding, the essential feature of a good cost accounting system that is missing, and suggest a remedy.

MODEL ANSWER

Finding

1. 46 pages, tyre treads

Essential feature missing

Informative and simple: utility sacrificed by unnecessary details

Remedy

A short, tailor-made report of what each depot manager needs

Finding

2. Diesel from memory

Essential feature missing

Accurate and authentic data

Remedy

Record issues when made, on authorised documents

Finding

3. Tyres treated differently

Essential feature missing

Uniformity and consistency in classification and treatment

Remedy

One classification for all depots, so results are comparable

Finding

4. Re-typing, no agreement

Essential feature missing

Integrated and inclusive

Remedy

Integrate with financial accounting (or reconcile through control accounts)

Finding

5. No new field for five months

Essential feature missing

Flexible and adaptive

Remedy

A system that can be amended for regulatory and reporting changes

Finding

6. Gut over reports

Essential feature missing

Trust on the system

Remedy

Management's active role in developing the system and a conviction to use its information

Worked example

Malwa Kulhad Works: the people problem

Tapri has acquired Malwa Kulhad Works, a family-run pottery. Ritu starts installing a costing system. The two founder-directors call it "Indore's spy in our factory"; the three accounts clerks fear the new system will replace them; the kiln supervisors refuse to fill in daily job cards ("we make pots, not paperwork"); and no one at the plant has worked in cost accounting before.

Required: State the practical difficulties Ritu faces in installing the costing system, and how she should overcome them.

MODEL ANSWER

Practical difficulties (ICAI, RTP September 2026 Q16(a)):

1. **Lack of top management support**: the founder-directors see the system as interference in their work and believe it is meant for keeping a check on their activities.
2. **Resistance from the cost accounting staff**: the clerks fear losing their jobs and importance after the new system is implemented.
3. **Non-co-operation from user departments**: the kiln supervisors will not provide data, as it adds to their responsibilities and their paperwork.
4. **Shortage of trained staff**: no one at the plant has cost accounting experience.

Steps to overcome them:

- Sell the idea to the founder-directors and convince them of the utility of the system (for example, the cost per kulhad and its price to Tapri's cafés).
- Overcome the clerks' resistance and the supervisors' non-co-operation by a behavioural approach, dealing with each group effectively.
- Give proper training to the staff at each level.

- Hold regular meetings with the cost staff, the kiln supervisors and the directors to clarify their doubts and misgivings.

Exam angle

In the exam

Where installation questions appeared

In the ICAI documents read for this lesson (new-scheme papers, RTPs and MTPs, May 2024 to September 2026, and some old-scheme ones):

- **Essential features of a good cost accounting system, 5 marks**: MTP Series II for May 2024 Q6(b) and MTP Series I for September 2026 Q6(b); in the old scheme, May 2022 Q6(a), MTP November 2019 Q6(a) and MTP Series II for November 2022 Q6(a). The chapter asks it too (SM 1.45, Q4).
- **Factors before installing, 5 marks**: "Enumerate the factors which are to be considered before installing a system of cost accounting in a manufacturing organization" (MTP Series II for May 2025 Q6(a)); "As a consultant ... how would you go about assessing the critical factors for designing and implementing a cost accounting system?" (MTP Series II for January 2025 Q6(a)). Both answers are the nine factors, in the Study Material's order, with its examples (the oil refinery, the lender's stock statement).

- **Practical difficulties and how to overcome them, 5 marks**: MTP Series II for September 2024 Q6(a); RTP September 2026 Q16(a).
- **Limitations of cost accounting**: RTP December 2021 (old scheme) Q15(b), answered with the three limitations above.
- No question in the papers read asked for the department's functions alone.

In the exam

Two lists for "essential features"

The suggested answers of the two newest MTPs (May 2024 Series II, September 2026 Series I) list: informative and simple; accuracy; support from management and subordinates; cost-benefit ("the cost of installing and operating the system should justify the results"); procedure (a carefully phased programme, prepared using network analysis); trust. The Study Material and the old-scheme May 2022 answer list the six in this lesson. Write the Study Material's six; support from management and cost-benefit may be added as further points if time allows.

Trick

The nine factors as four questions

Why (objective) → where (nature of business, organisational hierarchy) → what and how (product, production process) → fitting in (information synchronisation, maintenance of cost records, statutory compliances and audit, information attributes). Four questions, nine factors, in ICAI's order (a) to (i). A memory aid only: the Study Material does not group them.

Answer frame

"Discuss the essential features of a good cost accounting system" (5 marks)

Six headings, each with one line of ICAI's explanation: informative and simple (tailor-made, no unnecessary details); accurate and authentic; uniformity and consistency (benchmarking, comparability); integrated and inclusive (financial accounting, taxation, statistics, operational research); flexible and adaptive (technological, reporting, regulatory changes); trust on the system (active role of management).

"Factors to be considered before installing a costing system" (5 marks)

Factor	The line that earns the mark
(a) Objective	Fixing prices or establishing cost control
(b) Nature of business or industry	Methods follow the industry's needs (refinery: process-wise accounts)
(c) Organisational hierarchy	Top: corporate strategy; strategic: pricing, diversification; operational: standard quantities, idle time
(d) Knowing the product	By-products; perishables: marginal costing
(e) Knowing the production process	Scientific apportionment needs knowledge of the process
(f) Information synchronisation	Other departments' needs (stock statement to the bank, tax returns)
(g) Method of maintenance of cost records	Inter-locking or reconciliation through control accounts
(h) Statutory compliances and audit	Statutory records, cost accounting standards
(i) Information attributes	Complete, accurate, timely, relevant

In a case ("as a consultant"), apply each factor to the facts in one more line.

Three answers to "practical difficulties and how to overcome them"

Loses marks

"Staff do not like change, so train them."

One vague difficulty, one remedy.

Passes

The four difficulties named, with no remedies, or remedies with no difficulties.

Half the question answered.

Full marks

The four difficulties, each with its reason, then the four remedies: sell the idea, behavioural approach, training, regular meetings.

Common mistakes

⚠ **Common mistake**

Hall of Shame: "Install it first, study the business later"

Writing the essential features when the question asks for the factors to be considered *before installing*, or the other way round. **Fix:** factors (nine) come before installation: objective, nature of business, hierarchy, product, process, synchronisation, cost records, statutory compliances and audit, information attributes. Essential features (six) are qualities of the finished system.

⚠ **Common mistake**

Marginal costing for everything

Saying the nature of the product decides "marginal or absorption costing" in general. **Fix:** ICAI's point is specific: products with by-products need by-product accounting; for perishable or short shelf-life products, marginal costing is appropriate, to know the contribution and minimum selling price.

⚠ **Common mistake**

Limitations that are not ICAI's

Listing "cost accounting is inaccurate" or "it is not compulsory" as limitations. **Fix:** the Study Material's three are expensive, requirement of reconciliation, and duplication of work (SM 1.24).

Practice

Check yourself

1 Case scenario. Tapri's new mango-lassi line has a shelf life of two days. Ritu must recommend how its product cost information should be built so that Kesar knows the lowest price at which unsold stock can be cleared each evening. Which factor, and which treatment, does the Study Material point to?

A Organisational hierarchy; absorption costing

B Knowing the product; marginal costing, to know the contribution and the minimum price

C Information synchronisation; a monthly stock statement to the bank

D Method of maintenance of cost records; integrated accounts

2 The cost department of Tapri's premix plant reports packing cost under "consumables" in one month and under "packing materials" the next, so the months cannot be compared. Which essential feature of a good cost accounting system is missing?

- **A** Informative and simple
- **B** Flexible and adaptive
- **C** Uniformity and consistency
- **D** Trust on the system

3 Which of the following does the Study Material list as limitations of cost accounting? Select all that apply.

- **A** It is expensive
- **B** Reconciliation with financial accounts is required
- **C** Work is duplicated, as two sets of accounts are kept
- **D** Its reports are prepared only once a year

More than one answer may be right.

4 Which of these is the main function of cost and management accounting, according to the Study Material?

- **A** Collection and accumulation of cost for each element
- **B** Maintaining cost records required by statute
- **C** Preparing the annual financial statements
- **D** Provision of relevant information to management for decision making, through an MIS

5 Which of these is NOT one of the practical difficulties in installing a costing system that ICAI's RTP and MTP answers list?

- **A** Lack of top management support
- **B** Non-co-operation from user departments
- **C** Shortage of trained staff
- **D** The system is expensive, because analysis, allocation and absorption of overheads need considerable additional work

Answers: the answer key at the end of this lesson.

Recap

- Nine factors to study before installing: objective, nature of business or industry, organisational hierarchy, knowing the product, knowing the production process, information synchronisation, method of maintenance of cost records, statutory compliances and audit, and information attributes.
- Six essential features of a good system: informative and simple, accurate and authentic, uniformity and consistency, integrated and inclusive, flexible and adaptive, and trust on the system.
- Three limitations of cost accounting: it is expensive, it requires reconciliation with the financial accounts, and it duplicates work. So the first question is whether the system will be profitable to have.
- Five functions: collect cost by element; assign to cost objects; set budgets and standards, compare, report; information for decisions (MIS, the main function); evaluate responsibility centres.
- Perishables: marginal costing (contribution, minimum price). By-products: a system that accounts for them.
- Practical difficulties (ICAI's RTP and MTP answers): no top management support, staff resistance, user departments' non-co-operation, shortage of trained staff; remedies: sell the idea, behavioural approach, training, regular meetings.
- A system nobody trusts is a 46-page report nobody reads.

Answer key · Check yourself

1 **B** **Knowing the product; marginal costing, to know the contribution and the minimum price**

For perishable or short shelf-life products, marginal costing is appropriate to know the contribution and the minimum price at which the products could be sold (factor (d), knowing the product).

2 **C** **Uniformity and consistency**

Uniformity and consistency in classification, treatment and reporting of cost data is required for benchmarking and comparability of results.

3 **A** **It is expensive**; **B** **Reconciliation with financial accounts is required**; **C** **Work is duplicated, as two sets of accounts are kept**

The three limitations are expensive, requirement of reconciliation, and duplication of work. Cost reports are prepared as and when required, not once a year.

4 **D** **Provision of relevant information to management for decision making, through an MIS**

"The main function of Cost and Management Accounting is provision of relevant information to the management for decision making" (SM 1.11), through a management information system.

5 **D** **The system is expensive, because analysis, allocation and absorption of overheads need considerable additional work**

Being expensive is the first of the Study Material's three limitations of cost accounting (SM 1.24), not a practical difficulty of installation. ICAI's four difficulties are lack of top management support, resistance from cost accounting departmental staff, non-co-operation from user departments and shortage of trained staff.

1.3 Cost, financial and management accounting, and financial management

How cost accounting differs from financial accounting (seven bases) and from management accounting (six bases), and how all three relate to financial management.

In plain words

Two honest people, two different numbers

At Tapri's board meeting, Dev and Ritu present the delivery segment for 2025-26.

> **Dev:** The delivery segment earned ₹31.20 lakh. Audited, in the annual report, page 112.
>
> **Ritu:** And the Nagpur zone lost money on every order. Per unit, please: the annual report cannot show you that.
>
> **Kesar:** So which of you is lying?
>
> **Dev:** Neither. The present value of this argument is zero.

Both are right, because they keep different accounts for different jobs. **Financial accounting** tells shareholders, lenders and the government how the whole company (or a whole segment) performed, from historical records, usually once a year, in a set format. **Cost accounting** finds the cost of each **cost object**, a café, a cup, a delivery

run, for Tapri's own managers, using actual costs and costs fixed in advance (standards), whenever a report is needed and in whatever layout helps.

Management accounting is wider still. It takes cost accounts, financial accounts, statistics and other tools, adds qualitative information (how strong is BrewBucks' brand?), and looks forward, to help managers plan and decide. And under all three sits **financial management**: the decisions about where the money comes from, where it is invested and how much goes back to shareholders. Cost and management accounting is an application of it, for decision making.

Think of it like

The health check-up, the fitness tracker and the coach

Financial accounting is your annual health check-up: one report on the whole body, in a standard format, for anyone who needs to see it. Cost accounting is the fitness tracker on each arm and leg: it shows which knee is weak, every day, in whatever screen you choose. Management accounting is the coach who reads both, adds your diet and your rival's training plan, and draws up next season. Financial management is deciding whether the savings go into a gym membership or a bicycle.

Key idea

Financial accounting reports the financial performance of the entity, as a whole or by segment, to outsiders, from historical data, usually yearly, in a set format. Cost accounting ascertains the cost of each cost object for cost control and decisions, from historical and predetermined costs, as and when required. Management accounting is wider and forward-looking; all three serve financial management's decisions.

How it works

"Cost and Management Accounting as a discipline is interrelated and dependent on other disciplines of accounting" (SM 1.8). The Study Material compares it with three of them.

1. Cost accounting and management accounting (SM 1.8–1.9)

The two terms are often used synonymously, but there are a few differences. Management accounting is an open-ended discipline which enables managers to take informed decisions, and it takes inputs from cost accounts, financial accounts, statistical and operation management tools. In practice, cost accounting is concerned with accumulating and allocating costs to different cost objects, while management accounting is concerned with providing information to internal users for decision making (SM 1.11).

Difference between Cost Accounting and Management Accounting (SM 1.8–1.9)

Basis

(i) Nature

Cost Accounting

Records the quantitative aspect only

Management Accounting

Records both qualitative and quantitative aspects

Basis

(ii) Objective

Cost Accounting

Records the cost of producing a product and providing a service

Management Accounting

Provides information to management for planning and co-ordination

Basis

(iii) Area

Cost Accounting

Deals only with cost ascertainment

Management Accounting

Wider in scope: includes financial accounting, budgeting, taxation, planning etc.

Basis

(iv) Recording of data

Cost Accounting

Uses both past and present figures

Management Accounting

Focused on the projection of figures for the future

Basis

(v) Development

Cost Accounting

Related to the industrial revolution

Management Accounting

Related to the needs of the modern business world

Basis

(vi) Rules and regulation

Cost Accounting

Follows certain principles and procedures for recording the costs of different products

Management Accounting

Does not follow any specific rules and regulations

2. Cost accounting and financial accounting (SM 1.9–1.10)

"Cost accounting accumulates and ascertains costs for goods sold and inventories. It provides inputs to record costs in financial accounting system" (SM 1.9). So the cost accounts feed the financial accounts: the closing inventory and the cost of goods sold in Tapri's annual report start in Ritu's cost records.

Difference between Financial Accounting and Cost Accounting (SM 1.9–1.10)

Basis

(i) Objective

Financial Accounting

Provides information about the financial performance of an entity

Cost Accounting

Ascertainment of cost for the purpose of cost control and decision making

Basis

(ii) Nature

Financial Accounting

Classifies, records, presents and interprets transactions in monetary terms

Cost Accounting

Classifies, records, presents and interprets costs in a significant manner

Basis

(iii) Recording of data

Financial Accounting

Records historical data

Cost Accounting

Uses both historical and pre-determined costs

Basis

(iv) Users of information

Financial Accounting

Shareholders, creditors, financial analysts, government and its agencies etc.

Cost Accounting

Generally internal management; sometimes regulatory authorities also

Basis	Financial Accounting	Cost Accounting
(v) Analysis of cost and profit	Shows profit or loss of the organisation segment-wise or as a whole	Gives cost details for each cost object: product, process, job, operation, contract etc.

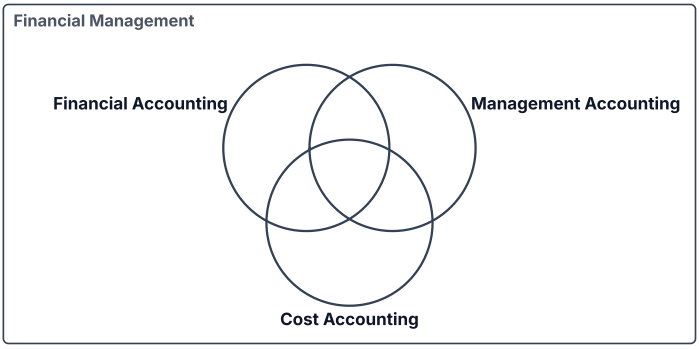
Basis	Financial Accounting	Cost Accounting
(vi) Time period	Financial statements prepared usually for a year	Reports and statements prepared as and when required

Basis	Financial Accounting	Cost Accounting
(vii) Presentation of information	A set format is used	In general, no set format is followed

3. Cost and management accounting and financial management (SM 1.10)

ICAI's sentence is short: "Cost and Management Accounting is an application of Financial Management for decision making purposes" (SM 1.10). Its diagram shows the relationship among all four.

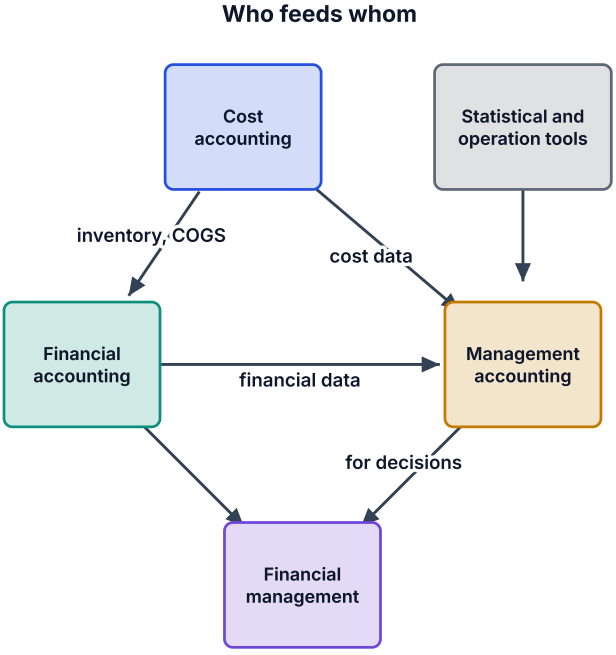
Cost, financial and management accounting within financial management



The three accounting disciplines overlap; financial management is the ground they all stand on. In the Study Material's figure (SM 1.10) the three circles sit on a broad arrow labelled Financial Management; here it is the box around them.

Read it with the two tables: the circles overlap because cost accounting feeds the financial accounts (inventories,

cost of goods sold) and management accounting draws on both. Where the information goes:



COGS: cost of goods sold. Cost accounting provides inputs to the financial accounts; management accounting takes inputs from cost accounts, financial accounts, statistical and operation management tools (SM 1.8–1.9); accounting is a necessary input into financial management (FM Study Material 1.21).

Worked examples

Worked example **The cafés segment: one profit, three cafés**

Tapri's financial statements for 2025-26 report a profit of ₹26,40,000 for the cafés segment (three cafés). Ritu's cost records for the same year show:

Café	Revenue (₹)	Total cost (₹)
Indore	96,00,000	72,00,000
Pune	72,00,000	61,20,000
Bhopal	48,00,000	56,40,000

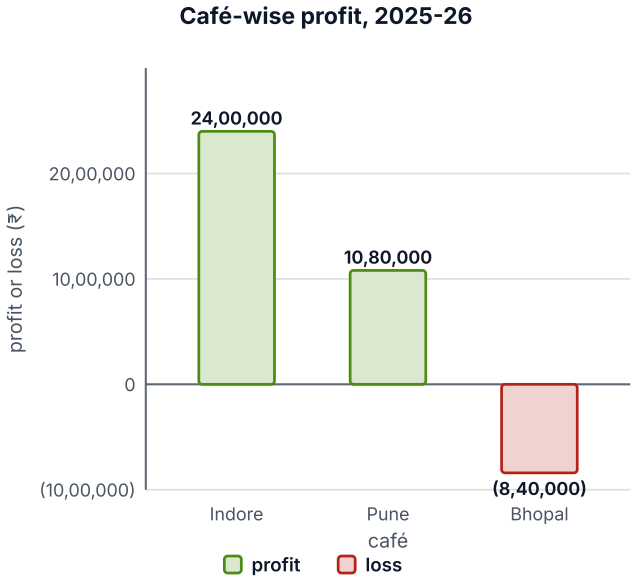
Ritu's records also give the cost per cup of each café every month, against a standard cost set at the start of the year.

Required:

(i) Calculate the profit or loss of each café, and check that the cost records agree with the segment profit.

(ii) Using the bases on which ICAI distinguishes financial accounting from cost accounting, explain why the board needs both sets of information.

THE WORKING



Green: profit; rose: loss. The segment's ₹26,40,000 hides a loss of ₹8,40,000 at Bhopal.

(i) Café-wise profit

Café	Revenue (₹)	Total cost (₹)	Profit / (Loss) (₹)
Indore	96,00,000	72,00,000	24,00,000
Pune	72,00,000	61,20,000	10,80,000
Bhopal	48,00,000	56,40,000	(8,40,000)
Total	**2,16,00,000**	**1,89,60,000**	**26,40,000**

The total agrees with the segment profit of ₹26,40,000 in the financial statements. Here the

two agree; in general the profit shown by the cost accounts and by the financial accounts may differ (purely financial items, notional costs, different treatment, stock valuation), which is why they are reconciled (reconciling cost and financial profit).

(ii) Why the board needs both

Basis	Financial accounting (the segment figure)	Cost accounting (Ritu's records)
Objective	Financial performance of the entity: the segment earned ₹26,40,000	Cost for control and decisions: why Bhopal loses, what to do
Recording of data	Historical data for the year	Historical and pre-determined (standard) costs per cup
Users	Shareholders, creditors, analysts, government	Internal management: the board and café managers

Basis

Analysis of cost and profit

Financial accounting (the segment figure)

Profit segment-wise or as a whole

Cost accounting (Ritu's records)

Cost details for each cost object: each café, each cup

Basis

Time period

Financial accounting (the segment figure)

Usually a year

Cost accounting (Ritu's records)

As and when required: monthly

Basis

Presentation

Financial accounting (the segment figure)

Set format

Cost accounting (Ritu's records)

No set format: café-wise, cup-wise

The financial statements tell the shareholders that the segment is profitable; only the cost records tell the board that Bhopal needs action. Neither replaces the other.

Worked example

Two requests, two disciplines

In the same week Ritu receives two requests:

1. From the Pune café manager: the cost per kulhad cup for September, built up from milk, tea, kulhad, gas and staff, and compared with its standard.
2. From Ananya Rao, Tapri's head of strategy: a five-year plan for entering twenty new cities, with projected sales, budgets and tax implications, and a note on how strongly customers in each city prefer BrewBucks' brand.

Required: With reference to the two requests, explain the differences between cost accounting and management accounting.

MODEL ANSWER

Though cost accounting and management accounting are used synonymously, there are a few differences (SM 1.8). The first request is cost accounting; the second is management accounting.

Basis

(i) Nature

Cost accounting (request 1)
Records the quantitative aspect only: rupees per cup

Management accounting (request 2)
Records both qualitative and quantitative aspects: projected sales and customer preference for BrewBucks' brand

Basis

(ii) Objective

Cost accounting (request 1)

Records the cost of providing a service: one cup

Management accounting (request 2)

Provides information to management for planning and co-ordination: the expansion plan

Basis

(iii) Area

Cost accounting (request 1)

Deals only with cost ascertainment

Management accounting (request 2)

Wider: budgeting, taxation and planning are all in the request

Basis

(iv) Recording of data

Cost accounting (request 1)

Uses past and present figures: September's actual cost

Management accounting (request 2)

Focused on projecting figures for the future: five years ahead

Basis

(v) Development

Cost accounting (request 1)

Related to the industrial revolution

Management accounting (request 2)

Related to the needs of the modern business world

Basis	Cost accounting (request 1)	Management accounting (request 2)
(vi) Rules and regulation	Follows certain principles and procedures for recording costs	Follows no specific rules and regulations

Management accounting will itself use the cost accounting figures, along with financial accounts and statistical tools, as its inputs.

Exam angle

In the exam

How ICAI examines these differences

In the ICAI documents read for this lesson (new-scheme papers, RTPs and MTPs, May 2024 to September 2026, and old-scheme papers back to 2019):

- **Cost accounting vs management accounting** was asked in six: RTP September 2025 Q18(a) ("Though Cost Accounting and Management Accounting are used synonymously, but there are a few differences. Enumerate any five differences"); and in the old scheme, November 2020 Q6(a) (5 marks), RTP May 2020 Q13(a), MTP December 2021 Series I Q6(b), RTP November 2022 Q15(c) and MTP May 2023 Series I Q6(b). The suggested answers give the

Study Material's six-row table, in the same order and nearly word for word.

- **Financial accounting vs cost accounting** and the link with **financial management** were not asked directly in the documents read; financial and cost accounts appear together in the reconciliation numericals instead. The tables in this lesson are the Study Material's own, so they are examinable as they stand.

Trick

Two sentences, two tables

Cost vs management accounting, six bases in ICAI's order: "**N**o **O**ther **A**ccountant **R**eads **D**ev's **R**ules": Nature, Objective, Area, Recording of data, Development, Rules and regulation.

Financial vs cost accounting, seven bases in ICAI's order: "**O**nly **N**ew **R**ecruits **U**se **A**ny **T**emplate **P**ossible": Objective, Nature, Recording of data, Users of information, Analysis of cost and profit, Time period, Presentation of information. (Cost reports follow no set format: any template is possible.)

Answer frame

"Distinguish between cost accounting and management accounting" (4–5 marks)

A three-column table, Basis | Cost Accounting | Management Accounting, with the six bases in ICAI's order: nature, objective, area, recording of data, development, rules and regulation. "Any five" means the first five done well; a sixth row costs little and covers a weak one.

"Distinguish between financial accounting and cost accounting"

The same table with seven bases: objective, nature, recording of data, users of information, analysis of cost and profit, time period, presentation of information. Write the key words examiners look for: "financial performance of an entity" against "cost control and decision making"; "historical" against "historical and pre-determined"; "segment-wise or as a whole" against "each cost object"; "usually for a year" against "as and when required"; "set format" against "no set formats".

"Explain the relationship of cost and management accounting with other disciplines"

1. Cost accounting provides inputs to financial accounting: cost of goods sold and inventories.
2. Management accounting takes inputs from cost accounts, financial accounts, statistical and operation management tools.

3. Cost and management accounting is an application of financial management for decision making purposes.
4. A labelled sketch: three overlapping circles (financial, management and cost accounting) on financial management.

Common mistakes

⚠ **Common mistake**

Hall of Shame: "Management accounting is cost accounting with a bigger salary"

Treating the two as the same and writing one table of cost vs financial accounting for a cost vs management accounting question. **Fix:** management accounting records qualitative as well as quantitative information, is wider in area (financial accounting, budgeting, taxation, planning), projects the future and follows no specific rules. Six bases, and none of them is "users" or "time period".

⚠ **Common mistake**

Predetermined costs in the financial accounts

Writing that financial accounting uses predetermined costs, or that cost accounting records only historical data. **Fix:** financial accounting records historical data; cost accounting uses both historical and pre-determined costs (standards and budgets are part of it).

⚠ **Common mistake**

The yearly cost report in a statutory format

Saying cost reports are prepared annually in a prescribed format for shareholders. **Fix:** cost reports and statements are prepared as and when required, in general with no set format, generally for internal management (sometimes for regulatory authorities also).

Practice

Check yourself

1 Case scenario. Ritu prepares a weekly café report in a layout she designed herself, comparing each café's actual cost per cup with a standard cost set before the year began, for Tapri's area managers. Which of the following correctly identifies it?

A Financial accounting: it reports financial performance in a set format

B Cost accounting: it is prepared as and when required, follows no set format and uses pre-determined costs

C Management accounting: it records only qualitative information

D Financial management: it is a financing decision

2 Which of the following statements about management accounting is correct, according to the Study Material?

- **A** It records only the quantitative aspect
- **B** It follows specific rules and regulations for recording costs of different products
- **C** It is wider in scope, as it includes financial accounting, budgeting, taxation, planning etc.
- **D** Its development is related to the industrial revolution

3 Tapri's financial statements show a profit of ₹18,00,000 for its kiosks segment of three kiosks. The cost records show a profit of ₹12,60,000 at the station kiosk and ₹9,40,000 at the airport kiosk; the bus-stand kiosk's revenue was ₹40,00,000. If the cost records agree with the segment profit, what was the bus-stand kiosk's total cost in rupees?

Answer: _____

4 Which of these are features of financial accounting, as distinct from cost accounting? Select all that apply.

- **A** It records historical data
- **B** A set format is used for presenting information
- **C** Statements are prepared usually for a year
- **D** It uses both historical and pre-determined costs
- **E** Its users include shareholders, creditors, financial analysts and government agencies

More than one answer may be right.

5 Complete ICAI's sentence: "Cost and Management Accounting is an application of ... for decision making purposes."

- **A** Financial Accounting
- **B** Financial Management
- **C** Statistical tools
- **D** Cost Audit

Answers: the answer key at the end of this lesson.

Recap

- Cost accounting provides inputs to financial accounting (costs of goods sold and inventories); management accounting takes inputs from cost accounts, financial accounts, statistical and operation management tools.
- Cost vs management accounting, six bases: nature (quantitative / both), objective (cost of product or service / planning and co-ordination), area (cost ascertainment / wider), recording (past and present / future), development (industrial revolution / modern business), rules (follows / none).
- Financial vs cost accounting, seven bases: objective, nature, recording of data (historical / historical and pre-determined), users (outsiders / internal management), analysis (segment or whole / each cost object), time period (usually a year / as and when required), presentation (set format / no set format).
- Cost and management accounting is an application of financial management for decision making purposes.
- A segment profit can hide a loss-making cost object: only the cost records show it. (The cafés' ₹26,40,000 hid Bhopal's loss of ₹8,40,000.)

Answer key · Check yourself

1 **B** **Cost accounting: it is prepared as and when required, follows no set format and uses pre-determined costs**

Weekly (as and when required), its own layout (no set format), standard costs (pre-determined) and internal users: all cost accounting features. Financial accounting uses historical data in a set format, usually yearly.

2 **C** **It is wider in scope, as it includes financial accounting, budgeting, taxation, planning etc.**

The other three describe cost accounting. Management accounting records qualitative and quantitative aspects, follows no specific rules, and its development is related to the needs of the modern business world.

3 **4400000**

Bus-stand result = ₹18,00,000 − ₹12,60,000 − ₹9,40,000 = a loss of ₹4,00,000. Cost = revenue + loss = ₹40,00,000 + ₹4,00,000 = ₹44,00,000. The segment profit hides the loss; the cost records show it.

4 **A** **It records historical data**; **B** **A set format is used for presenting information**; **C** **Statements are prepared usually for a year**; **E** **Its users include shareholders, creditors, financial analysts and government agencies**

Using both historical and pre-determined costs is a feature of cost accounting.

5 **B** **Financial Management**

SM 1.10: cost and management accounting is an application of financial management for decision making purposes; in ICAI's diagram financial management lies beneath all three accounting disciplines.

1.4 Cost units, cost centres, cost objects and responsibility centres

The cost object (what is costed), the cost unit (per what), the cost driver (what causes the cost), cost centres, and the four responsibility centres with how each is judged.

In plain words

"How much does the Pune café cost?"

Kesar asks it in the corridor. Ritu answers with three questions.

> **Kesar:** Simple question, Ritu. How much does the Pune café cost?
>
> **Ritu:** The café, or one cup from it? Per month or per cup? And who in Pune answers for which part of it?
>
> **Kesar:** You make everything sound like an exam.
>
> **Ritu:** Per unit, please. Then it never is one.

Ritu's three questions are this lesson's words.

- The thing whose cost you want, the café or a cup, is the **cost object**: anything for which a separate measurement of cost is required.
- The unit in which the cost is expressed, "per cup", "per tonne-km", "per machine hour", is the **cost unit**.
- What makes the cost go up, the number of cups, the number of delivery orders, is the **cost driver**.

- The person or department that answers for a part of the business is a **responsibility centre**. If it answers only for costs (the café kitchen), it is a **cost centre**; only for revenue (the billing counter), a **revenue centre**; for both, and so for profit (the café as a whole), a **profit centre**; and if it can also decide on capital investment (a whole subsidiary), an **investment centre**, judged on its return on investment.

Think of it like

The family wedding

A wedding is a cost object (so is the sangeet night on its own). The caterer quotes per plate: the cost unit. The number of guests drives almost every cost: the cost driver. Mama-ji runs the catering committee and must stay within its budget: a cost centre. The cousin collecting the shagun envelopes answers only for what comes in: a revenue centre. The tent house that rents out chairs and pays its own staff answers for both: a profit centre. And Papa, who can decide to buy the banquet hall instead of renting it, is the investment centre.

Key idea

The cost object is what you cost, the cost unit is what you cost it per, and the cost driver is what makes the cost rise. Responsibility centres say who answers for what: costs (cost centre), revenue (revenue centre), both (profit centre), or profit plus capital investment (investment centre).

How it works

Cost object: what is being costed (SM 1.19–1.20)

Definition

Cost object (SM 1.19)

Cost object is anything for which a separate measurement of cost is required. Cost object may be a product, a service, a project, a customer, a brand category, an activity, a department or a programme etc.

ICAI's examples (SM 1.20):

Cost object	Examples
Product	Smart phone, tablet computer, SUV car, book
Service	An airline flight from Delhi to Mumbai, a concurrent audit assignment, a utility bill payment facility
Project	Metro rail project, road projects
Activity	Quality inspection of materials, placing of orders
Process	Refinement of crudes in oil refineries, melting of billets or ingots in rolling mills
Department	Production department, finance and accounts, safety

The cost object remains in nucleus of cost classification and analysis of the cost behaviour: whether a cost is direct or indirect, fixed or variable, all depends on the cost object (SM 1.20). The café manager's salary is a direct cost of the Pune café, but an indirect cost of each cup. Classification is in elements and classification of cost.

Cost unit: per what (SM 1.20–1.21)

Definition

Cost unit (SM 1.20)

It is a unit of product, service or time (or combination of these) in relation to which costs may be ascertained or expressed.

Costs may be found per ton of steel, per ton-kilometre of a transport service or per machine hour. Sometimes a single order or contract is the cost unit; so is a **batch**, a group of identical items that keeps its identity through one or more stages of production. Cost units are usually units of physical measurement: number, weight, area, volume, length, time and value (SM 1.20).

Formula

Cost per unit

$$\text{Cost per cost unit} = \frac{\text{Total cost for the period}}{\text{Cost units in the period}}$$

The total cost is the cost of the cost object, the thing being costed (the fleet); the cost unit is the denominator (tonne-km).

ICAI's two lists (SM 1.20–1.21), worth knowing by heart because "state the cost unit" questions draw on them:

Industry or product	Cost unit basis
Automobile	Number
Cement	Ton / per bag etc.
Chemicals	Litre, gallon, kilogram, ton etc.
Power	Kilowatt-hour (kWh)
Steel	Ton
Transport	Passenger-kilometre
Gas	Cubic feet

CIMA industry sector	Cost unit
Brewing	Barrel
Brick-making	1,000 bricks
Coal mining	Tonne / ton
Electricity	Kilowatt-hour (kWh)
Engineering	Contract, job
Oil	Barrel, tonne, litre
Hotel / Catering	Room / meal
Professional services	Chargeable hour, job, contract
Education	Course, enrolled student, successful student
Hospitals	Patient day

CIMA activity	Cost unit
Credit control	Accounts maintained
Selling	Customer call, value of sales, orders taken
Materials storage / handling	Requisition unit issued / received, material movement, value issued / received
Personnel administration	Personnel record

Services often need a unit made of two measures (tonne-km, passenger-km, patient-day): a **composite cost unit**, built in service costing and composite cost units.

Cost driver: what causes the cost (SM 1.21–1.22)

Definition

Cost driver (SM 1.21–1.22)

A cost driver is a factor or variable which affects the level of cost. Generally, it is an activity which is responsible for cost incurrence; the level of activity or volume of production is an example. CIMA: "Factor influencing the level of cost", often used in Activity Based Costing to denote the factor which links activity resource consumption to product outputs; for example, the number of purchase orders would be a cost driver for procurement cost.

Examples: number of machine set-ups, number of purchase orders, hours spent on product inspection, number of tests performed (SM 1.22). A cost unit measures output; a cost driver explains why cost rises. How drivers charge overheads to products is in activities, cost pools and cost drivers.

Cost centres

A cost centre is where costs are collected and someone controls them. The Study Material uses the idea in two places:

1. **For collecting costs** (SM 4.12): a department may be sub-divided into cost centres for better cost control and performance evaluation. The principal object is to collect data on similar activities more conveniently; when costs are collected by cost centre, several items can be ascertained definitely and the element of estimation is reduced considerably. There are two main types, **machine** or **personnel**, depending on whether the work at the centre is done by machine or by people, and cost centres are sometimes given a **code number** for recording. Production and service departments, and how overheads reach them, are in classifying overheads, allocation and apportionment.
2. **As a responsibility centre** (SM 1.22–1.23): the responsibility centre held accountable for incurrence of costs which are under its control, of two types, standard and discretionary (below).

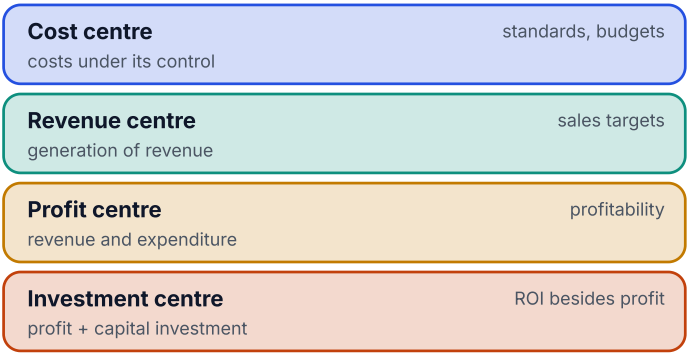
Responsibility centres (SM 1.22–1.23)

Definition

Responsibility centres (SM 1.22)

As the organisation grows, management delegates its responsibility and authority to various departments or persons, to have better control. These departments or persons are known as responsibility centres and are held responsible for performance in terms of expenditure, revenue, profitability and return on investment. Their performance is measured against set standards (input-output ratio, budgets etc.) and evaluated against organisational goals and performance targets.

The four responsibility centres (SM 1.22–1.23)

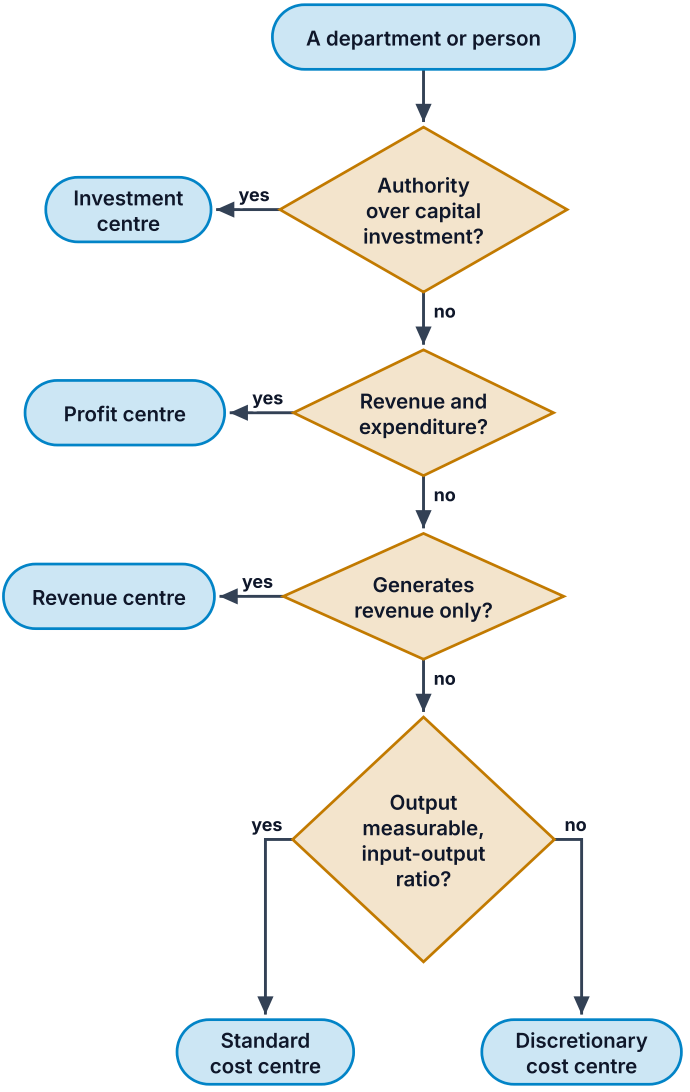


Middle: what each centre is accountable for; right: how its performance is measured. A profit centre answers for both costs and revenue; an investment centre adds capital investment decisions.

1. **Cost centres**: held accountable for incurrence of costs which are under their control; performance is measured against pre-determined standards or budgets. Two types:
    - (a) **Standard cost centre**: output is measurable and the input required for it can be specified. From a well-established study, standard units of input per unit of output are set; actual input cost is compared with standard cost, and any variance is analysed into controllable and uncontrollable cost. The manager must comply with the standard and is held responsible for adverse variances. The input-output ratio is clearly identifiable.
    - (b) **Discretionary cost centre**: output cannot be measured in financial terms, so the input-output ratio cannot be defined; the cost of inputs is compared with the budget allocated for the activity. Examples: research and development department, advertisement department.
2. **Revenue centres**: accountable for generation of revenue; a sales department, for example, is responsible for sales targets and revenue. Revenue centres do not control the expenditure they incur, though some selling expenditure such as sales commission is incurred by them.
3. **Profit centres**: responsible for both generation of revenue and incurrence of expenditure; since their managers are accountable for both, profitability is the basis for measuring performance. Example: decentralised branches of an organisation.

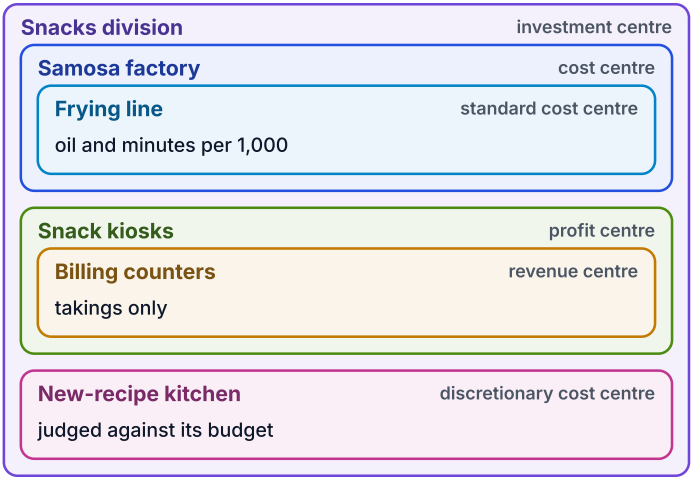
4. **Investment centres**: responsible for profitability and also have the authority to make capital investment decisions; performance is measured on the basis of **Return on Investment (ROI)** besides profit. Examples: Maharatna, Navratna and Miniratna companies of the Public Sector Undertakings of the Central Government.

Which responsibility centre is it?



Ask the widest question first. Anything that reaches the last question is accountable for costs only: a cost centre, standard or discretionary.

Responsibility centres inside one division



Smaller centres sit inside larger ones: the division answers for return on investment, each kiosk for its profit, each counter for its takings.

Worked examples

Worked example **Kesar's reorganisation memo**

After a reorganisation, Kesar wants six units of Tapri Tea Co. Ltd set up as responsibility centres. His memo describes each unit and ends with his own guesses.

1. **The batch-testing lab** at the premix plant tests every batch for moisture and strength. A work study fixed the reagents and technician minutes for each test; every month the actual cost of the tests done is compared with their standard cost, and the lab head explains any adverse variance.
2. **The flavour studio** experiments with new chai blends. In some years nothing it develops is launched; in one year a single blend became the best seller. Its spending is compared only with the amount sanctioned for it at the start of the year.
3. **The railway catering desk** used only to book orders from railway zones. Since April it also buys its own kulhads and ingredients, employs its own serving staff, sets its rates within limits fixed by head office, and is judged on the margin it earns.
4. **The franchise team** signs up franchise cafés against a target of 30 a year and is judged on the sign-ups and franchise fees it brings in. The franchisees bear the cost of setting up each café; the team's own expenses are its travel and a bonus for each sign-up.
5. **The Jaipur café cluster** has ₹6 crore of assets. Its head fixes menus and prices, hires staff and buys supplies, and is judged on the cluster's profit; any new café or equipment needs head-office approval.
6. **Tapri Cold Chain Pvt Ltd**, the subsidiary, has a board that approves its own capital budget for trucks and warehouses; it is judged on the return on the capital it employs as well as on its profit.

Kesar's guesses: "1 and 2 are both labs, so both discretionary cost centres. 3 sells, so a revenue centre. 4 earns fees and spends on travel, so a profit centre. 5 has ₹6 crore of assets, so an investment centre. 6 decides on its own trucks, so an investment centre."

Required: Identify the type of responsibility centre for each unit, with reasons, and state whether Kesar's guess is correct.

MODEL ANSWER

Responsibility centres are the departments or persons to whom management delegates responsibility and authority, held responsible for performance in terms of expenditure, revenue, profitability and return on investment (SM 1.22).

1. **Batch-testing lab: standard cost centre.** Its output (tests done) is measurable and the inputs required for it are specified from a work study; actual cost is compared with standard cost and the lab head is responsible for adverse variances. The input-output ratio is clearly identifiable. **Kesar's guess is wrong**: a cost centre is discretionary only when its output cannot be measured in financial terms; the word "lab" does not decide it.
2. **Flavour studio: discretionary cost centre.** Its output cannot be measured with certainty or co-related with the cost of its inputs, so no input-output ratio can be defined; the cost of inputs is compared with the budget allocated, as for

ICAI's research and development department. **Kesar's guess is correct.**

3. **Railway catering desk: profit centre.** Since April it is responsible both for generating revenue (orders, its own rates) and for incurring expenditure (kulhads, ingredients, staff), and profitability (its margin) is the basis of measuring its performance. **Kesar's guess is wrong**: a revenue centre does not have control over the expenditure it incurs (though some selling expenditure, such as commission, is incurred by it); this desk now controls its own costs. It has no authority over capital investment, so it is not an investment centre either.
4. **Franchise team: revenue centre.** It is accountable for generating revenue (sign-ups and franchise fees against a target) and does not answer for the cost of setting up the cafés; its own travel and sign-up bonus are selling expenditure of the kind revenue centres sometimes incur. **Kesar's guess is wrong**: incurring some selling expenditure does not make it a profit centre, because its performance is judged on revenue, not on profitability.
5. **Jaipur café cluster: profit centre.** Its head is accountable for both revenue (menus, prices) and expenditure (staff, supplies), and profitability measures its performance. **Kesar's guess is wrong**: the size of its assets does not decide the type. An investment centre needs the authority to make capital investment decisions, with

performance measured on return on investment, and here new cafés and equipment need head-office approval.

6. **Tapri Cold Chain Pvt Ltd: investment centre.** It is responsible for profitability and has the authority to make capital investment decisions (its own capital budget); its performance is measured on return on investment (ROI) besides profit. **Kesar's guess is correct.**

Worked example **Seven businesses, seven cost units**

Tapri runs the following activities:

(a) A cold-chain fleet carrying frozen samosas between cities.

(b) Tapri Academy, which trains baristas in eight-week courses.

(c) A rooftop solar plant supplying the premix plant with power.

(d) Café kitchens catering lunches for corporate clients.

(e) A pottery firing kulhads for the cafés.

(f) A credit control team looking after dealers' accounts.

(g) An engineering cell that builds and installs café counters, each to a separate order.

In March the fleet's total cost was ₹9,45,000 for 3,15,000 tonne-km. The Academy's cost for one course was ₹6,30,000; 140 students enrolled and 126 passed.

Required:

(i) Suggest a suitable cost unit for each activity.

(ii) Calculate the fleet's cost per tonne-km, and the Academy's cost per enrolled student and per successful student.

THE WORKING

(i) Cost units

Activity	Cost unit	Basis in the Study Material
(a) Cold-chain fleet	Tonne-km	Cost per ton-kilometre of a transport service
(b) Tapri Academy	Course, enrolled student, successful student	CIMA: education
(c) Rooftop solar plant	Kilowatt-hour (kWh)	Power / electricity
(d) Corporate catering	Meal	CIMA: hotel / catering

Activity	Cost unit	Basis in the Study Material
(e) Kulhad pottery	1,000 kulhads	Physical number, as brick-making uses 1,000 bricks
(f) Credit control	Accounts maintained	CIMA activity: credit control
(g) Counter-building cell	Job or contract	CIMA: engineering

(ii) Cost per unit

Cost object	Total cost (₹)	Cost units	Cost per unit (₹)
Fleet, March	9,45,000	3,15,000 tonne-km	3.00 per tonne-km
Academy course	6,30,000	140 enrolled students	4,500 per enrolled student
Academy course	6,30,000	126 successful students	5,000 per successful student

The same course has two honest costs per student: ₹4,500 for each one taught and ₹5,000 for each one who passed. The cost unit chosen decides which question the figure answers. How tonne-km

are counted (absolute or commercial) is in service costing.

Exam angle

In the exam

Where these terms appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **Responsibility centres, 5 marks each**: "Define Responsibility Centre and discuss the types" (September 2025 Q6(a)); "Explain the types of responsibility centres" (MTP Series I for September 2024 Q6(a), with standard and discretionary cost centres in the answer); "Explain responsibility centre and its types in brief" (MTP Series I for September 2025 Q6(a)).
- **Responsibility centres as a case, 5 marks**: identify the type for four departments, with reasons (MTP Series II for January 2026 Q3(b): standard cost centre, discretionary cost centre, revenue centre, investment centre); specify the type for five units (MTP Series I for May 2026 Q6(a): decentralised branches, a Central Government PSU, an e-commerce customer support department, a hotel's restaurant, an airline's reservation department).
- **Standard and discretionary cost centres**: RTP September 2024 Q19(d); in the old scheme, MTP

July 2021 Series I Q6(d) and MTP November 2023 Series II Q6(b).

- **Cost units, 4–5 marks**: "State cost unit of the following industry sector" for ten sectors (January 2025 Q6(b); the same list in MTP Series II for May 2026 Q6(a)); eight industries (MTP Series I for January 2026 Q6(c), an OR option); "Describe cost unit and give examples" (RTP September 2025 Q18(d)); in the old scheme, RTP November 2022 Q15(d) and MTP May 2023 Series I Q6(d).
- **Cost objects, 5 marks**: "Define cost objects and give examples of any four cost objects" (May 2023 Q6(a), old scheme), answered with the definition and ICAI's examples table.

In the exam

What the answers show

- The suggested answers on responsibility centres open with the Study Material's paragraph ("As the organization grows ...") before the four types. Write the definition first.
- In the case versions, each answer names the type and then gives the reason in the Study Material's words: "output is measurable", "input-output ratio cannot be defined", "authority to make capital investment decisions", "evaluated on ROI".
- The cost unit answers accept more than the Study Material's tables (January 2025: "per course, per student, per batch, per lecture" for education; "guest

days or room days, per item, per meal" for hotels), but the tables are the safe core.

- The chapter's own question "Discuss the different cost centres that an organization can have" is answered by "Please refer paragraph 11": the responsibility centres (SM 1.45–1.46).

Trick

C, R, P, I: what the manager answers for

Cost centre: costs. **R**evenue centre: revenue. **P**rofit centre: both, so profit. **I**nvestment centre: profit plus capital investment decisions, so ROI. Classify on what the manager is accountable for and has authority over, not on what the department happens to touch: a sales desk that pays commission is still a revenue centre.

Answer frame

"Define responsibility centre and discuss its types" (5 marks)

1. The definition: management delegates responsibility and authority to departments or persons, who are held responsible for performance in terms of expenditure, revenue, profitability and return on investment, measured against set standards.
2. **Cost centres**: accountable for costs under their control, measured against standards or budgets; (a) standard cost centre (measurable output, identifiable

input-output ratio, responsible for adverse variances); (b) discretionary cost centre (output not measurable in financial terms; inputs compared with the allocated budget; R&D, advertisement).

3. **Revenue centres**: accountable for revenue (sales department); may incur selling expenditure such as commission.
4. **Profit centres**: revenue and expenditure; judged on profitability (decentralised branches).
5. **Investment centres**: profitability plus authority over capital investment; judged on ROI besides profit (Maharatna, Navratna, Miniratna PSUs).

"Identify the responsibility centre, with reasons" (case)

Step	What to write
1	The type, in ICAI's name (standard cost centre, not "a production centre")
2	The fact in the case that decides it ("can buy trucks within its own capital budget")
3	The Study Material's test it meets ("authority to make capital investment decisions; measured on ROI")
4	Where a near-miss is likely, why not the next type ("not an investment centre: capital spending needs head-office approval")

"State the cost unit"

A two-column table, industry and cost unit, in the order the question lists them, using the Study Material's units; for a service, a composite unit where the service depends on two measures (tonne-km, passenger-km, patient day).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the cost unit that was a building

"Cost unit of a hotel: the hotel." That is the cost object. **Fix:** the cost unit is the unit of product, service or time in which cost is expressed: for a hotel, room or meal (the CIMA list in the Study Material; ICAI's January 2025 answer also accepts guest days or room days). Ask "per what?" and the answer is the cost unit.

⚠ **Common mistake**

Every branch an investment centre

Calling a branch an investment centre because it is big or makes a large profit. **Fix:** a branch whose manager answers for revenue and costs is a profit centre ("decentralised branches", SM 1.23). Only the authority to make capital investment decisions, with performance measured on ROI, makes it an investment centre.

⚠ **Common mistake**

The R&D lab on a standard cost

Treating the research or advertising department as a standard cost centre and blaming it for "adverse variances". **Fix:** where output cannot be measured in financial terms, no input-output ratio exists: it is a discretionary cost centre, judged against its allocated budget.

⚠ **Common mistake**

Cost driver or cost unit?

Writing "number of purchase orders" as the cost unit of the purchase department, or "tonne-km" as a cost driver. **Fix:** the cost driver is the factor that causes the cost (number of purchase orders for procurement cost); the cost unit is the measure the cost is expressed in (tonne-km for transport).

Practice

Check yourself

1 Case scenario. Tapri's Pune delivery hub packs and dispatches app orders. A time study fixed the packing material and minutes per order, the hub's actual cost is compared every week with the standard cost of the orders packed, and the hub manager answers for adverse variances. Head office sets the delivery charges and keeps all delivery revenue. Which of the following is correct?

- **A** It is a revenue centre, because every order it dispatches earns delivery revenue
- **B** It is a profit centre, because delivery charges are earned through its work
- **C** It is a standard cost centre, because its output is measurable and its input-output ratio is identifiable
- **D** It is a discretionary cost centre, because its spending is compared with a budget

2 A Tapri cold-chain truck cost ₹2,88,000 to run in October. It made 40 trips of 150 km each, carrying 12 tonnes on every trip (the load did not change during a trip). What was the cost per tonne-km, in rupees?

Answer: _____

3 Ritu reports: "The Diwali gift-box line cost ₹6,40,000 in October, which is ₹320 per box." In her sentence, the gift-box line and the box are respectively:

- **A** The cost unit and the cost object
- **B** The cost object and the cost unit
- **C** A cost centre and a cost driver
- **D** A cost driver and the cost unit

4 Which of these is a cost driver rather than a cost unit?

- **A** Tonne-km for a transport fleet
- **B** Kilowatt-hour for a power plant
- **C** Patient day for a hospital
- **D** Number of machine set-ups for set-up cost

5 Which of the following statements about responsibility centres agree with the Study Material? Select all that apply.

- **A** A revenue centre may incur selling expenditure, such as commission to salespersons
- **B** A discretionary cost centre has a clearly identifiable input-output ratio
- **C** An investment centre's performance is measured on return on investment besides profit
- **D** A profit centre has the authority to make capital investment decisions
- **E** The manager of a standard cost centre is held responsible for adverse cost variances

More than one answer may be right.

6 For better cost control, Ritu sub-divides the premix plant's packing department into cost centres: one for the four automatic pouch-filling machines and one for the six workers who hand-pack gift boxes. Which is correct?

A Both are personnel cost centres, because people operate the machines

B The first is a machine cost centre and the second a personnel cost centre

C Neither can be a cost centre: only a whole department can be one

D The first is a cost centre and the second a revenue centre

Answers: the answer key at the end of this lesson.

Recap

- Cost object: anything for which a separate measurement of cost is required (product, service, project, customer, brand category, activity, department, programme); the nucleus of cost classification.
- Cost unit: a unit of product, service or time (or a combination) in which costs are ascertained or expressed; usually a physical unit (number, weight, area, volume, length, time, value); an order, contract or batch can be one.

- Responsibility centres answer for expenditure, revenue, profitability or return on investment: cost centres (costs under their control), revenue centres (revenue), profit centres (revenue and expenditure, judged on profitability) and investment centres (profit plus capital investment decisions, judged on ROI besides profit).
- Standard cost centre: measurable output, identifiable input-output ratio. Discretionary cost centre: output not measurable in financial terms, inputs compared with the allocated budget (R&D, advertisement).
- Cost driver: a factor influencing the level of cost (number of set-ups, purchase orders, inspection hours, tests).
- Cost units to know: power and electricity kWh; steel ton; transport passenger-km (tonne-km for goods); brick-making 1,000 bricks; brewing barrel; hospitals patient day; education course, enrolled or successful student; credit control accounts maintained.
- Profit centre example: decentralised branches. Investment centre example: Maharatna, Navratna and Miniratna PSUs. (Papa buys the banquet hall; Mama-ji only stays within budget.)

Answer key · Check yourself

1 **C** **It is a standard cost centre, because its output is measurable and its input-output ratio is identifiable**

The hub answers only for costs under its control, its output (orders packed) is measurable and the inputs per order are specified: a standard cost centre. The revenue is earned, priced and kept by head office, so it is not the hub's responsibility; a discretionary cost centre is one whose output cannot be measured in financial terms.

2 **4**

Tonne-km = 12 tonnes × 150 km × 40 trips = 72,000. Cost per tonne-km = ₹2,88,000 ÷ 72,000 = ₹4.00. The truck (or its October operation) is the cost object; the tonne-km is the cost unit.

3 **B** **The cost object and the cost unit**

The gift-box line is the thing whose cost is measured separately: the cost object. The box is the unit in which the cost is expressed: the cost unit. A cost driver is what causes the cost to change, such as the number of set-ups.

4 **D** **Number of machine set-ups for set-up cost**

A cost driver is a factor that influences the level of cost; the number of machine set-ups is ICAI's own example. The other three are units in which cost is expressed.

5 **A** **A revenue centre may incur selling expenditure, such as commission to salespersons**; **C** **An investment centre's performance is measured on return on investment besides profit**; **E** **The manager of a standard cost centre is held responsible for adverse cost variances**

A discretionary cost centre is the one whose input-output ratio cannot be defined (its output is not measurable in financial terms). Authority over capital investment is what turns a profit centre into an investment centre.

6 **B** **The first is a machine cost centre and the second a personnel cost centre**

A department may be sub-divided into cost centres; the two main types are machine and personnel, depending on whether the work at the centre is done by machine or by people (SM 4.12). The principal object is to collect data on similar activities more conveniently.

1.5 Elements and classification of cost

Six ways ICAI sorts the same rupee of cost (element, function, behaviour, controllability, normality and managerial decision making), with the elements of cost and eighteen decision-making cost terms.

In plain words

One bill, five people, five questions

The Diwali gift-box line at Tapri's Indore factory has just closed its books for October. Ritu prints one list of every rupee it spent, and five people want five different answers from it.

> **Kesar:** Simple question. What did one gift box cost us?
>
> **Dev:** Not so simple. If we make 20% more boxes next month, which of these lines go up?
>
> **Ritu:** And the plant head wants to know which lines he will be blamed for. The flooded godown is a fourth question, and your new packing machine is a fifth.

Kesar's question needs the costs sorted by **what they are made of**: the cardboard and chai premix that go into each box (traceable to the box: **direct**) and the factory rent that no single box "uses" (**indirect**, an **overhead**). It also

needs them sorted by **what they were spent for**: making the box, running the office, selling it, or delivering it.

Dev's question sorts the same list by **behaviour**: rent stays put when output rises (**fixed**), cardboard rises with every box (**variable**), and electricity does a bit of both (**semi-variable**).

The plant head's question sorts it by **controllability**: he can influence the wastage of cardboard, not the head office's share of the rent. The flood sorts it by **normality**: an ordinary month's costs are normal; stock ruined by a flood is not. And the machine needs **decision-making costs**: what is already spent and gone (**sunk**), what cash the decision would actually cost (**out-of-pocket**), and what Tapri gives up by choosing it (**opportunity cost**).

Think of it like

The same student on five lists

In your coaching class you are on the roll-number list, the batch list, the attendance-defaulter list (hopefully not), the topper list and the fee list. You did not change; the teacher's question did. A rupee of cost is the same: classification does not change the rupee, it files it under whatever question management is asking today.

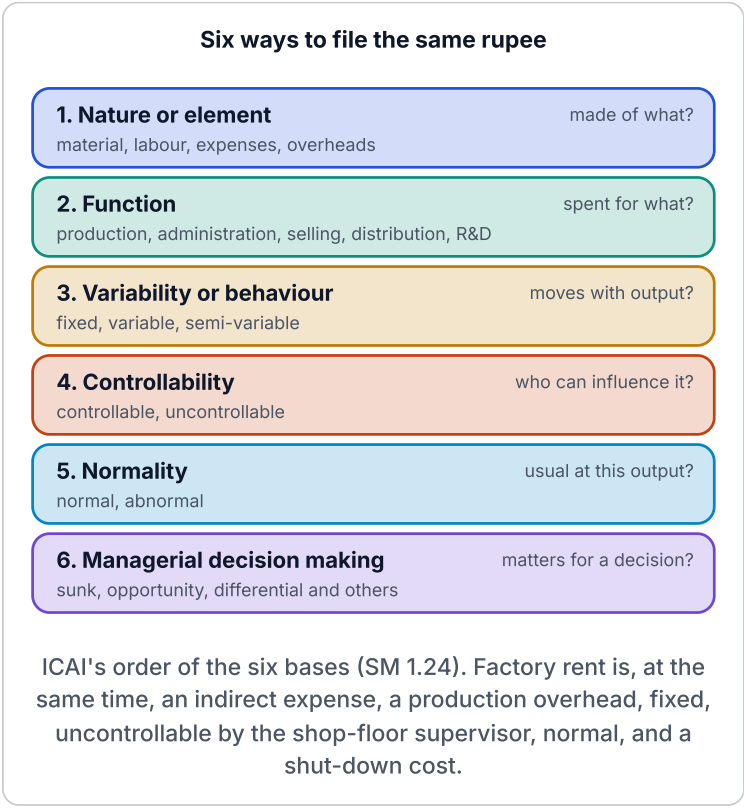
Key idea

Classification is grouping costs by a common characteristic. ICAI uses six bases, and every cost sits in all six at once: the one you use depends on the question being asked and on the cost object (a box, a café, a department) you are costing.

How it works

Classification of costs means the grouping of costs according to their common characteristics. ICAI's six

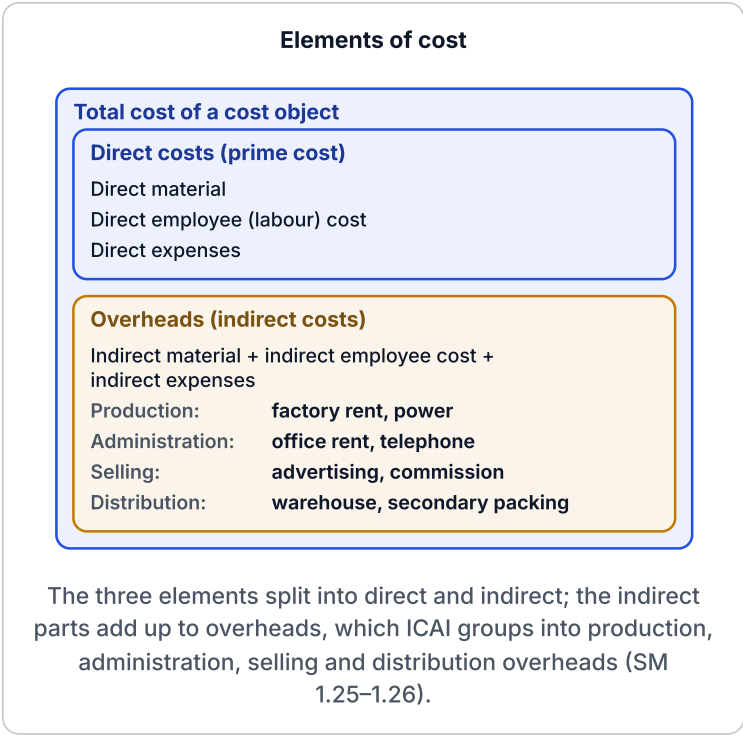
bases, in its order (SM 1.24):



ICAI's order of the six bases (SM 1.24). Factory rent is, at the same time, an indirect expense, a production overhead, fixed, uncontrollable by the shop-floor supervisor, normal, and a shut-down cost.

1. By nature or element: what the cost is made of

Under this classification, the total cost of a cost object is classified on the basis of the **elements of cost**: material, labour, other expenses and overheads (SM 1.24). Each of the first three is either direct or indirect, and the indirect parts together are the overheads.



The three elements split into direct and indirect; the indirect parts add up to overheads, which ICAI groups into production, administration, selling and distribution overheads (SM 1.25–1.26).

- **Material** is the cost of raw material required to make a product into finished goods. Materials present in the finished product (the cost object) or that can be economically identified in it are **direct materials**: cloth in dress making, materials purchased for a specific job. A material used in such small quantities that identifying it is not economically feasible is treated as **indirect**, and so are materials used for purposes ancillary to the business (SM 1.25).
- **Labour** is the wages paid to workers for converting raw materials into finished goods. Labour that can be economically identified or attributed wholly to a cost

object is **direct labour**: employees engaged on the actual production of the product (SM 1.26).

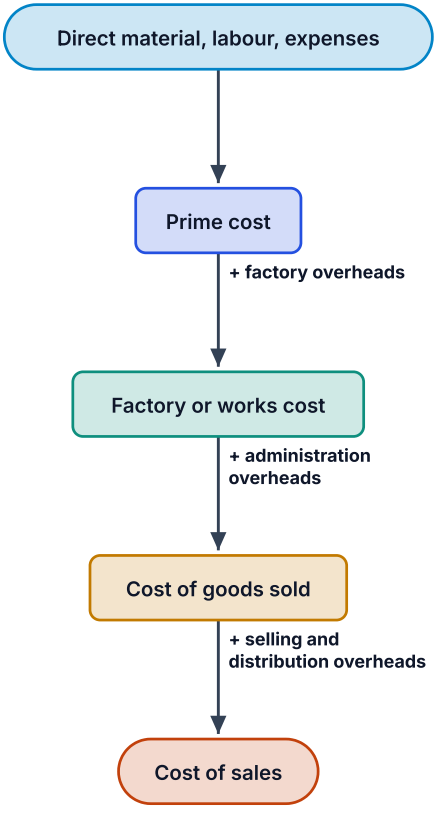
- **Other expenses** are all expenses other than material or labour incurred for a particular cost object, for example hire charges for some special machinery or the cost of defective work (SM 1.26). The Cost Sheet chapter calls these **direct expenses**, expenses that can be directly traced to a cost object in an economically feasible manner, and lists: cost of utilities such as power and fuel, royalty for production, hire charges of specific equipment, fee for technical assistance and know-how, amortised moulds, patterns and patents, product-specific design or drawing, and product-specific software (SM 6.4; see direct expenses).
- **Overheads** are the aggregate of indirect material costs, indirect labour costs and indirect expenses, subdivided into **production or works overheads** (incurred in and for running the factory: rent, power), **administration overheads** (management and administration: office rent, lighting, telephone), **selling overheads** (marketing: advertisement, salesmen's commission) and **distribution overheads** (dispatch: warehouse charges, secondary packing, loading) (SM 1.26).

The test for "direct" is **economic traceability to the cost object**, not the kind of item. Salt in a samosa is physically in the product, yet it is indirect, because weighing the pinch in each samosa costs more than it tells you.

2. By function: what the cost was spent for

Costs are divided according to the function for which they have been incurred: (i) production or manufacturing cost, (ii) administration cost, (iii) selling cost, (iv) distribution cost and (v) research and

development cost etc. (SM 1.26–1.27). The cost sheet is built on this classification:



ICAI's Chapter 1 chart of the flow of costs (SM 1.27). The specimen cost sheet (SM 6.8) adds stock adjustments and puts general administration overheads after cost of goods sold: see Formats and formulas.

The full format, with stocks of work-in-process and finished goods, primary and secondary packing and the split of administration overheads, is in the cost sheet for a manufacturer.

3. By variability or behaviour: does it move with output?

- **Fixed costs** are incurred for a period and, within certain output and turnover limits, tend to be unaffected by fluctuations in the level of activity: rent, insurance of factory building.
- **Variable costs** tend to vary with the volume of activity: direct material, direct labour.
- **Semi-variable costs** contain both fixed and variable components: telephone bills, gas and electricity (SM 1.27–1.28).

How each one behaves, step costs, and the five methods of splitting a semi-variable cost are in cost behaviour.

4. By controllability: who can influence it?

Definition

Controllable and uncontrollable costs

Controllable cost: cost that can be controlled, typically by a cost, profit or investment centre manager. Controllable costs incurred in a particular responsibility centre can be influenced by the action of the manager heading that responsibility centre: direct material, direct labour, direct expenses and some overheads are generally controllable by the shop floor supervisor or factory manager.

Uncontrollable cost: costs which cannot be influenced by the action of a specified member of an undertaking. The tool room's expenditure is controllable by the foreman in charge of the tool room, but its share apportioned to the machine shop is not controlled by the machine shop foreman. (SM 1.33)

The distinction is not very prominent and is sometimes left to individual judgment: no cost is uncontrollable; it is only in relation to a particular individual that a cost is controllable or uncontrollable (SM 1.33). Responsibility centres are in cost objects and responsibility centres.

5. By normality: is it usual at this output?

Definition

Normal and abnormal cost

Normal cost: the cost which is normally incurred at a given level of output under the conditions in which that level of output is normally attained.

Abnormal cost: the cost which is not normally incurred at a given level of output in the conditions in which that level of output is normally attained. It is charged to Costing Profit and Loss Account. (SM 1.33)

Normal costs belong in the cost of the product; abnormal ones are kept out of it, so that one bad month (a flood, a fire, a breakdown beyond the usual) does not inflate the cost per unit. The same idea returns as normal and abnormal loss in material losses and process accounts.

6. For managerial decision making: does it matter for this decision?

ICAI lists eighteen costs under this head, in this order (SM 1.34–1.36). The first column is what to memorise; the middle column is ICAI's meaning, shortened.

Cost

(a) Pre-determined

ICAI's meaning

Computed in advance before production starts, from the specification of all factors affecting cost

Tapri example

Cost of a new premix pack worked out before launch

Cost

(b) Standard

ICAI's meaning

Pre-determined from management's expected standard of efficient operation; basis for price fixation and for cost control through variance analysis

Tapri example

₹6.50 a cup of chai at standard usage and prices

Cost

(c) Marginal

ICAI's meaning

The amount by which aggregate costs change if output is increased or decreased by one unit

Tapri example

Cost of one more samosa pack

Cost

(d) Estimated

ICAI's meaning

Kohler: expected cost of manufacture or acquisition, computed from information available in advance; a prospective cost

Tapri example

Quoted cost of a wedding catering order

Cost

(e) Differential

ICAI's meaning

Change (increase or decrease) in total cost, variable and fixed, due to a change in activity level, technology, process or method: incremental or decremental cost

Tapri example

Extra cost of a second shift

Cost

(f) Imputed

ICAI's meaning

Notional costs with no cash outlay, such as interest on capital not actually paid; similar to opportunity costs

Tapri example

Notional interest on Kesar's own capital

Cost

(g) Capitalised

ICAI's meaning

Initially recorded as assets and subsequently treated as expenses

Tapri example

Installation of a new fryer added to its cost

Cost

(h) Product

ICAI's meaning

Costs of purchase and sale of goods (merchandise), or of acquiring and converting materials and other manufacturing inputs into finished product: variable manufacturing costs under marginal costing, total manufacturing costs under absorption costing

Tapri example

Materials, wages and factory overheads in a pack

Cost

(i) Opportunity

ICAI's meaning

Value of the sacrifice made, or benefit of opportunity forgone, in accepting an alternative

Tapri example

Interest lost on a fixed deposit withdrawn

Cost

(j) Out-of-pocket

ICAI's meaning

Portion of total cost involving cash outflow; short-run; avoidable if the proposal is not accepted

Tapri example

Cash paid for a new machine

Cost

(k) Shut down

ICAI's meaning

Costs that continue even when a plant is temporarily shut down: rent, rates, depreciation

Tapri example

Rent of a café closed for metro work

Cost

(l) Sunk

ICAI's meaning

Historical costs incurred in the past; no role in current decisions

Tapri example

Written-down value of the old fryer

Cost

(m) Absolute

ICAI's meaning

Cost of a product, process or unit in its totality, in absolute amount; the base for further analysis

Tapri example

Total cost of the gift-box line

Cost

(n) Discretionary

ICAI's meaning

Not tied to a clear cause-and-effect link between input and output; periodic decisions on the maximum outlay

Tapri example

Advertising, public relations, executive training

Cost

(o) Period

ICAI's meaning

Not assigned to products; charged against the revenue of the period: all non-manufacturing costs

Tapri example

Office salaries, selling and distribution

Cost

(p) Engineered

ICAI's meaning

A clear, usually observable, cause-and-effect link between inputs and outputs

Tapri example

Tea leaves per kilogram of premix

Cost

(q) Explicit

ICAI's meaning

Also called out-of-pocket; involve immediate payment of cash

Tapri example

Salaries, wages, postage, interest on loan

Cost

(r) Implicit

ICAI's meaning

No immediate cash payment; not recorded in the books; also called economic costs

Tapri example

Rent Kesar could earn on his own godown

⚠ **Careful**

Two pairs look alike. ICAI says imputed costs are "similar to" opportunity costs, and explicit costs are "also known as" out-of-pocket costs (SM 1.34–1.36). In an answer, use the term the question uses and give ICAI's meaning for it.

Formats and formulas

From a list of costs to a cost statement

ICAI questions hand you a jumbled list and ask you to classify it, then total it. File each item first by element (direct or indirect), then by function. The statement below follows the order of ICAI's specimen cost sheet (SM

6.8) for a month with no opening or closing stocks. *Reproduce this layout from memory.*

Particulars	Amount (₹)
Direct material	xxx
Direct employee (labour) cost	xxx
Direct expenses	xxx
Prime cost	**xxx**
Add: Works or factory overheads	xxx
Works or factory cost	**xxx**
Add: Administration overheads related to production	xxx
Cost of production (= cost of goods sold)	**xxx**
Add: General administration overheads	xxx
Add: Selling overheads	xxx
Add: Distribution overheads	xxx
Cost of sales	**xxx**

With stocks, quality control, research and development, recoveries and primary packing, the full format is in the cost sheet for a manufacturer. The Chapter 1 chart (SM 1.27) adds all administration overheads before cost of

goods sold; cost of sales comes out the same, but in a cost sheet answer keep general administration after cost of goods sold, as SM 6.7–6.8 does.

∑ **Formula**

The element identities

$$\text{Prime cost} = \text{Direct material} + \text{Direct employee cost} + \text{Direct expenses}$$

$$\text{Overheads} = \text{Indirect material} + \text{Indirect employee cost} + \text{Indirect expenses}$$

Total cost = prime cost + overheads, whichever way the overheads are grouped (SM 1.25–1.26).

∑ **Formula**

The functional build-up (SM 1.27)

$$\text{Works cost} = \text{Prime cost} + \text{Factory overheads}$$

$$\text{Cost of sales} = \text{Works cost} + \text{Administration} + \text{S\&D overheads}$$

S&D = selling and distribution.

Definition

Direct materials

Materials which are present in the finished product (cost object) or can be economically identified in the product. A direct material used in small quantities, where identifying the quantity is not economically feasible, is treated as indirect; materials used for purposes ancillary to the business are indirect materials. (SM 1.25)

Definition

Direct labour

Labour which can be economically identified or attributed wholly to a cost object: employees engaged on the actual production of the product or in carrying out the operations converting raw materials into finished product. (SM 1.26)

Definition

Overheads

The aggregate of indirect material costs, indirect labour costs and indirect expenses, subdivided into production or works, administration, selling and distribution overheads. (SM 1.26)

Worked examples

Worked example **The samosa factory's October list**

Tapri's Indore factory made 2,00,000 samosa packs in October. Its cost list for the month:

Item	Amount (₹)
(i) Potatoes, maida and peas	18,40,000
(ii) Piece-rate wages of the cooks	6,20,000
(iii) Royalty to a celebrity chef for the recipe, 50 paise a pack	1,00,000
(iv) Salt, spices and hing (tiny quantities)	45,000
(v) Factory supervisor's salary	1,20,000
(vi) Electricity for factory lighting, fans and canteen (fixed demand charge plus units)	2,30,000
(vii) Depreciation of the frying line (straight line)	1,50,000
(viii) Company secretary's salary	1,80,000
(ix) Distributors' commission, 3% of sales	1,35,000
(x) Diwali TV advertising	3,00,000
(xi) Cartons for dispatch (secondary packing)	80,000

Item	Amount (₹)
(xii) Fuel of delivery vans	1,10,000

There were no opening or closing stocks, and the company secretary's salary is general administration.

Required:

(i) Classify each item by element, by function and by behaviour.

(ii) Prepare a statement showing prime cost, works cost and cost of sales, in total and per pack.

(iii) Find the total of fixed, variable and semi-variable costs.

THE WORKING

(i) Classification

Item	Element	Function; behaviour
(i) Potatoes	Direct material	Production; variable
(ii) Cooks' wages	Direct labour	Production; variable
(iii) Royalty	Direct expense	Production; variable

Item	Element	Function; behaviour
(iv) Salt, spices	Indirect material	Production OH; variable
(v) Supervisor	Indirect labour	Production OH; fixed
(vi) Electricity	Indirect expense	Production OH; semi-variable
(vii) Depreciation	Indirect expense	Production OH; fixed
(viii) Secretary	Indirect labour	Admin. OH; fixed
(ix) Commission	Indirect expense	Selling OH; variable
(x) Advertising	Indirect expense	Selling OH; fixed
(xi) Cartons	Indirect material	Distribution OH; variable
(xii) Van fuel	Indirect expense	Distribution OH; variable

OH = overheads; Admin. = administration. Notes: (iii) royalty paid for production is a direct expense (SM 6.4); (iv) salt is in the product but used in quantities too small to identify economically, so it is indirect (SM 1.25); (vi) power used to run a process can be a direct expense when it is traced to the

product (SM 6.4), but lighting, fans and the canteen serve the whole factory, so this bill is a production overhead (SM 1.26); (xi) secondary packing is a distribution overhead (SM 1.26); (x) advertising is also a discretionary cost (SM 1.36).

(ii) Statement of cost for October (2,00,000 packs)

Particulars	Total (₹)	Per pack (₹)
Direct material	18,40,000	9.20
Direct employee cost	6,20,000	3.10
Direct expenses (royalty)	1,00,000	0.50
Prime cost	**25,60,000**	**12.80**
Add: Factory overheads (WN)	5,45,000	2.725
Works cost (= cost of goods sold)	**31,05,000**	**15.525**
Add: General administration overheads	1,80,000	0.90
Add: Selling overheads	4,35,000	2.175
Add: Distribution overheads	1,90,000	0.95

Particulars	Total (₹)	Per pack (₹)
Cost of sales	**39,10,000**	**19.55**

Working note: factory overheads = ₹45,000 + ₹1,20,000 + ₹2,30,000 + ₹1,50,000 = ₹5,45,000; selling = ₹1,35,000 + ₹3,00,000; distribution = ₹80,000 + ₹1,10,000. With no stocks and no administration related to production, cost of production and cost of goods sold both equal the works cost, ₹31,05,000; the company secretary's salary is general administration, added after cost of goods sold (SM 6.7).

(iii) By behaviour

Behaviour	Items	Amount (₹)
Variable	(i), (ii), (iii), (iv), (ix), (xi), (xii)	29,30,000
Fixed	(v), (vii), (viii), (x)	7,50,000
Semi-variable	(vi)	2,30,000
Total		**39,10,000**

The two totals agree: classification regroups the same ₹39,10,000, it never adds or loses a rupee.

Worked example

The old fryer and the new one

Tapri's samosa factory is deciding whether to replace its fryer.

- The old fryer cost ₹12,00,000 four years ago; its written-down value is ₹6,40,000 and a scrap dealer will pay ₹1,50,000 for it now.
- The new fryer costs ₹18,00,000 plus ₹60,000 for installation.
- Gas, oil and repairs cost ₹9,20,000 a year with the old fryer and would cost ₹6,80,000 with the new one.
- The net cash needed will come from a fixed deposit that earns 7% a year.
- The changeover shuts the factory for one month, during which rent (₹1,00,000), depreciation of the building (₹40,000), insurance (₹10,000) and the supervisor's salary (₹40,000) continue.
- Ritu's management report also charges notional interest at 10% on the ₹50,00,000 of capital Kesar has put into the factory; no such interest is paid.

Required:

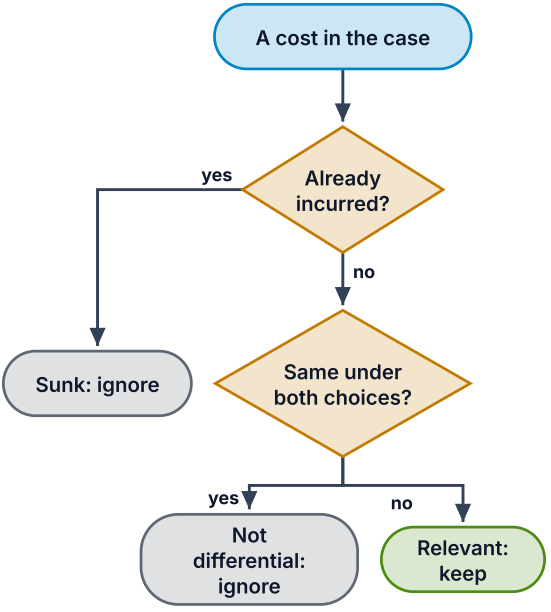
(i) Name the type of cost, as classified for managerial decision making, of each figure: the ₹6,40,000; the ₹60,000; the ₹18,60,000 paid for the new fryer; the interest on the fixed deposit; the month's rent, depreciation, insurance and salary; the notional interest; the change in operating cost.

(ii) Quantify the net cash outlay, the interest forgone in the first year, the change in annual operating cost and the shut-down cost of the changeover.

(iii) State which figures should be ignored in deciding whether to replace the fryer, with reasons.

THE WORKING

Does this cost matter for the decision?



Green: relevant costs, which change with the decision (out-of-pocket, differential, opportunity); grey: costs to leave out. The questions follow ICAI's meanings of sunk and differential cost (SM 1.34–1.35).

(i) Type of each cost

Figure
WDV ₹6,40,000

Type
Sunk

Reason
Historical cost; no role in the decision

Figure
Installation ₹60,000

Type
Capitalised

Reason
Added to the cost of the machine

Figure
₹18,60,000 for the new fryer

Type
Out-of-pocket (explicit)

Reason
Cash outflow, avoided if not accepted

Figure
Interest on the fixed deposit

Type
Opportunity

Reason
Benefit forgone by using the deposit

Figure
Rent, depreciation, insurance, salary in the closed month

Type
Shut-down

Reason
Continue while the plant is shut

Figure
Notional interest on capital
Type
Imputed (implicit)
Reason
Notional; no cash outlay, not in the books

Figure
Fall in operating cost
Type
Differential (decremental)
Reason
Change in total cost due to a change in method

(ii) Quantification

1. Net cash outlay = ₹18,00,000 + ₹60,000 – ₹1,50,000 (scrap received) = **₹17,10,000**.
2. Interest forgone in the first year = ₹17,10,000 × 7% = **₹1,19,700** (opportunity cost).
3. Decremental operating cost = ₹9,20,000 – ₹6,80,000 = **₹2,40,000 a year**.
4. Shut-down cost of the changeover = ₹1,00,000 + ₹40,000 + ₹10,000 + ₹40,000 = **₹1,90,000**.

(iii) Figures to ignore

- The written-down value of ₹6,40,000 and the book "loss" of ₹4,90,000 (₹6,40,000 – ₹1,50,000): sunk costs, already incurred; only the ₹1,50,000 scrap value is a future cash flow.
- The notional interest of ₹5,00,000 (₹50,00,000 × 10%): an imputed cost that is the same whether

or not the fryer is replaced.

- The shut-down costs of ₹1,90,000: rent, depreciation, insurance and the supervisor's salary continue whether the factory is open or shut, so they are incurred whether or not the fryer is replaced. They are not differential (and depreciation is not even a cash outflow). What the month's closure really costs is the contribution on the samosa packs that cannot be made that month, which the case does not give.

The relevant figures are the net outlay of ₹17,10,000, the interest forgone of ₹1,19,700 a year and the saving of ₹2,40,000 a year, plus any contribution lost during the closure; weighing them over the fryer's life is a capital budgeting exercise (relevant cash flows, Paper 6A).

Worked example **The frying supervisor's bad report**

Tapri's frying department is a cost centre. Its monthly report shows an adverse variance of ₹62,000 on "quality laboratory charges", the frying department's share of the quality laboratory's cost, apportioned on the basis of the number of tests. The quality laboratory has its own manager. Kesar wants the frying supervisor to explain the variance.

Comment on Kesar's view, with reference to the classification of costs by controllability.

MODEL ANSWER

Relevant classification. Costs are classified by controllability into controllable and uncontrollable costs. A controllable cost is one that can be influenced by the action of the manager heading the responsibility centre in which it is incurred; an uncontrollable cost is one which cannot be influenced by the action of a specified member of the undertaking. The share of a service department's expenditure apportioned to another department is not controlled by the head of the receiving department (SM 1.33).

Facts of the case. The ₹62,000 is the frying department's apportioned share of the quality laboratory's cost; the laboratory is run by its own manager.

Analysis. The quality laboratory's spending is controllable by the laboratory manager, not by the frying supervisor. For the frying supervisor it is an uncontrollable cost, except to the extent that the number of tests his department calls for (the apportionment basis) is within his influence.

Conclusion. Kesar's view is not correct. The variance in the laboratory's costs should be explained by the laboratory manager; the frying supervisor should be held responsible only for the costs he controls, such as direct material, direct labour and the department's own overheads. The

> frying department's report should show the apportioned charge separately as uncontrollable.

Exam angle

In the exam

Where classification appeared, May 2024 to September 2026 (and old-scheme papers)

In the ICAI documents read for this lesson, classification of cost was examined in 12 questions in 12 documents.

- **Classify a list, 5 marks.** January 2026 Q6(a): five items by function and variability (consumable stores, General Manager's salary, delivery van expenses, a salesman's fixed salary plus commission, rent of finished goods warehouse). September 2026 Q6(a): ten items "under appropriate functions of cost" (blueprint cost, hire charges of factory machinery for a product, one-time software licence for a particular job, overtime at a customer's request, specialised packing, royalty); suggested answers not yet published.
- **Decision-making costs.** RTP May 2025 Q1: five MCQs on one case (avoidable cost, sunk and shut-down costs, out-of-pocket cost, unexpired cost). RTP January 2026 Q19(a): "DISCUSS any five types of costs categorised based on its use in Managerial Decision Making". RTP May 2026 Q17(a): how seven of them (pre-determined, differential, product, out-

of-pocket, shut-down, sunk, implicit) influence decisions in a scenario.

- **Controllability, 4 marks.** MTP May 2024 Series I Q6(c): "DIFFERENTIATE" controllable and uncontrollable costs. Old scheme: "EXPLAIN the difference between controllable & uncontrollable costs" in MTP May 2023 Series II Q6(a), MTP November 2023 Series I Q6(a) and RTP May 2022 Q14(a) (5 marks each in the MTPs); MTP December 2021 Series II Q6(b) and RTP July 2021 Q13(c), "DISCUSS cost classification based on variability and controllability".
- **The examiner's complaint.** July 2021 (old scheme) Q2(a): examinees "were confused about the treatment of some items like cost of special drawing, hire charges paid for plant, carriage on return and general overheads" in a cost sheet.

In the exam

The traps ICAI sets

- **Direct costs hide in plain sight**: royalty for production, hire charges of equipment for a specific product or job, product-specific design or software are direct expenses (SM 6.4); overtime premium paid because a customer wants the job urgently is charged to that job directly (SM 3.18). They go in prime cost, not overheads.
- **Two answers can both be right on behaviour.** For delivery van expenses ICAI's January 2026 answer accepts "Semi-variable/Variable", and for a salesman paid a fixed salary plus commission, "Semi-variable". State your reading in a word.
- **Function words**: "Selling & Distribution/Distribution" was accepted for the warehouse rent. Use ICAI's five functions, not "miscellaneous".

Trick

Element first, then function

For each item ask two questions in order: (1) "Can it be traced economically to the cost object?" Yes: direct material, labour or expense, part of prime cost. No: overhead. (2) "For which function was it spent?" Factory, office, selling or delivery. The answer to (1) is about the cost object, so read the question for what the cost object is.

Answer frame

"Discuss any five types of costs used in managerial decision making"

Write each as the term, ICAI's meaning, and one example. The five most useful, in ICAI's order: **differential cost** (change in total cost due to a change in activity level, technology, process or method), **opportunity cost** (value of the sacrifice or benefit of the opportunity forgone), **out-of-pocket cost** (portion of total cost involving cash outflow; avoidable if the proposal is not accepted), **shut down cost** (costs that continue while a plant is temporarily shut down, such as rent, rates and depreciation) and **sunk cost** (historical cost that plays no role in current decisions, such as the written-down value of a machine being replaced) (SM 1.34–1.35).

"Differentiate controllable and uncontrollable costs"

Point	Meaning
Controllable cost	Can be influenced by the manager heading the responsibility centre
Uncontrollable cost	Cannot be influenced by the action of a specified member

Point

Usual examples

Controllable cost

Direct material, direct labour, direct expenses, some overheads

Uncontrollable cost

Share of a service department's cost apportioned to another department

Point

Responsibility

Controllable cost

The centre's manager is held accountable for it, as a cost centre is for costs under its control (SM 1.22)

Uncontrollable cost

The specified member cannot influence it, so it is not used to judge that member

Point

Key point

Controllable cost

No cost is uncontrollable for everyone; the classification is in relation to a particular individual (SM 1.33)

Uncontrollable cost

"Discuss cost classification based on variability and controllability"

1. Variability: fixed, variable and semi-variable costs, each with ICAI's meaning and an example (SM 1.27–1.28).
2. Controllability: controllable and uncontrollable costs, with ICAI's meaning and the tool-room example (SM 1.33).

3. A closing line: the distinction is sometimes left to judgment; no cost is uncontrollable, only in relation to a particular individual.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the royalty that ended up in overheads

Royalty paid per pack, hire charges of a special machine for one job, and design costs for one product are treated as factory overheads, so prime cost comes out too low and overheads too high. **Fix:** expenses that can be traced to the cost object in an economically feasible manner are direct expenses and belong in prime cost (SM 6.4).

⚠ **Common mistake**

The written-down value that sneaked into the decision

The old fryer's ₹6,40,000 written-down value (or the ₹4,90,000 "loss on sale") is counted as a cost of replacing it. **Fix:** it is a sunk cost, already incurred, and plays no role in the decision; only future costs that differ between the choices count, here the ₹1,50,000 scrap value.

⚠ **Common mistake**

"Uncontrollable" stamped on a cost for everyone

Writing "depreciation is an uncontrollable cost" without saying for whom. **Fix:** controllability is always in relation to a particular individual; the decision to buy the machine was controllable by someone (SM 1.33).

Practice

Check yourself

1 A Tapri café's monthly list shows direct material ₹4,00,000, direct wages ₹2,50,000, royalty paid per cup sold under a franchise recipe ₹30,000, hire of a special espresso machine for one corporate order ₹20,000, rent of the café ₹60,000 and office salaries ₹80,000. What is the prime cost in rupees?

Answer: _____

2 Case scenario. Tapri's Bhopal café has been closed for three months while metro work blocks its road. Kesar asks Ritu which of its costs are shut-down costs. Rent ₹75,000 a month, depreciation of fittings ₹20,000 a month and the security guard ₹15,000 a month continue; milk and tea purchases (₹1,40,000 a month when open) have stopped; ₹6,00,000 was spent on interiors two years ago. Which statement is correct?

A Shut-down costs are ₹2,50,000 a month, including milk and tea

B The ₹6,00,000 spent on interiors is a shut-down cost

C Shut-down costs are ₹1,10,000 a month: rent, depreciation and the guard

D There are no shut-down costs, because the café earns no revenue

3 Ritu reports costs under the heads "production, administration, selling, distribution and research and development". Which basis of classification is she using?

- **A** Nature or element
- **B** Variability or behaviour
- **C** Controllability
- **D** Function

4 Which of these are period costs? Select all that apply.

- **A** Salary of the company secretary
- **B** Wages of the cooks in the samosa factory
- **C** Commission paid to distributors
- **D** Fuel of the delivery vans

More than one answer may be right.

5 Kesar keeps an empty godown that a logistics firm would rent for ₹25,000 a month, and uses it to store gift boxes for six months instead. What is the opportunity cost of using it?

Answer: _____

6 The share of the head office's cost apportioned to the Pune café is, for the café manager:

- A A controllable cost, because it appears in the café's report
- B An uncontrollable cost, because she cannot influence it
- C An abnormal cost, to be charged to Costing Profit and Loss Account
- D Not a cost of the café at all

Answers: the answer key at the end of this lesson.

Recap

- Classification = grouping costs by common characteristics; six bases in ICAI's order: element, function, behaviour, controllability, normality, managerial decision making (SM 1.24).
- Elements: material, labour, other expenses (each direct or indirect) and overheads; overheads = indirect material + indirect labour + indirect expenses.
- Direct means economically traceable to the cost object; tiny quantities and ancillary materials are indirect even if physically present.
- Prime cost = direct material + direct labour + direct expenses; + factory overheads = works cost; with no

stocks, works cost = cost of goods sold; + general administration, selling and distribution overheads = cost of sales (SM 6.8).

- Functions: production, administration, selling, distribution, research and development.
- Controllability is always in relation to a particular individual; apportioned costs are uncontrollable for the receiving manager.
- Abnormal cost is charged to Costing Profit and Loss Account.
- Decisions: sunk costs are ignored; opportunity cost = benefit forgone; out-of-pocket = cash outflow, avoidable; shut-down costs continue while closed. (Same rupee, different list.)

Answer key · Check yourself

1 **700000**

Prime cost = direct material + direct wages + direct expenses = ₹4,00,000 + ₹2,50,000 + ₹30,000 + ₹20,000 = ₹7,00,000. The royalty and the hire charge are direct expenses; rent and office salaries are overheads.

2 **C Shut-down costs are ₹1,10,000 a month: rent, depreciation and the guard**

Shut-down costs are those that continue to be incurred while the plant is temporarily shut: ₹75,000 + ₹20,000 + ₹15,000 = ₹1,10,000 a month. Milk and tea have stopped, so they are not; the interiors are a sunk cost.

3 **D Function**

These are ICAI's five functions (SM 1.26–1.27). Element would be material, labour, expenses and overheads; behaviour would be fixed, variable and semi-variable.

4 **A Salary of the company secretary**; **C Commission paid to distributors**; **D Fuel of the delivery vans**

Period costs are not assigned to products but charged against the revenue of the period: all non-manufacturing costs, such as administration, selling and distribution (SM 1.36). The cooks' wages are a manufacturing cost, so a product cost.

5 **150000**

Opportunity cost is the benefit of the opportunity forgone: 6 × ₹25,000 = ₹1,50,000. It is not recorded in the books (an implicit cost), but it matters for the decision.

6 **B An uncontrollable cost, because she cannot influence it**

A cost apportioned from another centre cannot be influenced by the receiving centre's manager (SM 1.33). It is a normal cost of the café, just not one she controls.

1.6 Cost behaviour: fixed, variable, semi-variable and step costs

How each kind of cost moves when output moves, why fixed cost per unit falls, and ICAI's five methods of splitting a semi-variable cost into its fixed and variable parts.

In plain words

Kesar doubles the cups, Ritu does not double the costs

Tapri's Lucknow café sold 8,000 cups of chai in its first month. Kesar wants 16,000 next month and has already done the maths.

> **Kesar:** Costs were ₹2,04,400 for 8,000 cups. So 16,000 cups will cost ₹4,08,800. Simple.
>
> **Ritu:** Per unit, please. The rent will not double because you sell more chai.

Ritu splits the café's costs by how they behave when the number of cups changes.

- **Rent** of ₹90,000 a month stays the same at 8,000 cups or 16,000. A cost that does not move with output, within limits, is a **fixed cost**.
- **Milk, tea and the kulhad** cost ₹9 a cup: twice the cups, twice the cost. That is a **variable cost**.
- **Gas** has a cylinder rental of ₹6,000 a month whatever the sales, plus about 80 paise of gas a cup. Part fixed,

part variable: a **semi-variable cost**.

- **Baristas**: one barista can serve 4,000 cups a month and is paid ₹15,000. Two baristas cover 8,000 cups; the 8,001st cup needs a third. The cost is flat for a while, then jumps: a **step cost**.

So at 16,000 cups the café's costs are ₹3,12,800, not ₹4,08,800: the rent and the gas cylinder are not doubled, and the baristas step up from two to four. The rent per cup, meanwhile, halves.

Think of it like

The wedding caterer

A wedding caterer pays the same hall rent for 200 guests or 400 (fixed), one plate of food per guest (variable), a generator with a hiring charge plus diesel by the hour (semi-variable), and waiters in teams: one waiter for every 25 guests, so guest 226 brings a whole new waiter (step).

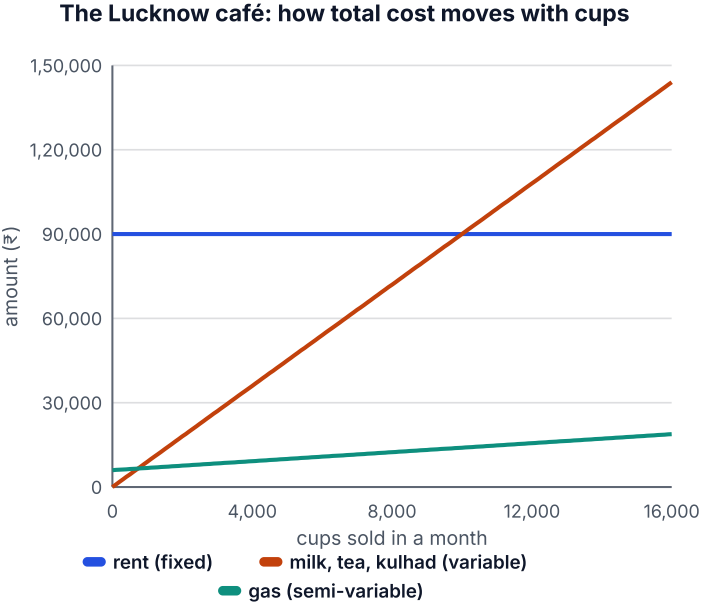
Key idea

Behaviour describes **totals** against activity within a range: fixed totals stay put, variable totals move in proportion, semi-variable totals do both, step totals stay flat and then jump. **Per unit**, the picture flips: fixed cost per unit falls as output rises, variable cost per unit stays the same.

How it works

Classification by **variability or behaviour** puts costs into three groups: fixed, variable and semi-variable (SM 1.27). The syllabus adds a fourth pattern, the step cost, which ICAI's problems use (below).

Three lines on one chart



Blue: rent, a flat line. Orange: milk, tea and kulhad at ₹9 a cup, a line from the origin. Teal: gas, ₹6,000 plus 80 paise a cup, a line that starts above zero and rises slowly.

- **Fixed costs** are the costs which are incurred for a period, and which, within certain output and turnover limits, tend to be unaffected by fluctuations in the

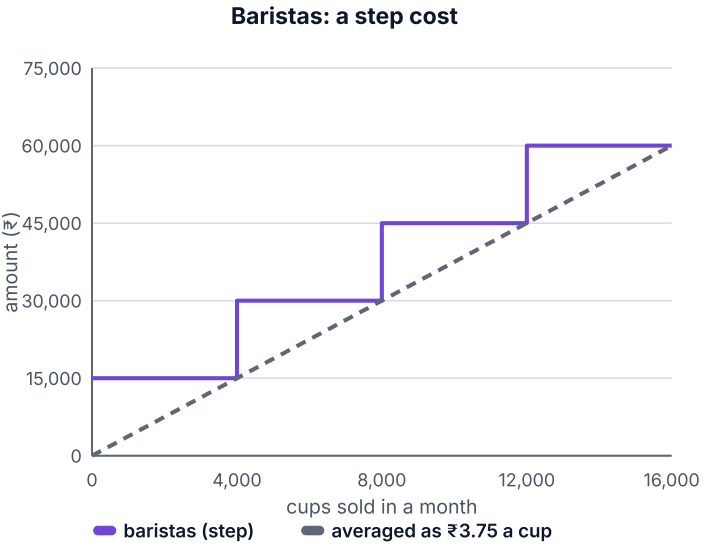
levels of activity (output or turnover). They do not tend to increase or decrease with the changes in output: rent, insurance of factory building (SM 1.27).

- **Variable costs** tend to vary with the volume of activity. Any increase in the activity results in an increase in the variable cost and vice-versa: direct material, direct labour (SM 1.28).
- **Semi-variable costs** contain both fixed and variable components and are thus partly affected by fluctuations in the level of activity: telephone bills, gas and electricity (SM 1.28).

Note the words "within certain output and turnover limits". Rent is fixed until the café needs a second floor; ICAI's problems write this as "fixed cost up to 80% capacity" plus an "additional fixed cost beyond 80% capacity" (May 2026 Q2(b)).

The staircase: step costs

A **step cost** stays fixed over a band of activity and jumps to a new level when activity crosses into the next band. The Study Material does not define it in Chapter 1, but its problems use it: "one supervisor for every twenty direct workers" (SM 15.38) and a semi-variable overhead of "₹20,00,000 per annum up to 50% capacity and an extra amount of ₹4,00,000 for every 25% increase in capacity or part thereof" (SM 8.18).



Violet: one barista at ₹15,000 for every 4,000 cups or part of it, flat within each band and then a jump. Grey dashed: the same cost wrongly averaged as a variable ₹3.75 a cup, too low just after every jump.

To cost a step, count the steps first, **rounding up**: 9,000 cups need 9,000 ÷ 4,000 = 2.25, so 3 baristas, ₹45,000. Buses for a college trip, supervisors per group of workers and block entrance fees by group size all work the same way (flexible budgets works ICAI's bus-trip problems).

Read the wording before you round. "For every 25% increase in capacity **or part thereof**" (SM 8.18) means whole steps: round up. "For every 20% increase in capacity **or proportionately**" (MTP May 2026 Series II Q5(a)) means pro rata: from 60% to 90% capacity adds 30

÷ 20 × ₹60,000 = ₹90,000, as ICAI's answer to that question computes it.

Per unit, the picture flips



Blue: rent per cup falls as cups rise (₹11.25 at 8,000 cups, ₹7.50 at 12,000, ₹5.63 at 16,000). Orange: variable cost per cup stays at ₹9.

This is why the cost per unit falls as a factory fills up, and why it rises sharply when output drops: the fixed cost is shared by fewer units.

Splitting a semi-variable cost: ICAI's five methods

A semi-variable cost must be split into its fixed and variable parts before it can be used for marginal costing, flexible budgets or decisions (SM 4.6–4.7; see classifying

overheads). ICAI gives five methods, in this order (SM 1.29):

1. **Graphical method.** Plot a large number of observations of total cost (Y-axis) against output (X-axis); draw by judgment a line of "best fit" passing through all or most of the points. Where the line cuts the Y-axis is the total fixed cost; a line from there parallel to the X-axis shows the fixed cost, and the variable cost at any output is the total cost minus this fixed cost (SM 1.29–1.30).

2. **High-low method.** The difference between the total cost at the highest and lowest volume is divided by the difference between the sales value at the highest and lowest volume; the quotient is the rate of variable cost in relation to sales value. ICAI's illustration: (₹72,000 − ₹60,000) ÷ (₹1,40,000 − ₹80,000) = 20% of sales, and fixed cost = ₹72,000 − 20% of ₹1,40,000 = ₹44,000 (SM 1.30). The same arithmetic works with units or hours as the measure of volume, giving a rate per unit or per hour.

3 **Analytical method.** An experienced cost accountant judges empirically what proportion of each semi-variable expense is variable (the **degree of variability**), say 70%. The fixed part then stays constant and only the variable part moves with production. Easy to apply, though the degree of variability is hard to estimate (SM 1.30–1.31).

4 **Comparison by period or level of activity.** Compare two levels of output with the expense at those levels; since the fixed element does not change, variable cost = change in the amount of expense ÷ change in the quantity of output (SM 1.31–1.32).

5 **Least squares method.** "This is the best method" (SM 1.32): a statistical method that finds a line of best fit for a number of observations, using y = mx + c. ICAI's illustration solves the equation at two levels of activity to get m and c (SM 1.32).

High-low and comparison by period use the same arithmetic on two observations; the difference is which two. High-low always takes the highest and lowest volumes; comparison takes any two periods or levels given.

Formats and formulas

Reproduce the high-low layout and these formulas from memory.

High-low working	Activity	Total cost (₹)
At the highest volume	xxx	xxx
At the lowest volume	xxx	xxx
Difference	**xxx**	**xxx**

Variable cost rate = difference in cost ÷ difference in activity; fixed cost = total cost at either level − variable rate × that level's activity. Check that both levels give the same fixed cost.

∑ **Formula**

Total cost within the relevant range

$$\text{Total cost} = \text{Fixed cost} + \text{Variable cost per unit} \times \text{Output}$$

Valid only "within certain output and turnover limits"; add any step or additional fixed cost separately.

∑ **Formula**

High-low method

$$\text{Variable cost rate} = \frac{\text{Cost at highest} - \text{cost at lowest}}{\text{Highest volume} - \text{lowest volume}}$$

Cost at highest and cost at lowest = total cost at the highest and at the lowest volume. The Study Material states it with sales value as the volume, so the rate is a percentage of sales (SM 1.30); with units or hours as the volume, the rate is per unit or per hour.

∑ **Formula**

Comparison by period or level of activity

$$\text{Variable cost per unit} = \frac{\text{Change in amount of expense}}{\text{Change in quantity of output}}$$

Fixed cost = expense at either level − variable cost per unit × output at that level (SM 1.31–1.32).

∑ **Formula**

Least squares method

$$y = mx + c$$

y = total cost, x = volume of output, m = variable element of cost per unit, c = total fixed cost (SM 1.32).

∑ Formula

Analytical method

$$\begin{aligned}\text{Variable part} &= \text{Degree of variability} \\ &\times \text{Semi-variable expense} \\ \text{New total} &= \text{Fixed part} \\ &+ \text{Variable part} \times \frac{\text{New output}}{\text{Old output}}\end{aligned}$$

∑ Formula

Step cost

$$\begin{aligned}&\text{Number of steps} \\ &= \frac{\text{Activity}}{\text{Activity per step}},\ \text{rounded up} \\ \text{Step cost} &= \text{Number of steps} \\ &\times \text{Cost of one step}\end{aligned}$$

Definition

Fixed costs

Costs which are incurred for a period, and which, within certain output and turnover limits, tend to be unaffected by fluctuations in the levels of activity (output or turnover). They do not tend to increase or decrease with the changes in output. (SM 1.27)

Definition

Variable costs

Costs which tend to vary with the volume of activity. Any increase in the activity results in an increase in the variable cost and vice-versa. (SM 1.28)

Definition

Semi-variable costs

Costs which contain both fixed and variable components and are thus partly affected by fluctuations in the level of activity. (SM 1.28)

Worked examples

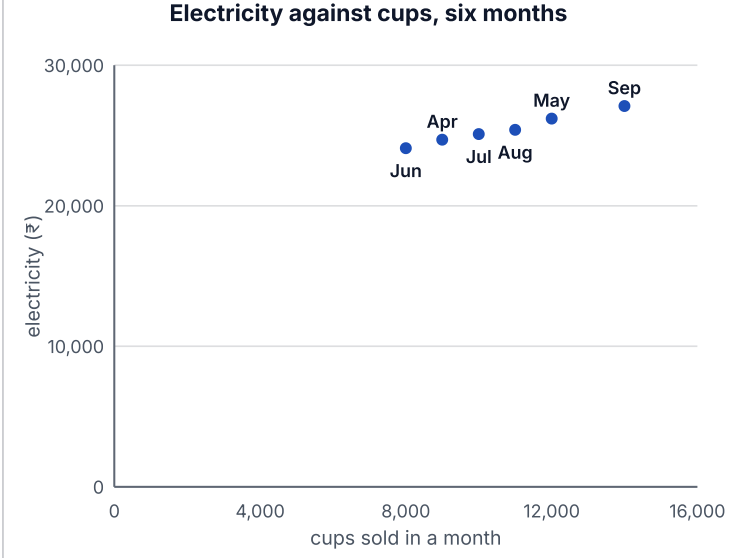
Worked example **The Lucknow café's electricity bill**

After six months, the Lucknow café's electricity bills

look like this:

Month	Cups sold	Electricity (₹)
April	9,000	24,700
May	12,000	26,200
June	8,000	24,100
July	10,000	25,100
August	11,000	25,400
September	14,000	27,100

Ritu has plotted them in the chart below:



One dot per month, labelled with the month.

Required:

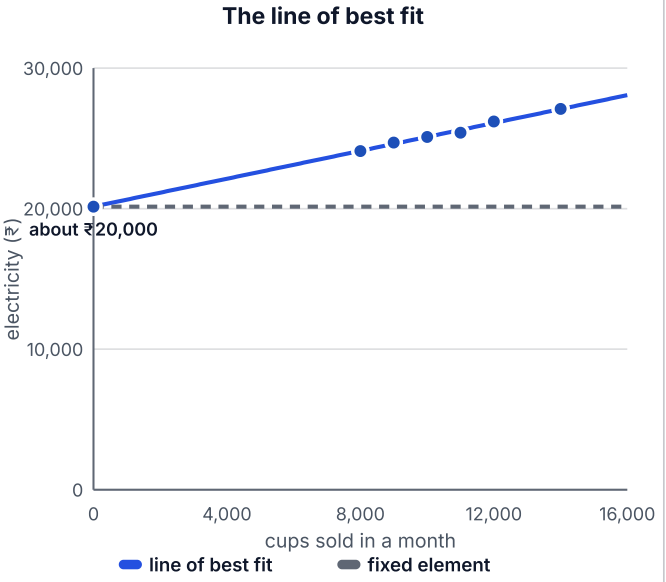
(i) Estimate the fixed element of the bill by the graphical method.

(ii) Segregate the bill into fixed and variable elements by the high-low method.

(iii) Segregate it by the comparison by period method, using July and August.

(iv) Estimate October's bill at 15,000 cups using (ii) and (iii), and comment on which estimate is more reliable.

THE WORKING



Blue: a line of best fit through the six dots; it cuts the Y-axis at about ₹20,000. Grey dashed: the fixed element, parallel to the X-axis. The gap between the two lines at any output is the variable cost.

(i) Graphical method. A line of best fit drawn through the six points cuts the Y-axis at about **₹20,000**: that is the fixed element. The variable element at any output is the total cost less ₹20,000; in September, ₹27,100 − ₹20,000 = ₹7,100 for 14,000 cups, about 50 paise a cup. (The line drawn here is the least squares line, ₹20,146 + ₹0.4957 a cup: see Under the hood.)

(ii) High-low method

Particulars	Cups	Electricity (₹)
Highest volume (September)	14,000	27,100
Lowest volume (June)	8,000	24,100
Difference	**6,000**	**3,000**

Variable cost = ₹3,000 ÷ 6,000 cups = **₹0.50 a cup**. Fixed cost = ₹27,100 − (14,000 × ₹0.50) = **₹20,100**. Check: ₹24,100 − (8,000 × ₹0.50) = ₹20,100.

(iii) Comparison by period, July and August

Variable cost = (₹25,400 − ₹25,100) ÷ (11,000 − 10,000) = ₹300 ÷ 1,000 = **₹0.30 a cup**. Fixed cost = ₹25,100 − (10,000 × ₹0.30) = **₹22,100**.

(iv) October at 15,000 cups

Method	Fixed + variable (₹)	Estimate (₹)
High-low	20,100 + 7,500	27,600
July and August	22,100 + 4,500	26,600

Comment. The high-low estimate of **₹27,600** is more reliable. July and August are only 1,000 cups apart, so a ₹100 quirk in one month's bill moves the variable rate by 10 paise a cup; high-low spreads

the comparison over the widest range, 6,000 cups. Both use only two observations, though; the line of best fit uses all six and gives ₹27,581 at 15,000 cups, almost the high-low figure. The extremes should still be checked for being normal months, since one unusual month at either end would distort high-low.

Worked example

The premix plant's overhead budget, with a staircase in it

Tapri's chai premix plant at Dewas has a capacity of 1,00,000 kg a month. Its overheads at two recent levels of activity were:

Overhead	60% (₹)	80% (₹)
Consumable stores	1,80,000	2,40,000
Power	2,10,000	2,60,000
Repairs and maintenance	96,000	1,12,000
Supervision	1,35,000	1,80,000
Depreciation and insurance	1,50,000	1,50,000
Plant office salaries	1,10,000	1,10,000

Supervisors are paid ₹45,000 a month each, and one supervisor is needed for every 20,000 kg of monthly

output or part of it.

Required:

(i) State the behaviour of each overhead, and segregate the semi-variable ones into fixed and variable elements.

(ii) Prepare an estimate of overheads at 70% and 90% capacity.

(iii) Calculate the overhead cost per kg at 60%, 70%, 80% and 90%, and comment.

THE WORKING

(i) Behaviour (comparison of the two levels: 60,000 kg and 80,000 kg, a difference of 20,000 kg)

Overhead	Behaviour	Fixed + variable
Consumable stores	Variable	₹3.00 a kg
Power	Semi-variable	₹60,000 + ₹2.50 a kg
Repairs	Semi-variable	₹48,000 + ₹0.80 a kg
Supervision	Step	₹45,000 a 20,000 kg band
Depreciation, insurance	Fixed	₹1,50,000
Office salaries	Fixed	₹1,10,000

The changes between the two levels are ₹60,000, ₹50,000, ₹16,000, ₹45,000, nil and nil.

Working notes:

1. Consumable stores: ₹60,000 ÷ 20,000 kg = ₹3 a kg; ₹1,80,000 ÷ 60,000 kg is also ₹3, so there is no fixed element.
2. Power: ₹50,000 ÷ 20,000 = ₹2.50 a kg; fixed = ₹2,10,000 − (60,000 × ₹2.50) = ₹60,000.
3. Repairs: ₹16,000 ÷ 20,000 = ₹0.80 a kg; fixed = ₹96,000 − (60,000 × ₹0.80) = ₹48,000.
4. Supervision: 60,000 kg needs 3 supervisors (₹1,35,000) and 80,000 kg needs 4 (₹1,80,000). It is a step cost, not a variable or semi-variable one: the number of supervisors is 60,000 ÷ 20,000 = 3 and 80,000 ÷ 20,000 = 4, rounded up for any part band.

(ii) and (iii) Overhead estimate

Overhead (₹)	70%	90%
Consumable stores	2,10,000	2,70,000
Power	2,35,000	2,85,000
Repairs	1,04,000	1,20,000
Supervision (WN 5)	1,80,000	2,25,000
Depreciation and insurance	1,50,000	1,50,000
Office salaries	1,10,000	1,10,000
Total	**9,89,000**	**11,60,000**

70% = 70,000 kg and 90% = 90,000 kg; each variable or semi-variable line is fixed part + rate × kg from (i), e.g. power at 70% = ₹60,000 + 70,000 × ₹2.50.

Working note 5: 70,000 ÷ 20,000 = 3.5, so 4 supervisors = ₹1,80,000; 90,000 ÷ 20,000 = 4.5, so 5 supervisors = ₹2,25,000.

Capacity (kg)	Overheads (₹)	Per kg (₹)
60% (60,000)	8,81,000	14.68
70% (70,000)	9,89,000	14.13
80% (80,000)	10,52,000	13.15
90% (90,000)	11,60,000	12.89

Comment. Overhead cost per kg falls as capacity utilisation rises, because the fixed elements (₹3,68,000 a month, including the fixed parts of power and repairs) are spread over more kilograms. The fall is uneven: at 70% and at 90% a new supervisor is needed for a part-filled band, so those levels carry a full extra ₹45,000. Treating supervision as variable at ₹2.25 a kg (₹45,000 ÷ 20,000) would have understated both estimates by ₹22,500.

Exam angle

In the exam

Where cost behaviour appeared, May 2024 to September 2026 (and old-scheme papers)

In the ICAI documents read for this lesson, cost behaviour and the split of semi-variable costs came up in 8 questions in 8 documents, almost always inside a bigger numerical.

- **Classify by variability, 5 marks.** January 2026 Q6(a): five items by function and variability.
- **Split from two periods.** MTP May 2025 Series II MCQs 6–7 (2 marks each): repairs of ₹1,40,000 for 18 trips per bus and ₹1,60,000 for 22 trips; the answer writes x + 18y = 1,40,000 and x + 22y = 1,60,000. Old scheme MTP November 2022 Series I Q2(b): a flexible budget at 75% and 100% capacity, each item to be split.

- **A given variable component.** May 2026 Q2(b) (7 marks): a semi-variable cost of ₹8,20,000 with a variable component of ₹10 a unit, plus an additional fixed cost beyond 80% capacity.
- **Steps.** May 2025 Q5(c) (7 marks): buses for a college trip and block entrance fees by group size.
- **Pro rata, not steps.** MTP May 2026 Series II Q5(a): a semi-variable overhead of ₹1,50,000 a year up to 60% capacity plus ₹60,000 "for every 20% increase in capacity or proportionately", and direct labour subject to a monthly minimum.
- **Theory.** Old scheme MTP December 2021 Series II Q6(b) and RTP July 2021 Q13(c): "DISCUSS cost classification based on variability and controllability".

In the exam

The traps

- A cost that changes between two levels is not automatically semi-variable. Test it: if the rate from the change also explains the total (₹1,80,000 ÷ 60,000 = ₹3, the same as the rate), it is purely variable.
- Supervisors, buses and batches are hired whole: round **up**, then cost the steps. But "or proportionately" means pro rata, not a step.
- "Fixed cost up to 80% capacity": use the higher fixed cost once the plan crosses 80%.
- The Study Material's high-low method uses **sales value**, not units: the variable cost comes out as a percentage of sales (SM 1.30). Read which measure of volume the question gives.

Trick

Fixed per unit is variable, variable per unit is fixed

Fixed cost per unit changes with every change in output; variable cost per unit stays the same. Condition: within the relevant range, with no change in prices or rates.

Answer frame

"Discuss cost classification based on variability"

1. Fixed costs: ICAI's definition ("incurred for a period … within certain output and turnover limits … unaffected by fluctuations in the levels of activity") and examples: rent, insurance of factory building.
2. Variable costs: definition ("tend to vary with the volume of activity") and examples: direct material, direct labour.
3. Semi-variable costs: definition ("contain both fixed and variable components") and examples: telephone bills, gas and electricity.
4. A sketch of each, cost on the Y-axis and output on the X-axis.

"Explain the methods of segregating semi-variable costs"

Name all five in ICAI's order (graphical, high-low, analytical, comparison by period or level of activity, least squares), then two or three lines on each with its formula, and close with ICAI's verdict that least squares is the best method (SM 1.29–1.32). A one-line numerical illustration for high-low earns the marks for "explain".

The numerical parts

Loses marks

"Power at 90% = ₹2,10,000 × 90 ÷ 60 = ₹3,15,000."

Treats a semi-variable cost as variable: the fixed ₹60,000 has been multiplied too.

Passes

"Variable rate ₹2.50 a kg, so power at 90% = ₹2,85,000."

Right figure, but the rate and the fixed part appear from nowhere.

Full marks

A working note for each item: change in cost ÷ change in output = rate; fixed = total − rate × output; check at the other level; then the estimate. Step costs shown as number of steps × cost per step, with the rounding stated.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the rent that doubled with the chai

Kesar's ₹4,08,800 for 16,000 cups: every cost scaled in proportion to output, ₹96,000 too high. **Fix:** split first. Only variable costs and the variable parts of semi-variable costs move with output; fixed costs stay, and step costs move only when a band is crossed.

⚠ **Common mistake**

The supervisor who became 2.25 rupees a kilogram

Running the high-low arithmetic on a step cost and costing supervisors at ₹2.25 a kg understates every level that is not exactly on a step: ₹1,57,500 instead of ₹1,80,000 at 70%. **Fix:** when the question says "one for every ..." or "for every 20% increase ... or part thereof", count the steps and round up ("or proportionately" is pro rata instead).

⚠ **Common mistake**

High-low with the wrong two months

Picking the months with the highest and lowest **cost** instead of the highest and lowest **activity**, or any two neighbouring months, and calling it high-low. **Fix:** high-low uses the highest and lowest volumes; comparison by period uses the two periods the question names.

Practice

Check yourself

1. A Tapri kiosk's total cost was ₹6,40,000 when sales were ₹9,00,000 (its highest month) and ₹4,80,000 when sales were ₹5,00,000 (its lowest). Using the high-low method, what is the fixed cost in rupees?

Answer: _____

2. Last month, Tapri's delivery cost was ₹4,80,000. Ritu judges that 40% of it is variable. In the monsoon, deliveries are expected to fall by 25%. What should the delivery cost be, in rupees?

Answer: _____

3 Case scenario. Tapri's cold-chain division runs 23 trucks. Each truck burns diesel by the kilometre; the depot rent is a fixed sum each month; and one refrigeration technician, paid a monthly salary, is employed for every 10 trucks or part of 10. Dev wants to buy 4 more trucks. How should Ritu describe the technicians' cost?

A Variable, because it rises with the number of trucks

B Fixed, because each technician is paid a fixed salary

C A step cost: fixed for each band of 10 trucks, rising by one salary when a new band starts

D Semi-variable, with a fixed part and a rate per kilometre

4 Within the relevant range, as output rises, what happens per unit?

- **A** Fixed cost per unit stays the same; variable cost per unit falls
- **B** Both fall
- **C** Fixed cost per unit rises; variable cost per unit stays the same
- **D** Fixed cost per unit falls; variable cost per unit stays the same

5 The premix plant's water bill is ₹84,000 when it makes 40,000 kg, of which ₹24,000 is a fixed charge. What should the bill be at 70,000 kg?

- **A** ₹1,47,000
- **B** ₹1,05,000
- **C** ₹1,29,000
- **D** ₹1,71,000

6 Which method of segregating semi-variable costs does the Study Material call the best?

- A Least squares method
- B High-low method
- C Analytical method
- D Graphical method

Answers: the answer key at the end of this lesson.

Recap

- Fixed: incurred for a period, unaffected by activity within limits (rent, insurance). Variable: varies with activity (direct material, direct labour). Semi-variable: both parts (telephone, gas, electricity) (SM 1.27–1.28).
- Step cost: flat within a band of activity, jumps at the next band; count steps and round up.
- Per unit: fixed cost per unit falls as output rises; variable cost per unit stays the same.
- Five methods in ICAI's order: graphical, high-low, analytical, comparison by period or level of activity, least squares (the best).
- High-low: variable rate = change in cost ÷ change in volume between the highest and lowest volumes; fixed = total − rate × volume; check at the other level.

- Comparison by period: change in expense ÷ change in output for the two periods given.
- Least squares: y = mx + c (m variable cost per unit, c total fixed cost).
- Split before you flex: never scale a whole semi-variable cost in proportion to output. (The rent does not double with the chai.)

Answer key · Check yourself

1 **280000**

Variable cost = (₹6,40,000 − ₹4,80,000) ÷ (₹9,00,000 − ₹5,00,000) = 40% of sales. Fixed cost = ₹6,40,000 − 40% × ₹9,00,000 = ₹2,80,000; check: ₹4,80,000 − ₹2,00,000 = ₹2,80,000.

2 **432000**

Analytical method. Variable = 40% × ₹4,80,000 = ₹1,92,000; fixed = ₹2,88,000, which does not fall. Monsoon month: ₹2,88,000 + ₹1,92,000 × 0.75 = ₹2,88,000 + ₹1,44,000 = ₹4,32,000. Cutting the whole cost by 25% (₹3,60,000) wrongly cuts the fixed part too.

3 **C** **A step cost: fixed for each band of 10 trucks, rising by one salary when a new band starts**

23 trucks need 3 technicians (23 ÷ 10 = 2.3, rounded up); 27 trucks still need 3. The cost is flat within each band of 10 trucks and jumps at the 31st truck: a step cost. It is not variable (it does not rise with each truck) and has no rate per kilometre.

4 **D** **Fixed cost per unit falls; variable cost per unit stays the same**

The fixed total is shared by more units, so fixed cost per unit falls; the variable cost per unit is the same for every unit.

5 **C** **₹1,29,000**

Variable rate = (₹84,000 − ₹24,000) ÷ 40,000 = ₹1.50 a kg; at 70,000 kg: ₹24,000 + ₹1,05,000 = ₹1,29,000. ₹1,47,000 scales the whole bill, fixed charge included; ₹1,05,000 forgets the fixed charge; ₹1,71,000 adds it on top of the scaled bill.

6 **A** **Least squares method**

"This is the best method to segregate semi-variable costs into its fixed and variable components" (SM 1.32): a statistical line of best fit, y = mx + c.

1.7 Cost control and cost reduction

Cost control keeps actual costs to the standard; cost reduction lowers the standard itself, permanently and without hurting quality. ICAI's steps, three-fold assumptions and five differences.

In plain words

The ₹6.50 cup and the ₹6.10 cup

Every kulhad of chai in Tapri's Indore cafés has a **standard cost** of ₹6.50: milk, tea, sugar and the kulhad, at the quantities and prices Ritu worked out in advance.

In August the actual cost came to ₹6.90. Ritu went looking and found that the new milk boilers were boiling over. The staff were trained, the spillage stopped, and September came back to ₹6.50. That is **cost control**: keeping actual costs in line with the standard that was set.

Then Ritu asked a different question: why ₹6.50 at all? With a three-year contract direct from a dairy co-operative and a kulhad mould that uses less clay for an equally strong cup, the cost became ₹6.10, month after month, with chai that tastes exactly the same. That is **cost reduction**: a real, permanent fall in the cost per unit without any loss of quality.

> **Kesar:** Easier way. Switch to milk powder: ₹5.20 a cup. Sab adjust ho jayega.
>
> **Ritu:** The customers will adjust too. They will walk to BrewBucks.

Kesar's milk powder lowers the cost, but it lowers the quality with it, so it is not cost reduction at all.

Think of it like

The household grocery bill

Keeping this month's groceries inside the ₹12,000 budget is control. Switching to a wholesale monthly order that buys the same groceries for ₹10,500, every month, is reduction. Buying watery dal to save money is neither: it is just a worse dinner.

Key idea

Cost control keeps costs to the standard (it accepts the standard). **Cost reduction** challenges the standard and lowers it, and counts only if the saving is in unit cost, permanent, and leaves utility and quality unaffected.

How it works

Both are objectives of a cost and management accounting system, listed by ICAI after ascertainment of cost and

determination of selling price (SM 1.4; the other objectives are in objectives and scope).

Cost control: four steps that repeat

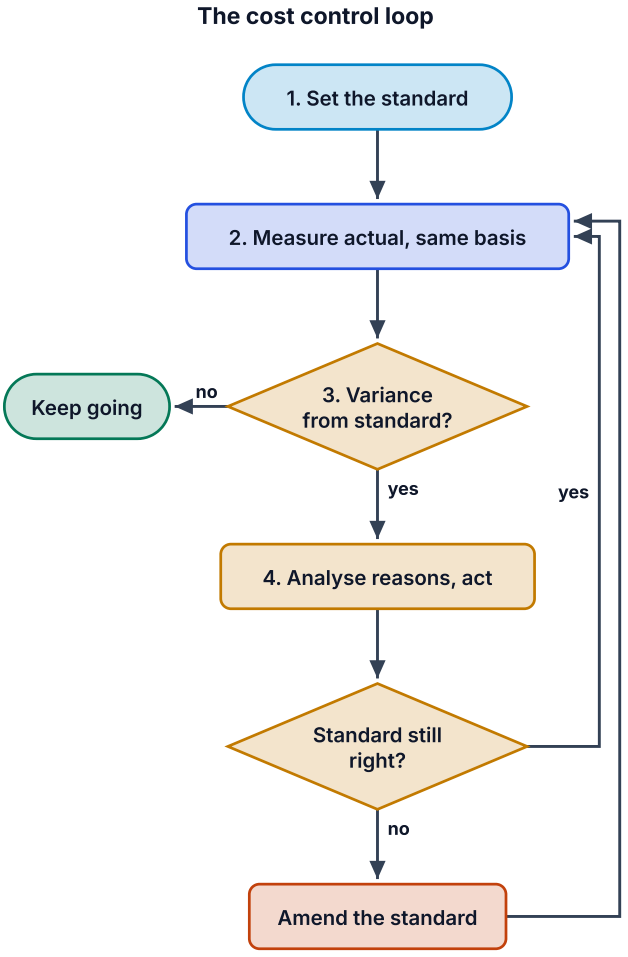
Cost control means maintaining discipline in expenditure. It ensures that expenditures are in consonance with predetermined set standard, and that any variation from these set standards is noted and reported on a continuous basis (SM 1.4). ICAI's four steps (SM 1.4–1.5):

1. **Determination of pre-determined standard or results.** Standard costs or performance targets for a cost object or a cost centre are set before production or service activity begins: the desired cost or result to be achieved.

2. **Measurement of actual performance.** The actual cost or result is measured **in the same manner in which the targets are set**: if targets are set operation-wise, actual costs are collected operation-wise, so that there is a common basis for comparison.

3. **Comparison of actual performance with set standard or target.** Any deviation (**variance**) is noted and reported to the appropriate person or authority.

4 **Analysis of variance and action.** The variance is analysed for its reasons and appropriate action is taken to ensure compliance in future. If necessary, the standards are amended to take developments into account.



ICAI's four steps (SM 1.4–1.5). Green: no variance, carry on. Amber: the variance is analysed and acted on. Orange: the standard is amended when developments make it out of date. Every path returns to measuring actual performance.

The techniques that do this in practice are **standard costing**, defined by CIMA as a "control technique that reports variances by comparing actual costs to pre-set standards so facilitating action through management by exception" (SM 13.2; see standard costing), and **budgetary control** (budgets and budgetary control).

Cost reduction: challenging the standard

> **Definition**
>
> **Cost reduction**
>
> "The achievement of real and permanent reduction in the unit cost of goods manufactured or services rendered without impairing their suitability for the use intended or diminution in the quality of the product." (SM 1.5)

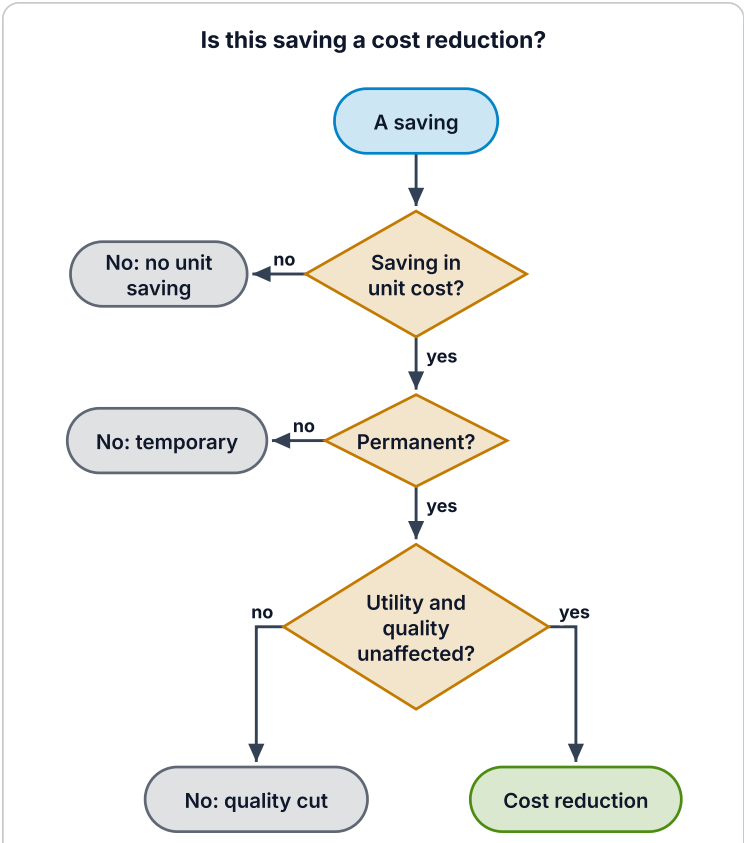
Cost reduction is an approach of management in which the cost of an object is believed to have scope for further reduction: no cost is termed as lowest, and every possibility of reduction is explored (SM 1.5). Two actions are taken:

- **(a) Value analysis.** Each activity is segmented to identify value added and non-value added activities, and all non-value added activities are eliminated without affecting the essential characteristics of the product or process. Value Chain Analysis, a strategic tool developed by Michael Porter, is one method of doing it.

- **(b) Continuous research and study** to find the most optimal way to manufacture a product or render a service (SM 1.5).

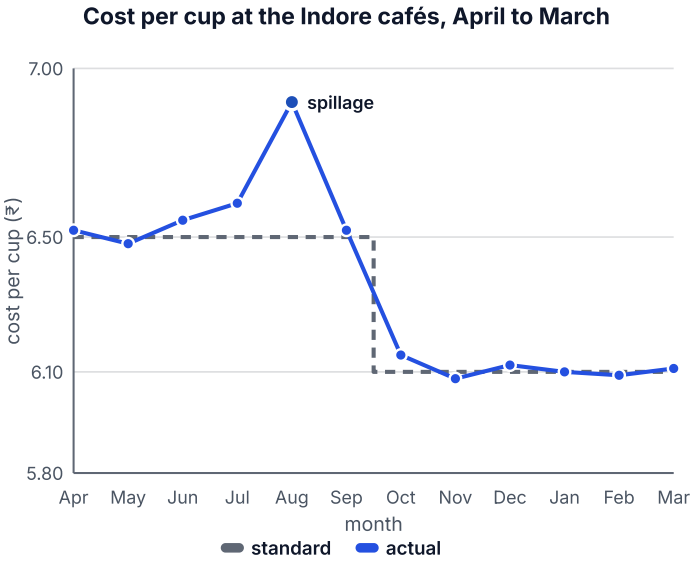
The definition rests on **three-fold assumptions** (SM 1.5–1.6):

1. There is a saving in unit cost.
2. Such saving is of permanent nature.
3. The utility and quality of the goods and services remain unaffected, if not improved.



The three-fold assumptions of SM 1.5–1.6 as three questions in ICAI's order. Green: only a saving that passes all three is cost reduction; grey: the three ways a saving fails.

How the two work together



Grey dashed: the standard, ₹6.50 until September and ₹6.10 from October. Blue: actual cost per cup. August's spillage variance is brought back by cost control; October's drop in the standard itself is cost reduction, after which control keeps actual cost near the new ₹6.10.

Cost reduction operates even when an efficient cost control system exists (SM 1.6): the cafés were already on standard in September when the cost was cut. Once a reduction is achieved, the standard is amended (step 4 of cost control), and control then guards the new, lower standard.

Cost control and cost reduction, side by side

ICAI's five points of difference, in its order (SM 1.6). *Reproduce this table from memory.*

	Cost control	Cost reduction
1. Aim	Maintaining the costs in accordance with the established standards	Reducing costs; challenges all standards and endeavours to improvise them continuously
2. Conditions	Seeks to attain lowest possible cost under existing conditions	Recognises no condition as permanent, since a change will result in lower cost
3. Emphasis	On past and present	On present and future
4. Function	A preventive function	A corrective function; operates even when an efficient cost control system exists
5. End	Ends when targets are achieved	Has no visible end; a continuous process

Worked examples

Worked example **Five savings on Ritu's desk**

Ritu has five savings to report for Tapri's café and delivery divisions. **State, with reasons**, whether each is cost control, cost reduction or neither.

(a) Route-planning software on the delivery app permanently cuts fuel per delivery by 12%; deliveries arrive as fast as before.

(b) A dairy offers 15% off milk for the monsoon months only, to clear its surplus; the milk is of the usual quality.

(c) The samosa counters switch to a cheaper frying oil; cost per samosa falls by 40 paise, and customers complain that the samosas turn soggy within an hour.

(d) A new pasteuriser uses the heat of boiled milk to pre-heat incoming milk, cutting gas per litre by 20% for good; the chai tastes the same.

(e) Gas used at the Bhopal kitchen was 18% above standard in July; a leaking regulator was found and replaced, and August's gas usage was on standard.

MODEL ANSWER

Relevant concept. Cost control maintains costs in accordance with established standards: actual performance is measured, compared with the standard, and variances are analysed and acted on (SM 1.4–1.5). Cost reduction is the achievement of

real and permanent reduction in the unit cost of goods manufactured or services rendered without impairing their suitability for the use intended or diminution in the quality of the product; it assumes a saving in unit cost, of a permanent nature, with utility and quality unaffected (SM 1.5–1.6).

(a) Cost reduction. The cost per delivery falls, the saving is permanent, and the service (speed of delivery) is unaffected: all three assumptions hold. It is the result of research into the most optimal way to render the service.

(b) Neither. The cost per cup falls during the monsoon, but the saving is not of a permanent nature, so it is not cost reduction; and it is not the result of controlling actual cost against a standard.

(c) Neither. There is a saving in unit cost, but the quality of the product is impaired, so it fails the third assumption. It is a lower-quality product, not a cost reduction.

(d) Cost reduction. The saving in gas per litre, and so per cup, is permanent and the product's quality is unaffected. The standard for gas per litre should now be revised downwards.

(e) Cost control. The variance between actual and standard gas usage was noted, analysed for its reason and corrected, bringing actual cost back to the existing standard. The standard itself did not change.

Worked example

The kulhad that got 60 paise cheaper

Tapri's kulhad unit at Khurja made 30,00,000 kulhads in 2025-26. Their cost fell from ₹4.00 to ₹3.40 each compared with the previous year. Ritu's analysis of the ₹0.60 fall:

Cause	Per kulhad (₹)
Clay prices fell this year after a bumper digging season; they are expected to return to normal next year	0.25
Research on kiln loading fits more kulhads in each firing; fuel per kulhad falls	0.20
Value analysis dropped a hand-polishing step that buyers never noticed; kulhads unchanged in use	0.10
Walls made thinner; breakage in the cafés has doubled	0.05
Total fall	**0.60**

Kesar wants the annual report to say "cost reduction of ₹18,00,000".

Required:

(i) Calculate the cost reduction per kulhad and in total, as ICAI defines it.

(ii) Comment on Kesar's figure.

THE WORKING

(i) Cost reduction

Cause	Per kulhad (₹)	Cost reduction?
Clay price fall	0.25	No: not permanent
Kiln research	0.20	Yes: permanent, quality same
Polishing dropped	0.10	Yes: non-value added step gone
Thinner walls	0.05	No: quality impaired

Cost reduction per kulhad = ₹0.20 + ₹0.10 = **₹0.30** (7.5% of ₹4.00). Total = ₹0.30 × 30,00,000 = **₹9,00,000**.

(ii) Comment. Kesar's ₹18,00,000 (₹0.60 × 30,00,000) is the total fall in cost, not cost reduction. ₹7,50,000 of it (₹0.25 × 30,00,000) comes from a temporary fall in clay prices and will reverse when prices return to normal: the saving is not of a permanent nature. ₹1,50,000 (₹0.05 ×

30,00,000) comes from thinner walls that have doubled breakage: the quality and suitability of the product are impaired, and the extra breakage in the cafés may well cost more than the saving. Only ₹9,00,000 is real and permanent reduction in unit cost without impairing quality: the result of continuous research (the kiln) and value analysis (the polishing step), the two actions ICAI names (SM 1.5). The thinner-wall change should be reversed.

Worked example

"Our control is so good, we do not need cost reduction"

At a board meeting, Kesar says: "Ritu's variance reports show every café on standard for six months. Cost control is working, so a cost reduction programme is a waste of time." **Comment.**

MODEL ANSWER

Relevant concept. Cost control aims at maintaining costs in accordance with established standards and seeks the lowest possible cost under existing conditions; it ends when targets are achieved. Cost reduction challenges all standards and endeavours to improvise them continuously; it recognises no condition as permanent, and it is a corrective function that operates even when an efficient cost

control system exists (SM 1.6). Under cost reduction no cost is termed as lowest (SM 1.5).

Facts of the case. Every café has been on standard for six months: cost control has achieved its targets.

Analysis. Being on standard shows only that actual costs equal the standards; it says nothing about whether the standards themselves are as low as they could be. Value analysis of each activity, to identify and eliminate non-value added activities, and continuous research into better methods may lower the standard cost per cup permanently without affecting quality.

Conclusion. Kesar's view is not correct. An efficient cost control system and a cost reduction programme are complementary: cost reduction operates even when cost control is efficient, and each reduction achieved becomes a revised standard that cost control then maintains.

Exam angle

In the exam

Where cost control and cost reduction appeared (new and old scheme)

In the ICAI documents read for this lesson, the distinction was asked 10 times in 10 documents, always as a descriptive question.

- **New scheme, 4–5 marks.** May 2024 Q6(a), "Distinguish between cost control and cost reduction" (5 marks); May 2026 Q6(b), the same words (5 marks); MTP May 2025 Series I Q6(d) and RTP January 2026 Q19(d): "Cost control emphasis is on past and present, while cost reduction emphasis is on present and future. DISCUSS some more differences" (4 marks in the MTP).
- **Old scheme.** December 2021 Q6(e) (5 marks); MTP July 2021 Series II Q6(a); MTP November 2023 Series II Q6(a); RTP December 2021 Q15(a); RTP May 2023 Q15(d); RTP November 2022 Q15(b).
- Every suggested answer read is ICAI's table from SM 1.6, in the same order. Where the question already gives one point, the suggested answer gives the other four (MTP May 2025).
- The chapter's own MCQs test the same table: "It ends when targets achieved" is true of cost control, and "lowest possible cost under best conditions" is the false statement, because ICAI's words are "under existing conditions" (SM 1.43–1.44).

Trick

Control looks back, reduction looks forward

Control: standard accepted, past and present, preventive, ends at the target. Reduction: standard challenged, present and future, corrective, never ends. Condition: these are ICAI's words; write them as given, including "preventive" for control and "corrective" for reduction, which are easy to swap.

Answer frame

"Distinguish between cost control and cost reduction" (5 marks)

A two-column table with ICAI's five points in ICAI's order: aim (maintain standards / challenge standards), conditions (lowest cost under existing conditions / no condition permanent), emphasis (past and present / present and future), function (preventive / corrective, operates even with efficient cost control), end (ends when targets achieved / no visible end, continuous). If time allows, add one line each: the definition of cost reduction and the four steps of cost control.

"DISCUSS some more differences" (when the question gives one)

Leave out the point already given (for example the emphasis) and write the other four. Repeating the given point earns nothing.

"Explain cost reduction" or "What are the assumptions of cost reduction?"

1. ICAI's definition, word for word.
2. The approach: no cost is termed as lowest.
3. The two actions: value analysis (value added and non-value added activities; Value Chain Analysis by Michael Porter) and continuous research and study.
4. The three-fold assumptions: saving in unit cost; permanent nature; utility and quality unaffected, if not improved.

Loses marks

"Cost control controls cost; cost reduction reduces cost."

Restates the names; no point of difference.

Passes

Three or four correct points in a list, in any order, some in the student's own words.

Correct, but misses points and ICAI's phrases.

Full marks

All five points in a two-column table, in ICAI's order, in ICAI's words ("existing conditions", "preventive", "corrective", "no visible end").

Common mistakes

⚠ **Common mistake**

Hall of Shame: the preventive and the corrective swapped

Writing "cost control is corrective, cost reduction is preventive", because control "corrects" variances. **Fix:** ICAI's table says cost control is a **preventive** function and cost reduction a **corrective** function that operates even when an efficient cost control system exists (SM 1.6), and the suggested answers of May 2024 and May 2026 print exactly these words. Write them as ICAI gives them, whatever your intuition says.

⚠ **Common mistake**

Calling every saving a cost reduction

A one-month discount, a fall in market price or a cheaper, worse material counted as cost reduction. **Fix:** test all three assumptions: a saving in unit cost, of a permanent nature, with utility and quality unaffected. Fail one, and it is not cost reduction.

⚠ **Common mistake**

"Under best conditions"

Writing that cost control seeks the lowest possible cost "under best conditions" or "ideal conditions". **Fix:** ICAI's words are "under **existing** conditions"; it is cost reduction that refuses to accept existing conditions as permanent.

Practice

Check yourself

1 Case scenario. In August, sugar cost per cup at Tapri's Indore cafés was ₹0.62 against a standard of ₹0.50. Ritu traced it to oversized scoops at two cafés, the scoops were replaced, and September's sugar cost was ₹0.50 a cup. Which statement is correct?

A It is cost reduction, because the cost per cup fell by ₹0.12

B It is both cost control and cost reduction

C It is cost control: a variance from the existing standard was analysed and corrected

D It is neither, because the standard was not changed

2 Ritu's two teams describe their work. Team P: "We take this year's recipes, suppliers and machines as they are, and keep every café's cost per cup at the standard set for them." Team Q: "We treat no recipe, supplier or machine as permanent, and test changes that would lower the cost per cup from next year on, at the same taste." Which description is correct?

- **A** P does cost reduction and Q does cost control, because Q works with next year's figures
- **B** P does cost control (lowest cost under existing conditions, emphasis on past and present); Q does cost reduction (no condition permanent, emphasis on present and future)
- **C** Both do cost control, because both compare costs with a standard
- **D** Both do cost reduction, because both aim at a lower cost per cup

3 The cost of a premix pack fell from ₹20.00 to ₹18.50. Of the fall, ₹0.90 comes from a permanent redesign of the blending process with no change in quality, and ₹0.60 from a supplier's one-year introductory discount. Output is 4,00,000 packs. What is the cost reduction for the year, as ICAI defines it, in rupees?

Answer: _____

4 According to ICAI, cost control is a _____ function and cost reduction is a _____ function.

- **A** corrective; preventive
- **B** preventive; corrective
- **C** preventive; preventive
- **D** corrective; continuous

5 Which of these are among ICAI's three-fold assumptions in the definition of cost reduction? Select all that apply.

- **A** There is a saving in unit cost
- **B** The saving is achieved within the budget period
- **C** The saving is of permanent nature
- **D** The utility and quality of the goods and services remain unaffected, if not improved

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Cost control: maintaining discipline in expenditure, so that expenditures are in consonance with predetermined standards and variations are noted and reported continuously.
- Four steps of cost control, in order: set the standard, measure actual on the same basis, compare and report the variance, analyse and act (amend the standard if necessary).
- Cost reduction: real and permanent reduction in unit cost without impairing suitability for the use intended or diminution in quality.

- Three-fold assumptions: saving in unit cost; permanent; utility and quality unaffected, if not improved.
- Two actions: value analysis (eliminate non-value added activities; Porter's Value Chain Analysis) and continuous research and study.
- Five differences: maintain vs challenge standards; existing conditions vs no condition permanent; past and present vs present and future; preventive vs corrective; ends at target vs no visible end.
- Reduction lowers the standard; control then guards the new standard. (The ₹6.50 cup became the ₹6.10 cup.)

Answer key · Check yourself

1 **C** **It is cost control: a variance from the existing standard was analysed and corrected**

Actual cost was brought back to the existing standard after the variance was analysed and acted on: cost control. Cost reduction would have lowered the ₹0.50 standard itself.

2 **B** **P does cost control (lowest cost under existing conditions, emphasis on past and present); Q does cost reduction (no condition permanent, emphasis on present and future)**

ICAI's table (SM 1.6): cost control maintains costs in accordance with established standards and seeks the lowest possible cost under existing conditions, with emphasis on past and present. Cost reduction recognises no condition as permanent and challenges the standards, with emphasis on present and future. Team Q also keeps the taste, so it meets the quality test of the definition.

3 **360000**

Only the permanent saving with quality unaffected counts: ₹0.90 × 4,00,000 = ₹3,60,000. The ₹0.60 discount is not of a permanent nature.

4 **B** **preventive; corrective**

ICAI's table: cost control is a preventive function; cost reduction is a corrective function (SM 1.6). Swapping them is the commonest slip.

5 **A There is a saving in unit cost; C The saving is of permanent nature; D The utility and quality of the goods and services remain unaffected, if not improved**

The three assumptions are a saving in unit cost, of permanent nature, with utility and quality unaffected (SM 1.5–1.6). Nothing ties cost reduction to a budget period: it has no visible end.

1.8 Methods and techniques of costing, and digital costing

Which method of costing fits which kind of output, the techniques that work with any method, how IT and ERP changed cost accounting, and what a digital costing system collects and delivers.

In plain words

One company, seven ways to collect costs

Ritu has drawn Tapri's businesses on the whiteboard, and Kesar is not happy.

> **Kesar:** Why do you need seven different cost sheets? Cost is cost. Make one format for everything.
>
> **Ritu:** A wedding order and a tanker of premix are different animals, Kesar-ji. One is finished once, to one customer's menu. The other never stops flowing.

The wedding catering order is costed **as a job**: every rupee spent on it is collected on that order. The Diwali gift boxes are made 5,000 identical boxes at a time, so each lot is costed **as a batch** and divided by 5,000. The chai premix flows through blending, roasting and packing, so costs are collected **stage by stage, process by process**. The new headquarters that Tapri's engineers are building for two years is costed **as a contract**. The cold-chain trucks sell a service, not a product, so they are costed per tonne-kilometre (**operating costing**). The

kulhad-making machine Tapri builds for its franchisees has parts made in batches and an assembly line at the end, so it needs **a mix of methods**. The way the output is produced decides the **method of costing**.

Then Dev joins in. He wants to see the premix cost split into fixed and variable, the plant's actual cost compared with the standard, and the full cost charged to every pack for valuing stock. These are not new methods: they are **techniques**, ways of looking at and using the costs, and any of them can be applied whichever method collected the costs.

Finally, Kesar's favourite: the delivery app. It now reports the cost of every order as it happens, links the plant with suppliers and customers, and predicts next week's demand. That is **digital costing**: the same costing ideas, with a different way of collecting, storing, analysing and reporting the data.

Think of it like

Containers and lenses

A method is the container you collect costs in: a job bag, a batch tray, a process bucket, a contract file. A technique is the lens you look at the contents through: marginal, standard, absorption. Any lens works on any container. Digital costing replaces the paper and the monthly wait with sensors, an ERP and real time.

Key idea

The **method** follows the nature of the output (job, batch, contract, process, single output, service, or a mix). A **technique** (uniform, marginal, standard, historical, absorption) is used with any method. **Digital costing** changes how cost data is collected, stored, analysed and reported, and links the entity with its suppliers, customers and market.

How it works

Methods of costing: what the output looks like

Different industries follow different methods of costing because of the differences in the nature of their work (SM 1.36). ICAI's list, in its order:

Method
Single or output costing

What it costs (ICAI)
The cost of a product that is the only one produced

Typical industries
Bricks, coal

Method
Batch costing

What it costs (ICAI)
An extension of job costing: each batch is a unit of cost; cost per unit = cost of the batch ÷ units in the batch

Typical industries
Small orders passed through in batches

Method

Job costing

What it costs (ICAI)

Cost of each job ascertained separately; work undertaken on receiving a customer's order

Typical industries

Printing press, motor workshop

Method

Contract costing

What it costs (ICAI)

Cost of each contract ascertained separately

Typical industries

Bridges, roads, buildings

Method

Process costing

What it costs (ICAI)

Cost of completing each stage of work, e.g. pulp, then paper from pulp; in mechanical operations, cost of each operation (**operation costing**)

Typical industries

Paper, sugar

Method

Operating costing

What it costs (ICAI)

Concerns rendering services

Typical industries

Transport, supply of water, retail trade

Method

Multiple costing

What it costs (ICAI)

A combination of two or more of the above, e.g. bicycle parts by job or batch costing and assembly by single or output costing

Typical industries

Bicycles

(SM 1.36–1.37.) ICAI summarises them by the nature of output (SM 1.37–1.38):

Nature of output

A series of processes

Method

Process or operation costing

Cost

For each process

Example

Sugar

Nature of output

Construction of building

Method

Contract costing

Cost

For each contract

Example

Real estate

Nature of output

Similar units of a single product, by a single process

Method

Unit or single or output costing

Cost

For the entire activity, averaged for the output

Example

Cold drinks

Nature of output

Rendering of services

Method

Operating costing

Cost

For all services

Example

Hospitals

Nature of output

Customer specifications: single unit

Method

Job costing

Cost

For each order, assignment or job

Example

Advertising

Nature of output

Multiple varieties of activities and processes

Method

Multiple costing

Cost

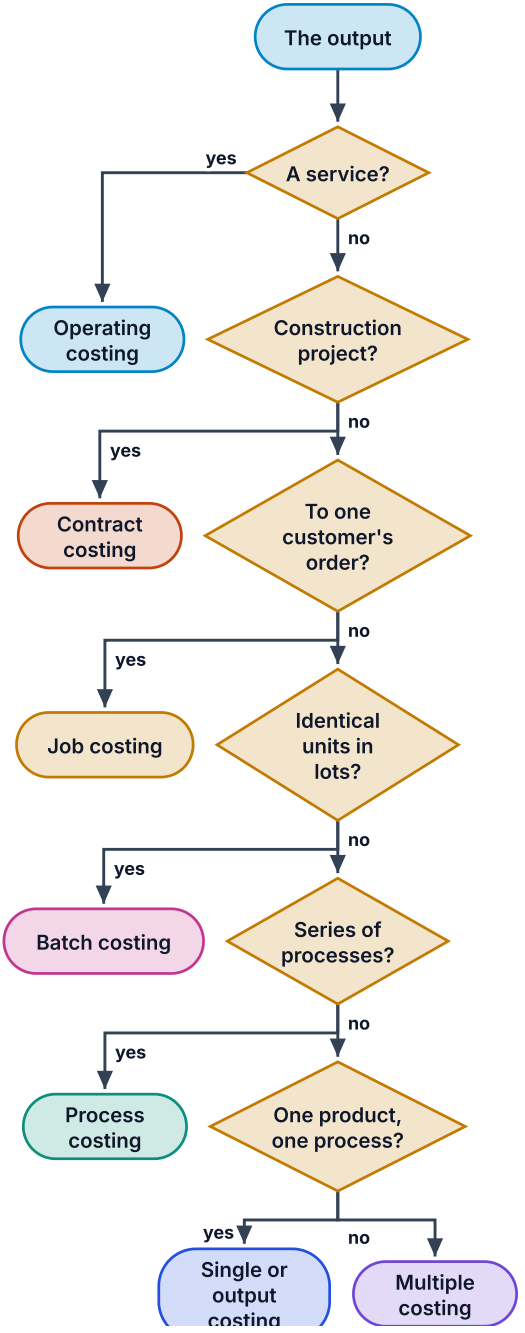
Combination of any method

Example

Car assembly

Industries also follow **Activity Based Costing**, an accounting methodology that assigns costs to activities rather than products or services: cost is identified with each cost-driving activity, which becomes the basis for apportioning costs over cost objects, jobs, products, customers or services (SM 1.38; see activity based costing).

Which method of costing?


The output
A service?
yes
no
Operating costing
Construction project?
no
yes
Contract costing
To one customer's order?
no
yes
Job costing
Identical units in lots?
no
yes
Batch costing
Series of processes?
no
yes
Process costing
One product, one process?
yes
no
Single or output costing
Multiple costing

A rough guide built from ICAI's descriptions (SM 1.36–1.38). Each method has its own colour. Multiple costing is for output that combines several of these, such as parts made in batches and assembled on a line.

Techniques of costing: how costs are looked at

"For ascertaining cost, following types of costing are usually used" (SM 1.38):

Technique	ICAI's description
Uniform costing	A number of firms in an industry agree among themselves to follow the same system of costing in detail, adopting common terminology for various items and processes
Marginal costing	The ascertainment of marginal cost by differentiating between fixed and variable costs; used to ascertain the effect of changes in volume or type of output on profit
Standard costing and variance analysis	Standard costs are pre-determined and subsequently compared with the recorded actual costs: a technique of cost ascertainment and cost control. It may be used with any method of costing, and is especially suitable for standardised goods of repetitive nature

Technique	ICAI's description
Historical costing	The ascertainment of costs after they have been incurred; limited utility. **Post costing**: after production is completed. **Continuous costing**: as soon as the job is completed, or even while the job is in progress
Absorption costing	Charging all costs, both variable and fixed, to operations, processes or products; differs from marginal costing, where fixed costs are excluded

(SM 1.38–1.39.) The two advantages of **uniform costing**: (i) the performance of each firm can be compared with that of another, or with the average performance in the industry; (ii) the cost of production of goods which is true for the industry as a whole can be determined, which is found useful when tax-relief or protection is sought from the Government (SM 1.38–1.39).

Information technology in cost accounting

The introduction of **Enterprise Resource Planning (ERP)** systems changed cost and management accounting dramatically, and Industry 4.0 puts more emphasis on digitisation and automation of business processes for better control over cost (SM 1.17). ICAI lists the impact of IT as follows (SM 1.17–1.18):

1. **Integration.** After ERPs, functional activities are integrated: a single entry into the accounting system provides custom-made reports for every purpose, and

reconciling the results of cost and financial accounts becomes simpler and less cumbersome.

2. **Paperless environment.** Documents like the Bill of Material, Material Requisition Note, Goods Received Note and labour utilisation report are no longer prepared in multiple copies; each department gets an e-copy from the system.
3. **Procurement and mobilisation.** Internet, intranet and extranet help: production gets materials from stores without a physical material requisition note, and purchase orders go to suppliers over the extranet. This enables a shift towards Just-in-Time (JIT) inventory management and production.
4. **Accurate and timely cost information.** Each cost centre and cost object is codified, and costs are assigned by code: accumulation and ascertainment are automated, and information can be customised job-wise, batch-wise, process-wise or cost centre-wise.
5. **Uniformity** in preparing reports, budgets and standards, irrespective of location, currency, language and regulations.
6. **Real-time variance reports** of cost and revenue, so management can take control measures immediately.
7. **Elimination of non-value added activities**, by monitoring and analysing each process closely.

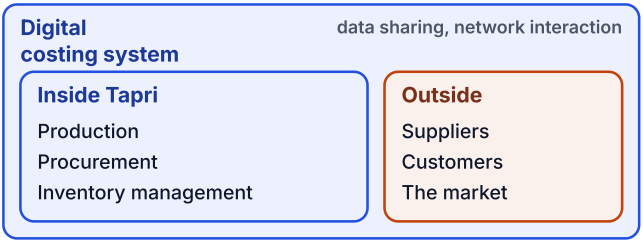
The digital costing system

Definition

Digital costing system

Like the conventional cost accounting system, a digital costing system collects data, classifies it and accounts for it to get and process information for decisions; the difference is the method of collection, medium of storage, forms of analysis and reporting. It links business functions such as production, procurement and inventory management with the digital costing systems of its suppliers, customers and the market, through data sharing and network interaction. (SM 1.18)

What a digital costing system links



SM 1.18: the system links the entity's own functions (blue) with the digital costing systems of its suppliers, customers and the market (orange).

It provides data to get six kinds of information (SM 1.18–1.19): (i) cost incurred on a cost object; (ii) data on time spent; (iii) data on resource consumption; (iv) data on current market price of the final product and raw

materials; (v) data on lead time and availability of materials; (vi) data on product demand and trend.

With **Artificial Intelligence (AI)** and **Machine Learning (ML)**, which help analyse **Big Data** and apprehend consumption and demand patterns, it delivers eight benefits (SM 1.19):

1. **Ascertainment of cost with certainty** on a cost object, which helps to analyse activities for cost allocation and apportionment.
2. **Analysis of time spent** on each activity, to study and formulate incentive plans.
3. **Material requirement planning** and scheduling of procurement; resource consumption data analysed for resource optimisation, zero wastage and Just-in-Time.
4. **Identification and elimination of non-value-added activities.**
5. **Setting standards and measuring variances on a real-time basis**, from resource consumption data.
6. **Estimating cost and setting standards on a Marked to Market (M2M) basis**, from current market prices of materials and consumables.

7 **Predicting market demand** by extrapolating data on customer behaviour, for budgets and production planning.

8 **Better analysis of cost behaviour**, which improves cost-benefit analysis and equips management for informed decision making.

Worked examples

Worked example **Six businesses, six methods**

Tapri asks you to recommend a method of costing and a cost unit for each of these operations, with reasons:

(a) A chai-masala unit: whole spices are cleaned, roasted, ground and packed, the output of each stage passing to the next as its input.

(b) A printing unit that prints menu cards to each café owner's own design.

(c) Chai biscuits, baked in batches of 20,000 identical packets.

(d) A cold-storage warehouse at Nagpur that Tapri's engineering wing is building for a client over 18 months.

(e) A chain of 12 highway motels.

(f) Electric delivery scooters: frames are made in the workshop in lots, motors are bought in, and each

scooter is assembled on a line.

MODEL ANSWER

Operation	Method	Cost unit
(a) Chai masala	Process costing	Kilogram
(b) Menu cards	Job costing	Each job
(c) Biscuits	Batch costing	Batch, then per packet
(d) Warehouse	Contract costing	Each contract
(e) Motels	Operating costing	Room-day
(f) Scooters	Multiple costing	Number of scooters

Reasons:

(a) The output passes through a series of processes (cleaning, roasting, grinding, packing); process costing ascertains the cost of completing each stage of work (SM 1.37). A product sold by weight is costed per kilogram, as chemicals are (SM 1.20).

(b) Work is undertaken on receiving each customer's order, to the customer's specification:

the cost of each job is ascertained separately.

(c) Identical units are made in lots; each batch is treated as a unit of cost, and cost per packet = cost of the batch ÷ number of packets in the batch (SM 1.36).

(d) Construction of a building: the cost of each contract is ascertained separately.

(e) The motels render a service: operating costing.

(f) Frames are costed by batch costing and the assembly by single or output costing; the combination is multiple costing, as in ICAI's bicycle example (SM 1.37).

Cost units: the Study Material's cost unit for hotels is the room or meal (SM 1.21); ICAI's suggested answers also accept guest days or room days (MTP May 2022 Series II, old scheme).

Worked example

"We already have accounting software"

Kesar tells the board: "Our accounting software gives a profit and loss account every month. A digital costing system linked to our suppliers and customers is an expensive toy."

Ritu has to reply. **Explain** how a digital costing system differs from the conventional one, and **describe five**

benefits it would bring Tapri.

MODEL ANSWER

What it is. Like the conventional cost accounting system, a digital costing system collects data, classifies it and accounts for it to get and process information for decisions. The difference is the method of collection, the medium of storage, and the forms of analysis and reporting. It links business functions such as production, procurement and inventory management with the digital costing systems of suppliers, customers and the market through data sharing and network interaction (SM 1.18). A monthly profit and loss account reports the results of the whole entity after the event; a digital costing system provides data on the cost of each cost object, time spent, resource consumption, current market prices, lead times and availability of materials, and product demand and trend.

Benefits for Tapri (with AI and machine learning analysing Big Data on consumption and demand; SM 1.19):

1. **Cost with certainty on each cost object**: the cost of each café, delivery route and premix batch, which also helps analyse activities for allocating and apportioning overheads.
2. **Material requirement planning**: milk, tea and kulhad procurement scheduled from consumption data, moving towards zero wastage and Just-in-Time.

3. **Identification and elimination of non-value-added activities** in the plant and the delivery chain.
4. **Standards and variances on a real-time basis**, so a cost overrun at a café is seen the same day, not at the month-end; and standards set on a Marked to Market basis from current prices of milk and tea.
5. **Prediction of market demand** by extrapolating customer behaviour data from the delivery app, for festive budgets and production planning.

Conclusion. The accounting software tells the board what profit was made; a digital costing system tells managers what each activity costs and why, while there is still time to act.

Worked example **Name the technique**

Identify the technique of costing in each case.

(i) Twelve chai-premix makers in an industry association agree to follow the same costing system in detail, with common terms for items and processes.

(ii) The cost of a menu-card printing job is worked out while the job is still on the press.

(iii) Each week the biscuit unit compares its actual flour usage and price with figures set in advance for efficient operation, and analyses the differences.

(iv) To judge a 20% price cut on festive packs, Ritu separates fixed and variable costs and studies how profit changes with volume.

(v) The motel chain charges all its costs, including the fixed rent and depreciation, to room-days.

(vi) After the season's last batch of mango chai is sold, Ritu works out what the season's production cost.

MODEL ANSWER

Case
(i)

Technique
Uniform costing

ICAI's description, applied
Firms in an industry follow the same system in detail with common terminology; it allows inter-firm comparison and an industry cost when tax relief or protection is sought

Case
(ii)

Technique
Historical costing: continuous costing

ICAI's description, applied
Cost ascertained as soon as the job is completed or even while it is in progress

Case
(iii)

Technique
Standard costing and variance analysis

ICAI's description, applied
Standard costs pre-determined and compared with recorded actual costs

Case
(iv)

Technique
Marginal costing

ICAI's description, applied
Fixed and variable costs differentiated to see the effect of a change in volume on profit

Case
(v)

Technique
Absorption costing

ICAI's description, applied
All costs, variable and fixed, charged to operations, processes or products

Case
(vi)

Technique
Historical costing: post costing

ICAI's description, applied
Cost ascertained after production is completed

(SM 1.38–1.39.)

Exam angle

In the exam

Where methods, techniques, IT and digital costing appeared (new and old scheme)

In the ICAI documents read for this lesson, these topics came up in 15 questions in 13 documents, all descriptive, usually 4 or 5 marks.

- **Choose the method, new scheme.** MTP January 2025 Series I Q6(d) (4 marks): identify the method from a description and give one example of an industry. MTP January 2026 Series I Q6(a) (5 marks): recommend the method for five businesses (cement, custom furniture, batch pharmaceuticals, packaged snacks, a bus company), "explaining your reasoning". MTP September 2025 Series II Q6(a) (5 marks): complete ICAI's nature-of-output table (method, cost, example).
- **Choose the method, old scheme.** November 2020 Q6(d) (ten industries), January 2021 Q6(d) (five industries), May 2022 Q6(e) (identify the method from five statements), MTP May 2022 Series II Q6(b) (method) and Q6(c) (unit of cost).
- **Techniques.** December 2021 Q6(a), old scheme (5 marks): "Briefly explain the techniques of costing".
- **IT and digital costing.** September 2024 Q6(a) (5 marks): "Describe any five benefits of the Digital Costing System". MTP May 2024 Series I Q6(b) (5 marks): "EXPLAIN the impact of Information Technology in Cost Accounting". Old scheme:

January 2021 Q6(e) ("Give any five examples of the impact"), MTP May 2023 Series II Q6(d), MTP November 2023 Series I Q6(d) and RTP May 2022 Q14(d) ("DISCUSS the impact").

- The suggested answers on IT and digital costing reproduce ICAI's lists (SM 1.17–1.19) almost word for word.

In the exam

Answers ICAI has accepted that look surprising

- Assembly of bicycles: "Unit/Single/Output/Multiple Costing"; car manufacturing: "Multiple Costing" (November 2020, old scheme). The Study Material's own bicycle example is multiple costing (SM 1.37): write multiple costing, with the reason.
- Biscuits: batch costing (November 2020); pharmaceuticals: batch costing (MTP January 2026); bakery: batch costing (MTP January 2025).
- Cement: "Unit/Output Costing" (MTP January 2026) and "Unit/Single/Output Costing" (November 2020).
- "Nature of the product is complex and method cannot be ascertained": multiple costing (May 2022, old scheme).

Trick

Read the output, not the industry name

The same industry can need different methods: custom furniture to a client's design is job costing, but identical chairs made in lots of 500 would be batch costing. Ask how the output is produced (to order, in lots, in a continuous flow, as a service, or a combination), then name the method. Condition: when the question names only the industry, use ICAI's standard pairing.

Answer frame

"State the method of costing and the suggested unit of cost"

A three-column table, one row per industry. ICAI's pairings from the Study Material and suggested answers:

Industry	Method	Cost unit
Transport	Operating (service) costing	Passenger-km, tonne-km
Power, electricity supply	Operating (service) costing	Kilowatt-hour (kWh)
Hospital	Operating (service) costing	Patient day

Industry	Method	Cost unit
Hotel	Operating (service) costing	Room, meal
Bridge, road, building	Contract costing	Each contract
Printing, interior decoration, advertising	Job costing	Each job
Bricks	Single or output costing	1,000 bricks
Coal	Single or output costing	Tonne
Cement	Single or output costing	Ton, per bag
Chemicals, oil refinery	Process costing	Litre, kilogram, ton; barrel
Biscuits, pharmaceuticals, bakery	Batch costing	Batch
Bicycles, car assembly	Multiple costing	Number

(Methods: SM 1.36–1.38 and the suggested answers in Exam angle; cost units: SM 1.20–1.21, and cost units.)

"Describe any five benefits of the Digital Costing System" (5 marks)

Open with a two-line definition (collects, classifies and accounts data like the conventional system; differs in collection, storage, analysis and reporting; links production, procurement and inventory with suppliers, customers and the market). Then five of the eight benefits, each with ICAI's key phrase: **cost with certainty** on a cost object; **time spent** for incentive plans; **material requirement planning** and JIT; **non-value-added activities** eliminated; **real-time** standards and variances; **Marked to Market** standards; **demand prediction** for budgets; **cost behaviour** for cost-benefit analysis.

"Explain the impact of IT in cost accounting" (5 marks)

ICAI's seven points in order: integration through ERP and simpler reconciliation; paperless documents; procurement over intranet and extranet, enabling JIT; codified, accurate and timely cost information, customised job-wise or process-wise; uniformity of reports, budgets and standards; real-time variance reports; elimination of non-value-added activities. Five well-explained points earn the marks.

Loses marks

"Bicycle: unit costing."

The industry name was read, not the output: parts and assembly need different methods.

Passes

"Bicycle: multiple costing."

Correct, but a bare name; an "explain your reasoning" question gives no marks for the reason.

Full marks

"Bicycle: multiple costing. The parts are costed by job or batch costing and the assembly by single or output costing; the combination is multiple costing. Cost unit: number of bicycles."

Common mistakes

⚠ **Common mistake**

Hall of Shame: marginal costing listed as a method

"Methods of costing: job, process, marginal, standard..." mixes the two lists. **Fix:** methods are single or output, batch, job, contract, process, operating and multiple costing (SM 1.36–1.37); uniform, marginal, standard, historical and absorption costing are techniques (SM 1.38–1.39).

⚠ **Common mistake**

Operating costing and operation costing treated as one

Writing "operation costing" for a bus company, or "operating costing" for a machine shop. **Fix:** operating costing is for concerns rendering services (transport, water supply, retail trade); operation costing is process costing applied to each mechanical operation (SM 1.37). ICAI's January 2021 (old scheme) answer printed "Operation/ Service Costing" for an airline; write the Study Material's term, operating (service) costing.

⚠ **Common mistake**

Digital costing answered as "computerised accounts"

An answer that says only "accounts are kept on a computer, so they are faster". **Fix:** use ICAI's points: links with suppliers, customers and the market; AI and machine learning on Big Data; real-time standards and variances; Marked to Market standards; demand prediction; elimination of non-value-added activities (SM 1.18–1.19).

Practice

Check yourself

1 Case scenario. Tapri's herbal-tea unit blends sachets in lots of 50,000 identical sachets. Each lot gets a lot number, its costs are collected separately, and the cost of a sachet is the lot's cost divided by 50,000. Dev asks Ritu which method of costing this is. Which answer is correct?

- **A** Job costing, because each lot is a separate order
- **B** Batch costing, because each lot of identical units is the unit of cost
- **C** Process costing, because blending is a process
- **D** Multiple costing, because several sachets are made

2 Which of these is a technique of costing, not a method?

- **A** Contract costing
- **B** Batch costing
- **C** Marginal costing
- **D** Multiple costing

3 A software firm, a cement plant and a chain of hospitals ask which method suits them. Which pairing is correct?

- **A** Cement plant: job costing
- **B** Hospitals: operating costing
- **C** Software firm: process costing
- **D** Cement plant: multiple costing

4 Which of these are benefits of a digital costing system as ICAI lists them? Select all that apply.

- **A** Setting standards and measuring variances on a real-time basis
- **B** Estimating cost and setting standards on a Marked to Market basis
- **C** Removing the need to classify cost data
- **D** Predicting market demand from data on customer behaviour

More than one answer may be right.

5 Ten kulhad makers in Khurja adopt the same costing system and common terms, so that each can compare its cost with the average of the group. Which technique is this, and which other advantage does ICAI give for it?

- **A** Standard costing; it controls cost through variances
- **B** Uniform costing; it gives a cost of production true for the industry as a whole, useful when seeking tax relief or protection from the Government
- **C** Marginal costing; it shows the effect of volume on profit
- **D** Multiple costing; it combines several methods

Answers: the answer key at the end of this lesson.

Recap

- Methods (SM 1.36–1.37): single or output, batch, job, contract, process (operation), operating, multiple. The method follows the nature of the output.
- Batch cost per unit = cost of the batch ÷ units in the batch; operation costing = process costing for mechanical operations.

- Techniques (SM 1.38–1.39): uniform, marginal, standard costing and variance analysis, historical (post, continuous), absorption. Standard costing can be used with any method.
- Uniform costing's two advantages: inter-firm comparison; an industry cost for tax relief or protection.
- IT impact (SM 1.17–1.18): ERP integration, paperless, extranet procurement and JIT, codified timely cost, uniformity, real-time variances, NVA elimination.
- Digital costing (SM 1.18): same functions, different collection, storage, analysis and reporting; links production, procurement and inventory with suppliers, customers and the market.
- Its benefits (SM 1.19): cost with certainty, time data for incentives, MRP and JIT, NVA elimination, real-time standards, M2M standards, demand prediction, cost behaviour analysis. (Containers, lenses, and now sensors.)

Answer key · Check yourself

1 **B Batch costing, because each lot of identical units is the unit of cost**

A group of identical items costed together, with cost per unit = cost of the batch ÷ units in the batch, is batch costing (SM 1.36). Job costing is for work to a customer's order; process costing collects cost stage by stage for a continuous flow.

2 **C Marginal costing**

Marginal costing is a technique (SM 1.39). Contract, batch and multiple costing are methods (SM 1.36–1.37).

3 **B Hospitals: operating costing**

Rendering of services, such as hospitals, uses operating costing (SM 1.38). Cement is costed by unit or output costing in ICAI's suggested answers. A software firm working to client orders would use job costing.

4 **A Setting standards and measuring variances on a real-time basis; B Estimating cost and setting standards on a Marked to Market basis; D Predicting market demand from data on customer behaviour**

SM 1.19 lists the first, second and fourth. A digital costing system still collects, classifies and accounts for data like a conventional one (SM 1.18); only the method of collection, storage, analysis and reporting differs.

5 **B Uniform costing; it gives a cost of production true for the industry as a whole, useful when seeking tax relief or protection from the Government**

Firms in an industry following the same system with common terminology is uniform costing. Its two advantages: inter-firm comparison, and an industry cost useful when tax relief or protection is sought (SM 1.38–1.39).

CHAPTER 2

Cost Sheet

2.1 Cost sheet for a manufacturer

The cost sheet stacks costs by function in ICAI's order: prime cost, works cost, cost of production, cost of goods sold and cost of sales, with stocks adjusted at the right step.

In plain words

Kesar's favourite word: miscellaneous

Tapri's chai premix sells for ₹150 a box, and Kesar wants to know what a box really costs. He hands Ritu a bag of bills: milk powder, wages, electricity, a supervisor's salary, the auditor's fee, a TV advertisement, foil pouches, cartons, diesel.

> **Kesar:** Total them and divide by boxes. Whatever does not fit, put under miscellaneous. Sab adjust ho jayega.
>
> **Ritu:** Per unit, please, but in layers. In one heap I cannot tell you whether the factory is expensive or the advertising is.

Ritu sorts every bill by the **function** it served: making the premix, running the company, selling it, or getting it to the shop. Then she stacks the piles in a fixed order. The costs that go straight into a box (milk powder, the mixing

team's wages, the royalty on the recipe) come to ₹80. The factory's indirect costs take it to ₹100. Quality checks, the production office and the foil pouch, less the scrap sold, take it to ₹110: the cost to *make* a box. The general office, selling and delivery add ₹20: ₹130 to *make and sell* it, and Kesar keeps ₹20.

That stacked statement is a **cost sheet**. Each subtotal has a name ICAI expects you to write, and each answers its own question: is the factory efficient (works cost), what does a finished box cost (cost of production), what does it cost in a customer's hands (cost of sales)?

Think of it like

The four-tier tiffin

A cost sheet is a steel tiffin carrier. The bottom tier is prime cost, the next adds the factory's overheads, the third what it takes to finish and pack a box, and the top the office, the salesman and the delivery boy. You can open any tier and see its total, but you build it from the bottom up, and nothing from the top tier may fall into the lower ones.

Key idea

A cost sheet classifies cost by function and builds it up in a fixed order: prime cost → works cost → cost of production → cost of goods sold → cost of sales. Each stock is adjusted at the tier where it sits: raw material in materials consumed, work-in-process in works cost, finished goods in cost of goods sold.

How it works

ICAI's cost sheet presents cost information "on the basis of functional classification" (SM 6.2). **Functional classification** divides costs by the function for which they are incurred: production (manufacturing), administration, selling, distribution, and research and development. They are grouped into four **cost heads**: prime cost, cost of production, cost of goods sold and cost of sales (SM 6.2).

Where each function lands in the cost sheet

Production direct costs, factory overheads	prime cost, works cost
Administration (production) production planning, factory office	cost of production
R&D (process) improving the process	cost of production
Administration (general) board, finance, audit	cost of sales
Selling, distribution ads, salesmen, delivery	cost of sales

Top to bottom in the order the cost sheet adds them; the right-hand column names the head each function joins.

Building the tiers, one step at a time

1 **Direct materials consumed** = opening stock of raw material + purchases − closing stock. Freight inwards and other procurement costs are added, trade discounts deducted, and duties and taxes added only if input tax credit is not available (SM 6.3). More in valuing material receipts.

2 **Direct employee (labour) cost**: wages, allowances, overtime, bonus, the employer's contribution to provident and similar funds, other benefits (SM 6.3–6.4).

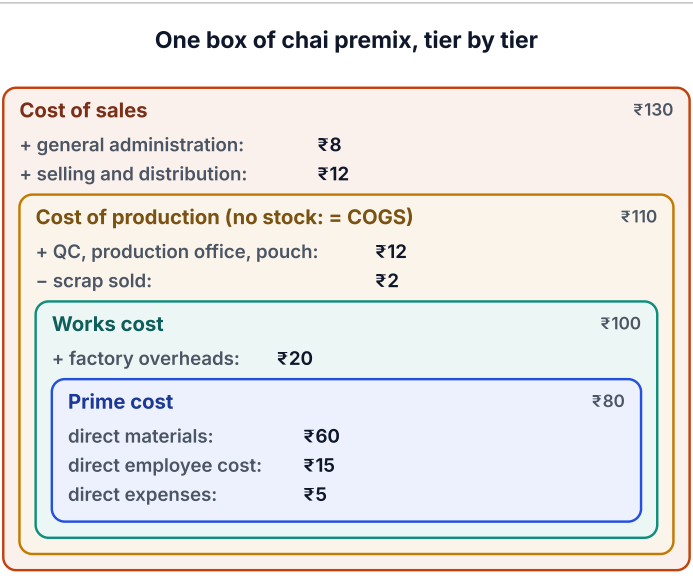
3 **Direct expenses**: power and fuel, royalty on production, hire of specific equipment, know-how fees, amortised moulds and patterns, product-specific design and software (SM 6.4). Steps 1 to 3 make **prime cost**.

4 **Works (factory) overheads** give **gross works cost**: consumable stores, depreciation, repairs and insurance of factory and plant, lease rent of production assets, supervisors, service departments such as the tool room (SM 6.5). Add opening and deduct closing work-in-process: **works (factory) cost**.

5 **Cost of production** = works cost + quality control + R&D (process related) + administration related to production − credit for recoveries (scrap, waste, by-products) + primary packing (SM 6.4–6.6).

6 **Cost of goods sold** = cost of production + opening − closing finished goods (SM 6.6).

7 **Cost of sales** = cost of goods sold + general administration + selling + secondary packing + distribution (SM 6.6–6.7); interest, where given, as a separate item (SM 6.10). Sales less cost of sales is **profit**.



Read from the innermost box outwards: each tier is the one inside it plus its own rows. Price ₹150 less cost of sales ₹130 leaves ₹20 a box.

The awkward items

Some bills look like costs and are not; some reduce cost. ICAI's rules, in ICAI's words (SM 6.5–6.10), are also the model answers to "explain the treatment of":

Item	Treatment in the cost sheet
Credit for recoveries	The realised or realisable value of scrap or waste is deducted, as it reduces the cost of production
Packing cost (primary)	Packing material essential to hold and preserve the product for its use by the customer: part of cost of production

Item	Treatment in the cost sheet
Joint products and by-products	Joint costs are allocated on a rational and consistent basis; the net realisable value of by-products is deducted from the cost of production
Quality control cost	The cost of resources consumed towards quality control procedures: part of cost of production
Abnormal costs	Where material and quantifiable (e.g. costs of a pandemic, employee costs of a sudden lockdown), they shall not form part of cost of production or acquisition or supply of goods or provision of service
Subsidy, grant, incentives	Reduced from the cost objects to which the amount pertains
Penalty, fine, damages, demurrage	Do not form part of cost
Interest and other finance costs	Not included in cost of production; presented as a separate item of cost of sales
GST and other duties on materials	Added to material cost only if input tax credit is not available or not availed

ICAI's suggested answers also leave out income tax and donations (RTP September 2026, Q8).

Why bother: the advantages of a cost sheet

ICAI lists five (SM 6.10): (i) the total cost figure as well as cost per unit of production; (ii) cost comparison; (iii) cost estimates required for submitting tenders; (iv) help in arriving at the selling price; (v) cost control by disclosing operational efficiency. The third and fourth are the next lesson, estimated cost sheets, tenders and quotations.

Formats and formulas

ICAI's specimen format for a manufacturing entity (SM 6.8–6.9), in ICAI's words and line order. *Reproduce it from memory: examiners mark the subtotals.* ICAI prints two amount columns, **Total Cost (₹)** and **Cost per unit (₹)**; one is shown here so the format reads on a phone.

Particulars	Total cost (₹)
1. Direct materials consumed: Opening Stock of Raw Material; Add: Additions/ Purchases; Less: Closing stock of Raw Material	xxx
2. Direct employee (labour) cost	xxx
3. Direct expenses	xxx
4. Prime Cost (1+2+3)	**xxx**
5. Add: Works/ Factory Overheads	xxx
6. Gross Works Cost (4+5)	**xxx**

Particulars	Total cost (₹)
7. Add: Opening Work in Process	xxx
8. Less: Closing Work in Process	(xxx)
9. Works/ Factory Cost (6+7−8)	**xxx**
10. Add: Quality Control Cost	xxx
11. Add: Research and Development Cost	xxx
12. Add: Administrative Overheads (relating to production activity)	xxx
13. Less: Credit for Recoveries/Scrap/By-Products/misc. income	(xxx)
14. Add: Packing cost (primary)	xxx
15. Cost of Production (9+10+11+12−13+14)	**xxx**
16. Add: Opening stock of finished goods	xxx
17. Less: Closing stock of finished goods	(xxx)
18. Cost of Goods Sold (15+16−17)	**xxx**
19. Add: Administrative Overheads (General)	xxx
20. Add: Marketing Overheads: Selling Overheads; Distribution Overheads	xxx
21. Cost of Sales (18+19+20)	**xxx**

Secondary packing sits with selling and distribution (SM 6.7); interest, if any, on its own line before cost of sales (SM 6.10). Then sales, with profit as the balancing figure (or profit first, then sales, when a margin is given).

Interactive version: PassBook app, lesson Cost sheet for a manufacturer

∑ **Formula**

Prime cost and works cost

$$\text{Prime cost} = \text{Direct materials} + \text{Direct employee cost} + \text{Direct expenses}$$

$$\text{Works cost} = \text{Prime cost} + \text{Factory overheads} + \text{Opening WIP} - \text{Closing WIP}$$

∑ **Formula**

Cost of production, cost of goods sold, cost of sales

$$\begin{aligned}\text{Cost of production} &= \text{Works cost} \\ &+\text{QC} + \text{R\&D} \quad +\text{Production admin} \\ &-\text{Recoveries} \quad +\text{Primary packing}\end{aligned}$$

$$\begin{aligned}\text{COGS} &= \text{Cost of production} \\ &+\text{Opening FG} - \text{Closing FG}\end{aligned}$$

$$\begin{aligned}\text{Cost of sales} &= \text{COGS} + \text{General admin} \\ &+\text{Selling} + \text{Secondary packing} \quad +\text{Distribution}\end{aligned}$$

QC = quality control; R&D = research and development for the process; production admin = administrative overheads related to production; recoveries = realised or realisable value of scrap, waste and by-products; FG = finished goods.

∑ **Formula**

Profit as a percentage of sales or of cost

$$\begin{aligned}\text{Profit} &= x\% \text{ of sales} \\ &= \frac{x}{100 - x} \text{ of cost of sales}\end{aligned}$$

20% on sales is 25% on cost; 25% on sales is 33⅓% on cost. Sales = cost of sales ÷ (1 – x%).

ICAI's examples under each head

The lists to classify from, in ICAI's words and order (SM 6.3–6.8). September 2026 Q6(a) asked exactly this for ten items.

Head	ICAI's examples
Direct material (additions to cost)	(a) freight inwards; (b) insurance and other expenditure directly attributable to procurement; (c) trade discounts or rebates (to be deducted); (d) duties and taxes (if input tax credit is not available or availed)
Direct employee (labour) cost	(a) wages and salary; (b) allowances and incentives; (c) payment for overtimes; (d) bonus / ex-gratia; (e) employer's contribution to welfare funds such as provident fund; (f) other benefits (medical, leave with pay, free or subsidised food, leave travel concession, provisions for retirement benefits)
Direct expenses	(a) utilities such as power and fuel, steam; (b) royalty paid or payable for production or provision of service; (c) hire charges for specific equipment; (d) fee for technical assistance and know-how; (e) amortised cost of moulds, patterns, patents; (f) product or service specific design or drawing; (g) product or service specific software; (h) other expenses directly related with production or provision of service

Head	ICAI's examples
Factory overheads	(a) consumable stores and spares; (b) depreciation of plant and machinery, factory building; (c) lease rent of production assets; (d) repair and maintenance of plant and machinery, factory building; (e) indirect employees cost related with production activities; (f) drawing and designing department cost; (g) insurance of plant and machinery, factory building, stock of raw material and WIP; (h) amortised cost of jigs, fixtures, tooling; (i) service department cost such as tool room, engineering and maintenance, pollution control
Selling overheads	(a) salary and wages of the sales department and of employees directly related with selling; (b) rent, depreciation, maintenance and other cost of the sales department; (c) advertisement, maintenance of website for online sales, market research
Distribution overheads	(a) salary and wages of employees engaged in distribution; (b) transportation and insurance related with distribution; (c) depreciation, hire charges, maintenance and other operating costs of distribution vehicles

Definition

Prime cost

Prime cost represents the total of direct materials costs, direct employee (labour) costs and direct expenses. (SM 6.3)

Definition

Primary and secondary packing cost

Primary packing cost: cost incurred on packing materials which are essential to hold and preserve the product for further processing or its use by consumer. Secondary packing cost: cost incurred on packing materials which is used to store, transport, and promote the product. (SM 6.20)

Definition

Cost of sales

It is the total cost of a product incurred to make the product available to the customer or consumer. It is the aggregate of cost of goods sold, administrative costs, marketing costs and other separate line items of cost which could not form part of cost of production. (SM 6.20)

Worked examples

Worked example

Samosa packs: a month at the Indore factory

Tapri's Indore factory made 50,000 packs of frozen samosas in October 2026 and sold 46,000 at ₹90 a pack. At 1 October it held 2,000 finished packs costing ₹1,10,000. Finished stock is issued on the first-in, first-out basis.

Particulars	Amount (₹)
Opening stock of raw materials	2,40,000
Purchases of raw materials	14,60,000
Freight inwards	40,000
Closing stock of raw materials	3,00,000
Direct wages	6,00,000
Power and fuel for the fryers	1,20,000
Royalty on production	50,000
Factory overheads: 5,000 machine hours at ₹86 an hour	
Opening work-in-process	60,000
Closing work-in-process	1,00,000

Particulars	Amount (₹)
Quality control laboratory	45,000
R&D on a faster frying process	25,000
Production office (planning and control)	80,000
Waste sold to a cattle-feed unit	20,000
Pouches that hold the samosas	1,70,000
General administration	2,30,000
Selling overheads	1,84,000
Outer cartons for the trucks	92,000
Distribution overheads	2,76,000

Required: Prepare a cost sheet for October 2026 showing, in total and per pack, (i) materials consumed, (ii) prime cost, (iii) works cost, (iv) cost of production, (v) cost of goods sold, (vi) cost of sales and profit.

THE WORKING

Working notes

1. Closing finished packs = 2,000 + 50,000 − 46,000 = 6,000. Under FIFO they come from October's production, at October's cost of production per pack.

2. Production heads are divided by the 50,000 packs produced; cost of goods sold and everything after it by the 46,000 packs sold.
3. Direct expenses = power and fuel ₹1,20,000 + royalty ₹50,000; factory overheads = 5,000 hours × ₹86 = ₹4,30,000.

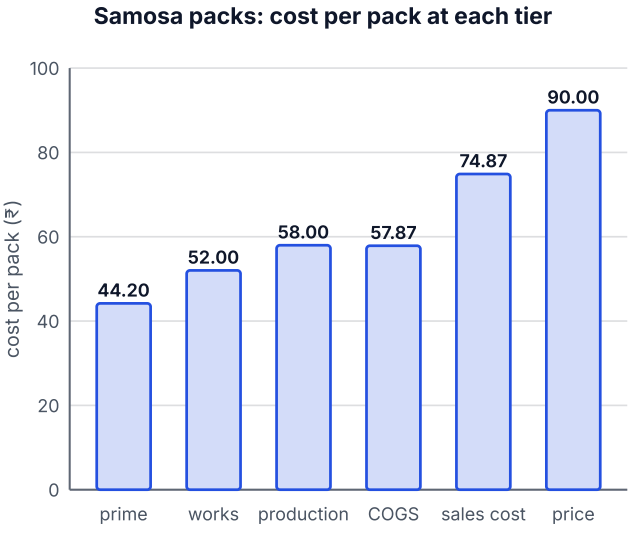
Cost sheet of Tapri Tea Co. Ltd, Indore factory, for October 2026 (output 50,000 packs; sold 46,000)

Particulars	Total (₹)	Per pack (₹)
Opening raw materials	2,40,000	
Add: Purchases	14,60,000	
Add: Freight inwards	40,000	
Less: Closing raw materials	(3,00,000)	
(i) Materials consumed	**14,40,000**	**28.80**
Direct wages	6,00,000	12.00
Direct expenses (WN 3)	1,70,000	3.40
(ii) Prime cost	**22,10,000**	**44.20**
Factory overheads (WN 3)	4,30,000	8.60

Particulars	Total (₹)	Per pack (₹)
Gross works cost	**26,40,000**	**52.80**
Add: Opening WIP	60,000	
Less: Closing WIP	(1,00,000)	
(iii) Works cost	**26,00,000**	**52.00**

Particulars	Total (₹)	Per pack (₹)
Works cost	**26,00,000**	**52.00**
Add: Quality control	45,000	0.90
Add: R&D (process)	25,000	0.50
Add: Production office	80,000	1.60
Less: Waste sold	(20,000)	(0.40)
Add: Primary packing	1,70,000	3.40
(iv) Cost of production	**29,00,000**	**58.00**
Add: Opening FG	1,10,000	
Less: Closing FG (WN 1)	(3,48,000)	
(v) Cost of goods sold	**26,62,000**	**57.87**

Particulars	Total (₹)	Per pack (₹)
Add: General administration	2,30,000	5.00
Add: Selling overheads	1,84,000	4.00
Add: Secondary packing	92,000	2.00
Add: Distribution	2,76,000	6.00
(vi) Cost of sales	**34,44,000**	**74.87**
Profit (balancing figure)	6,96,000	15.13
Sales (46,000 × ₹90)	**41,40,000**	**90.00**



Each tier adds a function. Cost of goods sold per pack dips below cost of production because 2,000 of the packs sold came from September's cheaper stock (₹55 each).

Cartons move the packs (secondary packing); pouches hold them (primary).

Interactive version: PassBook app, lesson Cost sheet for a manufacturer

Worked example

Chai premix plant: the intern's draft and Ritu's red pen

Tapri Premix Ltd runs the chai premix plant at Dewas. It produced 6,00,000 kg in 2025-26. Finished stock is

issued first-in, first-out, and the company prices to earn a profit of 20% on sales. Ritu's intern sorted the year's bills into cost heads as below and arrived at a cost of production of ₹6,72,90,000. Ritu sent it back: some bills sit under the wrong head, some are not costs at all, and two are at the wrong amount.

The intern's heads for 2025-26

Head, and the bills the intern put in it (₹)	Amount (₹)
Materials consumed: purchases 4,80,00,000; GST @ 18% on them 86,40,000 (input tax credit available); freight inwards 6,40,000; opening stock 24,00,000; less closing stock 30,00,000	5,66,80,000
Direct wages: wages 48,00,000; employer's PF and ESI 5,76,000; wages of workers idle in a sudden lockdown 1,20,000	54,96,000
Direct expenses: royalty for the drying process 2,40,000; power bill of the dryers 9,60,000; licence fee for an instant-dissolve formula 3,00,000 (the licence covers two years' production)	15,00,000
Works overheads: stores and spares 2,10,000; depreciation of plant 6,80,000, factory building 3,20,000 and delivery vans 2,60,000; supervisors 9,24,000; repairs of plant 1,80,000; interest on term loan 8,40,000	34,14,000

Head, and the bills the intern put in it (₹)	Amount (₹)
Work-in-process: opening 8,00,000 less closing 6,00,000	2,00,000
Cost of production (intern)	**6,72,90,000**
General administration: office depreciation 1,40,000; finance manager 10,80,000; managing director 36,00,000; audit fees 3,00,000; quality control laboratory 4,50,000; production planning and control manager 9,00,000; income tax 20,00,000; penalty for late filing of returns 50,000	85,20,000
Selling and distribution: sales manager 12,00,000; advertisement and website 18,00,000; running cost of delivery vans 9,60,000; foil sachets that hold the premix 16,50,000; outer cartons for dispatch 4,80,000	60,90,000
Other income, added to sales: spilt powder sold 1,20,000; State subsidy on the plant's power cost 1,60,000	2,80,000

Finished goods: 1 April 2025, 20,000 kg costing ₹19,00,000; 31 March 2026, 30,000 kg.

Required:

(i) Prepare the corrected statement of cost for 2025-26 showing prime cost, works cost, cost of production, cost of goods sold and cost of sales.

(ii) Calculate the sales, and the selling price per kg, that earn a profit of 20% on sales.

(iii) Reconcile the intern's cost of production with the corrected figure, stating the rule behind each adjustment.

THE WORKING

(i) Statement of cost of Tapri Premix Ltd for the year ended 31 March 2026 (output 6,00,000 kg)

Particulars	Amount (₹)
Purchases 4,80,00,000 + freight inwards 6,40,000 + opening stock 24,00,000 − closing stock 30,00,000	4,80,40,000
Materials consumed	**4,80,40,000**
Direct employee cost: wages 48,00,000 + PF and ESI 5,76,000	53,76,000
Direct expenses: royalty 2,40,000 + power net of subsidy 8,00,000 + licence amortised (3,00,000 ÷ 2) 1,50,000	11,90,000
Prime cost	**5,46,06,000**
Works overheads: stores 2,10,000; depreciation of plant 6,80,000 and factory building 3,20,000; supervisors 9,24,000; repairs 1,80,000	23,14,000

Particulars	Amount (₹)
Gross works cost	**5,69,20,000**
Add: Opening WIP 8,00,000; less: closing WIP 6,00,000	2,00,000
Works cost	**5,71,20,000**

Particulars	Amount (₹)
Works cost	**5,71,20,000**
Add: Quality control laboratory	4,50,000
Add: Production planning and control manager	9,00,000
Less: Spilt powder sold	(1,20,000)
Add: Primary packing (foil sachets)	16,50,000
Cost of production (₹100 a kg)	**6,00,00,000**
Add: Opening finished goods (20,000 kg)	19,00,000
Less: Closing finished goods (30,000 kg × ₹100)	(30,00,000)
Cost of goods sold (5,90,000 kg)	**5,89,00,000**
Administration: office depreciation 1,40,000 + finance manager 10,80,000 + MD 36,00,000 + audit fees 3,00,000	51,20,000

Particulars	Amount (₹)
Selling: sales manager 12,00,000 + advertisement and website 18,00,000	30,00,000
Secondary packing (outer cartons)	4,80,000
Distribution: van depreciation 2,60,000 + running cost 9,60,000	12,20,000
Interest on term loan (separate item)	8,40,000
Cost of sales	**6,95,60,000**

(ii) Sales and price. Profit 20% on sales = 25% on cost = ₹1,73,90,000; sales = ₹6,95,60,000 ÷ 80% = **₹8,69,50,000**. Kg sold = 20,000 + 6,00,000 − 30,000 = 5,90,000, so the price is ₹8,69,50,000 ÷ 5,90,000 = **₹147.37 a kg**.

(iii) Reconciliation of cost of production

Particulars	Amount (₹)
Cost of production as drafted by the intern	6,72,90,000
Less: GST with input tax credit available: not a cost (SM 6.3)	(86,40,000)
Less: Lockdown idle wages: abnormal cost, excluded (SM 6.9)	(1,20,000)

Particulars	Amount (₹)
Less: Power subsidy: reduced from the cost it pertains to (SM 6.9)	(1,60,000)
Less: Half the licence: amortised over two years' production (SM 6.4)	(1,50,000)
Less: Depreciation of delivery vans: distribution overhead (SM 6.8)	(2,60,000)
Less: Interest on term loan: separate item of cost of sales (SM 6.10)	(8,40,000)
Add: Quality control laboratory: part of cost of production (SM 6.5)	4,50,000
Add: Production planning and control manager: administration related to production (SM 6.6)	9,00,000
Less: Spilt powder sold: credit for recoveries (SM 6.6)	(1,20,000)
Add: Foil sachets: primary packing (SM 6.6)	16,50,000
Corrected cost of production	**6,00,00,000**

Working notes

1. Income tax (₹20,00,000) and the penalty (₹50,000, SM 6.9) are not costs: they leave the

intern's administration figure and appear nowhere in the statement.

2. Neither the scrap nor the subsidy is sales: one is a credit against cost of production, the other reduces power cost.
3. Under FIFO the closing 30,000 kg are this year's, at ₹6,00,00,000 ÷ 6,00,000 = ₹100 a kg.

Worked example

Kulhads at Khurja: the work-in-process under the chai

Kesar rested his cutting chai on the July 2026 ledger of Tapri's kulhad unit at Khurja, and the cup tipped over the stock page. This is what Ritu can still read. The unit made 4,00,000 kulhads in July and sold 3,90,000 at ₹6.96 each. Finished stock is issued on the first-in, first-out basis, and July's profit was one-sixth of sales.

Particulars (July 2026)	Amount (₹)
Clay purchased (invoice value)	12,50,000
Freight inwards on clay	50,000
Direct wages	3,20,000
Kiln fuel and power	1,20,000
Royalty for the glaze design	₹0.10 per kulhad produced

Particulars (July 2026)	Amount (₹)
Works overheads	4,000 kiln hours at ₹50 an hour
Quality control (leak and crack testing)	60,000
Administration overheads related to production	40,000
Broken kulhads sold as crushed clay	25,000
General administration overheads	1,32,000
Selling and distribution overheads	₹0.50 per kulhad sold

Stocks	1 July 2026 (₹)	31 July 2026 (₹)
Clay (raw material)	1,10,000	1,40,000
Work-in-process (dried, unfired)	80,000	smudged
Finished kulhads	1,35,000 (30,000 kulhads)	? (40,000 kulhads)

Required:

(i) Calculate July's cost of production, the cost per kulhad and the value of the closing finished kulhads.

(ii) Find the closing work-in-process.

(iii) Prepare the cost sheet for July from clay consumed to sales.

THE WORKING

Working notes

1. Sales = 3,90,000 × ₹6.96 = ₹27,14,400; profit = ⅙ × ₹27,14,400 = ₹4,52,400; cost of sales = ₹22,62,000.
2. Selling and distribution = 3,90,000 × ₹0.50 = ₹1,95,000. Cost of goods sold = ₹22,62,000 − ₹1,32,000 − ₹1,95,000 = **₹19,35,000**.
3. Closing finished kulhads = 30,000 + 4,00,000 − 3,90,000 = 40,000. Under FIFO they are July's, valued at July's cost per kulhad, which is unknown: closing stock = 40,000 ÷ 4,00,000 = 10% of the cost of production (CP).
4. Direct expenses = kiln fuel ₹1,20,000 + royalty 4,00,000 × ₹0.10 = ₹1,60,000 (SM 6.4). Works overheads = 4,000 × ₹50 = ₹2,00,000.

(i) Cost of production. Cost of goods sold = CP + opening stock − closing stock:

$$19{,}35{,}000 = \text{CP} + 1{,}35{,}000 - 0.10\,\text{CP}$$

$$0.90\,\text{CP} = 18{,}00{,}000 \qquad \text{CP} = 20{,}00{,}000$$

Cost of production **₹20,00,000**, or ₹20,00,000 ÷ 4,00,000 = **₹5 a kulhad**; closing finished kulhads 40,000 × ₹5 = **₹2,00,000**. Check: the 30,000 old kulhads (₹1,35,000) plus 3,60,000 new ones at ₹5 (₹18,00,000) = ₹19,35,000.

(ii) Closing work-in-process. Climb down from cost of production to works cost, and up from clay to gross works cost; the WIP closes the gap.

Particulars	Amount (₹)
Cost of production (part i)	20,00,000
Less: Quality control	(60,000)
Less: Administration related to production	(40,000)
Add: Crushed clay sold (deducted on the way up)	25,000
Works cost	**19,25,000**

Particulars	Amount (₹)
Clay consumed: 1,10,000 + 12,50,000 + 50,000 − 1,40,000	12,70,000
Direct wages	3,20,000
Direct expenses (WN 4)	1,60,000
Prime cost	**17,50,000**
Works overheads (WN 4)	2,00,000
Gross works cost	**19,50,000**

Closing WIP = gross works cost + opening WIP − works cost = ₹19,50,000 + ₹80,000 − ₹19,25,000 = **₹1,05,000**.

(iii) Cost sheet of the Khurja kulhad unit for July 2026 (output 4,00,000; sold 3,90,000)

Particulars	Amount (₹)
Opening clay 1,10,000 + purchases 12,50,000 + freight 50,000 − closing clay 1,40,000	12,70,000
Direct wages	3,20,000
Direct expenses (WN 4)	1,60,000
Prime cost	**17,50,000**

Particulars	Amount (₹)
Works overheads (WN 4)	2,00,000
Gross works cost	**19,50,000**
Add: Opening WIP 80,000; less: closing WIP 1,05,000	(25,000)
Works cost	**19,25,000**
Add: Quality control	60,000
Add: Administration related to production	40,000
Less: Crushed clay sold	(25,000)
Cost of production (₹5 a kulhad)	**20,00,000**
Add: Opening finished kulhads	1,35,000
Less: Closing finished kulhads	(2,00,000)
Cost of goods sold	**19,35,000**
Add: General administration	1,32,000
Add: Selling and distribution (WN 2)	1,95,000
Cost of sales (₹5.80 a kulhad)	**22,62,000**
Profit (⅙ of sales = 20% on cost of sales)	4,52,400
Sales (3,90,000 × ₹6.96)	**27,14,400**

The forward sheet landing on the sales register's ₹27,14,400 is the check that both climbs were right. Valuing the closing kulhads at the opening rate (₹4.50) instead of July's ₹5 would give a cost of production of ₹19,80,000 and a closing WIP of ₹1,25,000: the FIFO rule decides the answer.

Exam angle

In the exam

Where the manufacturer's cost sheet appeared, May 2024 to September 2026

In the documents read for this lesson, it was examined in seven of the eight past papers from May 2024 to September 2026 (all but May 2024), and in five MTP questions.

- **Descriptive, 7 to 9 marks.** A long list of bills, stocks and a margin: January 2025 Q2(a) (9 marks). Two models, office overheads at 20% of works cost: May 2025 Q2(a) (7 marks). Prime cost to profit with scrap: May 2026 Q2(a) (7 marks). Only the figures "as they will appear in the Cost Sheet": September 2026 Q5(a) (7 marks). Two months at 40% and 100% capacity: September 2024 Q2(a) (8 marks), worked in estimated cost sheets.
- **Working backwards, 5 to 9 marks.** Cost of production given, materials consumed as the balancing figure: January 2026 Q2(a) (9 marks). Cost of goods sold given, direct wages "being 180%

of factory overheads", production administration at 8% of factory overheads: MTP Series I September 2026 Q5(b) (5 marks). The Study Material's own Problem 1 (SM 6.23) is the same climb.

- **Classification and treatment, 4 to 5 marks.** Ten items by function: September 2026 Q6(a). Lockdown wages, a subsidy, interest on a term loan, GST with input tax credit: September 2026 Q6(c). The SM 6.5–6.10 items: MTP Series II May 2025 Q3(b); MTP Series I September 2024 Q2(b). Cost of production from a list of bills: MTP Series I September 2024 Q1(a).
- **Multiple choice, 2 marks each**, five on one case scenario walking down the tiers: September 2025 MCQs 1–5; MTP Series I May 2026 MCQs 6–10.

In the exam

The traps ICAI set

- "Inclusive of GST @ 18% (ineligible for ITC)" stays in material cost (January 2025); creditable GST comes out.
- Closing finished stock must be valued: units produced = units sold + closing − opening (January 2025, January 2026).
- "Profit is 20% on sales": add 25% of cost of sales, or divide it by 80%.
- "Direct labour cost ₹17,500 being 175% of works overheads" (SM 6.23): overheads = ₹17,500 ÷ 1.75 = ₹10,000, never ₹17,500 × 1.75.

Trick

FIFO stock in one line

Units produced = units sold + closing stock − opening stock; closing stock = closing units × (cost of production ÷ units produced). Condition: FIFO, or closing stock stated to be from this period's output.

Answer frame

The chapter's own theory questions (SM 6.22) and the MTP favourites. Write the points in this order.

Describe how costs are classified on the basis of function. Costs are divided according to the function for

which they have been incurred: (i) production / manufacturing cost; (ii) administration cost; (iii) selling cost; (iv) distribution cost; (v) research and development costs. In a cost sheet they are grouped into prime cost, cost of production, cost of goods sold and cost of sales.

Explain the treatment of administration overheads. (1) Only administrative overheads related to production are included in cost of production. (2) General administration overheads are not included in production cost; they are added to cost of goods sold to arrive at cost of sales. (3) General administration includes (SM 6.7): (a) depreciation and maintenance of building, furniture etc. of corporate or general management; (b) salary of administrative employees, accountants, directors, secretaries etc.; (c) rent, rates and taxes, insurance, lighting, office expenses etc.; (d) indirect materials: printing and stationery, office supplies etc.; (e) legal charges, audit fees, corporate office expenses like directors' sitting fees, remuneration and commission, meeting expenses etc.

State the advantages of a cost sheet. The five points of SM 6.10, in that order.

Explain the treatment of the following items. One line per item, in the words of the table in How it works.

Loses marks

One column of figures, closing stock at the cost of goods sold figure, GST added "to be safe", income tax in administration.

Passes

The right tiers in the right order, but stocks valued without showing units produced, and excluded items silently dropped.

Full marks

ICAI's line order, every subtotal named, working notes for units and closing stock, each excluded item listed with its reason.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the GST that sneaked into material cost

Ritu's intern added ₹86,40,000 of GST to the premix plant's materials "because we paid it". **Fix:** GST is material cost only if input tax credit is not available or not availed.

⚠ **Common mistake**

The managing director's salary in cost of production

General administration placed above cost of production inflates every closing stock. **Fix:** only administration related to production goes into cost of production; the rest comes after cost of goods sold.

⚠ **Common mistake**

The two kinds of packing swapped

Foil sachets in selling, outer cartons in cost of production. **Fix:** packing that holds and preserves the product is primary (cost of production); packing to store, transport or promote it is secondary (selling and distribution).

⚠ **Common mistake**

Closing stock valued at the wrong rate

The 6,000 closing samosa packs at cost of goods sold per pack (₹57.87), or at cost of sales per pack (₹74.87, carrying selling costs unsold stock never incurred). **Fix:** this period's cost of production per unit (FIFO), ₹58.

Practice

Check yourself

1. A Tapri unit's cost of production for the month is ₹12,00,000 for 20,000 units. Opening finished stock was 2,000 units costing ₹1,10,000 and 21,000 units were sold. Using FIFO, what is the cost of goods sold in rupees?

Answer: _____

2 Case scenario. Ritu is classifying the premix plant's bills: the salary of the chemist who tests each batch against food-safety standards; outer cartons for the trucks; interest on the working capital loan; wages paid to workers idle during a sudden State lockdown. Which treatment is correct?

A Chemist: factory overhead; cartons: primary packing; interest: works cost; idle wages: employee cost

B Chemist: quality control cost; cartons: secondary packing; interest: separate item of cost of sales; idle wages: excluded as abnormal cost

C Chemist: research and development; cartons: secondary packing; interest: excluded; idle wages: factory overhead

D Chemist: quality control cost; cartons: primary packing; interest: cost of production; idle wages: excluded

3 For August the premix plant shows: gross works cost ₹8,00,000; opening WIP ₹50,000; closing WIP ₹70,000; quality control ₹20,000; scrap sold ₹10,000; primary packing ₹30,000; secondary packing ₹15,000; general administration ₹40,000. What is the cost of production?

- **A** ₹8,75,000
- **B** ₹8,20,000
- **C** ₹8,60,000
- **D** ₹8,40,000

4 Cost of sales is ₹6,00,000 and Kesar wants a profit of 25% on sales. What should sales be?

- **A** ₹7,50,000
- **B** ₹8,00,000
- **C** ₹7,20,000
- **D** ₹7,80,000

5 Which of these are NOT part of any tier of the cost sheet? Select all that apply.

- **A** A fine paid for a pollution violation
- **B** Royalty paid to the owner of a production process
- **C** Income tax paid by the company
- **D** Insurance of the stock of raw materials and WIP
- **E** GST paid on purchases when input tax credit is available

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Functional classification: production, administration, selling, distribution, R&D (SM 6.2).
- Prime cost = direct materials consumed + direct employee cost + direct expenses.
- Works cost = prime cost + factory overheads + opening WIP − closing WIP.
- Cost of production = works cost + QC + R&D (process) + production administration − scrap and by-product credit + primary packing.

- Cost of goods sold = cost of production + opening − closing finished goods (closing at this period's cost of production per unit, FIFO).
- Cost of sales = cost of goods sold + general administration + selling + secondary packing + distribution; interest as a separate item.
- Out of cost: abnormal costs, penalties and fines, creditable GST, income tax; subsidies reduce the cost they relate to.
- x% profit on sales = x ÷ (100 − x) on cost. (The tiffin is built from the bottom up.)

Answer key · Check yourself

1 **1250000**

Closing units = 2,000 + 20,000 − 21,000 = 1,000, at ₹12,00,000 ÷ 20,000 = ₹60: ₹60,000. Cost of goods sold = ₹12,00,000 + ₹1,10,000 − ₹60,000 = ₹12,50,000. Check: the 2,000 old units (₹1,10,000) plus 19,000 new ones at ₹60 (₹11,40,000).

2 **B** **Chemist: quality control cost; cartons: secondary packing; interest: separate item of cost of sales; idle wages: excluded as abnormal cost**

Testing for compliance with standards is quality control (SM 6.5); cartons that help transport are secondary packing (SM 6.7); interest is a separate item of cost of sales (SM 6.10); lockdown wages are an abnormal cost (SM 6.9).

3 **B ₹8,20,000**

₹8,00,000 + ₹50,000 − ₹70,000 = works cost ₹7,80,000; + ₹20,000 − ₹10,000 + ₹30,000 = ₹8,20,000. Secondary packing and general administration come after cost of goods sold (₹8,75,000 adds them); ₹8,60,000 swaps the WIP signs; ₹8,40,000 adds the scrap instead of deducting it.

4 **B ₹8,00,000**

25% on sales is 33⅓% on cost: profit ₹2,00,000, sales ₹8,00,000. Adding 25% of cost gives ₹7,50,000, only 20% on sales.

5 **A A fine paid for a pollution violation**; **C Income tax paid by the company**; **E GST paid on purchases when input tax credit is available**

Fines are not cost (SM 6.9); income tax is left out in ICAI's answers; creditable GST is not a cost (SM 6.3). Production royalty is a direct expense, stock insurance a factory overhead (SM 6.4–6.5).

2.2 Estimated cost sheets, tenders and quotations

A quotation is a cost sheet for the future: the order's estimated direct costs, overheads at rates taken from past cost sheets, order-specific costs, then the profit margin converted correctly.

In plain words

The hotel chain wants a number by Friday

Sahyadri Hotels has invited tenders for 50,000 boxes of chai premix printed with its own logo. The quotation must reach them by Friday, and the order has not been made yet, so no bill for it exists.

> **Kesar:** Quote our list price, ₹120 a box, and let's go to lunch.
>
> **Ritu:** Per unit, please, for *this* order. The premix base is dearer, wages have gone up, the logo needs a printing plate, and you want 20% of the price, not of the cost.

Ritu builds a cost sheet that looks forward. She estimates what the order will consume (material at today's price, the hours of labour, the printing plate). For the overheads she cannot trace to one order (the factory's rent, the office, the sales team) she uses last year's cost sheet: what each overhead came to per labour hour or as a percentage of its base, after taking out anything that will not happen again. She applies those rates to the order's

own hours and costs, adds anything made for this order alone, and only then adds the profit.

That forward-looking statement is an **estimated cost sheet**, and the figure at the bottom is the **price to be quoted**. It is the cost sheet's third advantage in ICAI's list: it "facilitates the preparation of cost estimates required for submitting tenders" (SM 6.10).

Think of it like

The wedding halwai

A halwai quoting for a wedding of 500 guests does not wait for the wedding to happen. He prices the ghee, sugar and khoya at today's market rate, counts his karigars' days, adds a share of his shop's rent and lights in the proportion he has always found them to bear to wages, adds the tempo to the venue, and then his margin. Get any line wrong and he either loses the order or cooks at a loss.

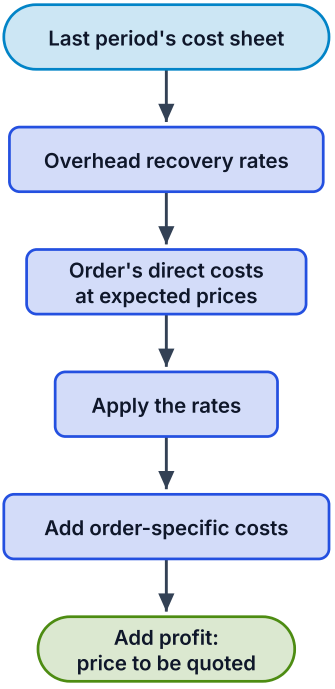
Key idea

Price to be quoted = the order's estimated direct costs + overheads at pre-determined recovery rates (from past cost sheets, on ICAI's bases) + costs specific to the order + profit. A margin "on selling price" must be converted before it is added to cost: 20% on price is 25% on cost.

How it works

An **estimated cost** is "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase" (Kohler, SM 1.34). Overheads cannot be traced to an order that has not been produced, so they are charged at **pre-determined rates**: "production overheads are to be determined in advance for fixing selling price, quote tender price and to formulate budgets" (SM 4.41).

From last year's cost sheet to a quoted price



Each rate is worked out on the base the question names, then applied to the same base for the order.

Six steps to a quotation

1. **Recovery rates from the past cost sheet.** Divide each overhead by its base: factory overheads by direct wages (or prime cost, or labour or machine hours); administration overheads by works (factory) cost; selling and distribution by cost of production (or per unit sold). The bases for administration include works cost, sales value, gross profit, quantity produced and conversion cost (SM 4.48); for selling, sales value, cost of goods sold, gross profit, or number of orders or units sold (SM 4.53). The Study Material calls sales value "ordinarily the most logical basis" for selling overheads and cost of production "not as satisfactory", yet ICAI's questions often fix the rate on cost of production (January 2025 Q4(b); MTP Series I September 2026 Q2(a)). Use the one the question states.

2. **Revise the rates for known changes.** "Factory overheads will fall 10%" changes the rate's numerator; a new base changes the denominator (RTP January 2026 Q13).

3. **Estimate the order's direct costs at expected prices**: material quantity × the new price, labour hours × the new rate, and direct expenses such as a special die or design for this order.

4 **Apply the rates to the same bases for the order**: factory overheads on the order's wages, administration on its works cost, selling and distribution on its cost of production.

5 **Add costs that belong to this order only**, such as delivery to the customer's premises or special packing, at their estimated amounts (January 2025 Q4(b)).

6 **Add the profit** and arrive at the price. Profit x% on selling price is x ÷ (100 − x) on cost; price = total cost ÷ (1 − x%).

When volume changes: fixed costs per unit do not stay put

A quotation for a different volume cannot use last year's cost *per unit* for everything. Variable cost per unit stays the same, fixed cost in total stays the same (so per unit it falls as volume rises), and a semi-variable cost must be split first: variable rate = change in cost ÷ change in output, fixed part = total cost − variable part (SM Illustration 2, 8.6–8.7; cost behaviour). ICAI's own problem prepares a cost sheet "at 30% and 100% capacity level" where selling the full output needs gifts, lucky draws and TV sponsorship (SM 6.24–6.25, 6.32–6.33).

Not to be confused with buying by tender

When Tapri *buys*, it invites quotations through a request for proposal (RFP) or a notification inviting tender (NIT) and selects the best one (SM 2.9–2.11): that is the procurement procedure. This lesson is the seller's side: pricing the quotation you send.

Formats and formulas

The estimated cost sheet follows the cost sheet's own order (cost sheet for a manufacturer), with each overhead line showing its rate and base. *Reproduce it from memory.*

Statement of estimated cost and price to be quoted	Amount (₹)
Direct materials (estimated quantity × expected price)	xxx
Direct wages (estimated hours × expected rate)	xxx
Direct expenses specific to the order	xxx
Prime cost	**xxx**
Add: Factory overheads (F% of direct wages)	xxx
Works / factory cost	**xxx**
Add: Administration overheads (A% of works cost)	xxx
Cost of production	**xxx**

Statement of estimated cost and price to be quoted	Amount (₹)
Add: Selling and distribution overheads (S% of cost of production)	xxx
Add: Costs specific to the order (delivery, special packing)	xxx
Total cost	**xxx**
Add: Profit (x% on price = x ÷ (100 − x) on cost)	xxx
Price to be quoted	**xxx**

∑ **Formula**

Overhead recovery rates from a past cost sheet

$$F = \frac{\text{Factory overheads}}{\text{Direct wages}} \times 100$$

$$A = \frac{\text{Admin overheads}}{\text{Works cost}} \times 100$$

$$S = \frac{\text{S\&D overheads}}{\text{Cost of production}} \times 100$$

S&D = selling and distribution. The base in each denominator is the one the question names; the same base is used for the order.

Σ **Formula**

Pre-determined overhead rate

$$\text{Pre-determined rate} = \frac{\text{Estimated overheads}}{\text{Budgeted units}}$$

Estimated (normal) overheads for the period ÷ budgeted number of units during the period (SM 4.41); with hours or wages as the denominator for an hour rate or a percentage rate.

Σ **Formula**

Price from a margin, and cost from a price

$$\text{Price} = \frac{\text{Total cost}}{1 - x\%}$$

$$\text{Total cost} = \frac{\text{Selling price}}{100\% + p\%}$$

x = profit as a percentage of selling price; p = profit as a percentage of total cost (SM 4.51).

Definition

Estimated cost

Kohler defines estimated cost as "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs. (SM 1.34)

Definition

Pre-determined cost

A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost. (SM 1.34)

Worked examples

Worked example

Sahyadri Hotels: last year's rates, cleaned, for this year's order

Tapri's premix division recorded the following for 2025-26 (no work-in-process):

Particulars, 2025-26	Amount (₹)
Direct materials consumed	1,50,00,000
Direct wages (30,000 labour hours at ₹200)	60,00,000
Factory overheads, including ₹3,00,000 spent repairing flood damage in the godown (not expected to recur)	48,00,000
Administration overheads	25,50,000
Selling and distribution overheads	22,44,000

Tapri recovers factory overheads at a rate per direct labour hour, administration overheads as a percentage of works cost and selling and distribution overheads as a percentage of cost of production, at last year's normal rates.

Sahyadri Hotels' tender is for 50,000 logo boxes in 2026-27. The boxes will hold 19,000 kg of premix base, and blending and filling lose 5% of the base put in (a normal loss). The base cost ₹60 a kg last year and is now 10% dearer. The order needs 4,000 labour hours; the wage rate has risen from ₹200 to ₹216 an hour. A printing plate for the logo costs ₹60,000.

Required:

(i) Calculate the overhead recovery rates for 2025-26.

(ii) Prepare a statement of estimated cost and the price to be quoted if Tapri wants a profit of 20% on the price.

(iii) Kesar asks why factory overheads are not simply charged at last year's percentage of wages. Show the price that basis would give, and say which basis suits this order.

THE WORKING

(i) Recovery rates, 2025-26

The flood repairs are an abnormal cost: a pre-determined rate is built on estimated *normal* overheads (SM 4.41), so they are left out of the base. Normal factory overheads = ₹48,00,000 –

₹3,00,000 = ₹45,00,000. Works cost = ₹1,50,00,000 + ₹60,00,000 + ₹45,00,000 = ₹2,55,00,000. Cost of production = ₹2,55,00,000 + ₹25,50,000 = ₹2,80,50,000.

Overhead	Calculation	Rate
Factory	45,00,000 ÷ 30,000 hours	₹150 an hour
Administration	25,50,000 ÷ 2,55,00,000	10% of works cost
Selling, distribution	22,44,000 ÷ 2,80,50,000	8% of cost of production

Working note: the order's materials. Input = 19,000 kg ÷ 95% = 20,000 kg, at ₹60 × 110% = ₹66 a kg: ₹13,20,000.

(ii) Statement of estimated cost and price to be quoted, Sahyadri Hotels order

Particulars	Amount (₹)
Direct materials (20,000 kg × ₹66)	13,20,000
Direct wages (4,000 hours × ₹216)	8,64,000
Direct expense: printing plate for the logo	60,000
Prime cost	**22,44,000**

Particulars	Amount (₹)
Factory overheads (4,000 hours × ₹150)	6,00,000
Works cost	**28,44,000**
Administration overheads (10% of works cost)	2,84,400
Cost of production	**31,28,400**
Selling and distribution (8% of cost of production)	2,50,272
Total cost	**33,78,672**
Profit (20% on price = 25% on cost)	8,44,668
Price to be quoted	**42,23,340**

That is ₹84.47 a box, against Kesar's ₹120. Check: ₹8,44,668 ÷ ₹42,23,340 = 20%. The plate is made for this order alone, so it is a direct expense (SM 6.4): it is in prime cost but carries no factory overhead, which is charged on labour hours.

(iii) Factory overheads on wages instead. Last year's rate = ₹45,00,000 ÷ ₹60,00,000 = 75% of wages, so the order would carry 75% × ₹8,64,000 = ₹6,48,000, ₹48,000 more. Works cost ₹28,92,000; administration ₹2,89,200; cost of production ₹31,81,200; selling and distribution ₹2,54,496; total cost ₹34,35,696; price ₹34,35,696 ÷ 80% = **₹42,94,620**, ₹71,280 higher.

The extra ₹48,000 comes only from the 8% rise in the wage rate: the order takes the same 4,000 hours of the factory's time either way. The labour hour rate "fully recognises the significance of the element of time in the incurring and absorption of manufacturing overhead expenses" and is an improvement on the percentage of direct wages (SM 4.32), so the hour rate suits this order. Had the flood repairs been left in, the hour rate would have been ₹160 and the quotation ₹40,000 of factory overheads too high.

Interactive version: PassBook app, lesson Estimated cost sheets, tenders and quotations

Worked example

The lost rate card: two old quotations, one new one

Tapri's kulhad unit makes custom-printed clay cups for hotel chains. It recovers administration overheads at 12% of factory cost, and factory overheads at a fixed percentage of direct wages that was set at the start of the year; the note recording that percentage is lost. Every quotation carries the same percentage of profit

on total cost, and that figure is in the same lost note. Two jobs quoted and completed last quarter:

Particulars	Job K-21 (₹)	Job K-22 (₹)
Direct materials	72,000	38,000
Direct wages	30,000	22,000
Price quoted	1,80,600	1,11,720

Required:

(i) Find the factory overhead recovery rate and the percentage of profit on total cost.

(ii) Prepare the cost sheet of each job.

(iii) Quote for Job K-23 (direct materials ₹56,000, direct wages ₹30,000) on the same basis, and state its profit as a percentage of the price.

THE WORKING

(i) The two missing figures

Let F = factory overheads as a fraction of direct wages and p = profit as a fraction of total cost. For each job:

Price = (materials + wages + F × wages) × 1.12 × (1 + p)

The factors 1.12 and (1 + p) are the same for both jobs, so they cancel in the ratio of the two prices:

$$\frac{1{,}80{,}600}{1{,}11{,}720} = \frac{1{,}02{,}000 + 30{,}000F}{60{,}000 + 22{,}000F}$$

₹1,80,600 : ₹1,11,720 = 215 : 133 (both divided by 840). Cross-multiplying:

215 × (60,000 + 22,000F) = 133 × (1,02,000 + 30,000F)

1,29,00,000 + 47,30,000F = 1,35,66,000 + 39,90,000F

7,40,000F = 6,66,000, so **F = 0.90**: factory overheads are **90% of direct wages**.

K-21: factory cost = ₹72,000 + ₹30,000 + ₹27,000 = ₹1,29,000; total cost = ₹1,29,000 × 1.12 = ₹1,44,480; profit = ₹1,80,600 − ₹1,44,480 = ₹36,120 = **25% of total cost**.

(ii) The two jobs

Particulars	K-21 (₹)	K-22 (₹)
Direct materials	72,000	38,000
Direct wages	30,000	22,000
Prime cost	**1,02,000**	**60,000**
Factory overheads (90% of wages)	27,000	19,800
Factory cost	**1,29,000**	**79,800**
Administration (12% of factory cost)	15,480	9,576
Total cost	**1,44,480**	**89,376**
Profit (25% of total cost)	36,120	22,344
Price quoted	**1,80,600**	**1,11,720**

K-22 lands exactly on its quoted ₹1,11,720: the check that F and p are right.

(iii) Job K-23

Particulars	Amount (₹)
Direct materials	56,000
Direct wages	30,000
Prime cost	**86,000**
Factory overheads (90% of ₹30,000)	27,000
Factory cost	**1,13,000**
Administration (12% of factory cost)	13,560
Total cost	**1,26,560**
Profit (25% of total cost)	31,640
Price to be quoted	**1,58,200**

Profit as a percentage of the price = ₹31,640 ÷ ₹1,58,200 = **20%**: 25% on cost is 20% on price. When both rates are unknown and the profit percentage is known, ICAI's method is two simultaneous equations in F and A (Illustration 9, SM 4.49–4.52); here the unknown, common mark-up drops out of the ratio instead.

Worked example

Samosa packs: four prices from the market survey

Tapri's samosa factory can make 80,000 packs a month in normal working hours. Costs per pack: materials ₹30, wages ₹12, variable factory overheads ₹6, distribution ₹4. Fixed costs a month: factory overheads ₹6,00,000, production administration ₹1,20,000, general administration ₹1,80,000, selling ₹2,40,000. Packs beyond 80,000 a month can be made only on overtime, at a premium of ₹6 a pack, and need a second blast freezer hired at ₹1,50,000 a month.

Dev's market survey gives the packs Tapri can sell a month at each price:

Price a pack (₹)	Packs a month
92	48,000
86	64,000
80	80,000
76	96,000

Required: Prepare a cost sheet for the month at each of the four levels showing works cost, cost of production, total cost and profit, in total and per pack, and advise Kesar which price to set.

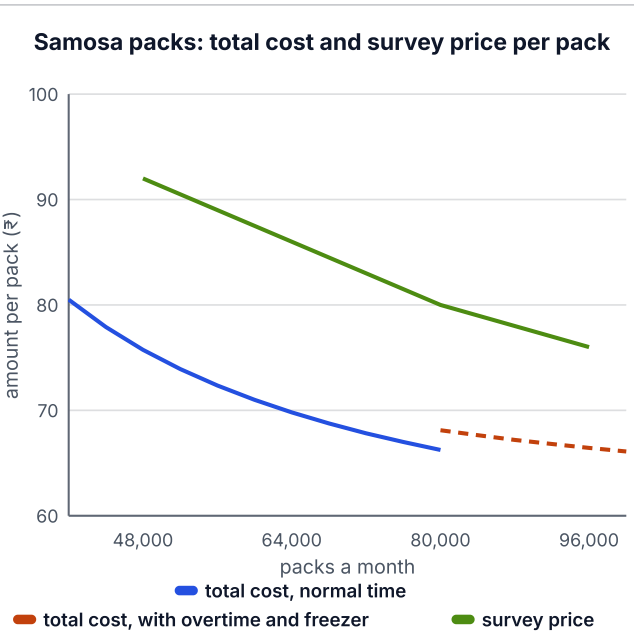
THE WORKING

Cost sheet for the month at each level

Particulars (₹)	48,000	64,000	80,000	96,000
Materials (₹30)	14,40,000	19,20,000	24,00,000	28,80,000
Wages (₹12)	5,76,000	7,68,000	9,60,000	11,52,000
Variable factory overheads (₹6)	2,88,000	3,84,000	4,80,000	5,76,000
Overtime premium (16,000 × ₹6)	—	—	—	96,000
Fixed factory overheads	6,00,000	6,00,000	6,00,000	6,00,000
Hired blast freezer	—	—	—	1,50,000
Works cost	**29,04,000**	**36,72,000**	**44,40,000**	**54,54,000**
Production administration	1,20,000	1,20,000	1,20,000	1,20,000
Cost of production	**30,24,000**	**37,92,000**	**45,60,000**	**55,74,000**
General administration	1,80,000	1,80,000	1,80,000	1,80,000
Selling	2,40,000	2,40,000	2,40,000	2,40,000
Distribution (₹4)	1,92,000	2,56,000	3,20,000	3,84,000
Total cost	**36,36,000**	**44,68,000**	**53,00,000**	**63,78,000**
Sales	44,16,000	55,04,000	64,00,000	72,96,000
Profit	**7,80,000**	**10,36,000**	**11,00,000**	**9,18,000**

Per pack (₹)	48,000	64,000	80,000	96,000
Works cost	60.50	57.38	55.50	56.81
Cost of production	63.00	59.25	57.00	58.06
Total cost	75.75	69.81	66.25	66.44
Selling price	92.00	86.00	80.00	76.00
Profit	16.25	16.19	13.75	9.56

The overtime premium is shown under factory overheads: it is paid because of the general volume of work, not for one batch.



Fixed costs of ₹11,40,000 a month spread thinner as volume rises (blue). Above 80,000 packs the freezer and the overtime premium lift cost per pack (orange, dashed). The survey price (green) falls faster than cost, so the gap between them narrows from ₹16.25 a pack at 48,000 to ₹9.56 at 96,000.

Advice. Set the price at ₹80 a pack: 80,000 packs earn the highest profit, ₹11,00,000 a month. Going to ₹76 adds 16,000 packs, but the ₹4 price cut on all 96,000 packs, the overtime premium and the freezer cost more than the extra packs earn, and profit falls to ₹9,18,000. Profit per pack is highest at ₹92, but total profit, not profit per pack, is what Kesar banks.

Exam angle

In the exam

Where quotations appeared, May 2024 to September 2026 (and the old scheme)

In the documents read for this lesson:

- **Bulk order priced from last year's rates, 6 marks:** January 2025 Q4(b): production overhead as a percentage of labour, marketing and distribution as a percentage of production cost, packaging and transport added, "Sales required for achieving a profit margin of 20% on the final sales amount".
- **Cost sheet at two activity levels, 8 marks:** September 2024 Q2(a): January at 40% and February at 100% capacity, with TV sponsorship, a hi-tea, a gift per dozen and lucky draw prizes.
- **Job cost sheet and price to be quoted:** MTP Series I September 2026 Q2(a) (7 marks, 20% on selling price, normal rectification); RTP January 2026 Q13 (revised overhead structure, same profit % on sales).
- **Old scheme:** May 2022 Q2(a) (10 marks: two jobs, find both rates, price a third job at 15% on selling price); RTPs May 2021 Q7, May 2022 Q8 and November 2023 Q7 (overhead rates from last year's actuals, price at 25% on sales); November 2019 Q5(a) (10 marks, lowest price for an extra order).
- **Multiple choice:** MTP Series I September 2026 MCQ 11 (2 marks), the relevant cost of material in a quotation.

In the exam

The traps ICAI set

- "Profit 20% on sales" or "on the final sales amount": divide total cost by 80%; never add 20% of cost.
- "Profit percentage on total cost" for the old jobs, then "on selling price" for the new one (May 2022): two different conversions in one question.
- Price changes: apply the new material price and wage rate before the overhead rates.
- Promotional costs (gifts, prizes, sponsorship) belong only to the level that needs them (September 2024).

Trick

Margin conversions you will use every paper

On price → on cost: 10% → 11.11%, 12.5% → 14.29%, 20% → 25%, 25% → 33.33%, 33⅓% → 50%. Rule: x on price = x ÷ (100 − x) on cost; p on cost = p ÷ (100 + p) on price. Always true, for any x below 100.

Answer frame

For a quotation question, ICAI's suggested answers earn marks in this order:

1. **Working note: recovery rates**, each with its formula and base ("Normal factory overheads ÷ direct labour hours = ₹150 an hour"; "Administration ÷ works cost × 100 = 10%").

2. **Working note: revised figures** (new material price, wage rate, revised overheads).
3. **Statement of estimated cost**, in cost sheet order, each overhead line showing "% of base".
4. **Profit line with the conversion stated** ("20% on price = 25% on cost").
5. **The final line in the question's words**: "Price to be quoted" or "Sales value", and a one-line conclusion ("Hence the price to be quoted is ₹42,23,340").

For theory, the point to write is ICAI's: pre-determined overhead rates are used because "production overheads are to be determined in advance for fixing selling price, quote tender price and to formulate budgets" (SM 4.41), and a cost sheet "facilitates the preparation of cost estimates required for submitting tenders" (SM 6.10).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the 20% that was added to cost

Total cost ₹33,78,672, "add 20% profit": ₹40,54,406. Sahyadri accepts, and Tapri earns only 16.67% of the price. **Fix:** 20% on price = 25% on cost; price = ₹33,78,672 ÷ 80% = ₹42,23,340.

⚠ **Common mistake**

Last year's material price in this year's quotation

Using 19,000 kg at last year's ₹60 (₹11,40,000) for materials, when the normal loss needs 20,000 kg put in and the price has risen 10% (₹13,20,000), understates every line below it. **Fix:** revise direct costs first (quantity with its normal loss, then the new price), then apply the rates.

⚠ **Common mistake**

The right rate on the wrong base

Administration at 10% of prime cost instead of works cost, or selling at 8% of works cost instead of cost of production. **Fix:** a rate is only valid on the base it was computed on; write the base on every overhead line.

⚠ **Common mistake**

Fixed cost per unit carried to a new volume

Using the 48,000-pack level's fixed cost per pack (₹23.75) for 80,000 packs gives ₹52 + ₹23.75 = ₹75.75 a pack, not ₹66.25, and makes ₹80 look like a loss-making price. **Fix:** at a new level, take fixed costs in total and divide by the new volume.

Practice

Check yourself

1. An order's total cost is estimated at ₹4,80,000. Tapri wants a profit of 25% on the price to be quoted. What price should it quote, in rupees?

Answer: _____

2. Last year's cost sheet shows direct wages ₹20,00,000, factory overheads ₹16,00,000, prime cost ₹50,00,000 and works cost ₹66,00,000. Factory overheads are recovered as a percentage of direct wages. A new order has prime cost ₹2,00,000 including wages of ₹50,000. What factory overheads should the quotation carry, in rupees?

Answer: _____

3 Case scenario. Dev reviews Ritu's quotation for a corporate client. It shows materials at this month's prices, factory overheads at last year's percentage of wages, administration at last year's percentage of works cost, a courier charge for this client only, and profit "at 20% of total cost" because the board wants a margin of 20% on price. Which correction is needed?

- **A** Administration should be charged as a percentage of prime cost
- **B** The courier charge should be absorbed into selling overheads at the normal rate
- **C** Profit should be 25% of total cost (total cost ÷ 80% gives the price)
- **D** Materials should be taken at last year's prices so that the rates stay consistent

4 A factory's monthly fixed costs are ₹6,00,000 and variable cost is ₹50 a unit. Last month it made 20,000 units at ₹80 a unit total cost. What is the total cost per unit at 30,000 units?

- **A** ₹80
- **B** ₹70
- **C** ₹75
- **D** ₹65

5 In an estimated cost sheet, which of these are added at their own estimated amount rather than through an overhead recovery rate? Select all that apply.

- **A** Delivery to this customer's premises
- **B** The factory's rent
- **C** A die made only for this order
- **D** The finance manager's salary

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- An estimated cost sheet is a cost sheet for the future; its bottom line is the price to be quoted (SM 6.10, advantage iii).
- Recovery rates from last period: factory overheads ÷ direct wages; administration ÷ works cost; selling and distribution ÷ cost of production (or the bases the question names).
- Revise direct costs to expected prices first, then apply each rate to the order's own base.
- Add order-specific costs (delivery, special dies or packing) at their own amounts.

- Price = total cost ÷ (1 − margin on price); x% on price = x ÷ (100 − x) on cost; total cost = price ÷ (100% + p% on cost).
- Two jobs with known prices give two equations in F and A; eliminate, then price the third job (SM 4.49–4.52). A common unknown mark-up cancels in the ratio of the two prices.
- At a new volume, fixed costs stay in total and change per unit; split semi-variable costs first.
- The halwai prices the wedding before the wedding.

Answer key · Check yourself

1 **640000**

Price = ₹4,80,000 ÷ 75% = ₹6,40,000; profit ₹1,60,000 is 25% of the price (33⅓% of cost). Adding 25% of cost gives ₹6,00,000, only 20% on price.

2 **40000**

Rate = 16,00,000 ÷ 20,00,000 = 80% of wages; 80% × ₹50,000 = ₹40,000. Using 32% of prime cost (16 ÷ 50) gives ₹64,000, the right figure on the wrong base.

3 **C** **Profit should be 25% of total cost (total cost ÷ 80% gives the price)**

A 20% margin on price is 25% on cost. The overhead bases are right as stated; an order-specific cost is added at its own amount; materials must be at expected prices.

4 **B ₹70**

₹50 + ₹6,00,000 ÷ 30,000 = ₹50 + ₹20 = ₹70. Keeping ₹80 treats the ₹30 fixed cost per unit as if it were variable.

5 **A Delivery to this customer's premises**; **C A die made only for this order**

Costs traceable to the order (delivery, a special die) are added directly. Rent and the finance manager's salary are overheads, charged through factory and administration rates.

2.3 Cost sheet for the service sector

A service's cost statement has no stocks to adjust; it is built by function or by variability (standing, running and maintenance charges), divided by the service unit, and priced on takings.

In plain words

A canteen has no closing stock of lunches

Tapri runs the canteen at an IT park in Pune: 1,600 lunches a day, 25 days a month. Kesar asks Ritu for the same cost sheet she makes for the samosa factory.

> **Kesar:** Same format, na? Prime cost, works cost, closing stock...
>
> **Ritu:** Per unit, please, and the unit is a meal. A lunch not eaten at one o'clock is not inventory, it is waste. And the rent and the cooks cost the same whether we serve 30,000 meals or 40,000.

So Ritu changes two things. First, there are no tiers for stock: a service cannot be stored, so there is no work-in-process or finished goods to carry forward. Second, she groups costs by how they behave. Rent, the manager's and cooks' salaries, insurance and the time-based depreciation of the kitchen are **standing charges**: fixed for the month. Food, gas, plates and the helpers paid per meal are **running charges**: they move with every meal. Repairs of the kitchen equipment are **maintenance charges**, partly fixed and partly variable. Totalled and

divided by 40,000 meals, they give a **cost per meal**, and the price per meal is built on it.

The same idea prices a training course (cost per batch, then per trainee), a software project (cost per project) or a loan department (cost per application): collect the period's costs, arrange them in a statement, express them per **unit of service**.

Think of it like

The dhaba's slate

A highway dhaba owner chalks three lines on a slate each month: what he pays even if no truck stops (rent, the cook's salary), what each plate costs him (atta, dal, gas), and the tandoor's repairs, which grow with use but never fall to zero. Divide the slate by plates served and he knows his cost per thali; add his margin and the tax he owes on every rupee he takes, and he knows his price.

Key idea

A service cost statement has no stock adjustments. It is prepared on functional classification or on variability (fixed or standing charges, variable or running charges, semi-variable or maintenance charges), and the total is divided by the service's cost unit (per meal, passenger-km, patient-day, project) to get cost per unit, then the price.

How it works

Service costing, "also known as operating costing" (SM 12.2), is used **internally** for support services such as a staff canteen, a boiler house or an IT department, and **externally** when the service is sold, as by a transporter, a hotel or an insurer (SM 12.2). Either way, "all costs incurred are collected, accumulated for a certain period or volume, recorded in the cost accounting system and then expressed in terms of a cost unit of service" (SM 12.2).

Why the manufacturer's format changes

ICAI gives four differences between service costing and product costing (SM 12.3):

1. **Tangibility**: services are intangible and cannot be stored, so there is no inventory for services.
2. **Cost units**: composite cost units are used to measure cost and express output (passenger-km, tonne-km, patient-day).
3. **Material vs employee cost**: employee cost is a bigger element than material cost.
4. **Traceability of costs**: indirect costs such as administration overheads are a significant proportion, as services depend heavily on support services and tracing costs may not be economically feasible.

The cost sheet chapter itself says so: the cost statement for services "is also prepared but the format and

presentation may differ as per the information requirement" (SM 6.9).

Two ways to arrange the statement

"The cost statement for services may be prepared either on the basis of functional classification as done for product costing or on the basis of variability" (SM 12.10). On variability it has three heads: **fixed costs or standing charges**, **variable costs or operating expenses**, and **semi-variable costs or maintenance expenses**; with no information about semi-variable costs, only fixed and variable heads are shown (SM 12.10).

The two layouts of a service cost statement

Statement of cost of a service — for a period

On variability (operating cost sheet)

A. Standing charges:	**fixed**
B. Running charges:	**variable**
C. Maintenance charges:	**semi-variable**

On function (as for a product)

Direct material:	**often small**
Direct employee cost:	**often largest**
Direct expenses:	**traced**
Overheads:	**apportioned**

Either layout ends in a total cost for the period, divided by the service units to give cost per unit. Neither has stock adjustments.

Two items that change sides

- **Depreciation**: "If related to effluxion of time or calculated on time basis, will be treated as fixed. However, if the depreciation is calculated on the basis of activity level or usage, it will be treated as variable cost" (SM 12.10).
- **Interest**: "Interest and finance charges shall be presented in the cost statement as a separate item of cost of sales. In general, interest is treated as fixed cost, unless otherwise given" (SM 12.10). ICAI's toll-plaza answer included interest in the month's total cost (September 2025 Q2(a)).

A head can also move with the facts: a driver's salary is a standing charge if paid monthly and a running charge if related to operations (SM 12.12).

From total cost to cost per unit and price

1. **Collect the period's costs** under the heads (or the functions) and total them.
2. **Count the service units** for the same period: meals served, passenger-km, patient-days, sessions, applications. Composite units are built in service costing and composite cost units.
3. **Cost per unit** = total cost ÷ total units, with a per-unit column beside each head.

4 **Price**: add profit, and any charge that is a percentage of the price or takings (a commission, a licence fee, a passenger tax). Since those are percentages of takings, solve for takings: Takings = costs + c% × Takings + p% × Takings (SM Illustration 3, 12.15–12.16).

The method of costing follows the service too: job costing suits customised software or healthcare, process costing utilities such as power or water, and joint-product costing bundled services such as telecom or events (SM 12.11).

Formats and formulas

Reproduce the operating cost sheet from memory. Each head (A standing charges, B running charges, C maintenance charges) is followed by its own total and a per-unit figure. A commission or tax on takings is shown either inside running charges (SM Illustration 3, 12.16) or on its own line after the A + B + C total (January 2025 Q4(a): "total cost before commission", then commission); either way its amount is solved from takings.

Operating cost sheet for ... (period)	Total (₹)	Per unit (₹)
A. Standing (fixed)		
Rent, insurance, taxes	xxx	
Salaries (time basis)	xxx	

Operating cost sheet for ... (period)	Total (₹)	Per unit (₹)
Depreciation (time basis)	xxx	
Interest, administration	xxx	
Total standing charges	**xxx**	**xxx**
B. Running (variable)		
Fuel, food, consumables	xxx	
Wages per trip or unit	xxx	
Depreciation (usage basis)	xxx	
Total running charges	**xxx**	**xxx**
C. Maintenance		
Repairs, tyres, spares	xxx	**xxx**
Total cost (A + B + C)	**xxx**	**xxx**
Add: Charges on takings	xxx	xxx
Add: Profit	xxx	xxx
Takings / price	**xxx**	**xxx**

On functional classification the heads are those of the manufacturer's cost sheet (direct material, direct employee cost, direct expenses, overheads), without the work-in-process and finished goods lines.

Σ Formula

Cost per unit of service

$$\text{Cost per unit} = \frac{\text{Total cost of the period}}{\text{Total service units of the period}}$$

Units: per meal, per passenger-km, per patient-day, per room-day, per project, per application (SM 12.3–12.4).

Σ Formula

Takings when charges are a percentage of takings

$$\text{Takings} = \frac{\text{Total cost}}{1 - c\% - p\%}$$

c = commission, licence fee or tax as a percentage of takings; p = profit as a percentage of takings. The equation behind it: T = cost + cT + pT (SM 12.16).

Definition

Statement of costs for service sectors (SM 12.10)

For preparing a statement of cost or a cost sheet for service sector, costs are usually collected and accumulated for a specified period viz. a month, quarter or a year. The cost statement may be prepared either on the basis of functional classification or on the basis of variability: (1) fixed costs or standing charges; (2) variable costs or operating expenses; (3) semi-variable costs or maintenance expenses.

Worked examples

Worked example **The IT-park canteen: cost per meal and price per meal**

Tapri's canteen serves 1,600 meals a day for 25 days a month. Monthly data:

Particulars	Amount
Kitchen rent	₹1,20,000 a month
Salaries of the manager and cooks (monthly)	₹2,40,000 a month
Kitchen equipment: cost ₹18,00,000, life 5 years, straight line	
Insurance	₹16,000 a month
Interest on the equipment loan	₹14,000 a month
Food materials	₹42 a meal
LPG	₹4 a meal
Plates and packing	₹3 a meal
Helpers, paid per meal served	₹5 a meal
Repairs and maintenance of equipment	₹20,000 a month + ₹1 a meal

The IT park charges a licence fee of 4% of takings, and Tapri wants a profit of 16% of takings.

Required:

(i) Prepare an operating cost sheet for the month showing standing, running and maintenance charges, in total and per meal.

(ii) Calculate the price to be charged per meal.

THE WORKING

Working note. Meals = 1,600 × 25 = 40,000. Depreciation = ₹18,00,000 ÷ 5 ÷ 12 = ₹30,000 a month (time basis, so a standing charge).

(i) Operating cost sheet, canteen, for the month (40,000 meals)

Particulars	Total (₹)	Per meal (₹)
Rent	1,20,000	
Salaries (monthly)	2,40,000	
Depreciation (time basis)	30,000	
Insurance	16,000	
Interest	14,000	
A. Standing charges	**4,20,000**	**10.50**
Food (₹42)	16,80,000	

Particulars	Total (₹)	Per meal (₹)
LPG (₹4)	1,60,000	
Plates and packing (₹3)	1,20,000	
Helpers (₹5)	2,00,000	
B. Running charges	**21,60,000**	**54.00**
C. Maintenance (20,000 + 40,000 × ₹1)	**60,000**	**1.50**
Total cost (A + B + C)	**26,40,000**	**66.00**

(ii) Price per meal

Let takings = T. T = ₹26,40,000 + 0.04T + 0.16T, so 0.80T = ₹26,40,000 and **T = ₹33,00,000**.

Particulars	Total (₹)	Per meal (₹)
Total cost	26,40,000	66.00
Licence fee (4% of ₹33,00,000)	1,32,000	3.30
Profit (16% of ₹33,00,000)	5,28,000	13.20
Takings	**33,00,000**	**82.50**

Price per meal: **₹82.50**. Adding 20% to cost would give ₹79.20 and leave Tapri short, because the fee

and the profit are fractions of the price, not of the cost.

Worked example **Tapri Chai Academy: the cost of a certified chaiwala**

Tapri runs a 6-week chai-making course in a rented studio kitchen. Each term the studio costs ₹2,40,000 and holds three batches (morning, afternoon and evening) of 20 trainees each. Every batch has its own trainer, paid ₹1,20,000 for the term, and its own set of utensils costing ₹30,000, which lasts 4 terms. Ingredients cost ₹3,000 for a trainee who completes the course. The Academy pays a food-safety exam and certificate fee of ₹1,500 for each trainee who completes. Experience shows that 2 trainees in every batch of 20 drop out in the third week, having used half their ingredients; fees are collected on enrolment and not refunded.

Required:

(i) Prepare a statement of cost for a term with three full batches, and find the cost per enrolled trainee and per certified trainee.

(ii) Calculate the fee per enrolled trainee that gives a margin of 20% on fees.

(iii) Kesar wants a fourth batch in the same studio at the fee found in (ii). Calculate the new cost per certified

trainee and the margin on fees.

THE WORKING

Working note: trainees. Enrolled 3 × 20 = 60; drop-outs 3 × 2 = 6; certified 54. Utensils = ₹30,000 ÷ 4 terms = ₹7,500 a batch a term (time basis).

(i) Statement of cost, one term, three batches

Particulars	Amount (₹)
Studio kitchen (shared by all batches)	2,40,000
Trainers (3 × ₹1,20,000)	3,60,000
Utensils (3 × ₹7,500)	22,500
Fixed cost of the term	**6,22,500**
Ingredients: 54 × ₹3,000 + 6 × ₹1,500	1,71,000
Exam and certificate fee (54 × ₹1,500)	81,000
Total cost of the term	**8,74,500**
Cost per enrolled trainee (÷ 60)	₹14,575.00
Cost per certified trainee (÷ 54)	**₹16,194.44**

(ii) Fee. Fees = ₹8,74,500 ÷ 80% = ₹10,93,125, collected from 60 enrolled trainees: **₹18,218.75**, say ₹18,219 a trainee. The fee is set per *enrolled* trainee

because drop-outs pay too; the cost per *certified* trainee is what each certificate actually costs.

(iii) A fourth batch

Particulars	3 batches (₹)	4 batches (₹)
Studio kitchen	2,40,000	2,40,000
Trainers	3,60,000	4,80,000
Utensils	22,500	30,000
Ingredients	1,71,000	2,28,000
Exam and certificate fee	81,000	1,08,000
Total cost	**8,74,500**	**10,86,000**
Certified trainees	54	72
Cost per certified trainee	**16,194.44**	**15,083.33**
Fees (₹18,219 × 60; × 80)	10,93,140	14,57,520
Margin on fees	**20.0%**	**25.5%**

The fourth batch brings its own trainer, utensils, ingredients and exam fees, but not a second studio: the ₹2,40,000 is now shared by 72 certificates instead of 54, so the cost per certified trainee falls

by ₹1,111.11 and the margin rises to ₹3,71,520 ÷ ₹14,57,520 = 25.5%.

Worked example **Tapri Tech: pricing the app by the hour**

Tapri's IT team costs ₹54,00,000 a year for its 6 developers, ₹24,00,000 for its 2 team leads and ₹18,00,000 for its project manager. Each person is paid for 2,000 hours a year, but 20% of those hours go on leave, training and waiting between projects, so only the remaining hours are productive. Office rent and utilities (₹17,28,000 a year) and administration and support staff (₹25,92,000 a year) are absorbed at a rate per productive hour of all nine staff.

A hotel chain wants an ordering app. The estimated timesheet is: developers 2,400 hours, team lead 640 hours, project manager 160 hours. Cloud servers rented for the project will cost ₹60,000, and two test phones will be hired at ₹3,000 a month each for the 4 months of the project.

Required:

(i) Calculate the cost per productive hour of each grade and the overhead rate per productive hour.

(ii) Prepare the project cost sheet and the price to quote at a margin of 25% on price.

THE WORKING

(i) Rates per productive hour

Productive hours a person = 2,000 × 80% = 1,600.

Grade	Annual cost (₹)	Productive hours	Rate per hour (₹)
Developers (6)	54,00,000	9,600	562.50
Team leads (2)	24,00,000	3,200	750.00
Project manager (1)	18,00,000	1,600	1,125.00

Overhead rate = (₹17,28,000 + ₹25,92,000) ÷ (9 × 1,600) = ₹43,20,000 ÷ 14,400 = **₹300 a productive hour**.

(ii) Project cost sheet, hotel ordering app

Particulars	Amount (₹)
Developers (2,400 hours × ₹562.50)	13,50,000
Team lead (640 hours × ₹750)	4,80,000
Project manager (160 hours × ₹1,125)	1,80,000
Direct employee cost	**20,10,000**
Overheads (3,200 hours × ₹300)	9,60,000
Cloud servers (direct expense)	60,000
Hire of test phones (2 × ₹3,000 × 4 months)	24,000
Total project cost	**30,54,000**
Profit (25% on price = 33⅓% on cost)	10,18,000
Price to be quoted	**40,72,000**

Charging each grade at its *paid* hours (₹54,00,000 ÷ 12,000 = ₹450 a developer-hour) would cost the project's hours below what they really cost, because the bench and leave hours must be paid for by the projects. Employee cost is about 66% of the project's cost (₹20,10,000 of ₹30,54,000), which is ICAI's third difference between service and product costing (SM 12.3). ICAI's own project cost sheet charges salaries by months on the project and overheads as a percentage of salaries (SM

Illustration 9, 12.35–12.36); a rate per productive hour suits a timesheet estimate.

Exam angle

In the exam

Where service cost statements appeared, May 2024 to September 2026

In the documents read for this lesson:

- **Training batch, 5 to 6 marks:** cost per batch, minimum fee for a full batch, then the fee or the fill needed for a margin on the fee: September 2024 Q1(a) (5 marks, TOEFL institute, 80% filled, 25% margin); January 2026 Q3(b) (6 marks, air-hostess institute, ₹14,800 fee, 20% margin, minimum 62.5% filled).
- **Cost per application, 8 marks:** January 2026 Q3(a), a bank's gold-loan and car-loan applications with overheads allocated by percentage.
- **Statement of cost and price on takings, 6 to 10 marks:** toll plaza, total monthly cost including interest, then toll rates at 20% profit on takings: September 2025 Q2(a) (10 marks). Bus fare with 10% commission on takings and 15% profit: January 2025 Q4(a) (8 marks). Operating cost statement with a passenger tax of 22% and profit of 25% on takings: MTP Series I September 2024 Q3(b) (8 marks); monthly operating cost sheet and fare: MTP Series II September 2026 Q2(b) (6 marks).

- **Theory:** "Differentiate between service costing and product costing": RTP September 2024 Q19(b). "What do you understand by operating costs? Describe its essential features": RTP May 2025 Q20(b).

In the exam

The traps ICAI set

- Charges and profit stated "on takings": solve T = cost + cT + pT; never add them to cost.
- Mixed periods (per month, per quarter, per annum) in one list.
- Depreciation on a time basis (standing charge) against an activity basis (running charge).
- A full batch versus a partly filled one: the fixed cost does not shrink with the batch.

Trick

The takings divisor

If commission and profit are c% and p% of takings, takings = cost ÷ (1 − c − p). For 10% and 15%: cost ÷ 0.75. Condition: both are percentages of takings, not of cost; a percentage of cost is added to cost first.

Answer frame

Differentiate between service costing and product costing. ICAI's four points, in ICAI's order (SM 12.3), as a table. ICAI's own answer (RTP September 2024 Q19(b)) states each point for services only; the product column is the contrast that makes it a "differentiate" answer.

Basis	Service costing	Product costing
Tangibility	Services are intangible and cannot be stored: no inventory	Products can be stored: inventories of WIP and finished goods
Cost units	Composite cost units (passenger-km, patient-day)	Simple units (per unit, per batch, per job)
Material vs employee cost	Employee cost is the major element	Material cost is usually the major element
Traceability	Indirect costs such as administration are significant; tracing may not be economically feasible	Larger share of costs traced directly

What do you understand by operating costs? Describe their features. ICAI's answer (RTP May 2025 Q20(b)): (1) operating costs are the costs incurred by undertakings which do not manufacture any product but provide a service, such as transport concerns, electricity

undertakings, hospitals and theatres; service costing is also known as operating costing (SM 12.2). (2) They are classified under three categories, with examples: operating and running charges (variable: petrol, diesel, lubricants), maintenance charges (semi-variable: tyres, repairs, spares) and fixed or standing charges (garage rent, insurance, road licence, depreciation, interest on capital, salary of the operating manager) (SM 12.10, 12.12). (3) The cost unit is composite, such as passenger-km or kWh (SM 12.3). Add internal and external applications (SM 12.2) if the marks allow.

Numerical answer, in order: working notes (units of service, period conversions, depreciation basis); the statement with A, B, C heads, totals and per-unit column; the takings equation; the final rate in the question's words ("fare per passenger-km", "fee per trainee").

Common mistakes

⚠ **Common mistake**

Hall of Shame: the licence fee charged on cost

Ritu's trainee priced the canteen meal at cost + 4% + 16% = ₹66 × 1.20 = ₹79.20. At that price the 4% fee and 16% profit come to ₹15.84, not the ₹13.20 left over. **Fix:** percentages of takings are solved for: ₹66 ÷ 0.80 = ₹82.50.

⚠ **Common mistake**

Closing stock in a service statement

Adding "opening and closing WIP" lines to a canteen or a training course. **Fix:** a service cannot be stored (SM 12.3); the statement runs straight from costs of the period to cost per unit.

⚠ **Common mistake**

Depreciation always fixed

A delivery van's depreciation charged "per km" goes in running charges, not standing. **Fix:** time basis = fixed; activity or usage basis = variable (SM 12.10).

⚠ **Common mistake**

The drop-outs forgotten

Costing the Academy's term for 60 completing trainees (60 × ₹4,500 of ingredients and exam fees) gives ₹8,92,500 and ₹14,875 a trainee, and divides by 60 certificates when only 54 are issued. **Fix:** cost what actually happens (54 completers, 6 half-used kits, 54 exam fees) and choose the divisor the question asks for: per enrolled trainee for the fee, per certified trainee for the cost of a certificate.

Practice

Check yourself

1. A tourist-bus operator's total operating cost for the month is ₹6,30,000. The driver's commission is 10% of takings and the operator wants a profit of 20% of takings. What must the month's takings be, in rupees?

Answer: _____

2. A day-care centre's standing charges are ₹3,60,000 a month and its running cost is ₹800 a child-day. It provided 600 child-days this month. What is its cost per child-day, in rupees?

Answer: _____

3 Case scenario. Tapri's delivery fleet is leased with a charge of ₹2 for every km run, plus a fixed monthly rent. Dev also notes that the vans' depreciation is calculated per km, and that the fleet manager is paid a fixed monthly salary. In the fleet's operating cost sheet, which treatment is correct?

A Depreciation and the lease charge per km are standing charges; the manager's salary is a running charge

B Depreciation per km and the ₹2 per km lease charge are running charges; the monthly rent and the manager's salary are standing charges

C All depreciation is a standing charge; the lease charge per km is a maintenance charge

D The manager's salary is excluded because it is administration

4 Which of these is one of ICAI's differences between service costing and product costing?

- **A** Services always use process costing
- **B** Services cannot be stored, so there is no inventory for services
- **C** Material cost is the largest element in services
- **D** Indirect costs are insignificant in services

5 In an operating cost sheet prepared on variability, which of these are standing charges? Select all that apply.

- **A** Insurance of the vehicle
- **B** Diesel
- **C** Interest on the loan for the vehicle, unless otherwise given
- **D** Driver's salary paid monthly
- **E** Tyres and spares

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- No inventory in services: no WIP or finished goods lines (SM 12.3).
- Two layouts: functional (as for products) or variability: standing (fixed), running (variable), maintenance (semi-variable) charges (SM 12.10).
- No information on semi-variable costs: show fixed and variable heads only.
- Depreciation: time basis fixed, usage basis variable. Interest: separate item of cost of sales, fixed unless otherwise given.
- Cost per unit = total cost of the period ÷ service units of the period (meal, passenger-km, patient-day, project).
- Commission, tax and profit on takings: takings = cost ÷ (1 − c − p).
- Training batches: fixed cost per batch, then per trainee; for a minimum fill, divide fixed cost by (fee × (1 − margin) − variable cost per trainee). A cost shared by several batches (one studio) falls per trainee as batches are added.
- A lunch not eaten at one o'clock is not inventory.

Answer key · Check yourself

1 **900000**

T = ₹6,30,000 + 0.10T + 0.20T, so 0.70T = ₹6,30,000 and T = ₹9,00,000. Adding 30% to cost gives ₹8,19,000.

2 **1400**

(₹3,60,000 + 600 × ₹800) ÷ 600 = ₹8,40,000 ÷ 600 = ₹1,400.

3 **B** **Depreciation per km and the ₹2 per km lease charge are running charges; the monthly rent and the manager's salary are standing charges**

Depreciation on an activity basis is variable (SM 12.10); a charge per km varies with distance. A fixed monthly rent and a monthly salary are standing charges.

4 **B** **Services cannot be stored, so there is no inventory for services**

SM 12.3: tangibility (no inventory), composite cost units, employee cost larger than material, and significant indirect costs. The other options reverse those points.

5 **A Insurance of the vehicle; C Interest on the loan for the vehicle, unless otherwise given; D Driver's salary paid monthly**

Insurance, interest (fixed unless otherwise given, SM 12.10) and a monthly salary are standing charges; diesel is a running charge; tyres and spares are maintenance charges (SM 12.12).

CHAPTER 3

Material Cost

3.1 Procurement procedure, GeM, ERP and stock verification

How material travels from the bill of materials to the store shelf and out again: every document and who prepares it, GeM, ERP, store records and surprise stock checks.

In plain words

Who ordered 40 kg of cardamom?

Monday, 9 a.m., Tapri's samosa factory in Indore. The line has stopped: no potatoes. The store holds 40 kg of cardamom nobody remembers asking for.

> **Kesar:** Potatoes zero, cardamom forty kilos. Who ordered cardamom?
>
> **Ritu:** You did, on the phone, from a wedding. No requisition, no purchase order, no goods received note. Nobody compared the rate, nobody inspected the bags, and accounts cannot pass the bill.
>
> **Kesar:** Sab adjust ho jayega.
>
> **Ritu:** Forty kilos of cardamom do not adjust. They sit.

Material is usually a factory's biggest cost, so every step happens only against a **document** from the department

that owns the step.

The **bill of materials** lists what a product needs. Production asks the store for material with a **material requisition**. When stock falls to its re-order level, the store keeper asks the buyers for more with a **purchase requisition**. The purchase department invites quotations and sends the chosen supplier a **purchase order**. Arriving goods are inspected and recorded on a **goods received note**; accounts check the bill against both papers.

Government buyers do the inviting and picking on **GeM**, the Government e-Marketplace. In an **ERP** system the documents become screens. And somebody independent of the store counts the shelves without warning: **stock verification**.

Think of it like

The wedding kitchen

A big Indian wedding kitchen runs on paper. The caterer's menu sheet says how much of everything 500 guests need (the bill of materials). The cook sends a slip to the storeroom for today's dal (the requisition). The storeroom tells the buyer what to fetch from the mandi (the purchase requisition), and the buyer confirms price and quantity with one vendor (the purchase order). At the gate someone checks the sacks before signing for them (the goods received note). And the bride's uncle counts the ghee tins at midnight, unannounced (stock verification).

Key idea

No material moves without the right document from the right department: bill of materials, material requisition, purchase requisition, purchase order, goods received note, then the bill passed against them. Someone independent checks the records against the shelves, without prior notice.

How it works

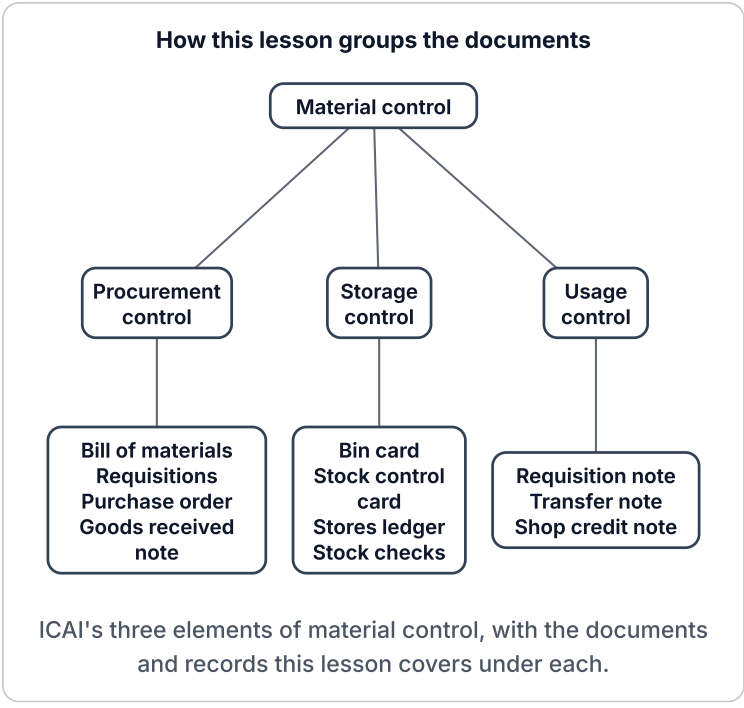
Material control: three gates

Material control is "a systematic control over the procurement, storage and usage of material so as to maintain an even flow of material" (SM 2.5). It involves

the efficient functioning of purchasing, receiving, inspection, storage and issuing of materials, maintenance of inventory records and stock audit (SM 2.6).

ICAI's five objectives of a system of material control (SM 2.4):

1. **Minimising interruption in production process**: every item that may be needed must be available, however small its cost.
2. **Optimisation of material cost**: price, ordering cost and holding cost together.
3. **Reduction in wastages** from deterioration due to defective or long storage, or from obsolescence.
4. **Adequate information**: reliable records for all items, which also help to detect losses and pilferage and to plan production.
5. **Completion of order in time**, which adds to the goodwill of the firm.

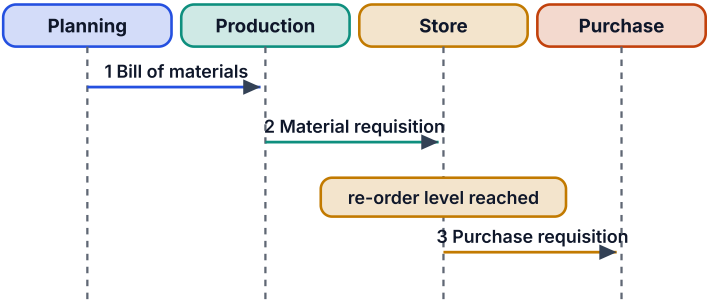


ICAI's three elements of material control, with the documents and records this lesson covers under each.

The procurement relay

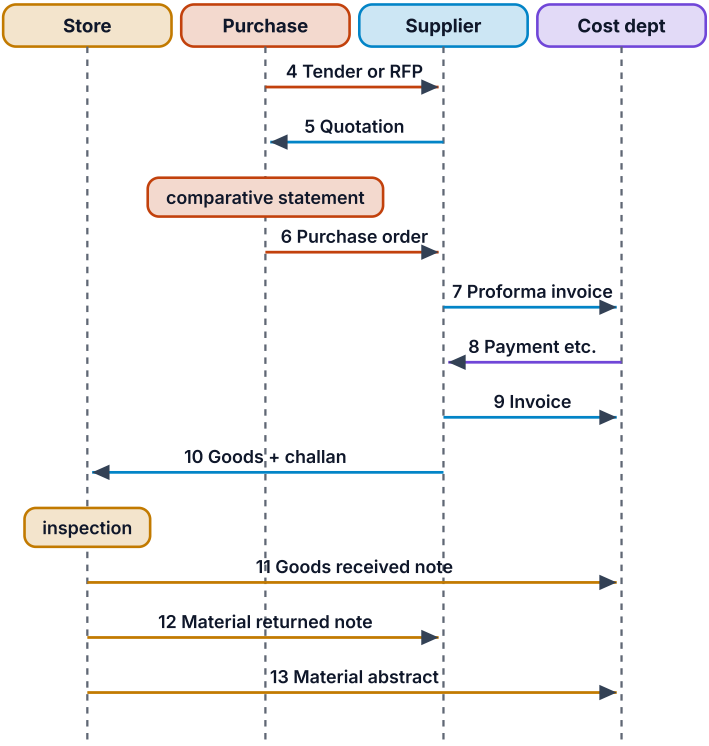
ICAI draws the procedure as 13 numbered documents passing between departments (SM 2.6; the names are illustrative). Read each picture top to bottom: first the request inside the factory, then the order out and the goods in.

Procurement procedure (1): asking for material



Planning is the engineering or planning department.

Procurement procedure (2): buying and receiving



Cost dept is the cost or accounting department; the store acts as the receiving department. The material returned note also goes to the cost department.

1 **Bill of materials** (materials specification list): "a detailed list specifying the standard quantities and qualities of materials and components required for producing a product or carrying out of any job", prepared by the engineering or planning department (SM 2.7).

2 **Material requisition note** (material requisition slip): "a voucher of authority used to get materials issued from store", generally prepared by the production department (SM 2.7).

3 **Purchase requisition**: the formal request to the purchasing department to purchase materials. Filled up by the **store keeper** for regular materials, the **departmental head** for special materials (SM 2.8).

4 **Inviting quotations**: an enquiry, **request for proposal (RFP)** or **notification inviting tender (NIT)**, offline or online, for example on GeM (SM 2.11).

5 **Selection of quotation**: a **comparative statement** weighing price, quantity, quality, delivery time, mode of transport, payment terms and the supplier's reputation (SM 2.11).

6 **Purchase order**: "a written request to the supplier to supply specified materials at specified rates and within a specified period" (SM 2.11).

7 **Receipt and inspection**: the receiving (store) department inspects the goods against the purchase order and issues a **goods received note**; goods not in good condition or not as ordered go back with a **material returned note** (SM 2.12).

8 **Checking and passing of bills**: accounts check the invoice for authenticity and arithmetic, the quantity against the goods received note and the price against the purchase order, then pass it for payment (SM 2.13).

Five questions before buying, and GeM

The purchase department answers **what** to buy, **when** (the re-order level), **how much** (the economic order quantity), **from where** and **at what price**: the lowest, **L1 bidder** is selected (SM 2.9–2.11). Where public money is spent, ICAI ties "from where" to the Preamble's "Equality of status and of opportunity": every supplier must get an equal chance in the tender.

Definition

Government e-Marketplace (GeM)

A dedicated e-market for different goods & services procured by Government Organisations / Departments / PSUs. It aims to enhance transparency, efficiency and speed in public procurement. It provides the tools of e-bidding, reverse e-auction and demand aggregation to facilitate the government users, achieve the best value for their money. The purchases through GeM by Government users have been authorised and made mandatory by Ministry of Finance. (SM 2.11)

Returns to the supplier: timing decides

Goods sent back travel with a **material returned note** (material outward return note). Returned **before** the receiving report is prepared, they never enter the report or the stores ledger: no adjustment is needed. Returned **after** the entry, the note supports the entry that takes their value out of the stores account (SM 2.12).

In the store and out of it

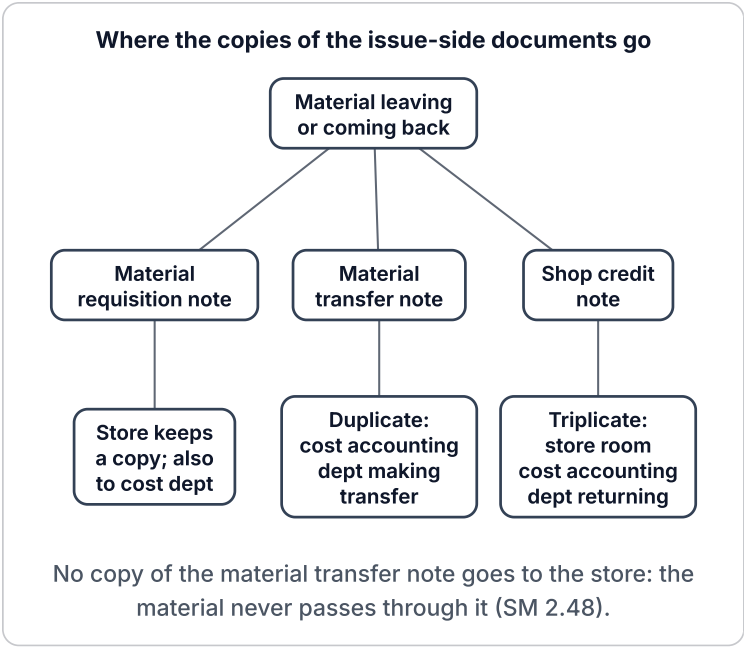
ICAI's seven duties of the store keeper (SM 2.18–2.19):

1. **General control over store**: checking quantities against the goods received note and arranging storage in appropriate places.
2. **Safe custody of materials** in the condition and environment that preserve their quality.

3. **Maintaining records** of quantity received, issued, in hand and transferred.
4. **Initiate purchase requisition** for regular items whenever stock approaches the re-order level.
5. **Maintaining adequate level of stock**, stopping further purchases as stock approaches the maximum level, and reserving material for a specific job when required.
6. **Issue of materials** only against an approved material requisition, with reference to the bill of materials.
7. **Stock verification and reconciliation** of book balances with physical stock at frequent intervals, checking irregular issues and pilferage.

The records are the **bin card** and **stock control card** (quantities, stores department) and the **stores ledger** (quantity and value, cost department).

Issues are made only against an authorised **material requisition note**, out of the earliest lot in hand. Surplus too bulky, heavy or brittle to go back may move straight to another job with a **material transfer note**; ICAI stresses that such job-to-job transfer is generally "irregular, if not improper" and should be made only in these circumstances (SM 2.47–2.48). Surplus that goes back to the store travels with a **shop credit note** (stores debit note), so the job is credited. From these three, the store keeper prepares the **material abstract**, the value charged to each product and to overheads (SM 2.71).



No copy of the material transfer note goes to the store: the material never passes through it (SM 2.48).

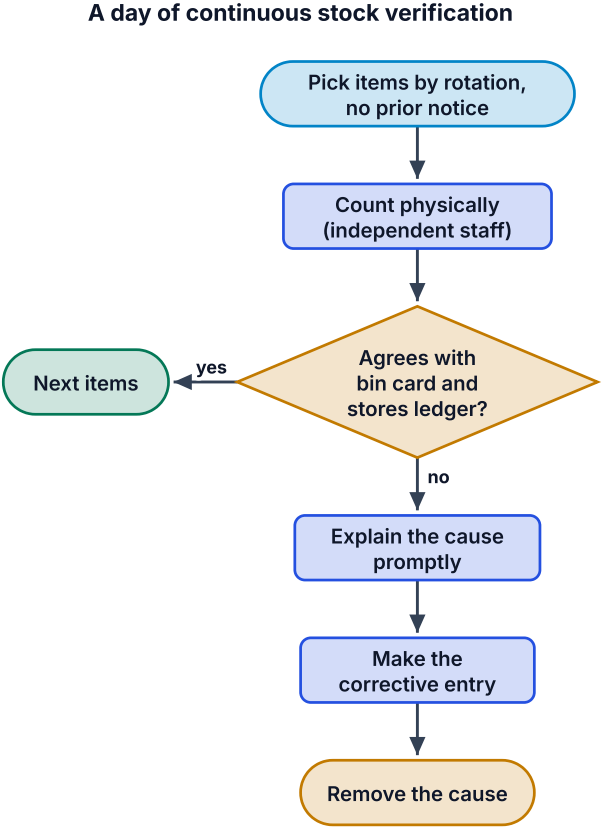
ERP: the same documents, fewer copies

With **Enterprise Resource Planning (ERP)** (SM 1.17–1.18; more in digital costing) one entry gives reports for every purpose; bills of material, requisitions and goods received notes are no longer made in multiple copies (each department takes an e-copy); and purchase orders reach suppliers over an extranet, a step towards just-in-time.

Stock verification: count without warning

"The checking of physical inventory is an essential feature of every sound system of material control" (SM 2.46). Under **continuous stock verification**, items are counted at appropriate intervals without prior notice, by

staff independent of the store and production (for example internal audit): "the element of surprise is essential". **Annual or periodic stock-taking** must cover every item in a given number of days, so production is shut down or the count is of a limited character.



Steps (d) to (g) of the perpetual inventory system (SM 2.45): count by rotation, explain, correct, remove the cause.

Formats and formulas

Asked as "distinguish" questions. *Reproduce the points, in this order, from memory.*

Bill of materials vs material requisition note (SM 2.8)

Bill of materials	Material requisition note
Prepared by the engineering or planning department.	Prepared by the production or other consuming department.
A complete schedule of component parts and raw materials required for a particular job or work order.	A document asking the store keeper to issue materials to the consuming department.
Often serves as a material requisition: it can replace material requisitions.	Cannot replace a bill of materials.
Can be used for the purpose of quotations.	Useful in arriving at historical cost only.
Helps in keeping a quantitative control on materials drawn through material requisitions.	Shows the material actually drawn from stores.

Bin card vs stores ledger (SM 2.21)

Bin card	Stores ledger
Maintained by the store keeper in the store.	Maintained in the cost accounting department.
Contains only quantitative details of material received, issued and returned to stores.	Contains information both in quantity and value.
Entries are made when the transaction takes place.	Always posted after the transaction.
Each transaction is individually posted.	Transactions may be summarised and then posted.
Inter-department transfers do not appear.	Material transfers from one job to another are recorded for costing purposes.

Document

Purchase requisition

Usually prepared by

Store keeper (regular); departmental head (special)

Copies

—

Document

Purchase order

Usually prepared by

Purchase manager or concerned officer

Copies

Store or indenting dept, receiving dept, cost accounting dept; one on file with the requisitions

Document

Goods received note

Usually prepared by

Receiving department

Copies

Quadruplicate: purchase, store or indenting, receiving, accounting

Document

Material transfer note

Usually prepared by

Not named by ICAI; made at the time of transfer

Copies

Duplicate: cost accounting dept, department making the transfer

Document

Shop credit note

Usually prepared by

Department returning the surplus

Copies

Triplicate: store room, cost accounting dept, department returning it

Definition

Stores ledger

A ledger containing a separate account for each item of material and component stocked in store giving details of the receipts, issues and balance both in terms of quantity and value. A subsidiary ledger to the main cost ledger, posted from goods received notes, material requisition notes etc. (SM 2.20, 2.73)

Formula

Consumption of materials in the financial accounts

$$\text{Consumption} = \text{Opening stock} + \text{Purchases} - \text{Closing stock}$$

Stock is taken at cost or net realisable value, whichever is less (SM 2.72).

Worked examples

Worked example

The samosa factory's paper trail

For a Diwali order, Tapri's samosa factory buys potatoes, a regular item:

(a) Planning lists the potatoes, maida and spices needed per batch of 1,000 samosas.

(b) Production asks the store for 1,200 kg of potatoes for Monday.

(c) After the issue, stock falls below the re-order level, so the store keeper asks the purchase department to buy 5,000 kg.

(d) The purchase department invites offers from farmers' co-operatives on its website.

(e) It tabulates the three offers and orders 5,000 kg from Malwa Kisan Co-operative at ₹22 a kg, in writing.

(f) 5,000 kg arrive with a challan and an invoice. At the gate 150 kg are rotten and sent straight back; 4,850 kg are accepted and recorded.

(g) Two days after the receipt was recorded, 100 kg more are found sprouted and sent back.

Required:

(i) Name the document at each of (a) to (g) and the department that prepares it.

(ii) How many copies of the receipt document at (f) are usually prepared, and who gets them?

(iii) Do the returns at (f) and (g) need the same treatment in the stores ledger?

(iv) For what quantity and amount should accounts pass the invoice, and against which documents is it checked?

MODEL ANSWER

(i) Documents

Step	Document	Prepared by
(a)	Bill of materials	Engineering or planning department
(b)	Material requisition note	Production department
(c)	Purchase requisition	Store keeper (regular material)
(d)	Request for proposal / enquiry	Purchase department
(e)	Comparative statement, then purchase order	Purchase department
(f)	Goods received note (4,850 kg); material returned note (150 kg)	Receiving (store) department
(g)	Material returned note (100 kg)	Receiving (store) department

(ii) Generally four (quadruplicate): purchase department, store or order indenting department, receiving department and accounting department.

(iii) No. The 150 kg went back before the receiving report was prepared: never in the stores ledger, no adjustment. The 100 kg went back after their entry, so the material returned note supports an entry excluding their value from the stores account.

(iv) Quantity is checked against the goods received note and price against the purchase order. Accepted quantity = 5,000 − 150 − 100 = **4,750 kg**; amount = 4,750 × ₹22 = **₹1,04,500**. The invoice for ₹1,10,000 is passed only for this amount; the difference of 250 kg (₹5,500) is taken up with the supplier.

Worked example **Bunty and the Saturday stock count**

Tapri's Dewas store holds about 1,400 items. Kesar proposes closing the store and packing line for four days every March for a full count. The store keeper offers: "Give me Friday the list of 20 items you will count on Saturday." Ritu proposes that the internal audit cell count a rotating set of items every day, without notice.

On one such count, 36 cartons of kulhads are short against the bin card. (Bunty, attending for the statutory auditors, had copied the bin card figure onto his count sheet before counting. Ms. Iyer was not amused.)

Required:

(i) Advise on the two proposals.

(ii) Comment on the store keeper's offer.

(iii) State what should follow the shortage.

MODEL ANSWER

(i) Kesar's proposal is annual stock-taking. Since all items have to be covered in a given number of days, either production has to be shut down to enable a thorough check, or the verification must be of a limited character (SM 2.46). Ritu's proposal, continuous stock verification, is advisable because:

1. Closure of normal functioning is not necessary.
2. Stock discrepancies are likely to be brought to notice and corrected much earlier.
3. The element of surprise has a sobering influence on the stores staff.
4. The stores auditor can watch the movement of items more closely, so the chances of obsolescence buying are reduced.
5. Final accounts can be ready quickly, and interim accounts are possible quite conveniently.

(ii) The offer defeats the purpose. Stock verification should be done by staff independent of the store and production staff, at appropriate intervals and without prior notice; the element of surprise is essential for effective control.

(iii) The cause of the discrepancy should be explained promptly, a corrective entry made and the cause removed (SM 2.45). The shortage is entered in the issue column of the stores ledger at the rate of the pricing method in use (SM 2.63). If its cause is abnormal (pilferage, mishandling) it is

charged to the Costing Profit and Loss Account; if normal, its cost is absorbed by the good units (SM 2.15).

Worked example **State with reasons: true or false**

State with reasons whether each statement is true or false. (5 marks)

(i) A bill of materials can serve as a material requisition, but a material requisition note cannot replace a bill of materials.

(ii) A purchase requisition for a special material not stocked as a regular item is filled up by the store keeper.

(iii) Material returned to the supplier before the receiving report is prepared must be adjusted in the stores ledger.

(iv) Purchases through GeM by Government users are optional.

(v) A purchase order is the store keeper's request to the purchase department to buy materials.

MODEL ANSWER

(i) True. A bill of materials is a complete schedule of the materials required for a particular job, so it often serves as a material requisition. A material

requisition note only shows the material actually drawn from stores and cannot replace a bill of materials.

(ii) False. The store keeper fills up the purchase requisition for regular materials; for special materials not stocked as regular items, the departmental head does.

(iii) False. Such material is not included in the receiving report and hence not shown in the stores ledger, so no adjustment in the books is necessary.

(iv) False. Purchases through GeM by Government users have been authorised and made mandatory by the Ministry of Finance.

(v) False. The store keeper's formal request to the purchasing department to purchase materials is the purchase requisition. A purchase order is a written request to the supplier to supply specified materials at specified rates and within a specified period, sent by the purchase department (SM 2.8, 2.11).

Exam angle

What ICAI asks from this topic

In the exam

Where these documents appeared, May 2024 to September 2026 (and the old scheme)

- **Bill of materials vs material requisition note**: MTP Series II for May 2025 Q6(b) (5 marks); September 2026 Q6(c) OR, "in terms of their contents and utility" (4 marks); old-scheme MTPs for November 2019 and November 2020, Q6(b).
- **Uses of the bill of materials** by department: September 2025 Q6(c) OR (4 marks); December 2021 (old scheme) Q4(c), with the definition (5 marks).
- **Bin card vs stores ledger**: old-scheme MTPs for May 2022 Q6(a) and May 2023 Q6(e).
- **True or false with reasons**: November 2023 (old scheme) Q6(b) (5 marks): "Material requisition note is prepared by the store keeper" and "Bin card and stores ledger are maintained by the purchasing department", both false.
- **Impact of IT on cost accounting** (ERP): MTP Series I for May 2024 Q6(b) (5 marks); RTP May 2024; January 2021 (old scheme) Q6(e).
- **GeM and stock verification**: in none of the documents read; they are in the syllabus and the Study Material, so expect a short note or an MCQ.

In the exam

The traps

- The store keeper prepares the **purchase** requisition (regular items), never the material requisition.
- Bin card: quantity, store keeper. Stores ledger: quantity and value, cost department.
- Two returns, two papers: to the **supplier**, a material returned note; back to the **store**, a shop credit note (which ICAI's summary also calls a material return note). Name the direction.
- The invoice: quantity against the goods received note, price against the purchase order.

Trick

Plan, Pull, Push, Promise, Proof

The first five documents in order: **Plan** (bill of materials), **Pull** from the store (material requisition), **Push** to the buyer (purchase requisition), **Promise** to the supplier (purchase order), **Proof** of receipt (goods received note). Holds for a regular item bought for stock; for a special item the purchase requisition comes from the departmental head.

Answer frame

"Distinguish" and "uses" questions

For "distinguish", use the two tables in *Formats and formulas*: five points each, starting with who prepares or

maintains it. For "uses of the bill of materials" (SM 2.7), open with the definition, then:

Department	Use
Marketing (purchase)	Materials are procured on the basis of the specifications mentioned in it.
Production	Production is planned according to the nature and volume of materials required; material requisition lists are prepared accordingly.
Stores	A reference document while issuing materials to the requisitioning department.
Cost / accounting	Used to estimate cost and profit; purchase, issue and usage are compared and verified against it.

"Write a note on continuous stock verification"

1. Definition: physical verification at appropriate intervals, without prior notice, by staff independent of the store and production; the element of surprise is essential.
2. The shortcoming of annual stock-taking: shutdown, or a count of limited character.
3. The five advantages, in ICAI's order (Example 2).

"State with reasons whether true or false"

Verdict first, then the provision or definition that decides it, then the correct position (Example 3). Five statements,

one mark each, as in November 2023 (old scheme) Q6(b).

Three answers to "Material requisition note is prepared by the store keeper. True or false?"

Loses marks

"False."

The reason carries the mark.

Passes

"False. It is prepared by the production department."

Correct, but says nothing of what the document is.

Full marks

"False. The material requisition note is a voucher of authority used to get materials issued from store; it is prepared by the production or other consuming department. The store keeper prepares the purchase requisition for regular materials."

Common mistakes

⚠ **Common mistake**

Hall of Shame: the store keeper who wrote his own permission slip

Writing that the store keeper prepares the material requisition note. If the person holding the stock also writes the authority to release it, there is no control. **Fix:** the consuming department prepares the material requisition; the store keeper issues against it.

⚠ **Common mistake**

The return that got the wrong paper

Sending shop-floor surplus back to the store "with a material returned note", or adjusting the stores ledger for goods rejected at the gate. **Fix:** to the supplier, a material returned note, with a books adjustment only if the goods were already in the receiving report; to the store, a shop credit note (stores debit note) in triplicate, made out by the returning department. ICAI's summary calls the note for returns to the store a "material return note", so always name the direction.

⚠ **Common mistake**

The surprise check that was announced

Writing that stock verification is done "by the store keeper at year end, after informing the store". **Fix:** continuous verification is by independent staff, at intervals, without notice.

Practice

Check yourself

1 Case scenario. Malwa District Hospital, a government hospital, needs 2,00,000 paper cups. Its purchase officer plans to phone three local vendors he knows and buy from the cheapest. The finance head objects. Which of the following is correct?

A The plan is fine: a phone enquiry of three vendors is a comparative statement.

B The hospital should buy through GeM, the e-market for government buyers with e-bidding, reverse e-auction and demand aggregation; it is mandatory for government users.

C GeM covers only services, so the officer should float a newspaper tender.

D The highest bidder should be selected, as a higher price signals better quality.

2 Case scenario. Ritu's intern is checking a supplier's bill for 4,000 kulhad sets at ₹22 each before accounts pass it for payment. Against which documents should the quantity and the price on the bill be checked?

- **A** Quantity against the purchase order; price against the goods received note
- **B** Quantity against the purchase requisition; price against the comparative statement
- **C** Quantity against the goods received note; price against the purchase order
- **D** Quantity against the bin card; price against the stores ledger

3 The goods received note is generally prepared in quadruplicate. Which of these does NOT get a copy?

- **A** Purchase department
- **B** Store or order indenting department
- **C** Accounting department
- **D** The supplier

4 Which of these receive a copy of a material transfer note? Select all that apply.

- **A** Cost accounting department
- **B** Department making the transfer
- **C** Store
- **D** Purchase department

More than one answer may be right.

5 Tapri's chai premix plant had raw material of ₹4,20,000 on 1 April, bought ₹18,60,000 in the year and held ₹3,90,000 on 31 March (all at cost, below net realisable value). What was the consumption of materials for the year, in rupees?

Answer: _____

6 "Entries in the bin card are made when the transaction takes place, and inter-department transfers appear in the bin card." Which parts are correct?

- **A** Both
- **B** Neither
- **C** Only the first
- **D** Only the second

Answers: the answer key at the end of this lesson.

Recap

- Material control: systematic control over procurement, storage and usage to keep an even flow of material.
- A supplier's invoice is passed after checking quantity against the goods received note and price against the purchase order.
- Bin card: quantity only, kept by the store keeper. Stores ledger: quantity and value, kept by the cost department.
- Documents in order: bill of materials (planning) → material requisition (production) → purchase requisition (store keeper, or departmental head for special items) → RFP / NIT → comparative statement → purchase order → goods received note or material returned note → bill passed.
- GeM: e-market for government buyers; e-bidding, reverse e-auction, demand aggregation; mandatory for government users; L1 bidder selected.
- Copies: GRN quadruplicate; material transfer note duplicate (no store copy); shop credit note triplicate.
- ERP: one entry for all departments; e-copies instead of multiple copies.
- Continuous stock verification: independent staff, intervals, no prior notice; annual counts mean

shutdown or a limited count. (Cardamom does not adjust.)

Answer key · Check yourself

1 **B** **The hospital should buy through GeM, the e-market for government buyers with e-bidding, reverse e-auction and demand aggregation; it is mandatory for government users.**

GeM is the e-market for goods and services procured by government buyers, mandatory for them. Phoning known vendors fails the equality-of-opportunity test; the L1 (lowest) bidder is selected.

2 **C** **Quantity against the goods received note; price against the purchase order**

The goods received note records what was actually received and accepted; the purchase order records the agreed rate (SM 2.13). The requisition only asked for a purchase, and the bin card and stores ledger record stock, not the supplier's terms.

3 **D** **The supplier**

The four copies go to the purchase, store or indenting, receiving and accounting departments.

4 **A Cost accounting department; B Department making the transfer**

Duplicate; the material never passes through the store.

5 **1890000**

Consumption = ₹4,20,000 + ₹18,60,000 – ₹3,90,000 = ₹18,90,000.

6 **C Only the first**

Bin card entries are made as each transaction takes place (the stores ledger is posted afterwards). Inter-department transfers do not appear in the bin card: the material does not pass through the store (SM 2.21).

3.2 Valuing material receipts

What one unit of purchased material really costs: which invoice items are added, deducted or kept out, and how normal and abnormal shortages change the rate.

In plain words

Kesar's ₹214.50 tea that cost ₹208.16

Tapri buys 1,000 kg of Assam tea. Kesar adds up every rupee on the paperwork, ₹2,14,500, and announces that tea costs ₹214.50 a kg.

> **Ritu:** Per usable kg, please. The ₹10,000 GST comes back to us as input credit. The ₹500 demurrage is a fine for keeping the truck waiting.
>
> **Kesar:** The forklift was at my nephew's wedding.
>
> **Ritu:** Then the forklift pays it, not the tea. And 20 kg were lost breaking open the chests, which happens every time.

Ritu's tea costs the price ₹2,00,000 plus the freight ₹4,000 that brought it to Indore: ₹2,04,000. That cost is carried by the 980 kg that can actually be used: **₹208.16 a kg**.

Working out this figure is **valuation of material receipts**: the cost at which each unit enters the stores ledger and,

later, every product that uses it. The rule of thumb has three parts:

- **In**: the price after trade discounts, plus every cost of bringing the material to the store (freight, transit insurance, customs duty, commission, containers that are not returned).
- **Out**: anything that is a finance reward (a cash discount for paying early), a tax that will be recovered (GST with input credit), or the price of a mistake (demurrage, fines, penalties).
- **Spread**: a normal shortage is shared by the good units; an abnormal one is written off.

Think of it like

Your new phone

What did your phone cost? The shop's price after its discount, plus the delivery charge and the transit insurance. Not the bank's cashback for paying by card, which rewards how you paid. Not the parking fine you collected while picking it up, which is your own mistake. And not the GST you will claim back as a business. If the box came with a charger that never worked, the phone's cost is still the whole bill: the good part carries it.

Key idea

Material cost = purchase price net of trade discounts, rebates, subsidies and recoverable taxes + the costs of bringing it to its present location and condition. Cash discount, demurrage, fines and penalties stay out. A normal shortage raises the cost of each good unit; an abnormal one goes to the Costing Profit and Loss Account.

How it works

"Ascertainment of cost of material purchased is called valuation of materials receipts. Cost of material includes cost of purchase net of trade discounts, rebates, duty draw-back, input credit availed, etc. and other costs incurred in bringing the inventories to their present location and condition" (SM 2.13).

Seven steps from invoice to rate per unit

1. **Start from the listed price** and deduct **trade** and **quantity discounts** if the invoice has not already done so.

2 **Taxes**: exclude **GST** when its credit is available (unless a question says otherwise, it is not part of cost); include it when it is not, for example for a buyer not registered under GST. Add **customs duty** on imports and **road or toll tax** paid by the buyer.

3 **Add the costs of getting it here**: freight inwards, insurance in transit, commission or brokerage, non-returnable containers, and for returnable containers only the part of the deposit not refunded.

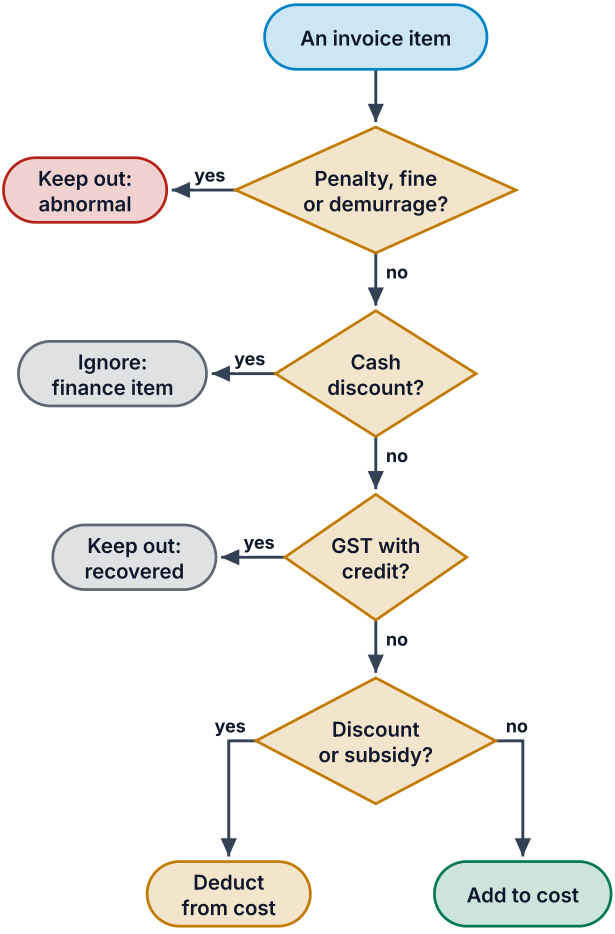
4 **Keep out** the **cash discount** (an interest and finance item, ignored even when received) and the abnormal costs: **demurrage**, **detention charges or fines** and any **penalty**.

5 **Deduct** any **subsidy, grant or incentive** received from the Government or other sources.

6 **Share costs** common to two materials on a suitable basis; ICAI's illustration splits freight in the ratio of quantity purchased (SM 2.16).

7 **Divide by the good quantity**: units purchased less the **normal shortage** (and any provision for deterioration). An **abnormal shortage** is valued at that rate and debited to the Costing Profit and Loss Account.

An item on the invoice: in or out?



Rose: abnormal costs (penalties, fines, demurrage, detention charges); grey: the cash discount and GST whose credit is available; amber: deductions; green: everything that brings the material to the store (price, freight, insurance, customs duty, toll tax, commission, containers not returned).

Shortages: normal or abnormal?

A **normal shortage** (breaking of bulk, evaporation, unavoidable conditions) is absorbed by the good units: the total cost stays, the quantity falls, the rate rises. An **abnormal shortage** (mishandling, pilferage, avoidable reasons) is not absorbed by the good units; its cost is debited to the Costing Profit and Loss Account (SM 2.15). ICAI's illustration also treats a **provision for further deterioration** as a reduction in usable quantity (SM 2.16).

Interactive version: PassBook app, lesson Valuing material receipts

Formats and formulas

ICAI's table of treatments (SM 2.13–2.15)

Learn the groups and the order: "state the treatment of" questions use exactly these words.

Item	Treatment
Discounts and subsidy	
(i) Trade discount	Deducted from the purchase price if not shown as deduction in the invoice.
(ii) Quantity discount	Like trade discount: deducted if not shown as deduction.
(iii) Cash discount	Not deducted; treated as interest and finance item. Ignored.

Item	Treatment
(iv) Subsidy / grant / incentives	Deducted from the cost of purchase.
Duties and taxes	
(v) Road tax / toll tax	Included in the cost of purchase if paid by the buyer.
(vi) GST	Excluded if credit is available; unless mentioned specifically, not part of cost.
(vii) Customs duty	Added to the purchase cost.
Penalty and charges	
(viii) Demurrage	Penalty imposed by the transporter for delay in loading or unloading: abnormal cost, not included.
(ix) Detention charges / fine	Imposed by a statutory authority for non-compliance: abnormal cost, not included.
(x) Penalty	Of any type, not included.
Other expenditures	
(xi) Insurance charges	Paid for protecting goods in transit: added.

Item	Treatment
(xii) Commission or brokerage	Added.
(xiii) Freight inwards	Added: directly attributable to procurement.
(xiv) Cost of containers	Non-returnable: added. Returnable and refunded: not considered. Refund less than amount paid: only the shortfall added.
(xv) Shortage	Normal: absorbed by good units. Abnormal: not absorbed, debited to Costing P&L.

The statement ICAI's answers use

Reproduce this layout: units in one column, rupees in the other.

Particulars	Units	Amount (₹)
Listed price of materials	xxx	xxx
Less: Trade discount		(xxx)
Add: GST (only if credit is not available)		xxx
Add: Toll tax, freight and insurance, commission		xxx
Add: Cost of returnable containers (deposit less refund)		xxx
Add: Other expenses		xxx
Total cost of material	**xxx**	**xxx**
Less: Normal shortage	(xxx)	—
Total cost of good units	**xxx**	**xxx**
Cost per unit		**xxx**

Notes below the statement explain each item left out (cash discount, demurrage, GST with credit).

∑ **Formula**

Cost per unit of material received

$$\text{Rate} = \frac{\text{Total cost of material}}{\text{Good units}}$$

Good units = units purchased − normal shortage. An abnormal shortage is valued at this rate and charged to the Costing Profit and Loss Account.

∑ **Formula**

"Other expenses at x% of total cost"

$$\text{OE} = C \times \frac{x}{100 - x}$$

OE = other expenses; C = the cost before them. The total cost includes the other expenses themselves, so 2% of total cost is 2/98 of the cost before them (ICAI's convention in MTP May 2024 and RTP September 2024).

Worked examples

Worked example **Two quotes for dried tulsi: which gives the lower ledger rate?**

Tapri needs 2,500 kg of dried tulsi leaves for its tulsi chai premix. Tapri is registered under GST, and GST at 5% (input tax credit available) is charged under both offers.

Particulars	Offer A: Rishikesh producer company	Offer B: Nashik trader
Price	₹340 per kg ex-works, less 5% trade discount	₹345 per kg delivered at Indore
Freight to Indore	₹21,000, paid by Tapri	Included
Transit insurance	₹3,500, paid by Tapri	Included
Packing	Returnable crates: deposit ₹12,000, refund ₹9,500 on return	Non-returnable sacks, ₹2,000 billed
Cash discount	2% if paid within 10 days	None
Normal loss of moisture in transit	2%	1%

Required:

(i) Calculate the cost per good kg at which the tulsi would enter the stores ledger under each offer.

(ii) Which offer should Tapri accept? State how the 2% cash discount affects your comparison.

THE WORKING

(i) Statement of cost of 2,500 kg

Particulars	Offer A (₹)	Offer B (₹)
Listed price: 2,500 kg × ₹340 / ₹345	8,50,000	8,62,500
Less: Trade discount @ 5%	(42,500)	—
	8,07,500	8,62,500
Add: Freight	21,000	—
Add: Transit insurance	3,500	—
Add: Crates (₹12,000 − ₹9,500) / sacks	2,500	2,000
Total cost of material	**8,34,500**	**8,64,500**
Good quantity: 2,500 kg less normal loss of 2% / 1%	2,450 kg	2,475 kg
Cost per good kg	**₹340.61**	**₹349.29**

Notes: (1) GST is excluded under both offers because input tax credit is available. (2) For returnable crates only the part of the deposit not refunded is a cost; non-returnable sacks are added in full. (3) The normal moisture loss reduces the quantity, not the cost, so the good kilograms carry it.

(ii) Offer A gives the lower stores ledger rate by ₹8.68 a kg (₹349.29 − ₹340.61), even though its loss in transit is higher and Tapri pays the freight and insurance itself. Accept Offer A.

The cash discount does not enter the comparison of material cost: it is an interest and finance item and is ignored in valuing the receipt, whether or not it is taken (SM 2.13). If Tapri pays within 10 days, the saving is a finance gain recorded separately; it makes Offer A more attractive still but does not change its ledger rate of ₹340.61.

Worked example

One truck, two materials: freight by weight, insurance by value

A dairy co-operative in Gujarat sends Tapri one truck carrying both materials:

Particulars	Sugar	Milk powder
Quantity (kg)	10,000	4,000
Price per kg (₹)	42	320

Tapri pays freight of ₹28,000 for the truck, charged by weight, and transit insurance of ₹17,000 on the value of the goods. GST is charged on both materials and input tax credit is available. On unloading, 100 kg of sugar have spilled from torn jute bags, which is usual for

sugar in jute. 40 kg of milk powder were stolen when the driver left the truck unattended overnight at a roadside dhaba.

Required:

(i) Compute the rate per kg of each material for the stores ledger.

(ii) Show the treatment of the 40 kg of milk powder.

THE WORKING

Working note: sharing the common costs

Common cost
Freight ₹28,000
Basis
Weight, 10,000 : 4,000 = 5 : 2
Sugar (₹)
20,000
Milk powder (₹)
8,000

Common cost
Insurance ₹17,000
Basis
Value, ₹4,20,000 : ₹12,80,000 = 21 : 64
Sugar (₹)
4,200
Milk powder (₹)
12,800

Each common cost is shared on the basis that drives it: the freight was charged by weight, the

insurance premium on the value insured.

(i) Statement showing the rate per kg

Particulars	Sugar (₹)	Milk powder (₹)
Purchase price	4,20,000	12,80,000
Add: Freight	20,000	8,000
Add: Transit insurance	4,200	12,800
Total cost (A)	**4,44,200**	**13,00,800**
Quantity carrying the cost (B)	9,900 kg	4,000 kg
Rate per kg (A ÷ B)	**44.87**	**325.20**

The sugar spilled from torn bags is a normal loss (breaking of bulk), so the 9,900 good kg carry the whole cost. The theft of milk powder is abnormal, so it is not absorbed: the rate is worked on all 4,000 kg.

(ii) The 40 kg stolen are an abnormal shortage, valued at the rate of the good units, 40 × ₹325.20 = **₹13,008**, and debited to the Costing Profit and Loss Account (SM 2.15). The stores ledger takes in 3,960 kg × ₹325.20 = ₹12,87,792.

Check: ₹4,44,200 + ₹12,87,792 + ₹13,008 = ₹17,45,000 = price ₹17,00,000 + freight ₹28,000 +

insurance ₹17,000.

Worked example

Diwali tins: the cost per good tin and the highest price Tapri can accept

For its Diwali premix, Tapri orders 25,000 printed tins from Sivakasi Tin Works. Tapri is registered under GST.

Particulars	Details
Listed price	₹32 per tin
Trade discount	12.5% of the listed price
Commission to the buying agent	2.5% of the listed price
GST	18% (input tax credit available)
Freight and transit insurance	₹24,000
Demurrage paid to the transporter for late unloading	₹3,500
Cash discount	2% if paid within 15 days; Tapri paid in 12 days

4% of the tins arrive dented, which is normal for printed tins in transit; they are scrapped with no value.

Required:

(i) Prepare a statement of the cost of the tins and calculate the cost per good tin.

(ii) Kesar's budget for the Diwali premix allows ₹28.75 per good tin. Other terms remaining the same, what is the highest listed price per tin Tapri can accept?

THE WORKING

(i) Statement of cost of printed tins

Particulars	Units	Amount (₹)
Listed price: 25,000 × ₹32	25,000	8,00,000
Less: Trade discount @ 12.5%		(1,00,000)
		7,00,000
Add: Commission to the buying agent @ 2.5% of listed price		20,000
Add: Freight and transit insurance		24,000
Total cost of material	**25,000**	**7,44,000**
Less: Normal shortage @ 4%	(1,000)	—
Total cost of good units	**24,000**	**7,44,000**
Cost per good tin		**31.00**

Notes:

1. GST of 18% is excluded: input tax credit is available.
2. Demurrage of ₹3,500 is a penalty imposed by the transporter: an abnormal cost, not included.
3. The cash discount is an interest and finance item and is ignored, though it was received.

4. The normal shortage reduces the units, not the cost.

(ii) Highest listed price within the budget

Let the listed price per tin be P. For each tin listed, Tapri pays P less 12.5% trade discount plus 2.5% commission = 0.90P; the freight of ₹24,000 does not depend on the price.

$$\frac{25{,}000 \times 0.90P + 24{,}000}{24{,}000} = 28.75$$

25,000 × 0.90P = (₹28.75 × 24,000) − ₹24,000 = ₹6,66,000, so P = ₹6,66,000 ÷ 22,500 = **₹29.60 per tin**.

Check: ₹7,40,000 listed − ₹92,500 trade discount + ₹18,500 commission + ₹24,000 freight = ₹6,90,000; ÷ 24,000 good tins = ₹28.75. Tapri must negotiate the listed price down from ₹32 to ₹29.60, or the trade discount up, to stay within budget.

Exam angle

What ICAI asks from this topic

In the exam

Where receipt valuation appeared, May 2024 to September 2026 (and the old scheme)

- **Full statement, cost per unit**: MTP Series I for May 2024 Q2(b) (5 marks): trade discount, GST without credit (unregistered buyer), cash discount, toll tax, freight, demurrage, commission, returnable containers, other expenses at 2% of total cost, 5% normal shortage. Old-scheme MTP Series I for November 2023 Q1(c): the same pattern.
- **Case scenario, five MCQs**: RTP September 2024 Q1: the cash discount and how it is treated, the other expenses, the GST, the total cost and the cost per good unit.
- **"State the treatment of" items, 5 marks**: May 2022 (old scheme) Q6(b): cash discount, IGST, demurrage, shortage, basic customs duty. January 2021 (old scheme) Q6(a): detention charges, demurrage, returnable containers, CGST, abnormal shortage.
- **The same items in a cost sheet**: September 2026 Q6(c) (4 marks): a State subsidy for eco-friendly raw materials, GST with input credit; MTP Series II for May 2025 Q3(b) (4 marks): subsidy, penalty and demurrage, interest; demurrage in the cost sheets of MTP Series I for January 2025 Q3(a) and RTP May 2026.

Interactive version: PassBook app, lesson Valuing material receipts

In the exam

The traps

- The cash discount was received: still ignored. The statement uses the price after trade discount, never after cash discount.
- GST is worked out on the price after trade discount (18% of ₹2,70,000, not ₹3,00,000).
- "Other expenses @ 2% of total cost" means 2/98 of the cost before them.
- A normal shortage is not deducted from cost: only the units fall.
- Read who the buyer is: "unregistered" decides the GST.

Trick

Price, Path, Penalty

Price net of trade discounts and subsidies: in. Every cost of the **path** to the store (freight, insurance, duty, toll, commission, containers kept): in. Every **penalty** for a mistake (demurrage, fines, detention) and every finance reward (cash discount): out. Holds for any purchase; taxes are the one exception to check, in only when their credit is not available.

Answer frame

"State the treatment of the following items

associated with the purchase of materials"

One line per item, ICAI's words, one mark each:

Item	Full-marks line
Cash discount	Not deducted from the purchase price; treated as an interest and finance item and ignored.
GST / IGST / CGST	Excluded from the cost of purchase if credit for the same is available; unless mentioned specifically, not part of cost.
Demurrage	A penalty imposed by the transporter for delay in loading or unloading; an abnormal cost, not included.
Basic customs duty	Paid on import of goods; added to the purchase cost.
Returnable containers	Not considered if refunded; if the refund is less than the amount paid, only the shortfall is added.
Shortage	Normal: absorbed by the good units. Abnormal: not absorbed; debited to the Costing Profit and Loss Account.

The numerical: three answers

Loses marks

"Cost per tin = ₹8,26,000 ÷ 25,000 = ₹33.04."

The supplier's bill (price after trade discount plus GST) divided by tins bought: GST in, the costs of the path out, the shortage ignored.

Passes

The right items, but the normal shortage taken out of cost as well as units: (₹7,44,000 − 1,000 × ₹29.76) ÷ 24,000 = ₹29.76.

Close, but the shortage is in the wrong column.

Full marks

ICAI's statement (units and rupees), commission on the listed price, the shortage reducing units only (₹31.00), and a numbered note for each item left out.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the cash discount that went shopping

Deducting the 2% cash discount because "we did pay early". It is a reward for prompt payment, a finance item. **Fix:** ignore it in material cost, whether or not it was received.

⚠ **Common mistake**

The demurrage that sneaked into stock

Adding demurrage (or a fine, or detention charges) as if it were freight. **Fix:** a penalty for delay or non-compliance is an abnormal cost; it never enters the cost of material.

⚠ **Common mistake**

The shortage that was taken out twice

Deducting the normal shortage's value from the total cost and then dividing by the good units too. **Fix:** keep the total cost and divide by the good units; the shortage's cost is spread over them.

Practice

Check yourself

1 Case scenario. Tapri buys 1,000 kg of jaggery from a farmers' co-operative at ₹60 a kg, with GST whose input credit is available. The supplier's credit note for a 5% quantity discount arrives a week after the invoice. The State pays Tapri a subsidy of ₹5 a kg for buying from co-operatives. Tapri pays freight of ₹2,000, and the transporter charges ₹500 demurrage because the godown opened late. Normal loss of moisture is 4%. Which cost per good kg is correct?

A ₹61.46

B ₹59.38

C ₹56.77

D ₹56.25

2 Which of these are added to the cost of material purchased? Select all that apply.

- **A** Freight inwards
- **B** Customs duty paid on import
- **C** Demurrage paid to the transporter
- **D** Insurance on the goods in transit
- **E** Cash discount not availed

More than one answer may be right.

3 A lot of material costs ₹1,47,000 before other expenses, which are 2% of total cost. What is the total cost of the lot, in rupees?

Answer: _____

4 Tapri deposits ₹8,000 for returnable drums and gets ₹6,500 back on returning them. What is added to the cost of material?

- **A** ₹8,000
- **B** ₹1,500
- **C** ₹6,500
- **D** Nothing, as the drums were returned

5 10,000 kg are bought for ₹5,00,000 in all. A normal shortage of 200 kg and an abnormal shortage of 300 kg occur. How much is debited to the Costing Profit and Loss Account? (Round to the nearest rupee.)

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Material cost: purchase price net of trade discounts, rebates and recoverable taxes, plus costs of bringing it to its present location and condition.
- Added: freight inwards, transit insurance, customs duty, toll tax paid by buyer, commission, non-returnable containers.
- Kept out: cash discount (finance item), demurrage, detention charges and fines, penalties.
- Deducted: trade and quantity discounts (if not already on the invoice), subsidies, grants and incentives.
- GST: out if credit is available (and when the question is silent); in for a buyer who cannot take credit.
- Returnable containers: only the shortfall of the refund is added.
- Rate = total cost ÷ (units − normal shortage); abnormal shortage at that rate to Costing P&L.

- "x% of total cost" = cost before it × x/(100 − x). (The forklift pays for the wedding.)

Answer key · Check yourself

1 **D** ₹56.25

Cost = ₹60,000 − ₹3,000 quantity discount (deducted even though it came on a separate credit note) + ₹2,000 freight − ₹5,000 subsidy = ₹54,000, carried by 960 good kg: ₹56.25. GST with credit and the demurrage stay out. ₹61.46 forgets the subsidy, ₹59.38 the quantity discount, and ₹56.77 adds the demurrage.

2 **A** **Freight inwards**; **B** **Customs duty paid on import**; **D** **Insurance on the goods in transit**

Freight inwards, customs duty and transit insurance bring the material to its present location and condition. Demurrage is a penalty (abnormal cost). A cash discount is a finance item, ignored either way.

3 **150000**

Other expenses = ₹1,47,000 × 2/98 = ₹3,000; total = ₹1,50,000. Taking 2% of ₹1,47,000 (₹2,940) is the reading ICAI does not use.

4 **B** ₹1,500

Only the shortfall between the amount paid and the refund is added: ₹8,000 − ₹6,500 = ₹1,500.

5 **15306**

Rate = ₹5,00,000 ÷ (10,000 − 200) kg = ₹51.0204 a kg; abnormal shortage = 300 × ₹51.0204 = ₹15,306. The normal shortage is absorbed by the good units; the abnormal one is valued at the same rate.

3.3 Pricing material issues: FIFO, LIFO and average methods

The same kilos bought at different prices: how FIFO, LIFO and the average methods price each issue and the closing stock, and how returns, transfers and shortages enter the stores ledger.

In plain words

Which sugar went to the café?

Tapri's store has two lots of sugar: 100 kg bought last month at ₹40 a kg, and 100 kg bought this week at ₹44. The Bhopal café asks for 120 kg.

> **Kesar:** So what did the café's sugar cost?
>
> **Ritu:** That depends on which kilos you say left the store.
>
> **Kesar:** Sugar is sugar!
>
> **Ritu:** Exactly. The kilos are identical, so the books need a rule.

The physical sugar is identical; only the price tags differ. A **method of pricing issues** is the rule that decides which price tag goes out with each issue and which stays on the closing stock.

Rule	The 120 kg issued	The 80 kg left
FIFO: oldest price first (100 at ₹40, 20 at ₹44)	₹4,880	₹3,520
LIFO: newest price first (100 at ₹44, 20 at ₹40)	₹5,200	₹3,200
Weighted average: one blended price, ₹8,400 ÷ 200 = ₹42	₹5,040	₹3,360

All three hand out the same ₹8,400 in total, split differently between what was used and what is left. When prices are rising, FIFO charges production the old, cheaper prices and leaves the dearer ones in stock; LIFO does the opposite; the average sits between them.

The record that carries these prices is the **stores ledger**: for every material, each receipt, issue and balance in quantity, rate and amount.

Think of it like

The ticket queue, the plate stack and the milk pot

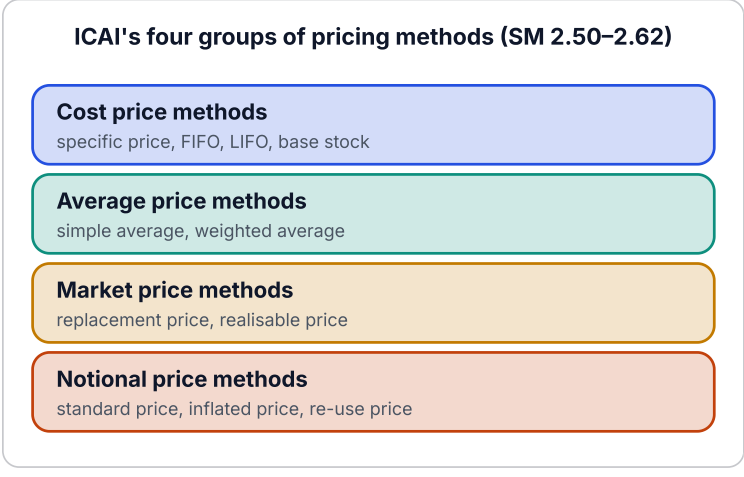
FIFO is the queue at a railway ticket counter: whoever came first is served first. LIFO is the stack of washed plates: the last one put on top is the first one picked up. Weighted average is the big milk pot at the café: every new can is poured in, and each glass served carries the same blended cost.

Key idea

A pricing method decides which purchase price each issue carries; the total of issues and closing stock is always the same. In rising prices, FIFO gives the lowest cost of issues and the highest closing stock, LIFO the reverse, and the weighted average falls in between.

How it works

Materials issued "should be priced at the value at which they are carried in stock", but lots are bought at different times, prices, discounts and taxes, so a method is needed (SM 2.49). The cost accountant chooses it considering: (1) the frequency of purchases, price fluctuations and their range; (2) the frequency of issues and relative quantity; (3) the nature of the cost accounting system; (4) the nature of business and the type of production process; (5) management policy on the valuation of closing stock.



The three you will compute

1. **First-in first-out (FIFO)**: issues are priced "in the order in which they are purchased": the items longest in stock go out first, so the closing stock is at the latest prices. ICAI calls it suitable when prices are falling (the charge to production is high while replacement cost is low); when prices rise, the charge to production lags behind replacement cost (SM 2.50).

2. **Last-in first-out (LIFO)**: the last lot purchased is issued first, until it is exhausted, then the earlier lot. In rising prices the cost of production is close to current prices and profits are not overstated. LIFO is not permitted under AS 2 and Ind AS 2; the Study Material keeps it for academic knowledge, and notes that it is not acceptable to the income tax authorities (SM 2.52–2.53).

3. **Weighted average price**: the issue price is the total cost of the material in stock divided by the total quantity, recomputed every time a new lot is received; issues between receipts go out at the same rate (SM 2.60–2.61). It smooths price fluctuations.

The rest, in one line each

- **Specific price**: material bought for a specific job is issued to it at its own purchase price; each lot is stored

and accounted separately (SM 2.50).

- **Base stock**: a minimum quantity is always held at the price of the first lot; the rest is priced by FIFO or LIFO, so it is not an independent method (SM 2.59).
- **Simple average**: total of the lot prices ÷ number of lots, ignoring quantities; a lot already fully issued is left out. Suitable when lots are uniform in size and prices do not fluctuate much (SM 2.59).
- **Replacement price**: the price at which an identical item could be bought now; useful in rising prices. **Realisable price**: the price at which the material could be sold (SM 2.61).
- **Standard price**: a predetermined rate, set from current prices, market trends, discounts and transport charges; it simplifies pricing and measures purchase efficiency (SM 2.62; material variances). **Inflated price**: raised to cover natural losses such as evaporation. **Re-use price**: for rejected or re-used material.

Returns, transfers and shortages in the stores ledger

Event	Entry in the stores ledger
Material **returned to the vendor**	Issue column. The price includes the invoice price plus freight, receiving and handling charges. Strictly it should go at the stores ledger price, but in practice the invoice price (the price of the lot returned) is used, as ICAI's solutions do (SM 2.69, 2.94); the gap between the two is charged as overhead or to an inventory adjustment account (SM 2.63).
Material **returned to stores** by a department	Receipt column at the price at which it was originally issued: held in suspense for the next requisition, or merged with stock as a fresh purchase at that price (SM 2.63).
Transfer between jobs or departments	No entry in the stock balance: the material does not pass through the store; the jobs' cost accounts are adjusted (SM 2.48, 2.93).
Replacement received from the vendor	A fresh receipt (SM 2.93).
Shortage found on physical verification	Issue column, at the rate of the method in use (SM 2.63).
Abnormal loss (pilferage, theft)	Issue column at the current rate; charged to Costing P&L (SM 2.69).

Event	Entry in the stores ledger
Normal loss (rusting, breakage, evaporation)	Quantity reduced, value kept: the good units absorb it and the rate rises (SM 2.69).

Interactive version: PassBook app, lesson Pricing material issues: FIFO, LIFO and average methods

Formats and formulas

The stores ledger

ICAI's layout: Date | Receipts (GRN or return note no., Qty, Rate, Amount) | Issues (requisition no., Qty, Rate, Amount) | Balance (Qty, Rate, Amount). *Rule these columns from memory; in this lesson each group is written as "qty @ rate = amount" so it reads on a phone.*

Date

Receipts

GRN no., qty @ rate = amount

Issues

MR no., qty @ rate = amount

Balance

qty @ rate = amount

∑ **Formula**

Weighted average price

$$\text{Rate} = \frac{\text{Total cost in stock}}{\text{Total units in stock}}$$

Recompute after every receipt (SM 2.61, 2.75).

∑ **Formula**

Simple average price

$$\text{Rate} = \frac{\text{Sum of the lot prices}}{\text{Number of lots}}$$

ICAI: total of unit price of each purchase ÷ total number of purchases. Lots already fully issued are left out of the average (SM 2.59, 2.75).

FIFO, LIFO and weighted average when prices are rising

Point	FIFO	LIFO	Weighted average
Cost of material issued	Lowest (old prices)	Highest (latest prices)	In between
Closing stock	Highest, near current price	Lowest, at oldest prices	In between
Profit shown	Highest	Lowest	In between

In falling prices all three rows swap between FIFO and LIFO.

Worked examples

Worked example **Elaichi after a bumper harvest: the three methods when prices fall**

Tapri's masala unit buys green cardamom (elaichi) for its masala chai premix. A bumper crop has pushed prices down all month. Transactions for 1 to 20 March:

Date	Particulars	Kg	Rate (₹)
1 Mar	Opening stock	300	2,400
4 Mar	Purchased	450	2,300
7 Mar	Issued	550	
11 Mar	Purchased	300	2,150
13 Mar	Issued	250	
16 Mar	Issued	150	
18 Mar	Purchased	300	2,000
20 Mar	Issued	250	

On 20 March the market price of cardamom is ₹2,000 a kg. Tapri values its stock at cost or market price, whichever is lower.

Required:

(i) Calculate the value of cardamom issued during 1 to 20 March and of the stock on 20 March under FIFO, LIFO and the weighted average method.

(ii) Under which method(s) must the stock on 20 March be written down to market price, and by how much?

(iii) Which of the three methods suits a period of falling prices such as this one? Give reasons.

THE WORKING

FIFO

Date
1 Mar
Receipts
Issues
Balance
300 @ 2,400 = 7,20,000

Date
4 Mar
Receipts
450 @ 2,300 = 10,35,000
Issues
Balance
300 @ 2,400 + 450 @ 2,300 = 17,55,000

Date

7 Mar

Receipts

Issues

300 @ 2,400 + 250 @ 2,300 = 12,95,000

Balance

200 @ 2,300 = 4,60,000

Date

11 Mar

Receipts

300 @ 2,150 = 6,45,000

Issues

Balance

200 @ 2,300 + 300 @ 2,150 = 11,05,000

Date

13 Mar

Receipts

Issues

200 @ 2,300 + 50 @ 2,150 = 5,67,500

Balance

250 @ 2,150 = 5,37,500

Date

16 Mar

Receipts

Issues

150 @ 2,150 = 3,22,500

Balance

100 @ 2,150 = 2,15,000

Date
18 Mar

Receipts
300 @ 2,000 = 6,00,000

Issues

Balance
100 @ 2,150 + 300 @ 2,000 = 8,15,000

Date
20 Mar

Receipts

Issues
100 @ 2,150 + 150 @ 2,000 = 5,15,000

Balance
150 @ 2,000 = 3,00,000

LIFO

Date
1 Mar

Receipts

Issues

Balance
300 @ 2,400 = 7,20,000

Date
4 Mar

Receipts
450 @ 2,300 = 10,35,000

Issues

Balance
300 @ 2,400 + 450 @ 2,300 = 17,55,000

Date

7 Mar

Receipts

Issues

450 @ 2,300 + 100 @ 2,400 = 12,75,000

Balance

200 @ 2,400 = 4,80,000

Date

11 Mar

Receipts

300 @ 2,150 = 6,45,000

Issues

Balance

200 @ 2,400 + 300 @ 2,150 = 11,25,000

Date

13 Mar

Receipts

Issues

250 @ 2,150 = 5,37,500

Balance

200 @ 2,400 + 50 @ 2,150 = 5,87,500

Date

16 Mar

Receipts

Issues

50 @ 2,150 + 100 @ 2,400 = 3,47,500

Balance

100 @ 2,400 = 2,40,000

Date
18 Mar
Receipts
300 @ 2,000 = 6,00,000
Issues
Balance
100 @ 2,400 + 300 @ 2,000 = 8,40,000

Date
20 Mar
Receipts
Issues
250 @ 2,000 = 5,00,000
Balance
100 @ 2,400 + 50 @ 2,000 = 3,40,000

Weighted average (rate recomputed after each receipt; the 13 and 16 March issues share one rate because no lot arrived between them)

Date
1 Mar
Receipts
Issues
Balance
300 @ 2,400.00 = 7,20,000

Date
4 Mar
Receipts
450 @ 2,300 = 10,35,000
Issues
Balance
750 @ 2,340.00 = 17,55,000

Date
7 Mar
Receipts
Issues
550 @ 2,340.00 = 12,87,000
Balance
200 @ 2,340.00 = 4,68,000

Date
11 Mar
Receipts
300 @ 2,150 = 6,45,000
Issues
Balance
500 @ 2,226.00 = 11,13,000

Date
13 Mar
Receipts
Issues
250 @ 2,226.00 = 5,56,500
Balance
250 @ 2,226.00 = 5,56,500

Date
16 Mar
Receipts
Issues
150 @ 2,226.00 = 3,33,900
Balance
100 @ 2,226.00 = 2,22,600

Date
18 Mar
Receipts
300 @ 2,000 = 6,00,000
Issues
Balance
400 @ 2,056.50 = 8,22,600

Date
20 Mar
Receipts
Issues
250 @ 2,056.50 = 5,14,125
Balance
150 @ 2,056.50 = 3,08,475

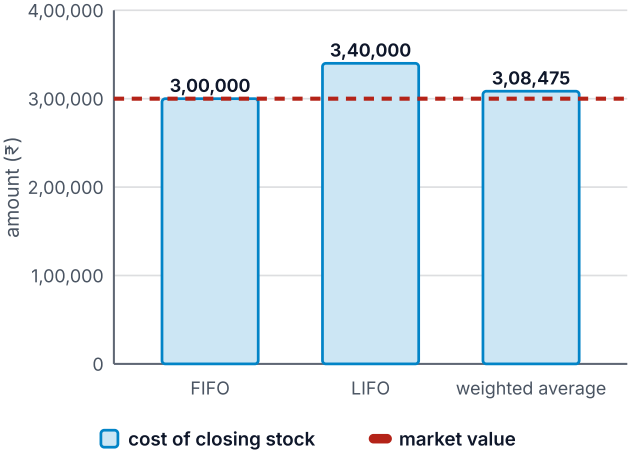
(i) Summary

Method	Cardamom issued (₹)	Stock on 20 Mar (₹)	Total (₹)
FIFO	27,00,000	3,00,000	30,00,000
LIFO	26,60,000	3,40,000	30,00,000
Weighted average	26,91,525	3,08,475	30,00,000

(ii) Write-down to market price. Market value of the 150 kg = 150 × ₹2,000 = ₹3,00,000.

Method	Cost of stock (₹)	Market value (₹)	Write-down (₹)
FIFO	3,00,000	3,00,000	Nil
LIFO	3,40,000	3,00,000	40,000
Weighted average	3,08,475	3,00,000	8,475

Cost of the 150 kg in stock on 20 March against its market value



Bars: the cost of the same 150 kg under each method. Dashed rose line: its market value. LIFO's stock still holds 100 kg of the opening lot at ₹2,400.

(iii) Method for falling prices. FIFO. The Study Material considers FIFO suitable in times of falling prices because the material cost charged to production is high while the replacement cost of

materials is low (SM 2.50), and under FIFO the closing stock is represented very closely at current market price (SM 2.51): here it needs no write-down. LIFO charges production the latest, lowest prices, so profit tends to rise, but the stock is left at the oldest, highest prices and has to be written down considerably to follow the principle of cost or market price, whichever is lower (SM 2.53): ₹40,000 here. The weighted average lies in between. In every method, issues plus closing stock equal opening stock plus purchases, ₹30,00,000.

Interactive version: PassBook app, lesson Pricing material issues: FIFO, LIFO and average methods

Worked example

Packing film in April: a weighted average ledger with a late return

Tapri's premix plant in Indore prices issues of laminated packing film (in rolls) by the weighted average method.

April transactions:

Date	Particulars	Rolls
1 Apr	Opening stock @ ₹150	1,600
3 Apr	Issued to Line A, MR 211	1,000
6 Apr	Received, GRN 41 (see note below)	3,000
9 Apr	Issued to Line B, MR 212	2,000
12 Apr	Transferred from Line B to Line A, MTN 5	100
14 Apr	Received, GRN 42, @ ₹164	2,000
16 Apr	Returned unused by Line A (against MR 211), MRR 7	150
21 Apr	Issued to Line B, MR 215	2,200
27 Apr	Issued to Line A, MR 218	1,200

GRN 41: invoice price ₹4,80,000 less 5% trade discount, plus GST (input tax credit is available) and freight of ₹12,000 for the lot. On 30 April the physical count shows 340 rolls; the 10 missing rolls were crushed by a forklift driven carelessly.

Required:

(i) Prepare the stores ledger for April.

(ii) Calculate the net cost of film charged to Line A and to Line B for April.

(iii) Line A's foreman wants the 150 returned rolls credited at the average rate standing on 16 April, "because that is what they are worth today". Comment, showing where the difference would go.

THE WORKING

Cost of GRN 41. ₹4,80,000 − 5% trade discount ₹24,000 + freight ₹12,000 = ₹4,68,000 (GST is left out because input tax credit is available) = ₹156 per roll.

Stores ledger: packing film (weighted average method)

Date	Receipts	Issues	Balance
1 Apr			1,600 @ 150.00 = 2,40,000
3 Apr		MR 211: 1,000 @ 150.00 = 1,50,000	600 @ 150.00 = 90,000
6 Apr	GRN 41: 3,000 @ 156.00 = 4,68,000		3,600 @ 155.00 = 5,58,000

Date	Receipts	Issues	Balance
9 Apr		MR 212: 2,000 @ 155.00 = 3,10,000	1,600 @ 155.00 = 2,48,000
14 Apr	GRN 42: 2,000 @ 164.00 = 3,28,000		3,600 @ 160.00 = 5,76,000
16 Apr	MRR 7: 150 @ 150.00 = 22,500		3,750 @ 159.60 = 5,98,500
21 Apr		MR 215: 2,200 @ 159.60 = 3,51,120	1,550 @ 159.60 = 2,47,380
27 Apr		MR 218: 1,200 @ 159.60 = 1,91,520	350 @ 159.60 = 55,860
30 Apr		Abnormal loss: 10 @ 159.60 = 1,596	340 @ 159.60 = 54,264

Working notes:

1. The return from Line A is a receipt at the price at which those rolls were issued on 3 April, ₹150, not at the ₹160 standing on 16 April. Because it

comes in below the current average, the rate is recomputed: ₹5,98,500 ÷ 3,750 = ₹159.60.

2. The transfer of 100 rolls from Line B to Line A does not pass through the store, so the stores ledger has no entry; the cost accounts move 100 × ₹155 (the rate at which Line B received them on 9 April) = ₹15,500 from Line B to Line A.
3. The 10 rolls crushed by a careless forklift are a shortage found on physical verification, entered in the issue column at the current rate. The cause is abnormal, so ₹1,596 goes to the Costing Profit and Loss Account.

Check: opening ₹2,40,000 + receipts ₹8,18,500 = issues ₹10,02,640 + abnormal loss ₹1,596 + closing ₹54,264 = ₹10,58,500.

(ii) Net cost of film charged to each line

Particulars	Line A (₹)	Line B (₹)
Issues (MR 211, 218 / MR 212, 215)	3,41,520	6,61,120
Less: returned to stores (MRR 7)	(22,500)	—
Add / (less): transfer MTN 5	15,500	(15,500)
Net cost of film	**3,34,520**	**6,45,620**

The two lines together carry ₹9,80,140 = issues ₹10,02,640 − return ₹22,500.

(iii) Comment. The foreman is not correct. Material returned to stores is entered in the receipt column at the price at which it was originally issued (SM 2.63), so Line A gets back exactly the ₹150 a roll it was charged. A credit at ₹160 would give Line A ₹1,500 more than it was ever charged (150 × ₹10), cutting its cost below what it actually used. The store's average would become ₹6,00,000 ÷ 3,750 = ₹160.00 instead of ₹159.60, so the ₹1,500 would be shifted to later issues and stock: 2,200 × ₹0.40 = ₹880 to Line B (MR 215), 1,200 × ₹0.40 = ₹480 back to Line A (MR 218) and 350 × ₹0.40 = ₹140 to the 350 rolls in stock on 27 April.

Worked example

Dev's claim: "one lot a month, so the method makes no difference"

Sugar at Tapri's sweets unit, January to March (no

opening stock on 1 January):

Date	Particulars	Kg	Rate (₹)
2 Jan	Purchased	800	40
10 Jan	Issued	300	
25 Jan	Issued	300	
3 Feb	Issued	200	
5 Feb	Purchased	900	42
15 Feb	Issued	500	
26 Feb	Issued	250	
4 Mar	Purchased	600	45
12 Mar	Issued	400	
28 Mar	Issued	300	

Dev tells the board: "We buy only one lot of sugar a month, so FIFO, LIFO and the weighted average must give the same value for every issue and for the closing stock. The choice of method makes no difference to us."

Required:

(i) Examine Dev's claim month by month. (Stores ledgers are not required.)

(ii) Calculate the value of the March issues and of the stock on 31 March under FIFO, LIFO and the weighted average method.

MODEL ANSWER

(i) Month by month. A method of pricing issues matters only when the store holds two or more lots at different prices at the time of an issue; the number of purchases in a month does not decide it.

- **January:** the only lot on hand is the 800 kg bought at ₹40 on 2 January. Both January issues (600 kg) can only be priced at ₹40, under every method: ₹24,000. The 200 kg left on 31 January are also at ₹40.
- **February:** the 3 February issue of 200 kg clears the January lot before the February lot arrives on 5 February, so again only one lot is ever on hand. Issues: 200 × ₹40 + 750 × ₹42 = ₹39,500 under every method, and 150 kg at ₹42 remain on 28 February. Dev is right for January and February, even though the stock was not nil at either month-end.
- **March:** the March lot (600 kg at ₹45) arrives on 4 March while 150 kg of the February lot (₹42) is still in the store. From here two lots are held together, so the methods give different figures. Dev's claim fails for March.

(ii) March issues and closing stock

Method	12 Mar issue (400 kg)	28 Mar issue (300 kg)	Stock on 31 Mar (50 kg)
FIFO	150 @ 42 + 250 @ 45 = ₹17,550	300 @ 45 = ₹13,500	50 @ 45 = ₹2,250
LIFO	400 @ 45 = ₹18,000	200 @ 45 + 100 @ 42 = ₹13,200	50 @ 42 = ₹2,100
Weighted average	400 @ 44.40 = ₹17,760	300 @ 44.40 = ₹13,320	50 @ 44.40 = ₹2,220

Weighted average rate on 4 March = (150 × ₹42 + 600 × ₹45) ÷ 750 = ₹33,300 ÷ 750 = ₹44.40.

March issues: FIFO ₹31,050, LIFO ₹31,200, weighted average ₹31,080. Under each method, March issues plus closing stock = ₹6,300 + ₹27,000 = ₹33,300. Conclusion: Dev's claim holds only while a single lot is on hand at each issue. Once two lots sit together, as in March, the method changes both the cost of issues and the closing stock.

Exam angle

What ICAI asks from this topic

In the exam

Where pricing of issues appeared, May 2024 to September 2026 (and the old scheme)

- **One-step MCQs, 2 marks**: January 2026 MCQ 14 (value of material consumed under LIFO, one issue against three lots); September 2026 MCQ 13 (value of one issue under FIFO after several receipts).
- **Inside a case scenario**: MTP Series I for September 2025, MCQs 1 to 5: closing raw material under FIFO when the price rose mid-period, feeding a cost sheet.
- **Full priced stores ledger, 10 marks**: May 2023 (old scheme) Q2(a), FIFO with month-wise purchases; MTP Series II for November 2022 (old scheme) Q5(b), weighted average with shortages reported by the stock verifier; RTP May 2020 (old scheme) Q1, weighted average with goods received and material returned notes, plus stock levels.
- **True or false**: November 2023 (old scheme) Q6(b): "Under LIFO, in falling prices, lower income is reported" (false: profit tends to rise) and "Simple average suits lots of different quantity with fluctuating prices" (false).

In the exam

The traps

- Under the weighted average, **recompute after every receipt**, including a return from a department; issues between receipts use the same rate.
- A transfer between jobs or departments is **not** a stores ledger entry.
- Shortages: ICAI's answers differ. The Study Material enters a verification shortage in the issue column at the method's rate (SM 2.63); one old MTP treated unexplained shortages as absorbed (quantity cut, value kept). If the cause is given, use it: abnormal to Costing P&L, normal absorbed. If not, state your assumption.
- Rates to two decimals; amounts = quantity × the rounded rate, as in ICAI's Illustration 14.

Trick

FIFO fills the cupboard with the latest prices

Whatever the dates, FIFO's closing stock is priced at the most recent purchases (as far back as needed to cover the quantity), and LIFO's at the oldest lots still standing. So for a FIFO closing stock you can skip the ledger: take the closing quantity from the latest receipts backwards. Holds as long as no returns or shortages intervene; LIFO's closing stock depends on the timing of issues, so it always needs the ledger.

Answer frame

"Which method would you prefer, and why?" (as in TYK 9, SM 2.96)

1. Compare the cost charged to each job and the closing stock under each method (a small table).
2. Rising prices: FIFO charges least and leaves the highest closing stock (higher profit); LIFO the reverse (lower profit).
3. The weighted average evens out fluctuations: issues and stock show greater uniformity.
4. Conclude with the method you prefer and why: ICAI's own answer prefers the weighted average for the reason in point 3. Add that LIFO is not permitted under AS 2 and Ind AS 2.

"When and why would you recommend LIFO?" (Illustration 12, SM 2.55)

When prices are rising: issues are close to current market price, so the cost of production matches current sales revenue; profit is not unduly high; in falling prices the products appear competitive; and in inflation it shows the correct profit and avoids paying undue taxes to some extent. Add that it is not permitted under AS 2 and Ind AS 2.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the average that never moved

Computing one average for the whole month (total purchases ÷ total quantity) and pricing every issue at it. That is not ICAI's weighted average method, which recomputes the rate after each receipt. **Fix:** a new rate after every receipt; issues carry the rate standing on their date.

⚠ **Common mistake**

The return priced at today's rate

Pricing a department's returned material at the current average instead of the rate at which it was issued, or pricing a return to the supplier at the ledger average when the question gives the lot. **Fix:** returns to stores at the original issue price; returns to the vendor at the price of that lot, as ICAI's solutions do.

⚠ **Common mistake**

The transfer that was booked twice

Showing a job-to-job transfer as an issue and a receipt in the stores ledger. **Fix:** no entry in the stores ledger; adjust the two jobs in the cost accounts.

Practice

Check yourself

1 Ghee at the sweets unit: opening stock 300 kg at ₹80; 2 June, issued 200 kg; 5 June, purchased 500 kg at ₹90; 9 June, issued 450 kg. What is the value of the stock left on 10 June under LIFO, in rupees?

Answer: _____

2 Case scenario. Ritu prices chai premix by the weighted average method. On 1 June the store holds 1,000 kg at ₹120; on 4 June it receives 3,000 kg at ₹128; on 6 June it issues 2,000 kg; on 8 June the Pune café returns 200 kg unused out of that issue. Which statement is correct?

- **A** The 6 June issue is priced at ₹124, the simple average of the two lots.
- **B** The 6 June issue is priced at ₹126 and the 8 June return is received at ₹126.
- **C** The 6 June issue is priced at ₹128, the latest price.
- **D** The 8 June return is received at ₹120, the oldest price.

3 In a period of rising prices, which method gives the lowest value of closing stock?

- **A** FIFO
- **B** LIFO
- **C** Weighted average
- **D** Replacement price

4 Which of these need NO entry in the stores ledger? Select all that apply.

- **A** Transfer of surplus material from Job 14 to Job 19
- **B** Return of surplus material from Job 14 to the store
- **C** Return of defective material to the supplier after the goods received note
- **D** Return of material to the supplier at the gate, before the receiving report

More than one answer may be right.

5 Lots bought in April: 100 units at ₹10, 300 at ₹12 and 100 at ₹17; no lot is yet fully issued. What is the issue price under the simple average method, in rupees?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- A pricing method decides which purchase price each issue carries; issues + closing stock = opening stock + receipts under every method.
- FIFO: oldest prices to issues, closing stock at latest prices; suits falling prices.
- LIFO: latest prices to issues, closing stock at oldest prices; suits rising prices; not permitted under AS 2 and Ind AS 2.
- Weighted average = total cost in stock ÷ total quantity, recomputed after every receipt; smooths fluctuations.
- Simple average = total of lot prices ÷ number of lots, ignoring quantities and fully issued lots.
- Rising prices: issues FIFO < weighted average < LIFO; closing stock FIFO > weighted average > LIFO.
- Returns to stores at the original issue price; returns to vendor in the issue column; transfers between jobs: no stores ledger entry.
- Shortages: abnormal in the issue column at the current rate, to Costing P&L; normal absorbed (quantity cut, value kept). (Sugar is sugar; price tags are not.)

Answer key · Check yourself

1 **12500**

LIFO prices each issue from the latest lot on hand at the time. On 2 June only the opening lot exists: 200 kg at ₹80 go out, leaving 100 kg at ₹80. On 9 June the 450 kg come from the 5 June lot at ₹90, leaving 50 kg of it. Stock: 100 × ₹80 + 50 × ₹90 = ₹12,500. Pricing all 150 kg at the oldest rate (₹12,000) ignores the issue made before the purchase arrived; FIFO would leave 150 × ₹90 = ₹13,500.

2 **B** **The 6 June issue is priced at ₹126 and the 8 June return is received at ₹126.**

Weighted average = (₹1,20,000 + ₹3,84,000) ÷ 4,000 kg = ₹126. A return from a department goes back at the price at which it was issued, ₹126. ₹124 ignores quantities; ₹128 is LIFO's first price.

3 **B** **LIFO**

LIFO issues the latest (dearest) lots first, so the closing stock is left at the oldest, cheapest prices. FIFO leaves the newest prices in stock.

4 **A Transfer of surplus material from Job 14 to Job 19; D Return of material to the supplier at the gate, before the receiving report**

A job-to-job transfer never passes through the store, and goods rejected before the receiving report never entered the stores ledger. A return to store is a receipt; a return to the supplier after the receiving report is shown in the issue column.

5 **13**

Simple average = (₹10 + ₹12 + ₹17) ÷ 3 = ₹13; quantities are ignored. The weighted average would be ₹6,300 ÷ 500 = ₹12.60.

3.4 Stock levels: re-order, minimum, maximum, average and danger levels

When to order, the least and most to hold, the stock normally carried and the emergency line: ICAI's stock-level formulas, the stock control chart and the traps in reverse questions.

In plain words

The café that ran out of chai on a Sunday

Tapri's Indore café uses 40 to 60 packs of chai premix a day, 50 on an ordinary day. The supplier takes 4 to 8 days to deliver, usually 6. One rainy Sunday the café ran dry by 11 a.m.

> **Kesar:** Just order more. Keep a lakh packs!
>
> **Ritu:** And pay rent, interest and expiry on a lakh packs. The question is not "more". It is *when* to order and *how much* to hold.
>
> **Kesar:** Then tell me when.
>
> **Ritu:** When the shelf is down to what the worst week can eat before the van arrives.

That worst case is the busiest usage (60 a day) for the longest delivery (8 days): 480 packs. When stock falls to 480, order. That is the **re-order level**. If things go normally (50 a day, 6 days), 300 packs are eaten while the order travels, so about 180 are still on the shelf when it

arrives: the **minimum level**, the cushion that should always be there.

If 700 packs arrive at a quiet time (40 a day) after the quickest delivery (4 days), only 160 have been used meanwhile, so stock could climb to 480 + 700 − 160 = 1,020: the **maximum level**, the ceiling beyond which money and space are wasted. Stock normally hovers around the **average level**: 180 + half of 700 = 530 packs. An emergency purchase from the local market takes 2 days, so the **danger level** is what normal usage eats in those 2 days, 50 × 2 = 100 packs: below it, only emergency issues are made.

Think of it like

The petrol gauge on a long drive

You do not refuel at the first pump, or when the tank is bone dry. You refuel when what is left covers the longest stretch to the next pump at your thirstiest speed (re-order level). You expect some petrol left when you get there (minimum level). The tank holds only so much (maximum level). And when the warning light comes on, you drive only to the pump (danger level).

Key idea

Order when stock falls to what the maximum usage could need over the maximum lead time (re-order level). The other levels follow from it: minimum = re-order level less normal usage during normal lead time; maximum = re-order level plus re-order quantity less the least usage during the shortest lead time.

How it works

Inventory control by setting quantitative levels answers seven questions (SM 2.22):

Level	Answers
Re-order stock level	When to order
Re-order quantity / EOQ	How much to order
Maximum stock level	Up to how much to stock
Minimum stock level	At least how much to stock
Average stock level	Stock normally kept
Danger stock level	Kept for emergency requirement
Buffer stock	To meet sudden demand

Re-order period is also called **lead time**: the time to get an order from the supplier to the stores (SM 2.22–2.23). **Re-order quantity** is the quantity for which the store

raises the purchase requisition; it should be the quantity at which ordering and carrying costs together are least, the economic order quantity (SM 2.23).

Work the levels in this order

1. **Fix the consumption rates**: maximum, minimum and average (normal) usage per day, week or month. If one is missing, use average = (maximum + minimum) ÷ 2, as ICAI's solutions do (SM 2.31).

2. **Fix the lead times**: maximum, minimum, average, and for emergency purchases. Average lead time = (maximum + minimum) ÷ 2 when not given, as in ICAI's solutions (SM 2.29, 2.100).

3. **Re-order level** = maximum consumption × maximum re-order period.

4. **Minimum level** = re-order level − (average consumption × average re-order period).

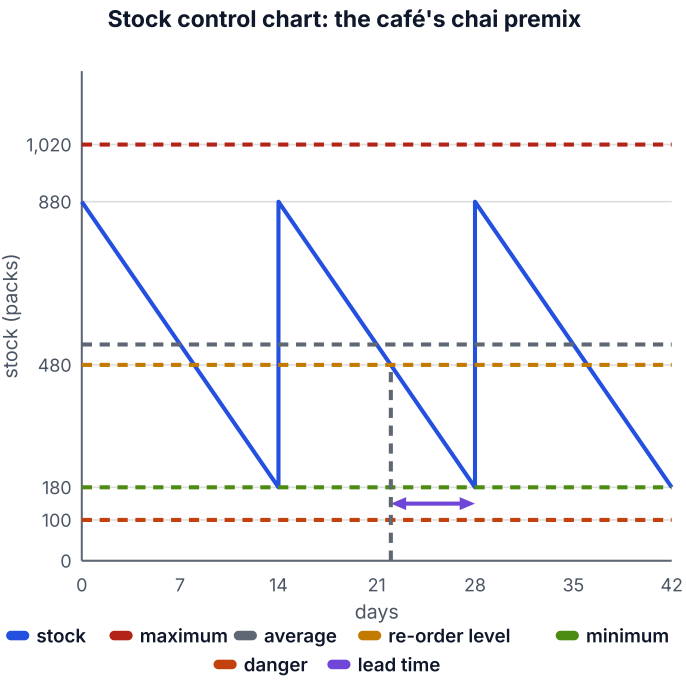
5. **Re-order quantity**: as given; if it is not given, compute the EOQ.

6. **Maximum level** = re-order level + re-order quantity − (minimum consumption × minimum re-order period).

7 **Average level** = minimum level + ½ re-order quantity, or (maximum + minimum) ÷ 2.

8 **Danger level** = average consumption × lead time for emergency purchases.

Keep every rate and time in the same unit: per day with days, per week with weeks.



Blue: stock, falling 50 packs a day and jumping by the 700-pack order. Dashed lines: maximum 1,020 (rose), average 530 (grey, minimum + ½ × 700), re-order level 480 (amber), minimum 180 (green), danger 100 (orange). The order is placed on day 22, when stock falls to 480 (grey dashed upright); after the normal 6-day lead time (violet arrow) it arrives when stock has fallen to the minimum.

On ICAI's chart (SM 2.28) the stock rises when materials are received and declines as they are consumed; at the re-order level the re-order quantity is ordered, and fresh supplies are normally received when stocks reach the minimum level. The time between the order and the receipt is the lead time.

Interactive version: PassBook app, lesson Stock levels: re-order, minimum, maximum, average and danger levels

Formats and formulas

Write each formula with its words first, then the figures, as ICAI's answers do. Below, usage = consumption and lead time = re-order period, as ICAI uses both pairs of words.

∑ **Formula**

Re-order level (ROL)

$$\text{ROL} = \text{Max. usage} \times \text{Max. lead time}$$

Alternatively ROL = minimum level + (average consumption × average re-order period) (SM 2.22–2.23).

∑ **Formula**

Minimum stock level

$$\text{Minimum} = \text{ROL} - (\text{Avg. usage} \times \text{Avg. lead time})$$

∑ **Formula**

Maximum stock level

$$\text{Maximum} = \text{ROL} + \text{ROQ}$$
$$-(\text{Min. usage} \times \text{Min. lead time})$$

Here the re-order quantity may be the EOQ (SM 2.27).

∑ **Formula**

Average stock level

$$\text{Average} = \text{Minimum} + \tfrac{1}{2}\,\text{ROQ} \qquad \text{or}$$
$$\frac{\text{Maximum} + \text{Minimum}}{2}$$

∑ **Formula**

Danger level

$$\text{Danger} = \text{Avg. usage} \qquad \times \text{Emergency lead time}$$

"Some time minimum consumption is also used" (SM 2.27).

Formula

Economic order quantity (when ROQ is not given)

$$\mathrm{EOQ} = \sqrt{\frac{2AO}{C}}$$

A = annual requirement, O = cost per order, C = carrying cost per unit per annum (SM 2.24). Details: EOQ and quantity discounts.

Definition

Re-order stock level

The level which lies between minimum and maximum levels in such a way that before the material ordered is received into the stores, there is sufficient quantity in hand to cover both normal and abnormal consumption situations; the level at which fresh order should be placed for replenishment of stock. (SM 2.22)

Definition

Minimum, maximum and danger levels (SM 2.27)

Minimum: the lowest level of material stock which must be maintained in hand at all times, so that there is no stoppage of production due to non-availability of inventory. **Maximum**: the highest level of quantity for any material which can be held in stock at any time; any quantity beyond it causes extra expenditure on funds, storage and obsolescence. **Danger**: the level at which normal issues of the raw material inventory are stopped and emergency issues only are made. **Buffer stock**: stock kept for contingency, used in case of sudden order.

Worked examples

Worked example

Two potato suppliers: what a faster truck is worth

Tapri's samosa factory uses potatoes at 1,600 to 2,400 kg a working day, 2,000 kg on average, for 300 working days a year. Potatoes are ordered 20,000 kg at a time. Holding a kilogram in cold storage for a year costs ₹5.

Particulars	Present supplier (Nashik)	Proposed supplier (Indore)
Lead time	6 to 10 days	1 to 3 days
Price	₹20 per kg delivered	₹20 per kg plus ₹500 per delivery

If stock runs low, potatoes can be bought from the local mandi within 2 days.

Required:

(i) For each supplier, calculate the re-order level, minimum level, maximum level and average stock level.

(ii) Calculate the danger level under the present supplier.

(iii) Advise Kesar whether to switch to the Indore supplier.

THE WORKING

Average lead time = (minimum + maximum) ÷ 2: Nashik 8 days, Indore 2 days.

(i) Stock levels (kg)

Level	Nashik	Indore
Re-order level = max. usage × max. lead time	2,400 × 10 = **24,000**	2,400 × 3 = **7,200**
Minimum = ROL − avg. usage × avg. lead time	24,000 − 2,000 × 8 = **8,000**	7,200 − 2,000 × 2 = **3,200**
Maximum = ROL + ROQ − min. usage × min. lead time	24,000 + 20,000 − 1,600 × 6 = **34,400**	7,200 + 20,000 − 1,600 × 1 = **25,600**
Average = minimum + ½ ROQ	8,000 + 10,000 = **18,000**	3,200 + 10,000 = **13,200**

(ii) Danger level = average usage × lead time for emergency purchases = 2,000 × 2 = **4,000 kg**.

(iii) Advice

Particulars	Amount (₹)
Fall in average stock: 18,000 − 13,200 = 4,800 kg × ₹5	24,000
Less: extra delivery charge: (6,00,000 kg ÷ 20,000 kg) = 30 deliveries × ₹500	15,000
Net annual saving from switching	**9,000**

Annual requirement = 2,000 kg × 300 days = 6,00,000 kg. The shorter lead time lowers every level that depends on it: the re-order level falls from 24,000 to 7,200 kg because far less stock must cover the wait. The Indore supplier saves ₹9,000 a year and should be preferred. With the alternative average, (maximum + minimum) ÷ 2, the averages are 21,200 and 14,400 kg and the saving is 6,800 × ₹5 − ₹15,000 = ₹19,000: the advice is the same.

Worked example

Working backwards from the danger level

Ritu has lost her working papers for chai premix at the central store. The stock register still shows:

Particulars	Packs
Re-order level	3,600
Minimum stock level	1,500
Maximum stock level	6,800
Danger level	600

Emergency purchases arrive in 2 days, and the danger level was set on average consumption. The maximum lead time is 9 days, and average lead time and average

consumption are the averages of their maximum and minimum.

Required:

(i) Calculate the average consumption per day and the average lead time.

(ii) Calculate the maximum and minimum consumption per day and the minimum lead time.

(iii) Calculate the re-order quantity Ritu was using, and the average stock level.

THE WORKING

(i) Danger level = average consumption × emergency lead time: 600 = average consumption × 2, so average consumption = **300 packs a day**.

Minimum level = ROL − (average consumption × average lead time): 1,500 = 3,600 − (300 × average lead time), so average lead time = 2,100 ÷ 300 = **7 days**.

(ii) ROL = maximum consumption × maximum lead time: 3,600 = maximum consumption × 9, so maximum consumption = **400 packs a day**.

Minimum consumption = 2 × average − maximum = 2 × 300 − 400 = **200 packs a day**.

Minimum lead time = 2 × average − maximum = 2 × 7 − 9 = **5 days**.

(iii) Maximum level = ROL + ROQ − (minimum consumption × minimum lead time): 6,800 = 3,600 + ROQ − (200 × 5), so ROQ = 6,800 − 3,600 + 1,000 = **4,200 packs**.

Average stock level = minimum + ½ ROQ = 1,500 + 2,100 = **3,600 packs**; or (6,800 + 1,500) ÷ 2 = 4,150 packs. State the formula you use.

Worked example

Diwali gift boxes: three items and one rack

For the Diwali season Tapri packs gift boxes at 500 to 900 boxes a day, 700 on average. Each box holds 2 tins of premix, 1 kulhad set and 1 greeting-card set.

Item	Per box	Re-order quantity	Lead time (days)
Premix tin	2	12,000	2 to 4
Kulhad set	1	8,000	5 to 9
Card set	1	20,000	10 to 14

Required:

(i) For each item, calculate the re-order level, the minimum level and the maximum level.

(ii) Calculate the average stock level of premix tins.

(iii) The godown has racks for only 12,000 kulhad sets. Is the present re-order quantity for kulhad sets acceptable? If not, find the largest re-order quantity that keeps the maximum level within the racks.

THE WORKING

Daily usage = per box × boxes a day (minimum / average / maximum): tins 1,000 / 1,400 / 1,800; kulhad sets 500 / 700 / 900; card sets 500 / 700 / 900. Average lead times: 3, 7 and 12 days.

(i) Levels (units)

Level
ROL = max. usage × max. lead time
Tins
7,200
Kulhad sets
8,100
Card sets
12,600

Level
Minimum = ROL – avg. usage × avg. lead time
Tins
3,000
Kulhad sets
3,200
Card sets
4,200

Level	Maximum = ROL + ROQ – min. usage × min. lead time
Tins	17,200
Kulhad sets	13,600
Card sets	27,600

Workings: tins 1,800 × 4; 7,200 − 1,400 × 3; 7,200 + 12,000 − 1,000 × 2. Kulhad sets 900 × 9; 8,100 − 700 × 7; 8,100 + 8,000 − 500 × 5. Card sets 900 × 14; 12,600 − 700 × 12; 12,600 + 20,000 − 500 × 10.

(ii) Average stock of tins = minimum + ½ ROQ = 3,000 + 6,000 = **9,000 tins**; or (17,200 + 3,000) ÷ 2 = 10,100 tins.

(iii) The maximum level of kulhad sets, 13,600, exceeds the 12,000 rack places, so an order of 8,000 can arrive when there is no room for it. Keeping the maximum level at 12,000:

$$12{,}000 = 8{,}100 + \text{ROQ} \qquad -(500 \times 5)$$

ROQ = 12,000 − 8,100 + 2,500 = **6,400 kulhad sets**. Order 6,400 sets at a time (or add rack space): the re-order and minimum levels do not change, because they do not depend on the order size.

Interactive version: PassBook app, lesson Stock levels: re-order, minimum, maximum, average and danger levels

Worked example

The Sunday stock-out: a post-mortem

The manager of Tapri's Bhopal café orders chai premix whenever stock falls to the re-order level, 1,200 packs at a time. Usage is 30 to 70 packs a day, 50 on average; the supplier takes 3 to 7 days, 5 on average. The manager set the re-order level at average usage × average lead time. In the week of the stock-out the order was placed on Monday morning when stock touched that level; customers took 70 packs every day, and the van came after 7 days.

Required:

(i) What re-order level did the manager use? How many packs could the café not serve before the van arrived?

(ii) Calculate the re-order level, minimum level and maximum level by ICAI's formulas.

(iii) With the re-order level in (ii), how much stock would be left when the van arrives in the worst week (70 packs a day, 7 days), a normal week (50 a day, 5 days) and the best week (30 a day, 3 days)? Relate each answer to a level in (ii).

THE WORKING

(i) The manager's re-order level = 50 × 5 = **250 packs**. In the worst week the café needed 70 × 7 = 490 packs before the van came. The 250 packs ran out after 250 ÷ 70 ≈ 3.6 days, and **240 packs** (490 − 250) could not be served.

(ii) ICAI's levels:

Level	Working	Packs
Re-order level	Max. usage × max. lead time = 70 × 7	**490**
Minimum level	ROL − avg. usage × avg. lead time = 490 − 50 × 5	**240**
Maximum level	ROL + ROQ − min. usage × min. lead time = 490 + 1,200 − 30 × 3	**1,600**

(iii) Stock left when the van arrives, starting from the re-order level of 490:

Week
Worst: 70 × 7

Used before delivery
490

Left on arrival
0

Level it matches
the re-order level covers exactly the worst case

Week

Normal: 50 × 5

Used before delivery

250

Left on arrival

240

Level it matches

minimum level

Week

Best: 30 × 3

Used before delivery

90

Left on arrival

400

Level it matches

400 + 1,200 delivered = 1,600, the maximum level

So the re-order level is built for the worst week, the minimum level is the stock normally left when supplies arrive, and the maximum level is the highest stock can reach, in the best week. A re-order level on averages protects only the normal week.

Exam angle

What ICAI asks from this topic

In the exam

Where stock levels appeared, May 2024 to September 2026

In the documents read for this lesson, stock levels appeared in 14 documents, in 11 of them together with the EOQ.

- **The full set, 5 to 8 marks**: MTP Series I for January 2025 Q3(b) and Series II Q4(a) (re-order, maximum, minimum and danger levels, 6 marks each; minimum consumption from the average; ROQ = EOQ); MTP Series I for May 2025 Q1(a) (three materials, usage per unit of product, 5 marks); MTP Series II for September 2025 Q3(a) (8 marks, with the cost of not ordering the EOQ); RTPs for September 2025 Q7 and September 2026 Q5.
- **Reverse working**: January 2025 Q1(c) (maximum and minimum consumption from given levels, 2 + 2 marks); May 2026 MCQ 11 (re-order quantity from the maximum level, 2 marks).
- **MCQs, 2 marks**: MTP Series I for January 2026 MCQ 11 and Series II MCQ 12, MTP Series I for May 2025 MCQ 13 (maximum level, some needing the EOQ first); MTP Series I for September 2026 MCQs 6 to 8 (ROQ, maximum and minimum levels of one material).
- **Perishable material**: RTP May 2024 Q1 and RTP May 2026 Q6 (maximum level capped by shelf life).
- **Safety stock**: May 2024 Q1(a) (ROL = safety stock + lead time consumption).
- **Theory**: MTP Series I for January 2026 Q6(b): explain any five methods of inventory control by

setting quantitative levels (5 marks).

In the exam

The traps

- ROQ not given: compute the EOQ first; it feeds the maximum and average levels (MTP May 2025 MCQ 13, MTP January 2025 Q3(b)).
- Only average and maximum consumption given: minimum = 2 × average − maximum.
- Units of time: a weekly rate with lead time in days (or 25 working days a month) must be converted first.
- Perishable material: the maximum level is the lower of the formula and the shelf-life consumption.
- The danger level uses average consumption ("sometimes minimum") and the **emergency** lead time, not the normal one.

Trick

ROL is "worst × worst", minimum is "ROL less normal", maximum is "ROL plus order less best"

Re-order level: maximum usage × maximum lead time. Minimum: ROL − average usage × average lead time. Maximum: ROL + ROQ − minimum usage × minimum lead time. Holds for ICAI's primary formulas; the alternative ROL (minimum + average × average) is the minimum formula turned around.

Answer frame

"Distinguish between re-order level and re-order quantity" (SM 2.22–2.23)

Re-order level	Re-order quantity
The stock level at which a fresh order is placed.	The quantity for which the purchase requisition is made.
Answers "when to order".	Answers "how much to order".
Maximum consumption × maximum re-order period.	Set considering the minimum level, re-order level, delivery time and cost; ideally the EOQ, where ordering and carrying costs together are minimum.
Lies between the minimum and maximum levels.	Added to the re-order level in the maximum-level formula.

"Explain the methods of inventory control by setting quantitative levels"

For each level: what it answers (the table in *How it works*), ICAI's definition, the formula. Five levels for five marks; end with a one-line link: all are set so that production is never interrupted (minimum, danger) and funds are not locked up (maximum).

Short notes: danger level; maximum and minimum levels

Definition in ICAI's words, formula, and the action: at the danger level normal issues stop and only emergency issues are made; near the maximum level the store keeper stops further purchasing; at the re-order level the store keeper initiates the purchase requisition (SM 2.18).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the re-order level built on a good day

Using average consumption × average lead time for the re-order level. On a busy week with a slow supplier, the café runs dry before the van arrives. **Fix:** ROL = maximum consumption × maximum re-order period (or minimum level + average × average).

⚠ **Common mistake**

The maximum level with the wrong subtraction

Subtracting average or maximum consumption, or the maximum lead time, in the maximum level. **Fix:** the maximum level subtracts the **minimum** consumption over the **minimum** lead time: the least that can be used before the order arrives.

⚠ **Common mistake**

The danger level on the normal lead time

Writing average consumption × average lead time for the danger level (that is part of the minimum level). **Fix:** use the lead time for **emergency** purchases.

Practice

Check yourself

1 Maida is used at 80 to 140 kg a working day, and the factory works 6 days a week. Deliveries take 1 to 2 weeks, and 3,000 kg are ordered at a time. What is the maximum stock level, in kg?

Answer: _____

2 Case scenario. At Tapri's Pune store the re-order level for milk powder is 1,500 kg, average consumption is 90 kg a day and the average lead time is 10 days. Ritu's report shows an average stock level of 1,700 kg, worked as minimum level + ½ re-order quantity. What re-order quantity is the store using?

- **A** 1,100 kg
- **B** 400 kg
- **C** 2,200 kg
- **D** 3,400 kg

3 Average consumption is 120 kg a day and maximum consumption 150 kg a day. The lead time is 8 to 12 days. What is the minimum stock level, in kg?

Answer: _____

4 Which formula gives the danger level?

- **A** Maximum consumption × maximum lead time
- **B** Re-order level − average consumption × average lead time
- **C** Average consumption × lead time for emergency purchases
- **D** Minimum level + ½ re-order quantity

5 Fresh paneer for the kathi-roll counters is used evenly at 250 kg a day and must be used within 5 days of receipt. Deliveries take 1 to 2 days. What is the largest order size, in kg, that keeps the maximum stock level within 5 days' consumption?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Re-order level = maximum consumption × maximum re-order period (or minimum level + average consumption × average period).
- Minimum level = re-order level − average consumption × average re-order period.

- Maximum level = re-order level + re-order quantity − minimum consumption × minimum re-order period.
- Average level = minimum + ½ ROQ, or (maximum + minimum) ÷ 2. Danger level = average consumption × emergency lead time.
- Re-order quantity not given: use the EOQ, $\sqrt{(2AO/C)}$.
- Missing minimum consumption: 2 × average − maximum. Same time units throughout.
- Perishable material: maximum level is the lower of the formula and shelf-life consumption.
- Safety stock: optimum where expected stock-out cost + carrying cost is least; ROL = safety stock + lead time consumption. (Kesar's lakh packs: no.)

Answer key · Check yourself

1 **4200**

Put both in days first: lead time 6 to 12 working days. ROL = 140 × 12 = 1,680 kg. Maximum = 1,680 + 3,000 − (80 × 6) = 4,200 kg. Mixing a daily rate with weeks (140 × 2) gives nonsense.

2 **C** **2,200 kg**

Minimum level = 1,500 − (90 × 10) = 600 kg. Then 1,700 = 600 + ½ ROQ, so ROQ = 2 × 1,100 = 2,200 kg. 1,100 kg forgets to double the half; 400 kg treats the re-order level as the minimum; 3,400 kg doubles the average and ignores the minimum level.

3 **600**

ROL = 150 × 12 = 1,800. Average lead time = 10 days. Minimum = 1,800 – (120 × 10) = 600 kg.

4 **C** **Average consumption × lead time for emergency purchases**

The danger level is where normal issues stop and only emergency issues are made: average (sometimes minimum) consumption × emergency lead time. The others are the re-order, minimum and average levels.

5 **1000**

Shelf-life limit = 250 × 5 = 1,250 kg. ROL = 250 × 2 = 500 kg. Maximum = 500 + ROQ – (250 × 1) = 250 + ROQ, which must not exceed 1,250, so ROQ = 1,000 kg. Ordering 1,250 kg forgets the 250 kg still on hand when the fastest delivery arrives.

3.5 Economic Order Quantity and quantity discounts

How much to order at a time: the order size at which ordering cost plus carrying cost is lowest, and how to test a supplier's quantity discount on total cost.

In plain words

One order a year, or one order a day?

Tapri's Indore blending unit uses 36,000 kg of Assam tea a year. The tea costs ₹300 a kg, and each order costs ₹600 to place (quotations, the purchase order, the truck, the quality check at the gate).

> **Kesar:** One order a year, one headache a year. Buy all 36,000 kg in April.
>
> **Ritu:** Then the godown holds 18,000 kg on average all year. Interest, rent, insurance and stale tea cost us 10% of the price, ₹30 a kg a year: ₹5,40,000, to save a few ₹600 purchase orders.
>
> **Kesar:** Fine. Order every three days.
>
> **Ritu:** That is 120 orders at ₹600: ₹72,000. Per order, please. Somewhere in between is cheapest.

Two costs pull in opposite directions. The **ordering cost** is paid every time an order is placed, so it falls as orders get bigger and fewer. The **carrying cost** is paid on the stock lying in the store, and since stock runs down from a full order to nothing, the store holds half an order on

average: bigger orders mean more stock and more carrying cost.

Order size, kg (orders a year)	Ordering cost (₹)	Carrying cost (₹)	Total (₹)
36,000 (1)	600	5,40,000	5,40,600
6,000 (6)	3,600	90,000	93,600
3,000 (12)	7,200	45,000	52,200
1,200 (30)	**18,000**	**18,000**	**36,000**
600 (60)	36,000	9,000	45,000
300 (120)	72,000	4,500	76,500

The cheapest order size, 1,200 kg, is the **Economic Order Quantity (EOQ)**: 30 orders a year, one every 12 days. Notice where it sits: the row where the two costs are equal.

Then a supplier says "order 6,000 kg at a time and I will knock 2% off the price". Now a third cost moves, the price itself. Ritu does not argue: she adds up purchase cost, ordering cost and carrying cost for each choice and takes the smaller total. (Here the 2% wins by ₹1,60,200: ₹2,16,000 off the price and ₹14,400 less ordering, against ₹70,200 more carrying, at 10% of the lower price.)

Think of it like

The vegetable run

Every trip to the mandi costs you an hour and an auto fare, so you do not go every morning. Buy a month's vegetables at once and half of them wilt in a fridge you could not shut. Most homes settle on two trips a week without any formula. EOQ is the same instinct done with numbers: as many trips as it takes to make trips plus storage cheapest.

Key idea

EOQ is the order size at which annual ordering cost plus annual carrying cost is lowest; in the basic model the two are equal there. A quantity discount is worth taking only if the total cost, purchase price included, is lower at the discount lot than at the EOQ.

How it works

Six steps from a purchase file to the EOQ

The **re-order quantity** is the quantity for which the stores department raises a purchase requisition. ICAI's rule for setting it: "the quantity should be where, the total of carrying cost and ordering cost is at minimum. For this purpose, an economic order quantity should be calculated" (SM 2.23). When to order is the re-order level; how much to order is the EOQ.

1. **Annual requirement (A)**: the material needed for a year, in the material's own unit. Monthly usage × 12. If you are given production, work back: units to produce × material per unit, grossed up for yield, less opening stock of the material where the question says purchases are net of it.

2. **Ordering cost per order (O)**: the costs of placing an order: inviting quotations, preparing purchase orders, procurement staff directly attributable, transportation and inspection (SM 2.23).

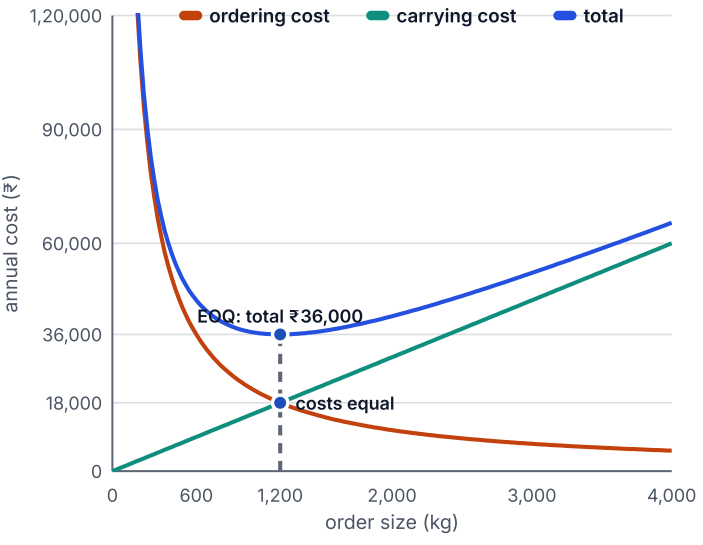
3. **Carrying cost per unit per annum (C)**: the cost of holding one unit for a year: cost of funds invested, storage, insurance, obsolescence (SM 2.23). Rates given as a % of price are added and multiplied by the price; a fixed ₹ per unit is added on top; monthly and quarterly rates are converted to a year.

4. **EOQ** $= \sqrt{2AO/C}$. For the tea: $\sqrt{2 \times 36{,}000 \times 600/30} = 1{,}200$ kg.

5. **Number of orders** = A ÷ EOQ = 30; **order interval** = days in the year ÷ number of orders = 360 ÷ 30 = 12 days.

6 **Annual costs**: ordering cost = (A ÷ Q) × O = ₹18,000; carrying cost = (Q ÷ 2) × C = ₹18,000; total inventory cost = purchase cost (A × price) + ordering cost + carrying cost = ₹1,08,00,000 + ₹36,000 = ₹1,08,36,000.

Assam tea: the costs of ordering and carrying, by order size



Orange: ordering cost, ₹2,16,00,000 ÷ order size, falls as orders grow. Teal: carrying cost, half the order × ₹30, a straight line. Blue: their total, lowest at 1,200 kg, exactly where the orange and teal lines cross.

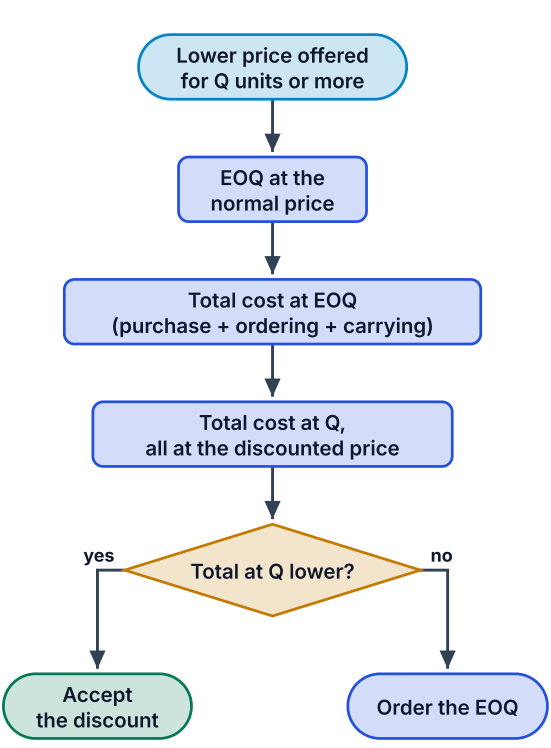
The bottom of the blue curve is flat. Ordering 600 kg or 2,400 kg (half or double the EOQ) costs ₹45,000, 25% more; ordering 20% below the EOQ costs 2.5% more, and 20% above it less than 2% more. So a supplier's pack size near the EOQ is fine, and a rounded EOQ changes little.

Interactive version: PassBook app, lesson Economic Order Quantity and quantity discounts

Quantity discounts: when the price moves too

A supplier offering a lower price for a bigger lot changes three costs at once: purchase cost falls (lower price), ordering cost falls (fewer orders), carrying cost rises (more stock, though each unit carries a little less where

carrying cost is a % of price). The test is always the **total cost**:



With several price breaks and the EOQ in the undiscounted band, compute the total at the EOQ and at the lowest quantity of each discount band, and pick the smallest. The Study Material's table (SM 2.90) has one row per order size tested.

ICAI's suggested answers end the comparison with a one-line advice: "The total cost of inventory is lower if EOQ is adopted. Hence, the company is advised not to accept the

quantity discount" (SM 2.87), or the reverse when the discount wins (May 2024 Q1(a)).

Formats and formulas

Reproduce the formula, the four assumptions and the comparison statement from memory.

∑ **Formula**

Economic Order Quantity

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A = annual requirement (the demand for raw material or input for a year); O = cost per order (the cost of placing an order for purchase); C = carrying cost per unit per annum (the cost of carrying average inventory on an annual basis) (SM 2.24).

∑ **Formula**

Carrying cost per unit per annum

$$C = r \times P + F$$

r = sum of the carrying cost rates per annum; P = price per unit; F = any fixed carrying cost per unit per annum. Storage 2%, interest 12% and obsolescence 6% on a ₹20 unit give C = 20% × ₹20 = ₹4 (SM 2.26). A storage cost of ₹2.5 a kg a month is ₹30 a year.

Σ **Formula**

Orders, interval and annual costs at order size Q

$$\text{Number of orders} = \frac{A}{Q}$$

$$\text{Order interval} = \frac{\text{Days in the year}}{\text{Number of orders}}$$

$$\text{Ordering cost} = \frac{A}{Q} \times O$$

$$\text{Carrying cost} = \frac{Q}{2} \times C$$

$$\text{Total cost} = A \times \text{price} \quad + \frac{A}{Q} \times O + \frac{Q}{2} \times C$$

Q ÷ 2 is the average inventory.

Definition

Economic Order Quantity (EOQ)

The size of an order for which total of ordering and carrying cost are minimum. (SM 2.23)

Definition

Ordering cost

Costs associated with the purchase or order of materials, such as cost to invite quotations, documentation works like preparation of purchase orders, employee cost directly attributable to the procurement of material, transportation and inspection cost. (SM 2.23)

Definition

Carrying cost

Costs for holding or carrying inventories in store, such as the cost of fund invested in inventories, cost of storage, insurance cost and obsolescence. (SM 2.23)

Assumptions underlying EOQ, in ICAI's order (SM 2.24):

1. Ordering cost per order and carrying cost per unit per annum are known and they are fixed.
2. Anticipated usage of material in units is known.
3. Cost per unit of the material is constant and is known as well.
4. The quantity of material ordered is received immediately, i.e. the lead time is zero.

The comparison statement for a quantity discount

ICAI prints it as two statements, "(a) When EOQ is ordered" and "(b) When quantity discount is accepted",

headed **Evaluation of Profitability of Different Options of Order Quantity**; side by side it reads:

Particulars	(a) At EOQ (₹)	(b) At discount lot (₹)
Purchase cost (A × price)	xxx	xxx
Ordering cost (A ÷ Q × O)	xxx	xxx
Carrying cost (Q ÷ 2 × C)	xxx	xxx
Total cost	**xxx**	**xxx**

End with "Advise – ...". With a schedule of price breaks, ICAI uses one row per order size (SM 2.90); on paper write the columns: annual requirement, order size, number of orders, cost of inventory, ordering cost, carrying cost, total cost.

Note

Rounding the number of orders

If A ÷ Q is not a whole number, the Study Material and recent ICAI answers round the number of orders up ("number of orders cannot be in decimals", SM 2.90; "22.60 Orders or 23 Orders", RTP September 2026 Q5; "rounded off to next whole number", MTP September 2026 Series I). When a question is silent, round up and say so in one line.

Worked examples

Worked example **Kulhads by the truckload**

Tapri's cafés use 7,20,000 kulhads a year, evenly at 60,000 a month. A kulhad costs ₹8. Carrying cost is interest at 12% a year on the cost of the stock plus ₹0.24 a kulhad a year for storage and breakage. Each order costs ₹1,080 to place and inspect. The potter delivers in his own trucks, each carrying at most 30,000 kulhads, and charges ₹4,800 a trip whether the truck is full or not. Assume 360 days in a year.

Required:

(i) Compute the EOQ, ignoring freight.

(ii) Compare the total annual cost of ordering, freight and carrying for orders of 36,000, 30,000 and 60,000 kulhads, and recommend the order size.

(iii) Compute the re-order level for the recommended order size if delivery takes 6 days from the date of order and a safety stock of 5 days' usage is kept.

THE WORKING

(i) EOQ. A = 7,20,000 kulhads; O = ₹1,080; C = 12% × ₹8 + ₹0.24 = ₹0.96 + ₹0.24 = ₹1.20 per kulhad per annum.

$$\text{EOQ} = \sqrt{\frac{2 \times 7{,}20{,}000 \times 1{,}080}{1.20}}$$

$$= \sqrt{1{,}29{,}60{,}00{,}000} = 36{,}000 \text{ kulhads}$$

(ii) Comparison of order sizes

Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Orders a year (7,20,000 ÷ order size)	20	24	12

Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Ordering cost (orders × ₹1,080)	21,600	25,920	12,960

Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Truck trips a year	40	24	24

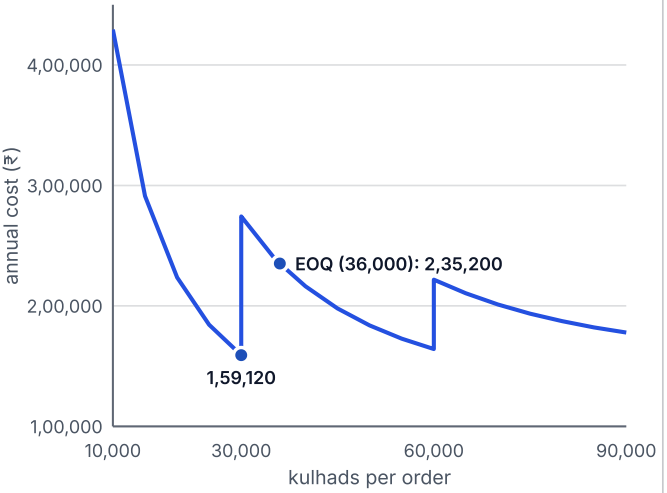
Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Freight (trips × ₹4,800)	1,92,000	1,15,200	1,15,200

Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Carrying cost (order size ÷ 2 × ₹1.20)	21,600	18,000	36,000

Particulars	36,000 (₹)	30,000 (₹)	60,000 (₹)
Total annual cost	**2,35,200**	**1,59,120**	**1,64,160**

An order of 36,000 needs two trucks (one full, one carrying only 6,000), so 20 orders make 40 trips. Orders in whole truckloads need only 24 trips a year whatever their size.

Annual cost of ordering, freight and carrying against order size



The cost jumps each time an order needs one more truck, so the cheapest sizes are whole truckloads; the EOQ, which ignores the trucks, sits just after a jump.

Recommendation: order **30,000 kulhads** (one full truck), 24 times a year. Freight is a cost per trip, not per kulhad, so the EOQ formula, which assumes a fixed cost per order, does not hold once a second truck is needed. Among whole truckloads, 30,000 has the lowest ordering plus carrying cost (₹43,920 against ₹48,960 for 60,000).

(iii) Re-order level = lead time consumption + safety stock = (2,000 a day × 6 days) + (2,000 × 5 days) = 12,000 + 10,000 = **22,000 kulhads** (daily usage = 7,20,000 ÷ 360).

Worked example

Sugar on a price ladder, and a godown that holds 1,500 quintals

Tapri's premix plant needs 12,000 quintals of sugar a year. Ordering cost is ₹1,500 per order and the holding cost is 10% of the purchase price per annum. The mill quotes:

Order size (quintals)	Price per quintal (₹)
Less than 600	4,000
600 and less than 1,200	3,960
1,200 and less than 2,400	3,940
2,400 and above	3,920

Tapri's godown holds at most 1,500 quintals. For a larger lot it must rent space in a warehouse at ₹200 a year for each quintal of the lot above 1,500.

Required:

(i) Compute the EOQ at the base price.

(ii) Compute the most economical purchase level.

(iii) State how your answer would change if the godown could hold any quantity.

THE WORKING

(i) EOQ at ₹4,000. C = 10% × ₹4,000 = ₹400.

$$\text{EOQ} = \sqrt{\frac{2 \times 12{,}000 \times 1{,}500}{400}}$$
$$= \sqrt{90{,}000} = 300 \text{ quintals}$$

300 quintals is within the "less than 600" slab, so it is a valid lot at ₹4,000.

(ii) Total cost at the EOQ and at each price break

Order size (quintals)	Orders a year	Price (₹)	Purchase cost (₹)
300	40	4,000	4,80,00,000
600	20	3,960	4,75,20,000
1,200	10	3,940	4,72,80,000
2,400	5	3,920	4,70,40,000

Order size (quintals)	Ordering (₹)	Carrying (₹)	Warehouse rent (₹)	Total cost (₹)
300	60,000	60,000	–	4,81,20,000
600	30,000	1,18,800	–	4,76,68,800
1,200	15,000	2,36,400	–	**4,75,31,400**
2,400	7,500	4,70,400	1,80,000	4,76,97,900

Ordering = orders × ₹1,500; carrying = order size ÷ 2 × 10% of the price; rent = (2,400 − 1,500) × ₹200. Total = purchase + ordering + carrying + rent.

The most economical purchase level is **1,200 quintals** (10 orders a year), with a total cost of ₹4,75,31,400.

(iii) Without the godown limit the rent disappears and 2,400 quintals costs ₹4,75,17,900, ₹13,500 less than 1,200 quintals: the most economical purchase level would then be 2,400 quintals (5 orders). The godown's limit is worth ₹1,80,000 of rent at 2,400 quintals, more than the ₹13,500 the bigger lot would otherwise save.

Interactive version: PassBook app, lesson Economic Order Quantity and quantity discounts

Worked example

Milk powder for the premix: the discount that the consultant forgot

For 2026-27 Tapri plans to sell 1,10,000 cartons of chai premix, with 6,000 cartons in stock at the start and 16,000 wanted at the end. A carton needs 2 kg of milk powder, and the input-output ratio is 5 : 4 (80% yield). 12,000 kg of milk powder are already in stock; purchases are to cover the rest. Milk powder costs ₹250 a kg; each order costs ₹2,500. Carrying costs: interest 1% per month and insurance 0.5% per quarter, both on the purchase price, and storage ₹5 per kg per annum.

At present Tapri orders monthly, in 12 equal lots, and its supplier gives a 1% discount on monthly orders. Dev's consultant says "Move to EOQ and you will save".

Required:

(i) Compute the annual purchase requirement of milk powder.

(ii) Compute the EOQ and the number of orders.

(iii) Compare the total annual cost of the present policy with the EOQ policy, and advise.

(iv) The supplier now withdraws the monthly discount and offers 1% only on quarterly orders. Advise whether to accept, and compute the minimum discount on quarterly orders that would make them worth accepting.

THE WORKING

(i) Purchase requirement

Particulars	Quantity
Sales	1,10,000 cartons
Add: Closing stock of cartons	16,000 cartons
Less: Opening stock of cartons	6,000 cartons
Production	**1,20,000 cartons**
Milk powder input (1,20,000 × 2 kg ÷ 80%)	3,00,000 kg
Less: Opening stock of milk powder	12,000 kg
Annual purchase requirement (A)	**2,88,000 kg**

(ii) EOQ

Carrying cost per kg per annum (C) = interest (1% × 12) + insurance (0.5% × 4) = 14% of ₹250 = ₹35, plus storage ₹5 = **₹40**.

$$\text{EOQ} = \sqrt{\frac{2 \times 2,88,000 \times 2,500}{40}}$$

$$= \sqrt{3,60,00,000} = 6,000 \text{ kg}$$

$$\text{Orders} = \frac{2,88,000}{6,000} = 48$$

(iii) Present policy against EOQ

Monthly lot = 2,88,000 ÷ 12 = 24,000 kg at ₹247.50; its carrying cost per kg = 14% of ₹247.50 + ₹5 = ₹39.65.

Particulars	EOQ, 6,000 kg (₹)	Monthly, 24,000 kg, 1% off (₹)
Purchase cost (2,88,000 × ₹250; × ₹247.50)	7,20,00,000	7,12,80,000
Ordering cost (48 × ₹2,500; 12 × ₹2,500)	1,20,000	30,000
Carrying cost (3,000 × ₹40; 12,000 × ₹39.65)	1,20,000	4,75,800
Total cost	**7,22,40,000**	**7,17,85,800**

Advise – the present policy costs ₹4,54,200 less: the 1% discount (₹7,20,000) is worth far more than the ordering and carrying costs EOQ would save. The consultant compared ordering and carrying cost only and forgot the price. Tapri should keep ordering monthly while the discount lasts.

(iv) 1% only on quarterly orders

Without the monthly discount, the choice is EOQ at ₹250 (total ₹7,22,40,000) against quarterly lots of 72,000 kg at ₹247.50: purchase ₹7,12,80,000 + ordering (4 × ₹2,500) ₹10,000 + carrying (36,000 × ₹39.65) ₹14,27,400 = **₹7,27,17,400**, which is ₹4,77,400 more. **Reject** the 1% offer and order the EOQ.

Minimum discount: let the quarterly price be p. Then

$$2{,}88{,}000p + 10{,}000 \qquad +36{,}000 \times (5 + 0.14p)$$
$$= 7{,}22{,}40{,}000 \qquad 2{,}93{,}040p = 7{,}20{,}50{,}000$$
$$p = ₹245.87$$

Discount = (₹250 − ₹245.87) ÷ ₹250 = 1.65%. A quarterly discount of **more than 1.65%** (say 1.66% or more) makes the quarterly lot cheaper than the EOQ; that is the counter-offer to make.

Exam angle

In the exam

Where EOQ appeared, May 2024 to September 2026

Of the 32 new-scheme Paper 4 documents (8 past papers, 8 RTPs and 16 MTPs, May 2024 to September 2026), 25 compute an EOQ somewhere, usually with a

second part (re-order level, a discount, total cost). The ones read in full for this lesson:

- **Descriptive, 3 to 6 marks.** EOQ, re-order level with a safety stock, and a 2% discount for a 30,000 lot, accepted: May 2024 Q1(a) (5 marks). EOQ and number of orders only: September 2025 Q5(b) (3 marks). EOQ from a market share and storage, interest and obsolescence rates, then the annual ordering cost: September 2026 Q1(b) (4 marks). The Study Material's 4% discount, rejected: MTP January 2026 Series II Q1(c) (4 marks). A discount schedule in rupees per order size, the re-order point with a safety stock, and the number of orders (one, two, four or five, or the EOQ's 20) that costs least: MTP January 2026 Series I Q4(b) (6 marks).
- **Multiple choice, 2 marks each, in case scenarios.** January 2026 MCQs 6–10: demand from a market share, EOQ of two materials whose carrying costs are given per month and per quarter, total annual cost at the EOQ and at another order size. Working backwards: annual demand from the EOQ (September 2024 MCQ 12), carrying cost per unit from the EOQ (January 2025 MCQ 15). MTP September 2024 Series I MCQs 1–3 (requirement from production and yield; quarterly ordering with a 2.5% discount against EOQ); MTP September 2026 Series I MCQs 6–10 (a re-order quantity 280 kg below the EOQ costs ₹244 a year more, orders "rounded off to next whole number").

- **Revision Test Papers** stretch it: a purchasing manager's bonus inside the comparison (May 2025 Q8); EOQ against a quarterly lot with a 5% volume discount (September 2025 Q7); a 1% offer rejected and the counter-offer discount solved for, at least 2.82% (September 2026 Q5).
- **Theory**: re-order quantity and EOQ inside "explain any 5 methods of inventory control by setting quantitative levels" (MTP January 2026 Series I Q6(b), 5 marks).

In the exam

The traps ICAI set

- Carrying cost given per month or per quarter: annualise before the formula (January 2026 MCQs 6–10; RTP September 2026 Q5, "1% per month plus ₹2 per unit per annum" is ₹14 on a ₹100 kg).
- Requirement in units of product, EOQ wanted in kg of material: multiply by material per unit and divide by the yield (MTP September 2024 Series I).
- In a discount comparison, carrying cost is recomputed at the discounted price when it is a % of price (SM 2.87: 4,000 × ₹384 × ½ × 20%).
- A safety stock does not enter the EOQ; ICAI's May 2024 comparison left it out of carrying cost under both options.

Trick

Two checks on any EOQ answer

At the EOQ, annual ordering cost = annual carrying cost, and their sum = EOQ × C. If your two figures differ (beyond rounding of the number of orders), the EOQ or one of the costs is wrong. Conditions: the basic model, with C a constant cost per unit per annum and no discount.

Answer frame

"Explain EOQ" or "Write a short note on EOQ"

1. Re-order quantity: the quantity for which the purchase requisition is made; it should be where the total of carrying cost and ordering cost is at minimum.
2. Definition: the size of an order for which the total of ordering and carrying cost is minimum.
3. Ordering cost, with ICAI's examples; carrying cost, with ICAI's examples.
4. Formula with A, O and C named.
5. The four assumptions, in order.
6. A sketch of the two cost curves and their total, lowest at the EOQ.

The numerical parts: three answers to the same price-ladder question

Loses marks

"Order 2,400 quintals: it has the lowest price."

The price alone; ordering, carrying and rent ignored.

Passes

EOQ computed and the totals compared at each price break, but the godown rent left out: "2,400 quintals at ₹4,75,17,900".

Method shown, one cost missing: the wrong advice.

Full marks

Data line (A, O, C with its build-up), formula, EOQ and its slab; a statement at the EOQ and every price break with purchase, ordering, carrying (at each slab's price) and rent; totals; "The most economical purchase level is 1,200 quintals", with the saving in rupees.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the carrying cost that forgot the price

"Interest 12%" plugged in as C = 0.12 + 0.24 = 0.36 gives an EOQ of about 65,727 kulhads instead of 36,000. **Fix:** C is rupees per unit per annum: 12% × ₹8 = ₹0.96, plus ₹0.24 = ₹1.20. Add every rate first, then multiply by the price, then add any fixed ₹ per unit.

⚠ **Common mistake**

The monthly figure that stayed monthly

Using 60,000 kulhads a month as A, or a 1% monthly interest rate as 1% a year. **Fix:** A and C must both be per annum: 60,000 × 12, and 1% × 12.

⚠ **Common mistake**

Carrying cost on the whole order

Charging carrying cost on Q instead of Q ÷ 2 doubles it. **Fix:** stock runs down from Q to nil, so the average inventory is Q ÷ 2 (plus nothing for a safety stock in ICAI's comparisons).

⚠ **Common mistake**

The discount compared without the price

Comparing only ordering plus carrying cost always "proves" EOQ is best, since EOQ minimises exactly those two. **Fix:** a discount changes the purchase cost, so the comparison must include A × price under each option.

Practice

Check yourself

1 Tapri's samosa factory uses 24,000 kg of refined flour a year. Each order costs ₹300 to place, flour costs ₹400 a kg and the carrying cost is 10% of the price per annum. What is the EOQ in kg?

Answer: _____

2 A material's EOQ is 800 units, the cost of placing an order is ₹160 and the carrying cost is ₹5 per unit per annum. What is the annual demand?

A 20,000 units

B 10,000 units

C 1,02,40,000 units

D 12.5 units

3 Case scenario. Tapri's cold-chain division buys 4,000 crates a year at ₹500 each. Ordering cost is ₹500 per order and carrying cost 20% of the price per annum, so it orders its EOQ of 200 crates. The supplier offers a 2% discount on orders of 1,000 crates, but the cold store holds only 600 crates, so Ritu would have to rent extra space for ₹15,000 a year. Which advice is correct?

- **A** Accept: total annual cost falls by ₹9,000
- **B** Accept: the ₹40,000 saved on price outweighs everything else
- **C** Reject: carrying cost rises by ₹39,000
- **D** Reject: total annual cost rises by ₹6,000

4 Which of these are carrying costs of inventory? Select all that apply.

- **A** Interest on the funds invested in inventory
- **B** Insurance of the stock in the godown
- **C** Cost of inviting quotations
- **D** Obsolescence of stored material
- **E** Inspection of material on receipt

More than one answer may be right.

5 A component's EOQ is 1,000 units when it costs ₹100 and carrying cost is 20% of the price per annum. The price rises to ₹400 and nothing else changes. What is the new EOQ?

- **A** 250 units
- **B** 500 units
- **C** 1,000 units
- **D** 2,000 units

Answers: the answer key at the end of this lesson.

Recap

- EOQ = √(2AO ÷ C): A annual requirement, O cost per order, C carrying cost per unit per annum.
- EOQ: the size of an order for which the total of ordering and carrying cost is minimum; at it, the two are equal.
- Ordering cost: quotations, purchase orders, procurement staff, transportation, inspection. Carrying cost: cost of funds, storage, insurance, obsolescence.
- C = (sum of annual rates × price) + fixed ₹ per unit; annualise monthly and quarterly rates first.
- Orders = A ÷ Q; ordering cost = A ÷ Q × O; carrying cost = Q ÷ 2 × C; total = A × price + both.

- Assumptions: costs known and fixed; usage known; price constant; lead time zero.
- Quantity discount: compare total cost including purchase cost, carrying cost at the discounted price; with price breaks (EOQ in the base band), test EOQ and each band's lowest quantity.
- Fractional orders: round up, and say so. (Two mandi trips a week, not one a month.)

Answer key · Check yourself

1 **600**

C = 10% × ₹400 = ₹40. EOQ = $\sqrt{}$(2 × 24,000 × 300 ÷ 40) = $\sqrt{}$3,60,000 = 600 kg.

2 **B** **10,000 units**

$EOQ^2 = 2AO \div C$, so $A = EOQ^2 \times C \div 2O$ = 6,40,000 × 5 ÷ 320 = 10,000. Leaving out the 2 gives 20,000; swapping O and C gives 1,02,40,000; not squaring the EOQ gives 12.5.

3 **D** **Reject: total annual cost rises by ₹6,000**

At EOQ: ₹20,00,000 + ₹10,000 ordering + ₹10,000 carrying = ₹20,20,000. At 1,000 crates: ₹19,60,000 + 4 × ₹500 (₹2,000) + 500 × 20% of ₹490 (₹49,000) + rent ₹15,000 = ₹20,26,000, ₹6,000 more. ₹9,000 forgets the rent; the price saving alone ignores ordering, carrying and rent; the ₹39,000 rise in carrying cost is real but is only one of the changes.

4 **A** **Interest on the funds invested in inventory**; **B** **Insurance of the stock in the godown**; **D** **Obsolescence of stored material**

Carrying costs are those of holding stock: cost of funds, storage, insurance, obsolescence. Quotations, purchase orders, transportation and inspection are ordering costs (SM 2.23).

5 **B** **500 units**

C rises fourfold (₹20 to ₹80), and the EOQ moves with $1 \div \sqrt{C}$: $1{,}000 \div \sqrt{4} = 500$. Dividing by 4 forgets the square root; "unchanged" forgets that C is a % of price.

3.6 ABC, FSN, HML, VED, JIT and perpetual inventory

Four ways to sort stock so that control goes where it matters (value, movement, criticality, unit price), the demand-pull JIT system, and the physical controls behind the records.

In plain words

Guarding the stirrers, losing the cardamom

Tapri's central store in Indore holds everything from Assam tea to wooden stirrers. Ramesh, the storekeeper, is proud of it.

> **Ramesh:** Fifty thousand stirrers, forty thousand paper cups, all counted every morning. Nothing goes missing on my watch.
>
> **Ritu:** And the cardamom sacks by the loading door, the ones that got damp in the rain?
>
> **Ramesh:** Only a few sacks, madam. The bulk items are what fill the store.
>
> **Ritu:** The stirrers are worth ₹30,000. Those "few sacks" are part of ₹3,00,000. Per rupee, please, not per piece.

Ramesh controls by bulk; Ritu wants control by value. That is **ABC analysis**: rank items by the money tied up in them. A handful of "A" items usually hold most of the

value and get tight control; a long tail of "C" items gets light control; "B" sits in between.

A store can also be sorted by how fast an item moves (fast, slow or non-moving: **FSN**), by how badly production suffers if it runs out (vital, essential or desirable: **VED**), or by the price of one unit (high, medium or low: **HML**). Or it can be emptied: under **Just in Time (JIT)** Tapri buys gift-box ribbon only when a Diwali order arrives. And the records need guarding too: **perpetual inventory** keeps bin cards and the stores ledger up to date, and **continuous stock verification** checks a few items every day, by surprise, against them.

Think of it like

The household budget

Nobody tracks matchboxes the way they track the rent and the school fees (ABC). The medicine cabinet is sorted differently: the asthma inhaler must never run out, the cough sweets can (VED). And the cook who buys vegetables only after fixing tonight's menu runs JIT.

Key idea

Each technique sorts stock on its own basis so that control effort goes where it matters: ABC by value, FSN by movement, VED by criticality, HML by unit price. JIT removes stock by letting demand pull material in. Perpetual inventory with continuous stock verification keeps the records true.

How it works

Four ways to control inventory

ICAI defines inventory control (after CIMA) as "the function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks" (SM 2.21). Management may control it on four bases (SM 2.22):

The four bases of inventory control (SM 2.22)

By setting quantitative levels — when and how much
re-order level, EOQ, maximum, minimum, danger level

On the basis of relative classification
where to look
ABC, FSN, VED, HML

Using ratio analysis — how well it moves
input-output ratio, inventory turnover

Physical control — what is really there
two bin, budgets, perpetual inventory, stock verification

This lesson covers relative classification, JIT and physical control; the levels, EOQ and ratios have their own lessons.

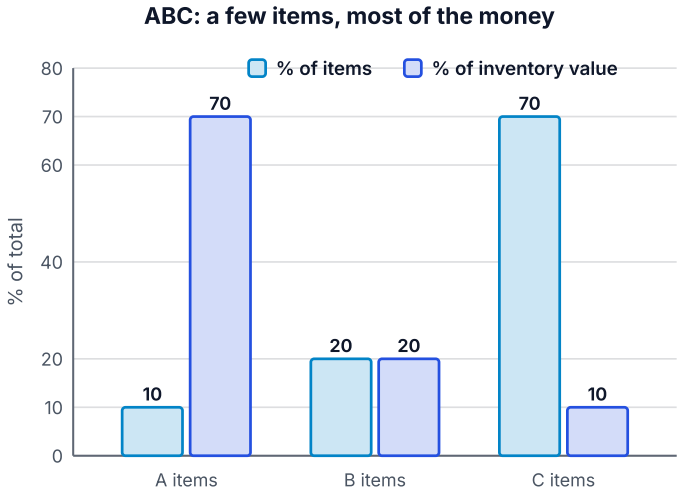
Levels and the re-order quantity are in stock levels and EOQ; the ratios in inventory turnover ratios; Paper 6A uses the same ABC analysis in inventory management.

Relative classification: four questions, four sortings

The Study Material's one-line bases (SM 2.35): ABC on the value and frequency of inventory; FSN on inventory turnover; VED on the importance of inventory; HML on the price of an item of inventory.

ABC analysis "exercises discriminating control over different items of inventory on the basis of the investment involved" (SM 2.35). The Study Material's typical split (SM 2.36):

- **A**: about 10% of the items, about 70% of the inventory value, because of high prices or heavy requirement or both. Controlled by a regular system that plans total requirements through budgets and fixes maximum, minimum and re-order levels.
- **B**: about 20% of the items, about 20% of the value. Orders placed after reviewing the situation periodically.
- **C**: about 70% of the items, about 10% of the value. No constant control; orders placed once in six months or a year, after ascertaining requirements, to economise on ordering and handling costs.



The Study Material's typical shares (SM 2.36). Sky blue: share of the items; blue: share of the value. A and C are mirror images, and the thresholds in a question are management's.

The steps of an ABC plan, as ICAI's answers lay them out:

1. Total cost of each item (units × unit cost), and its % of total units and of total cost.
2. **Rank** the items by total cost, highest first.
3. Put each item in A, B or C by the basis of selective control the question gives (a ₹ range or a % of total cost).
4. Present the **plan of ABC selective control**: items in rank order, with a total line for each category.

Advantages of ABC analysis (SM 2.39):

1. **Continuity in production**: no danger of interruption for want of materials, with minimum investment in inventories.
2. **Lower cost**: the cost of placing orders, receiving goods and maintaining stocks is minimised, especially if coupled with proper economic order quantities.
3. **Less attention required**: management time is saved, since attention is paid only to some of the items.
4. **Systematic working**: much of the purchase work can be systematised on a routine basis and handled by subordinate staff.

FSN (also called FNS: fast, normal and slow moving) classifies on frequency of usage, with a threshold range of inventory turnover decided by management (SM 2.40–2.41). **Fast moving**: placed nearer the store issue point, reviewed frequently for fresh orders. **Slow moving**: stored a little far, reviewed periodically for obsolescence, may be shifted to non-moving. **Non-moving**: kept for disposal, reported to management; an appropriate provision for loss may be created.

VED classifies on criticality for the production function and the final product, generally for **spare parts** (SM 2.41–2.42). **Vital**: unavailability can interrupt production and cause a production loss; strictly controlled by setting a re-order level. **Essential**: essential but not vital; unavailability may cause sub-standardisation and loss of efficiency; reviewed periodically, second priority.

Desirable: optional; unavailability causes no production or efficiency loss. In a hospital, life-saving, rare and critical drugs are vital (SM 2.42).

HML classifies on the cost of an individual item, "unlike ABC analysis where inventories are classified on the basis of overall value of inventory"; high-cost items get more priority (SM 2.42). With Tapri's ranges of ₹1,000 a unit and above (H) and below ₹50 (L), cardamom at ₹2,400 a kg is H, and a ₹6 kulhad is L even though 30,000 kulhads are a B item by value.

Slow-moving and non-moving stores

Reasons (SM 2.41): (i) failure of production management to communicate the updated requirement to the stores; (ii) technological upgradation (a new machine needs new material, or the material becomes obsolete); (iii) lack of periodic review of inventories. Remedies: timely identification, a minimum level or JIT, and turnover ratios reported against standards.

ICAI's suggested answer to "How is slow moving and non-moving item of stores detected and what steps are necessary to reduce such stocks?" (MTP May 2024 Series II Q4(b); also MTP January 2025 Series II Q6(c)):

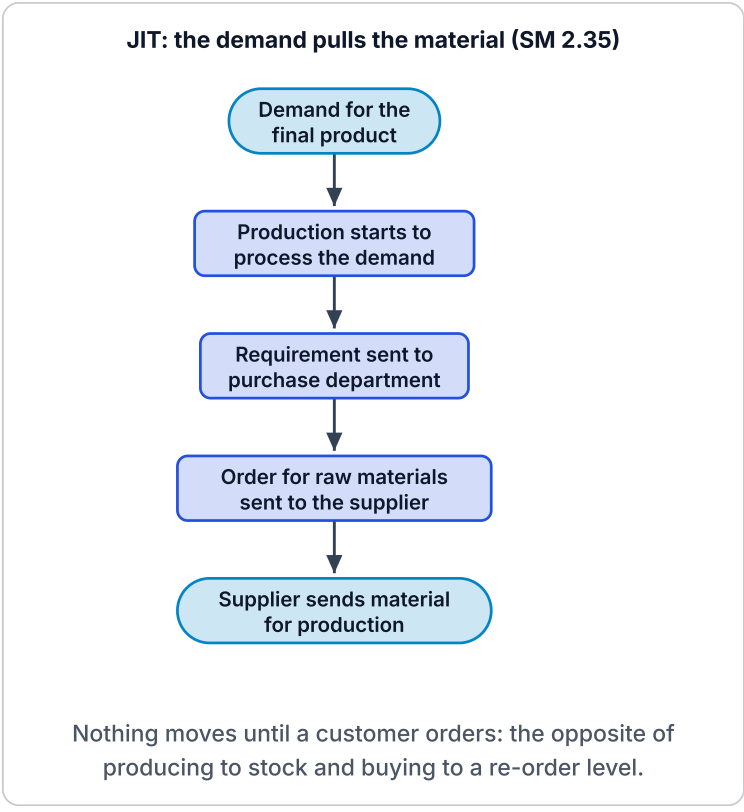
- **Detection**: (i) preparing and perusing periodic reports showing the status of different items of stores; (ii) calculating the inventory turnover period of various items in terms of number of days or months of consumption; (iii) computing the inventory turnover

ratio periodically, relating issues as a percentage of average stock held; (iv) implementing a well-designed information system.

- **Steps to reduce them**: (i) lay down proper procedure and guidelines for the disposal of non-moving items before they deteriorate further in value; (ii) diversify production to use up such materials; (iii) use these materials as substitutes in place of other materials.

Just in Time: let the order pull the material

JIT "is a system of inventory management with an approach to have zero inventories in stores. According to this approach material should only be purchased when it is actually required for production" (SM 2.35). It is based on two principles: (i) produce goods only when it is required, and (ii) the products should be delivered to customers at the time only when they want. It is also known as the **'Demand pull' or 'Pull through' system** of production: production starts after the order for the products is received.



Nothing moves until a customer orders: the opposite of producing to stock and buying to a re-order level.

Physical control: the records and the shelves

The fourth basis keeps the stock itself under control (SM 2.45–2.47):

- **Two bin system.** Each bin is divided into two parts: (I) a smaller part to stock the quantity equal to the minimum stock or even the re-ordering level, and (II) the other part to keep the remaining quantity. Issues are made out of the larger part; as soon as it becomes necessary to use quantity out of the smaller part, a fresh order is placed. The two bin system is

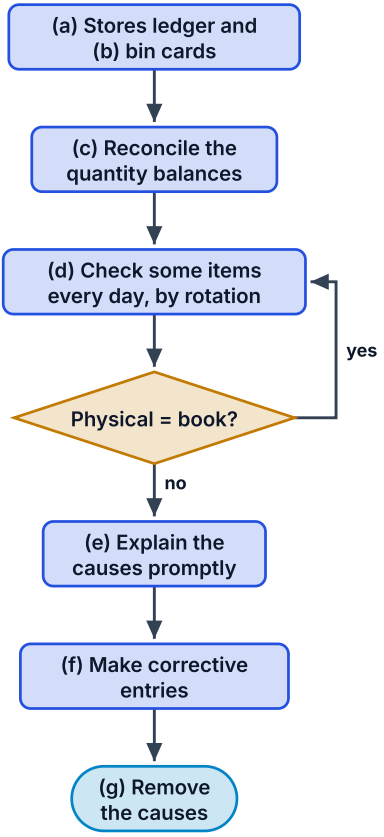
supplemental to the record of quantities on the bin card and the stores ledger card.

- **Establishment of a system of budgets.** To control investment in inventories, the exact quantity of each type of inventory and the time it is required are found by studying production plans and production schedules; an inventories requirement budget prepared on this basis discourages unnecessary investment in inventories.
- **Perpetual inventory records and continuous stock verification**, below.

Perpetual inventory "represents a system of records maintained by the stores department"; it comprises (i)

bin cards and (ii) the stores ledger (SM 2.45). Its success depends on the steps in the flow below:

Perpetual inventory: what its success depends on (SM 2.45)



How a discrepancy is valued and charged (normal or abnormal) is in the lesson on stock discrepancies.

Continuous stock verification is the physical verification of items, which may be done by the internal audit

department, independent of the store and production staff, at appropriate intervals and without prior notice: "the element of surprise is essential" (SM 2.46). It is a part of perpetual inventory (SM 2.77, 2.85). Valuing and booking the differences it finds is in stock discrepancies.

Annual stock-taking against continuous stock-taking

Annual stock-taking must cover all items in a given number of days: production shuts down, or the check is of a limited character (SM 2.46). **Continuous stock-taking** has these advantages (SM 2.47); who verifies, and how, is in stock verification:

1. Closure of normal functioning is not necessary.
2. Stock discrepancies are likely to be noticed and corrected much earlier.
3. It has a sobering influence on the stores staff because of the element of surprise.
4. The stores auditor can watch the movement of items more closely, so the chances of obsolescence buying are reduced.
5. Final accounts can be ready quickly, and interim accounts are possible quite conveniently.

The **advantages of perpetual inventory** (SM 2.46):

1. Physical stocks can be counted and book balances adjusted as and when desired, without waiting for the entire stock-taking to be done.

2. Quick compilation of the Profit and Loss Account for an interim period, due to prompt availability of stock figures.
3. Discrepancies are easily located, so corrective action can be taken promptly to avoid their recurrence.
4. A systematic review reveals the existence of surplus, dormant, obsolete and slow-moving materials, so that remedial measures may be taken in time.
5. Fixing the various stock levels and checking actual balances against them helps the storekeeper keep stocks within limits and raise purchase requisitions for the correct quantity at the appropriate time.

Formats and formulas

Reproduce both statements from memory: the column headings carry marks.

Statement of total cost and ranking

On paper ICAI's statement has seven columns: Item, Units, % of total units, Unit cost (₹), Total cost (₹), % of total cost, Ranking (SM 2.37). Here it is in two halves:

Item	Units	% of total units	Unit cost (₹)
1	xxx	xx.xx	xx
...	...	...	...
Total	**xxx**	**100**	

Item	Total cost (₹)	% of total cost	Ranking
1	xxx	xx.xx	x
...	...	...	...
Total	**xxx**	**100**	

Plan of ABC selective control

State the basis first ("₹ 50,000 & above: 'A' items ..."), then:

Ranking: item	% of total units	Cost (₹)	% of total cost
1: item ...	xx.xx	xxx	xx.xx
2: item ...	xx.xx	xxx	xx.xx
Total A (n items)	**xx.xx**	**xxx**	**xx.xx**
... B items, then **Total B**			
... C items, then **Total C**			
Grand total	**100**	**xxx**	**100**

ICAI writes the category in a last column next to each total line; here the total line names it.

Worked examples

Worked example **Two lists for one store: ABC by value, HML by unit price**

Tapri's central store gives you the annual consumption of eight items for 2025-26:

Item	Material	Units	Unit cost (₹)
1	Assam CTC tea (kg)	8,000	250.00
2	Milk powder (kg)	2,500	320.00
3	Saffron (g)	800	350.00
4	Kulhads	50,000	6.40
5	Sugar (kg)	8,000	50.00
6	Espresso machine gasket kits	50	1,600.00
7	Paper cups	60,000	1.20
8	Stirrers	1,20,000	0.40

The purchase manager proposes to "control the costly items" only, by HML analysis: unit cost ₹300 and above, H; ₹10 to below ₹300, M; below ₹10, L.

Required:

(i) Rank the items and draw a plan of ABC selective control on this basis: ₹5,00,000 and above, 'A' items; ₹1,00,000 to below ₹5,00,000, 'B' items; below ₹1,00,000, 'C' items.

(ii) Classify the items under the purchase manager's HML basis.

(iii) Identify the items on which the two classifications disagree, and advise on the control each needs.

THE WORKING

Statement of total cost and ranking

Item	% of total units	Total cost (₹)	% of total cost	Ranking
1	3.21	20,00,000	50.00	1
2	1.00	8,00,000	20.00	2
3	0.32	2,80,000	7.00	5
4	20.05	3,20,000	8.00	4
5	3.21	4,00,000	10.00	3
6	0.02	80,000	2.00	6
7	24.06	72,000	1.80	7
8	48.13	48,000	1.20	8
Total	**100.00**	**40,00,000**	**100.00**	

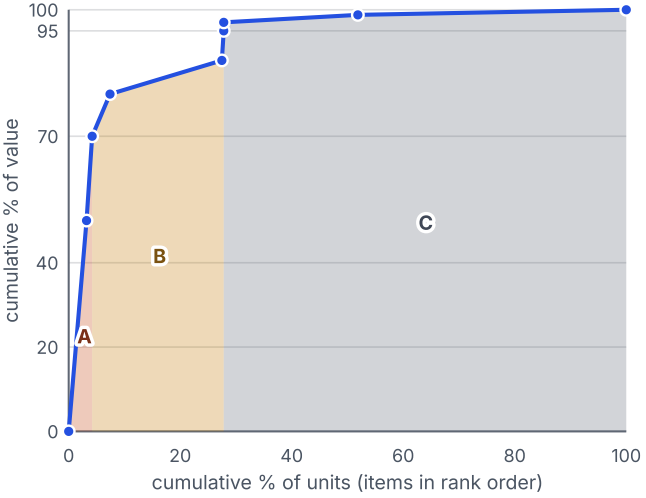
Total units = 2,49,350; percentages are rounded to two decimals.

(i) Plan of ABC selective control

Ranking: item	% of total units	Cost (₹)	% of total cost
1: item 1	3.21	20,00,000	50.00
2: item 2	1.00	8,00,000	20.00
Total A (2 items)	**4.21**	**28,00,000**	**70.00**

Ranking: item	% of total units	Cost (₹)	% of total cost
3: item 5	3.21	4,00,000	10.00
4: item 4	20.05	3,20,000	8.00
5: item 3	0.32	2,80,000	7.00
Total B (3 items)	**23.58**	**10,00,000**	**25.00**
6: item 6	0.02	80,000	2.00
7: item 7	24.06	72,000	1.80
8: item 8	48.13	48,000	1.20
Total C (3 items)	**72.21**	**2,00,000**	**5.00**
Grand total (8 items)	**100.00**	**40,00,000**	**100.00**

The store ranked by value: cumulative share of value against share of units



Orange band: the two A items, 4.21% of the units and 70% of the value. Amber band: B items. Grey band: the C items, 72.21% of the units for the last 5% of the value.

(ii) HML classification (on the cost of an individual item)

Class	Items
H (₹300 and above)	2 milk powder (₹320), 3 saffron (₹350 a g), 6 gasket kits (₹1,600)
M (₹10 to below ₹300)	1 tea (₹250), 5 sugar (₹50)
L (below ₹10)	4 kulhads (₹6.40), 7 paper cups (₹1.20), 8 stirrers (₹0.40)

(iii) Where the two lists disagree

Item
1 Assam tea
ABC
A
HML
M
Control needed
Half the store's value: budgets and fixed maximum, minimum and re-order levels, though its unit cost is only medium

Item
6 Gasket kits
ABC
C
HML
H
Control needed
Only 2% of the value: periodic ordering is enough on value; but as spares that can stop the espresso machines, check their criticality by VED

Item
4 Kulhads
ABC
B
HML
L
Control needed
8% of the value: review periodically before ordering, though each kulhad is cheap

Item
3 Saffron
ABC
B
HML
H
Control needed
7% of the value: periodic review; its high unit price also calls for safe custody

Milk powder (A, H), sugar (B, M), paper cups and stirrers (C, L) get similar priority under both methods.

Advice. HML classifies inventory on the basis of the cost of an individual item, unlike ABC, which uses the overall value of inventory (SM 2.42). Controlling by unit price alone would leave the tea, half the money in the store, under only moderate control, and spend tight control on gasket kits worth 2%. Use ABC to decide where control effort goes, with HML (and VED for spare parts) as a supplement.

Worked example

Fast, slow or non-moving? The Bhopal depot's turnover test

The Bhopal depot gives you the following for 2025-26:

Item	Consumed (₹)	Opening stock (₹)	Closing stock (₹)
Masala sachets	9,60,000	70,000	90,000
Kulhads	4,80,000	50,000	70,000
Gift tins	1,50,000	40,000	60,000
Ginger syrup (winter special)	18,000	24,000	12,000
Pouches with the old logo	Nil	45,000	45,000

Management's threshold range for FSN analysis: inventory turnover ratio of 6 or more, fast moving; below 6 but with some issues during the year, slow moving; no issue during the year, non-moving.

Required:

(i) Calculate the inventory turnover ratio and the number of days for which the average stock is held for each item (360 days in a year).

(ii) Classify the items under FSN analysis and state how each class is stored and controlled.

(iii) State possible reasons for the slow-moving and non-moving items and how they can be managed.

THE WORKING

(i) and (ii) Turnover and FSN class

Item	Average stock (₹)	Turnover ratio	Days held	Class
Masala sachets	80,000	12 times	30	Fast
Kulhads	60,000	8 times	45	Fast
Gift tins	50,000	3 times	120	Slow
Ginger syrup	18,000	1 time	360	Slow
Old-logo pouches	45,000	Nil	—	Non-moving

Average stock = (opening + closing) ÷ 2; turnover ratio = consumption ÷ average stock; days held = 360 ÷ turnover ratio.

FSN analysis classifies inventory on the basis of frequency of usage, using a threshold range of

inventory turnover decided by management (SM 2.40–2.41):

- **Fast moving** (sachets, kulhads): placed nearer to the store issue point; the stock is reviewed frequently for making fresh orders.
- **Slow moving** (gift tins, ginger syrup): stored a little farther away; the stock is reviewed periodically for obsolescence and may be shifted to the non-moving category.
- **Non-moving** (old-logo pouches): kept for disposal, reported to management, and an appropriate provision for loss may be created.

(iii) Reasons and management. The Study Material's reasons for slow-moving and non-moving stock fit these items: technological upgradation or existing material becoming obsolete (the rebranding made the old pouches useless); failure of production management to communicate the updated requirement to the stores (gift tins bought for a festival plan that changed); and lack of periodic review of inventories. Such stock is managed by careful observation and timely identification, by keeping stock near the minimum level or using the just-in-time approach (seasonal items such as the ginger syrup can be bought only against confirmed winter demand), and by reporting turnover ratios, actual against standard, to management (SM 2.41).

Worked example

Kesar's spare parts: vital, essential or desirable?

The samosa factory's maintenance store holds: (a) a compressor for the blast freezer, with no substitute and a three-week import lead time; (b) a spare belt for the packing conveyor (if it fails, packing continues by hand at half speed); (c) an LED display panel for the visitors' lobby; (d) gaskets for a freezer model scrapped last year, untouched for 14 months; and (e) printer ribbons for the batch-coding machine, issued every day. Kesar wants "the same control for everything".

Advise the factory on classifying these items and the control each needs.

MODEL ANSWER

Basis. Under VED analysis, inventories are classified on the basis of their criticality for the production function and the final product; generally this classification is done for spare parts. Under FSN analysis, inventories are classified on the basis of frequency of usage (inventory turnover) (SM 2.40–2.42).

Classification and control.

Item

(a) Freezer compressor

Class

Vital

Reason

Unavailability stops production

Control

Strictly controlled by a re-order level

Item

(b) Conveyor belt

Class

Essential

Reason

Loss of efficiency, not a stoppage

Control

Reviewed periodically; second priority

Item

(c) Lobby LED panel

Class

Desirable

Reason

Optional: no production or efficiency loss

Control

Lowest priority, after vital and essential

Item
(d) Old gaskets

Class
Non-moving

Reason
No issues in 14 months; machine scrapped

Control
Keep for disposal, report to management, provide for loss

Item
(e) Printer ribbons

Class
Fast moving

Reason
Issued daily

Control
Near the issue point; reviewed frequently

Conclusion. Uniform control is not advisable: it spends as much effort on the lobby panel as on the compressor whose absence stops the factory. Apply VED to the spare parts and FSN to all items, with control in proportion to criticality and movement.

Exam angle

In the exam

Where these techniques appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **ABC plan, 8 marks**: rank eight materials and draw the plan on a ₹ basis, with a storekeeper who "has taken proper care" of the bulk items while a few units of valuable ones deteriorated (September 2025 Q5(a)).
- **JIT, 5 marks**: "Which system of inventory management is known as 'Demand Pull' or 'Pull through' system of production? Explain. Also, specify the two principles on which this system is based." (May 2026 Q6(a)).
- **Slow and non-moving stores, 4 marks**: "HOW is slow moving and non-moving item of stores detected and WHAT steps are necessary to reduce such stocks?" (MTP May 2024 Series II Q4(b)); the same as a scenario, "HOW would you identify slow-moving and non-moving items, and WHAT strategies would you implement" (MTP January 2025 Series II Q6(c)).
- The chapter's own theory questions ask for ABC as a technique of inventory control, the detection and reduction of slow-moving and non-moving stores, and a short note on JIT (SM 2.78).
- **Old scheme**: an ABC plan on % of total inventory cost (July 2021 Q1(a), 5 marks); a short note on VED, on which the examiners found performance "below average" (July 2021 Q6(e)); JIT in the May 2026 words (November 2022 Q6(a)); "Under VED

analysis, inventories are classified on the basis of cost of individual items", true or false (false; November 2023 Q6(b)(ii)).

In the exam

What the suggested answers do

- ABC: the "Statement of Total Cost and Ranking", the basis restated, then the plan with a total line per category (September 2025 Q5(a); SM 2.37–2.39). A % basis that leaves the B band implicit is filled in ("7% to 14%", July 2021 Q1(a)).
- JIT: the definition, "production process actually starts after the order for the products is received", and the two principles, numbered (May 2026 Q6(a)).

Trick

One word per technique

ABC = **A**mount (value). **F**SN = **F**requency (movement). **V**ED = **V**ital to production (criticality). **H**ML = **H**ow much one unit costs (unit price). Condition: these are memory hooks for the basis only; write ICAI's full basis in the answer.

Answer frame

"Explain the concept of ABC analysis as a

technique of inventory control"

1. Basis: discriminating control over different items of inventory on the basis of the investment involved; categories by relative importance, their value and frequency of replenishment.
2. A: about 10% of items, about 70% of value; controlled by budgets and maximum, minimum and re-order levels.
3. B: about 20% of items, about 20% of value; orders after periodic review.
4. C: about 70% of items, about 10% of value; no constant control; orders once in six months or a year, to economise on ordering and handling costs.
5. The four advantages: continuity in production, lower cost, less attention required, systematic working.
6. A one-line example or a sketch of items against value.

"Which system is known as demand pull? Explain, and state its two principles" (5 marks)

1. Name it: Just in Time (JIT) inventory management, the 'Demand pull' or 'Pull through' system.
2. ICAI's definition: zero inventories in stores; material purchased only when actually required for production.
3. Working: production starts after the order for the products is received; the material requirement then goes to the purchase department.
4. The two principles, numbered, in ICAI's words.

5. The five-box chain from demand to supplier.

Distinguish ABC and HML (or VED)

Point

Basis

ABC

Investment involved: the overall value of the item's inventory, and frequency of replenishment

HML

Cost of an individual item, by a range of cost

VED

Criticality for the production function and the final product; generally for spare parts

Point

Categories

ABC

A, B, C

HML

High, medium, low cost

VED

Vital, essential, desirable

Point

Priority and control

ABC

A: budgets and maximum, minimum and re-order levels; B: periodic review; C: orders once in six months or a year

HML

High-cost items get more priority; medium and low cost, less

VED

Vital: strictly controlled by a re-order level; essential: periodic review, second priority

Three answers to "Write a short note on VED analysis"

Loses marks

"VED classifies stock on cost: very expensive, expensive, desirable."

The basis is wrong: the very statement ICAI set as a true-or-false trap (November 2023).

Passes

"Vital, Essential and Desirable: items classified by their importance for production."

Basis right; no meaning of each class, no control, no field of use.

Full marks

The basis (criticality for the production function and the final product), generally used for spare parts; each class with its effect on production and its control; the hospital example.

Common mistakes

⚠ **Common mistake**

Hall of Shame: ABC ranked by the number of units

Ranking by units makes the 1,20,000 stirrers the top item, though they are 1.20% of the value. **Fix:** rank by total cost (units × unit cost). The % of units column is shown to prove the point that A items are few, not to rank them.

⚠ **Common mistake**

ABC and HML swapped

Classifying the gasket kits as A items "because each costs ₹1,600". **Fix:** ABC uses the item's total inventory value; a high unit price makes it H under HML. In the example the kits are H under HML but C under ABC: 50 × ₹1,600 = ₹80,000, below the ₹1,00,000 limit.

⚠ **Common mistake**

VED on cost

Fix: VED is on criticality for production; a cheap seal without which the freezer stops is vital, an expensive lobby panel is desirable.

⚠ **Common mistake**

Perpetual inventory confused with annual stock-taking

Fix: perpetual inventory is the continuously updated records (bin cards and stores ledger), checked by continuous stock verification of a few items daily, by surprise; annual stock-taking counts everything once and is what it replaces.

Practice

Check yourself

1. A store has four items: P, 2,000 units at ₹150; Q, 500 units at ₹900; R, 12,000 units at ₹5; S, 300 units at ₹40. The basis is: ₹2,00,000 and above, A; ₹50,000 to ₹2,00,000, B; below ₹50,000, C. What is the total cost of the A items, in rupees?

Answer: _____

2 Case scenario. Ritu sets up a two bin system for kulhads at Tapri's Pune café: the smaller part of the bin holds 1,200 kulhads, the re-order level, and the larger part the rest. Which of the following is correct?

- **A** A fresh order is placed when the smaller part is empty; the bin card is no longer needed
- **B** A fresh order is placed when the smaller part is empty; the bin card and stores ledger are still kept
- **C** A fresh order is placed as soon as kulhads have to be taken from the smaller part; the bin card and stores ledger are still kept
- **D** A fresh order is placed as soon as kulhads have to be taken from the smaller part; the bin card is no longer needed

3 Under VED analysis, inventories are classified on the basis of:

- **A** The cost of an individual item
- **B** The frequency of usage
- **C** Their criticality for the production function and the final product
- **D** The value of the inventory held

4 Which of the following are the two principles on which JIT is based? Select all that apply.

- **A** Produce goods only when it is required
- **B** Order in lots of the economic order quantity
- **C** Deliver products to customers at the time only when they want them
- **D** Keep a buffer stock equal to the maximum lead time consumption

More than one answer may be right.

5 Which of these is NOT an advantage of continuous stock-taking?

- **A** Normal functioning need not be closed
- **B** Discrepancies come to notice and are corrected earlier
- **C** Every item is counted on the same day, so the store is checked in one go
- **D** The element of surprise has a sobering influence on the stores staff

Answers: the answer key at the end of this lesson.

Recap

- Four bases of inventory control: quantitative levels, relative classification, ratio analysis, physical control.
- ABC: by value and frequency; A about 10% of items and 70% of value, B about 20% and 20%, C about 70% and 10%; rank by total cost.
- FSN: by frequency of usage (inventory turnover); non-moving kept for disposal, with a provision for loss.
- VED: by criticality for production, mainly spare parts; vital items controlled by a re-order level.
- HML: by the cost of an individual item, unlike ABC's overall value.
- JIT: zero inventory, demand pull; two principles: produce only when required, deliver only when customers want.
- Perpetual inventory: bin cards and stores ledger, kept up to date and checked by continuous, surprise stock verification by rotation.
- Two bin: order when issues reach the smaller part. (Watch the rent, not the matchboxes.)

Answer key · Check yourself

1 **750000**

Values: P ₹3,00,000; Q ₹4,50,000; R ₹60,000; S ₹12,000. A = P + Q = ₹7,50,000 (of ₹8,22,000, 91.24%). R is B, S is C. Ranking by units would wrongly put R first.

2 **C** **A fresh order is placed as soon as kulhads have to be taken from the smaller part; the bin card and stores ledger are still kept**

Issues are made from the larger part, and a fresh order is placed as soon as it becomes necessary to use the smaller part, which holds the minimum stock or the re-order level. The two bin system is supplemental to the bin card and stores ledger, not a replacement (SM 2.45). Waiting until the smaller part is empty uses up the stock meant to cover the lead time.

3 **C** **Their criticality for the production function and the final product**

Cost of one item is HML, frequency of usage is FSN, value is ABC. VED is criticality, generally for spare parts.

4 **A** **Produce goods only when it is required**; **C** **Deliver products to customers at the time only when they want them**

JIT's principles: produce only when required, deliver only when customers want. EOQ lots and buffer stocks belong to stock levels, which JIT tries to remove.

5 **C** **Every item is counted on the same day, so the store is checked in one go**

Counting everything at once is annual stock-taking, whose shortcoming is that production must stop or the check is limited. Continuous stock-taking checks a few items every day, by rotation and without notice.

3.7 Inventory turnover ratios and stock discrepancies

How fast each material moves (inventory turnover ratio and days held), what the input-output ratio shows, and how shortages found on stock verification are valued and charged.

In plain words

A full godown is not a good godown

> **Kesar:** Our godown is full to the roof. Nobody can say Tapri ever runs out of anything.
>
> **Ritu:** Nobody can find our cash either. The elaichi in there is six months of use.

Ritu's test is the **inventory turnover ratio**: how many times in a year a material's average stock is used up and replaced. Last year the factory consumed ₹8,40,000 of cardamom while holding ₹4,20,000 on average: it turned over **2 times**, so the average stock sat for 365 ÷ 2 = **182.5 days**. Sugar worth ₹12,50,000 was consumed from an average stock of ₹1,00,000: **12.5 times**, about **29 days**. Sugar is fast moving; the cardamom is tying up money that could be in the bank.

Then comes the stock count. The stores ledger says 3,500 litres of cooking oil; the shelf holds 3,320.

Kesar: 180 litres? Sab adjust ho jayega.

Ritu: Sixty will: oil is lost every time a drum is decanted into tins. The other 120 went out of the gate without a slip. That one goes to the Costing Profit and Loss Account.

A difference between the book balance and the physical balance is a **stock discrepancy**. The shortage is entered as an issue at the stores ledger rate, and then split: a **normal** shortage (inherent, unavoidable) is part of the cost of the good units; an **abnormal** one (avoidable: theft, mishandling) is a loss kept out of product cost.

Think of it like

Table turns at a café

A café table that seats six groups an evening earns more than one occupied all night by a single cup of chai. Inventory turnover counts the table turns of your stock; days held is how long each group sits.

Key idea

The turnover ratio counts how many times a year the average stock is used up: a high ratio means fast-moving material, a low one over-investment and working capital locked up. A shortage found on verification is priced like an issue; a normal one stays in product cost, an abnormal one goes to the Costing Profit and Loss Account.

How it works

Ratio analysis as inventory control

Ratio analysis is the third basis of inventory control (SM 2.22). The Study Material gives two ratios (SM 2.42–2.43):

- **Input-output ratio**: the ratio of the quantity of input of material to production and the standard material content of the actual output. It compares actual consumption with standard consumption, "thus indicating whether the usage of material is favourable or adverse". If the samosa line used 10,500 kg of flour for output whose standard content is 10,000 kg, the ratio is 1.05: 5% more than standard, an adverse usage. The money value of that excess is the material usage variance.
- **Inventory turnover ratio**: computed for different items and compared. A high ratio indicates fast-moving material; a low ratio indicates over-investment and locking up of working capital (SM 2.42).

Five steps to the turnover ratio

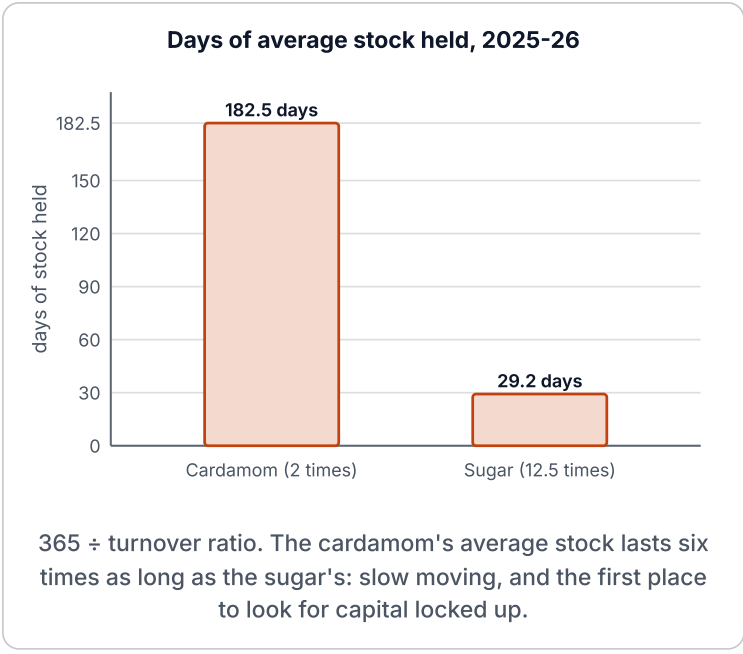
1. **Materials consumed** = opening stock + purchases − closing stock (SM 2.72). Cardamom: ₹3,60,000 + ₹9,60,000 − ₹4,80,000 = ₹8,40,000.

2. **Average stock** = ½ (opening stock + closing stock) = ½ (₹3,60,000 + ₹4,80,000) = ₹4,20,000.

3. **Inventory turnover ratio** = materials consumed ÷ average stock = 2 times.

4. **Average number of days of inventory holding** = 365 days (or 12 months) ÷ inventory turnover ratio = 182.5 days. Use 360 if the question says so.

5. **Compare** materials with each other, with a standard or with the industry, and comment: the material with fewer days is the faster moving.



365 ÷ turnover ratio. The cardamom's average stock lasts six times as long as the sugar's: slow moving, and the first place to look for capital locked up.

The ratio uses **cost**: cost of materials consumed over cost of average stock (SM 2.43). Paper 6A's inventory turnover uses cost of goods sold (or sales) over average inventory, and its raw material version is the same as this one (FM Study Material 3.16; activity ratios, Paper 6A).

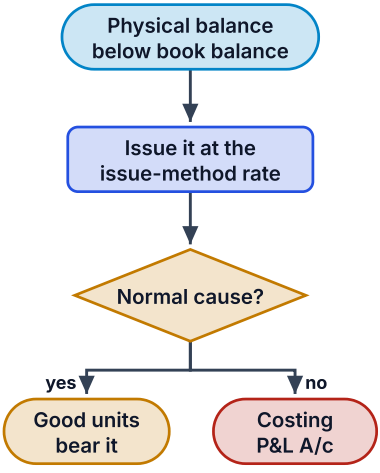
Stock discrepancies: when the shelf disagrees with the book

Under perpetual inventory the stores ledger and bin cards give a book balance at every moment, and continuous stock verification checks a few items a day against the shelves (perpetual inventory). When the physical balance differs, the Study Material's steps are: explain promptly the causes of the discrepancies, make

corrective entries wherever required, and remove the causes (SM 2.45). Some differences are errors in the records, which a corrective entry puts right. A real **shortage** is handled in two moves:

1. **Value it.** "Materials found short during physical verification should be entered in the issue column and valued at the rate as per the method adopted, i.e., FIFO or any other" (SM 2.63).
2. **Charge it** by its cause (SM 2.15, 2.69):

A shortage found on verification



Normal causes (breaking of bulk, evaporation, unavoidable conditions): the good units bear it, or factory overheads. Abnormal (mishandling, pilferage, avoidable reasons): Costing Profit and Loss Account (SM 2.15).

Shortage

Normal

Causes (SM 2.15, 2.69)

Breaking of bulk, evaporation, rusting, any unavoidable condition

Treatment

Absorbed by the good units: in the stores ledger the quantity is reduced and the value kept, so the rate per unit rises; in the cost ledger, charged to factory overheads

Shortage

Abnormal

Causes (SM 2.15, 2.69)

Mishandling, pilferage, misappropriation, any avoidable reason

Treatment

Valued at the issue rate and transferred to Costing Profit and Loss Account; not part of product cost

In the Study Material's Illustration 14, 900 missing micro-controllers (misappropriated) are issued at the weighted average rate of ₹282.04, ₹2,53,836, "transferred to Costing Profit & Loss A/c", and 300 rusted ones (normal) are "absorbed by good units": the balance becomes 6,600 units at ₹294.87, with the value unchanged (SM 2.69). The full treatment of material losses is in waste, scrap, spoilage and defectives.

Formats and formulas

Reproduce the statement and the formulas from memory.

∑ **Formula**

Inventory turnover ratio

$$\text{ITR} = \frac{\text{Cost of materials consumed}}{\text{Cost of average stock held}}$$

ITR = inventory turnover ratio; both terms for the same period. Average stock = ½ (opening stock + closing stock) (SM 2.43).

∑ **Formula**

Average number of days of inventory holding

$$\text{Days held} = \frac{365}{\text{ITR}}$$

Average number of days of inventory holding = 365 days (or 12 months) ÷ inventory turnover ratio (SM 2.43); 360 if the question fixes it.

∑ **Formula**

Materials consumed

$$\text{Consumption} = O + P - C$$

O = opening stock, P = purchases, C = closing stock of materials (SM 2.72).

∑ **Formula**

Input-output ratio

$$\text{Input-output ratio} = \frac{I}{S}$$

I = quantity of material input to production; S = standard material content of the actual output (SM 2.42). The Study Material sets no threshold, but the comparison reads directly: above 1, more material went in than the standard allows (adverse usage); below 1, less (favourable).

Statement of inventory turnover (ICAI's layout, SM 2.44)

Particulars	Material A (₹)	Material B (₹)
Opening stock	xxx	xxx
Add: Purchases	xxx	xxx
	xxx	xxx
Less: Closing stock	xxx	xxx
Materials consumed	**xxx**	**xxx**
Average inventory: (opening stock + closing stock) ÷ 2	xxx	xxx
Inventory turnover ratio (consumption ÷ average inventory)	x times	x times
Inventory turnover (number of days in a year ÷ IT ratio)	x days	x days

End with "Comments: Material A is moving faster than Material B" (SM 2.44).

A shortage in the cost ledger

Shortage	Debit	Credit
Normal	Factory Overhead Control A/c	Stores Ledger Control A/c
Abnormal	Costing Profit and Loss A/c	Stores Ledger Control A/c

The Study Material's cost-ledger illustration credits a shortage, "assumed normal", to stores ledger control and debits overhead control (SM 7.37).

Worked examples

Worked example **The samosa factory: three materials and a target for film**

The following data relate to three materials of Tapri's

samosa factory for 2025-26:

Particulars	Flour (₹)	Potatoes (₹)	Packaging film (₹)
Opening stock	2,40,000	36,000	3,00,000
Purchases during the year	19,20,000	14,48,000	6,00,000
Closing stock	1,60,000	44,000	4,20,000

Required:

(i) Calculate the inventory turnover ratio of each material and the number of days for which the average inventory is held (assume 360 days in a year).

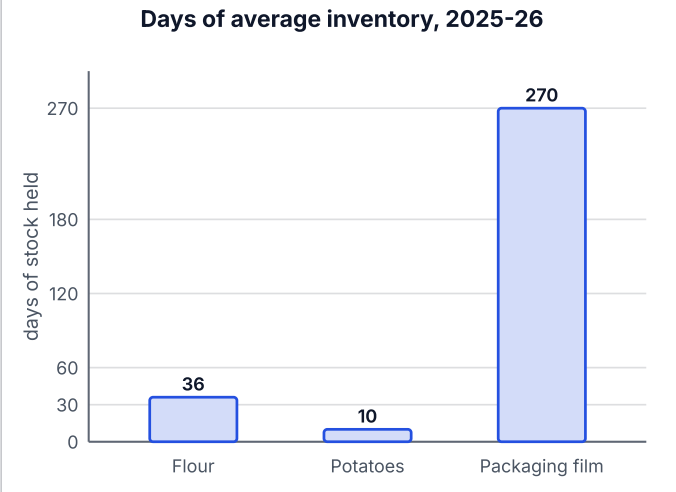
(ii) Comment on the movement of each material.

(iii) For 2026-27 the consumption of film is expected to rise by 25%. Kesar wants the average stock of film to be held for no more than 150 days. Taking the stock on 31 March 2026 as the opening stock, calculate the highest closing stock of film permissible on 31 March 2027 and the purchases of film this allows.

THE WORKING

(i) Statement of inventory turnover

Particulars	Flour (₹)	Potatoes (₹)	Film (₹)
Opening stock	2,40,000	36,000	3,00,000
Add: Purchases	19,20,000	14,48,000	6,00,000
	21,60,000	14,84,000	9,00,000
Less: Closing stock	1,60,000	44,000	4,20,000
Materials consumed	**20,00,000**	**14,40,000**	**4,80,000**
Average inventory	2,00,000	40,000	3,60,000
Inventory turnover ratio	10 times	36 times	1.33 times
Days of average inventory (360 ÷ ratio)	36 days	10 days	270 days



360 ÷ turnover ratio. Film's average stock lasts 270 days, more than seven times as long as flour's.

(ii) Comments

- **Potatoes** turn over 36 times (10 days): the fastest moving, as a perishable must be.
- **Flour** turns over 10 times (36 days): a regular, reasonably fast-moving material.
- **Packaging film** turns over only 1.33 times (270 days): slow moving, showing over-investment and working capital locked up. Its stock also rose by ₹1,20,000 during the year. Review it for obsolescence (a design change makes printed film useless) and cut the order size; see FSN analysis.

(iii) Film for 2026-27

Particulars	Amount (₹)
Consumption 2026-27: ₹4,80,000 × 125%	6,00,000
Required inventory turnover ratio: 360 ÷ 150 days	2.4 times
Highest average stock: ₹6,00,000 ÷ 2.4	2,50,000
Highest closing stock: (2 × ₹2,50,000) − opening ₹4,20,000	**80,000**
Purchases allowed: consumption ₹6,00,000 + closing ₹80,000 − opening ₹4,20,000	**2,60,000**

Even though film use rises by a quarter, purchases must fall from ₹6,00,000 to ₹2,60,000, because the year starts with ₹4,20,000 of film already in the store, about 252 days of the new year's use. Holding 150 days still leaves film far slower than flour; the excess has to be used up before fresh orders are placed.

Worked example

Flour for samosas: a faster turnover that is not good news

For 2025-26 Tapri's samosa factory made 18,00,000 samosas. The standard flour content is 40 g a samosa.

Flour issued to production was 75,600 kg. Flour stock was 6,000 kg on 1 April 2025 and 8,400 kg on 31 March 2026, and flour costs ₹40 a kg throughout.

Kesar is delighted: "Flour turned over 10 times last year. This year it is faster, so the store is more efficient."

Required:

(i) Calculate the input-output ratio for flour and state what it shows.

(ii) Calculate the inventory turnover ratio and the number of days for which the average stock is held (360 days in a year).

(iii) Calculate the flour purchased during the year, in kg and in rupees.

(iv) Comment on Kesar's statement.

THE WORKING

(i) Input-output ratio

Standard flour content of the actual output = 18,00,000 × 40 g = 72,000 kg.

$$\text{Input-output ratio} = \frac{75{,}600}{72{,}000} = 1.05$$

More flour went in than the standard allows: an adverse usage, 3,600 kg (₹1,44,000) above standard.

(ii) Inventory turnover

Particulars	kg
Flour consumed	75,600
Average stock: (6,000 + 8,400) ÷ 2	7,200
Inventory turnover ratio: 75,600 ÷ 7,200	10.5 times
Days of average stock: 360 ÷ 10.5	34.29 days

Both terms are in kg; at a constant ₹40 a kg the ratio in rupees is the same.

(iii) Purchases = consumption + closing stock − opening stock = 75,600 + 8,400 − 6,000 = **78,000 kg**, at ₹40 = **₹31,20,000**.

(iv) Comment. The ratio did rise, from 10 to 10.5 times, but not because the store improved. A high inventory turnover ratio shows fast-moving material; it does not show efficient usage, which is what the input-output ratio measures. Here consumption is inflated by 3,600 kg of excess usage. Had the line used only the standard 72,000 kg, the ratio on the same average stock would be 72,000 ÷ 7,200 = 10 times, exactly last year's. The faster turnover is wastage moving through the store, and the excess usage should be investigated at the samosa line, not praised in the store.

Worked example

Cooking oil: the count that came up 220 litres short

Tapri's samosa factory prices issues of cooking oil on the weighted average method. For October 2025:

Date	Transaction
Oct 1	Opening stock 2,000 litres at ₹150
Oct 5	Received 6,000 litres at ₹154
Oct 11	Issued 5,000 litres to the frying line
Oct 19	Received 5,000 litres at ₹157.80
Oct 26	Issued 4,000 litres to the frying line

The stores ledger shows 4,000 litres on 31 October; the physical count finds 3,780. Investigation shows:

1. Material requisition 318 of 28 October, for 150 litres issued to the snacks line, was never posted in the stores ledger.
2. 40 litres are the normal loss on decanting drums into tins.
3. A drum left uncapped in the rain spoiled 30 litres; no one reported it.

Required:

(i) Prepare the stores ledger for October, including the entries arising from the count.

(ii) Reconcile the book balance with the physical balance and state the treatment of each item.

(iii) Pass the entries in the cost ledger for the items found on the count.

THE WORKING

(i) Stores ledger, cooking oil (weighted average method)

On paper use ICAI's ten columns (date; receipts, issues and balance, each with quantity, rate and amount). Here each entry and its effect on the balance:

Date and entry	Litres	Rate (₹)	Amount (₹)	Balance: litres, ₹
Oct 1 opening	2,000	150.00	3,00,000	2,000; 3,00,000
Oct 5 receipt	6,000	154.00	9,24,000	8,000; 12,24,000 (₹153.00)
Oct 11 issue	5,000	153.00	7,65,000	3,000; 4,59,000
Oct 19 receipt	5,000	157.80	7,89,000	8,000; 12,48,000 (₹156.00)

Date and entry	Litres	Rate (₹)	Amount (₹)	Balance: litres, ₹
Oct 26 issue	4,000	156.00	6,24,000	4,000; 6,24,000
Oct 28 issue, MR 318 (posted on the count)	150	156.00	23,400	3,850; 6,00,600
Oct 31 abnormal shortage (issue)	30	156.00	4,680	3,820; 5,95,920
Oct 31 normal shortage (issue)	40	–	–	3,780; 5,95,920 (₹157.65)

Average rates: ₹12,24,000 ÷ 8,000 = ₹153.00; ₹12,48,000 ÷ 8,000 = ₹156.00. No receipt came after 19 October, so every later entry is at ₹156.00.

(ii) Reconciliation and treatment

Particulars	Litres
Balance as per stores ledger	4,000
Less: requisition 318 not posted (error in the records)	150
Less: abnormal shortage (uncapped drum)	30
Less: normal shortage (decanting)	40
Physical balance	**3,780**

- **Requisition 318, 150 litres**: not a shortage but an error in the records. The corrective entry posts the issue at ₹156.00, ₹23,400, and charges it to the snacks line, the job that used it (SM 2.45: explain the cause, make the corrective entry, remove the cause).
- **Abnormal shortage, 30 litres**: avoidable and unreported. Entered in the issue column at the weighted average rate, 30 × ₹156.00 = **₹4,680**, and transferred to Costing Profit and Loss Account.
- **Normal shortage, 40 litres**: absorbed by the good units. The quantity is written off and the value kept, so the 3,780 litres carry ₹5,95,920 at **₹157.65** a litre (₹5,95,920 ÷ 3,780).

(iii) Cost ledger

Entry	Dr (₹)	Cr (₹)
Work-in-Progress Control A/c (snacks line) Dr	23,400	
To Stores Ledger Control A/c		23,400
(Being requisition 318 omitted from the stores ledger, now posted)		
Costing Profit and Loss A/c Dr	4,680	
To Stores Ledger Control A/c		4,680
(Being abnormal shortage of 30 litres of oil written off)		

Convention: if the normal shortage is instead charged to production overheads, value it too: Factory Overhead Control A/c Dr ₹6,240 (40 × ₹156.00), To Stores Ledger Control A/c ₹6,240; the 3,780 litres then stay at ₹156.00 (₹5,89,680). Either way the normal loss ends up in product cost; say which you follow.

Exam angle

In the exam

Where these appeared

In the ICAI documents read for this lesson:

- **Inventory turnover, descriptive, 5 marks**: "calculate the quantity of each material purchased during the year and value of opening and closing stock", days held (360 days) and a comment, from average stock, the gap between opening and closing stock, and the turnover ratio (MTP May 2026 Series I Q3(a)). Old scheme: ratio, days and comments for two materials (December 2021 Q1(a)); ratio, days and "interpret the ratio ... if the industry inventory turnover rate is 10" (RTP December 2021 Q1).
- **Multiple choice, 2 marks each**: in one case scenario, the average price per unit, the days for which average inventory is held, and the "amount of abnormal loss during storage to be transferred to Costing Profit & Loss Account (based on average price)" (September 2024 MCQs 6, 9 and 10). The published key gives ₹5,100 and ₹2,04,000 without workings. ₹5,100 is what you get when the 80 units of normal wastage are absorbed by the good units: total cost of the opening stock and the year's purchases, ₹33,10,92,000, over 65,000 − 80 = 64,920 units; the 40 units of abnormal wastage at ₹5,100 are then ₹2,04,000. Spread over all 65,000 units the average would be ₹5,093.72, close to the ₹5,094 option.
- **Shortages in the cost ledger**: "Deficiency found in stock of Raw Material (Normal)", journalised as Factory overhead control A/c Dr, To Stores Ledger

Control A/c (January 2025 Q5(b), 4 marks); a deficiency "assumed as normal (alternatively can be treated as abnormal loss)" (MTP January 2025 Series I Q1(b)); "deficiencies found in stock taking" charged to the Costing Profit and Loss Account (RTP May 2026 Q11).

In the exam

The traps

- A deficiency with no word on its cause: ICAI's answers have gone both ways (SM 7.37 and January 2025: normal, to overheads; RTP May 2026: to Costing Profit and Loss). Write your assumption in one line.
- Consumption, not purchases, goes on top of the ratio: purchases ignore the change in stock.
- "Input-output ratio" also appears in process and joint-product questions as a yield, "100 : 90" (May 2026 Q3(a)): there it gives output from input, not a control ratio.

Trick

Ratio and days are two views of one number

Days × ratio = days in the year (365 or 360). A ratio of 8 is 45 days on 360; 73 days on 365 is a ratio of 5. Condition: the same year length on both sides.

Answer frame

"Explain inventory control by ratio analysis"

1. Input-output ratio: input of material ÷ standard material content of actual output; compares actual and standard consumption; shows favourable or adverse usage.
2. Inventory turnover ratio: cost of materials consumed ÷ cost of average stock; average stock = ½ (opening + closing).
3. Days of inventory holding = 365 days (or 12 months) ÷ the ratio.
4. Interpretation: high ratio, fast moving; low ratio, over-investment and locking up of working capital.
5. Use: compare materials (and with standards), reduce capital locked up and prevent over-stocking of slow-moving items.

"How are shortages found on physical verification treated?"

1. Valuation: entered in the issue column at the rate of the issue method adopted (FIFO or any other).
2. Normal shortage (breaking of bulk, evaporation, unavoidable): absorbed by good units, or charged to factory overheads.
3. Abnormal shortage (mishandling, pilferage, avoidable): transferred to Costing Profit and Loss Account.

4. Control: explain the causes promptly, make corrective entries, remove the causes.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the ratio built on purchases

Dividing purchases (₹6,00,000) by average stock gives the film a ratio of 1.67 instead of 1.33, and 216 days instead of 270. **Fix:** the numerator is materials consumed: opening + purchases − closing.

⚠ **Common mistake**

Closing stock in place of average stock

Using closing stock alone (₹4,20,000) as the denominator. **Fix:** average stock = ½ (opening + closing), unless the question gives the average.

⚠ **Common mistake**

The whole difference written off

Writing off the entire 220-litre gap between book and shelf to Costing Profit and Loss. **Fix:** first correct the records (the 150 litres of requisition 318 belong to the snacks line); of the real shortage only the abnormal 30 litres go to Costing Profit and Loss, while the normal 40 stay in product cost, absorbed by the good units or through overheads.

⚠ **Common mistake**

The shortage at the latest purchase price

Valuing the 30 spoiled litres at ₹157.80, the last invoice. **Fix:** a shortage is an issue: use the rate of the issue method, here the weighted average ₹156.00.

Practice

Check yourself

1. Masala tins at the Pune depot: consumption for the year was ₹9,60,000 and the opening stock ₹1,40,000. The average stock was held for 1.5 months. What was the closing stock, in rupees?

Answer: _____

2 Case scenario. At the year-end count of Tapri's Indore store, Assam tea is 80 kg short of the stores ledger: 50 kg lost by drying out, normal for the season, and 30 kg missing from a broken carton that was never reported. The weighted average rate is ₹300 a kg. Which of the following is correct?

A ₹24,000 to Costing Profit and Loss Account

B ₹9,000 to Costing Profit and Loss Account; ₹15,000 absorbed by the good units

C ₹15,000 to Costing Profit and Loss Account; ₹9,000 absorbed by the good units

D No entry: adjust the shortage when the next lot is received

3 A low inventory turnover ratio indicates:

A That the material is fast moving

B Over-investment and locking up of working capital in that material

C Efficient use of material against the standard

D A shortage on physical verification

4 The samosa line used 9,800 kg of flour for output whose standard flour content is 10,000 kg. What does the input-output ratio show?

- **A** 1.02: adverse usage
- **B** 0.98: favourable usage
- **C** 0.98: adverse usage
- **D** 1.02: favourable usage

5 Material found short on physical verification is valued at:

- **A** The invoice price of the latest purchase
- **B** The rate as per the method of pricing issues adopted (FIFO, weighted average or other)
- **C** The market price on the date of the count
- **D** The standard price, whatever the pricing method

Answers: the answer key at the end of this lesson.

Recap

- Materials consumed = opening stock + purchases − closing stock.

- Inventory turnover ratio = cost of materials consumed ÷ cost of average stock; average stock = ½ (opening + closing).
- Days of inventory holding = 365 days (or 12 months, or the 360 a question fixes) ÷ ratio.
- High ratio: fast moving. Low ratio: over-investment, working capital locked up.
- Input-output ratio = material input ÷ standard material content of actual output: above 1 adverse, below 1 favourable.
- Shortage on verification: an issue at the issue-method rate; normal (bulk breaking, evaporation, unavoidable) stays in product cost, abnormal (pilferage, mishandling, avoidable) to Costing Profit and Loss.
- Cost ledger: normal, Factory Overhead Control Dr; abnormal, Costing P&L Dr; Stores Ledger Control Cr. (Count the table turns, not the full tables.)

Answer key · Check yourself

1 **100000**

Inventory turnover ratio = 12 months ÷ 1.5 months = 8 times. Average stock = ₹9,60,000 ÷ 8 = ₹1,20,000 = (₹1,40,000 + closing) ÷ 2, so closing stock = ₹2,40,000 − ₹1,40,000 = ₹1,00,000.

2 **B** **₹9,000 to Costing Profit and Loss Account; ₹15,000 absorbed by the good units**

30 kg × ₹300 = ₹9,000 is abnormal (avoidable, unreported) and goes to Costing Profit and Loss. 50 kg × ₹300 = ₹15,000 is normal (evaporation-type loss) and stays in the cost of the good units. Writing off all ₹24,000 overstates the loss.

3 **B** **Over-investment and locking up of working capital in that material**

A low ratio means the average stock is used up slowly, so money sits in the store (SM 2.42). Fast moving is a high ratio; usage against standard is the input-output ratio.

4 **B** **0.98: favourable usage**

Input ÷ standard content of actual output = 9,800 ÷ 10,000 = 0.98: less material than standard, a favourable usage. Inverting the ratio gives 1.02.

5 **B** **The rate as per the method of pricing issues adopted (FIFO, weighted average or other)**

The Study Material: enter the shortage in the issue column at the rate as per the method adopted (SM 2.63).

3.8 Waste, scrap, spoilage, defectives and obsolescence

The five forms of material loss in ICAI's words, how each is treated when normal and when abnormal, and the spoiled and defective work rules ICAI keeps asking.

In plain words

Kesar's miscellaneous drawer

At the end of the festive rush Ritu walks the samosa factory with a clipboard. Flour dust lies under the sieves. A tub of dough trimmings waits for the cattle-feed dealer, who pays ₹10 a kg. A tray of samosas burnt black in a fryer that overheated. Two hundred samosas with torn corners, which a worker is re-folding with fresh dough. And in the godown, 1,50,000 printed pouches with the old logo, useless since the rebranding.

> **Kesar:** Burnt samosas, dust, old pouches. Put it all under miscellaneous expenses. Sab adjust ho jayega.
>
> **Ritu:** Some of it adjusts into the cost of the good samosas. The rest is a loss, and a loss does not hide inside a samosa's price.

Each pile is a different kind of loss. The dust is **waste**: material lost in storage or production and thrown away. The trimmings are **scrap**: discarded bits of the output with a small value. The burnt tray is **spoilage**: damaged

beyond economic repair. The torn samosas are **defectives**: below standard, but fixable with extra work. The pouches are **obsolescence**: material that lost its value because something replaced it.

Ritu sorts each pile by one question: was the loss **normal** (inherent in the operation, expected) or **abnormal** (avoidable, beyond the expected)? A normal loss is part of the cost of making good output, so the good samosas carry it, less whatever the loss sells for. An abnormal loss is kept out of product cost and goes to the **Costing Profit and Loss Account**. Obsolete material is always abnormal.

Think of it like

The tailor's floor

In a tailor's shop, the fluff and thread ends swept up each evening are waste; the cloth offcuts sold to a quilt-maker are scrap; the shirt cut wrong beyond saving is spoilage; the shirt with a crooked collar that is re-stitched is a defective; and last season's print nobody will buy is obsolete. A good tailor prices the expected offcuts into every shirt, but never the shirt a careless cutter ruined.

Key idea

Normal losses are inherent and are borne by the good units, net of any realisable value; abnormal losses are avoidable and are charged to the Costing Profit and Loss Account after crediting any realisable value. Obsolete material is a total loss, abnormal by definition, and never part of the cost of manufacture.

How it works

The five forms of loss

Loss of materials during handling, storage or process may occur as waste, scrap, spoilage, defectives or obsolescence (SM 2.64).

The five forms of material loss (SM 2.64–2.68)

Waste — value or none
lost in storage or production, and discarded

Scrap — nil or small value
output discarded without further treatment

Spoilage — disposed of
badly damaged, cannot be rectified economically

Defectives — rectified or sold lower
below quality standard: reworks or rejects

Obsolescence — always abnormal
loss in intrinsic value due to supersession

From input to output: waste is lost from the material, scrap comes off the output, spoilage and defectives are output that fails the standard, obsolescence is value lost while the material sits.

Definition

Waste

The portion of raw material which is lost during storage or production and discarded. The waste may or may not have any value. (SM 2.64)

Definition

Scrap

The materials that result from production which are discarded and disposed-off without further treatment. Generally, scrap has either no value or insignificant value; sometimes it may be reintroduced into the process as raw material. (SM 2.64)

Definition

Spoilage

Materials which are badly damaged in manufacturing operations, do not meet the specification required by the customers, cannot be rectified economically, and are taken out of the process to be disposed of in some manner without further processing. (SM 2.64)

Definition

Defectives

Those units or portions of production which do not meet the quality standards. Defectives that can be re-made to the standard with additional materials are **reworks** (repairs, reconditioning, refurbishing); those that cannot are **rejects**, which may be disposed of or recycled. (SM 2.65)

Defectives arise due to sub-standard materials, bad supervision, bad planning, poor workmanship,

inadequate equipment and careless inspection (SM 2.65).

Definition

Obsolescence

"The loss in the intrinsic value of an asset due to its supersession"; in simple words, the loss in value due to technological advancements. (SM 2.67)

The treatment, in ICAI's words

This table is the heart of every theory answer on the topic: learn it row by row.

Loss

Waste

Normal

Cost absorbed by good production units

Abnormal

Cost transferred to Costing Profit and Loss Account

Loss

Scrap

Normal

Cost borne by good units; income from its realisable value deducted from the cost

Abnormal

Scrap account charged with full cost and the job or process credited; profit or loss on realisation to Costing Profit and Loss Account

Loss

Spoilage

Normal

Included in costs: charged to the production order, or to production overhead so that it is spread over all products; any value realised credited to the order or the overhead account

Abnormal

Cost charged to Costing Profit and Loss Account

Loss

Defectives

Normal

Cost less realisable value charged to material cost of good production

Abnormal

Not included in material cost; treated as a loss after crediting realisable value, transferred to Costing Profit and Loss Account

Loss

Obsolescence

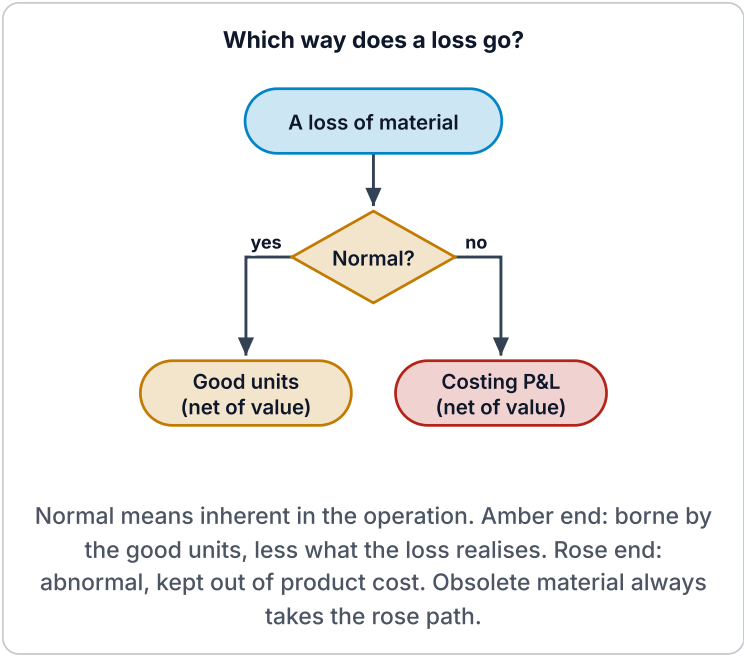
Normal

–

Abnormal

Always abnormal: a total loss; dispose of at the best available price; not part of the cost of manufacture

Sources: SM 2.64–2.68. One exception for spoilage: "When spoiled work is the result of rigid specification, the cost of spoiled work is absorbed by good production while the cost of disposal is charged to production overhead" (SM 2.65).



Normal means inherent in the operation. Amber end: borne by the good units, less what the loss realises. Rose end: abnormal, kept out of product cost. Obsolete material always takes the rose path.

Obsolescence: three ways material becomes useless

Materials may become obsolete (SM 2.67): (i) a spare part or component of a machine used in manufacturing that is now obsolete; (ii) material used for a product that has itself become obsolete; (iii) material replaced by another because of improved quality or a fall in price. In all three cases the value of the obsolete material held in stock is a total loss and immediate steps should be taken to dispose of it at the best available price; the loss is abnormal and does not form part of the cost of manufacture (SM 2.68).

Reclaiming what you can from defective units

For spoiled articles, ICAI's five steps to reclaim as much of the loss as possible (SM 2.66):

1. Send all defective units to a place fixed for the purpose.
2. Dismantle them.
3. Separate good and serviceable parts and take them back into stock.
4. Send parts that further work can make serviceable to the workshop, and take them into stock once the defects are removed.
5. Collect parts that cannot be made serviceable in one place, to be melted or sold off.

Printed forms should record the quantities at every stage (SM 2.66).

Spoiled and defective work in a job or batch

The Job Costing chapter treats the same losses by circumstance, and ICAI has asked this table three times since May 2024 (SM 9.6–9.7; job costing). **Spoiled work** is the quantity of production totally rejected that cannot be rectified; **defective work** is production not as perfect as

the saleable product but capable of being rectified, provided some additional expenditure is incurred.

Circumstance	Treatment (SM 9.7)
A percentage of defective work is allowed in a batch, as it cannot be avoided	Defectives within or near the normal limit: cost of rectification charged to the whole job and spread over the batch's output. Defectives substantially exceeding the normal: rectification of the excess written off as a loss in Costing Profit and Loss Account
Defect due to bad workmanship	Rectification is an abnormal cost, written off as a loss (unless recovered as a penalty from the workman); if management provided for a proportion of such defectives as unavoidable, rectification up to that extent is a normal cost charged to the batch
Defect due to the Inspection Department wrongly accepting incoming material of poor quality	Rectification charged to that department, not to the cost of manufacture of the batch; being abnormal, written off to Costing Profit and Loss Account

Where the defectives are inherent in the manufacturing process, the rectification cost may be charged to the specific jobs in which they arose; if they cannot be identified with jobs, it may be treated as factory

overheads. Abnormal defectives are written off to the Costing Profit and Loss Account (SM 9.10).

Formats and formulas

Reproduce the cost-per-good-unit formula and the spoilage entry from memory.

∑ **Formula**

Cost of a good unit when there is a normal loss

$$\text{Cost per good unit} = \frac{T - R}{I - N}$$

T = total cost; R = realisable value of the normal loss (scrap value less cost to sell); I = input units; N = normal loss units. An abnormal loss unit is valued at the same cost as a good unit (SM 10.7–10.9).

∑ **Formula**

Splitting a spoilage between normal and abnormal

$$\text{Abnormal} = \text{Actual} - \text{Normal spoilage}$$

Any realisable value of the spoiled work is apportioned between the two in the ratio of their quantities (SM 9.10).

Costing journal for spoilage in a job (SM 9.10):

Account	Dr	Cr
Material (Stores) Control A/c, with the realisable value	xxx	
Overhead Control A/c, with the normal spoilage less its share of realisable value	xxx	
Costing Profit and Loss A/c, with the abnormal spoilage less its share of realisable value	xxx	
To Work-in-Progress Control A/c, with the total cost of spoiled work		xxx

Worked examples

Worked example

Kulhads from the kiln: two jobs and a failed thermostat

Tapri's kulhad unit fires clay cups in batches for café orders. Some cracking in the kiln is inherent: the normal rate of spoilage is 4% of the kulhads fired. Cracked

kulhads are sold to a nursery as pot filler at ₹1 each. October's two jobs:

Job	Kulhads fired	Manufacturing cost (₹)	Cracked
K1	12,000	72,000	360
K2	15,000	90,000	1,350

The kiln's thermostat failed during Job K2.

Required:

(i) Separate the normal and abnormal spoilage of each job and record them in the costing journal, charging normal spoilage to production overhead.

(ii) Show how the entry for Job K2 changes if normal spoilage is charged to the job itself, and calculate K2's cost per good kulhad under each treatment.

THE WORKING

Working note: spoilage of each job (cost per kulhad fired = ₹72,000 ÷ 12,000 = ₹90,000 ÷ 15,000 = ₹6)

Particulars	K1	K2
Normal spoilage allowed (4% of kulhads fired)	480	600
Actual spoilage	360	1,350
Normal spoilage (kulhads)	360	600
Abnormal spoilage (kulhads)	Nil	750
Cost of normal spoilage @ ₹6 (₹)	2,160	3,600
Cost of abnormal spoilage @ ₹6 (₹)	Nil	4,500

K1's spoilage is within the normal rate, so all of it is normal. The realisable value is known per kulhad (₹1), so each part is credited with its own units: K1 ₹360; K2 normal ₹600 and abnormal ₹750.

(i) Costing journal (normal spoilage charged to production overhead)

Account	Dr (₹)	Cr (₹)
Material Control A/c (realisable value)	360	
Overhead Control A/c (₹2,160 – ₹360)	1,800	
To Work-in-Progress Control A/c (Job K1)		2,160

Account	Dr (₹)	Cr (₹)
(Being normal spoilage of Job K1)		
Material Control A/c (realisable value)	1,350	
Overhead Control A/c (₹3,600 – ₹600)	3,000	
Costing Profit and Loss A/c (₹4,500 – ₹750)	3,750	
To Work-in-Progress Control A/c (Job K2)		8,100
(Being spoilage of Job K2: normal to overheads, abnormal to Costing P&L)		

(ii) Normal spoilage charged to the job (Job K2)

Account	Dr (₹)	Cr (₹)
Material Control A/c (realisable value)	1,350	
Costing Profit and Loss A/c (₹4,500 − ₹750)	3,750	
To Work-in-Progress Control A/c (Job K2)		5,100
(Being abnormal spoilage written off and value realised from normal spoilage credited to the job)		

The cost of normal spoilage less its value realised, ₹3,000, now stays in Job K2 instead of going to Overhead Control. The abnormal entry is the same under both treatments: an abnormal loss never reaches a product.

Treatment

Normal spoilage to production overhead

Cost left in K2 (₹)

90,000 − 8,100 = 81,900

Good kulhads

13,650

Cost per good kulhad (₹)

6.00

Treatment

Normal spoilage to the job

Cost left in K2 (₹)

90,000 − 5,100 = 84,900

Good kulhads

13,650

Cost per good kulhad (₹)

6.22

Charging production overhead spreads the normal spoilage over all products through the overhead rate; charging the job puts it on the order that caused it (SM 2.65).

Worked example

Dough sheets: normal trimmings and an abnormal tub of scrap

The samosa factory puts 10,000 kg of refined flour, costing ₹48.40 a kg, into a month's production of dough sheets (consider material cost only). Trimmings are normally 4% of input and sell as scrap at ₹10 a kg. This month a mis-set cutter raised the trimmings to 700 kg, all sold at ₹10 a kg.

Required:

(i) Calculate the material cost per kg of good dough sheets.

(ii) Calculate the normal and abnormal scrap, and show the treatment of each, with the abnormal scrap

account.

THE WORKING

(i) Cost per kg of good output

Normal scrap = 4% × 10,000 = 400 kg, realisable value 400 × ₹10 = ₹4,000.

$$\text{Cost per kg} = \frac{4{,}84{,}000 - 4{,}000}{10{,}000 - 400} = \frac{4{,}80{,}000}{9{,}600} = ₹50$$

(ii) Normal and abnormal scrap

Actual scrap 700 kg − normal 400 kg = **300 kg abnormal**; good output = 10,000 − 700 = 9,300 kg.

- **Normal scrap (400 kg)**: its cost is borne by the good units; the ₹4,000 realised is deducted from the cost. It is why each good kg costs ₹50 and not ₹48.40.
- **Abnormal scrap (300 kg)**: the scrap account is charged with its full cost at the good-unit rate, 300 × ₹50 = ₹15,000, and production is credited; the loss on realisation goes to Costing Profit and Loss Account.

Abnormal Scrap Account

Dr.	kg	₹
To Production A/c	300	15,000
Total	**300**	**15,000**

Cr.	kg	₹
By Cash	300	3,000
By Costing P&L A/c (loss)		12,000
Total	**300**	**15,000**

Check: good output 9,300 × ₹50 = ₹4,65,000; + abnormal scrap ₹15,000 + normal scrap realised ₹4,000 = ₹4,84,000, the material cost. The same arithmetic, with labour and overheads added, is the process account of normal and abnormal process loss.

Worked example **Five piles, five treatments**

Advise Tapri's cost accountant, with reasons, on the treatment in cost accounts of each of the following at

the samosa factory:

(a) Flour lost as dust in sieving, about 1% of input every month.

(b) A hotel chain's order required every samosa to weigh 60 g, give or take 1 g; 3% of production failed the specification and was sold off cheaply, and disposing of it cost ₹2,000.

(c) The batch defect rate allowed for frying is 2%; this month 7% of a batch had to be re-fried, after a new worker set the wrong temperature.

(d) Rectification of a batch made from poor-quality potatoes that the Inspection Department had wrongly accepted.

(e) 1,50,000 printed pouches with the old logo, unusable after the rebranding.

MODEL ANSWER

(a) Normal waste. Waste is the portion of raw material lost during storage or production and discarded. A steady 1% loss in sieving is inherent in the operation, so it is normal waste: its cost is absorbed by the good production units.

(b) Spoilage from rigid specification. Spoilage is material badly damaged or not meeting the customer's specification that cannot be rectified economically. When spoiled work is the result of a rigid specification, the cost of the spoiled work is

absorbed by good production, while the cost of disposal (₹2,000) is charged to production overhead (SM 2.65).

(c) Defectives, partly normal. A normal rate of 2% has been established. The cost of rectifying defectives up to that limit is charged to the whole batch and spread over its output; the 5% substantially above the normal, caused by bad workmanship, is abnormal: the cost of rectifying it is written off as a loss in the Costing Profit and Loss Account (SM 9.7).

(d) Defect due to the Inspection Department. The cost of rectification is charged to the Inspection Department and is not considered part of the cost of manufacture of the batch; being abnormal, it is written off to the Costing Profit and Loss Account (SM 9.7).

(e) Obsolescence. The pouches are material used for a design that has been superseded. The value of obsolete material in stock is a total loss; the pouches should be disposed of immediately at the best available price, and the loss (cost less the amount realised) is an abnormal loss, charged to the Costing Profit and Loss Account. It does not form part of the cost of manufacture (SM 2.67–2.68).

Exam angle

In the exam

Where these appeared, May 2024 to September 2026

In the ICAI documents read for this lesson:

- **Spoiled and defective work, 4 marks**: define both and give the treatment in three circumstances (a percentage of defectives allowed; bad workmanship; the Inspection Department accepting poor material): January 2025 Q5(c) (1 + 3 marks); May 2026 Q6(c), the OR option; MTP September 2025 Series II Q6(c). Three times in the period, always with the same table.
- **Waste and scrap, 5 marks**: "Distinguish between Waste and Scrap. Discuss the treatment of normal and abnormal scrap in Cost Accounts" (May 2024 Q6(b)).
- **Defectives, 5 marks**: "EXPLAIN Defectives unit and its treatment. Also EXPLAIN the process of Reclamation of loss from defective units" (MTP September 2025 Series I Q6(b)).
- **Scrap against defectives**: wooden pieces damaged in cutting and sold as waste, against tables with rough edges corrected with extra labour (RTP May 2026 Q17(b)); the answer is ICAI's four-point table and both treatments.
- **Multiple choice, 2 marks**: the abnormal loss during storage, at the average price, to be transferred to

Costing Profit and Loss Account (September 2024 MCQ 10).

In the exam

How the suggested answers earn the marks

They reproduce the Study Material: the definition first, then "Normal –" and "Abnormal –" in ICAI's wording, and distinctions as a numbered two-column table (May 2024 Q6(b); RTP May 2026 Q17(b)). The defective-work answer is the three-row circumstance table, word for word (January 2025, May 2026).

Trick

"Inherent goes in, avoidable goes out"

Normal = inherent in the operation: cost stays in the good units, less what the loss sells for. Abnormal = avoidable or beyond the normal: cost goes to Costing Profit and Loss, after what it sells for. Conditions: two exceptions to remember, rigid-specification spoilage (absorbed by good production, disposal cost to production overhead) and obsolescence (always abnormal).

Answer frame

Distinguish between waste and scrap (SM 2.64, 2.66)

ICAI's suggested answer to May 2024 Q6(b) puts the two definitions first, then the Study Material's three points:

Waste	Scrap
1. The portion of raw material which is lost during storage or production and discarded	1. The output which is discarded and disposed of without further treatment
2. Connected with raw material or inputs to the production process	2. The loss connected with the output
3. May be visible or invisible	3. Generally identifiable and has physical substance
4. Generally has no recoverable value	4. Termed as by-products and has small recoverable value

Distinguish between scrap and defectives (SM 2.67)

Scrap	Defectives
1. Loss connected with the output	1. Connected with the output as well as the input
2. Not intended, but cannot be eliminated due to the nature of material or process	2. Not intended, but can be eliminated through a proper control system
3. Generally not used or rectified	3. Can be used after rectification
4. Insignificant recoverable value	4. Sold at a lower value than the good one

Spoilage against defectives (SM 2.67): spoilage cannot be repaired or reconditioned, while defectives can be rectified and transferred back to standard production or to the seconds; the problem in accounting for defective work is the cost of rectification or rework.

Three answers to "Discuss the treatment of normal and abnormal scrap"

Loses marks

"Normal scrap is absorbed; abnormal scrap goes to P&L."

No realisable value, no scrap account, no credit to the job.

Passes

"Normal: cost borne by good units, realisable value deducted. Abnormal: written off to Costing P&L."

Normal is right; abnormal skips the scrap account at full cost.

Full marks

The definition first. Normal: cost of scrap borne by good units, income from its realisable value deducted from the cost. Abnormal: scrap account charged with full cost, job or process credited, profit or loss on realisation to Costing Profit and Loss Account.

Common mistakes

⚠ **Common mistake**

Hall of Shame: Kesar's miscellaneous drawer

Lumping all losses into one "miscellaneous" figure in the cost sheet. **Fix:** split each loss into normal (stays in product cost) and abnormal (Costing Profit and Loss Account); an abnormal loss inside product cost overstates the cost of every good unit and hides the inefficiency.

⚠ **Common mistake**

Abnormal scrap valued at its sale price

Crediting production with only the ₹3,000 the abnormal trimmings fetched. **Fix:** the scrap account is charged with the **full cost** (300 kg × ₹50 = ₹15,000); production is credited with that, and only the ₹12,000 shortfall on realisation goes to Costing Profit and Loss.

⚠ **Common mistake**

Obsolete stock carried at cost "until it is used"

Fix: obsolete material is a total loss: dispose of it at the best available price and charge the loss to Costing Profit and Loss; it is never part of the cost of manufacture.

⚠ **Common mistake**

Every defect written off

Writing off all rectification costs as abnormal. **Fix:** rectification within the normal rate of defectives (or the proportion management provided for) is a normal cost charged to the batch; only the excess, or defects caused by the Inspection Department's error, are written off.

Practice

Check yourself

1. A batch of 2,000 samosas allows a normal rate of defectives of 2%. This month 150 samosas came out with torn corners and were re-folded at a rectification cost of ₹3 each. Defectives this far above the normal rate are treated as abnormal. How much rectification cost is written off to the Costing Profit and Loss Account, in rupees?

Answer: _____

2 Case scenario. After Tapri's rebranding, 1,50,000 pouches printed with the old logo, costing ₹1.20 each, are sold as waste paper for ₹15,000. Kesar wants to add their cost to the cost of the new pouches "since packaging is packaging". Which of the following is correct?

A Add ₹1,80,000 to the cost of the new pouches, spread over this year's production

B Charge the net loss of ₹1,65,000 to the Costing Profit and Loss Account as an abnormal loss

C Charge ₹1,65,000 to factory overhead as a normal loss of the packing department

D Keep the pouches in stock at cost until a use is found for them

3 Case scenario. Decanting bulk ghee into 1-litre tins normally loses 0.5% of the ghee, which clings to the drums. In March a worn valve on the decanting machine raised the loss to 1.2%. Which treatment is correct?

A Absorb the whole 1.2% in the cost of the good tins, since decanting losses are inherent

B Charge the whole 1.2% to Costing Profit and Loss Account, since the loss exceeded the normal rate

C Charge 0.5% to Costing Profit and Loss Account and absorb 0.7% in the good tins

D Absorb 0.5% in the cost of the good tins and charge the cost of the other 0.7% to Costing Profit and Loss Account

4 Which of these statements are true of defectives, as distinct from scrap? Select all that apply.

- **A** They can be used after rectification
- **B** The loss is connected with the output as well as the input
- **C** They generally have insignificant recoverable value
- **D** They cannot be eliminated because of the nature of the material or process

More than one answer may be right.

5 A customer's rigid weight specification makes 3% of a samosa order fail and be scrapped. How is it treated?

- **A** The whole cost of the spoiled work goes to Costing Profit and Loss Account
- **B** The cost of spoiled work is absorbed by good production; the cost of disposal is charged to production overhead
- **C** The cost of spoiled work is charged to production overhead; the cost of disposal to Costing Profit and Loss Account
- **D** The spoiled work is carried in stock at full cost

Answers: the answer key at the end of this lesson.

Recap

- Five forms: waste (lost, discarded), scrap (discarded output, small value), spoilage (beyond economic repair), defectives (below standard: reworks or rejects), obsolescence (value lost to supersession).
- Normal loss: borne by good units, net of realisable value. Abnormal loss: Costing Profit and Loss Account, after realisable value.
- Scrap: normal, realisable value deducted from cost; abnormal, scrap account at full cost, job credited, loss on realisation to Costing P&L.
- Spoilage: normal, charged to the production order or production overhead; abnormal, Costing P&L; rigid specification, absorbed by good production with disposal cost to production overhead.
- Defectives: normal, cost less realisable value to material cost of good production; abnormal, Costing P&L after crediting realisable value.
- Defective work by circumstance: allowed percentage, to the batch (excess written off); bad workmanship, written off; Inspection Department's error, charged to it and written off.
- Obsolescence: always abnormal, total loss, dispose at the best price. (The tailor prices in the offcuts, never the ruined shirt.)

Answer key · Check yourself

1 **330**

Normal defectives = 2% × 2,000 = 40; their rectification (40 × ₹3 = ₹120) is charged to the batch and spread over its output. The excess 110 samosas cost 110 × ₹3 = ₹330 to rectify, written off to Costing Profit and Loss (SM 9.7). Writing off all 150 (₹450) ignores the normal rate.

2 **B** **Charge the net loss of ₹1,65,000 to the Costing Profit and Loss Account as an abnormal loss**

Material that has become obsolete is a total loss; it is disposed of at the best price and the loss (₹1,80,000 – ₹15,000) is abnormal, not part of the cost of manufacture (SM 2.68).

3 **D** **Absorb 0.5% in the cost of the good tins and charge the cost of the other 0.7% to Costing Profit and Loss Account**

Only the loss inherent in the operation (0.5%) is normal and borne by the good units. The extra 0.7% arose from an avoidable cause, the worn valve, so it is abnormal and goes to Costing Profit and Loss. Exceeding the normal rate does not make the normal part abnormal.

4 **A They can be used after rectification; B The loss is connected with the output as well as the input**

Defectives can be rectified, are connected with input and output, can be eliminated through proper control, and sell below the good units. Insignificant value and "cannot be eliminated" describe scrap (SM 2.67).

5 **B The cost of spoiled work is absorbed by good production; the cost of disposal is charged to production overhead**

The Study Material's rule for spoilage caused by rigid specification (SM 2.65). Sending it to Costing P&L treats an expected consequence of the order as abnormal.

CHAPTER 4

Employee Cost and Direct Expenses

4.1 Attendance, payroll and the elements of wages

How a factory records who came and what they worked on, turns that into a payroll, and works out what one productive hour of an employee really costs.

In plain words

₹75 on the napkin, much more on the cost sheet

Tapri's samosa factory in Indore is pricing a big airline order, and Kesar has done the labour cost on the back of a napkin.

> **Kesar:** Ramesh gets ₹15,000 a month and is paid for 2,400 hours a year. ₹1,80,000 ÷ 2,400 is ₹75 an hour. Done.
>
> **Ritu:** Per productive hour, please. Add our share of his PF and ESI, his Diwali bonus and the canteen. Then take out his 20 days of paid leave and the time lost in tea breaks and machine set-ups. Ramesh costs Tapri nearly half as much again as your napkin says.
>
> **Kesar:** Put the difference in miscellaneous expenses.
>
> **Ritu:** Then every airline samosa is priced at a loss.

Ritu's figure is a fraction. On top is everything Tapri spends because Ramesh works there: wages, employer's contributions, bonus, benefits in kind. That total is his **employee cost**. Below are the hours he can actually work: hours paid for, less paid leave, less unavoidable idle time. The result is the rate at which his work is charged to the samosas.

Three records get there. **Time-keeping** records when Ramesh came and left: his attendance time. **Time booking** records what he did with it: 3½ hours on one batch, 4 on another, half an hour waiting for dough. The **payroll** turns attendance into pay; costing uses the time booked to charge each batch.

Think of it like

The iceberg under the payslip

A payslip is the tip of an iceberg: basic pay and dearness allowance. Below the water, just as real to the employer: PF and ESI contributions, bonus, paid holidays, the canteen, medical care. Costing measures the whole iceberg, then divides it only by the hours that do useful work.

Key idea

Time-keeping says how long an employee was at work; time booking says what that time was spent on. The cost of an employee hour for costing is the whole employee cost (monetary and non-monetary) divided by the effective hours, not the payslip divided by the hours paid.

How it works

Five departments and one worker

Employee cost covers wages and salary, allowances and incentives, overtime, the employer's contribution to provident fund and other welfare funds, and benefits such as leave with pay and subsidised food (SM 3.3). The part traceable to a cost object economically is **direct**; the rest, **indirect**, goes to overheads.

Control does not mean a small wage bill or low wages for each employee: ICAI's aim is to keep the wages per unit of output as low as possible, by paying in a way that encourages efficiency (SM 3.4). Five departments share the job (SM 3.4–3.5):

Department	Functions
Personnel	(i) On a requisition from a department, searches for the skills and qualification required; (ii) ensures recruits have them; (iii) arranges training for new recruits and workshops for existing employees; (iv) maintains personal and job records; (v) evaluates performance from time to time
Engineering and Work Study	(i) Prepares plans and specifications for each job; (ii) training and guidance; (iii) supervises production; (iv) time and motion studies; (v) job analysis; (vi) job evaluation
Time-keeping	(i) Attendance records (time-keeping); (ii) time spent on various jobs (time booking)
Payroll	(i) Prepares the payroll; (ii) disburses salary and wages
Cost Accounting	(i) Accumulates and classifies employee costs; (ii) analyses and allocates them to cost centres or cost objects

The gate and the job card

Time-keeping is the correct recording of employees' attendance time; even piece workers are recorded, for

discipline and an adequate rate of production (SM 3.6–3.7). Its **objectives** are: preparation of payrolls; calculating overtime; ascertaining and controlling employee cost; ascertaining idle time; disciplinary purposes; and overhead distribution (SM 3.7).

The **methods** are manual or mechanical/automated (SM 3.7–3.8):

Method	How it works	Note
Attendance register (manual)	Arrival and departure written in a register	Simple, cheap, for small firms; wrong and back-dated entries in collusion with the time-keeper
Metal disc / token (manual)	Numbered token recorded by the time-keeper	Almost obsolete; errors, proxy attendance
Punch card (automated)	Card inserted or waved at a reader	No time-keeper; minimises recording error and time manipulation

Method

Bio-metric (automated)

How it works

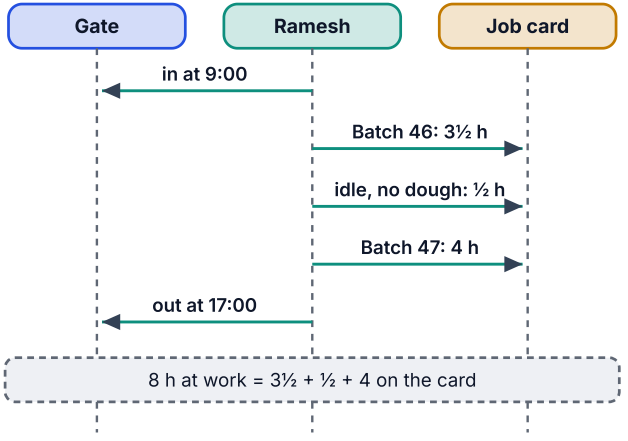
Fingerprint, face or retina matched to a database

Note

Reduces manipulation and proxy; costly for small firms

Time booking records each activity of the employee on a **time (or job) card**, for **costing** (the cost of each job), **measuring efficiency** (actual against standard time) and **fixing responsibility** (variances split into controllable and uncontrollable causes) (SM 3.9). Job time plus idle time must equal attendance time: the two records check each other (SM 3.10).

One day of Ramesh: the gate record and the job card



The gate record is time-keeping (attendance time); the job card is time booking (jobs and idle time). The two must agree.

From attendance sheet to bank transfer

The payroll procedure has five steps (SM 3.11–3.12):

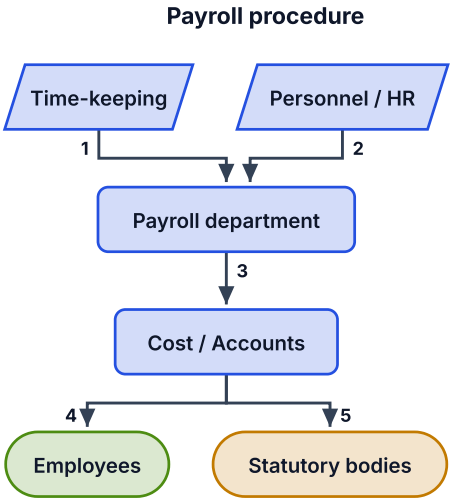
1. **Time and attendance details** from the time-keeping department: days or hours (time rate), units (piece rate); with time booking, payroll works out overtime and bonus.

2. **List of employees and other details** from personnel/HR: employees on roll and their rates, so that no unauthorised or bogus employee is paid.

3. **Computation of wages and other incentives** by payroll, which sends pay slips to the cost/accounting department.

4. **Payment to the employees** by cost/accounting, after statutory deductions (employee's PF and ESI, TDS on salary).

5. **Deposit of all statutory liabilities**: the deductions and the employer's contributions go to the statutory bodies.



ICAI's five steps (SM 3.11): 1 time and attendance, 2 employee details, 3 wage and salary sheet, 4 payment after deductions, 5 deposit of deductions and contributions.

What is generally deducted from the payroll (SM 3.12–3.13):

Statutory deductions

1. Provident fund: the employee's contribution.
2. Employee State Insurance (ESI): the employee's contribution.
3. Tax deducted at source (TDS), when the salary exceeds the maximum exemption limit.
4. Professional tax, a state-level tax on carrying on a business, profession or service.

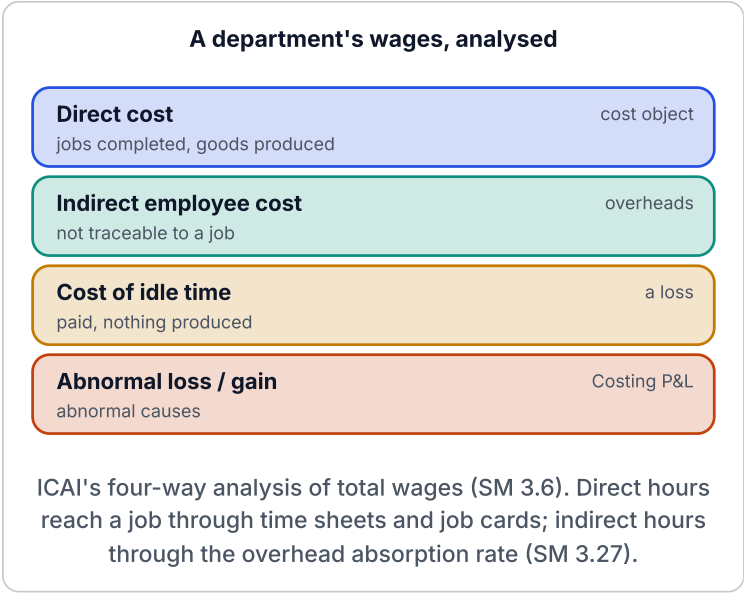
Other deductions

1. Voluntary contribution to provident fund: an employee who so desires may contribute over and above the contribution payable by the employer.
2. Contribution to any benevolent fund, on the employee's request.
3. Loan deductions: instalments of loans taken.
4. Other advances and dues: festival advance, unadjusted advances.

Where the wage bill goes

Cost accounting analyses each department's total wages into four parts (SM 3.6), by cost centre, job and work

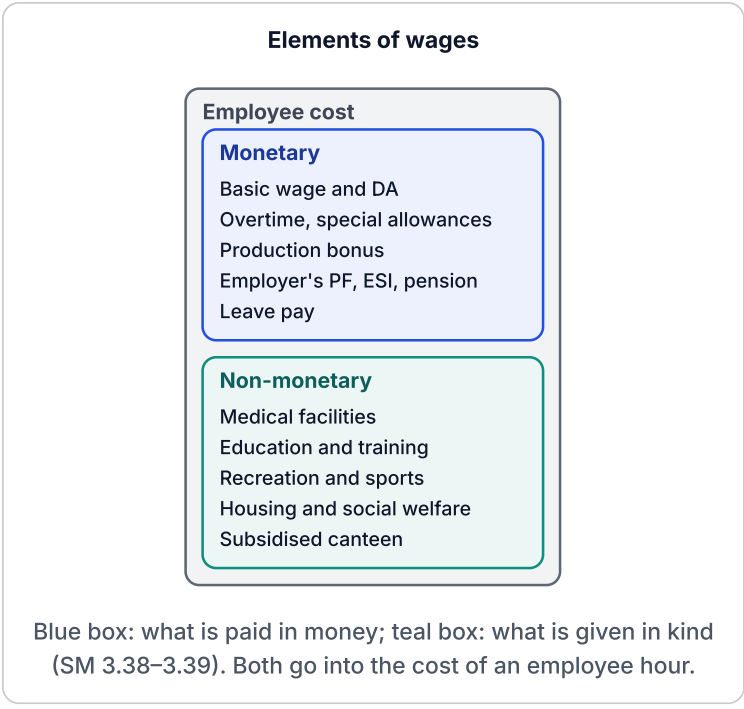
order, on a **wage analysis sheet** built from wage sheets, time cards, piece-work cards and job cards (SM 3.26):



ICAI's four-way analysis of total wages (SM 3.6). Direct hours reach a job through time sheets and job cards; indirect hours through the overhead absorption rate (SM 3.27).

The iceberg under the payslip

Strictly, wages include the **non-monetary benefits** received by virtue of employment: medical facilities; educational and training facilities; recreational and sports facilities; housing and social welfare; and the cost of subsidised canteen and co-operative societies. Their money value is part of the wage cost per worker (SM 3.38).



Blue box: what is paid in money; teal box: what is given in kind (SM 3.38–3.39). Both go into the cost of an employee hour.

In the monetary part (SM 3.39), the **basic wage** pays for work done, in hours attended or units produced. **Dearness allowance (DA)** covers the rise in the cost of living: a percentage of basic or a fixed amount, revised with the cost of living index as agreed with the union. **Overtime allowance** pays extra hours at the rates of the Factories Act; special allowances (working conditions, tool maintenance) are wages too. **Production bonus** rewards output above standard.

> ⚠ **Careful**
>
> The Study Material states that under the Payment of Bonus Act a worker is entitled to a compulsory bonus of 8.33% of wages earned in the year or ₹100, whichever is greater, and up to 20% of wages depending on profits (SM 3.39). Verify against the Study Material and Statutory Update for your attempt.

An hour for costing is not an hour for paying

For **indirect workers**, all wages and benefits are overheads. For **direct workers** there are two choices (SM 3.40, 3.42):

1. Treat the payments other than basic wages and DA, and the non-monetary benefits, as overheads; or
2. Build them into the rate charged to jobs: total wages and additional payments ÷ **effective hours**. ICAI calls this the ideal method; the rate is for costing only, not for paying the worker.

Holiday and leave wages are either departmental overheads (kept apart from "worked for wages" in the payroll) or built into the inflated rate, which is possible only for direct workers (SM 3.41). A **night shift allowance** is overheads: it is not for any particular job (SM 3.42). Idle time and overtime premium have rules of their own: idle time and overtime.

Formats and formulas

Definition

Employee (labour) cost

Benefits paid or payable to the employees of an entity, whether permanent or temporary, for the services rendered by them. Employee cost includes payments made in cash or kind. (SM 3.3)

Definition

Direct and indirect employee cost

Direct: benefits paid or payable to the employees which can be attributed to a cost object in an economically feasible manner. Indirect: benefits paid or payable to the employees which cannot be directly attributable to a particular cost object in an economically feasible manner. (SM 3.3)

Definition

Time-keeping and time booking

Time-keeping refers to recording and keeping of the employees' attendance time. Time booking is recording the details of work done and the time spent by an employee on each job or process. (SM 3.55)

∑ **Formula**

Rate per hour for costing (direct workers)

$$\text{Rate per hour} = \frac{\text{All benefits}}{\text{Effective hours}}$$

ICAI: total estimated monetary benefits and cost of non-monetary benefits ÷ (budgeted direct employee hours – normal idle time) (SM 3.42). Leave on full pay also comes off the hours.

ICAI's layout for the effective hourly cost (SM 3.15, Illustration 1). *Reproduce it from memory.*

Statement showing computation of effective hourly cost	Per month (₹)	Per annum (₹)
(A) Earnings of the employee:		
Basic pay	xxx	xxx
Dearness allowance	xxx	xxx
Bonus	xxx	xxx
Employer's contribution to provident fund	xxx	xxx
Other allowances and benefits	xxx	xxx
Total	**xxx**	**xxx**
(B) Effective working hours (working note)		xxx hours
(C) Effective hourly cost (A ÷ B)		**₹ xx**

When the question asks for both the pay and the cost, ICAI keeps two statements (SM 3.19–3.20, Illustration 3):

Statement of earnings	Statement of employee cost
Basic wages	Gross wages (excluding overtime)
Add: Dearness allowance, overtime, allowances	Add: Employer's contribution to PF
Gross wages earned	Add: Employer's contribution to ESI
Less: Employee's contribution to PF	**Employee cost**
Less: Employee's contribution to ESI	÷ Normal working hours
Net wages earned (what is paid)	**Ordinary wage rate per hour**

The employee's own contributions are inside his gross wages, merely withheld; only the employer's are added to cost. The cost statement leaves overtime out because ICAI allocates the overtime wages separately, to the job on which the overtime was worked (Job Y in Illustration 3); the ordinary wages are shared over the jobs in the ratio of time booked.

Worked examples

Worked example **The real price of Ramesh's hour**

Ramesh Yadav is a fryer operator at Tapri's Indore samosa factory, one of its 250 workers. His terms are:

Particulars	Amount
Basic pay	₹12,000 p.m.
Dearness allowance	25% of basic pay
Diwali bonus	one month's basic pay and DA
Employer's contribution to PF	12% of basic pay and DA
Employer's contribution to ESI	3% of basic pay and DA
Canteen subsidy and medical scheme	₹22,50,000 a year for all 250 workers
Working days	300 a year of 8 hours, including 20 days of leave on full pay
Normal idle time (set-ups, oil changes, tea breaks)	140 hours a year
Deducted from his pay	his own PF contribution of 12% of basic pay and DA, and professional tax of ₹200 a month

Required:

(i) Prepare a statement showing the effective hourly cost of Ramesh, per month and per annum.

(ii) In October, Ramesh's job card shows 150 hours on Batch 47, an order of frozen samosas for an airline. Calculate the employee cost to be charged to Batch 47.

(iii) Calculate Ramesh's monthly net pay, and state why it is not his employee cost.

THE WORKING

Working notes

1. DA = 25% × ₹12,000 = ₹3,000 a month; basic pay and DA = ₹15,000 a month.
2. Employer's PF = 12% × ₹15,000 = ₹1,800 a month; employer's ESI = 3% × ₹15,000 = ₹450 a month.
3. Bonus = ₹15,000 a year = ₹1,250 a month. Canteen and medical = ₹22,50,000 ÷ 250 = ₹9,000 a year = ₹750 a month.
4. Effective working hours:

Particulars	Hours
Working days (300 × 8 hours)	2,400
Less: Leave on full pay (20 × 8 hours)	160
Hours available	2,240
Less: Normal idle time	140
Effective working hours	**2,100**

(i) Statement showing computation of effective hourly cost of Ramesh

Particulars	Per month (₹)	Per annum (₹)
Basic pay	12,000	1,44,000
Dearness allowance	3,000	36,000
Bonus	1,250	15,000
Employer's contribution to PF	1,800	21,600
Employer's contribution to ESI	450	5,400
Canteen subsidy and medical scheme	750	9,000
(A) Employee cost	**19,250**	**2,31,000**
(B) Effective working hours (WN 4)		2,100 hours
(C) Effective hourly cost (A ÷ B)		**₹110**

(ii) Batch 47 = 150 hours × ₹110 = **₹16,500**.

(iii) Net pay = basic pay and DA ₹15,000 − his PF ₹1,800 − professional tax ₹200 = **₹13,000** a month. His own PF and the professional tax are part of his gross wages, only withheld and passed on, so they are not added again. Employee cost is what Tapri bears: gross wages, employer's contributions, bonus and benefits in kind.

> The napkin method (₹1,80,000 ÷ 2,400 hours) gives ₹75: ₹35 short on every hour charged to the airline's batch.

Worked example **November's wage analysis in the blending section**

The blending section of Tapri's premix plant has six blenders (direct workers) and one supervisor. For November:

1. Each blender earns basic pay of ₹16,000 and dearness allowance of 25% of basic pay for 200 normal hours. The employer contributes 12% to PF and 3% to ESI, both on basic pay.
2. Two blenders worked 50 overtime hours in all, paid at double the rate of basic pay and DA: 30 hours on Job J-3, at the request of a hotel chain that agreed to bear the extra cost, and 20 hours on Job J-1, for general production. No contributions are payable on overtime.
3. Three blenders on the night shift received a night shift allowance of ₹4,500 in all.
4. The supervisor's employee cost, contributions included, was ₹36,000.
5. The blenders' job cards show (including the overtime hours): J-1 600 hours, J-2 350 hours, J-3 300 hours.

Required:

(i) Calculate the employee cost of a blender for the month and the ordinary rate per hour for costing.

(ii) Prepare the analysis of November's wages: the amount charged to each job and to the section's overheads, giving the reason for each treatment.

THE WORKING

(i) Employee cost of one blender

Particulars	Amount (₹)
Basic pay	16,000
Dearness allowance (25% of basic)	4,000
Employer's contribution to PF (12% of basic)	1,920
Employer's contribution to ESI (3% of basic)	480
Employee cost for 200 normal hours	**22,400**
Ordinary rate per hour (÷ 200)	**112**

The blenders' own PF and ESI contributions are deducted from their pay; they are not added to the cost.

Working note: overtime. Rate of basic pay and DA = ₹20,000 ÷ 200 = ₹100 an hour; overtime is paid

at ₹200 an hour: ₹100 for the hour and a premium of ₹100. Normal hours on the jobs: J-1 600 − 20 = 580; J-2 350; J-3 300 − 30 = 270; total 1,200 = 6 × 200.

(ii) Analysis of November's wages

Item	Normal hours at ₹112 (580; 350; 270)
J-1 (₹)	64,960
J-2 (₹)	39,200
J-3 (₹)	30,240
Section overheads (₹)	–

Item	Overtime hours at ₹100 (20; –; 30)
J-1 (₹)	2,000
J-2 (₹)	–
J-3 (₹)	3,000
Section overheads (₹)	–

Item

Premium, customer's request: 30 × ₹100

J-1 (₹)

–

J-2 (₹)

–

J-3 (₹)

3,000

Section overheads (₹)

–

Item

Premium, general production: 20 × ₹100

J-1 (₹)

–

J-2 (₹)

–

J-3 (₹)

–

Section overheads (₹)

2,000

Item

Night shift allowance

J-1 (₹)

–

J-2 (₹)

–

J-3 (₹)

–

Section overheads (₹)

4,500

Item
Supervisor (indirect worker)
J-1 (₹)
–
J-2 (₹)
–
J-3 (₹)
–
Section overheads (₹)
36,000

Item
Total
J-1 (₹)
66,960
J-2 (₹)
39,200
J-3 (₹)
36,240
Section overheads (₹)
42,500

Check: ₹66,960 + ₹39,200 + ₹36,240 + ₹42,500 = ₹1,84,900 = six blenders' ₹1,34,400 + overtime ₹10,000 + night shift allowance ₹4,500 + supervisor ₹36,000.

Reasons. The blenders' hours booked on each job are direct employee cost of that job, at the ordinary rate that includes the employer's contributions. The overtime premium worked at the customer's request is charged to that job; overtime premium for general production is an overhead of the section (SM 3.18). The night shift allowance is not for any

particular job, so it is an overhead (SM 3.42), and the pay of an indirect worker such as the supervisor is wholly overheads (SM 3.40).

Worked example **The 6,500 hours no job received**

Tapri's chai premix plant at Pithampur records attendance on punch cards. In September its 250 direct workers were paid for 52,000 hours, but their job cards account for only 45,500 hours, so the job costs came out lower than the payroll. The workers fill one job card per job, and time not spent on a job is not recorded anywhere. Ritu's sample of the missing 6,500 hours shows:

Use of the time	Hours
Tea breaks and machine changeovers	3,250
Waiting for blend that the stores issued late	1,300
Cleaning and minor maintenance of the machines	1,950

The cost of a direct worker's hour is ₹110. Kesar wants to know what went wrong and what should change. **Advise.**

MODEL ANSWER

Time keeping and time booking. Time keeping records the total time an employee spends in the factory; time booking breaks that time up over the jobs on which it was spent (SM 3.6). Time booking is used for costing, for measuring efficiency against standard time and for fixing responsibility for variances (SM 3.9). The total of each employee's job time and idle time must equal his attendance time (SM 3.10). At Pithampur the punch cards keep time correctly, but the booking misses 6,500 hours (12.5% of the hours paid), costing ₹7,15,000 (6,500 × ₹110).

What the missing hours are, and their treatment

Use of the time	Hours	Cost (₹)	Treatment
Tea breaks and changeovers	3,250	3,57,500	Normal idle time: part of the cost of production, recovered through the rate charged to jobs

Use of the time

Waiting for blend issued late

Hours

1,300

Cost (₹)

1,43,000

Treatment

Abnormal idle time: Costing Profit and Loss Account, shown as controllable against the stores department

Use of the time

Cleaning and minor maintenance

Hours

1,950

Cost (₹)

2,14,500

Treatment

Indirect work: production overheads

Use of the time

Total

Hours

6,500

Cost (₹)

7,15,000

Treatment

Because none of this was booked, ₹7,15,000 of the payroll reached neither the jobs nor the overheads, and the job costs and the overhead rates were both understated.

What should change.

1. Each worker should book every hour of attendance: job time on the job, idle time with its cause, and indirect work against the overhead account.
2. With many short jobs, a time card for each employee for each day or week, showing all his jobs and idle time, makes the reconciliation with attendance time easy; one card per job makes the job's cost easy to compute but scatters each worker's time over many cards (SM 3.10).
3. The cost accounting department should reconcile the hours booked with the hours paid every week, and analyse the wages into direct cost, overheads and idle time (SM 3.6), before the job costs are used for pricing.

Exam angle

In the exam

Where these topics appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, the topic came up in 9 questions.

- **Theory, 4 marks.** Objectives of time keeping and time booking: MTP September 2024 Series II Q6(c). Factors for controlling employee costs: September 2025 Q6(c). Proxy attendance and payroll errors, "discuss the features of the Bio-Metric Attendance System": May 2026 Q6(c).

- **Numerical, 4–5 marks.** Effective hourly cost: MTP September 2025 Series I Q3(c). Labour cost per man-day, DA on a cost of living index: MTP January 2026 Series I Q1(a). Cost per hour, then abnormal idle time and overtime premium: MTP September 2026 Series I Q1(a). Earnings per day, effective rate and wages charged to a job: RTP January 2026 Q9.
- **Inside other questions.** Production bonus and leave wages in a machine hour rate: RTP September 2024 Q9. Employer's EPF and ESI in direct employee cost of a cost sheet: RTP September 2026 Q8.

In the exam

The traps ICAI set

- Leave on full pay and normal idle time come off the hours, not the cost; abnormal idle time stays in the hours and its cost is charged separately (MTP September 2026 Series I Q1(a)).
- Read the base of each percentage: "10% of (i), (ii) and (iii)" puts PF on basic, DA and leave salary together (MTP January 2026 Series I Q1(a)).
- A welfare cost for all employees is divided per head first.

Trick

Cost on top, effective hours below

Rate for costing = everything the employer bears ÷ hours that can be worked. What is withheld *from* the worker's wages (his PF, ESI, TDS) is already in the top line; paid time not worked (leave, normal idle time) comes off the bottom line. Condition: the inflated-rate method for direct workers; under the overheads method only basic and DA go on top.

Answer frame

The theory questions ICAI asks

Objectives of time-keeping and time booking (SM 3.7, 3.9): the six objectives of time-keeping as listed above, then the three purposes of time booking (costing, efficiency, responsibility), one line each.

Distinguish between time-keeping and time booking

Point	Time-keeping	Time booking
Records	Attendance time: arrival and departure	Time spent on each job, operation or process
Shows	Total time spent in the factory	Break-up of that time over jobs and idle time
Used for	Payroll, overtime, discipline, idle time	Job cost, efficiency, fixing responsibility
Record	Attendance register, token, punch card, bio-metric device	Time (job) card

Distinguish between direct and indirect employee cost (SM 3.3–3.4)

Direct employee cost	Indirect employee cost
Paid to employees directly engaged in the production process	Paid to employees not directly engaged in the production process
Easily identified and allocated to the cost unit	Apportioned on some appropriate basis
Varies with the volume of production	May not vary with the volume of production

Factors for controlling employee cost (SM 3.5–3.6), in ICAI's order: (i) assessment of manpower requirements; (ii) control over time-keeping and time-booking; (iii) time

and motion study; (iv) control over idle time and overtime; (v) control over employee turnover; (vi) wage and incentive systems; (vii) job evaluation and merit rating; (viii) employee productivity. Four are lessons of their own: idle time and overtime, turnover, incentive systems, labour efficiency.

A factory with proxy attendance: features of the bio-metric attendance system (May 2026 Q6(c)). Write the features in ICAI's words, then apply them to the facts:

1. Attendance is marked by recognising an employee on the basis of physical and behavioural traits.
2. Each employee's unique identity (fingerprint, face, retina image) is kept in a database and matched when attendance is marked.
3. It includes fingerprint recognition, face recognition and time and attendance tracking technology.
4. It reduces the risk of time manipulation and proxy attendance.
5. Its set-up and maintenance cost may make it unsuitable for small organisations (SM 3.8).
6. Apply it: a large factory with proxy attendance justifies the cost; payroll should also pay only the employees on roll listed by HR (SM 3.11).

The numerical: three answers to the same question

Loses marks

"₹2,31,000 ÷ 2,400 hours = ₹96.25 an hour."

Leave and idle time left in the hours; no working notes.

Passes

"Rate = ₹2,31,000 ÷ 2,100 hours = ₹110."

Right, but the ₹2,31,000 and 2,100 hours come from nowhere: no method marks if one is wrong.

Full marks

ICAI's statement, each element per month and per annum, a working note for effective hours, the rate, and any assumption stated.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the PF counted twice

Adding the employee's own PF (deducted from his pay) as well as the employer's. For Ramesh that adds ₹21,600 a year and turns ₹110 an hour into ₹120.29. **Fix:** the employee's contribution is already inside his gross wages; add only the employer's contribution.

⚠ **Common mistake**

The paid holiday that worked a shift

Dividing the annual cost by all 2,400 paid hours. Ramesh is paid for 20 days of leave but works none of them, so the rate comes out at ₹96.25 instead of ₹110 and every job is under-costed. **Fix:** leave on full pay and normal idle time come out of the hours; their cost stays in the numerator, spread over the hours actually worked.

⚠ **Common mistake**

The night shift that billed the night's job

Charging the blenders' ₹4,500 night shift allowance to the job that ran at night. **Fix:** it is not for any particular job, so it is overheads (SM 3.42).

Practice

Check yourself

1 A Tapri café supervisor costs ₹2,85,600 a year in wages, employer's contributions and benefits. She is paid for 290 days of 8 hours, of which 18 days are leave on full pay, and normal idle time is 136 hours a year. What is her employee cost per hour for costing, in rupees?

Answer: _____

2 Case scenario. Tapri's Bhopal unit, 40 workers, keeps an attendance register. Ritu finds back-dated entries made with the time-keeper's help and wants a method that needs no time-keeper and suits a small budget. Which method fits best?

- **A** Metal disc or token method
- **B** Attendance register with a second time-keeper
- **C** Punch card attendance
- **D** Bio-metric attendance system

3 Which of these is an "other" (non-statutory) deduction from the payroll?

- **A** Employee's contribution to ESI
- **B** Professional tax
- **C** Contribution to a benevolent fund
- **D** Tax deducted at source on salary

4 A worker's basic pay and DA are ₹20,000 a month. He contributes 12% to PF and the employer contributes 12%. What is the monthly employee cost to the employer, counting only these items, in rupees?

Answer: _____

5 Which of these does ICAI treat as overheads rather than as part of a job's direct employee cost? Select all that apply.

- **A** Night shift allowance paid to a direct worker
- **B** All wages and benefits of indirect workers
- **C** Basic wages of a direct worker for hours booked on the job
- **D** Holiday and leave wages, when the firm does not inflate the wage rate

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Employee cost: benefits paid or payable to employees, permanent or temporary, in cash or kind; direct if traceable to a cost object economically, else indirect (overheads).
- Aim of control: low wages per unit of output, not a low wage bill. Eight factors, from manpower assessment to employee productivity.
- Time-keeping = attendance time (six objectives); time booking = time on each job (costing, efficiency, responsibility). Job time + idle time = attendance time.

- Payroll in five steps: time and attendance, employee details, wage sheet, payment after deductions, deposit with statutory bodies. HR's list stops bogus employees.
- Wages include non-monetary benefits: medical, education and training, recreation, housing and welfare, subsidised canteen.
- Rate for costing = (monetary + non-monetary benefits) ÷ (budgeted hours − normal idle time); leave days come off the hours too.
- Add the employer's PF and ESI; never the employee's. Night shift allowance: overheads. (Measure the iceberg, not the tip.)

Answer key · Check yourself

1 **140**

Effective hours = 2,320 − 144 − 136 = 2,040; rate = ₹2,85,600 ÷ 2,040 = ₹140. Dividing by all 2,320 paid hours gives ₹123.10.

2 **C** **Punch card attendance**

A punch card needs no time-keeper and minimises recording error and time manipulation (SM 3.8). Tokens are almost obsolete and allow proxies; a second time-keeper keeps the collusion risk; bio-metric costs may not suit a small unit.

3 **C Contribution to a benevolent fund**

ICAI lists PF, ESI, TDS and professional tax as statutory deductions; voluntary PF, benevolent fund, loan instalments and other advances are other deductions (SM 3.12–3.13).

4 **22400**

₹20,000 + employer's 12% (₹2,400) = ₹22,400. The worker's own ₹2,400 is withheld from the ₹20,000, not added to it; ₹24,800 counts it twice.

5 **A Night shift allowance paid to a direct worker**; **B All wages and benefits of indirect workers**; **D Holiday and leave wages, when the firm does not inflate the wage rate**

Night shift allowance is not for a particular job (SM 3.42); indirect workers' pay is all overheads (SM 3.40); holiday and leave wages are overheads unless built into an inflated rate (SM 3.41). Booked basic wages are direct.

4.2 Idle time and overtime

Paid hours that produce nothing and extra hours that cost extra: what causes each, where ICAI charges it, and how to compute the charge to a job.

In plain words

The mixer that broke and the order that could not wait

Diwali week at Tapri's samosa factory. On Monday the dough mixer breaks down and twelve fryer operators stand around for two hours, on full pay. On Thursday an airline wants 50,000 extra samosas by Saturday, so the same operators stay till midnight and are paid double for the extra hours.

> **Kesar:** Charge the two lost hours to the Diwali batch. Sab adjust ho jayega.
>
> **Ritu:** The batch did not break the mixer. A breakdown is abnormal: its cost goes to Costing Profit and Loss, not into the samosas. The tea breaks and machine set-ups are normal: those stay in the cost.
>
> **Kesar:** And the double pay on Thursday?
>
> **Ritu:** The airline asked for the rush, so the extra half goes to the airline's job.

Ritu has just used the two rules of this lesson. **Idle time** is time paid for in which nothing is produced. If it cannot

be avoided in the normal course of business (walking from the gate, set-ups, lunch), it is **normal idle time** and part of the cost of production. If it comes from something abnormal (a breakdown, a power cut, no material, a strike), it is **abnormal idle time** and charged to Costing Profit and Loss.

Overtime is work beyond normal hours, paid at a higher rate. The extra over the normal rate, the **overtime premium**, is charged according to *why* the overtime was worked: to the job if the customer wanted it, to overheads if general production needed it, to the department at fault if another department caused it, and to Costing Profit and Loss if a flood or earthquake did.

Think of it like

The wedding caterer

A caterer's staff are paid while they wait for the baraat to arrive: every wedding has that wait, so it is built into the price per plate (normal idle time). If the generator fails for three hours, that is not the guests' doing, and the caterer swallows the loss (abnormal idle time). If the bride's father asks for dinner to run past midnight, he pays the late-night surcharge (overtime premium charged to the job).

Key idea

Normal idle time and the normal-rate part of overtime are costs of production; abnormal idle time is a loss for Costing Profit and Loss. The overtime premium follows the reason the overtime was worked.

How it works

Paid, but producing nothing

Idle time is the time during which no production is carried out because the worker remains idle but is paid: the difference between the time paid and the time booked. The time paid includes holidays, paid leave and allowable rest (SM 3.13). Its two kinds, with ICAI's causes and treatment (SM 3.13–3.14):

Meaning

Normal idle time
Cannot be avoided or reduced in the normal course of business

Abnormal idle time
Arises from abnormal factors

Causes

Normal idle time

Time lost between factory gate and place of work; interval between one job and another; setting-up time for the machine; normal rest time, lunch break

Abnormal idle time

Lack of coordination; power failure, breakdown of machines; non-availability of raw materials, strikes, lockouts, poor supervision, fire, flood

Treatment

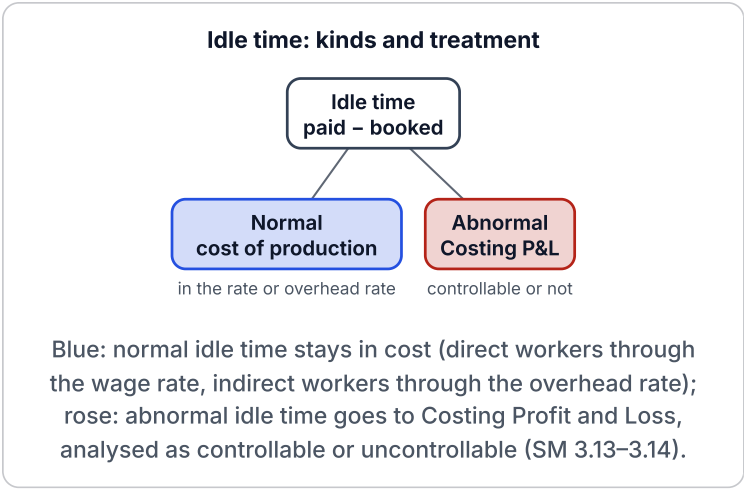
Normal idle time

Part of the cost of production: for direct workers an allowance in the standard hours or standard rate; for indirect workers, in the overhead rate

Abnormal idle time

Not part of production cost: a separate item in the Costing Profit and Loss Account

Abnormal idle time is further split into **controllable** (time that could have been put to productive use had management been more alert and efficient) and **uncontrollable** (lost to causes management cannot control, such as a breakdown of machines or a flood). The cost of each category is shown by cause, so that responsibility can be fixed. Management should aim to eliminate controllable idle time and, in the long run, to reduce even normal idle time (SM 3.14).



Blue: normal idle time stays in cost (direct workers through the wage rate, indirect workers through the overhead rate); rose: abnormal idle time goes to Costing Profit and Loss, analysed as controllable or uncontrollable (SM 3.13–3.14).

How the allowance for normal idle time gets into the rate: divide the wages by the hours that can be worked (paid hours less normal breaks), not by all the paid hours. Every job then carries a little of the normal idle time, and abnormal idle hours are charged to Costing Profit and Loss at the same rate (SM 3.16, Illustration 2). Building the full hourly rate is in attendance, payroll and the elements of wages.

Overtime: the extra hours and the extra half

Overtime is work done beyond normal working hours. The **overtime payment** has two elements: wages for the overtime hours at the normal rate, and the **premium** (extra) payment. The overtime rate is usually double the normal rate; the extra over the normal rate is the **overtime premium** (SM 3.17).

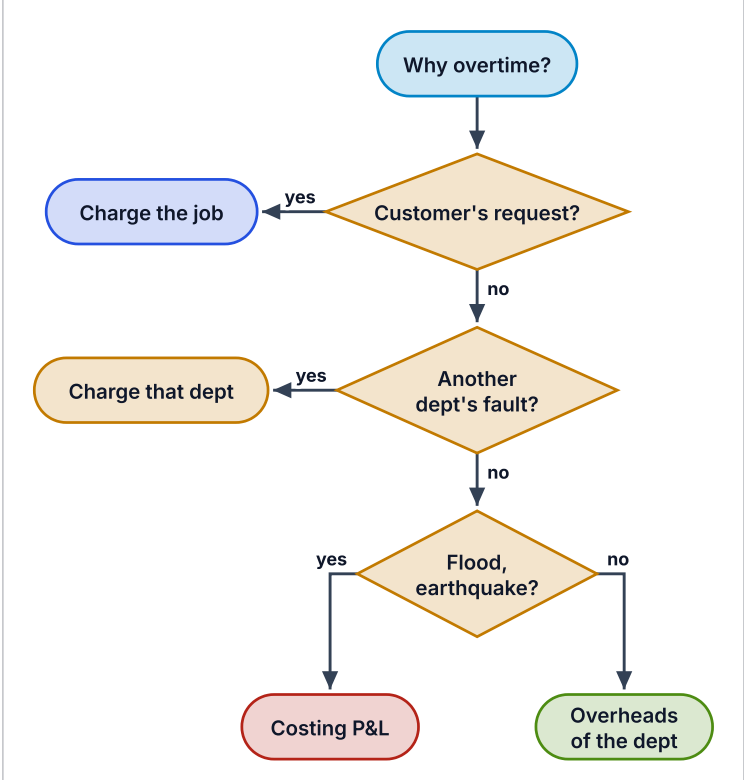
The rate may be fixed by the entity or by statute, but the premium may not be less than the statute requires. The Study Material quotes the Factories Act, 1948: "Where a worker works in a factory for more than nine hours in any day or for more than forty-eight hours in any week, he shall, in respect of overtime work, be entitled to wages at the rate of twice his ordinary rate of wages." For piece-rate workers, the time rate is deemed to be the daily average of their full-time earnings on the same or identical job in the month before (SM 3.17).

> ⚠ **Careful**
>
> The Factories Act provisions are quoted as the Study Material gives them (SM 3.17). Verify against the Study Material and Statutory Update for your attempt.

Occasional overtime is a healthy sign: the firm has the optimum capacity and is using it fully. **Persistent overtime is a bad sign**: either the firm needs more capacity in men and machines, or the workers have got into the habit of postponing ordinary work to the evening to earn overtime wages (SM 3.17–3.18).

Who pays the extra half

The normal-rate part of overtime wages is charged like any other wages. The **premium** goes where its cause sends it (SM 3.18):



ICAI's four treatments of overtime premium (SM 3.18); the last branch covers general production programmes and urgent orders. When overtime is worked regularly as a policy, for example because of a shortage of workers, ICAI charges all jobs at an inflated wage rate instead (SM 3.25).

ICAI prints the causes and the treatments side by side (SM 3.18); learn both lists in this order. Each treatment names the situation it applies to.

Causes of overtime	Treatment of overtime premium
1. The customer may agree to bear the entire charge of overtime because of urgency of work	1. Overtime at the desire of the customer: premium charged to the job directly
2. To make up a shortfall in production due to some unexpected development	2. Overtime to cope with general production programmes or to meet urgent orders: overhead cost of the department or cost centre that works overtime
3. To make up a shortfall in production due to some fault of management	3. Overtime in a department due to the fault of another department: charged to the latter department
4. To secure an out-turn in excess of normal output, to take advantage of an expanding market or rising demand	4. Overtime on account of abnormal conditions such as flood or earthquake: not charged to cost, but to Costing Profit and Loss Account

Overtime as a policy. If overtime is worked regularly throughout the year, for example because workers are short, the premium is a normal part of employee cost. ICAI then charges every job at an **average inflated wage**

rate: the year's total wages including premium, divided by the total hours (SM 3.25, Illustration 5).

Formats and formulas

Definition

Idle time

The time for which the employer pays but obtains no direct benefit or for no productive purpose. (SM 3.55)

Definition

Overtime premium

The rate for overtime work is higher than the normal time rate; the extra amount so paid over the normal rate is called overtime premium. (SM 3.17)

Definition

Ordinary rate of wages (Factories Act, as quoted in the SM)

The basic wages plus such allowances, including the cash equivalent of the advantage accruing through the concessional sale to workers of food grains and other articles, as the worker is for the time being entitled to, but does not include a bonus and wages for overtime work. (SM 3.17)

∑ **Formula**

Idle time and overtime

$$\text{Idle time} = \text{Time paid} - \text{Time booked}$$

$$\text{OT pay} = \text{OT hours} \times \text{normal rate} + \text{premium}$$

Premium = overtime hours × (overtime rate − normal rate).

∑ **Formula**

Average inflated wage rate

$$\text{Inflated rate} = \frac{\text{Total wages}}{\text{Total hours}}$$

Total wages = normal-time wages + overtime wages at the overtime rates (premium included). Used when overtime is a regular policy (SM 3.25).

ICAI's layout for the wages of a period (SM 3.16, Illustration 2). *Reproduce it from memory.*

Allocation of wages in cost accounting	Hours × rate	Amount (₹)
Allocated to Job X	hours booked × rate	xxx
Allocated to Job Y	hours booked × rate	xxx
Charged to Costing Profit and Loss A/c	abnormal idle hours × rate	xxx
Total (equals the wages paid)		**xxx**

Rate = wages of the period ÷ effective hours (paid hours less normal breaks). For overtime questions, ICAI shows each situation separately: employee cost chargeable to the job, and the premium as overhead, to the job, to another department or to Costing Profit and Loss (SM 3.25–3.26).

Worked examples

Worked example **The packing department's idle-time report**

Tapri's packing department has 20 packers, each paid for 8 hours a day on 25 days in September at ₹90 an

hour. Ritu's time records show these hours not booked to any job:

Cause	Hours
Tea breaks and hand-washing allowed by the hygiene rules	250
Changing the packing machine over between products	150
Waiting for cartons that the stores department issued late	90
A grid power failure across the industrial area	60
A sealing machine breakdown after its servicing was skipped by the maintenance department	50

All other hours were booked to jobs.

Required:

(i) Classify each cause as normal or abnormal idle time and, where abnormal, as controllable or uncontrollable.

(ii) Calculate the cost of each category at the wage rate paid.

(iii) Show how September's wages are charged in cost accounts.

(iv) State how the report should present the abnormal idle time.

THE WORKING

Working notes

1. Hours paid = 20 × 8 × 25 = **4,000**; wages = 4,000 × ₹90 = **₹3,60,000**.
2. Normal idle time = 250 + 150 = **400 hours**; abnormal idle time = 90 + 60 + 50 = **200 hours**.
3. Hours booked to jobs = 4,000 − 400 − 200 = **3,400**; effective hours (paid less normal idle time) = 4,000 − 400 = **3,600**.
4. Rate for costing = ₹3,60,000 ÷ 3,600 = **₹100 an hour**.

(i) and (ii) Classification and cost (at the rate paid, ₹90)

Cause
Tea breaks and hand-washing
Classification
Normal
Hours
250
Cost (₹)
22,500

Cause
Machine changeovers (set-up time)
Classification
Normal
Hours
150
Cost (₹)
13,500

Cause
Normal idle time
Classification
Hours
400
Cost (₹)
36,000

Cause
Cartons issued late by stores
Classification
Abnormal, controllable
Hours
90
Cost (₹)
8,100

Cause
Breakdown after skipped servicing
Classification
Abnormal, controllable
Hours
50
Cost (₹)
4,500

Cause
Grid power failure
Classification
Abnormal, uncontrollable
Hours
60
Cost (₹)
5,400

Cause
Abnormal idle time
Classification
Hours
200
Cost (₹)
18,000

Rest time and set-up time are ICAI's own examples of normal idle time; lack of coordination (the late cartons), breakdown of machines and power failure are its abnormal causes (SM 3.13–3.14). The late cartons and the skipped servicing could have been avoided by the departments concerned, so they are controllable; the grid failure was outside Tapri's control.

(iii) Charging September's wages

Particulars	Hours × rate	Amount (₹)
Jobs (cost of production)	3,400 × ₹100	3,40,000
Costing Profit and Loss A/c (abnormal idle time)	200 × ₹100	20,000
Total		**3,60,000**

Normal idle time is not shown separately: it is inside the ₹100 rate, which recovers the ₹36,000 over the 3,600 effective hours (₹34,000 on the jobs and ₹2,000 on the abnormal hours). Valued at the rate

paid in (ii), abnormal idle time is ₹18,000; charged at the costing rate in (iii), ₹20,000. State the rate used in a working note, as ICAI's answers do.

(iv) Presentation. Abnormal idle time is shown in the Costing Profit and Loss Account, analysed by cause into controllable and uncontrollable (SM 3.14). The 90 hours are reported against the stores department and the 50 hours against the maintenance department, so that responsibility is fixed; the 60 hours of grid failure are reported as uncontrollable.

Worked example

October's overtime: who pays the premium?

Tapri's packing unit pays a basic rate of ₹96 an hour. Overtime carries a premium of 50% on weekdays and

100% on Sundays. October's overtime register shows:

Reason for the overtime	Weekday hours	Sunday hours
(a) A hotel chain asked for its hamper order, Job J-77, three days early and agreed to bear the extra cost	60	20
(b) The general Diwali rush across all orders	240	40
(c) The samosa factory sent its output a day late through an error in its own planning	80	–
(d) A cloudburst flooded the unit and the lost output was made up	–	60

Job J-77 also took 500 normal hours.

Required:

(i) Calculate the overtime premium for each reason and state where it is charged.

(ii) Calculate the employee cost of Job J-77.

(iii) If overtime were worked regularly as a policy, the unit would charge every job at its average inflated wage rate: in the last six months it paid ₹1,15,00,000, premiums included, for 1,15,000 hours. Calculate the employee cost of Job J-77 on that basis.

THE WORKING

Working note. Premium per hour: weekdays 50% × ₹96 = ₹48; Sundays 100% × ₹96 = ₹96.

(i) Overtime premium by reason

Reason	Premium (₹)	Charged to
(a) Customer's request: 60 × ₹48 + 20 × ₹96	4,800	Job J-77
(b) General production: 240 × ₹48 + 40 × ₹96	15,360	Overheads of the packing unit
(c) Another department's fault: 80 × ₹48	3,840	The samosa factory
(d) Flood: 60 × ₹96	5,760	Costing Profit and Loss Account
Total premium	**29,760**	

The 500 overtime hours themselves are paid at the basic ₹96 and charged to the jobs worked on, like normal hours (SM 3.18).

(ii) Employee cost of Job J-77

Particulars	Amount (₹)
Normal and overtime hours at basic rate: (500 + 80) × ₹96	55,680
Add: Overtime premium at the customer's request	4,800
Employee cost of Job J-77	**60,480**

(iii) On the inflated rate. Average inflated wage rate = ₹1,15,00,000 ÷ 1,15,000 = ₹100 an hour. Job J-77 = 580 hours × ₹100 = **₹58,000**. Under a policy of regular overtime the premium is part of employee cost and every job shares it through the average rate, so J-77 would bear ₹4 an hour of average premium instead of its own ₹4,800 (SM 3.25).

Exam angle

In the exam

Where idle time and overtime appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, the topic came up in 11 questions.

- **Theory, 4 marks.** "Describe briefly idle time and explain the treatment" for a set-up (direct worker), a lunch break (indirect worker) and a breakdown: May 2024 Q6(c), repeated in MTP May 2026 Series II

Q6(c). Causes of normal and abnormal idle time: MTP January 2025 Series II Q4(b). Treatment of idle time and overtime wages: MTP September 2025 Series I Q4(b). Effect of overtime on productivity: MTP May 2025 Series I Q2(b).

- **Numerical, 4–6 marks.** Inflated wage rate and the job's cost in four overtime situations, including a flood, after first finding the basic rate from 50 workers' effective hours (8 hours less a 30-minute lunch break, 160 days): September 2024 Q2(b) (6 marks). The same in two situations, premium stated as 75% and 125%: MTP May 2025 Series I Q1(c). Abnormal idle time and overtime premium per employee: MTP September 2026 Series I Q1(a). Overtime at twice basic and DA, allocated to jobs: RTP May 2025 Q9. Daily overtime after nine hours, holiday work: RTP January 2026 Q9.
- **Multiple choice, 2 marks.** Cost of normal and abnormal idle time in a 48-hour week: January 2025 MCQ 13. Allocation of a week's wages with an abnormal idle time charge: MTP September 2025 Series II MCQ 14.

In the exam

The traps ICAI set

- **Rate or premium?** Illustration 5 gives overtime "rates" of 175% and 225% of basic; MTP May 2025 Series I Q1(c) gives a "premium" of 75% and 125%: the same thing. Premium = rate − 100%.

- **Only the premium moves.** In the irregular, customer and flood cases, the job keeps all hours at the basic rate; only the premium goes elsewhere.
- **Which time is abnormal?** "Unaccounted time attributable to power failure" (January 2025 MCQ 13) is abnormal; the allowed 15 minutes a day for maintenance is normal.
- **Which rate?** The cost of idle time with a rate per hour given is hours × that rate (January 2025 MCQ 13); an allocation of the week's wages uses wages ÷ effective hours for the jobs and the abnormal hours alike (MTP September 2025 Series II MCQ 14: ₹2,640 ÷ 44 hours = ₹60).
- One MTP answer (September 2026 Series I Q1(a)(iv)) words the customer case differently; write the Study Material's rule: a customer's request is charged to the job.

Trick

Follow the cause

Ask "Who or what made this time necessary?" The normal course of business: cost of production. The customer: the job. Another department: that department. Nature or abnormal events: Costing Profit and Loss. Condition: overtime worked regularly as a policy is the exception; it is spread through the inflated rate.

Answer frame

"Describe idle time and its treatment in the following situations"

Define idle time first (time paid but no production; time paid – time booked; normal or abnormal), then a table, as ICAI's suggested answer to May 2024 Q6(c) does:

Situation	Idle time	Treatment
Setting-up time for a direct worker's machine	Normal	Part of cost of production; allowed for in standard hours or rate
Lunch break of an indirect worker	Normal	Considered in computing the overhead rate
Time lost by a machine breakdown	Abnormal	Separate item in the Costing Profit and Loss Account

"Discuss the effect of overtime payment on productivity"

1. From the SM (3.17–3.18): occasional overtime is healthy (optimum capacity, fully used); persistent overtime is a bad sign (more capacity needed, or work habitually postponed to earn overtime).
2. ICAI's suggested answer (MTP May 2025 Series I Q2(b)) adds that overtime raises the cost of production: the premium is an extra payment; efficiency during

overtime may fall; workers may slacken in normal hours to earn more; lower output and higher premium raise cost.

"Discuss the accounting treatment of idle time and overtime wages"

The chapter's own Theoretical Question 1, set as MTP September 2025 Series I Q4(b) (4 marks). ICAI's suggested answer gives the normal idle time treatment and the four premium treatments; add abnormal idle time to make the answer complete. In this order:

1. **Normal idle time**: part of the cost of production; for direct workers an allowance in the standard hours or standard rate, for indirect workers considered in computing the overhead rate (SM 3.13).
2. **Abnormal idle time**: not part of production cost; a separate item in the Costing Profit and Loss Account, analysed into controllable and uncontrollable by cause (SM 3.14).
3. **Overtime premium** (SM 3.18): at the desire of the customer, charged to the job directly; for general production programmes or urgent orders, overhead cost of the department or cost centre that works overtime; due to the fault of another department, charged to that department; on account of abnormal conditions such as flood or earthquake, not charged to cost but to Costing Profit and Loss Account. The overtime hours at the normal rate are charged like ordinary wages.

4. **Overtime worked regularly as a policy**: the premium is part of employee cost, and jobs are charged at the average inflated wage rate (SM 3.25).

The numerical: three answers to reason (b) of October's overtime

Loses marks

"Overheads: 240 × ₹144 + 40 × ₹192 = ₹42,240."

The whole overtime wage sent to overheads: the basic part belongs to the jobs.

Passes

"Packing overheads ₹15,360."

Right figure, but no rates, no premium working, no reason.

Full marks

Premium per hour as a working note (₹48 and ₹96), the premium 240 × ₹48 + 40 × ₹96 = ₹15,360, the basic part charged to the jobs, and the treatment named with its reason: overtime for general production is an overhead of the department that works it.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the 75% premium paid at 75%

Reading "overtime premium of 75%" as an overtime rate of 75% of basic, so overtime is paid *less* than normal time. **Fix:** a premium is the extra; the overtime rate is 175% of basic.

⚠ **Common mistake**

The whole overtime wage sent to overheads

Charging all 240 × ₹144 to overheads for the Diwali rush. **Fix:** the overtime hours are still the jobs' hours at the normal rate; only the premium (240 × ₹48) goes to overheads.

⚠ **Common mistake**

The power cut spread over the jobs

Dividing the packing department's ₹3,60,000 by the 3,400 booked hours (₹105.88 an hour) so that every job carries the late cartons, the breakdown and the power failure. **Fix:** divide by the effective hours (3,600), and charge the 200 abnormal hours separately to Costing Profit and Loss.

⚠ **Common mistake**

Normal idle time written off

Sending tea breaks or set-up time to Costing Profit and Loss. **Fix:** normal idle time cannot be avoided in the normal course of business; it is part of the cost of production.

Practice

Check yourself

1. A packer is paid ₹70 an hour for a 48-hour week. Normal idle time is 2 hours; she booked 41 hours to jobs; the rest was lost waiting for a delayed truck. What is the cost of abnormal idle time at the rate paid, in rupees?

Answer: _____

2 Case scenario. In November Tapri's packing department works 300 overtime hours because the samosa factory sent its output two days late after a mix-up in its own planning. Ritu must book the ₹13,500 overtime premium. Which treatment is correct?

- **A** Charge it to the jobs packed in those 300 hours
- **B** Treat it as an overhead of the packing department
- **C** Charge it to the samosa factory, the department at fault
- **D** Charge it to the Costing Profit and Loss Account

3 A worker's ordinary rate is ₹60 an hour, her normal day is 8 hours, and her employer pays twice the ordinary rate only for hours beyond nine in a day. What is her pay for an 11-hour day, in rupees?

Answer: _____

4 Which of these are normal idle time? Select all that apply.

- **A** Time lost walking from the factory gate to the place of work
- **B** Time lost because raw material ran out
- **C** Setting-up time for the machine
- **D** Time lost in a lockout

More than one answer may be right.

5 Overtime is worked irregularly to meet production requirements. 100 overtime hours are paid at ₹150 against a normal rate of ₹100. How much goes to factory overheads, in rupees?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Idle time = time paid − time booked: paid, nothing produced.
- Normal idle time (gate to workplace, between jobs, set-ups, rest and lunch): cost of production; direct workers through the rate, indirect workers through the overhead rate.

- Abnormal idle time (lack of coordination, power failure, breakdown, no material, strikes, lockouts, fire, flood): Costing Profit and Loss, analysed as controllable and uncontrollable.
- Overtime payment = overtime hours at normal rate + premium; usually double the normal rate.
- Premium: customer's request → job; general production or urgent orders → department overheads; another department's fault → that department; abnormal conditions → Costing P&L. Regular policy → inflated rate for all jobs.
- The SM's Factories Act quote: more than nine hours a day or 48 a week, twice the ordinary rate.
- Occasional overtime healthy; persistent overtime a bad sign. (Whoever caused the wait pays for it.)

Answer key · Check yourself

1 **350**

Abnormal idle time = 48 − 2 − 41 = 5 hours; 5 × ₹70 = ₹350. Counting the 2 normal hours as abnormal gives ₹490.

2 **C Charge it to the samosa factory, the department at fault**

Overtime worked in one department because of the fault of another is charged to the latter (SM 3.18). Overheads of the packing department would be right for general production needs; Costing P&L is for abnormal conditions such as floods.

3 **780**

9 hours at ₹60 = ₹540, plus 2 overtime hours at ₹120 = ₹240: ₹780. Treating all 3 hours beyond 8 as overtime gives ₹840: a different rule (RTP January 2026 Q9 style), which must be stated in the question.

4 **A Time lost walking from the factory gate to the place of work**; **C Setting-up time for the machine**

ICAI's normal causes: gate to workplace, interval between jobs, set-up time, rest and lunch (SM 3.13). Shortage of material and lockouts are abnormal (SM 3.14).

5 **5000**

Only the premium: 100 × (₹150 − ₹100) = ₹5,000. The other ₹10,000 is the job's normal wages for those hours. ₹15,000 sends the whole overtime wage to overheads.

4.3 Employee turnover: measurement, causes, effects and cost

How fast a workforce changes, measured three ways; why people leave; what it does to cost; and how to compute the profit forgone because of it.

In plain words

The baristas who went to BrewBucks

BrewBucks opened in Pune last spring, and Tapri's kulhad cafés started losing baristas. By March, 60 of them had left.

> **Kesar:** People come, people go. Hire 60 more. Sab adjust ho jayega.
>
> **Ritu:** Each one who leaves costs us twice. While the post is empty, the counter sells less tea. When the new barista comes, we pay for recruiting and training, and for a month she is slow and breaks kulhads.
>
> **Kesar:** How bad is it, in one number?
>
> **Ritu:** Sixty left out of an average of 500 on the rolls: a separation rate of 12% for the year.

That number is the **employee (labour) turnover**: the rate at which the composition of the workforce changes. ICAI measures it three ways, always against the **average number of employees on roll**: by the people who **left**

(separation method), by the people **hired to fill their places** (replacement method), or by both together (flux method). Hiring for expansion is not replacement: a new café's staff did not take anyone's place.

People leave for **personal** reasons (a better job, health, family), for **unavoidable** reasons (the business is seasonal, the plant moves, discipline), and for **avoidable** reasons that management can fix (poor pay, poor conditions, a bad supervisor, no training or promotion). High turnover raises the cost of production; its cost is the **contribution lost** on the hours that could not be worked, plus the money spent on settling, recruiting and training.

Think of it like

The railway coach

A Mumbai local stops at a station. The passengers who get off are the separations. Those who board to take their seats are the replacements. If the railway adds a new coach, the people who fill it are new joinings, not replacements. Turnover measures the churn in the existing coaches; adding a coach is growth.

Key idea

Turnover = movement out, in, or both, divided by the average number on roll. It costs the contribution on the productive hours lost plus the costs of leaving, hiring and training; preventive spending is worth it up to the optimum level of turnover.

How it works

One measure, three ways

ICAI defines employee turnover as the rate of change in the composition of employee force during a specified period measured against a suitable index. The index or norm may be the usual turnover in the industry or locality, or the firm's own rate for a past period (SM 3.45). The three methods (SM 3.45–3.46):

1. **Replacement method**: employees actually replaced ÷ average number on roll × 100. New employees appointed for expansion are not replacements.
2. **Separation method**: employees separated (resigned, retired, discharged) ÷ average number on roll × 100.
3. **Flux method**: separations plus replacements ÷ average number on roll × 100.

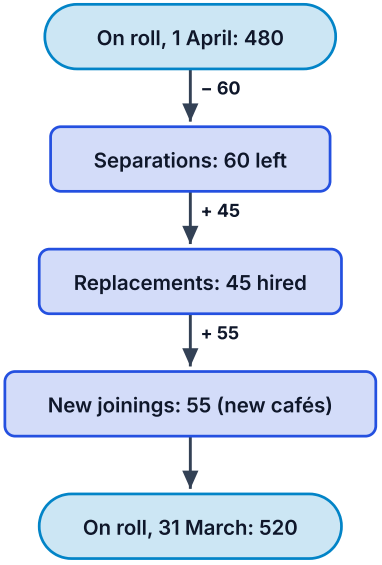
New recruitment. Recruits for expansion are generally left out, but since they too change the composition of the workforce, some management accountants include them. All workers joining, including replacements, are **accessions**, and the flux rate may then be (separations +

replacements + new joinings) ÷ average, or (separations + accessions) ÷ average (SM 3.46).

The **average number on roll** = (number at the beginning + number at the end) ÷ 2. A rate for a period shorter than a year is converted to an **equivalent annual rate**: rate for the period × 365 ÷ days in the period; for a quarter ICAI multiplies by 4 (SM 3.46–3.48).

Tapri's cafés over the year, with the figures from the story:

Tapri's café staff: who left, who replaced, who was added



480 − 60 + 45 + 55 = 520; average on roll (480 + 520) ÷ 2 = 500. Accessions = 45 + 55 = 100.

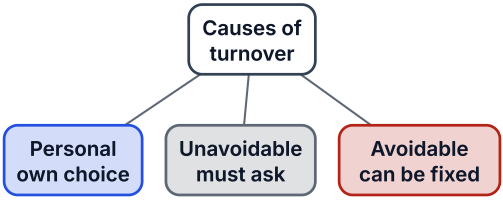
Method	Working	Rate
Separation	60 ÷ 500 × 100	12%
Replacement	45 ÷ 500 × 100	9%
Flux (separations + replacements)	(60 + 45) ÷ 500 × 100	21%
Flux with accessions	(60 + 100) ÷ 500 × 100	32%

Interactive version: PassBook app, lesson Employee turnover: measurement, causes, effects and cost

Why they leave

ICAI groups the causes under three heads (SM 3.48–3.49):

Causes of employee turnover



Blue: personal; grey: unavoidable; rose: avoidable, the head that needs management's continuous attention (SM 3.48–3.49).

(a) Personal causes: personal reasons which induce or compel an employee to leave:

1. Change of jobs for betterment.
2. Premature retirement due to ill health or old age.
3. Domestic problems and family responsibilities.
4. Discontent over the jobs and working environment.

(b) Unavoidable causes: management is obliged to ask one or more employees to leave:

1. Seasonal nature of the business.
2. Shortage of raw material, power; slack market for the product.
3. Change in the plant location.
4. Disability, making a worker unfit for work.
5. Disciplinary measures.

(c) Avoidable causes: they need management's attention on a continuous basis:

1. Dissatisfaction with job, remuneration, hours of work, working conditions.
2. Strained relationship with management, supervisors or fellow workers.
3. Lack of training facilities and promotional avenues.
4. Lack of recreational and medical facilities.
5. Low wages and allowances.

Under personal causes the employee leaves at will, and the first three are hard to remedy; the fourth, discontent,

can be overcome by a healthy working environment, with officers making sure subordinates work in healthy conditions. Unavoidable causes oblige management to ask employees to leave. Avoidable causes need management's attention continuously, and proper and timely action can reduce turnover from them appreciably (SM 3.48–3.49).

What it does to cost

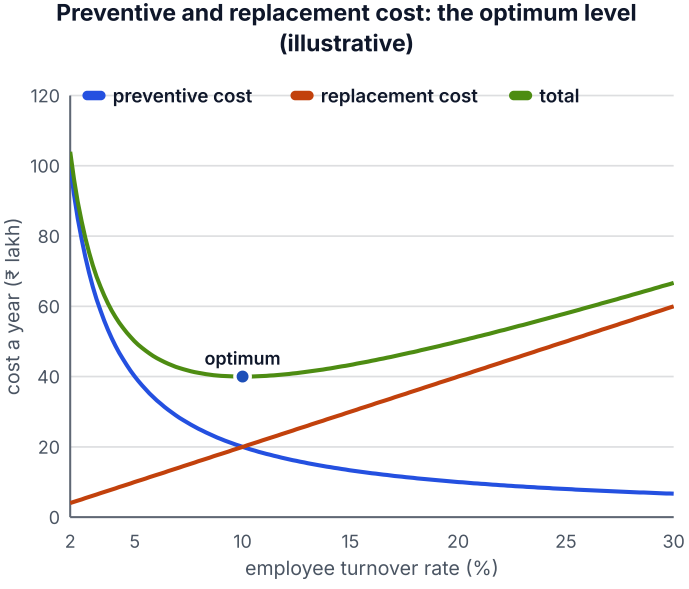
High employee turnover increases the cost of production because (SM 3.50):

1. the even flow of production is disturbed;
2. the efficiency of new workers is low, and even experienced newcomers are less productive at first;
3. the cost of training and induction increases;
4. new workers cause more breakage of tools, wastage of materials and so on;
5. the cost of recruitment and training increases.

Two types of cost go with it (SM 3.50):

- **Preventive costs**, incurred to keep turnover as low as possible: (i) medical benefits; (ii) employee welfare such as pension; (iii) other benefits aimed at retaining employees.
- **Replacement costs**, which arise because of turnover: recruitment, training and induction, abnormal breakage and scrap, and extra wages and overheads due to the inefficiency of new workers.

High turnover means high replacement costs, and only adequate preventive costs keep turnover low, so each company must work out its **optimum level** of turnover from its personnel policies and the behaviour of the two costs (SM 3.50):



Illustrative curves, not ICAI figures. Blue: preventive cost falls as more turnover is tolerated; orange: replacement cost rises with turnover; green: their total is lowest at the optimum level.

The profit forgone

The cost of turnover in rupees is the **profit forgone**: the contribution that the lost productive hours would have

earned, plus the costs incurred because of turnover (SM 3.51–3.52, Illustration 15). The steps:

1. **Actual productive hours** = actual hours worked − unproductive training hours.
2. **Productive hours lost** = potential productive hours lost (vacancies not filled in time) + unproductive training hours.
3. **Sales lost** = sales ÷ actual productive hours × productive hours lost.
4. **Contribution lost** = sales lost × P/V ratio (contribution and the P/V ratio).
5. **Profit forgone** = contribution lost + settlement, recruitment, selection and training costs (+ rectification of defective output, when given).

When the data are in units instead (the chapter's case scenario of J Ltd), output lost = potential hours lost ÷ hours per unit + training hours' output lost through lower productivity, and contribution lost = units lost × contribution per unit (SM 3.61–3.62, 3.68).

Formats and formulas

Definition

Employee (labour) turnover

The rate of change in the composition of employee force during a specified period measured against a suitable index. (SM 3.45)

Formula

Replacement and separation methods

$$\text{Replacement} = \frac{\text{Employees replaced}}{\text{Average on roll}} \times 100$$

$$\text{Separation} = \frac{\text{Employees separated}}{\text{Average on roll}} \times 100$$

(SM 3.45–3.46)

Formula

Flux method

$$\text{Flux} = \frac{\text{Separated} + \text{Replaced}}{\text{Average on roll}} \times 100$$

With accessions (all joining, including replacements): (separations + accessions) ÷ average on roll × 100, or (separations + replacements + new joinings) ÷ average on roll × 100 (SM 3.46).

∑ **Formula**

Average on roll and the equivalent annual rate

$$\text{Average} = \frac{\text{At beginning} + \text{At end}}{2}$$

$$\text{Annual rate} = \frac{\text{Rate for period} \times 365}{\text{Days in period}}$$

(SM 3.46–3.47)

ICAI's layout for the cost of turnover (SM 3.52, Illustration 15). *Reproduce it from memory.*

Computation of profit forgone on account of employee turnover	Amount (₹)
Contribution foregone (working note)	xxx
Settlement cost due to leaving	xxx
Recruitment cost	xxx
Selection cost	xxx
Training costs	xxx
Cost of rectification of defective units (if given)	xxx
Profit foregone	**xxx**

Worked examples

Worked example **Riders and cooks: two turnover rates and a retention plan**

Tapri's HR team gives these figures for 2025-26:

Particulars	Delivery riders	Kitchen staff
On roll on 1 April 2025	380	150
On roll on 31 March 2026	420	150
Left and discharged during the year	114	12
Replaced during the year	104	12
Joined for new posts (expansion)	50	–

Each rider who leaves costs Tapri ₹9,000 to replace (recruitment, induction, uniform and road-safety training) and ₹8,000 of contribution on deliveries lost while the post is vacant. HR proposes a rider retention plan (medical cover and a loyalty bonus) costing ₹6,00,000 a year, which is expected to cut rider separations by one-third; every leaver would still be replaced.

Required:

(i) Calculate the separation rate, the replacement rate and the flux rate (on separations and replacements, and including all accessions) for each group.

(ii) Comment on the comparison.

(iii) Evaluate the retention plan.

(iv) For next year HR targets a rider flux rate (separations + replacements) of 30%, with the same average on roll and every leaver replaced. How many riders may leave?

THE WORKING

Working note: average on roll = (beginning + end) ÷ 2: riders (380 + 420) ÷ 2 = **400**; kitchen staff (150 + 150) ÷ 2 = **150**. Accessions of riders = 104 replacements + 50 new posts = 154.

(i) Employee turnover rates

Method	Riders	Kitchen staff
Separation: 114 ÷ 400; 12 ÷ 150	28.50%	8.00%
Replacement: 104 ÷ 400; 12 ÷ 150	26.00%	8.00%
Flux, separations + replacements: 218 ÷ 400; 24 ÷ 150	54.50%	16.00%
Flux, separations + accessions: 268 ÷ 400; 24 ÷ 150	67.00%	16.00%

The 50 riders hired for new posts are not replacements (SM 3.45); they enter only the flux rate that includes all accessions.

(ii) Comment. More than a quarter of the riders left during the year, against one in twelve of the kitchen staff. The riders' turnover is the one that disturbs the flow of work and keeps raising recruitment and training costs, so it needs attention to its avoidable causes (pay, working conditions, relations with supervisors). The kitchen's rates are low and steady; its separations were all replaced, so separation, replacement and both flux rates move together.

(iii) Retention plan

Particulars	Amount (₹)
Separations avoided: 114 ÷ 3 = 38	
Replacement cost saved: 38 × ₹9,000	3,42,000
Contribution no longer lost: 38 × ₹8,000	3,04,000
Saving	**6,46,000**
Less: Cost of the plan (preventive cost)	6,00,000
Net benefit	**46,000**

The plan is worthwhile, but only just: its preventive cost of ₹6,00,000 buys ₹6,46,000 of lower replacement cost and lost contribution, and the rider separation rate falls to 76 ÷ 400 = 19%. If it cuts separations by less than 36 (₹6,00,000 ÷ ₹17,000 = 35.3), it does not pay. This is the trade-off between preventive and replacement costs that decides a company's optimum level of turnover (SM 3.50).

(iv) Target. Flux = (S + R) ÷ 400 × 100 = 30%, so S + R = 120; with every leaver replaced, R = S, so **S = 60 riders** (a separation rate of 15%).

Worked example

The samosa factory's vacancies

Kesar wants to know what last year's turnover at the samosa factory cost before he approves any retention plan. Ritu's records show:

1. 40 direct workers left during the year, and all were replaced. On average a post stayed vacant for 20 working days of 8 hours before the replacement joined.
2. Each recruit spent 10 days of 8 hours in training, half of which time produced nothing.
3. A direct worker fries 4 trays an hour; each tray earns a contribution of ₹45, and all lost output could have been sold.
4. Recruits wasted materials worth ₹1,80,000 during training.
5. Settlement cost was ₹4,000 for each worker leaving; recruitment and selection cost ₹6,500 and training and induction ₹5,000 for each recruit.

Required:

(i) Calculate the productive hours lost on account of employee turnover.

(ii) Calculate the profit forgone last year on account of employee turnover, and the profit forgone per worker who left.

THE WORKING

(i) Productive hours lost

Particulars	Hours
Vacancies: 40 posts × 20 days × 8 hours	6,400
Unproductive training hours: 40 recruits × 10 days × 8 hours × 50%	1,600
Productive hours lost	**8,000**

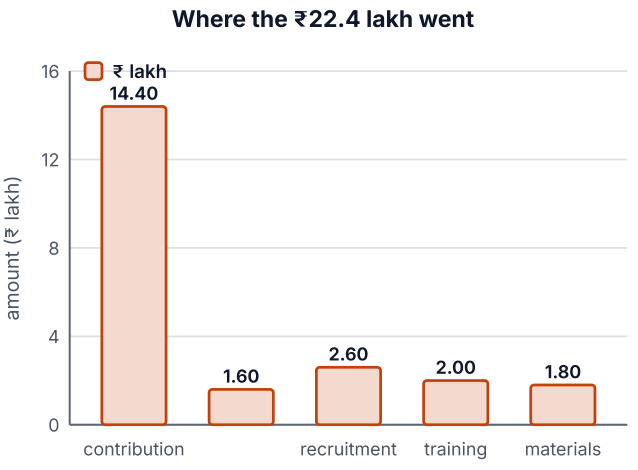
The productive half of the training hours is not lost: it produced output.

Working note: contribution lost = 8,000 hours × 4 trays × ₹45 = 32,000 trays × ₹45 = **₹14,40,000**.

(ii) Computation of profit forgone on account of employee turnover

Particulars	Amount (₹)
Contribution foregone (working note)	14,40,000
Settlement cost due to leaving (40 × ₹4,000)	1,60,000
Recruitment and selection cost (40 × ₹6,500)	2,60,000
Training and induction cost (40 × ₹5,000)	2,00,000
Materials wasted by recruits	1,80,000
Profit foregone	**22,40,000**

Profit forgone per worker who left = ₹22,40,000 ÷ 40 = **₹56,000**.



Nearly two-thirds of the profit forgone is contribution on hours never worked, not the visible costs of hiring and training.

The visible costs of turnover (settlement, recruitment, training, wasted materials) are ₹8,00,000; the contribution lost on vacant posts and unproductive training is almost twice as large. Any retention plan costing less than about ₹56,000 for each departure it prevents would pay for itself.

Exam angle

In the exam

Where employee turnover appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, employee turnover was examined in 11 questions.

- **Numerical from the rates, 4–5 marks.** Average on roll, recruited and joined, and left, from flux, replacement and separation rates for a quarter: May 2026 Q4(b) (5 marks); RTP September 2024 Q8. Replacement and flux rates from a separation rate: May 2025 Q1(c) (4 marks). The same as an MCQ: MTP January 2025 Series I MCQ 11.
- **Separation rate from opening numbers and movements**, 2 marks: September 2024 MCQ 14.
- **Profit forgone, 6 marks**: May 2024 Q5(a), with unproductive training hours, potential hours lost and rectification of defectives. Output and contribution lost in units, as five MCQs on one case: RTP May 2024 Q2.
- **Theory, 4–5 marks.** Two types of employee cost from turnover, with two productivity factors: MTP May 2026 Series I Q6(b). Two cost impacts and two measures: MTP September 2025 Series II Q6(c). Remedial steps to minimise labour turnover: MTP January 2026 Series II Q6(b) (5 marks); MTP May 2024 Series II Q6(c) (4 marks).

In the exam

The traps ICAI set

- **Accessions or replacements?** "Recruited and joined" is accessions: replacements plus new joinings (Illustration 14; May 2026 Q4(b)).
- **Productive hours, not all hours.** Only the unproductive part of training hours is lost (40% in May 2024 Q5(a)); sales per hour is sales ÷ actual productive hours.
- **Contribution, not sales.** The lost sales are multiplied by the P/V ratio before the turnover costs are added.
- **Annualise.** A quarter's rate × 4; any other period × 365 ÷ days.

Trick

One count unlocks the rest

The three methods share one denominator. Find the average on roll from whichever method's count is given (count ÷ rate), then every other count is rate × average. Condition: all the rates are for the same period and use the same average.

Interactive version: PassBook app, lesson Employee turnover: measurement, causes, effects and cost

Answer frame

"Discuss the two types of cost associated with labour turnover"

ICAI's suggested answers (MTP May 2026 Series I Q6(b); MTP September 2025 Series II Q6(c)) reproduce SM 3.50: (a) **preventive costs**, defined, with the three items (medical benefits, welfare such as pension, other retention benefits); (b) **replacement costs**, defined, with the items (recruitment, training and induction, abnormal breakage and scrap, extra wages and overheads from the inefficiency of new workers); then the conclusion that high turnover means high replacement costs and only adequate preventive costs keep it low, so each firm works out its optimum level.

"Discuss the remedial steps to minimise labour turnover"

The Study Material says proper and timely management action can appreciably reduce turnover from avoidable causes (SM 3.49) and lists the preventive costs (SM 3.50). ICAI's suggested answers to the MTP questions (January 2026 Series II Q6(b); May 2024 Series II Q6(c)) give these steps; write them with one line each:

1. **Exit interview** with each outgoing employee, to find out why he is leaving.
2. **Job analysis and evaluation**, to know the requirements of each job.

3. A **scientific system of recruitment, placement and promotion**.
4. A **healthy atmosphere**, with education, medical and housing facilities for workers.
5. A **committee for settling workers' grievances**.

Then tie them to the avoidable causes they remove (dissatisfaction, strained relations, no training or promotion, no recreational or medical facilities, low wages).

"Causes" and "effects"

Write the three heads with ICAI's items in ICAI's order (the lists above), adding the remark on remedies under each head. For effects, the five ways high turnover raises the cost of production (SM 3.50).

The numerical: three answers to the profit-forgone question

Loses marks

"Profit forgone = settlement + recruitment + training = ₹6,20,000."

Only the visible costs: the contribution on 8,000 lost hours and the wasted materials are missing.

Passes

"Contribution lost ₹14,40,000; profit forgone ₹22,40,000."

Right, but the hours working is invisible: no method marks if a figure slips.

Full marks

Working notes for the productive hours lost (vacancies and unproductive training) and the contribution lost (or, when sales and a P/V ratio are given, sales per productive hour × P/V ratio); then ICAI's statement with each turnover cost on its own line, ending in the profit foregone.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the opening headcount as the denominator

Dividing the 60 café separations by the 480 on roll at the start (12.5%) instead of the average of 500 (12%). **Fix:** every method divides by the average number on roll, (beginning + end) ÷ 2.

⚠ **Common mistake**

The new café's staff counted as replacements

Adding the 55 baristas hired for the new cafés to the 45 replacements, giving a replacement rate of 20%. **Fix:** employees appointed for expansion are not replacements (SM 3.45); they enter only the flux rate with accessions.

⚠ **Common mistake**

All training hours written off

Treating all 80 training hours of each recruit as lost (3,200 hours). **Fix:** only the unproductive half (1,600 hours) is lost; the rest produced output.

⚠ **Common mistake**

A month's rate reported as the year's

Giving a 30-day month's 2% separation rate as the year's rate. **Fix:** rate × 365 ÷ days in the period (24.33% here), or a quarterly rate × 4.

Practice

Check yourself

1 A Tapri kitchen had 220 workers at the start of the year. During the year 50 left and 10 were replaced; no one was hired for expansion. What is the separation rate, in per cent?

Answer: _____

2 Case scenario. In one year Tapri's cafés lose staff for these reasons: some move with their families to another city, a seasonal kiosk at a hill station closes for the monsoon, and several leave after constant quarrels with one supervisor. Ritu must tell the board which losses management can reduce by timely action. Which is correct?

- **A** The family moves, a personal cause
- **B** The kiosk closure, an unavoidable cause
- **C** The quarrels with the supervisor, an avoidable cause
- **D** None of them: turnover is outside management's control

3 Average on roll 400; during the period 30 separated, 20 were replaced and 25 joined for expansion. What is the flux rate including accessions, in per cent?

Answer: _____

4 A factory's separation rate for a 30-day month is 2%. What is the equivalent annual rate, in per cent, to one decimal?

Answer: _____

5 Which of these is a preventive cost of employee turnover rather than a replacement cost?

- **A** Training and induction of new recruits
- **B** Pension and other welfare for existing employees
- **C** Abnormal breakage and scrap by new workers
- **D** Recruitment advertising for vacancies

Answers: the answer key at the end of this lesson.

Recap

- Employee turnover: the rate of change in the composition of the workforce in a period, against an

index (industry, locality, or own past rate).

- Replacement = replaced ÷ average on roll; separation = separated ÷ average; flux = (separated + replaced) ÷ average; all × 100.
- Expansion hires are not replacements; accessions = all joinings including replacements; flux with accessions = (separations + accessions) ÷ average.
- Average on roll = (beginning + end) ÷ 2; annual rate = period rate × 365 ÷ days (a quarter × 4).
- Causes: personal, unavoidable, avoidable (management can act on the avoidable ones).
- Costs: preventive (medical, welfare, retention benefits) and replacement (recruitment, training, breakage, inefficiency); find the optimum level.
- Profit forgone = (productive hours lost × sales per productive hour × P/V ratio) + settlement, recruitment, selection, training and rectification costs. (Count who got off the coach, not the new coach.)

Answer key · Check yourself

1 **25**

Closing = 220 − 50 + 10 = 180; average = (220 + 180) ÷ 2 = 200; separation rate = 50 ÷ 200 × 100 = 25%. Dividing by the opening 220 gives 22.73%.

2 **C The quarrels with the supervisor, an avoidable cause**

Strained relationship with supervisors is an avoidable cause, which proper and timely management action can reduce (SM 3.49). Domestic reasons are personal and hard to remedy; the seasonal nature of business is unavoidable.

3 **18.75**

Accessions = 20 + 25 = 45; (30 + 45) ÷ 400 × 100 = 18.75%. The flux rate without new joinings is (30 + 20) ÷ 400 = 12.5%.

4 **24.3 to 24.4**

2% × 365 ÷ 30 = 24.33%. Multiplying by 12 gives 24%, close, but ICAI's formula uses days.

5 **B Pension and other welfare for existing employees**

Preventive costs are incurred to keep turnover low: medical benefits, welfare like pension, other retention benefits. The other three arise because employees have left (SM 3.50).

4.4 Time rate, piece rate and the Halsey and Rowan premium plans

Paying for time, for output, or for time plus a share of the time saved: the Halsey and Rowan formulas, which pays more when, and what each does to labour cost per unit.

In plain words

Shabnam finishes early

A lot of 400 premix boxes at Tapri's packing line is allowed 8 hours. Shabnam packs it in 6. She is paid ₹100 an hour.

> **Kesar:** She worked 6 hours, so she gets ₹600. Simple.
>
> **Ritu:** Then why would she ever hurry? Pay her by the box and she gets 8 hours' worth, ₹800, and we gain nothing from her speed.
>
> **Shabnam:** Somewhere in between would be fair, Sir.
>
> **Ritu:** That is what the premium plans do. Halsey gives her half the 2 hours saved: ₹700. Rowan gives her 2/8 of her time wages as bonus: ₹750.

Those are the four ideas of this lesson. A **time rate** pays for the hours, whatever the output. A **piece rate** pays for the output, whatever the hours. A **premium bonus** plan fixes a **time allowed** (standard time) for the job, guarantees the time wages, and shares the **time saved** between worker and employer. Under the **Halsey plan**

the worker gets wages for half the time saved. Under the **Rowan plan** the bonus is the fraction of her time wages that the time saved is of the time allowed.

The two premium plans pay exactly the same when the job takes half the time allowed. If she saves less than that, Rowan pays more; if she saves more, Halsey pays more. Rowan's bonus is largest at that halfway point and then falls, so racing does not pay under it.

Think of it like

The saved-time cake

Each time a worker beats the standard, she bakes a cake: the wages of the hours saved. Halsey always cuts it in half. Rowan gives her a slice the size of the share of the allowed time she actually worked: generous when she saves a little, stingy when she saves a lot. Under Rowan, the bigger the cake, the smaller her share of it.

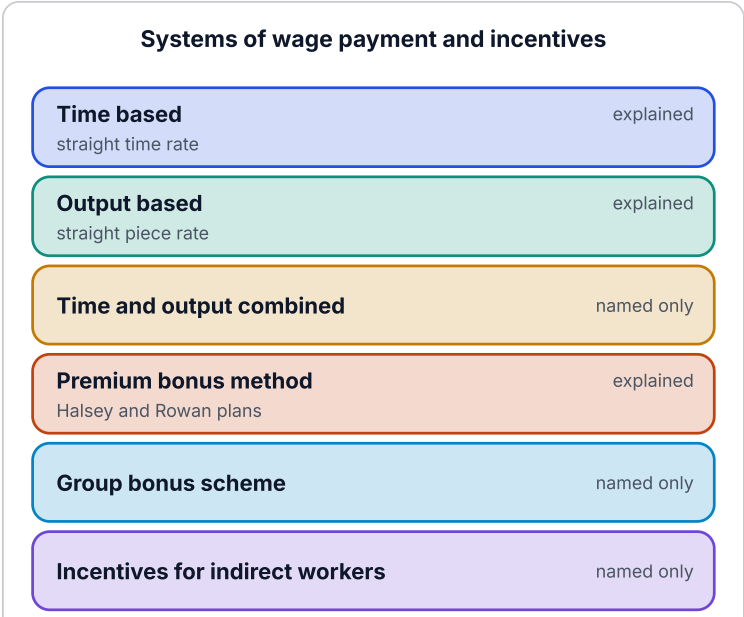
Key idea

Premium plans guarantee time wages and share the time saved. Halsey: 50% of the time saved, at the time rate. Rowan: (time saved ÷ time allowed) × time wages. Equal when time taken is 50% of time allowed; Rowan pays more for moderate efficiency, Halsey for high efficiency.

How it works

Six ways to pay, three to master

ICAI classifies wage systems under six heads (SM 3.27); the Study Material for January 2027 explains only the time rate, the piece rate and, under the premium bonus method, the Halsey and Rowan plans:



ICAI's six heads (SM 3.27). Only the time rate, the piece rate and the Halsey and Rowan premium plans are worked in this edition.

Straight time rate (SM 3.27–3.28): wages = time worked (including normal idle time) × rate for the time. It suits employees (i) whose services cannot be directly or tangibly measured, such as general helpers, supervisory

and clerical staff; (ii) engaged in highly skilled jobs; (iii) where the pace of output is independent of the operator, as in automatic chemical plants.

Straight piece rate (SM 3.28): each operation, job or unit is a piece with a piece rate; wages = number of units produced × rate per unit, independent of the time taken.

Premium bonus method (SM 3.28): a standard time is set for the job. The worker is guaranteed his day wages (except in the Barth system) if his output is below or up to standard; if he finishes in less than the standard time, the time saved is shared between employee and employer.

Halsey: half the saving

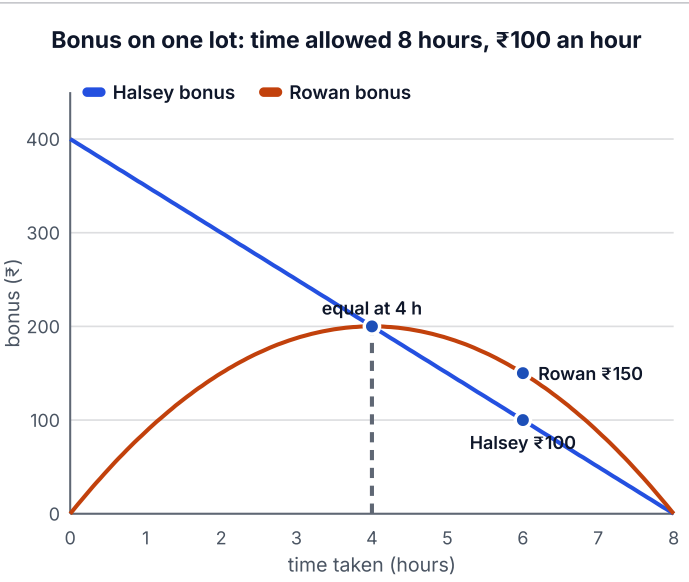
If there is no saving, the worker gets his day rate, even if he exceeds the standard time. If he saves time, he gets a bonus of 50% of the wages of the time saved; the employer keeps the other 50%. It is also called the Halsey fifty percent plan (SM 3.28).

Halsey: advantages	Halsey: disadvantages
Time rate is guaranteed, with a chance to earn more by producing more	Incentive not as strong as piece rate: the harder he works, the less he gets per piece
Equitable: the employer gets a direct return for improving methods and equipment	The sharing principle may not be liked by employees

Rowan: a share that shrinks as the saving grows

The bonus is that proportion of the time wages which the time saved bears to the time allowed (SM 3.29).

Rowan: advantages	Rowan: disadvantages
Fool-proof: a worker can never double his earnings, even with bad rate setting	A bit complicated
Encourages moderately efficient workers: better return for moderate efficiency than Halsey	Weak incentive at high production, where time saved is more than 50% of time allowed
The sharing principle appeals to the employer as equitable	The sharing principle is not generally welcomed by employees



Blue: Halsey bonus rises steadily as time taken falls. Orange: Rowan bonus is largest at 4 hours, half the time allowed, and falls on either side. Right of 4 hours (saving under 50%) Rowan pays more; left of it Halsey does. Shabnam's 6 hours are marked.

Which plan pays more

ICAI proves it (SM 3.33, Illustration 9): Halsey bonus = 50% × (SH − AH) × R and Rowan bonus = (AH ÷ SH) × (SH − AH) × R, so the two are equal when the time taken is 50% of the time allowed.

Time taken (AH) compared with time allowed (SH)	Pays more
AH more than 50% of SH (time saved less than 50%)	Rowan
AH exactly 50% of SH	Equal
AH less than 50% of SH (time saved more than 50%)	Halsey
AH equal to or more than SH	Neither: time wages only, guaranteed

Interactive version: PassBook app, lesson Time rate, piece rate and the Halsey and Rowan premium plans

Efficiency rating

Efficiency compares the time taken with the standard time: a worker taking no more than standard is efficient (SM 3.43). The procedure is: (1) determine standard time by time and motion or work study, for a heterogeneous group with contingency allowances; (2) record each worker's output and time taken; (3) compute the rating. It is needed where pay is by results, and for the employee requirement budget: Tapri's budget of 18,000 standard hours at an expected efficiency of 90% needs 18,000 ÷ 90% = 20,000 hours (SM 3.43–3.44, Example 1).

Employee productivity = standard time for doing actual work ÷ actual time taken. It improves by employing workers with the right skill, placing the right person in

the right job, training young and old workers, avoiding excess or shortage of employees, and work study for fixing wages and simplifying and standardising work (SM 3.44–3.45).

Formats and formulas

SH = time allowed (standard hours), AH = time taken (actual hours), R = time rate per hour.

∑ **Formula**

Time rate and piece rate

$$\text{Wages} = \text{Time worked} \times \text{Rate}$$

$$\text{Wages} = \text{Units} \times \text{Rate per unit}$$

(SM 3.28)

∑ **Formula**

Halsey premium plan

$$\text{Earnings} = AH \times R$$

$$+50\% \times (SH - AH) \times R$$

(SM 3.28)

∑ **Formula**

Rowan premium plan

$$\text{Earnings} = AH \times R \qquad + \frac{SH - AH}{SH} \times AH \times R$$

(SM 3.30)

∑ **Formula**

Effective hourly rate and cost per unit

$$\text{Effective rate} = \frac{\text{Total earnings}}{\text{Hours worked}}$$

$$\text{Cost per unit} = \frac{\text{Total earnings}}{\text{Units}}$$

∑ **Formula**

Efficiency

$$\text{Efficiency } (\%) = \frac{\text{Time allowed}}{\text{Time taken}} \times 100$$

Time allowed = time allowed as per standard for the work done (SM 3.43).

ICAI's comparative statement (RTP September 2026 Q6; the chapter's Practical Problem 2 gives the same figures scheme by scheme, SM 3.65–3.67). *Reproduce it from memory.*

Method	Hours worked	Earnings (₹)	Units	Cost per unit (₹)
Time rate	xx	xxx	xx	x.xx
Piece rate	xx	xxx	xx	x.xx
Halsey	xx	xxx	xx	x.xx
Rowan	xx	xxx	xx	x.xx

Worked examples

Worked example

One lot, three operators: what each scheme costs Tapri

A lot of 600 trays of samosas at Tapri's frying line is allowed 30 hours. Three operators, Rekha, Imran and Joseph, each fried one such lot in a week, taking 30, 24 and 12 hours. The time rate is ₹100 an hour, and the piece rate is set at the standard (20 trays an hour). Factory overheads are absorbed at ₹150 per hour worked.

Required:

(i) Calculate each operator's earnings and effective hourly rate under the time rate, the straight piece rate, the Halsey plan (50%) and the Rowan plan.

(ii) Calculate the conversion cost (wages plus overheads) per tray of each lot under each scheme.

(iii) Comment on the results from Tapri's point of view and from the operators'.

THE WORKING

Working notes

1. Piece rate = ₹100 ÷ 20 = **₹5 a tray**; piece earnings = 600 × ₹5 = ₹3,000 for every operator.
2. Time saved = 30 − time taken: Rekha nil, Imran 6 hours, Joseph 18 hours.
3. Halsey bonus = 50% × time saved × ₹100: nil, ₹300, ₹900.
4. Rowan bonus = time saved ÷ time allowed × time wages: nil; 6/30 × ₹2,400 = ₹480; 18/30 × ₹1,200 = ₹720.

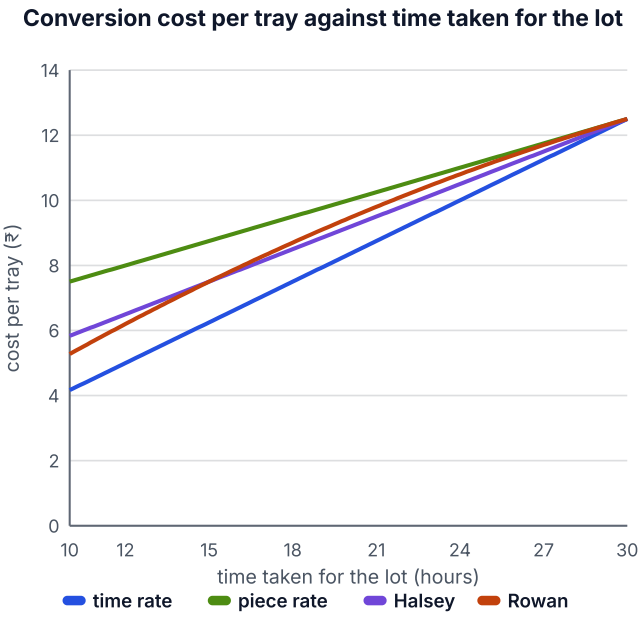
(i) Earnings (₹) and effective hourly rate (₹)

Scheme	Rekha (30 h)	Imran (24 h)	Joseph (12 h)
Time rate	3,000 / 100.00	2,400 / 100.00	1,200 / 100.00
Piece rate	3,000 / 100.00	3,000 / 125.00	3,000 / 250.00
Halsey	3,000 / 100.00	2,700 / 112.50	2,100 / 175.00
Rowan	3,000 / 100.00	2,880 / 120.00	1,920 / 160.00

Effective hourly rate = earnings ÷ hours actually worked.

(ii) Conversion cost per tray (₹) = (earnings + hours × ₹150) ÷ 600

Scheme	Rekha	Imran	Joseph
Overheads (30, 24, 12 hours × ₹150)	4,500	3,600	1,800
Time rate	12.50	10.00	5.00
Piece rate	12.50	11.00	8.00
Halsey	12.50	10.50	6.50
Rowan	12.50	10.80	6.20



All four schemes cost ₹12.50 a tray when the lot takes the 30 hours allowed. The quicker the lot, the cheaper each tray under every scheme, because overheads are absorbed per hour. Piece rate (green) passes the whole saving in wages to the operator; the premium plans share it; Rowan (orange) costs more than Halsey (violet) above 15 hours and less below it.

(iii) Comment. For Tapri every quick lot is cheaper, whichever scheme is used, because each hour saved also saves ₹150 of overheads: Joseph's lot costs ₹6.20 to ₹8.00 a tray against Rekha's ₹12.50. The time rate looks cheapest, but it gives the operators no reason to be quick: Joseph earns only ₹1,200 for the fastest lot. Piece rate gives the operators the entire gain in wages and is the

dearest scheme for quick lots. The premium plans share the gain. Imran, who took 80% of the time allowed, earns more under Rowan (₹2,880) than under Halsey (₹2,700); Joseph, who took 40%, earns more under Halsey (₹2,100) than under Rowan (₹1,920), since the two plans pay the same only when the time taken is half the time allowed.

Interactive version: PassBook app, lesson Time rate, piece rate and the Halsey and Rowan premium plans

Worked example

Two jobs, one painter: find the rates from the cost records

Gita, a painter at Tapri's kulhad workshop, is paid under the Rowan plan. Her job cards show only the time allowed, the time taken and the conversion cost (her

wages plus factory overheads, which are absorbed at a rate per hour actually worked):

Job
P: 400 festival kulhads
Time allowed
10 hours
Time taken
8 hours
Conversion cost (₹)
1,792

Job
Q: 1,000 corporate kulhads
Time allowed
20 hours
Time taken
12 hours
Conversion cost (₹)
2,976

Required:

(i) Compute Gita's normal wage rate per hour and the overhead absorption rate per hour.

(ii) Prepare a statement of the conversion cost of each job under the Rowan plan and as it would have been under the Halsey plan (50%), and comment.

THE WORKING

Step 1. Let R be the wage rate per hour and H the overhead rate per hour.

Step 2. Job P: wages = 8R + (10 − 8) ÷ 10 × 8R = 8R + 1.6R = 9.6R; overheads = 8H. So

9.6R + 8H = ₹1,792 ... (i)

Step 3. Job Q: wages = 12R + (20 − 12) ÷ 20 × 12R = 12R + 4.8R = 16.8R; overheads = 12H. So

16.8R + 12H = ₹2,976 ... (ii)

Step 4. (i) × 1.5 gives 14.4R + 12H = ₹2,688. Subtracting from (ii): 2.4R = ₹288, so **R = ₹120 an hour**. From (i): 8H = ₹1,792 − 9.6 × ₹120 = ₹640, so **H = ₹80 an hour**.

(ii) Statement of conversion cost

Particulars	Job P, Rowan (₹)	Job P, Halsey (₹)	Job Q, Rowan (₹)	Job Q, Halsey (₹)
Time wages (8 or 12 hours × ₹120)	960	960	1,440	1,440
Bonus	192	120	576	480
Overheads (8 or 12 hours × ₹80)	640	640	960	960
Conversion cost	**1,792**	**1,720**	**2,976**	**2,880**

Bonus: Rowan 2/10 × ₹960 and 8/20 × ₹1,440; Halsey 50% × 2 × ₹120 and 50% × 8 × ₹120.

Comment. Gita took 80% and 60% of the time allowed, more than half in both jobs, so the Rowan plan pays her more than Halsey would: ₹72 more on Job P and ₹96 more on Job Q. Overheads are the same under both plans, because they depend on the hours worked, not on the plan.

Worked example

Kesar's 20% promise: how many trays does each plan need?

Tapri's kulhad potters work 48 hours a week at a guaranteed ₹60 an hour. At their previous performance a tray of kulhads took 4 hours, and this is taken as the time allowed. To meet Diwali demand Kesar has promised that an incentive scheme will raise each potter's weekly earnings by 20%. Ritu must choose between the Halsey plan (50%) and the Rowan plan.

Required:

(i) Calculate the minimum number of trays a potter must make in a week for each plan to deliver the promised 20% rise.

(ii) Ritu expects 16 trays a potter a week. At that output, calculate each plan's weekly earnings and effective hourly rate, the potter's efficiency, and the labour cost per tray, and advise which plan keeps the promise.

(iii) At what weekly output would the two plans pay the same?

THE WORKING

Working note. Weekly time wages = 48 × ₹60 = **₹2,880**; the promise needs **₹3,456** (₹2,880 × 120%). Let SH be the time allowed for the week's output (trays × 4 hours).

(i) Minimum output

Plan
Halsey
Condition for a 20% rise
50% × (SH − 48) × ₹60 ≥ 20% × ₹2,880, so SH − 48 ≥ 19.2
Time allowed needed
67.2 hours
Trays needed
16.8, so **17**

Plan
Rowan
Condition for a 20% rise
(SH − 48) ÷ SH ≥ 20%, so SH ≥ 48 ÷ 0.8
Time allowed needed
60 hours
Trays needed
15

Under Rowan the bonus is (SH − 48) ÷ SH × time wages, so a 20% rise needs time saved of 20% of the time allowed.

(ii) At 16 trays a week (time allowed 64 hours, time saved 16 hours)

Particulars	Halsey	Rowan
Time wages (₹)	2,880	2,880
Bonus (₹): 50% × 16 × ₹60; 16/64 × ₹2,880	480	720
Weekly earnings (₹)	**3,360**	**3,600**
Rise over time wages	16.67%	25.00%
Effective hourly rate (₹): ÷ 48 hours	70.00	75.00
Labour cost per tray (₹): ÷ 16	210.00	225.00
Saving per tray against ₹240 (4 hours × ₹60) (₹)	30.00	15.00

Efficiency = time allowed ÷ time taken × 100 = 64 ÷ 48 × 100 = **133.33%**.

Advice. At the expected 16 trays only the Rowan plan keeps Kesar's promise (a rise of 25%); Halsey gives 16.67% and would need 17 trays. Ritu should adopt the **Rowan plan**: it meets the promise from 15 trays a week and still saves Tapri ₹15 a tray. Halsey would cost Tapri less, but it breaks the promise.

(iii) The plans pay the same when the time taken is 50% of the time allowed: SH = 96 hours, that is **24**

trays a week (efficiency 200%). Below that output Rowan pays more; above it Halsey does.

Exam angle

In the exam

Where these plans appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, the Halsey and Rowan plans were examined in 10 questions.

- **Numerical, 4–7 marks.** Earnings under both plans and which suits the worker: January 2025 Q2(b) (5 marks). Remuneration with time taken from clock times: January 2026 Q1(c) (4 marks). Effective hourly rates, increase in efficiency, saving in labour cost per unit and advice against a promised 15% rise for 128 workers: May 2025 Q2(b) (7 marks). Normal wage rate and material cost from equal factory costs: September 2025 Q2(b) (5 marks). Time rate, piece work, Rowan and Halsey for one operator, with a comparative statement: RTP September 2026 Q6.
- **Theory, 5 marks.** A suitable incentive scheme for a factory whose rate-setting department is new: January 2026 Q6(b).
- **Multiple choice, 2 marks.** Equal total earnings under Rowan and Halsey, find one worker's rate: September 2024 MCQ 13. Case scenario on

replacing overtime with a Rowan scheme: MTP September 2024 Series I.

In the exam

The traps ICAI set

- "Time saved" is SH − AH; Rowan multiplies time saved ÷ **time allowed** by **time wages** (AH × R), not by time saved × R.
- The promise test (May 2025 Q2(b)): compare total wages under each plan with present earnings × 1.15; only the plan that meets it is advisable.
- Efficiency is time allowed ÷ time taken (122.07% in May 2025 Q2(b)), not the other way round.
- Round only at the end: ICAI shows ₹1,38,854.40 and ₹35.424 an hour in full.

Trick

The 50% line

Work out AH ÷ SH first. Above 50%: Rowan pays more. Exactly 50%: equal, and Rowan's bonus is at its maximum. Below 50%: Halsey pays more. Conditions: same rate and same standard for both plans, and Halsey at 50%.

Answer frame

"Describe how wages are calculated under the Rowan and Halsey systems"

For each plan: the standard time and the guarantee; the bonus rule in ICAI's words; the formula; a two-line example; and one advantage and one disadvantage.

"Distinguish between the Halsey and Rowan plans"

Point	Halsey	Rowan
Bonus	50% of time saved × rate	Time saved ÷ time allowed × time wages
Worker's share of the saving	Fixed at 50%	Equals AH ÷ SH: falls as the saving grows
Highest bonus	Keeps rising with time saved	At time taken = 50% of time allowed
Earnings can double?	Yes: hourly earnings reach twice the rate once time taken is a third of time allowed or less (loose standards)	Never, even with bad rate setting (SM 3.30)
Better for	Highly efficient workers (saving above 50%)	Moderately efficient workers (saving below 50%)

Point	Halsey	Rowan
Simplicity	Simple	A bit complicated

"When is the time rate system preferred?"

ICAI's three cases (SM 3.27): services that cannot be directly or tangibly measured (helpers, supervisory and clerical staff); highly skilled jobs; output whose pace is independent of the operator (automatic chemical plants).

The numerical: three answers to the same question

Loses marks

"Rowan (Imran) = 24 × 100 + 6/30 × 6 × 100 = ₹2,520."

Time saved × rate in place of time wages: wrong formula, no marks for method.

Passes

"Rowan = ₹2,880, Halsey = ₹2,700."

Right figures, but no time allowed, no time saved, no formula.

Full marks

Working notes for time allowed and time saved, each formula written once in symbols, figures substituted, results with units, and the comparison or advice the question asks for.

Common mistakes

⚠ **Common mistake**

Hall of Shame: Rowan's bonus built on the time saved

Writing Rowan bonus = time saved ÷ time allowed × **time saved** × rate. For Imran that gives 6/30 × 6 × ₹100 = ₹120 instead of ₹480. **Fix:** the fraction multiplies the **time wages**, time taken × rate.

⚠ **Common mistake**

The negative bonus

When time taken exceeds time allowed, deducting a "bonus" from day wages. **Fix:** premium plans guarantee day wages; with no time saved the bonus is nil (SM 3.28).

⚠ **Common mistake**

The effective rate on the wrong hours

Dividing total earnings by the time allowed. **Fix:** the effective hourly rate is earnings ÷ hours actually worked (Joseph under Halsey: ₹2,100 ÷ 12 = ₹175, not ÷ 30 = ₹70).

⚠ **Common mistake**

Calendar hours for a clock-time question

Counting 1 June 9 a.m. to 4 June noon as 75 hours. **Fix:** count only working hours in each day.

Practice

Check yourself

1 Time allowed 10 hours, time taken 6 hours, rate ₹80 an hour. What are the earnings under the Rowan plan, in rupees?

Answer: _____

2 Time allowed 10 hours, time taken 4 hours, rate ₹80 an hour. What is the effective hourly rate under the Halsey plan (50%), in rupees?

Answer: _____

3 Case scenario. Tapri's cold-chain unit discovers that its new standard for a packing job is 12 hours, though the job really needs about 8. An operator paid ₹90 an hour finishes it in 4 hours. Under which plan does the loose standard cost Tapri less, and what does the operator earn under it?

- **A** Halsey: ₹720
- **B** Rowan: ₹600
- **C** Rowan: ₹720
- **D** Both plans: ₹600

4 Salma is on the Rowan plan and usually takes 75% of the time allowed. Tapri offers her a switch to the Halsey plan (50%), and she plans to speed up to 40% of the time allowed. Which statement is correct?

- **A** Rowan pays her more both now and after she speeds up
- **B** Halsey pays her more both now and after she speeds up
- **C** Rowan pays her more now; Halsey would pay her more after she speeds up
- **D** Both plans pay her the same, because her rate does not change

5 A worker's output needs 36 standard hours and she took 40 hours. What is her efficiency, in per cent?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Time rate: time worked × rate; for unmeasurable services, highly skilled jobs, operator-independent output.
- Piece rate: units × rate per unit, independent of time.
- Premium plans: standard time set, time wages guaranteed, time saved shared.
- Halsey: AH × R + 50% × (SH − AH) × R.
- Rowan: AH × R + (SH − AH) ÷ SH × AH × R; the worker can never double his earnings.
- Equal when AH = 50% of SH; Rowan more above it (moderate efficiency), Halsey more below it. Rowan's bonus is greatest at 50%.
- Efficiency % = time allowed ÷ time taken × 100. (Halsey halves the cake; Rowan slices it by time worked.)

Answer key · Check yourself

1 **672**

Time wages = 6 × ₹80 = ₹480; bonus = 4/10 × ₹480 = ₹192; earnings ₹672. Using time saved instead of time taken gives 4/10 × 4 × 80 = ₹128 and ₹608.

2 **140**

Earnings = 4 × 80 + 50% × 6 × 80 = ₹320 + ₹240 = ₹560; ₹560 ÷ 4 hours = ₹140 an hour. Rowan would give 320 + 6/10 × 320 = ₹512, less, because time taken is below 50% of time allowed.

3 **B** **Rowan: ₹600**

Halsey: 4 × ₹90 + 50% × 8 × ₹90 = ₹720. Rowan: 4 × ₹90 + 8/12 × 4 × ₹90 = ₹360 + ₹240 = ₹600. Rowan's bonus is a fraction of time wages and the hourly earnings can never reach twice the rate, so a mistake in rate setting costs less under it (SM 3.30).

4 **C** **Rowan pays her more now; Halsey would pay her more after she speeds up**

The plans pay the same when the time taken is 50% of the time allowed. At 75% (time saved 25%) her Rowan share of the saving is 75% against Halsey's 50%; at 40% (time saved 60%) her Rowan share falls to 40%, so Halsey pays more.

5 **90**

Efficiency = time allowed as per standard ÷ time taken × 100 = 36 ÷ 40 × 100 = 90%. Inverting it gives 111.11%.

4.5 Direct expenses

Costs other than direct material and direct employee cost that can be traced to one product, job or service: what they are, how they are measured, and where they sit in prime cost.

In plain words

The bills only one product ran up

Every Diwali Kesar launches a new product. This year it is a bottled Kesar-Elaichi Cold Chai, and the accounts team has a pile of bills that belong to nothing else.

> **Kesar:** Royalty to the chef who created the recipe, a bottling line hired for the season, the label design, a software licence for the QR codes. Put the lot in miscellaneous.
>
> **Ritu:** Each one exists only because of this chai. They are its direct expenses: part of its prime cost, next to the milk, the bottles and the line workers' wages.
>
> **Kesar:** And the courier who took one sample bottle to Mumbai for ₹300?
>
> **Ritu:** That one is traceable, but too small to matter. It can go to overheads.

A **direct expense** is a cost that is neither direct material nor direct employee cost but can still be traced to one cost object in an economically feasible manner: the royalty, the hire of a specific machine, a design made for

one product, software bought for one service. It is added to direct material and direct employee cost to give **prime cost**.

Two practical rules come with it. A **one-time or lump-sum** payment (a design, a licence) is spread over the output that will benefit from it, not charged to the first batch. And an amount too small to matter can be treated as overheads.

Think of it like

The party caterer's bill

For a wedding dinner the caterer buys the paneer (direct material) and pays the cooks (direct employee cost). He also hires a tandoor only for that wedding and pays a fee for a special dessert recipe: those are direct expenses of that dinner. The rent of his kitchen and his accountant's salary serve every function he caters: those are overheads.

Key idea

Direct expenses are the third part of prime cost: costs other than direct material and direct employee cost that are directly traceable, in an economically feasible way, to a product, job or service. Measure them net of rebates, with non-creditable taxes; spread lump sums over the output that benefits; treat trivial amounts as overheads.

How it works

What counts

ICAI's definition: expenses other than direct material cost and direct employee cost, which are incurred to manufacture a product or for provision of service and can be directly traced in an economically feasible manner to a cost object (SM 3.52). ICAI's examples, in its order (SM 3.52):

1. Royalty paid or payable for production or provision of service;
2. Hire charges paid for hiring **specific** equipment;
3. Cost for product or service specific design or drawing;
4. Cost of product or service specific software;
5. Other expenses directly related with the production of goods or provision of service.

The list is not exhaustive: any other expense directly attributable to the production or service is also a direct expense (SM 3.53). The Cost Sheet chapter gives a longer list, in this order (SM 6.4):

1. Cost of utilities such as power and fuel, steam etc.;
2. Royalty paid or payable for production or provision of service;
3. Hire charges paid for hiring specific equipment;
4. Fee for technical assistance and know-how;
5. Amortised cost of moulds, patterns, patents etc.;
6. Cost for product or service specific design or drawing;

7. Cost of product or service specific software;
8. Other expenses directly related with the production of goods or provision of service.

ICAI's answers also put job charges paid to job workers and the fee of a technician hired for the job under direct expenses (SM 3.67, Practical Problem 3; SM 6.17).

How it is measured

Direct expenses are measured at **invoice or agreed price net of rebate or discount**, but **including** duties and taxes for which input credit is not available, commission and other directly attributable costs (SM 3.53).

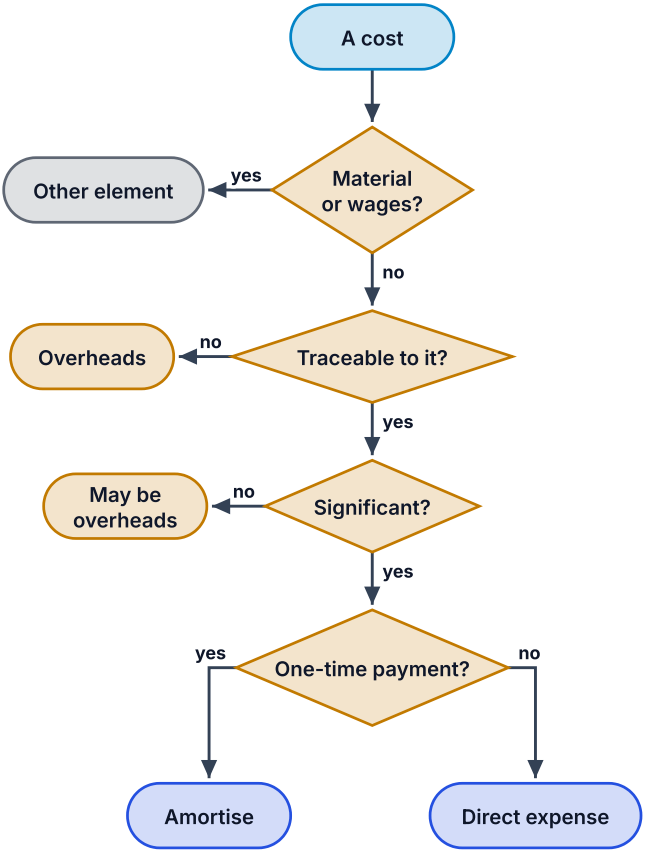
Sub-contracting. Goods made by job workers independent of the principal are measured at the agreed price. Where the principal supplies some materials to the job workers, the value of those materials and other incidental expenses are added to the job charges (SM 3.53).

Where it goes

Direct expenses form part of the **prime cost** of the product or service to which they can be traced (SM 3.53). Two exceptions:

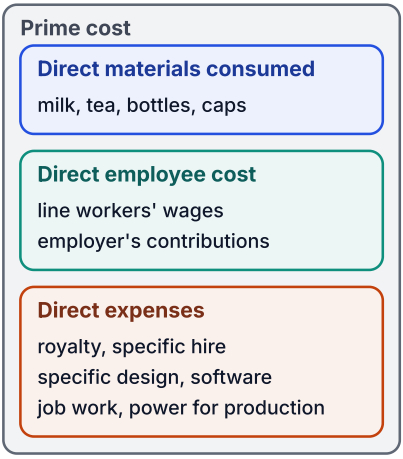
- **Lump-sum or one-time payment**: amortised over the estimated production volume or the benefit derived.
- **Insignificant amount** (not material): may be treated as part of overheads.

Is it a direct expense?



The tests of SM 3.52–3.53 in order, from the sky-blue start: is it direct material or direct employee cost (another element of prime cost); is it traceable to the cost object in an economically feasible manner; is it material in amount; is it a lump sum. The two blue end boxes are direct expenses, charged in full or amortised over the estimated output or benefit.

Where direct expenses sit



Prime cost = direct materials + direct employee cost + direct expenses. Factory overheads are added after it to give works cost: see the cost sheet.

The full cost sheet, from prime cost to cost of sales, is in the cost sheet for a manufacturer; the classification of costs by element is in elements and classification of cost.

Formats and formulas

Definition

Direct expenses

Expenses other than direct material cost and direct employee cost, which are incurred to manufacture a product or for provision of service and can be directly traced in an economically feasible manner to a cost object. (SM 3.52)

∑ **Formula**

Amortising a one-time payment

$$\text{Per unit} = \frac{\text{One-time payment}}{\text{Estimated output}}$$

Charge for the period = charge per unit × units produced in the period. Use the estimated benefit instead of units when the benefit is not measured in output (SM 3.53).

∑ **Formula**

Measuring a direct expense

$$\text{Cost} = \text{Price} - \text{Rebate}$$

$$+\text{Non-creditable taxes} + \text{Commission}$$

Price = invoice or agreed price; add any other directly attributable cost (SM 3.53).

ICAI's layout for a classification question (SM 3.54, Illustration 16). *Reproduce it from memory, with notes for every item left out.*

Calculation of direct expenses	Job A (₹)	Job B (₹)	Job C (₹)
Product blueprint cost	xxx	—	—
Hire charges for machinery	—	xxx	—
Licence fee for software	—	—	xxx
Total direct expenses	**xxx**	**xxx**	**xxx**

Notes: office and administration cost is overheads; salary of office attendants is office and administration cost; salary of the marketing manager is selling overheads (SM 3.54).

Worked examples

Worked example **The kulhad orders: checking a draft of direct expenses**

Tapri's kulhad unit completed three corporate orders in March: K1 (printed kulhads for an airline), K2 (a wedding order) and K3 (an artist's designer range). A trainee

prepared this list of the direct expenses he charged to the job cards:

No.
(1)

Item
Rent of the kulhad workshop, shared equally

Charged to
K1, K2, K3

Amount (₹)
60,000

No.
(2)

Item
Design of K1's logo stamp

Charged to
K1

Amount (₹)
45,000

No.
(3)

Item
GST on (2), for which input credit is available

Charged to
K1

Amount (₹)
8,100

No.
(4)

Item
Kiln hired for K2 and K3: used 20 hours for K2 and 40 hours for K3

Charged to
K2

Amount (₹)
54,000

No.
(5)

Item
Glazing by an outside job worker for K1: charges ₹30,000, glaze supplied by Tapri ₹8,000, freight to the job worker ₹2,000

Charged to
K1

Amount (₹)
40,000

No.
(6)

Item
Commission to the salesman who won K2

Charged to
K2

Amount (₹)
15,000

No.
(7)

Item
Courier of one sample to the airline

Charged to
K1

Amount (₹)
400

He did not charge the royalty of ₹6 a kulhad payable to the artist whose designs K3 uses; K3 made 8,000 kulhads. The glaze in (5) has already been charged to K1 as direct material.

Required:

(i) Examine each item and state its correct treatment.

(ii) Prepare the corrected statement of direct expenses for each job.

THE WORKING

(i) Examination of the draft

No.	Correct treatment
(1)	Wrong: the workshop serves every job and cannot be traced to one economically; rent is a factory overhead
(2)	Correct: designed for K1 alone and traceable to it

No.	Correct treatment
(3)	Wrong: GST for which input credit is available is not part of the cost (SM 3.53)
(4)	Wrong split: the hire is traceable to K2 and K3 by hours used: K2 ₹54,000 × 20 ÷ 60 = ₹18,000; K3 ₹36,000
(5)	Wrong amount: job work is measured at the agreed price plus the value of materials supplied and incidental expenses (SM 3.53), but the glaze is already in K1's direct material; adding it again counts it twice. Direct expense = ₹30,000 + ₹2,000 = ₹32,000
(6)	Wrong: commission to sales staff is a selling overhead, not a cost of making the kulhads
(7)	Traceable to K1 but insignificant: it may be treated as part of overheads (SM 3.53)
Royalty	Omitted: a royalty for production is a direct expense of K3: 8,000 × ₹6 = ₹48,000

(ii) Corrected statement of direct expenses

Particulars

Design of the logo stamp (2)

K1 (₹)

45,000

K2 (₹)

–

K3 (₹)

–

Particulars

Glazing job work: ₹30,000 + freight ₹2,000 (5)

K1 (₹)

32,000

K2 (₹)

–

K3 (₹)

–

Particulars

Hire of kiln, 20 and 40 hours (4)

K1 (₹)

–

K2 (₹)

18,000

K3 (₹)

36,000

Particulars

Royalty, 8,000 × ₹6

K1 (₹)

–

K2 (₹)

–

K3 (₹)

48,000

Particulars	Total direct expenses
K1 (₹)	77,000
K2 (₹)	18,000
K3 (₹)	84,000

The draft charged ₹1,13,500 to K1, ₹89,000 to K2 and ₹20,000 to K3 (₹2,22,500 in all); the corrected total is ₹1,79,000. The workshop rent goes to factory overheads, the commission to selling overheads and the courier charge to overheads; the GST is recovered as input credit.

Worked example

Cold Chai: measuring, amortising and building prime cost

In October, Tapri's new bottling unit produced 80,000 bottles of Kesar-Elaichi Cold Chai. Its records show:

No.	Item	Amount
(i)	Direct materials consumed	₹9,60,000
(ii)	Wages of bottling line workers	₹3,20,000
(iii)	Employer's contribution to PF on those wages	₹38,400

No.	Item	Amount
(iv)	Royalty to the recipe's creator	₹2 per bottle produced
(v)	Hire of a bottling line used only for this product: invoice ₹1,50,000 less 4% trade discount, plus GST at 18% on which input credit is available	
(vi)	Commission to the agent who arranged the hire	₹6,000
(vii)	Label and bottle-mould design for the product, paid once: ₹6,00,000 plus GST of ₹1,08,000 on which input credit is not available; expected to serve 11,80,000 bottles	
(viii)	Power for the bottling line, separately metered	₹42,000
(ix)	Rent of the factory building	₹80,000

Required:

(i) Calculate the direct expenses for October.

(ii) Prepare a statement of prime cost, in total and per bottle.

THE WORKING

Working notes

1. Royalty = 80,000 × ₹2 = **₹1,60,000**.
2. Hire of bottling line = ₹1,50,000 − 4% (₹6,000) = ₹1,44,000. GST is left out because input credit is available; the agent's commission of ₹6,000 is directly attributable and added: **₹1,50,000**.
3. Design is a one-time payment, amortised: (₹6,00,000 + ₹1,08,000 non-creditable GST) ÷ 11,80,000 bottles = ₹0.60 a bottle; October's charge = 80,000 × ₹0.60 = **₹48,000**.
4. Power for the line is separately metered, so traceable to this product: **₹42,000**.
5. Factory rent serves the whole building: factory overheads, not prime cost.

(i) Direct expenses for October

Particulars	Amount (₹)
Royalty (WN 1)	1,60,000
Hire charges of bottling line (WN 2)	1,50,000
Design, amortised (WN 3)	48,000
Power for the bottling line (WN 4)	42,000
Total direct expenses	**4,00,000**

(ii) Statement of prime cost for 80,000 bottles

Particulars	Total (₹)	Per bottle (₹)
Direct materials consumed	9,60,000	12.00
Direct employee cost (₹3,20,000 + ₹38,400)	3,58,400	4.48
Direct expenses	4,00,000	5.00
Prime cost	**17,18,400**	**21.48**

The factory rent of ₹80,000 joins the factory overheads below prime cost.

Exam angle

In the exam

Where direct expenses appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, direct expenses were examined in 8 questions, most often inside a cost sheet.

- **Classification on its own.** "Calculate direct expenses for the month" from power and fuel, factory wages, job workers' bill, royalty, technician's fee, administrative overheads and sales commission: RTP September 2026 Q7. The same idea as an MCQ (2 marks): MTP May 2026 Series I MCQ 11.

- **Inside cost sheets, 7–9 marks.** Hire of specific equipment at a rate per labour hour and fees for technical assistance as direct expenses, lease rent of production assets as factory overhead: January 2026 Q2(a). Job charges paid to job workers: January 2025 Q2(a). Royalty paid for production: MTP September 2024 Series II Q3(a) (7 marks). Consumable materials and power and fuel as direct expenses: RTP September 2026 Q8. Hire charges for a plant marked as indirect: RTP January 2025 Q5.
- **Case scenario MCQs.** Power, diesel for the generator and hire of machines used directly in production inside prime cost; car hire with GST under reverse charge without credit added to the cost: MTP May 2024 Series I, case scenario 2.

In the exam

The traps ICAI set

- "Specific equipment" or a job's own machine is direct; a plant used by all is overheads.
- GST or duty **with** input credit stays out; **without** credit it is part of the cost.
- Sales commission, marketing manager's salary and office attendants' salary are never direct expenses (SM 3.54, 3.67).
- A rate per hour or per unit (hire per labour hour, royalty per unit) is multiplied by the hours or units of the period.

Answer frame

"What are direct expenses? How are they measured and treated in cost accounts?"

1. **Definition** in ICAI's words (SM 3.52).
2. **Examples**: royalty; hire of specific equipment; product or service specific design or drawing; product or service specific software; other directly related expenses. The list is not exhaustive.
3. **Measurement**: invoice or agreed price net of rebate or discount, including duties and taxes without input credit, commission and other directly attributable costs; job work at agreed price plus materials supplied and incidental expenses.
4. **Treatment**: part of prime cost; lump sums amortised over estimated production or benefit; insignificant amounts as overheads.

A classification question

Answer in ICAI's layout: a table of the direct expenses by job or product, with the working of each figure (hours × rate, packs × royalty, job work plus materials), then numbered notes giving the treatment of every item left out ("administrative overhead is indirect expense"; "commission paid to sales staff comes under selling expenses"), as the Suggested Answers do. The notes carry marks: a bare total of direct expenses earns little.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the design charged to the launch month

Charging Cold Chai's whole ₹7,08,000 design to October's 80,000 bottles (₹8.85 a bottle). **Fix:** a one-time payment is amortised over the estimated production that benefits: ₹0.60 a bottle.

⚠ **Common mistake**

GST that will come back

Adding 18% GST to the bottling line's hire when input credit is available. **Fix:** only duties and taxes for which input credit is **not** available are part of the cost (SM 3.53).

⚠ **Common mistake**

Factory wages under direct expenses

Listing wages paid to factory workers as a direct expense because they are "direct". **Fix:** they are direct employee cost, the second element of prime cost, together with the employer's contributions on them (SM 3.67; RTP September 2026 Q8): see attendance, payroll and the elements of wages.

⚠ **Common mistake**

The sales commission that became prime cost

Treating commission paid to sales staff as a direct expense of the product sold. **Fix:** it is a selling overhead (SM 3.67); the commission that *is* part of a direct expense is one paid to acquire that expense, like the agent's fee for hiring the bottling line.

Practice

Check yourself

1 Tapri pays ₹3,60,000 once for a mould expected to make 2,40,000 kulhads. In November it makes 18,000 kulhads. What direct expense is charged for the mould in November, in rupees?

Answer: _____

2 A product-specific software licence is invoiced at ₹2,00,000 less a 5% rebate, plus a tax of 18% on the net amount for which input credit is not available. A commission of ₹4,000 was paid to the reseller. At what amount is the direct expense measured, in rupees?

Answer: _____

3 Case scenario. For Job 17, a hotel chain's private-label chai, Ritu finds these costs: wages of the blending staff, a royalty per pack to the hotel for its brand name, the salary of Tapri's marketing manager, and rent of the blending unit used for all products. Which of them is a direct expense of Job 17?

- **A** Wages of the blending staff
- **B** Royalty per pack paid to the hotel
- **C** Salary of the marketing manager
- **D** Rent of the blending unit

4 Which of these are part of the measured cost of a direct expense? Select all that apply.

- **A** Commission paid to acquire it
- **B** GST on it for which input credit is available
- **C** Duty on it for which input credit is not available
- **D** The trade discount allowed on the invoice

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Direct expenses: costs other than direct material and direct employee cost, incurred for a product or service and traceable to it in an economically feasible manner.
- ICAI's examples: royalty; hire of specific equipment; product- or service-specific design or drawing; specific software; other directly related expenses (not exhaustive). The Cost Sheet chapter adds utilities such as power and fuel, technical know-how fees and amortised moulds and patterns (SM 6.4).
- Measured at invoice or agreed price net of rebate or discount, plus taxes without input credit, commission and other directly attributable costs.
- Job work: agreed price, plus materials supplied by the principal and incidental expenses.
- Part of prime cost; one-time payments amortised over estimated production or benefit; insignificant amounts may be overheads.
- Not direct expenses: factory wages (employee cost), administration, sales commission, marketing salaries, general plant hire. (If only this product ran up the bill, it is a direct expense.)

Answer key · Check yourself

1 **27000**

₹3,60,000 ÷ 2,40,000 = ₹1.50 a kulhad; 18,000 × ₹1.50 = ₹27,000. Charging all ₹3,60,000 to November is the launch-month mistake.

2 **228200**

₹2,00,000 − ₹10,000 = ₹1,90,000; tax ₹34,200 (no credit, so included); commission ₹4,000: ₹2,28,200. Leaving the tax out (₹1,94,000) would be right only if input credit were available.

3 **B Royalty per pack paid to the hotel**

Royalty for production is ICAI's first example of a direct expense (SM 3.52). The wages are direct employee cost, the marketing manager's salary is a selling overhead, and the rent serves every product, so it is a factory overhead.

4 **A Commission paid to acquire it; C Duty on it for which input credit is not available**

Measured at invoice or agreed price net of rebate or discount, including duties and taxes without input credit, commission and other directly attributable costs (SM 3.53).

CHAPTER 5

Overheads: Absorption Costing

5.1 Classifying overheads, allocation and apportionment

What overheads are, the four ways ICAI classifies them, and the five-stage journey from estimate to absorption, with allocation and apportionment told apart.

In plain words

The ₹15.78 lakh heap called "miscellaneous"

Tapri's samosa factory in Indore spent ₹15.78 lakh last month on things that are not potatoes, flour or the wages of the people shaping samosas.

> **Kesar:** Rent, gas, the supervisor, the canteen... put it all under "miscellaneous". Sab adjust ho jayega.
>
> **Ritu:** Per unit, please. How much of this heap belongs to one pack of samosas? "Miscellaneous" cannot tell me.

The potatoes in a pack can be weighed and charged to that pack. The factory rent cannot: no pack "uses" ₹3 of rent that anyone can point to. Costs like the rent, the supervisor's salary and the fryer lubricant are **overheads**: costs that cannot be conveniently traced to

any one unit of output, because they are incurred for output in general.

Ritu sorts the heap twice.

First, **by kind**: what the cost is for (the factory, the office, selling or delivery), how it moves with output (fixed, variable or in between), what it is made of (indirect material, indirect employee cost, indirect expenses) and whether a manager can control it.

Second, **by place**: she gives each cost to a department. A cost that belongs wholly to one department goes there in full: the gas read from the Frying department's own meter is **allocated** to Frying. A cost that several departments share is split in fair proportions: the factory rent is **apportioned** by the floor area each department occupies. Then the costs of the departments that only help (Stores, Maintenance) are passed on to the departments that make the product (**re-apportionment**), and finally each production department's total is charged to the packs it makes (**absorption**).

Think of it like

The flat-share bills

Four friends share a flat. Each one's mobile bill is that person's alone: allocation. The rent is split by the size of each room, and the cook's salary by how many meals each eats: apportionment, each on its own fair basis. A bill nobody can trace gets split anyway, because someone has to pay it; the only question is how fairly.

Key idea

A cost is **allocated** when the whole of it belongs to one cost centre, and **apportioned** when it is shared and must be split on a fair basis. The link between the expense and the cost centre decides which, not the kind of expense: power on its own meter is allocated, power on a common meter is apportioned.

How it works

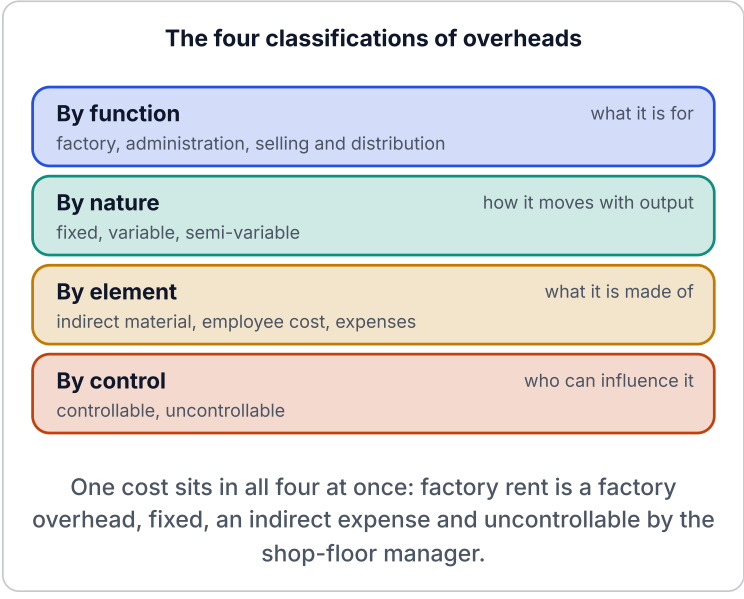
What counts as an overhead

Overheads are the expenditure which cannot be conveniently traced to or identified with any particular cost unit; they are incurred for output generally and not for a particular work order, e.g. wages of watch and ward staff, heating and lighting of the factory (SM 4.2). They are the aggregate of **indirect material**, **indirect employee cost** and **indirect expenses** (SM 1.26), and they arise in the factory, in administration and in selling and

distribution. In a manufacturing concern they often exceed direct wages or direct materials, and at times both together, so ignoring them is "a grave mistake" (SM 4.2).

Four ways to sort the heap

ICAI classifies overheads four ways, in this order (SM 4.3–4.5):



By function

Class

Factory (manufacturing, production)

ICAI's description

Indirect cost of manufacturing, from procurement of materials to completion of the finished product

ICAI's examples

Stock keeping, repairs of plant, depreciation of factory building, indirect labour, cost of primary packing, insurance of plant

Class

Office and administrative

ICAI's description

General management and administration, not related directly to production, selling, distribution or R&D

ICAI's examples

Office salaries, office repairs and depreciation, postage and stationery, lease rental on operating lease, accounts and audit

Class

Selling and distribution

ICAI's description

Selling: sale of products and sales management. Distribution: making the product available for sale in the market

ICAI's examples

Salesmen's commission, advertisement, sales office; delivery vans, transit insurance, warehouse and cold storage, secondary packing

By nature

Class	ICAI's description	ICAI's examples
Fixed	Incurred for a period; within output and turnover limits, unaffected by changes in activity	Salary of permanent employees, depreciation, interest on capital, insurance
Variable	Vary with the volume of activity	Indirect materials, power and fuel, lubricants, tools and spares
Semi-variable	Contain fixed and variable components	Electricity, water, telephone and internet

By element

Class	ICAI's description	ICAI's examples
Indirect materials	Do not normally form part of the finished product	Lubricants, cotton waste, stores of service departments
Indirect employee cost	Cannot be allocated but can be apportioned or absorbed	Foreman's and supervisor's salary, administration staff
Indirect expenses	Cannot be directly, conveniently and wholly allocated to cost centres	Rates and taxes, insurance, depreciation, advertisement

By control

Class
Controllable

ICAI's description
Can be controlled by appropriate managerial influence and policies

ICAI's examples
Materials, wages and salary, power and fuel

Class
Uncontrollable

ICAI's description
Cannot be controlled even with appropriate managerial influence

ICAI's examples
Rates and taxes, depreciation, interest on borrowings

Why split fixed from variable

The primary objective of splitting semi-variable expenses into fixed and variable is to ascertain **marginal costs**. It also helps in (SM 4.6–4.7):

1. **Controlling expenses**: fixed costs are policy costs, hard to reduce; variable costs are controlled by allowances related to the output achieved.
2. **Preparation of budget estimates**: the flexible budget estimates costs at each level of activity (worked in Example 2 below).
3. **Decision making**: pricing in a recession or for export, make or buy, shut down or continue. Marginal costing,

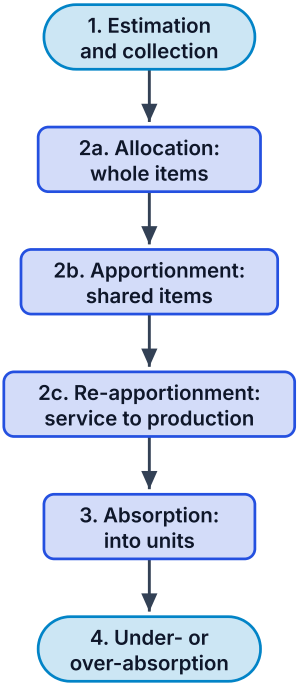
CVP and break-even analysis all rest on this split (marginal cost and contribution).

Four steps, five stages, from estimate to unit

Actual overheads are known only after the accounts close, but cost sheets are needed now. So overheads are estimated in advance and charged to products at **pre-determined rates**, and the difference is adjusted at year end (SM 4.7–4.8).

The Study Material tells this journey twice. Its procedure for manufacturing overheads has **four steps**: estimation and collection, assignment, absorption, and treatment of under- or over-absorption (SM 4.8–4.10). Its chart of the steps for the distribution of overheads has **five stages**: estimation, allocation, apportionment, re-apportionment and absorption (SM 4.11). Same road: the chart splits assignment into its three parts and stops before the year-end adjustment.

The journey of a factory overhead



Allocation takes whole items to one department, apportionment splits shared items on a fair basis, re-apportionment moves service departments' totals to production departments, and absorption charges them to units at a pre-determined rate. Stages 2a and 2b together are the primary distribution; 2c is the secondary distribution. ICAI groups 2a to 2c as assignment.

1 **Estimation and collection.** Estimate overheads from past figures adjusted for known future changes. Sources: (a) invoices, (b) stores requisitions, (c) the wage analysis book, (d) journal entries (SM 4.8). Collection is usually through **standing orders**: a sanction for indirect expenses under various heads, each head given a number that continues from year to year (SM 4.11–4.12).

2 **Assignment.** The guiding principle is the traceability of the overhead in an economically feasible manner, on one of two principles: **cause and effect** (the cause is the process, operation or activity, the effect is the incurrence of cost) or **benefit received** (overheads apportioned in proportion to the benefit each cost object receives) (SM 4.8). Assignment has three parts: (a) **allocation**, (b) **apportionment**, (c) **re-apportionment** (primary distribution does the first two, secondary distribution the third).

3 **Absorption.** Each production department's overheads are recovered from its output. Variable manufacturing overheads are absorbed on actual production; fixed manufacturing overheads on normal capacity (SM 4.10). Rates and methods: absorption rates and the machine hour rate.

4 **Under- or over-absorption.** There is bound to be some difference between the actual and the absorbed overheads; it is adjusted keeping in view its reasons (SM 4.10) (under- and over-absorption).

The whole exercise serves two purposes: (a) to charge products with an **equitable portion** of factory overheads, and (b) to charge them **immediately** as a job is completed, without waiting for actual figures (SM 4.10).

Departments first: departmentalisation

A factory has **production departments**, which make the product, and **service departments**, which do not take part in production directly but provide auxiliary services to other cost centres: engineering, quality control, laboratory, canteen, stores, time office, dispensary (SM 4.9). Collecting overheads department by department, or by smaller **cost centres** (machine or personnel), is **departmentalisation** (cost centres). Its advantages (SM 4.12–4.13):

1. **Better estimation of expenses**: departmental expenses can be estimated almost exactly.
2. **Better control**: you can see which department overspent.
3. **Ascertainment of cost for each department**: a job is charged only for the departments it passes through.
4. **Suitable method of costing**: e.g. batch costing in a parts department, single output costing in assembly.

Formats and formulas

Write the definitions in ICAI's words; they are asked as MCQs and as "distinguish" questions.

Definition

Overheads

Overheads represent expenses that have been incurred in providing certain ancillary facilities or services which facilitate or make possible the carrying out of the production process; by themselves these services are not of any use. (SM 4.63)

Definition

Cost allocation

The term "allocation" refers to assignment or allotment of an entire item of cost to a particular cost center or cost unit. (SM 4.63)

Definition

Cost apportionment

Apportionment implies "the allotment of proportions of items of cost to cost centres or departments". (SM 4.9)

Definition

Re-apportionment

The process of assigning service department overheads to production departments is called reassignment or re-apportionment. (SM 4.9)

Definition

Absorption

The process of recovering overheads of a department or any other cost center from its output is called recovery or absorption. (SM 4.10)

Formula

Budgeted overhead at a new output

$$\text{Budget} = F + SF + (V + SV) \times \frac{Q_1}{Q_0}$$

F = fixed overheads; SF = fixed part of the semi-variable overheads; V = variable overheads and SV = variable part of the semi-variable overheads, both at the old output; Q_0 = old output, Q_1 = new output. Split the semi-variable cost first; only the variable parts move with output (SM 4.6–4.7). Splitting methods are in cost behaviour.

Worked examples

Worked example **Kesar's "miscellaneous" ledger**

The samosa division's "miscellaneous" head for March holds these items:

Item	Amount (₹)
Rent of the factory building	1,20,000
Lubricants and cotton waste for the fryers	8,000
Salary of the factory supervisor	45,000
Printed pouches in which each pack is sealed	64,000
Corrugated cartons for dispatch to distributors	38,000
Salaries of head-office accounts staff	90,000
Audit fees	25,000
Salesmen's commission	52,000
Instagram campaign for the peri-peri samosa	75,000
Delivery van diesel and driver	41,000
Transit insurance	6,000
Cold storage rent at the city warehouse	30,000

Item	Amount (₹)
Depreciation of the office building	18,000
Directors' remuneration	1,00,000
Diwali gift tins designed to attract buyers	22,000
Rent of the sales office	20,000

Required: Calculate the (i) production, (ii) administration, (iii) selling and (iv) distribution costs.

THE WORKING

(i) Production cost

Item	Amount (₹)
Rent of the factory building	1,20,000
Lubricants and cotton waste (indirect material)	8,000
Salary of the factory supervisor	45,000
Printed pouches (primary packing)	64,000
Total	**2,37,000**

(ii) Administration cost

Item	Amount (₹)
Salaries of head-office accounts staff	90,000
Audit fees	25,000
Depreciation of the office building	18,000
Directors' remuneration	1,00,000
Total	**2,33,000**

(iii) Selling cost

Item	Amount (₹)
Salesmen's commission	52,000
Instagram campaign (advertisement)	75,000
Diwali gift tins (fancy packing)	22,000
Rent of the sales office	20,000
Total	**1,69,000**

(iv) Distribution cost

Item	Amount (₹)
Corrugated cartons for dispatch (secondary packing)	38,000
Delivery van diesel and driver	41,000
Transit insurance	6,000
Cold storage rent at the city warehouse	30,000
Total	**1,15,000**

Check: 2,37,000 + 2,33,000 + 1,69,000 + 1,15,000 = ₹7,54,000, the whole ledger.

Notes on the three packing items (SM 4.60): primary packing, which protects the product or makes it convenient to handle, is part of production cost; packing to carry the product from factory to customer is distribution cost; fancy packing to attract customers is advertising, so a selling overhead.

Worked example

Kesar budgets April by "just adding 25%"

In March the samosa factory packed 80,000 packs, and

its overheads were:

Overhead	Amount (₹)
Fixed	6,00,000
Variable	4,80,000
Semi-variable (30% fixed)	3,00,000
Total	**13,80,000**

April's output is expected to be 1,00,000 packs, 25% more, so Kesar's budget simply adds 25% to everything: ₹13,80,000 × 1,00,000 ÷ 80,000 = ₹17,25,000.

Required:

(i) Prepare the overhead budget for April.

(ii) Calculate the overhead per pack in March and in April, and comment on Kesar's figure.

THE WORKING

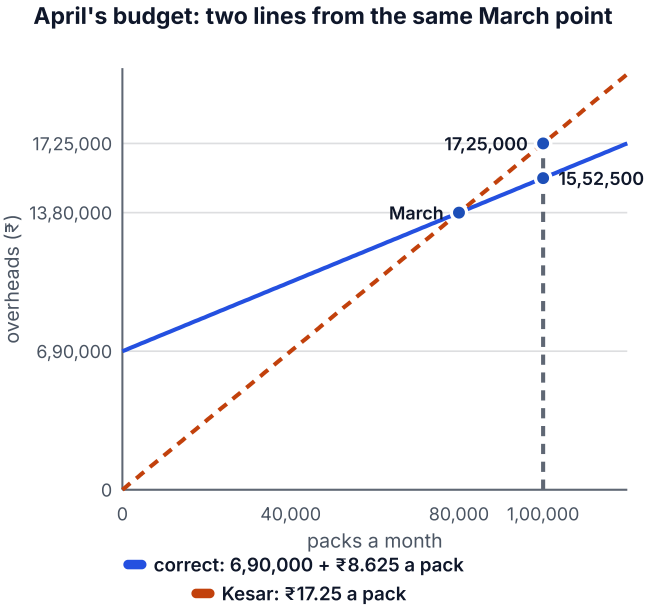
Working note: split the semi-variable cost. Fixed part = 30% × ₹3,00,000 = ₹90,000; variable part = ₹2,10,000.

(i) Overhead budget for April (1,00,000 packs)

Overhead	Working	Amount (₹)
Fixed	unchanged	6,00,000
Variable	4,80,000 × 1,00,000 ÷ 80,000	6,00,000
Semi-variable, fixed part	unchanged	90,000
Semi-variable, variable part	2,10,000 × 1,00,000 ÷ 80,000	2,62,500
Total		**15,52,500**

(ii) Overhead per pack: March ₹13,80,000 ÷ 80,000 = **₹17.25**; April ₹15,52,500 ÷ 1,00,000 = **₹15.53** (15.525).

Comment. Only the variable parts grow with output; the ₹6,90,000 of fixed overhead is spread over more packs, so the cost per pack falls. Kesar's ₹17,25,000 treats every rupee as variable and overstates the budget by ₹1,72,500: the "wrong budgeting" the Study Material warns about (SM 4.7).



Blue: the budget line with the fixed ₹6,90,000 kept fixed; orange dashed: Kesar's line through the origin. They meet at March's 80,000 packs and part at April's 1,00,000.

Worked example

Allocate or apportion? Eight items in the samosa factory

The factory has two production departments, Frying and Packing, and two service departments, Stores and Maintenance. For each item, state whether it is allocated or apportioned, and to which department or on what kind of basis.

1. Gas for the fryers, read from Frying's own gas meter.

2. Salary of the works manager, who runs the whole factory.
3. Depreciation of the pouch-sealing machine installed in Packing.
4. Rent of the factory building, which houses all four departments.
5. Wages of the storekeepers.
6. Electricity for lighting, on one meter for the whole factory.
7. Electricity for lighting the Packing hall, which has its own sub-meter.
8. Subsidy to the canteen used by workers of all four departments.

MODEL ANSWER

Item	Treatment	Department or basis
1. Gas, own meter	Allocated	Frying
2. Works manager's salary	Apportioned	Technical estimate of time (managerial salaries, SM 4.14)
3. Depreciation of the sealing machine	Allocated	Packing

Item	Treatment	Department or basis
4. Factory rent	Apportioned	Floor area (or volume)
5. Storekeepers' wages	Allocated	Stores, a service department
6. Lighting, common meter	Apportioned	Light points, area or metered units
7. Lighting, Packing's sub-meter	Allocated	Packing
8. Canteen subsidy	Apportioned	Number of workers

Items 6 and 7 are the same kind of expense with opposite treatment: as the Study Material puts it, the allocation or apportionment of an expense is not dependent on its nature, but on the relationship between the expense and the cost centre (SM 4.16). The bases are chosen in primary distribution.

Exam angle

In the exam

How ICAI asks the classification and the stages

In the documents read for this lesson:

- **MCQs**: the chapter's own Test Your Knowledge asks which term means allotment of whole items of cost (cost allocation), where primary packing belongs (production cost), what directors' remuneration is (administration overhead) and which is not a functional class (fixed overhead) (SM 4.65).
- **Classify a list, 6 marks**: MTP January 2025 Series II Q3(b) gave twelve items and asked for administration, selling and distribution cost separately; the answer is three short tables, one per function.
- **Theory**: "Distinguish between cost allocation and cost absorption" (RTP November 2023, old scheme, Q14(e); the chapter's question 4); "Discuss the difference between allocation and apportionment" (the chapter's question 7); controllable versus uncontrollable costs (MTP May 2024 Series I Q6(c), 4 marks).
- In cost sheets, primary packing was added to arrive at the cost of production (MTP January 2025 Series I, Q3(a)).

In the exam

Traps

- A cost with its own meter, its own machine or its own staff is allocated, whatever its nature.
- Secondary packing is distribution; fancy packing is selling; primary packing is production.
- A service department's own wages (storekeepers, maintenance fitters) are allocated to that service department, never treated as direct wages of a product.

Trick

A-for-All, P-for-Part

Allocation takes **a**ll of one item to one centre; a**P**portionment takes **p**arts of one item to several centres. Works whenever the question gives a single expense; for a list, decide item by item.

Answer frame

Distinguish between allocation and apportionment (SM 4.16): write it as a table, then the two points below it.

Point	What is charged
Allocation	Whole items of cost identifiable with one department (e.g. indirect wages of each department obtained separately)
Apportionment	Proportions of an item of cost (e.g. the cost of a service shared by the departments that avail it)

Point	Nature of the process
Allocation	A direct process of charging expenses to cost centres
Apportionment	An indirect process: the appropriate portion borne by each department must be identified

- The relationship between the expense and the cost centre, not its nature, decides whether an expense is allocated or apportioned.
- Allocation is a much wider term than apportionment.

Distinguish between cost allocation and cost absorption: allocation is the allotment of a whole item of cost to a cost centre or cost unit; absorption is the process of absorbing all overheads allocated to or apportioned over a cost centre or production department into the units produced. Allocation gets costs **into** cost centres; absorption gets them **out**, into products.

Advantages of departmentalisation (four points, SM 4.12–4.13) and **advantages of separating fixed and variable overheads** (the objective plus three points, SM

4.6–4.7) are written as numbered points with one line each, in ICAI's order.

Common mistakes

⚠ **Common mistake**

Hall of Shame: "power is always apportioned"

Students learn "power: horse power × machine hours" and apply it even when a department has its own meter. **Fix:** check the facts first; a metered or exclusively used cost is allocated in full, and only the shared remainder is apportioned.

⚠ **Common mistake**

The 25% that ate the budget

Increasing every overhead in proportion to output, as Kesar did, overstated April by ₹1,72,500. **Fix:** split semi-variable costs, keep the fixed parts fixed, and move only the variable parts.

⚠ **Common mistake**

Cartons in the factory, pouches on the truck

Primary and secondary packing swapped. **Fix:** ask what the packing does. Protects or holds the product: production. Carries it to the customer: distribution. Attracts the customer: selling.

Practice

Check yourself

1 Which of these classifies overheads by their behaviour (nature)?

- **A** Factory, administration, selling and distribution
- **B** Fixed, variable and semi-variable
- **C** Indirect material, indirect employee cost and indirect expenses
- **D** Controllable and uncontrollable

2 The Pune café's electricity was ₹50,000 last month for 20,000 cups, 40% of it fixed. What is the budgeted electricity for 25,000 cups (₹)?

Answer: _____

3 Case scenario. Tapri's cold-chain hub has three departments under one roof. The walk-in cold room in the Dairy section has its own electricity sub-meter, which read ₹18,000 for May; the rest of the hub's electricity, ₹42,000, is on the main meter. Ritu must charge Dairy for electricity. Which treatment is correct?

- **A** Apportion ₹60,000 among the three departments by floor area
- **B** Allocate ₹18,000 to Dairy and apportion ₹42,000 among the three departments
- **C** Allocate ₹60,000 to Dairy, the biggest user
- **D** Apportion ₹18,000 among the three departments by horse power

4 Which of these are uncontrollable overheads in ICAI's examples? Select all that apply.

- **A** Rates and taxes
- **B** Power and fuel
- **C** Interest on borrowings
- **D** Depreciation

More than one answer may be right.

5 In April Tapri's premix plant spent ₹40,000 on the sealed pouches each pack is filled into, ₹25,000 on gift sleeves designed to attract buyers, ₹30,000 on corrugated cartons to carry packs to distributors, and ₹12,000 on wooden crates that an export customer asked for on its order. How much of this is distribution overhead (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Overheads: costs not conveniently traceable to a cost unit; the aggregate of indirect material, indirect employee cost and indirect expenses.
- Four classifications, in ICAI's order: function (factory, administration, selling and distribution), nature (fixed, variable, semi-variable), element, control.
- Splitting fixed and variable serves marginal costing, then control, budget estimates and decisions; budget only the variable parts in proportion to output.
- Stages: estimation and collection (standing orders) → allocation → apportionment → re-apportionment → absorption → under- or over-absorption.
- Allocation: whole item to one cost centre; apportionment: proportions of an item on a fair basis (cause and effect or benefit received).

- The relationship with the cost centre decides allocation or apportionment; allocation is the wider term.
- Variable manufacturing overheads are absorbed on actual production, fixed ones on normal capacity.
- Primary packing: production; secondary: distribution; fancy: selling. ("Miscellaneous" is not a function.)

Answer key · Check yourself

1 **B Fixed, variable and semi-variable**

Behaviour means how a cost responds to output: fixed, variable, semi-variable. The first option is by function, the third by element, the fourth by control.

2 **57500**

Fixed ₹20,000 stays; variable ₹30,000 × 25,000 ÷ 20,000 = ₹37,500. Total ₹57,500. Scaling the whole ₹50,000 by 25% gives ₹62,500.

3 **B Allocate ₹18,000 to Dairy and apportion ₹42,000 among the three departments**

A sub-metered cost belongs wholly to one cost centre, so it is allocated; only the common-meter amount is shared. The relationship with the cost centre, not the nature of electricity, decides.

4 **A** Rates and taxes; **C** Interest on borrowings; **D** Depreciation

ICAI lists rates and taxes, depreciation and interest on borrowings as uncontrollable, and materials, wages and salary, and power and fuel as controllable (SM 4.5).

5 **30000**

Only the transport cartons are distribution overhead (₹30,000). The pouches are primary packing, part of production cost; the gift sleeves are fancy packing, a selling (advertising) overhead; special packing at a customer's request is charged to that customer's order (SM 4.60).

5.2 Primary distribution of factory overheads

How every factory overhead reaches a department, production or service: allocate whole items, apportion shared ones on ICAI's bases, and total the primary distribution summary.

In plain words

Equal shares, said Kesar

Ritu has the samosa factory's overhead budget for a month: ₹15.78 lakh, already sorted by kind (classifying overheads). Now each rupee must reach a department. The factory has two **production departments**, Frying and Packing, and two **service departments**, Stores and Maintenance, which only help the other two.

> **Kesar:** Divide by four. Equal shares, nobody fights.
>
> **Ritu:** Frying has ten fryers that run day and night; Packing has sixty workers and one sealing machine. Equal shares would make Packing pay Frying's power bill.

So Ritu goes item by item. Some items belong wholly to one department: the storekeepers' wages go to Stores, the fryer lubricants to Frying. Those are **allocated**. The rest are shared, and for each one she picks a yardstick that measures how much each department benefits from it or causes it: rent by **floor area**, supervision by **number of workers**, depreciation of plant by **capital value**, machine

power by **horse power × machine hours**. Each shared item is split in the ratio of its own yardstick: that is **apportionment**.

When every item has been placed, she adds up each department's column. The result, every factory overhead sitting in a production or service department, is the **primary distribution**. The service departments' totals will be passed on to Frying and Packing in the next step, the secondary distribution.

Think of it like

The housing society's maintenance bill

A housing society bills each flat on several yardsticks at once: water by the flat's own meter, security by the number of flats, the garden by carpet area, the lift by the floors it serves. Nobody would split the water bill by carpet area. Each bill is shared by whatever measures who uses it, and that is exactly how each factory overhead finds its basis.

Key idea

Primary distribution charges every factory overhead to every department that benefits from it, **production and service alike**: whole items by allocation, shared items by apportionment, each shared item on the basis that best measures the benefit received or the cause of the cost.

How it works

Five steps to a primary distribution summary

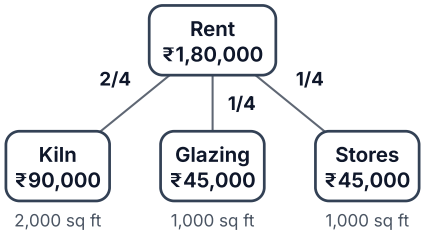
1. **Lay out the data**: the overheads, and the facts about each department the bases need (floor area, workers, capital values, horse power, machine hours, light points).

2. **Allocate** every item identifiable with one department, service departments included: the storekeepers' wages and the stores they consume go to Stores. Direct materials and direct wages of the production departments are prime cost, not overheads: they stay out.

3. **Choose a basis** for each shared item, on the principle of benefit received or cause and effect (SM 4.8), using ICAI's bases below.

4. **Reduce the basis to a ratio**, over only the departments that share the cost, and split the item: share = item × the department's units of the base ÷ the total units.

5. **Total each column**. A department's primary total = its allocated overheads + its apportioned overheads. Check that the columns add up to the total overheads.



Tapri's kulhad unit: rent apportioned on floor area, 2,000 : 1,000 : 1,000 = 2 : 1 : 1. Each department's share is the rent times its fraction of the total area.

ICAI's bases of apportionment

The same basis cannot be used for every item, because the benefit of each service to a department has to be measured differently (SM 4.13). The Study Material suggests these (SM 4.14):

Overhead cost	Basis of apportionment
1. Rent and other building expenses; lighting and heating (conditioning); fire precaution service; air-conditioning	Floor area, or volume of department
2. Perquisites; labour welfare expenses; time keeping; personnel office; supervision	Number of workers
3. Compensation to workers; holiday pay; ESI and PF contribution; perquisites	Direct wages

Overhead cost	Basis of apportionment
4. General overhead	Direct labour hours, or direct wages, or machine hours
5. Depreciation of plant and machinery; repairs and maintenance of plant and machinery; insurance of stock	Capital values
6. Power or steam consumption; internal transport; managerial salaries	Technical estimates
7. Lighting expenses	Number of light points, or area, or metered units
8. Electric power (machine operation)	Horse power of machines, or number of machine hours, or value of machines, or units consumed
9. Material handling; stores overhead	Weight, volume, value or units of materials

Perquisites appear in two rows, under number of workers and under direct wages: pick one and state which you use.

Interactive version: PassBook app, lesson Primary distribution of factory overheads

Beyond benefit received: three other bases

The benefit received by the department is generally the principal criterion, but the Study Material names three other bases that may be used (SM 4.15–4.16):

1. **Analysis or survey of existing conditions**: when the benefit cannot be seen without analysing the expenditure, e.g. lighting distributed on the number of light points in each department.
2. **Ability to pay**: a principle of taxation; expenses distributed in proportion to the income of the paying department, e.g. a sales organisation's cost spread over products by their sales. Simple, but it may be inequitable: the charge may have no relation to the effort of selling, so easy-selling lines bear the largest share when they should bear the lowest.
3. **Efficiency or incentive**: overheads distributed on pre-determined levels of production or sales. A department that exceeds the level bears less overhead per unit and shows a greater profit; one that falls short bears more.

A concern may use one criterion or all of them for different phases of its activity. ICAI prints its note on choosing a base under the table of bases for service departments, but the four tests are a sound check on any basis: economy, practicability, equitability and reliability are the matters of consideration for selection of the base (SM 4.17).

Formats and formulas

ICAI calls this statement the Primary Distribution Summary, the Overhead Distribution Summary or the Overhead Distribution Statement. Reproduce the layout from memory.

Item of cost	Basis	Total (₹)	Production depts (₹)	Service depts (₹)
Allocated: indirect material	Actual (allocation)	xxx	xxx	xxx
Indirect wages	Actual (allocation)	xxx	xxx	xxx
Apportioned: rent	Floor area (ratio)	xxx	xxx	xxx
Depreciation of plant	Capital value (ratio)	xxx	xxx	xxx
Power	H.P. × machine hours (ratio)	xxx	xxx	xxx
...	...	...	...	...
Total		**xxx**	**xxx**	**xxx**

In an answer, each production and service department gets its own column. ICAI's answers print the ratio in brackets beside the item or its basis, e.g. "Factory rent, floor area (6 : 3 : 1)", and some split the statement into

allocated overheads and a "schedule of apportioned expenses" (SM 4.80–4.81). Both layouts are ICAI's own; whichever you use, show every basis and its ratio.

∑ **Formula**

A department's share of a shared overhead

$$\text{Share} = \text{Overhead} \times \frac{\text{Its base}}{\text{Total base}}$$

"Its base" is the department's units of the base (square metres, workers, rupees of capital value); "total base" is the total of the sharing departments only, those that benefit: a rented block's rent is not shared with a department housed in an owned building.

∑ **Formula**

Compound base for machine power

$$\text{Power share} \propto \text{HP} \times \text{machine hours}$$

Used in ICAI's Illustration 4 (SM 4.24–4.25); a 50 HP machine running 6,000 hours draws 3,00,000 HP-hours.

∑ **Formula**

Department's primary total

$$\text{Total} = \text{Allocated} + \text{Apportioned}$$

Check: the primary totals of all departments add up to the total factory overheads.

Definition

Service departments

Service departments are those departments which do not directly take part in the production of goods or providing services. Such departments provide auxiliary services across the entity and render services to other cost centres and in some cases to outside parties; e.g. engineering, quality control and assurance, laboratory, canteen, stores, time office, dispensary. (SM 4.9)

Worked examples

Worked example

The samosa factory's overhead budget

Tapri's samosa factory has production departments Frying and Packing and service departments Stores and Maintenance. Its overhead budget for a month:

Allocated overheads

Item (₹)	Frying	Packing	Stores	Maintenance
Indirect material	1,10,000	64,500	15,500	50,000
Indirect wages	72,000	96,000	54,000	1,08,000

Overheads to be apportioned

Item	Amount (₹)
Factory rent	2,40,000
Fire insurance of buildings	36,000
Lighting	30,000
Power (machines)	1,80,000
Depreciation of plant and machinery	3,00,000
Insurance of plant and machinery	30,000
Supervision	1,20,000
Canteen subsidy	72,000

Survey data

Particulars	Frying	Packing	Stores	Maintenance
Floor area (sq m)	3,000	1,500	1,000	500
Light points	40	30	20	10
Horse power of machines	50	20	5	25
Machine hours	6,000	3,000	3,000	3,000
Capital value of plant (₹ lakh)	60	15	5	20
Number of workers	40	60	10	10

Stores works from a godown the company owns. The factory rent is for the main block, which houses the other three departments; the fire insurance covers both buildings.

Required:

(i) Prepare a primary distribution summary, showing the basis of apportionment of each item.

(ii) Calculate how much power Frying would bear if power were apportioned on horse power alone, and say which basis is better here.

THE WORKING

Working note 1: bases, ratios and totals

Item	Basis	Ratio (F : P : S : M)	Total (₹)
Factory rent	Floor area of the main block	6 : 3 : – : 1	2,40,000
Fire insurance of buildings	Floor area, both buildings	6 : 3 : 2 : 1	36,000
Lighting	Light points	4 : 3 : 2 : 1	30,000
Power	H.P. × machine hours	20 : 4 : 1 : 5	1,80,000
Depreciation of plant	Capital value	12 : 3 : 1 : 4	3,00,000
Insurance of plant	Capital value	12 : 3 : 1 : 4	30,000
Supervision	Number of workers	4 : 6 : 1 : 1	1,20,000
Canteen subsidy	Number of workers	4 : 6 : 1 : 1	72,000

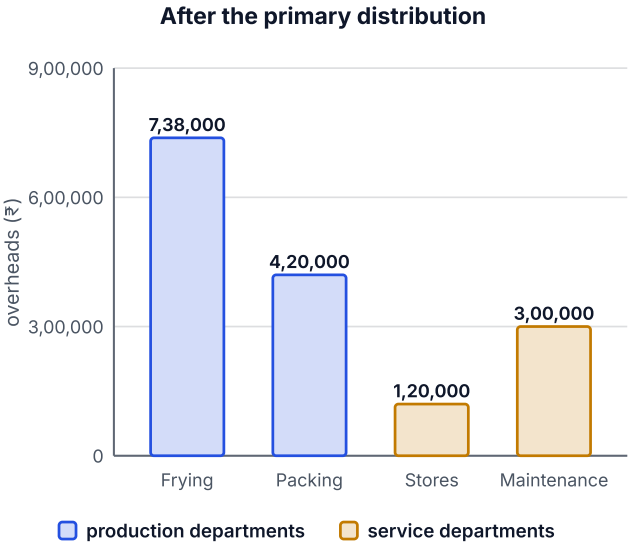
H.P. × machine hours: 50 × 6,000 = 3,00,000; 20 × 3,000 = 60,000; 5 × 3,000 = 15,000; 25 × 3,000 =

75,000; ratio 20 : 4 : 1 : 5.

(i) Primary distribution summary

Item of cost (₹)	Frying	Packing	Stores	Maintenance
Indirect material (allocated)	1,10,000	64,500	15,500	50,000
Indirect wages (allocated)	72,000	96,000	54,000	1,08,000
Factory rent	1,44,000	72,000	–	24,000
Fire insurance of buildings	18,000	9,000	6,000	3,000
Lighting	12,000	9,000	6,000	3,000
Power	1,20,000	24,000	6,000	30,000
Depreciation of plant	1,80,000	45,000	15,000	60,000
Insurance of plant	18,000	4,500	1,500	6,000
Supervision	40,000	60,000	10,000	10,000
Canteen subsidy	24,000	36,000	6,000	6,000
Total	**7,38,000**	**4,20,000**	**1,20,000**	**3,00,000**

Check: 7,38,000 + 4,20,000 + 1,20,000 + 3,00,000 = ₹15,78,000 = allocated ₹5,70,000 + apportioned ₹10,08,000.



Blue: production departments; amber: service departments. The amber ₹4,20,000 still has to reach Frying and Packing through the secondary distribution.

(ii) Power on horse power alone: 50 : 20 : 5 : 25 = 10 : 4 : 1 : 5, so Frying would bear ₹1,80,000 × 10 ÷ 20 = **₹90,000**, against ₹1,20,000 on H.P. × machine hours. Power is consumed only while machines run, and Frying's fryers run twice as long as any other department's machines, so **H.P. × machine hours** is the better measure of the benefit received. ICAI's suggested answer to November 2023 (old scheme) Q3(a) used horse power alone and showed H.P. × machine hours as an alternative, so state the basis you use.

The ₹1,20,000 of Stores and ₹3,00,000 of Maintenance are re-apportioned in the secondary

distribution.

Interactive version: PassBook app, lesson Primary distribution of factory overheads

Worked example

Kesar wants "ability to pay"

Tapri's Indore sales office costs ₹6,00,000 a month and sells three lines. Kesar wants the cost spread over the lines in proportion to their sales; Ritu has a survey of how the sales staff actually spent their hours.

Line	Sales for the month (₹)	Sales staff hours
Chai premix	60,00,000	1,200
Frozen samosas	30,00,000	1,800
Kulhad cups	10,00,000	3,000

Required:

(i) Apportion the sales office cost on ability to pay and on the survey of effort.

(ii) Comment on the two bases.

THE WORKING

(i) Apportionment

Line	Ability to pay: sales (6 : 3 : 1) (₹)	Survey: hours (2 : 3 : 5) (₹)
Chai premix	3,60,000	1,20,000
Frozen samosas	1,80,000	1,80,000
Kulhad cups	60,000	3,00,000
Total	**6,00,000**	**6,00,000**

(ii) Comment. Ability to pay distributes the expense in proportion to the income of each line. It is simple to apply but may be inequitable, because the charge has no relation to the actual effort of selling: chai premix, which takes a fifth of the staff's time, would bear 60% of the cost, while kulhad cups, which take half of it, would bear 10%. Easy-selling lines bear the largest proportion when they should bear the lowest (SM 4.15). The survey of existing conditions measures the benefit each line receives, so it should be used for costing; the sales basis makes the cups look more profitable than they are.

Worked example

Bunty's draft: one basis for everything

Tapri's kulhad unit has two production departments, Kiln and Glazing, and one service department, Stores. Bunty, attending for Iyer & Associates, finds that the unit's draft overhead statement has apportioned four items on the number of workers alone:

Item	Amount (₹)
Air-conditioning (only the kiln control room and the glazing hall are air-conditioned)	48,000
Holiday pay of production workers	36,000
Insurance of stock	18,000
Canteen subsidy	30,000

Particulars	Number of workers
Kiln	30
Glazing	20
Stores	10

Particulars	Air-conditioned volume (cubic metres)
Kiln	200
Glazing	600
Stores	nil

Particulars	Direct wages (₹)
Kiln	2,40,000
Glazing	1,20,000
Stores	nil

Particulars	Average value of stock held (₹)
Kiln	2,00,000
Glazing	1,00,000
Stores	9,00,000

Required:

(i) Show the draft apportionment on the number of workers.

(ii) Re-apportion each item on the basis the Study Material gives, and show the effect on each department's total.

THE WORKING

(i) Draft: number of workers (3 : 2 : 1)

Item (₹)	Kiln	Glazing	Stores
Air-conditioning	24,000	16,000	8,000
Holiday pay	18,000	12,000	6,000
Insurance of stock	9,000	6,000	3,000
Canteen subsidy	15,000	10,000	5,000
Total	**66,000**	**44,000**	**22,000**

(ii) Corrected apportionment

Item	Basis (SM 4.14, 4.62)	Ratio (K : G : S)
Air-conditioning	Volume of department, only the spaces air-conditioned	1 : 3 : nil
Holiday pay	Direct wages	2 : 1 : nil
Insurance of stock	Capital values: the value of stock held	2 : 1 : 9
Canteen subsidy	Number of workers	3 : 2 : 1

Item (₹)	Kiln	Glazing	Stores
Air-conditioning	12,000	36,000	–
Holiday pay	24,000	12,000	–
Insurance of stock	3,000	1,500	13,500
Canteen subsidy	15,000	10,000	5,000
Total	**54,000**	**59,500**	**18,500**
Change from the draft	(12,000)	15,500	(3,500)

The total of ₹1,32,000 is unchanged; only its spread moves. Number of workers was right for the canteen subsidy alone. Air-conditioning benefits only the cooled spaces, by their volume, so Stores bears none and Glazing's large hall bears most; holiday pay follows the wages it is paid with; stock insurance follows the value insured, most of which sits in Stores. On the draft, Glazing was undercharged by ₹15,500 and every job passing through it would have been undercosted.

Exam angle

In the exam

How ICAI asks primary distribution

In the documents read for this lesson, primary distribution came up in 9 of them, almost always as the first part of a longer overhead question:

- **Full questions, 8 to 12 marks**, where primary distribution is followed by the secondary distribution and an absorption rate: MTP January 2025 Series I Q2(a) (12 marks, five departments, then repeated distribution and an hourly rate), MTP May 2025 Series I Q5(b) (8 marks), MTP May 2026 Series I Q3(b) (9 marks, "overhead distribution statement with supporting schedules"), RTP September 2025 Q9 ("overhead analysis sheet, showing the bases of apportionment"), RTP January 2026 Q10 (allocation and primary distribution over functional departments), and in the old scheme November 2020 Q2(b) and November 2023 Q3(a) (10 marks).
- **Case-scenario MCQs**: RTP September 2026 case scenario 1 built five MCQs on a primary and a step distribution.
- **Bases only, 4 marks**: MTP January 2026 Series I Q6(c) listed the overheads of ICAI's table and asked for the basis of each; the answer was the table itself.

In the exam

Traps

- The ratio is marked: write "Floor area (6 : 3 : 1)", not only "floor area".
- Power: horse power alone or H.P. × machine hours; the November 2023 (old scheme) suggested answer showed both. Say which you used.
- A department in an owned building does not share the rent of a rented one.
- Service departments' material and wages are allocated to them, even when the question heads them "direct wages" (MTP January 2025 Series I Q2(a) gave the service departments direct wages of ₹12,00,000 and ₹1,56,000; the answer allocated both, basis "Actual"). A production department's direct wages never enter the statement, although they may serve as a basis (indirect wages and sundries on direct wages in the same answer).

Trick

Ratio first, rupees second

Cut each base to its smallest whole-number ratio (3,000 : 1,500 : 500 = 6 : 3 : 1), then share = amount × part ÷ sum of parts. Conditions: the ratio must be exact (1,240 : 1,600 : 1,200 = 31 : 40 : 30, not "1 : 1.3 : 1"), and the same base gives the same ratio for every item on it, so compute it once.

Answer frame

"Suggest a basis of apportionment for each of the following": a two-column table (overhead | basis), one line per item, in ICAI's words from the bases table; where ICAI gives alternatives ("floor area, or volume of department"), write both.

"Explain the other bases of apportioning overheads" (SM 4.15–4.16):

1. Benefit received is generally the principal criterion; other bases that may be used are the three below.
2. Analysis or survey of existing conditions, with the light points example.
3. Ability to pay: income of the paying department; simple but may be inequitable; easy-selling lines bear the most.
4. Efficiency or incentive: pre-determined levels; exceeding them lowers the overhead per unit and raises the department's profit.
5. A concern may use one criterion, or all of them for different activities.

Common mistakes

⚠ **Common mistake**

Hall of Shame: Kesar's equal shares

₹15,78,000 ÷ 4 = ₹3,94,500 each makes Packing pay for Frying's fryers and Stores for Packing's workers. **Fix:** allocate what can be allocated, then give every shared item its own basis from ICAI's table.

⚠ **Common mistake**

The godown that paid rent

Apportioning the main block's rent over all four departments by floor area (3,000 : 1,500 : 1,000 : 500) charges Stores ₹40,000 for a building it does not use. **Fix:** a cost is shared only by the departments that benefit from it: 6 : 3 : 1 over Frying, Packing and Maintenance.

⚠ **Common mistake**

The service departments that vanished

Putting rent, power and supervision only in the production columns, "because only they make samosas". **Fix:** primary distribution covers all departments; service departments take their share now and pass it on in the secondary distribution.

⚠ **Common mistake**

Prime cost in the overhead statement

Adding the Packing line's direct wages or the potatoes to the distribution summary. **Fix:** only indirect items belong there. A service department's wages are indirect for the product, so they are allocated to it.

Practice

Check yourself

1 Supervision of ₹84,000 is shared by four departments with 30, 45, 9 and 6 workers. What is the share of the department with 45 workers (₹)?

Answer: _____

2 On which basis does the Study Material suggest apportioning depreciation of plant and machinery?

- A Floor area
- B Number of workers
- C Capital values
- D Direct wages

3 Case scenario. A year after the rent diagram above, Tapri's kulhad unit has moved Stores (1,000 sq ft) into a shed the company has bought. The rented main shed still houses Kiln (2,000 sq ft) and Glazing (1,000 sq ft), and its rent is now ₹1,20,000 a month. The question gives no notional rent. What is Stores' share of the rent?

- **A** Nil
- **B** ₹30,000
- **C** ₹40,000
- **D** ₹24,000

4 Which of these appear in the samosa factory's primary distribution summary? Select all that apply.

- **A** Wages of the storekeepers
- **B** Wages of the workers on the Packing line who fill the pouches
- **C** Depreciation of the fryers
- **D** Canteen subsidy
- **E** Potatoes used in the filling

More than one answer may be right.

5 Power of ₹90,000 drives the machines of two departments: X has 20 HP running 1,000 hours, Y 10 HP running 4,000 hours. What is X's share on H.P. × machine hours (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Primary distribution: every factory overhead to every department, production and service, by allocation or apportionment.
- Allocate whole items first, including service departments' own material and wages; prime cost stays out.
- Each shared item gets its own basis: building costs by floor area; supervision, welfare and time keeping by workers; ESI, PF and holiday pay by direct wages; depreciation and repairs of plant by capital values; machine power by horse power or H.P. × machine hours; lighting by light points.
- Share the item only among the departments that benefit (a rented block's rent skips the owned godown).
- Other bases: survey of existing conditions, ability to pay (simple, may be inequitable), efficiency or incentive.

- Base selection: economy, practicability, equitability and reliability.
- Show the basis and the ratio for every item, and check the column totals against total overheads. (Equal shares are not a basis.)

Answer key · Check yourself

1 **42000**

Supervision goes on the number of workers: 30 + 45 + 9 + 6 = 90, so ₹84,000 × 45 ÷ 90 = ₹42,000.

2 **C Capital values**

Depreciation of plant and machinery, its repairs and maintenance, and insurance of stock go on capital values (SM 4.14).

3 **A Nil**

Rent is shared only by the departments that use the rented shed: 2 : 1 between Kiln (₹80,000) and Glazing (₹40,000). ₹30,000 spreads it over all 4,000 sq ft.

4 **A Wages of the storekeepers; C Depreciation of the fryers; D Canteen subsidy**

Only overheads are distributed. Storekeepers' wages are indirect (allocated to Stores); depreciation and the canteen subsidy are apportioned. Packing-line wages and potatoes are direct wages and direct material: prime cost.

5 **30000**

20 × 1,000 = 20,000 and 10 × 4,000 = 40,000, so X bears ₹90,000 × 20,000 ÷ 60,000 = ₹30,000. On horse power alone X would bear ₹60,000: the commonest slip when machine hours are given.

5.3 Secondary distribution: direct, step, repeated and simultaneous methods

How service departments' overheads reach the production departments: the direct method, the step ladder, and the reciprocal methods (repeated distribution, trial and error, simultaneous equations).

In plain words

Two departments that keep serving each other

After the primary distribution, the samosa factory's Stores holds ₹1,20,000 of overheads and Maintenance ₹3,00,000. Neither makes a single samosa, so their costs must move to Frying and Packing before any pack can carry them. The catch: Maintenance repairs Stores' racks and cold room, and Stores issues spares to Maintenance. They serve each other.

> **Bunty:** Easy. Send Stores to Frying and Packing, send Maintenance to Frying and Packing. Done.
>
> **Ritu:** Then the spares Stores gives Maintenance count for nothing. Fine if they are small.
>
> **Kesar:** And if they are big?
>
> **Ritu:** Then the costs go back and forth until there is nothing left to send.

Moving service department costs to production departments is **re-apportionment**, or **secondary**

distribution. There are three ways to do it, and they differ only in how they treat the services that service departments give each other:

- The **direct method** ignores them: each service department's cost goes straight to the production departments.
- The **step method** counts them one way: service departments are closed one at a time, and a closed department receives nothing back.
- The **reciprocal service method** counts them both ways, by passing the costs back and forth until they run out (**repeated distribution**, or **trial and error**), or by solving two equations in one go (**simultaneous equations**).

Think of it like

Ping-pong with a melting ball

Stores and Maintenance play ping-pong with a ball of ice. Every time one of them hits it, 80% melts onto the production departments' side of the table and only 20% reaches the other player. After a few rallies the ball is gone. You can play every rally (repeated distribution) or work out at once where every drop fell (simultaneous equations): the puddles are the same. The direct method pretends the ball never crosses the net; the step method lets it cross once.

Key idea

The methods differ only in how they treat services between service departments: the direct method ignores them, the step method recognises them one way, and the reciprocal methods recognise them both ways. All the reciprocal methods give the same answer.

How it works

The bases for service departments

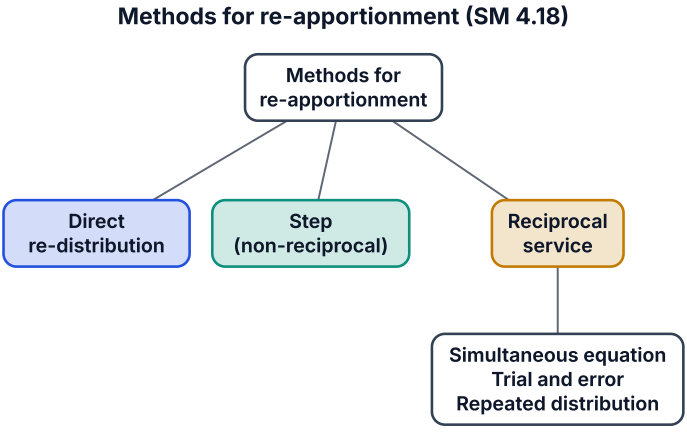
Each service department's cost is re-apportioned on the measure of the service it gives. The Study Material suggests these bases (SM 4.17):

Cost of the service department	Basis
1. Maintenance and repair shop; 2. Planning and progress; 3. Tool room	Direct labour hours, machine hours, direct labour wages, asset value × hours worked
4. Canteen and welfare; 5. Hospital and dispensary; 6. Personnel department	Number of direct workers, number of employees etc.
7. Time-keeping	Number of cards punched, number of employees
8. Computer section	Computer hours, specific allocation to departments

Cost of the service department	Basis
9. Power house (electric lighting cost)	Floor area, cubic content, number of electric points, wattage
10. Power house (electric power cost)	Horse power, kWh, horse power × machine hours, kWh × machine hours
11. Stores department	Number of requisitions, weight or value of materials issued
12. Transport department	Crane hours, truck hours, truck mileage, truck tonnage, truck ton-hours, tonnage handled, number of packages of standard size
13. Fire protection	Capital values
14. Inspection	Inspection hours

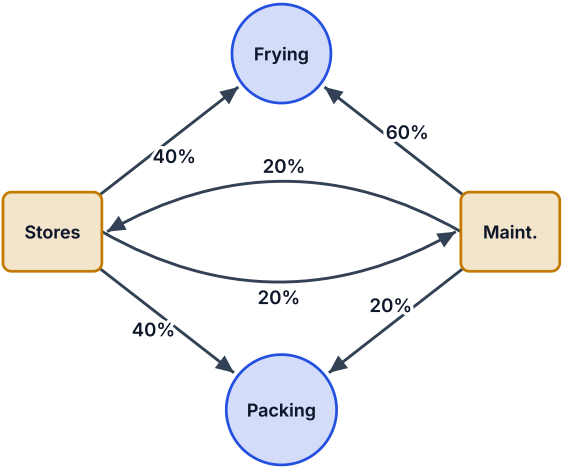
ICAI's notes to the table: (1) repairs included in the repair shop's cost, building maintenance included in the maintenance shop's cost and the like should be apportioned on capital values; (2) economy, practicability, equitability and reliability are the matters of consideration for selection of the base.

Three methods, one family tree



The reciprocal service method has three ways of handling the services that service departments render to each other.

For the samosa factory, the services are (percentages of each service department's work):



Amber squares: service departments; blue circles: production departments. Maint.: Maintenance. Each arrow is the share of a service department's work done for the department it points to. Example 1 works all four methods on these figures.

Method 1: direct re-distribution

Service department costs are apportioned over the **production departments only, ignoring the services rendered by one service department to the other** (SM 4.18). The percentages for the production departments are scaled up to 100%: Stores' 40% and 40% become 1 : 1; Maintenance's 60% and 20% become 3 : 1. Simple, but the services between service departments are lost: the bigger

they are, the further the result drifts from the reciprocal answer (Example 1 measures the drift).

Method 2: the step method (non-reciprocal)

This method gives cognizance to the services rendered by one service department to another, so a **sequence** must be chosen (SM 4.19–4.20):

1. Start with the service department that serves the **largest number of other departments** (service and production). Apportion its cost to all the departments it serves.
2. Then take the department serving the next largest number, with its own cost plus the share it has received, and apportion it to the departments **not yet closed**.
3. Continue until the last service department, whose cost goes to the production departments only.
4. A closed department receives nothing back. When a basis is a count (employees, requisitions), leave out the department's own figure and those of departments already closed.

The Study Material's summary words the rule as "the department that renders service to the maximum number of other service departments" (SM 4.64). Both

wordings give the same order whenever every service department serves all the production departments, as in this lesson's examples.

The Study Material also notes the view of some authors that the service department with the **largest amount of cost** should be distributed first (SM 4.20). PassBook's suggestion: use it to break a tie in the number of departments served. When the question states the order ("in the same order viz. R and S"), follow it.

Method 3: the reciprocal service method

Where two or more service departments render services to each other, these inter-departmental services are given due weight (SM 4.21). Three ways:

- **(a) Simultaneous equation method.** Write each service department's total cost as its own cost plus its share of the other's total, solve, and apportion the totals to the production departments on the given percentages.
- **(b) Trial and error method.** Apportion one service department's cost to the other service department only, then the other's cost plus that share back to the first, and repeat until the amount becomes negligible. Add up each service department's amounts to get its total cost, and apportion the totals to the production departments. It gives the same result as the simultaneous equation method (SM 4.23).

- **(c) Repeated distribution method.** Distribute each service department's cost to the other service and production departments on the agreed percentages, and repeat the process until the service departments' figures are exhausted or reduced to too small a figure (SM 4.26).

Interactive version: PassBook app, lesson Secondary distribution: direct, step, repeated and simultaneous methods

Formats and formulas

ICAI heads the statement Secondary Overhead Distribution Statement (SM 4.19), Secondary Distribution Summary, or Re-distribution of Service Department's Expenses. Reproduce the layout from memory. In an answer each department has its own column; the direct and simultaneous equation methods may show the production departments only, the step and repeated distribution methods need the service departments' columns too.

Particulars	Production depts (₹)	Service depts (₹)
Overheads as per primary distribution	xxx	xxx
Service department S1 apportioned (ratio)	xxx	(xxx)
Service department S2 apportioned (ratio)	xxx	(xxx)
... repeated, for the repeated distribution method	...	...
Total	**xxx**	**Nil**

Definition

Secondary distribution

The re-apportionment of the service department cost to the production department is known as secondary distribution. (SM 4.16)

Definition

Direct re-distribution method

Under this method service department costs are apportioned over the production departments only, ignoring the services rendered by one service department to the other. (SM 4.64)

Definition

Step method or non-reciprocal method

This method gives cognizance to the service rendered by service department to another service department. The sequence here begins with the department that renders service to the maximum number of other service departments. (SM 4.64)

Definition

Reciprocal service method

These methods are used when different service departments render services to each other, in addition to rendering services to production departments; the various service departments have to share overheads of each other. (SM 4.64)

Formula

Direct method: share of a production department

$$\text{Share} = \text{Service cost} \times \frac{\text{Its \%}}{\text{Production \%}}$$

"Its %" is the production department's share of the service; "production %" is the total of the production departments' shares (80% for Stores in the samosa factory).

∑ **Formula**

Simultaneous equations for two service departments

$$S_1 = a + p\,S_2 \qquad S_2 = b + q\,S_1$$

Substituting the second equation in the first:

$$S_1 = \frac{a + p\,b}{1 - p\,q}$$

a, b: the service departments' costs after primary distribution; p: the share of S_2's work done for S_1; q: the share of S_1's work done for S_2; S_1, S_2: their total costs including reciprocal services. Then $S_2 = b + q\,S_1$. Apportion S_1 and S_2 on the production departments' percentages only.

∑ **Formula**

The check that saves marks

$$\text{Production shares} = a + b$$

Under every method, the production departments receive exactly the service departments' primary totals ($a + b$). The solved totals $S_1 + S_2$ exceed this figure, because they include the services the departments render to each other.

Worked examples

Worked example **One factory, four answers**

Tapri's samosa factory has these overheads after primary distribution: Frying ₹7,38,000, Packing ₹4,20,000, Stores ₹1,20,000, Maintenance ₹3,00,000. The services rendered by the service departments are:

Service department	Frying	Packing	Stores	Maintenance
Stores	40%	40%	–	20%
Maintenance	60%	20%	20%	–

Frying works 6,000 machine hours and Packing 7,500 direct labour hours in the month.

Required: Re-apportion the service departments' overheads to Frying and Packing using

(i) the direct re-distribution method;

(ii) the step method, apportioning Maintenance first;

(iii) the repeated distribution method;

(iv) the simultaneous equation method.

(v) Using (iv), calculate the machine hour rate of Frying and the labour hour rate of Packing.

(vi) Comment on the results.

THE WORKING

(i) Direct re-distribution method

Working note: production percentages scaled to 100%: Stores 40 : 40 = 1 : 1; Maintenance 60 : 20 = 3 : 1.

Particulars	Frying (₹)	Packing (₹)
Overheads as per primary distribution	7,38,000	4,20,000
Stores (1 : 1)	60,000	60,000
Maintenance (3 : 1)	2,25,000	75,000
Total	**10,23,000**	**5,55,000**

(ii) Step method, Maintenance first

Particulars	Frying	Packing	Stores	Maintenance
Primary distribution	7,38,000	4,20,000	1,20,000	3,00,000
Maintenance (60 : 20 : 20)	1,80,000	60,000	60,000	(3,00,000)
Stores (40 : 40 = 1 : 1)	90,000	90,000	(1,80,000)	–
Total	**10,08,000**	**5,70,000**	**–**	**–**

Maintenance is closed, so Stores' 20% for Maintenance is ignored and its ₹1,80,000 goes to Frying and Packing only.

(iii) Repeated distribution method (amounts rounded to the rupee; Stores taken first)

Particulars	Frying	Packing	Stores	Maintenance
Primary distribution	7,38,000	4,20,000	1,20,000	3,00,000
Stores (40 : 40 : – : 20)	48,000	48,000	(1,20,000)	24,000
Maintenance (60 : 20 : 20 : –)	1,94,400	64,800	64,800	(3,24,000)
Stores	25,920	25,920	(64,800)	12,960
Maintenance	7,776	2,592	2,592	(12,960)
Stores	1,037	1,037	(2,592)	518
Maintenance	311	104	103	(518)
Stores	41	41	(103)	21
Maintenance	13	4	4	(21)
Stores	2	2	(4)	–
Total	**10,15,500**	**5,62,500**	**–**	**–**

The melting ball: what is still with the service departments



After the first pass, each pass leaves only 20% of what it moved with the other service department; by the fourth pass less than ₹3,000 is left.

(iv) Simultaneous equation method

Let S and M be the total overheads of Stores and Maintenance.

$$S = 1{,}20{,}000 + 0.20M \qquad M = 3{,}00{,}000 + 0.20S$$

Substituting M in the first equation:

$$\begin{aligned} S &= 1{,}20{,}000 + 0.20\,(3{,}00{,}000 + 0.20S) \\ &= 1{,}80{,}000 + 0.04S \end{aligned}$$

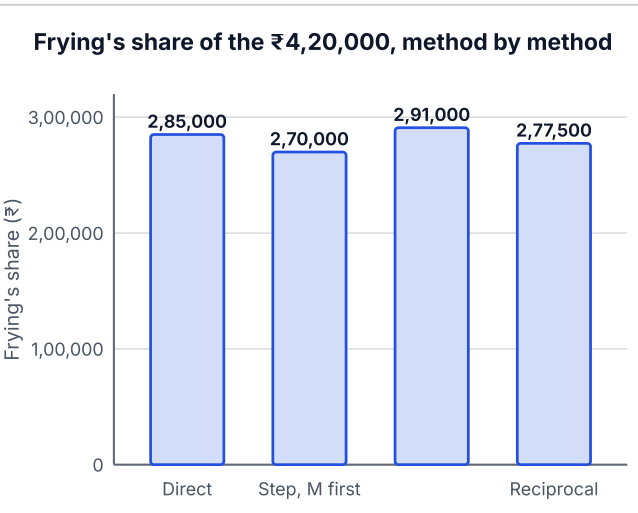
$$0.96S = 1{,}80{,}000 \qquad S = ₹1{,}87{,}500$$

M = ₹3,00,000 + 0.20 × ₹1,87,500 = **₹3,37,500**.

Particulars	Frying (₹)	Packing (₹)
Overheads as per primary distribution	7,38,000	4,20,000
Stores (40% and 40% of ₹1,87,500)	75,000	75,000
Maintenance (60% and 20% of ₹3,37,500)	2,02,500	67,500
Total	**10,15,500**	**5,62,500**

Check: ₹2,77,500 + ₹1,42,500 = ₹4,20,000, the service departments' primary total. The repeated distribution gave the same figures.

(v) Overhead rates: Frying ₹10,15,500 ÷ 6,000 machine hours = **₹169.25 per machine hour**; Packing ₹5,62,500 ÷ 7,500 labour hours = **₹75.00 per labour hour**.



Step, S first: Stores ₹1,20,000 to Frying, Packing and Maintenance (40 : 40 : 20), then Maintenance's ₹3,24,000 at 3 : 1. The reciprocal bar is the same for the repeated, trial and error and simultaneous equation methods.

(vi) Comment. The reciprocal methods recognise the services Stores and Maintenance render each other in full, so ₹2,77,500 is Frying's fair share. The direct method ignores those services and overcharges Frying by ₹7,500 (a rate of ₹170.50); the step method recognises them in one direction only, and its result depends on the order: ₹2,70,000 with Maintenance first, ₹2,91,000 with Stores first. Where service departments serve each other this much (20% each way), the Study Material's point applies: the inter-departmental services are to be

given due weight, and only a reciprocal method does that in full.

Worked example

Chai premix plant: the canteen and the fitters

Tapri's chai premix plant has production departments Blending and Pouching and service departments Canteen and Maintenance. Overheads after primary distribution: Blending ₹4,60,000, Pouching ₹3,10,000, Canteen ₹1,20,000, Maintenance ₹1,80,000.

Particulars	Blending	Pouching	Canteen	Maintenance
Number of employees	45	75	8	30
Maintenance hours	1,800	600	–	–
Machine hours	5,000	–	–	–
Direct labour hours	–	10,000	–	–

The canteen serves the employees of all departments; Maintenance does no work for the canteen.

Required:

(i) State the order in which the step method should close the service departments, with the reason.

(ii) Re-apportion the service departments' overheads by the step method.

(iii) Calculate a machine hour rate for Blending and a labour hour rate for Pouching.

THE WORKING

(i) Order. The canteen serves three other departments (Blending, Pouching and Maintenance); Maintenance serves two (Blending and Pouching). The step method begins with the department that serves the largest number of other departments, so **Canteen first, then Maintenance**.

(ii) Step distribution

Working note: the canteen's basis is the number of employees of the departments it serves, leaving out its own 8: 45 : 75 : 30 = 3 : 5 : 2.

Particulars	Blending	Pouching	Canteen	Maintenance
Primary distribution	4,60,000	3,10,000	1,20,000	1,80,000
Canteen (3 : 5 : – : 2)	36,000	60,000	(1,20,000)	24,000
Maintenance, hours (1,800 : 600 = 3 : 1)	1,53,000	51,000	–	(2,04,000)
Total	**6,49,000**	**4,21,000**	**–**	**–**

Check: 6,49,000 + 4,21,000 = ₹10,70,000, the total overheads.

(iii) Rates: Blending ₹6,49,000 ÷ 5,000 = **₹129.80 per machine hour**; Pouching ₹4,21,000 ÷ 10,000 = **₹42.10 per labour hour**.

Because Maintenance gives no service to the canteen, nothing is lost by closing the canteen first: here the step method recognises every service and gives the same result as a reciprocal method would.

Interactive version: PassBook app, lesson Secondary distribution: direct, step, repeated and simultaneous methods

Worked example

The power house that also needs testing: services in units

Tapri's kulhad unit has production departments Kiln, Glazing and Painting, and two service departments, the Quality lab and the Power house, which serve each other. Their expenses after primary distribution are:

Quality lab ₹76,200; Power house ₹2,52,000. The services were measured in units:

Service	Power house: kWh supplied
Kiln	80,000
Glazing	60,000
Painting	40,000
Quality lab	20,000
Power house	10,000 (own use)

Service	Quality lab: tests carried out
Kiln	300
Glazing	300
Painting	150
Quality lab	–
Power house	250

Required:

(i) Convert the services into percentages for apportionment.

(ii) Apportion the service departments' expenses to the production departments by the simultaneous equation method.

(iii) Calculate the full cost of a kWh of power and comment on an offer from the grid to supply power at ₹1.30 a kWh.

THE WORKING

(i) Percentages. The Power house's own use of 10,000 kWh is not a service to any other department, so it is left out: the base is the 2,00,000 kWh supplied to others (80 : 60 : 40 : 20 = 40% : 30% : 20% : 10%). The Quality lab's 1,000 tests give 30% : 30% : 15% : – : 25%.

(ii) Simultaneous equations. Let Q and H be the total expenses of the Quality lab and the Power house.

$$Q = 76{,}200 + 0.10H \qquad H = 2{,}52{,}000 + 0.25Q$$

$$\begin{aligned} Q &= 76{,}200 + 0.10\,(2{,}52{,}000 + 0.25Q) \\ &= 1{,}01{,}400 + 0.025Q \end{aligned}$$

$$0.975Q = 1{,}01{,}400 \qquad Q = ₹1{,}04{,}000$$

$$H = 2{,}52{,}000 + 0.25 \times 1{,}04{,}000 \qquad = ₹2{,}78{,}000$$

Particulars	Quality lab (30%, 30%, 15% of ₹1,04,000)
Kiln (₹)	31,200
Glazing (₹)	31,200
Painting (₹)	15,600

Particulars	Power house (40%, 30%, 20% of ₹2,78,000)
Kiln (₹)	1,11,200
Glazing (₹)	83,400
Painting (₹)	55,600

Particulars	Total
Kiln (₹)	**1,42,400**
Glazing (₹)	**1,14,600**
Painting (₹)	**71,200**

Check: ₹1,42,400 + ₹1,14,600 + ₹71,200 = ₹3,28,200 = ₹76,200 + ₹2,52,000.

(iii) Cost of a kWh. The Power house's total cost, including its share of the Quality lab, is ₹2,78,000 for 2,00,000 kWh supplied: **₹1.39 a kWh**. That is the full cost, not the cost that buying from the grid would save. Comparing a service department's cost with the price of the same service from outside is one way of controlling it (SM 4.49), but before accepting ₹1.30 Ritu must find how much of the ₹2,52,000 would actually stop (fuel, operators' wages) and how much would continue (depreciation of the generator set, a share of rent), and that the Quality lab's tests of the Power house would no longer be needed.

Interactive version: PassBook app, lesson Secondary distribution: direct, step, repeated and simultaneous methods

Exam angle

In the exam

How ICAI asks secondary distribution

In the documents read for this lesson, secondary distribution came up in 15 of them:

- **Simultaneous equations alone, 3–4 marks**: September 2025 Q2(b) (two service and three production departments, 4 marks); September 2026 Q1(c) ("Use simultaneous equation method and find out the total expenses of FC", 3 marks).

- **Inside a full overhead question, 8–12 marks**, after the primary distribution and before an hourly rate: MTP January 2025 Series I Q2(a) (repeated distribution), MTP May 2025 Series I Q5(b) and MTP May 2026 Series I Q3(b) (simultaneous equations), RTP September 2025 Q9 (direct method, "ignoring the apportionment of service department costs among service departments"), RTP May 2025 Q10 (service costs to two production departments in given ratios, then pre-determined rates), RTP January 2026 Q10 (administration and general management re-apportioned in sequence), and in the old scheme November 2020 Q2(b) (simultaneous, with a note that repeated distribution is acceptable) and November 2023 Q3(a) (step method, 10 marks).
- **All three methods on one set of data, 12 marks**: MTP September 2025 Series II Q2(a).
- **Case-scenario MCQs**: RTP September 2026 case scenario 1 (step method).
- **Bases, 4–5 marks**: January 2025 Q6(c) (OR, 4 marks) and, in the old scheme, May 2023 Q6(d) (5 marks): "Suggest any one basis of re-apportionment of service department overheads" for eight or ten service departments; the answers were ICAI's table.
- **Theory, 4 marks**: MTP January 2025 Series II Q6(d) (OR), "Discuss in brief three main methods of allocating support departments costs to operating departments"; the chapter's own questions 2 and 8.

In the exam

Traps

- **Direct method**: scale the production percentages to 100% (MTP September 2025 used 4,500 : 2,250 hours, not 60% : 30%).
- **Own count out**: a service department's own employees, hours or requisitions are left out of its basis under every method (MTP May 2026 Series I Q3(b), simultaneous equations: General Plant spread on employees 20 : 12 : 8 = 5 : 3 : 2, leaving out its own 4).
- **Step method**: closed departments are left out too (November 2023 apportioned R's cost "in the ratio of number of employees (10 : 30 : 4)", leaving out R's own 6, then S's cost on machine hours 40 : 10, leaving out R's and S's).
- **Simultaneous equations**: apportion the solved totals on the production percentages only; the production departments must receive exactly the service departments' primary total.
- Round consistently (ICAI's answers usually round to the rupee) and show the check line.

Trick

The closed-form shortcut

For two service departments: $S_1 = (a + pb) \div (1 - pq)$. Kulhad unit: (76,200 + 0.10 × 2,52,000) ÷ (1 − 0.10 × 0.25) = 1,01,400 ÷ 0.975 = ₹1,04,000. Condition: exactly two service departments; still write both equations in the answer, because the method is what earns the marks.

Answer frame

"Explain the methods of re-apportionment of service department expenses" (SM theory questions 2 and 8; MTP January 2025 Series II Q6(d)): the points ICAI's answer gives, in order.

1. **Direct re-distribution method**: service department costs are apportioned over the production departments only, ignoring the services rendered by one service department to the other.
2. **Step method (non-reciprocal method)**: gives cognizance to the services rendered by one service department to another; the cost of the service department that serves the maximum number of departments is apportioned first, then the next, until the last service department's cost is apportioned to production departments only.
3. **Reciprocal service method**: recognises that service departments may render services to each other and

gives these inter-departmental services due weight. The methods available are (a) the simultaneous equation method, (b) the repeated distribution method and (c) the trial and error method, each in one line.

If time allows, add a two-line illustration (two service departments that serve each other) under the reciprocal method.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the step method's boomerang

After closing Maintenance, the student apportions Stores' ₹1,80,000 as 40 : 40 : 20, sending ₹36,000 back to the closed Maintenance column, which then never empties. **Fix:** a closed department receives nothing; the last department's cost goes to production departments only (here 1 : 1).

⚠ **Common mistake**

The direct method that lost 20%

Taking Stores' 40% and 40% literally distributes only ₹96,000 of ₹1,20,000. **Fix:** in the direct method, scale the production percentages to 100% (1 : 1); the service departments must end at nil.

⚠ **Common mistake**

Distributing the wrong totals

Solving S = ₹1,87,500 and M = ₹3,37,500, then apportioning ₹1,20,000 and ₹3,00,000 anyway; or apportioning the solved totals on all four percentages, so that the 20% shares go round a second time. **Fix:** apportion the solved totals on the production percentages only, and check that the production departments received exactly ₹4,20,000.

⚠ **Common mistake**

The canteen that fed itself

Apportioning the canteen on all 158 employees (45 : 75 : 8 : 30) leaves ₹6,076 with the canteen. **Fix:** a service department's own count is left out of its basis: 45 : 75 : 30.

Practice

Check yourself

1. Service department X costs ₹76,000 and Y ₹1,00,000 after primary distribution. X receives 20% of Y's services and Y 20% of X's. What is X's total cost under the simultaneous equation method (₹)?

Answer: _____

2 Which method ignores the services that service departments render to each other?

- **A** Direct re-distribution method
- **B** Step method
- **C** Repeated distribution method
- **D** Trial and error method

3 Case scenario. Tapri's cold-chain hub has three service departments. The Time office serves four other departments, Stores serves three and the Boiler house serves two. Ritu uses the step method and the question gives no order. In which order should she close them?

- **A** Boiler house, Stores, Time office
- **B** Time office, Stores, Boiler house
- **C** Stores, Time office, Boiler house
- **D** In the order of the largest cost first, whatever the services

4 Compared with the simultaneous equation method, the trial and error method gives:

- **A** a higher cost for the production department with the most machine hours
- **B** the same result
- **C** a lower total, because the last small amounts are ignored
- **D** the same result as the direct method

5 Stores costs ₹50,000 and serves Frying 50%, Packing 30% and Maintenance 20%. Under the direct method, what is Frying's share (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Secondary distribution: re-apportioning service department costs to production departments, on bases such as machine or direct labour hours (maintenance), number of employees (canteen, personnel), requisitions or value of materials issued (stores), kWh (power house), capital values (fire protection).
- Direct method: production departments only, inter-service services ignored, production percentages

scaled to 100%.

- Step method: start with the department serving the largest number of others (some authors: largest cost first); a closed department receives nothing; the last goes to production only.
- Reciprocal service method: simultaneous equations, trial and error, repeated distribution; all give the same result.
- Two service departments: $S_1 = a + pS_2$ and $S_2 = b + qS_1$; apportion the solved totals on production percentages only.
- Check: production departments receive exactly the service departments' primary total.
- Then the production totals ÷ machine or labour hours give the absorption rates. (The ball always melts.)

Answer key · Check yourself

1 **100000**

X = 76,000 + 0.2Y and Y = 1,00,000 + 0.2X, so X = (76,000 + 20,000) ÷ (1 − 0.04) = ₹1,00,000 (and Y = ₹1,20,000).

2 **A** **Direct re-distribution method**

The direct method apportions service costs over the production departments only. The step method recognises them one way; the other two are reciprocal methods.

3 **B** **Time office, Stores, Boiler house**

The Study Material's rule: begin with the department that serves the largest number of other departments, then the next largest. "Largest cost first" is only a view of some authors that the Study Material mentions, and no costs are given here: the rule in the text decides.

4 **B** **the same result**

Both recognise the reciprocal services in full; the Study Material says they give the same result (SM 4.23). Any difference is rounding.

5 **31250**

The direct method ignores Maintenance and scales the production shares to 100%: 50 : 30 = 5 : 3, so ₹50,000 × 5 ÷ 8 = ₹31,250. Taking 50% (₹25,000) leaves part of Stores undistributed.

5.4 Absorption rates and the machine hour rate

How a production department's overheads reach each job or unit: ICAI's six bases, four kinds of rate, the machine hour rate statement, and the capacity a rate is built on.

In plain words

Same fryer, same 20 hours, three times the overhead

The Indore samosa factory has finished its secondary distribution: Frying now holds ₹10,15,500 of overheads for the month (secondary distribution). Two orders went through Frying this month, each needing 20 hours on the fryer line: a paneer samosa order whose filling cost ₹30,000, and an aloo order whose filling cost ₹10,000.

> **Kesar:** Easy. Frying's overheads are half its materials this month. Add 50% of material cost to each order.
>
> **Ritu:** Then paneer carries ₹15,000 of Frying's overhead and aloo ₹5,000. Did the fryer burn three times the gas for paneer?
>
> **Kesar:** Same fryer, same 20 hours...
>
> **Ritu:** Then the same overhead. Per hour, please: ₹169.25 a fryer hour, ₹3,385 each.

Charging a production department's overheads to the jobs and units that pass through it is **absorption** (or recovery). It needs an **overhead rate**: the department's overheads divided by a **base**, a measure of how much each job used the department, such as its direct materials, direct wages, labour hours, machine hours or units of output.

Most factory overheads (rent, supervision, power, depreciation) pay for facilities by the hour, so the base that follows them best is usually time: **machine hours** where machines do the work, **labour hours** where people do it. Actual overheads are known only after the books close, so the rate is fixed in advance from budgets: a **pre-determined rate**. Its fixed part is built on **normal capacity**, the output the factory achieves on average.

Think of it like

The mall car park

A car park charges by the hours your car occupies a slot, not by the price of the car: a Mercedes and a Maruti parked for two hours pay the same. Factory overheads are the rent of the factory's facilities, so they too are best charged by the hours each job occupies them. One tariff for the whole mall (a blanket rate) is fair only if every car spends the same time on every floor; otherwise each floor needs its own meter (departmental rates).

Key idea

An absorption rate is a cost centre's overheads divided by the base that best measures its use. Time bases follow most overheads (machine hours where machines dominate, labour hours where labour does). The rate is pre-determined from budgets, with fixed overheads on normal capacity.

How it works

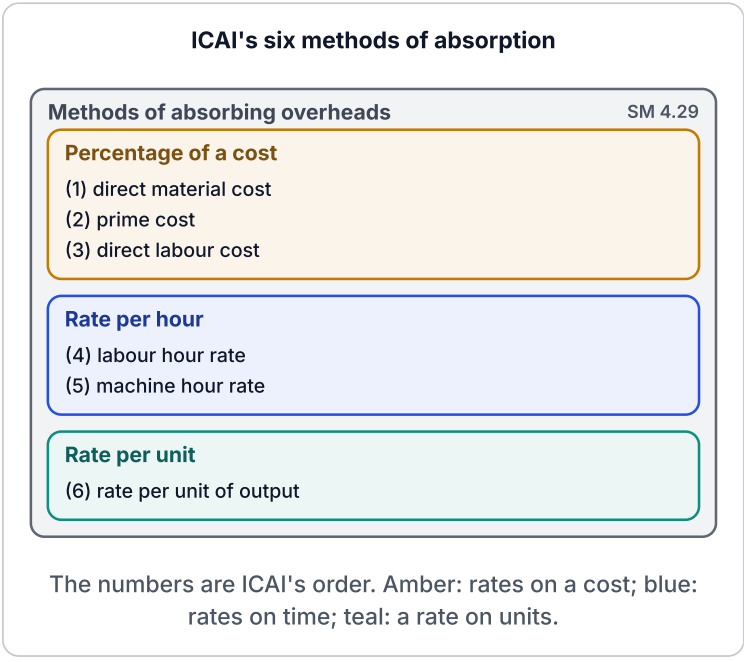
What a good method must achieve

The method chosen for charging overheads to products or jobs should ensure (SM 4.29):

1. that the total amount charged (recovered) in a period does not differ materially from the actual expenses of the period; and
2. that the amount charged to individual jobs or products is equitable. For factory overhead this means: (a) the time spent on each job is considered; (b) jobs done by skilled and by unskilled workers are distinguished; (c) jobs done by manual labour and by machines are distinguished.

It should also be convenient to use and give uniform results from period to period; any change should reflect a change in the underlying situation, such as machines replacing human labour.

Six methods in three families



The numbers are ICAI's order. Amber: rates on a cost; blue: rates on time; teal: a rate on units.

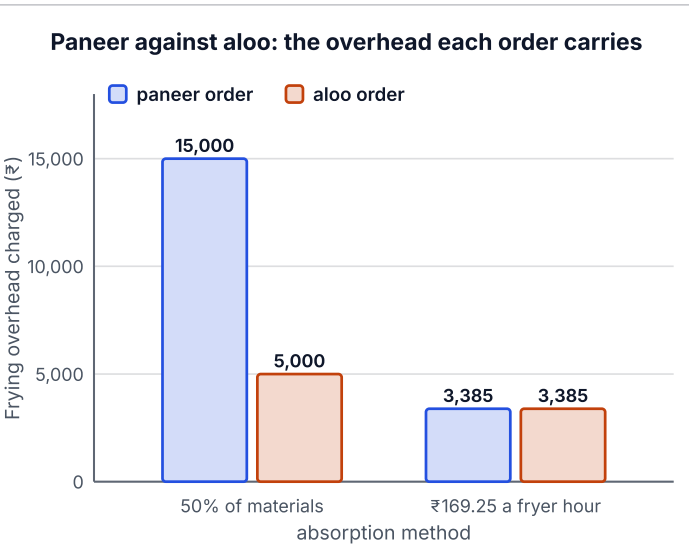
Method	Base	ICAI's view (SM 4.30–4.35)
1. % of direct material cost	Budgeted direct material cost	Generally unsuitable; may be used for material handling and stores overheads
2. % of prime cost	Prime cost	Generally unsuitable
3. % of direct labour cost	Direct wages	Simple; recognises time only indirectly
4. Labour hour rate	Direct labour hours	Suits operations without large use of machinery
5. Machine hour rate	Machine hours	Usually the best where machines are the main factor of production
6. Rate per unit of output	Units produced	The simplest; suits one uniform product

Why the first two mislead

ICAI's example (SM 4.30–4.31): materials ₹2,00,000, labour ₹1,00,000, factory overheads ₹90,000 give 45% of materials or 30% of prime cost. A job with materials ₹10,000 and labour ₹7,000 costs ₹21,500 by the first and ₹22,100 by the second: the method alone changes the

works cost. The four reasons the two methods are "generally considered to be unsuitable" (SM 4.31–4.32):

1. Manufacturing overheads are mostly a **function of time**: a job that uses the facilities longer costs more. The first method ignores time altogether and the second largely.
2. Overheads are related to neither prime cost nor material cost, except to a very small extent: two jobs with the same operational time but materials of different prices would bear different overheads. Adding labour to material (prime cost) only compounds the error.
3. Material prices fluctuate widely, so overheads charged on them fluctuate violently from period to period.
4. They ignore the skill of the workers and whether machines were used.



Blue: the paneer order; orange: the aloo order. Both used the fryer for 20 hours. On materials paneer carries three times the overhead; on fryer hours they carry the same.

Direct labour cost and labour hours

The **percentage of direct labour cost** (SM 4.32):

Advantages	Disadvantages
(i) Simple and economical to apply	(i) Inaccuracies, because the time factor is not given full importance
(ii) The time factor is recognised, even if indirectly	(ii) No allowance where machinery is used to some extent
(iii) Total expenses recovered will not differ much from the estimate, since total wages are not likely to fluctuate much	(iii) It does not provide for varying skills of workers

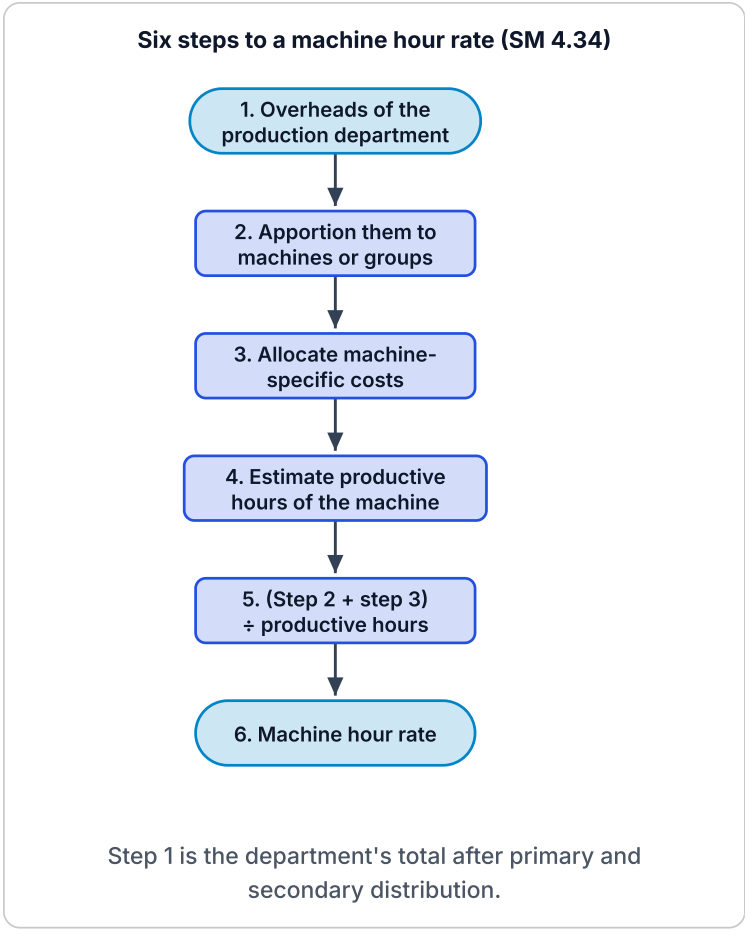
The **labour hour rate** is an improvement on the percentage of direct wages: it fully recognises time, and is admirably suited to operations that do not involve any large use of machinery. It can be calculated for each category of workers (SM 4.32–4.33). ICAI's example: ₹90,000 of overheads over 1,50,000 labour hours is ₹0.60 an hour, so a job taking 795 labour hours absorbs ₹477.

The machine hour rate

A **machine hour rate** is the cost of running a machine for an hour to produce goods; each machine or group of

machines is treated as a cost centre (SM 4.33). Two ways to build it:

- **Direct machine hour rate**: only the expenses directly or immediately connected with operating the machine (power, depreciation, repairs and maintenance, insurance), divided by the machine's estimated operational hours.
- **Comprehensive machine hour rate**: also a share of the department's other manufacturing expenses (supervision, shop cleaning and lighting, consumable stores, shop general labour, rent and rates), so that none is left out of production cost. Alternatively those may be absorbed on a productive labour hour rate. Sometimes the **machine operator's wages** are added too, for a comprehensive rate of working the machine for one hour.



The costs are then split (SM 4.34–4.35):

- **Standing charges** (fixed): remain constant whether or not the machine runs: insurance premium, rent of the premises, supervisor's salary, depreciation if it relates to the effluxion of time.
- **Machine expenses** (variable): vary with operation: power, consumables (lubricants, oils), repairs and

maintenance, depreciation if it relates to activity.

Advantages of the machine hour rate	Disadvantages
(1) Where machines are the main factor of production, usually the best method of charging machine operating expenses	(1) Additional data on machines' operating time must be recorded and maintained
(2) Under-absorption of machine overheads indicates the extent to which the machines were idle	(2) A separate rate for each machine or group means further work, where a general department rate may be suitable
(3) Particularly advantageous where one operator attends several machines, or several operators work one machine	

The **rate per unit of output** is the simplest of all: amount of overheads ÷ number of units (SM 4.35).

Four kinds of rate

Rate (SM 4.36–4.37)	Computed as	Use
Normal rate (also called actual rate)	Actual overheads ÷ actual base	Known only after the period
Pre-determined rate	Budgeted overheads ÷ budgeted base	Fixed in advance; the usual rate
Blanket rate	Total overheads for the factory ÷ total units of the base for the factory	One single rate for the whole factory
Departmental rate	Overheads of a department or cost centre ÷ its base	A separate rate for each department

A pre-determined rate may be estimated on three bases: (1) the previous year's or period's figure, assuming production and overheads stay constant or change proportionately; (2) estimated expenses and anticipated volume of production (₹10,000 over 4,000 units is ₹2.50 a unit); (3) the normal volume of the business.

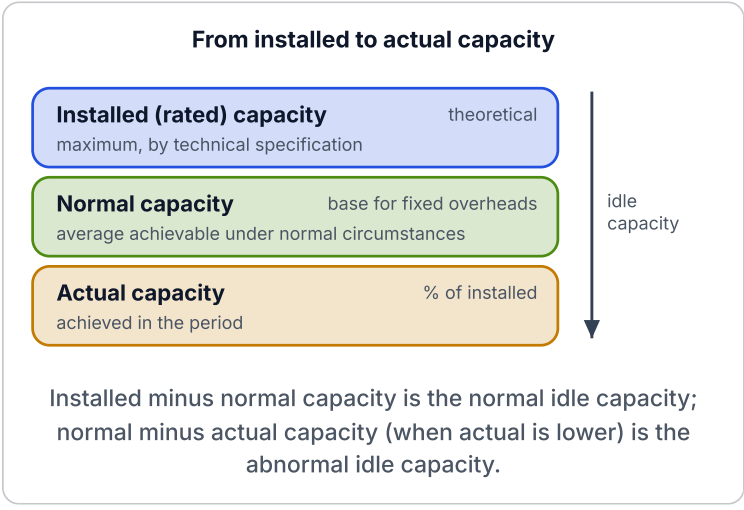
A **blanket rate** should be applied only (1) where one major product is produced, or (2) where several products are produced but (a) all products pass through all departments and (b) all are processed for the same length of time in each department. Where these

conditions do not exist, departmental rates should be used: where product lines vary or machinery is used to a varying degree in different departments (SM 4.37).

Interactive version: PassBook app, lesson Absorption rates and the machine hour rate

Capacity: what the denominator assumes

A budgeted base is a capacity estimate, and ICAI defines five capacity concepts (SM 4.58–4.59):



- **Installed (rated) capacity**: the maximum capacity of producing goods or providing services, set by technical specification or evaluation; also called theoretical capacity, it cannot be achieved in normal operating circumstances.
- **Practical capacity** (operating, net or available capacity): ICAI calls it the actually utilised capacity of

a plant; it allows for repairs, maintenance, minor breakdowns, idle time, set-up time, normal delays, Sundays and holidays, stock taking. Generally 80% to 90% of the rated capacity, and also used as a base for determining overhead rates.

- **Normal capacity**: the volume achieved or achievable on an average over a period under normal circumstances, after the reduction for planned maintenance.
- **Actual capacity**: achieved in a given period, presented as a percentage of installed capacity.
- **Idle capacity**: the part that cannot be effectively utilised. **Normal idle capacity** = installed − normal capacity; **abnormal idle capacity** = normal − actual capacity, where actual is lower.

Variable manufacturing overheads are absorbed on actual production, fixed manufacturing overheads on **normal capacity** (SM 4.10). How idle capacity costs are treated is in under- and over-absorption.

Formats and formulas

Reproduce the machine hour rate statement and the hours working from memory: ICAI's answers use exactly this layout.

∑ **Formula**

Percentage of direct material cost; of prime cost

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Budgeted direct material cost}} \times 100$$

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Prime cost}} \times 100$$

∑ **Formula**

Percentage of direct labour cost

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Direct labour cost}} \times 100$$

∑ **Formula**

Labour hour rate and machine hour rate

$$\text{Labour hour rate} = \frac{\text{Production overheads of the department}}{\text{Direct labour hours}}$$

$$\text{Machine hour rate} = \frac{\text{Overheads of the machine}}{\text{Productive machine hours}}$$

Both are rupees per hour. (SM 4.33 prints the labour hour formula with "× 100"; its own example, ₹90,000 ÷ 1,50,000 hours = ₹0.60 an hour, applies no × 100, so write the rate in rupees per hour.)

∑ **Formula**

Rate per unit, normal and pre-determined rates

$$\text{Per unit} = \frac{\text{Overheads}}{\text{Number of units}}$$

$$\text{Normal} = \frac{\text{Actual overheads}}{\text{Actual base}}$$

$$\text{Pre-determined} = \frac{\text{Budgeted overheads}}{\text{Budgeted base}}$$

∑ **Formula**

Blanket and departmental rates

$$\text{Blanket} = \frac{\text{Total overheads for the factory}}{\text{Total units of base for the factory}}$$

$$\text{Departmental} = \frac{\text{Overheads of the department}}{\text{Corresponding base}}$$

The machine hour rate statement

Computation of machine hour rate	Per period (₹)	Per hour (₹)
(A) Standing charges		
Rent, lighting, supervision (shares)	xxx	
Insurance; depreciation (time basis)	xxx	
Operator's wages (if included)	xxx	
Total standing charges ÷ productive hours	**xxx**	**xxx**
(B) Machine expenses		
Power, consumables, repairs; depreciation (activity basis)	xxx	xxx
Machine hour rate (A) + (B)	**xxx**	**xxx**

Productive hours when a machine needs an operator

Hours (all operators, for the period)	Hours
Normal available hours × months × operators	xxx
Less: absenteeism without pay × operators	(xxx)
Paid hours (operators' wages are on these)	**xxx**
Less: leave with pay × operators	(xxx)
Less: normal unavoidable idle time × operators	(xxx)
Effective (productive) working hours	**xxx**

Normal capacity (SM 4.58)

Normal capacity	Units
Installed capacity	xxx
Adjustments for: (i) time lost to scheduled preventive or planned maintenance; (ii) number of shifts, machine hours or man hours; (iii) holidays, normal shut-down days, normal idle time; (iv) normal time lost in batch change-over	(xxx)
Normal capacity	**xxx**

Definition

Absorption

The process of recovering overheads of a department or any other cost centre from its output is called recovery or absorption. (SM 4.10)

Definition

Machine hour rate

Machine hour rate implies cost of running a machine for an hour to produce goods. (SM 4.33)

Definition

Blanket overhead rate

Blanket overhead rate refers to the computation of one single overhead rate for the whole factory, as distinguished from the departmental overhead rate, a separate rate for each individual cost centre or department. (SM 4.37)

Definition

Normal capacity

The volume of production or services achieved or achievable on an average over a period under normal circumstances taking into account the reduction in capacity resulting from planned maintenance. (SM 4.58)

Worked examples

Worked example **One fryer, one shift or two**

Tapri's continuous fryer in the Frying department works one shift today. The budget for a year:

Item	Data
Depreciation	₹2,00,000 a year for obsolescence, plus ₹50 per running hour for wear
Electricity	demand charge for the fryer's connected load ₹6,000 a month; energy 25 units per running hour at ₹9 a unit
Gas to heat the oil	₹120 per running hour
Oil top-up and filters	₹60 per running hour
Share of the department's rent	₹1,20,000 a year
Share of the supervisor's salary	₹1,80,000 a year for each shift worked
Insurance	₹40,000 a year
Operator, wholly on the fryer (included in the rate)	₹3,00,000 a year for each shift

One shift gives 2,000 productive hours a year. A second shift would give 3,800 productive hours in all, with a second operator and a night allowance of ₹60,000 a year, and a second supervisor share.

Required:

(i) Calculate the comprehensive machine hour rate for one shift, showing standing charges and machine expenses separately.

(ii) Calculate the rate if a second shift is worked.

(iii) An order needs 50 fryer hours. Calculate the overhead charged to it under each rate, and comment.

THE WORKING

Working note: what is standing and what is running. Items that accrue with time, whether or not the fryer runs, are standing charges: time-based depreciation, the electricity demand charge, rent, supervision, insurance and the operator. Items that arise only while it runs are machine expenses: wear-based depreciation, energy, gas, oil and filters.

(i) Computation of machine hour rate, one shift (2,000 hours)

Particulars	Per annum (₹)	Per hour (₹)
(A) Standing charges		

Particulars	Per annum (₹)	Per hour (₹)
Depreciation, obsolescence (time basis)	2,00,000	
Electricity demand charge (₹6,000 × 12)	72,000	
Share of rent	1,20,000	
Share of supervisor's salary	1,80,000	
Insurance	40,000	
Operator's wages	3,00,000	
Total standing charges	**9,12,000**	**456.00**
(B) Machine expenses (per running hour)		
Depreciation, wear		50.00
Energy (25 units × ₹9)		225.00
Gas		120.00
Oil top-up and filters		60.00
Total machine expenses		**455.00**
Machine hour rate (A) + (B)		**911.00**

(ii) Two shifts (3,800 hours). Standing charges = ₹9,12,000 + second operator ₹3,00,000 + night allowance ₹60,000 + second supervisor share ₹1,80,000 = ₹14,52,000; per hour = ₹14,52,000 ÷ 3,800 = ₹382.11. Machine expenses stay ₹455.00 an hour. Rate = **₹837.11**.

(iii) Order of 50 fryer hours. One shift: 50 × ₹911.00 = **₹45,550**. Two shifts: 50 × ₹837.11 = **₹41,855.50**.

Comment. Only the standing charges per hour fall (₹456.00 to ₹382.11), because the time-based costs are spread over more hours; the running costs of ₹455 an hour do not change with the shift pattern. The lower rate holds only if the second shift's hours are actually used: the rate is set on the hours expected, and if the second shift ran half empty the standing charges would be under-absorbed (see under- and over-absorption).

Worked example

Two catering bids: the blanket rate that won the wrong order

Tapri's catering kitchen has two departments. Monthly

budget:

Budget	Cooking (combi-ovens)	Assembly (plating by hand)
Factory overheads (₹)	9,60,000	2,40,000
Direct wages (₹)	1,60,000	6,40,000
Machine hours	8,000	nil
Direct labour hours	4,000	32,000

Tapri absorbs factory overheads by a blanket rate, the percentage of total factory overheads to total direct wages, and quotes factory cost plus 10%. This week it bid for two orders, against a rival quoting ₹55,000 for the wedding buffet and ₹45,000 for the corporate lunch boxes:

Order	Direct materials (₹)	Cooking: machine hours	Cooking: wages (₹)	Assembly: labour hours	Assembly: wages (₹)
W: wedding buffet	30,000	20	800	400	8,000
C: corporate lunch boxes	25,000	120	2,000	60	1,200

Required:

(i) Compute the blanket rate and the factory cost and quotation of each order on it.

(ii) Recommend a rate for each department, and recompute the factory costs and quotations.

(iii) Explain what the blanket rate did to each bid.

THE WORKING

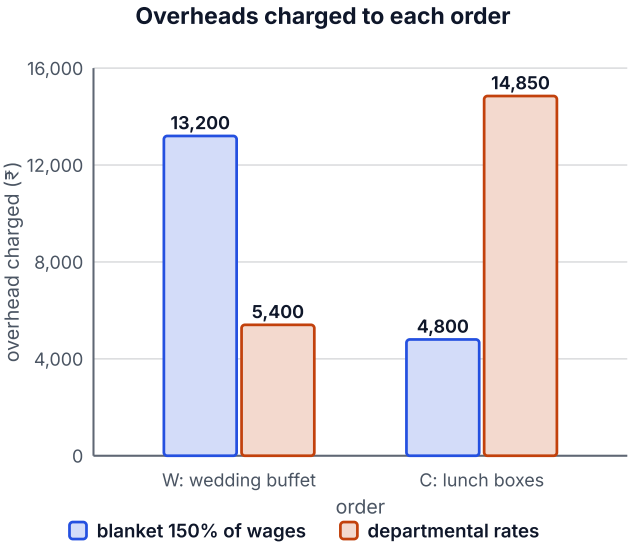
(i) Blanket rate

$$\text{Blanket rate} = \frac{9{,}60{,}000 + 2{,}40{,}000}{1{,}60{,}000 + 6{,}40{,}000} \times 100 = 150\% \text{ of direct wages}$$

(ii) Departmental rates. Cooking is machine-intensive (overheads six times its wages, mostly the ovens), so a machine hour rate: ₹9,60,000 ÷ 8,000 = **₹120 per machine hour**. Assembly is labour-intensive, so a labour hour rate: ₹2,40,000 ÷ 32,000 = **₹7.50 per labour hour**.

Particulars	W: blanket (₹)	W: departmental (₹)	C: blanket (₹)	C: departmental (₹)
Direct materials	30,000	30,000	25,000	25,000
Direct wages	8,800	8,800	3,200	3,200
Factory overheads (WN)	13,200	5,400	4,800	14,850
Factory cost	**52,000**	**44,200**	**33,000**	**43,050**
Quotation (factory cost + 10%)	57,200	48,620	36,300	47,355

Working note: blanket: W 150% × ₹8,800 = ₹13,200; C 150% × ₹3,200 = ₹4,800. Departmental: W 20 × ₹120 + 400 × ₹7.50 = ₹2,400 + ₹3,000 = ₹5,400; C 120 × ₹120 + 60 × ₹7.50 = ₹14,400 + ₹450 = ₹14,850.



Blue: blanket rate; orange: departmental rates. The blanket rate follows wages, so it loads the plating-heavy buffet and barely touches the oven-heavy lunch boxes.

(iii) What the blanket rate did. It assumes every order spends the same time in each department, which these two do not: W is mostly hand plating, C mostly oven time. On the blanket rate Tapri quoted ₹57,200 for W, above the rival's ₹55,000, and lost an order whose true factory cost is only ₹44,200. It quoted ₹36,300 for C and won an order whose true factory cost is ₹43,050: a loss of ₹6,750 on it. With departmental rates Tapri would quote ₹48,620 for W (and win it) and ₹47,355 for C (above the rival's ₹45,000; even matching ₹45,000 would leave only ₹1,950 over factory cost). The blanket-rate

conditions are not met here, so departmental rates should be used.

Interactive version: PassBook app, lesson Absorption rates and the machine hour rate

Worked example

Seven sealing machines and a fifth operator

Tapri's premix plant has 7 identical pouch-sealing machines; a machine runs only while an operator is wholly engaged on it, and there are 4 operators. For each operator, for the quarter (3 months):

Hours per operator for the quarter	Hours
Normal available (200 a month)	600
Absenteeism, without pay	12
Leave, with pay	18
Normal unavoidable idle time	10

Operators are paid ₹800 for an 8-hour day, plus a production bonus of 10% of wages. Power is ₹20 per

running hour. The other overheads of the sealing section for the quarter are:

Overheads for the quarter	Amount (₹)
Supervision	45,000
Lighting	9,000
Depreciation (10% a year on ₹16,00,000)	40,000
Repairs and maintenance (4% a year on ₹16,00,000)	16,000
Insurance (₹48,000 a year)	12,000
General management expenses allocated	18,000
Total	**1,40,000**

Required:

(i) Calculate the effective machine hours of the section for the quarter.

(ii) Calculate the comprehensive machine hour rate.

(iii) Kesar proposes to hire a fifth operator on the same terms "to make every hour cheaper". Calculate the new rate and comment.

THE WORKING

(i) Effective hours. The machines can run only while operators work, so the hours come from the

operators, not from the 7 machines.

Hours for the quarter	Per operator	4 operators
Normal available	600	2,400
Less: absenteeism without pay	(12)	(48)
Paid hours	**588**	**2,352**
Less: leave with pay	(18)	(72)
Less: normal idle time	(10)	(40)
Effective machine hours	**560**	**2,240**

(ii) Comprehensive machine hour rate (4 operators)

Particulars	Amount (₹)
Operators' wages: paid hours 2,352 × ₹100 (₹800 ÷ 8)	2,35,200
Production bonus (10% of wages)	23,520
Power (2,240 running hours × ₹20)	44,800
Other overheads of the section	1,40,000
Total overheads for the quarter	**4,43,520**
Machine hour rate (₹4,43,520 ÷ 2,240)	**₹198.00**

Wages are paid on paid hours (leave with pay is paid but not worked), while the rate is spread over effective hours only.

(iii) Five operators. Paid hours = 5 × 588 = 2,940; effective hours = 5 × 560 = 2,800.

Particulars	Amount (₹)
Operators' wages (2,940 × ₹100)	2,94,000
Production bonus (10%)	29,400
Power (2,800 × ₹20)	56,000
Other overheads	1,40,000
Total	**5,19,400**
Machine hour rate (₹5,19,400 ÷ 2,800)	**₹185.50**

Comment. The rate falls by ₹12.50 an hour only because the ₹1,40,000 of other overheads is spread over 2,800 hours instead of 2,240; every effective hour still carries ₹115.50 of wages and bonus and ₹20 of power. The section's total cost rises by ₹75,880. A lower rate is not a saving: the fifth operator pays only if there are orders for the extra 560 machine hours whose contribution, before these sealing costs, exceeds ₹75,880. With 4 operators the 7 machines are used for only 2,240 of a possible 4,200 hours; the constraint is operators, not machines.

Exam angle

In the exam

How ICAI asks absorption rates

In the ten documents read for this lesson, absorption rates came up as follows:

- **Comprehensive machine hour rate, 5–7 marks**: PYQ September 2024 Q5(a) (7 marks: three machines, expenses shared on workers, floor space and kW rating, hours from a calendar at 80% capacity); PYQ January 2026 Q5(a) (6 marks, reverse: the rate is given, find the annual insurance; operators limit the hours); old scheme January 2021 Q1(b) (5 marks: 8 machines, 6 operators).
- **Single-machine rates, 4–6 marks**: MTP May 2025 Series II Q1(c) (4 marks, set-up time productive and unproductive); MTP September 2024 Series I Q4(a) (6 marks, maintenance hours excluded, set-up time productive).
- **Revised pre-determined rates**: RTP May 2026 Q8 (three CNC machines, price and consumption increases, interest excluded).
- **Case-scenario MCQs, 2 marks each**: PYQ May 2026 MCQs 6–10, machine hour rates with and without a hired AI system, for the firm and for three jobs. PYQ September 2026 MCQ 15 tested the capacity levels (a brochure figure is installed capacity; normal idle capacity = installed − normal).
- **Blanket versus departmental, 7–14 marks**: MTP September 2024 Series II Q4 (3 + 4 + 3 + 4 marks: blanket rate on wages, suggest methods, price the job, under- or over-recovery); PYQ September 2026 Q5(b) (2 + 3 + 2 marks: blanket on labour hours, a

recommended method, the under- or over-recovery of one department).

- **Theory**: old scheme January 2021 Q6(c), "Explain blanket overhead rate and departmental overhead rate ... State the conditions required for the application of blanket overhead rate"; MTP January 2025 Series II Q1(a)(v), arguments for and against departmental rates.

In the exam

Traps

- Absenteeism without pay cuts paid hours; leave with pay and idle time cut working hours but not wages (PYQ January 2026, SM practical problem 5).
- In those questions the lost hours are per operator **for the whole period**: ICAI multiplied them by operators only, not by months.
- Annual, quarterly, half-yearly and monthly figures mixed in one question: convert all to the period of the rate.
- "Recommend a method" earns marks for the reason: machine time predominant, machine hour rate; labour predominant, labour hour rate.

Trick

The car park test

Ask what the department mainly uses up: machine time (machine hour rate), people's time (labour hour rate), or nothing that differs between products (one product: rate per unit). Conditions: when the question prescribes a base (a percentage of direct wages, say), use it; a percentage of material cost suits only material handling and stores overheads (SM 4.32).

Answer frame

"Explain blanket and departmental overhead rates; state the conditions for a blanket rate" (5 marks):

1. Blanket rate: one single overhead rate for the whole factory; formula.
2. Departmental rate: one rate for each department or cost centre; formula.
3. Blanket rate conditions: (1) only one major product; (2) several products, but (a) all pass through all departments and (b) all are processed for the same length of time in each.
4. Otherwise departmental rates: product lines varied, or machinery used to a varying degree.

"Arguments for and against departmental absorption rates" (MTP January 2025): for: costs more precise than blanket rates; they facilitate variance analysis and cost control; stock and work-in-process valuation easier and

more precise. Against: setting up and monitoring them can be time-consuming and expensive.

"Direct and comprehensive machine hour rate": definition of MHR; direct (expenses directly connected with the machine ÷ operational hours); comprehensive (plus a share of the department's other expenses, sometimes operator's wages); standing charges and machine expenses; the six steps.

Three answers to "compute the machine hour rate"

Loses marks

"₹9,12,000 ÷ 2,000 = ₹456 an hour."

Standing charges only: the running costs are left out, and nothing shows which items were treated how.

Passes

The right rate, ₹911, but every item listed in one column with no workings.

Correct, yet a slip in one item would leave no method marks.

Full marks

Working notes first (which items accrue with time and which with running, productive hours), then ICAI's statement: standing charges and machine expenses, each with its working, an hourly rate for each, and the total.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the maintenance hours that made the fryer cheaper

Dividing the year's expenses by every hour the factory is open, planned maintenance and idle hours included, gives too low a rate and undercharges every order. **Fix:** the denominator is productive hours: leave out planned maintenance always, set-up time when it is unproductive, and idle time.

⚠ **Common mistake**

Paying for the absentee

Charging wages on effective hours (2,240 × ₹100) or on available hours (2,400 × ₹100). **Fix:** wages are paid on paid hours: available less absenteeism without pay (2,352), because leave with pay is paid but not worked.

⚠ **Common mistake**

The blanket that covered a machine shop

Using one factory-wide rate on wages when one department is machine-paced and another labour-paced: the job's overheads follow its wages, not its use of machines. **Fix:** check the blanket-rate conditions; if products do not spend the same time in each department, compute departmental rates.

⚠ **Common mistake**

A month of rent, a year of insurance

Adding a monthly rent to an annual depreciation to a quarterly power bill. **Fix:** convert every item to the period of the rate before adding, and write the conversion in the working.

Practice

Check yourself

1 A packing machine's standing charges are ₹36,000 a month and it runs 180 productive hours a month. It uses 12 units of power an hour at ₹7.50 a unit and ₹10 of consumables an hour. What is its machine hour rate (₹)?

Answer: _____

2 Case scenario. Tapri's batter-grinding department runs four large wet grinders, each tended by one worker. Its monthly overheads of ₹3,40,000 are 85% power and depreciation of the grinders. The month's budget is 4,000 grinder hours, 4,800 direct labour hours and direct wages of ₹2,40,000. Which absorption rate should Ritu use?

- **A** ₹70.83 per direct labour hour
- **B** 141.67% of direct wages
- **C** ₹85 per machine hour
- **D** ₹72.25 per machine hour

3 For which limited purpose does ICAI accept the percentage of direct material cost as an absorption base?

- **A** Factory overheads of a machine-intensive department
- **B** Material handling and stores overheads
- **C** Administration overheads
- **D** Overheads of a department with skilled workers

4 The Shaping department has 25 workers working 24 days a month, 8 hours a day; normal idle time is 5% of the hours. Its budgeted overheads are ₹2,28,000 a month. What is its labour hour rate (₹)?

Answer: _____

5 When is a single blanket overhead rate appropriate? Select all that apply.

- **A** Only one major product is produced
- **B** Several products pass through all departments and spend the same time in each
- **C** Machinery is used to a varying degree in different departments
- **D** Product lines are varied but the factory is small

More than one answer may be right.

6 Which of these are standing charges in a machine hour rate? Select all that apply.

- **A** Insurance premium of the machine
- **B** Share of the department's rent
- **C** Power consumed while the machine runs
- **D** Depreciation charged on the basis of time
- **E** Lubricating oil

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Absorption = recovering a cost centre's overheads from its output, at overheads ÷ base.
- ICAI's six methods: % of direct materials, % of prime cost, % of direct labour cost, labour hour rate, machine hour rate, rate per unit of output. The first two are generally unsuitable (time, price swings, skill and machines ignored).
- Labour hour rate: little machinery. Machine hour rate: machines the main factor; each machine a cost centre.
- MHR = (standing charges + machine expenses) ÷ productive hours; comprehensive adds the department's other expenses, sometimes operator's wages.

- Rates: normal (actual ÷ actual), pre-determined (budget ÷ budget), blanket (one for the factory), departmental (one per department).
- Blanket only for one major product, or products that pass through every department for the same time; otherwise departmental.
- Fixed overheads on normal capacity, variable on actual production. Normal idle = installed − normal; abnormal idle = normal − actual.
- Wages on paid hours, rate on effective hours. (Same fryer hours, same overhead.)

Answer key · Check yourself

1 **300**

Standing charges ₹36,000 ÷ 180 = ₹200; power 12 × ₹7.50 = ₹90; consumables ₹10. Total ₹300 an hour.

2 **C** **₹85 per machine hour**

The department's overheads are mostly the cost of running the grinders, so machine time is the predominant factor: ₹3,40,000 ÷ 4,000 = ₹85 per machine hour. ₹72.25 spreads only the power and depreciation (₹2,89,000) and leaves the rest unabsorbed; the labour-based rates charge orders by the workers' time, not the grinders'.

3 **B Material handling and stores overheads**

SM 4.32: percentage of materials cost may be used for the limited purpose of absorbing material handling and store overheads. For factory overheads in general it ignores time, skill and the use of machines.

4 **50**

Hours = 25 × 24 × 8 = 4,800, less normal idle time of 5% (240) = 4,560 productive labour hours. Rate = ₹2,28,000 ÷ 4,560 = ₹50 an hour. Dividing by 4,800 (₹47.50) ignores the idle time that no job can be charged with.

5 **A Only one major product is produced; B Several products pass through all departments and spend the same time in each**

SM 4.37: one major product, or several products that all pass through all departments for the same length of time in each. Varied product lines or machinery used to a varying degree call for departmental rates.

6 **A Insurance premium of the machine; B Share of the department's rent; D Depreciation charged on the basis of time**

Standing charges stay the same whether or not the machine runs (SM 4.34). Power and lubricants vary with operation: machine expenses.

5.5 Under- and over-absorption of overheads

Why absorbed overheads never equal actual ones, how to measure the gap after removing abnormal items, and ICAI's treatments: Costing P&L, supplementary rates and, in its answers, carry forward.

In plain words

The ₹66,350 that nobody can find

In March, Ritu fixed Frying's rate at ₹169.25 a fryer hour: budgeted overheads of ₹10,15,500 over 6,000 budgeted hours (absorption rates). In April the fryer worked 5,800 hours, so the samosa orders absorbed 5,800 × ₹169.25 = ₹9,81,650. The bills for April came to ₹10,48,000.

> **Kesar:** We charged the samosas ₹9,81,650 and spent ₹10,48,000. Where did ₹66,350 go?
>
> **Ritu:** Nowhere. We under-absorbed it: the rate recovered less than we spent.
>
> **Kesar:** Put it in miscellaneous. Sab adjust ho jayega.
>
> **Ritu:** Where it goes depends on why it happened.

Overheads are charged to products at a **pre-determined rate**, fixed before the period from estimates. The **absorbed** overhead is that rate times the actual base (hours, units). It almost never equals the **actual** overhead. When less is absorbed than spent, the

overhead is **under-absorbed** (a debit balance left in the overhead account); when more is absorbed than spent, it is **over-absorbed** (a credit balance). ICAI also calls either one an **overhead variance**.

The gap has two possible causes: the overheads were not what was estimated, or the activity was not. What to do with it depends on its size and its reason. A small gap, or one caused by something abnormal (faulty planning, inefficiency), goes to the **Costing Profit and Loss Account**. A large gap caused by an estimate that was wrong for unavoidable reasons (a gas price rise nobody could foresee) is spread back over the products with a **supplementary rate**, to the goods sold and to the goods still in stock.

Think of it like

The hostel mess advance

A hostel mess collects ₹3,000 from each student at the start of the month, from the expected grocery bill and head count. If vegetable prices rise and the true cost is ₹3,200 a head, the mess asks every student for ₹200 more, including those who have already gone home: that is a supplementary rate over cost of sales and stock. If the cook burns 20 kg of rice, the warden pays from the mess fund, not the students: an abnormal loss to Costing P&L. If the difference is ₹5 a head, nobody chases it.

Key idea

Under- or over-absorption = actual overheads – absorbed overheads (actual base × pre-determined rate), after removing items that are not this period's normal overheads. The abnormal part and any small balance go to Costing P&L; a large balance from an unavoidable estimating error is spread by a supplementary rate over cost of sales, finished goods and work-in-progress.

How it works

Why the two never match

Production overheads are determined in advance to fix selling prices, quote tenders and prepare budgets, so the rate is pre-determined, and the actual rate will rarely coincide with it (SM 4.41–4.42):

1. Overheads as estimated, but more units produced than budgeted: the actual rate is lower, so **over-absorption**; fewer units: **under-absorption**.
2. Units as budgeted, but actual overheads more than estimated: **under-absorption**; less: **over-absorption**.
3. Both change in different proportions: under- or over-absorption, depending on the figures.
4. Both change in the same proportion: **neither**.

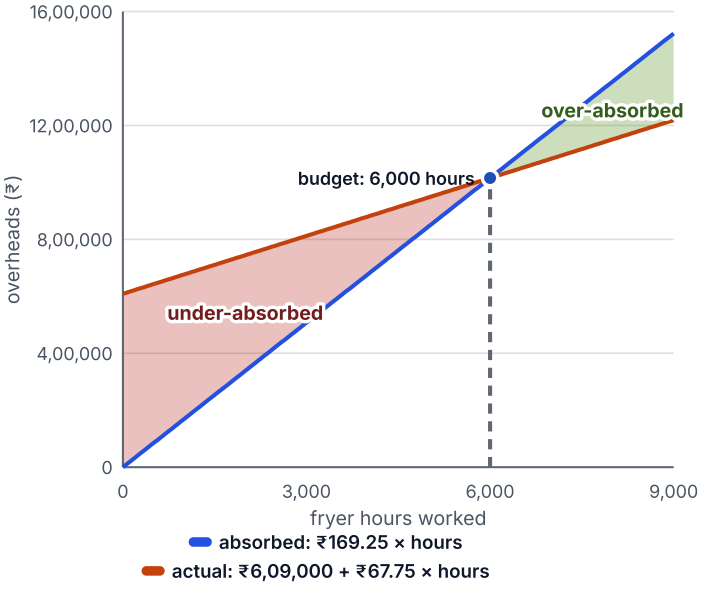
When the absorbed amount is less than the actual amount it is called **under-absorption**; when more, **over-**

absorption. ICAI's table, at a pre-determined rate of ₹100 ÷ 100 units = ₹1 (SM 4.42–4.43):

Budget ₹ / units	Actual ₹ / units	Absorbed (₹)	Actual – absorbed (₹)	Result
100 / 100	110 / 100	100	10	Under-absorption
100 / 100	90 / 100	100	−10	Over-absorption
100 / 100	100 / 90	90	10	Under-absorption
100 / 100	100 / 110	110	−10	Over-absorption
100 / 100	90 / 90	90	0	Neither
100 / 100	110 / 110	110	0	Neither
100 / 100	110 / 90	90	20	Under-absorption
100 / 100	90 / 110	110	−20	Over-absorption

Even when every rupee of overhead behaves exactly as budgeted, a change in activity alone opens a gap, because

the fixed part of the rate is recovered only at the budgeted level:



Blue: overhead absorbed at ₹169.25 a fryer hour; orange: overheads incurred (fixed ₹6,09,000 plus ₹67.75 an hour). Below the budgeted 6,000 hours part of the fixed cost is not recovered (rose); above it, more than the actual is recovered (green).

What to do with the balance

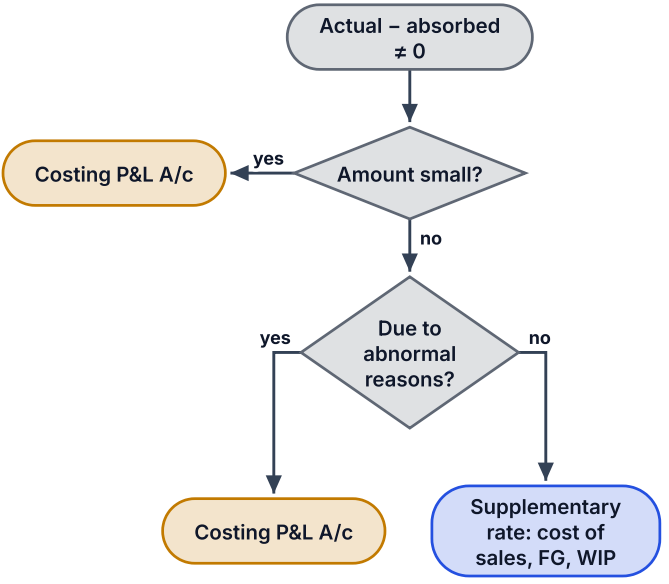
As regards the treatment of the balances (SM 4.43–4.45):

- If the balances are **small**, they are transferred to the **Costing Profit and Loss Account**; product costs are

not increased or reduced, as they would be representing normal cost.

- If the difference is **large and due to wrong estimation** (estimation wrong for unavoidable reasons), the cost of products manufactured should be adjusted, as otherwise cost figures would convey a misleading impression. Such adjustments take the form of **supplementary rates**.
- Over- or under-recovery due to **abnormal reasons** (such as abnormal over- or under-capacity utilisation) is transferred to the **Costing Profit and Loss Account**.

Treatment of under- or over-absorption



Amber: written off to the Costing Profit and Loss Account; blue: spread over cost of sales, finished goods and work-in-progress, for a large balance from an estimate that was wrong for unavoidable reasons. When a question splits the balance by reasons, each part takes its own branch.

Six steps through a question

1. **Clean the actual overheads.** ICAI's answers deduct items that are not this period's normal production overheads: wages paid for a strike period under an award, an amount paid under a labour court award or court order (assumed non-recurring), expenses of the previous year booked in the current year, obsolete stores written off (SM practical problems 7 and 11; PYQ May 2024). Normal overtime wages stay in (MTP May 2025 Series II MCQ 12).

2. **Absorbed overheads** = actual base × pre-determined rate: actual hours, never budgeted hours.

3. **The balance**: net actual − absorbed. Positive: under-absorbed (debit); negative: over-absorbed (credit).

4. **Split by reason**: the part due to defective planning, factory inefficiency or other abnormal causes goes to Costing P&L; the part due to normal cost increases is spread.

5. **Supplementary rate** = the part to spread ÷ units produced, counting work-in-progress in equivalent completed units.

6 **Distribute**: finished goods stock (units × rate), work-in-progress (equivalent units × rate), cost of sales (units sold × rate); debit for under-absorption, credit for over-absorption.

Idle capacity costs

The cost of capacity left idle (installed, normal and abnormal idle capacity are defined in absorption rates) is a common reason for under-absorption. ICAI's three treatments (SM 4.59):

Cause of idle capacity cost	Treatment
(a) Unavoidable reasons: repairs, maintenance, changeover of job	A supplementary overhead rate: the cost is charged to the production capacity utilised
(b) Avoidable reasons: faulty planning, power failure	Charged to the Costing Profit and Loss Account
(c) Seasonal factors	Charged to the cost of production by inflating overhead rates

The machine hour rate has a built-in alarm here: under-absorption of machine overheads indicates the extent to which machines were idle (SM 4.35).

Formats and formulas

Reproduce the calculation statement and the accounting table from memory.

∑ **Formula**

Absorbed overheads and the balance

$$\text{Absorbed} = \text{Actual base} \times \text{Pre-determined rate}$$

$$\text{Under } (+) \text{ or over } (-) = \text{Actual overheads} - \text{Absorbed}$$

Actual base: the actual hours or units of the period. Actual overheads: after removing the abnormal and other-period items.

∑ **Formula**

Supplementary rate (SM 4.44)

$$\text{Supplementary rate} = \frac{\text{Under- or over-absorbed overheads}}{\text{Units produced}}$$

Units produced = finished units (sold and in stock) + equivalent completed units in work-in-progress.

Calculation of under- or over-absorption

Particulars	Amount (₹)
Actual production overheads incurred	xxx
Less: wages for strike period; previous year's expenses; obsolete stores written off	(xxx)
Net production overheads actually incurred (A)	**xxx**
Overheads absorbed (actual base × pre-determined rate) (B)	xxx
Under-absorption (A − B) or over-absorption (B − A)	**xxx**

The accounting (SM 4.44)

Account

1. Stock of finished goods A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Units of finished stock × supplementary rate

Account

2. Stock of semi-finished goods (WIP) A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Equivalent completed units × supplementary rate

Account

3. Cost of sales A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Units sold × supplementary rate

The journal entry for an under-absorption, as in ICAI's suggested answer to old-scheme December 2021 Q5(b):

Journal entry	Dr (₹)	Cr (₹)
Work-in-progress Control A/c Dr.	xxx	
Finished Goods Control A/c Dr.	xxx	
Cost of Sales A/c Dr.	xxx	
Costing Profit and Loss A/c Dr. (abnormal part)	xxx	
To Overhead Control A/c		xxx

For an over-absorption the entry is reversed: Overhead Control A/c is debited, and the three accounts (and Costing P&L for any abnormal part) are credited.

Definition

Under- and over-absorption

When the absorbed amount is less than the actual amount it is called under-absorption. Similarly, when the absorbed amount is more than the actual amount it is called over-absorption. Over-absorption is represented by a credit balance in the accounts and under-absorption by a debit balance; either may be termed overhead variance. (SM 4.42)

Worked examples

Worked example **The fryer line's short year: hours lost, prices risen**

The samosa factory set its production overhead rate for 2025-26 on a normal capacity of 80,000 machine hours: budgeted fixed overheads ₹72,00,000 (₹90 an hour) plus variable overheads of ₹30 an hour, a pre-

determined rate of ₹120 per machine hour. The year's actuals:

Particulars	Data
Machine hours worked	68,000
Fixed overheads incurred (an insurance premium rose)	₹73,09,000
Variable overheads incurred (gas and oil at ₹32 an hour)	₹21,76,000

The 12,000 hours lost against normal capacity were: 8,000 hours in the monsoon, when demand always falls; 2,500 hours of a strike that followed the management's delay in signing a wage agreement; and 1,500 hours for the planned changeover to a new oil-filtration system. Output was 2,75,000 equivalent cartons: 2,20,000 sold, 40,000 in finished stock and 30,000 in work-in-progress, 50% complete.

Required:

(i) Calculate the under- or over-absorption for the year.

(ii) Analyse it by cause and state the treatment of each part.

(iii) Calculate the supplementary rate and apportion the amount to be spread.

THE WORKING

(i) Under-absorption

Particulars	Amount (₹)
Actual overheads (₹73,09,000 + ₹21,76,000) (A)	94,85,000
Absorbed (68,000 hours × ₹120) (B)	81,60,000
Under-absorption (A – B)	**13,25,000**

(ii) Analysis by cause and treatment

Cause
Seasonal idle capacity (monsoon)

Working
8,000 × ₹90

Amount (₹)
7,20,000

Treatment (SM 4.45, 4.59)
Seasonal: charged to the cost of production (by inflating overhead rates)

Cause
Idle capacity, strike from management's delay

Working
2,500 × ₹90

Amount (₹)
2,25,000

Treatment (SM 4.45, 4.59)
Avoidable: Costing Profit and Loss Account

Cause

Idle capacity, planned changeover

Working

1,500 × ₹90

Amount (₹)

1,35,000

Treatment (SM 4.45, 4.59)

Unavoidable: supplementary rate on the capacity utilised

Cause

Rise in variable overhead prices

Working

68,000 × (₹32 – ₹30)

Amount (₹)

1,36,000

Treatment (SM 4.45, 4.59)

Normal increase in costs: supplementary rate

Cause

Rise in fixed overheads

Working

₹73,09,000 – ₹72,00,000

Amount (₹)

1,09,000

Treatment (SM 4.45, 4.59)

Normal increase in costs: supplementary rate

Cause

Total

Working

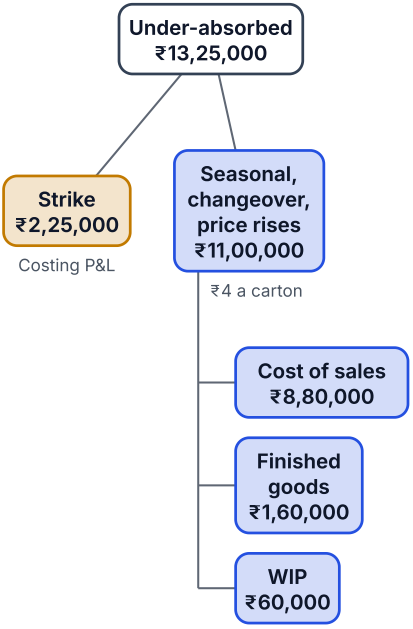
Amount (₹)

13,25,000

Treatment (SM 4.45, 4.59)

Each hour short of normal capacity leaves ₹90 of fixed overhead unabsorbed; the variable rate recovers its cost on every hour actually worked, so lost hours affect only the fixed part.

Where the ₹13,25,000 goes



Amber: the avoidable part, written off; blue: the parts charged to production through the supplementary rate.

(iii) Supplementary rate = ₹11,00,000 ÷ 2,75,000 equivalent cartons = **₹4 a carton** (work-in-progress counts as 30,000 × 50% = 15,000).

Apportionment of ₹11,00,000	Equivalent cartons	Amount (₹)
Cost of sales (2,20,000 × ₹4)	2,20,000	8,80,000
Finished goods (40,000 × ₹4)	40,000	1,60,000
Work-in-progress (15,000 × ₹4)	15,000	60,000
Total	**2,75,000**	**11,00,000**

Profit falls by ₹11,05,000 this year (₹8,80,000 through cost of sales and ₹2,25,000 written off); stocks rise by ₹2,20,000, which reaches profit next year as the stock is sold.

Worked example

Which capacity for the rate? Kesar's slow-year proposal

Tapri's fryer line has fixed overheads of ₹60,00,000 a year and variable overheads of ₹25 per machine hour. Its capacities are: practical capacity 96,000 machine hours; normal capacity (the average expected over the next few years) 75,000; this year's budget, a slow year, 60,000. The line actually worked 64,000 hours, and actual overheads were as budgeted for those hours.

> **Kesar:** This year will be slow, so set the rate on 60,000 hours. Then the rate recovers everything.

Required:

(i) Calculate the pre-determined rate on each of the three capacities, and the under- or over-absorption each gives this year.

(ii) Advise which capacity the rate should be based on, and state the treatment of this year's balance on that basis.

THE WORKING

Actual overheads = ₹60,00,000 + 64,000 × ₹25 = **₹76,00,000**.

(i) Rates and absorption

Capacity used for the rate	Practical, 96,000 hours
Fixed rate (₹)	62.50
Total rate (₹)	87.50
Absorbed: 64,000 × rate (₹)	56,00,000
(Under) / over (₹)	(20,00,000)

Capacity used for the rate

Normal, 75,000 hours

Fixed rate (₹)

80.00

Total rate (₹)

105.00

Absorbed: 64,000 × rate (₹)

67,20,000

(Under) / over (₹)

(8,80,000)

Capacity used for the rate

This year's budget, 60,000 hours

Fixed rate (₹)

100.00

Total rate (₹)

125.00

Absorbed: 64,000 × rate (₹)

80,00,000

(Under) / over (₹)

4,00,000

(ii) Advice. Set the rate on **normal capacity**, ₹105 an hour. The Study Material defines normal capacity as the production achieved or achievable on an average over a period under normal circumstances, taking into account planned maintenance (SM 4.58), and a pre-determined rate on normal capacity keeps the difference between actual and absorbed overheads to a minimum over the years (SM 4.66, MCQ 9).

- Kesar's rate of ₹125 would make every carton fried this year carry ₹20 an hour more fixed

overhead than in a normal year, only because fewer cartons are being made. Product costs and quotations would rise in the very year demand is weak, and the rate would change every year with volume.

- A rate on practical capacity (₹87.50) assumes a level of activity the line does not normally reach, so it under-absorbs year after year (₹20,00,000 this year) and undercosts every product.

On the normal-capacity rate, this year's ₹8,80,000 is entirely fixed overhead left unabsorbed by 11,000 hours of capacity below normal (75,000 − 64,000 = 11,000 × ₹80). Under-recovery due to abnormal under-capacity utilisation is transferred to the **Costing Profit and Loss Account** (SM 4.45); it is the cost of idle capacity, not a cost of the cartons made.

Worked example

The kulhad unit's overhead control account

For the quarter ended 30 June, the kulhad unit's cost ledger shows these production overheads: indirect materials issued ₹3,20,000; indirect wages ₹5,50,000; depreciation of kilns ₹2,40,000; power and other expenses paid ₹3,60,000. Overheads are absorbed at ₹150 per kiln hour, and the kilns worked 9,000 hours.

Of the balance, ₹30,000 arose from a two-day power failure caused by an unpaid electricity bill; the rest is due to normal cost increases and is to be spread by a supplementary rate. Output was 18,000 equivalent units: 14,000 sold, 2,000 in finished stock and 4,000 in work-in-progress, 50% complete.

Required:

(i) Prepare the Production Overhead Control Account for the quarter.

(ii) Pass the journal entry to close its balance.

THE WORKING

Working notes

1. Actual overheads = ₹3,20,000 + ₹5,50,000 + ₹2,40,000 + ₹3,60,000 = ₹14,70,000; absorbed = 9,000 × ₹150 = ₹13,50,000; **under-absorbed ₹1,20,000**.
2. Power failure from an unpaid bill: avoidable, ₹30,000 to Costing P&L (SM 4.59). Normal part ₹90,000 ÷ 18,000 equivalent units = **₹5 a unit**: cost of sales 14,000 × ₹5 = ₹70,000; finished goods 2,000 × ₹5 = ₹10,000; work-in-progress (4,000 × 50%) 2,000 × ₹5 = ₹10,000.

(i) Production Overhead Control Account

Production Overhead Control Account

Dr.	₹
To Stores Ledger Control A/c (indirect materials)	3,20,000
To Wages Control A/c (indirect wages)	5,50,000
To Cost Ledger Control A/c (depreciation)	2,40,000
To Cost Ledger Control A/c (power and other expenses)	3,60,000
Total	**14,70,000**

Cr.	₹
By Work-in-Progress Control A/c (absorbed: 9,000 × ₹150)	13,50,000
By Costing P&L A/c (power failure)	30,000
By Cost of Sales A/c (supplementary)	70,000
By Finished Goods Control A/c (supplementary)	10,000
By Work-in-Progress Control A/c (supplementary)	10,000
Total	**14,70,000**

The debit side collects the actual overheads; the credit side shows what was absorbed. The difference before closing, ₹1,20,000 on the debit side, is the under-absorption; the last four credits close it.

(ii) Journal entry

Particulars	Dr (₹)	Cr (₹)
Costing Profit and Loss A/c Dr.	30,000	
Cost of Sales A/c Dr.	70,000	
Finished Goods Control A/c Dr.	10,000	
Work-in-Progress Control A/c Dr.	10,000	
To Production Overhead Control A/c		1,20,000
(Being under-absorbed overheads closed: the avoidable part written off, the normal part spread by a supplementary rate of ₹5 a unit)		

Exam angle

In the exam

How ICAI asks under- and over-absorption

In the 13 documents read for this lesson it came up as follows:

- **Compute and treat, 6–10 marks**: PYQ May 2024 Q4(b) (6 marks: obsolete stores excluded, 20% defective planning, work-in-progress 40%

complete); old scheme December 2021 Q5(b) (10 marks: strike-period wages and previous-year expenses excluded, 40% factory inefficiency, journal entry).

- **From a trial balance, 5 marks**: MTP January 2026 Series I Q1(b), apportioned on values.
- **Complete the accounting table, 3 marks**: PYQ September 2025 Q5(c), the Dr/Cr and amount columns of SM 4.44.
- **Case-scenario MCQs**: RTP May 2024 case scenario (rate, absorbed amount, over-absorption after four exclusions, supplementary rate, the share of work-in-progress), the Study Material's own case scenario 1.
- **Single MCQs, 2 marks**: MTP May 2025 Series II MCQ 12 and MTP September 2026 Series II MCQ 15 (strike wages out, overtime in, one-third to Costing P&L); MTP May 2024 Series I MCQ 6 and MTP September 2026 Series I MCQ 14 (absorbed on actual hours); RTP September 2024 MCQ 3.
- **Inside larger questions**: MTP January 2025 Series II Q1(a)(iv); MTP September 2024 Series II Q4(d) (4 marks, department-wise, blanket against departmental).
- **Theory**: MTP May 2024 Series I Q6(a) (5 marks, the three treatments); MTP May 2025 Series II Q4(c) (2 marks) and old-scheme RTP November 2023, idle capacity costs.

In the exam

Traps

- Absorbed = **actual** hours × rate. Budgeted overheads minus actual overheads is not the under-absorption: in MTP September 2026 Series I MCQ 14 that shortcut leads to the ₹5,000 distractor, while ICAI's answer is ₹8,800 under-absorbed.
- Exclude strike-period wages, court-award payments, previous-year expenses and obsolete stores before comparing; keep normal overtime.
- Work-in-progress enters the supplementary rate at equivalent completed units (30,000 cartons at 50% = 15,000).
- "Transferred to Costing P&L" in an MCQ usually means only the abnormal share of the balance.

Trick

Sign check

Under-absorption: the accounts are short of cost, so debit the products (and Costing P&L) and credit Overhead Control. Over-absorption: the reverse. Condition: a debit balance on the overhead account is always under-absorption.

Answer frame

"Explain the treatment of over- and under-absorption

of overheads in cost accounts" (5 marks):

1. Meaning: absorbed at pre-determined rates; actual differs; under = absorbed less than actual (debit balance), over = absorbed more than actual (credit balance).
2. Small balances: Costing P&L, product costs not disturbed, as they represent normal cost.
3. Large balances due to wrong estimation for unavoidable reasons: supplementary rate = under/over-absorbed ÷ units produced, applied to finished goods, WIP (equivalent units) and cost of sales.
4. Abnormal reasons (abnormal over- or under-capacity utilisation, defective planning): Costing P&L.
5. Carry forward to the subsequent year (ICAI's answer to MTP May 2024 Series I Q6(a)).

"How would you treat idle capacity costs?" (2 marks in MTP May 2025 Series II Q4(c)): the three causes and treatments in the table above, in ICAI's order: unavoidable (supplementary rate), avoidable (Costing P&L), seasonal (inflated overhead rates).

Three answers to the fryer-line question

Loses marks

"Under-absorption ₹13,25,000; charge to Costing P&L."

No analysis by cause, no supplementary rate.

Passes

The ₹13,25,000 is right and split by cause, but the supplementary rate divides by finished cartons only.

Work-in-progress missed: the apportionment fails.

Full marks

The calculation statement; the analysis by cause (hours lost × fixed rate, price rises), each part's treatment stated with its reason; the working note for equivalent units and the rate; the apportionment table; the journal entry if asked.

Common mistakes

⚠ **Common mistake**

Hall of Shame: budget minus actual

A rate of ₹30 an hour was budgeted on ₹7,20,000 for 24,000 hours; the factory worked 23,100 hours and spent ₹7,05,000. Comparing budgeted with actual overheads (₹15,000 "over") ignores the hours actually worked. **Fix:** absorbed = 23,100 actual hours × ₹30 = ₹6,93,000, against actual ₹7,05,000: ₹12,000 under-absorbed.

⚠ **Common mistake**

The strike that was charged to samosas

Leaving strike-period wages, previous-year expenses and obsolete stores inside actual overheads inflates the under-absorption and spreads abnormal costs into stock. **Fix:** deduct them first, and name each one in the statement.

⚠ **Common mistake**

The forgotten half-made cartons

Dividing the part to be spread only by finished cartons (2,60,000) gives ₹4.23 a carton and leaves work-in-progress at the old cost. **Fix:** add work-in-progress at equivalent completed units: 2,75,000 cartons, ₹4 each.

⚠ **Common mistake**

Everything to Costing P&L

Writing off a large, normal under-absorption leaves every product undercosted and stock undervalued. **Fix:** only small balances and abnormal parts go to Costing P&L; a large normal balance is spread by a supplementary rate.

Practice

Check yourself

1 Budgeted overheads were ₹5,40,000 for 18,000 machine hours. The factory worked 17,000 machine hours and incurred ₹5,30,000 of overheads. What is the result?

- **A** ₹10,000 over-absorbed
- **B** ₹10,000 under-absorbed
- **C** ₹20,000 under-absorbed
- **D** ₹20,000 over-absorbed

2 Case scenario. Tapri's kulhad unit absorbed ₹16,80,000 of overheads. Its actual overheads of ₹18,40,000 include ₹60,000 of wages paid for a strike period under an award and ₹25,000 of normal overtime premium. Ritu finds that one-fourth of the under-absorption was due to faulty planning. How much of the under-absorption should be charged to the Costing Profit and Loss Account?

- **A** ₹40,000
- **B** ₹25,000
- **C** ₹18,750
- **D** ₹75,000

3 How does ICAI treat idle capacity cost caused by seasonal factors?

- **A** Charged to the Costing Profit and Loss Account
- **B** Charged to the cost of production by inflating overhead rates
- **C** Recovered by a supplementary rate on the capacity utilised
- **D** Carried forward to the next year

4 Which items do ICAI's answers deduct from actual production overheads before computing the under-absorption? Select all that apply.

- **A** Wages paid for a strike period under an award
- **B** Expenses of the previous year booked in the current year
- **C** Obsolete stores written off
- **D** Normal overtime wages of indirect workers

More than one answer may be right.

5 A normal under-absorption of ₹54,000 is to be spread by a supplementary rate. 16,000 units were completed (12,000 sold) and 4,000 units are in progress, 50% complete. How much is charged to work-in-progress (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Absorbed = actual base × pre-determined rate; under = actual > absorbed (debit balance); over = absorbed > actual (credit balance); both are "overhead variance".
- Causes: overheads differ from estimate, activity differs from budget, or both in different proportions; a proportional change in both gives neither.
- First remove strike wages, court awards, previous-year expenses and obsolete stores; keep normal overtime.
- Small balance: Costing P&L. Abnormal part: Costing P&L. Large normal part: supplementary rate.
- Supplementary rate = balance ÷ units produced (WIP at equivalent units); applied to finished goods, WIP and cost of sales.
- Idle capacity: unavoidable, supplementary rate; avoidable, Costing P&L; seasonal, inflated overhead rates.

- ICAI's answers add a third treatment: carry forward to the next year.
- The mess asks every student for the vegetable price rise, not for the burnt rice.

Answer key · Check yourself

1 **C** **₹20,000 under-absorbed**

Rate = ₹5,40,000 ÷ 18,000 = ₹30; absorbed = 17,000 × ₹30 = ₹5,10,000; actual ₹5,30,000 exceeds it by ₹20,000: under-absorbed. ₹10,000 compares budgeted with actual overheads and ignores the hours.

2 **B** **₹25,000**

Net actual = ₹18,40,000 − ₹60,000 = ₹17,80,000; under-absorption = ₹1,00,000; one-fourth = ₹25,000. ₹40,000 forgets to remove the strike wages; ₹18,750 also removes the overtime, which is a normal cost; ₹75,000 is the part spread by a supplementary rate.

3 **B** **Charged to the cost of production by inflating overhead rates**

SM 4.59: seasonal idle capacity cost is charged to cost of production by inflating overhead rates. A supplementary rate is for unavoidable causes such as repairs and changeovers; Costing P&L for avoidable causes such as faulty planning.

4 **A Wages paid for a strike period under an award; B Expenses of the previous year booked in the current year; C Obsolete stores written off**

The first three are abnormal or belong to another period. Normal overtime is a normal cost of the period and stays in (MTP May 2025 Series II MCQ 12).

5 **6000**

Equivalent units = 16,000 + 4,000 × 50% = 18,000; rate = ₹54,000 ÷ 18,000 = ₹3; work-in-progress = 2,000 × ₹3 = ₹6,000.

5.6 Administration, selling, distribution and R&D overheads

How the costs of running, selling and delivering are accounted for, absorbed and controlled, and how ICAI treats R&D, bad debts, interest, depreciation, carriage and night shift allowance.

In plain words

Who pays for the boardroom chai?

Tapri's head office in Indore costs lakhs every month: the board, finance, legal, the audit fee. The sales team, the Diwali advertising and the dealers' commission cost more. The delivery vans and godowns cost more again. None of it fries a samosa.

> **Kesar:** The factory makes the samosas. Why should a samosa pay for the board's chai?
>
> **Ritu:** Because without the board there is no factory, no price list and no samosa.
>
> **Kesar:** Then add 20% to every pack and be done. Or put it all in miscellaneous.
>
> **Ritu:** The first is one of ICAI's three methods. The second is not.

Administration overheads are the costs of general management and of secretarial, accounting and administrative services, which cannot be related directly to production, marketing, research or development.

Selling overheads are incurred to promote marketing and sales (salesmen, advertising), and **distribution overheads** to deliver and despatch goods sold (vans, transit insurance, godowns). They sit below works cost in the cost sheet, and each has its own question: how much of it each product should carry, and how to keep it from growing.

Administration costs are mostly fixed and arise from management's policy; selling and distribution costs depend on markets, competitors and customers. Both are hard to tie to a unit and hard to control. ICAI's answers are a choice of methods and bases for absorbing them, and three tools for control: comparison with the past, budgets and standards. **Research and development** costs go wherever the research points: to production, to administration, to selling, to one product, or to Costing P&L if the research fails.

Think of it like

The school principal's salary

A school's teachers are its production cost. Who bears the principal's salary? The school can split it between the classrooms and the admissions office (apportion between production and sales), treat it as a school-wide expense charged to no class (Costing P&L), or add it to each class as a separate charge on its fees or its number of students (a separate addition to cost). The admission brochures are selling costs, and the school bus is distribution.

Key idea

Administration overheads are apportioned between production and sales, charged to Costing P&L, or added to cost as a separate charge on a base such as works cost; selling and distribution overheads are best absorbed as an estimated amount per unit (fixed costs by benefit, plus variable costs). Both are controlled by comparison, budgets and standards. R&D follows the purpose of the research.

How it works

Administration overheads

Definition

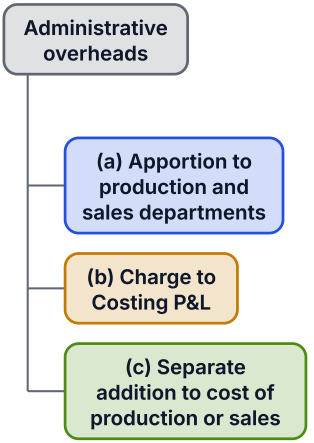
Administrative overhead

"The sum of those costs of general management and of secretarial accounting and administrative services, which cannot be directly related to the production, marketing, research or development functions of the enterprise." (CIMA Terminology, SM 4.46)

They are the expenses of formulating policy, directing the organisation and controlling its operations, and are collected and classified like factory overheads. ICAI gives

three methods of accounting for them, in this order (SM 4.47–4.48):

Three ways to account for administrative overheads



Under (a) they merge into production and selling and distribution overheads; under (b) they never reach product cost; under (c) they stay a separate function with their own rate.

Method (SM 4.47–4.48)

(a) Apportion between production and sales departments

Why

Incurred for the benefit of both, so each bears a share; they lose their identity and merge with production and selling and distribution overheads

Disadvantages, or bases

(1) Suitable bases are difficult to find; (2) a lot of clerical work; (3) not justified to apportion only over production and sales when equally important departments, like finance, exist

Method (SM 4.47–4.48)

(b) Charge to Costing Profit and Loss Account

Why

They concern policy formulation, not production or selling directly; a suitable basis is difficult to find; they are fixed costs

Disadvantages, or bases

(1) Cost of products is understated; (2) excluding them from product cost is against sound accounting principle

Method (SM 4.47–4.48)

(c) A separate addition to cost of production or sales

Why

Administration is a separate function like production and sales, charged to each job or product with the cost of the other functions

Disadvantages, or bases

Bases used: (i) works cost; (ii) sales value or quantity; (iii) gross profit on sales; (iv) quantity produced; (v) conversion cost, etc.

Control. Mostly fixed, arising from management policy and generally non-controllable, they should still not be allowed to grow disproportionately. Three methods (SM 4.48–4.49):

1. **Classification and analysis by administrative departments according to their functions, compared with the results accomplished.** Each department's expenses are collected under standing orders and compared with the previous period in relation to accomplishment. This gives limited control when activity is not constant, so absorption rates may also be compared from period to period: the extent of over-

or under-absorption reveals efficiency. A service department's cost may be compared with the price of the same service from outside, to decide whether to continue it or entrust the work to outsiders.

2. **Control through budgets**: monthly or annual administration budgets for each department, compared with actuals; variances analysed and responsibility assigned.
3. **Control through standards**: standards of performance for each administrative activity, compared with actual performance.

Selling and distribution overheads

Selling overheads are the expenses of promoting the marketing and sales of different products; distribution expenses relate to delivery and despatch of goods sold. Many concerns group them as one class, **selling and distribution overheads**, collected through standing

orders (SM 4.53). The usual absorption bases, with ICAI's comments (SM 4.53–4.54):

Basis	ICAI's comment
(a) Sales value of goods	Ordinarily the most logical basis: there is some connection between sales and the expenses incurred to achieve them
(b) Cost of goods sold	Not as satisfactory: it may have no direct relationship with selling and distribution cost
(c) Gross profit on sales	Gives a larger share to goods with a large margin: follows the principle of "ability to pay"; may not reflect costs or efforts incurred
(d) Number of orders or units sold	One of the usual bases

The best method, an estimated amount per unit. Separate fixed expenses from variable ones. Apportion the fixed expenses according to the benefit each product derives, then divide by the units sold or likely to be sold; add the variable expenses per unit, worked out for each product: packaging, freight outwards, insurance in

transit, commission to salesmen, rebate to customers (SM 4.54). ICAI's bases for the fixed ones:

Fixed expense	Basis of apportionment (SM 4.54)
Salaries of the sales department and of salesmen	Estimated time devoted to the sale of each product
Advertisement	Actual amount for each product; common expenses (an exhibition) on each product's advertisement expenditure
Show room expenses	Average space occupied by each product
Rent of finished goods godowns; expenses on own delivery vans	Average quantities delivered during a period
No suitable basis	Proportion of sales of the products

Control is difficult, for three reasons (SM 4.55):

1. Their incidence depends mainly on **external factors** beyond management's control: distance of the market, extent and nature of competition, terms of sale.
2. They depend on **customers' behaviour**, likes, dislikes and tastes, so controlling them may lose customers.
3. Being **policy costs**, they are not amenable to control.

In spite of this, three methods may be used: **(a) comparison with past performance** (the expenses, or the expenses as a percentage of sales, against the previous period; suitable for small concerns); **(b) budgetary control** (a budget split into fixed and variable, if necessary a flexible budget for different levels of sales, actuals compared and variances acted on); **(c) standard costing** (standards for salesmen, territories, products, set in relation to the standard sales volume; actuals compared and variances enquired into).

Research and development expenses

Definition

Research expenses; development expenses

Research expenses: "the expenses of searching for new or improved products, new application of materials, or new or improved methods." Development expenses: "the expenses of the process which begins with the implementation of the decision to produce a new or improved product." (Terminology, SM 4.62)

Where an R&D expense goes (SM 4.62–4.63)

Research and development expenses

Cost of production, a separate item
research in the methods of production

The particular product
development costs of that product:
deferred revenue expenses, recovered per unit
once production is fully established

Administration overhead
research relating to administration
fundamental research, not for a specific product

Selling and distribution overhead
market research

General overhead
routine research on new or improved methods,
or on improving existing products;
if substantial, deferred over the benefit period

Costing P&L, excluded from costs
research that proves a failure

Each box is a destination, and its lines the research that goes there. Blue and green: product cost; amber, orange and violet: overheads; rose: written off.

Other items: what ICAI says

Item (SM 4.59–4.62)	Treatment
Interest and financing charges: interest for the use of non-equity funds and the incidental cost of arranging them (interest on borrowings, financing charges of finance leases, cash discount allowed to customers); not imputed costs	Presented in the cost statement as a **separate item of cost of sales**
Depreciation: "the diminution in the intrinsic value of an asset due to use and/or the lapse of time"	Traced to the cost object to the extent economically feasible; otherwise assigned on (i) cause and effect or (ii) benefit received
Bad debts	No unanimity. One view: excluded from cost, being financial losses. Another: part of selling and distribution overheads when they arise in the normal course of trading. Extraordinarily large bad debts are not included in cost accounts
Carriage and cartage	On direct material: cost of direct material; on indirect material (stores): factory overheads; on finished goods: distribution overhead

Item (SM 4.59–4.62)	Treatment
Night shift allowance	Usually caused by pressure of work beyond normal capacity: a production overhead, recovered as such (as direct wages it would make night jobs costlier than day jobs). For a specific customer's order: charged to that order. Due to abnormal circumstances: Costing P&L

Packing (primary, secondary, special and fancy), fringe benefits, removal and re-erection of machines, canteen, welfare and training expenses are tabulated in classifying overheads.

Formats and formulas

Reproduce the reverse-calculation steps and the product-wise statement from memory.

∑ **Formula**

Administration and selling rates on a base

$$\text{Admin. rate} = \frac{\text{Admin. overheads}}{\text{Works cost}} \times 100$$

$$\text{S\&D rate} = \frac{\text{S\&D overheads}}{\text{Sales value}} \times 100$$

Works cost is factory cost. The base is the one the question names: works cost, cost of production, sales value, units.

∑ **Formula**

Estimated selling and distribution expense per unit (SM 4.54)

$$\text{S\&D per unit} = \frac{\text{Fixed S\&D of the product}}{\text{Units sold}} + \text{Variable per unit}$$

Fixed S&D of the product: the fixed expenses apportioned to it on their bases. Units sold: sold or likely to be sold. Variable per unit: packaging, freight outwards, insurance in transit, commission, rebate.

∑ **Formula**

Working back from a selling price

$$\text{Total cost} = \frac{\text{Price}}{1 + p_c}$$

$$\text{Total cost} = \text{Price} \times (1 - p_s)$$

p_c = profit as a fraction of cost; p_s = profit as a fraction of the selling price. With factory overheads at F% of direct wages and administration at A% of factory cost: factory cost = prime cost + F × direct wages; total cost = factory cost × (1 + A).

Product-wise statement of profit (ICAI's layout, SM 4.57)

Particulars	Product A (₹)	Product B (₹)
Sales	xxx	xxx
Less: Variable costs: cost of goods sold, commission, packing, stationery	xxx	xxx
Contribution	**xxx**	**xxx**
Less: Fixed costs, each apportioned on its basis (ratio shown)	xxx	xxx
Profit or loss	**xxx**	**xxx**
Percentage of profit or loss on sales	**x%**	**x%**

ICAI's statement also has a Total column; write it first.

Worked examples

Worked example **Administration overheads: works cost, conversion cost or units?**

Tapri's Indore factory makes two products. Per unit and

for the year:

Particulars	Chai premix (per box)	Frozen samosa (per pack)
Direct materials (₹)	60	20
Direct wages (₹)	10	25
Factory overheads (₹)	15	35
Units produced and sold	2,00,000	1,00,000

Administration overheads of ₹22,00,000 (accounts, payroll, HR and the general office) are treated as a separate addition to the cost of production. Ritu must choose the base, and the factory prices at cost of production plus 20%.

Required:

(i) Calculate the administration overhead per unit of each product if it is absorbed on (a) works cost, (b) conversion cost, (c) quantity produced.

(ii) Calculate the cost of production and the price of each product under each base.

(iii) Advise on the base, given that most of the administration work is payroll and HR for the people in the factory and the office.

THE WORKING

Working note: the three bases

Base

Works cost (₹85 / ₹80 a unit)

Chai premix (₹)

1,70,00,000

Frozen samosa (₹)

80,00,000

Total (₹)

2,50,00,000

Rate

8.8%

Base

Conversion cost: wages + factory overheads (₹25 / ₹60 a unit)

Chai premix (₹)

50,00,000

Frozen samosa (₹)

60,00,000

Total (₹)

1,10,00,000

Rate

20%

Base

Quantity produced (units)

Chai premix (₹)

2,00,000

Frozen samosa (₹)

1,00,000

Total (₹)

3,00,000

Rate

₹7.33 a unit

Rates: ₹22,00,000 ÷ ₹2,50,00,000 = 8.8%; ₹22,00,000 ÷ ₹1,10,00,000 = 20%; ₹22,00,000 ÷ 3,00,000 = ₹7.33.

(i) and (ii) Administration overhead, cost of production and price per unit

Base	Premix: admin (₹)	Premix: cost / price (₹)	Samosa: admin (₹)	Samosa: cost / price (₹)
(a) Works cost, 8.8%	7.48	92.48 / 110.98	7.04	87.04 / 104.45
(b) Conversion cost, 20%	5.00	90.00 / 108.00	12.00	92.00 / 110.40
(c) Quantity produced	7.33	92.33 / 110.80	7.33	87.33 / 104.80

Each base recovers the full ₹22,00,000. For example, on works cost: 2,00,000 × ₹7.48 +

1,00,000 × ₹7.04 = ₹14,96,000 + ₹7,04,000.

(iii) Advice. The Study Material lists works cost, sales value or quantity, gross profit on sales, quantity produced and conversion cost among the bases for treating administration overheads as a separate addition (SM 4.48). The base should follow what causes the overhead. Here the work is mostly payroll and HR for people, and the samosa line employs far more labour (₹25 of wages a pack against ₹10 a box). Works cost loads the material-heavy premix with administration it does not cause, because ₹60 of its ₹85 is material; a rate per unit treats a box and a pack as making the same demand. Conversion cost follows the labour and factory activity that the administration actually serves, so it is the most suitable of the three: the samosa's price rises to ₹110.40 and the premix's falls to ₹108.00.

Worked example **Three channels, one advertising budget**

Tapri sells through its own cafés, a food-delivery app

and supermarkets. For 2025-26:

Particulars	Cafés	Delivery app	Supermarkets
Sales (₹)	80,00,000	60,00,000	40,00,000
Cost of goods sold (₹)	40,00,000	30,00,000	26,00,000
Orders or invoices	2,00,000	75,000	4,000

Selling and distribution costs:

Cost	Amount and nature
App platform commission	20% of app sales (variable)
Delivery packaging	₹8 per app order (variable)
Card and UPI charges	1% of café and app sales (variable)
Freight to supermarkets	₹100 per invoice (variable)
Supermarket merchandisers	₹4,80,000 (fixed, supermarkets only)
Promotions run with the app	₹3,00,000 (fixed, app only)
Brand advertising and sales office	₹12,60,000 (fixed, common to all, apportioned on sales value)

> **Ananya:** App orders bring ₹80 each against ₹40 in the cafés. The app is our best channel.

Required:

(i) Prepare a statement of contribution and profit by channel.

(ii) Calculate the contribution per ₹100 of sales and per order of each channel.

(iii) An extra ₹5,00,000 of advertising is expected to add ₹25,00,000 of sales to whichever channel it is spent on, with the same cost structure. Advise where to spend it, and comment on Ananya's view.

THE WORKING

(i) Statement of profit by channel

Particulars	Cafés (₹)	App (₹)	Supermarkets (₹)
Sales	80,00,000	60,00,000	40,00,000
Less: Cost of goods sold	40,00,000	30,00,000	26,00,000
Less: App commission (20%)	—	12,00,000	—

Particulars	Cafés (₹)	App (₹)	Supermarkets (₹)
Less: Delivery packaging (75,000 × ₹8)	—	6,00,000	—
Less: Card and UPI charges (1%)	80,000	60,000	—
Less: Freight (4,000 × ₹100)	—	—	4,00,000
Contribution	**39,20,000**	**11,40,000**	**10,00,000**

Particulars	Cafés (₹)	App (₹)	Supermarkets (₹)
Less: Specific fixed costs	—	3,00,000	4,80,000
Contribution after specific fixed costs	**39,20,000**	**8,40,000**	**5,20,000**
Less: Common fixed costs (sales value, 4 : 3 : 2)	5,60,000	4,20,000	2,80,000
Profit	**33,60,000**	**4,20,000**	**2,40,000**

Total profit = ₹40,20,000 (contribution ₹60,60,000 – specific ₹7,80,000 – common ₹12,60,000).

(ii) Contribution ratios

Measure	Cafés	App	Supermarkets
Sales per order (₹)	40	80	1,000
Contribution per ₹100 of sales (₹)	49.00	19.00	25.00
Contribution per order (₹)	19.60	15.20	250.00

(iii) Where to spend ₹5,00,000

Channel	Extra contribution on ₹25,00,000 of sales (₹)	Less: advertising (₹)	Gain / (loss) (₹)
Cafés (49%)	12,25,000	5,00,000	7,25,000
App (19%)	4,75,000	5,00,000	(25,000)
Supermarkets (25%)	6,25,000	5,00,000	1,25,000

Spend it on the cafés. Ananya's measure, sales per order, ignores the costs each channel's sales carry: the app's 20% commission and packaging leave only ₹19 of every ₹100 of app sales as contribution, the lowest of the three, and its extra sales would not even pay for the advertising. A channel is judged by contribution per rupee of sales (and its specific fixed costs), not by the size of its orders.

Worked example

The kulhad unit's odd items: where does each go?

Tapri's kulhad unit made and sold 6,00,000 kulhads in 2025-26. Its records show direct materials ₹9,00,000, direct wages ₹6,00,000, factory overheads ₹4,80,000,

administration overheads ₹2,40,000 and routine selling and distribution overheads ₹1,50,000. In addition:

(a) A night shift allowance of ₹60,000, paid because Diwali demand exceeded the kiln's normal day capacity.

(b) A night shift allowance of ₹24,000, paid only to finish a hotel chain's order of 40,000 kulhads on time.

(c) ₹4,50,000 spent on developing a new glaze; production with it will be fully established next year, and 15,00,000 glazed kulhads are expected over three years.

(d) ₹1,20,000 of research on a biodegradable lid that failed.

(e) ₹90,000 on fancy printed sleeves to attract buyers.

(f) ₹1,20,000 on cartons to carry kulhads safely from the unit to the cafés.

(g) ₹80,000 for removing and re-erecting the kiln, which had been sited wrongly when the unit was planned.

(h) ₹60,000 for training new potters, at the normal level for the unit.

Required:

(i) State the treatment of each item (a) to (h).

(ii) Prepare a cost sheet for the year showing works cost, cost of production, cost of sales and cost per

kulhad, with the hotel chain's order costed separately.

THE WORKING

(i) Treatment of each item

Item	Treatment (SM 4.60–4.63)
(a) Night shift, general pressure of work	Production overhead, recovered as such
(b) Night shift for one customer's order	Charged directly to that order
(c) Development of a new glaze	Deferred revenue expense; recovered from next year at ₹4,50,000 ÷ 15,00,000 = ₹0.30 a kulhad; nothing charged this year
(d) Failed research	Excluded from costs; Costing Profit and Loss Account
(e) Fancy sleeves to attract buyers	Advertising: selling overhead
(f) Cartons to transport the product	Distribution overhead
(g) Re-erection due to faulty planning	Abnormal: Costing Profit and Loss Account

Item	Treatment (SM 4.60–4.63)
(h) Normal training of factory workers	Part of the cost of production (production overhead)

(ii) Cost sheet for 2025-26 (6,00,000 kulhads)

Particulars	Amount (₹)	Per kulhad (₹)
Direct materials	9,00,000	1.50
Direct wages	6,00,000	1.00
Prime cost	**15,00,000**	**2.50**
Factory overheads	4,80,000	
Night shift allowance, general (a)	60,000	
Training of potters (h)	60,000	
Works cost	**21,00,000**	**3.50**
Administration overheads	2,40,000	0.40
Cost of production	**23,40,000**	**3.90**
Selling and distribution: routine	1,50,000	
Selling: fancy sleeves (e)	90,000	

Particulars	Amount (₹)	Per kulhad (₹)
Distribution: transport cartons (f)	1,20,000	
Cost of sales	**27,00,000**	**4.50**

Hotel chain's order of 40,000 kulhads: 40,000 × ₹4.50 = ₹1,80,000, plus its own night shift allowance (b) ₹24,000 = **₹2,04,000**, or ₹5.10 a kulhad. The other 5,60,000 kulhads cost ₹4.50 each.

Total cost for the year, including the hotel order's own charge, = ₹27,24,000. Kept out of the year's costs: failed research ₹1,20,000 and re-erection ₹80,000 (₹2,00,000 to Costing P&L), and the development cost ₹4,50,000 carried forward as deferred revenue expenditure.

Exam angle

In the exam

How ICAI asks these overheads

In the eight documents read for this lesson:

- **Reverse calculation of recovery rates, 2–10 marks**: PYQ January 2026 MCQ 12 (one job, find the factory overhead percentage, 2 marks); old scheme May 2022 Q2(a) (two jobs, rates, statement, quote for a

third job at a margin on selling price, 10 marks); old scheme MTP November 2023 Series II Q4(b) (Illustration 9's data).

- **Last year's rates applied to an order, 6–8 marks**: PYQ January 2025 Q4(b) (production overhead on labour cost, marketing and distribution on production cost, price at 20% on sales, 6 marks); MTP May 2024 Series II Q4(a) (administration and selling and distribution as percentages of works cost, 8 marks); PYQ September 2026 Q2(b) (administration as a percentage of factory cost for a special order, 7 marks).
- **Theory, 4–5 marks**: PYQ January 2025 Q6(c) (4 marks), "Explain the methods that can be used for controlling Selling and Distribution Overheads"; MTP September 2025 Series I Q6(d) (OR, 4 marks), "Research and Development Expenses and its treatment in costing"; old scheme MTP December 2021 Series I Q6(d) (5 marks), fringe benefits and bad debts. The chapter's own question 3 asks the problems of controlling selling and distribution overheads.
- **Product-wise profitability with selling and distribution costs on drivers**: RTP January 2025 Q10 (packing per parcel, commission on sales, stationery per invoice, fixed costs apportioned; examine a recommendation to discontinue a product), the pattern of Illustration 10.

In the exam

Traps

- "Profit on total cost" divides by (1 + p); "profit on selling price" multiplies by (1 – p), or divides the cost by (1 – p).
- Read the base: administration on factory cost (works cost) is not administration on prime cost or on cost of production.
- Special packing at a customer's request is charged to that order (SM 4.60). Order-specific costs given separately are added outside the percentages: ICAI's answer to PYQ January 2025 Q4(b) added the order's packaging and transportation after the percentage-based overheads, and PYQ September 2026 Q2(b) gives delivery and installation at the customer's site in the same way.
- Bad debts: the chapter's own MCQ (SM 4.66, question 7) answers "selling overhead" (SM 4.80).
- For S&D control ICAI's answer gives three methods, each with one line of how it works.

Trick

P-C-F, top-down

Work a reverse question from the top: Price → total Cost → Factory cost → factory overheads. Condition: the profit is stated on cost or on price, and each overhead on a single base.

Answer frame

"Discuss the problems of controlling selling and distribution overheads" (SM question 3), then **"methods"** (PYQ January 2025 Q6(c), 4 marks):

1. Problems: (i) external factors (distance of market, competition, terms of sale) beyond management's control; (ii) dependence on customers' behaviour and tastes, so control may lose customers; (iii) policy costs, not amenable to control.
2. Methods: (a) comparison with past performance, also as a percentage of sales; suitable for small concerns; (b) budgetary control, fixed and variable, a flexible budget if needed, variances acted on; (c) standard costing, standards for salesmen, territories, products, variances enquired into.

"Explain the methods of accounting administrative overheads": the three methods in ICAI's order, each with its reason and its disadvantages (the table in How it works); for (c), the five bases.

"Explain research and development expenses and their treatment in costing" (MTP September 2025, 4 marks): the two Terminology definitions; production-method research in cost of production; administration research in administration overhead; market research in selling and distribution; development of a particular product charged to it as deferred revenue expenditure, recovered per unit when production is fully established; routine general research to general overhead (deferred if

substantial); fundamental research to administration; failed research to Costing P&L.

"Treatment of bad debts in cost accounts": no unanimity; view 1, excluded as financial losses; view 2, part of selling and distribution overheads in the normal course of trading; extraordinarily large bad debts excluded.

Three answers to the three-channels question

Loses marks

"App orders are the largest consumer orders (₹80), so spend the ₹5,00,000 on the app."

Sales per order says nothing about the costs each rupee of sales carries.

Passes

A profit statement with the common costs deducted before contribution, and the advice taken from the profit line.

Right channel by luck; no contribution measure, no incremental working.

Full marks

The statement in layers (variable costs, contribution, specific fixed costs, apportioned common costs); contribution per ₹100 of sales and per order; the incremental table for the ₹5,00,000; a recommendation with its reason.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the margin that moved

Taking "20% on selling price" as 20% on cost prices a job costing ₹1,15,200 at ₹1,38,240 instead of ₹1,44,000. **Fix:** profit on price means cost is 80% of price: price = cost ÷ 0.80.

⚠ **Common mistake**

Administration on the wrong base

Applying the 20% conversion-cost rate to the premix's works cost of ₹85 charges ₹17 of administration to a box instead of ₹5. **Fix:** apply each rate to the base it was computed on, and write the base next to every percentage in the statement, as ICAI does.

⚠ **Common mistake**

Dropping the product that "makes a loss"

A product or channel shows a loss only after common fixed costs apportioned to it, which stay if it goes; in the channel example even the weakest channel earns ₹5,20,000 after its own fixed costs. **Fix:** look at contribution and at which fixed costs would really be saved.

⚠ **Common mistake**

Interest inside the factory

Adding interest on borrowings to works cost or cost of production. **Fix:** interest and financing charges are a separate item of cost of sales (SM 4.60).

Practice

Check yourself

1 A factory absorbs administration overheads of ₹6,00,000 on conversion cost, which is budgeted at ₹30,00,000. A job has direct materials of ₹40,000, direct wages of ₹25,000 and factory overheads of ₹35,000. How much administration overhead is charged to the job (₹)?

Answer: _____

2 Case scenario. Tapri's lab spends ₹18 lakh this year developing a sugar-free chai premix, a single new product whose production will be fully established next year. Bunty, on an audit visit from Iyer & Associates, suggests charging it to the Costing Profit and Loss Account. Which treatment is correct?

A Charge it to the Costing Profit and Loss Account this year

B Add it to administration overheads

C Charge it to the premix as deferred revenue expenditure, recovered per unit once production is established

D Add it to selling and distribution overheads

3 Which basis does ICAI call ordinarily the most logical for absorbing selling overheads?

A Cost of goods sold

B Sales value of goods

C Gross profit on sales

D Units produced

4 Why is control of selling and distribution overheads difficult? Select all that apply.

- **A** Their incidence depends on external factors such as competition and distance of markets
- **B** Controlling them may lose customers, whose tastes they depend on
- **C** They are policy costs, not amenable to control
- **D** They are mostly variable with production volume

More than one answer may be right.

5 How does ICAI treat a night shift allowance paid because work is beyond normal capacity, and one paid to finish a particular customer's order?

- **A** Both as direct wages of the jobs worked at night
- **B** The first as production overhead; the second charged to that order
- **C** Both to the Costing Profit and Loss Account
- **D** The first to Costing P&L; the second as production overhead

Answers: the answer key at the end of this lesson.

Recap

- Administration overhead (CIMA): costs of general management and secretarial, accounting and administrative services not directly related to production, marketing, research or development.
- Three methods: apportion to production and sales; charge to Costing P&L; separate addition on works cost, sales, gross profit, quantity produced or conversion cost.
- Control of administration: departmental analysis against results (and absorption rates), budgets, standards.
- S&D bases: sales value (most logical), cost of goods sold, gross profit, orders or units. Best: estimated amount per unit (fixed by benefit + variable).
- S&D control is hard (external factors, customers, policy costs); use past comparison, budgets, standard costing.
- R&D: production methods, cost of production; administration or fundamental, administration overhead; market research, S&D; one product, deferred and charged to it; failure, Costing P&L.
- Interest: separate item of cost of sales. Bad debts: two views; very large ones excluded. Carriage follows what it carries.

- Reverse questions: Price → Cost → Factory cost → factory overheads. (The principal's salary still has to land somewhere.)

Answer key · Check yourself

1 **12000**

Rate = ₹6,00,000 ÷ ₹30,00,000 = 20% of conversion cost. The job's conversion cost = ₹25,000 + ₹35,000 = ₹60,000, so administration = ₹12,000. Applying the 20% to the job's works cost (₹1,00,000) gives ₹20,000: a rate must be applied to the base it was computed on.

2 **C** **Charge it to the premix as deferred revenue expenditure, recovered per unit once production is established**

SM 4.63: development costs for a particular product are charged directly to that product, usually as deferred revenue expenses recovered as a cost per unit when production is fully established. Costing P&L is for research that fails.

3 **B** **Sales value of goods**

SM 4.53: sales value, because there is some connection between sales and the expenses incurred to achieve them. Cost of production may have no direct relationship; gross profit follows "ability to pay".

4 **A Their incidence depends on external factors such as competition and distance of markets; B Controlling them may lose customers, whose tastes they depend on; C They are policy costs, not amenable to control**

ICAI's three reasons (SM 4.55). They vary with sales and with policy, not with production volume.

5 **B The first as production overhead; the second charged to that order**

SM 4.62: general pressure beyond normal capacity makes it a production overhead; a specific customer's order bears its own night shift cost; abnormal circumstances send it to Costing P&L.

CHAPTER 6

Activity Based Costing

6.1 Activities, cost pools, cost drivers and the cost hierarchy

Why one labour-hour rate miscosts products, and how ABC traces overheads to activities, collects them in cost pools and charges cost objects through cost drivers at four levels of the cost hierarchy.

In plain words

The festival special that looked cheap

Tapri's chai premix plant in Indore makes one big product and one tiny one. Kadak Classic runs in long batches, 1,20,000 packs a month. Kesar Pista Royale, Kesar's Diwali launch, runs in 30 small batches of 200 packs. Every batch needs the machines stripped, cleaned and reset and three lab tests, and its saffron and pistachio arrive in small, frequent orders.

> **Kesar:** Ritu, Royale carries only ₹12 of overhead a pack. Let us cut its price and sell it all year.
>
> **Ritu:** That ₹12 comes from charging overhead by labour hours, and Royale uses very few. Its overheads come from set-ups, orders and tests. Per unit, please, but per set-up first.

The traditional system collects overheads in a few cost centres and spreads them by direct labour hours or machine hours. It assumes that a product which takes longer causes more overhead. But a set-up, a purchase order or a lab test happens once per batch, however few hours the batch takes. So Kadak Classic, with most of the labour hours, quietly pays for Royale's set-ups.

Activity based costing (ABC) fixes this in two moves. First it finds the **activities** that cause overhead (setting up machines, purchasing, testing, blending) and collects the cost of each in its own **cost pool**. Then it charges each product for the activities it uses, measured by a **cost driver**: the number of set-ups, purchase orders or tests, or machine hours. Done this way, Royale's overhead is ₹42 a pack, not ₹12, and Kadak Classic's falls from ₹4.80 to ₹3.30.

Activities come in levels. Some happen for every unit (blending), some for every batch (set-ups), some for every product (designing Royale's recipe), and some just keep the plant running (rent, security). This is the **cost hierarchy**, and it tells you which kind of driver to look for.

Think of it like

The group dinner bill

Six friends split a restaurant bill by how long each sat at the table. The one who lingered over a single cutting chai pays the most; the one who ordered five starters and two desserts in twenty minutes pays the least. Time at the table is not what drives the bill: the orders are. ABC splits the bill by the orders. Every item has a price (the driver rate), and each friend pays for what they ordered (the driver quantity).

Key idea

Overheads are caused by activities, not by labour hours. ABC traces overheads to activities first, then charges each cost object for the activities it actually uses, through cost drivers.

How it works

Why one labour-hour rate misfires

In the traditional system overheads are grouped under cost centres and absorbed on direct labour hours, machine hours or volume (absorption rates). ICAI is careful: not every traditional system is inaccurate. "The main cause of inaccuracy is in the calculation of the overhead rate itself", which assumes "that products that take longer to make, generate more overheads" (SM 5.2).

Four factors prompted the development of ABC (SM 5.2):

1. Growing overhead costs because of increasingly automated production.
2. Increasing market competition, which necessitated more accurate product costs.
3. Increasing product diversity to secure economies of scope and increased market share.
4. Decreasing costs of information processing because of continual improvements and increasing application of information technology.

ABC is particularly needed (its **usefulness or suitability**, SM 5.3) in these situations:

1. **High amount of overhead**: production overheads are high and form significant costs.
2. **Wide range of products**: there is diversity in the product range, or multiple products.
3. **Presence of non-volume related activities**: activities such as material handling, inspection and set-up are significant and the traditional system cannot be applied. ABC also identifies non-value-adding activities that might be a focus for attention or elimination.
4. **Stiff competition**: cost must be computed accurately and the price fixed to the market. ABC also helps reduce cost by identifying non-value-adding activities.

The words ICAI uses

Definition

Activity Based Costing

An accounting methodology that assigns costs to activities rather than products or services. This enables resources and overhead costs to be more accurately assigned to products and services that consume them. ABC is a technique which involves identification of cost with each cost driving activity and making it as the basis for apportionment of costs over different cost objects/jobs/products/customers or services. (SM 5.3)

CIMA defines it as "an approach to the costing and monitoring of activities which involves tracing resource consumption and costing final outputs. Resources are assigned to activities, and activities to cost objects based on consumption estimates. The latter utilise cost drivers to attach activity costs to outputs" (SM 5.4).

Definition

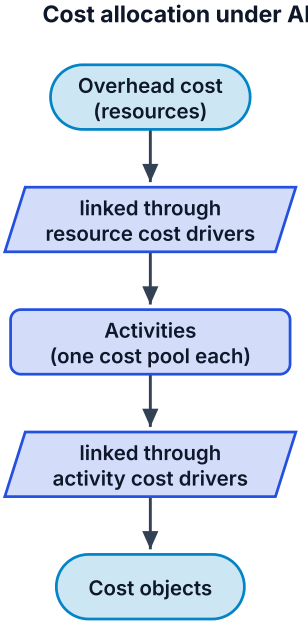
Terms used in ABC (SM 5.4)

- (i) **Activity**: an event that incurs cost.
- (ii) **Cost object**: an item for which cost measurement is required, e.g. a product or a customer.
- (iii) **Cost driver**: a factor that causes a change in the cost of an activity. Two categories: a **resource cost driver** is a measure of the quantity of resources consumed by an activity, used to assign the cost of a resource to an activity or cost pool; an **activity cost driver** is a measure of the frequency and intensity of demand placed on activities by cost objects, used to assign activity costs to cost objects.
- (iv) **Cost pool**: a group of various individual cost items that have the same cause and effect relationship, e.g. machine set-up.

Two stages, and where ABC differs

Traditional absorption costing relates overheads first to cost centres (production and service) and then to products. ABC relates them first to activities, grouped into cost pools, and then to cost objects. "The two processes are, therefore, very similar, but the first stage is different, as ABC uses activities instead of functional departments" (SM 5.6). A department holds many activities whose costs behave differently, and activities run across departments: a purchase requisition is raised in the factory or stores, not in the purchasing

department. The costs inside one activity share a driver, so ABC gives a more realistic picture of the way in which costs behave.



Stage one: resource cost drivers measure the share of each resource an activity consumes. Stage two: activity cost drivers measure how much of each activity a cost object (a product, a customer, a channel) demands. Based on the figure at SM 5.10.

Five stages, on Ritu's premix numbers

ICAI's stages (SM 5.9–5.10), applied to the plant's monthly overheads of ₹6,48,000:

1. **Identify the activities.** A traditional system may use up to about fifteen cost centres; ABC could use 200 activities, but about 40 is practical. Ritu picks four: machine set-ups, purchasing, quality inspection and blending.

2. **Relate the overheads to the activities that caused them**, support and primary, using resource cost drivers. This creates the **cost pools** ("cost buckets"). Set-up crew wages ₹1,20,000 and changeover cleaning materials ₹30,000 make the set-up pool of ₹1,50,000; purchase salaries ₹90,000 and order software ₹18,000 make purchasing ₹1,08,000; lab staff ₹84,000 and test kits ₹36,000 make inspection ₹1,20,000; blender power ₹1,62,000 and depreciation ₹1,08,000 make blending ₹2,70,000.

3. **Spread support activities over the primary activities** on a base that reflects their use (the plant has none here; Worked example 1 has one).

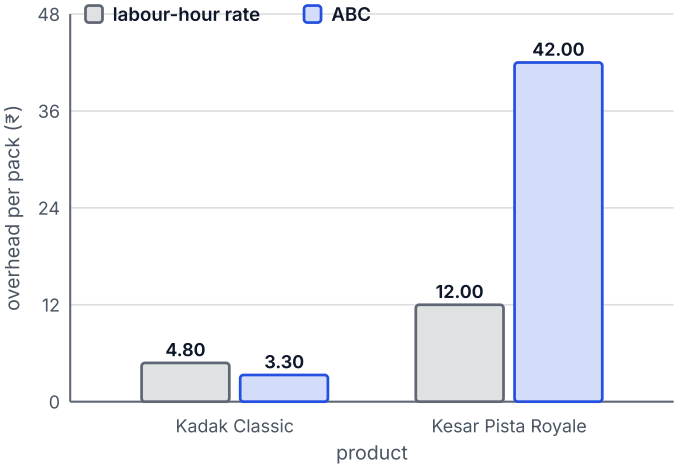
4. **Determine the activity cost drivers** by asking what causes each activity to incur cost: 50 set-ups, 90 purchase orders, 150 inspections, 4,500 machine hours.

5 **Calculate the activity cost driver rates** and multiply them by what each cost object uses: ₹3,000 a set-up, ₹1,200 an order, ₹800 an inspection, ₹60 a machine hour.

Activity (driver used: Classic; Royale)	Kadak Classic (₹)	Kesar Pista Royale (₹)
Set-ups at ₹3,000 (20; 30)	60,000	90,000
Purchasing at ₹1,200 (40; 50 orders)	48,000	60,000
Inspection at ₹800 (60; 90)	48,000	72,000
Blending at ₹60 (4,000; 500 machine hours)	2,40,000	30,000
Overhead under ABC	**3,96,000**	**2,52,000**
Packs made	1,20,000	6,000
Overhead per pack under ABC	**3.30**	**42.00**
Per pack at ₹80 per labour hour (7,200; 900 hours)	4.80	12.00

The single rate is ₹6,48,000 ÷ 8,100 labour hours = ₹80. Both methods charge the same ₹6,48,000 in total; ABC only moves ₹1,80,000 from Kadak Classic to Royale.

Overhead per pack: one labour-hour rate against ABC



Grey: overhead per pack at ₹80 per direct labour hour; blue: under ABC. The small-batch product was carrying less than a third of the overhead it causes.

Interactive version: PassBook app, lesson Activities, cost pools, cost drivers and the cost hierarchy

The cost hierarchy

These levels of activity were first identified by Cooper (1990), and "the categories of activities help to determine the type of activity cost driver required" (SM 5.8):

The cost hierarchy (SM 5.8–5.9)

Unit level varies with units produced	machine hours, units
Batch level varies with batches made	set-ups, orders
Product level supports each product	number of products
Facility level sustains the whole plant	no product driver

Left: what the activity's cost varies with; right: the kind of activity cost driver to look for.

Level

Unit level

Meaning (SM 5.8–5.9)

Consumption of resources can be identified with the number of units produced

ICAI's examples

Indirect materials or consumables used in proportion to units; inspection or testing of every item (or every 100th item)

Level

Batch level

Meaning (SM 5.8–5.9)

Performed each time a batch is produced; cost varies with the number of batches, but is common (fixed) for all units within the batch

ICAI's examples

Material ordering for every batch; machine set-up between batches; inspecting the first item in every batch

Level

Product level

Meaning (SM 5.8–5.9)

Performed to support different products in the product line

ICAI's examples

Designing the product; producing parts specifications; keeping technical drawings up to date

Level

Facility level

Meaning (SM 5.8–5.9)

Cannot be directly attributed to individual products; necessary to sustain the manufacturing process; common and joint to all products

ICAI's examples

Maintenance of buildings; plant security

Inspection appears twice: testing every 100th item is unit level, testing the first item of every batch is batch level. The level follows what the cost varies with, not the activity's name.

Formats and formulas

Reproduce the formula and both driver tables from memory: ICAI asks for examples of cost drivers as 5-mark parts.

∑ **Formula**

Activity cost driver rate (SM 5.10)

$$\text{Activity cost driver rate} = \frac{\text{Total cost of activity}}{\text{Activity driver}}$$

The activity driver is the total quantity of the driver for the period (all set-ups, all orders). Overhead charged to a cost object = the sum, over the activities it uses, of driver rate × driver quantity it consumed.

Cost drivers by business function (SM 5.4–5.5):

Business function	Cost drivers
Research and development	Number of research projects; personnel hours on a project
Design of products, services and procedures	Number of products in design; number of parts per product; number of engineering hours
Customer service	Number of service calls; number of products serviced; hours spent on servicing products
Marketing	Number of advertisements; number of sales personnel; sales revenue
Distribution	Number of units distributed; number of customers

Cost drivers by activity cost pool in a production department (SM 5.13):

Activity cost pool	Related cost driver
Ordering and receiving materials	Number of purchase orders
Setting up machines	Number of set-ups
Machining	Machine hours
Assembling	Number of parts
Inspecting and testing	Number of tests
Painting	Number of parts
Supervising	Direct labour hours

Worked examples

Worked example **The samosa factory: from the ledger to driver rates**

Tapri's samosa factory makes Classic Aloo (1,50,000 packs a month) and Cocktail Mini (30,000 packs). It absorbs production overheads at a single rate per direct labour hour; Ritu proposes ABC with four primary activities (set-ups, material receiving, quality testing,

frying) and one support activity (maintenance). The month's overheads are:

Overhead	Amount (₹)	How it is used
Supervisors' salaries	2,40,000	Time: set-ups 40%, receiving 20%, testing 40%
Stores and receiving staff wages	1,20,000	Receiving only
Lab staff and test kits	1,80,000	Testing only
Power	3,00,000	kWh: frying 80%, set-ups 20%
Depreciation of fryers	2,10,000	Frying only
Maintenance department	1,50,000	150 hours on set-ups, 600 on frying

Driver data	Classic Aloo	Cocktail Mini
Set-ups	15	45
Material deliveries received	90	150
Quality tests	75	225
Machine hours (frying)	6,000	1,600
Direct labour hours	18,000	6,000

Required:

(i) Calculate the cost of each activity pool and the activity cost driver rates.

(ii) Calculate the overhead per pack of each product under ABC and under the present labour-hour rate.

(iii) State which product is under-costed, by how much, and why.

THE WORKING

Working note 1: cost pools (stage 2 by resource cost drivers, stage 3 for maintenance in the ratio 150 : 600)

Overhead	Set-ups (₹)	Receiving (₹)	Testing (₹)	Frying (₹)
Supervisors (40 : 20 : 40)	96,000	48,000	96,000	–
Stores and receiving staff	–	1,20,000	–	–
Lab staff and test kits	–	–	1,80,000	–
Power (20 : 80)	60,000	–	–	2,40,000
Depreciation of fryers	–	–	–	2,10,000
Maintenance (1 : 4)	30,000	–	–	1,20,000
Cost pool	**1,86,000**	**1,68,000**	**2,76,000**	**5,70,000**

Check: the pools add up to ₹12,00,000, the ledger total.

Working note 2: activity cost driver rates

Activity	Cost pool (₹)	Driver	Rate (₹)
Set-ups	1,86,000	60 set-ups	3,100 per set-up
Receiving	1,68,000	240 deliveries	700 per delivery
Testing	2,76,000	300 tests	920 per test
Frying	5,70,000	7,600 machine hours	75 per machine hour

(ii) Overhead charged

Activity	Classic Aloo (₹)	Cocktail Mini (₹)
Set-ups (15; 45 × ₹3,100)	46,500	1,39,500
Receiving (90; 150 × ₹700)	63,000	1,05,000
Testing (75; 225 × ₹920)	69,000	2,07,000
Frying (6,000; 1,600 × ₹75)	4,50,000	1,20,000
Overhead under ABC	**6,28,500**	**5,71,500**
Per pack under ABC	**4.19**	**19.05**
Present rate: ₹12,00,000 ÷ 24,000 hours = ₹50 per hour	9,00,000	3,00,000
Per pack at the present rate	6.00	10.00

(iii) Cocktail Mini is under-costed by ₹9.05 a pack (₹2,71,500 in total) and Classic Aloo over-costed by ₹1.81 a pack (the same ₹2,71,500). Cocktail Mini uses only a quarter of the labour hours, but three-quarters of the set-ups and tests and most of the deliveries: batch-level costs that a labour-hour rate spreads by hours. The ABC costs appear to be more precise.

Worked example

Kesar's draft of the cost hierarchy for the cafés

Ritu is extending ABC to Tapri's kulhad cafés. Kesar, keen to help, has placed ten café activities in the cost hierarchy himself:

Activity	Kesar's level
(1) Steaming milk for each cup served	Batch
(2) Brewing a 20-litre urn of chai, about every 90 minutes	Unit
(3) Tasting the first cup from every urn	Unit
(4) Developing the recipe and menu card for the monsoon special	Facility
(5) Each café's annual food-safety licence fee	Product
(6) The café manager's salary	Facility
(7) Receiving the milk and tea-leaf delivery, three times a week	Facility
(8) The kulhad (clay cup) used for each serving	Unit
(9) Keeping the allergen sheet of each menu item up to date	Batch
(10) Cleaning the café after closing	Unit

Required:

(i) State whether each of Kesar's levels is correct. Where it is not, give the correct level with the reason.

(ii) Suggest an activity cost driver for each activity that has one.

MODEL ANSWER

As per the Study Material, unit-level activities consume resources in proportion to the number of units produced; batch-level activities are performed each time a batch is produced, and their cost varies with the number of batches but is common to all units in the batch; product-level activities support the different products in the line; facility-level activities cannot be directly attributed to individual products and sustain the whole process (SM 5.8–5.9). Here a cup is the unit, an urn is the batch and a menu item is the product.

Activity
(1) Steaming milk

Kesar
Wrong

Correct level
Unit

Reason
Done for every cup

Cost driver
Number of cups

Activity

(2) Brewing an urn

Kesar

Wrong

Correct level

Batch

Reason

Done once per urn, shared by all its cups

Cost driver

Number of urns brewed

Activity

(3) Tasting the first cup

Kesar

Wrong

Correct level

Batch

Reason

Once per urn, like inspecting the first item of a batch

Cost driver

Number of urns (tastings)

Activity

(4) Monsoon recipe and menu card

Kesar

Wrong

Correct level

Product

Reason

Supports one product in the line, like designing a product

Cost driver

Number of new menu items

Activity

(5) Food-safety licence

Kesar

Wrong

Correct level

Facility

Reason

Sustains the café; no product or batch causes it

Cost driver

None

Activity

(6) Café manager's salary

Kesar

Correct

Correct level

Facility

Reason

Common to everything the café sells

Cost driver

None

Activity

(7) Receiving deliveries

Kesar

Wrong

Correct level

Batch

Reason

Cost varies with the number of deliveries, like material ordering per batch

Cost driver

Number of deliveries received

Activity

(8) Kulhad per serving

Kesar

Correct

Correct level

Unit

Reason

One for every cup

Cost driver

Number of cups

Activity

(9) Allergen sheets

Kesar

Wrong

Correct level

Product

Reason

Kept for each menu item, like keeping technical drawings up to date

Cost driver

Number of menu items

Activity

(10) Cleaning after closing

Kesar

Wrong

Correct level

Facility

Reason

Done once a day, whatever is sold

Cost driver

None

Kesar has two of the ten right. The test is what the cost varies with, not the activity's name: tasting is an inspection, but tasting the first cup of every urn is batch level; tasting every cup would be unit level.

Worked example

One ABC system this year: where should it go?

Ritu can install ABC in only one Tapri unit this year and must explain why the others can wait or need not have it at all.

Unit

(a) Premix plant

Products

40 packs and flavours

Overheads in total cost

38%

Non-volume activities

Set-ups and lab tests for every batch

Market

Price war with BrewBucks

Unit

(b) Catering division

Products

120 event menus a year

Overheads in total cost

25%

Non-volume activities

Menu design, tastings and a site visit for each event

Market

Competitive tenders, priced at cost plus

Unit

(c) Kulhad workshop

Products

2 hand-made cups

Overheads in total cost

4%

Non-volume activities

Negligible

Market

Sells everything to Tapri's cafés at a fixed price

Unit

(d) Sachet line

Products

3 flavours on one line

Overheads in total cost

20%

Non-volume activities

Each flavour uses set-ups, tests and machine hours in the same proportion as its labour hours

Market

Stable

Required: Advise Ritu, unit by unit, using ICAI's situations in which ABC is useful and its limitations, and state where ABC should be installed first.

MODEL ANSWER

ABC is particularly useful where overheads are high, the product range is wide, non-volume related activities are significant, and competition is stiff (SM 5.3); its limitations are that it is more expensive than the traditional system, is not helpful to small organisations, may not be applied to organisations with limited products, and its drivers can be difficult to select (SM 5.18).

(a) Premix plant: install ABC first. All four situations are present: overheads are 38% of cost, there are 40 products, set-ups and lab tests are batch-level activities that a volume rate spreads wrongly, and the BrewBucks price war needs accurate costs for pricing.

(b) Catering division: a strong case, next in line. A wide range of menus, significant non-volume activities for each event, and cost-plus tenders in a competitive market. It comes second only because its overheads (25%) are smaller and the plant's pricing problem is more urgent.

(c) Kulhad workshop: not recommended. A small organisation with limited products and overheads of only 4%: the expense of ABC would not be justified,

and a fixed internal price leaves no pricing decision for ABC to improve.

(d) Sachet line: not recommended. Its flavours use every activity in the same proportion as their labour hours, so activity driver rates would charge each flavour the same overhead as the present labour-hour rate. With no distortion to correct, ABC would only add cost.

Conclusion: install ABC in the premix plant this year, plan the catering division next, and keep the traditional rate in the workshop and on the sachet line.

Exam angle

In the exam

How ICAI asked this, May 2024 to September 2026 (and old-scheme 2022–23)

In the documents read for this lesson, the concepts of this lesson were asked in 10 questions:

- **"Discuss the usefulness/suitability of ABC"**: January 2026 Q6(c), 4 marks. The suggested answer is SM 5.3's four headed situations, word for word.
- **Classify cost pools by the cost hierarchy**: May 2025 Q6(b), 5 marks, ten pools. ICAI's answer: facility level for human resources, maintenance of buildings, plant security and the floor manager's salary; batch

level for purchasing, machine set-up and the receiving department; product level for designing the product; "batch-level/product level" for parts management and "batch-level/unit level" for quality control.

- **Give examples of cost drivers**: for five cost pools (quality control, R&D, machine maintenance, employee training, customer service) in RTP January 2026 Q19(b), RTP September 2026 Q16(d) and MTP May 2025 Series I Q6(b) (5 marks); for business functions in old-scheme November 2022 Q6(c); for activity cost pools in old-scheme MTP November 2022 Series I Q6(d).
- **Meaning and categories of cost driver, with drivers for three functions**: old-scheme November 2023 Q6(e), 5 marks.
- **Recommend ABC or not** in four situations: MTP September 2025 Series II Q5(b), 5 marks.
- **"Explain how an ABC system would be developed"**, inside a 10-mark costing question: MTP January 2026 Series I Q2(a)(ii); the answer lists the stages before the numbers.

In the exam

The traps

- A cost pool is classified by what its cost varies with: ICAI's own answer moved quality control between batch and unit level. State the assumption.
- "Cost driver" questions want the two categories by name: resource cost driver and activity cost driver.
- ICAI's examples for quality control include "product units produced"; for employee training, "number of training hours" and "employees trained" (RTP January 2026 answer).

Trick

One more what?

To place an activity in the hierarchy, ask: its cost rises if we make one more what? One more unit: unit level. One more batch: batch level. One more product in the range: product level. None of these: facility level. Condition: judge the activity as the question describes it, not by its name.

Answer frame

"Distinguish between ABC and traditional

absorption costing" (SM 5.7)

Point

1. Grouping

Activity based costing

Overheads related to activities and grouped into activity cost pools

Traditional absorption costing

Overheads related to cost centres or departments

Point

2. Realism

Activity based costing

Costs related to activities, hence more realistic

Traditional absorption costing

Costs related to cost centres, hence not realistic of cost behaviour

Point

3. Drivers

Activity based costing

Activity-wise cost drivers determined

Traditional absorption costing

Time (hours) assumed to be the only cost driver in all departments

Point

4. Rates

Activity based costing

Activity-wise recovery rates; no single overhead recovery rate

Traditional absorption costing

Multiple rates (one per department) or a single overhead recovery rate

Point

5. Cost objects

Activity based costing

Customers, products, services, departments, etc.

Traditional absorption costing

Cost units: products, jobs or hours

Point

6. Control

Activity based costing

Essential activities simplified, unnecessary ones eliminated: aids cost control

Traditional absorption costing

Cost centres cannot be eliminated: not suitable for cost control

Other questions, in ICAI's order

- **Usefulness or suitability** (4 marks): the four headed situations of SM 5.3, each with one line of explanation.
- **Cost driver**: definition; the two categories with their uses; then two drivers for each function asked, from the SM 5.4–5.5 table.
- **How product costs are determined in ABC, or how an ABC system is developed**: the five stages in order (identify activities; relate overheads to activities to create cost pools using resource cost drivers; spread support activities over primary activities; determine activity cost drivers; calculate driver rates and charge cost objects), then the numbers.
- **Levels of activities**: the four levels, each with ICAI's meaning and one example.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the inspection that was batch level because it said "inspection"

Bunty learned "inspection = batch level" and applied it to a factory that tests every pack. **Fix:** classify by what the cost varies with. Testing every unit is unit level; testing the first item of each batch is batch level (SM 5.8).

⚠ **Common mistake**

Charging products with a resource cost driver

Charging the supervisors' salaries straight to Classic Aloo and Cocktail Mini by the time supervisors spend on each product, skipping the cost pools. **Fix:** a resource cost driver (supervisors' time on set-ups, receiving and testing) takes a resource to the activities: stage one. Products are then charged through activity cost drivers (set-ups, tests, deliveries) at each activity's rate.

⚠ **Common mistake**

Dividing a pool by one product's driver

Taking ₹1,86,000 ÷ 45 set-ups for Cocktail Mini. **Fix:** the rate is the pool ÷ the total driver for the period (60 set-ups), then multiplied by each product's own quantity. Check: the products' charges add back to the pool.

⚠ **Common mistake**

"ABC increased the overheads"

Fix: ABC redistributes the same total; under-costed and over-costed amounts are equal and opposite. If your totals differ from the ledger, a pool or a rate is wrong.

Practice

Check yourself

1 Tapri's dispatch activity costs ₹2,88,000 a month and is driven by the number of dispatch orders: 960 in the month, of which the Elaichi premix used 140. How much dispatch cost is charged to the Elaichi premix (₹)?

Answer: _____

2 Case scenario. At Tapri's premix plant the cost of the purchasing department (salaries, software) is first shared between "ordering spices" and "ordering packing material" by the number of buyers' hours spent on each. Each product is then charged ₹1,200 for every purchase order raised for it. Which statement is correct?

- **A** Buyers' hours are the activity cost driver; purchase orders are the resource cost driver
- **B** Buyers' hours are the resource cost driver; purchase orders are the activity cost driver
- **C** Both are activity cost drivers, because both are measured in numbers
- **D** Both are resource cost drivers, because both relate to the purchasing department

3 In a plant that tests the first piece of every production run, at which level of the cost hierarchy does that testing sit?

- **A** Unit level
- **B** Batch level
- **C** Product level
- **D** Facility level

4 Which of the following are situations in which ICAI says ABC is particularly useful? Select all that apply.

- **A** Production overheads are high and form significant costs
- **B** Non-volume related activities such as set-ups and inspection are significant
- **C** Overheads are small and the firm makes a single product
- **D** The firm faces stiff competition and must price accurately

More than one answer may be right.

5 Under a labour-hour rate a product carries ₹4,50,000 of overhead; under ABC it carries ₹3,20,000. Total overhead is ₹10,00,000 under both. Which is correct?

- **A** ABC has reduced total overhead by ₹1,30,000
- **B** The product was over-costed by ₹1,30,000, and other products were under-costed by the same total
- **C** The product was under-costed by ₹1,30,000
- **D** The difference is an under-absorption to be written off

Answers: the answer key at the end of this lesson.

Recap

- ABC assigns overheads to activities, then to cost objects according to their use of activities (SM 5.3).
- Two stages: resources to activities by **resource cost drivers**; activities to cost objects by **activity cost drivers**.
- Activity: an event that incurs cost. Cost pool: costs with the same cause and effect relationship. Cost driver: a factor that causes a change in the cost of an activity.
- Activity cost driver rate = total cost of activity ÷ activity driver.
- Hierarchy: unit (varies with units), batch (with batches), product (supports each product), facility (sustains the plant).
- Useful when overheads are high, products many, non-volume activities significant, competition stiff.
- Same total overhead, spread differently: small-batch products often turn out under-costed by a labour-hour rate. (Split the dinner bill by the orders.)

Answer key · Check yourself

1 **42000**

Rate = ₹2,88,000 ÷ 960 = ₹300 per dispatch order; Elaichi: 140 × ₹300 = ₹42,000.

2 **B Buyers' hours are the resource cost driver; purchase orders are the activity cost driver**

Buyers' hours measure the resources each activity consumes (stage one: resources to activities). Purchase orders measure the demand each product places on the activity (stage two: activities to cost objects).

3 **B Batch level**

Its cost varies with the number of runs (batches) and is common to all units in the run. Testing every item would be unit level.

4 **A Production overheads are high and form significant costs**; **B Non-volume related activities such as set-ups and inspection are significant**; **D The firm faces stiff competition and must price accurately**

SM 5.3 lists high overhead, a wide range of products, non-volume related activities and stiff competition. A single product with small overheads is the case where ABC is not worth its cost.

5 **B The product was over-costed by ₹1,30,000, and other products were under-costed by the same total**

ABC only redistributes the same ₹10,00,000. The product was charged ₹1,30,000 more than the activities it used, so others carried ₹1,30,000 too little.

6.2 Product costs under ABC versus traditional absorption

Building ICAI's two cost statements side by side (one overhead rate against activity driver rates), reading the cost distortion, costing unused capacity and measuring customer-level operating income.

In plain words

The price war that Ritu's spreadsheet explained

BrewBucks has launched an instant premix at ₹640 a carton, and Tapri Everyday, priced at ₹675, is losing shelf space. Kesar prices every product at cost plus 25%, and his costs come from one overhead rate per machine hour.

> **Kesar:** Everyday is our biggest seller and still we cannot match their price? Masala Gold, our fancy one, sells beautifully. Make more Masala Gold.
>
> **Ritu:** Masala Gold sells beautifully because we sell it below its cost. Per unit, please: let me rerun the costs activity by activity.

Ritu's rerun leaves the **prime cost** alone: materials and labour are traced to each product directly, under any system. Only the **production overheads** move. The old system spread them by machine hours. Everyday uses most of the machine hours, so it carried most of the overhead. ABC charges them by what each product actually demands: set-up hours, quality tests, material moves, recipe changes, and machine hours only for the

machine costs. Masala Gold is made in 40 tiny runs: half the factory's production runs and more than half its quality tests are its, though it is less than 7% of the cartons.

The gap between the two costs is the **cost distortion**. Everyday turns out to be **over-costed**, so its cost-plus price was too high and BrewBucks walked in. Masala Gold is **under-costed** by so much that every carton sold at ₹975 costs more than ₹975 to make. The over-costing and the under-costing cancel out: the factory's total overhead is the same either way. Worked example 1 rebuilds Ritu's spreadsheet line by line.

The same driver rates do two more jobs. Set on the **capacity** of each activity (the inspections the lab could do, not the ones it did), they show the **cost of unused capacity** separately instead of loading it onto products. And charged to customers instead of products (orders, visits, deliveries), they show which customers really earn money.

Think of it like

The housing society bill

A housing society splits its ₹6 lakh maintenance bill by flat area. The ground-floor family that never uses the lift, the gym or a parking slot pays the same per square foot as the fourteenth-floor family with two cars and a gym habit. Charging per lift trip, parking slot and gym hour is ABC; the ground floor was over-costed and the fourteenth floor under-costed, by the same total. And if the generator was sized for 100 flats but only 80 are occupied, the cost of the empty 20 is reported as idle capacity, not quietly added to the bills of the 80.

Key idea

Prime cost does not change; only overheads are re-spread. Cost distortion = traditional cost − ABC cost: positive means over-costed, negative means under-costed, and across all products the distortion adds up to zero.

How it works

Building the two statements

1. **Prime cost per unit**: direct materials, direct labour and direct expenses, the same under both systems.

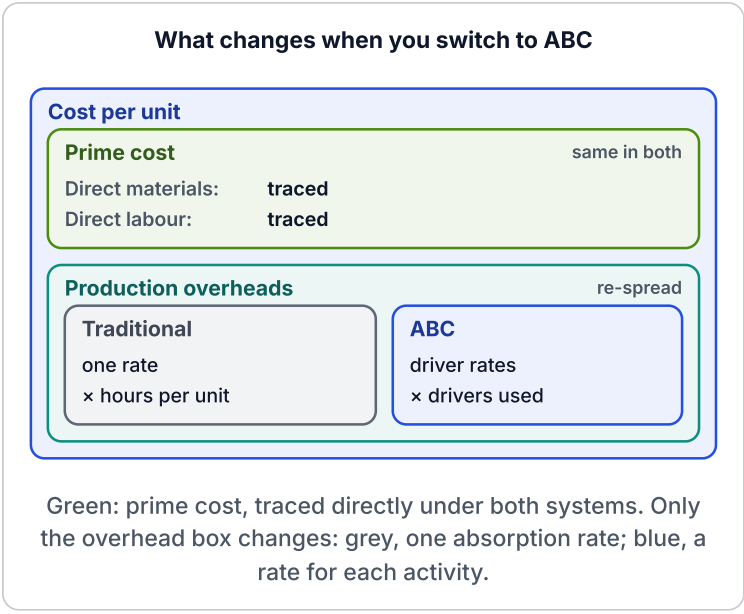
2 **Traditional overhead**: absorption rate = budgeted overheads ÷ budgeted base (direct labour hours, machine hours or units, as the question says); overhead per unit = rate × the base per unit (SM 5.15). See absorption rates.

3 **ABC driver totals**: turn the data into totals for each product. Per-unit data × units produced; batches = units ÷ batch size; per-batch data × batches (SM 5.45). Where the question gives percentages or ratios, first split the overheads into the activity pools (the five stages).

4 **Driver rates and charges**: rate = cost pool ÷ total driver; each product's overhead = rate × its driver quantity; per unit = ÷ units produced.

5 **Compare**: cost distortion per unit = traditional cost − ABC cost. Check: the distortions × units add up to zero.

6 **Use the cost**: price (cost-plus), profit, product mix. When production and sales differ, closing stock is valued at each system's cost per unit, so each product's profit differs between the systems (May 2026 Q5(a)).



Green: prime cost, traced directly under both systems. Only the overhead box changes: grey, one absorption rate; blue, a rate for each activity.

Rates on capacity, and the cost of unused capacity

When a firm "calculates activity cost rates based on cost driver capacity" (SM 5.22), the rate is the activity's cost ÷ the driver quantity the activity can supply. Products are charged only for what they used, and the rest is reported as the **cost of unused capacity**: ICAI's Illustration 2 charges power at ₹4 per kWh on a capacity of 50,000 kWh, products use 45,000, and the unused 5,000 kWh cost ₹20,000 (SM 5.23).

ICAI lists the factors management considers in choosing a capacity level to compute the budgeted fixed overhead rate (SM 5.23):

1. Effect on product costing and capacity management.
2. Effect on pricing decisions.

3. Effect on performance evaluation.
4. Effect on financial statements.
5. Regulatory requirements.
6. Difficulties in forecasting.

Costing customers and channels

"The activity driver rate can be used not only to identify cost of products ... but it can also be used for costing other cost objects such as customers/customer segments and distribution channels" (SM 5.10). Each customer's revenue is reduced by the discount it receives, its cost of goods sold, and the cost of the activities it makes the company perform: orders, deliveries, visits, merchandising, collections. What is left is the **customer-level operating income**.

Formats and formulas

Reproduce both statements from memory. ICAI's suggested answers show the driver rate and the quantity in brackets on every overhead line.

Statement of cost per unit: traditional absorption (SM 5.16, 5.44)

Particulars	Product A (₹)	Product B (₹)
Direct materials	xxx	xxx
Direct labour (hours × rate)	xxx	xxx
Production overheads (hours × absorption rate)	xxx	xxx
Cost per unit	**xxx**	**xxx**

Statement of cost: activity based costing (SM 5.17, 5.44–5.45)

Particulars	Product A (₹)	Product B (₹)
Direct materials	xxx	xxx
Direct labour	xxx	xxx
Set-up costs @ ₹... per set-up (set-ups)	xxx	xxx
Machine costs @ ₹... per machine hour (hours)	xxx	xxx
Inspection costs @ ₹... per inspection (inspections)	xxx	xxx
Total cost	**xxx**	**xxx**
Cost per unit (total cost ÷ units)	**xxx**	**xxx**

Working notes before the statement: driver totals (batches, orders, inspections, hours) and the driver rate table (cost pool, driver, quantity, rate).

∑ **Formula**

Activity cost driver rate and the product's overhead

$$\text{Rate} = \frac{\text{Cost of activity pool}}{\text{Total quantity of the driver}}$$

$$\text{Overhead}_{\text{product}} = \sum \text{Rate} \times \text{Driver quantity used}$$

On a capacity basis the denominator is the driver's capacity, not its actual use.

∑ **Formula**

Cost distortion

$$\text{Cost distortion per unit} = \text{Traditional cost} - \text{ABC cost}$$

Both costs are per unit. Positive: over-costed by the traditional system; negative: under-costed. ICAI prints the sign: "(–) 40.42" for the under-costed product (SM 5.26).

∑ **Formula**

Cost of unused capacity

$$\begin{aligned}&\text{Cost of unused capacity}\\&= \text{Cost of activity} - \text{Cost allocated}\\&= \text{Unused driver units} \times \text{Rate}\end{aligned}$$

Cost allocated = the driver units products actually used × the capacity rate.

Customer-level operating income (SM 5.54–5.55; May 2024 Q2(a))

Particulars	Customer A (₹)
Revenues at list price (units × list price)	xxx
Less: Discount	xxx
Revenue at actual price	**xxx**
Less: Cost of goods sold	xxx
Gross margin	**xxx**
Less: Customer-level operating activity costs (orders, visits, delivery, expediting)	xxx
Customer-level operating income	**xxx**
Operating income as a % of revenue (at actual price)	x%

Worked examples

Worked example **Three premix lines: one machine-hour rate against ABC**

Tapri's premix plant makes three products. At present all production overheads of the quarter, ₹32,00,000, are absorbed at ₹40 per machine hour. Data for the quarter:

Particulars	Everyday	Elaichi Premium	Masala Gold
Production (cartons)	20,000	8,000	2,000
Prime cost per carton (₹)	420	500	700
Machine hours per carton	3	2	2
Production runs	20	20	40
Set-up hours per run	6	4	3
Quality tests per run	4	3	4
Material moves per run	3	2	2
Recipe and pack changes	1	3	7

Ritu traces the same ₹32,00,000 to five activities:

Activity	Cost (₹)	Cost driver
Machine running	12,80,000	Machine hours
Set-ups	6,40,000	Set-up hours
Quality testing	4,80,000	Number of tests
Material handling	3,60,000	Number of material moves
Recipe and pack development	4,40,000	Number of recipe and pack changes

The quarter's entire output is sold.

Required:

(i) Calculate the cost per carton of each product under the present machine-hour rate.

(ii) Calculate the cost per carton of each product using activity based costing.

(iii) Calculate the cost distortion per carton and in total, stating which products are under-costed and why.

(iv) Kesar prices every product at traditional cost plus 25%. Calculate the profit per carton under ABC.

(v) Kesar now proposes runs of 100 cartons for Masala Gold, with set-up hours, tests and material moves per run unchanged. Assuming the activity driver rates do

not change, calculate Masala Gold's new ABC cost per carton, state whether its ₹975 price covers it, and what else must change.

THE WORKING

Working note 1: driver totals

Driver	Everyday	Elaichi Premium	Masala Gold	Total
Machine hours	60,000	16,000	4,000	80,000
Set-up hours (runs × per run)	120	80	120	320
Tests (runs × per run)	80	60	160	300
Moves (runs × per run)	60	40	80	180
Recipe and pack changes	1	3	7	11

Working note 2: cost driver rates

Activity	Cost (₹)	Driver quantity	Rate (₹)
Machine running	12,80,000	80,000 machine hours	16 per machine hour
Set-ups	6,40,000	320 set-up hours	2,000 per set-up hour
Quality testing	4,80,000	300 tests	1,600 per test
Material handling	3,60,000	180 moves	2,000 per move
Recipe and pack development	4,40,000	11 changes	40,000 per change

(i) Cost per carton: machine-hour rate

Particulars	Everyday (₹)	Elaichi Premium (₹)	Masala Gold (₹)
Prime cost	420	500	700
Production overheads (3, 2, 2 hours × ₹40)	120	80	80
Cost per carton	**540**	**580**	**780**

(ii) Cost: activity based costing

Particulars
Prime cost

Everyday (₹)
84,00,000

Elaichi Premium (₹)
40,00,000

Masala Gold (₹)
14,00,000

Particulars
Machine running @ ₹16 (60,000; 16,000; 4,000)

Everyday (₹)
9,60,000

Elaichi Premium (₹)
2,56,000

Masala Gold (₹)
64,000

Particulars	Everyday (₹)	Elaichi Premium (₹)	Masala Gold (₹)
Set-ups @ ₹2,000 (120; 80; 120 hours)	2,40,000	1,60,000	2,40,000
Quality testing @ ₹1,600 (80; 60; 160)	1,28,000	96,000	2,56,000
Material handling @ ₹2,000 (60; 40; 80)	1,20,000	80,000	1,60,000
Development @ ₹40,000 (1; 3; 7)	40,000	1,20,000	2,80,000

Particulars
Total cost
Everyday (₹)
98,88,000
Elaichi Premium (₹)
47,12,000
Masala Gold (₹)
24,00,000

Particulars
Cost per carton
Everyday (₹)
494.40
Elaichi Premium (₹)
589.00
Masala Gold (₹)
1,200.00

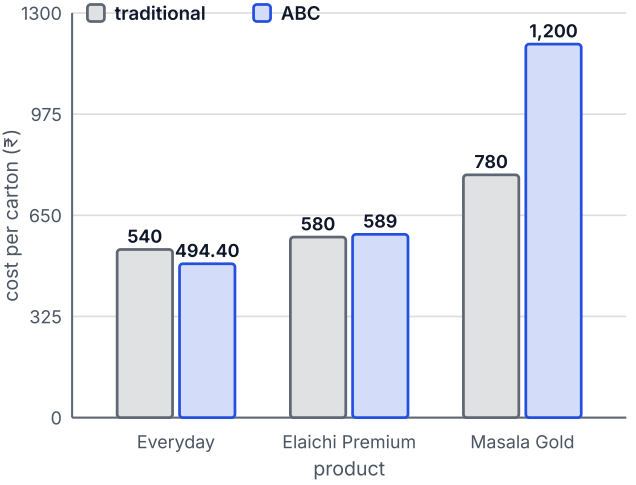
Check: the ABC overheads, ₹14,88,000 + ₹7,12,000 + ₹10,00,000 = ₹32,00,000.

(iii) Cost distortion (traditional − ABC)

Particulars	Everyday	Elaichi Premium	Masala Gold
Per carton (₹)	+45.60	(−) 9.00	(−) 420.00
Total (₹)	+9,12,000	(−) 72,000	(−) 8,40,000
Result	Over-costed	Under-costed	Under-costed

The total distortion is nil: ₹9,12,000 = ₹72,000 + ₹8,40,000. Masala Gold is under-costed because machine hours ignore the batch-level and product-level work it causes: it has half the runs and more than half the tests of the plant, and 7 of the 11 recipe and pack changes, on less than 7% of the cartons.

Cost per carton: machine-hour rate against ABC



Grey: traditional cost; blue: ABC cost. Masala Gold's 40 small runs and 7 recipe changes carry costs that machine hours never saw.

(iv) Profit per carton under ABC

Particulars	Everyday (₹)	Elaichi Premium (₹)	Masala Gold (₹)
Price (traditional cost × 125%)	675.00	725.00	975.00
Less: ABC cost	494.40	589.00	1,200.00
Profit / (loss) per carton	**180.60**	**136.00**	**(225.00)**

Everyday earns ₹180.60, not the ₹135 the old cost suggested: it can meet BrewBucks' ₹640 and still earn ₹145.60 a carton. Total profit for the quarter (₹42,50,000) is the same under both systems because all output is sold; the difference is in which product earns it.

(v) Masala Gold in runs of 100 cartons

Runs = 2,000 ÷ 100 = 20.

Activity	Driver	Amount (₹)
Machine running	4,000 hours × ₹16	64,000
Set-ups	20 × 3 = 60 hours × ₹2,000	1,20,000
Quality testing	20 × 4 = 80 tests × ₹1,600	1,28,000
Material handling	20 × 2 = 40 moves × ₹2,000	80,000
Recipe and pack development	7 changes × ₹40,000	2,80,000
Overheads		**6,72,000**

New ABC cost per carton = ₹700 + ₹6,72,000 ÷ 2,000 = ₹700 + ₹336 = **₹1,036**. The ₹975 price still loses ₹61 a carton. Halving the runs halves the batch-level costs (₹3,28,000 saved), but the product-level cost of ₹2,80,000 (₹140 a carton) does not move with batch size. To cover cost at ₹975, development cost must fall by at least ₹61 × 2,000 = ₹1,22,000, that is by 4 changes (3 changes a quarter, giving a cost of ₹956), or the price must rise; ABC cost plus 25% is ₹1,295. The Activity Based Costs appear to be more precise.

Interactive version: PassBook app, lesson Product costs under ABC versus traditional absorption

Worked example

Punjabi Jumbo is dropped: rates on use or on capacity?

The packing hall of Tapri's samosa factory has three activities whose costs are fixed for the year:

Activity	Cost driver	Capacity	Annual cost (₹)
Packing machine	Machine hours	12,000 hours	9,60,000
Quality inspection	Number of inspections	800	4,80,000
Dispatch	Number of dispatch orders	2,500	3,75,000

Until last year Ritu set the rates on the drivers **actually used**, so the whole cost went to products. Last year's use:

Product
Classic Aloo (5,00,000 packs)
Machine hours
5,000
Inspections
350
Dispatch orders
900

Product
Cocktail Mini (2,00,000 packs)
Machine hours
3,000
Inspections
250
Dispatch orders
600

Product
Punjabi Jumbo
Machine hours
1,600
Inspections
150
Dispatch orders
500

This year Punjabi Jumbo is discontinued. Classic Aloo and Cocktail Mini will use exactly what they used last year; the hall and its costs stay as they are.

Required:

(i) Calculate the activity rates (a) on actual use last year, (b) on actual use this year and (c) on cost driver capacity.

(ii) Calculate the packing-hall cost per pack of Classic Aloo and Cocktail Mini under each set of rates.

(iii) Under capacity-based rates, calculate this year's cost of unused capacity for each activity and reconcile with the hall's total cost.

(iv) Tapri prices at cost plus a markup. Advise which rates should be used, with reference to the factors management considers in choosing a capacity level.

THE WORKING

(i) Activity rates

Activity	(a) Last year's use	(b) This year's use	(c) Capacity
Packing (₹ per hour)	9,60,000 ÷ 9,600 = 100.00	9,60,000 ÷ 8,000 = 120.00	9,60,000 ÷ 12,000 = 80.00
Inspection (₹ each)	4,80,000 ÷ 750 = 640.00	4,80,000 ÷ 600 = 800.00	4,80,000 ÷ 800 = 600.00
Dispatch (₹ per order)	3,75,000 ÷ 2,000 = 187.50	3,75,000 ÷ 1,500 = 250.00	3,75,000 ÷ 2,500 = 150.00

(ii) Cost charged to each product

Particulars	(a) Last year (₹)	(b) This year (₹)	(c) Capacity (₹)
Classic Aloo: packing	5,00,000	6,00,000	4,00,000
Classic Aloo: inspection	2,24,000	2,80,000	2,10,000
Classic Aloo: dispatch	1,68,750	2,25,000	1,35,000
Classic Aloo: total	**8,92,750**	**11,05,000**	**7,45,000**
Per pack (÷ 5,00,000)	**1.79**	**2.21**	**1.49**
Cocktail Mini: packing	3,00,000	3,60,000	2,40,000
Cocktail Mini: inspection	1,60,000	2,00,000	1,50,000
Cocktail Mini: dispatch	1,12,500	1,50,000	90,000

Particulars	(a) Last year (₹)	(b) This year (₹)	(c) Capacity (₹)
Cocktail Mini: total	**5,72,500**	**7,10,000**	**4,80,000**
Per pack (÷ 2,00,000)	**2.86**	**3.55**	**2.40**

Under (a) Punjabi Jumbo carried the remaining ₹3,49,750 (₹1,60,000 + ₹96,000 + ₹93,750).

(iii) Cost of unused capacity this year (capacity-based rates)

Activity	Unused capacity	Amount (₹)
Packing	12,000 − 8,000 = 4,000 hours × ₹80	3,20,000
Inspection	800 − 600 = 200 × ₹600	1,20,000
Dispatch	2,500 − 1,500 = 1,000 × ₹150	1,50,000
Total cost of unused capacity		**5,90,000**

Reconciliation: allocated ₹7,45,000 + ₹4,80,000 = ₹12,25,000; add unused capacity ₹5,90,000 =

₹18,15,000, the total cost of the three activities (₹9,60,000 + ₹4,80,000 + ₹3,75,000).

(iv) Advice. Rates on actual use put the whole ₹18,15,000 on whatever products remain. Nothing about Classic Aloo or Cocktail Mini has changed, yet their costs per pack rise by about 24% (₹1.79 to ₹2.21 and ₹2.86 to ₹3.55), because the cost of the 4,000 idle machine hours, 200 idle inspections and 1,000 idle dispatch orders is loaded onto them. Rates on cost driver capacity should be used:

1. **Effect on product costing and capacity management**: each product is charged only for the capacity it uses (₹1.49 and ₹2.40 a pack, the same as before Jumbo left), and the ₹5,90,000 of unused capacity is reported to management, who can sell it, redeploy it or reduce it.
2. **Effect on pricing decisions**: cost-plus prices on actual-use rates would rise for the remaining products, which may lose volume and push the rates up again.
3. **Effect on performance evaluation**: the managers of Classic Aloo and Cocktail Mini are not charged for a capacity decision they did not make.

The other factors ICAI lists are the effect on financial statements, regulatory requirements and difficulties in forecasting (SM 5.23).

Worked example

Should the kirana stores go to a distributor?

Tapri sells its premix at a list price of ₹500 a carton through three channels; the cost of goods sold is ₹380 a carton. Its customer-level activities and their cost driver rates are: order processing ₹100 per order; van delivery ₹250 per delivery drop; in-store merchandising ₹300 per merchandiser hour; payment collection ₹150 per collection visit. Last year:

Particulars	Supermarkets	Kirana stores	Café franchisees
Cartons sold	15,000	9,000	6,000
Discount on list price	12%	4%	8%
Orders	60	1,800	300
Delivery drops	60	1,800	150
Merchandising hours	600	900	Nil
Collection visits	Nil	900	60

Malwa Traders, a distributor, offers to take over the whole kirana channel. It would buy 10,000 cartons a year at 15% off list price, order weekly (52 orders, each

delivered in one drop to its godown) and do all merchandising and collection itself.

Required:

(i) Calculate the customer-level operating income of each channel and the operating income as a percentage of revenue, and rank the channels.

(ii) Evaluate Malwa Traders' proposal and advise, stating the condition on which the activity cost savings depend.

THE WORKING

Working note: customer-level activity costs

Activity
Orders @ ₹100 (60; 1,800; 300)
Supermarkets (₹)
6,000
Kirana stores (₹)
1,80,000
Café franchisees (₹)
30,000

Activity
Delivery drops @ ₹250 (60; 1,800; 150)
Supermarkets (₹)
15,000
Kirana stores (₹)
4,50,000
Café franchisees (₹)
37,500

Activity

Merchandising @ ₹300 (600; 900; –)

Supermarkets (₹)

1,80,000

Kirana stores (₹)

2,70,000

Café franchisees (₹)

–

Activity

Collection @ ₹150 (–; 900; 60)

Supermarkets (₹)

–

Kirana stores (₹)

1,35,000

Café franchisees (₹)

9,000

Activity

Total

Supermarkets (₹)

2,01,000

Kirana stores (₹)

10,35,000

Café franchisees (₹)

76,500

(i) Customer-level operating income

Particulars	Supermarkets (₹)	Kirana stores (₹)	Café franchisees (₹)
Revenues at list price	75,00,000	45,00,000	30,00,000

Particulars	Supermarkets (₹)	Kirana stores (₹)	Café franchisees (₹)
Less: Discount	9,00,000	1,80,000	2,40,000
Revenue at actual price	**66,00,000**	**43,20,000**	**27,60,000**
Less: Cost of goods sold (₹380)	57,00,000	34,20,000	22,80,000
Gross margin	**9,00,000**	**9,00,000**	**4,80,000**
Less: Customer-level activity costs	2,01,000	10,35,000	76,500
Customer-level operating income	**6,99,000**	**(1,35,000)**	**4,03,500**
Operating income as % of revenue	10.59%	(3.13%)	14.62%
Rank	**II**	**III**	**I**

(ii) Kirana channel through Malwa Traders

Particulars	Amount (₹)
Revenue at list price (10,000 × ₹500)	50,00,000
Less: Discount 15%	7,50,000
Revenue at actual price	**42,50,000**
Less: Cost of goods sold (10,000 × ₹380)	38,00,000
Gross margin	**4,50,000**
Less: Orders 52 × ₹100 and drops 52 × ₹250	18,200
Customer-level operating income	**4,31,800**
Operating income as % of revenue	10.16%

The channel moves from a loss of ₹1,35,000 to an income of ₹4,31,800, an improvement of ₹5,66,800. The kirana stores have the smallest discount and the same gross margin as the supermarkets, but they buy 5 cartons a drop, so 1,800 orders, 1,800 drops, 900 merchandiser hours and 900 collection visits swallow it. The distributor gives away 11 more points of discount and still leaves more, because it removes almost all the activities.

Advice: accept the proposal, on one condition. The activity costs are the cost of resources used (vans, order clerks, merchandisers, collectors). The saving of ₹10,35,000 − ₹18,200 = ₹10,16,800 is real only if

> Tapri releases or redeploys those resources; if the vans and merchandisers stay on the payroll with nothing to do, the cost remains and becomes unused capacity. Tapri should also check that Malwa Traders can sell the 10,000 cartons it promises.

Interactive version: PassBook app, lesson Product costs under ABC versus traditional absorption

Exam angle

In the exam

How ICAI examined ABC computations, May 2024 to September 2026

In the documents read for this lesson, ABC computations appeared in 10 questions, usually 8 to 10 marks:

- **Traditional against ABC product cost**: September 2024 Q3(a), 8 marks: overheads first distributed to activities in given ratios (technical staff 1 : 2 : 2, machine costs 2 : 3), then driver rates; May 2026 Q5(a), 10 marks: total sales and profit of two products under both systems, with closing stock; MTP January 2026 Series I Q2(a), 10 marks: machine-hour rate against ABC with allocation percentages, plus "explain how an ABC system would be developed".

- **Capacity-based rates and unused capacity**: May 2025 Q3(a), 8 marks, four activities; the inspection capacity was fully used, so its unused cost was nil.
- **Service sectors**: September 2025 Q3(a), 8 marks: cost per project in three sectors, then the most profitable sector on profitability % of fees.
- **Customer profitability**: May 2024 Q2(a), 8 marks: customer-level operating income, % of revenue at actual price, and rank; September 2026 Part I MCQs 1–5, a five-MCQ case (cost of visits, transport, activity cost as % of revenue net of discount, operating income).
- **Pricing**: MTP January 2025 Series II Q2(a), 10 marks: cost and quoted price at cost plus 50% under labour hours and ABC, then pricing strategies; RTP May 2026 Q9: profit per printer under the current basis and ABC, and the benefits of ABC.
- **MCQ**: MTP September 2025 Series II MCQ 6, ABC cost per unit of three products.

In the exam

The traps

- Driver data per batch or per unit: multiply up to totals first (the SM's Practical Problem 1).
- Overheads given as ledger items with sharing ratios: distribute them to the activities before computing rates (September 2024 Q3(a)).
- "Calculates activity cost rates based on cost driver capacity": divide by capacity, not by use (May 2025).
- Percentages of revenue: on revenue at actual price, net of discount (May 2024; September 2026 MCQ 3 says "net of discount").
- State the comment ICAI expects: which product is under- or over-costed, and why; ICAI's own line is "The Activity Based Costs appear to be more precise" (SM 5.17).

Trick

Check the zero

Total overhead under ABC must equal the traditional total, so the distortions × units must add up to zero (and with capacity rates, allocated cost + unused capacity cost = total activity cost). If they do not, a driver total or a rate is wrong. Condition: both systems absorb the same overheads; a question that gives a different total for ABC must be read as given.

Answer frame

A full-marks numerical answer, in ICAI's order

1. **Working notes**: driver totals (batches, orders, inspections, hours); the split of overheads into pools if percentages or ratios are given; the driver rate table (pool, driver basis, quantity, rate).
2. **Statement 1**: cost per unit under the traditional method (rate shown).
3. **Statement 2**: cost under ABC with "@ ₹ rate (quantities)" on each overhead line; total cost and cost per unit.
4. **Comparison**: cost distortion with its sign, under- or over-costed.
5. **Comment**: reason for the difference (overheads borne by each product differ; batch-level and product-level activities) and the decision asked (price, mix, customer action).

Loses marks

"Masala Gold ABC cost ₹1,200, so it is under-costed."

No rates, no driver totals: a slip anywhere loses every method mark.

Passes

Driver rates and the ABC statement, but the per-run tests were charged as if they were totals, and the distortion has no sign.

One slip in a working note spoils every figure after it.

Full marks

Working notes for driver totals and rates, both statements, the distortion with "(–)" for under-costed products and the check that it nets to zero, and a two-line comment on pricing.

Common mistakes

⚠ **Common mistake**

Hall of Shame: per-run data used as if it were the total

"Quality tests per run: 4" for Masala Gold, charged as 4 tests × ₹1,600 = ₹6,400. **Fix:** tests = 40 runs × 4 = 160; charge = ₹2,56,000. Where a question gives a batch size instead of the number of runs, batches = units ÷ batch size first.

⚠ **Common mistake**

The rate on actual use when the question says capacity

Dividing ₹9,60,000 by the 8,000 hours used gives ₹120 an hour and hides ₹3,20,000 of idle packing capacity inside Classic Aloo's and Cocktail Mini's costs. **Fix:** divide by capacity (12,000 hours: ₹80) and report the unused capacity separately.

⚠ **Common mistake**

Dividing ABC overhead by units sold

Cost per unit is total cost ÷ units **produced**. Units sold matter only for sales, cost of goods sold and closing stock.

⚠ **Common mistake**

Percentages on list-price revenue

Operating income as a percentage of revenue uses revenue at actual price, after discount, as ICAI's answers do. **Fix:** supermarkets ₹6,99,000 ÷ ₹66,00,000 = 10.59%, not ÷ ₹75,00,000 = 9.32%.

Practice

Check yourself

1. A product's cost is ₹612 a unit under a single labour-hour rate and ₹655 a unit under ABC. 5,000 units are made. What is the total cost distortion in rupees? (Enter a negative number for under-costing.)

Answer: _____

2. Tapri's lab can perform 1,200 tests a year at a cost of ₹7,20,000; rates are based on capacity. Products used 1,050 tests. What is the cost of unused capacity (₹)?

Answer: _____

3 Case scenario. Under ABC, Tapri's samosa factory charges set-ups at ₹4,000 per set-up, one set-up per batch. Party Pack (20,000 packs a month) is made in batches of 250 packs, so it carries ₹16 of set-up cost per pack. Kesar wants that figure down to ₹8 a pack, with the rate per set-up unchanged. What batch size does Party Pack need?

- **A** 1,000 packs
- **B** 500 packs
- **C** 250 packs
- **D** 125 packs

4 Under ABC, a company finds Product X over-costed by ₹3 a unit on 40,000 units, and Product Y under-costed by ₹12 a unit. If these are its only products and total overheads are unchanged, how many units of Y were made?

- **A** 1,20,000
- **B** 40,000
- **C** 10,000
- **D** 4,80,000

5 Which of the following are correct about the comparison of traditional absorption costing and ABC? Select all that apply.

- **A** Direct materials and direct labour per unit are the same under both
- **B** ABC always shows a higher total cost for the factory
- **C** With capacity-based rates, allocated cost plus the cost of unused capacity equals the activity's total cost
- **D** When all output is sold, total profit is the same under both

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Prime cost is the same under both systems; only production overheads move.
- Traditional: overhead per unit = absorption rate × base per unit. ABC: driver rate × driver quantity per product ÷ units produced.
- Turn per-batch and per-unit data into totals first; batches = units ÷ batch size.
- Cost distortion = traditional − ABC: + over-costed, (−) under-costed; the total distortion is nil.

- Capacity-based rates: allocated + cost of unused capacity = activity cost; unused capacity = unused units × rate.
- Customer-level operating income = revenue at actual price − cost of goods sold − customer-level activity costs; % on revenue at actual price.
- Small-batch, low-volume products tend to be under-costed by volume rates (SM 5.26). (The ground floor was paying the fourteenth floor's bills.)

Answer key · Check yourself

1 **-215000**

Distortion = traditional − ABC = ₹612 − ₹655 = −₹43 a unit; × 5,000 = −₹2,15,000: the product is under-costed by ₹2,15,000.

2 **90000**

Rate = ₹7,20,000 ÷ 1,200 = ₹600 per test. Unused 150 tests × ₹600 = ₹90,000 (= ₹7,20,000 − 1,050 × ₹600).

3 **B** 500 packs

Set-up cost per pack = ₹4,000 ÷ batch size, so ₹8 needs 4,000 ÷ 8 = 500 packs a batch (40 batches a month instead of 80). The monthly volume does not enter: set-ups are batch-level, so the cost per pack depends on the batch size, not on how many packs are made. 1,000 packs would give ₹4; 125 packs, ₹32.

4 **C** **10,000**

The distortions net to zero: 40,000 × ₹3 = ₹1,20,000 = units of Y × ₹12, so Y = 10,000 units.

5 **A** **Direct materials and direct labour per unit are the same under both**; **C** **With capacity-based rates, allocated cost plus the cost of unused capacity equals the activity's total cost**; **D** **When all output is sold, total profit is the same under both**

Only overheads are re-spread; the total is unchanged, so with no closing stock total profit is unchanged too. Unused capacity is reported separately, so the pieces add back to the activity's cost.

6.3 Activity based management and budgeting; benefits and limits

What ABC is good for and where it struggles; how ABM uses activity costs to cut non-value-added work, re-engineer, benchmark and measure; and how ABB runs ABC backwards to budget resources.

In plain words

Kesar finds the "useless" department

Ritu's first ABC report for Tapri's cold-chain and café business lists what each activity costs: brewing, delivering, moving milk crates from the receiving dock to the back store and then to the front store, re-delivering orders that came back, setting up the kulhad kiln for every new design.

> **Kesar:** Ritu, this is brilliant. Which activity adds no value? Delete it. Start with the accounts department; it has never made a single cup of chai.
>
> **Ritu:** The customer will not pay for accounts, but the law will not let you skip them. The crate moving and the re-deliveries, though: nobody pays for those, and they cost us ₹4 lakh a month.

That is **activity based management (ABM)**: using the activity costs that ABC produces to manage the activities themselves. An activity the customer is willing to pay for, like brewing chai to order, is **value-added**. One that does not improve the product or service but adds cost, like

moving materials around or setting up machines for each run, is **non-value-added**: the first target for cutting, streamlining or elimination. ABM also redesigns whole processes, compares one café's activity costs with another's, and measures how efficiently each activity is performed.

Activity based budgeting (ABB) runs ABC backwards. ABC starts from the money spent and traces it to products. ABB starts from next year's planned work (how many deliveries, how many orders, how many support calls), works out the activities that work will need, and then the resources and money those activities will cost.

ABC itself has a price. It gives more accurate costs, a logical basis for overheads, better pricing and a spotlight on waste. But it is more expensive than the traditional system, is not much help to a small organisation or one with few products, and choosing the right driver is not always easy.

Think of it like

The fitness tracker

A fitness tracker tells you what each activity burns: 300 calories for the walk, 40 for the meeting you could have skipped (ABC). The diet and exercise plan you build from those numbers is ABM: keep the walk, drop the pointless meeting. Planning next month's meals from next month's training schedule, work first and food second, is ABB. And a tracker costs money: a person who sits all day at one desk may not need one.

Key idea

ABC measures what activities cost; ABM manages the activities (keep value-added, remove or reduce non-value-added); ABB budgets by working back from the planned quantity of work to the resources it needs.

How it works

Benefits and limits of ABC

The main **advantages** of ABC (SM 5.17–5.18), in ICAI's order:

1. More accurate costing of products/services.
2. Overhead allocation is done on a logical basis.
3. It enables better pricing policies by supplying accurate cost information.
4. It utilizes unit cost rather than just total cost.

5. It helps to identify non-value added activities, which facilitates cost reduction.
6. It is helpful to organizations with multiple products.
7. It highlights problem areas which require the attention of management.

The main **limitations** (SM 5.18):

1. It is more expensive, particularly in comparison with the traditional costing system.
2. It is not helpful to small organizations.
3. It may not be applied to organizations with limited products.
4. Selection of the most suitable cost driver may not be easy, or may be difficult or complicated.

Requirements in ABC implementation

"A number of distinct practical stages are required in the ABC implementation" (SM 5.18):

1 **Staff training**: the co-operation of the workforce is critical to successful implementation; staff training should create awareness of the purpose of ABC.

2 **Process specification**: informal but structured interviews with key members of personnel identify the different stages of the production process, the commitment of resources to each, processing times and bottlenecks.

3 **Activity definition**: activities must be defined clearly at an early stage to manage problems effectively. There might be overloading of information from the new data, but it is needed in codification.

4 **Activity driver selection**: a cost driver is selected for each activity.

5 **Assigning cost**: a single representative activity driver can be used to assign costs from the activity pools to the cost objects.

These are the stages of *installing* ABC in an organisation. Do not confuse them with the five calculation stages of identifying activities, cost pools and driver rates.

ABC as a decision-making tool

ICAI's five ways (SM 5.19):

1. ABC along with some other cost management technique can be used to improve the performance and profitability of an organisation.
2. Wholesale distributors gain in decision-making by correlating costs to activities; the introduction of a new product or vendor can be better decided.
3. Decisions on facility and resource expansion, such as relocating or opening a new distribution centre: the ABC model identifies the specific cost elements

targeted (savings in freight or logistics may offset the new facility).

4. Decision support for human resources: activity costs can be associated with an individual, so new levels of financial performance can be determined (branch management, sales).
5. Companies that price on a cost plus markup basis find ABC very relevant for determining competitive prices.

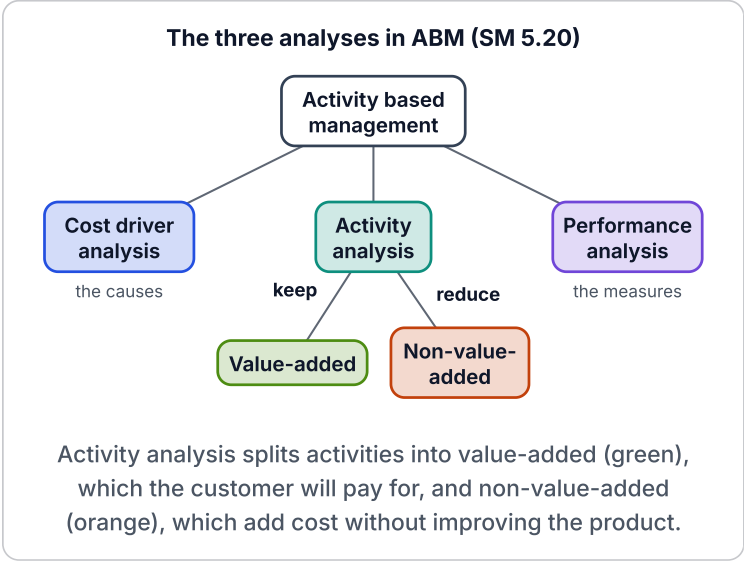
Activity based management

Definition

Activity Based Management (ABM)

The term Activity Based Management (ABM) is used to describe the cost management application of ABC. The use of ABC as a costing tool to manage costs at activity level is known as Activity Based Management (ABM). ABM is a discipline that focuses on the efficient and effective management of activities as the route to continuously improving the value received by customers. ABM utilizes cost information gathered through ABC. (SM 5.19)

The types of analysis involved in ABM (SM 5.20):



Activity analysis splits activities into value-added (green), which the customer will pay for, and non-value-added (orange), which add cost without improving the product.

1. **Cost driver analysis**: the factors that cause activities to be performed need to be identified in order to manage activity costs; cost driver analysis identifies these causal factors.
2. **Activity analysis**: (a) **value-added activities (VA)** are indispensable in order to complete the process, and customers are usually willing to pay (in some way) for them, e.g. polishing furniture by a furniture manufacturer; (b) **non-value-added activities (NVA)** represent work that is not valued by the external or internal customer; they do not improve the quality or function of a product or service, but they can adversely affect costs and prices, e.g. moving materials and machine set-up for a production run.

3. **Performance analysis**: identifying appropriate measures to report the performance of activity centres or other organisational units, consistent with each unit's goals and objectives.

ABM can be used in business in four ways (SM 5.20–5.21):

1. **Cost reduction**: ABM identifies costs against activities and finds opportunities to streamline or reduce the costs or eliminate the entire activity, especially if there is no value added.
2. **Business process re-engineering**: examining business processes and making substantial changes to how the organisation currently operates. ABM is a powerful tool for measuring business performance, determining the cost of business output, and identifying opportunities to improve process efficiency and effectiveness.
3. **Benchmarking**: comparing ABC-derived activity costs of one segment of the company with those of other segments. It requires uniformity in the definition of activities and the measurement of their costs.
4. **Performance measurement**: focusing on activity performance to face competitors and manage costs, by monitoring the efficiency and effectiveness of activities.

ICAI's examples of performance measures (SM 5.21):

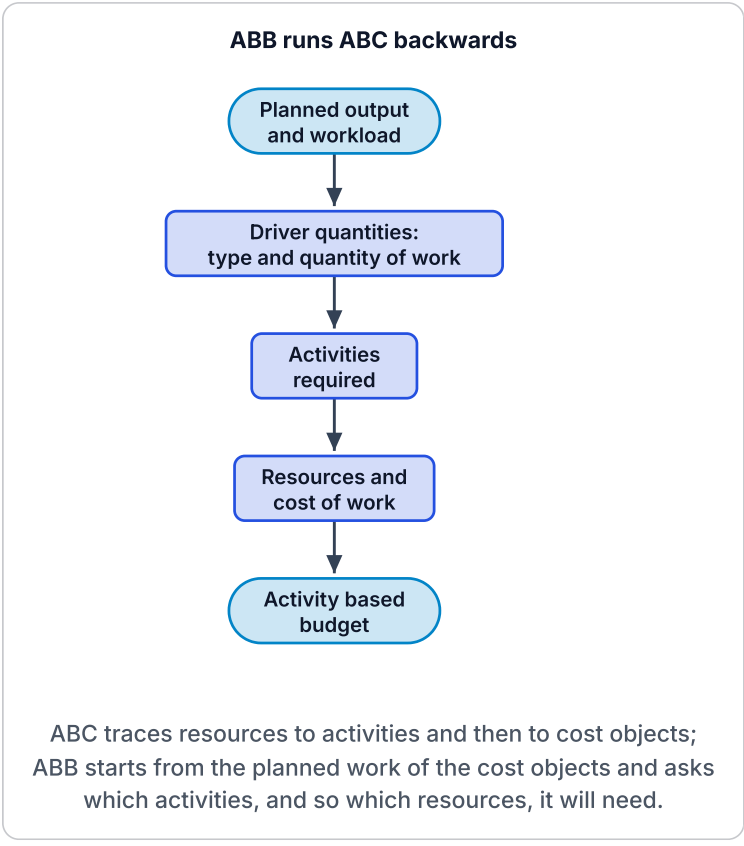
Area	Measure
Quality of purchased component	Zero defects
Quality of output	% yield
Customer awareness	Orders; number of complaints

Activity based budgeting

Definition

Activity Based Budgeting (ABB)

Activity based budgeting analyses the resource input or cost for each activity. It provides a framework for estimating the amount of resources required in accordance with the budgeted level of activity. Actual results can be compared with budgeted results to highlight, in financial and non-financial terms, those activities with major discrepancies from budget, for potential reduction in the supply of resources. It is a planning and control system which seeks to support the objectives of continuous improvement. It means planning and controlling the expected activities of the organisation to derive a cost-effective budget that meets forecast workload and agreed strategic goals. ABB is the reversing of the ABC process to produce financial plans and budgets. (SM 5.21)



ABC traces resources to activities and then to cost objects; ABB starts from the planned work of the cost objects and asks which activities, and so which resources, it will need.

The three **key elements** of ABB (SM 5.21): type of work to be done; quantity of work to be done; cost of work to be done.

Its **benefits** (SM 5.22):

1. ABB can enhance the accuracy of financial forecasts and increase management understanding.
2. When automated, ABB can rapidly and accurately produce financial plans and models based on varying levels of volume assumptions.

3. ABB eliminates much of the needless rework created by traditional budgeting techniques.

Formats and formulas

Reproduce ICAI's budget statement layout (Illustration 4, SM 5.28–5.30): working note of budgeted activity costs first, then rates, then the cost per unit of each product.

∑ **Formula**

Budgeted activity cost

$$\text{Budgeted activity cost} = \text{Fixed part} + \text{Present variable part} \times \text{Growth factor}$$

The growth factor is given ("to double": 2; "to increase by 80%": 1.8), or is budgeted level ÷ present level when the variable part moves with the driver. Stepped costs are added step by step ("₹1 lakh more for every one lakh more statements").

∑ **Formula**

Resources required for the budgeted work

$$\text{Resource units required} = \frac{\text{Budgeted driver quantity}}{\text{Capacity of one resource unit}} \text{ (rounded up to whole units)}$$

People, vehicles and machines come in whole units, so the budget carries some spare capacity: capacity provided minus capacity used, valued at the capacity rate, is the cost of unused capacity (see capacity-based rates).

∑ **Formula**

Budgeted activity rate and product cost

$$\text{Budgeted activity rate} = \frac{\text{Budgeted activity cost}}{\text{Budgeted driver quantity}}$$

$$\text{Budgeted cost per unit} = \frac{\sum \text{Rate} \times \text{Driver quantity}}{\text{Budgeted units of product}}$$

Statement showing budgeted cost per unit of the product (layout of SM 5.28–5.29, split for phones)

Activity	Budgeted cost (₹)	Activity driver	Budgeted driver units	Rate (₹)
Activity 1	xxx	Number of ...	xxx	xxx
Total budgeted cost	**xxx**			

Activity	Product A (₹)	Product B (₹)	Product C (₹)
Activity 1 (rate × driver units)	xxx	xxx	xxx
Budgeted cost	**xxx**	**xxx**	**xxx**
Units of product (budget)	xxx	xxx	xxx
Budgeted cost per unit	**xxx**	**xxx**	**xxx**

Worked examples

Worked example **TapriNow: budget the work first, then the riders**

In Indore, TapriNow, Tapri's delivery app, offers three services: home delivery, office subscriptions and café pre-orders (collected at the counter, so never delivered). Dev wants next year's budget built activity

by activity from the planned workload, not from this year's department totals. The plan for next year:

Service	Planned volume	Work needed
Home delivery	60,000 orders	Per order: 1 app order, 1 delivery
Office subscriptions	100 office accounts	Per account per month: 24 deliveries, 1 standing app order
Café pre-orders	30,000 orders	Per order: 1 app order

Customer support: home delivery raises 1 contact for every 20 orders (6 minutes each); each office account raises 2 contacts a month (10 minutes each); café pre-orders raise 1 contact for every 50 orders (4 minutes each).

Resources and their costs:

1. **Deliveries**: riders are hired on annual contracts at ₹2,16,000 each, in whole numbers; a rider can make at most 7,200 deliveries a year. Fuel and rider incentives cost ₹15 per delivery.
2. **Order processing**: platform charges of ₹2,73,600 a year (fixed) plus a payment-gateway fee of ₹2 per app order.
3. **Customer support**: outsourced at ₹5 per minute plus a fixed platform fee of ₹44,400 a year.

4. **Office account management**: one key-account executive (₹4,80,000 a year) looks after at most 50 office accounts.

Required:

(i) Budget the quantity of work: the driver quantity each service needs from each activity.

(ii) Determine the resources required and the budgeted cost of each activity.

(iii) Compute the budgeted activity rates (charge rider cost at its rate per delivery of rider capacity) and the budgeted cost per home order, per office account and per café pre-order. Show the cost of unused rider capacity separately and comment on it.

THE WORKING

(i) Working note 1: budgeted quantity of work

Driver	Home	Office	Café	Total
Deliveries	60,000	28,800	–	88,800
App orders	60,000	1,200	30,000	91,200
Support minutes	18,000	24,000	2,400	44,400
Office accounts	–	100	–	100

Office: deliveries 100 × 24 × 12 = 28,800; app orders 100 × 12 = 1,200; support 100 × 2 × 12 × 10 = 24,000 minutes. Home support: 60,000 ÷ 20 × 6 = 18,000 minutes. Café support: 30,000 ÷ 50 × 4 = 2,400 minutes.

(ii) Working note 2: resources required and activity budgets

Activity	Resources and cost	Budgeted cost (₹)
Deliveries	Riders: 88,800 ÷ 7,200 = 12.33, so 13 riders × ₹2,16,000 = ₹28,08,000; fuel and incentives 88,800 × ₹15 = ₹13,32,000	41,40,000
Order processing	₹2,73,600 + 91,200 × ₹2	4,56,000
Customer support	44,400 × ₹5 + ₹44,400	2,66,400
Office account management	100 ÷ 50 = 2 executives × ₹4,80,000	9,60,000
Total		**58,22,400**

(iii) Budgeted activity rates

Activity	Driver	Rate (₹)
Deliveries	Deliveries	45.00
Order processing	App orders	5.00
Customer support	Support minutes	6.00
Office account management	Office accounts	9,600.00

Delivery rate: rider capacity cost ₹2,16,000 ÷ 7,200 = ₹30 per delivery, plus ₹15 fuel and incentives. Order processing: ₹4,56,000 ÷ 91,200. Support: ₹2,66,400 ÷ 44,400. Accounts: ₹9,60,000 ÷ 100.

Budgeted cost statement

Activity	Home (₹)	Office (₹)	Café (₹)
Deliveries @ ₹45	27,00,000	12,96,000	–
Order processing @ ₹5	3,00,000	6,000	1,50,000
Customer support @ ₹6	1,08,000	1,44,000	14,400
Account management @ ₹9,600	–	9,60,000	–
Budgeted cost	**31,08,000**	**24,06,000**	**1,64,400**
Units of service	60,000 orders	100 accounts	30,000 orders
Budgeted cost per unit	**51.80**	**24,060**	**5.48**

Unused rider capacity: 13 riders × 7,200 = 93,600 deliveries of capacity against 88,800 planned; 4,800 × ₹30 = **₹1,44,000**, reported separately.

Check: ₹31,08,000 + ₹24,06,000 + ₹1,64,400 + ₹1,44,000 = ₹58,22,400, the total of the activity budgets.

Comment: The budget starts from the type and quantity of work each service needs and only then costs the resources, which is activity based

budgeting. Riders come in whole numbers: twelve riders give 86,400 deliveries, so the 13th rider is needed for the last 2,400 deliveries and two-thirds of that rider's capacity is spare. Charging the services only for the capacity they use keeps an office account at ₹24,060 a year (₹2,005 a month) and shows management the ₹1,44,000 it is paying for and not using: the spare 4,800 deliveries could serve 16 more office accounts (288 deliveries a year each) without another rider.

Worked example **Ananya's activity review at the kulhad cafés**

Ananya Rao reviews five activities at Tapri's kulhad cafés:

(a) Brewing chai to each customer's order.

(b) Moving milk crates from the receiving dock to the back store, and later to the front store.

(c) Hand-painting the festival designs on the kulhads that customers buy as gifts.

(d) Resetting the kulhad kiln for every new design run.

(e) Re-delivering app orders returned because the address was incomplete.

She then proposes: (1) to compare the cost per delivery of the Pune and Indore cafés, defined and measured

the same way; (2) to redesign the whole order-to-doorstep process; (3) to report each café's complaints per thousand orders every month.

Required:

(i) Classify each activity as value-added or non-value-added, with reasons.

(ii) Identify the use of ABM in each of Ananya's proposals.

MODEL ANSWER

(i) As per the Study Material, value-added activities are indispensable in order to complete the process, and customers are usually willing to pay for them; non-value-added activities represent work not valued by the external or internal customer, do not improve the quality or function of the product or service, but can adversely affect costs and prices (SM 5.20).

Activity	Classification	Reason
(a) Brewing to order	Value-added	Indispensable to the product; the customer pays for it
(b) Moving milk crates	Non-value-added	Moving materials improves nothing (ICAI's own example)

Activity	Classification	Reason
(c) Hand-painting gift kulhads	Value-added	Customers pay for the design, like polishing furniture
(d) Resetting the kiln	Non-value-added	Machine set-up for a production run (ICAI's own example); reduce by longer runs
(e) Re-delivering returned orders	Non-value-added	Repeats work without adding value; remove its cause

(ii) (1) **Benchmarking**: comparing ABC-derived activity costs of one segment with another, which requires uniform definition and measurement of activities. (2) **Business process re-engineering**: examining a business process and making substantial changes to how it operates. (3) **Performance measurement**: monitoring the efficiency and effectiveness of activities ("customer awareness: number of complaints" is ICAI's own measure). The NVA activities in (i) are targets for the fourth use, **cost reduction**.

Exam angle

In the exam

How ICAI asked this, in the documents read

In the documents read for this lesson, this lesson's topics were asked in 6 questions:

- **ABM**: old-scheme May 2023 Q6(c), 5 marks: "What is meant by Activity Based Management (ABM) and discuss how ABM can be used in the business?" The suggested answer is the SM definition followed by the four uses (cost reduction, business process re-engineering, benchmarking, performance measurement), each with ICAI's sentence.
- **Benefits of ABC**: RTP May 2026 Q9(c), after a printer-costing numerical: more accurate costs and better decisions; improved cost management through activity cost management; more accurate profitability analysis; and "the implementation of ABC will enable the company to adopt activity-based budgeting".
- **Implementation**: RTP September 2025 Q18(b): "suggest the key practical stages involved in successfully implementing an ABC system"; the answer is SM 5.18's five stages, word for word.
- **ABB numericals**: old-scheme MTP May 2022 Series II Q3(b) and MTP November 2023 Series II Q2(a), 10 marks each: a bank's four activities budgeted from fixed, doubling, tripling and stepped costs, then the cost per account, the pattern of SM Illustration 4.
- **Whether ABC is worth it**: MTP September 2025 Series II Q5(b), 5 marks: ICAI's answer refused ABC for a one-product firm whose overhead is mainly

depreciation, and for a labour-intensive firm with a low break-even point (see the concept lesson).

The Study Material's own questions ask for the benefits, the limitations, the practical applications, ABM and how ABC helps it, and ABB with its key elements (SM 5.33–5.34), and its MCQs 7 to 10 test the same lists.

In the exam

The traps

- "Reduction of prime cost" is **not** a benefit of ABC (SM MCQ 7): ABC re-spreads overheads; prime cost is traced anyway.
- "Borrowing funds" is not a step in installing ABC (SM MCQ 8).
- ICAI's NVA examples are **moving materials** and **machine set-up for a production run**; its VA example is **polishing furniture** by a furniture manufacturer. Quote them.
- ABB's three key elements are type, quantity and cost of work (SM MCQ 10: "all of the above").

Trick

C-B-B-P: what ABM does in business

Cut costs, **B**PR, **B**enchmark, **P**erformance measurement, in ICAI's order. Condition: these are the four uses of ABM in business (SM 5.20–5.21); the three analyses (cost driver, activity, performance) are a separate list.

Answer frame

"What is ABM? How can it be used in business?" (5 marks)

1. **Meaning**: the cost management application of ABC; the use of ABC to manage costs at activity level; a discipline focusing on the efficient and effective management of activities as the route to continuously improving the value received by customers; it utilises ABC information.
2. **Cost reduction**: identify costs against activities; streamline, reduce or eliminate, especially where no value is added.
3. **Business process re-engineering**: examine processes and make substantial changes; measure performance and cost of output; find efficiency improvements.
4. **Benchmarking**: compare ABC-derived activity costs across segments; needs uniform definition and measurement.
5. **Performance measurement**: monitor the efficiency and effectiveness of activities to face competitors and

manage costs.

If the question asks for the analyses instead: cost driver analysis; activity analysis (VA with ICAI's example, NVA with ICAI's examples); performance analysis.

Other lists, in ICAI's order

Question	Points (SM page)
Benefits (advantages) of ABC	Seven, from "more accurate costing" to "highlights problem areas" (5.17–5.18)
Limitations of ABC	Expensive; not for small organisations; not for limited products; driver selection difficult (5.18)
Requirements in implementing ABC	Staff training; process specification; activity definition; activity driver selection; assigning cost (5.18)
Practical applications	Decision-making tool (five ways); ABM; ABB (5.19–5.22)
ABB: meaning and key elements	Definition ending "reversing of the ABC process"; type, quantity and cost of work (5.21)
Benefits of ABB	Accuracy of forecasts; rapid plans at varying volumes when automated; less needless rework (5.22)

Loses marks

"ABM means managing activities to reduce cost."

One idea, no definition, no uses.

Passes

The definition and two uses, cost reduction and benchmarking, one line each.

Half the points.

Full marks

ICAI's definition with "efficient and effective management of activities" and "value received by customers", then all four uses with one explanatory sentence each.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the accounts department declared non-value-added

Kesar's logic: the café customer does not pay for it, so delete it. **Fix:** classify with ICAI's definition and examples. NVA is work "not valued by the external or internal customer" that does not improve the quality or function of the product but adds cost, such as moving materials or set-ups; the answer then says reduce, streamline or eliminate, not "delete everything the customer does not see". The accounts department serves internal users (management, lenders, the board) and the law requires it, so it is not NVA merely because no chai drinker sees it; ABM can still make it cheaper.

⚠ **Common mistake**

Mixing up the three lists

Writing "staff training, process specification" when asked how ABC is used in business, or "benchmarking" when asked for the stages of implementation. **Fix:** implementation has five stages (5.18); ABM in business has four uses (5.20–5.21); ABM analysis has three types (5.20).

⚠ **Common mistake**

Budgeting the variable part on the whole present cost

"Half fixed, the variable portion to three times": taking ₹5,00,000 × 3. **Fix:** fixed ₹2,50,000 + variable ₹2,50,000 × 3 = ₹10,00,000 (SM 5.29).

⚠ **Common mistake**

Hall of Shame: the budget that hired 12.33 riders

Dividing 88,800 deliveries by 7,200 per rider and budgeting 12.33 riders. **Fix:** ABB estimates the resources the planned work requires, and riders, analysts and vans come in whole units: round up, budget the cost of the whole resource, and show the spare capacity (here 4,800 deliveries, ₹1,44,000) separately instead of hiding it in the services' rates.

Practice

Check yourself

1. Tapri's quality lab budgets 52,000 test-minutes of work for next year. One lab analyst provides at most 15,000 test-minutes a year and costs ₹3,00,000; analysts are hired in whole numbers. Test kits cost ₹10 per test-minute. What is the budgeted cost of the testing activity (₹)?

Answer: _____

2 Case scenario. At Tapri's samosa factory Ananya finds that the frying line is stopped and re-set 45 times a month for different flavours, and that trays are carried twice between floors before packing. She proposes larger runs and a single-floor layout. Which of the following is correct?

A Both activities are value-added, because the samosas cannot be made without them

B Both are non-value-added; her proposals are cost reduction and business process re-engineering under ABM

C Only the carrying of trays is non-value-added; set-ups are value-added

D Both are non-value-added; her proposals are an example of activity based budgeting

3 Case scenario. Ritu writes one line on ABC for each of two Tapri units: a two-person kulhad workshop that makes one product with small overheads, and the premix plant that makes 40 products with frequent set-ups. Which pair of lines is consistent with ICAI's lists of the advantages and limitations of ABC?

A Workshop: "ABC will reduce your prime cost"; plant: "ABC is too expensive for any firm with many products"

B Workshop: "ABC is helpful to organisations with multiple products"; plant: "ABC is not helpful to small organisations"

C Workshop: "ABC may not be applied to organisations with limited products"; plant: "ABC is helpful to organisations with multiple products"

D Workshop: "ABC gives more accurate costing, so it must be installed"; plant: "the most suitable cost driver is always easy to select"

4 Ritu prepares next year's budget for dispatch at the premix plant. Which of her steps follow activity based budgeting? Select all that apply.

- **A** Forecast the number of dispatch orders from the planned sales of each product
- **B** Take this year's dispatch department total and add 8% for inflation
- **C** Work out how many packers and vans those dispatch orders will need
- **D** Cost those packers and vans to arrive at the dispatch activity's budget

More than one answer may be right.

5 Which of these is a stage in implementing ABC rather than a use of ABM?

- **A** Benchmarking
- **B** Process specification
- **C** Performance measurement
- **D** Business process re-engineering

Answers: the answer key at the end of this lesson.

Recap

- ABC advantages: more accurate costing; logical overhead allocation; better pricing; unit cost; identifies NVA activities; helps multi-product firms; highlights problem areas.
- ABC limitations: more expensive; not helpful to small organisations; not for limited products; driver selection difficult.
- Implementing ABC: staff training, process specification, activity definition, activity driver selection, assigning cost.
- ABM: the cost management application of ABC; analyses: cost driver, activity (VA/NVA), performance.
- ABM in business: cost reduction, business process re-engineering, benchmarking, performance measurement.
- VA example: polishing furniture. NVA examples: moving materials, machine set-up for a production run.
- ABB: the reversing of the ABC process; key elements: type, quantity and cost of work; benefits: accurate forecasts, rapid plans at varying volumes, less needless rework. (The accounts department stays.)

Answer key · Check yourself

1 **1720000**

Analysts required = 52,000 ÷ 15,000 = 3.47, so 4 analysts × ₹3,00,000 = ₹12,00,000; kits 52,000 × ₹10 = ₹5,20,000; total ₹17,20,000. Budgeting 3.47 analysts (₹10,40,000) forgets that people come in whole units; the fourth analyst's spare capacity is part of the budget.

2 **B** **Both are non-value-added; her proposals are cost reduction and business process re-engineering under ABM**

ICAI's own NVA examples are machine set-up for a production run and moving materials. Reducing them is ABM cost reduction, and redesigning the layout re-engineers the process. ABB is budgeting, not process change.

3 **C** **Workshop: "ABC may not be applied to organisations with limited products"; plant: "ABC is helpful to organisations with multiple products"**

ICAI lists "may not be applied to organisations with limited products" and "not helpful to small organisations" among the limitations (SM 5.18), and "helpful to organisations with multiple products" among the advantages (SM 5.17–5.18). ABC re-spreads overheads, so it does not reduce prime cost; and ICAI names driver selection as a difficulty, not as easy.

4 **A Forecast the number of dispatch orders from the planned sales of each product; C Work out how many packers and vans those dispatch orders will need; D Cost those packers and vans to arrive at the dispatch activity's budget**

ABB reverses ABC: it starts from the planned work (the type and quantity of work: dispatch orders), works out the resources that work needs, and then the cost of work (SM 5.21). Adding a percentage to the department's present total is the traditional incremental approach that ABB moves away from.

5 **B Process specification**

Process specification (structured interviews to map stages, resources, times and bottlenecks) is the second of the five implementation stages; the others are uses of ABM in business.

CHAPTER 7

Cost Accounting Systems

7.1 Non-integrated and integrated accounting systems

How a separate cost ledger keeps its double entry through the Cost Ledger Control Account, the control accounts and journal entries ICAI expects, and how integrated accounts replace that account with real assets and liabilities in one set of books.

In plain words

Two sets of books for one samosa

At Tapri's samosa factory, Dev's finance team keeps the financial books: bank, debtors, creditors, fixed assets, share capital, interest, bad debts. Ritu keeps a separate **cost ledger** with only what concerns the product: materials, wages, overheads, work-in-process, finished goods and cost of sales.

> **Kesar:** Ritu, your ledger has no bank account. How do you pay for potatoes?
>
> **Ritu:** I don't; Dev does. When his books buy potatoes, mine debit the Stores Ledger Control Account and credit one account that stands for everything outside my ledger: the Cost Ledger Control Account.
>
> **Kesar:** One account for the whole outside world? Sab adjust ho jayega in that.
>
> **Ritu:** That is exactly its job.

This is the **non-integrated system**: two separate sets of books. The cost ledger leaves out purely financial items (interest, bad debts, income from anything other than selling the product) and balance sheet items (fixed assets, debtors, creditors). Every entry that would need one of them uses the **Cost Ledger Control Account** instead, so the cost ledger still balances on its own. Because the two sets of books treat some items differently, they show two different profits, which must later be reconciled.

The **integrated system** keeps cost and financial accounts in the same set of books. The Cost Ledger Control Account disappears: the real accounts (bank, debtors, creditors, provision for depreciation, fixed assets, share capital) take its place. One set of books, one profit, nothing to reconcile.

Think of it like

The embassy

The Cost Ledger Control Account is an embassy. Inside the cost ledger it represents the whole country of the financial books: every rupee arriving from outside (a purchase, wages paid, an expense) enters through it, sales leave through it, and the costing profit is finally sent back to it. Integrated accounts merge the two countries, so no embassy is needed.

Key idea

In a non-integrated system the cost ledger is made self-balancing by the Cost Ledger Control Account, which stands in for every account kept only in the financial books. Integrated accounts use the real accounts instead, in one set of books, so no reconciliation is needed.

How it works

The non-integrated system

> **Definition**
>
> **Non-integrated accounting system**
>
> A system of accounting under which separate ledgers are maintained for both cost and financial accounts; also known as the cost ledger accounting system. The cost accounts restrict themselves to recording only those transactions which relate to the product or service being supplied. (SM 7.2)

It excludes expenses like interest and bad debts, and revenue or income from "other than the sale of product or service". It has fewer accounts than the financial books because purchases, expenses and balance sheet items like fixed assets, debtors and creditors are excluded; the excluded items are represented by the **Cost Ledger Control Account** (SM 7.2–7.3). With no common books, the two profits differ and must be reconciled (reconciling cost and financial profit).

The important **ledgers** maintained under the non-integrated system (SM 7.3):

1. **Cost ledger**: the principal ledger of the cost department, in which impersonal accounts are recorded; made self-balancing by maintaining a control account for each subsidiary ledger.

2. **Stores ledger**: an account for each item of stores, posted from invoices, goods received notes and material requisitions; shows receipt, issue and balance in quantity and value.
3. **Work-in-process ledger** (job ledger): accounts of unfinished jobs and processes; all material costs, wages and overheads for each job are posted to its job account.
4. **Finished goods ledger**: an account for each finished product or completed job; a transfer to stock is credited in the work-in-process ledger and debited here.

The principal accounts

What each control account in the cost ledger records (SM 7.3–7.5):

Account
1. Cost Ledger Control A/c (General Ledger Adjustment A/c)
Debited with
Sales; costing loss
Credited with
All items of expenditure; costing profit (from Costing P&L)

Account
2. Stores Ledger Control A/c
Debited with
Purchase of materials; abnormal gain (to Costing P&L)
Credited with
Issues; abnormal loss to Costing P&L

Account

3. Wages Control A/c

Debited with

Total wages paid, direct and indirect

Credited with

Direct wages to WIP; indirect wages to the overhead controls; abnormal idle time wages to Costing P&L

Account

4. Production Overhead Control A/c

Debited with

Indirect material, employee cost and expenses (carriage inward etc.)

Credited with

Overhead recovered (absorbed); the difference to Overhead Adjustment A/c

Account

5. Work-in-Process Control A/c

Debited with

Direct materials, direct employee cost, direct expenses, production overhead recovered

Credited with

Finished goods completed and transferred

Account

6. Administrative Overhead Control A/c

Debited with

Overheads incurred

Credited with

Overhead recovered (to Finished Goods if production-related, else to Cost of Sales); the difference to Overhead Adjustment A/c

Account

7. Finished Goods Control A/c

Debited with

Goods from WIP; administration overhead recovered if production-related

Credited with

Cost of Sales A/c

Account

8. Selling and Distribution Overhead Control A/c

Debited with

Overheads incurred

Credited with

Overheads recovered; the difference usually to Overhead Adjustment A/c

Account

9. Cost of Sales A/c

Debited with

Finished goods sold; general administration and S&D overhead recovered

Credited with

Balance to Sales A/c or Costing P&L A/c

Account

10. Costing Profit & Loss A/c

Debited with

Cost of sales; under-absorbed overheads; abnormal losses

Credited with

Sales; over-absorbed overheads; abnormal gains; the net profit or loss goes to the Cost Ledger Control A/c

Account

11. Overhead Adjustment A/c

Debited with

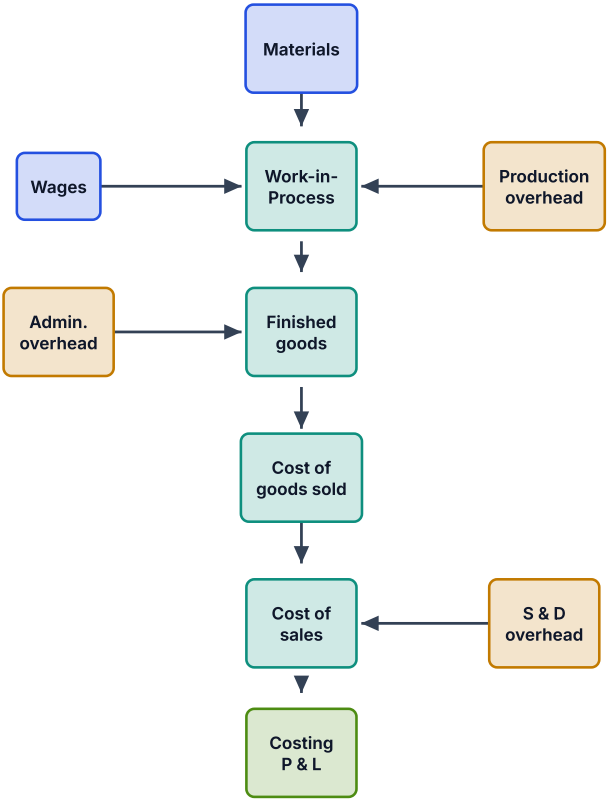
Under-recovery of overhead

Credited with

Over-recovery of overhead; net balance to Costing P&L

The balances mean something. Stores Ledger Control shows the total of all individual stores accounts; WIP Control the total of the jobs in process; Finished Goods Control the goods unsold; the **Cost Ledger Control A/c** the net total of all the balances of the impersonal accounts (SM 7.4–7.5). The Overhead Adjustment A/c is sometimes dispensed with, and under- or over-absorbed overheads are transferred straight to the Costing P&L A/c (SM 7.5). Abnormal idle time and abnormal losses follow idle time and under- and over-absorption.

Non-integrated accounting system: how costs flow through the control accounts



Each box is a control account. Blue: materials and wages; amber: the three overhead controls; teal: the product's path from WIP to cost of sales; green: the Costing P&L. Administration overhead joins finished goods when it relates to production; general administration joins cost of sales instead. Based on the flowchart at SM 7.9.

The journal entries

ICAI's scheme of entries in the cost books (SM 7.6–7.9). "CLC" is the Cost Ledger Control A/c.

Transaction	Debit	Credit
(a) Materials purchased (cash or credit)	Material Control, then Stores Ledger Control (or Stores Ledger Control directly)	CLC (Material Control in the second step)
(b) Purchased for a special job	WIP Ledger Control	CLC
(c) Returned to vendor	CLC	Stores Ledger Control
(d) Direct material issued; indirect material issued	WIP Control; Production OH Control	Stores Ledger Control
(e) Returned from shop to stores; transferred Job 1 to Job 2	Stores Ledger Control; Job 2	WIP Control; Job 1
(f) Issued for repairs	Production OH Control	Stores Ledger Control
(g) Wages paid	Wages Control	CLC
(h)–(j) Indirect wages: production, administration, selling	Production, Administration, S&D OH	Wages Control
(k) Direct expenses for a job	Job (WIP Control)	CLC

Transaction	Debit	Credit
(l)–(m) Overheads incurred; carriage inward	Production, Administrative, S&D OH Control	CLC
(n) Production overhead recovered	WIP Ledger Control	Production OH Control
(o)–(p) Administration recovered from finished goods; S&D recovered from sales	Cost of Sales	Administrative OH; S&D OH Control
(q) Under-recovery of overheads	Costing P&L	the overhead control
(r) Over-recovery of overheads	the overhead control	Costing P&L
(s) Sales	CLC	Costing P&L
(t) Profit; (u) loss	Costing P&L; CLC	CLC; Costing P&L

Every entry that involves the outside world, purchases, payments, sales, the final profit, touches the CLC. Entries between two cost accounts never do.

Entry (o) as printed debits Cost of Sales with the administrative overhead recovered. When the administrative overhead relates to production, principal

account (6) sends the recovery to the Finished Goods Control A/c instead, and Illustration 2 does exactly that: ₹52,900 to Finished Goods, ₹14,800 to Cost of Sales (SM 7.4, 7.17). Follow the question's wording ("allocated to production", "absorbed in cost of sales").

The integrated system

> **Definition**
>
> **Integrated (integral) accounts**
>
> A system of accounting whereby cost and financial accounts are kept in the same set of books. There are no separate sets of books for costing and financial records. Integrated accounts meet fully the information requirement for costing as well as for financial accounts. (SM 7.19)

Advantages (SM 7.20):

1. **No need for reconciliation**: there is only one figure of profit.
2. **Less efforts**: one set of books means a significant saving in efforts.
3. **Less time consuming**: no delay in obtaining information, as it is provided from books of original entry.
4. **Economical process**: it is based on the concept of "centralisation of accounting function".

Essential pre-requisites (SM 7.20):

1. The management's decision about the extent of integration of the two sets of books: some integrate up to prime cost or factory cost, others fully.
2. A suitable coding system serving the purposes of both financial and cost accounts.
3. An agreed routine for the treatment of provision for accruals, prepaid expenses and other adjustments for interim accounts.
4. Perfect coordination between the staff responsible for the financial and cost aspects of the accounts, and efficient processing of accounting documents.

There is no separate cost ledger, but there are subsidiary ledgers: besides the customers' ledger and the purchase ledger, a stores ledger, a stock ledger and a job ledger (SM 7.20).

Features (SM 7.21): (a) complete analysis of cost and sales is kept; (b) complete details of all payments in cash are kept; (c) complete details of all assets and liabilities are kept, and the system does not use a notional account to represent all impersonal accounts. The Cost Ledger Control A/c is replaced by the **bank account, receivables (debtors), payables (creditors), provision for depreciation, fixed assets and share capital** accounts (SM 7.21).

Formats and formulas

Reproduce the two-sided control account and the cost-ledger trial balance from memory: ICAI asks for both, and for "what each balance represents".

Stores Ledger Control Account (format)

Dr.	₹
To Balance b/d	xxx
To Cost Ledger Control A/c (purchases)	xxx
To WIP Control A/c (returns from shop)	xxx

Cr.	₹
By WIP Control A/c (direct issues)	xxx
By Production OH Control A/c (indirect issues)	xxx
By Cost Ledger Control A/c (returns to vendor)	xxx
By Balance c/d	xxx

Trial balance of the cost ledger: debit balances are Stores Ledger Control, WIP Control and Finished Goods Control (plus any overhead control left open with an under-absorbed balance, and the Cost of Sales A/c if it has not been closed to Costing P&L, as in Illustration 1, SM

7.14); the credit balance is the Cost Ledger Control A/c (plus any over-absorbed balance). The Cost Ledger Control balance equals the net of all the others.

Missing figures in incomplete control accounts (SM 7.28–7.30, 7.51–7.53):

To find	Use
Materials purchased	Payments to creditors + closing creditors − opening creditors
Cost of goods completed	Cost of goods sold + closing finished goods − opening finished goods
Overhead applied	Hours worked × predetermined rate (budgeted overheads ÷ budgeted hours)
Closing WIP	Materials + labour hours × wage rate + labour hours × overhead rate, for the unfinished jobs
Materials consumed	Balancing figure of the WIP Control A/c
Closing stores	Opening stores + purchases − materials consumed

Worked examples

Worked example

The samosa factory's November: what the cost ledger leaves out

Tapri's samosa factory keeps a cost ledger not integrated with the financial accounts. Balances on 1 November: Stores Ledger Control ₹3,10,000 (Dr), WIP Control ₹95,000 (Dr), Finished Goods Control ₹1,40,000 (Dr) and Cost Ledger Control ₹5,45,000 (Cr). November's transactions, as Dev's team recorded them:

Particulars	₹
Materials purchased on credit	4,20,000
Materials issued: to production ₹3,60,000; to factory maintenance ₹15,000	3,75,000
Materials returned from the shop floor to stores	8,000
Potatoes in the cold store spoiled when its compressor failed (abnormal loss)	12,000
Wages paid: direct workers ₹1,90,000 (including ₹10,000 for abnormal idle time in a two-day power failure); indirect factory staff ₹50,000; sales staff ₹20,000	2,60,000
Hire of a special fryer for Job S-15	18,000

Particulars	₹
Overheads incurred: factory ₹1,02,000; administration (all related to production) ₹48,000; selling and distribution ₹30,000	1,80,000
Overheads absorbed: production ₹1,70,000 (in WIP); administration ₹50,000 (in finished goods); selling and distribution ₹46,000 (on sales)	2,66,000
Goods completed and transferred to finished stock (works cost)	6,90,000
Cost of goods sold	7,10,000
Sales	9,60,000
Interest received on fixed deposits ₹9,000; bad debts written off ₹7,000; income tax paid ₹40,000; a new packing machine bought ₹2,50,000	

Under- or over-absorbed overheads are transferred to the Costing P&L A/c.

Required:

(i) Pass the journal entries in the cost ledger, and state which transactions it does not record.

(ii) Prepare the Costing P&L A/c and the Cost Ledger Control A/c.

(iii) Prepare the trial balance as on 30 November.

THE WORKING

(i) Journal entries in the cost ledger

Particulars	Dr. (₹)	Cr. (₹)
Stores Ledger Control A/c Dr. / To Cost Ledger Control A/c (purchases)	4,20,000	4,20,000
WIP Control A/c Dr. / To Stores Ledger Control A/c (direct issues)	3,60,000	3,60,000
Production OH Control A/c Dr. / To Stores Ledger Control A/c (maintenance)	15,000	15,000
Stores Ledger Control A/c Dr. / To WIP Control A/c (returns from shop)	8,000	8,000
Costing P&L A/c Dr. / To Stores Ledger Control A/c (abnormal loss)	12,000	12,000
Wages Control A/c Dr. / To Cost Ledger Control A/c (wages paid)	2,60,000	2,60,000
WIP Control A/c Dr. / To Wages Control A/c (direct, productive)	1,80,000	1,80,000
Costing P&L A/c Dr. / To Wages Control A/c (abnormal idle time)	10,000	10,000

Particulars	Dr. (₹)	Cr. (₹)
Production OH Control A/c Dr. / To Wages Control A/c (indirect)	50,000	50,000
S&D OH Control A/c Dr. / To Wages Control A/c (sales staff)	20,000	20,000
WIP (Job S-15) Control A/c Dr. / To Cost Ledger Control A/c (fryer hire)	18,000	18,000
Production OH Control A/c Dr. / To Cost Ledger Control A/c	1,02,000	1,02,000
Administrative OH Control A/c Dr. / To Cost Ledger Control A/c	48,000	48,000
S&D OH Control A/c Dr. / To Cost Ledger Control A/c	30,000	30,000
WIP Control A/c Dr. / To Production OH Control A/c (absorbed)	1,70,000	1,70,000
Finished Goods Control A/c Dr. / To WIP Control A/c	6,90,000	6,90,000
Finished Goods Control A/c Dr. / To Administrative OH Control A/c	50,000	50,000

Particulars	Dr. (₹)	Cr. (₹)
Cost of Sales A/c Dr. / To Finished Goods Control A/c	7,10,000	7,10,000
Cost of Sales A/c Dr. / To S&D OH Control A/c	46,000	46,000
Cost Ledger Control A/c Dr. / To Costing P&L A/c (sales)	9,60,000	9,60,000
Costing P&L A/c Dr. / To Cost of Sales A/c	7,56,000	7,56,000
Production OH Control A/c Dr. / To Costing P&L A/c (over-absorbed)	3,000	3,000
Administrative OH Control A/c Dr. / To Costing P&L A/c (over-absorbed)	2,000	2,000
Costing P&L A/c Dr. / To S&D OH Control A/c (under-recovered)	4,000	4,000
Costing P&L A/c Dr. / To Cost Ledger Control A/c (profit)	1,83,000	1,83,000

Not recorded in the cost ledger: the interest received, the bad debts and the income tax (purely financial items, SM 7.2), and the packing machine (a fixed asset, a balance sheet item, SM 7.3). The

machine reaches the cost ledger later, as depreciation in production overheads.

Workings: production overhead incurred = 15,000 + 50,000 + 1,02,000 = ₹1,67,000, absorbed ₹1,70,000, so ₹3,000 over-absorbed; administration ₹48,000 against ₹50,000 absorbed, ₹2,000 over; selling and distribution 20,000 + 30,000 = ₹50,000 against ₹46,000 recovered, ₹4,000 under. Cost of sales = ₹7,10,000 + ₹46,000 = ₹7,56,000.

(ii) Accounts

Costing Profit & Loss A/c

Dr.	₹
To Cost of Sales A/c	7,56,000
To Stores Ledger Control A/c (abnormal loss)	12,000
To Wages Control A/c (abnormal idle time)	10,000
To S&D OH Control A/c (under-recovered)	4,000
To Cost Ledger Control A/c (profit)	1,83,000
Total	**9,65,000**

Cr.	₹
By Cost Ledger Control A/c (sales)	9,60,000
By Production OH Control A/c (over-absorbed)	3,000
By Administrative OH Control A/c (over-absorbed)	2,000
Total	**9,65,000**

Cost Ledger Control A/c

Dr.	₹
To Costing P&L A/c (sales)	9,60,000
To Balance c/d	6,46,000
Total	**16,06,000**

Cr.	₹
By Balance b/d	5,45,000
By Stores Ledger Control A/c	4,20,000
By Wages Control A/c	2,60,000
By WIP Control A/c (fryer hire)	18,000
By Production OH Control A/c	1,02,000
By Administrative OH Control A/c	48,000
By S&D OH Control A/c	30,000
By Costing P&L A/c (profit)	1,83,000
Total	**16,06,000**

(iii) Trial balance as on 30 November

Particulars	Dr. (₹)	Cr. (₹)
Stores Ledger Control A/c	3,51,000	
WIP Control A/c	1,25,000	
Finished Goods Control A/c	1,70,000	
Cost Ledger Control A/c		6,46,000
Total	**6,46,000**	**6,46,000**

Workings: stores = 3,10,000 + 4,20,000 + 8,000 − 3,60,000 − 15,000 − 12,000; WIP = 95,000 + 3,60,000 + 1,80,000 + 18,000 + 1,70,000 − 8,000 − 6,90,000; finished goods = 1,40,000 + 6,90,000 + 50,000 − 7,10,000. The interest received, the bad debts and the income tax are in Dev's books only, one reason the two profits will need reconciling.

Worked example

The kulhad workshop's first integrated month

Tapri's kulhad workshop moved to integrated accounts on 1 April 2026, but its cost clerk posted April in the old non-integrated cost ledger out of habit. Ritu has his

entries that touched the Cost Ledger Control A/c (CLC), with notes on how each was settled:

Entry posted by the clerk

Stores Ledger Control Dr. / To CLC (materials purchased)

₹

2,80,000

How it was settled

70% on credit, the rest by cheque

Entry posted by the clerk

CLC Dr. / To Stores Ledger Control (returned to suppliers)

₹

10,000

How it was settled

Credit suppliers

Entry posted by the clerk

Wages Control Dr. / To CLC (April wages)

₹

1,50,000

How it was settled

₹1,35,000 paid by bank; ₹15,000 of employees' PF deducted, payable in May

Entry posted by the clerk

Factory OH Control Dr. / To CLC

₹

90,000

How it was settled

Rent ₹30,000 paid; power ₹24,000 unpaid on 30 April; depreciation of kilns ₹36,000

Entry posted by the clerk

Administrative OH Control Dr. / To CLC

₹

40,000

How it was settled

Paid by bank

Entry posted by the clerk

S&D OH Control Dr. / To CLC

₹

35,000

How it was settled

₹20,000 paid; ₹15,000 is April's share of a three-month advertising contract paid in March

Entry posted by the clerk

CLC Dr. / To Costing P&L (sales)

₹

6,00,000

How it was settled

80% on credit, the rest received in the bank

In April the workshop also received ₹3,20,000 from debtors and paid ₹1,50,000 to creditors. The bank balance on 1 April was ₹2,00,000.

Required:

(i) Pass the journal entries that the integrated accounts record in place of each of the clerk's entries.

(ii) Pass the entries for the receipts from debtors and the payments to creditors, and say why the clerk's ledger had none.

(iii) Prepare the Bank A/c for April.

THE WORKING

(i) Integrated entries in place of the CLC

Particulars	Dr. (₹)	Cr. (₹)
Stores Ledger Control A/c Dr.	2,80,000	
To Payables (Creditors) A/c (70%)		1,96,000
To Bank A/c		84,000
Payables (Creditors) A/c Dr. / To Stores Ledger Control A/c (returns)	10,000	10,000
Wages Control A/c Dr.	1,50,000	
To Bank A/c		1,35,000
To Employees' PF Payable A/c		15,000
Factory OH Control A/c Dr.	90,000	
To Bank A/c (rent)		30,000
To Outstanding Expenses A/c (power)		24,000
To Provision for Depreciation A/c (kilns)		36,000

Particulars	Dr. (₹)	Cr. (₹)
Administrative OH Control A/c Dr. / To Bank A/c	40,000	40,000
S&D OH Control A/c Dr.	35,000	
To Bank A/c		20,000
To Prepaid Expenses A/c (advertising)		15,000
Receivables (Debtors) A/c Dr. (80%)	4,80,000	
Bank A/c Dr.	1,20,000	
To Sales A/c		6,00,000

Every credit to the CLC has become a credit to a real account: the bank, a creditor, a liability (PF payable, outstanding power), the provision for depreciation or an asset used up (the prepaid advertising). This is the agreed routine for accruals and prepaid expenses that integration needs (SM 7.20), and the replacement of the CLC by real accounts (SM 7.21). Entries between two cost accounts (issues, absorption, transfers to finished goods and cost of sales) are the same in both systems and are not repeated here.

(ii) Cash movements with no cost

Particulars	Dr. (₹)	Cr. (₹)
Bank A/c Dr. / To Receivables (Debtors) A/c	3,20,000	3,20,000
Payables (Creditors) A/c Dr. / To Bank A/c	1,50,000	1,50,000

They change neither cost nor revenue: they only settle balances that exist in the financial books, so a non-integrated cost ledger never records them (SM 7.2–7.3). Integrated accounts keep "complete details of all payments in cash" (SM 7.21), so they appear.

(iii) Bank A/c for April

Bank Account

Dr.	₹
To Balance b/d	2,00,000
To Sales A/c (cash sales)	1,20,000
To Receivables A/c	3,20,000
Total	**6,40,000**

Cr.	₹
By Stores Ledger Control A/c	84,000
By Wages Control A/c	1,35,000
By Factory OH Control A/c	30,000
By Administrative OH Control A/c	40,000
By S&D OH Control A/c	20,000
By Payables A/c	1,50,000
By Balance c/d	1,81,000
Total	**6,40,000**

Worked example

The crashed laptop: rebuild September from the profit

The cost clerk's laptop at Tapri's premix plant crashed on 1 October. The backup holds only these figures for September (non-integrated cost ledger):

Particulars	₹
Sales	6,00,000
Costing profit for September	72,000

Particulars	₹
Production overhead under-absorbed, transferred to Costing P&L (the only other item in it)	6,000
Production overheads incurred, all posted (including ₹9,000 of indirect materials from stores)	1,41,000
Finished Goods Control: 1 September; 30 September	80,000; 1,04,000
WIP Control, 1 September	40,000
Stores Ledger Control: 1 September; 30 September	70,000; 66,000
Creditors for materials, 1 September (from Dev's books)	55,000
Payments to creditors in September (from Dev's books)	2,10,000

Production overheads are absorbed at 75% of direct wages. On 30 September two batches were unfinished; direct materials of ₹14,000 and direct wages of ₹8,000 had been charged to them. All materials are bought on credit; stores issued only direct materials and the indirect materials above. The Cost of Sales A/c received nothing but finished goods sold.

Required: Find (a) the cost of goods sold; (b) the cost of goods completed; (c) the production overheads absorbed and the direct wages; (d) the closing WIP; (e)

the direct materials consumed; (f) the materials purchased; (g) the creditors for materials on 30 September.

THE WORKING

Work down from the Costing P&L, then through the finished goods, overhead, WIP and stores accounts, and only at the end into Dev's creditors.

Item	Working	₹
(a) Cost of goods sold	Sales 6,00,000 − profit 72,000 − under-absorbed 6,000	5,22,000
(b) Cost of goods completed	5,22,000 + closing FG 1,04,000 − opening FG 80,000	5,46,000
(c) Overheads absorbed	Incurred 1,41,000 − under-absorbed 6,000	1,35,000
(c) Direct wages	Absorbed ÷ 75% = 1,35,000 ÷ 0.75	1,80,000
(d) Closing WIP	Materials 14,000 + wages 8,000 + overheads 75% × 8,000	28,000
(e) Materials consumed	Balancing figure in WIP Control (below)	2,19,000

Item	Working	₹
(f) Materials purchased	Balancing figure in Stores Ledger Control (below)	2,24,000
(g) Creditors, 30 September	55,000 + purchases 2,24,000 − payments 2,10,000	69,000

WIP Control A/c: credits: finished goods ₹5,46,000 + balance c/d ₹28,000 = ₹5,74,000; debits: balance b/d ₹40,000 + wages ₹1,80,000 + overheads ₹1,35,000 + materials consumed (balancing figure) ₹2,19,000 = ₹5,74,000.

Stores Ledger Control A/c: credits: WIP ₹2,19,000 + production overheads ₹9,000 + balance c/d ₹66,000 = ₹2,94,000; debits: balance b/d ₹70,000 + purchases (balancing figure) ₹2,24,000 = ₹2,94,000.

The under-absorbed ₹6,000 does double duty: it explains the gap between sales less profit and cost of goods sold, and it turns overheads incurred into overheads absorbed, from which the wages follow. Taking the ₹1,41,000 incurred as absorbed would give wages of ₹1,88,000 and every later figure wrong.

Exam angle

In the exam

How ICAI asked this, May 2024 to September 2026

In the documents read for this lesson, cost accounting systems (other than the reconciliation) appeared in 6 questions:

- **Journal entries, non-integrated**: September 2025 Q4(b), 5 marks: materials purchased on credit, materials issued, direct wages allocated, factory overheads over-absorbed, administration overheads under-absorbed. The answer showed both forms of the purchase entry (via Material Control, or straight to Stores Ledger Control).
- **Ledgers**: MTP May 2024 Series I Q3(a), 4 marks: "What are the important ledgers to be maintained under non-integrated accounting system?" (the four ledgers of SM 7.3).
- **Pre-requisites**: MTP September 2024 Series I Q6(c), 4 marks: "What are the essential pre-requisites for integrated accounts?"
- **Reconciliation under integration**: RTP May 2024 Q13(a): "Is reconciliation of cost accounts and financial accounts necessary in case of integrated accounting system?" ICAI's answer: one set of books, so no.
- **Integrated control accounts**: RTP May 2025 Q12: Amy Ltd's batch costing fully integrated with the financial accounts; prepare six accounts from Stores

Ledger Control to Costing P&L, with non-productive time and a physical WIP value.

- **MCQ**: RTP September 2025 MCQ 4: the journal entry for overheads incurred (Production and Administrative OH Control Dr. / To Cost Ledger Control A/c).

In the exam

The traps

- Under a non-integrated system nothing is credited to Bank or Creditors: the credit goes to the Cost Ledger Control A/c (SM MCQ 5: purchases are debited to Material or Stores Ledger Control).
- Materials bought for a special job go straight to WIP (job) A/c, and returns to vendor are a debit to the Cost Ledger Control A/c (SM MCQs 6 and 7).
- Notional costs may be included in non-integrated accounts (SM MCQ 2); the cost ledger can carry them because it is separate.
- Over-absorbed: debit the overhead control, credit Costing P&L; under-absorbed: the reverse (September 2025 Q4(b)).

Trick

Does it cross the border?

For each transaction ask: does it involve anything kept only in the financial books (cash, bank, debtors, creditors, sales, the final profit)? If yes, one side of the entry is the Cost Ledger Control A/c in a non-integrated system, or that real account in an integrated one. If no, both sides are cost accounts and the entry is identical in both systems. Condition: the cost ledger in question records the item at all (purely financial items such as interest are not recorded in the cost ledger).

Answer frame

"Distinguish between integrated and non-integrated accounting systems"

Point

Books

Non-integrated system

Separate ledgers for cost and financial accounts (SM 7.2)

Integrated system

Cost and financial accounts in the same set of books (SM 7.19)

Point

Linking account

Non-integrated system

Cost Ledger Control A/c (General Ledger Adjustment A/c) completes the double entry (SM 7.3)

Integrated system

No notional account; bank, debtors, creditors, provision for depreciation, fixed assets and share capital accounts used instead (SM 7.21)

Point

Coverage

Non-integrated system

Only transactions relating to the product or service; excludes purely financial items and balance sheet items (SM 7.2–7.3)

Integrated system

Complete analysis of cost and sales; complete details of cash payments, assets and liabilities (SM 7.21)

Point

Profit

Non-integrated system

Two profits; reconciliation necessary (SM 7.2)

Integrated system

One profit; no reconciliation (SM 7.2, 7.20)

Point

Effort

Non-integrated system

Two sets of books to maintain

Integrated system

Less effort, less time, economical: centralisation of accounting function (SM 7.20)

Other questions, in ICAI's order

- **Ledgers under non-integrated accounts** (4 marks): cost ledger, stores ledger, work-in-process ledger, finished goods ledger, each with ICAI's one-line description.
- **Advantages of integrated accounts**: no need for reconciliation; less efforts; less time consuming; economical process.
- **Essential pre-requisites**: extent of integration; suitable coding system; agreed routine for accruals, prepaid expenses and adjustments; perfect coordination between cost and financial staff.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the bank account in the cost ledger

"Wages Control A/c Dr. / To Bank A/c" in a non-integrated answer. **Fix:** the cost ledger has no bank account; the credit is the Cost Ledger Control A/c. Bank appears only when the books are integrated.

⚠ **Common mistake**

Receipts from debtors in the cost books

Recording receipts from debtors or payments to creditors in a non-integrated cost ledger. **Fix:** these touch only financial accounts and are not recorded at all; in integrated accounts they are Bank Dr. / To Receivables, and Payables Dr. / To Bank.

⚠ **Common mistake**

Under- and over-absorption the wrong way round

Crediting the overhead control when overhead was over-absorbed. **Fix:** over-absorbed means the control has a credit balance (absorbed > incurred); close it by debiting the control and crediting Costing P&L. Under-absorbed: Costing P&L Dr. / To the control.

⚠ **Common mistake**

Non-productive time charged to WIP

Fix: non-productive (idle) time of direct workers is a production overhead: debit Production OH Control, not WIP (SM 7.27); abnormal idle time goes to Costing P&L (SM 7.4).

Practice

Check yourself

1 A cost ledger (non-integrated) has these closing balances: Stores Ledger Control ₹1,85,000 (Dr), WIP Control ₹62,000 (Dr), Finished Goods Control ₹1,48,000 (Dr) and Production Overhead Control ₹9,000 (Cr, over-absorbed, not yet transferred). What is the balance of the Cost Ledger Control A/c (₹)?

Answer: _____

2 Case scenario. Ritu's cost ledger is not integrated with Dev's financial books. Chutney sachets worth ₹7,000, issued to Job S-21, are not needed there and are moved straight to Job S-22. How is the transfer recorded?

- **A** Stores Ledger Control A/c Dr. / To WIP Control A/c ₹7,000, then WIP Control A/c Dr. / To Stores Ledger Control A/c ₹7,000
- **B** Cost Ledger Control A/c Dr. / To WIP Control A/c ₹7,000
- **C** WIP Control A/c Dr. / To Cost Ledger Control A/c ₹7,000
- **D** Job S-22 A/c Dr. / To Job S-21 A/c ₹7,000 in the job ledger; no entry in the control accounts

3 Which of the following replace the Cost Ledger Control A/c in an integrated system? Select all that apply.

- **A** Bank account
- **B** Payables (creditors) account
- **C** Provision for depreciation account
- **D** Overhead Adjustment account

More than one answer may be right.

4 Production overheads incurred are ₹2,38,000 and absorbed ₹2,60,000; the balance is transferred to Costing P&L. Which entry is correct?

- **A** Costing P&L A/c Dr. / To Production OH Control A/c ₹22,000
- **B** Production OH Control A/c Dr. / To Costing P&L A/c ₹22,000
- **C** WIP Control A/c Dr. / To Production OH Control A/c ₹22,000
- **D** Production OH Control A/c Dr. / To Cost Ledger Control A/c ₹22,000

5 Which is NOT an essential pre-requisite of integrated accounts in the Study Material?

- **A** A decision about the extent of integration
- **B** A suitable coding system
- **C** A Cost Ledger Control A/c to make the cost ledger self-balancing
- **D** Perfect coordination between cost and financial staff

Answers: the answer key at the end of this lesson.

Recap

- Non-integrated: separate cost ledger; records only product-related transactions; excludes purely financial items and balance sheet items.
- Cost Ledger Control A/c (General Ledger Adjustment A/c): credited with all expenditure, debited with sales, receives the costing profit; its balance is the net of all impersonal account balances.
- Four ledgers: cost ledger, stores ledger, work-in-process (job) ledger, finished goods ledger.
- Over-absorbed: overhead control Dr. / To Costing P&L; under-absorbed: the reverse; Overhead Adjustment A/c may be dispensed with.
- Integrated: one set of books; bank, debtors, creditors, provision for depreciation, fixed assets, share capital

replace the CLC.

- Advantages: no reconciliation, less effort, less time, economical. Pre-requisites: extent of integration, coding system, agreed routine for accruals, coordination of staff.
- Incomplete records: purchases from creditors; goods completed from finished goods; materials consumed as the WIP balancing figure. (No bank account in the embassy.)

Answer key · Check yourself

1 **386000**

The CLC balances the ledger: Dr balances 1,85,000 + 62,000 + 1,48,000 = 3,95,000, less the 9,000 credit balance = ₹3,86,000, a credit balance.

2 **D** **Job S-22 A/c Dr. / To Job S-21 A/c ₹7,000 in the job ledger; no entry in the control accounts**

Both jobs are inside the WIP Control A/c, so its total does not change: "materials transferred between batches will not affect the Control Accounts" (SM 7.27), and the entry is Job 2 Dr. / To Job 1 (SM 7.6). Routing it through stores invents a return that never happened, and the Cost Ledger Control A/c is touched only by transactions with the outside world.

3 **A Bank account; B Payables (creditors) account; C Provision for depreciation account**

SM 7.21 lists bank, receivables, payables, provision for depreciation, fixed assets and share capital. The Overhead Adjustment A/c is a cost account found in both systems.

4 **B Production OH Control A/c Dr. / To Costing P&L A/c ₹22,000**

Absorbed exceeds incurred by ₹22,000: over-absorbed, a credit balance in the control. Close it by debiting the control and crediting Costing P&L.

5 **C A Cost Ledger Control A/c to make the cost ledger self-balancing**

The Cost Ledger Control A/c belongs to the non-integrated system; integrated accounts have no separate cost ledger and use the real accounts (bank, debtors, creditors) instead. The other three are among ICAI's four pre-requisites (SM 7.20).

7.2 Reconciling cost and financial profit

Why the cost books and the financial books show different profits, the four causes ICAI lists, and how to reconcile them item by item in a reconciliation statement or a Memorandum Reconciliation Account.

In plain words

Two profits for one year

At the year-end review Kesar holds two sheets of paper: Ritu's Costing Profit and Loss Account and Dev's financial statements. Both are for the premix division, and they show two different profits.

> **Kesar:** Same division, same premix, same year. Put the difference in miscellaneous expenses and let us go for lunch.
>
> **Ritu:** Miscellaneous is where differences go to hide, Kesar-ji. Give me ten minutes and I will show you every rupee of it.
>
> **Dev:** And the present value of your lunch falls with every minute we argue.

The two profits differ because, under a non-integrated system, the two sets of books record some things differently:

1. **Items only in the financial books**: interest on the term loan, income tax, donations, dividends received,

a profit or loss on selling an old delivery van. They are not costs or revenues of making premix, so the cost books ignore them.

2. **Items only in the cost books**: **notional** charges that nobody pays, like rent for Tapri's own building or interest on the owners' capital, added so that product costs are comparable.
3. **Items treated differently**: depreciation by a different method, or stores issued at LIFO in the cost books.
4. **Different stock values**: the financial books value stock at cost or market price, whichever is lower; the cost books at cost.

On top of these, the cost books charge overheads at predetermined rates. If they absorbed less than was actually spent (**under-absorption**), the costing profit is higher than the real one; if more (**over-absorption**), it is lower.

A **reconciliation** starts from one profit and adds or subtracts each difference until it reaches the other. Nothing is changed in either set of books: the statement only explains the gap.

Think of it like

The bank passbook

You did this in Class 11. Your cash book says ₹12,000 in the bank; the passbook says ₹10,500. Neither is wrong: a cheque has not been cleared, the bank charged a fee you have not recorded. You list each item and add or subtract it until the two agree. ICAI says the reconciliation of cost and financial profit "is similar to those which are followed for reconciling bank balance as per bank ledger with the balance as shown in bank statement" (SM 7.31).

Key idea

Each difference changes one profit and not the other. For every item ask: did it make the financial profit higher or lower than the costing profit? Starting from the costing profit, add what made the financial profit higher and subtract what made it lower.

How it works

Why reconcile, and how

"When the cost and financial accounts are kept separately, it is imperative that these should be reconciled to make the cost accounts reliable" (SM 7.31). For this, the financial accounts should analyse expenses the same way as the cost accounts, so that the differences can be located. The differences are collected in a **Memorandum Reconciliation Account** (or a

reconciliation statement): items charged in one set of accounts but not in the other, or charged in excess, are added to or subtracted from the profit of one set until it equals the other (SM 7.31).

Procedure for reconciliation (SM 7.33), in three steps:

1. Ascertainment of profit as per financial accounts.
2. Ascertainment of profit as per cost accounts.
3. Reconciliation of both the profits.

When it can be avoided (SM 7.33): when the cost and financial accounts are integrated, there is no need for a separate reconciliation statement; integration means that the same set of accounts fulfils the requirement of both.

The causes of difference, in ICAI's order

Causes of differences in financial and cost accounts (SM 7.31–7.32)

Why the two profits differ

1. Financial accounts only

(a) Purely financial expenses

interest on loans; share and debenture issue costs
capital losses; loss on sale of assets
income tax, donations, subscriptions
share transfer office

(b) Purely financial income

interest and dividends received
profit on sale of assets
transfer fees; rent receivable

2. Cost accounts only (notional)

rent of own premises; interest on capital
proprietor's salary; depreciation on
fully depreciated assets

3. Treated differently

LIFO in cost accounts; other depreciation methods

4. Varying basis of valuation

financial: cost or market, lower; cost: at cost

The four causes in ICAI's order. The illustrations add one more source of difference: under- and over-absorbed overheads, because the cost books charge overheads at predetermined rates and the financial books at actual.

In ICAI's words (SM 7.31–7.32):

1. **Items included in financial accounts only.** (a) *Purely financial expenses*: (i) interest on loans or bank mortgages; (ii) expenses and discounts on issue of shares, debentures etc.; (iii) other capital losses, i.e. loss by fire not covered by insurance etc.; (iv) losses on the sales of fixed assets and investments; (v) income tax, donations, subscriptions; (vi) expenses of the company's share transfer office, if any. (b) *Purely financial income*: (i) interest received on bank deposits, loans and investments; (ii) dividends received; (iii) profits on the sale of fixed assets and investments; (iv) transfer fee received; (v) rent receivables.
2. **Items included in cost accounts only (notional expenses)**: (i) charges in lieu of rent where premises are owned; (ii) interest on capital at notional figure though not incurred; (iii) salary for the proprietor at notional figure though not incurred; (iv) notional depreciation on the assets fully depreciated for which book value is nil.
3. **Items whose treatment is different in the two sets of accounts**: cost accounting serves management's decisions and control, financial accounting conforms to external reporting requirements. LIFO is not recommended by Accounting Standards for financial reporting but may be adopted in cost accounts (pricing material issues); cost accounts may use a different method of depreciation.

4. **Varying basis of valuation**: in financial accounts stock is valued at cost or market price, whichever is lower; in cost accounts only at cost.

Which way each item moves

Starting from the **profit as per cost accounts**, going to the profit as per financial accounts:

Item	Financial profit is	From cost profit
Expense or loss in financial accounts only (interest paid, income tax, donations, write-offs, loss on sale of assets)	Lower	Less
Income in financial accounts only (interest and dividends received, profit on sale of assets, transfer fees)	Higher	Add
Notional cost charged in cost accounts only (rent of own premises, interest on capital)	Higher	Add
Overheads under-absorbed in cost accounts (actual exceeded the charge)	Lower	Less
Overheads over-absorbed in cost accounts	Higher	Add
Depreciation (or any expense) over-charged in cost accounts	Higher	Add

Item	Financial profit is	From cost profit
Closing stock over-valued in cost accounts	Lower	Less
Closing stock under-valued in cost accounts	Higher	Add
Opening stock over-valued in cost accounts	Higher	Add
Opening stock under-valued in cost accounts	Lower	Less
Excess material consumption shown in cost accounts	Higher	Add

Starting from the **profit as per financial accounts**, every sign reverses. A **loss** is a negative profit: "add" reduces it, "less" increases it (May 2026 Q1(c): net loss as per cost accounts ₹4,20,000; add ₹2,10,000; less ₹2,50,000; net loss as per financial accounts ₹4,60,000).

This is not the marginal-versus-absorption reconciliation of marginal and absorption costing, which explains a difference in stock valuation between two costing methods: here two sets of books are compared.

Interactive version: PassBook app, lesson Reconciling cost and financial profit

Formats and formulas

Reproduce both layouts from memory. Use the one the question names; when it only says "reconcile", ICAI's answers use the statement (SM Illustrations 6 to 8).

Reconciliation statement (SM 7.35–7.36)

Particulars	₹	₹
Profit as per cost accounts		xxx
Add: Overheads over-absorbed in cost accounts	xxx	
Add: Income in financial accounts only	xxx	
Add: Notional expenses charged in cost accounts	xxx	xxx
		xxx
Less: Overheads under-absorbed in cost accounts	xxx	
Less: Expenses in financial accounts only	xxx	
Less: Over-valuation of closing stock in cost accounts	xxx	(xxx)
Profit as per financial accounts		**xxx**

Memorandum Reconciliation Account, starting from a costing profit: the costing profit and the additions are credited, the deductions debited, and the financial profit

is the balancing figure on the debit side. Starting from a costing **loss**, the loss is debited and the financial result is the balancing figure on the credit side (SM 7.61):

Memorandum Reconciliation Account (costing profit as base)

Dr.	₹
To Deductions (under-absorption, financial-only expenses, over-valued closing stock)	xxx
To Net profit as per financial accounts (balancing figure)	xxx
Total	**xxx**

Cr.	₹
By Net profit as per cost accounts	xxx
By Additions (over-absorption, financial-only income, notional costs)	xxx
Total	**xxx**

Interactive version: PassBook app, lesson Reconciling cost and financial profit

∑ **Formula**

The identity behind every reconciliation

$$\text{Financial profit} = \text{Cost profit} + \text{Additions} - \text{Deductions}$$

Additions are the items that make the financial profit higher than the cost profit; deductions, those that make it lower. A balancing item (one missing difference) is found by solving this identity for it (May 2025 Q4(a)).

Worked examples

Worked example **The premix division's two books, side by side**

Dev's financial accounts show a net profit of ₹12,00,000 for Tapri's premix division for 2025-26. Ritu's cost accounts are not yet closed, but she has the figures each set of books used for the items on which they differ:

Item	Cost accounts (₹)	Financial accounts (₹)
Raw materials consumed (cost books issue at LIFO, financial books at FIFO)	48,60,000	47,90,000

Item	Cost accounts (₹)	Financial accounts (₹)
Works overheads (absorbed in cost; actual in financial)	18,40,000	19,15,000
Administration overheads (absorbed in cost; actual in financial)	6,20,000	6,02,000
Depreciation of plant	4,20,000	3,80,000
Rent of the division's own building	2,40,000	—
Interest on term loan	—	1,10,000
Interest received on fixed deposits	—	36,000
Opening finished stock	5,40,000	5,10,000
Closing finished stock (financial: lower of cost and net realisable value)	7,25,000	6,90,000

Everything else is the same in both books.

Required:

(i) Prepare a statement reconciling the two profits, starting from the profit as per financial accounts, and find the profit as per cost accounts.

(ii) Explain why the materials item appears at all, when both books record the same purchases.

THE WORKING

(i) Statement of reconciliation of profit as per financial accounts with profit as per cost accounts

For each item: did it make the *cost* profit higher or lower than the financial profit? Higher: add. Lower: deduct.

Particulars	₹	₹
Net profit as per financial accounts		**12,00,000**
Add: Works overheads under-absorbed in cost accounts (19,15,000 – 18,40,000)	75,000	
Interest on term loan (financial accounts only)	1,10,000	
Closing stock valued higher in cost accounts (7,25,000 – 6,90,000)	35,000	2,20,000
		14,20,000
Less: Materials consumed higher in cost accounts (48,60,000 – 47,90,000)	70,000	

Particulars	₹	₹
Administration overheads over-absorbed in cost accounts (6,20,000 – 6,02,000)	18,000	
Depreciation charged higher in cost accounts (4,20,000 – 3,80,000)	40,000	
Notional rent of own building (cost accounts only)	2,40,000	
Interest received on fixed deposits (financial accounts only)	36,000	
Opening stock valued higher in cost accounts (5,40,000 – 5,10,000)	30,000	(4,34,000)
Net profit as per cost accounts		**9,86,000**

(ii) The materials item is an item "treated differently" (SM 7.32): both books bought the same materials, but the cost books priced issues at LIFO, which management may adopt for cost accounting though the Accounting Standards do not accept it for financial reporting (SM 7.32). In a period of rising prices LIFO charges the latest, dearer purchases to production, so the cost books show

₹70,000 more consumed and ₹70,000 less in closing raw-material stock; the costing profit is lower by that amount.

Check by the stock items: a higher closing stock raises a profit, so the cost books' extra ₹35,000 makes the costing profit higher (add); a higher opening stock lowers a profit, so their extra ₹30,000 makes it lower (deduct).

Worked example

Mehra Kulhad Works: both profits from one year's records

Kesar's cousin Rohit Mehra runs Mehra Kulhad Works, a proprietary firm that supplies Tapri with kulhads. Its

financial records for 2025-26:

Particulars	₹
Sales (4,00,000 kulhads at ₹4.50)	18,00,000
Clay and glaze consumed	5,60,000
Direct wages	3,20,000
Factory overheads	3,90,000
Administration overheads (all related to production)	1,10,000
Selling and distribution overheads	1,30,000
Bank charges and interest on overdraft	24,000
Rent received for sub-letting part of the godown	36,000
Closing work-in-process	40,000

There was no opening stock. 4,40,000 kulhads were made; the 40,000 unsold include 8,000 of a discontinued design that can be sold only at ₹1.50 each, and the financial books value closing finished stock at the lower of cost and net realisable value. The cost records absorb factory overheads at ₹90 a kiln hour (4,000 kiln hours were worked), administration overheads at 10% of works cost and selling and distribution overheads at ₹0.30 a kulhad sold; they also charge a notional salary of ₹1,80,000 for Rohit's own

work as general administration. Work-in-process is valued at the same amount in both books.

Required:

(i) Prepare a cost sheet and find the profit as per the cost records.

(ii) Prepare the profit and loss account as per the financial records.

(iii) Reconcile the two profits.

THE WORKING

(i) Cost sheet (output 4,40,000 kulhads; sold 4,00,000)

Particulars	₹
Clay and glaze consumed	5,60,000
Direct wages	3,20,000
Prime cost	**8,80,000**
Add: Factory overheads (4,000 kiln hours × ₹90)	3,60,000
Gross works cost	**12,40,000**
Less: Closing work-in-process	(40,000)
Works cost	**12,00,000**

Particulars	₹
Add: Administration overheads (10% of works cost)	1,20,000
Cost of production (₹3 a kulhad)	**13,20,000**
Less: Closing finished stock (40,000 × ₹3)	(1,20,000)
Cost of goods sold	**12,00,000**
Add: Selling and distribution (4,00,000 × ₹0.30)	1,20,000
Add: Notional salary of the proprietor	1,80,000
Cost of sales	**15,00,000**
Profit	**3,00,000**
Sales	**18,00,000**

(ii) Profit and loss account

Closing finished stock in the financial books = 32,000 × ₹3 + 8,000 × ₹1.50 = ₹96,000 + ₹12,000 = ₹1,08,000.

Profit & Loss Account (financial records)

Dr.	₹
To Clay and glaze consumed	5,60,000
To Direct wages	3,20,000
To Factory overheads	3,90,000
To Administration overheads	1,10,000
To Selling & distribution overheads	1,30,000
To Bank charges and interest	24,000
To Net profit	4,50,000
Total	**19,84,000**

Cr.	₹
By Sales	18,00,000
By Closing stock: finished goods	1,08,000
By Closing stock: work-in-process	40,000
By Rent received	36,000
Total	**19,84,000**

(iii) Statement of reconciliation

Particulars	₹	₹
Profit as per cost records		**3,00,000**
Add: Administration overheads over-absorbed (1,20,000 − 1,10,000)	10,000	
Notional salary of the proprietor (cost records only)	1,80,000	
Rent received (financial records only)	36,000	2,26,000
		5,26,000
Less: Factory overheads under-absorbed (3,90,000 − 3,60,000)	30,000	
Selling and distribution under-absorbed (1,30,000 − 1,20,000)	10,000	
Bank charges and interest (financial records only)	24,000	
Closing stock valued higher in cost records (1,20,000 − 1,08,000)	12,000	(76,000)
Profit as per financial records		**4,50,000**

A proprietor's notional salary is one of ICAI's items "included in cost accounts only" (SM 7.32): Rohit's financial profit includes the reward for his own work, while his cost sheet charges it so that the kulhad's cost is comparable with a company that pays a manager.

Worked example

Tapri Cold Chain: a costing profit that became a loss

Tapri Cold Chain Pvt Ltd, Tapri's cold-chain subsidiary, shows a net profit of ₹1,20,000 in its cost accounts for

the year. A scrutiny of both sets of books reveals:

Particulars	₹
Depreciation of reefer trucks: the cost accounts recover ₹6 per km for the 1,50,000 km run; the financial accounts charge 20% on the trucks' written-down value of ₹50,00,000	
Operating overheads over-recovered in cost accounts	18,000
Penalty paid to a hotel chain for late deliveries	45,000
Interest on vehicle loans	96,000
Profit on sale of an old reefer truck	30,000
Notional interest on capital charged in cost accounts	60,000
Closing stock of fuel and spares: cost accounts 62,000; financial accounts (lower of cost and net realisable value) 58,000	

Required:

(i) Calculate the depreciation difference.

(ii) Prepare a Memorandum Reconciliation Account and find the result as per the financial accounts.

THE WORKING

(i) Depreciation. Cost accounts: 1,50,000 km × ₹6 = ₹9,00,000. Financial accounts: 20% × ₹50,00,000 =

₹10,00,000. Depreciation is under-recovered in the cost accounts by ₹1,00,000: a different method of depreciation in the two books (SM 7.32).

(ii) Memorandum Reconciliation Account

Memorandum Reconciliation Account

Dr.	₹
To Depreciation under-recovered in cost accounts	1,00,000
To Penalty for late deliveries (financial accounts only)	45,000
To Interest on vehicle loans (financial accounts only)	96,000
To Closing stock over-valued in cost accounts (62,000 − 58,000)	4,000
Total	**2,45,000**

Cr.	₹
By Net profit as per cost accounts	1,20,000
By Operating overheads over-recovered in cost accounts	18,000
By Profit on sale of reefer truck (financial accounts only)	30,000
By Notional interest on capital (cost accounts only)	60,000
By Net loss as per financial accounts (balancing figure)	17,000
Total	**2,45,000**

The financial accounts show a **net loss of ₹17,000**. The deductions (₹2,45,000) exceed the costing profit and the additions (₹2,28,000), so the balancing figure falls on the credit side: a loss, not a profit. Check: ₹1,20,000 + ₹1,08,000 − ₹2,45,000 = −₹17,000.

A penalty is not a cost (SM 6.9), and interest on loans is a purely financial charge (SM 7.31), so the cost accounts rightly leave both out; together with the depreciation that the per-km rate under-recovered, they turn the costing profit into a financial loss.

Exam angle

In the exam

How ICAI asked this, May 2024 to September 2026

In the documents read for this lesson, reconciliation was asked in 13 questions, including 7 of the 8 past papers from May 2024 to September 2026 (September 2025 asked the ledgers instead):

- **Statement from a list of differences**: May 2024 Q3(b), 7 marks (from cost profit, plus "when is reconciliation not required"); September 2024 Q1(b), 5 marks (from financial profit to cost profit); May 2025 Q4(a), 8 marks (from a cost loss, with a balancing item, "excess of material consumption in cost accounts", and closing stock at ₹41,000 in cost against market ₹80 × 500 units); May 2026 Q1(c), 4 marks (net loss as base, then the financial loss); September 2026 Q3(a): from cost profit, "maintaining the serial order given above" (answers not yet published).
- **Both profits from raw data**: January 2026 Q4(b), 7 marks: cost sheet with fixed overheads absorbed on normal capacity, then the statement; MTP May 2026 Series II Q3(b), 8 marks: both profits with opening finished stock at ₹125 a unit in cost accounts and an abnormal inventory loss excluded from costs, then the statement.
- **Memorandum Reconciliation Account**: MTP January 2026 Series I Q3(b), 5 marks (loss base); MTP January 2026 Series II Q1(a), 5 marks (loss

base, opening and closing stock both valued differently); MTP May 2026 Series I Q1(b), 5 marks (the same data again).

- **Theory**: January 2025 Q6(a), 5 marks (3 + 2): the three steps and when reconciliation can be avoided; MTP May 2024 Series II Q4(c), 2 marks: when is it not required; old-scheme November 2022 Q6(b): classify items as cost accounts or financial accounts.

In the exam

The traps

- "Notional rent of own premises charged in cost accounts" is **added** to the costing profit to reach the financial profit (May 2024); starting from the financial profit it is deducted (September 2024).
- Stock "valued at ₹41,000 in cost accounts whereas market price ₹80 per unit" for 500 units (May 2025): the financial books take ₹40,000, so cost over-values by ₹1,000: deduct.
- Read the base: "taking net loss as per cost accounts as base" (May 2025), "to arrive at net profit/loss as per cost accounts" (September 2024: start from financial profit, reverse every sign).
- ICAI's MCQ: costing loss ₹5,600 with administrative overhead under-absorbed ₹600 gives a financial loss of ₹6,200 (SM 7.45).

Trick

The "who got poorer" test

For every item ask: which book shows the lower profit because of it? If it is the financial book (an expense only Dev records, an overhead Ritu under-charged, closing stock Ritu valued higher), the item is deducted from the cost profit. If it is the cost book (a notional charge, an over-absorption, closing stock Ritu valued lower, an income only Dev records), it is added. Conditions: the base is the profit as per cost accounts (from the financial profit, reverse it), and opening stock works the other way round: opening stock Ritu valued higher makes the cost book poorer.

Answer frame

Theory questions, in ICAI's order

- **Why is reconciliation necessary?** When cost and financial accounts are kept separately, they must be reconciled to make the cost accounts reliable; the financial accounts should analyse expenses as the cost accounts do, so that differences and their reasons can be located; the differences are collected in a Memorandum Reconciliation Account (SM 7.31).
- **Reasons for disagreement** (or "list the financial expenses not included in cost"): the four causes, with ICAI's sub-lists (SM 7.31–7.32).
- **Procedure and when it can be avoided** (January 2025, 3 + 2 marks): the three steps; then integrated accounts,

where the same set of accounts fulfils both requirements (SM 7.33).

A numerical answer

1. Heading: "Statement of Reconciliation of profit as per cost accounts and financial accounts" (or the account, if asked).
2. Start from the base the question names, with its sign.
3. "Add:" items with a working in brackets for every computed difference (₹19,15,000 − ₹18,40,000); subtotal.
4. "Less:" items, the same way; total in brackets.
5. End on the other profit, labelled, and check it against the figure given.

Loses marks

A list of items with plus and minus signs and a final figure, no labels, no workings.

The examiner cannot see why the stock item was deducted.

Passes

The statement with Add and Less, but "Depreciation 1,60,000" and "1,45,000" both entered as separate items.

Half the depreciation logic is wrong.

Full marks

Base labelled; each item named with "(cost accounts only)", "under-absorbed", "over-valued in cost accounts"; computed differences shown; subtotals; final profit labelled and matching.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the notional rent that was deducted

"Notional rent is an expense, so less." **Fix:** it was charged only in the cost books, so the cost profit is the lower one; going to the financial profit, add it back.

⚠ **Common mistake**

Entering both depreciation figures

Deducting ₹4,20,000 and adding ₹3,80,000 as two items for the premix division. **Fix:** only the difference matters: the cost accounts charged ₹40,000 more, so the costing profit is ₹40,000 lower; from the financial profit, deduct ₹40,000 (from the costing profit, add it).

⚠ **Common mistake**

Closing stock and opening stock with the same sign

Fix: closing stock over-valued in cost accounts: deduct (it inflated the cost profit); opening stock over-valued in cost accounts: add (it inflated the cost of goods sold). The MTP January 2026 Series II answer debits "under valuation of opening stock in cost accounts", a deduction.

⚠ **Common mistake**

Losing the minus sign of a loss

Writing Tapri Cold Chain's balancing figure of ₹17,000 on the debit side as a "profit as per financial accounts". **Fix:** a loss is a negative profit. With the costing profit as base, the balancing figure goes on whichever side is short: on the debit side it is a profit, on the credit side (deductions larger than the base and the additions) it is a loss. A costing loss as base sits on the debit side.

Practice

Check yourself

1 Profit as per cost accounts is ₹3,80,000. Factory overheads were over-absorbed by ₹25,000, interest on debentures paid was ₹40,000, and notional interest on capital of ₹30,000 was charged in the cost accounts. What is the profit as per financial accounts (₹)?

Answer: _____

2 Case scenario. Dev's financial accounts show a profit of ₹9,10,000. Ritu's cost accounts valued closing stock at ₹2,80,000; the financial accounts valued it at ₹2,65,000, being market price. Starting from the financial profit, how is this item treated to reach the profit as per cost accounts?

- **A** Deduct ₹15,000, because closing stock is over-valued in the cost accounts
- **B** Add ₹15,000, because the cost accounts' higher closing stock gives them a higher profit
- **C** Add ₹2,80,000 and deduct ₹2,65,000 as two items
- **D** Ignore it, because stock valuation does not affect profit

3 Which of the following are included in the financial accounts only? Select all that apply.

- **A** Interest received on bank deposits
- **B** Notional depreciation on a fully depreciated machine
- **C** Expenses on issue of debentures
- **D** Loss by fire not covered by insurance

More than one answer may be right.

4 Costing profit is ₹2,10,000; administrative overheads were under-absorbed by ₹14,000 and selling overheads over-absorbed by ₹6,000. Nothing else differs. What is the financial profit?

- **A** ₹2,30,000
- **B** ₹2,02,000
- **C** ₹2,18,000
- **D** ₹1,90,000

5 In which situation is a reconciliation of cost and financial profit not needed?

A When the cost accounts use LIFO

B When there are no notional costs

C When the cost and financial accounts are integrated

D When overheads are absorbed at actual rates

Answers: the answer key at the end of this lesson.

Recap

- Reconciliation is needed when cost and financial accounts are kept separately (non-integrated); not when they are integrated.
- Procedure: profit as per financial accounts; profit as per cost accounts; reconcile the two.
- Causes, in ICAI's order: financial-only items (expenses and income); cost-only notional items; items treated differently; varying basis of stock valuation (financial: lower of cost or market; cost: at cost).
- From cost profit: add over-absorption, notional costs, financial-only income, cost over-charges, closing stock under-valued in cost; deduct the opposites.
- Opening stock works the other way round from closing stock.

- From financial profit, reverse every sign; a loss is a negative profit.
- Memorandum Reconciliation Account: base on its side, the other result as the balancing figure. (Miscellaneous expenses is not a reconciling item.)

Answer key · Check yourself

1 **395000**

₹3,80,000 + ₹25,000 (over-absorption) + ₹30,000 (notional, cost only) − ₹40,000 (financial only) = ₹3,95,000.

2 **B Add ₹15,000, because the cost accounts' higher closing stock gives them a higher profit**

Higher closing stock means higher profit. The cost books valued it ₹15,000 higher, so the cost profit is ₹15,000 above the financial one: from the financial profit, add. (From the cost profit it would be deducted.)

3 **A Interest received on bank deposits; C Expenses on issue of debentures; D Loss by fire not covered by insurance**

Interest received is purely financial income; issue expenses and uninsured fire losses are purely financial expenses (SM 7.31–7.32). Notional depreciation appears in the cost accounts only.

4 **B** ₹2,02,000

Under-absorption means actual overheads exceeded the cost charge, so the financial profit is lower: −₹14,000. Over-absorption is the reverse: +₹6,000. ₹2,10,000 − ₹14,000 + ₹6,000 = ₹2,02,000.

5 **C** **When the cost and financial accounts are integrated**

With integrated accounts the same set of books serves both purposes, so there is only one profit (SM 7.33). The other options remove only one cause of difference.

CHAPTER 8

Unit and Batch Costing

8.1 Unit costing

Where every unit of output is identical, cost is collected element-wise for a period and divided by the units produced; the statement of cost per unit then prices any output level.

In plain words

Ten lakh identical kulhads

Tapri's kulhad unit at Khurja turns out clay cups by the lakh, and every kulhad is the same: same clay, same mould, same hours in the kiln.

> **Kesar:** Ritu, I want a cost card for every kulhad. Each one is special to me.
>
> **Ritu:** Per unit, please, but one unit is like the next. Add up the month's clay, wages and kiln, divide by kulhads made, and every cup has its cost.

That is **unit costing**. When the output is identical, and each unit needs identical cost, there is nothing to trace to one cup rather than another. So costs are collected element by element for a period (a month, four weeks), arranged in a statement of cost, and divided by the number of units produced. Cement, paper, steel, mining

and breweries work this way: nobody keeps a separate cost card for one bag of cement.

The interesting questions come next. What happens when some cups crack in the kiln? And what price should Tapri quote if next month's output is higher, when the rent and the supervisor's salary do not rise with it? For that Ritu splits the costs by behaviour, because a cost per unit is only true at the output it was worked out for.

Think of it like

The sugarcane-juice cart

The juice-wallah outside the station sells only one thing: a glass of sugarcane juice. At night he adds up the cane, ice, lemons, the helper's wage and the cart's rent, and divides by glasses sold. He does not need a card for each glass. But if he expects twice the crowd during Navratri, he knows the cane and ice will double and the cart's rent will not, so his cost per glass will fall.

Key idea

Unit costing applies where the output is identical and each unit requires identical cost: cost per unit = total cost of production ÷ number of units produced. To price a different level of output, keep variable cost per unit, keep fixed cost in total, and split semi-variable costs first.

How it works

Industries fall into two groups (SM 8.2): those doing **job work** to customers' specifications, costed by job, contract and batch costing; and **continuous or process-type** industries making uniform products to standard specifications, costed by process costing, single output costing and operating costing.

Definition

Unit costing

Unit costing is that method of costing where the output produced is identical and each unit of output requires identical cost. Unit costing is synonymously known as single or output costing, but these are sub-division of unit costing method. (SM 8.3)

It is followed by industries "which produce single output or few variants of a single output" (SM 8.3): **paper, cement, steel works, mining, breweries** (SM 8.3).

How the costs are collected

Where each element of cost is collected from (SM 8.3–8.4)

Materials material requisition notes, accumulated for the period	cost sheet
Direct employee cost job time cards, idle time cards, valued at rates	cost sheet
Indirect employee cost payroll, posted to standing order numbers	overheads
Overheads standing order numbers; apportioned to departments	absorbed
Cost per unit total cost ÷ units produced	statement

Top to bottom in the order the costs are gathered; service department overheads are transferred to production departments before absorption.

1. **Materials**: the cost of materials issued for production is collected from material requisition notes and accumulated for a period or volume of activity (SM 8.3).
2. **Employee cost**: direct employee cost is collected from job time cards or sheets; the time booked and the idle time cards are valued at appropriate rates. Indirect employee cost comes from the payroll books and is posted to standing order or expense code numbers (SM 8.4).

3. **Overheads** are collected under standing order numbers (selling and distribution against cost account numbers), apportioned to service and production departments, the service departments' costs transferred to production departments, and the production overheads applied to products on a realistic basis: machine hour, labour hour, percentage of direct wages, percentage of direct materials (SM 8.4). The absorption bases are in absorption rates and the machine hour rate.
4. **Statement of cost per unit**: the cost sheet with a per-unit column, prime cost to cost of sales (cost sheet for a manufacturer).

∑ **Formula**

Cost per unit

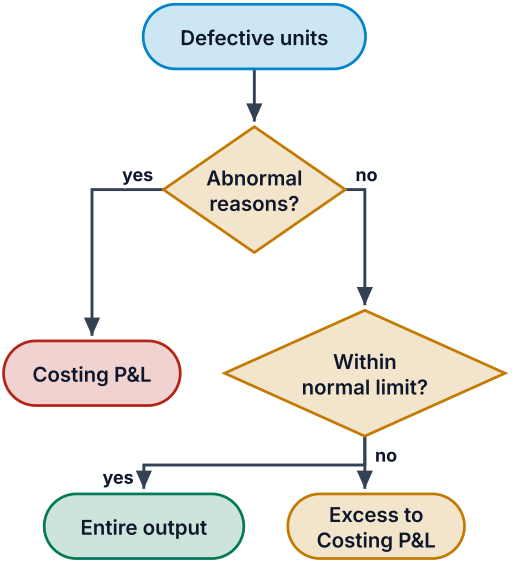
$$\text{Cost per unit} = \frac{\text{Total cost of production}}{\text{Number of units produced}}$$

(SM 8.3)

When some units come out defective

ICAI's rule for spoiled and defective work in unit costing (SM 8.4):

Spoiled and defective work in unit costing



Normal defectives within the limit are charged to the entire output, raising the cost of the good units; abnormal ones, and the excess over the normal limit, are written off in the Costing Profit and Loss Account (Costing P&L).

Circumstances	Treatment (SM 8.4)
Loss due to normal reasons, within the established normal rate	Cost of rectification or loss charged to the entire output
Normal reasons, but defectives substantially exceed the normal limit	Cost of rectification or loss beyond normal limits written off in Costing Profit and Loss Account
Loss due to abnormal reasons	Treated as abnormal cost; written off as loss in Costing Profit and Loss Account

The wider treatment of spoilage and defectives is in waste, scrap, spoilage and defectives.

Pricing a different level of output

A unit cost is only true at its own output. To find the cost and price at another level (SM Illustration 2, 8.6–8.7):

1. **Variable costs** (direct material, direct labour) stay the same per unit: divide any week's cost by its output.

2. **Semi-variable costs** rise with output but not in proportion: variable rate = change in cost ÷ change in output; fixed part = total cost − output × variable rate (cost behaviour).

3 **Fixed costs** stay the same in total at the new level.

4 **Price**: profit as a percentage of selling price is converted to a percentage of cost (20% of sales = 25% of cost).

For a whole year's **production programme** (some months at 50% capacity, others at 80%), cost each period at its own level, including any **stepped** semi-variable cost ("₹4,00,000 for every 25% increase in capacity or part thereof"), add the desired profit to the total, and divide by total units for the **average selling price** (SM 8.18–8.19, 8.23).

∑ **Formula**

Splitting a semi-variable cost

$$\text{Variable cost per unit} = \frac{\text{Change in cost}}{\text{Change in output}}$$

$$\text{Fixed part} = \text{Total cost} - \text{Output} \times \text{Variable rate}$$

∑ **Formula**

Average selling price for a production programme

$$\text{Average selling price} = \frac{\text{Total cost of all periods} + \text{Desired profit}}{\text{Total units of all periods}}$$

Formats and formulas

ICAI's statement of cost per unit (SM Illustration 1, 8.5). *Reproduce it from memory.*

Statement of cost per unit (units produced: ...)	Cost per unit (₹)	Amount (₹)
Raw materials consumed	xxx	xxx
Direct wages	xxx	xxx
Prime cost	**xxx**	**xxx**
Add: Manufacturing overheads	xxx	xxx
Works cost	**xxx**	**xxx**
Add: Office overheads	xxx	xxx
Cost of goods sold	**xxx**	**xxx**
Add: Selling overheads	xxx	xxx
Cost of sales / total cost	**xxx**	**xxx**
Add: Profit (balancing figure)	xxx	xxx
Sales	**xxx**	**xxx**

ICAI labels the line after office overheads "cost of goods sold": in SM Illustration 1 all 10,000 units are produced and sold, so it is also the cost of production. With stock, show cost of production first and adjust finished goods as in the full cost sheet.

Worked examples

Worked example

Khurja kulhads: a month's cost per cup, with cracks

In July 2026 Tapri's kulhad unit made 2,00,000 kulhads. Firing cracks are normal up to 2% of kulhads made; this month 5,000 cracked, and cracked kulhads have no value. Clay and colour consumed cost ₹2,40,000, direct wages ₹1,80,000, the kiln ran 600 hours at a rate of ₹250 an hour, and the production office cost ₹57,200. All the good kulhads were sold at ₹4.50 each, and selling overheads are ₹0.40 a kulhad sold.

Required:

(i) Prepare a statement of cost and find the cost per good kulhad.

(ii) Show the treatment of the cracked kulhads.

(iii) Calculate the cost of sales and the profit, in total and per kulhad sold.

THE WORKING

Working note. Normal cracks = 2% × 2,00,000 = 4,000; abnormal = 5,000 − 4,000 = 1,000. Normal output = 2,00,000 − 4,000 = 1,96,000 kulhads (the 1,95,000 good ones and the 1,000 abnormal ones). The cost of the normal cracks is charged to the entire output (SM 8.4), so cost per unit = cost of production ÷ 1,96,000.

(i) Statement of cost, July 2026

Particulars	Amount (₹)
Clay and colour consumed	2,40,000
Direct wages	1,80,000
Prime cost	**4,20,000**
Kiln overheads (600 hours × ₹250)	1,50,000
Works cost	**5,70,000**
Production office overheads	57,200
Cost of production (2,00,000 made)	**6,27,200**
Cost per kulhad of normal output (₹6,27,200 ÷ 1,96,000)	₹3.20

(ii) Cracked kulhads. The 4,000 normal cracks carry no separate charge: their cost is in the ₹3.20. The 1,000 abnormal cracks, 1,000 × ₹3.20 = **₹3,200**, are written off to the Costing Profit and Loss Account. Good kulhads carry 1,95,000 × ₹3.20 = **₹6,24,000**.

(iii) Cost of sales and profit

Particulars	Total (₹)	Per kulhad (₹)
Cost of 1,95,000 good kulhads	6,24,000	3.20
Selling overheads (₹0.40)	78,000	0.40
Cost of sales	**7,02,000**	**3.60**
Profit	1,75,500	0.90
Sales (₹4.50)	**8,77,500**	**4.50**

Spreading the cost over all 2,00,000 kulhads made would give ₹3.136 and hide the cracks; spreading it over the 1,95,000 good ones alone (₹3.216) would push the abnormal cracks into stock and sales. Valuing the abnormal ones at the cost of normal output is the process-costing convention (normal and abnormal loss).

Worked example

Dewas premix plant: one price for a year of shifts

Tapri's Dewas plant makes a single product, a 1 kg box of chai premix, to order (nothing is made for stock). One shift can make at most 10,000 boxes a month, and a

shift is either run for the whole month or not at all. The order book for 2026-27:

Months	Boxes ordered a month
April to September (6 months)	10,000
October to January (4 months)	28,000
February and March (2 months)	18,000

Costs:

- Direct material ₹42 and direct labour (piece rate) ₹14 a box.
- Each shift run in a month costs ₹1,20,000 for that month (shift supervisor, canteen and shift allowance).
- Plant fixed overheads (rent, depreciation, plant manager) ₹36,00,000 a year.
- Power is semi-variable. The bill was ₹2,30,000 in a month when 10,000 boxes were made and ₹4,10,000 in a month when 25,000 boxes were made.

Kesar wants one selling price for the whole year that earns a profit of 20% on the selling price.

Required:

(i) Prepare a statement of cost for each part of the year and determine the average selling price per box.

(ii) Calculate the cost per box in each part of the year, and explain the difference.

THE WORKING

Working notes

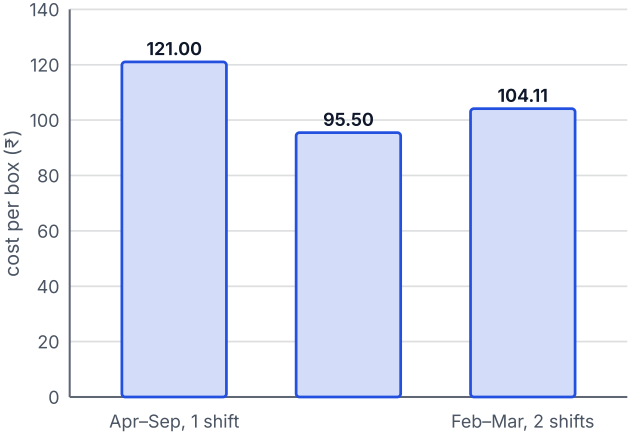
1. Boxes: 10,000 × 6 = 60,000; 28,000 × 4 = 1,12,000; 18,000 × 2 = 36,000. Total 2,08,000.
2. Shifts needed (a part shift cannot be run): April to September 10,000 ÷ 10,000 = 1 shift; October to January 28,000 ÷ 10,000 = 2.8, so 3 shifts; February and March 18,000 ÷ 10,000 = 1.8, so 2 shifts. Shift costs: ₹1,20,000 × 1 × 6 = ₹7,20,000; ₹1,20,000 × 3 × 4 = ₹14,40,000; ₹1,20,000 × 2 × 2 = ₹4,80,000.
3. Power: variable rate = (₹4,10,000 − ₹2,30,000) ÷ (25,000 − 10,000) = ₹12 a box; fixed part = ₹2,30,000 − 10,000 × ₹12 = ₹1,10,000 a month.
4. Plant fixed overheads are spread by months: 6/12, 4/12 and 2/12 of ₹36,00,000.

(i) Statement of cost for 2026-27

Particulars	Apr–Sep (₹)	Oct–Jan (₹)	Feb–Mar (₹)
Boxes	60,000	1,12,000	36,000
Direct material (₹42)	25,20,000	47,04,000	15,12,000
Direct labour (₹14)	8,40,000	15,68,000	5,04,000
Power: variable (₹12)	7,20,000	13,44,000	4,32,000
Power: fixed (WN 3)	6,60,000	4,40,000	2,20,000
Shift costs (WN 2)	7,20,000	14,40,000	4,80,000
Plant fixed overheads	18,00,000	12,00,000	6,00,000
Total cost	**72,60,000**	**1,06,96,000**	**37,48,000**

Particulars	Year (₹)
Total cost (72,60,000 + 1,06,96,000 + 37,48,000)	2,17,04,000
Add: Profit (20% on selling price = 25% of cost)	54,26,000
Sales required	**2,71,30,000**
Average selling price (÷ 2,08,000 boxes)	**₹130.43**

(ii) Cost per box by period



₹72,60,000 ÷ 60,000; ₹1,06,96,000 ÷ 1,12,000; ₹37,48,000 ÷ 36,000. Variable cost is ₹68 a box throughout; the fixed costs per box fall as monthly output rises.

Cost per box (₹)	Apr–Sep	Oct–Jan	Feb–Mar
Variable (₹42 + ₹14 + ₹12)	68.00	68.00	68.00
Plant fixed and fixed power	41.00	14.64	22.78
Shift costs	12.00	12.86	13.33
Total	**121.00**	**95.50**	**104.11**

Variable cost is ₹68 a box in every period. The plant's fixed overheads (₹3,00,000 a month) and the fixed part of power (₹1,10,000 a month) do not change with output, so they are spread over 10,000, 28,000 and 18,000 boxes a month and fall from ₹41.00 to ₹14.64 a box. Shift costs are fixed per shift, not per box: a full shift costs ₹12.00 a box, but October to January's third shift and February and March's second shift each make only 8,000 boxes, so the shift cost rises to ₹12.86 and ₹13.33 a box. No single period's cost is the year's cost, which is why the price is averaged over the whole programme.

Exam angle

In the exam

Where unit costing appeared, May 2024 to September 2026 (and the old scheme)

In the documents read for this lesson:

- **Theory, 4 to 5 marks:** "Describe Unit Costing and Batch Costing. Give three examples of industries for each method where these are used": May 2024 Q6(c) (4 marks); the same with "explain the costing procedure in batch costing": MTP Series II May 2026 Q6(b) (5 marks). Old scheme: "Describe Unit Costing. What kind of industries follow this method of costing?": MTP Series I November 2022 Q6(c).
- **Average selling price for a production programme, 5 to 10 marks (old scheme):** January 2021 Q3(b) (10 marks: 50%, 75% and 100% capacity in parts of the year, a semi-variable cost rising in steps, then whether to accept offers of ₹130 and ₹129 against the computed ₹128.45); MTP Series II December 2021 Q1(c) and MTP Series II November 2022 Q2(b) (10 marks), both "determine the average selling price at which each unit should be quoted".
- **Single-output cost statements** in the new scheme are set as cost sheet questions, such as May 2026 Q2(a) (cotton yarn, 1,500 kg): see the manufacturer's cost sheet.

In the exam

The traps ICAI set

- Capacity given at one level ("1,80,000 units at 60% capacity"): find 100% first (3,00,000), then each period's units.
- Semi-variable "for every 20% increase in the capacity or part thereof": 50% to 75% is two steps, not one (MTP Series II December 2021 and November 2022).
- Fixed costs are annual: take the months' share, not the full year, for each period.

Trick

Units in a period

Units = annual capacity at 100% × months ÷ 12 × capacity %. Condition: the capacity is stated per annum; convert a monthly or a partial-capacity figure to 100% per annum first.

Answer frame

Describe unit costing and batch costing, with three examples of industries for each (May 2024 Q6(c), 4 marks). ICAI's suggested answer:

1. **Unit costing**: the method of costing where the output produced is identical and each unit of output requires identical cost; costs are collected and analysed

element-wise, and cost per unit = total cost ÷ number of units produced. Used in paper, cement, steel works, mining, breweries.

2. **Batch costing**: a type of specific order costing where articles are manufactured in predetermined lots, known as batches; cost per unit in a batch = total cost of the batch ÷ units produced in that batch. ICAI's answer named pharmaceutical or drug industries, readymade garments and electronic parts of TV and radio sets (batch costing).

For an average-price question, the marks are in: a working note for units by period, a working note for each step of the semi-variable cost, a statement of cost by period with a total column, profit added, and the average price with its division shown; then the advice ("accept, as ₹130 is above the required ₹128.45").

Common mistakes

⚠ **Common mistake**

Hall of Shame: one cost per unit for every output

Kesar's ₹121 a box (the one-shift cost of April to September) used for the whole year makes the price ₹121 × 1.25 = ₹151.25 instead of ₹130.43, and loses orders to competitors. **Fix:** at each level, take fixed costs in total; only variable costs stay the same per unit.

⚠ **Common mistake**

The semi-variable cost treated as fixed (or as variable)

Factory overheads of ₹31,000 at 1,200 units taken as ₹25.83 a unit at 2,000 units gives ₹51,667 instead of ₹35,000 (SM Illustration 2). **Fix:** split it: ₹5 a unit plus ₹25,000.

⚠ **Common mistake**

Annual fixed cost charged in full to each period

Putting the whole ₹36,00,000 of plant fixed overheads into each part of the year. **Fix:** months ÷ 12 of the annual figure for each period; a cost given per month (the shift cost, the fixed part of power) is multiplied by the months instead.

⚠ **Common mistake**

Abnormal defectives left in the cost of good units

Dividing the whole cost by the 1,95,000 good kulhads (₹3.216) makes the customers pay for the abnormal cracks. **Fix:** cost of abnormal defectives goes to the Costing Profit and Loss Account (SM 8.4).

Practice

Check yourself

1. A cement plant's statement of cost for April shows a total cost of production of ₹1,26,00,000 for 42,000 tonnes produced, with no defective output. What is the cost per tonne, in rupees?

Answer: _____

2. Factory overheads were ₹48,000 at 4,000 units and ₹54,000 at 6,000 units. What would they be at 7,000 units, in rupees?

Answer: _____

3 Case scenario. A steel re-rolling mill, a cement plant and a brewery each make a single standard product in continuous runs. A furniture maker builds each piece to the customer's drawing. Ritu must recommend a costing method for the first three. Which is correct?

- **A** Job costing, because each production run is a separate order
- **B** Unit (single or output) costing, because the output is identical and each unit requires identical cost
- **C** Contract costing, because the plants are large
- **D** Batch costing, because every product is made in lots to a customer's specification

4 Semi-variable cost is ₹10,00,000 a year up to 50% capacity, plus ₹2,00,000 for every 20% increase in capacity or part thereof. What is it at 85% capacity?

- **A** ₹13,50,000
- **B** ₹12,00,000
- **C** ₹14,00,000
- **D** ₹13,00,000

5 In unit costing, which treatments of spoiled and defective work are correct? Select all that apply.

- **A** Normal defectives within the established rate: cost charged to the entire output
- **B** Defectives beyond the normal limit: the excess written off to Costing Profit and Loss Account
- **C** Abnormal defectives: cost charged to the entire output
- **D** Loss due to abnormal reasons: treated as abnormal cost and written off to Costing Profit and Loss Account

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Unit costing: output identical, each unit requires identical cost; also called single or output costing (SM 8.3).
- Cost per unit = total cost of production ÷ number of units produced.
- Industries: paper, cement, steel works, mining, breweries.

- Costs collected element-wise: materials from requisition notes, labour from time cards, overheads under standing order numbers, apportioned and absorbed (SM 8.3–8.4).
- Defectives: normal within limit to the entire output; excess and abnormal to Costing Profit and Loss Account (SM 8.4).
- New output level: variable per unit same, fixed in total same, semi-variable split by change in cost ÷ change in output.
- Production programme: cost each period at its own capacity (steps "or part thereof" round up), add profit, divide by total units.
- One juice cart, one product, one division.

Answer key · Check yourself

1 **300**

Cost per unit = total cost of production ÷ units produced = ₹1,26,00,000 ÷ 42,000 = ₹300 a tonne (SM 8.3).

2 **57000**

Variable rate = ₹6,000 ÷ 2,000 = ₹3 a unit; fixed = ₹48,000 − 4,000 × ₹3 = ₹36,000. At 7,000 units: ₹36,000 + ₹21,000 = ₹57,000.

3 **B Unit (single or output) costing, because the output is identical and each unit requires identical cost**

ICAI names paper, cement, steel works, mining and breweries for unit costing (SM 8.3). Job costing suits the furniture maker, whose pieces are unique.

4 **C ₹14,00,000**

50% to 85% is 35%: one full 20% step and part of another, so two steps: ₹10,00,000 + 2 × ₹2,00,000 = ₹14,00,000. Counting only completed steps gives ₹12,00,000.

5 **A Normal defectives within the established rate: cost charged to the entire output; B Defectives beyond the normal limit: the excess written off to Costing Profit and Loss Account; D Loss due to abnormal reasons: treated as abnormal cost and written off to Costing Profit and Loss Account**

SM 8.4: normal within limits to the entire output; the excess beyond normal limits and all losses due to abnormal reasons to Costing Profit and Loss Account.

8.2 Batch costing and the economic batch quantity

Batch costing makes the batch the cost object and divides its cost by the units in it; the economic batch quantity is the batch size at which set-up and holding costs together are lowest.

In plain words

The grinder that has to be cleaned

Tapri sells jars of chai masala to a supermarket chain, 1,20,000 jars a year, taken evenly. Each production run means cleaning the grinder, changing the spice settings and test-grinding: ₹2,400 a run, however many jars follow. Every jar waiting in the store costs ₹4 a year to keep (space, insurance, money locked up).

> **Kesar:** Easy. Make all 1,20,000 jars in one run. One cleaning, done.
>
> **Ritu:** And rent a godown for a year's stock. Per unit, please: that is ₹2,40,000 a year of holding cost.
>
> **Kesar:** Then a small run every working morning.
>
> **Ritu:** And pay for the cleaning 250 times a year. That is ₹6,00,000.

The jars of one run are made together, in a **batch**, and they are identical, so their cost is worked out for the batch and divided by the jars in it: that is **batch costing**. The argument between Kesar and Ritu is about the size of

the batch. Bigger batches mean fewer set-ups but more stock to hold; smaller batches mean less stock but more set-ups. Somewhere between the two extremes the total is lowest: here at 12,000 jars a run, ten runs a year, ₹24,000 of set-ups and ₹24,000 of holding cost. That batch size is the **economic batch quantity (EBQ)**.

Think of it like

The atta for the week

A household that kneads atta once a week saves effort but keeps a big ball of dough in the fridge, going stale. One that kneads before every roti has fresh dough and an aching back. Most kneading families settle on a ball for a day or two: the batch where effort and staleness together hurt least.

Key idea

In batch costing the batch is the cost object: cost per unit = cost of the batch ÷ units in the batch. The economic batch quantity is the batch size at which set-up costs and inventory holding costs together are at a minimum: EBQ = $\sqrt{(2DS \div C)}$.

How it works

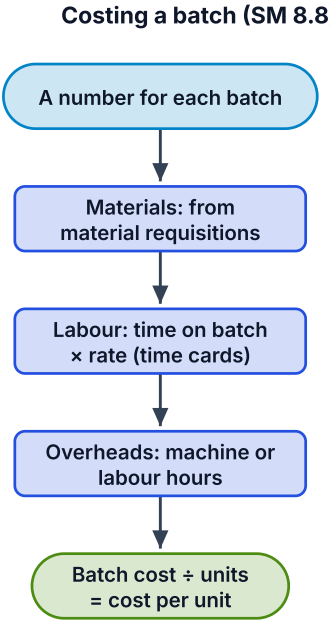
Definition

Batch costing

Batch Costing is a type of specific order costing where articles are manufactured in predetermined lots, known as batch. Under this costing method, the cost object for cost determination is a batch for production rather output as seen in unit costing method. (SM 8.8)

A batch "consists of certain number of units which are processed simultaneously" (SM 8.8). Inputs accumulate until the minimum batch size is reached, and the reasons for batch production "may be either technical or economical or both": making one pen of a design for one customer would be too costly, while 10,000 pens of the same design cut the cost "to a sizeable extent" (SM 8.8). Each run carries a minimum level of expenditure, on workers, supervision and setting up the machine, "irrespective of number of units produced" (SM 8.8).

The costing procedure



Material, labour and overheads are booked against the batch number; set-up costs are part of the batch's cost.

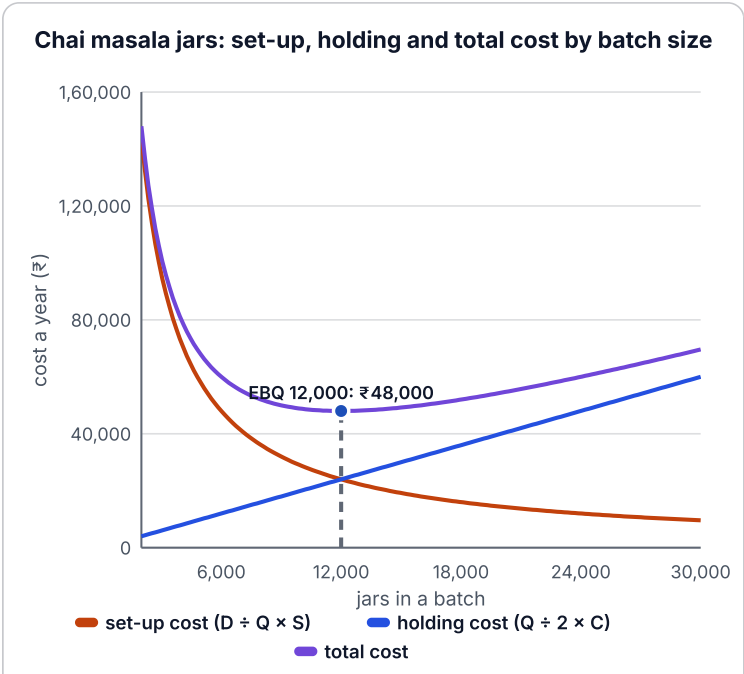
1. **One number is allotted for each batch** (SM 8.8).
2. **Material cost** for the batch comes from the material requisitions for the batch.
3. **Labour cost** = time spent on the batch by direct workers (from time cards or job tickets) × the rate.
4. **Overheads** are absorbed on a suitable basis such as machine hours or direct labour hours. When a batch order is repeated each month, the overhead rate is the

month's overheads ÷ the month's labour hours, applied to the batch's hours (SM Illustration 4, 8.9–8.11).

5. **Cost per unit** = total cost of the batch ÷ units in the batch; set-up cost is added per batch, whatever the number of units (SM Illustration 3, 8.8–8.9).

Choosing the batch size

The total production cost under batch production has two main costs: **machine set-up costs** and **inventory holding costs** (SM 8.11). A larger batch means fewer set-ups but more units in stock; a smaller one means lower holding costs "but only with higher set up costs" (SM 8.11).



Set-up cost (orange) falls as the batch grows; holding cost (blue) rises; their total (violet) is lowest at 12,000 jars, where the two are equal at ₹24,000 each.

The EBQ "may be determined by calculating the total cost for a series of possible batch sizes and checking which batch size gives the minimum cost", or by a formula "similar to determination of Economic Order Quantity (EOQ)" (SM 8.12), the twin of EOQ in material cost.

Job costing and batch costing

ICAI's three differences, in ICAI's order (SM 8.15):

No.	Job costing	Batch costing
1	Method of costing used for non-standard and non-repetitive products produced as per customer specifications and against specific orders	Homogeneous products produced in a continuous production flow in lots
2	Cost determined for each job	Cost determined in aggregate for the entire batch and then arrived at on per unit basis
3	Jobs are different from each other and independent of each other; each job is unique	Products produced in a batch are homogeneous and lack of individuality

More on jobs in job costing.

Formats and formulas

∑ **Formula**

Economic batch quantity

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

D = annual demand for the product; S = setting up cost per batch; C = carrying cost per unit of production, per annum (SM 8.12). If C is given per month, multiply by 12; if as a percentage, apply it to the cost of making one unit (SM Illustrations 5 and 6).

∑ **Formula**

Number of runs, interval and the costs at any batch size Q

$$\text{Runs} = \frac{D}{Q} \qquad \text{Interval (days)} = \frac{365}{\text{Runs}}$$

$$\text{Set-up cost} = \frac{D}{Q} \times S \qquad \text{Holding cost} = \frac{Q}{2} \times C$$

Average inventory is half the batch (SM 8.13). At the EBQ the two costs are equal.

Reproduce this statement from memory (SM Illustration 3, 8.9):

Statement of cost per batch and per order	Per batch (₹)	Order (₹)
Direct material cost	xxx	xxx
Direct wages	xxx	xxx
Set-up cost	xxx	xxx
Add: Production overheads	xxx	xxx
Total production cost	**xxx**	**xxx**
Add: Administration, selling and distribution	xxx	xxx
Total cost	**xxx**	**xxx**
Add: Profit	xxx	xxx
Selling price	**xxx**	**xxx**

Selling price per unit = selling price per batch ÷ units in a batch.

Definition

Economic batch quantity (EBQ)

Economic batch quantity is the size of a batch where total cost of set-up and holding costs are at minimum. (SM 8.11)

Worked examples

Worked example

Nankhatai for a wedding: the order that does not fill the last tray

Tapri's bakery oven takes at most 400 nankhatai in one load, and every load bakes for 1.5 oven hours whether it is full or not. A caterer orders 6,100 nankhatai. Costs:

- Flour, ghee and sugar ₹4.20 a piece; packing ₹0.60 a piece.
- Bakers' wages ₹320 a load; oven set-up and cleaning ₹300 a load.
- Oven overheads are absorbed at ₹240 an oven hour.
- Administration, selling and distribution: 10% of the production cost of the order.
- Tapri wants a profit of 20% on the selling price.

Required:

(i) Determine the number of loads, the cost of the order, its selling price and the selling price of one nankhatai.

(ii) The caterer asks what one nankhatai would cost him if he raised the order to 6,400. Calculate it and explain the difference.

THE WORKING

Working note: loads. 6,100 ÷ 400 = 15.25, so **16 loads**: 15 full loads and one of 100 pieces. A part load still needs a set-up, a bakers' shift and 1.5

oven hours. An order of 6,400 also needs 16 loads (6,400 ÷ 400 = 16, all full).

Statement of cost and selling price

Particulars	6,100 pieces (₹)	6,400 pieces (₹)
Flour, ghee and sugar (₹4.20 a piece)	25,620	26,880
Packing (₹0.60 a piece)	3,660	3,840
Bakers' wages (16 loads × ₹320)	5,120	5,120
Oven set-up (16 loads × ₹300)	4,800	4,800
Oven overheads (16 × 1.5 hours × ₹240)	5,760	5,760
Total production cost	**44,960**	**46,400**
Administration, selling and distribution (10%)	4,496	4,640
Total cost	**49,456**	**51,040**
Profit (20% on price = 25% on cost)	12,364	12,760
Selling price of the order	**61,820**	**63,800**
Selling price per nankhatai	**₹10.13**	**₹9.97**

(i) 16 loads; cost of the order ₹49,456; selling price ₹61,820, or ₹61,820 ÷ 6,100 = **₹10.13** a nankhatai.

(ii) At 6,400 pieces the price is ₹63,800 ÷ 6,400 = **₹9.97** a nankhatai. The 300 extra pieces add only ₹1,980 to the bill (₹6.60 each), because they fill the sixteenth load, which was already being set up, staffed and fired for 100 pieces. Only the per-piece costs rise (₹4.80 of ingredients and packing, ₹6.60 with the 10% administration charge and the profit on it); the per-load costs of ₹980 a load (₹320 + ₹300 + ₹360) stay at 16 loads.

Worked example

Printed kulhads: three batches, three prices

In August Tapri's printing line ran three batch orders of printed kulhads. The machine makes 1,200 passes an hour, and each colour is one pass of every kulhad in the batch.

Batch	Customer	Kulhads	Colours	Price a kulhad (₹)
B-81	Hotel chain	14,400	1	4.50
B-82	Wedding planner	3,000	4	6.50
B-83	College fest	6,000	2	5.50

Costs:

- Plain kulhads (issued on material requisitions) ₹3.00 each; ink ₹0.10 a kulhad for each colour; packing ₹0.30 a kulhad.
- A new screen and make-ready for each colour of each batch: ₹1,000.
- Operators' wages ₹250 a machine hour.
- Printing department overheads for August, ₹1,12,000 for 320 machine hours worked, are absorbed on machine hours.

Required:

(i) Calculate the cost and the profit or loss per kulhad of each batch, and the profit of the month's three orders.

(ii) The wedding planner wants a repeat of the same four-colour design at ₹6.50 a kulhad. Calculate the smallest batch at which the order recovers its full cost.

THE WORKING

Working notes

1. Machine hours = kulhads × colours ÷ 1,200: B-81 14,400 × 1 ÷ 1,200 = 12; B-82 3,000 × 4 ÷ 1,200 = 10; B-83 6,000 × 2 ÷ 1,200 = 10.
2. Overhead rate = ₹1,12,000 ÷ 320 = ₹350 a machine hour.
3. Screens = colours × ₹1,000: ₹1,000; ₹4,000; ₹2,000.

(i) Statement of cost and profit of each batch

Particulars	B-81 (₹)	B-82 (₹)	B-83 (₹)	Total (₹)
Plain kulhads	43,200	9,000	18,000	70,200
Ink	1,440	1,200	1,200	3,840
Screens and make-ready	1,000	4,000	2,000	7,000
Operators' wages	3,000	2,500	2,500	8,000
Overheads (₹350 an hour)	4,200	3,500	3,500	11,200
Packing	4,320	900	1,800	7,020
Total cost	**57,160**	**21,100**	**29,000**	**1,07,260**
Sales	64,800	19,500	33,000	1,17,300
Profit / (loss)	**7,640**	**(1,600)**	**4,000**	**10,040**

Per kulhad (₹)	B-81	B-82	B-83
Cost (total cost ÷ kulhads)	3.97	7.03	4.83
Price	4.50	6.50	5.50
Profit / (loss)	**0.53**	**(0.53)**	**0.67**

The three orders together earn ₹10,040. The wedding batch loses ₹0.53 a kulhad: four screens

(₹4,000) spread over only 3,000 kulhads add ₹1.33 a kulhad, and four passes take four times the machine time of a one-colour kulhad.

(ii) Smallest repeat batch. Cost per kulhad other than screens = ₹3.00 + 4 × ₹0.10 + ₹0.30 + 4 passes ÷ 1,200 × (₹250 + ₹350) = ₹3.70 + ₹2.00 = ₹5.70. The screens, ₹4,000, are a cost per batch. Full cost is recovered when ₹4,000 ÷ batch size = ₹6.50 − ₹5.70 = ₹0.80, so the batch must be at least ₹4,000 ÷ ₹0.80 = **5,000 kulhads**. Check: 5,000 × ₹5.70 + ₹4,000 = ₹32,500 = 5,000 × ₹6.50.

Worked example

Kulhad cartons for a café chain: is a quick-change kit worth it?

Tapri supplies 72,000 cartons of printed kulhads a year to a café chain, at a steady daily rate. Each production run needs a change of print screens: 3 hours of a printer's time at ₹250 an hour, and 6 cartons of test prints that are scrapped (a carton costs ₹100 to make). Cartons in stock are carried at interest of 12% a year on their cost of making, plus storage of ₹3 a carton a year.

A supplier offers a quick-change screen kit on hire at ₹12,000 a year. With it a change takes 2 hours and scraps only 1 carton of test prints.

Required:

(i) Determine the set-up cost per run, the carrying cost per carton per annum, the economic batch quantity, the number of runs a year and the interval between runs (365 days).

(ii) Calculate the annual set-up cost and holding cost at the EBQ.

(iii) Advise whether Tapri should hire the kit.

THE WORKING

(i) EBQ without the kit

Set-up cost per run S = 3 × ₹250 + 6 × ₹100 = ₹750 + ₹600 = **₹1,350**. Carrying cost C = 12% × ₹100 + ₹3 = **₹15** a carton a year.

$$\text{EBQ} = \sqrt{\frac{2 \times 72{,}000 \times 1{,}350}{15}}$$

$$= \sqrt{1{,}29{,}60{,}000} \qquad = 3{,}600 \text{ cartons}$$

Runs = 72,000 ÷ 3,600 = **20 a year**; interval = 365 ÷ 20 = **18.25 days**.

(ii) Costs at the EBQ. Set-up 20 × ₹1,350 = **₹27,000**; holding 3,600 ÷ 2 × ₹15 = **₹27,000**; total ₹54,000. The two are equal, which is the check that 3,600 is right.

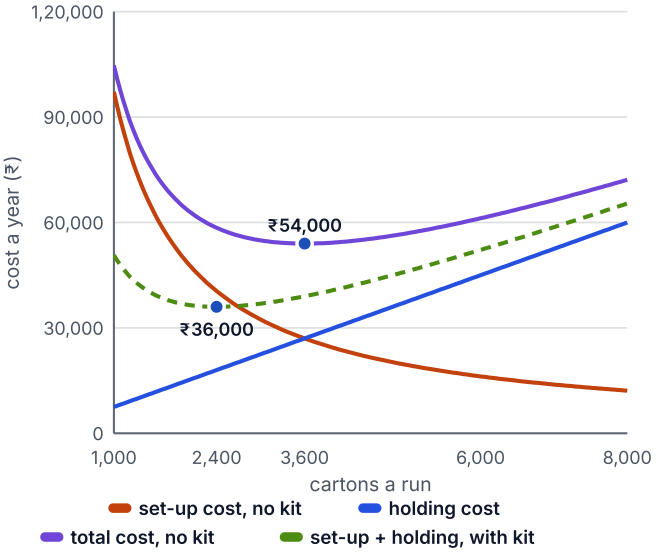
(iii) With the kit. S = 2 × ₹250 + 1 × ₹100 = ₹600; C is unchanged at ₹15.

$$\text{EBQ} = \sqrt{\frac{2 \times 72{,}000 \times 600}{15}} = \sqrt{57{,}60{,}000}$$
$$= 2{,}400 \text{ cartons}$$

Runs = 72,000 ÷ 2,400 = 30 a year; interval = 365 ÷ 30 = 12.17 days.

Particulars	Without kit	With kit
Run size (EBQ), cartons	3,600	2,400
Runs a year	20	30
Set-up cost (runs × S), ₹	27,000	18,000
Holding cost (run ÷ 2 × ₹15), ₹	27,000	18,000
Kit hire, ₹	—	12,000
Total cost a year, ₹	**54,000**	**48,000**

Cartons: total cost a year by run size, with and without the kit



Without the kit the total (violet) is lowest at 3,600 cartons; with it the set-up and holding total (green, dashed) is lowest at 2,400 cartons, before the ₹12,000 hire.

Advice: hire the kit. Set-up and holding costs fall from ₹54,000 to ₹36,000 a year, a saving of ₹18,000 against the hire of ₹12,000: a net saving of **₹6,000** a year, provided the run size is cut to the new EBQ of 2,400. Kept at 3,600 cartons a run, the kit would save only ₹15,000 (20 × ₹600 + ₹27,000 = ₹39,000), so the net gain would fall to ₹3,000. A lower set-up cost always means a smaller economic batch: the EBQ moves with the square root of S ($\sqrt{}$(600 ÷ 1,350) = 2/3).

Exam angle

In the exam

Where batch costing and EBQ appeared, May 2024 to September 2026 (and the old scheme)

In the documents read for this lesson:

- **Batch cost and price, 5 marks:** May 2026 Q1(a) (a batch of 20 chairs: cost of the batch and selling price per chair at 25% on selling price); January 2025 Q1(b) (batches of 2,000 with a 3% rejection rate and scrap value: contribution per good unit and monthly profit).
- **EBQ multiple choice, 2 marks each**, in seven of the sixteen MTP papers: Series I May 2026 MCQ 15 (frames: monthly demand, carrying cost 10% of unit cost, answer 800); Series II May 2026 MCQ 14; Series I September 2025 MCQ 14; Series I January 2026 MCQ 14; Series I September 2024 MCQ 11; Series I May 2024 MCQ 7 and Series I May 2025 MCQ 4 (optimum run size, with holding cost per month). A batch-cost MCQ besides: Series II September 2024 MCQ 15 (total cost of a batch with a set-up cost).
- **EBQ descriptive:** MTP Series II September 2024 Q1(c) (4 marks: vaccines, carrying cost per month, extra cost of 15,000 a run); old-scheme May 2023 Q1(a) (5 marks: EBQ of 13,500, then the extra ₹12,000 of a 7,500-unit policy); MTP Series II September 2025 Q2(b) (2 marks: rented storage per

square metre); RTP May 2024 and RTP September 2026 (the SM soft-drink case, EBQ per bottle size).

- **Theory, 4 to 5 marks:** "how batch size affects set-up costs and holding costs" and "how to determine EBQ": MTP Series I May 2026 Q6(c); the number of batches at which set-up and carrying cost are optimum: MTP Series I September 2026 Q6(c); describe batch costing with examples and its costing procedure: May 2024 Q6(c), MTP Series II May 2026 Q6(b).

In the exam

The traps ICAI set

- Carrying cost per month (multiply by 12) or as a percentage of the unit's production cost.
- Monthly demand: annualise before the formula (500 a month × 12 in SM Illustration 5).
- "Extra cost of the present policy": total cost at the policy minus total cost at the EBQ, both set-up and holding.
- Rejections in a batch: cost is spread over the good units only; scrap value reduces cost (January 2025).

Trick

The equal-cost check

At the EBQ, annual set-up cost equals annual holding cost (D ÷ Q × S = Q ÷ 2 × C, which solves to the formula). If your two figures differ, recheck C's period. Condition: the simple EBQ model, with average inventory taken as half the batch.

Answer frame

In batch costing, state how the economic batch quantity is determined (SM 8.18, Q3; MTP Series I May 2026 Q6(c)).

1. In batch production the lot size is critical to the least cost of operation; total production cost comprises two main costs: machine set-up costs and inventory holding costs.
2. If the batch size is higher, set-up costs decline because fewer set-ups are needed, but units in inventory rise and holding costs go up; if it is lower, holding costs fall but set-up costs rise.
3. EBQ is the size of a batch where the total of set-up and holding costs is at a minimum: on a graph, where the two cost lines intersect.
4. It is found by calculating total cost for a series of batch sizes and picking the lowest, or by the formula EBQ = √(2DS ÷ C), with D, S and C defined.

Describe batch costing, with examples, and its costing procedure (MTP Series II May 2026 Q6(b)): the SM 8.8 definition; cost per unit = cost of the batch ÷ units in it; examples (ICAI's May 2024 answer: pharmaceutical or drug industries, readymade garments, electronic parts of TV and radio sets; the SM's own: pens, bakery items); then the procedure (one number per batch, materials from requisitions, labour from time cards, overheads on machine or labour hours).

Distinguish job costing and batch costing: ICAI's three points in the table in How it works.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the monthly carrying cost used as annual

Holding cost "₹1 per bearing per month" put straight into the formula gives an EBQ $\sqrt{12}$ times too large (17,527 instead of 5,060 in SM Illustration 6). **Fix:** C is per unit per annum: ₹1 × 12 = ₹12.

⚠ **Common mistake**

Monthly demand in an annual formula

500 units a month used as D. **Fix:** D = annual demand, 500 × 12 = 6,000 (SM Illustration 5). Keep D and C over the same period.

⚠ **Common mistake**

Set-up cost spread per unit

Charging the per-load costs per piece: (₹300 + ₹320 + ₹360) ÷ 400 = ₹2.45 a nankhatai × 6,100 = ₹14,945, instead of 16 loads × ₹980 = ₹15,680. The part load of 100 still needs a full set-up, a full bakers' shift and 1.5 oven hours. **Fix:** a cost per batch is counted by batches, and a part batch is a whole batch.

⚠ **Common mistake**

The extra cost of a policy measured on one cost only

Comparing only set-up costs, a run of 30,000 jars of chai masala (4 runs, ₹9,600) looks ₹14,400 cheaper than the EBQ of 12,000 jars (10 runs, ₹24,000). **Fix:** compare total cost, set-up plus holding: holding rises from ₹24,000 to 15,000 × ₹4 = ₹60,000, so the policy costs ₹21,600 more.

Practice

Check yourself

1. Annual demand is 36,000 units, the set-up cost is ₹500 a batch and the carrying cost is ₹10 per unit per annum. What is the economic batch quantity, in units?

Answer: _____

2 Case scenario. Dev reviews the masala unit: demand 1,20,000 jars a year, set-up ₹2,400 a run, holding ₹4 a jar a year. The production head wants 24,000 jars a run to "cut set-up cost in half". Which statement is correct?

- **A** The proposal saves ₹12,000 a year because set-up cost falls from ₹24,000 to ₹12,000
- **B** The proposal raises total cost from ₹48,000 to ₹60,000 a year, because holding cost doubles to ₹48,000
- **C** The EBQ is 24,000 jars, so the proposal is optimal
- **D** Set-up cost does not change with batch size

3 A batch of 250 units needs materials of ₹6,000, wages of ₹2,000 and a set-up of ₹1,500; production overheads are 60% of wages. What is the production cost per unit, in rupees?

Answer: _____

4 Which of these is one of ICAI's differences between job costing and batch costing?

- **A** Jobs are unique and independent of each other; products in a batch are homogeneous and lack individuality
- **B** Job costing determines cost in aggregate for many jobs and then per unit
- **C** Batch costing is used for non-standard products made to customer specifications
- **D** Batch costing never includes set-up costs

5 Holding cost is given as ₹0.50 per unit per month and set-up cost as ₹1,200 a run, with monthly demand of 2,000 units. Which inputs give the correct EBQ? Select all that apply.

- **A** D = 24,000 units a year
- **B** C = ₹0.50
- **C** C = ₹6 per unit per annum
- **D** S = ₹1,200

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Batch costing: a type of specific order costing; articles made in predetermined lots; the batch is the cost object (SM 8.8).
- Procedure: a number per batch; materials from requisitions; labour from time cards; overheads on machine or labour hours; cost per unit = batch cost ÷ units.
- Set-up cost is incurred per batch, whatever its size.
- EBQ = size of batch where total set-up and holding costs are minimum: √(2DS ÷ C), D annual, C per unit per annum.
- Runs = D ÷ EBQ (ICAI rounds up); interval = 365 ÷ runs; at EBQ, set-up cost = holding cost.
- Extra cost of a policy = total cost at the policy − total cost at the EBQ.
- Job vs batch (SM 8.15): unique jobs vs homogeneous lots; cost per job vs cost per batch then per unit; individuality vs none.
- Knead the atta for a day or two, not for every roti.

Answer key · Check yourself

1 **1897.3 to 1897.4**

EBQ = √(2 × 36,000 × 500 ÷ 10) = √36,00,000 = 1,897.37 units (round to 1,897 in an answer).

2 **B** **The proposal raises total cost from ₹48,000 to ₹60,000 a year, because holding cost doubles to ₹48,000**

At 24,000 a run: 5 runs × ₹2,400 = ₹12,000 set-up; 12,000 average × ₹4 = ₹48,000 holding; total ₹60,000 against ₹48,000 at the EBQ of 12,000.

3 **42.8**

Production cost = 6,000 + 2,000 + 1,500 + 1,200 = ₹10,700; ÷ 250 = ₹42.80.

4 **A** **Jobs are unique and independent of each other; products in a batch are homogeneous and lack individuality**

SM 8.15: job costing suits non-standard, non-repetitive products to customer specification, with cost found for each job; batch costing suits homogeneous products in lots, with cost found for the batch and then per unit.

5 **A** **D = 24,000 units a year**; **C** **C = ₹6 per unit per annum**; **D** **S = ₹1,200**

D and C must be annual: 2,000 × 12 = 24,000 and ₹0.50 × 12 = ₹6. Then EBQ = $\sqrt{}$(2 × 24,000 × 1,200 ÷ 6) = 3,098 units.

CHAPTER 9

Job Costing

9.1 Job costing

Each customer order is its own cost object: ICAI's job cost sheet, how materials, labour and overheads reach it, the thirteen control-account entries, and how job costing differs from process costing.

In plain words

Per job, please

Tapri Kiosk Works, the company's workshop in Indore, builds kulhad kiosks, café counters and tea carts, each to a franchisee's own drawing. This month it is finishing a steel tea cart with a glass hood for an airport franchisee and a plain wooden kiosk for a mall.

> **Kesar:** A kiosk is a kiosk. Charge both the same as last time.
>
> **Ritu:** The airport cart has twice the steel and three times the welding. Per job, please.
>
> **Bunty:** So every kiosk gets its own bill?
>
> **Ritu:** Every kiosk gets its own number, and every cost carries that number.

When every order is different, the order itself is the cost unit. Each job gets a **job number**. Materials are drawn from stores against that number, workers book their

hours against it, and overheads are added at the factory's rates. The costs pile up on the job's own cost sheet, the **job cost card**, and when the job is finished the card shows what it cost and what it earned. This is **job costing**, the main method of **specific order costing**.

It is the opposite of costing a stream of identical products, where costs are collected for a period and averaged over the output (unit and process costing). Here nothing is averaged: two kiosks never share a cost sheet, and the work-in-progress at any date is simply the jobs still on the shop floor.

Think of it like

The tailor's order book

A tailor writes a slip for every customer: cloth used, hours of stitching, buttons and lining, a share of the shop's rent. When the suit is ready the slip shows its cost, and the price follows. A ready-made shirt factory cannot do that for each shirt: it adds up the month's costs and divides by the shirts made. The tailor uses job costing; the factory uses process or unit costing.

Key idea

In job costing the job is the cost unit: materials (stores requisitions), labour (job time cards) and overheads (pre-determined rates) are collected by job number on a job cost sheet, and the job's cost and profit are known when it is completed.

How it works

What ICAI means by job costing

Definition

Job costing

"The category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or contract." (CIMA London, SM 9.2)

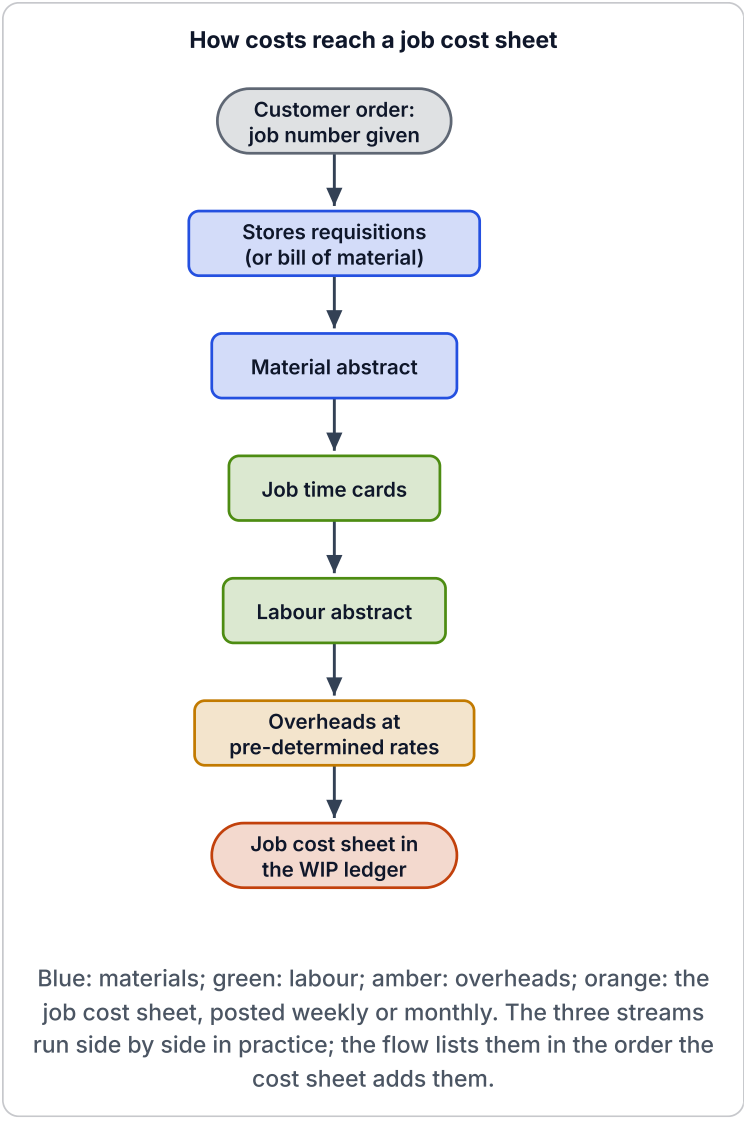
Costs are collected and accumulated according to jobs, contracts, products or work orders; each job or unit of production is a separate entity for costing. The method also applies where production is in batches, batch production being of the same character as job order production and differing mainly in the size of orders (SM 9.2; batch costing).

Principles. Job costing may be regarded as the principal method of costing, since the basic object of all costing is the analysis and ascertainment of the cost of each unit of

production, to control and regulate cost, and to determine profitability. Its principles hold for printing, furniture, hardware, ship-building, heavy machinery, interior decoration, repairs and similar work (SM 9.2).

Process of job costing (SM 9.2): prepare a separate cost sheet for each job; record the materials issued for the job and the employee costs incurred (on the basis of the bill of material and time cards respectively); when the job is completed, add overhead charges to ascertain the total expenditure.

Suitability (SM 9.3): when jobs are executed for different customers according to their specifications; when no two orders are alike and each needs special treatment; where work-in-progress differs from period to period with the number of jobs in hand.



Blue: materials; green: labour; amber: overheads; orange: the job cost sheet, posted weekly or monthly. The three streams run side by side in practice; the flow lists them in the order the cost sheet adds them.

Collecting materials (SM 9.4–9.5)

- Direct materials must be traced to the specific job through a **separate stores requisition** for each job or

work order. A **bill of material** is the basis for the requisitions; if the whole quantity in it is drawn in one lot or in instalments, the bill itself may serve as the requisition.

- Priced requisitions are entered in a **material abstract** (analysis book), which collects direct materials by job and indirect materials by standing order or expense code. Job totals are posted to the job cost sheets in the **Work-in-Progress ledger**, usually weekly or monthly; indirect materials go to standing orders in the **Overhead Expenses ledger**.
- **Special material** purchased for a particular job is charged direct to that job as soon as it is bought, without passing through the Stores Ledger.
- **Surplus material** that cannot immediately and economically be used on another job is returned to stores with a **stores debit note or shop credit**, and the job is credited. If it is used on another job instead, a **material transfer note** shows both jobs, and the WIP ledger is adjusted.

Collecting labour (SM 9.5)

All direct labour is booked against jobs on **job time cards** or sheets. Idle time is booked against standing order numbers, on the job time card or on a separate **idle time card**. Time booked is valued at the appropriate rates and entered in a **labour abstract**, then posted to the job cost cards weekly or monthly; idle time costs and other indirect labour go to standing orders in the Overhead Expenses ledger. Overtime premium paid because the

customer wanted the job early may be charged to that job directly (SM 3.18; idle time and overtime).

Collecting overheads (SM 9.6)

Manufacturing overheads are collected under standing order numbers, and selling and distribution overheads against cost account numbers. They are apportioned to service and production departments, service departments' costs transferred to production departments, and the production overheads applied to jobs on a realistic basis: machine hours, labour hours, percentage of direct wages, percentage of direct materials (absorption rates). Different methods give a different works overhead and so a different job cost. Because overheads cannot be traced to any particular job, the best that can be done is to apply a suitable overhead rate to each job: essentially an arbitrary method.

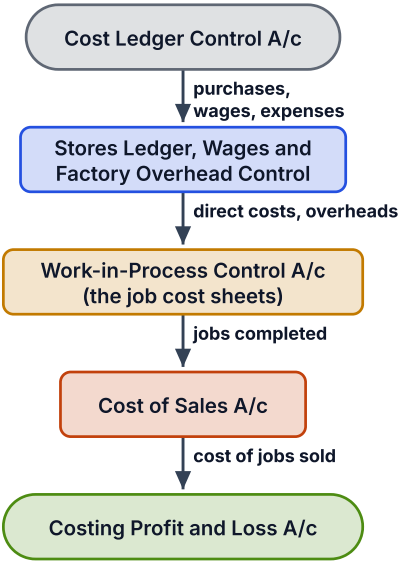
Spoiled and defective work

Spoiled work is production totally rejected that cannot be rectified; **defective work** is production not as perfect as the saleable product but capable of being rectified at additional cost (SM 9.6). The cost of rectifying a normal rate of defectives is charged to the job and spread over its output, and rectification of the defectives above normal is written off to Costing P&L. Defects from bad workmanship are an abnormal cost, written off unless recovered as a penalty from the workman (or treated as normal to the extent management provided for them).

Defects from the inspection department wrongly accepting poor incoming material are charged to that department and written off to Costing P&L (SM 9.7). Defectives inherent in the process may be charged to the jobs in which they arise, or treated as factory overheads if they cannot be identified with jobs (SM 9.10). The full circumstance-by-circumstance treatment and the spoilage journal entry are in spoilage and defectives.

Accounting: the entries in control accounts (SM 9.8–9.9)

Where a job's costs travel in the cost ledger



Each arrow is a debit to the lower account and a credit to the upper one. Sales are credited to Costing P&L against Cost Ledger Control; the balance of Costing P&L is the profit or loss.

The thirteen entries are in Formats and formulas. Integrated and non-integrated ledgers are explained in non-integrated and integrated accounts.

Advantages and disadvantages (SM 9.13)

Advantages	Disadvantages
1. The details of cost of material, labour and overhead of every job are available to control	1. Costly and laborious
2. The profitability of each job can be derived	2. A lot of clerical work, so more chance of error
3. It facilitates production planning	3. Not suitable in inflationary conditions
4. Budgetary control and standard costing can be applied	4. Previous cost records become meaningless if market conditions change
5. Spoilage and defectives can be identified and responsibility fixed	

Job costing against process costing (SM 9.13–9.14)

	Job costing	Process costing
(i)	A job is carried out or a product produced by specific orders	Production is a continuous flow and the product is homogeneous
(ii)	Costs are determined for each job	Costs are compiled on a time basis: for an accounting period, for each process or department

	Job costing	Process costing
(iii)	Each job is separate and independent of other jobs	Products lose their individual identity in the continuous flow
(iv)	Each job has a number and costs are collected against it	The unit cost of a process is an average cost for the period
(v)	Costs are computed when the job is completed, by adding all costs against it	Costs are calculated at the end of the period: total cost ÷ output of the process
(vi)	Production is not continuous and jobs differ, so more managerial attention is needed for control	Production is standardised and stable, so control is comparatively easier

Job against batch costing is set out in batch costing; process accounts in process costing.

Formats and formulas

Reproduce the job cost sheet and the control-account entries from memory.

Job cost sheet (ICAI's format, SM 9.3–9.4)

Header	
Description; Blue Print No.; Material No.; Reference No.	Job No.; Quantity; Date of delivery; Date commenced; Date finished

Date / Reference	Details	Material (₹)	Labour (₹)	Overhead (₹)
		xxx	xxx	xxx
Total		**xxx**	**xxx**	**xxx**

Summary of costs	Estimated (₹)	Actual (₹)
Direct material cost	xxx	xxx
Direct wages	xxx	xxx
Production overhead	xxx	xxx
Production cost	**xxx**	**xxx**
Administration and selling and distribution overheads	xxx	xxx
Total cost	**xxx**	**xxx**
Profit / loss	xxx	xxx
Selling price	**xxx**	**xxx**

Beside the summary: for the job; units produced; cost per unit; remarks; prepared by; checked by.

Entries in control accounts (SM 9.8–9.9)

No.	For	Entry
1	Purchase of materials	Stores Ledger Control A/c Dr. To Cost Ledger Control A/c
2	Direct materials issued to jobs	Work-in-Process Control A/c Dr. To Stores Ledger Control A/c
3	Return of direct materials from jobs	Stores Ledger Control A/c Dr. To Work-in-Process Control A/c
4	Return of materials to suppliers	Cost Ledger Control A/c Dr. To Stores Ledger Control A/c
5	Indirect materials	Factory Overhead Control A/c Dr. To Stores Ledger Control A/c
6	Wages paid	Wages Control A/c Dr. To Cost Ledger Control A/c
7	Direct wages incurred on jobs	Work-in-Process Control A/c Dr. To Wages Control A/c
8	Indirect wages	Factory Overhead Control A/c Dr. To Wages Control A/c

No.	For	Entry
9	Any indirect expense paid	Factory Overhead Control A/c Dr. To Cost Ledger Control A/c
10	Overhead charged to jobs	Work-in-Process Control A/c Dr. To Factory Overhead Control A/c
11	Total cost of jobs completed	Cost of Sales A/c Dr. To Work-in-Progress Control A/c
12	Balance of Cost of Sales transferred	Costing Profit and Loss A/c Dr. To Cost of Sales A/c
13	Sales value of jobs completed	Cost Ledger Control A/c Dr. To Costing Profit and Loss A/c

General Ledger Adjustment Account is another name of the Cost Ledger Control Account. After entry 13 the balance of Costing P&L is the profit or loss; the Cost Ledger Control balance is carried forward, and a trial balance can be drawn from the balances (SM 9.9).

∑ **Formula**

Pricing a job

$$\text{Price} = \text{Total cost} \times (1 + m_c)$$

$$\text{Price} = \frac{\text{Total cost}}{1 - m_s}$$

m_c = profit as a fraction of total cost; m_s = profit as a fraction of the selling price (20% on price is 25% on cost).

Worked examples

Worked example **October at Tapri Kiosk Works: fixed prices and the WIP account**

Tapri Kiosk Works had two jobs in progress on 1 October; their job cards showed factory cost to date of

₹1,24,000 (K-201) and ₹86,500 (K-202). October's documents show:

Document	Job	Amount (₹)
Requisition R-71	K-201	22,000
Requisition R-72	K-202	31,000
Requisition R-73	K-203	48,000
Requisition R-74	K-204	19,000
Shop credit SC-9: part of R-72 returned to stores	K-202	5,000
Material transfer note MT-4: part of R-73 sent to K-204	K-203 to K-204	6,000

Direct labour hours booked:

Job	Fabrication hours	Finishing hours
K-201	60	40
K-202	90	30
K-203	120	–
K-204	40	20

Fabrication labour is paid ₹110 an hour and finishing ₹90. Overheads are applied at ₹150 per direct labour hour in fabrication and ₹100 in finishing. A further 25

fabrication hours were lost waiting for a steel delivery that the supplier sent a week late. K-201 and K-202 were completed and delivered in October at the fixed prices agreed with the franchisees, ₹2,10,000 and ₹1,50,000. Administration and selling overheads are 8% of factory cost.

Required:

(i) Calculate October's cost of each job and the factory cost of the completed jobs.

(ii) Calculate the profit or loss on each completed job.

(iii) Prepare the Work-in-Process Control Account for October and reconcile its closing balance with the job cards.

THE WORKING

Working note 1: materials. The shop credit takes ₹5,000 off K-202; the transfer note moves ₹6,000 from K-203 to K-204 (credit K-203, debit K-204). Net materials: K-201 ₹22,000; K-202 ₹26,000; K-203 ₹42,000; K-204 ₹25,000; total ₹1,15,000 (requisitions ₹1,20,000 less ₹5,000 returned).

Working note 2: labour and overheads

Job	Labour (₹)	Overheads (₹)
K-201: 60 × ₹110 + 40 × ₹90; 60 × ₹150 + 40 × ₹100	10,200	13,000
K-202: 90 × ₹110 + 30 × ₹90; 90 × ₹150 + 30 × ₹100	12,600	16,500
K-203: 120 × ₹110; 120 × ₹150	13,200	18,000
K-204: 40 × ₹110 + 20 × ₹90; 40 × ₹150 + 20 × ₹100	6,200	8,000
Total	**42,200**	**55,500**

The 25 hours lost waiting for the late steel are abnormal idle time: their wages (25 × ₹110 = ₹2,750) go to the Costing Profit and Loss Account, not to any job.

(i) Cost of jobs

Particulars	K-201 (₹)	K-202 (₹)	K-203 (₹)	K-204 (₹)
Cost to 30 September	1,24,000	86,500	–	–
Materials (WN 1)	22,000	26,000	42,000	25,000
Labour (WN 2)	10,200	12,600	13,200	6,200
Overheads (WN 2)	13,000	16,500	18,000	8,000
Factory cost	**1,69,200**	**1,41,600**	**73,200**	**39,200**
Status	Completed	Completed	In progress	In progress

(ii) Profit or loss on completed jobs

Particulars	K-201 (₹)	K-202 (₹)
Factory cost	1,69,200	1,41,600
Administration and selling (8%)	13,536	11,328
Total cost	**1,82,736**	**1,52,928**
Fixed price	2,10,000	1,50,000
Profit / (loss)	**27,264**	**(2,928)**

K-202 was priced below its cost: with a fixed price, every hour or rupee of material above the estimate comes out of the profit.

(iii) Work-in-Process Control Account for October

Dr.	Amount (₹)
To Balance b/d	2,10,500
To Material Control A/c (₹1,20,000 – ₹5,000)	1,15,000
To Wages Control A/c	42,200
To Factory Overhead Control A/c	55,500
Total	**4,23,200**

Cr.	Amount (₹)
By Cost of Sales A/c (K-201 and K-202 at factory cost)	3,10,800
By Balance c/d	1,12,400
Total	**4,23,200**

The returned material is credited to the job and debited back to stores, so only ₹1,15,000 reaches the account; the transfer between K-203 and K-204 changes the job cards but not the control account.

Reconciliation: closing balance ₹1,12,400 = K-203 ₹73,200 + K-204 ₹39,200, the two open job cards.

Worked example

The airport tea cart, Job K-118: estimate, extras and actual cost

Tapri Kiosk Works quoted ₹1,56,800 for Job K-118, a steel tea cart for an airport franchisee. The estimate was: materials ₹80,000; welding 100 hours and carpentry 50 hours; total cost ₹1,25,440; profit 20% of the price. The contract says that changes the customer asks for after the quote are billed at their direct cost plus 25%, and overtime premium worked at the customer's request is billed at cost. The actual figures:

Particulars	Details
Direct materials issued from stores	₹86,000
Direct materials returned to stores (shop credit)	₹3,000
Glass hood bought for the job, added at the customer's request after the quote	₹18,000
Design fee for the hood, paid to an outside architect	₹4,000

Particulars	Details
Welding labour	120 hours at ₹100, of which 20 hours were overtime worked at the customer's request to deliver early; overtime premium is 50% of the normal rate
Carpentry labour	50 hours at ₹80
Factory overheads	Welding ₹150 and carpentry ₹60 per labour hour
Rectification of panels rejected at inspection, normal for this type of job	₹2,500 (materials and labour)
Administration, selling and distribution overheads	12% of works cost

> **Kesar:** The airport wanted it early and changed the design. Do we eat all of that?
>
> **Ritu:** Read the contract first; then cost the job as it was actually built.

Required:

(i) Prepare the job cost sheet of Job K-118, showing prime cost, works cost and total cost.

(ii) Calculate the amount to be billed and the profit on the job.

(iii) Reconcile the actual profit with the profit in the estimate.

THE WORKING

(i) Job cost sheet, Job K-118

Particulars	Amount (₹)	Amount (₹)
Direct materials issued	86,000	
Less: returned to stores	3,000	
Add: glass hood bought for the job	18,000	1,01,000
Direct labour: welding 120 × ₹100	12,000	
Overtime premium at the customer's request: 20 × ₹50	1,000	
Carpentry 50 × ₹80	4,000	17,000
Direct expenses: design fee		4,000
Prime cost		**1,22,000**
Factory overheads: welding 120 × ₹150 + carpentry 50 × ₹60		21,000

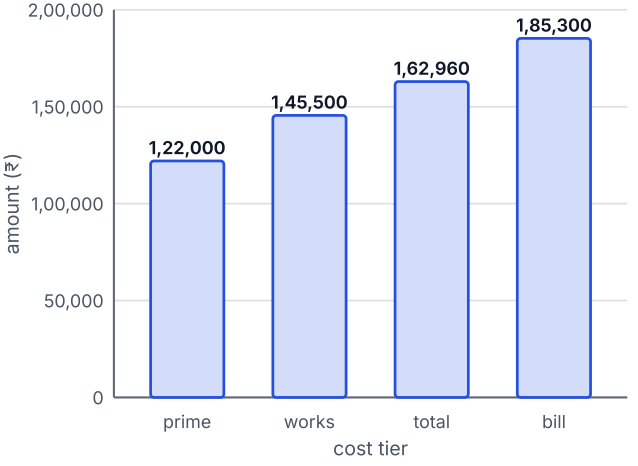
Particulars	Amount (₹)	Amount (₹)
Rectification of defective work (normal)		2,500
Works cost		**1,45,500**
Administration, selling and distribution (12%)		17,460
Total cost		**1,62,960**

The returned material is credited to the job; the hood bought for it is charged direct, without passing through stores (SM 9.5). The overtime premium was incurred at the customer's desire, so it is charged to the job directly (SM 3.18). The rectification is normal for this kind of job, so the job bears it.

(ii) Amount to be billed and profit

Particulars	Amount (₹)
Quoted price	1,56,800
Extras: hood and design fee (₹22,000 × 125%)	27,500
Extras: overtime premium at cost	1,000
Amount to be billed	**1,85,300**
Less: Total cost	1,62,960
Profit	**22,340**

Job K-118: the cost tiers and the bill



Each bar adds one layer: factory overheads and rectification, then administration, selling and distribution, then the profit in the bill.

(iii) Reconciliation of profit

Particulars	Amount (₹)
Profit in the estimate (₹1,56,800 − ₹1,25,440)	31,360
Add: margin earned on the extras: ₹28,500 billed − (₹23,000 cost + 12% overheads ₹2,760)	2,740
Less: overruns not billable: materials ₹3,000 (₹83,000 against ₹80,000), 20 extra welding hours × (₹100 + ₹150) = ₹5,000, rectification ₹2,500, and 12% overheads on them ₹1,260	(11,760)
Actual profit	**22,340**

The profit fell from 20% of the price to 12.06% of the bill because of costs the contract does not let Tapri recover: extra materials, extra welding hours and rework.

Exam angle

In the exam

How ICAI asks job costing

In the ten documents read for this lesson:

- **Job cost sheet and price, 6–7 marks**: MTP September 2026 Series I Q2(a) (materials returned,

direct expenses, rectification of normal defectives charged to the job, administration on works cost, selling on cost of production, 20% on price; part (iii) the abnormal case, 7 marks); MTP May 2025 Series II Q4(a) (6 marks) and RTP January 2026 Q13 (RTPs carry no marks): last year's rates, a new job at the same profit on sales with revised overheads; PYQ September 2026 Q2(b) (an aircraft component order at last year's rates plus a 10% safety margin, 7 marks); RTP May 2025 Q13 (departmental overheads from last year's profit and loss account).

- **Case-scenario MCQ**: RTP September 2026 case scenario, full production cost of Job AMS-101.
- **Theory, 3–5 marks**: PYQ September 2025 Q6(b), "List the advantages of Job Costing" (5 marks; the answer was ICAI's five points word for word); MTP January 2026 Series I Q2(b), "Distinguish between Job Costing and Process Costing" (4 marks); PYQ September 2024 Q6(c), "Define Job costing and explain differences between job and batch costing" (4 marks, the CIMA definition first); PYQ May 2025 Q5(b), job against batch costing (3 marks).
- The spoiled and defective work table (SM 9.7) was set as 4-mark theory in PYQ January 2025 Q5(c), PYQ May 2026 (an OR option) and MTP September 2025 Series II Q6(c); it is taught in spoilage and defectives.

In the exam

Traps

- A shop credit removes the material from the job; a transfer note moves it to the other job. Neither stays where it was first requisitioned.
- Special material bought for a job goes straight to the job, not through stores.
- Waiting time, breakdowns and idle time are indirect labour, booked to standing orders: do not add them to a job. ICAI's answer to SM practical problem 1 assumes they are covered by the overhead rate.
- "Same rate as last month" means the percentage computed from last month's abstract, applied to this month's labour only.

Trick

The three streams test

Before adding up a job, check that every figure came through one of three doors: a requisition (materials), a time card (labour) or a pre-determined rate (overheads). Anything else (special purchases, direct expenses, customer-requested overtime) must be traceable to the job by its own document. Condition: abnormal items never enter the job.

Answer frame

"Describe job costing, giving examples of industries" (SM question 1):

1. CIMA definition, in quotation marks.
2. Costs collected by jobs, contracts, products or work orders; each job a separate entity; also applies to batches.
3. Process: a separate cost sheet for each job; materials and labour from the bill of material and time cards; overheads added on completion.
4. Suitability: customers' own specifications; no two orders alike; WIP varies with jobs in hand.
5. Industries: printing, furniture, hardware, ship-building, heavy machinery, interior decoration, repairs.

"List the advantages of job costing" (PYQ September 2025 Q6(b), 5 marks): ICAI's five, in order: cost details of every job for control; profitability of each job; production planning; budgetary control and standard costing can be applied; spoilage and defectives identified and responsibility fixed.

"Distinguish between job costing and process costing" (4 marks): the six points of the compare table, each as a pair, in ICAI's order.

Three answers to the October question

Loses marks

K-202 charged with the whole of R-72 (₹31,000) and K-203 with all of R-73, the waiting hours added to the jobs.

The documents were not read.

Passes

Returns and transfer handled and the job costs right, but no Work-in-Process Control Account, so the closing balance is never checked against the open job cards.

Right costs, half the answer.

Full marks

Working notes for the materials, labour and overheads; a job-wise factory cost table as ICAI lays it out; the profit on each fixed price; the control account reconciled with the job cards; the treatment of the waiting hours stated.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the material that was charged twice

Charging R-73's ₹6,000 to K-203 as requisitioned and again to K-204 after the transfer. **Fix:** a material transfer note moves the cost: credit the transferor job, debit the transferee job; a shop credit returns it to stores.

⚠ **Common mistake**

Waiting time on the customer's bill

Adding the 25 hours lost waiting for the late steel (₹2,750) to the jobs. **Fix:** abnormal idle time goes to the Costing Profit and Loss Account; normal idle and waiting time, booked to standing orders, is factory overhead already covered by the overhead rate.

⚠ **Common mistake**

Overtime premium in the overhead pool

Spreading the airport cart's ₹1,000 premium over all jobs. **Fix:** overtime at the customer's desire is charged to that job directly. Overtime to cope with general production programmes or urgent orders is overhead of the department that works it; overtime caused by another department's fault is charged to that department; overtime from abnormal conditions goes to Costing P&L (SM 3.18).

⚠ **Common mistake**

The abnormal defect the customer paid for

Charging rectification from a machine fault to the job. **Fix:** abnormal rectification goes to Costing P&L; only the normal rate of defectives is charged to the job.

Practice

Check yourself

1 A job's direct materials are ₹50,000 and direct labour ₹20,000. Factory overheads are 80% of direct labour, administration overheads 10% of factory cost, and the company wants a profit of 20% on the selling price. What price should it quote (₹)?

Answer: _____

2 Case scenario. At Tapri Kiosk Works, steel sheets issued to Job K-104 are not needed there, and the shop supervisor sends them straight to Job K-105 on the next bay instead of returning them to stores. Which document must record this?

A A shop credit (stores debit note)

B A material transfer note

C A fresh bill of material

D A material requisition for K-105

3 Which journal entry records the total cost of jobs completed?

A Work-in-Process Control A/c Dr. To Cost of Sales A/c

B Cost of Sales A/c Dr. To Work-in-Progress Control A/c

C Costing Profit and Loss A/c Dr. To Work-in-Process Control A/c

D Finished Goods A/c Dr. To Factory Overhead Control A/c

4 Which of these is a feature of job costing, not process costing?

- **A** The unit cost is an average cost for the period
- **B** Costs are compiled for each process for an accounting period
- **C** Costs are computed when the job is completed, by adding all costs against its number
- **D** Products lose their individual identity in a continuous flow

5 Which are among ICAI's advantages of job costing? Select all that apply.

- **A** The profitability of each job can be derived
- **B** It facilitates production planning
- **C** It is economical and needs little clerical work
- **D** Spoilage and defectives can be identified and responsibility fixed

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Job costing (CIMA): where work consists of separate contracts, jobs or batches, each authorised by a specific order or contract; each job a separate cost unit.
- Suits customers' own specifications, orders that are never alike, WIP varying with jobs in hand: printing, furniture, ship-building, interior decoration, repairs.
- Materials: a stores requisition per job (or the bill of material); special purchases direct; returns by shop credit; transfers by material transfer note.
- Labour: job time cards; idle time to standing orders. Overheads: pre-determined rates, "essentially an arbitrary method".
- Thirteen entries: purchases, issues, returns, wages, overheads into WIP Control; completed jobs to Cost of Sales; then to Costing P&L; sales against Cost Ledger Control.
- Normal rectification is charged to the job; abnormal goes to Costing P&L; customer-requested overtime premium goes to the job.
- Job against process: specific orders vs continuous flow; cost per job on completion vs average cost per period. (A kiosk is never just a kiosk.)

Answer key · Check yourself

1 **118250**

Factory cost = ₹50,000 + ₹20,000 + ₹16,000 = ₹86,000; total cost = ₹86,000 × 1.10 = ₹94,600; price = ₹94,600 ÷ 0.80 = ₹1,18,250.

2 **B** **A material transfer note**

SM 9.5: surplus used on another job without returning to stores is recorded on a material transfer note naming both jobs, so the WIP ledger is adjusted. A shop credit is for material returned to stores.

3 **B** **Cost of Sales A/c Dr. To Work-in-Progress Control A/c**

SM 9.9, entry 11. The balance of Cost of Sales then goes to Costing P&L (entry 12).

4 **C** **Costs are computed when the job is completed, by adding all costs against its number**

SM 9.13–9.14: the other three describe process costing.

5 **A** The profitability of each job can be derived; **B** It facilitates production planning; **D** Spoilage and defectives can be identified and responsibility fixed

SM 9.13. Job costing is costly and laborious, with a lot of clerical work: those are its disadvantages.

CHAPTER 10

Process and Operation Costing

10.1 Process accounts, normal loss, abnormal loss and abnormal gain

How costs are charged to process accounts and averaged over units, why normal loss raises the cost of good units, and how abnormal loss and gain are valued and closed.

In plain words

10,000 kg in, 9,300 kg out, and one very annoyed founder

Tapri's chai premix begins in a roasting drum in Indore. Tea leaves go in, roasted leaves come out and move on to blending. Every month, less comes out than went in.

> **Kesar:** We fed 10,000 kg into the roaster and only 9,300 kg came out. Who is eating my tea?
>
> **Ritu:** The drum, mostly. Roasting dries the leaves and the sieve takes out dust: about 5% every month, so 500 kg is expected. The other 200 kg went on the floor when the conveyor jammed on the 14th.
>
> **Kesar:** Then put all 700 kg into the cost per kg. Sab adjust ho jayega.
>
> **Ritu:** The 500 kg, yes. The 200 kg, no. A jammed conveyor is not a cost of roasting tea.

A **process** is one stage of production (roasting, blending, packing). All its cost is collected in its **process account** and averaged over the units it produces; its output is the next process's input.

The 500 kg that roasting always loses is **normal loss**: part of making roasted tea, so the good kilograms carry its cost. Dhabas buy the dust at ₹20 a kg, and that sale reduces the cost.

The 200 kg beyond the expected loss is **abnormal loss**: valued like a good kilogram and sent to the Costing Profit and Loss Account, so a jammed conveyor does not quietly raise the cost of every packet. In a month when only 400 kg is lost, the 100 kg Tapri did not expect to keep is **abnormal gain**, valued the same way and credited to that account.

Think of it like

The chai pan

A litre of chai gives 900 ml in cups; the rest is steam. Each cup's cost includes that steam: normal loss. If the pan boils over while you are on the phone, the 200 ml on the stove is the cost of the phone call, not of chai (abnormal loss). If a lid one day gets you 950 ml, the extra 50 ml is the lid's credit (abnormal gain). Tomorrow's cup is still costed on the normal 900 ml.

Key idea

Cost is averaged over the **normal output** (input minus normal loss). Good units, abnormal loss and abnormal gain are all valued at that one normal cost per unit. Only the normal loss is absorbed by the good units, and only its scrap value comes back to the process.

How it works

Where ICAI uses process costing

Process costing is "a method of Cost Accounting whereby costs are charged to processes or operations and averaged over units produced" (SM 10.2), with a separate account for each process. ICAI's examples: steel, paper, medicines, soaps, chemicals, rubber, vegetable oil, paints and varnish.

Industries using it normally have one or more of these **features** (SM 10.2, in ICAI's order):

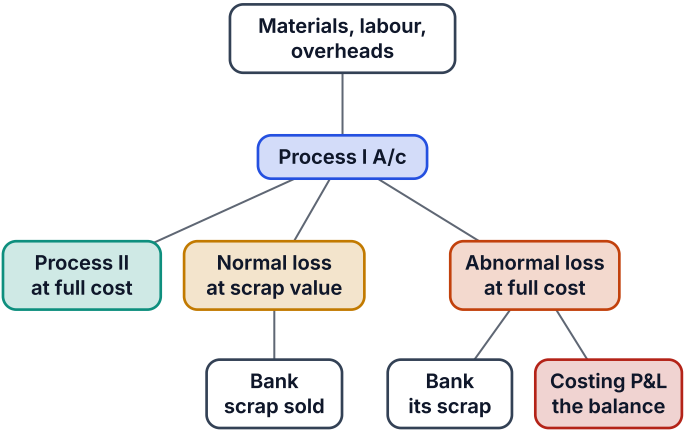
1. Each plant or factory is divided into a number of processes, cost centres or departments, and each such division is a stage of production or a process.
2. Manufacturing is carried on continuously by one or more processes run sequentially, selectively or simultaneously.
3. The output of one process becomes the input of another process.

4. The end product usually is of like units, not distinguishable from one another.
5. The identity of any particular lot of output cannot be traced to any lot of input materials (no bag of sugar can be traced to a particular lot of sugarcane).
6. Production may give rise to joint products and by-products.

What a process account collects

Each process is debited with its **materials** (from the second process on, the cost transferred from the previous process plus any material added), its **employee cost**, its **direct expenses** (depreciation, repairs, maintenance, insurance of its own plant) and its **production overheads** (rent, power, lighting, gas, water), apportioned on a suitable basis, usually at a pre-determined rate (SM 10.3).

Where a process account's cost comes from and where it goes



Good units go on to Process II (or finished stock) at full cost. Normal loss leaves only at its scrap value, which the bank receives. Abnormal loss leaves at the full cost of a good unit: its scrap goes to the bank, the rest to the Costing P&L.

Five steps for any process account with losses

The roaster in June: 10,000 kg of tea at ₹4,00,000, wages ₹1,25,000, overheads ₹1,02,500 (total ₹6,27,500); normal loss 5% of input, sold at ₹20 a kg; output 9,300 kg.

1 **Normal loss in units** = input × normal loss %: 10,000 × 5% = 500 kg. Its **value** is the scrap's realisable value less any cost of selling it: 500 × ₹20 = ₹10,000.

2 **Normal output** = input − normal loss = 9,500 kg. Actual output below it is **abnormal loss** (9,500 − 9,300 = 200 kg); above it, **abnormal gain**.

3 **Normal cost per unit** = (total cost − scrap value of normal loss) ÷ normal output = (₹6,27,500 − ₹10,000) ÷ 9,500 = ₹65 a kg.

4 **Value** the output and the abnormal units at that one rate: 9,300 × ₹65 = ₹6,04,500 to blending; 200 × ₹65 = ₹13,000 abnormal loss. With the ₹10,000 of normal loss, the credits total ₹6,27,500.

5 **Close the loss accounts.** The dust sale clears the normal loss account. The abnormal loss account is credited with its own 200 kg of scrap (₹4,000) and sends the balance, ₹9,000, to the Costing P&L.

Interactive version: PassBook app, lesson Process accounts, normal loss, abnormal loss and abnormal gain

July: the month the roaster behaved

Same input and costs, but 9,600 kg came out: 400 kg lost against 500 kg expected. The extra 100 kg is **abnormal gain**, at the same ₹65: "the cost of abnormal gain is computed on the basis of normal production" (SM 10.9).

- The process account is **debited** with the abnormal gain, 100 kg, ₹6,500; blending receives 9,600 × ₹65 =

₹6,24,000.

- The process is still credited with the **normal** loss, 500 kg × ₹20 = ₹10,000, although only 400 kg of dust exists: normal loss is valued on the estimated figure (SM 10.10).
- The **abnormal gain account** gives up the dust that was never there to sell: it is debited with 100 kg × ₹20 = ₹2,000 (to the normal loss account), and only the balance, ₹4,500, is credited to the Costing P&L. "The Costing P&L Account is credited only for actual gain amount" (SM 10.11). The normal loss account's ₹10,000 is then cleared by that ₹2,000 and ₹8,000 from the 400 kg of dust sold.

The roaster in June and July, in kilograms

June: 10,000 kg in

Normal loss

kg:	500
scrap:	₹10,000

Normal output @ ₹65

To blending

kg:	9,300
₹:	6,04,500

Abnormal loss

kg:	200
₹:	13,000

July: 10,000 kg in

Expected loss 500 kg

Dust

kg:	400
sold:	₹8,000

Abnormal gain

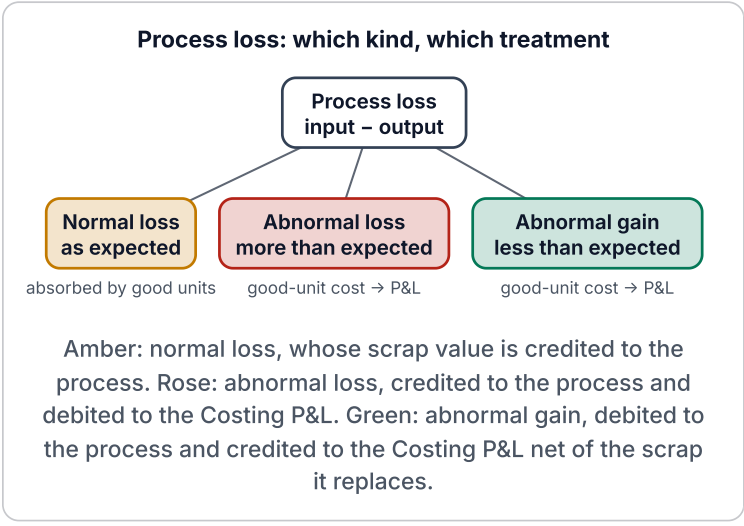
kg:	100
₹:	6,500

Normal output @ ₹65

kg:	9,500
₹:	6,17,500

June: abnormal loss is carved out of the normal output, so it costs what a good kilogram costs. July: abnormal gain is carved out of the expected loss; it leaves the drum as good tea, so blending receives 9,500 + 100 = 9,600 kg at ₹65.

Three kinds of loss, three treatments



Amber: normal loss, whose scrap value is credited to the process. Rose: abnormal loss, credited to the process and debited to the Costing P&L. Green: abnormal gain, debited to the process and credited to the Costing P&L net of the scrap it replaces.

Formats and formulas

ICAI writes each account as a T-account, debit side on the left and credit side on the right, with a units column beside every amount column. On a phone the credit side drops below the debit side, and in the worked examples the longer process accounts always show it below; on paper, write them side by side. *Reproduce these formats from memory.*

Process I Account

Dr. Particulars	Units	Amount (₹)
To Materials (in Process II: To Process I A/c)	xxx	xxx
To Direct wages	–	xxx
To Direct expenses	–	xxx
To Production overheads	–	xxx
To Abnormal gain A/c (units × normal cost)	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Normal loss A/c (normal units × scrap rate)	xxx	xxx
By Abnormal loss A/c (units × normal cost)	xxx	xxx
By Process II A/c (units transferred × normal cost)	xxx	xxx
Total	**xxx**	**xxx**

Units and rupees must both balance. Only one of the two abnormal lines appears in a month.

Abnormal Loss Account

Dr. Particulars	Units	Amount (₹)
To Process I A/c	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Bank (abnormal units × scrap rate)	xxx	xxx
By Costing P&L A/c (balance)	–	xxx
Total	**xxx**	**xxx**

Abnormal Gain Account

Dr. Particulars	Units	Amount (₹)
To Normal Loss A/c (gain units × scrap rate)	xxx	xxx
To Costing P&L A/c (balance)	–	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Process I A/c	xxx	xxx
Total	**xxx**	**xxx**

Normal Loss Account

Dr. Particulars	Units	Amount (₹)
To Process I A/c (normal loss units × scrap rate)	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Abnormal Gain A/c (gain units × scrap rate), if any	xxx	xxx
By Bank or Cost Ledger Control A/c (scrap actually sold)	xxx	xxx
Total	**xxx**	**xxx**

∑ **Formula**

Normal cost per unit

$$\text{Normal output} = \text{Input units} - \text{Normal loss units}$$

$$\text{Cost per unit} = \frac{\text{Total cost} - \text{Scrap value}}{\text{Normal output}}$$

Scrap value is that of the normal loss. ICAI's words: (Total cost − Realisable value of normal loss) ÷ (Total input units − Normal loss units). The rate values units transferred, abnormal loss and abnormal gain (SM 10.7–10.10).

∑ **Formula**

Value of normal loss

$$\begin{aligned}\text{Value of normal loss} &= \text{Normal loss units} \\ &\times (\text{Realisable value} - \text{Cost to sell})\end{aligned}$$

ICAI: "Scrap realisable value less cost to sale" (SM 10.6).

∑ **Formula**

Abnormal units

$$\begin{aligned}&\text{Abnormal loss (gain)} \\ &= \text{Normal output} - \text{Actual output}\end{aligned}$$

Positive: abnormal loss; negative: abnormal gain.

∑ **Formula**

What reaches the Costing P&L

$$\begin{aligned}\text{Net amount} &= \text{Abnormal units} \\ &\times (\text{Cost per unit} - \text{Scrap rate})\end{aligned}$$

A debit to the Costing P&L for an abnormal loss, a credit for an abnormal gain (ICAI's income statement, SM 10.15).

Definition

Process loss

The loss of material arising during the course of a processing operation, equal to the difference between the input quantity of the material and its output. (SM 10.5)

Definition

Normal process loss

The loss of material which is inherent in the nature of work. It can be reasonably anticipated from the nature of the material, nature of operation, experience and technical data; it is unavoidable because of the nature of the material or the process. It also includes units withdrawn from the process for test or sampling. (SM 10.6)

Definition

Abnormal process loss

The loss in excess of the pre-determined loss (normal process loss). It may occur due to carelessness of workers, a bad plant design or operation, sabotage etc.; it cannot be estimated in advance, but it can be kept under control by taking suitable measures. (SM 10.8)

Definition

Abnormal gain

An unexpected gain in production under normal conditions: the difference between actual and expected loss, or actual and expected production, when the loss is less than the normal figure. It arises from over-estimation of process loss, improved efficiency of workers, better technology etc. (SM 10.9)

Worked examples

Worked example

The kulhad kiln and the 300 extra cracks

Tapri fires its own clay kulhads. In October, 25,000 raw kulhads were loaded into the kiln. Costs charged to the Kiln process were: clay and glaze ₹2,00,000, kiln workers' wages ₹1,10,000, fuel and kiln overheads ₹99,000. Normal breakage is 4% of input, and broken pieces are sold to a brick maker at ₹1 each. 23,700 good kulhads were transferred to the Packing process. There was no opening or closing stock.

Required:

(i) Prepare the Kiln Process Account.

(ii) Prepare the Normal Loss Account and the Abnormal Loss Account.

(iii) Kesar wants the extra breakage "adjusted in the cost of the good kulhads". Calculate the cost per kulhad on his basis, and state why cost accounts do not follow it.

THE WORKING

Working notes

1. Normal loss = 4% × 25,000 = 1,000 kulhads; scrap value 1,000 × ₹1 = ₹1,000.
2. Normal output = 25,000 − 1,000 = 24,000. Actual output 23,700, so abnormal loss = 300 kulhads.
3. Total cost = ₹2,00,000 + ₹1,10,000 + ₹99,000 = ₹4,09,000.
4. Cost per kulhad = (₹4,09,000 − ₹1,000) ÷ 24,000 = **₹17**.

(i) Kiln Process Account (debit side)

Dr. Particulars	Units	Amount (₹)
To Clay and glaze	25,000	2,00,000
To Wages	–	1,10,000
To Fuel and kiln overheads	–	99,000
Total	**25,000**	**4,09,000**

Kiln Process Account (credit side)

Cr. Particulars	Units	Amount (₹)
By Normal loss A/c (1,000 × ₹1)	1,000	1,000
By Abnormal loss A/c (300 × ₹17)	300	5,100
By Packing process A/c (23,700 × ₹17)	23,700	4,02,900
Total	**25,000**	**4,09,000**

(ii) Normal Loss Account

Dr.	Units	(₹)
To Kiln Process A/c	1,000	1,000
Total	**1,000**	**1,000**

Cr.	Units	(₹)
By Bank (broken pieces sold)	1,000	1,000
Total	**1,000**	**1,000**

Abnormal Loss Account

Dr.	Units	(₹)
To Kiln Process A/c	300	5,100
Total	**300**	**5,100**

Cr.	Units	(₹)
By Bank (300 × ₹1)	300	300
By Costing P&L A/c	–	4,800
Total	**300**	**5,100**

(iii) Kesar's basis. All 1,300 broken kulhads treated as normal: (₹4,09,000 − ₹1,300) ÷ 23,700 = **₹17.2025** a kulhad. The extra ₹0.2025 on 23,700 kulhads is ₹4,800, exactly the net abnormal loss, now hidden in Packing and in stock.

The cost of an abnormal process loss is not treated as a part of the cost of the product; it is debited to the Costing Profit and Loss Account. Charging it to good units would overstate their cost and stock, and hide an avoidable loss that management must investigate and control.

Worked example

Samosas through two processes: a good month in Shaping, a bad one in Frying

Tapri's samosa factory makes frozen cocktail samosas in two processes. In **Shaping** (Process I) dough and filling are shaped into raw samosas; torn samosas are sold back as dough scrap. In **Frying** (Process II) the samosas are fried and blast-frozen; samosas that break in the fryer are sold to a namkeen maker as crumbs. Fried samosas go to finished stock. For December:

Particulars	Shaping	Frying
Dough and filling for 50,000 samosas	₹3,25,000	–
Frying oil and gas	–	₹1,26,100
Direct wages	₹1,11,000	₹1,05,000
Machine hours	300	600
Normal loss (on input)	4%	5%
Scrap value per samosa lost	₹2	₹4
Output (samosas)	48,400	45,500

Factory overheads of ₹2,16,000 for the month are apportioned to the two processes on machine hours. Shaping's output all goes to Frying; frying oil adds no units. 44,000 samosas were sold to Tapri's cafés at ₹28

each. There was no work-in-process and no opening finished stock.

Required:

(i) Prepare the Shaping Process Account and the Frying Process Account.

(ii) Prepare the Abnormal Gain Account, the Abnormal Loss Account and the Normal Loss Account of Shaping.

(iii) Prepare the Finished Stock Account and an Income Statement.

(iv) Kesar says the extra samosas in Shaping and the broken ones in Frying "more or less cancel out". Calculate the net effect of the two abnormal items on profit, and explain why it is not nil.

THE WORKING

Working notes

Particulars	Shaping	Frying
Input (samosas)	50,000	48,400
Normal loss	2,000	2,420
Normal output	48,000	45,980
Actual output	48,400	45,500
Abnormal (gain) / loss	(400)	480
Scrap value of normal loss (₹)	4,000	9,680

1. Factory overheads: ₹2,16,000 ÷ 900 machine hours = ₹240 an hour; Shaping 300 × ₹240 = ₹72,000; Frying 600 × ₹240 = ₹1,44,000.
2. Shaping cost = ₹3,25,000 + ₹1,11,000 + ₹72,000 = ₹5,08,000. Cost per samosa = (₹5,08,000 − ₹4,000) ÷ 48,000 = **₹10.50**.
3. Frying cost = ₹5,08,200 (from Shaping) + ₹1,26,100 + ₹1,05,000 + ₹1,44,000 = ₹8,83,300. Cost per samosa = (₹8,83,300 − ₹9,680) ÷ 45,980 = **₹19.00**.

(i) Shaping Process Account (debit side)

Dr. Particulars	Units	Amount (₹)
To Dough and filling	50,000	3,25,000
To Direct wages	–	1,11,000
To Factory overheads (300 × ₹240)	–	72,000
To Abnormal gain A/c (400 × ₹10.50)	400	4,200
Total	**50,400**	**5,12,200**

Shaping Process Account (credit side)

Cr. Particulars	Units	Amount (₹)
By Normal loss A/c (2,000 × ₹2)	2,000	4,000
By Frying process A/c (48,400 × ₹10.50)	48,400	5,08,200
Total	**50,400**	**5,12,200**

Frying Process Account (debit side)

Dr. Particulars	Units	Amount (₹)
To Shaping process A/c	48,400	5,08,200
To Frying oil and gas	–	1,26,100
To Direct wages	–	1,05,000
To Factory overheads (600 × ₹240)	–	1,44,000
Total	**48,400**	**8,83,300**

Frying Process Account (credit side)

Cr. Particulars	Units	Amount (₹)
By Normal loss A/c (2,420 × ₹4)	2,420	9,680
By Abnormal loss A/c (480 × ₹19)	480	9,120
By Finished stock A/c (45,500 × ₹19)	45,500	8,64,500
Total	**48,400**	**8,83,300**

(ii) Abnormal Gain Account

Dr.	Units	(₹)
To Normal loss A/c, Shaping (400 × ₹2)	400	800
To Income statement (Costing P&L)	–	3,400
Total	**400**	**4,200**

Cr.	Units	(₹)
By Shaping process A/c	400	4,200
Total	**400**	**4,200**

Abnormal Loss Account

Dr.	Units	(₹)
To Frying process A/c	480	9,120
Total	**480**	**9,120**

Cr.	Units	(₹)
By Bank (480 × ₹4)	480	1,920
By Income statement (Costing P&L)	–	7,200
Total	**480**	**9,120**

Normal Loss Account (Shaping)

Dr.	Units	(₹)
To Shaping process A/c	2,000	4,000
Total	**2,000**	**4,000**

Cr.	Units	(₹)
By Abnormal gain A/c	400	800
By Bank (torn samosas sold)	1,600	3,200
Total	**2,000**	**4,000**

Only 1,600 samosas tore, so only ₹3,200 of dough scrap was sold; the abnormal gain account makes good the other ₹800. Frying's normal loss account

(₹9,680) is cleared by the sale of its 2,420 normal broken samosas.

(iii) Finished Stock Account

Dr.	Units	(₹)
To Frying process A/c	45,500	8,64,500
Total	**45,500**	**8,64,500**

Cr.	Units	(₹)
By Cost of sales (44,000 × ₹19)	44,000	8,36,000
By Balance c/d (1,500 × ₹19)	1,500	28,500
Total	**45,500**	**8,64,500**

Income Statement

Dr. Particulars	Amount (₹)
To Cost of sales	8,36,000
To Abnormal loss {480 × (₹19 − ₹4)}	7,200
To Net profit	3,92,200
Total	**12,35,400**

Cr. Particulars	Amount (₹)
By Sales (44,000 × ₹28)	12,32,000
By Abnormal gain {400 × (₹10.50 − ₹2)}	3,400
Total	**12,35,400**

(iv) Net effect. Abnormal gain credited ₹3,400 less abnormal loss debited ₹7,200 = **₹3,800 less profit**. It is not nil because each abnormal unit is valued at the normal cost per unit of the process in which it arises, less that process's scrap value: a Shaping gain unit is worth ₹10.50 − ₹2 = ₹8.50, while a Frying loss unit has absorbed Shaping's cost and Frying's cost and is worth ₹19 − ₹4 = ₹15 (and 480 units were lost against 400 gained). Each item is

shown separately so that the process responsible for the abnormal loss can be investigated; netting them would hide it.

Interactive version: PassBook app, lesson Process accounts, normal loss, abnormal loss and abnormal gain

Exam angle

In the exam

Where process losses appeared in the documents read for this lesson

- **Descriptive, 5 to 10 marks.** Process account and income statement with an accident loss in rupees and containers: January 2026 Q2(b) (5 marks). Two processes, part of the output sold, abnormal loss and gain accounts and the Costing P&L: MTP Series I for May 2025 Q3(a) (10 marks); the same pattern over three processes in RTP September 2025 Q13. Process I and the abnormal gain account: December 2021 (old scheme) Q1(d) (5 marks).
- **Multiple choice, 2 marks each.** Value transferred to Process II: September 2025 MCQ 14. Normal loss % and abnormal gain worked backwards: May 2025 MCQ 15. Abnormal gain or loss in Process II: RTP May 2025 MCQ 5. Abnormal loss in a FIFO case: September 2024 MCQ 3.
- **Theory, 5 marks.** "How will you treat normal loss, abnormal loss and abnormal gain in process costing?

Explain.": May 2023 (old scheme) Q6(e).

In the exam

The traps ICAI set

- **Work it backwards** (May 2025 MCQ 15): normal loss = input − output + abnormal gain; then its percentage of input.
- **An accident given in rupees** (January 2026 Q2(b)): "other expenses ₹37,350 including an abnormal loss of ₹12,500 due to an accident". The suggested answer charged only ₹24,850 to the process and put the ₹12,500 straight into the income statement. Oil and containers each had a normal and an abnormal part, in their own columns.
- **Part of the output sold, part transferred** (MTP May 2025 Q3(a), RTP September 2025 Q13): both at the same cost per unit.
- The paper's own instruction: "Working notes should form part of the answer."

Interactive version: PassBook app, lesson Process accounts, normal loss, abnormal loss and abnormal gain

Trick

Three checks before you move on

Units balance; rupees balance; good units, abnormal loss and abnormal gain carry the same cost per unit. When one fails, the normal loss is usually on the wrong base or at actual instead of normal units. Holds for any process account, with or without scrap value.

Answer frame

Theory: "How will you treat normal loss, abnormal loss and abnormal gain in process costing?"

ICAI's suggested answer (May 2023 Q6(e), old scheme) is the Study Material's three treatment paragraphs, in this order:

1. **Normal loss.** The cost of normal process loss is absorbed by good units produced under the process. The amount realised by the sale of normal process loss units is credited to the process account.
2. **Abnormal loss.** The cost of an abnormal process loss unit is equal to the cost of a good unit. The total cost of abnormal process loss is credited to the process account from which it arises; it is not treated as a part of the cost of the product, and is debited to the Costing Profit and Loss Account.
3. **Abnormal gain.** The process account under which abnormal gain arises is debited with the abnormal gain and credited to the abnormal gain account, which

is closed by transfer to the Costing Profit and Loss Account. The cost of abnormal gain is computed on the basis of normal production.

Open each heading with one line on the loss's meaning (normal: inherent, anticipated; abnormal: beyond the pre-determined loss, controllable). For "What is process costing? State its features.": ICAI's definition, the six features in ICAI's order, three industries.

The numerical parts: three answers to the same question

Loses marks

A process account with "By Abnormal loss 300, ₹5,100" and no working: if ₹17 is wrong, nothing earns method marks.

Passes

Working notes and the process account, but no abnormal loss account: the ₹4,800 to the Costing P&L never appears.

Full marks

Working notes, the process account with units and amounts on both sides, and the loss and gain accounts closed to bank and the Costing P&L.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the abnormal loss that went to the Costing P&L at the wrong rate

Valuing the roaster's 200 kg at ₹6,27,500 ÷ 10,000 = ₹62.75 (the average over all input) or at the ₹20 scrap rate. **Fix:** abnormal loss costs what a good unit costs, (total cost − scrap value of normal loss) ÷ normal output = ₹65. The scrap rate enters later, in the abnormal loss account.

⚠ **Common mistake**

The abnormal gain that forgot the dust it replaced

Crediting the full ₹6,500 of July's abnormal gain to the Costing P&L. **Fix:** the abnormal gain account is debited with the scrap value of the gain units (100 × ₹20 = ₹2,000, to the normal loss account); only ₹4,500 reaches the Costing P&L.

⚠ **Common mistake**

Normal loss credited at the actual loss

In July, crediting the process with 400 kg × ₹20 = ₹8,000 because only 400 kg was lost. **Fix:** the process account always credits the **normal** loss, 500 kg × ₹20 = ₹10,000; the abnormal gain account handles the 100 kg difference.

Practice

Check yourself

1 A Tapri blending process received 4,000 kg costing ₹2,47,200 in all. Normal loss is 5% of input. The dust sells at ₹18 a kg, but carting it to the buyer costs ₹2 a kg. The process lost 300 kg in all this month. What value is transferred to the next process, to the nearest rupee?

Answer: _____

2 Case scenario. Ritu reviews the Packing process for March: 12,000 units were input at a total cost of ₹5,78,400. Normal loss is 4% of input, saleable at ₹5 a unit. Output was 11,700 units. Kesar asks what this process puts into the Costing Profit and Loss Account for March. Which is correct?

- **A** ₹9,000 credited to the Costing P&L
- **B** ₹8,100 credited to the Costing P&L
- **C** ₹900 debited to the Costing P&L
- **D** ₹8,100 debited to the Costing P&L

3 In a month with an abnormal gain, at what quantity is the normal loss credited to the process account?

- **A** The actual loss units
- **B** The normal (expected) loss units
- **C** The normal loss units less the abnormal gain units, at good-unit cost
- **D** No normal loss is credited when there is an abnormal gain

4 Which of these are credited to the Abnormal Loss Account? Select all that apply.

- **A** The scrap value realised from the abnormal loss units
- **B** The cost of the abnormal loss units at the normal cost per unit
- **C** The balance transferred to the Costing Profit and Loss Account
- **D** The scrap value of the normal loss units

More than one answer may be right.

5 Units withdrawn from a process for testing or sampling are treated as which of the following?

- **A** Normal process loss
- **B** Abnormal process loss
- **C** Abnormal gain
- **D** A by-product

Answers: the answer key at the end of this lesson.

Recap

- Process costing: costs charged to processes and averaged over units produced.
- Normal loss = input × normal %; its scrap value (realisable value less cost to sell) is credited to the process.
- Cost per unit = (total cost − scrap value of normal loss) ÷ (input − normal loss units).
- Abnormal loss (gain) = normal output − actual output (or the reverse), valued at the same cost per unit.
- Abnormal loss: credited to the process; scrap credited in the abnormal loss account; balance debited to the Costing P&L.
- Abnormal gain: debited to the process; the abnormal gain account gives up the scrap value of the gain units; balance credited to the Costing P&L.

- The process is credited with the normal loss on the expected figure, even in a month of abnormal gain.
- Units and rupees must both balance. (Steam is a cost of chai; a boil-over is not.)

Answer key · Check yourself

1 **237578 to 237580**

Value of normal loss = 200 kg × (₹18 − ₹2) = ₹3,200: realisable value less cost to sell (SM 10.6). Cost per kg = (₹2,47,200 − ₹3,200) ÷ 3,800 = ₹64.2105. Output = 4,000 − 300 = 3,700 kg, so ₹2,37,579 is transferred; the other 100 kg lost are abnormal loss of ₹6,421. Valuing the dust at ₹18 gives ₹2,37,189.

2 **B** **₹8,100 credited to the Costing P&L**

Normal loss 480 units (scrap ₹2,400); cost per unit = ₹5,76,000 ÷ 11,520 = ₹50. Output beats the normal 11,520 by 180 units: abnormal gain ₹9,000, less the ₹900 of scrap those units would have fetched. ₹8,100 is credited; ₹9,000 forgets the scrap.

3 **B** **The normal (expected) loss units**

Normal loss is valued on the estimated figure even when the actual loss is less (SM 10.10); the abnormal gain account adjusts the difference.

4 **A The scrap value realised from the abnormal loss units; C The balance transferred to the Costing Profit and Loss Account**

Debited with the abnormal loss at good-unit cost; credited with its scrap and the balance to the Costing P&L. Normal loss scrap goes to the process account.

5 **A Normal process loss**

ICAI's definition of normal process loss "also includes units withdrawn from the process for test or sampling" (SM 10.6).

10.2 Equivalent units: FIFO and weighted average

How partly finished units become equivalent finished units for each cost element, and how the FIFO and weighted average methods value output, losses and work-in-process.

In plain words

5,000 kg of achaar that is sort of finished

Every Diwali, Kesar launches something. This year it is Tapri mango achaar. Mango, oil and spices go into the curing vats on day one; the curing (labour, power, the warm room) takes six weeks. On 31 October, 20,000 kg have been jarred and sent to packing, and 5,000 kg are still in the vats, 60% cured.

> **Kesar:** Count the vats as finished. Or don't count them. Whichever gives the nicer cost per kg.
>
> **Ritu:** Neither. All the mango is in those vats, so for materials they are 5,000 full kilograms. The curing is 60% done, so for labour and overheads they count as 3,000.
>
> **Kesar:** So half-pickled achaar is pickled achaar at a discount?
>
> **Ritu:** Per unit, please. It is 3,000 equivalent kilograms of curing.

That is the whole idea of **equivalent units**: units in process are converted into the number of finished units the same work would have made, separately for each **cost element** (material, labour, overheads).

There is a twist. On 1 October, 6,000 kg from September were already in the vats, 40% cured. Ritu can treat them in two ways. **FIFO** (first in, first out) finishes the old vats first and counts only October's 60% of their curing; their September cost stays separate. **Weighted average** pours September's cost into October's pot and treats all 20,000 finished kilograms alike.

Think of it like

Four glasses, two glasses of chai

Four glasses half-filled with chai: you have bought four glasses (material, all there) but poured only two glasses of chai (the work, half done). Equivalent units ask the same question twice, once per cost element.

Key idea

Equivalent units = units × percentage of work done, worked out separately for each cost element. FIFO counts only this period's work (opening work-in-process gets only its remaining percentage) and divides only this period's cost; weighted average adds the opening work-in-process cost to this period's and counts every completed unit as 100%.

How it works

Why work-in-process needs equivalent units

In continuous manufacture, a period's cost is spent on three groups at once: the opening work-in-process (WIP) still being finished, the units started and finished, and the closing WIP. Dividing total cost by completed units would load the closing WIP's cost onto finished goods. WIP could be valued on an actual basis (materials used plus the actual labour and expenses on the unfinished units), but ICAI notes that its accuracy "cannot be satisfactory"; the alternative is to convert partly finished units into **equivalent finished units** (SM 10.15–10.16).

$$\begin{aligned}&\text{Equivalent completed units}\\&= \text{Actual units in process}\\&\times \text{Percentage of work completed}\end{aligned}$$

ICAI's own example: 200 units 25% done equal 50 completed units, and their cost is the cost of 50 finished units (SM 10.16).

The five steps of a production cost report

ICAI prepares a **production cost report** for each process at the end of each period (SM 10.18–10.19):

1. **Physical flow**: units in (opening WIP, units introduced or transferred in) and units out (completed, losses, closing WIP). The two sides must be equal.

2. **Equivalent units for each cost element**, from the extent of work done on each. Materials put in at the start make closing WIP 100% complete for materials.

3. **Total cost for each element** for the period (and, under weighted average, the opening WIP's cost too), less the scrap value of normal loss.

4. **Cost per equivalent unit** = step 3 ÷ step 2, element by element.

5. **Assign the cost** to units completed and transferred, abnormal loss and closing WIP; post them to the process account.

The achaar vats, both ways

October: opening WIP 6,000 kg (materials 100%, curing 40%), cost ₹7,20,000 materials and ₹1,93,000 curing. Introduced 19,000 kg: materials ₹24,70,000 (mango prices

are up), curing ₹18,54,000. Completed 20,000 kg; closing WIP 5,000 kg (materials 100%, curing 60%).

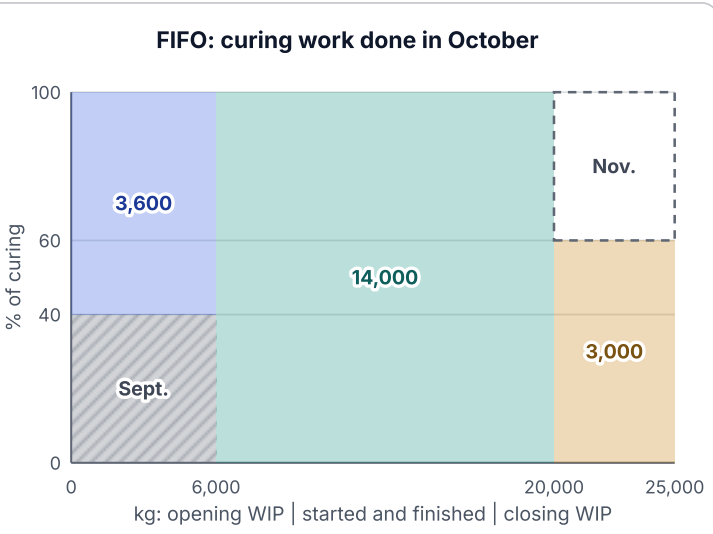
Equivalent units, FIFO

Output	Kg	Materials	Curing
Opening WIP	6,000	0	3,600 (60%)
Started and finished	14,000	14,000	14,000
Closing WIP	5,000	5,000	3,000 (60%)
Total	**25,000**	**19,000**	**20,600**

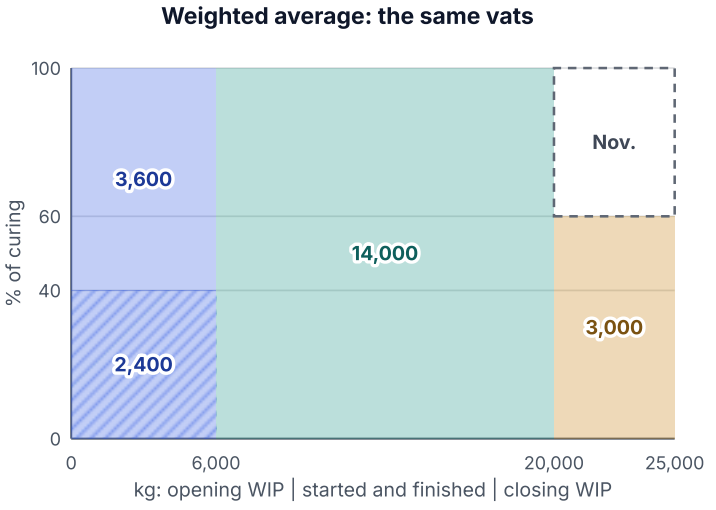
Equivalent units, weighted average

Output	Kg	Materials	Curing
Completed	20,000	20,000	20,000
Closing WIP	5,000	5,000	3,000 (60%)
Total	**25,000**	**25,000**	**23,000**

Under weighted average every completed kilogram counts in full, whenever its work was done.



Each block is kg (width) × % of curing done (height), so its area is equivalent units. Blue: finishing the opening WIP; teal: started and finished; amber: closing WIP. Grey hatched: September's work, outside FIFO's count; dashed: November's. Coloured area = 20,600.



The hatched blue block is September's 40% of the opening WIP: weighted average counts it, together with its September cost of ₹1,93,000. Coloured area = 23,000 = 20,000 completed + 3,000 in closing WIP.

Cost per equivalent unit

- FIFO, October's cost only: materials ₹24,70,000 ÷ 19,000 = ₹130; curing ₹18,54,000 ÷ 20,600 = ₹90.
- Weighted average, September's cost added: materials (₹7,20,000 + ₹24,70,000) ÷ 25,000 = ₹127.60; curing (₹1,93,000 + ₹18,54,000) ÷ 23,000 = ₹89.

Element	FIFO (₹)	Weighted average (₹)
Materials	130.00	127.60
Curing	90.00	89.00
Total	**220.00**	**216.60**

Where the ₹52,37,000 goes

Particulars	FIFO (₹)	Weighted average (₹)
Opening WIP (₹9,13,000 + 3,600 × ₹90)	12,37,000	–
Started and finished (14,000 × ₹220)	30,80,000	–
Completed (20,000 × ₹216.60)	–	43,32,000
Transferred to packing	**43,17,000**	**43,32,000**
Closing WIP (5,000 materials, 3,000 curing)	9,20,000	9,05,000
Total	**52,37,000**	**52,37,000**

With mango prices rising, FIFO's closing WIP carries October's higher cost: under FIFO "the closing stock of work in process is valued at current cost" (SM 10.19). Weighted average spreads September's cheaper cost over everything. The two equivalent-unit totals differ by

exactly the opening WIP's work done last period (6,000 for materials, 2,400 for curing).

Interactive version: PassBook app, lesson Equivalent units: FIFO and weighted average

Losses in the statement

ICAI's footnotes to the statement of equivalent units (SM 10.17):

- **Normal loss**: no equivalent unit is calculated. Its cost is absorbed by the good units, and its scrap value is deducted from the cost (usually from materials).
- **Abnormal loss**: equivalent units at its stage of completion for each element, which the question gives (80% for labour and overheads in SM Problem 4). A loss found when the process ends is 100% complete, as in the Study Material's case scenario (SM 10.49); some questions say so outright ("any loss beyond normal loss is considered 100% complete", MTP Series II for January 2026).
- **Abnormal gain**: treated as 100% complete for all elements, whatever the percentage of completion, and deducted.
- **Finished output** under FIFO = units completed and transferred less opening WIP; under weighted average = all units completed and transferred.

Formats and formulas

Statement of equivalent production

ICAI's statement has ten columns: input details and units, output particulars and units, then the percentage and units for materials, labour and overheads. Below, the input side comes first and the percentages are left out to fit a phone; on paper, write a % column before each element's units. *Reproduce it from memory.*

Input details	Units
Opening WIP	xxx
Units introduced (or transferred in)	xxx
Total	**xxx**

Output particulars	Units	Material	Labour	Overheads
Opening WIP (1)	xxx	xxx	xxx	xxx
Finished output (2)	xxx	xxx	xxx	xxx
Normal loss (3)	xxx	–	–	–
Abnormal loss (4)	xxx	xxx	xxx	xxx
Less: Abnormal gain (5)	(xxx)	(xxx)	(xxx)	(xxx)
Closing WIP (4)	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

(1) FIFO only: units × (100% − % done before). (2) FIFO: units completed less opening WIP; weighted average: all units completed; 100%. (3) No equivalent units. (4) Units × % complete. (5) 100% for every element.

Statement of cost per equivalent unit

Particulars
Cost of opening WIP (weighted average only)
Materials
xxx
Labour
xxx
Overheads
xxx
Total
xxx

Particulars
Cost incurred during the period
Materials
xxx
Labour
xxx
Overheads
xxx
Total
xxx

Particulars
Less: Scrap value of normal loss
Materials
(xxx)
Labour
–
Overheads
–
Total
(xxx)

Particulars

Total cost (A)

Materials

xxx

Labour

xxx

Overheads

xxx

Total

xxx

Particulars

Equivalent units (B)

Materials

XXX

Labour

XXX

Overheads

XXX

Total

Particulars

Cost per equivalent unit (A ÷ B)

Materials

xxx

Labour

xxx

Overheads

xxx

Total

xxx

Statement of evaluation (also called apportionment or distribution of cost)

FIFO	Weighted average
(i) Opening WIP: cost b/f + equivalent units to complete × cost per EU	(i) Units completed and transferred × cost per EU
(ii) Units started and completed × cost per EU	(ii) Abnormal loss (or gain), element-wise
(iii) Abnormal loss (or gain), element-wise	(iii) Closing WIP, element-wise
(iv) Closing WIP, element-wise	

The process account then shows units and amounts as in process accounts, with opening WIP on the debit side and closing WIP on the credit side.

∑ **Formula**

Equivalent units, FIFO

$$\begin{aligned}\text{EU} = {} & \text{Opening WIP} \\ & \times(100\% - \%\text{ done before}) \\ & + \text{Started and completed} \\ & + \text{Abnormal loss} \times \% \qquad - \text{Abnormal gain} \\ & + \text{Closing WIP} \times \%\end{aligned}$$

One line per cost element.

∑ **Formula**

Equivalent units, weighted average

$$\begin{aligned}\text{EU} = {} & \text{Units completed} && + \text{Abnormal loss} \times \% \\ & - \text{Abnormal gain} && + \text{Closing WIP} \times \%\end{aligned}$$

∑ **Formula**

Cost per equivalent unit

$$\begin{gathered}\text{Cost to divide, FIFO} \\ = \text{Period's cost} - \text{Scrap of normal loss}\end{gathered}$$

$$\begin{gathered}\text{Cost to divide, weighted average} \\ = \text{Opening WIP cost} \\ + \text{Period's cost} - \text{Scrap of normal loss}\end{gathered}$$

$$\text{Cost per EU} = \frac{\text{Cost to divide}}{\text{Equivalent units}}$$

Scrap of normal loss is its realisable value. Work each cost element separately.

Formula

The bridge between the two

$$\text{EU (weighted average)} - \text{EU (FIFO)} = \text{Opening WIP} \quad \times \% \text{ done last period}$$

Holds for each cost element; a quick check on both statements.

Definition

Equivalent units

Equivalent units or equivalent production units means converting the incomplete production units into their equivalent completed units. (SM 10.16)

Definition

FIFO method

Units completed and transferred are taken from both opening WIP and freshly introduced inputs. The cost to complete the opening WIP and the other completed units are calculated separately; the cost of opening WIP is added to the cost of completing it. Closing WIP is valued at current cost. (SM 10.19)

Definition

Weighted average method

The cost of opening work-in-process and the cost of the current period are aggregated, and the aggregate cost is divided by output in terms of completed units. Equivalent production consists of the work-load already contained in opening WIP and the work-load of the current period. (SM 10.21)

Worked examples

Worked example

FIFO: cold-brew chai, and the caps go on last

Tapri's cold-brew chai is steeped, filled into 1-litre bottles and capped in one continuous process. For

November:

Particulars	Litres	Amount (₹)
Opening WIP	3,000	2,10,000
Introduced: brew ingredients	22,000	14,40,560
Bottles and caps		4,69,040
Labour		4,40,400
Overheads		3,30,300
Lost in process	1,180	
Closing WIP	2,500	

The brew ingredients go in at the start. Labour and overheads are incurred evenly and are always at the same stage, so treat them together as conversion. Each litre is bottled and capped at the very end of the process. Opening WIP was 40% complete for conversion and closing WIP 70%; neither had been bottled. The rest was transferred to finished stock.

Normal loss is 4% of the litres introduced this month. The loss beyond normal came from a leaking valve in the steeping tank and was found when that brew was half way through conversion. Lost brew is sold to a biogas plant at ₹5 a litre, and its value is deducted from the cost of brew ingredients.

Required: Using the FIFO method,

(i) Prepare the statement of equivalent production.

(ii) Calculate the cost per equivalent unit for each element.

(iii) Calculate the cost of abnormal loss, closing WIP and the litres transferred.

(iv) Prepare the Process Account and the Abnormal Loss Account.

THE WORKING

Working notes: transferred = 3,000 + 22,000 − 1,180 − 2,500 = 21,320 litres, of which 3,000 came from opening WIP and 18,320 were started and completed. Normal loss = 4% × 22,000 = 880 litres, so abnormal loss = 1,180 − 880 = 300 litres. Bottles go on only at the end: every litre transferred gets one this month (the opening WIP's 3,000 included), and no litre lost or in closing WIP has one.

(i) Statement of equivalent production (FIFO)

Output	Litres	Brew	Bottles	Conversion
From opening WIP	3,000	–	3,000	1,800 (60%)
Started and completed	18,320	18,320	18,320	18,320
Normal loss	880	–	–	–
Abnormal loss	300	300	–	150 (50%)
Closing WIP	2,500	2,500	–	1,750 (70%)
Total (= 3,000 + 22,000)	**25,000**	**21,120**	**21,320**	**22,020**

(ii) Cost per equivalent unit

Element	Cost (₹)	EU	Cost per EU (₹)
Brew (14,40,560 − 880 × ₹5)	14,36,160	21,120	68.00
Bottles and caps	4,69,040	21,320	22.00
Conversion (4,40,400 + 3,30,300)	7,70,700	22,020	35.00
Total	**26,75,900**		**125.00**

(iii) Statement of evaluation

Particulars	Working	Amount (₹)
Opening WIP: cost b/f		2,10,000
Add: bottling it	3,000 × ₹22	66,000
Add: completing its conversion	1,800 × ₹35	63,000
Started and completed	18,320 × ₹125	22,90,000
Transferred to finished stock (21,320 litres)		**26,29,000**
Abnormal loss	300 × ₹68 + 150 × ₹35	25,650
Closing WIP	2,500 × ₹68 + 1,750 × ₹35	2,31,250

(iv) Process Account (debit side)

Dr. Particulars	Litres	Amount (₹)
To Opening WIP	3,000	2,10,000
To Brew ingredients	22,000	14,40,560
To Bottles and caps	–	4,69,040
To Labour	–	4,40,400
To Overheads	–	3,30,300
Total	**25,000**	**28,90,300**

Process Account (credit side)

Cr. Particulars	Litres	Amount (₹)
By Normal loss A/c (880 × ₹5)	880	4,400
By Abnormal loss A/c	300	25,650
By Finished stock A/c	21,320	26,29,000
By Closing WIP c/d	2,500	2,31,250
Total	**25,000**	**28,90,300**

Abnormal Loss Account

Dr.	Litres	(₹)
To Process A/c	300	25,650
Total	**300**	**25,650**

Cr.	Litres	(₹)
By Bank (300 × ₹5)	300	1,500
By Costing P&L A/c	–	24,150
Total	**300**	**25,650**

Treating the bottles like the brew, 100% for every litre in the process, would put bottles into the closing WIP and the abnormal loss that were never filled.

Worked example

Weighted average: bhujia in the fryer, two trays of WIP

Tapri's bhujia is made in two processes, Mixing and then Frying. Frying receives dough from Mixing at

Mixing's cost and adds its own labour and overheads. For January, Frying shows:

Particulars	Kg	Amount (₹)
Opening WIP: cost from Mixing	3,000	1,29,000
Opening WIP: labour (60% done)		19,500
Opening WIP: overheads (50% done)		16,500
Received from Mixing	37,000	16,31,000
Labour		4,59,550
Overheads		5,30,700

Transferred to Packing: 35,700 kg. Closing WIP: 2,800 kg in two lots, 1,800 kg that was 75% complete for labour and 60% for overheads, and 1,000 kg that had just entered the fryer, 30% complete for labour and 20% for overheads. Normal loss is 5% of total input (opening WIP plus kg received); burnt crumbs sell at ₹25 a kg, credited against the cost received from Mixing.

Required: Using the weighted average method, prepare:

(i) The statement of equivalent production.

(ii) The statement of cost.

(iii) The statement of distribution of cost.

(iv) The Frying Process Account and the Abnormal Gain Account.

THE WORKING

Working notes: total input 3,000 + 37,000 = 40,000 kg; normal loss 5% = 2,000 kg; actual loss = 40,000 − 35,700 − 2,800 = 1,500 kg. The loss is 500 kg less than normal: an **abnormal gain** of 500 kg, 100% complete for every element. Dough from Mixing is Frying's material: 100% complete for every kg in Frying, both WIP lots included. Weighted average does not use the opening WIP's 60% and 50%: its cost is pooled with January's, and its units count among the completed units.

(i) Statement of equivalent production

Output	Kg	From Mixing	Labour	Overheads
Completed and transferred	35,700	35,700	35,700	35,700
Normal loss	2,000	–	–	–
Closing WIP, lot 1	1,800	1,800	1,350 (75%)	1,080 (60%)
Closing WIP, lot 2	1,000	1,000	300 (30%)	200 (20%)
Less: Abnormal gain	(500)	(500)	(500)	(500)
Total	**40,000**	**38,000**	**36,850**	**36,480**

(ii) Statement of cost

Particulars	From Mixing	Labour	Overheads	Total
Opening WIP (₹)	1,29,000	19,500	16,500	1,65,000
January (₹)	16,31,000	4,59,550	5,30,700	26,21,250
Less: Scrap of normal loss (₹)	(50,000)	–	–	(50,000)
Total cost (₹)	**17,10,000**	**4,79,050**	**5,47,200**	**27,36,250**
Equivalent units	38,000	36,850	36,480	
Cost per EU (₹)	**45.00**	**13.00**	**15.00**	**73.00**

(iii) Statement of distribution of cost

Particulars	Working	Amount (₹)
Completed and transferred	35,700 × ₹73	26,06,100
Closing WIP, lot 1	1,800 × ₹45 + 1,350 × ₹13 + 1,080 × ₹15	1,14,750
Closing WIP, lot 2	1,000 × ₹45 + 300 × ₹13 + 200 × ₹15	51,900
Closing WIP		**1,66,650**
Abnormal gain	500 × ₹73	36,500

Check: ₹26,06,100 + ₹1,66,650 − ₹36,500 = ₹27,36,250, the total cost.

(iv) Frying Process Account (debit side)

Dr. Particulars	Kg	Amount (₹)
To Opening WIP	3,000	1,65,000
To Mixing process A/c	37,000	16,31,000
To Labour	–	4,59,550
To Overheads	–	5,30,700
To Abnormal gain A/c	500	36,500
Total	**40,500**	**28,22,750**

Frying Process Account (credit side)

Cr. Particulars	Kg	Amount (₹)
By Normal loss A/c (2,000 × ₹25)	2,000	50,000
By Packing A/c	35,700	26,06,100
By Closing WIP c/d	2,800	1,66,650
Total	**40,500**	**28,22,750**

Abnormal Gain Account

Dr.	Kg	(₹)
To Normal loss A/c (500 × ₹25)	500	12,500
To Costing P&L A/c	–	24,000
Total	**500**	**36,500**

Cr.	Kg	(₹)
By Frying process A/c	500	36,500
Total	**500**	**36,500**

Worked example

The two printouts: rebuild the month from equivalent units

The ERP at Tapri's chai-concentrate plant crashed while printing October's production cost report. Before it died it printed two summary lines, one for the FIFO report used by the plant and one for the weighted average report used by the head office:

Equivalent units	Materials	Conversion
Weighted average	10,800	9,900
FIFO	9,800	9,500

The FIFO report's cost per equivalent unit was ₹150 for materials and ₹100 for conversion. Ritu also knows:

- Opening WIP: 1,000 litres, valued at ₹1,69,300 (materials ₹1,39,200, conversion ₹30,100). Its stage of completion for conversion is not on record.
- All materials are added at the start of the process.
- Closing WIP: 1,500 litres, 40% complete for conversion.
- Normal loss is 2% of the litres introduced during the month, has no realisable value and is found at the end of the process. There was no abnormal loss or gain.

Required: Calculate

(i) The stage of completion of the opening WIP for conversion.

(ii) The litres introduced, the normal loss and the litres completed and transferred.

(iii) The cost of materials and of conversion added during October.

(iv) The value of output transferred and of closing WIP under FIFO.

(v) The cost per equivalent unit, the output transferred and the closing WIP under weighted average, and why the two closing WIP values differ.

(vi) Also prepare the Process Account under FIFO.

THE WORKING

(i) Stage of opening WIP. The two methods differ only in the opening WIP's work done last period: EU (weighted average) − EU (FIFO) = opening WIP × % done last period. Conversion: 9,900 − 9,500 = 400 = 1,000 × **40%**, so 60% was done in October. (Materials: 10,800 − 9,800 = 1,000 = 1,000 × 100%, as materials go in at the start.)

(ii) Litres. Under weighted average the materials EU are completed units plus closing WIP at 100%: completed = 10,800 − 1,500 = **9,300 litres**. Under FIFO they are units started and completed plus closing WIP: started and completed = 9,800 − 1,500 = 8,300 (= 9,300 − 1,000). Normal loss has no

equivalent units, so FIFO's 9,800 is the litres introduced less the 2% normal loss: litres introduced = 9,800 ÷ 98% = **10,000**, normal loss = **200 litres**.

The month's litres: what went in equals what came out

In	
Opening WIP:	1,000
Introduced:	10,000
Total:	11,000

Out	
Finished goods:	9,300
Normal loss:	200
Closing WIP:	1,500
Total:	11,000

Introduced is found from FIFO's materials EU: 98% of it is 8,300 + 1,500 = 9,800 litres, so 10,000 litres went in.

Statement of equivalent production (FIFO), which reproduces the printout:

Output	Litres	Materials	Conversion
From opening WIP	1,000	–	600 (60%)
Started and completed	8,300	8,300	8,300
Normal loss	200	–	–
Closing WIP	1,500	1,500	600 (40%)
Total (= 1,000 + 10,000)	**11,000**	**9,800**	**9,500**

(iii) Costs added in October (FIFO divides only the period's cost): materials 9,800 × ₹150 = **₹14,70,000**; conversion 9,500 × ₹100 = **₹9,50,000**.

(iv) FIFO evaluation

Particulars	Working	Amount (₹)
Opening WIP b/f		1,69,300
Completing it	600 × ₹100	60,000
Started and completed	8,300 × ₹250	20,75,000
Output transferred (9,300 litres)		**23,04,300**
Closing WIP	1,500 × ₹150 + 600 × ₹100	2,85,000

(v) Weighted average. Materials (₹1,39,200 + ₹14,70,000) ÷ 10,800 = ₹149; conversion (₹30,100 + ₹9,50,000) ÷ 9,900 = ₹99.

Particulars	Working	Amount (₹)
Output transferred	9,300 × ₹248	23,06,400
Closing WIP	1,500 × ₹149 + 600 × ₹99	2,82,900
Total		**25,89,300**

Both methods distribute the same ₹25,89,300 (₹1,69,300 + ₹14,70,000 + ₹9,50,000). FIFO's closing WIP is ₹2,100 higher because it is valued at October's cost (₹150 and ₹100), while weighted

average blends in the opening WIP's cheaper September cost (₹139.20 a litre for materials, ₹30,100 ÷ 400 = ₹75.25 per EU for conversion). Under FIFO the closing stock of work in process is valued at current cost (SM 10.19).

(vi) Process Account (FIFO) (debit side)

Dr. Particulars	Litres	Amount (₹)
To Opening WIP	1,000	1,69,300
To Materials	10,000	14,70,000
To Conversion (labour and overheads)	–	9,50,000
Total	**11,000**	**25,89,300**

Process Account (FIFO) (credit side)

Cr. Particulars	Litres	Amount (₹)
By Normal loss A/c (no realisable value)	200	–
By Finished goods A/c	9,300	23,04,300
By Closing WIP c/d	1,500	2,85,000
Total	**11,000**	**25,89,300**

Exam angle

In the exam

Where equivalent units appeared in the documents read for this lesson

- **Rebuild the month from the cost per EU** (quantity introduced, normal and abnormal loss, costs added, closing WIP, output): the Study Material's case scenario (SM 10.37), reproduced as the case-scenario MCQs of MTP Series I for May 2024; MTP Series I for May 2026 Q4(b) (7 marks); September 2026 Q4(b) (5 marks).
- **Full FIFO statement with losses**: September 2024 Case Scenario I, MCQs 1–5 (equivalent units, cost per unit, abnormal loss, transfer, closing WIP); RTP September 2024 Q14 (abnormal gain).
- **Weighted average**: MTP Series I for January 2026 Q4(a) (8 marks, abnormal loss 80% complete); RTP May 2025 Q14, repeated as RTP January 2026 Q14 (two processes, missing figures); MTP Series II for January 2026 Q2(a) (8 marks).
- **Two processes, no opening WIP, packing added at the end of Process II**: RTP September 2026 Q11, the Study Material's Problem 5 with new figures.
- **No opening WIP**: May 2025 Q5(a) (4 marks: equivalent units, cost per EU, closing WIP and finished output).
- **Single MCQs, 2 marks**: FIFO conversion EU, September 2026 MCQ 11 and MTP Series II for

September 2025 MCQ 13; units completed worked back from FIFO EU, RTP May 2026 MCQ 4.

Interactive version: PassBook app, lesson Equivalent units: FIFO and weighted average

In the exam

What the suggested answers do

- Present three statements in this order: equivalent production, cost per EU element by element, evaluation (distribution) of cost; then the process account.
- Deduct the scrap value of normal loss from materials when the question says so (September 2024: "deducted from cost of material introduced").
- Show abnormal gain as a "Less" row at 100% (RTP September 2024 Q14).

Trick

Same EU totals, both ways

Units out (including normal loss) must equal units in, on every statement. And EU (weighted average) − EU (FIFO) = opening WIP × % done last period, element by element. Both checks hold for any data; if either fails, a row is in the wrong place.

Answer frame

"Explain briefly the procedure for the valuation of work-in-process"

The chapter's own theory question (SM 10.31); ICAI's answer is paragraph 4 (SM 10.15–10.16):

1. The average cost per unit is total cost ÷ units produced, but in continuous manufacture the period's cost covers opening WIP, closing WIP and completed units.
2. So the cost of WIP at the beginning and at the end must be ascertained to find the cost of each completed unit.
3. WIP holds units at every stage of completion, which makes valuation difficult.
4. It can be valued on an actual basis (materials used and actual labour and expenses), but the accuracy is not satisfactory.
5. The alternative converts partly finished units into equivalent finished units; then name the two methods, FIFO and weighted average.

"Explain equivalent units"

Definition (converting incomplete units into equivalent completed units); the estimate of percentage completion for each element, which "should be as accurate as possible"; the formula; ICAI's example of 200 units at 25% = 50 completed units (SM 10.16).

"Distinguish between FIFO and weighted average"

Point	FIFO	Weighted average
Opening WIP	Work and cost kept separate; only the remaining work is counted	Units and cost taken in full
Cost divided	Current period only	Opening WIP cost + current period
Completed units	Opening WIP and fresh units costed separately	All treated as equally complete, one rate
Closing WIP	At current cost	At the average cost

The Study Material's one-line version: "units of opening work in process and their cost are taken in full under average method while under FIFO method only the remaining work done now is considered" (SM 10.22).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the opening WIP counted at the work already done

Counting the achaar's opening WIP as 6,000 × 40% = 2,400 equivalent kg under FIFO. **Fix:** the 40% was done in September; FIFO counts October's remaining 60%, 3,600 equivalent kg.

⚠ **Common mistake**

Every litre transferred at this month's rate

Valuing FIFO output as 9,300 × ₹250 = ₹23,25,000. **Fix:** under FIFO the opening WIP brings its own b/f cost: ₹1,69,300 + ₹60,000 to complete + 8,300 × ₹250 = ₹23,04,300.

⚠ **Common mistake**

Equivalent units for the normal loss

Giving the 2,000 kg of normal loss equivalent units, which lowers the cost per unit and leaves its cost nowhere. **Fix:** normal loss has no equivalent units; only its scrap value is deducted from the cost.

⚠ **Common mistake**

The abnormal gain that was added

Adding the 500 kg of abnormal gain to equivalent units. **Fix:** abnormal gain is deducted, at 100% for every element; the units column still balances because the normal loss row is in full.

Practice

Check yourself

1. Tapri's tomato chutney line adds tomatoes at the start and the spice mix when a batch is 50% cooked; conversion is incurred evenly. In March, opening WIP was 4,000 kg, 30% cooked; 30,000 kg were completed and transferred; closing WIP was 5,000 kg, 60% cooked. There were no losses. What are the equivalent units for conversion under the weighted average method?

Answer: _____

2 Same data. Under FIFO, what are the equivalent units for conversion and for the spice mix?

- **A** Conversion 30,200; spice mix 35,000
- **B** Conversion 31,800; spice mix 31,000
- **C** Conversion 31,800; spice mix 35,000
- **D** Conversion 33,000; spice mix 35,000

3 Case scenario. Mango prices rose sharply this month at the achaar plant; the opening WIP in the vats carries last month's lower cost, and this month's cost per equivalent unit is higher for both materials and curing. Dev asks Ritu which of the following is correct for this month.

- **A** FIFO values closing WIP higher than weighted average, because it uses only this month's cost per equivalent unit
- **B** Weighted average values closing WIP higher, because it includes the opening WIP's cost
- **C** Both methods give the same closing WIP, because the equivalent units of closing WIP are the same
- **D** FIFO transfers more cost to packing than weighted average

4 Which statements about the statement of equivalent production are correct? Select all that apply.

- **A** Normal loss is given no equivalent units
- **B** Abnormal gain is treated as 100% complete for all elements and deducted
- **C** Abnormal loss is always treated as 100% complete
- **D** Opening WIP is shown separately only under FIFO

More than one answer may be right.

5 Under weighted average, opening WIP materials cost ₹40,000, materials introduced ₹3,20,000 and the scrap value of normal loss ₹10,000. Equivalent units for materials are 14,000. What is the cost per equivalent unit for materials, in rupees?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Equivalent units = units × % of work done, separately for each cost element.

- Five steps: physical flow, EU per element, cost per element, cost per EU, assign costs.
- FIFO: opening WIP gets only its remaining %; divide only the period's cost; opening WIP's b/f cost is added to its completion cost; closing WIP at current cost.
- Weighted average: all completed units at 100%; divide opening WIP cost + period cost.
- EU (weighted average) − EU (FIFO) = opening WIP × % done last period.
- Normal loss: no EU, scrap value deducted. Abnormal loss: at its stage of completion. Abnormal gain: 100%, deducted.
- Order of statements: equivalent production, cost per EU, evaluation, process account. (Half-pickled achaar is 3,000 equivalent kilograms of curing.)

Answer key · Check yourself

1 **33000**

All 30,000 kg completed count at 100%, plus closing WIP 5,000 × 60% = 3,000: 33,000. Under weighted average the opening WIP's 30% does not enter the count.

2 **C Conversion 31,800; spice mix 35,000**

Conversion: opening WIP 4,000 × 70% still to do = 2,800, plus 26,000 started and completed, plus closing 5,000 × 60% = 3,000: 31,800 (30,200 counts the 30% done last month; 33,000 is the weighted average). Spice goes in at the 50% point, so the opening WIP (30% cooked) receives all of it this month (4,000) and the closing WIP (60% cooked) already has it (5,000): 4,000 + 26,000 + 5,000 = 35,000. 31,000 forgets the opening WIP's spice.

3 **A FIFO values closing WIP higher than weighted average, because it uses only this month's cost per equivalent unit**

FIFO values closing WIP at current cost (SM 10.19), which is higher when prices rise. In the achaar example FIFO's closing WIP is ₹9,20,000 against ₹9,05,000, and it transfers less (₹43,17,000 against ₹43,32,000).

4 **A Normal loss is given no equivalent units**; **B Abnormal gain is treated as 100% complete for all elements and deducted**; **D Opening WIP is shown separately only under FIFO**

Abnormal loss takes its own stage of completion (80% in SM Problem 4); the other three are ICAI's footnotes (SM 10.17).

5 **25**

(₹40,000 + ₹3,20,000 − ₹10,000) ÷ 14,000 = ₹25.

10.3 Inter-process profit and operation costing

Why processes may transfer output at a profit, how the cost and profit columns keep unrealised profit out of stock, and what operation costing (hybrid costing) is.

In plain words

Every process wants a trophy

Kesar has turned Tapri's Roasting and Blending processes into profit centres. Roasting now "sells" its roasted leaves to Blending at a mark-up, and Blending "sells" the premix to the finished-goods godown at another.

> **Kesar:** Roasting made ₹1,20,000 this month, Blending ₹1,80,000. Bonus for everyone!
>
> **Ritu:** On paper. Roasting sold to Blending, and Blending is us. Whatever premix is still sitting in the godown, nobody outside has paid for.
>
> **Dev:** So the profit inside that stock has a present value of exactly zero. Finally, something I don't need to discount.

When a process passes its output on at a **transfer price** (cost plus a profit) instead of at cost, the difference is **inter-process profit**. It lets each process show how it performs on its own. But the business earns a profit only when goods are sold to an outsider. Profit locked inside

the closing stocks of the processes and of finished goods is **unrealised**, so the accounts carry a separate profit column to keep it apart from cost.

Operation costing is the other half of this lesson. Tapri's kulhad unit makes Plain kulhads for its cafés and glazed Royal kulhads for the Diwali gift box. Both go through the same moulding and firing; only the clay and glaze differ. So materials are collected batch by batch, as in job costing, while the moulding and firing costs are collected by operation, as in process costing: a **hybrid**.

Think of it like

The car you sold to your brother

You buy a car for ₹4 lakh and sell it to your brother for ₹5 lakh. You show a profit of ₹1 lakh; the family is not one rupee richer. Only when your brother sells it to a stranger for ₹6 lakh does the family gain ₹2 lakh, and if the car is still in his garage, your ₹1 lakh is still "in the stock".

Key idea

A transfer at a profit shows each process's profit, but profit is earned only on sale outside the business. The inter-process profit that sits in closing stocks is unrealised; the profit column keeps it out of the cost of stock.

How it works

What ICAI means by inter-process profit

To control cost and to measure performance, different processes may be designated as separate profit centres. "The output of one process is transferred to the next process not at cost but at market value or cost plus a percentage of profit. The difference between cost and the transfer price is known as inter-process profits." (SM 10.23)

ICAI's advantages and disadvantages, in ICAI's order (SM 10.23–10.24):

Advantages	Disadvantages
1. Comparison between the cost of output and its market price at the stage of completion is facilitated.	1. The use of inter-process profits involves complication.
2. Each process is made to stand by itself as to the profitability.	2. The system shows profits which are not realised because of stock not sold out.

From a profit percentage to a transfer price

Read whether the profit is a percentage of the **transfer price** or of **cost**:

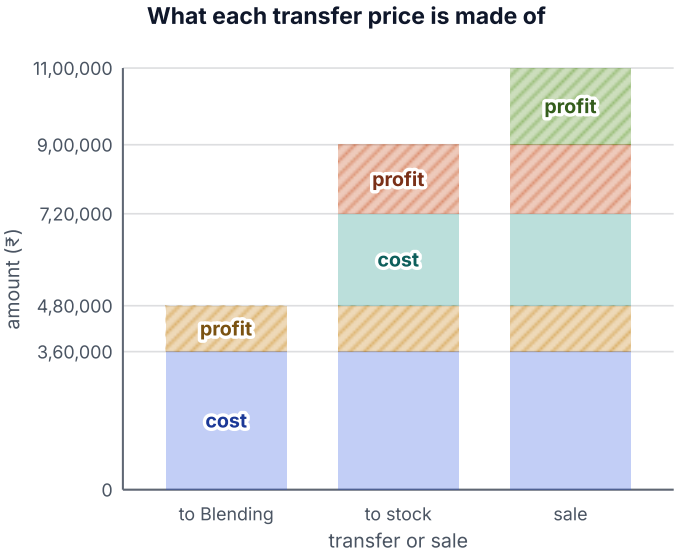
- "25% profit on the transfer price": transfer price = cost ÷ 75%, so ₹3,60,000 becomes ₹4,80,000 and the profit is ₹1,20,000.

- "Cost plus 25%" (25% on cost): transfer price = cost × 125%, so ₹3,60,000 becomes ₹4,50,000.

The same profit can be written both ways: 20% of the transfer price is 25% of cost, 25% of the transfer price is 33⅓% of cost, and one-third of cost is one-quarter of the transfer price.

No stocks: every rupee of profit is realised

Roasting's cost is ₹3,60,000 and it transfers at 25% profit on the transfer price: ₹4,80,000. Blending adds ₹2,40,000 of its own cost (total ₹7,20,000) and transfers at 20% profit on the transfer price: ₹9,00,000. The premix sells for ₹11,00,000.



Blue: Roasting's cost; teal: Blending's cost; hatched: profit (amber Roasting's, orange Blending's, green the profit on sale). Of the ₹9,00,000 at which finished stock receives the premix, ₹3,00,000 is the two processes' own profit.

The three profits, ₹1,20,000 + ₹1,80,000 + ₹2,00,000 = ₹5,00,000, equal sales less the real cost (₹11,00,000 − ₹3,60,000 − ₹2,40,000). This month nothing was left in stock anywhere, so all of it is realised.

With stocks: three columns, five steps

Every process account and the finished stock account get three columns on each side: **Total**, **Cost** and **Profit** (SM 10.24–10.26, Illustration 6).

1. **Debit** the opening stock (its cost and, separately, the inter-process profit the question says it contains), the transfer from the previous process (its cost and profit columns come from that process's credit side), direct materials, direct wages and any direct expenses. Subtotal: **prime cost**, in all three columns.

2. **Closing stock** valued at prime cost (the instruction in Illustration 6 and in every ICAI question with stocks read for this lesson) carries the same share of profit as the prime cost: cost element = closing stock × prime cost (cost column) ÷ prime cost (total column).

3. **Add overheads** to reach total cost. In the Total column, total cost less closing stock is what the output transferred carries; gross it up to the transfer price.

4. **Profit** = transfer price − (total cost − closing stock), both in the Total column. It is debited in the process account (Total and Profit columns) and credited to the Costing P&L.

5 **Finished stock** valued at the price at which it is received from the last process: its cost element = closing stock × cost of the transfer ÷ transfer price (SM 10.26). The sales line then shows, in its cost column, the true cost of goods sold, and in its profit column, the **total profit realised**.

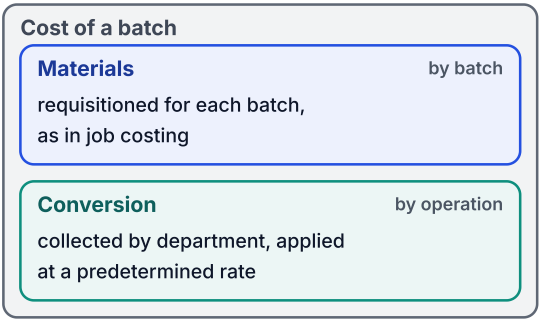
The arithmetic of those accounts gives one check (it follows from ICAI's Illustration 6, SM 10.24–10.26, where ₹52,250 + ₹9,750 − ₹4,425 = ₹57,575):

$$\text{Realised profit} = \text{Profits shown} + \text{Profit in opening stocks} - \text{Profit in closing stocks}$$

Operation costing

ICAI's paragraph 8 (SM 10.26): this product costing system is used when an entity produces more than one variant of the final product using different materials but with similar conversion activities. It is also known as a **hybrid product costing system**, because material costs are accumulated by job order or batch, while conversion costs (labour and overheads) are accumulated by department and assigned to products by process costing methods. Conversion costs are applied using a predetermined rate based on budgeted conversion costs. ICAI's examples: ready-made garments, shoe making, jewellery. The chapter summary adds that operation costing is a refinement of process costing, concerned

with the cost of each operation rather than of the process (SM 10.27).



Blue: the materials route, which differs for every variant. Teal: the conversion route, the same operations for every variant. Their sum is the batch cost; divided by the units, the cost per unit.

Formats and formulas

Process account showing the profit element

Write it as ICAI does, with Total, Cost and Profit on both sides, debit on the left and credit on the right. Four columns of lakh amounts do not fit side by side on a screen, so here the credit side sits below the debit side. *Reproduce it from memory.*

Process II Account (debit side)

Dr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	xxx	xxx	xxx
Transferred from Process I	xxx	xxx	xxx
Direct materials	xxx	xxx	–
Direct wages	xxx	xxx	–
Direct expenses, if any	xxx	xxx	–
Prime cost	**xxx**	**xxx**	**xxx**
Overheads	xxx	xxx	–
Total cost	**xxx**	**xxx**	**xxx**
Profit (to Costing P&L)	xxx	–	xxx
Total	**xxx**	**xxx**	**xxx**

Process II Account (credit side)

Cr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Finished stock A/c (transfer)	xxx	xxx	xxx
Closing stock	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**

Finished Stock Account: debit opening stock, the transfer from the last process and the profit (balancing figure, total and profit columns); credit the Costing P&L A/c with sales and the closing stock, each split into cost and profit.

∑ **Formula**

Transfer price

Profit as a fraction p of the transfer price:

$$\text{TP} = \frac{\text{Cost transferred}}{1 - p}$$

Profit as a fraction q of cost:

$$\text{TP} = \text{Cost transferred} \times (1 + q)$$

Cost transferred = total cost – closing stock.

∑ **Formula**

Converting a profit percentage

$$p \text{ on TP} = \frac{p}{1 - p} \text{ on cost}$$

$$q \text{ on cost} = \frac{q}{1 + q} \text{ on TP}$$

20% on TP = 25% on cost; 1/3 on cost = 1/4 on TP.

Σ **Formula**

Cost element of closing stock in a process (stock at prime cost)

$$\text{Cost of closing stock} = \text{Closing stock} \times \frac{\text{Prime cost (cost)}}{\text{Prime cost (total)}}$$

The cost and total columns of prime cost in the process account (SM 10.25).

Σ **Formula**

Cost element of closing finished stock

$$\text{Cost of closing stock} = \text{Closing stock} \times \frac{\text{Cost of the transfer}}{\text{Transfer price}}$$

When the question values finished stock at the price received from the last process (SM 10.26).

Definition

Inter-process profits

The output of one process is transferred to the next process not at cost but at market value or cost plus a percentage of profit; the difference between cost and the transfer price is known as inter-process profits. (SM 10.23)

Definition

Operation costing

A product costing system used when an entity produces more than one variant of the final product using different materials but with similar conversion activities; also known as a hybrid product costing system. (SM 10.26)

Worked examples

Worked example

Roasting at market value, Blending at cost plus: a godown full of profit

Tapri's chai premix passes through Roasting (Process I) and Blending (Process II) before it reaches finished stock. Kesar has made both processes profit centres. Roasting transfers its roasted leaves at their market value, ₹75 a kg (what a tea trader charges for the same leaves); Blending transfers the premix to finished stock at cost plus 20%. For December:

- **Roasting**: opening stock ₹48,000; direct materials ₹2,64,000; direct wages ₹96,000; factory overheads ₹72,000; closing stock ₹30,000; 8,000 kg of roasted leaves transferred to Blending.
- **Blending**: opening stock ₹90,000, of which ₹15,000 is inter-process profit; direct materials (sugar, milk powder, spices) ₹2,10,000; direct wages ₹60,000;

factory overheads ₹48,000; closing stock ₹96,000; 12,000 kg of premix transferred to finished stock.

- **Finished stock**: opening stock 2,000 kg, ₹1,90,000, of which ₹52,000 is inter-process profit; 12,800 kg sold to distributors at ₹112.50 a kg.

Stock in processes is valued at prime cost. Finished stock is valued at the price at which it is received from Blending.

Required:

(i) Prepare the Process I, Process II and Finished Stock Accounts showing the profit element at each stage.

(ii) Calculate the profit actually realised in December and reconcile it with the profits shown in the three accounts.

(iii) Using ICAI's advantages and disadvantages of inter-process profit, state what the market-value transfer tells Kesar about Roasting, and what the accounts overstate.

THE WORKING

Working notes

1. Process I: transfer at market value = 8,000 kg × ₹75 = ₹6,00,000. Cost of the output transferred = ₹4,80,000 − ₹30,000 = ₹4,50,000 (₹56.25 a kg); profit = ₹1,50,000. There is no percentage to gross up: the market value is the transfer price (SM 10.23).

2. Process II closing stock at prime cost: cost = ₹96,000 × ₹7,95,000 ÷ ₹9,60,000 = ₹79,500; profit ₹16,500.
3. Process II: total cost less closing stock = ₹10,08,000 − ₹96,000 = ₹9,12,000; transfer price = ₹9,12,000 × 120% = ₹10,94,400 (₹91.20 a kg); profit = ₹1,82,400.
4. Finished stock closing: 12,000 + 2,000 − 12,800 = 1,200 kg at the price received from Process II, ₹91.20, = ₹1,09,440. Cost = ₹1,09,440 × ₹7,63,500 ÷ ₹10,94,400 = ₹76,350; profit ₹33,090.

Process I Account (debit side)

Dr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	48,000	48,000	–
Direct materials	2,64,000	2,64,000	–
Direct wages	96,000	96,000	–
Prime cost	**4,08,000**	**4,08,000**	**–**
Factory overheads	72,000	72,000	–
Total cost	**4,80,000**	**4,80,000**	**–**
Profit (WN 1)	1,50,000	–	1,50,000
Total	**6,30,000**	**4,80,000**	**1,50,000**

Process I Account (credit side)

Cr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Process II A/c (WN 1)	6,00,000	4,50,000	1,50,000
Closing stock	30,000	30,000	–
Total	**6,30,000**	**4,80,000**	**1,50,000**

Process II Account (debit side)

Dr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	90,000	75,000	15,000
Process I A/c	6,00,000	4,50,000	1,50,000
Direct materials	2,10,000	2,10,000	–
Direct wages	60,000	60,000	–
Prime cost	**9,60,000**	**7,95,000**	**1,65,000**
Factory overheads	48,000	48,000	–
Total cost	**10,08,000**	**8,43,000**	**1,65,000**
Profit (WN 3)	1,82,400	–	1,82,400
Total	**11,90,400**	**8,43,000**	**3,47,400**

Process II Account (credit side)

Cr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Finished stock A/c (WN 3)	10,94,400	7,63,500	3,30,900
Closing stock (WN 2)	96,000	79,500	16,500
Total	**11,90,400**	**8,43,000**	**3,47,400**

The transfer's cost column is the balance of the cost column (₹8,43,000 – ₹79,500), and its profit column the balance of the profit column.

Finished Stock Account (debit side)

Dr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	1,90,000	1,38,000	52,000
Process II A/c	10,94,400	7,63,500	3,30,900
Profit (balancing figure)	2,65,040	–	2,65,040
Total	**15,49,440**	**9,01,500**	**6,47,940**

Finished Stock Account (credit side)

Cr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Costing P&L A/c (sales, 12,800 × ₹112.50)	14,40,000	8,25,150	6,14,850
Closing stock (WN 4)	1,09,440	76,350	33,090
Total	**15,49,440**	**9,01,500**	**6,47,940**

(ii) Profit realised and reconciliation

Realised profit = sales − actual cost of goods sold = ₹14,40,000 − ₹8,25,150 = **₹6,14,850**, the profit column of the sales line.

Particulars	Amount (₹)
Profits shown: Process I ₹1,50,000 + Process II ₹1,82,400 + Finished stock ₹2,65,040	5,97,440
Add: Inter-process profit in opening stocks, now realised (₹15,000 + ₹52,000)	67,000
Less: Inter-process profit in closing stocks, unrealised (₹16,500 + ₹33,090)	(49,590)
Profit realised	**6,14,850**

(iii) What the transfer tells Kesar. Roasting's leaves cost ₹56.25 a kg against a market value of ₹75 a kg. The transfer at market value facilitates the comparison between the cost of output and its market price at the stage of completion, and makes Roasting stand by itself as to profitability: it earns ₹1,50,000 (₹18.75 a kg), and a cost above ₹75 a kg would tell Kesar that buying roasted leaves is cheaper than roasting them. What the accounts overstate: the system shows profits which are not realised because stock is not sold out (₹49,590 of the ₹5,97,440 shown is locked in closing stocks whose true cost is ₹30,000, ₹79,500 and ₹76,350), and the three-column accounts are the complication ICAI lists as the other disadvantage (SM 10.23–10.24).

Worked example **Plain and Royal kulhads on one line**

Tapri's kulhad unit makes two variants on the same line. Plain kulhads (for its cafés) use ordinary clay; Royal kulhads (glazed, for the Diwali gift box) use fine clay and glaze. Both pass through the same two operations, Moulding and Firing, whose costs are applied per cup at predetermined rates.

- Budget for November: Moulding conversion cost ₹1,80,000 and Firing ₹2,40,000, each for 60,000 cups.

- Two of November's batches: P-11, 10,000 Plain kulhads, materials ₹20,000; R-04, 4,000 Royal kulhads, materials ₹26,000.
- Actual for November, all batches together: 62,000 cups passed through both operations; Moulding cost ₹1,92,200 and Firing ₹2,38,000.

Required:

(i) Calculate the predetermined conversion rate per cup for each operation.

(ii) Calculate the cost of each batch and the cost per kulhad of each variant.

(iii) Calculate the conversion cost under- or over-applied in each operation for November.

(iv) State why this is operation costing and not pure process costing.

THE WORKING

(i) Predetermined rates: Moulding ₹1,80,000 ÷ 60,000 = **₹3** a cup; Firing ₹2,40,000 ÷ 60,000 = **₹4** a cup; conversion ₹7 a cup.

(ii) Batch costs

Particulars	P-11 Plain (₹)	R-04 Royal (₹)
Materials (by batch)	20,000	26,000
Moulding (10,000 and 4,000 × ₹3)	30,000	12,000
Firing (10,000 and 4,000 × ₹4)	40,000	16,000
Cost of batch	**90,000**	**54,000**
Cost per kulhad	**9.00**	**13.50**

(iii) Conversion cost applied and actual

Operation	Applied (₹)	Actual (₹)	Difference (₹)
Moulding (62,000 × ₹3)	1,86,000	1,92,200	6,200 under-applied
Firing (62,000 × ₹4)	2,48,000	2,38,000	10,000 over-applied

They are dealt with like under- or over-absorbed overheads: under- and over-absorption.

(iv) The two variants use different materials but similar conversion activities. Materials are accumulated batch-wise, as in job or batch costing; conversion costs are accumulated by operation

(department) and applied at a predetermined rate based on budgeted conversion cost, as in process costing. That combination is ICAI's hybrid product costing system, operation costing (SM 10.26).

Exam angle

In the exam

Where inter-process profit and operation costing appeared in the documents read

- **Case scenario, 5 MCQs of 2 marks**: May 2026 Case Scenario I (profit of the first process, prime cost including inter-process profit, profit in closing stock, transfer price, total profit realised); RTP January 2025 Q2 (transfer prices of three processes at cost plus 25% and 33⅓%, cost value of a transfer and of closing stock).
- **Descriptive**: three-column accounts with stocks, MTP Series I for September 2025 Q2 (14 marks); process accounts and total profit with no stocks, MTP Series II for January 2025 Q3(a) (6 marks) and MTP Series II for May 2024 Q1(a) (5 marks).
- **Theory, 5 marks**: "What is inter-process profit? State its advantages and disadvantages", MTP Series I for May 2022 (old scheme) Q6(e); "Describe operation costing with two examples of industries", RTP November 2020 (old scheme) Q15(d). Both are also the chapter's own theory questions (SM 10.31).

In the exam

The traps

- **Which base?** "30% profit on the transfer price" (May 2026) means TP = cost ÷ 70%; "1/3rd of cost" (MTP September 2025) means TP = cost × 4/3, which is 1/4 of the TP.
- **"Total profit"** with no stocks (MTP January 2025): the sum of the process profits and the profit on sale, which equals sales less the actual costs.
- **"Total profit realised"** with stocks (May 2026 MCQ 5): the profit column of the sales line, not the sum of the profits shown.

Trick

Percentages on cost and on transfer price

p% on the transfer price is p ÷ (100 − p) on cost; q% on cost is q ÷ (100 + q) on the transfer price. So 20% ↔ 25%, 25% ↔ 33⅓%, 30% ↔ 42.86%. Holds for any percentage, because transfer price = cost + profit.

Answer frame

"What is inter-process profit? State its advantages and disadvantages."

ICAI's suggested answer (MTP May 2022 Q6(e), old scheme) is the Study Material's paragraph 7, nearly word

for word:

1. **Context**: to control cost and to measure performance, processes are designated as separate profit centres.
2. **Definition**: the output of one process is transferred to the next not at cost but at market value or cost plus a percentage of profit; the difference between cost and the transfer price is known as inter-process profits.
3. **Advantages**: comparison between the cost of output and its market price at the stage of completion is facilitated; each process is made to stand by itself as to the profitability.
4. **Disadvantages**: the use of inter-process profits involves complication; the system shows profits which are not realised because of stock not sold out.

"Operation costing is defined as refinement of process costing. Explain it."

The chapter's own theory question (SM 10.31); ICAI's answers point to paragraph 8, and RTP November 2020's suggested answer reproduces it with two industries:

1. Used when an entity produces more than one variant of the final product using different materials but similar conversion activities.
2. Also known as a hybrid product costing system: material costs accumulated by job order or batch; conversion costs (labour and overheads) accumulated by department and assigned by process costing methods.

3. Conversion costs applied at a predetermined rate based on budgeted conversion costs.
4. Example: Deluxe and Regular grades of one product, with different raw materials but the same conversion activities.
5. Industries: ready-made garments, shoe making, jewellery. The refinement: it determines the cost of each operation rather than of the whole process (SM 10.27).

Common mistakes

⚠ **Common mistake**

Hall of Shame: 25% on the transfer price added to cost

Taking Roasting's transfer price as ₹3,60,000 × 125% = ₹4,50,000 when the question says 25% profit on the transfer price. **Fix:** profit on transfer price means TP = cost ÷ (1 − 25%) = ₹4,80,000; "cost plus 25%" is the one that multiplies by 125%.

⚠ **Common mistake**

Closing stock of Process II carried entirely at cost

Putting all ₹96,000 into the cost column. **Fix:** Process II's prime cost holds ₹1,65,000 of inter-process profit out of ₹9,60,000, so the closing stock holds the same share: ₹16,500 profit, ₹79,500 cost.

⚠ **Common mistake**

Overheads in the ratio

Splitting the closing stock with total cost (₹8,43,000 ÷ ₹10,08,000) when stock is valued at prime cost. **Fix:** use the ratio of the column the stock is valued at: prime cost, ₹7,95,000 ÷ ₹9,60,000.

⚠ **Common mistake**

Adding up the process profits and calling it the business's profit

Reporting ₹5,97,440 as December's profit. **Fix:** profit realised is ₹6,14,850: add the inter-process profit released from opening stocks and take out what is locked in closing stocks.

Practice

Check yourself

1 A process transfers output costing ₹2,40,000 to the next process at a profit of 20% on the transfer price. What is the transfer price, in rupees?

Answer: _____

2 Case scenario. Blending's prime cost for the month is ₹5,00,000, of which ₹1,00,000 is inter-process profit received with transfers from Roasting. Blending's closing stock is valued at prime cost at ₹50,000. Dev wants to know how much unrealised profit sits in it. Which of the following is correct?

- **A** Nil, because stock is valued at prime cost
- **B** ₹10,000
- **C** ₹12,500
- **D** ₹50,000

3 A process's output is transferred at "cost plus one-third". What percentage of the transfer price is profit?

- **A** 33⅓%
- **B** 25%
- **C** 20%
- **D** 40%

4 Which of these are ICAI's advantages of using inter-process profit? Select all that apply.

- **A** Comparison of the cost of output with its market price at the stage of completion is facilitated
- **B** Each process is made to stand by itself as to profitability
- **C** It removes unrealised profit from stock
- **D** It simplifies the cost accounts

More than one answer may be right.

5 In operation costing, material costs and conversion costs are accumulated how?

- **A** Both by department, using process costing
- **B** Both by job or batch
- **C** Materials by job or batch; conversion by department, applied at a predetermined rate
- **D** Materials by department; conversion by job or batch

Answers: the answer key at the end of this lesson.

Recap

- Inter-process profit = transfer price − cost of the output transferred (market value or cost plus a percentage).
- Advantages: cost compared with market price at each stage; each process stands by itself as to profitability. Disadvantages: complication; unrealised profit shown when stock is not sold.
- On transfer price: TP = cost ÷ (1 − p). On cost: TP = cost × (1 + q).
- Three columns (Total, Cost, Profit) on both sides; stock at prime cost split in the ratio of the prime cost columns; finished stock split in the ratio of the last transfer.
- Realised profit = profits shown + profit in opening stocks − profit in closing stocks = sales − actual cost of goods sold. (Selling to your brother is not a profit until he sells to a stranger.)
- Operation costing: variants with different materials and similar conversion; materials by batch, conversion by department at a predetermined rate; hybrid; garments, shoes, jewellery.

Answer key · Check yourself

1 **300000**

TP = ₹2,40,000 ÷ (1 − 20%) = ₹3,00,000; the profit of ₹60,000 is 20% of it (and 25% of cost).

2 **B ₹10,000**

The stock carries the prime cost's share of profit: ₹50,000 × ₹1,00,000 ÷ ₹5,00,000 = ₹10,000. ₹12,500 divides by the cost (₹4,00,000) instead of the total.

3 **B 25%**

Cost 3, profit 1, transfer price 4: profit is 1/4 = 25% of the transfer price.

4 **A Comparison of the cost of output with its market price at the stage of completion is facilitated; B Each process is made to stand by itself as to profitability**

ICAI's two advantages are the first two. Its disadvantages are the opposite of the last two: the system involves complication and shows profits not realised because stock is not sold out.

5 **C Materials by job or batch; conversion by department, applied at a predetermined rate**

That split is why ICAI calls operation costing a hybrid product costing system (SM 10.26).

CHAPTER 11

Joint Products and By-Products

11.1 Apportioning joint costs: physical units, sales value and net realisable value

One process, several products, one bill. How ICAI's methods share the joint cost up to the split-off point, and why the method changes each product's cost and profit.

In plain words

One factory, three teas, one bill

Tapri's tea factory near Jorhat buys green leaf from the gardens around it. Every kilo goes through the same withering, rolling, fermenting and drying. Only at the end, at the sorting sieve, does the made tea fall into three grades: whole **Leaf**, **Brokens** and **Dust**. In March 2026 the factory spent ₹1,20,00,000 and made 10,000 kg of Leaf, 20,000 kg of Brokens and 30,000 kg of Dust, all sold at the Guwahati auction.

> **Kesar:** Ritu, the accounts say Dust lost ₹30 lakh. Shut the Dust line. Sab adjust ho jayega.
>
> **Ritu:** There is no Dust line, Kesar-ji. Same leaf, same dryer. Dust falls through the sieve whether we like it or not.
>
> **Kesar:** Then who decided it lost money?
>
> **Ritu:** Whoever split the ₹1.2 crore by weight.

The three grades are **joint products**: several products of real value that come out of one process at the same time. The ₹1,20,00,000 is their **joint cost**, spent before the grades can be told apart. The point where they become separately identifiable (here, the sieve) is the **split-off point**.

Nobody can say how much of the dryer's gas was "for" the Dust, so the joint cost cannot be traced; it can only be **apportioned** on some rational basis. Shared by weight, every kilo carries ₹200, and Dust, which fetches only ₹100 a kilo, shows a ₹30 lakh loss. Shared by what each grade sells for (₹60 lakh, ₹60 lakh and ₹30 lakh, together ₹1.5 crore), every grade carries 80% of its own sales value, and every grade makes a 20% profit. The factory's total profit, ₹30 lakh, is the same both ways.

Think of it like

The wedding biryani

A caterer cooks one giant handi of biryani; leg pieces, rice and gravy all come out of the same pot. Charge the cost of the handi by weight and the plain rice costs as much per kilo as the leg pieces, so the rice counter looks like a loss-maker. Charge it by what each portion sells for and every counter carries cost in proportion to what it earns. The handi cost the same either way; only the slicing changed.

Key idea

A joint cost cannot be traced to any one joint product; it can only be apportioned on a rational basis. The method (by quantity, by sales value, by net realisable value) changes each product's cost, profit and inventory value, never the total joint cost, so a product's "loss" after apportionment proves nothing by itself.

How it works

The vocabulary, in ICAI's words

Definition

Joint products

Two or more products separated in the course of the same processing operation usually requiring further processing, each product being in such proportion that no single product can be designated as a major product. In other words, two or more products of equal importance, produced simultaneously from the same process, with each having a significant relative sale value. Example: gasoline, fuel oil, lubricants, paraffin, coal tar, asphalt and kerosene from crude petroleum. (SM 11.2)

Definition

Split-off point

A point in a production process where joint products emerging from the process get separately identifiable. Joint cost is incurred up to this point only; any cost incurred after the split-off point is a product-specific cost, borne by the product concerned. (SM 11.3)

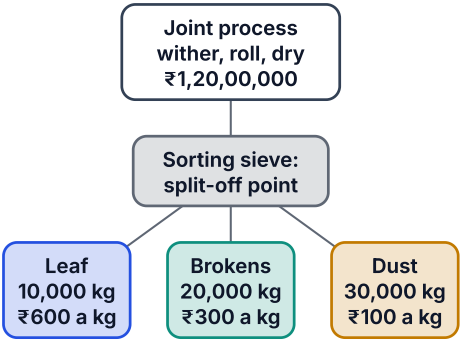
Definition

Joint costs

The expenditures incurred up to the point of separation, i.e. the split-off point: the resources spent for producing more than one product from processing a common input. They include raw material, labour, power, fuel, depreciation and overhead costs towards the production of the joint products. (SM 11.4)

A product of small value that comes out incidentally (molasses beside sugar) is a **by-product**, with its own treatment: treatment of by-products. **Co-products** are two or more products which are contemporary but do not necessarily emerge from the same material in the same process, such as wheat and gram grown on two separate farms (SM 11.3); each co-product is treated as a separate cost object for costing purposes (SM 11.21–11.22, MCQ 9).

The tea factory: one joint process, three joint products

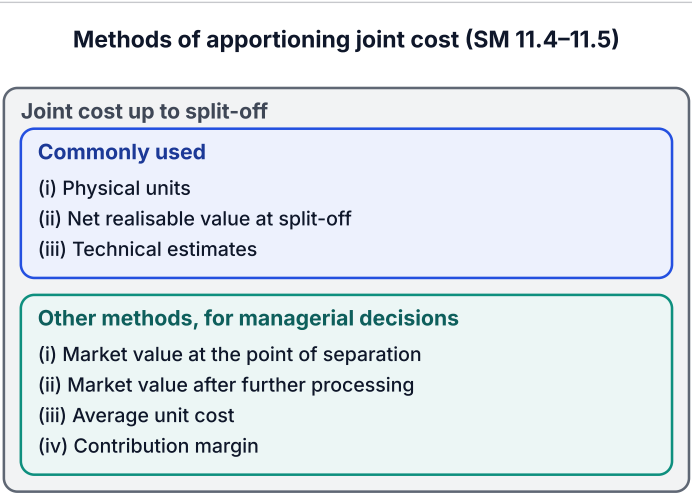


Everything above the split-off point is joint cost; everything below it belongs to one product. Each grade shows the kg made and its auction price.

Why the apportionment matters

ICAI gives three reasons (SM 11.4): it affects (a) the **valuation of closing inventory**; (b) the **pricing of products**; and (c) the **profit or loss on the sale of different products**. Because the relations between materials, processes and joint products are complex and unobservable, the cost of each joint product cannot be correctly identified; it can only be apportioned by using some rational method.

ICAI's seven methods, in ICAI's two groups



ICAI's list and order. Blue box: the three commonly used methods; teal box: the four other methods managers may use for decisions.

In ICAI's order, first the three commonly used methods, then the four others:

1. **Physical units.** Share the joint cost in the ratio of output at split-off in a common unit (kg, litres, numbers). Any loss in the joint process is shared on the same basis. Products in different units must first be converted to a common unit; if they cannot be, the method cannot be applied. Its main defect: it gives equal importance and value to all the joint products (SM 11.5).

2 **Net realisable value (NRV) at split-off.** From each product's final sales value deduct its post split-off processing cost and its directly attributable selling and distribution expenses (freight, royalty, commission), and a profit margin where one is given; share the joint cost in the ratio of what is left (SM 11.6–11.7).

3 **Technical estimates.** Apportion the joint cost on the basis of technical estimates; used when the result of the other methods does not match the resources consumed by the joint products, or their realisable values are not readily available (SM 11.8).

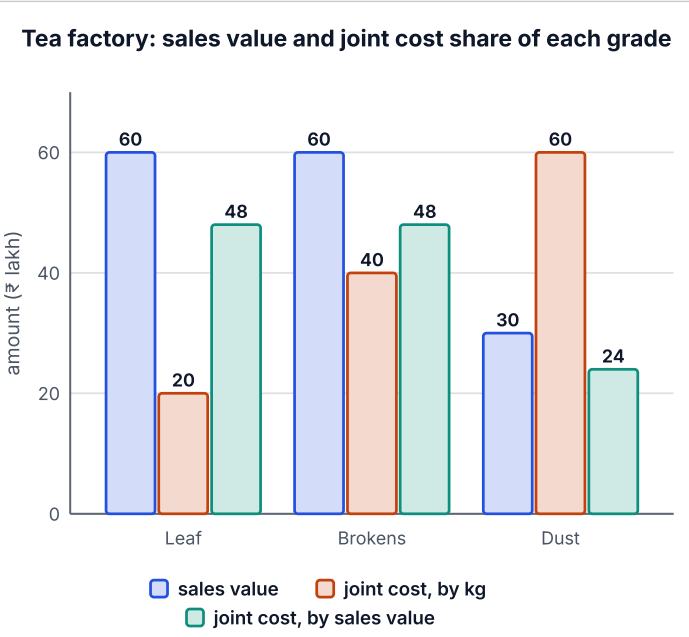
4 **Market value at the point of separation** (sales value at split-off). Share the joint cost in the ratio of each product's quantity produced × its selling price at split-off. ICAI's shortcut is the **multiplying factor**: joint cost ÷ total sales value × 100, applied to each product's sales value (SM 11.8).

5 **Market value after further processing.** Share it in the ratio of final sales values of the finished products (SM 11.9). Unfair when further processing costs are disproportionate or not all products are processed further (SM 11.10).

6 **Average unit cost.** Divide the joint cost by the total units produced and charge every unit that average. It gives the same answer as physical units (SM 11.10).

7 **Contribution margin.** Split the joint cost into variable and fixed. Share the variable part on physical units; add any variable cost after split-off; deduct the total from each product's sales to get its contribution; share the fixed part in the ratio of contributions (SM 11.11).

Weight against value, on the tea factory's numbers



Blue: each grade's sales value. Orange: its share of the ₹120 lakh by weight (₹200 a kg); Dust's ₹60 lakh share is twice its sales. Teal: its share by sales value (80% of each grade's sales).

Interactive version: PassBook app, lesson Apportioning joint costs: physical units, sales value and net realisable value

Which method when: ICAI's circumstances table

Method	When it is used (SM 11.11)
Physical unit	Sale price of all the products is uniform
NRV at split-off	Realisable value at split-off is not known
Technical estimates	NRV result does not match the resources consumed
Market value at separation	Further processing costs are disproportionate
Market value after processing	Further costs proportionate; all products processed
Average unit cost	Units produced have the same unit

Formats and formulas

Reproduce the NRV statement and the formulas from memory; they are the working notes ICAI's answers start with.

∑ **Formula**

Physical units (and average unit cost)

$$\text{Share of a product} = \frac{\text{Joint cost}}{\text{Total units}} \times \text{Units of the product}$$

Units are the quantities **produced** at split-off, in one common unit.

∑ **Formula**

Sales value at split-off: the multiplying factor

$$\text{Multiplying factor} = \frac{\text{Joint cost}}{\text{Total sales revenue}} \times 100$$

Total sales revenue = Σ (quantity produced × market price at split-off). Joint cost of a product = its sales value at split-off × multiplying factor (SM 11.8).

∑ **Formula**

Net realisable value

$$\text{NRV} = \text{Final sales} - \text{Profit margin}$$
$$-\text{S\&D costs} - \text{Post split-off cost}$$

Joint cost of a product = total joint cost × its NRV ÷ total NRV. Final sales value = units after processing × selling price. S&D = directly attributable selling and distribution costs.

NRV statement (ICAI's pro-forma, SM 11.7):

Particulars	Sales value (units × final price)
Product A (₹)	xxx
Product B (₹)	xxx
Product C (₹)	xxx

Particulars
Less: Profit margin
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Less: Selling and distribution costs
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Less: Post split-off cost
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Net realisable value
Product A (₹)
xxx
Product B (₹)
xxx
Product C (₹)
xxx

Particulars	Joint cost apportioned (NRV ratio)
Product A (₹)	XXX
Product B (₹)	XXX
Product C (₹)	XXX

The profit margin line is used only when the question gives a margin; most questions give none, and the line drops out.

∑ **Formula**

Contribution margin method

$$\text{Contribution} = \text{Sales} - \text{Variable cost}$$

Variable cost = variable joint cost shared on physical units + variable cost after split-off. Fixed joint cost of a product = total fixed joint cost × its contribution ÷ total contribution. Its joint cost = its variable share + its fixed share.

∑ **Formula**

Constant gross-margin percentage NRV method

$$\text{Gross margin \%} = \frac{\text{Final sales} - \text{All costs}}{\text{Final sales}} \times 100$$

All costs = joint cost + separable (post split-off) costs; final sales = final sales value of the whole production.

Joint cost of a product = its final sales value × (1 – gross margin %) – its separable costs, so every product earns the same gross margin percentage (SM 11.38–11.39, Case Scenario 1).

Worked examples

Worked example **The Dust that "loses" money**

From April 2026 Tapri stopped auctioning tea: every grade is now packed and sold under its own brand. For May 2026 the factory's accountant has apportioned the joint cost by weight, at ₹150 a kg, and sent Kesar the figures below.

Per kg (₹)	Leaf	Brokens	Dust
Selling price after packing	650	420	200
Packing and blending	120	50	80
Freight and commission	30	20	20
Joint cost, by weight	150	150	150
Output (kg)	8,000	16,000	24,000

Kesar reads it as "Dust loses money; stop making Dust".

Required:

(i) Prepare a statement of profit or loss of each grade on the accountant's figures.

(ii) Apportion the joint cost by the net realisable value method and prepare the statement again.

(iii) Comment on Kesar's reading.

THE WORKING

(i) Profit or loss on the present basis

Per kg (₹)	Leaf	Brokens	Dust
Selling price	650	420	200
Less: Packing and blending	120	50	80
Less: Freight and commission	30	20	20
Less: Joint cost	150	150	150
Profit/(loss) per kg	**350**	**200**	**(50)**

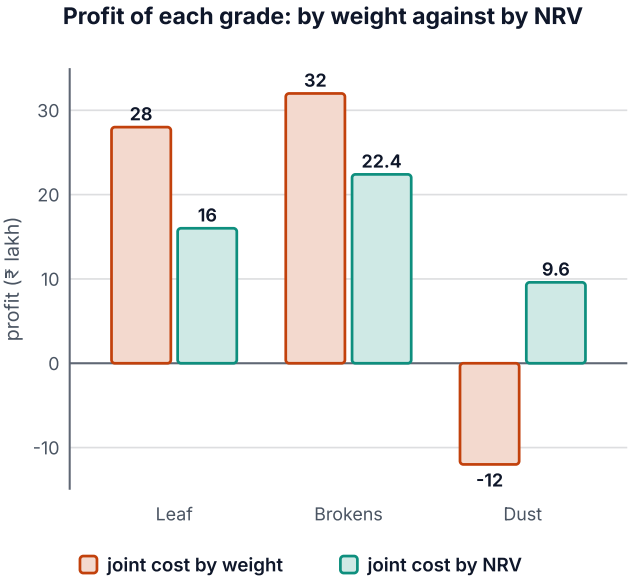
Particulars	Leaf	Brokens	Dust	Total
Output (kg)	8,000	16,000	24,000	48,000
Profit/(loss) (₹)	**28,00,000**	**32,00,000**	**(12,00,000)**	**48,00,000**

(ii) Net realisable value method

Total joint cost = ₹150 × 48,000 kg = **₹72,00,000**.

Particulars (₹)	Leaf	Brokens	Dust	Total
Sales value	52,00,000	67,20,000	48,00,000	1,67,20,000
Less: Packing and blending	9,60,000	8,00,000	19,20,000	36,80,000
Less: Freight and commission	2,40,000	3,20,000	4,80,000	10,40,000
NRV	**40,00,000**	**56,00,000**	**24,00,000**	**1,20,00,000**
Joint cost (5 : 7 : 3)	24,00,000	33,60,000	14,40,000	72,00,000
Profit	**16,00,000**	**22,40,000**	**9,60,000**	**48,00,000**

Joint cost per kg on the NRV basis: Leaf ₹300, Brokens ₹210, Dust ₹60.



Orange: profit with joint cost shared by weight; teal: shared by NRV. Both sets add up to ₹48 lakh; only the split between the grades moves.

(iii) Comment. The loss on Dust comes from the basis of apportionment, not from the product. By weight, Dust carries ₹150 a kg of joint cost although its NRV is only ₹100 a kg. Apportioned by NRV, every grade is profitable: Dust earns ₹9,60,000. The total profit is ₹48,00,000 under both methods. Dust is a joint product: it cannot be stopped without stopping the joint process, and it contributes its NRV of ₹24,00,000 that would be lost if it were thrown away. Kesar's reading is not correct. (Whether each grade should be packed or sold

loose is the separate question of selling at split-off or processing further.)

Worked example **Cocoa butter and powder: when the butter price falls**

Tapri's hot chocolate has to beat BrewBucks, so Tapri presses its own cocoa. In June 2026 its cocoa unit pressed cocoa mass into 8,000 kg of cocoa butter and 12,000 kg of cocoa powder, both sold at the split-off point: butter at ₹850 a kg and powder at ₹350 a kg. The joint costs were ₹50,00,000 of variable cost (cocoa mass, labour and power, which vary with the kilos pressed) and ₹10,00,000 of fixed overheads.

Required:

(i) Apportion June's joint cost by the physical units method and by the contribution margin method, and show the profit of each product under each.

(ii) In July a glut of cocoa butter from BrewBucks' supplier pushes the butter price down to ₹700 a kg; quantities, costs and the powder price do not change. Repeat (i) for July and explain why the powder's profit changes under one method but not under the other.

THE WORKING

(i) June

Physical units method (8,000 : 12,000 kg = 2 : 3; joint cost ₹60,00,000, ₹300 a kg)

Particulars (₹)	Butter	Powder	Total
Sales value (8,000 × ₹850; 12,000 × ₹350)	68,00,000	42,00,000	1,10,00,000
Joint cost (2 : 3)	24,00,000	36,00,000	60,00,000
Profit	**44,00,000**	**6,00,000**	**50,00,000**

Contribution margin method

Particulars (₹)	Butter	Powder	Total
A. Sales value	68,00,000	42,00,000	1,10,00,000
B. Variable joint cost (2 : 3)	20,00,000	30,00,000	50,00,000
C. Contribution (A − B)	48,00,000	12,00,000	60,00,000
D. Fixed joint cost (48 : 12 = 4 : 1)	8,00,000	2,00,000	10,00,000
Profit (C − D)	**40,00,000**	**10,00,000**	**50,00,000**

Joint cost apportioned: butter ₹28,00,000 and powder ₹32,00,000 (against ₹24,00,000 and ₹36,00,000 by physical units).

(ii) July (butter sales 8,000 × ₹700 = ₹56,00,000)

Particulars (₹)	
Physical units: joint cost (unchanged, 2 : 3)	
Butter	24,00,000
Powder	36,00,000
Total	60,00,000

Particulars (₹)	Physical units: profit
Butter	32,00,000
Powder	6,00,000
Total	38,00,000

Particulars (₹)	Contribution (₹56,00,000 – ₹20,00,000; ₹42,00,000 – ₹30,00,000)
Butter	36,00,000
Powder	12,00,000
Total	48,00,000

Particulars (₹)	Fixed joint cost (36 : 12 = 3 : 1)
Butter	7,50,000
Powder	2,50,000
Total	10,00,000

Particulars (₹)	Contribution margin: profit
Butter	28,50,000
Powder	9,50,000
Total	38,00,000

Explanation. Total profit falls by ₹12,00,000 under both methods, because butter's sales fall by that much and the joint cost does not change. The physical units method shares the joint cost by kilos only, so the whole fall stays with butter and the powder's profit is unchanged at ₹6,00,000. The contribution margin method shares the fixed cost by contribution: butter's contribution falls from ₹48,00,000 to ₹36,00,000, so it takes a smaller share of the fixed overheads (₹7,50,000) and ₹50,000 more moves to the powder, whose profit falls to ₹9,50,000 although nothing about the powder changed. Neither apportionment should decide whether to keep a joint product: the joint cost is incurred for both together.

Interactive version: PassBook app, lesson Apportioning joint costs: physical units, sales value and net realisable value

Worked example

Cream, milk powder and the stock in the cold room

Tapri's dairy at Dewas separates whole milk into cream and skimmed milk. The cream is sold at the split-off point to ice-cream makers at ₹360 a kg. There is no market for Tapri's skimmed milk at the split-off point, so it is dried into skimmed milk powder (SMP), sold to bakeries at ₹350 a kg. For the quarter ended 30 June 2026:

Particulars	Cream (kg)	SMP (kg)
Sold	22,000	20,000
Opening stock	1,500	2,400
Closing stock	3,500	4,000

Joint cost up to the split-off point was ₹97,20,000 and drying cost ₹32,40,000. Opening stocks were valued at ₹3,90,000 (cream) and ₹6,96,000 (SMP). Stocks are valued on the FIFO basis, and there was no work in progress.

Required:

(i) Calculate the quantity of each product produced in the quarter.

(ii) Apportion the joint cost by the net realisable value method.

(iii) Calculate the cost of closing inventory and the cost of goods sold of each product.

(iv) Apportion the joint cost by the constant gross-margin percentage NRV method.

THE WORKING

(i) Quantity produced = sold + closing stock − opening stock

Particulars (kg)	Cream	SMP
Sold	22,000	20,000
Add: Closing stock	3,500	4,000
Less: Opening stock	(1,500)	(2,400)
Produced	**24,000**	**21,600**

(ii) NRV method

Particulars (₹)	Cream	SMP	Total
Final sales value of production	86,40,000	75,60,000	1,62,00,000
Less: Drying cost	–	32,40,000	32,40,000
NRV	**86,40,000**	**43,20,000**	**1,29,60,000**
Joint cost (2 : 1)	64,80,000	32,40,000	97,20,000

Cost per kg: cream ₹64,80,000 ÷ 24,000 = **₹270**; SMP (₹32,40,000 + ₹32,40,000) ÷ 21,600 = **₹300**.

(iii) Closing inventory and cost of goods sold

Both selling prices are above cost, so closing stocks are valued at cost.

Particulars (₹)	Cream	SMP
Opening stock	3,90,000	6,96,000
Add: Cost of production	64,80,000	64,80,000
Less: Closing stock	(9,45,000)	(12,00,000)
Cost of goods sold	**59,25,000**	**59,76,000**

Closing stock: cream 3,500 × ₹270 = ₹9,45,000; SMP 4,000 × ₹300 = ₹12,00,000. Check (FIFO):

cream sold = 1,500 kg × ₹260 + 20,500 kg × ₹270 = ₹59,25,000.

(iv) Constant gross-margin percentage NRV method

Gross margin = ₹1,62,00,000 − (₹97,20,000 + ₹32,40,000) = ₹32,40,000, which is **20%** of the final sales value.

Particulars (₹)	Cream	SMP	Total
Final sales value	86,40,000	75,60,000	1,62,00,000
Less: Gross margin (20%)	17,28,000	15,12,000	32,40,000
Cost of goods available	69,12,000	60,48,000	1,29,60,000
Less: Separable cost	–	32,40,000	32,40,000
Joint cost	**69,12,000**	**28,08,000**	**97,20,000**

Here each product earns the same 20% gross margin: cream costs ₹288 a kg and SMP ₹280 a kg. Under the NRV method, cream's margin is 25% (₹270 against ₹360) and SMP's 14.29% (₹300 against ₹350).

Exam angle

In the exam

Where joint cost apportionment appeared, May 2024 to September 2026

In the new-scheme Paper 4 documents from May 2024 to September 2026, joint cost apportionment was examined in 7 of the 8 past papers (September 2024 asked by-product treatment instead), in all 8 RTPs and in most MTPs, often with a further-processing part. The 22 documents read for this lesson (7 past papers, 6 RTPs, 9 MTPs):

- **Descriptive, 4 to 8 marks.** Physical units and contribution margin method: May 2024 Q1(b) (5 marks). NRV to answer a "the biggest product is a loss" complaint: May 2025 Q1(b) (2 + 3 marks). Sales value at split-off with an input-output ratio: May 2026 Q3(a) (8 marks). Sales value at split-off after a by-product credit: January 2026 Q1(b) (5 marks). NRV with closing inventories and cost of goods sold: MTP September 2025 Series II Q4(a) (7 marks), RTP May 2026 Q14. Sales value at split-off, cost per kg and closing stock: RTP January 2026 Q15 and RTP September 2025 Q14. Departments with overheads to apportion first: MTP May 2025 Series II Q5(a) (8 marks). NRV, then an offer for the intermediate product: MTP January 2026 Series I Q1(c) (4 marks). NRV with a by-product from a finishing process: RTP May 2025 Q15.

- **Multiple choice, 2 marks.** NRV with a 25% margin on cost: January 2025 MCQ 12. Contribution margin ratio: September 2025 MCQ 12. Units produced from sales and stocks: RTP May 2026 MCQ 5, RTP September 2025 MCQ 5. Physical units after normal loss and abnormal gain: RTP May 2025 MCQ 6. NRV on production, not sales: RTP January 2025 MCQ 6, MTP May 2026 Series I MCQ 14. Whole case scenarios of five MCQs comparing methods: MTP January 2025 Series II (MCQs 6–10), MTP September 2026 Series II (MCQs 1–5).
- **Theory, 2 to 4 marks.** "Mention the circumstances" for each method: MTP January 2026 Series II Q6(c) (4 marks). "Describe briefly" market value after further processing and NRV: MTP May 2025 Series I Q2(c) (2 marks). Market value at separation and NRV: RTP May 2024 Q13(d), MTP May 2024 Series I Q6(d). How NRV is applied when one product needs refining: September 2026 Q6(b)(ii).

In the exam

The traps ICAI set

- Quantities **sold** given with opening and closing stocks: the base is quantity **produced** = sold + closing − opening (RTP May 2026 MCQ 5; RTP January 2025 MCQ 6, whose wrong option (a) uses units sold).
- Joint cost not given as one figure: add material, conversion cost and overheads first (May 2026 Q3(a): raw material plus Department I conversion cost), or apportion factory overheads to the departments first (MTP May 2025 Series II Q5(a)).
- Selling price "assuming a net margin of 25% on cost": deduct the margin before the post split-off cost (January 2025 MCQ 12).
- Contribution margin method: variable cost on physical units, fixed cost on contribution (May 2024 Q1(b); September 2025 MCQ 12).

Trick

Check the total

Whatever the method, the shares must add back to the total joint cost, and (with no closing stock) the product profits must add to the same total profit. Condition: the same joint cost, and no by-product credit taken in one method but not the other.

Answer frame

"Describe briefly how joint costs are apportioned under ..."

The chapter's own theory question (SM 11.23, Q4). Write each method as a definition, the basis and one limitation, in this order:

1. **Average unit cost**: total process cost up to separation ÷ total units produced; all products get a uniform cost per unit; if used for pricing, all products may have more or less the same price.
2. **Contribution margin**: joint cost split into variable and fixed; variable apportioned on physical quantities, plus variable cost after split-off; contribution = sales − variable cost; fixed apportioned in the ratio of contributions.
3. **Market value at the point of separation**: joint cost apportioned in the ratio of sales value at split-off (multiplying factor = joint cost ÷ total sales value × 100); difficult when the market value at separation is not available; useful when further processing costs are disproportionate.
4. **Market value after further processing**: ratio of final sales values; unfair when further processing costs are disproportionate or not all products are processed further.
5. **Net realisable value**: from final sales value deduct estimated profit margin, selling and distribution

expenses and post split-off costs; joint cost apportioned in the ratio of NRV.

"Mention the circumstances ..."

Reproduce ICAI's circumstances table from How it works, one line a method, in ICAI's words.

The numerical parts: three answers

Loses marks

"Cream ₹64,80,000; SMP ₹32,40,000."

Bare figures: no NRV, no ratio, no units produced. A slip anywhere earns nothing.

Passes

The NRV table with the ratio, but NRV worked on units sold.

Right method, wrong base: the method earns marks, but every figure after it is wrong.

Full marks

Working note for quantity produced; the NRV statement in ICAI's layout; the ratio stated; each share; a check that the shares add to the joint cost; a one-line comment where asked.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the joint cost that followed the sales invoice

Apportioning on units **sold** when the question gives stocks. The joint cost was spent on everything produced, including what still sits in the cold room. **Fix:** produced = sold + closing stock − opening stock, then apportion.

⚠ **Common mistake**

NRV that forgot to subtract anything

Taking the final sales values as the "NRV" (that is the market value after further processing method). In the tea factory it would give Dust ₹48,00,000 ÷ ₹1,67,20,000 of the joint cost instead of ₹24,00,000 ÷ ₹1,20,00,000. **Fix:** NRV = final sales value − post split-off cost − attributable selling and distribution costs (− profit margin if given).

⚠ **Common mistake**

The contribution margin method done in one step

Sharing the whole joint cost in the ratio of contributions, or working contribution without the variable joint cost. **Fix:** variable joint cost on physical units first; contribution = sales − variable cost; only the fixed joint cost follows the contribution ratio.

⚠ **Common mistake**

Kilos and litres added together

Adding 4,000 kg and 3,000 litres for the physical units method. **Fix:** convert to a common unit first (litres to kg through the density, for example); if no common unit exists, the method cannot be applied (SM 11.5).

Practice

Check yourself

1. Joint cost of ₹9,60,000 is apportioned on units produced. Product P: sold 30,000 kg, opening stock 2,000 kg, closing stock 6,000 kg. Product Q: sold 44,000 kg, no opening stock, closing stock 2,000 kg. What joint cost (₹) goes to P?

Answer: _____

2 Case scenario. Ritu is choosing a method for Tapri's spice unit. Market prices at the split-off point are known for all three products, and their further processing costs differ hugely: one is only bagged, another is roasted, ground and packed. According to ICAI's circumstances, which method suits this case?

A Physical units method

B Market value after further processing

C Market value at the point of separation

D Average unit cost method

3 Joint cost is ₹6,00,000. Product M has a final sales value of ₹6,00,000 and further processing cost of ₹3,00,000; product N ₹4,00,000 and ₹1,00,000. What joint cost goes to M under the NRV method?

A ₹3,60,000

B ₹3,00,000

C ₹2,40,000

D ₹4,50,000

4 Joint variable cost ₹2,40,000 and joint fixed cost ₹1,20,000 produce X 3,000 kg (sold at ₹100) and Y 5,000 kg (sold at ₹60). Under the contribution margin method, what total joint cost (₹) goes to X?

Answer: _____

5 Which of these methods charge every unit of every joint product the same joint cost per unit? Select all that apply.

- **A** Physical units method
- **B** Average unit cost method
- **C** Net realisable value method
- **D** Market value at the point of separation

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Joint products: two or more products of equal importance, produced simultaneously from the same process, each with a significant relative sale value (SM 11.2).
- Joint cost is everything up to the split-off point; after it, costs belong to the product concerned.

- Apportionment affects inventory valuation, pricing and product profit; the cost cannot be identified, only apportioned on a rational basis.
- Commonly used: physical units, NRV at split-off, technical estimates. Other: market value at separation, market value after further processing, average unit cost, contribution margin.
- Multiplying factor = joint cost ÷ total sales value at split-off × 100.
- NRV = final sales − profit margin − selling and distribution costs − post split-off cost.
- Contribution margin method: variable joint cost on units, fixed joint cost on contribution.
- Base is units produced (sold + closing − opening). The handi cost the same; only the slicing changed.

Answer key · Check yourself

1 **408000**

Produced: P = 30,000 + 6,000 − 2,000 = 34,000 kg; Q = 44,000 + 2,000 = 46,000 kg; total 80,000 kg. P's share = ₹9,60,000 × 34,000 ÷ 80,000 = ₹4,08,000. Using units sold gives ₹3,89,189.

2 **C Market value at the point of separation**

ICAI uses market value at the point of separation "where further processing costs are incurred disproportionately" (SM 11.11). Final market values would favour the product with heavy processing; physical units and average unit cost suit uniform prices or a common unit.

3 **B ₹3,00,000**

NRV: M = ₹3,00,000, N = ₹3,00,000, so 1 : 1 and ₹3,00,000 each. ₹3,60,000 shares on final sales values (6 : 4), the market value after further processing method; ₹4,50,000 shares on further processing costs.

4 **160000**

Variable cost on kg 3 : 5: X ₹90,000, Y ₹1,50,000. Contribution: X ₹3,00,000 − ₹90,000 = ₹2,10,000; Y ₹3,00,000 − ₹1,50,000 = ₹1,50,000; ratio 7 : 5. Fixed to X = ₹1,20,000 × 7/12 = ₹70,000. Total ₹90,000 + ₹70,000 = ₹1,60,000.

5 **A Physical units method; B Average unit cost method**

Both divide the joint cost by total units, so every unit carries the same amount; ICAI notes they give the same outcome (SM 11.10). The value-based methods charge more to the products that earn more.

11.2 Sell at split-off or process further

At the split-off point the joint cost is already spent. Process a joint product further only if the extra revenue beats the extra cost; ICAI's statement and its traps.

In plain words

Kesar discovers "value addition"

Tapri's tea factory can sell each grade loose at the Guwahati auction, or pack it under the Tapri brand. Kesar has just come back from a seminar.

> **Kesar:** Value addition, Ritu! Pack everything. Packed tea always sells for more.
>
> **Ritu:** Brokens fetch ₹300 a kg at auction and ₹420 packed. Dust fetches ₹100 at auction and ₹180 in sachets.
>
> **Kesar:** So both go up by more than ₹70. Pack them!
>
> **Ritu:** Per unit, please. Packing and marketing cost ₹100 a kg for Brokens and ₹90 a kg for Dust.

Packing earns Brokens ₹120 a kg more and costs ₹100 a kg: ₹20 a kg better off, so Brokens should be packed. Packing earns Dust ₹80 a kg more and costs ₹90 a kg: ₹10 a kg worse off, so Dust should go to the auction loose.

The extra revenue from processing further is the **incremental revenue** (final price minus the split-off price); the extra cost it causes is the **incremental cost**. Notice what is missing: the ₹200 a kg of joint cost that each grade carried in the books. It was spent in the dryer before the sieve, and it is the same whether the tea is packed or not, so it cannot change the answer.

Think of it like

The tender-coconut vendor

The vendor has already paid ₹40 for the coconut and sold its water. Should he also crack it open and scoop out the malai for a customer who will pay ₹10 more? That depends only on the ₹10 and on his minute of work. The ₹40 he paid this morning is the same whichever he does.

Key idea

At the split-off point the joint cost is already spent and is the same whichever way you go, so it is irrelevant. Process a product further only if its incremental revenue (sales after further processing minus sales value at split-off) exceeds its incremental cost (every cost the further processing adds, less any it saves).

How it works

Six steps to the decision

1. **Sales value at split-off**: the quantity available at split-off × the split-off price (or the price offered for the intermediate product).

2. **Sales value after further processing**: the quantity that comes out of the further process × the final price. A loss or yield in further processing reduces the quantity: 4,000 units with a 2% loss leave 3,920 to sell.

3. **Incremental revenue** = step 2 − step 1.

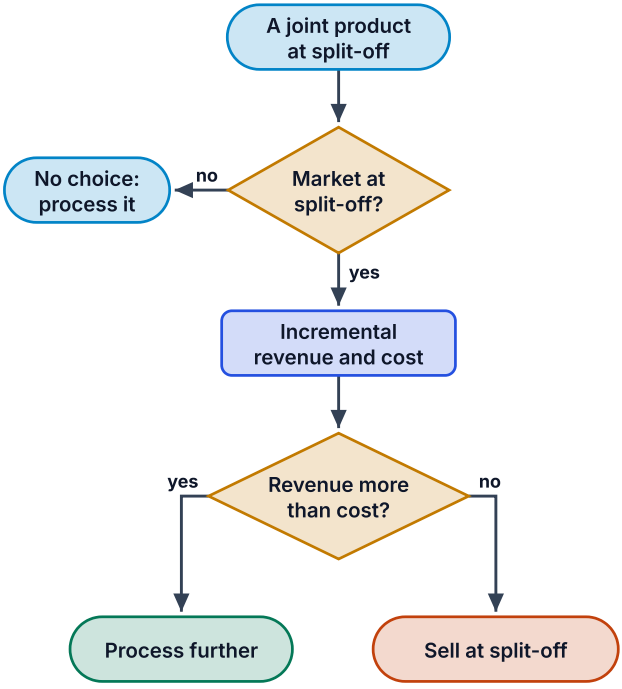
4. **Incremental cost**: the further processing cost (materials, labour, variable overheads, and fixed costs that arise only because of the processing), plus any additional selling, packing or marketing costs, less any selling expenses saved.

5. **Incremental profit or loss** = step 3 − step 4.

6. **Decide**: positive, process further; negative, sell at split-off. ICAI writes the conclusion in this form: It is advisable to further process only product X and Z and to sell product Y at the point of separation (SM 11.36).

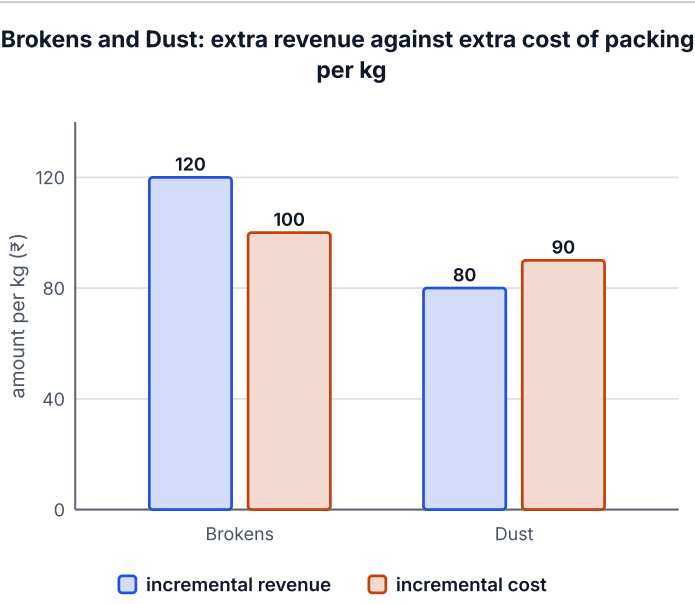
This is the relevance principle of short-term decisions: a cost is relevant only if (a) it is a future cost and (b) it differs under the options under consideration (SM 14.33). The joint cost fails (b); further processing costs pass both. "Sale or further processing" is one of ICAI's short-term decisions related with excess demand (SM 14.36); the other decisions are in make or buy, shut down and other short-term decisions.

Sell at split-off or process further?



Green: process further; orange: sell at split-off. Joint costs appear nowhere in the path. A product with no market at split-off has no choice to make; a tie is decided on factors other than profit.

The tea factory's two grades, side by side



Blue: what packing adds to the price; orange: what packing and marketing cost. Brokens' blue bar is taller (pack them, ₹20 a kg better); Dust's orange bar is taller (sell loose, ₹10 a kg better).

What goes into the decision

Item	In the decision?	Why
Joint cost up to split-off	No	Spent; same either way
Share of joint cost, any method	No	A slice of the same cost

Item	In the decision?	Why
Further processing materials, labour, variable overheads	Yes	Arise only if processed
Fixed cost of a department used only for it	Yes	Avoided if sold at split-off
General overhead apportioned to it	No	Continues either way
Extra selling, packing, marketing	Yes	Arise only if processed
Selling expenses saved	Yes, deduct	Avoided by processing
Loss in further processing	Yes	Fewer units to sell

ICAI counted the whole conversion cost of a department "used exclusively" for the product, fixed overhead included, and deducted the selling expenses saved (May 2026 Q3(a), suggested answer); it added the further marketing cost to the further processing cost (January 2026 Q1(b)).

Interactive version: PassBook app, lesson Sell at split-off or process further

Formats and formulas

Reproduce the incremental statement and the plan statement from memory; ICAI's suggested answers use exactly these layouts.

∑ **Formula**

Incremental profit from further processing

$$\text{Incremental profit} = (\text{Sales after processing} - \text{Sales at split-off}) - \text{Incremental cost}$$

Sales after processing = output after any loss × final price. Sales at split-off = quantity at split-off × split-off price. Process further if the result is positive.

∑ **Formula**

Lowest final price that justifies further processing

$$\text{Minimum final price} = \frac{\text{Sales value at split-off} + \text{Incremental cost}}{\text{Output after further processing}}$$

At this price the incremental profit is nil.

Statement of incremental profit or loss (per product; per unit or in total):

Particulars	Product X (₹)	Product Y (₹)
Sales value after further processing	xxx	xxx
Less: Sales value at split-off	(xxx)	(xxx)
Incremental revenue	**xxx**	**xxx**
Less: Further processing cost	(xxx)	(xxx)
Less: Further marketing cost	(xxx)	(xxx)
Add: Selling expenses saved	xxx	xxx
Incremental profit/(loss)	**xxx**	**xxx**

Statement of profit under a production plan (SM 11.31–11.33; one column a product, then a total):

Line	Particulars
A	Sales revenue
B	Apportioned joint costs
C	Further processing cost
D	Total cost (B + C)
E	Excess of sales revenue over cost (A − D)
F	Fixed cost
G	Profit (E − F)

Prepare it for "none processed further", "all processed further" and "the suggested plan" when asked. Line B is the same in all three plans, which is why the plans differ by exactly the incremental profits.

Worked examples

Worked example **Two masalas, one sachet line**

Tapri's spice unit grinds whole spices in a joint process into 20,000 kg of coarse garam masala and 30,000 kg of fine chai masala a month, at a joint cost of ₹55,00,000. At the split-off point coarse masala sells at ₹150 a kg and fine masala at ₹120 a kg. Either can

instead be blended and packed in branded sachets on the sachet line:

Particulars	Coarse masala	Fine masala
Sachet price (₹ per kg of sachets)	240	200
Yield in sachets (per kg put in)	95%	90%
Variable blending and packing cost (₹ per kg put in)	40	30
Sachet-line hours (per kg put in)	0.10	0.05

The sachet line can run 2,500 hours a month; its fixed cost of ₹1,50,000 a month is incurred whether it runs or not.

Required:

(i) Apportion the joint cost on the basis of sales value at the split-off point.

(ii) Calculate the incremental profit of putting each masala through the sachet line, per kg and per sachet-line hour.

(iii) Recommend how the sachet line's hours should be used, and calculate the incremental profit of your plan. Compare it with giving priority to the masala with the higher incremental profit per kg.

(iv) Calculate the lowest sachet price per kg at which putting coarse masala through the line is still worthwhile.

THE WORKING

(i) Joint cost on sales value at split-off

Particulars	Sales value at split-off (₹) (20,000 × ₹150; 30,000 × ₹120)
Coarse	30,00,000
Fine	36,00,000
Total	66,00,000

Particulars	Joint cost apportioned (5 : 6) (₹)
Coarse	25,00,000
Fine	30,00,000
Total	55,00,000

(ii) Incremental profit

Per kg put in (₹)	Coarse	Fine
Sachet revenue (95% × ₹240; 90% × ₹200)	228	180
Less: Revenue given up at split-off	150	120
Incremental revenue	**78**	**60**
Less: Incremental cost (blending and packing)	40	30
Incremental profit per kg	**38**	**30**
Sachet-line hours per kg	0.10	0.05
Incremental profit per line-hour	**380**	**600**

The joint cost and the line's fixed cost are the same whatever is decided, so neither enters the comparison.

(iii) Use of the sachet line. The line's hours are the limiting factor, so the masalas are ranked by incremental profit per line-hour: fine masala first.

Plan

Recommended: fine first

Fine masala

All 30,000 kg (1,500 hours)

Coarse masala

10,000 kg (the remaining 1,000 hours)

Incremental profit (₹)

9,00,000 + 3,80,000 = **12,80,000**

Plan

Coarse first (higher profit per kg)

Fine masala

10,000 kg (the remaining 500 hours)

Coarse masala

All 20,000 kg (2,000 hours)

Incremental profit (₹)

3,00,000 + 7,60,000 = 10,60,000

It is advisable to put all the fine masala and 10,000 kg of the coarse masala through the sachet line and to sell the other 10,000 kg of coarse masala at the point of separation. This earns ₹2,20,000 a month more than giving priority to coarse masala, whose higher profit per kg uses twice the line time.

(iv) Lowest sachet price for coarse masala = (split-off revenue given up + incremental cost) ÷ yield = (₹150 + ₹40) ÷ 0.95 = **₹200 a kg** of sachets. At that price the line adds nothing; at ₹240 it adds ₹38 a kg put in.

Worked example **Four grades and a packing line**

Tapri's tea factory produced the following in October 2026, at a joint cost of ₹1,00,80,000. Each grade can be sold at the auction or packed under the Tapri brand. In packing, 5% of the Fannings is lost as fluff with no value; the other grades lose nothing.

Grade	Output (kg)	Auction price (₹/kg)	Packed price (₹/kg)	Packing and marketing (₹ per kg put in)
Leaf	6,000	500	800	160
Brokens	18,000	300	420	100
Fannings	12,000	200	260	50
Dust	24,000	100	180	90

The packing line also has fixed costs of ₹9,00,000 a month, which would be saved if the line were closed and everything sold at the auction.

Required:

(i) Calculate the incremental profit or loss of packing each grade, before the packing line's fixed costs.

(ii) Decide which grades to pack and whether the packing line is worth running at all; calculate the factory's profit under your plan and if everything were sold at the auction.

(iii) Kesar wants every grade packed "because packed tea always sells for more". Show the effect on profit compared with your plan.

THE WORKING

(i) Incremental profit or loss of packing

Grade

Leaf: 6,000 × (₹800 – ₹500)

Incremental revenue (₹)

18,00,000

Incremental cost (₹)

9,60,000

Incremental profit/(loss) (₹)

8,40,000

Grade

Brokens: 18,000 × (₹420 – ₹300)

Incremental revenue (₹)

21,60,000

Incremental cost (₹)

18,00,000

Incremental profit/(loss) (₹)

3,60,000

Grade

Fannings: 11,400 × ₹260 – 12,000 × ₹200

Incremental revenue (₹)

5,64,000

Incremental cost (₹)

6,00,000

Incremental profit/(loss) (₹)

(36,000)

Grade

Dust: 24,000 × (₹180 – ₹100)

Incremental revenue (₹)

19,20,000

Incremental cost (₹)

21,60,000

Incremental profit/(loss) (₹)

(2,40,000)

Fannings packed = 12,000 × 95% = 11,400 kg. Incremental cost = kg put in × packing and marketing cost a kg.

(ii) Plan. Only Leaf and Brokens add to profit, so only they should be packed. Together they add ₹8,40,000 + ₹3,60,000 = ₹12,00,000, which covers the line's fixed costs of ₹9,00,000 and leaves ₹3,00,000: the line is worth running. Leaf alone would not keep it going (₹8,40,000 < ₹9,00,000): Brokens' modest ₹20 a kg is what makes the line pay.

Particulars (₹)	Everything at the auction	Recommended plan
Sales: Leaf	30,00,000	48,00,000
Sales: Brokens	54,00,000	75,60,000
Sales: Fannings	24,00,000	24,00,000
Sales: Dust	24,00,000	24,00,000
Total sales	**1,32,00,000**	**1,71,60,000**
Less: Joint cost	1,00,80,000	1,00,80,000
Less: Packing and marketing (Leaf ₹9,60,000; Brokens ₹18,00,000)	–	27,60,000

Particulars (₹)	Everything at the auction	Recommended plan
Less: Fixed costs of the packing line	–	9,00,000
Profit	**31,20,000**	**34,20,000**

It is advisable to pack only Leaf and Brokens and to sell Fannings and Dust at the auction.

(iii) Every grade packed. Packing Fannings and Dust as well adds their incremental losses of ₹36,000 and ₹2,40,000: profit falls to ₹34,20,000 – ₹2,76,000 = **₹31,44,000**. Packed tea does sell for more, but for Fannings and Dust the extra price does not cover the extra cost (and the fluff lost in packing Fannings).

Exam angle

In the exam

Where the decision appeared, May 2024 to September 2026

In the new-scheme Paper 4 past papers, RTPs and MTPs from May 2024 to September 2026, the sell-or-process-further decision for a joint product was examined in 10 documents (3 past papers, 3 RTPs, 4 MTPs), always after a joint cost apportionment in the same question. (MTP September 2026 Series I also

asks it for a single intermediate product in a process costing case.)

- **Descriptive, as one part of a 4 to 8-mark question.** Product B with a 2% loss in further processing: May 2024 Q1(b)(ii) ("Explain the impact on profitability"). Input-output ratios, an exclusive department and selling expenses saved: May 2026 Q3(a)(ii) (8 marks for the whole question). Further processing plus further marketing cost: January 2026 Q1(b)(iii). Per-kg incremental statement: RTP January 2025 Q14(ii), RTP January 2026 Q15(iv), RTP September 2025 Q14(iv). Current policy against the better plan, with profit statements: MTP May 2025 Series II Q5(a)(iii)–(iv) (8 marks). An offer for the intermediate product: MTP January 2026 Series I Q1(c)(ii) (4 marks).
- **Multiple choice, 2 marks.** Which product loses after further processing, total profit, and the profit of the optimum plan: MTP September 2026 Series II MCQs 1, 2, 4 and 5. Incremental revenue of processing Halogen into Vinyl: MTP January 2025 Series II MCQ 10.

In the exam

The wording traps ICAI used

- "Loss in further processing is 2%": incremental revenue uses 98% of the units at the final price against 100% at the split-off price (May 2024 Q1(b)(ii); ICAI showed the answer both as a profit comparison, ₹22,500 against ₹21,060, and as an incremental loss of ₹1,440).
- Two input-output ratios in a row: the output of Department I feeds Department II (May 2026 Q3(a)).
- "Further marketing cost (per kg)" sits next to the further processing cost: both are incremental (January 2026 Q1(b)).
- "Selling expense" for the product at split-off and after processing: only the difference counts (May 2026 Q3(a)).

Interactive version: PassBook app, lesson Sell at split-off or process further

Trick

Per kg first

When prices and costs are given per kg and there is no loss in further processing, work the decision per kg (final price – split-off price – further cost per kg) and multiply once at the end, as RTP January 2025 Q14 does. Condition: no loss or yield change in further processing and no fixed further processing cost; otherwise work in totals.

Answer frame

The "advise" part: how ICAI words it

1. A statement of incremental revenue, incremental cost and incremental profit or loss, product by product (per unit or total).
2. The working notes for quantities after any loss or yield.
3. One sentence of advice in ICAI's form: "It is advisable to further process only product ___ and to sell product ___ at the point of separation."
4. Where asked, the profit statement of the suggested plan, with the increase over the present plan.

Three answers to "Should the fine masala go into sachets?"

Loses marks

"Yes, sachets sell for ₹200 a kg against ₹120."

Compares prices only: no costs, no loss in processing, no figures.

Passes

"Profit after sachets = ₹54,00,000 − ₹30,00,000 − ₹9,00,000 = ₹15,00,000, so pack."

Right decision, but drags in the joint cost and gives no incremental figure to compare with.

Full marks

Yield working (27,000 kg of sachets); incremental revenue ₹18,00,000; incremental cost ₹9,00,000; incremental profit ₹9,00,000 (₹30 a kg, ₹600 a line-hour); the ranking by line-hours; the advice sentence; joint cost stated as irrelevant.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the joint cost that sneaked back in

Deducting the product's share of joint cost from the further-processing revenue and deciding on what is left, or dropping a product because it shows a loss after its share of joint cost. **Fix:** the joint cost is the same whether the product is sold at split-off or processed further; compare only incremental revenue with incremental cost.

⚠ **Common mistake**

The 5% that vanished

Multiplying the final price by the quantity at split-off when the further process loses part of it. In the tea factory that turned the Fannings' loss of ₹36,000 into a profit of ₹1,20,000. **Fix:** final sales = output **after** the loss × final price; split-off sales = quantity **before** the process × split-off price.

⚠ **Common mistake**

Comparing the final sales with the further processing cost alone

"Fine masala sachets bring ₹54,00,000 and cost ₹9,00,000 to make, so pack." This forgets the ₹36,00,000 the fine masala would have brought at split-off. **Fix:** incremental revenue is the *difference* between the two sales values; the split-off sales given up are part of the comparison.

⚠ **Common mistake**

Forgetting the costs on either side of the factory gate

Leaving out further marketing or packing costs, or the selling expenses the processed product saves. **Fix:** every cost that changes with the choice goes in, with its sign.

Practice

Check yourself

1 A joint product gives 10,000 kg at split-off, saleable at ₹45 a kg. Processed further, 5% is lost and the rest sells at ₹60 a kg; further processing costs ₹1,00,000 plus packing of ₹2 per kg of output. What is the incremental profit or loss (₹) from processing further? (Enter a loss as a negative number.)

Answer: _____

2 Case scenario. Tapri's cocoa unit can sell 4,000 kg of natural cocoa powder at the split-off point at ₹360 a kg, or alkalise it into 3,600 kg of Dutch cocoa selling at ₹450 a kg. Alkalising costs ₹1,60,000, and packing ₹20 per kg of Dutch cocoa. Under the NRV method the powder carries ₹7,00,000 of joint cost. Ritu must advise Kesar. Which advice is correct?

- **A** Process: Dutch cocoa brings ₹16,20,000 against ₹14,40,000 for the powder
- **B** Process: Dutch cocoa earns ₹6,88,000 after its joint cost and further costs
- **C** Sell the powder at split-off: processing loses ₹52,000
- **D** Indifferent: the ₹7,00,000 joint cost is the same either way

3 After apportioning the joint cost, product Z shows a loss of ₹20,000 when processed further. Its incremental revenue from further processing is ₹90,000 and its further processing cost ₹60,000. What should be done?

- **A** Stop producing Z
- **B** Sell Z at split-off, because it shows a loss
- **C** Process Z further
- **D** Close the joint process

4 Which of these belong in the incremental cost of processing a joint product further? Select all that apply.

- **A** Materials and labour used in the further process
- **B** The product's share of joint cost under the NRV method
- **C** Salary of a supervisor engaged only for the further process
- **D** A share of general factory rent that is paid either way
- **E** Additional marketing cost of the processed product

More than one answer may be right.

5 A product gives 10,000 kg at split-off worth ₹40 a kg. Further processing costs ₹1,40,000 and loses 10% of the weight. What is the lowest final price per kg (₹) at which processing further is worthwhile?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Joint cost is spent before the split-off point and is the same whatever is decided after it: irrelevant to sell-or-process-further.
- Incremental revenue = sales after further processing − sales value at split-off.
- Incremental cost = further processing cost + extra selling, packing, marketing − selling expenses saved; fixed costs only if they arise because of the processing.
- Process further if incremental revenue > incremental cost; otherwise sell at the split-off point.
- A loss in further processing cuts the output: final sales use the units after the loss.
- An offer for the product at split-off replaces the split-off price in the comparison.
- ICAI's advice: "It is advisable to further process only product ___ and to sell product ___ at the point of separation."

- The coconut cost ₹40 either way; only the malai's ₹10 and the minute matter.

Answer key · Check yourself

1 **1000**

Output 9,500 kg. Incremental revenue = 9,500 × ₹60 − 10,000 × ₹45 = ₹5,70,000 − ₹4,50,000 = ₹1,20,000. Incremental cost = ₹1,00,000 + 9,500 × ₹2 = ₹1,19,000. Incremental profit ₹1,000: just worth it. Ignoring the loss would give ₹30,000.

2 **C** **Sell the powder at split-off: processing loses ₹52,000**

Incremental revenue = 3,600 × ₹450 − 4,000 × ₹360 = ₹16,20,000 − ₹14,40,000 = ₹1,80,000; incremental cost = ₹1,60,000 + 3,600 × ₹20 = ₹2,32,000; incremental loss ₹52,000. The first option ignores the costs; the second measures Dutch cocoa's profit after joint cost instead of comparing the two choices; the fourth is true of the joint cost but ignores the incremental figures.

3 **C** **Process Z further**

Processing adds ₹90,000 − ₹60,000 = ₹30,000. The ₹20,000 "loss" includes Z's share of the joint cost, which does not change with the decision; Z cannot be stopped without stopping the joint process.

4 **A** **Materials and labour used in the further process**; **C** **Salary of a supervisor engaged only for the further process**; **E** **Additional marketing cost of the processed product**

A cost belongs if it arises only because of the further processing. The joint cost and the general rent are the same under both choices.

5 **60**

Output 9,000 kg. Minimum price = (₹4,00,000 + ₹1,40,000) ÷ 9,000 = ₹60 a kg. Dividing by 10,000 kg gives ₹54, a price at which processing would lose ₹54,000.

11.3 Treatment of by-products

What makes a by-product, ICAI's four ways of valuing it, and its three treatments in cost accounts: small value, considerable value, and value only after further processing.

In plain words

Kesar finds money on the floor

Every evening at Tapri's tea factory the sorting room is swept. The sweepings (fibre, stalks and fluff) are cleaned, bagged and sold to a caffeine-extraction plant in Guwahati. A month's sweepings fetch about ₹1 lakh. A month's tea fetches about ₹70 lakh.

> **Kesar:** Free money! Put it in miscellaneous income.
>
> **Ritu:** It is not free, Kesar-ji. It came out of the same green leaf, and the cleaning and freight cost us.
>
> **Kesar:** Fine. Then miscellaneous income, minus cleaning.
>
> **Ritu:** That is one of the ways ICAI allows. The question is which way is right for us.

The sweepings are a **by-product**: something recovered incidentally while making the main products, worth little compared with them. Nobody runs a tea factory for the fluff. So the fluff is not given its own share of the joint cost the way Leaf and Brokens are. Instead, what it brings in, after the costs of cleaning and selling it (its **net**

realisable value), either lowers the cost of the main products or is shown as other income in the Costing Profit and Loss Account.

That holds only while the by-product stays small. If caffeine prices tripled and the sweepings started earning as much as the Brokens, they would have to be treated as a **joint product** and carry a share of the joint cost.

Think of it like

The barber's floor

A barber sweeps up the hair and sells it to a wig-maker. Nobody opens a salon for the hair; the wig money just makes each haircut a little cheaper to provide. If wigs became so profitable that the hair earned as much as the haircuts, the barber would have to start costing the hair as a product of its own.

Key idea

A by-product is recovered incidentally and is of small value compared with the main product. Its realisation, net of the costs of processing and selling it, is credited (to the cost of the main product, or to the Costing Profit and Loss Account when small) instead of the by-product carrying a share of joint cost; once its value is considerable, it is treated as a joint product.

How it works

What ICAI calls a by-product

Definition

By-products

Products recovered from material discarded in a main process, or from the production of some major products, where the material value is to be considered at the time of severance from the main product. In a nutshell, a product recovered incidentally from the material used in the manufacture of main or desired products, having either a net realisable value or a usable value which is relatively insignificant in comparison with the saleable value of the main or desired products. By-products may be further processed to increase their realisable value. (SM 11.2–11.3)

ICAI's examples: **molasses** in the manufacture of sugar; **tar, ammonia and benzole** obtained on carbonisation of coal; **glycerin** obtained in the manufacture of soap (SM 11.3). Its MCQs add **oil cakes** beside edible oils (SM 11.21, MCQ 7; key (b) on SM 11.29).

Joint product or by-product?

Point	Joint products	By-products
Importance	Of equal importance	Of small economic value
How produced	Simultaneously	Incidentally, in addition to the main products
Cost up to split-off	Joint cost apportioned on a rational basis	Realisation credited: deducted from cost, or to Costing P&L
Examples	Petrol, kerosene, lubricants from crude oil	Molasses, glycerin, tar

The first two rows are ICAI's own two points of distinction (SM 11.3); the third follows from its methods (SM 11.4, 11.15).

Four ways to put a value on a by-product

ICAI lists these methods for accounting for by-products and arriving at the cost of production of the main product, in this order (SM 11.15–11.16):

1. **Net realisable value method.** The realisation on the disposal of the by-product is deducted from the total cost of production to arrive at the cost of the main product: what is realised from molasses reduces the cost of sugar. If the by-product needs further processing to be saleable, deduct those expenses from its realisation and credit only the **net** realisation. Keep separate accounts for (a) further processing of the by-product and (b) selling, distribution and administration expenses attributable to it.

2. **Standard cost in technical estimates.** Value the by-product at a standard cost, set by averaging past costs and making technical estimates of how much of the original raw material goes into the main product and how much forms the by-product (or another consistent basis). Used where the by-product is not saleable in the condition in which it emerges, or comparative prices of similar products are not available.

3 **Comparative price.** Value it at the price of a similar or an alternative material. A blast furnace that makes steel for car bodies also gives gas used in the factory: value the gas at what a gas company would have charged.

4 **Re-use basis.** A by-product that can be put back into the process as part of its input is valued at the same value as the materials introduced into that process.

Three treatments in the cost accounts

ICAI's answer to "Discuss the treatment of by-product cost in cost accounting" has three cases (SM 11.17):

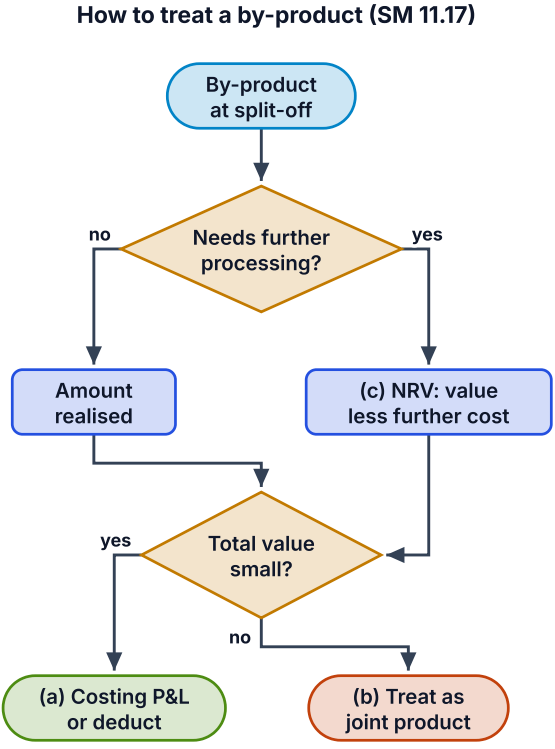
(a) When they are of small total value, the amount realised from their sale may be dealt with in either of two ways:

1. The sales value of the by-products may be credited to the **Costing Profit and Loss Account** and no credit be given in the cost accounts; the credit is treated either as **miscellaneous income** or as **additional sales revenue**.
2. The sale proceeds may be treated as **deductions from the total costs**: deducted either from the **production cost** or from the **cost of sales**.

(b) When they are of considerable total value, they may be regarded as joint products rather than as by-products.

The costs up to the point of separation are apportioned over the by-products and joint products on a logical basis: relative market values, physical output (at the point of split-off) or ultimate selling prices (if sold).

(c) Where they require further processing, the net realisable value of the by-product at the split-off point is its realisable value less the further processing cost. If that total is small, treat it as in (a); if considerable, as in (b).



Case (c) is the yes-branch: work out the NRV at split-off first, then treat it as (a) or (b). Green: case (a), small value; orange: case (b), considerable value.

Interactive version: PassBook app, lesson Treatment of by-products

Where the credit sits in the cost sheet

In a manufacturer's cost sheet the credit appears as "Less: Credit for Recoveries/Scrap/By-Products/misc. income" between works cost and cost of production (SM 6.9); in

ICAI's words, "in case of by-products, the net realisable value of by-products is deducted from the cost of production" (SM 6.6). The full layout is in the cost sheet for a manufacturer.

Formats and formulas

Reproduce these two working notes from memory; ICAI's suggested answers open with one of them.

∑ **Formula**

Net realisable value of a by-product

$$\text{NRV} = \text{Sales value} - \text{Further processing cost} - \text{Selling, distribution, admin. cost}$$

The admin. cost counted is the administration expense attributable to the by-product (SM 11.16).

∑ **Formula**

Net joint cost to apportion over the main products

$$\text{Net joint cost} = \text{Joint cost} - \text{NRV of by-product}$$

Then apportion the net joint cost over the joint products by the method the question names (apportioning joint costs).

Calculation of net joint cost	Amount (₹)
Joint costs	xxx
Less: Net realisable value of by-product	(xxx)
Net joint costs to be apportioned	**xxx**

∑ **Formula**

Joint cost carried by a by-product, when an estimated profit is given

$$\text{Joint cost of by-product} = \text{Sales} - \text{Estimated profit} - \text{Selling expenses} - \text{Cost after separation}$$

The main product bears the rest: total joint cost – joint cost of the by-products. The by-product is left with exactly its estimated profit.

Statement of joint cost allocated	By-product 1 (₹)	By-product 2 (₹)
Sales value	xxx	xxx
Less: Estimated profit	(xxx)	(xxx)
Cost of sales	**xxx**	**xxx**
Less: Estimated selling expenses	(xxx)	(xxx)
Cost of production	**xxx**	**xxx**
Less: Cost after separation	(xxx)	(xxx)
Joint cost allocated	**xxx**	**xxx**

This is the layout of ICAI's suggested answers to MTP September 2024 Series I Q3(a) and MTP September 2025 Series I Q3(a), and the NRV pro-forma of SM 11.7 with its profit-margin line applied to a by-product.

Worked examples

Worked example **Tea, and the fluff on the floor**

Tapri's tea factory spent ₹57,51,000 up to the sorting sieve in April 2026. Leaf and Brokens are joint products, sold at the split-off point. The sweepings are a by-product, saleable only after cleaning and bagging. The management deducts the net realisable value of the by-

product from the joint cost and apportions the rest on the sales value at the split-off point.

Particulars	Output (kg)
Leaf	5,000
Brokens	15,000
Sweepings	3,550

Particulars	Price at split-off (₹ per kg)
Leaf	520
Brokens	300
Sweepings	–

Particulars	Cleaning and bagging (₹ per kg)
Leaf	–
Brokens	–
Sweepings	5

Particulars	Freight to buyer (₹ per kg)
Leaf	–
Brokens	–
Sweepings	3

Particulars	Selling price after cleaning (₹ per kg)
Leaf	–
Brokens	–
Sweepings	28

Required:

(i) Calculate the net realisable value of the sweepings and the net joint cost to be apportioned.

(ii) Apportion the net joint cost between Leaf and Brokens.

(iii) Prepare a statement of profit if the joint products are sold at the split-off point.

(iv) Kesar wants the sweepings' money credited to the Costing Profit and Loss Account as miscellaneous income instead. Show the effect.

THE WORKING

(i) Net realisable value of the by-product

Particulars	Per kg (₹)
Selling price after cleaning	28
Less: Cleaning and bagging	5
Less: Freight	3
Net realisable value	**20**

NRV of sweepings = 3,550 kg × ₹20 = **₹71,000**.

Calculation of net joint cost	Amount (₹)
Joint costs	57,51,000
Less: Net realisable value of by-product	71,000
Net joint costs to be apportioned	**56,80,000**

(ii) Apportionment on sales value at split-off

Particulars	Leaf	Brokens	Total
Output (kg)	5,000	15,000	
Price at split-off (₹)	520	300	
Sales value at split-off (₹)	26,00,000	45,00,000	71,00,000
Net joint cost (26 : 45) (₹)	**20,80,000**	**36,00,000**	**56,80,000**

Multiplying factor = ₹56,80,000 ÷ ₹71,00,000 × 100 = 80%. Joint cost per kg: Leaf ₹416, Brokens ₹240.

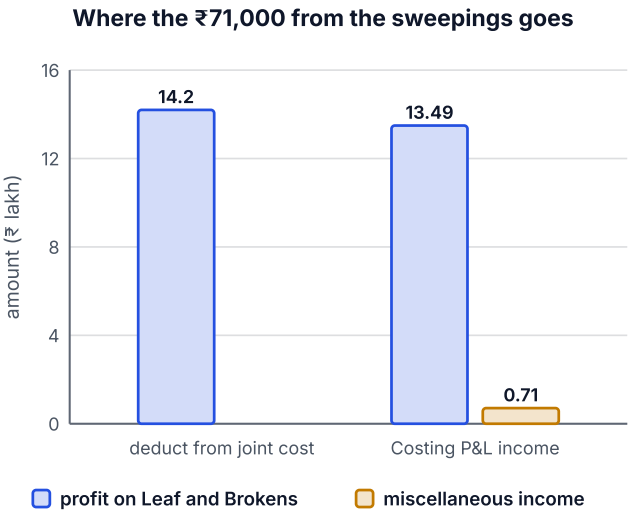
(iii) Profit if sold at the split-off point

Particulars (₹)	Leaf	Brokens	Total
Sales value at split-off	26,00,000	45,00,000	71,00,000
Less: Net joint cost	20,80,000	36,00,000	56,80,000
Profit	**5,20,000**	**9,00,000**	**14,20,000**

(iv) Credit to the Costing Profit and Loss Account instead

No credit is given in the cost accounts, so the whole ₹57,51,000 is apportioned (26 : 45). The sweepings appear in the Costing Profit and Loss Account at ₹71,000: sales of ₹99,400 (3,550 kg × ₹28) less cleaning, bagging and freight of ₹28,400.

Particulars (₹)	Leaf	Brokens	Total
Joint cost	21,06,000	36,45,000	57,51,000
Profit on the products	4,94,000	8,55,000	13,49,000
Add: Miscellaneous income			71,000
Total profit			**14,20,000**



Blue: profit shown on the two joint products; amber: the sweepings shown as miscellaneous income. Either way the total is ₹14.20 lakh; only the product costs differ.

Both are treatments ICAI allows for a by-product of small total value (SM 11.17). Kesar's choice raises the cost of Leaf by ₹26,000 and of Brokens by ₹45,000 (₹5.20 and ₹3 a kg) and shows ₹71,000 as miscellaneous income; total profit is unchanged at ₹14,20,000. The deduction method is the one the cost sheet uses ("Less: Credit for Recoveries/Scrap/By-Products", SM 6.9), so the cost of the main products, used for pricing and inventory, is shown net of what the by-product recovers.

Interactive version: PassBook app, lesson Treatment of by-products

Worked example

Sugar, molasses in the tank and the price of sugar

Malwa Sugar Mills Ltd, near Indore, supplies the sugar for Tapri's chai premix and wants to agree next season's price with Tapri. For the 2025-26 crushing season, the cost up to the point of separation was ₹3,20,70,000. Sugar is the main product; molasses (sold to distilleries) and bagasse (sold as boiler fuel) are by-products, valued by the reverse cost method.

Particulars	Molasses	Bagasse
Output (tonnes)	450	3,000
Selling price (₹ per tonne)	10,000	1,800
Cost after separation (₹)	3,00,000	6,00,000
Estimated selling expenses (% of sales)	10%	5%
Estimated net profit (% of sales)	20%	15%

The mill produced 1,000 tonnes of sugar, all of which Tapri buys. At the end of the season 90 tonnes of molasses were still in the mill's tank; all the bagasse was sold. Sugar's selling expenses are 5% of sales.

Required:

(i) Calculate the joint cost to be allocated to each by-product, and the cost of the molasses in the tank at the end of the season.

(ii) Calculate the joint cost per tonne of sugar.

(iii) Calculate the price per tonne the mill must charge for sugar to earn a profit of 20% on sales.

(iv) Prepare a statement of the season's profit by product at that price.

THE WORKING

(i) Joint cost allocated to the by-products (on the whole output)

Particulars (₹)	Molasses	Bagasse
Sales value (450 × ₹10,000; 3,000 × ₹1,800)	45,00,000	54,00,000
Less: Estimated profit (20%; 15%)	9,00,000	8,10,000
Cost of sales	**36,00,000**	**45,90,000**
Less: Estimated selling expenses (10%; 5%)	4,50,000	2,70,000
Cost of production	**31,50,000**	**43,20,000**
Less: Cost after separation	3,00,000	6,00,000
Joint cost allocated	**28,50,000**	**37,20,000**

Cost of production of molasses = ₹31,50,000 ÷ 450 = ₹7,000 a tonne, so the 90 tonnes in the tank are valued at **₹6,30,000**.

(ii) Joint cost of sugar = ₹3,20,70,000 − ₹28,50,000 − ₹37,20,000 = ₹2,55,00,000, that is **₹25,500 a tonne**.

(iii) Price of sugar. Cost + selling expenses (5% of price) + profit (20% of price) = price, so cost = 75% of price: ₹25,500 ÷ 75% = **₹34,000 a tonne**.

(iv) Statement of profit for the season

Particulars (₹)

A. Sales (1,000 × ₹34,000; 360 × ₹10,000; 3,000 × ₹1,800)

Sugar

3,40,00,000

Molasses

36,00,000

Bagasse

54,00,000

Particulars (₹)

Cost of production of the quantity sold

Sugar

2,55,00,000

Molasses

25,20,000

Bagasse

43,20,000

Particulars (₹)

Selling expenses (5%; 10%; 5%)

Sugar

17,00,000

Molasses

3,60,000

Bagasse

2,70,000

Particulars (₹)	B. Cost of sales
Sugar	2,72,00,000
Molasses	28,80,000
Bagasse	45,90,000

Particulars (₹)	Profit (A – B)
Sugar	**68,00,000**
Molasses	**7,20,000**
Bagasse	**8,10,000**

Molasses sold: 360 tonnes × ₹7,000 = ₹25,20,000. Overall profit = **₹83,30,000**. Check: total sales ₹4,30,00,000 − joint cost ₹3,20,70,000 − costs after separation ₹9,00,000 − selling expenses ₹23,30,000 + closing stock of molasses ₹6,30,000 = ₹83,30,000.

Each by-product earns exactly its estimated percentage of profit on what it sells; the reverse cost method gives it a cost of production, which is also the basis for valuing its unsold stock.

Worked example

Three leftovers at the tea factory

Tapri's tea factory sells about ₹3 crore of tea a month. Its processing leaves three by-products:

(a) Fluff swept from the sorting floor, which a compost maker collects for about ₹10,000 a month.

(b) Tea seeds, which an oil mill buys for about ₹45 lakh a month once Tapri has dried them; drying costs about ₹5 lakh a month.

(c) Stalks and coarse waste, which the factory now burns in its own dryer furnace in place of firewood that would cost about ₹40,000 a month.

Kesar asks Ritu how each should be treated in the cost accounts. Discuss.

MODEL ANSWER

By-products are products recovered from material discarded in a main process, or from the production of some major products, and their value is relatively small compared with the main product. Their treatment depends on their value and on whether they are sold or used inside the business.

(a) Fluff: small total value. At about ₹10,000 against ₹3 crore of tea, its value is small. As per the Study Material, the amount realised from a by-product of small total value may be dealt with in either of two ways: (1) its sales value may be

credited to the Costing Profit and Loss Account, as miscellaneous income or as additional sales revenue, with no credit in the cost accounts; or (2) the sale proceeds may be deducted from the total costs, either from the production cost or from the cost of sales (SM 11.17). Either is permissible here.

(b) Tea seeds: further processing. Where a by-product requires further processing, its net realisable value at the split-off point is found by subtracting the further processing cost from its realisable value (SM 11.17): ₹45 lakh – ₹5 lakh = about ₹40 lakh a month, more than an eighth of the tea's sales. That is a considerable amount, so the seeds are treated as by-products of considerable total value: they may be regarded as joint products rather than as by-products, and the cost up to the point of separation is apportioned over tea and seeds on a logical basis: relative market values, physical output at the point of split-off, or ultimate selling prices (SM 11.17). The drying cost and the selling, distribution and administration expenses of the seeds are collected in separate accounts (SM 11.16). Had the seeds' net value been small, they would be treated as in (a).

(c) Stalks: used inside the business. They are not sold, so there is no realisation to credit. Under the **comparative price** method, a by-product is valued at the price of a similar or alternative material (SM 11.16): the stalks are worth the ₹40,000 of firewood they replace. That value is credited to the cost of

the main process and charged to the dryer furnace, so the cost of tea is not overstated and the furnace's fuel cost is not understated. (Had the stalks been fed back into the same process as input, the **re-use basis** would value them at the value of the materials introduced into that process.)

Exam angle

In the exam

Where by-products appeared, May 2024 to September 2026

In the 15 new-scheme Paper 4 documents (May 2024 to September 2026) read for this lesson that examined by-products (3 past papers, 4 RTPs, 8 MTPs):

- **Theory, 4 or 5 marks.** "Discuss the treatment of by-products in cost accounting": September 2024 Q6(c) (4 marks, with an OR on job costing), MTP September 2026 Series II Q6(a) (5 marks); only the small-value case: RTP January 2025 Q19(c); the same three cases for buttermilk as a by-product of butter: MTP September 2024 Series II Q6(b) (5 marks). The NRV method of apportioning joint costs to by-products: RTP September 2024 Q19(a). Molasses, and a by-product that needs further processing: September 2026 Q6(b)(i) and (iii) (answers not yet published). The cost-sheet treatment of joint products and by-products, one of

four items: MTP September 2024 Series I Q2(b), MTP September 2026 Series I Q5(b) (4 marks each).

- **Numerical, 5 or 6 marks.** By-product NRV credited before apportioning the joint cost: January 2026 Q1(b) (5 marks), RTP January 2025 Q14. Main product with two by-products valued at sales less estimated profit, selling expenses and cost after separation: MTP September 2024 Series I Q3(a) (6 marks), MTP September 2025 Series I Q3(a) (5 marks). A by-product credited at sales less a margin on cost: RTP September 2024 Q15. A by-product with estimated profit and selling expenses shared in proportion to sales: RTP September 2026 Q12. A by-product from a finishing process: RTP May 2025 Q15.
- **Multiple choice, 2 marks.** Cost of the main product after the joint cost of two by-products: MTP May 2025 Series I MCQ 14 and MTP May 2026 Series II MCQ 13 (the same question). Net joint cost after a by-product's NRV, then physical units: MTP January 2025 Series I MCQ 15.

In the exam

How the marks are earned

The theory answers ICAI publishes are the Study Material's paragraph 5, case by case, in its order: small total value (two ways), considerable total value, requiring further processing. The numerical answers open with a working note: either "Calculation of net joint costs to be apportioned" (joint costs less NRV of the by-product) or a statement of joint cost allocated to each by-product.

Trick

Small, Big, Busy

Small value: credit Costing P&L or deduct from cost. **B**ig value: treat as a joint product. **B**usy (needs further processing): find the NRV at split-off, then decide small or big. Condition: this is ICAI's order of the three cases (SM 11.17); write them in that order.

Answer frame

"Distinguish between joint products and by-products"

Write the compare table from How it works: ICAI's two points (importance; simultaneous against incidental production) first, then cost treatment and one example on each side.

"Discuss the treatment of by-product cost in cost accounting"

1. **When they are of small total value**: (1) credit the sales value to the Costing Profit and Loss Account, no credit in the cost accounts, as miscellaneous income or additional sales revenue; (2) deduct the sale proceeds from the total costs, from the production cost or the cost of sales.
2. **When they are of considerable total value**: regard them as joint products; apportion the costs up to the point of separation on a logical basis (relative market values, physical output at split-off, or ultimate selling prices).
3. **Where they require further processing**: NRV at split-off = realisable value − further processing cost; small, treat as in 1; considerable, as in 2.

"Discuss the NRV method of apportioning joint costs to by-products"

1. The realisation on disposal of the by-product is deducted from the total cost of production to arrive at the cost of the main product (molasses reduces the cost of sugar).
2. If the by-product needs additional processing, the expenses are deducted from its realisation, and only the net realisation is deducted from the cost of production.
3. Separate accounts collect the additional expenses on (a) further processing of the by-product and (b) its

selling, distribution and administration.

Three answers to the numerical part

Loses marks

"Sugar's joint cost = ₹3,20,70,000 − ₹99,00,000 = ₹2,21,70,000."

Credits the by-products at their full sales value.

Passes

Deducts profit and cost after separation but forgets the selling expenses: sugar's joint cost comes to ₹2,47,80,000.

Right idea, one line missing.

Full marks

ICAI's statement for each by-product (sales, less estimated profit, cost of sales, less selling expenses, cost of production, less cost after separation, joint cost allocated); sugar by difference; profitability statement with a check.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the by-product credited at its price tag

Deducting the by-product's full sales value from the joint cost, ignoring its further processing, freight or selling costs. **Fix:** credit only the net realisable value: sales value – further processing – attributable selling, distribution and administration costs (SM 11.16).

⚠ **Common mistake**

A by-product with a share of the joint cost on physical units

Adding the sweepings' 3,550 kg to Leaf and Brokens and sharing the joint cost over 23,550 kg. That treats a by-product as a joint product. **Fix:** a by-product of small value is credited at its NRV; only a by-product of considerable value is treated as a joint product, and then by the basis the question gives.

⚠ **Common mistake**

The estimated profit left inside the by-product's cost

Working joint cost of the by-product as sales – selling expenses – cost after separation, without deducting the estimated profit. **Fix:** when the question gives an estimated profit on the by-product, deduct it first; the by-product keeps that profit and only the rest reduces the main product's joint cost.

Practice

Check yourself

1. A joint process costs ₹8,40,000 and yields main products A (30,000 kg) and B (10,000 kg) and a by-product (20,000 kg) that sells at ₹6 a kg after further processing of ₹1 a kg and selling costs of ₹1 a kg. The NRV of the by-product is deducted from the joint cost and the rest is apportioned on physical units. What joint cost (₹) goes to A?

Answer: _____

2 Case scenario. Tapri's premix plant sells the powder swept from its filling lines to a cattle-feed maker for about ₹12,000 a month, against premix sales of ₹2 crore a month. Ritu lists the treatments ICAI allows for it. Which of these is NOT one of ICAI's treatments for a by-product of small total value?

- **A** Credit the sales value to the Costing Profit and Loss Account as miscellaneous income
- **B** Credit it to the Costing Profit and Loss Account as additional sales revenue
- **C** Deduct the sale proceeds from the cost of production
- **D** Apportion part of the joint cost to it on the basis of physical output

3 A by-product sells for ₹50,000. Its cost after separation is ₹8,000, selling expenses ₹5,000, and its estimated profit is 20% of sales. How much of the joint cost does it carry?

- **A** ₹37,000
- **B** ₹27,000
- **C** ₹50,000
- **D** ₹40,000

4 Which of these are methods ICAI lists for apportioning joint cost to by-products? Select all that apply.

- **A** Net realisable value method
- **B** Standard cost in technical estimates
- **C** Comparative price
- **D** Re-use basis
- **E** Contribution margin method
- **F** Average unit cost method

More than one answer may be right.

5 A by-product can be fed back into the process it came from as part of that process's input. At what value should it be taken?

- **A** Its selling price to outside buyers
- **B** The price of a similar material bought from outside
- **C** The value of the materials introduced into that process
- **D** Nil, since it is reused internally

Answers: the answer key at the end of this lesson.

Recap

- By-product: recovered incidentally from the material used for the main product; value relatively insignificant against the main product (molasses, tar, glycerin).
- Joint products: equal importance, produced simultaneously; by-products: small economic value, produced incidentally.
- ICAI's four methods for by-products: NRV; standard cost in technical estimates; comparative price; re-use basis.
- NRV of a by-product = sales value − further processing − attributable selling, distribution and administration costs.
- Small total value: credit Costing P&L (miscellaneous income or additional sales) or deduct from production cost or cost of sales.
- Considerable total value: treat as a joint product and apportion the joint cost.
- Needs further processing: NRV at split-off first, then small or considerable.
- With an estimated profit: by-product's joint cost = sales − profit − selling expenses − cost after separation; the main product bears the rest. The barber still sells haircuts.

Answer key · Check yourself

1 **570000**

NRV = 20,000 × (₹6 − ₹1 − ₹1) = ₹80,000. Net joint cost = ₹8,40,000 − ₹80,000 = ₹7,60,000. A = ₹7,60,000 × 30,000 ÷ 40,000 = ₹5,70,000. The by-product's kilos are not part of the physical base.

2 **D Apportion part of the joint cost to it on the basis of physical output**

Small total value: credit Costing P&L (as miscellaneous income or additional sales revenue) or deduct from production cost or cost of sales (SM 11.17). Sharing the joint cost on physical output is for a by-product of considerable total value, treated as a joint product.

3 **B ₹27,000**

₹50,000 − ₹10,000 profit − ₹5,000 selling − ₹8,000 after separation = ₹27,000. ₹37,000 forgets the estimated profit; ₹40,000 deducts only the profit; ₹50,000 credits the full sales value.

4 **A Net realisable value method; B Standard cost in technical estimates; C Comparative price; D Re-use basis**

ICAI's four methods for by-products (SM 11.15–11.16). Contribution margin and average unit cost are methods for joint products.

5 **C The value of the materials introduced into that process**

Re-use basis: the value put on the by-product is the same as that of the materials introduced into the process (SM 11.16). The price of a similar material is the comparative price method.

CHAPTER 12

Service Costing

12.1 Service costing and composite cost units

A service has no physical unit to count, so its cost unit often joins two measures (tonne-km, patient-day); absolute and commercial bases count them, and weights put different grades on one scale.

In plain words

A truck, three towns and one silly question

Tapri's refrigerated truck leaves the Indore depot at dawn with 12 tonnes of chai premix and frozen samosas. It drops 8 tonnes at Dewas, 40 km away, carries the other 4 tonnes 40 km on to Ujjain, and comes back 60 km to Indore with 2 tonnes of kulhads bought from Ujjain's potters (bulky, not heavy). The trip cost ₹22,800.

> **Kesar:** So what does one trip cost us? Done. ₹22,800.
>
> **Ritu:** Per unit, please. A trip with 12 tonnes for 40 km and a trip with 1 tonne for 300 km are both "one trip". So is a kilometre run empty.
>
> **Kesar:** Then count tonnes!
>
> **Ritu:** A tonne carried to Dewas and a tonne carried to Bhopal are not the same work either.

A service cannot be stacked in a godown and counted like samosa packs, so Ritu needs a **cost unit**: the unit of service in which cost is measured. For a goods truck the work done depends on two things at once, how much is carried and how far, so she joins them into one unit: the **tonne-km**, one tonne carried one kilometre. A unit made of two measures is a **composite cost unit**. A bus uses the **passenger-km**, a hospital the **patient-day**.

There are two ways to count the truck's tonne-km. The careful way takes each stretch of the trip on its own: 12 tonnes × 40 km, then 4 × 40, then 2 × 60, and adds them up: 760 tonne-km. That is the **absolute** basis. The quick way takes the average load over the three stretches, (12 + 4 + 2) ÷ 3 = 6 tonnes, and multiplies it by the whole distance, 140 km: 840 tonne-km. That is the **commercial** basis. Divide the ₹22,800 by either total and Kesar has a cost per tonne-km: ₹30 on the absolute basis, about ₹27.14 on the commercial basis.

One more problem: services come in grades. A sleeper berth and an ordinary seat on Tapri's overnight bus use the same driver, the same diesel and the same road, but nobody pays the same for both. So each grade gets a **weight** (a berth counts as more than one seat) and every grade is converted into one common unit, an **equivalent service unit**.

Think of it like

The railway porter

A porter at Indore station charges by bags and by platforms walked. Three bags across two platforms is six "bag-platforms"; one bag across five platforms is five. Counting only bags, or only platforms, would cheat him or you. If he drops one bag at platform 2 and carries the rest on to platform 6, the honest count goes stretch by stretch (absolute); the quick count multiplies his average number of bags by the platforms walked (commercial). And the first-class trunk counts as two bags: that is a weight.

Key idea

When one measure cannot capture what a service does, the cost unit joins two (a composite unit: tonne-km, passenger-km, patient-day). Absolute basis: add up load × distance leg by leg. Commercial basis: average load × total distance. Different grades of a service are put on one scale by weights (equivalent service units).

How it works

Service costing, "also known as operating costing" (SM 12.2), collects all the costs of a period, records them and expresses them "in terms of a cost unit of service" (SM 12.2). It is used for in-house support services (a staff canteen, a boiler house, a vehicle fleet, the IT

department) and for services sold to outside customers (transport, hotels, banks, insurers, IT companies). How it differs from product costing, and how its cost statement is laid out (standing, running and maintenance charges), is in the service-sector cost sheet.

The hard part: what is one unit of service?

ICAI names the "specific issue" with service costing: "the difficulty in defining a realistic cost unit that represents a suitable measure of the service provided"; the unit must be defined carefully and "frequently, a composite cost unit may be deemed more appropriate", like a hotel's "Occupied Room Days" (SM 12.3). The Study Material's list (SM 12.3–12.4):

Service industry	Unit of cost (examples)
Transport services	Passenger-km (public transport); quintal-km or tonne-km (goods carriage)
Electricity supply	Kilowatt-hour (kWh)
Hospital	Patient per day, room per day or per bed, per operation
Canteen	Per item, per meal
Cinema	Per ticket
Hotels	Guest days or room days

Service industry	Unit of cost (examples)
Bank or financial institutions	Per transaction, per service (per letter of credit, per application, per project)
Educational institutes	Per course, per student, per batch, per lecture
IT and ITES	Per project, per module
Insurance	Per policy, per claim, per TPA

"The costing should be comprehensive enough to show the effects like off-season and peak-season demand, full time, part time, etc." (SM 12.4). One vehicle can also be costed per day maintained, per day operated, per hour and per km; each answers a different question (Worked example 2).

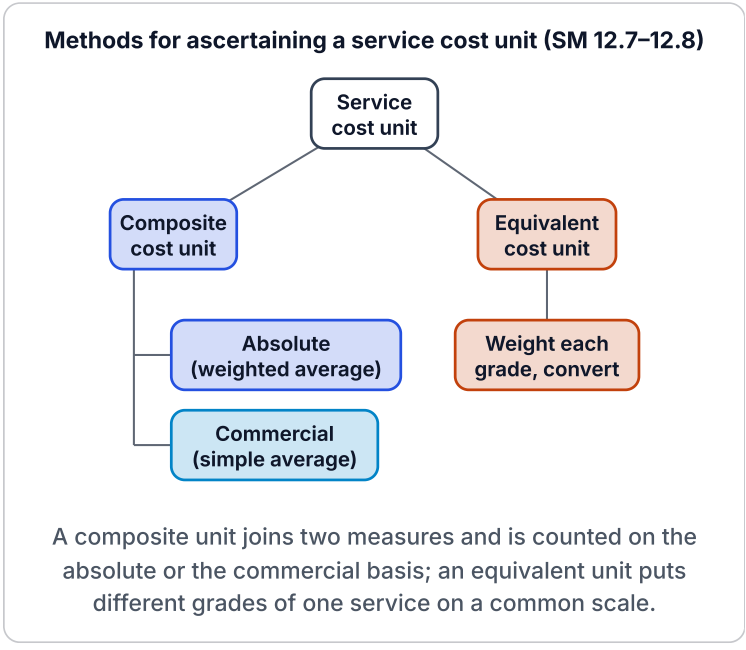
Key performance indicators

Key Performance Indicators (KPIs) are "the quantitative and qualitative factors which are commonly used to assess the performance of an organization which are important to achieve its goal" (SM 12.4). A cost or revenue per KPI lets a service compare its performance against industry standards; ICAI's example is telecom's **Average Return per User (ARPU)**, the average revenue generated

from a user of its services. The Study Material's list (SM 12.4–12.6):

Industry	KPIs
Transportation	Number of shipments; truck turnaround rate; lead time (order cycle time); on-time and in-full (OTIF)
Hotel	Cost per occupied room (CPOR); occupancy rate; revenue per available room (RevPAR)
Hospitals and health care	Bed occupancy rate; staff-to-patient ratio; average treatment charge
IT and ITES	Gross burn rate; customer acquisition cost (CAC); customer lifetime value (CLV); monthly recurring revenue (MRR); churn rate; cost per feature
Telecom	ARPU; subscriber acquisition cost (SAC); network operating cost; gross revenue retention (GRR)
Education	Instructional costs; administrative costs per student; tuition costs; student-to-faculty ratio
Insurance	Average cost per claim; components of claim costs (CCC); cost per quote; administrative costs per policy; average policy size

Two methods for ascertaining a service cost unit



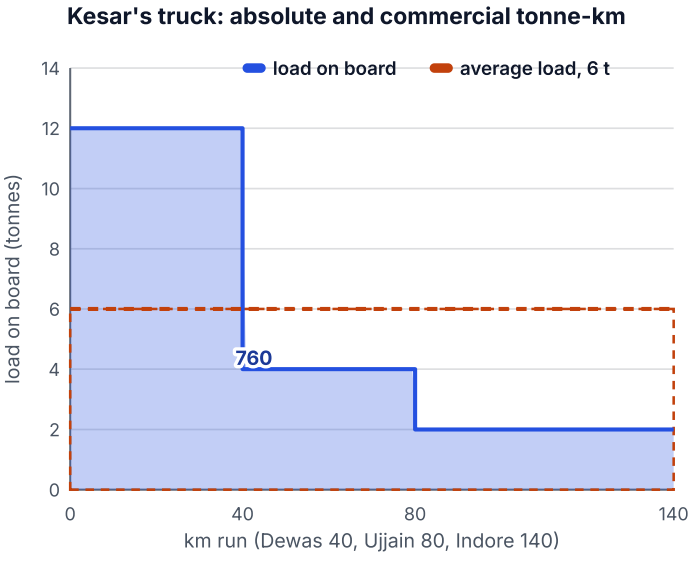
A **composite cost unit** combines two measurement units, "for example, a public transportation undertaking would measure the operating cost per passenger per kilometer" (SM 12.7). It may be computed in two ways, and in both, weight is given to a **qualitative** factor as well as to the **quantitative** one. In goods transport, distance is the quantitative factor (the variable cost depends on it); the weight carried does not affect the variable cost and is the qualitative factor (SM 12.7–12.8).

1. **List the legs.** Every stretch between two stops, with its distance and the load actually on board (zero for an empty run).

2. **Absolute (weighted average) basis**: "a summation of the products of qualitative and quantitative factors" (SM 12.7). Multiply load × distance for each leg and add: Kesar's truck, 12 × 40 + 4 × 40 + 2 × 60 = 760 tonne-km.

3. **Commercial (simple average) basis**: "the product of average qualitative and total quantitative factors" (SM 12.7). Average the loads of the legs, then multiply by the total distance: (12 + 4 + 2) ÷ 3 × 140 = 6 × 140 = 840 tonne-km.

4. **Cost per unit** = total cost of the trip or period ÷ the units: ₹22,800 ÷ 760 = ₹30.00 per absolute tonne-km; ₹22,800 ÷ 840 = ₹27.14 per commercial tonne-km.



Absolute tonne-km are the blue shaded area under the load line: 480 + 160 + 120 = 760. Commercial tonne-km are the orange dashed rectangle, average load 6 t × 140 km = 840. They differ because the longest leg carried the lightest load.

The two bases agree when every leg is the same length or carries the same load; otherwise they usually differ. The absolute basis follows the work actually done; the commercial basis is quicker and treats a short leg and a long leg alike.

Equivalent service units

When two or more grades of a service use common resources, each grade is assigned a weight and converted into equivalent units, which makes the grades equivalent and comparable (SM 12.8). ICAI's example is a hotel whose Deluxe suite is let at 2.5 times the Standard suite

and its Luxurious suite at twice the Deluxe suite, all using the same amount of overheads (SM 12.8–12.10):

1. Find the units of each grade actually used: rooms × days (× occupancy).
2. Multiply by the weight, taking one grade as the base (weight 1): Deluxe 2.5, Luxurious 2.5 × 2 = 5.
3. Add up the equivalent units and divide the cost (or the takings needed) by them: that is the rate for the base grade; each other grade pays rate × weight.

Either the cheapest or the dearest grade may be the base. ICAI converts the same suites both ways, into 1,35,000 equivalent Standard room-days or 27,000 equivalent Luxurious room-days (SM 12.9–12.10); because the weights keep the same ratios, either base gives the same rents.

Formats and formulas

Reproduce the working-note layout from memory. ICAI's suggested answers lay the absolute tonne-km out leg by

leg (MTP Series I for May 2024 Q2(a)):

Leg	Distance (km)	Load (tonnes)	Tonne-km
A to B	d_1	W_1	$d_1 \times W_1$
B to C	d_2	W_2	$d_2 \times W_2$
C to A	d_3	W_3	$d_3 \times W_3$
Total	**Σd**		**Absolute tonne-km**

∑ **Formula**

Absolute (weighted average) tonne-km

$$\text{Absolute} = (W_1 \times D_1) + (W_2 \times D_2) + \cdots + (W_n \times D_n)$$

W = weight carried on a leg, D = the distance of that leg (SM 12.7).

∑ **Formula**

Commercial (simple average) tonne-km

$$\text{Commercial} = (D_1 + D_2 + \cdots + D_n) \times \frac{W_1 + W_2 + \cdots + W_n}{n}$$

n = the number of legs. Total distance × the simple average of the loads (SM 12.7).

∑ **Formula**

Cost per composite unit

$$\text{Cost per tonne-km} = \frac{\text{Total cost}}{\text{Total tonne-km}}$$

Total cost and total tonne-km of the same period. The same for passenger-km, patient-days or room-days.

∑ **Formula**

Equivalent service units

$$\text{Equivalent units} = \sum(\text{units} \times \text{weight})$$

$$\text{Grade rate} = \text{Base rate} \times \text{weight}$$

Definition

Composite cost unit (SM 12.7)

Sometime two measurement units are combined together to know the cost of service or operation. These are called composite cost units. Examples are tonne-km, quintal-km, passenger-km, patient-day.

Definition

Equivalent service unit (SM 12.57)

To calculate cost or pricing of two or more different grades of services which use common resources, each grade of service is assigned a weight and converted into equivalent units. Converting services into equivalent units makes different grades of services equivalent and comparable.

Definition

Service costing (SM 12.57)

The application of cost concepts in the ascertainment of the cost of providing services. It is also known as operating costing, as it relates to the operating of a service.

Worked examples

Worked example **Tapri's Pune milk-run: who pays for the truck?**

Tapri's Pune depot runs a shared truck on a round trip that drops and picks up goods for five consignments:

Consignment
A: premix for the Lonavala café
Tonnes
6
Loaded at
Pune
Unloaded at
Lonavala

Consignment
B: premix for the Panvel café
Tonnes
8
Loaded at
Pune
Unloaded at
Panvel

Consignment
C: frozen samosas for Bhiwandi
Tonnes
4
Loaded at
Pune
Unloaded at
Bhiwandi

Consignment
D: kulhads for the Bhiwandi godown
Tonnes
5
Loaded at
Lonavala
Unloaded at
Bhiwandi

Consignment
E: packaging material for Pune
Tonnes
8
Loaded at
Bhiwandi
Unloaded at
Pune

The distances are Pune to Lonavala 65 km, Lonavala to Panvel 45 km, Panvel to Bhiwandi 35 km and Bhiwandi to Pune 145 km. The round trip cost ₹51,150.

Required:

(i) Compute the absolute tonne-km and the commercial tonne-km of the trip.

(ii) Calculate the cost per tonne-km on each basis.

(iii) Charge each consignment its share of the trip's cost, and show why the absolute basis is the one to use for this.

THE WORKING

(i) Working note: load on board, leg by leg

Leg	Distance (km)	Load on board (tonnes)	Tonne-km
Pune to Lonavala	65	6 + 8 + 4 = 18	1,170
Lonavala to Panvel	45	18 − 6 + 5 = 17	765
Panvel to Bhiwandi	35	17 − 8 = 9	315
Bhiwandi to Pune	145	8	1,160
Total	**290**		**3,410**

Absolute tonne-km = **3,410**.

Commercial tonne-km = total distance × average load = 290 km × (18 + 17 + 9 + 8) ÷ 4 = 290 × 13 = **3,770**.

(ii) Cost per tonne-km

$$\text{Absolute} = \frac{51{,}150}{3{,}410} = ₹15.00$$

$$\text{Commercial} = \frac{51{,}150}{3{,}770} = ₹13.57$$

(iii) Each consignment's share

Consignment	Tonnes × km carried	Tonne-km	Share at ₹15 (₹)
A	6 × 65	390	5,850
B	8 × (65 + 45)	880	13,200
C	4 × (65 + 45 + 35)	580	8,700
D	5 × (45 + 35)	400	6,000
E	8 × 145	1,160	17,400
Total		**3,410**	**51,150**

The consignments' own tonne-km add up to the trip's absolute tonne-km (3,410), so charging each at the absolute rate of ₹15 recovers the whole ₹51,150. The commercial figure (3,770) is not the sum of anything the consignments carried: charging them ₹13.57 a tonne-km would recover only 3,410 × ₹13.57 = ₹46,266, ₹4,884 short. The commercial basis is a quick measure of the trip as a whole; the absolute basis follows each tonne for the distance it actually travelled.

Worked example

The chiller van: one van, three meters

Tapri's refrigerated van supplies the Indore cafés and, when needed, stands at events as a cold store with its refrigeration running. For September (30 days):

Particulars	Data
Km run	3,900
Hours the refrigeration unit ran	520
Driver's salary (monthly)	₹22,000
Insurance; road tax and permit; garage rent (for the month)	₹2,500; ₹1,500; ₹5,000
Interest on the vehicle loan (for the month)	₹3,000
Van	cost ₹14,40,000; residual value ₹2,40,000 after 5 years (time basis)
Diesel for driving	₹96 a litre; 6 km a litre
Tyres and servicing	₹2 a km
Refrigeration unit	burns 1.5 litres of diesel an hour; servicing ₹12 an hour

Kesar has promised the van to the Indore food festival for 2 days: it will drive 40 km in all and run its refrigeration for 36 hours as a cold store at the venue.

Required:

(i) Prepare the van's cost statement for September, and find the standing charges per day maintained, the running charges per km and the refrigeration charges per hour.

(ii) Calculate the cost per km run if all the month's costs are spread over the km.

(iii) Price the festival job at a profit of 20% on the price using the three rates, and again using the single cost per km. Comment.

THE WORKING

Working notes

1. Depreciation (time basis, a standing charge) = (₹14,40,000 − ₹2,40,000) ÷ (5 × 12) = ₹20,000 a month.
2. Driving diesel = ₹96 ÷ 6 km = ₹16 a km; with tyres and servicing, ₹18 a km.
3. Refrigeration = 1.5 litres × ₹96 + ₹12 = ₹156 an hour.

(i) Cost statement of the chiller van, September

Particulars	Amount (₹)	Rate
Driver's salary	22,000	
Insurance	2,500	
Road tax and permit	1,500	
Garage rent	5,000	
Depreciation (WN 1)	20,000	
Interest	3,000	
A. Standing charges	**54,000**	**₹1,800 a day maintained**
B. Running charges, driving (3,900 km × ₹18)	**70,200**	**₹18 a km**
C. Refrigeration charges (520 hours × ₹156)	**81,120**	**₹156 an hour**
Total cost (A + B + C)	**2,05,320**	

Standing charges ÷ 30 days maintained = ₹1,800 a day.

(ii) Single rate. ₹2,05,320 ÷ 3,900 km = **₹52.65 a km**.

(iii) The festival job

Particulars	Three rates (₹)	Single rate (₹)
Standing charges (2 days × ₹1,800)	3,600	
Driving (40 km × ₹18)	720	
Refrigeration (36 hours × ₹156)	5,616	
All costs at ₹52.65 a km (40 km)		2,106
Cost of the job	**9,936**	**2,106**
Profit (20% on price = 25% on cost)	2,484	526
Price	**12,420**	**2,632**

Comment. The single cost per km assumes every cost moves with distance. The festival job hardly moves the van but keeps it for two days and runs its refrigeration for 36 hours, so a per-km price would recover less than a quarter of its cost. A service with more than one cost driver needs a cost unit for each: here the day for standing charges, the km for driving and the refrigeration hour for the chiller.

Worked example

Chai Yatra Sleeper: seats and berths on one bus

Tapri's overnight bus runs Indore to Mumbai and back, 600 km each way, one single trip a night in each direction, for 30 nights a month. Seats and berths share the same bus, driver, diesel and tolls.

Class	Number	Average occupancy	Fare relative to a seat
Seater	20	70%	1
Lower sleeper berth	15	90%	1.6 times
Upper sleeper berth	15	80%	1.4 times

The monthly operating cost is ₹21,22,200, and Tapri wants a profit of 25% on takings.

Required:

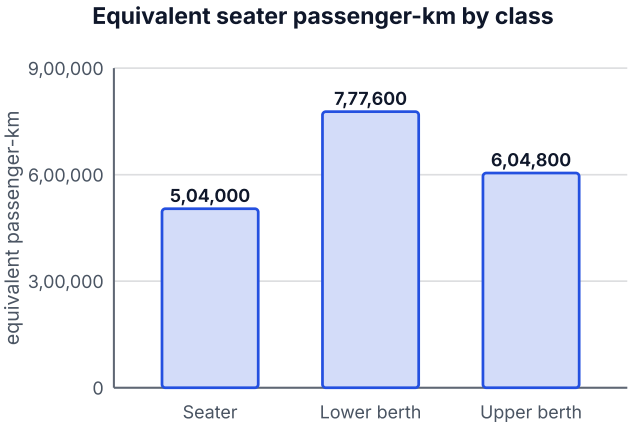
(i) Calculate the fare per passenger-km and the fare for the full 600 km in each class, taking the seat as the base.

(ii) Check the fares with the lower berth as the base.

THE WORKING

Working note: equivalent seater passenger-km
(60 single trips × 600 km)

Class	Passenger-km	Weight	Equivalent seater passenger-km
Seater	20 × 70% × 60 × 600 = 5,04,000	1	5,04,000
Lower berth	15 × 90% × 60 × 600 = 4,86,000	1.6	7,77,600
Upper berth	15 × 80% × 60 × 600 = 4,32,000	1.4	6,04,800
Total			**18,86,400**



Lower berths carry fewer passenger-km than seats (4,86,000 against 5,04,000) but count 1.6 times each, so they carry the largest share of the 18,86,400 equivalent units.

Takings = ₹21,22,200 ÷ (1 − 0.25) = **₹28,29,600**.

(i) Fare per seater passenger-km = ₹28,29,600 ÷ 18,86,400 = **₹1.50**; lower berth ₹1.50 × 1.6 = **₹2.40**; upper berth ₹1.50 × 1.4 = **₹2.10** a passenger-km. For the full 600 km: seat **₹900**, lower berth **₹1,440**, upper berth **₹1,260**.

(ii) Check with the lower berth as the base: equivalent lower-berth passenger-km = 5,04,000 ÷ 1.6 + 4,86,000 × 1 + 4,32,000 × 1.4 ÷ 1.6 = 3,15,000 + 4,86,000 + 3,78,000 = 11,79,000; lower-berth fare = ₹28,29,600 ÷ 11,79,000 = ₹2.40 a passenger-km, and the others follow at ₹1.50 and ₹2.10. Takings check: 5,04,000 × ₹1.50 + 4,86,000 × ₹2.40 +

4,32,000 × ₹2.10 = ₹7,56,000 + ₹11,66,400 + ₹9,07,200 = ₹28,29,600.

Exam angle

In the exam

Where these ideas appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, this lesson's ideas came up in 9 documents:

- **Absolute and commercial tonne-km, MCQ or 4 marks**: a lorry dropping part of its load at a middle station (May 2025 MCQ 13; MTP Series I for May 2026 Q1(c), 4 marks). Both follow SM Illustration 1.
- **One vehicle, several cost units, 5 marks**: cost per day maintained, day operated, hour, km and commercial tonne-km (MTP Series II for September 2024 Q1(b)).
- **Theory, 5 marks**: "Explain the methods for ascertaining service cost unit" (MTP Series I for January 2025 Q6(b)); the suggested answer reproduces SM 12.7–12.8: the composite unit, both bases with their formulas, the cinema seats, and the definition of the equivalent unit.
- **Cost units table, 4 to 5 marks**: "State cost unit of the following industry sector" for ten industries (January 2025 Q6(b), 5 marks); the same for eight industries (MTP Series I for January 2026 Q6(c), 4 marks, an

OR option); the January 2025 list again (MTP Series II for May 2026 Q6(a), 5 marks); "Describe cost unit and give examples" (RTP September 2025 Q18(d)).

- **Theory on operating costs**: RTP May 2025 Q20(b), where the features include "the cost unit used is composite like passenger-mile; kilowatt-hour".

In the exam

The traps ICAI set

- The middle station: the load falls after the first drop, so each leg has its own load (May 2025 MCQ 13: 15 tonnes, then 10, then 8 on the return).
- Rounding the average: MTP Series I for May 2026 Q1(c) writes 22.667 MT × 750 km = 17,000 MT-km "(approx.)"; keep the fraction, (30 + 18 + 20) ÷ 3 = 68/3, and 68/3 × 750 is exactly 17,000.
- Answer options that swap the two figures: one option gives absolute and commercial the wrong way round (May 2025 MCQ 13, options (B) and (C)).

Trick

Which basis is bigger?

Absolute − commercial = $\Sigma(d_i - \bar{d})(w_i - \bar{w})$, where $\bar{d}$ and $\bar{w}$ are the average distance and the average load of the legs. So when the longer legs carry the lighter loads (the sum is negative), commercial > absolute; when the longer legs carry the heavier loads (the sum is positive), absolute > commercial; when all legs are equal in length or in load, the sum is zero and the two are equal. Use it to check an answer option in seconds. Condition: both bases computed on the same legs, the commercial average being the simple average of the legs' loads.

Answer frame

"Explain the methods for ascertaining service cost unit" (5 marks)

ICAI's answer (MTP Series I for January 2025 Q6(b)) follows the Study Material in this order:

1. **Composite cost unit**: two measurement units combined to know the cost of a service or operation; examples tonne-km, quintal-km, passenger-km, patient-day.
2. It may be computed in two ways: absolute (weighted average) basis and commercial (simple average) basis; in both, weightage is given to qualitative factors as well as quantitative factors.

3. **Absolute basis**: the summation of the products of qualitative and quantitative factors, Σ(weight carried × distance) for each leg; cinema seat classes are weighted the same way.
4. **Commercial basis**: the product of the average qualitative factor and the total quantitative factor, total distance × $(W_1 + W_2 + \ldots + W_n) \div n$.
5. Distance is the quantitative factor (the variable cost depends on it); weight carried is the qualitative factor.
6. **Equivalent cost unit / equivalent service unit**: to calculate the cost or price of two or more grades of a service using common resources, each grade is assigned a weight and converted into equivalent units, which makes the grades equivalent and comparable. (A two-line example, such as the hotel suites, may be added.)

"State the cost unit of ..."

Write a two-column table, industry and cost unit, in the order the question lists them, using the Study Material's units (ICAI's January 2025 suggested answer copies them, "patient per day, room per day or per bed, per operation etc." for hospitals): transport, passenger-km and tonne-km; electricity, kWh; hospital, patient per day, room per day or per bed, per operation; hotel, guest days or room days; canteen, per meal; education, per student or per course; bank, per transaction or application; IT, per project or module; insurance, per policy or claim. Units for manufacturing industries (brick-making, brewing, coal) come from the cost units lesson.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the distance-weighted "commercial" average

Bunty averages the milk-run's loads by distance: 3,410 tonne-km ÷ 290 km = 11.76 t, then 11.76 × 290 = 3,410, and proudly finds commercial = absolute. A distance-weighted average always gives back the absolute figure. **Fix:** the commercial basis uses the simple average of the legs' loads, (18 + 17 + 9 + 8) ÷ 4 = 13 t, so 3,770 tonne-km.

⚠ **Common mistake**

The load that never fell

Using the starting load for the whole trip: 12 t × 80 km for Kesar's truck to Ujjain, although 8 t came off at Dewas. **Fix:** list the legs and the load on board after each stop.

⚠ **Common mistake**

Equivalent units built on capacity

Weighting every seat and berth on the sleeper bus (20 + 15 × 1.6 + 15 × 1.4 = 65 a trip) when occupancy differs by class (the occupied figure is 52.4). **Fix:** weight the units actually sold or occupied: seats × occupancy × trips × km, then × weight.

⚠ **Common mistake**

"Per bed" when the question asks for a composite unit

The Study Material's own MCQ asks for the *composite* cost unit of a hospital and offers per patient, per patient-day, per day and per bed. Per bed, per patient and per day are single measures: valid cost units in a "state the cost unit" table, wrong here. **Fix:** a composite unit joins two measures: patient-day (and for a hotel, room-day or guest-day).

Practice

Check yourself

1 A van starts with 9 tonnes, drops 3 tonnes after 30 km, carries the rest another 50 km, and returns empty over 80 km. What are its absolute tonne-km?

Answer: _____

2 Case scenario. A Tapri truck leaves Bhopal empty to collect kulhad clay. It loads 6 tonnes at Vidisha (60 km), adds 4 tonnes at Sagar (120 km further) and returns full 180 km to Bhopal. Ritu asks her intern for the commercial tonne-km, counting every leg as ICAI's formula does. Which figure is correct?

- **A** 1,920 tonne-km
- **B** 2,520 tonne-km
- **C** 2,880 tonne-km
- **D** 1,440 tonne-km

3 Which of the following is a composite cost unit?

- **A** Per meal served in a canteen
- **B** Per ticket sold by a cinema
- **C** Per passenger-km run by a bus operator
- **D** Per loan application processed by a bank

4 A guest house has Standard and Suite rooms; a Suite is let at 2.5 times a Standard room. In a year 5,400 Standard room-days and 1,200 Suite room-days are let, and the takings needed are ₹50,40,000. What is the rent of a Suite per room-day, in rupees?

Answer: _____

5 Which statements about absolute and commercial tonne-km are correct? Select all that apply.

- **A** The absolute basis adds weight carried × distance for each leg
- **B** The commercial basis multiplies the total distance by the simple average of the loads
- **C** The two bases always give the same figure on a round trip
- **D** In both bases, distance is the quantitative factor on which the variable cost depends

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Service costing = operating costing: costs of a period expressed per cost unit of service.
- The specific issue is defining a realistic cost unit; often a composite unit is more appropriate (hotel: occupied room-days).
- Composite cost unit: two measurement units combined (tonne-km, quintal-km, passenger-km, patient-day).
- Absolute (weighted average): Σ(weight × distance), leg by leg.
- Commercial (simple average): total distance × (W_1 + ... + W_n) ÷ n.
- Equivalent service units: weight each grade, convert to a base, divide, then base rate × weight; any grade can be the base.
- KPIs assess performance: CPOR, occupancy and RevPAR for hotels; OTIF and truck turnaround for transport; ARPU for telecom.
- Count what was carried, stretch by stretch. (The porter knows.)

Answer key · Check yourself

1 **570**

9 × 30 + 6 × 50 + 0 × 80 = 270 + 300 + 0 = 570 tonne-km.

2 **A 1,920 tonne-km**

Loads on the three legs are 0, 6 and 10 tonnes; average (0 + 6 + 10) ÷ 3 = 5.33 t × 360 km = 1,920. 2,520 is the absolute figure (0 + 720 + 1,800); 2,880 leaves the empty leg out of the average (8 t × 360); 1,440 divides the loads by 4 instead of 3 legs.

3 **C Per passenger-km run by a bus operator**

A composite unit combines two measurement units: passengers and kilometres. The other three are single units, though all are valid service cost units (SM 12.3–12.4).

4 **1500**

Equivalent Standard room-days = 5,400 + 1,200 × 2.5 = 8,400. Standard rent = ₹50,40,000 ÷ 8,400 = ₹600; Suite = ₹600 × 2.5 = ₹1,500.

5 **A The absolute basis adds weight carried × distance for each leg; B The commercial basis multiplies the total distance by the simple average of the loads; D In both bases, distance is the quantitative factor on which the variable cost depends**

They agree when the legs are equal in length or in load, and usually differ otherwise (5,600 against 5,760 in SM Illustration 1). The Study Material calls distance the quantitative factor and weight carried the qualitative factor (SM 12.7–12.8).

12.2 Transport costing: passenger-km and tonne-km

Costing a bus or a truck: kilometres run, passenger-km or tonne-km, the operating cost sheet in standing, running and maintenance charges, and a fare built on takings.

In plain words

The Chai Yatra trial month

Kesar has a new idea for the pilgrims who travel from Indore to the Mahakal temple at Ujjain, 55 km away: a bus with hot chai served at every seat. The trial bus has 40 seats and makes 4 round trips a day for 25 days.

> **Kesar:** The bus cost us ₹2,97,000 this month. Ritu, what do I charge a ticket?
>
> **Ritu:** Per unit, please. And for a bus the unit is one passenger carried one kilometre.
>
> **Kesar:** Why not per seat?
>
> **Ritu:** An empty seat earns nothing, Kesar-ji. It still burns diesel.

Ritu counts the work in three steps. **Kilometres run**: 55 km × 2 (there and back) × 4 round trips × 25 days = 11,000 km. **Passenger-km**: on average 30 of the 40 seats were filled, so the bus carried 30 passengers over every one of those kilometres: 11,000 × 30 = 3,30,000 passenger-km.

Cost per passenger-km: ₹2,97,000 ÷ 3,30,000 = 90 paise. A pilgrim travelling 55 km costs Tapri 55 × ₹0.90 = ₹49.50.

That is only the cost. The ticket must also cover anything charged as a share of the ticket money (the conductor's commission, a passenger tax) and Kesar's profit, so Ritu works out the total ticket money needed, the **takings**, and divides it by the passenger-km too.

The cost is grouped the way the bus behaves. Insurance, the road tax, the garage rent and the monthly salaries are paid whether or not the bus moves: **standing charges**. Diesel and oil rise with every kilometre: **running charges**. Repairs, tyres and spares wear out with use but never fall to zero: **maintenance charges**. A truck is costed the same way, with the **tonne-km** (one tonne carried one kilometre) in place of the passenger-km.

Think of it like

The shared auto at the bus stand

A shared auto-rickshaw driver knows his day's cost: CNG, his EMI, the permit. He does not divide it by the seats in his auto but by the passengers he actually carries and how far he takes them. A full auto on a long route makes every rupee of cost cheap per passenger; a half-empty one makes it dear, though the CNG bill is the same.

Key idea

Cost per passenger-km (or tonne-km) = total operating cost of the period ÷ passenger-km (or tonne-km) of the same period, where passenger-km = kilometres run × passengers actually carried. The fare is built on takings: cost plus everything charged as a percentage of takings, divided by the passenger-km, then × the length of the journey.

How it works

Transport organisations are of two kinds, goods transport and passenger transport (SM 12.11). Their cost units are composite, built in service costing and composite cost units:

Definition

Tonne-km and passenger-km (SM 12.11)

The cost unit for a goods transport organisation is the tonne-kilometre: the cost of carrying one tonne of goods over a distance of one kilometre. The cost unit for a passenger transport organisation is the passenger-kilometre: the cost of carrying one passenger over a distance of one kilometre.

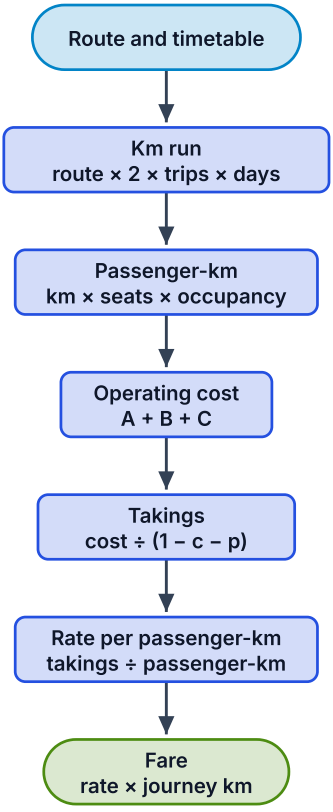
The three heads of a transport cost sheet

ICAI's "suggestive" heads and items, in ICAI's order (SM 12.12):

Head	Items
Standing charges or fixed costs: constant irrespective of the distance travelled	Insurance; licence fees; salary to driver, conductor, cleaners, etc. if paid on a monthly basis; garage costs, including garage rent; depreciation (if related to efflux of time); taxes; administration expenses
Variable costs or running costs: generally associated with the distance travelled	Petrol and diesel; lubricant oils; wages to driver, conductor, cleaners, etc. if related to operations; depreciation (if related to activity); any other variable costs identified
Semi-variable costs or maintenance costs	Repairs and maintenance; tyres; spares

"The heads for a cost may change as per the situation or condition": a driver's salary is a standing charge or a running cost "depending on the situation and nature of his employment" (SM 12.12). Interest is a separate item of cost and fixed unless otherwise given (SM 12.10).

From a timetable to a fare



Trips are round trips, for every vehicle; A, B and C are the standing, running and maintenance charges; c and p are the commission (or passenger tax) and the profit as fractions of takings. For a truck, read tonne-km for passenger-km.

1. **Kilometres run** in the period. A round trip covers the route twice. ICAI's working note for buses: number of buses × distance × number of days × number of trips, with "6 single trips (3 round trips)" written out (SM 12.15).

2. **Passenger-km** = kilometres run × passengers actually carried per trip (seats × occupancy). ICAI's version in one line: buses × distance × seating capacity × used capacity × days × trips (SM 12.13).

3. **Operating cost** for the same period under the three heads. Convert every item to that period first: insurance at 3% of the bus's cost, a monthly garage rent × 12, repairs per annum.

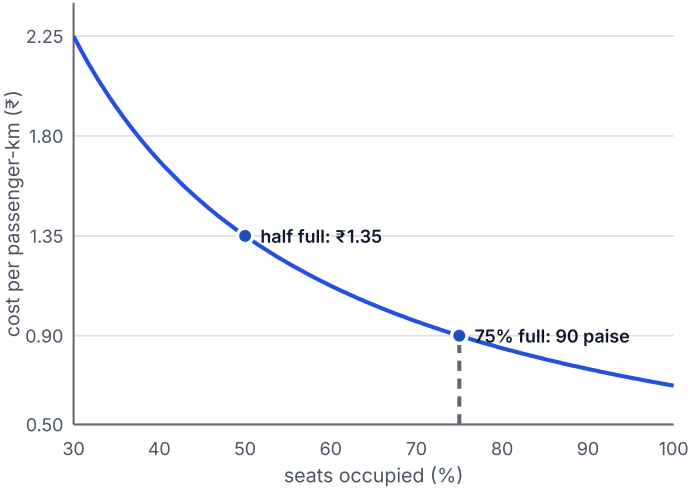
4. **Takings.** Commission, passenger tax and profit stated as a percentage of takings are solved, not added: Takings = cost + c × Takings + p × Takings (SM Illustration 3, 12.16).

5. **Rate per passenger-km** = takings ÷ passenger-km; **fare** for a journey = rate × its kilometres. Round as the question says (ICAI's September 2026 paper asked for the next higher rupee).

Why empty seats are expensive

Once the timetable is fixed, the bus runs the same kilometres and spends the same money whether it is full or half-empty; only the passenger-km change.

Chai Yatra trial bus: cost per passenger-km against occupancy



₹2,97,000 spread over 11,000 km × 40 seats × occupancy. Halving the passengers doubles the cost of each passenger-km; the cost of the bus does not move.

Goods transport

A truck is costed under the same three heads, per tonne-km. When it drops part of its load on the way, count the tonne-km leg by leg on the **absolute** basis or as total

distance × average load on the **commercial** basis (both bases). Two rules from ICAI's lorry illustration (SM 12.22):

- The **actual load carried** is used for tonne-km, even when it attracts a fine for overloading.
- The **penalty for overloading** is abnormal: it is not an operating cost of the lorry but is debited to the Costing Profit and Loss Account, so it comes off the operating profit to give the net profit or loss.

Formats and formulas

Reproduce this statement from memory. It follows SM Illustrations 2 and 3 (12.14–12.16) and the January 2025 suggested answer; add a "per month" column when the question mixes periods.

Operating cost statement for … (period)	Total (₹)	Per passenger-km (₹)
A. Standing charges		
Insurance, taxes, licence, garage rent	xxx	
Salaries paid monthly	xxx	
Depreciation (time basis)	xxx	
Interest, administration, stationery	xxx	
Total standing charges	**xxx**	**x.xxxx**

Operating cost statement for … (period)	Total (₹)	Per passenger-km (₹)
B. Running charges		
Petrol or diesel, lubricants	xxx	
Wages per trip; depreciation (usage basis)	xxx	
Total running charges	**xxx**	**x.xxxx**
C. Maintenance charges		
Repairs, tyres, spares	xxx	x.xxxx
Total cost (A + B + C)	**xxx**	**x.xxxx**
Add: Commission, passenger tax (on takings)	xxx	x.xxxx
Add: Profit (on takings)	xxx	x.xxxx
Takings	**xxx**	**x.xxxx**

ICAI shows commission inside running charges in SM Illustration 3, and on a line after "total cost before commission" in the January 2025 suggested answer; either way its amount is solved from takings.

∑ **Formula**

Kilometres run and passenger-km

$$\text{Km run} = \text{Vehicles} \times \text{Route km} \times 2 \times \text{Round trips a day} \times \text{Days}$$

$$\text{Passenger-km} = \text{Km run} \times \text{Seats} \times \text{Occupancy}$$

∑ **Formula**

Fuel cost

$$\text{Diesel cost} = \frac{\text{Km run}}{\text{Km per litre}} \times \text{Price per litre}$$

∑ **Formula**

Takings when charges and profit are on takings

$$T = \text{Cost} + cT + pT \qquad T = \frac{\text{Cost}}{1 - c - p}$$

c = commission or passenger tax, p = profit, each as a fraction of takings (SM 12.16).

∑ **Formula**

Rate per passenger-km and fare

$$\text{Rate} = \frac{\text{Takings}}{\text{Passenger-km}}$$

$$\text{Fare} = \text{Rate} \times \text{Journey km}$$

Rate = rate per passenger-km; for goods, rate per tonne-km = takings ÷ tonne-km.

Worked examples

Worked example **Chai Yatra: a pilgrim bus with a stop at Dewas**

After the trial, Tapri runs 2 buses on the Indore–Dewas–Ujjain route: Indore to Dewas 35 km, Dewas to Ujjain 45 km. Each bus has 50 seats and makes 2 round trips a

day, 30 days a month. The conductors' counts give the average passengers on board on each stage:

Stage	Distance (km)	Passengers on board
Indore to Dewas	35	40
Dewas to Ujjain	45	25
Ujjain to Dewas	45	28
Dewas to Indore	35	44

Monthly costs:

Particulars	Data
Each bus	cost ₹36,00,000; residual value ₹6,00,000 after 10 years; straight line, time basis
Insurance and road tax	₹12,000 a month per bus
Garage rent	₹20,000 a month for both buses
Driver and conductor	₹25,000 and ₹18,000 a month per bus
Supervisor	₹30,000 a month
Diesel	₹96 a litre; 4 km a litre
Lubricants	₹2 a km
Repairs, tyres and spares	₹3 a km

The crews get a commission of 5% of takings, and Tapri wants a profit of 20% of takings. Fares are charged per passenger-km of the stage travelled.

Required:

(i) Calculate the kilometres run and the passenger-km for the month.

(ii) Prepare an operating cost statement for the month, in total and per passenger-km.

(iii) Calculate the rate per passenger-km and the fares from Indore to Dewas, from Dewas to Ujjain and from Indore to Ujjain.

THE WORKING

Working notes

1. Round trips = 2 buses × 2 × 30 days = 120; each covers 35 + 45 + 45 + 35 = 160 km. **Km run = 120 × 160 = 19,200 km.**
2. Passenger-km per round trip, stage by stage = 35 × 40 + 45 × 25 + 45 × 28 + 35 × 44 = 1,400 + 1,125 + 1,260 + 1,540 = 5,325. **Passenger-km = 120 × 5,325 = 6,39,000.**
3. Depreciation per bus = (₹36,00,000 − ₹6,00,000) ÷ 10 ÷ 12 = ₹25,000 a month (time basis: a standing charge).
4. Diesel = 19,200 ÷ 4 = 4,800 litres × ₹96 = ₹4,60,800.

Operating cost statement of Chai Yatra for the month (6,39,000 passenger-km)

Particulars	Total (₹)	Per passenger-km (₹)
Depreciation (2 × ₹25,000)	50,000	
Insurance and road tax (2 × ₹12,000)	24,000	

Particulars	Total (₹)	Per passenger-km (₹)
Garage rent	20,000	
Crew salaries (2 × ₹43,000)	86,000	
Supervisor	30,000	
A. Standing charges	**2,10,000**	**0.3286**
Diesel (WN 4)	4,60,800	
Lubricants (19,200 × ₹2)	38,400	
B. Running charges	**4,99,200**	**0.7812**
C. Maintenance (19,200 × ₹3)	**57,600**	**0.0901**
Total cost (A + B + C)	**7,66,800**	**1.2000**
Commission (5% of takings)	51,120	0.0800
Profit (20% of takings)	2,04,480	0.3200
Takings	**10,22,400**	**1.6000**

Per passenger-km figures are rounded to four decimals. **Working note 5: takings.** T = ₹7,66,800 +

0.05T + 0.20T, so 0.75T = ₹7,66,800 and T = ₹10,22,400.

(iii) Fares

$$\text{Rate} = \frac{10{,}22{,}400}{6{,}39{,}000} = ₹1.60 \text{ per passenger-km}$$

Indore to Dewas (35 km) **₹56**; Dewas to Ujjain (45 km) **₹72**; Indore to Ujjain (80 km) **₹128**.

The load changes at Dewas, so the passenger-km are counted stage by stage, as tonne-km are on the absolute basis. Taking the average load of the four stages (34.25) over all 19,200 km would give 6,57,600 passenger-km and a rate of ₹1.5547: the long Dewas–Ujjain stages carry the fewest passengers, so the shortcut overstates the work done and the fares would fall short of the takings needed.

Worked example

Tapri Cold Chain: the empty way home

Tapri Cold Chain Pvt Ltd, Tapri's cold-chain subsidiary, runs a 10-tonne refrigerated truck between Indore and Bhopal (190 km). In May it made 12 round trips. Every outward trip carried 9 tonnes of premix and frozen snacks; on 4 return trips it brought back 6 tonnes of a

Bhopal dairy's butter, and on the other 8 it came back empty.

Costs for May	Amount
Standing charges (driver's salary, insurance, permits, depreciation on time basis, interest)	₹72,000
Maintenance	₹20,000 + ₹1 a km
Diesel	₹95 a litre; 4 km a litre
Fuel of the refrigeration unit (run only when loaded)	₹6 a loaded km

Required:

(i) Calculate the absolute and the commercial tonne-km for May, and explain why they agree.

(ii) Prepare the cost statement for May and find the cost per tonne-km.

(iii) Calculate the freight per tonne-km that would give a profit of 20% on the freight.

(iv) The dairy offers 6 tonnes of butter on each of the 8 empty return trips at ₹3 a tonne-km. Loading does not change the truck's diesel use. Should the offer be accepted, though ₹3 is below the cost per tonne-km?

THE WORKING

(i) Tonne-km

Journey	Working	Tonne-km
Indore to Bhopal	12 × 190 km × 9 t	20,520
Bhopal to Indore, loaded	4 × 190 km × 6 t	4,560
Bhopal to Indore, empty	8 × 190 km × 0 t	0
Absolute tonne-km		**25,080**

Commercial tonne-km = total distance × average load = (12 × 2 × 190 = 4,560 km) × (12 × 9 + 4 × 6 + 8 × 0) ÷ 24 journeys = 4,560 × 5.5 t = **25,080**. They agree because every journey is the same length (190 km): the two bases differ only when long and short legs carry different loads. The empty journeys stay in the average with a load of zero.

(ii) Cost statement for May

Particulars	Amount (₹)
A. Standing charges	72,000
B. Maintenance (₹20,000 + 4,560 km × ₹1)	24,560
Diesel (4,560 km ÷ 4 × ₹95)	1,08,300
Refrigeration fuel ((12 + 4) × 190 = 3,040 loaded km × ₹6)	18,240
C. Running charges	1,26,540
Total cost	**2,23,100**

Cost per tonne-km = ₹2,23,100 ÷ 25,080 = **₹8.90**.

(iii) Freight. Profit 20% on freight: ₹8.8955 ÷ 0.80 = **₹11.12 a tonne-km**.

(iv) The backhaul offer

Per backhaul trip	Amount (₹)
Freight (6 t × 190 km × ₹3)	3,420
Extra cost: refrigeration fuel (190 km × ₹6)	1,140
Gain	**2,280**

Accept: the 8 trips add ₹18,240 to May's profit. The standing charges, the diesel and the maintenance

are incurred whether the truck comes back full or empty, so only the refrigeration fuel is extra. With the backhaul the month carries 34,200 tonne-km for ₹2,32,220, and the cost per tonne-km falls to ₹6.79. The full cost per tonne-km is the right base for the regular freight rate, not for pricing a load the truck can carry at almost no extra cost.

Worked example

Own, lease or outsource: deliveries to the cafés

Tapri delivers premix and snacks to its 40 Indore cafés: 2,400 drops a month over 3,600 km. Dev compares three ways of doing it:

(i) Own a van: cost ₹12,00,000, residual value ₹3,00,000 after 5 years (time basis); insurance and road tax ₹3,000 a month; driver ₹22,000 and helper ₹14,000 a month; diesel ₹96 a litre at 8 km a litre; repairs, tyres and servicing ₹3 a km.

(ii) Lease a van with its driver: ₹60,000 a month including 3,000 km, and ₹8 for every km above that; Tapri pays the diesel and its own helper.

(iii) Use a logistics partner at ₹42 a drop plus a fuel surcharge of 5%.

Required:

(i) Work out the cost a month and per drop of each option at the present level, and rank them.

(ii) From next year Kesar expects 3,000 drops a month over 4,500 km. Re-rank the options, and find the drops a month at which owning and outsourcing cost the same (km per drop unchanged).

THE WORKING

Working notes

1. Own van, fixed a month: depreciation (₹12,00,000 − ₹3,00,000) ÷ 60 = ₹15,000; insurance and road tax ₹3,000; driver ₹22,000; helper ₹14,000: **₹54,000**. Variable: diesel ₹96 ÷ 8 = ₹12 a km + repairs ₹3 = **₹15 a km**.
2. Lease: excess km at 3,600 km = 600 × ₹8 = ₹4,800; diesel ₹12 a km.
3. Partner: ₹42 × 105% = **₹44.10 a drop**.

(i) Present level: 2,400 drops, 3,600 km

Particulars (₹)	(i) Own	(ii) Lease	(iii) Partner
Fixed charges	54,000	60,000	–
Excess km	–	4,800	–
Diesel and repairs (3,600 km)	54,000	43,200	–
Helper	(in fixed)	14,000	–
Drops × ₹44.10	–	–	1,05,840
Cost a month	**1,08,000**	**1,22,000**	**1,05,840**
Cost per drop	**45.00**	**50.83**	**44.10**
Rank	**2**	**3**	**1**

(ii) Next year: 3,000 drops, 4,500 km

Particulars (₹)	(i) Own	(ii) Lease	(iii) Partner
Fixed charges	54,000	60,000	–
Excess km (1,500 × ₹8)	–	12,000	–

Particulars (₹)	(i) Own	(ii) Lease	(iii) Partner
Diesel and repairs (4,500 km)	67,500	54,000	–
Helper	(in fixed)	14,000	–
Drops × ₹44.10	–	–	1,32,300
Cost a month	**1,21,500**	**1,40,000**	**1,32,300**
Cost per drop	**40.50**	**46.67**	**44.10**
Rank	**1**	**3**	**2**

Indifference point. A drop needs 3,600 ÷ 2,400 = 1.5 km, so the own van's variable cost is 1.5 × ₹15 = ₹22.50 a drop against the partner's ₹44.10. They cost the same when ₹54,000 + ₹22.50D = ₹44.10D, so

$$D = \frac{54{,}000}{44.10 - 22.50} = 2{,}500 \text{ drops a month}$$

Below 2,500 drops the partner is cheaper; above it the van's fixed ₹54,000 is spread thin enough for owning to win. The cost unit here is the drop, not

> the km: the question is what each delivery costs, and only the own van's cost moves with km.

Exam angle

In the exam

Where transport costing appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, transport costing came up in 12 documents:

- **Bus fare on takings, 6 to 8 marks**: January 2025 Q4(a) (8 marks: 4 round trips, 10% commission shared by driver and conductor, 15% profit on takings); MTP Series I for September 2024 Q3(b) (8 marks: annual statement, passenger tax 22% and profit 25% on takings, one-way fare); MTP Series I for September 2025 Q5(b) (mini bus, profit 20% of revenue, 43 paise per passenger-km); MTP Series II for September 2026 Q2(b) (6 marks: monthly cost sheet, 10% commission, 25% profit on takings).
- **A fleet for a year, 7 marks**: September 2026 Q3(b): 8 buses, two under servicing for a month each, profit one-third of revenue, the fare rounded off to the next higher rupee (suggested answers not yet published). A city fleet of 25 buses of two sizes, 5 of them off the road each day for repairs, costed per passenger-km: RTP September 2026 Q13.
- **Goods vehicles, up to 9 marks**: absolute ton-km of six routes with up and down loads, depreciation per

km (MTP Series I for May 2024 Q2(a), 9 marks); five MCQs on a truck with a 75% minimum-load condition (MTP Series II for January 2025); five MCQs on a fleet's fixed and variable costs and the freight rate per ton-km (RTP January 2026).

- **MCQs, 2 marks each**: cost per passenger-km for 4 buses (May 2026 MCQ 14); ton-km of a truck (MTP Series I for September 2024 MCQ 14); a round trip that takes 2 days (MTP Series I for May 2025 MCQ 12); a bus case with break-even passengers per trip, and a ferry fare in nautical miles (MTP Series II for May 2025, MCQs 6–10 and 13).

In the exam

The traps ICAI set

- **What is a trip?** In MTP Series II for May 2025 the Mumbai–Pune buses made "20 round trips", and the cost "per trip" meant per round trip of 300 km; in SM Illustration 3, "3 up and down trips" are 6 single trips. Read the question twice.
- **Days on the road**: a round trip that takes 2 days halves the number of trips (MTP Series I for May 2025 MCQ 12).
- **Months**: two buses under servicing run 11 months, while the salaries and the depreciation are given as annual totals for the fleet (September 2026 Q3(b)): kilometres and diesel follow the months run, fixed costs do not.

- **Commission shared by two people** is still one percentage of takings: 10% shared equally is 5% each, 10% in all (January 2025 Q4(a)).
- **Depreciation on the straight-line method**: the January 2025 suggested answer showed it among running charges; SM 12.10 treats time-based depreciation as fixed. The total, and so the fare, is the same either way; follow the Study Material unless the question says otherwise.

Trick

The fare shortcut

One-way fare = takings ÷ passengers carried in the period, because takings ÷ passenger-km × route km = takings ÷ (passenger-km ÷ route km). Use it to check the rate-per-passenger-km route. Condition: every passenger travels the full route.

Answer frame

The numerical answer, in ICAI's order

1. **Working note: kilometres run**, with the trips written out ("3 round trips = 6 single trips").
2. **Working note: passenger-km or tonne-km** (absolute or commercial as asked, legs in a table).
3. **Working notes for items that need arithmetic**: diesel, depreciation, the takings equation.

4. **The operating cost statement**: A standing, B running, C maintenance, each with its total and per-unit figure; then commission or tax, profit, takings.
5. **The answer in the question's words**: "cost per passenger-km", "fare per passenger-km", "one-way fare per passenger", rounded as instructed; abnormal items such as a penalty shown below the operating profit.

"Classify these costs of a transport undertaking"

Use ICAI's three heads and their items (SM 12.12): **standing** (insurance, licence fees, monthly salaries, garage costs and rent, time-based depreciation, taxes, administration); **running** (petrol and diesel, lubricant oils, wages related to operations, activity-based depreciation); **maintenance** (repairs and maintenance, tyres, spares). Where an item can move (a salary, depreciation), state the basis you assume.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the bus that only went one way

"2 round trips × 80 km" gives 160 km a day for a Chai Yatra bus; it actually runs 320. **Fix:** a round trip covers the route twice; write "2 round trips = 4 single trips" in the working note, as ICAI does.

⚠ **Common mistake**

Seats in place of passengers

Multiplying Chai Yatra's 19,200 km by the 50 seats gives 9,60,000 "passenger-km" and a rate of ₹1.065, a third too low. **Fix:** passenger-km use passengers actually carried, stage by stage (6,39,000).

⚠ **Common mistake**

The fine that became a running cost

Leaving a penalty for overloading a lorry in its running charges raises every tonne-km cost. **Fix:** a penalty is abnormal; take it out of the operating cost and deduct it after the operating profit (SM 12.22).

⚠ **Common mistake**

The full cost per tonne-km used to refuse a backhaul

Turning down the dairy's butter at ₹3 a tonne-km "because the truck costs ₹8.90" throws away ₹2,280 a trip: the truck comes back from Bhopal anyway. **Fix:** the full cost per tonne-km prices the regular freight; a load on a journey that will be run anyway needs to cover only its extra cost (here the refrigeration fuel).

Practice

Check yourself

1 A Tapri staff shuttle runs 3 buses on a 30 km route, 4 round trips a day for 26 days. The month's takings must be ₹9,82,800, and the one-way fare is fixed at ₹45. How many passengers must each single trip carry on average?

Answer: _____

2 Case scenario. Tapri's airport coach makes 8 single trips a day for 30 days. On an average trip it carries 30 full-fare passengers, 10 children at half fare and 10 senior citizens at 60% of the full fare. The month's operating cost is ₹5,90,400, and Tapri wants a profit of 20% of takings. Ritu must fix the full fare per trip. Which is correct?

- **A** ₹60.00
- **B** ₹61.50
- **C** ₹72.00
- **D** ₹75.00

3 How is a penalty paid for overloading a lorry treated in ICAI's transport costing?

- **A** As a running charge, because it arises from a journey
- **B** As a maintenance charge
- **C** Excluded from the operating cost and debited to the Costing Profit and Loss Account
- **D** Spread over the tonne-km of the month

4 A Tapri truck based at Indore drives empty 40 km to a dairy at Dewas, carries 9 tonnes from Dewas to Bhopal (150 km), and returns empty to Indore (190 km). It does this circuit 20 times in a month, and the month's cost is ₹2,70,000. What is the cost per absolute tonne-km, in rupees?

Answer: _____

5 Which of these are standing charges of a bus operator under ICAI's classification? Select all that apply.

- **A** Garage rent
- **B** Conductor's salary paid monthly
- **C** Lubricant oil
- **D** Depreciation related to the efflux of time
- **E** Tyres

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Goods transport: tonne-km; passenger transport: passenger-km (one tonne or one passenger carried one km).
- Km run = vehicles × route km × 2 × round trips × days; passenger-km = km run × seats × occupancy.
- Standing (insurance, licence, monthly salaries, garage, time depreciation, taxes, administration); running (fuel, lubricants, wages per operation, activity depreciation); maintenance (repairs, tyres, spares).
- Interest: a separate item, fixed unless otherwise given; heads can change with the facts (a driver's salary).

- Takings = cost ÷ (1 − c − p) when commission, passenger tax and profit are on takings.
- Rate per passenger-km = takings ÷ passenger-km; fare = rate × journey km.
- Actual load counts for tonne-km even when fined; the fine itself is abnormal and goes to the Costing P&L.
- An empty seat still burns diesel.

Answer key · Check yourself

1 **35**

Km run = 3 × 30 × 2 × 4 × 26 = 18,720. Rate = ₹45 ÷ 30 km = ₹1.50 a passenger-km, so the takings need ₹9,82,800 ÷ ₹1.50 = 6,55,200 passenger-km; ÷ 18,720 km = 35 passengers a trip.

2 **D** **₹75.00**

Equivalent full-fare passengers a trip = 30 + 10 × 0.5 + 10 × 0.6 = 41; × 240 trips = 9,840. Takings = ₹5,90,400 ÷ 0.80 = ₹7,38,000; full fare = ₹7,38,000 ÷ 9,840 = ₹75 (children ₹37.50, seniors ₹45). ₹60 recovers only the cost; ₹61.50 treats all 50 as full-fare passengers; ₹72 adds 20% to cost instead of taking 20% of takings.

3 **C Excluded from the operating cost and debited to the Costing Profit and Loss Account**

SM 12.22: it is abnormal expenditure, not part of the operating cost of the lorry; it is deducted from the operating profit to arrive at the net profit or loss.

4 **10**

Absolute tonne-km = 20 × 9 t × 150 km = 27,000; the empty legs add kilometres and cost but no tonne-km. ₹2,70,000 ÷ 27,000 = ₹10 a tonne-km.

5 **A Garage rent; B Conductor's salary paid monthly; D Depreciation related to the efflux of time**

SM 12.12: garage costs, monthly salaries and time-based depreciation are standing charges; lubricants are running costs; tyres are maintenance costs.

12.3 Hotels, hospitals, power, IT and other services

The service chapter's other industries: room-days for hotels, patient-days for hospitals, kWh for power, toll rates for BOT roads, projects for IT, students for schools, policies for insurers, applications for banks.

In plain words

Kesar's week of new ventures

Tapri has grown far beyond chai. In one week Kesar asks Ritu to price a hill-station inn at Pachmarhi, the dialysis centre that Tapri's charitable trust runs in Jabalpur, the captive power plant at the samosa factory, the toll road near Indore in which Tapri has taken a stake, an ordering app and a term-insurance plan for the delivery riders.

> **Kesar:** Six businesses, six headaches. One formula for all, na? Total cost divided by customers.
>
> **Ritu:** The idea is the same, Kesar-ji. The unit changes. A room is sold by the night, a hospital bed by the patient-day, a dialysis machine by the session, the power plant by the unit of electricity, a toll road by the vehicle.
>
> **Kesar:** And the app?
>
> **Ritu:** By the project, built out of the months your developers spend on it.

So every service gets the unit it is sold in: a **room-day** (one room let for one day) for a hotel, a **patient-day** for a hospital, a **kWh** (one kilowatt-hour, the "unit" on an electricity bill) for a power plant, a **vehicle** passing the plaza for a toll road, a **project** for an IT company, a **student** or a **batch** for a school, a **policy** for an insurer, an **application** for a bank.

Then three moves, in every business. Count the units **actually** delivered: an inn with 40 rooms that is 40% full in winter sells 16 rooms a night, not 40. Split the cost into what stays fixed and what moves with each unit. Divide, then add the profit and anything charged as a share of takings. Where some units are worth more (a deluxe room, off-season half rent, a truck against a car at the toll plaza), give them weights, exactly as in equivalent service units.

Think of it like

One recipe, many kitchens

Ritu's dal recipe is the same in every kitchen; only the measure changes: a katori at a dhaba, a kilo at a wedding, a degchi at a langar. Service costing is one recipe (count the units used, total the cost, divide, add the margin) cooked with a different measure in each industry.

Key idea

Each service has its own cost unit (room-day, patient-day, kWh, vehicle, project, student, policy, application). Count the units actually used, not the capacity; give weights to grades and seasons; divide the period's cost, then build the price on takings.

How it works

Hotels and lodges

Hotels are run on commercial basis, so cost is computed "to fix the price of various services provided by the hotel and to find out the profit or loss" (SM 12.22). The cost unit "may be Guest-day or Room Day", and the estimated occupancy rate at different points of time, peak season or lean season, is taken into account (SM 12.23).

1. **Room-days occupied** for each type and season: rooms × occupancy × days.

2. **Equivalent room-days**: weight each type (a suite let at twice a Standard room) and each season (off-season at half rent) to one base: full-rent Standard room-days (SM Illustrations 6 and 7, 12.23–12.28).

3 **Total cost of the year**: fixed costs (salaries, depreciation, repairs) plus costs per occupied room-day (room attendants paid on occupancy, laundry).

4 **Takings**, solving for any rent or commission charged as a percentage of takings and for the profit on takings; then rent per equivalent room-day × each weight.

Hospitals

A hospital has departments for **out-patients**, **in-patients**, **medical services** (X-ray, scanning), **general services** (catering, laundry, power house) and **miscellaneous services** (transport, dispensary) (SM 12.28). Their units (SM 12.28):

Department	Unit of cost
Out-patient	Per out-patient
In-patient	Per room-day
Scanning	Per case
Laundry	Per 100 items laundered

Cost segregation (SM 12.28–12.29): **fixed costs** are based on timelines and do not depend on the services provided (staff salaries, depreciation on building and equipment); **variable costs** vary with the level of services rendered

(laundry, food supplied to patients, power). ICAI's critical care unit (CCU) illustration treats doctors paid on the basis of the number of patients attended and the time spent as a variable cost, works out the contribution per patient-day and finds a break-even point in patient-days (SM Illustration 8, 12.29–12.31), the method of break-even analysis.

IT and ITES

IT and IT-enabled services organisations are highly labour intensive and provide services to outside customers or internally (captive consumption); employee cost is a significant part of their operating cost, and the direct employee cost is traceable to the services rendered; their significant overheads ("service overheads") are allocated or apportioned to cost units (SM 12.31). The jobs undertaken are **projects**, each unique in size, functionality, duration and staffing (SM 12.32):

1. The project is broken into activities and their dependencies; the project schedule is developed from them.
2. The skill level and duration of each activity are identified: **effort estimation**. Scheduling and effort estimation are normally done together.
3. The manpower needed is identified, and effort is costed **per person-day, person-week or person-month** (SM 12.33).

Direct manpower (software engineers, functional consultants, business analysts; project leaders; project manager; program manager) is traceable and a direct cost. **Support manpower** (quality assurance, testing, version control, staffing manager) is a direct cost if its time can be identified with the project, otherwise an overhead apportioned on a suitable basis (SM 12.32–12.33). The parameters of total cost (SM 12.34–12.35):

Parameter	Direct to the project when	Otherwise
A. Hardware and software	Identifiable with the project	Service overhead
B. Travel and training	Incurred for the project	Service overhead (Java training useful across many projects)
C. Effort costs	Identified with the project: direct unless otherwise specified	–

Effort cost is not just salary: it includes a share of the cost of running the organisation (office space, heating and lighting; support staff; networking and communications; central facilities such as a library; social security and employee benefits). In short, salary plus part of common overhead (SM 12.34–12.35). A project cost sheet built this way is worked in the service-sector cost sheet.

Toll roads

Highway economic analysis quantifies a scheme's costs and benefits over a time horizon, compares the "with" and "without" project conditions, and tests the project by sensitivity analysis (SM 12.36–12.37). The project cost has two components:

Costs of a toll road project (SM 12.37–12.38)

Toll road project

Capital costs — during construction

- Preliminary and pre-operative expenses
- Land acquisition
- Materials
- Labour
- Overheads in actual construction
- Contingency allowance
- Interest during construction period

Operating and maintenance costs — once operational

- Toll collection expenses
- Administrative expenses for day-to-day operation
- Maintenance: routine and periodic
- Interest for servicing term loans

Interest appears in both: during construction it is capital cost; once the road is operational, interest for servicing term loans is an operating cost.

Annual operating cost covers operating the tollbooths, administration, emergency services, communications, security and other operating costs. **Maintenance** is annual (routine: the wearing surface, railings, roadside furniture; patching potholes, sealing cracks, edge repair,

surface renewal) or periodic (overlays or wearing coats, painting of railings) (SM 12.37–12.38).

Definition

Build-Operate-Transfer (BOT) (SM 12.38)

BOT is an option for the Government to outsource public projects to the private sector. With BOT, the private sector designs, finances, constructs and operates the facility and eventually, after a specified concession period, the ownership is transferred to the Government. Therefore, BOT can be seen as a developing technique for infrastructure projects by making them amenable to private sector participation.

The **toll rate** should have a direct relation with the benefits users gain from the road: fuel savings, improvement in travel time and good riding quality. The Study Material's formula, "with rounding off to nearest multiple of five": `User Fee = Total Distance × Toll Rate per km` (SM 12.38–12.39). The Study Material's own worked answers still show the exact toll (₹24.35 a vehicle in Illustration 10, SM 12.40; ₹19.06 for a two-wheeler in Practical Problem 1, SM 12.67), as does ICAI's September 2025 suggested answer: compute the exact rate first, and round only when the question asks.

Educational institutions

Schools, colleges and training institutes, "Not-for profit" or "For profit", need costing for cost-social benefit

analysis, allocation of funds and budgeting (zero-based budgeting) and performance evaluation (SM 12.41).

	Items (SM 12.41–12.42)
One-time fees	Admission fee, development fee, annual fee
Recurring fees	Tuition, laboratory, computer and internet, library, training, amenities, sports, extracurricular activities
Other incomes	Transport, hostel, mess and canteen, normally on a no-profit-no-loss basis
Operational costs	Teaching and non-teaching salaries; laboratory maintenance; computer maintenance and internet; building maintenance; repairs of equipment; administration; finance charges
Cost centres	Primary or direct (Civil, Mechanical Engineering departments); service (laboratory, library, sports); students' self-supporting services (transport, hostel and mess); administration (research and improvement, examination)

Government and aided institutes may not be permitted to collect fees such as capitation and development fees, and are not free to fix fees beyond a prescribed limit. Research costs are collected in a cost centre and set off against research revenue, any balance going to the course cost centres as "Research costs"; publication costs are allocated to the publication department (SM 12.41–12.42).

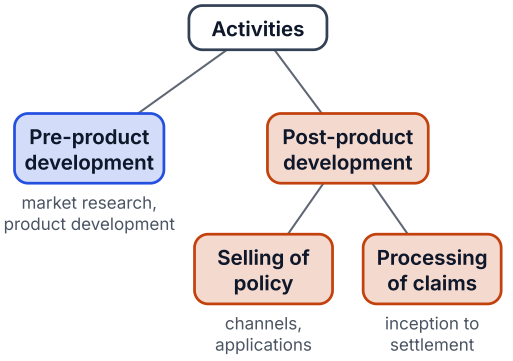
Training-batch fees are worked in the service-sector cost sheet.

Insurance companies

Insurers are life or non-life (general); their products include life policies with or without maturity benefits, general insurance (health, fire, property, travel) and other services (re-insurance, fund management) (SM 12.47). **Income**: premium; commission on re-insurance; fund administration fee and return on investment. **Expenditure**, direct to a policy or product: commission paid to agents, claim settlement, cost of valuation, premium for re-insurance, legal and other costs; indirect: actuarial fees, market and product development costs, administration, asset management (SM 12.48).

The cost object may be a product, a policy, a department or region, or an agent, and **activity based costing** is used for direct product profitability and customer (policy) profitability analysis (SM 12.48). Costs are traced to the activities that caused them, then reassigned to cost objects by cost drivers: hours spent on processing applications and claims, number of applications, policies, claims (SM 12.48–12.49).

Activities of an insurance company (SM 12.48–12.49)



Pre-product development covers market research and product development (coverage, conditions, premium, contract, policy forms, sales channel). Claims processing runs from claim inception and estimation to settlement and legal action.

ICAI's illustration groups a policy's costs into marketing and sales support, operations, IT and support functions, then finds the **cost per policy** and the **cost per rupee of insured value** (SM Illustration 12, 12.49–12.50).

Financial institutions

Manpower cost, other than interest and finance charges, is one of the largest single cost components (SM 12.51). The objectives of cost measurement (SM 12.51): understand profitability by products and by customers; establish a mechanism for pricing products by identifying product-level and activity-level unit costs; understand productivity issues and their relationship with strategic goals. ABC helps to identify profitability by

product and by customer and to build product-level costs from activity-level unit costs; profitability improves by concentrating on more profitable products and high-margin customers (SM 12.52). The cost of processing a loan application = (direct professional labour + the share of the loan department's overheads applicable to that work) ÷ applications processed (SM Illustration 13, 12.52–12.53).

Power houses

Power houses generating electricity or steam use "Powerhouse Costing"; the cost unit for electricity is the **kilowatt-hour (kWh)**, commonly known as a "Unit" (SM 12.53). The heads (SM 12.54):

Head	Items
Standing charges or fixed costs	Rent, rates and taxes; insurance; depreciation; salaries if paid on time (monthly) basis; administration expenses
Variable costs or running costs	Fuel charges; water charges; wages or labour charges if paid on the basis of production; any other variable costs
Semi-variable or maintenance costs	Meters; furnaces; service materials; tools

Formats and formulas

Reproduce the hotel working note and the hospital statement from memory.

Type and season	Occupied room-days	Weight	Equivalent room-days
Standard, season	rooms × % × days	1	xxx
Standard, off-season	rooms × % × days	½	xxx
Deluxe, season	rooms × % × days	w	xxx
Total			**xxx**

Statement of profitability (hospital)	Amount (₹)
Income (patient-days × rate)	xxx
Less: Variable costs (doctors paid per patient, food, medicines, laundry, bed hire)	xxx
Contribution	**xxx**
Less: Fixed costs (rent, salaries, fixed repairs, administration)	xxx
Profit	**xxx**

∑ **Formula**

Rent per room-day when charges and profit are on takings

$$R \times E = \text{Cost} + (c + p) \times R \times E$$

$$R = \frac{\text{Cost}}{E \times (1 - c - p)}$$

E = equivalent base room-days, R = rent of the base room-day, c = rent, commission or fee as a fraction of takings, p = profit as a fraction of takings (SM 12.25).

∑ **Formula**

Hospital break-even point

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{Contribution per patient-day}}$$

The BEP comes out in patient-days; ÷ (beds × days) gives it as a percentage of capacity.

∑ **Formula**

Toll rate

$$\text{Base toll} = \frac{\text{Total cost} + \text{Profit}}{\text{Equivalent vehicles}}$$

$$\text{User fee} = \text{Distance} \times \text{Rate per km}$$

Base toll: the toll of the vehicle type with weight 1. Equivalent vehicles: each type × its weight or toll ratio (SM 12.38–12.40, 12.66–12.67). User fee: the Study Material's formula, rounded off to the nearest multiple of five (SM 12.38–12.39); ICAI's worked answers show the exact toll, so round only when the question asks.

∑ **Formula**

Power, insurance and banks

$$\text{Cost per kWh} = \frac{\text{Total cost}}{\text{kWh generated}}$$

$$\text{Cost per policy} = \frac{\text{Total cost}}{\text{Policies sold}}$$

$$\text{Cost per ₹ insured} = \frac{\text{Total cost}}{\text{Insured value}}$$

$$\text{Per application} = \frac{\text{Labour} + \text{Overhead share}}{\text{Applications}}$$

Cost per ₹ insured = cost per rupee of insured value. Per application: direct professional labour plus the share of the department's overheads applicable to that work (SM 12.50, 12.53).

Worked examples

Worked example

Kulhad Inn, Pachmarhi: rooms and a banquet hall under one roof

Tapri's Kulhad Inn at Pachmarhi has 30 Standard rooms and 6 Family suites; a Family suite is let at twice the rent of a Standard room. The building also holds a banquet hall that is run as a separate business. Of the building's 12,000 sq ft, the rooms wing takes 9,000 sq ft and the banquet hall 3,000 sq ft. Occupancy for the year:

Standard rooms 75%, Family suites 50%. Take 360 days in a year.

Costs for the year	Amount
Building costs shared by the rooms and the banquet hall: depreciation ₹12,00,000; repairs ₹3,60,000; security and gardening ₹4,80,000; property tax and insurance ₹2,40,000	₹22,80,000
Rooms only: front-office and housekeeping salaries	₹15,00,000
Rooms only: depreciation of room furniture	₹3,00,000
Room attendants, paid on occupancy	₹150 per occupied room-day
Laundry and toiletries	₹100 per occupied room-day
Power for the rooms	₹150 per occupied room-day

A booking app takes a commission of 10% of room takings, and Tapri wants a profit of 20% of room takings.

Required:

(i) Calculate the rent per room-day of a Standard room and of a Family suite.

(ii) Kesar's first draft charged all the building costs to the rooms. Calculate the Standard rent it would give, and comment.

THE WORKING

Working note 1: room-days

Rooms	Occupied room-days	Weight	Equivalent Standard room-days
Standard	30 × 75% × 360 = 8,100	1	8,100
Family suite	6 × 50% × 360 = 1,080	2	2,160
Total	**9,180**		**10,260**

Working note 2: the rooms' share of building costs. The building costs are common to the two businesses and are apportioned on floor area: rooms 9,000 ÷ 12,000 = 75% of ₹22,80,000 = ₹17,10,000; the banquet hall bears the other ₹5,70,000.

Working note 3: statement of cost of the rooms for the year

Particulars	Amount (₹)
Building costs (75%, WN 2)	17,10,000
Front-office and housekeeping salaries	15,00,000
Depreciation of room furniture	3,00,000
Room attendants (9,180 × ₹150)	13,77,000
Laundry and toiletries (9,180 × ₹100)	9,18,000
Power (9,180 × ₹150)	13,77,000
Total cost	**71,82,000**

(i) Rent. Let R be the rent of a Standard room-day. Takings = 10,260R = ₹71,82,000 + (0.10 + 0.20) × 10,260R, so 7,182R = ₹71,82,000 and **R = ₹1,000**. Takings ₹1,02,60,000; commission ₹10,26,000; profit ₹20,52,000.

Rent per room-day (₹)	
Standard room	1,000
Family suite (× 2)	2,000

Check: 8,100 × ₹1,000 + 1,080 × ₹2,000 = ₹1,02,60,000. Costs paid per occupied room-day use the 9,180 room-days actually occupied; only the rent uses the 10,260 equivalent ones.

(ii) All building costs on the rooms. Cost = ₹71,82,000 + ₹5,70,000 = ₹77,52,000; takings = ₹77,52,000 ÷ 0.70 = ₹1,10,74,286; Standard rent = ₹1,10,74,286 ÷ 10,260 = **₹1,079.36**, and a Family suite ₹2,158.73. The rooms would carry ₹79.36 a room-day of the banquet hall's building costs, overpricing the rooms against nearby hotels and making the banquet hall look more profitable than it is. A cost common to two services is apportioned on a suitable basis (here floor area) before either is priced.

Worked example

Narmada Seva Trust: the dialysis centre and its third shift

Tapri's charitable trust runs a dialysis centre in a hired building in Jabalpur. It has 10 dialysis machines; each machine completes 2 sessions in a shift, the centre works 2 shifts a day for 26 days a month, and 90% of the session slots are used. The trust recovers ₹2,200 a session. Monthly data:

Particulars	Amount
Rent	₹1,20,000 a month

Particulars	Amount
Machines	10 at ₹9,00,000 each; life 5 years, straight line, time basis
Annual maintenance contract for the machines	₹50,000 a month
Technicians (3 a shift)	₹30,000 a month each
Nurses (2 a shift)	₹35,000 a month each
Administration expenses allocated	₹60,000 a month
Nephrologist, paid per session supervised	₹300 a session
Dialyser, tubing and needles	₹650 a session
Fluids and drugs	₹250 a session
Power and treated water	₹100 a session

Required:

(i) Calculate the cost per session and the profit per session for the month.

(ii) Find the break-even point in sessions a month, and as a percentage of the two-shift capacity.

(iii) A third shift would need 3 more technicians and 2 more nurses, and 60% of its slots are expected to be

used. Calculate its effect on the month's profit and the sessions it needs to break even on its own.

THE WORKING

Working note: sessions. Capacity = 10 machines × 2 sessions × 2 shifts × 26 days = 1,040 a month; sessions = 90% × 1,040 = **936**. Depreciation = 10 × ₹9,00,000 ÷ (5 × 12) = ₹1,50,000 a month.

Statement of profitability for the month

Particulars	Amount (₹)
Income (936 × ₹2,200)	20,59,200
Nephrologist (936 × ₹300)	2,80,800
Dialyser, tubing and needles (936 × ₹650)	6,08,400
Fluids and drugs (936 × ₹250)	2,34,000
Power and treated water (936 × ₹100)	93,600
Total variable costs (₹1,300 a session)	**12,16,800**
Contribution (₹900 a session)	**8,42,400**
Rent	1,20,000
Depreciation of machines	1,50,000
Annual maintenance contract	50,000

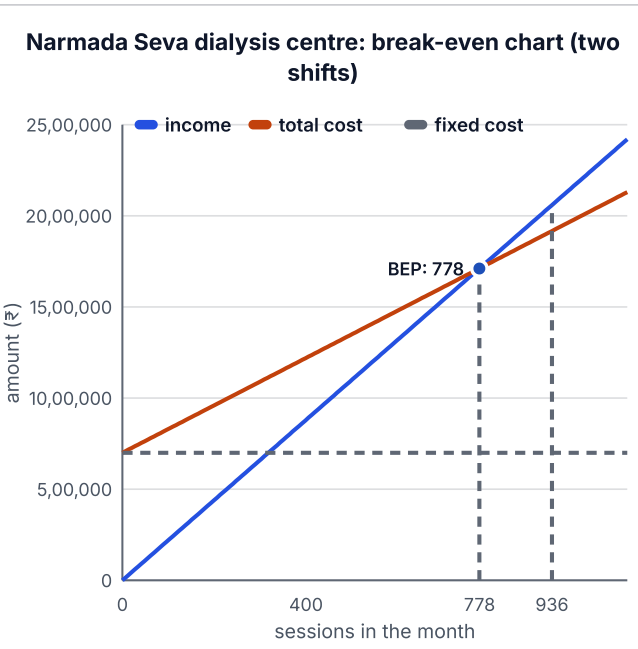
Particulars	Amount (₹)
Technicians (6 × ₹30,000)	1,80,000
Nurses (4 × ₹35,000)	1,40,000
Administration allocated	60,000
Total fixed costs	**7,00,000**
Profit	**1,42,400**

(i) Cost per session = ₹1,300 + ₹7,00,000 ÷ 936 = **₹2,047.86**; profit per session = ₹1,42,400 ÷ 936 = **₹152.14**.

(ii)

$$\text{BEP} = \frac{7{,}00{,}000}{900} = 777.8, \text{ say } 778 \text{ sessions}$$

$$\frac{777.8}{1{,}040} = 74.79\%$$



Blue: income at ₹2,200 a session; orange: total cost (₹7,00,000 fixed plus ₹1,300 a session). They cross at about 778 sessions; the month's 936 leave a margin of safety of 158 sessions.

(iii) A third shift. Capacity 10 × 2 × 26 = 520 sessions; 60% used = 312 sessions. Extra contribution 312 × ₹900 = ₹2,80,800; extra fixed cost 3 × ₹30,000 + 2 × ₹35,000 = ₹1,60,000; extra profit **₹1,20,800**, so the month's profit rises to ₹2,63,200. The rent, machines, maintenance contract and administration do not change: the shift's own break-even is ₹1,60,000 ÷ ₹900 = 177.8, say **178 sessions** (34.19% of its 520 slots). Staff hired per shift are a stepped fixed cost: fixed while

the shift runs, added in a block when a shift is opened.

Worked example **Malwa Expressway: tolls by the kilometre**

Malwa Expressway Ltd, in which Tapri holds a stake, operates an 80 km access-controlled highway near Indore under BOT. Vehicles take a ticket where they enter and pay by the kilometre where they leave. The expected traffic for a year:

Vehicle	Trips a year	Average distance a trip (km)	Toll per km relative to a car
Cars and jeeps	30,00,000	50	1
Buses and LCVs	6,00,000	70	1.6
Trucks	4,00,000	80	3

Annual costs: toll operations and administration ₹4.80 crore; routine maintenance ₹6.00 crore; periodic maintenance (overlays, painting of railings) ₹3.60 crore; interest for servicing term loans ₹18.24 crore. The project cost of ₹720 crore is to be recovered evenly

over the 24-year concession period. The company wants a profit of 25% on total cost.

Required:

(i) Calculate the total annual cost and the takings required.

(ii) Calculate the toll per km for each type of vehicle.

(iii) Calculate the toll each type pays for the full 80 km and for a 46 km trip, rounded to the nearest multiple of ₹5 (user fee = distance × toll rate per km).

THE WORKING

(i) Annual cost and takings

Particulars	Amount (₹ crore)
Toll operations and administration	4.80
Routine maintenance	6.00
Periodic maintenance	3.60
Interest for servicing term loans	18.24
Recovery of project cost (₹720 crore ÷ 24 years)	30.00
Total annual cost	**62.64**
Profit (25% on total cost)	15.66
Takings required	**78.30**

(ii) Working note: equivalent car-km. Each type's vehicle-km × its weight:

Vehicle	Vehicle-km	Weight	Equivalent car-km
Cars and jeeps	30,00,000 × 50 = 15,00,00,000	1	15,00,00,000
Buses and LCVs	6,00,000 × 70 = 4,20,00,000	1.6	6,72,00,000
Trucks	4,00,000 × 80 = 3,20,00,000	3	9,60,00,000
Total			**31,32,00,000**

Car toll per km = ₹78,30,00,000 ÷ 31,32,00,000 = **₹2.50**; buses and LCVs ₹2.50 × 1.6 = **₹4.00**; trucks ₹2.50 × 3 = **₹7.50** per km.

Check: 15,00,00,000 × ₹2.50 + 4,20,00,000 × ₹4.00 + 3,20,00,000 × ₹7.50 = ₹37.50 crore + ₹16.80 crore + ₹24.00 crore = ₹78.30 crore.

(iii) User fee

Vehicle	80 km (₹)	46 km, exact (₹)	46 km, rounded (₹)
Car or jeep	200	115.00	115
Bus or LCV	320	184.00	185
Truck	600	345.00	345

The weights are applied to vehicle-km, not to trips: a truck pays three times a car for each kilometre, and trucks also travel further on average. Pricing per trip at the average distance of all vehicles would overcharge short car trips and undercharge long truck trips.

Exam angle

In the exam

Where these industries appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, these industries came up in 13 documents (12 of the new scheme, 1 of the old):

- **Hotels, 2 to 7 marks**: hill-station hotel with off-season half rent and lighting by months (MTP Series I for January 2026 Q5(b), 7 marks); tariff per room-day and break-even occupancy in the normal season

(MTP Series II for May 2024 Q5(b), 6 marks); Premium rooms at double the Deluxe rent, profit 20% of revenue (RTP September 2024 MCQ 5 and MTP Series II for September 2026 MCQ 12); single and double rooms, attendants ₹125 per room-day, profit 25% on takings (RTP May 2025 Q16).

- **Toll roads, 5 to 10 marks**: monthly cost of a plaza and tolls for three vehicle types in a ratio, 20% profit over takings (September 2025 Q2(a), 10 marks); "Explain BOT and classify six expenses" into capital or operating and maintenance cost (September 2024 Q4(b), 5 marks); a BOT highway's project cost per day of the concession period and tolls by vehicle weights (RTP September 2024 Q16).
- **Banks, 2 to 8 marks**: gold-loan and car-loan applications, solving for the salary that equalises their cost (January 2026 Q3(a), 8 marks); home-loan and education-loan applications (MTP Series I for September 2025 MCQ 12; MTP Series II for January 2025 Q3(c), 2 marks).
- **Insurance, MCQs**: cost per rupee of insured value (RTP September 2024 MCQ 6; MTP Series I for May 2026 MCQ 13); a policy's activity costs (MTP Series I for January 2025 MCQ 14).
- **Hospitals**: no numerical in the new-scheme documents read; the old-scheme January 2021 paper Q5(a) (10 marks) asked profit per patient-day and the break-even patient-days, as SM Illustration 8 does. Hospitals, hotels and electricity also appear in the cost-units table questions (composite cost units).

- **IT and power**: no numerical in the documents read; power appears only as "kWh" in cost-units questions.
- **Education and training, 6 marks**: an air-hostess training institute's cost per batch, minimum fee per trainee and minimum batch size (January 2026 Q3(b)); this pattern is worked in the service-sector cost sheet.

In the exam

The traps ICAI set

- **A cost traceable to one service is not apportioned.** In MTP Series II for January 2025 Q3(c), ₹11,000 of legal advice for an education loan was charged in full, and only 30% of the rest of the department's overheads.
- **Interest**: capital cost during construction, an operating cost once the road runs (September 2024 Q4(b)).
- **"20% profit margin over total takings"** means takings = cost ÷ 0.80, not cost × 1.20 (September 2025 Q2(a)).
- **Off-season half rent** is a weight of ½ on the off-season room-days, not half the occupancy.

Trick

The equivalent-unit shortcut for any grading

Whenever prices of grades stand in fixed ratios (rooms, seasons, vehicle classes, seat classes), total takings ÷ Σ(units × ratio) gives the base price in one line. Condition: the ratios are of prices (rents, tolls, fees), and the units are those actually sold.

Answer frame

"Explain the BOT approach and classify these expenses"

1. **BOT**: an option for the Government to outsource public projects to the private sector; the private sector designs, finances, constructs and operates the facility and, after the specified concession period, ownership is transferred to the Government. Therefore, BOT can be seen as a developing technique for infrastructure projects by making them amenable to private sector participation (ICAI's September 2024 answer gives exactly this paragraph for the 2 marks).
2. **A two-column table**, the expenses in the question's order, each against its class (SM 12.37–12.38; September 2024 Q4(b)):

Expense	Classification
Preliminary and pre-operative expenses; land acquisition	Capital cost
Materials; labour; overheads of actual construction	Capital cost
Contingency allowance; interest during construction period	Capital cost
Toll collection expenses; day-to-day administration	Operating and maintenance cost
Routine and periodic maintenance (overlays, painting of railings)	Operating and maintenance cost
Interest for servicing term loans	Operating and maintenance cost

Short notes on any one industry

Use the Study Material's skeleton for that industry: what the service is; its **cost unit**; its **income** heads (education, insurance); its **cost heads** or segregation (fixed and variable for hospitals; standing, running and maintenance for power; capital and O&M for toll roads; direct and indirect for insurers); the **method** (projects and effort for IT; ABC for insurers and banks; cost centres for schools). The numerical answer runs: units actually delivered (working note) → statement of cost → takings → rate per unit.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the inn that was always full

Bunty divides Kulhad Inn's room takings of ₹1,02,60,000 by 36 rooms × 360 days = 12,960 room-days and gets a rent of ₹791.67 for every room on every day. Occupancy was 75% and 50%, and a Family suite pays double. **Fix:** room-days occupied, then weights: 10,260 equivalent room-days and ₹1,000.

⚠ **Common mistake**

Per-room costs on equivalent room-days

Charging room attendants on the 10,260 equivalent room-days. **Fix:** costs paid per occupied room-day use the 9,180 room-days actually occupied; equivalents are only for sharing out the rent.

⚠ **Common mistake**

Doctors on a per-patient fee shown as fixed

In the Narmada Seva dialysis centre, moving the nephrologist's ₹2,80,800 for the month into fixed costs gives a contribution of ₹1,200 a session and a break-even point of about 818 sessions instead of 778. **Fix:** pay that varies with patients attended or sessions supervised is variable (SM Illustration 8).

⚠ **Common mistake**

The training course charged to one project

Java training useful to many projects charged in full to the project in hand. **Fix:** travel and training incurred for a project are direct; anything useful across projects is service overhead (SM 12.34).

Practice

Check yourself

1 Tapri's captive power plant generated 12,00,000 kWh in the year, but 5% of it was used by the plant itself (auxiliary consumption), so only the rest reached the samosa factory. Costs: supervision, administration and depreciation ₹30,00,000; operating staff on fixed shifts ₹18,00,000; repairs ₹5,40,000; biomass fuel ₹1.25 per kWh generated. What is the cost per kWh delivered to the factory, in rupees?

Answer: _____

2 Case scenario. Kavach Life Ltd sells two term plans to Tapri's people. The Rider plan cost ₹1,20,00,000 for the year and sold 500 policies with a total insured value of ₹600 crore. The Café plan cost ₹90,00,000 and sold 600 policies insured for ₹300 crore in all. Dev compares the two plans. Which statement is correct?

- **A** The Rider plan costs less per policy and less per rupee of insured value
- **B** The Rider plan costs less per policy but more per rupee of insured value
- **C** The Rider plan costs more per policy but less per rupee of insured value
- **D** The Café plan costs less on both measures

3 A bank's loan processors are paid ₹8,00,000 a month; their timesheets show 60% of their hours on home-loan applications and 40% on car-loan applications. The department's other overheads, ₹5,00,000 a month, are shared in the ratio of applications processed. In the month 300 home-loan and 700 car-loan applications were processed. What is the cost per home-loan application, in rupees?

Answer: _____

4 Which of these are capital costs of a toll road project? Select all that apply.

- **A** Land acquisition
- **B** Contingency allowance
- **C** Interest during the construction period
- **D** Toll collection expenses
- **E** Periodic painting of railings
- **F** Interest for servicing term loans after the road opens

More than one answer may be right.

5 In an IT company, Java training given to software engineers that will be useful in several Java projects is treated as:

- **A** A direct cost of the project running when the training was given
- **B** Service overhead, apportioned over projects
- **C** Effort cost of the trainees
- **D** A capital cost

Answers: the answer key at the end of this lesson.

Recap

- Hotels: room-day or guest-day; occupancy by season; weights for room types and off-season rent; rent = cost ÷ (equivalent room-days × (1 − c − p)).
- Hospitals: OP per out-patient, IP per room-day, scanning per case, laundry per 100 items; fixed vs variable; BEP in patient-days.
- IT and ITES: projects; effort estimation; direct vs support manpower; hardware, travel and training direct if identifiable, else service overhead; effort cost = salary + part of common overhead.
- Toll roads: capital costs (land, materials, labour, construction overheads, contingency, interest during construction) vs O&M (toll collection, administration, routine and periodic maintenance, interest for servicing term loans); BOT; user fee = distance × rate per km, rounded to the nearest multiple of five when the question asks.
- Education: one-time, recurring and other income; four kinds of cost centre.
- Insurance: ABC with pre- and post-product development activities; cost per policy and per rupee of insured value.
- Banks: cost per application = (direct labour + share of overheads applicable) ÷ applications processed.
- Power: kWh; standing (rent, insurance, depreciation, time salaries, administration), running (fuel, water,

production wages), maintenance (meters, furnaces, tools). Same recipe, different katori.

Answer key · Check yourself

1 **6**

Fuel = 12,00,000 × ₹1.25 = ₹15,00,000; total = ₹30,00,000 + ₹18,00,000 + ₹5,40,000 + ₹15,00,000 = ₹68,40,000. kWh delivered = 12,00,000 × 95% = 11,40,000; ₹68,40,000 ÷ 11,40,000 = ₹6.00. Dividing by the kWh generated (₹5.70) understates what the factory's power really costs.

2 **C** **The Rider plan costs more per policy but less per rupee of insured value**

Per policy: ₹1,20,00,000 ÷ 500 = ₹24,000 against ₹90,00,000 ÷ 600 = ₹15,000. Per rupee of insured value: ₹1.2 crore ÷ ₹600 crore = ₹0.002 against ₹0.9 crore ÷ ₹300 crore = ₹0.003. The two cost units rank the plans differently, so state which one is asked (SM 12.50).

3 **2100**

Salaries: 60% × ₹8,00,000 = ₹4,80,000. Overheads: ₹5,00,000 × 300 ÷ 1,000 = ₹1,50,000. (₹4,80,000 + ₹1,50,000) ÷ 300 = ₹2,100. Using 60% for the overheads as well gives ₹2,600; sharing the salaries by applications gives ₹1,300: each cost goes on its own basis.

4 **A Land acquisition; B Contingency allowance; C Interest during the construction period**

SM 12.37–12.38: capital costs are incurred during construction; toll collection, periodic maintenance and interest for servicing term loans are operating and maintenance costs.

5 **B Service overhead, apportioned over projects**

SM 12.34: travel and training incurred for a project are allocated to it; when not identifiable with one project, they are service overhead.

CHAPTER 13

Standard Costing

13.1 Standards, their types and standard costing

What a standard cost is, the four types of standards, how quantity and price standards are set, controllable and favourable variances, and the advantages and criticism of standard costing.

In plain words

The ₹27,000 question

Tapri's Pune café sends Kesar its October bills.

> **Kesar:** ₹2,07,000 on milk this month. Is that good or bad?
>
> **Ritu:** Per unit, please. The café sold 20,000 cups. Our recipe card says 150 ml of milk a cup at ₹60 a litre: ₹9 a cup, ₹1,80,000 for 20,000 cups.
>
> **Kesar:** So we are ₹27,000 over. Who drank it?
>
> **Ritu:** That is exactly the question a standard is for.

Ritu's ₹9 is a **standard cost**: what a cup *should* cost, worked out in advance from a quantity (150 ml) and a price (₹60 a litre). With it, the bill splits into what 20,000 cups should have cost and a **variance** of ₹27,000 to be explained. The milk price was exactly the standard, so

the extra 450 litres went into cups: the new barista's generous pour.

Standard costing is the routine around that idea: set standards, record actual costs, compare, investigate the differences, act. A manager cannot read every bill; with standards, only the significant variances reach the desk. That is **management by exception**.

How strict should it be? A standard for the best barista on her best day is never met; one copied from last year's average approves last year's waste.

Think of it like

The map app's arrival time

The map app tells you when you should arrive: a standard. Arrive late and you ask why: a jam on the highway (not your fault) or a wrong turn (yours). The time for empty roads at 3 a.m. is an ideal standard: nobody makes it at 6 p.m. Today's time, with today's traffic, is a current standard. Your first year's time, kept unchanged to see how much worse traffic gets each year, is a basic standard.

Key idea

A standard cost is a planned cost, set in advance in quantities and prices, for the actual output. Standard costing compares actual cost with it, reports the variances and investigates the significant ones. The standard that controls cost best is one that good, efficient work can attain.

How it works

Two definitions to write word for word

Definition

Standard costing

"Control technique that reports variances by comparing actual costs to pre-set standards so facilitating action through management by exception." (CIMA Official Terminology, SM 13.2)

Definition

Standard cost

"The planned unit cost of the product, component or service produced in a period. The standard cost may be determined on a number of bases. The main use of standard costs is in performance measurement, control, stock valuation and in the establishment of selling prices." (CIMA Official Terminology, SM 13.2) So a standard cost is a **planned cost**, **determined on a base or number of bases**.

Standards are established to evaluate the performance of a **responsibility centre**. They are also used to value inventory where actual figures are not reliably available, and to determine selling prices, particularly for quotations (SM 13.2). Standard costing is preferred for these reasons (SM 13.2–13.3):

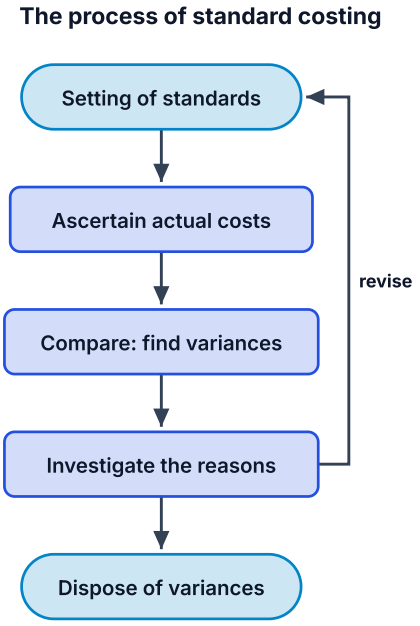
1. **Prediction of future cost for decision making**: set after considering present conditions and future possibilities, a standard cost is the future cost for estimating a proposed project, order or activity.
2. **Provide target to be achieved**: a cost the responsibility centre should not cross; performance is measured against it and variances are reported for action.
3. **Used in budgeting and performance evaluation**: budgets are set from standard costs, so managers do not compromise quality to fulfil the budgeted

quantity, and variances can be traced to the responsible department or person.

4. **Interim profit measurement and inventory valuation**: for interim reports, before the accounts are closed, standard costs are deducted from revenue to arrive at the profit.

The process of standard costing (SM 13.5)

1. **Setting of standards** to be achieved.
2. **Ascertainment of actual costs** (books, invoices, wage sheets).
3. **Comparison of actual cost with standard cost**: variances are determined.
4. **Investigate the reasons for variances**; evaluate performance and act.
5. **Disposition of variances**: transferred to the relevant accounts (the Costing Profit and Loss Account) as per the accounting method adopted.



The five steps of SM 13.5. The loop: significant, persistent uncontrollable variances mean the standard itself may need revision (SM 13.12).

Four types of standards (SM 13.3–13.4)

1. **Ideal standards**: the level of performance attainable when prices for material and labour are most favourable, when the highest output is achieved with the best equipment and layout, and when maximum efficiency in the utilisation of resources results in maximum output with minimum cost. They are criticised on three grounds: (a) being unattainable, no one would take them seriously; (b) the variances disclosed, being variances from ideal standards,

would not indicate the extent to which they could have been reasonably and practically avoided; (c) there would be no logical method of disposing of these variances.

2. **Normal standards**: standards that may be achieved under normal operating conditions. Normal activity is "the number of standard hours which will produce at normal efficiency sufficient good to meet the average sales demand over a term of years". They are difficult to set because they require a degree of forecasting. Their variances are deviations from normal efficiency, normal sales volume or normal production volume; if actual performance is abnormal, large variances may result and necessitate revision of standards.
3. **Basic or bogey standards**: used only when they are likely to remain constant over a long period; a base year is chosen for comparison, as statisticians use price indices. They are set on a long-term basis, seldom revised, and well suited to businesses with a small range of products and long production runs. As they do not represent what should be attained in the present period, current standards should also be prepared. Variances are not calculated: the actual cost is expressed as a percentage of basic cost, the current cost likewise, and the two percentages are compared, then compared with those of previous periods to establish the trend.
4. **Current standards**: management's anticipation of what actual costs will be for the current period: the costs the business will incur if the anticipated prices

are paid and the usage corresponds to that believed necessary to produce the planned output. Their variances represent the degree of efficiency in usage of the factors of production, variation in prices paid for materials and services, and difference in the volume of production.

The standard most effective in the control of costs represents an attainable level of good performance (SM 13.52). In the Study Material's MCQs, the standard "attainable under favourable conditions" is the **theoretical standard** and the one "most suitable from cost control point of view" the **expected standard** (SM 13.55, key 13.67); its current-standards paragraph speaks of variances "arising from expected standards" (SM 13.4).

Setting up standard costs (SM 13.5–13.11)

Standard cost is set on the basis of management's estimation, from the technical specifications of the engineering department or another expert such as the production engineer, considering historical data, the current production plan and expected future conditions. Standards are set for each element of cost (material, labour, variable and fixed overheads) and for sales quantity and value (budgeted sales), **in both physical and monetary terms**.

PHYSICAL STANDARDS (SM 13.6–13.9)

Standard quantities and hours, to secure economies in scale of production and to set selling prices for

quotations. The industrial engineering department sets them, considering:

- the company's operating plan, i.e. budgets;
- the final output to be produced;
- material specification, in quantity and quality, provided by the engineering department;
- the proportion of materials to be used where there are multiple inputs;
- the method of production: fully automated, semi-automated or manual;
- the skill set and availability of workers;
- working conditions and internal factors;
- external factors (Labour Law, Factories Act, Government policy).

Procedure of setting material quantity standards (SM 13.6–13.7):

1. **Standardisation of products**: what to produce, which type, and how much, from the production plan and customers' orders.
2. **Product study**: by the engineering department or product consultants: how to produce it, the pre-requisites, which materials, market acceptance.
3. **Preparation of specification list**: types (quality), quantities, substitutes and proportions of materials, the process and conditions, allowing for expected wastage.

4. **Test runs**: sample runs under specified conditions; deviations from the specification are noted and the list updated.

Procedure of setting labour time standards (SM 13.7–13.8):

1. **Standardisation of product and product study**, as for material.
2. **Labour specification**: the types of labour and the labour time, based on past records, allowing for normal wastage of time.
3. **Standardisation of methods**: selection of proper machines, and the proper sequence and method of operations.
4. **Manufacturing layout**: a plan of operations for each product.
5. **Time and motion study**: the best way to do the job and the standard time an average worker takes, allowing for the learning effect.
6. **Training and trial**: workers are trained; the time in the trial run is noted.

Overheads time or quantity standards (SM 13.8): variable overheads may be based on direct material quantity or labour hours, generally labour time worked; fixed overhead time is based on budgeted production volume.

Problems faced while setting physical standards (SM 13.8–13.9):

1. A new product line needing inexperienced workers: the standard time must allow for inexperience.
2. New technology and sophisticated machines: output and efficiency cannot be estimated precisely until conditions settle.
3. Product diversification: new tools (a copper part now made in bronze) need new standard times.
4. Materials not available as per specification: the standards may not be achievable.
5. The choice of the type of standard: maximum efficiency, attainable good performance or average past performance; likewise for waste standards, unavoidable wastage only or the past average (the first worked example below).

PRICE OR RATE STANDARDS (SM 13.9–13.11)

Set on (a) the actual average or mean price expected during the coming period, say one year, or (b) the normal prices expected during a cycle of seasons, possibly several years.

Material price standards (purchasing department), if price fluctuations are small, are based on:

- (a) the stock of materials on hand and the prices at which it is held;
- (b) the prices at which orders for future deliveries have already been placed;

- (c) the minimum support price fixed by the appropriate authority;
- (d) anticipated fluctuation in price levels.

For unsystematic fluctuations:

- (a) month to month: a year's average price, corrected for known secular changes;
- (b) seasonal: the weighted average of the likely prices if the year's requirement is bought at one time, or of the whole year's prices if buying is spread;
- (c) year to year: an estimate of next year's price from a statistical study.

Wage rate standards: the type of labour is the most important factor; rates are set on the time taken per unit of production, the time or piece rate prevailing in the industry, the wage agreement with the workers' union, and the law of the area (minimum wages, bonus).

Overhead expense standards: a budgeted output (from practical capacity and anticipated sales demand), fixed and variable budgets for it, and a flexible budget if production is seasonal or fluctuates.

Types and classification of variances (SM 13.11–13.13)

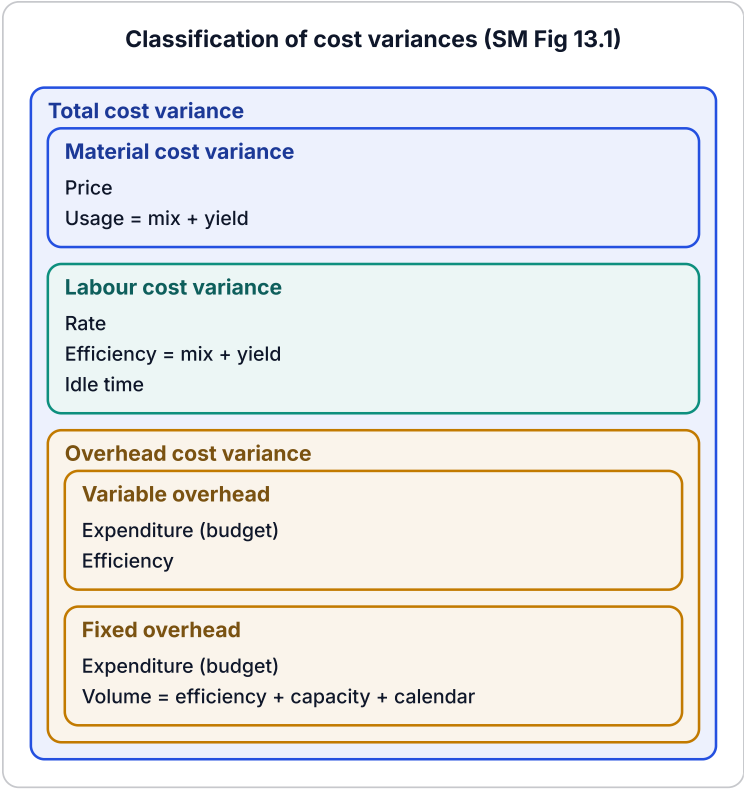
Controllable variances can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently; **uncontrollable variances** occur due to conditions beyond its control and

cannot be controlled even with all preventive measures in place. Responsibility centres are answerable for all adverse variances which could have been controlled. Controllability is a subjective matter and varies from situation to situation; if uncontrollable variances are significant and persistent, the standard may need revision.

Favourable variances are profitable to the company, **adverse** ones cause loss. For costs, favourable means actual below standard; for sales, actual above budget. They are written F and A, whatever the mathematical sign.

Variances are broadly classified into **revenue variances** (actual sales against budgeted sales) and **cost variances**.

The cost variances, as in the Study Material's Fig 13.1 (SM 13.13), each with its own lesson:



See material variances, labour variances, variable overhead variances and fixed overhead variances.

Formats and formulas

ICAI's numerical questions open with a **standard cost card** (September 2024 Q4(a): "The standard cost card per pack is as follows"). *Reproduce the layout from memory.*

Standard cost card: one tray of 50 frozen samosas	Quantity × price	Amount (₹)
Direct material: dough and filling	2.5 kg × ₹80 per kg	200
Direct labour	0.25 hour × ₹160 per hour	40
Variable overhead	0.25 hour × ₹40 per hour	10
Fixed overhead	0.25 hour × ₹80 per hour	20
Standard cost per tray		**270**

∑ **Formula**

Standard cost of actual output and the variance

Standard cost of actual output = standard cost per unit × actual output.

Cost variance = standard cost of actual output − actual cost.

Positive: F; negative: A. For 6,000 trays: 6,000 × ₹270 = ₹16,20,000, whatever the budget said.

∑ **Formula**

Reading a basic standard

$$\frac{\text{Actual cost}}{\text{Basic cost}} \times 100$$

The current standard cost is expressed the same way; the two percentages are compared with each other and with earlier periods (SM 13.4).

ICAI's glossary (SM 13.52–13.53)

The chapter summary's definitions, in its order:

Term	ICAI's definition
Standard costing	A technique which uses standards for costs and revenues for the purposes of control through variance analysis
Standard price	A predetermined price fixed on the basis of a specification of a product or service and of all factors affecting that price
Standard time	The total time in which a task should be completed at standard performance
Variance	A divergence from the predetermined rates, expressed ultimately in money value, generally used in standard costing and budgetary control systems
Variance analysis	The analysis of variances arising in a standard costing system into their constituent parts

Term	ICAI's definition
Revision variance	The difference between the original standard cost and the revised standard cost of actual production
Basic standard	A standard fixed for a fairly long period
Current standard	A standard fixed for a short period
Estimated cost	An estimate of what the cost is likely to be during a given period of time
Ideal cost	A cost which should be incurred during a period under ideal conditions

Worked examples

Worked example **Three wastage allowances for one samosa dough**

Every 1,000 samosas from Tapri's Indore factory hold 40 kg of finished dough. Kneading, sheeting and trimming waste some dough, and the production head offers Ritu three standards of dough input per 1,000 samosas:

1. Only the wastage that cannot be avoided with the best sheeter and a fully trained team: 41 kg.
2. The wastage a trained team achieves with good effort on the present sheeter: 42 kg.

3. The average wastage of the last three years: 44 kg.

Dough costs ₹60 per kg, and November's price paid was the same. In November the factory made 2,50,000 samosas and used 10,600 kg of dough.

Required:

(i) Calculate the standard dough cost per samosa on each basis.

(ii) Calculate November's material usage variance on each basis, indicating whether it is favourable (F) or adverse (A).

(iii) Kesar prices a canteen contract for 5,00,000 samosas at the standard dough cost plus ₹3.50 a samosa for all other costs and profit. Show the price per samosa on each basis, and what each price would lose or leave on the table if November's dough usage repeats.

(iv) Advise Ritu which basis to adopt.

THE WORKING

(i) and (ii) Standard quantity for 2,50,000 samosas = 250 × kg per 1,000. Actual dough cost = 10,600 × ₹60 = ₹6,36,000, or ₹2.544 a samosa. The price paid equals the standard, so the whole variance is a usage variance = (SQ − AQ) × SP.

Basis	SQ (kg)	Standard cost (₹)	Per samosa (₹)	Usage variance (₹)
Unavoidable wastage only	10,250	6,15,000	2.46	21,000 (A)
Attainable good performance	10,500	6,30,000	2.52	6,000 (A)
Past average wastage	11,000	6,60,000	2.64	24,000 (F)

(iii) The canteen contract

Basis	Price (₹)	Actual vs standard (₹)	On 5,00,000 (₹)
Unavoidable wastage only	5.96	0.084 over	42,000 lost
Attainable good performance	6.02	0.024 over	12,000 lost
Past average wastage	6.14	0.096 under	48,000 extra

At November's usage the dough actually costs ₹2.544 a samosa. "Lost" is dough cost the price does not recover, which comes out of the ₹3.50 meant for other costs and profit; "extra" is cost built into the price that November's kitchen did not incur.

(iv) Advice

- **Unavoidable wastage only (an ideal standard)**: the lowest cost for pricing (₹2.46), but a large adverse variance whenever that wastage is missed, and a contract priced on it under-recovers ₹42,000; it does show how far performance is off the best (SM 13.9).
- **Attainable good performance**: the variance is small (₹6,000 (A)), close to nil as the Study Material expects of this basis (SM 13.9), and it is the type of standard most effective in the control of costs (SM 13.52).
- **Past average wastage**: the ₹24,000 "favourable" variance only says November beat a past average that includes past waste, and the ₹6.14 price may lose the contract to a rival.

Ritu should adopt **attainable good performance**: 42 kg of dough per 1,000 samosas.

Worked example **Kesar's perfect samosa standard**

Kesar insisted the samosa factory's standards be "the best we have ever done: best prices, best machines, zero waste", and wants the production head to explain every adverse variance. November's report shows:

1. Material price variance ₹1,20,000 (A): the purchase department bought potatoes in small lots at the last

minute instead of under the annual contract.

2. Labour rate variance ₹45,000 (A): the State Government revised statutory minimum wages from 1 November.
3. Labour efficiency variance ₹38,000 (A): workers hired for the Diwali rush had not been trained.
4. Idle time variance ₹22,000 (A): a flood in the industrial area cut power for a day.

Required:

(i) Comment on Kesar's choice of standards.

(ii) Classify each variance as controllable or uncontrollable and state who is responsible.

(iii) Advise whether any standard should be revised.

MODEL ANSWER

(i) Kesar has set **ideal standards**: the level of performance attainable when prices for material and labour are most favourable, the highest output is achieved with the best equipment and layout, and maximum efficiency in the utilisation of resources results in maximum output with minimum cost. Such standards are criticised because (a) being unattainable, no one takes them seriously; (b) the variances do not indicate the extent to which they could have been reasonably and practically avoided; and (c) there is no logical method of disposing of them. The type of standard most effective in the control of costs represents an

attainable level of good performance, and standards should be set on that basis.

(ii) Controllable variances can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently; uncontrollable variances occur due to conditions beyond its control. Responsibility centres are answerable for all adverse variances which could have been controlled.

Variance
Material price ₹1,20,000 (A)

Nature
Controllable: the buying method is within the purchase department's control

Responsibility
Purchase department

Variance
Labour rate ₹45,000 (A)

Nature
Uncontrollable: a minimum wage fixed by law, beyond the organisation's control

Responsibility
Not the production head's; the personnel manager negotiates rates, not statutory minimums

Variance
Labour efficiency ₹38,000 (A)

Nature
Controllable: an untrained team of workers

Responsibility
Production manager or foreman

Variance
Idle time ₹22,000 (A)

Nature
Uncontrollable: abnormal idle time from a flood

Responsibility
None; shown separately as abnormal idle time

The production head is answerable only for the efficiency variance.

(iii) If uncontrollable variances are significant and persistent, the standard may need revision. The higher minimum wage is permanent, and wage rate standards are set with regard to the law of the area, so the wage rate standard should be revised; the flood is a one-time event. The ideal standards should be replaced by standards of attainable good performance.

Exam angle

In the exam

Where this theory appeared

In the ICAI documents read for this lesson, the theory of standards and standard costing was asked in 8: 5 new-scheme (May 2024 to September 2026) and 3 old-scheme.

- **"Calculation of variances in standard costing is not an end in itself, but a means to an end." DISCUSS**: MTP January 2025 Series II Q2(b) (4 marks), MTP

September 2025 Series I Q6(c) (4 marks), MTP May 2026 Series II Q4(b) (5 marks).

- **Controllable and uncontrollable variances**: September 2024 Q6(b) (two of five definitions, 5 marks); RTP September 2024 Q19(c); old-scheme MTP July 2021 Series I Q6(c).
- **Old scheme, 5 marks each**: steps in setting labour time standards (December 2021 Q6(c)); criticism of standard costing (May 2022 Q6(d)).
- The Study Material's own questions: the process of setting standards, the types of standards, how the material usage standard is set (SM 13.57).

In the exam

What the marks were given for

The suggested answers reproduce the Study Material's lists in its order and words (September 2024 Q6(b) quotes SM 13.11–13.12); May 2022 gives each criticism with its answer.

Trick

Ideal, Normal, Basic, Current: "I Never Buy Chai"

The four types of standards in the Study Material's order: **I**deal, **N**ormal, **B**asic (bogey), **C**urrent.

Answer frame

The Study Material's own theory questions (SM 13.57)

- **"Discuss the types of standards"**: the four types in ICAI's order (ideal, normal, basic, current), each with its meaning and the points listed under it above; end with the type most effective for cost control.
- **"Discuss the process of setting standards"**: (1) set on management's estimation, for each element of cost, in physical and monetary terms; (2) physical standards: the factors considered, the material quantity procedure (four steps), the labour time procedure (six steps), the overhead time standards, and the problems faced; (3) price or rate standards: the two bases, material price standards, wage rate standards, overhead expense standards.
- **"How is the material usage standard set?"**: the four steps of the material quantity procedure, each with a line of explanation.
- **"Discuss the process of standard costing"**: the five steps of SM 13.5, each with a line of explanation.

"Distinguish between controllable and uncontrollable variances"

Point	Controllable	Uncontrollable
Meaning	Can be controlled under normal operating conditions if the responsibility centre takes preventive measures and acts prudently	Due to conditions beyond the responsibility centre's control; cannot be controlled even with all preventive measures
Responsibility	Answerable for adverse variances	Not answerable
Standard	Corrective action on performance	If significant and persistent, the standard may need revision

"Calculation of variances is not an end in itself, but a means to an end"

ICAI's MTP answer, in order: standard costing stresses what the cost should be; the variance may be favourable or unfavourable; knowing the figures is not enough, their origin and causes must be traced; a detailed probe,

particularly into controllable variances, helps management ascertain (i) the amount of the variance, (ii) the factors or causes of its occurrence, (iii) the responsibility to be laid on executives and departments, and (iv) the corrective actions to obviate or reduce it; success depends on how quickly and effectively corrective action is taken; information is the means and the action taken on it is the end.

Advantages of standard costing (SM 13.49–13.51)

1. A basis for **measuring operating performance and cost control**: only exceptional variances are reported, a signal for prompt action.
2. It **aids price fixing**: stable standard costs are a basis for the selling price.
3. It facilitates **evaluation of jobs and introduction of incentives**.
4. It facilitates the **estimation of the cost of new products** with greater accuracy.
5. A basis for **inventory valuation**; material stock can be recorded in quantities only.
6. **Measurement of profit**: it eliminates variations in profit due to changes in stock values from one period to another.
7. Use in **planning, budgeting and decision making**.
8. **Standardisation of products, operations and processes**.
9. **Objectives and targets for each level of management**, defining managers' responsibilities.

10. A **uniform basis for comparison**; the standard hour compares dissimilar products.
11. **Maximum use of working capital, plant facilities and current assets**, as wastage and idle time are closely controlled.

"Discuss the criticism of standard costing" (SM 13.51–13.52)

The Study Material answers each criticism but the last, so they do not invalidate the usefulness of the system (SM 13.51). Write both halves:

Criticism	Answer
(i) Variation in price: prices are hard to estimate precisely	Forecasting covers fluctuations to some extent; uncontrollable variances are isolated and dealt with separately
(ii) Varying levels of output: standard costs not realised if the standard output is missed	True only in some jobbing industries; forecasting estimates output reasonably; prime cost is unaffected; variance analysis measures idle time
(iii) Changing standard of technology	Does not affect the system: standards are revised; cost reduction and control remain its cardinal feature

Criticism	Answer
(iv) Attitude of technical people: used to physical standards, misled by standard costs	Educated through orientation courses: not insurmountable
(v) Mix of products: a pre-determined combination is presupposed	Standards are set for a short period and revised
(vi) Level of performance: too strict or too liberal (theoretical maximum, attainable good, average past performance)	Select a suitable type: attainable good performance
(vii) Cannot reflect the true value in exchange	True standards analyse economic and technical factors into quantities and prices reflecting optimum operations: realistic measures of the sacrifices involved
(viii) Fixation of standards may be costly: high skill needed, hard for small concerns	No answer given

Common mistakes

⚠ **Common mistake**

Hall of Shame: the standard cost of the budgeted output

The café budgeted 24,000 cups (₹2,16,000 of milk) and sold 20,000: "₹9,000 favourable" hides a ₹27,000 overspend. **Fix:** the variance is always against the standard cost of the **actual** output, 20,000 × ₹9.

⚠ **Common mistake**

Calling every adverse variance someone's fault

A variance from a flood or a change in law is uncontrollable: blaming the production head is wrong, and so is ignoring it. **Fix:** classify first; centres answer only for adverse variances they could have controlled; significant, persistent uncontrollable ones may mean revising the standard.

⚠ **Common mistake**

Basic standards producing variances

"The basic standard variance is ₹x." With basic standards, variances are not calculated (SM 13.4). **Fix:** express actual and current costs as percentages of basic cost and compare; keep current standards for the variances.

Practice

Check yourself

1 Tapri's standard cost card shows ₹270 per tray of samosas. In March the factory made 6,000 trays at an actual cost of ₹16,50,000; the budget was 6,500 trays. What is the cost variance (adverse), in rupees?

Answer: _____

2 Case scenario. Ritu proposes standards for Tapri's new cold-chain depot. Kesar wants "the best we could ever do"; Dev wants "the average of the last five years"; the plant head wants "what a trained crew can achieve with good effort". Which proposal does the Study Material treat as the most effective for cost control?

- **A** Kesar's: an ideal standard
- **B** Dev's: an average past performance standard
- **C** The plant head's: attainable good performance
- **D** A basic standard kept unchanged from the depot's first year

3 Under which type of standard are variances NOT calculated, the actual cost being expressed as a percentage instead?

- **A** Ideal standard
- **B** Normal standard
- **C** Basic standard
- **D** Current standard

4 A factory's labour rate variance is adverse because the State Government raised statutory minimum wages; it is significant and will recur every month. What does the Study Material suggest?

- **A** Hold the personnel manager responsible
- **B** Hold the production manager responsible
- **C** Treat it as uncontrollable and consider revising the standard
- **D** Ignore it, as standards are never revised

5 Which of these are steps in setting labour time standards? Select all that apply.

- **A** Time and motion study
- **B** Manufacturing layout
- **C** Test runs of the specification list
- **D** Training and trial

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Standard cost: "the planned unit cost of the product, component or service produced in a period"; set in quantities and prices.
- Standard costing: compare actual costs with pre-set standards; report variances; manage by exception.
- Process: set standards, ascertain actual costs, compare, investigate, dispose of variances (Costing P&L).
- Types: ideal (three criticisms), normal, basic or bogey (no variances; percentages of basic cost), current. Attainable good performance controls cost best.
- Material steps: standardisation, product study, specification list, test runs. Labour steps: product

study, labour specification, methods, layout, time and motion study, training and trial.

- Price standards: the average price expected in the coming period, or the normal price over a cycle of seasons. Material price (small fluctuations): stock held and its price, orders placed, minimum support price, anticipated fluctuation.
- Controllable vs uncontrollable; F means actual cost below standard; significant, persistent uncontrollable variances: the standard may need revision. (The map app does not blame you for the flood.)

Answer key · Check yourself

1 **30000**

Standard cost of actual output = 6,000 × ₹270 = ₹16,20,000; variance = ₹30,000 (A). The budget plays no part.

2 **C** **The plant head's: attainable good performance**

The standard most effective in the control of costs represents an attainable level of good performance (SM 13.52). Ideal standards are not taken seriously; past averages build in past inefficiency.

3 **C** **Basic standard**

With basic standards, actual and current costs are expressed as percentages of basic cost and compared (SM 13.4).

4 **C Treat it as uncontrollable and consider revising the standard**

Uncontrollable, significant and persistent: the standard may need revision (SM 13.12).

5 **A Time and motion study**; **B Manufacturing layout**; **D Training and trial**

Test runs belong to material quantity standards (SM 13.7); the other three are labour time steps (SM 13.7–13.8).

13.2 Material variances

Why the material bill differs from the standard: price, usage, and usage split into mix and yield, with standard loss, purchases against consumption, and the checks ICAI expects.

In plain words

Kesar's sweeter recipe

Tapri's premix plant in Indore blends sugar and milk powder into a "Lite" chai premix. The recipe card says 60 kg of sugar and 40 kg of milk powder go in, and 90 kg of premix comes out: 10 kg are lost in sieving and dust. One month Kesar has an idea.

> **Kesar:** Sugar is ₹40 a kilo, milk powder ₹250. Put in more sugar, less milk powder. Cheaper premix, same taste!
>
> **Ritu:** The month's material cost came in ₹3,100 below the standard.
>
> **Kesar:** Told you. Sab adjust ho jayega.
>
> **Ritu:** Per unit, please. That ₹3,100 is three different stories, and one of them is bad news.

Ritu's first question is: **what should the material for this month's output have cost?** Not the output Kesar planned: the output the plant actually made. Standard quantities at standard prices for it give the **standard cost of actual output**. The gap between it and the actual cost is the

material cost variance: **favourable** (F) when actual cost is lower, **adverse** (A) when higher.

Then she splits the gap into causes. **Price**: the purchase team paid more or less per kg than the standard price. **Usage**: the plant used more or fewer kg than the standard allows for the output. Usage has two causes of its own: a changed **mix**, the proportions of the recipe (Kesar's extra sugar), and a changed **yield**, the output the inputs gave (Kesar's sugary blend lost more in sieving).

This month the cheaper mix saved ₹10,500, the poorer yield cost ₹12,400, and the purchase team saved ₹5,000 on prices: net ₹3,100 favourable. Kesar's idea lost more than it saved; the buyers rescued him.

Think of it like

The wedding caterer's bill

A caterer quoted for 500 plates of biryani from 50 kg of rice and 25 kg of chicken, and the bill came in higher. Chicken may have cost more per kg (price). The kitchen may have used more kg than 500 plates need (usage), either by putting in more chicken against rice, the costlier ingredient (mix), or by burning a pot so that the same pots fed fewer plates (yield).

Key idea

Compare the actual cost with the standard cost of the **actual output**, then explain the gap: price (what each kg cost) and usage (how many kg), with usage split into mix (the proportions) and yield (what the inputs produced). The parts must add back to the whole.

How it works

Material cost variance is the difference between the standard cost of materials used and the actual cost of materials: the difference between the standard material cost of the actual production volume and the actual cost of material (SM 13.14). It arises from a price different from the standard price, a consumption different from the standard consumption, or both, so it is analysed into a **material price variance** and a **material usage variance**.

The Lite premix month, step by step. Standard: sugar 60 kg at ₹40 and milk powder 40 kg at ₹250 for 90 kg of premix (standard loss 10% of input). Actual: 1,710 kg of premix from 1,250 kg of sugar at ₹42 and 750 kg of milk powder at ₹240.

Six steps from a recipe card to five variances

1. **Standard quantity for actual output (SQ).** 90 kg of output needs 100 kg of input, so 1,710 kg needs 1,710 × 100 ÷ 90 = 1,900 kg: sugar 60% = 1,140 kg, milk powder 40% = 760 kg. Standard cost of actual output = 1,140 × ₹40 + 760 × ₹250 = **₹2,35,600**.

2. **Material cost variance (MCV)** = standard cost − actual cost = ₹2,35,600 − (1,250 × ₹42 + 750 × ₹240 = ₹2,32,500) = **₹3,100 (F)**.

3. **Material price variance (MPV)**: the price difference on the actual quantity. Sugar (₹40 − ₹42) × 1,250 = ₹2,500 (A); milk powder (₹250 − ₹240) × 750 = ₹7,500 (F); total **₹5,000 (F)**. The purchase department is answerable for it.

4. **Material usage variance (MUV)**: the quantity difference at standard price. Sugar (1,140 − 1,250) × ₹40 = ₹4,400 (A); milk powder (760 − 750) × ₹250 = ₹2,500 (F); total **₹1,900 (A)**. The production department is answerable for it.

5 **Split usage with the revised standard quantity (RSQ)**: the actual total input, 2,000 kg, in the standard 60 : 40 proportion: sugar 1,200 kg, milk powder 800 kg. **Mix variance** = (RSQ − AQ) × SP: sugar (1,200 − 1,250) × ₹40 = ₹2,000 (A), milk powder (800 − 750) × ₹250 = ₹12,500 (F), total **₹10,500 (F)**. **Yield variance** = (SQ − RSQ) × SP: sugar (1,140 − 1,200) × ₹40 = ₹2,400 (A), milk powder (760 − 800) × ₹250 = ₹10,000 (A), total **₹12,400 (A)**.

6 **Check**: mix + yield = usage (₹10,500 F + ₹12,400 A = ₹1,900 A), and price + usage = cost (₹5,000 F + ₹1,900 A = ₹3,100 F).

The whole analysis is four amounts in a row, each with one thing changed from the one before. Learn the ladder and the formulas follow:

The material variance ladder, Lite premix month

(1) SQ × SP = ₹2,35,600
standard cost of actual output

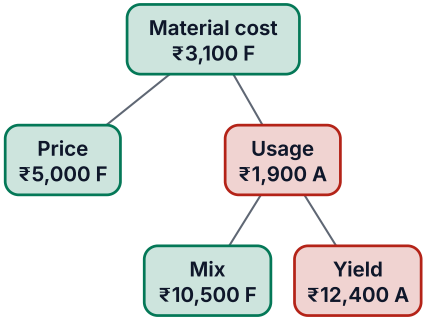
(2) RSQ × SP = ₹2,48,000 — yield (1) – (2): ₹12,400 (A)
actual total input in the standard mix

(3) AQ × SP = ₹2,37,500 — mix (2) – (3): ₹10,500 (F)
actual quantities at standard prices

(4) AQ × AP = ₹2,32,500 — price (3) – (4): ₹5,000 (F)
actual cost

Each rung changes one thing from the rung above: total quantity (yield), proportions (mix), price (price). A positive difference is favourable, a negative one adverse. Yield + mix = usage, (1) – (3) = ₹1,900 (A); all three = cost variance, (1) – (4) = ₹3,100 (F).

Lite premix: the variance tree



Green: favourable (F); rose: adverse (A). Each pair of children adds up to its parent.

Interactive version: PassBook app, lesson Material variances

The vocabulary, in ICAI's words

Term	Meaning (SM 13.17)
Standard quantity (SQ)	Quantity of inputs to be used to produce actual output
Actual quantity (AQ)	Quantity of inputs actually used to produce actual output
Revised standard quantity (RSQ)	The actual total quantity of inputs, if used in standard proportion

A **mix variance** only arises when two or more inputs are used to produce a product (SM 13.16). The **yield variance** is also called the **material sub-usage variance** (and, in the check, the "revised usage" variance); it may arise

from sub-standard quality of materials, inefficiency of workers or wrong processing (SM 13.17).

Formats and formulas

Every material variance answer starts from a working table with the standard for actual output on one side and the actual on the other. *Reproduce the formulas and the table from memory.* A variance computed as standard minus actual is favourable when positive and adverse when negative; write F or A after every figure (SM 13.12).

∑ **Formula**

Material cost variance

$$\text{MCV} = (SQ \times SP) - (AQ \times AP)$$

SQ = standard quantity for actual output, SP = standard price, AQ = actual quantity, AP = actual price. In words: standard cost of actual output – actual cost (SM 13.14).

∑ **Formula**

Material price variance

$$\text{MPV} = AQ \times (SP - AP)$$
$$= (SP \times AQ) - (AP \times AQ)$$

AQ here is the quantity **purchased**; if purchases are not given, take it as equal to the quantity consumed (SM 13.15). The Study Material calls the quantity used "also correct", and its own worked answers use it (Illustration 4, Practical Problem 6): when purchases and consumption differ, state the basis you use (Under the hood).

∑ **Formula**

Material usage variance

$$\text{MUV} = SP \times (SQ - AQ)$$
$$= (SQ \times SP) - (AQ \times SP)$$

AQ here is the quantity **used** (SM 13.16).

∑ **Formula**

Material mix variance

$$\text{MMV} = SP \times (RSQ - AQ)$$
$$= (RSQ \times SP) - (AQ \times SP)$$

∑ Formula

Material yield (sub-usage) variance

$$\begin{aligned}\text{MYV} &= SP \times (SQ - RSQ) \\ &= (SQ \times SP) - (RSQ \times SP)\end{aligned}$$

∑ Formula

Standard quantity with a standard loss, and the revised standard quantity

$$SQ = \text{Actual output} \times \frac{\text{Std input}}{\text{Std output}}$$

$$RSQ_{\text{A}} = \text{Total AQ} \times \frac{SQ_{\text{A}}}{\text{Total SQ}}$$

Std input and std output: the standard batch's input and output. Then split SQ by the standard mix: with a 15% standard loss, 1,700 kg of output needs 1,700 × 100 ÷ 85 = 2,000 kg of input (SM 13.23).

∑ Formula

The checks

$$\text{MCV} = \text{MPV} + \text{MUV}$$

$$\text{MUV} = \text{MMV} + \text{MYV}$$

The first holds when the quantity purchased equals the quantity consumed (SM 13.17).

The working table

Material	SQ × SP (₹)	RSQ × SP (₹)	AQ × SP (₹)	AQ × AP (₹)
A	xxx	xxx	xxx	xxx
B	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

Yield = column 1 − column 2; mix = column 2 − column 3; price = column 3 − column 4; usage = column 1 − column 3; cost = column 1 − column 4. Show the quantities (SQ, RSQ, AQ) and prices in a working note above it.

Worked examples

Worked example **The tea that went into the godown**

Tapri's premix plant uses Assam CTC tea. The standard is 0.25 kg of tea per carton of premix at ₹400 per kg. During November the plant made 12,000 cartons. It purchased 3,500 kg of tea for ₹13,65,000 and used 3,100 kg; there was no opening stock.

Required:

(i) Calculate the material cost variance.

(ii) Calculate the material price variance at the time of purchase.

(iii) Calculate the material usage variance.

(iv) Explain why (ii) and (iii) do not add up to (i), and show the price variance that does.

THE WORKING

Working notes

1. Standard quantity for actual output = 12,000 × 0.25 kg = **3,000 kg**.
2. Actual price = ₹13,65,000 ÷ 3,500 kg = **₹390 per kg**.

(i) MCV = (SQ × SP) − (AQ × AP) = (3,000 × ₹400) − (3,100 × ₹390) = ₹12,00,000 − ₹12,09,000 = **₹9,000 (A)**.

(ii) MPV = (SP − AP) × AQ purchased = (₹400 − ₹390) × 3,500 = **₹35,000 (F)**.

(iii) MUV = (SQ − AQ used) × SP = (3,000 − 3,100) × ₹400 = **₹40,000 (A)**.

(iv) ₹35,000 (F) + ₹40,000 (A) = ₹5,000 (A), not ₹9,000 (A). The price variance at purchase includes ₹4,000 (F) on the 400 kg still in stock (400 × ₹10),

which is not part of the cost of material used. On the quantity consumed:

Particulars	Working	Amount (₹)
Price variance on material used	(₹400 − ₹390) × 3,100	31,000 (F)
Usage variance	(3,000 − 3,100) × ₹400	40,000 (A)
Material cost variance		**9,000 (A)**

Convention: the Study Material prefers the purchase basis because it reports a price variance as soon as it arises (SM 13.15); the check MCV = MPV + MUV then holds only on the consumption basis. State the basis you use.

Worked example

Three ingredients, a standard loss and a cheaper mix

Tapri's masala chai premix is blended from three ingredients. The standard mix for 100 kg of input and

the standard prices are:

Ingredient	Standard mix	Standard price (₹ per kg)
Sugar	50%	40
Milk powder	30%	260
Tea and masala	20%	450

A standard loss of 10% of input is expected. In December, 45,900 kg of premix was produced from:

Ingredient	Quantity used (kg)	Actual price (₹ per kg)
Sugar	27,000	42
Milk powder	15,800	255
Tea and masala	9,200	460

Purchases equalled consumption.

Required:

(i) Calculate the material cost, price, usage, mix and yield variances, for each ingredient and in total, indicating whether each is favourable (F) or adverse (A).

(ii) Verify your answers.

(iii) Comment on the result in two sentences.

THE WORKING

Working notes

1. Standard input for 45,900 kg of output = 45,900 × 100 ÷ 90 = 51,000 kg; split 50 : 30 : 20.
2. Total actual input = 27,000 + 15,800 + 9,200 = 52,000 kg; RSQ = 52,000 split 50 : 30 : 20.

Ingredient	SQ (kg)	RSQ (kg)	AQ (kg)	SP / AP (₹)
Sugar	25,500	26,000	27,000	40 / 42
Milk powder	15,300	15,600	15,800	260 / 255
Tea and masala	10,200	10,400	9,200	450 / 460
Total	**51,000**	**52,000**	**52,000**	

Ingredient	SQ × SP (₹)	RSQ × SP (₹)	AQ × SP (₹)	AQ × AP (₹)
Sugar	10,20,000	10,40,000	10,80,000	11,34,000
Milk powder	39,78,000	40,56,000	41,08,000	40,29,000
Tea and masala	45,90,000	46,80,000	41,40,000	42,32,000
Total	**95,88,000**	**97,76,000**	**93,28,000**	**93,95,000**

(i) Variances. Cost = (SQ × SP) − (AQ × AP); price = (SP − AP) × AQ; usage = (SQ − AQ) × SP; mix = (RSQ − AQ) × SP; yield = (SQ − RSQ) × SP. From the table above:

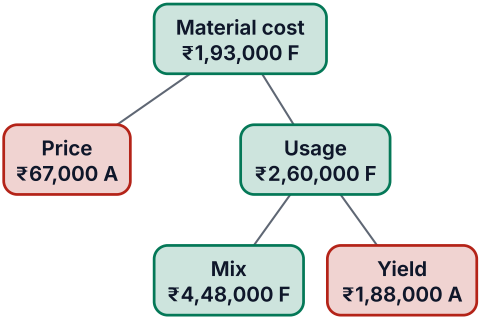
Variance (₹)	Sugar	Milk powder	Tea and masala	Total
Cost	1,14,000 (A)	51,000 (A)	3,58,000 (F)	**1,93,000 (F)**
Price	54,000 (A)	79,000 (F)	92,000 (A)	**67,000 (A)**
Usage	60,000 (A)	1,30,000 (A)	4,50,000 (F)	**2,60,000 (F)**
Mix	40,000 (A)	52,000 (A)	5,40,000 (F)	**4,48,000 (F)**
Yield	20,000 (A)	78,000 (A)	90,000 (A)	**1,88,000 (A)**

(ii) Check

- MUV = MMV + MYV: ₹2,60,000 (F) = ₹4,48,000 (F) + ₹1,88,000 (A).

- MCV = MPV + MUV: ₹1,93,000 (F) = ₹67,000 (A) + ₹2,60,000 (F).

Masala premix: the variance tree



Green: favourable (F); rose: adverse (A). The favourable total comes from the mix.

(iii) Comment. The favourable cost variance comes entirely from the mix: 1,200 kg less of tea and masala, the costliest input, than the standard proportion. The yield variance shows the price of that mix, 900 kg less output than 52,000 kg should give (46,800 kg), and the taste of the premix should be checked before the "saving" is welcomed.

Interactive version: PassBook app, lesson Material variances

Worked example

Bunty's chai-stained standard cost card

On the audit visit to the premix plant, Bunty spilt chai over the standard cost card of the elaichi-adrak premix, a blend of ginger powder and cardamom. The standard mix and the standard loss are illegible; the standard prices survived. The plant's records for October and Ritu's variance summary are intact:

Particulars	Ginger powder	Cardamom
Standard price (₹ per kg)	200	1,600
Actual input (kg)	340	60
Actual price (₹ per kg)	205	1,650

Actual output was 324 kg. Ritu's summary shows a total material usage variance of ₹8,800 (F) and a total material yield variance of ₹19,200 (A). Purchases equalled consumption.

Required:

(i) Find the standard mix and the standard loss as a percentage of input.

(ii) Calculate the material cost, price, usage, mix and yield variances for each material and in total.

(iii) Compare the actual loss with the standard loss, and comment on the month in two sentences.

THE WORKING

(i) Recovering the standard

1. Mix = usage – yield: ₹8,800 (F) – ₹19,200 (A) = 8,800 + 19,200 = **₹28,000 (F)**.
2. In total, mix = (total AQ × standard cost per kg of the standard mix) – Σ(AQ × SP). Σ(AQ × SP) = 340 × ₹200 + 60 × ₹1,600 = ₹68,000 + ₹96,000 = ₹1,64,000. So 400 × c – ₹1,64,000 = ₹28,000 and c = ₹1,92,000 ÷ 400 = **₹480 per kg of input**.
3. If ginger is a fraction g of the standard mix: ₹200g + ₹1,600(1 – g) = ₹480, so ₹1,400g = ₹1,120 and g = 0.80. **Standard mix: ginger 80%, cardamom 20%.**
4. Yield = (total SQ – total AQ) × standard cost per kg of input (SM 13.71): (SQ – 400) × ₹480 = –₹19,200, so total SQ = 400 – 40 = **360 kg** of input for 324 kg of output. Standard loss = 36 kg ÷ 360 kg = **10% of input**.

(ii) Variances

SQ (360 kg at 80 : 20): ginger 288 kg, cardamom 72 kg. RSQ (400 kg at 80 : 20): ginger 320 kg, cardamom 80 kg.

Variance (₹)	Ginger powder	Cardamom	Total
Cost: (SQ × SP) − (AQ × AP)	12,100 (A)	16,200 (F)	**4,100 (F)**
Price: (SP − AP) × AQ	1,700 (A)	3,000 (A)	**4,700 (A)**
Usage: (SQ − AQ) × SP	10,400 (A)	19,200 (F)	**8,800 (F)**
Mix: (RSQ − AQ) × SP	4,000 (A)	32,000 (F)	**28,000 (F)**
Yield: (SQ − RSQ) × SP	6,400 (A)	12,800 (A)	**19,200 (A)**

Workings: cost, ginger 288 × ₹200 − 340 × ₹205 = ₹57,600 − ₹69,700; cardamom 72 × ₹1,600 − 60 × ₹1,650 = ₹1,15,200 − ₹99,000. Check: MCV = MPV + MUV = ₹4,700 (A) + ₹8,800 (F) = ₹4,100 (F); MUV = MMV + MYV = ₹28,000 (F) + ₹19,200 (A) = ₹8,800 (F), the figures Ritu recorded.

(iii) Loss and comment. Actual loss = 400 − 324 = **76 kg** (19% of input) against a standard loss of 10% × 400 = **40 kg**: 36 kg of output short. At the standard cost of ₹1,72,800 ÷ 324 kg of output, 36 kg is worth ₹19,200, the yield variance again. The ₹4,100 (F) total hides a blend with 20 kg less cardamom than the standard proportion (a ₹28,000 mix saving) that lost 36 kg of output, and dearer

purchases of both spices; the aroma of October's premix should be tested before anyone calls it a saving.

Exam angle

In the exam

Where material variances appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, material variances were asked in 23 of them: 4 past papers, 7 RTPs and 12 MTPs.

- **Descriptive, 4 to 9 marks**, usually with labour variances alongside. Full analysis with mix and yield and labour idle time: September 2024 Q4(a) (9 marks). Price and usage from budget and actual totals: January 2026 Q4(a) (7 marks). Actual price worked back from the price variance: May 2026 Q1(b) (5 marks). Three-material mix and yield: RTP January 2025 Q16. Missing figures for a two-material mix: RTP May 2025 Q17 and MTP May 2026 Series I Q2(a) (8 marks); RTP January 2026 Q16. Purchases larger than consumption: MTP January 2026 Series II Q5(a) (11 marks, "based on actual material consumption").
- **Multiple choice, 2 marks each**: price and mix variances from the cost, usage and yield variances (September 2025 MCQ 11); usage variance (RTP January 2026 MCQ 3); actual price from the price

variance (MTP September 2025 Series I MCQ 13); mix, yield, price and cost for two fabrics with a normal loss (MTP September 2024 Series II MCQs 6–8).

In the exam

The wording traps ICAI used

- "Purchased" and "consumed" given separately: the rule is purchases unless the question says otherwise (RTP September 2025 Case Scenario 1 does it on purchases; MTP January 2026 Series II Q5(a) asks for consumption), but the Study Material's own Novell & Co. answer uses consumption: state the basis and give the other figure in a note.
- "Standard loss 10% of input" or "yield 80%": scale the input up from the actual output before anything else.
- A material usage variance "adjusted for abnormal loss" (MTP January 2026 Series I Q3(a)) takes the abnormal loss out of the actual quantity; follow the question's instruction and say so.
- "Indicate the nature of variance": every figure needs its (F) or (A).

Trick

Read the ladder backwards for missing figures

Any two of cost, price and usage give the third; any two of usage, mix and yield give the third. Condition: purchases equal consumption, or all the variances are on the same quantity. September 2025 MCQ 11 is just this: MPV = MCV − MUV and MMV = MUV − MYV, with F as plus and A as minus.

Answer frame

A full-marks material variance answer

1. **Working notes first**: the standard quantity for actual output (show the standard-loss scaling), the actual quantity (opening stock + purchases − closing stock where stocks are given), the actual price, and the RSQ.
2. **The working table**: SQ × SP, RSQ × SP, AQ × SP, AQ × AP for each material, with totals.
3. **Each variance**: the formula in symbols, then each material's figure with (F) or (A), then the total.
4. **The check lines**: MUV = MMV + MYV and MCV = MPV + MUV; if purchases differ from consumption, state the basis of the price variance and show the consumption-basis figure, on which the second check closes.
5. **A one-line comment** when the question asks for one: which variance drives the total, and who is responsible.

"State the reasons for, and the responsibility for, material variances"

Variance

Price

Reasons (Study Material)

Purchase price different from standard; some causes within the purchase department's control, some beyond it

Responsibility

Purchase department, for adverse variances from factors it controls

Variance

Usage

Reasons (Study Material)

Consumption different from standard: a different proportion (mix) or a different yield

Responsibility

Production department

Variance

Mix

Reasons (Study Material)

Proportion of inputs actually used different from the standard mix; only with two or more inputs

Responsibility

Production department (mix is part of usage)

Variance

Yield

Reasons (Study Material)

Sub-standard quality of materials, inefficiency of workers, wrong processing

Responsibility

Production department (yield is part of usage)

Three answers to the same question

Loses marks

"Mix variance = ₹4,48,000."

No formula, no material-wise figures, no F or A.

Passes

"Mix variance = (RSQ − AQ) × SP = ₹4,48,000 (F)."

Formula and nature, but no RSQ shown: one slip in it costs every later mark.

Full marks

Working notes (SQ with the 10% loss, RSQ), the table, each material's mix figure with F or A, the total, and the check MUV = MMV + MYV.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the standard quantity for the output Kesar planned

Taking SQ from the budgeted output instead of the actual output. Kesar planned 1,800 kg of Lite premix and the plant made 1,710 kg: an SQ of 2,000 kg from the plan turns the usage variance of ₹1,900 (A) into a "saving" of ₹10,500 (F) that never happened. **Fix:** SQ is always the standard quantity for the **actual** output; the budget plays no part in material variances.

⚠ **Common mistake**

The revised standard quantity built from the standard total

RSQ splits the **actual total input** (52,000 kg) in the standard proportion, not the standard total (51,000 kg). Built from 51,000 kg, the RSQ equals the SQ, the yield variance vanishes and the mix variance swallows the whole usage variance. **Fix:** RSQ = total AQ × standard proportion; its total must equal the total AQ.

⚠ **Common mistake**

The standard loss forgotten

For 45,900 kg of output, SQ = 45,900 kg of input ignores the 10% loss. **Fix:** standard input = actual output × standard input ÷ standard output (here × 100 ÷ 90 = 51,000 kg).

⚠ **Common mistake**

Price variance on the wrong quantity, without a word

Using consumption when purchases are given (or purchases when the question asks for consumption), silently. **Fix:** purchases unless told otherwise (SM 13.15); state the basis, and give the consumption-basis figure in a note so that MPV + MUV reconciles to MCV.

Practice

Check yourself

1 Tapri's kulhad unit has a standard of 0.5 kg of clay per kulhad at ₹12 per kg. In March it made 40,000 kulhads, purchased 24,000 kg of clay at ₹11.50 per kg and used 21,000 kg. What is the material price variance at the time of purchase, in rupees (favourable)?

Answer: _____

2 Case scenario. Ritu's notes on a two-spice blend for November: the standard input allowed for the actual output is 2,000 kg; the actual input was 2,080 kg; the standard mix costs ₹500 per kg of input; the material mix variance is ₹6,000 (F) and the material price variance ₹9,000 (A). Purchases equal consumption. What is the material cost variance?

- **A** ₹49,000 (A)
- **B** ₹55,000 (A)
- **C** ₹25,000 (A)
- **D** ₹43,000 (A)

3 A unit's budget was 1,000 units; it produced 900. The standard is 2 kg per unit at ₹50 per kg, and it used 1,850 kg. What is the material usage variance?

- **A** ₹7,500 (F)
- **B** ₹2,500 (A)
- **C** ₹2,500 (F)
- **D** ₹5,000 (A)

4 When can a material mix variance arise?

- **A** Whenever the actual price differs from the standard price
- **B** Whenever there is a standard loss in processing
- **C** Only when two or more materials are used to make the product
- **D** Only when purchases differ from consumption

5 Which of these can cause an adverse material yield variance? Select all that apply.

- **A** Sub-standard quality of materials
- **B** Inefficiency of workers
- **C** A rise in the market price of the material
- **D** Wrong processing

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- MCV = (SQ × SP) − (AQ × AP): standard cost of **actual** output minus actual cost.

- MPV = AQ × (SP − AP), on the quantity **purchased** (consumed if purchases are not given); the purchase department's.
- MUV = SP × (SQ − AQ), on the quantity **used**; the production department's.
- MMV = SP × (RSQ − AQ); MYV (sub-usage) = SP × (SQ − RSQ); RSQ = actual total input in the standard proportion.
- Checks: MUV = MMV + MYV; MCV = MPV + MUV when purchases equal consumption.
- Standard loss: SQ = actual output × standard input ÷ standard output, then split by the standard mix.
- Ladder: SQ × SP → RSQ × SP → AQ × SP → AQ × AP gives yield, mix, price. (Kesar's sweeter recipe: mix favourable, yield adverse.)

Answer key · Check yourself

1 **12000**

MPV = (SP − AP) × quantity purchased = (₹12 − ₹11.50) × 24,000 = ₹12,000 (F). On the 21,000 kg used it would be ₹10,500 (F).

2 **D ₹43,000 (A)**

Yield = (total SQ − total AQ) × standard cost per kg of input = (2,000 − 2,080) × ₹500 = ₹40,000 (A). Usage = mix + yield = ₹6,000 (F) + ₹40,000 (A) = ₹34,000 (A). Cost = price + usage = ₹9,000 (A) + ₹34,000 (A) = ₹43,000 (A). ₹49,000 leaves out the mix variance, ₹55,000 treats it as adverse, and ₹25,000 treats the price variance as favourable.

3 **B ₹2,500 (A)**

SQ for the actual 900 units = 1,800 kg; MUV = (1,800 − 1,850) × ₹50 = ₹2,500 (A). ₹7,500 (F) uses the budgeted 2,000 kg, the commonest mistake.

4 **C Only when two or more materials are used to make the product**

The mix variance compares actual with standard proportions, so it needs at least two inputs (SM 13.16).

5 **A Sub-standard quality of materials**; **B Inefficiency of workers**; **D Wrong processing**

The Study Material's three causes of a yield variance are sub-standard materials, inefficient workers and wrong processing (SM 13.17). A price rise shows up in the price variance.

13.3 Labour variances, including idle time

Why the wage bill differs from the standard: rate on hours paid, idle time at the standard rate, efficiency on hours worked, and efficiency split into mix (gang) and yield.

In plain words

The night the lights went out

The night shift at Tapri's samosa factory in Indore fries in batches of 100 samosas. The standard for a batch is 2 hours of a skilled fryer and 3 hours of a helper. One October night a five-hour power cut stops the ten fryers and ten helpers on duty, and they wait, paid, in the dark.

> **Kesar:** We paid twenty people to sit in the dark? Cut their wages!
>
> **Ritu:** The law and the union will both disagree. Per unit, please: the shift still made 400 batches in October. Let me show you where the ₹26,120 went.

Ritu starts where the material lesson starts: the **standard labour cost of the actual output**, the hours 400 batches should take at the standard rates, ₹2,80,000. The wages actually paid were ₹3,06,120. The gap, ₹26,120 adverse, is the **labour cost variance**, and it has three causes.

Rate: the fryers were paid ₹210 an hour against a standard of ₹200, the helpers ₹96 against ₹100. **Idle time**:

100 hours were paid but not worked because of the power cut. **Efficiency**: the hours actually worked differed from the hours 400 batches should take. Efficiency itself has two causes: the **mix** of the gang (more fryers, the costlier grade, than the 2 : 3 standard) and the **yield** of the hours worked (the output those hours gave).

The power cut alone cost ₹15,000: a hundred idle hours at the standard rates. Cutting wages would not have bought back the electricity.

Think of it like

The taxi on hourly hire

You hire a cab by the hour. The bill can rise because the driver charged more per hour than agreed (rate), because you paid for an hour stuck at a closed toll plaza (idle time), or because the trip took longer than it should (efficiency). And if the company sent a senior driver at a higher rate for a job a junior could do, that is a change in the mix.

Key idea

Compare the wages paid with the standard cost of the **actual output**, then explain the gap: rate on the hours **paid**, idle time on the hours paid but not worked, at the standard rate, and efficiency on the hours **worked**, split into mix and yield. The parts add back to the whole.

How it works

Labour cost variance is the difference between the standard labour cost and the actual labour cost incurred for the production achieved (SM 13.25). It arises from a rate different from the standard rate or from hours different from the standard hours, and is divided into a **labour rate variance**, a **labour efficiency variance** and a **labour idle time variance**.

The samosa night shift, step by step. Standard per batch: skilled 2 hours at ₹200, helper 3 hours at ₹100. Actual: 400 batches; skilled 900 hours paid at ₹210, helpers 1,220 hours paid at ₹96; 50 hours of each grade lost to the power cut.

Six steps from a wage sheet to six variances

1. **Standard hours for actual output (SH).** 400 batches × 2 hours = 800 skilled hours; × 3 hours = 1,200 helper hours. Standard cost = 800 × ₹200 + 1,200 × ₹100 = **₹2,80,000**.

2. **Labour cost variance (LCV)** = standard cost − actual cost = ₹2,80,000 − (900 × ₹210 + 1,220 × ₹96 = ₹3,06,120) = **₹26,120 (A)**.

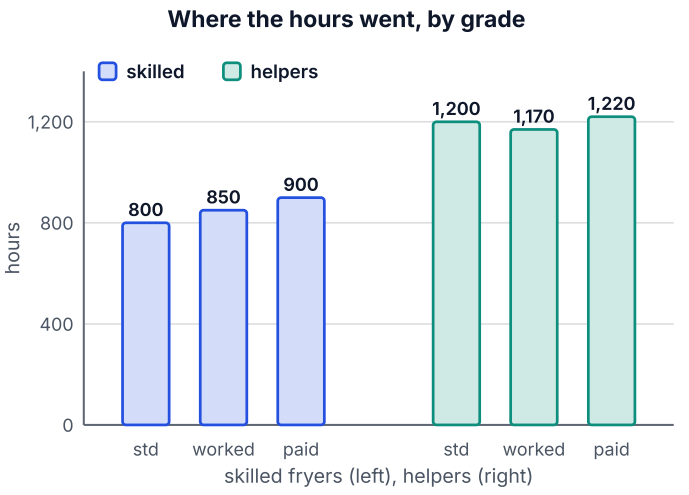
3 **Labour rate variance (LRV)**: the rate difference on the hours **paid**. Skilled (₹200 − ₹210) × 900 = ₹9,000 (A); helpers (₹100 − ₹96) × 1,220 = ₹4,880 (F); total **₹4,120 (A)**.

4 **Idle time variance (ITV)**: the idle hours at the standard rate. Skilled 50 × ₹200 = ₹10,000 (A); helpers 50 × ₹100 = ₹5,000 (A); total **₹15,000 (A)**.

5 **Labour efficiency variance (LEV)**: standard hours against hours **worked** (900 − 50 = 850 and 1,220 − 50 = 1,170), at the standard rate. Skilled (800 − 850) × ₹200 = ₹10,000 (A); helpers (1,200 − 1,170) × ₹100 = ₹3,000 (F); total **₹7,000 (A)**.

6 **Split efficiency with the revised standard hours (RSH)**, the 2,020 hours worked in the standard 2 : 3 ratio: skilled 808, helpers 1,212. **Mix (gang) variance** = (RSH − AH worked) × SR: skilled (808 − 850) × ₹200 = ₹8,400 (A), helpers (1,212 − 1,170) × ₹100 = ₹4,200 (F), total **₹4,200 (A)**. **Yield variance** = (SH − RSH) × SR: skilled (800 − 808) × ₹200 = ₹1,600 (A), helpers (1,200 − 1,212) × ₹100 = ₹1,200 (A), total **₹2,800 (A)**.

Check: rate + idle time + efficiency = ₹4,120 (A) + ₹15,000 (A) + ₹7,000 (A) = ₹26,120 (A), the cost variance; mix + yield = ₹4,200 (A) + ₹2,800 (A) = ₹7,000 (A), the efficiency variance.



Blue: skilled fryers; teal: helpers. Worked minus standard hours drives the efficiency variance; paid minus worked hours (50 for each grade) is the idle time.

The variances form a ladder like the material ladder, with one more rung for the idle hours:

The labour variance ladder, samosa night shift

(1) SH × SR = ₹2,80,000
standard cost of actual output

(2) RSH × SR = ₹2,82,800 yield (1) – (2): ₹2,800 (A)
hours worked, in the standard mix

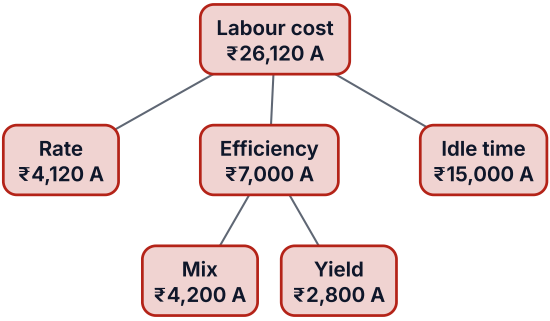
(3) AH worked × SR = ₹2,87,000
mix (2) – (3): ₹4,200 (A)
hours worked, actual mix

(4) AH paid × SR = ₹3,02,000
idle (3) – (4): ₹15,000 (A)
hours paid, at standard rates

(5) AH paid × AR = ₹3,06,120 rate (4) – (5): ₹4,120 (A)
actual wages

Each rung changes one thing from the rung above. A positive difference is favourable, a negative one adverse. Efficiency = (1) – (3) = ₹7,000 (A); cost variance = (1) – (5) = ₹26,120 (A).

Samosa night shift: the variance tree



Rose: adverse (A). The three children add up to the cost variance; mix and yield add up to efficiency.

Interactive version: PassBook app, lesson Labour variances, including idle time

Who answers for which variance

- **Rate**: labour rates are generally influenced by external factors beyond the organisation's control; the **personnel manager** is responsible for rate negotiation (SM 13.26).
- **Efficiency**: it may arise from the ability of the workers, an inappropriate team of workers, or inefficiency of the production manager or foreman; the **production manager or foreman** can be held responsible for the adverse variance that could otherwise have been controlled (SM 13.26).
- **Idle time**: abnormal idle time should be analysed into controllable and uncontrollable causes (power failure,

breakdown, lack of materials, strikes) so that responsibility can be fixed (SM 3.14).

Formats and formulas

Reproduce the formulas and the five-rung ladder from memory. SH = standard hours for actual output, SR = standard rate, AR = actual rate, RSH = revised standard hours. Hours **paid** and hours **worked** are different numbers whenever there is idle time: write the subscript every time.

∑ **Formula**

Labour cost variance

$$\text{LCV} = (SH \times SR) - (AH_{\text{paid}} \times AR)$$

The standard labour cost minus the actual labour cost incurred for the production achieved (SM 13.25).

∑ **Formula**

Labour rate variance

$$\text{LRV} = AH_{\text{paid}} \times (SR - AR)$$

The difference between the standard and actual rate per hour for the actual hours **paid** (SM 13.25).

∑ **Formula**

Labour idle time variance

$$\text{ITV} = SR \times (AH_{\text{paid}} - AH_{\text{worked}})$$
$$= SR \times \text{Idle hours}$$

The difference between the actual hours paid and the actual hours worked, at standard rate (SM 13.27). Idle hours cannot be negative, so this variance is either nil or adverse.

∑ **Formula**

Labour efficiency variance

$$\text{LEV} = SR \times (SH - AH_{\text{worked}})$$

The difference between the standard hours specified for actual production and the actual hours **worked**, at standard rate (SM 13.26). The Study Material's middle line of this formula marks AH with the asterisk it uses for hours paid, but its bracket form, its words and its footnote say hours worked, and Illustrations 6 and 7 use hours worked (SM 13.29–13.32): use hours worked.

∑ **Formula**

Labour mix (gang) variance

$$\text{LMV} = SR \times (RSH - AH_{\text{worked}})$$

∑ **Formula**

Labour yield (sub-efficiency) variance

$$\text{LYV} = SR \times (SH - RSH)$$

Also called the labour revised-efficiency variance (SM 13.26–13.27).

∑ **Formula**

Revised standard hours and the checks

RSH of a grade = total hours worked × (that grade's SH ÷ total SH).

$$\text{LCV} = \text{LRV} + \text{ITV} + \text{LEV}$$

$$\text{LEV} = \text{LMV} + \text{LYV}$$

With no idle time, LCV = LRV + LEV (SM 13.28).

Term	Meaning (SM 13.37)
Standard hours (SH)	Hours required to produce the actual output
Actual hours (AH)	Actual hours taken to produce the actual output
Revised standard hours (RSH)	If actual labour hours worked were worked by standard mix (combination) of labour
Actual yield (AY)	Actual hours worked
Standard yield (SY)	Actual hours if labour worked in standard ratio
Standard labour cost (SLC)	Standard labour cost for actual output

The working table

Grade	SH × SR (₹)	RSH × SR (₹)	AH worked × SR (₹)	AH paid × SR (₹)
Skilled	xxx	xxx	xxx	xxx
Unskilled	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

Add a column for AH paid × AR (the actual wages) in a second table or a working note: yield = column 1 − 2, mix = 2 − 3, idle time = 3 − 4, rate = 4 − actual wages.

Worked examples

Worked example

The cold-chain crew that beat the clock

Tapri's cold-chain depot loaded 8,000 crates of frozen samosas in October. The standard is 0.5 hour per crate at ₹120 an hour. The crew worked 3,200 hours, with no idle time. It is paid ₹120 an hour plus a bonus under the Halsey premium plan: 50% of the time saved against the standard time for the month's output, at the time rate.

Required:

(i) Calculate the crew's wages for October and the actual rate per hour.

(ii) Calculate the labour cost, rate and efficiency variances, and verify them.

(iii) Calculate the standard and the actual labour cost per crate.

(iv) Kesar wants to move the crew to the Rowan premium plan. Calculate the labour rate variance and the actual labour cost per crate under it, and explain why an incentive bonus appears as an adverse rate variance.

THE WORKING

Working notes

1. Standard hours for 8,000 crates = 8,000 × 0.5 = **4,000 hours**; time saved = 4,000 − 3,200 = 800 hours. Labour efficiency = SH ÷ AH × 100 = 4,000 ÷ 3,200 × 100 = 125% (the efficiency ratio, SM 15.68).
2. Halsey earnings = time taken × time rate + 50% of time saved × time rate (SM 3.28).

(i) Wages = 3,200 × ₹120 + 50% × 800 × ₹120 = ₹3,84,000 + ₹48,000 = **₹4,32,000**. Actual rate = ₹4,32,000 ÷ 3,200 = **₹135 an hour**.

(ii)

- LRV = (SR − AR) × AH = (₹120 − ₹135) × 3,200 = **₹48,000 (A)**.
- LEV = (SH − AH) × SR = (4,000 − 3,200) × ₹120 = **₹96,000 (F)**.
- LCV = (SH × SR) − (AH × AR) = ₹4,80,000 − ₹4,32,000 = **₹48,000 (F)**. Check: ₹96,000 (F) + ₹48,000 (A) = ₹48,000 (F).

(iii) Standard labour cost per crate = 0.5 × ₹120 = **₹60.00**; actual = ₹4,32,000 ÷ 8,000 = **₹54.00**.

(iv) Rowan plan. Bonus = time saved ÷ time allowed × time taken × rate (SM 3.29–3.30) = 800 ÷ 4,000 × 3,200 × ₹120 = ₹76,800. Wages = ₹3,84,000 + ₹76,800 = ₹4,60,800; actual rate = ₹144 an hour.

Particulars	Halsey	Rowan
Bonus (₹)	48,000	76,800
Actual rate per hour (₹)	135	144
Labour rate variance (₹)	48,000 (A)	76,800 (A)
Labour efficiency variance (₹)	96,000 (F)	96,000 (F)
Labour cost variance (₹)	48,000 (F)	19,200 (F)
Actual labour cost per crate (₹)	54.00	57.60

The standard rate is the time rate, so any bonus paid for speed lifts the actual rate above it and shows as an adverse rate variance; the speed itself shows as the favourable efficiency variance. Under the Halsey plan the bonus is always half the saving at the time rate, so here the rate variance is exactly 50% of the efficiency variance and Tapri keeps the other half. Under the Rowan plan, with time saved only 20% of time allowed, the crew takes 80% of the saving (₹76,800 of ₹96,000) and Tapri keeps ₹19,200. Read alone, the adverse rate variance would wrongly blame rate negotiation: it is the agreed price of the incentive.

Worked example

Diwali gift boxes: the night the conveyor stopped

Kesar's Diwali gift-box line has a standard packing team of 1 skilled packer (₹180 an hour), 2 semi-skilled packers (₹120 an hour) and 3 unskilled helpers (₹80 an hour), which packs 30 boxes an hour. In October the line packed 45,000 boxes. One night the conveyor broke down and the line stood still for 12 hours while 4 skilled, 8 semi-skilled and 12 unskilled workers were on duty; the clock cards record the stoppage as idle time. The wage sheet for October:

Grade	Hours paid	Wages paid (₹)
Skilled	1,748	3,23,380
Semi-skilled	2,916	3,44,088
Unskilled	4,144	3,48,096
Total	**8,808**	**10,15,564**

Required:

(a) Calculate the labour cost, rate, idle time, efficiency, mix and yield variances, grade-wise and in total, indicating whether each is favourable (F) or adverse (A), and verify them.

(b) The supervisor ran the line with more skilled packers than the standard team because the Diwali

boxes need ribbon work. Kesar says the change cost Tapri money. Using the variances, say who is right, and name who answers for the rate and the efficiency variances.

THE WORKING

Working notes

1. Team hours for 45,000 boxes = 45,000 ÷ 30 = 1,500. SH = team hours × workers of each grade: skilled 1,500, semi-skilled 3,000, unskilled 4,500; total 9,000 (ratio 1 : 2 : 3).
2. Idle hours = workers on duty × 12: skilled 48, semi-skilled 96, unskilled 144 (288). Hours worked = hours paid − idle: 1,700, 2,820 and 4,000 (8,520).
3. Actual rates = wages ÷ hours paid: ₹185, ₹118 and ₹84.
4. RSH = 8,520 hours worked in the ratio 1 : 2 : 3: 1,420, 2,840 and 4,260.

Grade	SH × SR (₹)	RSH × SR (₹)	AH worked × SR (₹)	AH paid × SR (₹)
Skilled	2,70,000	2,55,600	3,06,000	3,14,640
Semi-skilled	3,60,000	3,40,800	3,38,400	3,49,920
Unskilled	3,60,000	3,40,800	3,20,000	3,31,520
Total	**9,90,000**	**9,37,200**	**9,64,400**	**9,96,080**

The actual wages (AH paid × AR) are in the wage sheet: total ₹10,15,564.

(a) Variances. Cost = (SH × SR) − (AH paid × AR); rate = (SR − AR) × AH paid; idle time = idle hours × SR; efficiency = (SH − AH worked) × SR; mix = (RSH − AH worked) × SR; yield = (SH − RSH) × SR.

Variance (₹)	Skilled	Semi-skilled	Unskilled	Total
Cost	53,380 (A)	15,912 (F)	11,904 (F)	**25,564 (A)**
Rate	8,740 (A)	5,832 (F)	16,576 (A)	**19,484 (A)**
Idle time	8,640 (A)	11,520 (A)	11,520 (A)	**31,680 (A)**
Efficiency	36,000 (A)	21,600 (F)	40,000 (F)	**25,600 (F)**
Mix	50,400 (A)	2,400 (F)	20,800 (F)	**27,200 (A)**
Yield	14,400 (F)	19,200 (F)	19,200 (F)	**52,800 (F)**

Check

- LCV = LRV + ITV + LEV: ₹25,564 (A) = ₹19,484 (A) + ₹31,680 (A) + ₹25,600 (F).
- LEV = LMV + LYV: ₹25,600 (F) = ₹27,200 (A) + ₹52,800 (F).

(b) Who is right. The supervisor, on the gang. The richer gang shows as an adverse mix variance of ₹27,200, but the line packed 45,000 boxes with 8,520 hours worked against a standard of 9,000: a yield variance of ₹52,800 (F), so the gang change earned a net favourable efficiency variance of ₹25,600. The adverse total comes from the conveyor (idle time ₹31,680 (A), to be traced to its cause, here maintenance) and from wage rates (₹19,484 (A)). The personnel manager is responsible for rate negotiation and the production manager or foreman for efficiency (SM 13.26).

Worked example

The café fit-out gang: who was on site?

Tapri hires a contractor's gang to fit out its new kulhad cafés. The contract fixes the gang at 4 masons and 6 helpers and the hourly rates at ₹260 for a mason and ₹96 for a helper; Tapri's standard rates are ₹250 and ₹100, for the same gang of 4 : 6. Last week's site register and the contractor's bill were lost in the rain. Ritu's notes survive: 10 workers were paid for a 48-hour week; the whole gang stood idle for 2 hours when the tiles came late; the work done was 400 standard hours; and the labour mix (gang) variance was ₹6,900 (A).

Required:

(i) Find the number of masons and helpers on site.

(ii) Calculate the labour cost, rate, idle time, efficiency and yield variances, indicating whether each is favourable (F) or adverse (A), and verify them.

(iii) Did the contractor keep to the agreed gang, and what did the change cost Tapri on the week's bill?

THE WORKING

Working notes

1. Hours paid = 10 × 48 = 480; idle = 10 × 2 = 20; hours worked = 460.
2. RSH = 460 hours worked in the ratio 4 : 6: masons 184, helpers 276. SH = 400 standard hours in the same ratio: masons 160, helpers 240.

(i) The gang. Let the masons' hours worked be M, so the helpers' are 460 − M. Mix = (RSH − AH worked) × SR:

$$(184 - M) \times 250 + (276 - (460 - M)) \times 100 = -6{,}900$$

$$27{,}600 - 150M = -6{,}900 \qquad M = 230$$

Each worker worked 46 hours, so masons = 230 ÷ 46 = **5** and helpers = (460 − 230) ÷ 46 = **5**.

(ii) Variances. Masons: paid 240 hours, worked 230, idle 10; helpers the same.

Variance (₹)

Cost: (SH × SR) − (AH paid × AR)

Masons

22,400 (A)

Helpers

960 (F)

Total

21,440 (A)

Variance (₹)

Rate: (SR − AR) × AH paid

Masons

2,400 (A)

Helpers

960 (F)

Total

1,440 (A)

Variance (₹)

Idle time: idle hours × SR

Masons

2,500 (A)

Helpers

1,000 (A)

Total

3,500 (A)

Variance (₹)

Efficiency: (SH − AH worked) × SR

Masons

17,500 (A)

Helpers

1,000 (F)

Total

16,500 (A)

Variance (₹)

Mix: (RSH − AH worked) × SR (given)

Masons

11,500 (A)

Helpers

4,600 (F)

Total

6,900 (A)

Variance (₹)

Yield: (SH − RSH) × SR

Masons

6,000 (A)

Helpers

3,600 (A)

Total

9,600 (A)

Workings, masons: cost 160 × ₹250 − 240 × ₹260 = ₹40,000 − ₹62,400; efficiency (160 − 230) × ₹250; yield (160 − 184) × ₹250. Check: ₹1,440 (A) + ₹3,500 (A) + ₹16,500 (A) = ₹21,440 (A); ₹6,900 (A) + ₹9,600 (A) = ₹16,500 (A).

(iii) No: the contractor sent 5 masons and 5 helpers, one helper replaced by a mason. At standard rates the change is the mix variance of ₹6,900 (A); on the bill it costs 48 hours × (₹260 − ₹96) = ₹7,872 more than the agreed gang (₹85,440 against ₹77,568). The extra mason did not speed the work up: the gang still produced only 400 standard hours from 460 hours worked (yield ₹9,600 (A)).

Interactive version: PassBook app, lesson Labour variances, including idle time

Exam angle

In the exam

Where labour variances appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, labour variances were asked in 23 of them: 5 past papers, 5 RTPs and 13 MTPs.

- **Descriptive, 5 to 9 marks**, usually with material variances in the same question. Three grades with abnormal idle time, analysed into mix, efficiency, yield and rate: September 2026 Q4(a) (9 marks; suggested answers not yet published). Two grades, 5% of time paid lost for an abnormal reason: September 2024 Q4(a) (9 marks). Actual rate and standard hours worked back from a rate variance and a labour efficiency of 106.8%: January 2025 Q5(a) (6 marks). Rate and efficiency from budget and actual totals: January 2026 Q4(a) (7 marks); May 2026 Q1(b) (5 marks). Idle time excluded from efficiency: MTP January 2026 Series I Q3(a).
- **Multiple choice, 2 marks each**: actual wage rate per unit when the rate variance is ₹7,500 (A) and efficiency nil (January 2026 MCQ 11, and MTP May 2024 Series I MCQ 3); net hours worked from an idle time variance (RTP September 2024 Case Scenario 2); actual labour cost after 500 idle hours from a

power outage (MTP May 2026 Series II MCQ 12); efficiency and yield variances of a two-grade gang (MTP September 2024 Series II MCQ 9).

In the exam

The wording traps ICAI used

- "Hours paid (including abnormal idle time)" against "hours worked": the rate variance uses hours paid, the efficiency variance hours worked. RTP September 2024 Case Scenario 2 called the 30,000 hours paid "actual working hours", and its answer took the efficiency variance on them, though it also found 2,000 idle hours; on the Study Material's formula, with the 28,000 hours worked (SM 13.26), it is ₹15,000 (A), not ₹25,000 (A). Write hours worked; in an MCQ whose options match only the hours-paid figure, that is the examiner's reading.
- A rate given for only one grade in a two-grade question (September 2024 Q4(a): semi-skilled paid at standard): that grade's rate variance is nil, not missing.
- "Labour efficiency 106.8%" is standard hours ÷ actual hours × 100 (January 2025 Q5(a)).

Trick

One rung per cause

Hours paid × actual rate → hours paid × standard rate (rate) → hours worked × standard rate (idle time) → revised standard hours × standard rate (mix) → standard hours × standard rate (yield). Each step changes one thing. Condition: every grade's idle hours are known; with none, rungs 3 and 4 are the same amount.

Answer frame

A full-marks labour variance answer

1. **Working notes first**: SH for actual output (per grade), hours paid, idle hours and hours worked per grade, actual rates, RSH from total hours worked.
2. **The working table**: SH × SR, RSH × SR, AH worked × SR, AH paid × SR, AH paid × AR per grade, with totals.
3. **Each variance**: formula in symbols with "paid" or "worked" written on the hours, grade-wise figures with (F) or (A), total.
4. **The checks**: LCV = LRV + ITV + LEV; LEV = LMV + LYV.
5. **A comment** when asked: which grade or cause drives the total, and who is responsible (rate: personnel manager; efficiency: production manager or foreman; idle time: by its cause).

Three answers to the same question

Loses marks

"Efficiency variance = (SH − hours paid) × SR."

Hours paid used for efficiency, so the idle hours are counted twice; no grade-wise figures.

Passes

"Efficiency variance = (SH − AH worked) × SR = ₹25,600 (F)."

Right formula and total, but no grade-wise figures and no check against the cost variance.

Full marks

Working notes, the five-amount table, each variance grade-wise with F or A and in total, and both checks.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the idle hours counted twice

Taking the efficiency variance on hours paid and then adding an idle time variance as well. In the gift-box example, efficiency on hours paid gives ₹9,90,000 − ₹9,96,080 = ₹6,080 (A); with the ₹31,680 (A) of idle time and the ₹19,484 (A) rate variance on top, the variances add up to ₹57,244 (A), not the ₹25,564 (A) cost variance, and a favourable efficiency of ₹25,600 has turned adverse. **Fix:** efficiency on hours **worked**; idle time on hours paid − hours worked.

⚠ **Common mistake**

Rate variance on hours worked

The workers were paid for every hour, idle or not, so the rate difference applies to all the hours paid. **Fix:** LRV = (SR − AR) × AH **paid**.

⚠ **Common mistake**

Idle time valued at the actual rate

The idle time variance is at the **standard** rate; the difference between actual and standard rate on the idle hours is already inside the rate variance. **Fix:** ITV = idle hours × SR.

⚠ **Common mistake**

RSH from hours paid

Revised standard hours split the hours **worked** in the standard ratio; built from hours paid, the mix and yield no longer add up to the efficiency variance. **Fix:** RSH total = total hours worked.

Practice

Check yourself

1 A packing team's standard rate is ₹90 an hour. In a week it was paid for 2,400 hours and worked 2,250. What is the idle time variance (adverse), in rupees?

Answer: _____

2 Case scenario. Tapri's two central kitchens each made 6,000 trays of samosas in March, against a standard of 0.25 hour per tray at ₹160 an hour, with no idle time. Indore paid ₹168 an hour and worked exactly the standard hours. Bhopal paid ₹160 an hour and worked 1,575 hours. Kesar asks Ritu which kitchen did worse. Which statement is correct?

A Indore's labour cost variance is ₹12,000 (A) and Bhopal's is nil, because Bhopal paid the standard rate

B Both kitchens have a labour efficiency variance of ₹12,000 (A)

C Both have a labour cost variance of ₹12,000 (A) and a labour cost of ₹42 a tray, but Indore's is a rate variance (personnel manager) and Bhopal's an efficiency variance (production manager or foreman)

D Bhopal's labour cost is ₹40 a tray, the standard, because it paid the standard rate

3 Standard hours for the output are 1,000 at ₹100 an hour. Workers were paid for 1,100 hours and worked 1,050; the rest was abnormal idle time. What is the labour efficiency variance?

- **A** ₹5,000 (A)
- **B** ₹10,000 (A)
- **C** ₹5,000 (F)
- **D** ₹15,000 (A)

4 According to the Study Material, who is responsible for labour rate negotiation, and so for the labour rate variance?

- **A** The production manager or foreman
- **B** The personnel manager
- **C** The purchase department
- **D** The cost accountant

5 Which of these statements are correct? Select all that apply.

- **A** The idle time variance is valued at the standard rate.
- **B** The labour mix (gang) variance arises from a change in the combination of skilled, semi-skilled and unskilled workers.
- **C** The labour rate variance is computed on the hours worked.
- **D** Labour efficiency variance = labour mix variance + labour yield variance.

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- LCV = (SH × SR) − (AH paid × AR): standard cost of the **actual** output minus the wages paid.
- LRV = AH **paid** × (SR − AR); the personnel manager negotiates rates.
- ITV = idle hours × SR = SR × (AH paid − AH worked): nil or adverse.
- LEV = SR × (SH − AH **worked**); the production manager or foreman answers for it.

- LMV (gang) = SR × (RSH − AH worked); LYV (sub-efficiency) = SR × (SH − RSH); RSH = hours worked in the standard ratio.
- Checks: LCV = LRV + ITV + LEV; LEV = LMV + LYV.
- Normal idle time is built into the standard; the idle time variance is for abnormal idle hours. (Cutting wages does not bring the electricity back.)

Answer key · Check yourself

1 **13500**

Idle time variance = (hours paid − hours worked) × standard rate = 150 × ₹90 = ₹13,500 (A).

2 **C** **Both have a labour cost variance of ₹12,000 (A) and a labour cost of ₹42 a tray, but Indore's is a rate variance (personnel manager) and Bhopal's an efficiency variance (production manager or foreman)**

SH = 6,000 × 0.25 = 1,500 hours; standard cost ₹2,40,000 (₹40 a tray). Indore: 1,500 × ₹168 = ₹2,52,000; LRV = (₹160 − ₹168) × 1,500 = ₹12,000 (A); LEV nil. Bhopal: 1,575 × ₹160 = ₹2,52,000; LRV nil; LEV = (1,500 − 1,575) × ₹160 = ₹12,000 (A). Same ₹42 a tray, different causes and different people responsible (SM 13.26).

3 **A ₹5,000 (A)**

LEV = (SH − AH worked) × SR = (1,000 − 1,050) × ₹100 = ₹5,000 (A). ₹10,000 (A) uses hours paid and double-counts the 50 idle hours, which belong in the idle time variance (₹5,000 A).

4 **B The personnel manager**

Labour rates are generally influenced by external factors, but the personnel manager is responsible for rate negotiation (SM 13.26). The production manager or foreman answers for efficiency.

5 **A The idle time variance is valued at the standard rate.**; **B The labour mix (gang) variance arises from a change in the combination of skilled, semi-skilled and unskilled workers.**; **D Labour efficiency variance = labour mix variance + labour yield variance.**

Rate variance is on hours paid; idle time at standard rate; the mix variance comes from the gang's composition; mix + yield = efficiency (SM 13.26–13.28).

13.4 Variable overhead variances

Why variable overheads differ from the standard for the output made: a rate paid per hour (expenditure) and hours used (efficiency), worked on actual hours worked.

In plain words

The fryers that ran late

Tapri's samosa factory in Indore fries in batches. Power, gas and consumable stores for the fryers rise and fall with the hours the fryers run: they are **variable overheads**, the indirect costs that vary with the level of production. Ritu's standard says a batch takes 2 fryer hours, and every fryer hour costs ₹60 of variable overheads.

> **Kesar:** Gas got cheaper this month. The overhead per hour fell from ₹60 to ₹59. We saved, no?
>
> **Ritu:** Per unit, please. We made 4,000 batches, which should take 8,000 hours. The fryers ran 8,400.
>
> **Kesar:** Four hundred hours. Sab adjust ho jayega.
>
> **Ritu:** It did not. Four hundred hours at ₹60 is ₹24,000.

Ritu's question is the same one she asks for material and labour: **what should the variable overheads for this month's output have cost?** 8,000 standard hours at ₹60 is ₹4,80,000. The fryers actually cost ₹4,95,600 (8,400 hours

at ₹59), so the **variable overhead cost variance** is ₹15,600 adverse.

Two things explain it. The rate: each of the 8,400 hours cost ₹1 less than the standard, a saving of ₹8,400 (the **expenditure variance**). The hours: the fryers ran 400 hours longer than 4,000 batches need, and each of those hours carried ₹60 of overheads, a loss of ₹24,000 (the **efficiency variance**). Cheaper gas saved ₹8,400; slow frying lost ₹24,000.

Think of it like

The cab billed by the minute

A cab that bills ₹10 a minute should cost ₹300 for a 30-minute trip. If the bill is ₹352, either the rate per minute went up (surge pricing: the expenditure variance), or the driver took the long route and the ride lasted 32 minutes (the efficiency variance), or both. You cannot blame the surge for the long route.

Key idea

Compare the actual variable overheads with the standard variable overheads for the **actual output**, then split the gap: the rate per hour, on the hours actually worked (expenditure), and the hours worked against the standard hours, at the standard rate (efficiency). The two must add back to the cost variance.

How it works

Variable overheads vary with the level of production. If they consist of indirect materials, they vary with the direct material used; if they depend on the hours worked, they vary with labour hours or machine hours; **if nothing is mentioned, take labour hours as the basis** (SM 13.36). The calculation is similar to labour cost variance, and the cost variance is divided into two parts: the **variable overhead expenditure variance** (the formula chart also calls it the **spending** variance, SM 13.36, and Fig 13.1 the "expenditure or **budget** variance", SM 13.13) and the **variable overhead efficiency variance** (SM 13.36).

Five steps from a rate card to two variances

The fryer month: standard 2 hours per batch at ₹60 per hour; actual 4,000 batches, 8,400 hours worked, ₹4,95,600 of variable overheads.

1. **Standard rate per hour (SR)** = budgeted variable overheads ÷ budgeted hours (SM 13.42). Here it is given: ₹60.

2. **Standard hours for actual output (SH)** = 4,000 batches × 2 hours = 8,000 hours. The standard variable overheads for actual production, SH × SR = 8,000 × ₹60 = **₹4,80,000**, are the **recovered** (absorbed) overheads.

3 **Variable overhead cost variance** = standard variable overheads for actual production − actual variable overheads = ₹4,80,000 − ₹4,95,600 = **₹15,600 (A)**.

4 **Expenditure (spending) variance** = standard variable overheads for actual hours − actual variable overheads = (8,400 × ₹60) − ₹4,95,600 = ₹5,04,000 − ₹4,95,600 = **₹8,400 (F)**.

5 **Efficiency variance** = standard variable overheads for production − standard variable overheads for actual hours = ₹4,80,000 − ₹5,04,000 = **₹24,000 (A)**, which is (8,000 − 8,400) × ₹60. Check: ₹8,400 (F) + ₹24,000 (A) = ₹15,600 (A).

The analysis is three amounts in a row, each changing one thing from the one before:

The variable overhead ladder, fryer month

(1) SH × SR = ₹4,80,000
standard VOH for actual output (recovered)

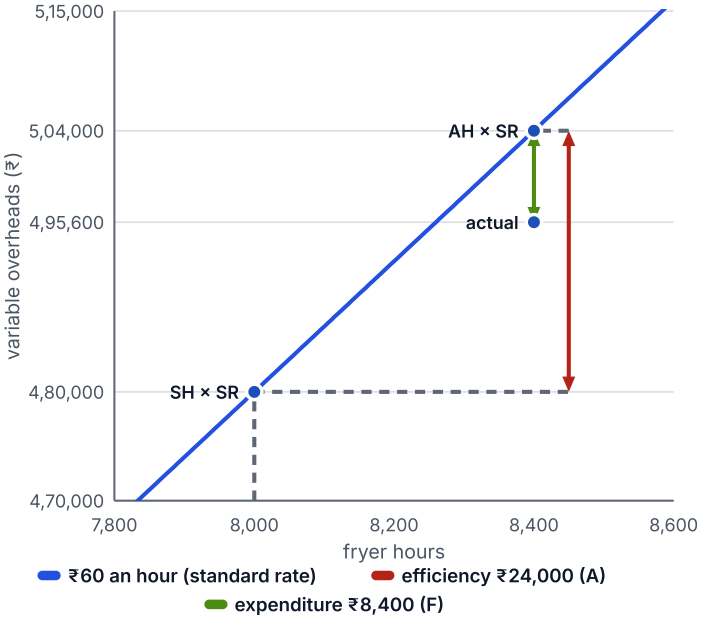
(2) AH × SR = ₹5,04,000
efficiency (1) – (2): ₹24,000 (A)
standard VOH for actual hours worked

(3) Actual VOH = ₹4,95,600
expenditure (2) – (3): ₹8,400 (F)
8,400 hours at ₹59

Rung (2) changes the hours, rung (3) the rate. A positive difference is favourable, a negative one adverse. Both together: (1) – (3) = cost variance ₹15,600 (A).

On a chart, the standard rate is a straight line from the origin: every hour adds ₹60 (the chart shows a close-up). The two variances are two moves away from the point where the actual output should have landed.

Fryer month: the standard rate line and the two variances



Blue: the standard rate line. The output should have used 8,000 hours (₹4,80,000); it used 8,400, which on the standard line is ₹5,04,000 (rose arrow: efficiency, adverse). The actual cost sat ₹8,400 below the line (green arrow: expenditure, favourable).

Interactive version: PassBook app, lesson Variable overhead variances

The vocabulary, in ICAI's words

Term	Meaning (SM 13.37, 13.42)
Standard hours (SH)	Hours required to produce the actual output
Actual hours (AH)	Actual hours taken to produce the actual output: in variable overhead formulas, the **actual hours worked**
Standard variable overheads for actual production	SH × SR: also called the recovered (absorbed) variable overheads
Standard variable overheads for actual hours	AH × SR: also called the standard overhead

Idle hours are paid but not worked. The variable overhead formulas use the hours **worked** (SM 13.36), and variable overheads are "generally" based on labour time worked (SM 13.8). So idle hours enter the labour idle time variance (labour variances, including idle time) and stay out of AH here: the Study Material has no idle time variance for variable overheads.

Formats and formulas

Every variable overhead answer starts with two working notes: the standard rate (per hour, and per unit if needed) and the standard hours for actual output. *Reproduce the formulas and the working notes from memory.* A variance

computed as standard minus actual is favourable when positive and adverse when negative; write (F) or (A) after every figure (SM 13.12).

∑ **Formula**

Variable overhead cost variance

$$\text{Cost} = (SH \times SR) - \text{Actual VOH}$$

SH = standard hours for actual output, SR = standard variable overhead rate per hour. In words: standard variable overheads for actual production – actual variable overheads (SM 13.36). With a rate per unit: actual output × standard rate per unit – actual variable overheads.

∑ **Formula**

Variable overhead expenditure (spending) variance

$$\text{Expenditure} = (SR - AR) \times AH$$
$$= (SR \times AH) - (AR \times AH)$$

AH = actual hours worked, AR = actual variable overhead rate per hour (actual variable overheads ÷ AH). In words: standard variable overheads for actual hours – actual variable overheads (SM 13.36).

∑ **Formula**

Variable overhead efficiency variance

$$\text{Efficiency} = (SH - AH) \times SR$$
$$= (SH \times SR) - (AH \times SR)$$

In words: standard variable overheads for production – standard variable overheads for actual hours (SM 13.36).

∑ **Formula**

The working notes and the check

$$SR = \frac{BV}{BH} \qquad SH = \frac{BH}{BO} \times AO$$

BV = budgeted variable overheads, BH = budgeted hours, BO = budgeted output, AO = actual output. A rate per unit is BV ÷ BO (SM 13.42). Check: cost variance = expenditure variance + efficiency variance (SM 13.78).

The Study Material's summary names the three amounts **recovered overhead** (SH × SR), **standard overhead** (AH × SR) and **actual overhead**: cost = recovered – actual, expenditure = standard – actual, efficiency = recovered – standard (SM 13.54).

The working table

Particulars	Hours	Rate (₹)	Amount (₹)
Recovered: standard hours for actual output × SR	SH	SR	xxx
Standard: actual hours worked × SR	AH	SR	xxx
Actual variable overheads	AH	AR	xxx

Efficiency = row 1 − row 2; expenditure = row 2 − row 3; cost = row 1 − row 3.

Worked examples

Worked example **The packing line and the power cut**

Tapri's premix plant packs cartons on a labour-paced line. Variable overheads are absorbed on direct labour hours. For October:

Particulars	Budget	Actual
Production (cartons)	12,000	11,400
Standard time per carton	1.5 hours	–
Variable overheads (₹)	3,60,000	3,57,000
Labour hours paid	–	18,000

Of the hours paid, 500 were idle when the power failed for two shifts.

Required:

(i) Calculate the variable overhead cost variance.

(ii) Calculate the variable overhead expenditure and efficiency variances.

(iii) Verify your answer, and show what goes wrong if the 18,000 hours paid are used.

THE WORKING

Working notes

1. Budgeted hours = 12,000 × 1.5 = 18,000 hours. Standard rate = ₹3,60,000 ÷ 18,000 = **₹20 per hour** (₹30 per carton).
2. Standard hours for actual output = 11,400 × 1.5 = **17,100 hours**.
3. Actual hours worked = 18,000 − 500 idle = **17,500 hours**; actual rate = ₹3,57,000 ÷ 17,500 = ₹20.40 per hour.

Particulars	Hours	Rate (₹)	Amount (₹)
Recovered (SH × SR)	17,100	20.00	3,42,000
Standard (AH × SR)	17,500	20.00	3,50,000
Actual	17,500	20.40	3,57,000

(i) VOH cost variance = ₹3,42,000 − ₹3,57,000 = **₹15,000 (A)**. (Check by units: 11,400 × ₹30 = ₹3,42,000.)

(ii) Expenditure variance = (SR × AH) − actual = ₹3,50,000 − ₹3,57,000 = **₹7,000 (A)**. Efficiency variance = (SH − AH) × SR = (17,100 − 17,500) × ₹20 = **₹8,000 (A)**.

(iii) Check: ₹7,000 (A) + ₹8,000 (A) = ₹15,000 (A), the cost variance. With the 18,000 hours paid, the standard for actual hours becomes ₹3,60,000: the expenditure variance turns into ₹3,000 (F) and the

efficiency variance into ₹18,000 (A). The total still checks, which is why the error survives a careless check. The power failure stopped the line, so no variable overheads were incurred in those 500 hours; the SM's formulas use hours worked.

Worked example

Kiln hours or labour hours? The intern picks a base

Tapri's kulhad unit fires clay cups in trays of 100. Its variable overheads (gas for the kilns and power for their fans) rise and fall with the hours the kilns fire, not with the hours the potters work. The standard cost card per tray shows direct labour 3 hours at ₹80 per hour, kiln time 0.5 kiln-hour, and variable overheads of ₹120 per kiln-hour. In November, 2,600 trays were made in 8,000 labour hours worked and 1,250 kiln-hours, and variable overheads came to ₹1,62,500.

The draft prepared by Ritu's intern absorbs the variable overheads on labour hours instead, at ₹60 per tray ÷ 3 hours = ₹20 per labour hour.

Required:

(i) Calculate the variable overhead cost, expenditure and efficiency variances on kiln-hours, and the actual rate per kiln-hour.

(ii) Calculate the same variances as the intern's labour-hour basis would show.

(iii) Calculate the labour efficiency variance, and advise Ritu which variable overhead analysis to report.

THE WORKING

Working notes. Standard variable overheads per tray = 0.5 × ₹120 = ₹60. Standard kiln-hours for actual output = 2,600 × 0.5 = **1,300**; standard labour hours = 2,600 × 3 = **7,800**.

(i) On kiln-hours (SR ₹120)

Particulars	Kiln-hours	Rate (₹)	Amount (₹)
Recovered (SH × SR)	1,300	120.00	1,56,000
Standard (AH × SR)	1,250	120.00	1,50,000
Actual	1,250	130.00	1,62,500

- Cost = ₹1,56,000 − ₹1,62,500 = **₹6,500 (A)**.
- Expenditure = ₹1,50,000 − ₹1,62,500 = **₹12,500 (A)**; actual rate = ₹1,62,500 ÷ 1,250 = **₹130 per kiln-hour**.
- Efficiency = (1,300 − 1,250) × ₹120 = **₹6,000 (F)**.

Check: ₹12,500 (A) + ₹6,000 (F) = ₹6,500 (A).

(ii) On the intern's labour hours (SR ₹20)

- Expenditure = 8,000 × ₹20 − ₹1,62,500 = ₹1,60,000 − ₹1,62,500 = **₹2,500 (A)**.

- Efficiency = (7,800 − 8,000) × ₹20 = **₹4,000 (A)**.
- Cost = 2,600 × ₹60 − ₹1,62,500 = **₹6,500 (A)**, the same total.

(iii) Labour efficiency = (7,800 − 8,000) × ₹80 = **₹16,000 (A)**.

Variance (₹)	Kiln-hour basis	Labour-hour basis
Expenditure	12,500 (A)	2,500 (A)
Efficiency	6,000 (F)	4,000 (A)
Cost	**6,500 (A)**	**6,500 (A)**

Advice: report the kiln-hour analysis. The cost variance does not depend on the base, but the split does. The overheads are incurred when the kilns fire, and the kilns fired 50 hours less than standard for 2,600 trays (fuller loads), a ₹6,000 (F) efficiency; each kiln-hour cost ₹10 more than standard, a ₹12,500 (A) expenditure that points to the gas price or the burners. The intern's base charges the overheads to the potters' extra 200 hours, which did not burn any gas, and hides the rise in the cost per kiln-hour. The Study Material bases the variable overhead variances on labour hours or machine hours, whichever the overheads depend on, and takes labour hours only "if nothing is mentioned specifically" (SM 13.36); here the overheads are stated to vary with kiln-hours.

Exam angle

In the exam

Where variable overhead variances appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, variable overhead variances were asked in 14 of them: 4 past papers, 2 RTPs and 8 MTPs. Most of them set it beside fixed overhead or labour variances in the same question.

- **Descriptive, 4 to 8 marks.** Actual hours from the efficiency variance, then the actual rate and the cost variance: September 2025 Q1(c) (4 marks), RTP May 2026 Q15. Cost variance on units only: May 2025 Q4(b) (6 marks, with fixed overheads). Cost variance with material and labour from budget and actual totals: January 2026 Q4(a) (7 marks). Budgeted hours after normal down time: MTP September 2025 Series II Q1(c) (4 marks). Rate split out of a total overhead rate: MTP September 2026 Series II Q4(a) (8 marks). Cost variance next to material and labour variances from a standard cost card: MTP May 2024 Series I Q5(a) (6 marks), MTP September 2024 Series I Q4(b) (8 marks), MTP January 2026 Series I Q3(a) (9 marks, with idle time in the labour data).
- **Multiple choice, 2 marks each**: standard rate from the efficiency variance (January 2025 MCQ 14); expenditure and efficiency (MTP January 2026 Series I MCQ 12); efficiency nil and a two-tier actual rate (MTP January 2025 Series I MCQs 3–4); cost

variances from budget totals (MTP May 2025 Series II MCQ 11, MTP September 2026 Series II MCQ 14); actual overheads from the two variances (RTP September 2025 Case Scenario 1); semi-variable split (MTP May 2024 Series II Case Scenario, the Study Material's own Case Scenario 1).

In the exam

The wording traps ICAI used

- "Variable overheads are based on direct labour hours" plus a given efficiency variance: solve for actual hours first (RTP May 2026 Q15; SM Practical Problem 15).
- "Actual production expressed in standard hours" is SH itself (MTP September 2025 Series II Q1(c)).
- "₹50 per hour for the first 10,000 hours, ₹60 for additional hours" (MTP January 2025 Series I): the actual overheads are a sum of two slabs; the expenditure variance still uses the standard ₹50 on all actual hours.
- "Indicate the nature of variance": every figure needs its (F) or (A).

Trick

Three amounts, three subtractions

Write SH × SR, AH × SR and actual in a column. Top minus middle is efficiency, middle minus bottom is expenditure, top minus bottom is cost. Condition: AH is the hours worked, and SR is the standard variable rate, never the actual rate.

Answer frame

A full-marks variable overhead answer

1. **Working notes**: the standard rate per hour (and per unit), the standard hours for actual output, the actual hours worked (paid − idle), and the actual rate if needed.
2. **The three amounts**: recovered (SH × SR), standard (AH × SR), actual.
3. **Each variance**: formula in symbols, figures, result with (F) or (A).
4. **The check**: expenditure + efficiency = cost.
5. **A one-line comment** if asked: the efficiency variance comes from the hours worked against the standard hours (when overheads are absorbed on labour hours, the same hours as the labour efficiency variance); the expenditure variance from the variable overheads spent per hour worked against the standard rate.

Three answers to the same question

Loses marks

"Efficiency variance = ₹8,000."

No formula, no hours, no (A).

Passes

"Efficiency = (SH − AH) × SR = (17,100 − 18,000) × ₹20 = ₹18,000 (A)."

Formula and nature, but on hours paid: the method mark survives, the figure does not.

Full marks

Working notes for SR, SH and hours worked (18,000 − 500), the three amounts, both variances with (F) or (A), and the check against the cost variance.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the budget that stood in for the standard

"VOH cost variance = budgeted ₹3,60,000 − actual ₹3,57,000 = ₹3,000 (F)." The budget is for 12,000 cartons; the line made 11,400. **Fix:** the cost variance compares actual overheads with the standard for the **actual output**: 11,400 × ₹30 = ₹3,42,000, a variance of ₹15,000 (A).

⚠ **Common mistake**

The idle hours that ran up the gas bill

Using the 18,000 hours paid instead of the 17,500 worked in the packing-line example makes the expenditure variance ₹3,000 (F) instead of ₹7,000 (A) and the efficiency variance ₹18,000 (A) instead of ₹8,000 (A): ₹10,000 (500 idle hours × ₹20) changes sides, and the check still balances. **Fix:** the variable overhead formulas use actual hours worked (SM 13.36); idle hours belong to the labour idle time variance only.

⚠ **Common mistake**

The actual rate inside the efficiency variance

(SH − AH) × AR mixes two variances. **Fix:** efficiency is always at the **standard** rate; the actual rate appears only in the expenditure variance.

Practice

Check yourself

1 In June the premix line's variable overheads cost ₹21.50 for each of the 6,400 labour hours worked, against a standard of ₹20 an hour; the output was worth 6,000 standard hours. What is the variable overhead expenditure variance, in rupees (adverse)?

Answer: _____

2 Case scenario. Ritu's budget for the chai premix line was 8,000 units in 16,000 labour hours, with variable overheads of ₹2,40,000. The line made 7,500 units in 15,600 hours, and variable overheads came to ₹2,40,240. Dev reads the draft prepared by Ritu's intern: "Variable overheads were only ₹240 over budget." Which of these is correct?

- **A** Variable overhead cost variance ₹240 (A)
- **B** Variable overhead expenditure variance ₹6,240 (A); efficiency variance ₹9,000 (A)
- **C** Variable overhead expenditure variance ₹9,000 (A); efficiency variance ₹6,240 (A)
- **D** Variable overhead cost variance ₹15,240 (A); efficiency variance ₹6,000 (F)

3 Labour efficiency variance is ₹12,000 (A) at a standard wage rate of ₹60 per hour. Variable overheads are absorbed on the same labour hours at ₹15 per hour. What is the variable overhead efficiency variance?

- **A** ₹48,000 (A)
- **B** ₹12,000 (A)
- **C** It cannot be found without the actual variable overheads
- **D** ₹3,000 (A)

4 Why does labour idle time create no variable overhead variance in the Study Material's formulas?

- **A** Because idle time is always abnormal and goes to costing profit and loss
- **B** Because the variable overhead rate is lower than the wage rate
- **C** Because variable overhead variances use actual hours worked, and idle hours are not worked
- **D** Because idle time is included in the standard hours

5 Which data are needed to calculate the variable overhead expenditure variance? Select all that apply.

- **A** The actual hours worked
- **B** The standard variable overhead rate per hour
- **C** The actual variable overheads
- **D** The standard hours for actual output

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- VOH cost variance = (SH × SR) − actual VOH: standard for **actual output** minus actual.
- Expenditure (spending) = (SR − AR) × AH = (SR × AH) − actual VOH.
- Efficiency = (SH − AH) × SR: hours, at the **standard** rate.
- Check: expenditure + efficiency = cost.
- SR = budgeted VOH ÷ budgeted hours; SH = budgeted hours ÷ budgeted output × actual output.
- AH = hours **worked**; idle hours give a labour idle time variance, not a VOH variance.

- Basis: labour hours unless the question says machine hours or material (SM 13.36). No hours given: cost variance only, on a rate per unit.
- Fryer month: cheaper gas ₹8,400 (F), slow frying ₹24,000 (A), net ₹15,600 (A).

Answer key · Check yourself

1 **9600**

Expenditure = (SR − AR) × AH = (₹20 − ₹21.50) × 6,400 = ₹9,600 (A), the same as AH × SR − actual = ₹1,28,000 − ₹1,37,600. The 400 extra hours are the efficiency variance, (6,000 − 6,400) × ₹20 = ₹8,000 (A); the expenditure variance is measured on the hours actually worked, not the standard hours.

2 **B** **Variable overhead expenditure variance ₹6,240 (A); efficiency variance ₹9,000 (A)**

SR = ₹2,40,000 ÷ 16,000 = ₹15; SH = 7,500 × 2 = 15,000 hours. Expenditure = 15,600 × ₹15 − ₹2,40,240 = ₹6,240 (A); efficiency = (15,000 − 15,600) × ₹15 = ₹9,000 (A); cost = 7,500 × ₹30 − ₹2,40,240 = ₹15,240 (A). The intern's ₹240 compares actual with the budget for 8,000 units.

3 **D** **₹3,000 (A)**

The labour variance gives 12,000 ÷ 60 = 200 extra hours; at ₹15 they cost ₹3,000 (A). Actual overheads are needed for the expenditure variance, not for efficiency.

4 **C Because variable overhead variances use actual hours worked, and idle hours are not worked**

The formulas use AH = actual hours worked (SM 13.36). Idle hours are paid, so they enter the labour idle time variance; the Study Material defines no idle time variance for variable overheads. Idle time is not always abnormal (normal idle time exists), and it is never part of the standard hours for output.

5 **A The actual hours worked**; **B The standard variable overhead rate per hour**; **C The actual variable overheads**

Expenditure = (SR × AH) – actual variable overheads. Standard hours for actual output enter the efficiency and cost variances, not the expenditure variance.

13.5 Fixed overhead variances

Why absorbed fixed overheads differ from actual: spending against the budget (expenditure) and output against the budget (volume), with volume split into efficiency, capacity and calendar.

In plain words

The factory that worked every hour and still fell short

Tapri's samosa factory has fixed overheads of ₹6,00,000 a month: rent, supervisors' salaries, depreciation of the fryers. They stay the same whatever the factory makes. To put them into the cost of each batch, Ritu fixes a rate in advance from the budget: 15,000 batches a month (30,000 machine hours in 25 days), so **₹40 a batch**, or ₹20 a machine hour. Each batch made carries ₹40; that is the **absorbed** fixed overhead.

In October the factory worked 24 days (Diwali took one), made 14,100 batches, and spent ₹5,88,000.

> **Kesar:** We spent ₹12,000 less than the budget, and the fryers ran all 30,000 budgeted hours. Why is the fixed overhead report ₹24,000 adverse?
>
> **Ritu:** Per unit, please. Each batch carries ₹40. The budget expected 15,000 batches; we made 14,100. The 900 missing batches could not carry their ₹40.
>
> **Kesar:** Diwali. Sab adjust ho jayega.
>
> **Ritu:** Diwali took one day. Slow frying took the rest.

The 14,100 batches absorbed 14,100 × ₹40 = ₹5,64,000, against the ₹5,88,000 spent: the **fixed overhead cost variance** is ₹24,000 adverse. Two questions explain it. **Did we spend what the budget said?** ₹6,00,000 budgeted, ₹5,88,000 spent: ₹12,000 favourable, the **expenditure variance**. **Did we make what the budget said?** ₹5,64,000 absorbed against ₹6,00,000 budgeted: ₹36,000 adverse, the **volume variance**, fixed overhead left unrecovered because output fell short.

Volume has three causes. **Calendar**: one working day lost, ₹24,000 (A). **Capacity**: in the 24 days the fryers ran 30,000 hours, 1,200 more than 24 days were budgeted for: ₹24,000 (F). **Efficiency**: those 30,000 hours produced only what 28,200 standard hours should: ₹36,000 (A). The longer shifts made up for Diwali; slow frying made the month adverse.

Think of it like

The leased auto-rickshaw

Raju leases an auto for ₹30,000 a month and plans 25 days × 10 hours × 2 trips an hour = 500 trips, so each fare must carry ₹60 of lease. If the owner raises the lease, that is expenditure. A city bandh that shuts him out for a day is calendar. Driving 8 hours instead of 10 on working days is capacity. Traffic that allows 1.5 trips an hour is efficiency. Every trip he does not make leaves ₹60 of lease unrecovered: the volume variance.

Key idea

Fixed overheads are absorbed on output at a rate fixed from the budget. The cost variance (absorbed – actual) splits into **expenditure** (budgeted – actual: spending) and **volume** (absorbed – budgeted: output), and volume splits into **efficiency**, **capacity** and **calendar**. A fixed overhead variance is an under- or over-absorption.

How it works

The recovery of fixed overheads depends on capacity utilisation: a company that produces less than the projected utilisation cannot recover all the budgeted fixed overheads (SM 13.38). With fixed overheads of ₹10,000, the cost per unit is ₹1.25 at 8,000 units and ₹0.91 at 11,000 (SM 13.39), so standard costing fixes a

predetermined rate on a predetermined capacity. ICAI's own words: overhead variance is nothing but under or over-absorption of overhead (SM 13.40).

ICAI's definitions, in its order (SM 13.40–13.41):

- **Fixed overhead cost variance**: the difference between actual fixed overhead and absorbed fixed overhead. It is divided into two parts:
- (A) **Fixed overhead expenditure variance**: the difference between the actual fixed overhead incurred and budgeted fixed overhead. Fig 13.1 calls it the "expenditure or **budget** variance" (SM 13.13).
- (B) **Fixed overhead volume variance**: variance in fixed overhead which arises due to the volume of production. It is further divided into three:
  - (a) **Efficiency variance**: the difference between fixed overhead absorbed and standard fixed overhead.
  - (b) **Capacity variance**: the difference between standard fixed overhead and budgeted overhead.
  - (c) **Calendar variance**: arises due to the difference in number of actual working days and the standard working days.

When the calendar variance is computed, the capacity variance is modified to a **revised capacity variance** = (actual hours − revised budgeted hours) × standard fixed rate per hour (SM 13.41). Revised budgeted hours are the budgeted hours for the days actually worked: budgeted hours ÷ budgeted days × actual days (SM 13.76).

Seven steps through the Diwali month

Budget: ₹6,00,000; 25 days; 30,000 machine hours; 15,000 batches. Actual: ₹5,88,000; 24 days; 30,000 hours; 14,100 batches.

1. **Rates** (SM 13.42): per hour ₹6,00,000 ÷ 30,000 = ₹20; per batch ₹6,00,000 ÷ 15,000 = ₹40; per day ₹6,00,000 ÷ 25 = ₹24,000. Standard time = 30,000 ÷ 15,000 = 2 hours a batch.

2. **Standard hours for actual output** = 14,100 × 2 = 28,200. **Absorbed** = 28,200 × ₹20 = **₹5,64,000** (also 14,100 × ₹40).

3. **Standard fixed overhead** = actual hours × SR = 30,000 × ₹20 = **₹6,00,000**. **Revised budgeted fixed overhead** = (30,000 ÷ 25 × 24 = 28,800 hours) × ₹20 = **₹5,76,000**.

4. **Cost variance** = absorbed − actual = ₹5,64,000 − ₹5,88,000 = **₹24,000 (A)**. **Expenditure** = budgeted − actual = ₹6,00,000 − ₹5,88,000 = **₹12,000 (F)**. **Volume** = absorbed − budgeted = ₹5,64,000 − ₹6,00,000 = **₹36,000 (A)**.

5. **Efficiency** = (SH − AH) × SR = (28,200 − 30,000) × ₹20 = **₹36,000 (A)**.

6 **Revised capacity** = (AH – revised budgeted hours) × SR = (30,000 – 28,800) × ₹20 = **₹24,000 (F)**. **Calendar** = (actual days – budgeted days) × rate per day = (24 – 25) × ₹24,000 = **₹24,000 (A)**.

7 **Checks** (SM 13.42): volume = efficiency + capacity + calendar: ₹36,000 (A) + ₹24,000 (F) + ₹24,000 (A) = ₹36,000 (A). Cost = expenditure + volume: ₹12,000 (F) + ₹36,000 (A) = ₹24,000 (A).

Without the calendar split the capacity variance is (30,000 – 30,000) × ₹20 = nil: the factory did work its budgeted hours. The calendar variance shows the lost day, the revised capacity variance the longer shifts that made up for it.

The analysis is five amounts in a row, each changing one thing from the one before:

The fixed overhead ladder, Diwali month

(1) SH × SR = ₹5,64,000
absorbed: standard hours for actual output

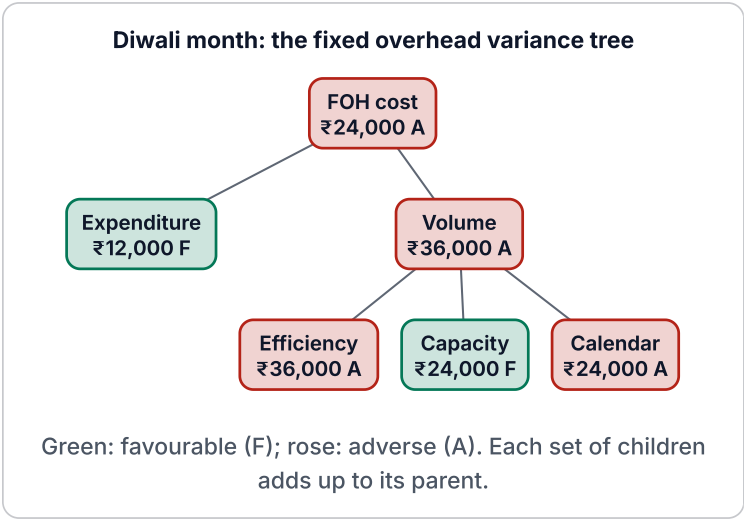
(2) AH × SR = ₹6,00,000
efficiency (1) – (2): ₹36,000 (A)
standard fixed overhead for actual hours

(3) RBH × SR = ₹5,76,000
capacity (2) – (3): ₹24,000 (F)
revised budget: budgeted hours for 24 days

(4) BH × SR = ₹6,00,000 calendar (3) – (4): ₹24,000 (A)
budgeted fixed overhead

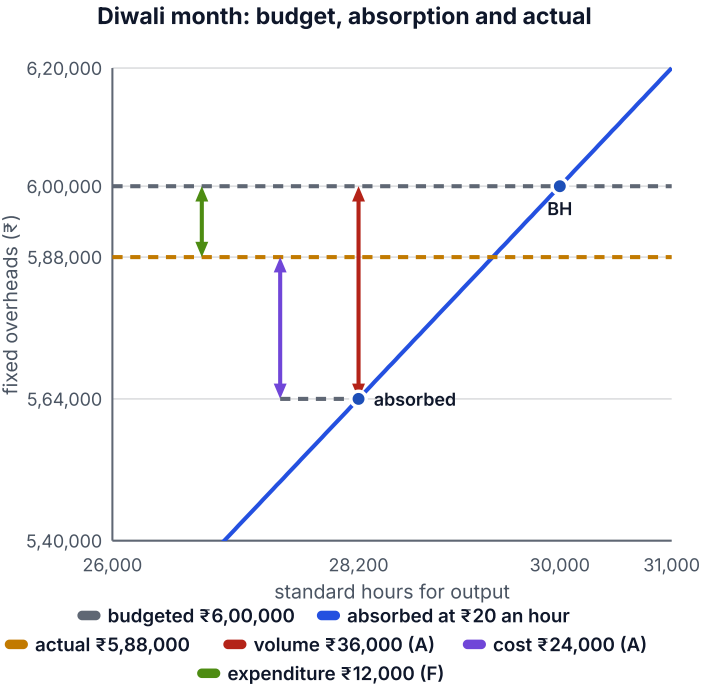
(5) Actual = ₹5,88,000
expenditure (4) – (5): ₹12,000 (F)
fixed overhead incurred

Rungs (2) to (4) change the hours, rung (5) the spending. Volume = (1) – (4) = ₹36,000 (A); cost = (1) – (5) = ₹24,000 (A). RBH: revised budgeted hours.



Green: favourable (F); rose: adverse (A). Each set of children adds up to its parent.

On a chart, the budget is a flat line (fixed cost), while absorption is a line from the origin rising ₹20 with every standard hour. They meet only at the budgeted 30,000 hours; anywhere else, the gap between them is the volume variance.



Grey dashed: the budget; blue: absorption at ₹20 a standard hour; amber dashed: actual spending. At 28,200 standard hours the factory absorbed ₹5,64,000. Rose arrow: volume (absorbed below budget); violet arrow: cost (absorbed below actual); green arrow: expenditure (actual below budget). At BH, the budgeted 30,000 hours, absorption would equal the budget.

Interactive version: PassBook app, lesson Fixed overhead variances

Formats and formulas

Reproduce the formulas and the working-note table from memory. SH = standard hours for actual output, AH =

actual hours, BH = budgeted hours, RBH = revised budgeted hours, SR = standard fixed overhead rate per hour, BF = budgeted fixed overheads, AF = actual fixed overheads, AD and BD = actual and budgeted working days. Each variance is the first amount minus the second: positive is (F), negative is (A).

∑ **Formula**

Fixed overhead cost variance

$$\text{Cost} = (SH \times SR) - AF$$

Absorbed (recovered) fixed overheads – actual fixed overheads. Absorbed = SH × SR, or actual output × standard rate per unit (SM 13.41–13.42).

∑ **Formula**

Expenditure and volume variances

$$\text{Expenditure} = BF - AF$$

$$\text{Volume} = (SH \times SR) - BF$$

Budgeted – actual; absorbed – budgeted. BF = BH × SR (SM 13.41).

∑ **Formula**

Efficiency, capacity and calendar variances

$$\text{Efficiency} = SR\,(SH - AH)$$

$$\text{Capacity} = SR\,(AH - BH)$$

$$\text{Calendar} = SR_{\text{day}} \times (AD - BD)$$

SR per day = standard fixed overhead rate per day. With a calendar variance, use the revised capacity variance = SR (AH − RBH) (SM 13.41).

∑ **Formula**

Rates and revised budgeted hours

$$SR = \frac{BF}{BH} \qquad RBH = \frac{BH}{BD} \times AD$$

Rate per unit = BF ÷ budgeted output; rate per day = BF ÷ BD. Revised budgeted fixed overheads (the Study Material also calls them "possible fixed overheads") = RBH × SR = BF ÷ BD × AD (SM 13.42, 13.76, 13.87).

∑ **Formula**

Verification

$$\text{Cost} = \text{Expenditure} + \text{Volume}$$

And volume = efficiency + capacity + calendar (SM 13.42).

The summary in the Study Material uses the names **recovered**, **standard**, **budgeted** and **actual** overhead: cost = recovered − actual, expenditure = budgeted − actual, volume = recovered − budgeted, efficiency = recovered − standard, capacity = standard − budgeted (SM 13.54).

The working notes

ICAI's solutions open with a Budget / Actual table (Illustration 11, SM 13.45):

Working note	Budget	Actual
Fixed overheads (₹)	xxx	xxx
Working days	xxx	xxx
Hours	xxx	xxx
Output (units)	xxx	xxx

Then: standard hours for actual output, standard rate per hour, per unit and per day, and the revised budgeted hours.

Worked examples

Worked example **Two cup sizes, one kiln down and a Sunday shift**

Tapri's kulhad unit fires its cups in a shed of 6 kilns. The monthly budget: 24 working days, each kiln firing 16 hours a day, and fixed overheads (kiln-shed rent, depreciation, supervisors) of ₹6,91,200. Standard kiln time is 1.5 kiln-hours for 1,000 small kulhads and 2.4 kiln-hours for 1,000 large ones.

In November the shed worked 25 days, the last one a Sunday added for wedding orders. One kiln was under repair for 5 of those days; every other kiln fired its full 16 hours each day. Output was 7,00,000 small and 5,20,000 large kulhads, and actual fixed overheads were ₹7,02,000.

Required:

(i) Calculate the standard kiln-hours for the actual output.

(ii) Calculate the fixed overhead efficiency, capacity and calendar variances.

(iii) Calculate the fixed overhead volume, expenditure and cost variances, verify them, and say what each part of the volume variance tells Ritu about November.

THE WORKING

Working notes

Particulars	Budget	Actual
Fixed overheads (₹)	6,91,200	7,02,000
Working days	24	25
Kiln-hours	6 × 16 × 24 = 2,304	6 × 16 × 25 − 16 × 5 = 2,320

1. Standard rate = ₹6,91,200 ÷ 2,304 = **₹300 per kiln-hour**; rate per day = ₹6,91,200 ÷ 24 = **₹28,800**.
2. Revised budgeted hours = 2,304 ÷ 24 × 25 = **2,400**.

(i) Standard hours for actual output = 700 × 1.5 + 520 × 2.4 = 1,050 + 1,248 = **2,298 kiln-hours**. With two products, standard hours (not units) are the common measure of output.

(ii)

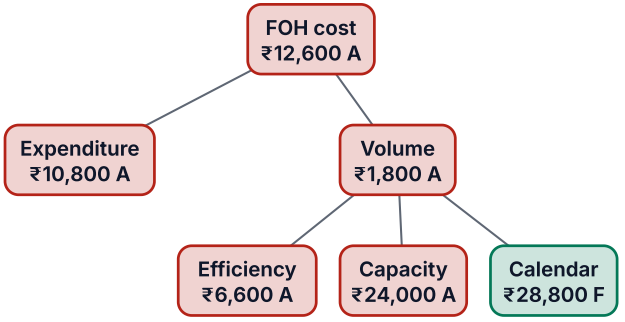
- Efficiency = SR (SH − AH) = ₹300 × (2,298 − 2,320) = **₹6,600 (A)**.
- Revised capacity = SR (AH − RBH) = ₹300 × (2,320 − 2,400) = **₹24,000 (A)**.
- Calendar = (actual days − budgeted days) × rate per day = (25 − 24) × ₹28,800 = **₹28,800 (F)**.

(iii)

- Volume = absorbed − budgeted = (2,298 × ₹300 = ₹6,89,400) − ₹6,91,200 = **₹1,800 (A)**.
- Expenditure = budgeted − actual = ₹6,91,200 − ₹7,02,000 = **₹10,800 (A)**.
- Cost = absorbed − actual = ₹6,89,400 − ₹7,02,000 = **₹12,600 (A)**.

Verification: efficiency + capacity + calendar = ₹6,600 (A) + ₹24,000 (A) + ₹28,800 (F) = ₹1,800 (A), the volume variance; expenditure + volume = ₹10,800 (A) + ₹1,800 (A) = ₹12,600 (A), the cost variance.

November in the kiln shed: variance tree



Green: favourable (F); rose: adverse (A). The Sunday shift more than paid for the broken kiln; slow firing and overspending made the month adverse.

What the three parts say: the **calendar** variance is the extra Sunday (one day more than budgeted); the **revised capacity** variance is the kiln under repair, 16 hours × 5 days = 80 kiln-hours below

what 25 days should give, × ₹300 = ₹24,000; the **efficiency** variance is the 22 kiln-hours fired beyond the standard for the cups made. Without the calendar split, capacity = ₹300 × (2,320 − 2,304) = ₹4,800 (F), and ₹4,800 (F) + ₹6,600 (A) = ₹1,800 (A), the same volume variance: the Sunday and the breakdown simply net off. Use one set or the other, never the unrevised capacity with the calendar variance.

Worked example

A flexible budget, a boiler repair and the October report

Tapri's premix plant prepares an annual flexible budget of overheads. It plans 1,80,000 cartons a year (its normal capacity), and works 25 days a month:

Annual activity	1,50,000 cartons	1,80,000 cartons
Total overheads (₹)	33,00,000	36,90,000

In October a boiler repair shut the plant for 3 days, and it made 13,800 cartons. October's overheads came to ₹3,08,600; of these, rent, salaries, insurance and depreciation (the fixed heads) were ₹1,16,000 and the rest varied with output.

Required:

(i) Separate the budgeted overheads into fixed and variable, and find the standard rates.

(ii) Calculate the variable overhead cost variance.

(iii) Calculate the fixed overhead cost, expenditure, volume and calendar variances, and explain the part of the volume variance the calendar does not explain.

(iv) Calculate the total overhead cost variance.

THE WORKING

Working notes

1. Variable rate = change in overheads ÷ change in activity = (₹36,90,000 − ₹33,00,000) ÷ (1,80,000 − 1,50,000) = **₹13 per carton**. Annual fixed overheads = ₹36,90,000 − 1,80,000 × ₹13 = **₹13,50,000**.
2. Fixed rate = ₹13,50,000 ÷ 1,80,000 = **₹7.50 per carton**. Monthly budget = ₹13,50,000 ÷ 12 = **₹1,12,500**; per day = ₹1,12,500 ÷ 25 = **₹4,500**.
3. Actual fixed = ₹1,16,000; actual variable = ₹3,08,600 − ₹1,16,000 = **₹1,92,600**.
4. Days worked = 25 − 3 = 22. Revised budgeted ("possible") fixed overheads = ₹1,12,500 ÷ 25 × 22 = **₹99,000**.

(i) Variable ₹13 a carton; fixed ₹13,50,000 a year, ₹7.50 a carton.

(ii) VOH cost variance = 13,800 × ₹13 − ₹1,92,600 = ₹1,79,400 − ₹1,92,600 = **₹13,200 (A)**.

(iii)

- Absorbed fixed overheads = 13,800 × ₹7.50 = ₹1,03,500.
- Cost = ₹1,03,500 − ₹1,16,000 = **₹12,500 (A)**.
- Expenditure = ₹1,12,500 − ₹1,16,000 = **₹3,500 (A)**.
- Volume = ₹1,03,500 − ₹1,12,500 = **₹9,000 (A)**.
- Calendar = (22 − 25) × ₹4,500 = **₹13,500 (A)**, which is ₹99,000 − ₹1,12,500.

Check: ₹3,500 (A) + ₹9,000 (A) = ₹12,500 (A). The rest of the volume variance is ₹1,03,500 − ₹99,000 = **₹4,500 (F)**: in 22 days the budget expected 15,000 ÷ 25 × 22 = 13,200 cartons and the plant made 13,800, 600 more, × ₹7.50. The question gives no hours, so this cannot be split into efficiency and capacity.

(iv) Total overhead cost variance = ₹13,200 (A) + ₹12,500 (A) = **₹25,700 (A)** = (₹1,79,400 + ₹1,03,500) − ₹3,08,600.

Two traps in this layout: the monthly budget is one-twelfth of the annual fixed amount, and a flexible budget (flexible budgets) at two levels is split like a semi-variable cost (change in cost ÷ change in activity) before any rate is computed.

Interactive version: PassBook app, lesson Fixed overhead variances

Worked example

Bunty works backwards from three variances

On an audit visit, Bunty has only these figures for the kulhad unit's machine shop for June: actual fixed overheads ₹3,10,500 for 14,400 actual machine hours; fixed overhead expenditure variance ₹10,500 (A); capacity variance ₹12,000 (A); efficiency variance ₹9,000 (A). Standard machine time is 0.75 hour per batch of cups, and no days are given.

Required: Calculate:

(i) the budgeted fixed overheads;

(ii) the standard rate per machine hour and the budgeted machine hours;

(iii) the standard hours for actual output, the actual output and the budgeted output;

(iv) the volume and cost variances, and the overhead to be written off to Costing Profit and Loss Account.

THE WORKING

(i) Expenditure = budgeted − actual, so budgeted = ₹3,10,500 − ₹10,500 = **₹3,00,000**.

(ii) Capacity = SR × AH − SR × BH = SR × 14,400 − ₹3,00,000 = −₹12,000, so SR × 14,400 = ₹2,88,000 and **SR = ₹20 per machine hour**. Budgeted hours = ₹3,00,000 ÷ ₹20 = **15,000**.

(iii) Efficiency = SR (SH − AH): ₹20 × (SH − 14,400) = −₹9,000, so SH = 14,400 − 450 = **13,950 hours**. Actual output = 13,950 ÷ 0.75 = **18,600 batches**; budgeted output = 15,000 ÷ 0.75 = **20,000 batches**.

(iv) Volume = SR (SH − BH) = ₹20 × (13,950 − 15,000) = **₹21,000 (A)**; check: capacity ₹12,000 (A) + efficiency ₹9,000 (A). Cost = absorbed − actual = 13,950 × ₹20 − ₹3,10,500 = ₹2,79,000 − ₹3,10,500 = **₹31,500 (A)**; check: expenditure ₹10,500 (A) + volume ₹21,000 (A). The ₹31,500 is the fixed overhead under-absorbed in June, written off to the Costing Profit and Loss Account.

Exam angle

In the exam

Where fixed overhead variances appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, fixed overhead variances were asked in 16 of them: 4 past papers, 3 RTPs and 9 MTPs.

- **Descriptive, 4 to 10 marks.** Days, man-hours per day and output per man-hour, then expenditure,

volume, calendar and cost: May 2026 Q3(b) (6 marks). Units and days only, with variable overheads: May 2025 Q4(b) (6 marks). Cost variance with the variable overhead working-back: September 2025 Q1(c). Capacity and efficiency after normal down time: MTP September 2025 Series II Q1(c) (4 marks). Working back from given variances to budgeted output, capacity and efficiency: MTP May 2025 Series II Q3(a) (10 marks), MTP September 2026 Series II Q4(a) (8 marks). Cost variance next to material and labour: MTP May 2024 Series I Q5(a), MTP September 2024 Series I Q4(b), MTP January 2026 Series I Q3(a).

- **Multiple choice, 2 marks each**: volume variance from units (January 2025 MCQ 11; RTP January 2026 MCQ 3); budget worked back from the expenditure variance (RTP September 2024 Case Scenario 2); actual from the expenditure variance (RTP September 2025 Case Scenario 1); cost, volume and expenditure with an extra quality-control cost in the actual (MTP January 2025 Series I MCQs 1, 2, 5); actual overheads when the cost variance is nil (MTP January 2025 Series II MCQ 15); semi-variable split, a festival holiday and a calendar variance (MTP May 2024 Series II Case Scenario, the Study Material's Case Scenario 1); cost variances from totals (MTP May 2025 Series II MCQ 11, MTP September 2026 Series II MCQ 14).
- **Theory**: "Discuss the various types of fixed overhead variances" is one of the chapter's own questions (SM 13.57, answer: paragraph 7.4). The

chapter's Practical Problem 7 asks for the production volume variance and the overhead expenses variance (Under the hood).

- **Old scheme** (practice): 10-mark questions in November 2020 Q3(a) (working back), January 2021 (down time), December 2021 Q5(a) (semi-variable, holiday) and November 2023 Q5(a) (hours from the variable overhead efficiency variance).

In the exam

The wording traps ICAI used

- "Output (budgeted) per man-hour" and "actual output per man-hour" (May 2026 Q3(b)): build budgeted and actual hours and units first; budgeted 25 × 12,000 = 3,00,000 hours.
- "The company could have operated for 20 days but there was a festival holiday" (SM Case Scenario 1): actual days 19; possible fixed overheads = budget ÷ 20 × 19.
- An annual budget with a monthly actual: divide by 12 before anything else.
- "Fixed overhead budget variance" (old-scheme January 2021, Premier Industries) is the expenditure variance: Fig 13.1 names it "Expenditure or Budget Variance" (SM 13.13).
- "Production was at 60% capacity" (MTP September 2026 Series II Q4(a)): budgeted hours = standard hours ÷ 60%.

Trick

Five amounts, six subtractions

Write SH × SR, AH × SR, RBH × SR, BH × SR and actual in a column. Neighbours give efficiency, capacity, calendar and expenditure; first minus fourth is volume; first minus last is cost. Condition: SR is the standard fixed rate per hour from the budget, and RBH is used only when days differ.

Answer frame

"Discuss the various types of fixed overhead variances" (theory)

1. **Why they arise**: the recovery of fixed overheads depends on capacity utilisation; producing less than the projected level leaves part of the budgeted fixed overheads unrecovered (SM 13.38).
2. **Fixed overhead cost variance**: the difference between actual fixed overhead and absorbed fixed overhead; overhead variance is an under- or over-absorption.
3. **Expenditure variance**: actual fixed overhead incurred against budgeted fixed overhead.
4. **Volume variance**: arises due to the volume of production; absorbed against budgeted. It shows the cost of failing to produce at the planned activity, or the benefit of producing above it (SM 13.39).
5. Volume splits into **efficiency** (absorbed against standard fixed overhead), **capacity** (standard fixed

overhead against budgeted) and **calendar** (actual against standard working days).

6. When the calendar variance is computed, the capacity variance is revised to (actual hours − revised budgeted hours) × standard rate.
7. Verification: cost = expenditure + volume; volume = efficiency + capacity + calendar.
8. The formula chart, or a tree.

Three answers to the same question

Loses marks

"Capacity variance = ₹4,800 (F); calendar variance = ₹28,800 (F)."

The extra Sunday is counted twice, and the volume no longer checks.

Passes

"Capacity = ₹300 × (2,320 − 2,400) = ₹24,000 (A)."

Correct, but 2,400 comes from nowhere: no working note for revised budgeted hours.

Full marks

The Budget / Actual working-note table; SH, SR, rate per day and RBH; each variance with formula and (F) or (A); both verification lines.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the day lost twice

Computing the calendar variance and then the capacity variance on the original budgeted hours. The kiln shed's Sunday sits inside both: ₹4,800 (F) capacity + ₹28,800 (F) calendar + ₹6,600 (A) efficiency = ₹27,000 (F), against a true volume variance of ₹1,800 (A). **Fix:** with a calendar variance, capacity is SR (AH – revised budgeted hours) (SM 13.41).

⚠ **Common mistake**

The expenditure variance measured against absorption

"Expenditure = absorbed ₹5,64,000 – actual ₹5,88,000 = ₹24,000 (A)." That is the cost variance. **Fix:** expenditure compares actual with the **budget** (₹6,00,000), because fixed overheads are not expected to move with output; only absorption does.

⚠ **Common mistake**

The annual budget in a monthly question

Using the annual ₹13,50,000 as October's fixed budget gives an expenditure variance of ₹12,34,000 (F) instead of ₹3,500 (A). **Fix:** split the budget into fixed and variable first, then convert it to the period of the actuals (÷ 12).

Practice

Check yourself

1 Tapri's kulhad unit absorbs fixed overheads at ₹15 a tray, on a monthly budget of ₹6,00,000. Its fixed overhead volume variance for March was ₹45,000 (A). How many trays did it make?

Answer: _____

2 Case scenario. Dev reviews the samosa factory's May report. Budget: 25 days, 800 machine hours a day, fixed overheads ₹4,00,000. Actual: 23 days, 19,000 machine hours, output worth 18,500 standard hours. The report prepared by Ritu's intern shows a calendar variance of ₹32,000 (A). Which capacity variance belongs beside it?

- **A** ₹20,000 (A)
- **B** ₹12,000 (F)
- **C** ₹10,000 (A)
- **D** ₹32,000 (F)

3 Case scenario. The samosa factory budgets fixed overheads of ₹3,60,000 for 18,000 machine hours. In August it worked 17,000 hours, its output was worth 16,500 standard hours, and it spent ₹3,48,000 on fixed overheads. The report prepared by Ritu's intern says "Fixed overhead expenditure variance ₹18,000 (A)". Which statement is correct?

- **A** The intern is right: the factory overspent by ₹18,000
- **B** The expenditure variance is ₹30,000 (A): absorbed less budgeted
- **C** The expenditure variance is ₹8,000 (A): the budget for 17,000 hours less actual
- **D** The expenditure variance is ₹12,000 (F); the intern's ₹18,000 (A) is the cost variance

4 Which variance is "the difference between standard fixed overhead and budgeted overhead" in the Study Material?

- **A** Efficiency variance
- **B** Capacity variance
- **C** Expenditure variance
- **D** Volume variance

5 Which of these make up the fixed overhead volume variance? Select all that apply.

- **A** Efficiency variance
- **B** Capacity variance
- **C** Calendar variance
- **D** Expenditure variance

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- FOH cost variance = absorbed (SH × SR) − actual: an under- or over-absorption.
- Expenditure = budgeted − actual (spending). Volume = absorbed − budgeted (output).
- Volume = efficiency SR (SH − AH) + capacity SR (AH − BH) + calendar (actual days − budgeted days) × rate per day.
- With a calendar variance, capacity = SR (AH − RBH); RBH = BH ÷ budgeted days × actual days.
- Checks: cost = expenditure + volume; volume = efficiency + capacity + calendar.
- Rates: per hour BF ÷ BH, per unit BF ÷ budgeted output, per day BF ÷ budgeted days; monthly budget = annual ÷ 12.

- Five amounts: SH × SR, AH × SR, RBH × SR, BH × SR, actual. (Diwali month: worked every budgeted hour, still ₹24,000 (A).)

Answer key · Check yourself

1 **37000**

Budgeted output = ₹6,00,000 ÷ ₹15 = 40,000 trays. Volume = (actual − budgeted output) × rate, so actual − 40,000 = −₹45,000 ÷ ₹15 = −3,000: 37,000 trays. The ₹45,000 is the fixed overhead the 3,000 missing trays could not carry.

2 **B** **₹12,000 (F)**

SR = ₹4,00,000 ÷ 20,000 = ₹20; revised budgeted hours = 800 × 23 = 18,400. Revised capacity = ₹20 × (19,000 − 18,400) = ₹12,000 (F). ₹20,000 (A) uses the original 20,000 hours and counts the two lost days again; ₹10,000 (A) is the efficiency variance. Check: ₹10,000 (A) + ₹12,000 (F) + ₹32,000 (A) = ₹30,000 (A) = 18,500 × ₹20 − ₹4,00,000.

3 **D The expenditure variance is ₹12,000 (F); the intern's ₹18,000 (A) is the cost variance**

SR = ₹3,60,000 ÷ 18,000 = ₹20. Expenditure = budgeted − actual = ₹3,60,000 − ₹3,48,000 = ₹12,000 (F). The intern compared actual with absorbed (16,500 × ₹20 = ₹3,30,000): that is the cost variance, ₹18,000 (A). ₹30,000 (A) is the volume variance; ₹8,000 (A) compares actual with the standard fixed overhead for actual hours, which no fixed overhead variance does.

4 **B Capacity variance**

Capacity = standard fixed overhead (AH × SR) − budgeted (BH × SR) (SM 13.41). Efficiency compares absorbed with standard; expenditure compares budgeted with actual; volume compares absorbed with budgeted.

5 **A Efficiency variance**; **B Capacity variance**; **C Calendar variance**

Volume = efficiency + capacity + calendar (SM 13.42). Expenditure is the other half of the cost variance.

13.6 Reconciling budgeted and actual profit, and accounting for variances

How every variance adds up to the gap between standard and actual cost, and between budgeted and actual profit; and where the variances go in the books.

In plain words

The bridge Dev wanted on one slide

Tapri's masala chai premix plant budgeted 10,000 tins for March, at ₹320 a tin and a standard cost of ₹260: a budgeted profit of **₹6,00,000**. The month closed at **₹5,37,600**.

> **Kesar:** Only ₹62,400 short. Sab adjust ho jayega.
>
> **Dev:** The board wants one slide: why ₹62,400? Discounted at 12%, it is less, of course.
>
> **Ritu:** It is one month, Dev. Per unit, please. We sold 500 tins fewer, at ₹10 more a tin, and the costs ran over the standard.

Ritu's slide is a **reconciliation**: a bridge from the budgeted profit to the actual profit in which every rupee of the gap has one cause, and the causes add up to the gap.

- **Volume.** 9,500 tins were sold, not 10,000. Each tin not sold took away its standard contribution of ₹120 (price ₹320 less variable cost ₹200): **₹60,000 adverse**.

- **Selling price.** The 9,500 tins sold at ₹330, ₹10 above the standard: **₹95,000 favourable**.
- **Costs.** The cost of 9,500 tins came in above what the standard allows for 9,500 tins: the material, labour and variable overhead variances and the fixed overhead expenditure variance, measured as in the earlier lessons, total **₹97,400 adverse**.

−₹60,000 + ₹95,000 − ₹97,400 = −₹62,400. The bridge closes.

The second question is bookkeeping. In a standard costing system stocks sit in the books at standard cost, so the variances are not hidden inside stock values: they are **disposed of**, normally by transfer to the **Costing Profit and Loss Account** (SM 13.5).

Think of it like

The train's delay report

The Indore express reaches Delhi 47 minutes late. The report: late start, 10 minutes; signal at Kota, 25; slow track near Mathura, 15; 3 minutes made up on the last stretch. Each piece has an owner, and the pieces add to exactly 47. A report that says only "47 minutes late" helps nobody, and one whose pieces add to 52 is wrong.

Key idea

A reconciliation explains the whole gap, budget to actual or standard to actual, as a list of variances, each measured with everything else at standard; they must add to the gap. Variances are then disposed of, normally to the Costing Profit and Loss Account, because stocks are carried at standard cost.

How it works

Standard costing starts with setting standards and ends with reporting variances to management for corrective action (SM 13.2). Its process has five steps, the last of which is the disposition of variances (SM 13.5):

1. **Setting of standards**: the standards to be achieved.
2. **Ascertainment of actual costs** for each component of cost, from books of account, material invoices, wage sheets, charge slips.
3. **Comparison of actual cost with standard cost**: variances are determined.
4. **Investigation of the reasons for variances**: performance is evaluated and appropriate actions are taken.

5 **Disposition of variances**: variances are disposed of by transferring them to the relevant accounts (Costing Profit and Loss Account) as per the accounting method (plan) adopted.

Bridge 1: standard cost of actual output to actual cost

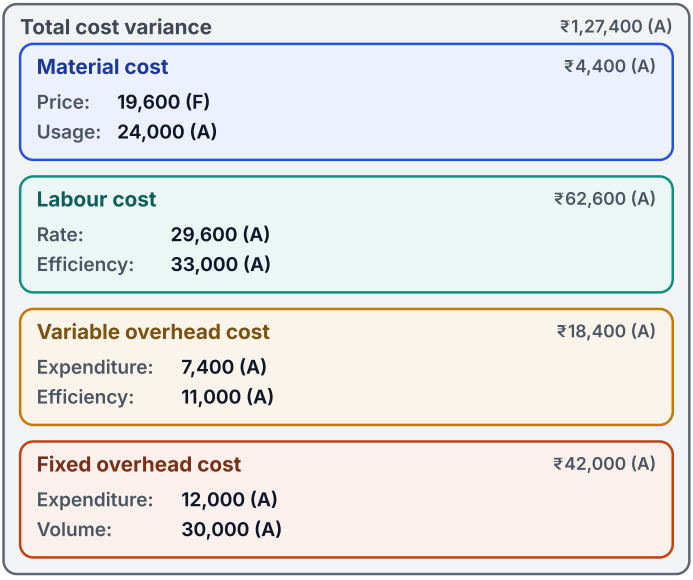
Variances are classified as **revenue variances**, actual sales against budgeted sales, and **cost variances** (SM 13.12). The total cost variance divides into material, labour and overhead cost variances, and overheads into variable and fixed (SM 13.13, Fig 13.1). So the first reconciliation is:

Standard cost of actual output ± cost variances = actual cost.

The masala tins: standard cost of 9,500 tins at ₹260 = ₹24,70,000; actual cost ₹25,97,400 (material ₹7,64,400,

labour ₹9,17,600, variable overheads ₹3,03,400, fixed overheads ₹6,12,000). Total cost variance ₹1,27,400 (A):

Masala tins: the total cost variance, element by element



Each box is one element of cost, its total at the right of the title bar and its two parts below. The four elements add to ₹1,27,400 (A), the gap between ₹24,70,000 standard and ₹25,97,400 actual.

Each element was computed exactly as in its own lesson (material, labour, variable overheads, fixed overheads), with the standard: 2 kg of material at ₹40, 1.5 labour hours at ₹60, variable overheads ₹20 and fixed overheads ₹40 per labour hour (₹6,00,000 budgeted for 15,000 hours); actual 19,600 kg bought and used at ₹39, and 14,800 hours at ₹62. (When purchases and consumption

differ, the price variance basis changes the actual cost the bridge reaches: Under the hood.)

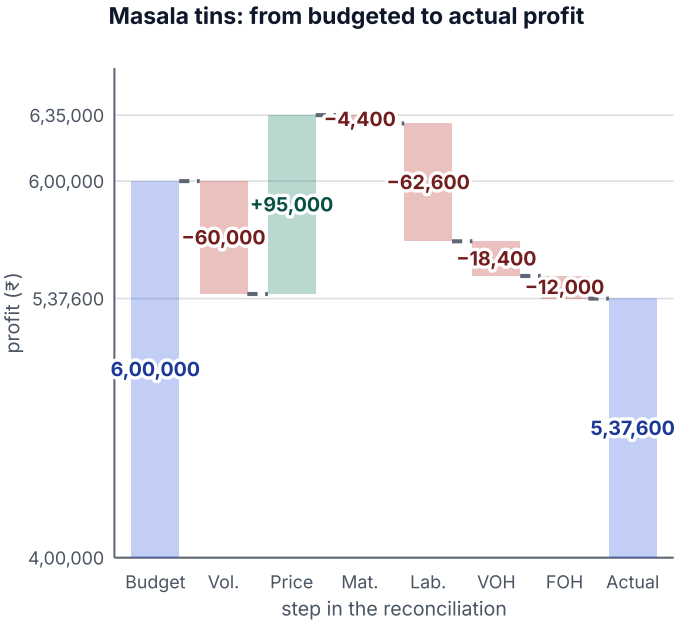
Bridge 2: budgeted profit to actual profit

The Study Material's own budget-against-actual comparison (SM 15.92) goes through a **flexible budget**: the budget recast for the actual volume at standard prices and costs (flexible budgets). Three columns then do all the work: the master budget (10,000 tins), the flexible budget (9,500 tins) and the actual (9,500 tins). In the exam they are one table (Format 1 below); here it is split in two so that it reads on a phone.

₹	Master (10,000)	Flexible (9,500)
Sales	32,00,000	30,40,000
Material	8,00,000	7,60,000
Labour	9,00,000	8,55,000
VOH	3,00,000	2,85,000
Contribution	**12,00,000**	**11,40,000**
FOH	6,00,000	6,00,000
Profit	**6,00,000**	**5,40,000**

₹	Actual (9,500)	Variance from flexible
Sales	31,35,000	95,000 (F)
Material	7,64,400	4,400 (A)
Labour	9,17,600	62,600 (A)
VOH	3,03,400	18,400 (A)
Contribution	**11,49,600**	**9,600 (F)**
FOH	6,12,000	12,000 (A)
Profit	**5,37,600**	**2,400 (A)**

- **Master budget → flexible budget** (₹6,00,000 → ₹5,40,000): the effect of **volume** alone, 500 tins × ₹120 standard contribution = ₹60,000 (A). The fixed overheads stay at ₹6,00,000 in both: a flexible budget flexes only what varies.
- **Flexible budget → actual** (₹5,40,000 → ₹5,37,600): the selling price (₹95,000 (F)) and the cost variances. Fixed overheads appear only as the expenditure variance, ₹12,000 (A). So the cost variances here total ₹97,400 (A): Bridge 1's ₹1,27,400 (A) less the fixed overhead volume variance of ₹30,000 (A), which this bridge counts inside the volume step (see Under the hood).



The vertical axis starts at ₹4,00,000. Blue: budgeted and actual profit; green: a step that raises profit; rose: a step that lowers it. Vol. = volume ₹60,000 (A); Price ₹95,000 (F); Mat. = material ₹4,400 (A); Lab. = labour ₹62,600 (A); VOH ₹18,400 (A); FOH = fixed overhead expenditure ₹12,000 (A).

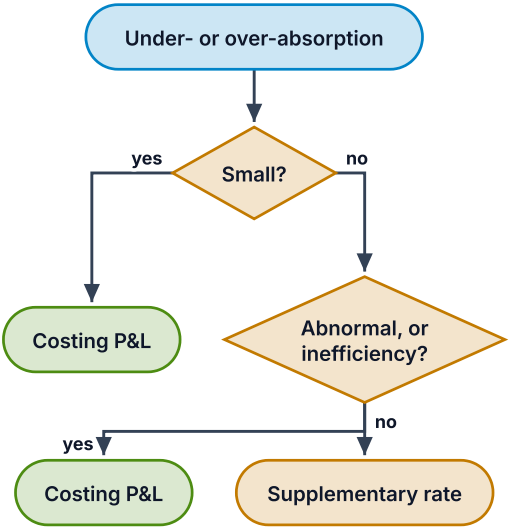
The Study Material notes that for revenue (sales) variances the signs reverse: favourable when actual is more than budgeted (SM 13.12). It does not compute named sales variances for your attempt, and none of the ICAI questions read for this lesson asks for them by name, so label the two revenue lines by what they measure: the effect of volume and the effect of selling price.

Accounting for variances: where they go

For variances in general, the Study Material says that they are transferred to the relevant accounts (Costing Profit and Loss Account) as per the plan adopted (SM 13.5), and that controllable variances are best transferred to the Costing Profit and Loss Account (MCQ 8, SM 13.56). The detailed rules are given for overheads, whose variance "is nothing but under or over-absorption" (SM 13.40), in under- and over-absorption of overheads (SM 4.42–4.46):

The variance is	Treatment (Study Material)
Small	Transfer to the Costing Profit and Loss Account; product costs are not changed, as they represent normal cost (SM 4.44)
Large, from a wrong estimate made for unavoidable reasons (e.g. a rise in the price of indirect material)	Adjust product costs with a **supplementary rate**, spread over cost of sales, finished goods and work-in-progress (SM 4.44–4.46)
From abnormal reasons, or factory inefficiency	Transfer to the Costing Profit and Loss Account (SM 4.45–4.46)
Controllable (any element)	Best disposed of by transfer to the Costing Profit and Loss Account (SM 13.56, MCQ 8, answer 13.67)

Disposing of a variance



Green: written off to the Costing Profit and Loss Account in the period. Amber: a large amount from an unavoidable wrong estimate, spread by a supplementary rate over cost of sales, finished goods and work-in-progress (SM 4.44–4.46).

Note

The flowchart in the Study Material reads differently

The flowchart printed on SM 4.43 labels its second question "Due to wrong estimation and abnormal reasons" and sends a "yes" to the Costing P&L. The text that follows (SM 4.44–4.45) and Illustration 8 (SM 4.45–4.46) send a large difference from an estimate that was wrong for unavoidable reasons to a supplementary rate, and only abnormal causes and inefficiency to the Costing P&L. The flow above follows the text; write the text's rule in an answer.

In the books, an under-absorption (an adverse overhead variance) is a debit balance and an over-absorption a credit balance (SM 4.42); the Costing Profit and Loss Account is debited with cost of sales, under-absorbed overheads and abnormal losses, and credited with sales, over-absorbed overheads and abnormal gains (SM 7.5). Because stocks are valued at standard cost, a standard costing system "eliminates any variations in profit due to changes in stock values from one period to another" (SM 13.50).

Two more rules from the Study Material: responsibility centres are answerable for the adverse variances they could have controlled, and if uncontrollable variances are significant and persistent, the standard may need revision (SM 13.12). The difference between the original standard cost and the revised standard cost of actual production is the **revision variance** (SM 13.53).

Formats and formulas

Reproduce these layouts from memory. Each variance is written with (F) or (A); in a profit bridge, (F) is added and (A) deducted.

Format 1: Flexible budget compared with actual (SM 15.92)

₹	Master	Flex.	Actual	Var.
Sales	xxx	xxx	xxx	xxx
Each variable cost	xxx	xxx	xxx	xxx
Contribution	**xxx**	**xxx**	**xxx**	**xxx**
Fixed overheads	xxx	xxx	xxx	xxx
Profit	**xxx**	**xxx**	**xxx**	**xxx**

Master = master budget at budgeted volume; Flex. = flexible budget at actual volume, at standard cost; Var. = variance, actual against flexible, each with (F) or (A). The Study Material's answer also shows the standard per unit beside the flexible budget, and a total variable cost line above contribution. In the answer book write the column heads in full.

Format 2: Statement reconciling budgeted profit and actual profit

Particulars	Amount (₹)
Budgeted profit	xxx
Less / Add: effect of volume (units × standard contribution)	xxx
Flexible budget profit	**xxx**
Add / Less: effect of selling price	xxx
Add / Less: material price and usage variances	xxx
Add / Less: labour rate, idle time and efficiency variances	xxx
Add / Less: variable overhead expenditure and efficiency variances	xxx
Add / Less: fixed overhead expenditure variance	xxx
Actual profit	**xxx**

ICAI prints no pro-forma for this statement: this layout is the variance column of Format 1 set out as a statement, with the volume step added at the top and each cost variance split into its parts. Whatever the layout, start at the budgeted profit, show every step with its (F) or (A), and end at the actual profit.

Format 3: Income statement showing the actual gross margin

ICAI's layout in MTP January 2026 Series II and MTP September 2026 Series I:

Particulars	Amount (₹)
Sales	xxx
Less: Standard cost of goods sold	xxx
Standard gross margin	**xxx**
Less: Adverse variances (each listed)	xxx
Add: Favourable variances (each listed)	xxx
Actual gross margin	**xxx**

∑ **Formula**

The reconciliation identities

$$AC = SC - NV$$

AC = actual cost; SC = standard cost of actual output; NV = net cost variance = MCV + LCV + VOCV + FOCV, the material, labour, variable overhead and fixed overhead cost variances, with (F) as plus and (A) as minus (SM 13.13).

Interactive version: PassBook app, lesson Reconciling budgeted and actual profit, and accounting for variances

Formula

The two revenue steps

$$\text{Volume} = (AU - BU) \times c$$

$$\text{Price} = (AP - SP) \times AU$$

AU, BU = actual and budgeted units sold; c = standard contribution per unit; AP, SP = actual and standard selling price. Volume is the master budget profit against the flexible budget profit.

Definition

Controllable and uncontrollable variances

Controllable variances are those which can be controlled under the normal operating conditions if a responsibility centre takes preventive measures and acts prudently. **Uncontrollable variances** are those which occur due to conditions which are beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place. (SM 13.11–13.12)

Definition

Revision variance

The difference between the original standard cost and the revised standard cost of actual production. (SM 13.53)

Worked examples

Worked example

Rusk packs: flexible budget and the profit bridge

Tapri's bakery line makes elaichi rusk packs. Standard cost per pack: material 0.4 kg at ₹50 per kg; labour 0.2 hour at ₹75 per hour; variable overheads ₹25 per labour hour. Budgeted fixed overheads are ₹1,80,000 a month, and the standard selling price is ₹60. The master budget for May was 30,000 packs, made and sold. Actual results for May:

Particulars	Actual
Packs made and sold	27,000
Sales	₹16,74,000
Material used: 11,000 kg	₹5,39,000
Labour: 5,600 hours	₹4,31,200
Variable overheads	₹1,44,200
Fixed overheads	₹1,77,000

Required:

(i) Prepare the flexible budget for May and compare it with the actual results.

(ii) Calculate the material, labour and variable overhead variances and the fixed overhead expenditure variance.

(iii) Prepare a statement reconciling the budgeted profit with the actual profit.

THE WORKING

Working notes: standard variable cost per pack = ₹20 + ₹15 + ₹5 = ₹40; standard contribution = ₹60 − ₹40 = **₹20** a pack. Actual price ₹16,74,000 ÷ 27,000 = ₹62; actual material price ₹49 per kg; actual wage rate ₹77 per hour. Standard quantity for actual output = 10,800 kg; standard hours = 5,400.

(i) Flexible budget compared with actual. In the answer book this is one table (Format 1); it is split in two here.

₹	Master (30,000)	Flexible (27,000)
Sales	18,00,000	16,20,000
Material	6,00,000	5,40,000
Labour	4,50,000	4,05,000
VOH	1,50,000	1,35,000
Contribution	**6,00,000**	**5,40,000**
FOH	1,80,000	1,80,000
Profit	**4,20,000**	**3,60,000**

₹	Actual (27,000)	Variance from flexible
Sales	16,74,000	54,000 (F)
Material	5,39,000	1,000 (F)
Labour	4,31,200	26,200 (A)
VOH	1,44,200	9,200 (A)
Contribution	**5,59,600**	**19,600 (F)**
FOH	1,77,000	3,000 (F)
Profit	**3,82,600**	**22,600 (F)**

(ii) Variances

- Material price = (₹50 − ₹49) × 11,000 = **₹11,000 (F)**; usage = (10,800 − 11,000) × ₹50 = **₹10,000 (A)**; cost ₹1,000 (F).
- Labour rate = (₹75 − ₹77) × 5,600 = **₹11,200 (A)**; efficiency = (5,400 − 5,600) × ₹75 = **₹15,000 (A)**; cost ₹26,200 (A).
- VOH expenditure = 5,600 × ₹25 − ₹1,44,200 = **₹4,200 (A)**; efficiency = (5,400 − 5,600) × ₹25 = **₹5,000 (A)**; cost ₹9,200 (A).
- FOH expenditure = ₹1,80,000 − ₹1,77,000 = **₹3,000 (F)**.

(iii) Statement reconciling budgeted profit and actual profit

Particulars	(₹)	(₹)
Budgeted profit		4,20,000
Less: effect of volume (3,000 packs × ₹20)		60,000
Flexible budget profit		**3,60,000**
Add: effect of selling price (27,000 × ₹2)		54,000
Add: material price variance	11,000	
Less: material usage variance	(10,000)	1,000
Less: labour rate variance	(11,200)	
Less: labour efficiency variance	(15,000)	(26,200)
Less: VOH expenditure variance	(4,200)	
Less: VOH efficiency variance	(5,000)	(9,200)
Add: FOH expenditure variance		3,000
Actual profit		**3,82,600**

Check: ₹16,74,000 − (₹5,39,000 + ₹4,31,200 + ₹1,44,200 + ₹1,77,000) = ₹3,82,600. Profit is

₹37,400 below budget: the lost volume (₹60,000) and the labour and overhead overruns outweighed the higher price.

Worked example **Samosa packs: the freezer empties, the store fills**

Tapri's samosa factory uses standard costing, and carries both its stores and its finished packs at standard cost. Standard cost per frozen pack: filling and dough 0.6 kg at ₹50; labour 0.2 hour at ₹90; factory overheads absorbed at ₹60 per standard labour hour. The selling price is ₹85 a pack. December opened with 6,000 packs in the freezer. During December:

Particulars	Actual
Packs made	40,000
Packs sold	44,000
Filling and dough bought: 26,000 kg	₹12,74,000
Filling and dough used	24,400 kg
Labour: 8,200 hours	₹7,46,200
Factory overheads	₹4,70,000

The price variance is taken on the quantity purchased.

Required:

(i) Calculate the material price and usage variances, the labour rate and efficiency variances and the factory overhead cost variance.

(ii) Prepare an income statement for December showing the standard and the actual gross margin.

(iii) Value the closing stocks of materials and of finished packs, and verify the actual gross margin from the actual costs.

THE WORKING

Working notes: standard cost per pack = ₹30 + ₹18 + ₹12 = **₹60**. For 40,000 packs made: standard quantity 24,000 kg, standard hours 8,000. Actual price ₹12,74,000 ÷ 26,000 = ₹49 per kg; actual wage rate ₹7,46,200 ÷ 8,200 = ₹91 per hour. Closing packs = 6,000 + 40,000 − 44,000 = 2,000; closing material = 26,000 − 24,400 = 1,600 kg.

(i)

- Material price (on purchases) = (₹50 − ₹49) × 26,000 = **₹26,000 (F)**; usage = (24,000 − 24,400) × ₹50 = **₹20,000 (A)**.
- Labour rate = (₹90 − ₹91) × 8,200 = **₹8,200 (A)**; efficiency = (8,000 − 8,200) × ₹90 = **₹18,000 (A)**.
- Factory overhead = absorbed (8,000 × ₹60 = ₹4,80,000) − actual ₹4,70,000 = **₹10,000 (F)**.

Net variance: ₹26,000 + ₹10,000 − ₹20,000 − ₹8,200 − ₹18,000 = **₹10,200 (A)**.

(ii) Income statement for December

Particulars	Amount (₹)
Sales (44,000 × ₹85)	37,40,000
Less: Standard cost of goods sold (44,000 × ₹60)	26,40,000
Standard gross margin	**11,00,000**
Less: Adverse variances: material usage 20,000; labour rate 8,200; labour efficiency 18,000	46,200
Add: Favourable variances: material price 26,000; factory overhead 10,000	36,000
Actual gross margin	**10,89,800**

The 44,000 packs sold are 6,000 from November's freezer and 38,000 of December's 40,000; at standard cost every pack costs ₹60, whichever month made it.

(iii) Stocks and verification

Closing material = 1,600 kg × ₹50 = **₹80,000** at standard price (the ₹1,600 saved on those 1,600 kg is already in the ₹26,000 price variance). Closing

packs = 2,000 × ₹60 = **₹1,20,000**; opening packs were 6,000 × ₹60 = ₹3,60,000.

Particulars	Amount (₹)
Materials bought	12,74,000
Less: Closing material at standard price	(80,000)
Labour	7,46,200
Factory overheads	4,70,000
Add: Opening packs at standard	3,60,000
Less: Closing packs at standard	(1,20,000)
Cost charged against December's sales	**26,50,200**

Actual gross margin = ₹37,40,000 − ₹26,50,200 = **₹10,89,800**, as in (ii). Every variance of December, including the price saving on material still in the store, goes to December's Costing Profit and Loss Account (SM 13.5); neither stock carries any of it.

Worked example **Day shift, night shift, and a plant "on standard"**

Ritu's standard for a carton of chai premix: material 0.5 kg at ₹40; labour 0.75 hour at ₹24; variable overheads

₹8 per labour hour. Fixed overheads are budgeted at ₹1,00,000 for the month on 8,000 cartons and absorbed per carton. February's actuals:

Particulars	Day shift	Night shift
Cartons made	4,600	3,800
Material used (kg)	2,250	2,050
Labour hours worked	3,600	2,700
Wage rate paid (₹ an hour)	24	26
Variable overheads (₹, metered)	27,000	22,950

The purchase department bought exactly the 4,300 kg used, at ₹39.50 a kg. Fixed overheads for the plant were ₹97,000.

> **Kesar:** The plant report nets to almost nothing. We are on standard. Next.

Required:

(i) Calculate the material usage, labour and variable overhead variances for each shift, the material price variance, and the fixed overhead expenditure and volume variances.

(ii) Reconcile each shift's standard cost of output with its actual cost at standard material price, and the plant's standard cost of actual output with its actual cost.

(iii) Comment on Kesar's conclusion.

THE WORKING

Working notes: standard per carton: material ₹20, labour ₹18, variable overheads ₹6 (variable ₹44), fixed overheads ₹1,00,000 ÷ 8,000 = ₹12.50; total ₹56.50. Standard quantity and hours: day 2,300 kg and 3,450 hours; night 1,900 kg and 2,850 hours.

(i) Variances

Variance (₹)	Day shift	Night shift
Material usage (SQ − AQ) × ₹40	2,000 (F)	6,000 (A)
Labour rate (₹24 − AR) × AH	nil	5,400 (A)
Labour efficiency (SH − AH) × ₹24	3,600 (A)	3,600 (F)
VOH expenditure (AH × ₹8 − actual)	1,800 (F)	1,350 (A)
VOH efficiency (SH − AH) × ₹8	1,200 (A)	1,200 (F)
Total	**1,000 (A)**	**7,950 (A)**

Plant: material price = (₹40 − ₹39.50) × 4,300 = **₹2,150 (F)**, the purchase department's; fixed overhead expenditure = ₹1,00,000 − ₹97,000 = **₹3,000 (F)**; volume = (8,400 − 8,000) × ₹12.50 = **₹5,000 (F)**.

(ii) Reconciliations

Particulars (₹)	Day shift	Night shift
Standard variable cost of output (× ₹44)	2,02,400	1,67,200
Add: net adverse variance (table above)	1,000	7,950
Actual at standard material price	**2,03,400**	**1,75,150**

Check: day 2,250 × ₹40 + 3,600 × ₹24 + ₹27,000 = ₹2,03,400; night 2,050 × ₹40 + 2,700 × ₹26 + ₹22,950 = ₹1,75,150.

Particulars (plant)	Amount (₹)
Standard cost of actual output (8,400 × ₹56.50)	4,74,600
Add: adverse variances (6,000 + 5,400 + 3,600 + 1,350 + 1,200)	17,550
Less: favourable variances (2,150 + 2,000 + 1,800 + 3,600 + 1,200 + 3,000 + 5,000)	(18,750)
Actual cost	**4,73,400**

Check: 4,300 × ₹39.50 + ₹86,400 + ₹70,200 + ₹27,000 + ₹22,950 + ₹97,000 = ₹4,73,400.

(iii) Comment. The plant's ₹1,200 (F) is the sum of ₹18,750 favourable and ₹17,550 adverse, and the plant report hides three problems. The two labour efficiency variances cancel (₹3,600 (A) and ₹3,600 (F)), so the plant shows none, yet the day shift took 150 hours more than standard. The night shift wasted 150 kg of material (₹6,000 (A)), more than the purchase department saved. And the night shift's ₹2 an hour over standard is a night allowance the standard never included: an uncontrollable variance for the shift supervisor which, if significant and persistent, calls for a revision of the standard (SM 13.12). The volume and price gains belong to other people. Variances are reported to the responsibility centre that could control them, one by

one; a net figure for the plant fixes nobody's responsibility (SM 13.12).

Exam angle

In the exam

How ICAI asks for the reconciliation, May 2024 to September 2026

In the ICAI documents read for this lesson, an income statement, a budget-against-actual comparison or a working-back from variances was asked in 7 of them, and the theory around variances in 4 more.

- **Income statement with the actual gross margin**: standard gross margin, less adverse, plus favourable variances, with production 51,200 units and sales 50,000: MTP January 2026 Series II Q5(a) (11 marks) and MTP September 2026 Series I Q4 (14 marks).
- **Working back from a variance report to actual costs**: RTP September 2025 Case Scenario 1 (ALZO Toys, five MCQs); MTP May 2025 Series II Q3(a) (10 marks, to actual output, overheads absorbed and budgeted output); MTP September 2026 Series II Q4(a) (8 marks); RTP September 2024 Case Scenario 2 (actual output from the usage variance).
- **Budget against actual, flexed to actual volume**: January 2026 Q4(a) (7 marks): budget and actual sales, material, labour and variable overheads, variances on the flexed standard. The Study

Material's own Practical Problem 6 of Budgets (SM 15.79, solution 15.92) is the full comparison.

- **Theory**: "Calculation of variances in standard costing is not an end in itself, but a means to an end. Discuss." (MTP January 2025 Series II Q2(b), 4 marks; MTP September 2025 Series I Q6(c), 4 marks; MTP May 2026 Series II Q4(b), 5 marks). "Define controllable and uncontrollable variance" (September 2024 Q6(b), with other terms). Disposal of controllable variances: the chapter's own MCQ 8 (SM 13.56).
- **Named sales variances** and a profit "operating statement": in none of them.

In the exam

The traps ICAI used

- MTP September 2026 Series I printed the absorbed factory overhead as actual hours × rate (₹30,000 (A)); the same question in MTP January 2026 Series II used standard hours for actual output × rate (₹24,000 (A)). The Study Material's absorbed overhead is standard hours for actual output × rate (SM 13.42): follow it.
- "Stocks of raw materials and finished goods are valued at a predetermined standard cost" (RTP September 2025 Case Scenario 1): the suggested answer takes the material price variance on the quantity **purchased** (the rule of SM 13.15), and the variances go to the period, not into stock.

- "CALCULATE all material variances based on actual material consumption" (MTP January 2026 Series II, MTP September 2026 Series I): purchases exceed consumption, and the question sets the purchase rule aside; on consumption the variances reconcile to the actual cost of the material used. On purchases they would reconcile to purchases less closing stock at standard price (Under the hood).

Trick

Signs in a bridge

Start from the budget (or the standard) and walk to the actual: add (F), deduct (A). If the last line is not the actual figure, a variance is missing or has the wrong sign. Condition: every variance is measured once, on one basis (contribution or profit, purchase or consumption).

Answer frame

"Calculation of variances is not an end in itself, but a means to an end. Discuss."

The points ICAI's suggested answers give (MTP January 2025 Series II Q2(b); MTP September 2025 Series I Q6(c); MTP May 2026 Series II Q4(b)):

1. The crux of standard costing lies in variance analysis: standard costs are predetermined and compared with

actual costs; the difference between standard cost and actual cost of actual output is the variance.

2. A variance is favourable when actual cost is less than standard, unfavourable (adverse) when more.
3. The figures alone do not give control: their origin and causes must be traced to take remedial steps.
4. A detailed probe, particularly into **controllable** variances, helps management ascertain (i) the amount of the variance, (ii) the causes of its occurrence, (iii) the responsibility of executives and departments, (iv) the corrective action to reduce or remove it.
5. Success depends on how quickly and effectively corrective action is taken: the variance is information, the means; action on it is the end.

"How are variances disposed of?"

Point	Source
Disposition is the last step of the standard costing process: variances are transferred to the relevant accounts (Costing P&L) as per the plan adopted	SM 13.5
Small variances, and those from abnormal causes or inefficiency (controllable): Costing P&L	SM 4.44–4.46; MCQ 8
Large variances from an unavoidable wrong estimate: supplementary rate over cost of sales, finished goods and WIP	SM 4.44–4.46
Stocks remain at standard cost, so profit is not distorted by stock values	SM 13.50
Significant, persistent uncontrollable variances: revise the standard (revision variance)	SM 13.12, 13.53

Three answers to "Reconcile the budgeted profit with the actual profit"

Loses marks

"Budgeted ₹4,20,000 – actual ₹3,82,600 = ₹37,400 (A)."

A difference is not a reconciliation.

Passes

A list of the cost variances that adds to ₹34,400 (A) of variable costs and ₹3,000 (F) of fixed overheads.

The volume and price effects are missing: the list explains the flexible budget, not the budget.

Full marks

Working notes; the flexible budget; budgeted profit → volume → flexible budget profit → price → each cost variance with (F)/(A) → actual profit; the check against actual sales less actual costs.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the budget that praised the material buyer

Comparing the master budget with the actual: material ₹8,00,000 budgeted against ₹7,64,400 actual looks like ₹35,600 (F). It is only a smaller volume. **Fix:** compare costs with the **flexible budget** (standard cost of actual output): ₹7,60,000, a variance of ₹4,400 (A) (SM 15.92).

⚠ **Common mistake**

The volume counted at profit, the volume variance forgotten

Taking volume at standard profit (₹30,000 (A) for the masala tins) and then only the fixed overhead expenditure variance overstates actual profit by ₹30,000. **Fix:** either volume at standard contribution with fixed overhead expenditure only, or volume at standard profit with the full fixed overhead variance.

⚠ **Common mistake**

Variances parked in the closing stock

Valuing the 2,000 unsold samosa packs, or the 1,600 kg left in the store, at actual cost "to be fair" moves part of the variances into stock. **Fix:** in a standard costing system stock is at standard cost and the variances go to the Costing Profit and Loss Account of the period (SM 13.5, 13.50), unless the case for a supplementary rate is made.

⚠ **Common mistake**

Netting the report

"Net variance ₹1,200 (F): no action." **Fix:** investigate each variance by responsibility centre; a small net hides large offsetting causes (the two shifts' labour efficiency variances cancel exactly), and responsibility is fixed variance by variance (SM 13.12).

Practice

Check yourself

1 Tapri's kulhad unit budgeted a profit of ₹4,00,000. The effect of lower volume was ₹30,000 (A), the effect of selling price ₹20,000 (F), and the cost variances added to ₹45,000 (A). What was the actual profit, in rupees?

Answer: _____

2 Case scenario. Dev's board slide for the cold-chain division: budget 20,000 deliveries at ₹100, standard variable cost ₹60 a delivery, fixed overheads ₹5,00,000 (absorbed at ₹25 a delivery). Actual: 18,000 deliveries at ₹102; the cost variances, including a fixed overhead expenditure variance but no volume variance, total ₹22,000 (A). Which actual profit should the slide show?

A ₹2,34,000

B ₹2,84,000

C ₹3,14,000

D ₹1,98,000

3 The kulhad unit had no opening stock, made 30,000 trays and sold 27,500. The standard cost is ₹48 a tray, and the month's variances net to ₹9,000 (A). At what value, in rupees, does the closing stock go into the books under standard costing?

Answer: _____

4 Ritu's intern wants to add an adverse labour efficiency variance, caused by poor supervision, to the cost of the closing stock. Which treatment does the Study Material support?

A. Spread it over closing stock and cost of sales with a supplementary rate

B. Transfer it to the Costing Profit and Loss Account

C. Carry it forward to next month's standard

D. Treat it as a revision variance

5 A large under-absorption of factory overheads arose because the price of indirect material rose after the standards were set. Under a supplementary rate, which accounts share it? Select all that apply.

- **A** Cost of sales
- **B** Finished goods stock
- **C** Work-in-progress
- **D** Costing Profit and Loss Account

More than one answer may be right.

6 Case scenario. The standard material cost of the rusk line's output for June is ₹5,40,000 (10,800 kg at ₹50). Ritu bought 12,000 kg at ₹49 and used 11,000 kg; the store carries material at standard price. Ritu's intern takes the price variance on the quantity purchased, as SM 13.15 says. Which actual figure do the standard cost and his two material variances reconcile to?

A ₹5,39,000, the actual cost of the 11,000 kg used

B ₹5,88,000, the actual cost of the purchases

C ₹5,38,000, the purchases less the closing stock at standard price

D ₹5,40,000, because the price and usage variances cancel

Answers: the answer key at the end of this lesson.

Recap

- Standard cost of actual output ± cost variances = actual cost; net cost variance = MCV + LCV + VOCV + FOCV. Price variance on purchases (SM 13.15): the actual cost reached is purchases − closing stock at standard price.
- Budgeted profit → (volume, at standard contribution) → flexible budget profit → (selling price) → (each cost

variance) → actual profit. Add (F), deduct (A).

- Volume at standard profit instead: then include the fixed overhead volume variance. Never mix the two.
- Compare costs with the flexible budget, never the master budget.
- Disposition (SM 13.5): to the Costing Profit and Loss Account, where small, abnormal and controllable variances go.
- Large variance from an unavoidable wrong estimate: supplementary rate over cost of sales, finished goods and WIP (SM 4.44–4.46).
- Stocks at standard cost; significant persistent uncontrollable variances → revise the standard (revision variance).
- The masala tins: −₹60,000 volume + ₹95,000 price − ₹97,400 costs = −₹62,400. The bridge closes to the rupee, or it is not a bridge.

Answer key · Check yourself

1 **345000**

₹4,00,000 − ₹30,000 + ₹20,000 − ₹45,000 = ₹3,45,000. Favourable steps are added, adverse ones deducted.

2 **A ₹2,34,000**

Budgeted profit = 20,000 × ₹40 − ₹5,00,000 = ₹3,00,000. Volume at contribution: 2,000 × ₹40 = ₹80,000 (A); price 18,000 × ₹2 = ₹36,000 (F); costs ₹22,000 (A): ₹2,34,000. ₹2,84,000 takes volume at the ₹15 standard profit without adding the ₹50,000 fixed overhead volume variance; ₹3,14,000 forgets volume; ₹1,98,000 forgets the price.

3 **120000**

Closing stock = 30,000 − 27,500 = 2,500 trays × ₹48 = ₹1,20,000, at standard cost. The ₹9,000 (A) goes to the Costing Profit and Loss Account of the month (SM 13.5); none of it is carried in the stock.

4 **B Transfer it to the Costing Profit and Loss Account**

Poor supervision is controllable inefficiency: controllable variances are best transferred to the Costing Profit and Loss Account (SM 13.56, MCQ 8), as inefficiency is in Illustration 8 (SM 4.46). Supplementary rates are for large differences from unavoidable wrong estimates.

5 **A Cost of sales; B Finished goods stock; C Work-in-progress**

A supplementary rate distributes the amount over the units of the period: cost of sales, finished goods and work-in-progress (SM 4.44–4.46). The Costing Profit and Loss Account takes small and abnormal amounts.

6 **C ₹5,38,000, the purchases less the closing stock at standard price**

Price on purchases = 12,000 × (₹50 − ₹49) = ₹12,000 (F); usage = (10,800 − 11,000) × ₹50 = ₹10,000 (A). ₹5,40,000 + ₹10,000 − ₹12,000 = ₹5,38,000 = ₹5,88,000 − 1,000 kg × ₹50. ₹5,39,000 is the consumption-basis answer (price ₹11,000 (F)); the ₹1,000 between them is the price variance on the 1,000 kg still in the store.

CHAPTER 14

Marginal Costing

14.1 Marginal cost, contribution and the P/V ratio

What one more unit really costs, why fixed costs are charged to the period, how contribution pays fixed cost and profit, and the P/V ratio that links contribution to sales.

In plain words

The ₹95 jar that was never ₹95

Tapri's cookie unit in Indore bakes 20,000 jars a month and sells them at ₹100. Kesar has just seen the cost report.

> **Kesar:** Each jar costs us ₹95? Then a hotel order at ₹90 is a loss. Cancel it. And next month, when we make 30,000 jars, why does the report say ₹85? Did the butter get cheaper?
>
> **Ritu:** Per unit, please, but the right unit. Flour, butter, the jar and the gas cost ₹65 a jar. The other ₹30 is this month's ₹6,00,000 of rent, oven lease and salaries, spread over 20,000 jars. Spread it over 30,000 and it becomes ₹20. The butter did not change; the spreading did.

The ₹65 is the **marginal cost** of a jar: what the total cost goes up by when one more jar is made. It is the same at 20,000 jars and at 30,000. The ₹6,00,000 is a **fixed cost**: it is the cost of the month, not of any jar, and it stays the same however many jars are baked.

So every jar sold at ₹100 leaves ₹35 after its own marginal cost. That ₹35 is its **contribution**: it contributes first to paying the month's fixed cost and then to profit. Twenty thousand jars contribute ₹7,00,000, the fixed cost takes ₹6,00,000, and ₹1,00,000 is profit. And the hotel order at ₹90 is not a loss at all: if the oven has spare capacity and the fixed cost does not change, each of those jars still leaves ₹25.

Ritu has one more number: ₹35 out of ₹100 is 35%. Every rupee of cookie sales leaves 35 paise of contribution. That is the **P/V ratio**, and it gives the contribution from any sales figure without counting jars.

Keeping the two kinds of cost apart, valuing products and stock at variable cost only and charging fixed costs to the period, is **marginal costing**.

Think of it like

The auto-rickshaw and the meter

An auto driver pays ₹600 a day to rent his auto whether he drives one passenger or thirty. Each ride costs him only the CNG it burns. Ask him what one more ride costs and he says "the gas", not "₹600 divided by today's rides", a number that changes every evening. The CNG is his marginal cost, fare minus CNG is the ride's contribution, and the rent is the day's fixed cost, which the rides pay off before he takes anything home.

Key idea

Marginal cost is what one more unit adds to total cost: its variable cost. Price minus marginal cost is contribution, which pays the period's fixed costs first and becomes profit only after they are covered. The P/V ratio is contribution as a share of sales.

How it works

Five ideas, in the order ICAI builds them

1 **Split every cost by behaviour.** Variable costs change with output; fixed costs stay the same in total; semi-variable costs are split into both parts first (cost behaviour). Absorption costing classifies by function instead (factory, office, selling).

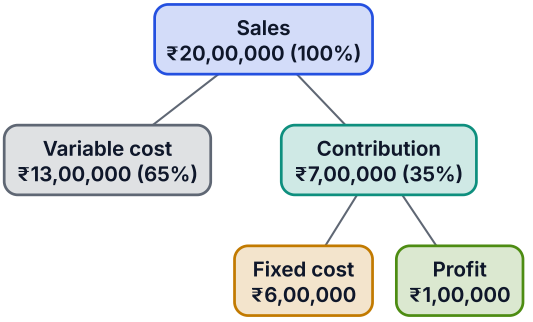
2 **Marginal cost** is the variable cost of one more unit: "the sum of prime cost and variable overhead" (SM 14.3). The cookie jar: ₹65, at any volume within the range.

3 **Product costs and period costs.** Only variable manufacturing costs are **product (inventoriable) costs**: they go into the product and the value of closing stock and WIP. All fixed costs, manufacturing or not, are **period costs**: charged to the profit and loss account of the period in which they are incurred.

4 **Contribution** = sales − all variable costs, manufacturing and non-manufacturing. It is a fund for fixed costs and profit: contribution − fixed cost = profit; if fixed cost is larger, the shortfall is the loss.

5 **P/V ratio** (contribution to sales ratio) = contribution ÷ sales × 100: the share of each rupee of sales left to cover fixed costs and profit. It also works from two periods: change in contribution (or in profit) ÷ change in sales × 100.

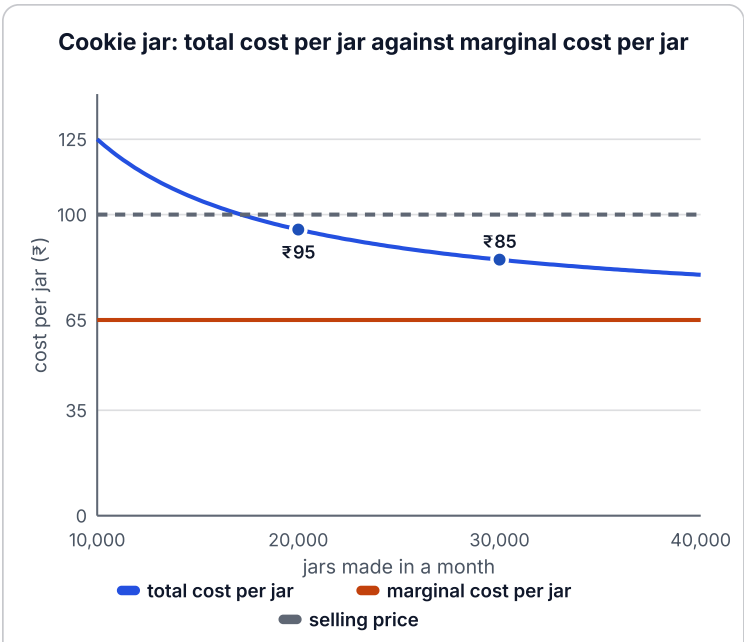
Where the cookie unit's ₹20,00,000 of sales goes



20,000 jars at ₹100. Sales split into variable cost and contribution (the P/V ratio is the 35%); contribution splits into fixed cost and profit. This is the marginal cost equation S − V = C = F + P.

Why the "cost per jar" kept changing

When each jar carries a share of fixed cost, its cost moves with volume: ₹95 at 20,000 jars, ₹85 at 30,000. ICAI's point (SM 14.6): costing can say an item costs ₹20 one day and ₹18 the next only because volume changed, which makes decisions difficult. Marginal cost does not move.



Blue: total cost per jar (₹65 plus ₹6,00,000 spread over the jars) falls as volume rises. Orange: marginal cost stays at ₹65. The grey dashed line is the ₹100 price; the gap between it and the orange line is the ₹35 contribution.

Marginal cost is a unit-level idea

ICAI warns that "direct costing" and "marginal costing" are used as synonyms, but costs behave at different levels

(SM 14.3–14.5). Tapri fills chai premix boxes on a machine that is set up once for every run of 500 boxes:

Chai premix: the levels at which costs change

Unit level	materials, labour, variable overhead: ₹60 a box	changes with every box
Batch level	machine set-up: ₹2,400 a run of 500 boxes	changes with every run
Product level	depreciation of the premix filling machine: ₹30,000	changes only if the product is dropped
Department level	apportioned fixed overheads: ₹2,00,000	does not change with the product

Only the unit-level ₹60 is the marginal cost of a box. The other levels are direct to a batch, a product or a department, but they do not move with one more box.

Going from 9,999 to 10,000 boxes adds ₹60. Going from 10,000 to 10,001 starts a 21st run and adds ₹2,460: that is the **incremental cost** of the extra box (one more batch), while its **marginal cost** is still ₹60. Incremental cost can be for one unit or a whole change in volume or process; **differential cost** is the difference between the costs of two levels or alternatives.

What moves the P/V ratio

The ratio depends only on price and variable cost per unit (SM Illustration 9):

Change, on its own	P/V ratio
Physical sales volume rises or falls	no change
Fixed cost rises or falls	no change
Variable cost per unit falls	rises
Selling price per unit rises	rises
Contribution margin falls	falls
Price and variable cost per unit both rise by the same percentage	no change
Price rises 10%, volume falls 10%	rises (price raises it; volume cannot move it)
Angle of incidence widens	rises

Interactive version: PassBook app, lesson Marginal cost, contribution and the P/V ratio

Formats and formulas

ICAI's **Cost and Profit Statement under Marginal Costing** (SM 14.7). *Reproduce the headings and their order from memory.*

Particulars	Amount (₹)	Amount (₹)
Revenue (A)		xxx
Product cost:		
– Direct materials	xxx	
– Direct employee (labour)	xxx	
– Direct expenses	xxx	
– Variable manufacturing overheads	xxx	
Product (inventoriable) costs (B)	**xxx**	**xxx**
Product contribution margin {A – B}		**xxx**
– Variable administration overheads	xxx	
– Variable selling and distribution overheads	xxx	xxx
Contribution margin (C)		**xxx**
Period cost (D):		
Fixed manufacturing expenses	xxx	
Fixed non-manufacturing expenses	xxx	xxx
Profit / (loss) {C – D}		**xxx**

The short form is the **marginal cost statement**: Sales, less variable cost, = contribution, less fixed cost, = profit (SM 14.13).

∑ **Formula**

The marginal cost equation

$$S - V = C = F \pm P$$

S = sales value, V = variable cost, C = contribution, F = fixed cost, P = profit (+) or loss (−) (SM 14.13). If C = F there is no profit or loss; at zero output the loss equals F.

∑ **Formula**

Contribution

$$\text{Contribution} = \text{Sales revenue} - \text{Total variable cost}$$
$$= \text{Fixed cost} + \text{Profit}$$

Variable cost here means all of it, manufacturing and non-manufacturing (variable selling commission included).

∑ Formula

P/V ratio (contribution to sales ratio)

$$\text{P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$
$$= \frac{\text{Change in contribution or profit}}{\text{Change in sales}} \times 100$$

The change form is valid when price, variable cost per unit and fixed cost are the same in both periods.

∑ Formula

Transpositions

$$C = S \times \text{P/V ratio} \qquad S = \frac{C}{\text{P/V ratio}}$$

$$\text{Variable cost ratio} = 100\% - \text{P/V ratio}$$

The first two are ICAI's (SM 14.14). The third follows because sales are variable cost plus contribution: a P/V ratio of 40% means variable cost is 60% of sales.

∑ Formula

Fixed cost from any period

$$F = C - P \qquad = (S \times \text{P/V ratio}) - P$$

Definition

Marginal cost

The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit. (SM 1.34) In Chapter 14: the incremental cost of production for producing one additional unit, measured by the total variable cost attributable to one additional unit. (SM 14.2–14.3)

Definition

Marginal costing

A costing system where products or services and inventories are valued at variable costs only. It does not take consideration of fixed costs. Also known as direct costing. (SM 14.3) As a technique: the ascertainment of marginal cost by differentiating between fixed and variable costs; it is used to ascertain the effect of changes in volume or type of output on profit. (SM 1.39)

Definition

Contribution

Contribution or contribution margin is the difference between sales revenue and total variable costs irrespective of manufacturing or non-manufacturing. (SM 14.8)

Definition

Period cost

Costs which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All fixed costs, either manufacturing or non-manufacturing, are recognised as period costs in marginal costing. (SM 14.9)

Definition

P/V ratio

This ratio shows the proportion of sales available to cover fixed costs and profit. (SM 14.13)

Worked examples

Worked example

Chai premix: two levels, one statement

Tapri's chai premix line sells boxes at ₹250. Ritu has the

cost records at two levels of activity:

Particulars	60,000 boxes (₹)	80,000 boxes (₹)
Direct materials	48,00,000	64,00,000
Direct employee cost	18,00,000	24,00,000
Direct expenses (royalty)	3,00,000	4,00,000
Manufacturing overheads	33,00,000	36,00,000
Administration overheads	9,00,000	10,00,000
Selling and distribution overheads	16,00,000	18,00,000

Required:

(i) Separate each overhead into its fixed and variable parts.

(ii) Prepare the Cost and Profit Statement under marginal costing for a budget of 90,000 boxes, and state the contribution per box and the P/V ratio.

(iii) Kesar notes that the total cost per box is ₹211.67 at 60,000 boxes and ₹195 at 80,000. Calculate the marginal cost per box and explain the difference.

THE WORKING

(i) Working note 1: splitting the overheads
(variable part = change in cost ÷ change in boxes; fixed part = total − variable part)

Overhead	Change in cost (₹)	Variable per box (₹)	Fixed (₹)
Manufacturing	3,00,000	15	24,00,000
Administration	1,00,000	5	6,00,000
Selling and distribution	2,00,000	10	10,00,000

Each change is over 20,000 boxes; e.g. manufacturing ₹33,00,000 − 60,000 × ₹15 = ₹24,00,000. Materials (₹80), employee cost (₹30) and royalty (₹5) per box are the same at both levels: wholly variable.

(ii) Cost and Profit Statement under marginal costing, 90,000 boxes

Particulars	Amount (₹)	Amount (₹)
Revenue (90,000 × ₹250) (A)		2,25,00,000
Product cost:		
Direct materials (₹80)	72,00,000	

Particulars	Amount (₹)	Amount (₹)
Direct employee cost (₹30)	27,00,000	
Direct expenses (₹5)	4,50,000	
Variable manufacturing overheads (₹15)	13,50,000	
Product (inventoriable) costs (B)		**1,17,00,000**
Product contribution margin {A – B}		**1,08,00,000**
Variable administration overheads (₹5)	4,50,000	
Variable selling and distribution overheads (₹10)	9,00,000	13,50,000
Contribution margin (C)		**94,50,000**
Period cost (D):		
Fixed manufacturing expenses	24,00,000	

Particulars	Amount (₹)	Amount (₹)
Fixed non-manufacturing expenses (₹6,00,000 + ₹10,00,000)	16,00,000	40,00,000
Profit {C – D}		**54,50,000**

Contribution per box = ₹94,50,000 ÷ 90,000 = **₹105** (₹250 – ₹145). P/V ratio = ₹105 ÷ ₹250 × 100 = **42%**. The product cost per box, the value of a box in closing stock under marginal costing, is ₹130: variable administration and selling costs are not inventoried.

(iii) Marginal cost and total cost per box

Boxes	Variable cost at ₹145 (₹)	Fixed cost (₹)	Total cost (₹)	Per box (₹)
60,000	87,00,000	40,00,000	1,27,00,000	211.67
80,000	1,16,00,000	40,00,000	1,56,00,000	195.00

The marginal cost is **₹145 a box** at both levels (₹80 + ₹30 + ₹5 + ₹15 + ₹5 + ₹10). Total cost per box falls only because the same ₹40,00,000 of fixed cost is spread over more boxes: the extra 20,000 boxes added ₹29,00,000, exactly 20,000 × ₹145. The fixed cost belongs to the period and is recovered out of total contribution.

Worked example

Monsoon pakora combo: what a 10% price cut really costs

Tapri's app sells a monsoon pakora combo at ₹30. Last season it sold 2,40,000 combos at a P/V ratio of 25%, with fixed costs of ₹12,00,000. BrewBucks has launched a cheaper snack box, and Dev proposes a 10% price cut for the coming season, variable cost per combo unchanged.

Required:

(i) Calculate the variable cost and contribution per combo, and last season's profit.

(ii) Calculate the new P/V ratio after the price cut.

(iii) Calculate the sales value and the number of combos needed to keep last season's total contribution, and the percentage increase in each.

(iv) Kesar expects volume to rise by only 30% after the cut. Calculate the profit.

(v) Ritu's alternative is to drop the price cut and hold the P/V ratio at 25%. If oil prices push the variable cost per combo up by 6%, calculate the selling price needed.

THE WORKING

(i) Variable cost = 75% of ₹30 = **₹22.50**; contribution = **₹7.50** a combo.

Particulars	Amount (₹)
Sales (2,40,000 × ₹30)	72,00,000
Less: Variable cost (2,40,000 × ₹22.50)	54,00,000
Contribution	**18,00,000**
Less: Fixed cost	12,00,000
Profit	**6,00,000**

(ii) New price = ₹27; contribution = ₹27 − ₹22.50 = ₹4.50; P/V ratio = 4.50 ÷ 27 × 100 = **16.67%** (1/6).

(iii) Keeping the contribution of ₹18,00,000

$$\text{Sales value} = \frac{C}{\text{P/V ratio}} = \frac{18{,}00{,}000}{1/6} = ₹1{,}08{,}00{,}000$$

$$\text{Combos} = \frac{18{,}00{,}000}{4.50} = 4{,}00{,}000$$

Sales value must rise by **50%** (₹72,00,000 to ₹1,08,00,000) and the number of combos by **66.67%** (2,40,000 to 4,00,000). Check: 4,00,000 × ₹27 = ₹1,08,00,000.

(iv) Volume up 30%: 3,12,000 combos × ₹4.50 = ₹14,04,000 contribution, less ₹12,00,000 fixed cost = profit of **₹2,04,000**, against ₹6,00,000 last season.

(v) New variable cost = ₹22.50 × 1.06 = ₹23.85. To keep a P/V ratio of 25%, variable cost must stay 75% of price:

$$\text{Price} = \frac{23.85}{75\%} = ₹31.80$$

The price rises by the same 6% as the variable cost. Left at ₹30, the P/V ratio would fall to 20.5%.

Comment. The 10% price cut removes 40% of each combo's contribution (₹7.50 to ₹4.50), so two-thirds more combos are needed just to stand still; any smaller rise in volume reduces profit.

Worked example

Three cafés, one equation: fill in the blanks

Tapri's cafés in Bhopal, Ujjain and Gwalior sell the same menu at the same prices, with the same variable cost

per order, so they share one P/V ratio. Ritu's half-finished statement for 2025-26:

Particulars	Bhopal (₹)	Ujjain (₹)	Gwalior (₹)
Sales	48,00,000	?	?
Variable cost	28,80,000	?	?
Contribution	?	15,20,000	?
Fixed cost	12,00,000	?	9,60,000
Profit / (loss)	?	3,20,000	(40,000)

Kesar wants to close the Gwalior café "because it makes a loss". Its fixed cost of ₹9,60,000 includes ₹3,20,000 of head-office cost allocated to it, which the company will incur whether or not Gwalior stays open; the rest of its fixed cost would be saved.

Required:

(i) Complete the statement and state the P/V ratio.

(ii) Advise Kesar on closing the Gwalior café, showing the effect on the company's total profit.

THE WORKING

(i) Working note: the P/V ratio. Bhopal's contribution = ₹48,00,000 − ₹28,80,000 = ₹19,20,000, so P/V ratio = ₹19,20,000 ÷ ₹48,00,000 × 100 = **40%**, common to all three cafés.

Each blank comes from the marginal cost equation S − V = C = F ± P:

- **Bhopal:** profit = C − F = ₹19,20,000 − ₹12,00,000 = ₹7,20,000.
- **Ujjain:** sales = C ÷ P/V ratio = ₹15,20,000 ÷ 40% = ₹38,00,000; variable cost = ₹38,00,000 − ₹15,20,000 = ₹22,80,000; fixed cost = C − P = ₹15,20,000 − ₹3,20,000 = ₹12,00,000.
- **Gwalior:** contribution = F − loss = ₹9,60,000 − ₹40,000 = ₹9,20,000; sales = ₹9,20,000 ÷ 40% = ₹23,00,000; variable cost = ₹13,80,000.

Particulars	Bhopal (₹)	Ujjain (₹)	Gwalior (₹)	Total (₹)
Sales	48,00,000	38,00,000	23,00,000	1,09,00,000
Variable cost	28,80,000	22,80,000	13,80,000	65,40,000
Contribution	**19,20,000**	**15,20,000**	**9,20,000**	**43,60,000**
Fixed cost	12,00,000	12,00,000	9,60,000	33,60,000
Profit / (loss)	**7,20,000**	**3,20,000**	**(40,000)**	**10,00,000**

(ii) Closing Gwalior

Particulars	Amount (₹)
Contribution lost	(9,20,000)
Fixed cost saved: ₹9,60,000 − ₹3,20,000 head-office cost that continues	6,40,000
Fall in the company's profit	**(2,80,000)**

Company profit would fall from ₹10,00,000 to ₹7,20,000. Gwalior's "loss" of ₹40,000 arises only after it is charged ₹3,20,000 of head-office cost that does not go away; its contribution covers its own fixed cost and adds ₹2,80,000 towards the head office. Do not close it on this basis: judge a unit by its contribution, not by its profit after allocated fixed costs. Closure decisions in full are in decisions with marginal costing.

Exam angle

How ICAI tests the basics

In the exam

What the papers asked, May 2024 to September 2026

In the documents read for this lesson, the basics came up in 12 places in 9 documents, always as a step inside a bigger question or as theory.

- **Contribution per unit as the first step, 2–5 marks.** January 2025 Q1(b) (contribution per good unit after a 3% rejection with scrap, then monthly profit: 3 + 2 marks). September 2025 MCQs 6–10 (one case: absorption cost of production, total variable cost, contribution per unit, BES, profit at 10% lower sales; 2 marks each). RTP January 2025 MCQ 7 (variable cost from the average cost at two volumes).

- **Splitting costs first.** RTP May 2024 Q11 (variable portions as percentages of cost of goods sold and of cost of sales, then P/V ratio, BES and profit); MTP Series II for May 2025, MCQ case (total variable and fixed cost "as per marginal costing income statement").
- **Theory, 4–5 marks.** RTP January 2025 Q19(a) "DISCUSS advantages of Marginal Costing"; MTP Series II for September 2026 Q6(c) "Elaborate the practical application of Marginal Costing" (4 marks); "DISCUSS any five types of costs categorised based on its use in Managerial Decision Making", where marginal, differential and product cost are answer points (MTP Series I for May 2025 Q6(a), MTP Series II for September 2026 Q6(b), 5 marks each; RTP January 2026 Q19(a)); RTP May 2026 Q17(a), the same costs in a decision scenario.

In the exam

The traps

- Contribution deducts variable administration and selling costs too (September 2025 MCQ 7: total variable cost included the variable general and administration and the variable selling overheads).
- "Profit as a % of sales" targets: MTP Series II for September 2026 Q5(b)(d) solved fixed cost ÷ (P/V ratio − profit %).

Trick

P/V ratio does the counting for you

Profit at any sales = sales × P/V ratio − fixed cost: no units needed. A target profit of p% of sales needs sales = fixed cost ÷ (P/V ratio − p%). Conditions: one product (or a constant mix); price, variable cost per unit and fixed cost unchanged.

Trick

The same percentage up, the same P/V ratio

Price and variable cost per unit moved by the same percentage leave the P/V ratio unchanged (SM Illustration 9, item vii). So the price that keeps a P/V ratio is new variable cost ÷ (1 − P/V ratio).

Answer frame

"Discuss the advantages of marginal costing" (ICAI's seven, in ICAI's order)

1. **Simplified pricing policy:** marginal cost per unit is constant in the short run, so firm pricing decisions can be taken.
2. **Proper recovery of overheads:** no under- or over-recovery of fixed overheads absorbed at predetermined rates.
3. **Shows realistic profit:** stock and WIP at marginal cost, fixed expenses written off as period cost.

4. **How much to produce:** break-even analysis shows the effect of more or less production on profit.
5. **More control over expenditure:** actual variable expenses compared with budget through variances.
6. **Helps in decision making:** make or buy, discontinuance of a product, replacement of machines.
7. **Short-term profit planning:** through break-even charts.

"Discuss the limitations of marginal costing" (ICAI's seven)

1. **Difficulty in classifying fixed and variable elements:** most expenses are neither wholly fixed nor wholly variable.
2. **Dependence on key factors:** contribution is no guide to optimum profitability unless linked with the key factor.
3. **Scope for low profitability:** sales staff may mistake marginal cost for total cost and sell at a loss.
4. **Faulty valuation:** fixed overheads cannot be wholly excluded from WIP, particularly in large contracts.
5. **Unpredictable nature of cost:** fixed cost and variable cost per unit do change in reality.
6. **Ignores time factor and investment:** two jobs with the same marginal cost may take different time and machines.

7. **Understating of WIP:** stocks and work-in-progress are understated.

"Elaborate the practical application of marginal costing"

ICAI's suggested answer (MTP Series II for September 2026, 4 marks) has four points: **pricing policy** (constant marginal cost permits firm short-term pricing decisions), **decision making** (make or buy, discontinuance, replacement), **ascertaining realistic profit** (stock at marginal cost, fixed cost as period cost) and **determination of production level** (break-even analysis). The Study Material's own answer key to "Discuss the practical application of marginal costing" points instead to its "facts about marginal costing" (SM 14.6, under the hood): a fuller answer adds them after the four points.

"Explain the characteristics of marginal costing" (ICAI's six)

(1) All costs classified into fixed and variable, semi-variable costs split; (2) variable costs are the cost of the product; (3) finished goods and WIP at marginal cost only, variable selling and distribution excluded; (4) fixed costs are period costs; (5) prices set with reference to marginal cost and contribution; (6) profitability of departments and products judged by contribution.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the fixed overhead that sneaked into marginal cost

Writing "marginal cost per box = ₹145 + ₹40,00,000 ÷ 90,000 = ₹189.44". That is the total cost per box at one volume, and it changes with every volume. **Fix:** marginal cost is prime cost plus variable overheads; fixed costs are deducted from total contribution, never from a unit.

⚠ **Common mistake**

Contribution that forgot the salesman

Stopping at sales minus variable *manufacturing* cost (₹1,08,00,000 in the premix example) and calling it contribution. That is ICAI's **product contribution margin**. **Fix:** contribution is after *all* variable costs: deduct variable administration and selling costs too (₹94,50,000).

⚠ **Common mistake**

Closing stock valued with the selling commission in it

Valuing a box of premix in closing stock at ₹145 under marginal costing. **Fix:** stock is valued at variable manufacturing cost only (₹130); variable selling and distribution costs are excluded from inventory.

Practice

Check yourself

1 Tapri's kulhad ice-cream has a variable cost ratio of 65% and a contribution of ₹14 a cup. What is its selling price per cup (₹)?

Answer: _____

2 Case scenario. Tapri's cold-chain division had sales of ₹50,00,000 at a P/V ratio of 30% and fixed costs of ₹9,00,000. Dev tells the board that sales will rise by 20% next year with prices, variable cost per unit and fixed costs unchanged, and "so profit will rise by 20% too". Which statement is correct?

- **A** Profit rises by 20%, from ₹6,00,000 to ₹7,20,000
- **B** Profit rises by 50%, from ₹6,00,000 to ₹9,00,000
- **C** Profit stays at ₹6,00,000, because the P/V ratio does not change
- **D** The P/V ratio rises to 36%, so profit becomes ₹12,60,000

3 Under marginal costing, which of these is a period cost?

- **A** Direct materials of the units produced
- **B** Variable factory power
- **C** Rent of the factory building
- **D** Royalty paid per unit produced

4 A product has a P/V ratio of 30%. By what percentage must its selling price rise, variable cost per unit unchanged, to lift the P/V ratio to 40%? (Two decimals.)

Answer: _____

5 A product sells at ₹500; direct materials ₹180, direct labour ₹70, variable factory overheads ₹50, variable selling commission ₹20, and fixed factory overheads absorbed at ₹60 per unit. What is its contribution per unit?

- **A** ₹200
- **B** ₹180
- **C** ₹120
- **D** ₹140

Answers: the answer key at the end of this lesson.

Recap

- Marginal cost = the variable cost of one more unit (prime cost + variable overheads); it is the same at every volume within the range.
- Marginal costing values products and stock at variable (manufacturing) cost only; all fixed costs are period costs charged to the period's profit and loss account.
- Contribution = sales − all variable costs = fixed cost + profit. Marginal cost equation: S − V = C = F ± P.
- Product contribution margin = revenue − variable manufacturing costs; contribution margin also deducts variable administration and selling costs.
- P/V ratio = contribution ÷ sales × 100, or change in contribution (profit) ÷ change in sales × 100; C = S × P/V; variable cost ratio = 100% − P/V.
- The P/V ratio moves only with price and variable cost per unit; volume and fixed cost leave it alone; equal % changes in both leave it alone too.
- Unit total cost wobbles with volume (₹95 at 20,000 jars, ₹85 at 30,000); marginal cost does not.

Answer key · Check yourself

1 **40**

P/V ratio = 100% − 65% = 35%. Price = contribution ÷ P/V ratio = ₹14 ÷ 35% = ₹40.

2 **B Profit rises by 50%, from ₹6,00,000 to ₹9,00,000**

Contribution = sales × 30%: ₹15,00,000 now, ₹18,00,000 at sales of ₹60,00,000. Fixed cost stays ₹9,00,000, so profit goes from ₹6,00,000 to ₹9,00,000, a 50% rise. The extra ₹3,00,000 of contribution is all profit; volume does not move the P/V ratio.

3 **C Rent of the factory building**

All fixed costs, manufacturing or non-manufacturing, are period costs under marginal costing. Materials, variable power and a per-unit royalty are variable manufacturing (product) costs.

4 **16.6 to 16.7**

Take the price as ₹100: variable cost is ₹70. At a P/V ratio of 40%, ₹70 must be 60% of the new price, so the price = ₹70 ÷ 60% = ₹116.67, a rise of 16.67%. Adding 10 percentage points to the price (₹110) gives a P/V ratio of only 36.36%.

5 **B ₹180**

Contribution = price − all variable costs, manufacturing and non-manufacturing: ₹500 − (₹180 + ₹70 + ₹50 + ₹20) = ₹180. ₹200 leaves out the variable commission (it is the product contribution margin); ₹140 and ₹120 also deduct the absorbed fixed overhead of ₹60, which is a period cost, never part of contribution.

14.2 Break-even point, margin of safety and angle of incidence

The volume at which contribution has just paid the fixed cost, in units and rupees; how far sales can fall before a loss; and what the angle of incidence shows.

In plain words

The café with a queue and a loss

Tapri's new kulhad café in Pune has a queue every evening and a loss for its first month.

> **Kesar:** We sold 12,000 cups and still lost money? The queue reaches the parking lot!
>
> **Ritu:** Per unit, please. A cup sells for ₹30; milk, tea, kulhad and gas cost ₹18. Each cup leaves ₹12, so 12,000 cups left ₹1,44,000. Rent, salaries and the machine lease are ₹1,80,000 a month however many cups we sell.

The ₹12 each cup leaves is its **contribution**: the price minus the costs that rise with every cup (the **variable costs**). Before it becomes profit, contribution must pay the costs that stay the same whatever the volume (the **fixed costs**).

At 15,000 cups the contributions add up to exactly ₹1,80,000: no profit, no loss. That volume is the **break-even point**; from cup 15,001, every cup's ₹12 is profit. Kesar expects 24,000 cups next month, so 9,000 cups are

his **margin of safety**: sales could fall that far before the café is back in a loss.

On a chart, the sales line crosses the total cost line at the break-even point and the two then open apart like scissors. The angle at which they cross is the **angle of incidence**: the wider it opens, the faster profit piles up.

Think of it like

The bucket and the glass

Fixed cost is a bucket that must be full before anything reaches your glass, and every sale pours its contribution into it. Break-even is the moment the bucket is full; after that every pour overflows into the glass, which is profit. The margin of safety is how much pouring you could lose and still fill the bucket. The angle of incidence is the size of each pour.

Key idea

Contribution pays the fixed cost first; only what is left over is profit. The break-even point is the volume at which the fixed cost has just been paid, and the margin of safety is how far expected sales sit above it.

How it works

Cost-volume-profit (CVP) analysis studies how cost, selling price and volume decide profit. A generally used method of it, **break-even analysis**, finds the volume

where total cost equals total sales revenue (the narrow sense) or the profit at any level of sales (the broad sense) (SM 14.14).

Six steps from a price tag to a profit figure

1. **Contribution per unit** = selling price − variable cost per unit: ₹30 − ₹18 = ₹12 a cup. Divided by the price it gives the **P/V ratio** (really a contribution-to-sales ratio): ₹12 ÷ ₹30 = 40%. Both are built in the lesson on contribution and the P/V ratio.

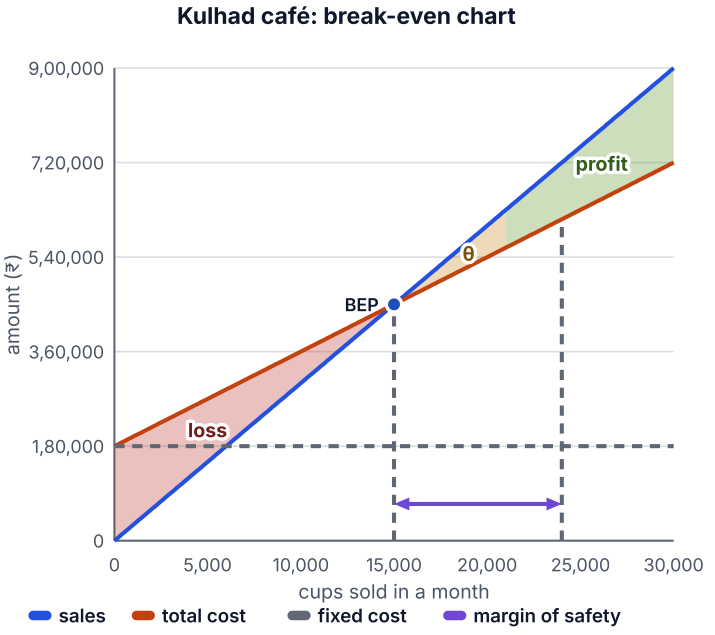
2. **Break-even point (BEP)** = fixed cost ÷ contribution per unit: ₹1,80,000 ÷ ₹12 = 15,000 cups. In rupees, the **break-even sales (BES)** are 15,000 × ₹30 = ₹4,50,000, or ₹1,80,000 ÷ 40%.

3. **A target profit** is added to the fixed cost before dividing: a profit of ₹60,000 needs (₹1,80,000 + ₹60,000) ÷ ₹12 = 20,000 cups.

4. **Margin of safety (MOS)** = expected sales − break-even sales: 24,000 − 15,000 = 9,000 cups, or ₹2,70,000. As a share of expected sales, 9,000 ÷ 24,000 = 37.5%: the **MOS ratio**.

5 **Profit comes straight from the margin of safety**, because only the cups above break-even earn it: 9,000 × ₹12 = ₹1,08,000, which is also ₹2,70,000 × 40%.

6 **On a break-even chart** (costs and revenues up the side, activity along the bottom), the sales line starts at the origin and the total cost line at the fixed cost. They cross at the BEP, and the angle at which they cross, the **angle of incidence**, shows the rate at which profit is earned once the break-even point is reached.



Blue: sales; orange: total cost; grey dashed: fixed cost. The lines cross at the BEP, 15,000 cups (₹4,50,000). Rose: the loss area; green: the profit area; θ (amber) marks the angle of incidence. The violet arrow is the margin of safety, from 15,000 to the expected 24,000 cups.

The **cash break-even point** uses only the fixed costs payable in cash (depreciation and other non-cash fixed costs are left out), so it is never above the BEP.

Interactive version: PassBook app, lesson Break-even point, margin of safety and angle of incidence

What moves the break-even point

Each row changes one thing; the P/V column is ICAI's (Illustration 9, SM 14.29).

Change, all else equal	P/V ratio	BEP	Margin of safety	Angle of incidence
Selling price per unit rises	rises	falls	rises	wider
Variable cost per unit rises	falls	rises	falls	narrower
Fixed cost rises	no change	rises	falls	no change
Units sold rise	no change	no change	rises	no change

The angle depends only on how steep the two lines are, which price and variable cost per unit decide. More fixed cost lifts the whole total cost line: the BEP moves right, the angle stays.

Interactive version: PassBook app, lesson Break-even point, margin of safety and angle of incidence

Formats and formulas

Every break-even answer starts from the marginal cost statement or one of these formulas. *Reproduce the*

statement and the formulas from memory.

Marginal cost statement	Amount (₹)
Sales	xxx
Less: Variable cost	xxx
Contribution	**xxx**
Less: Fixed cost	xxx
Profit	**xxx**

∑ **Formula**

The marginal cost equation

$$S - V = C = F \pm P$$

S = sales, V = variable cost, C = contribution, F = fixed cost, P = profit or loss (SM 14.13). At the BEP, P = 0, so C = F.

∑ **Formula**

Break-even point in units

$$\text{BEP (units)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}}$$

∑ **Formula**

Break-even sales in rupees

$$\text{BES} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{\text{Fixed cost}}{\text{Contribution}} \times \text{Sales}$$

P/V ratio = contribution ÷ sales × 100.

∑ **Formula**

Cash break-even point

$$\text{Cash BEP (units)} = \frac{\text{Cash fixed costs}}{\text{Contribution per unit}}$$

ICAI: the break-even point "calculated only with those fixed costs which are payable in cash" (SM 14.15); depreciation and other non-cash fixed costs are left out.

∑ **Formula**

Sales for a target profit

$$\text{Units} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{Contribution per unit}}$$

$$\text{Sales value} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{P/V ratio}}$$

∑ **Formula**

Margin of safety

$$\text{MOS} = \text{Total sales} - \text{BES} \qquad = \frac{\text{Profit}}{\text{P/V ratio}}$$

Total sales are the actual or budgeted sales; in units, units sold – BEP units.

∑ **Formula**

Margin of safety ratio and profit

$$\text{MOS ratio} = \frac{\text{Total sales} - \text{BES}}{\text{Total sales}} \times 100$$

$$\text{Profit} = \text{MOS} \times \text{P/V ratio}$$

∑ **Formula**

P/V ratio from two periods

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100$$

Valid when price, variable cost per unit and fixed cost are the same in both periods. A loss is a negative profit.

Which formula from which data

You are given	Find first	Then
Price, variable cost per unit, fixed cost	Contribution per unit	BEP = fixed cost ÷ contribution per unit
Sales and profit of two periods	P/V = change in profit ÷ change in sales	Fixed cost = contribution − profit; BES = fixed cost ÷ P/V
Profit and P/V ratio	MOS = profit ÷ P/V	BES = sales − MOS
Fixed cost, P/V ratio, MOS ratio	BES = fixed cost ÷ P/V	Sales = BES ÷ (1 − MOS ratio)
MOS in units and in ₹, BEP units	Price = MOS (₹) ÷ MOS (units)	Sales = (MOS + BEP) units × price

Definition

Break-even point (BEP)

The level of sales where an entity neither earns profit nor incurs loss. BEP is indicated in both quantity and monetary value terms. (SM 14.49)

Definition

Margin of safety (MOS)

The margin between sales and the break-even sales is known as margin of safety. It can either be indicated in quantitative or monetary terms. (SM 14.49)

Definition

Angle of incidence

This angle is formed by the intersection of sales line and total cost line at the break-even point. This angle shows the rate at which profit is earned once the break-even point is reached. (SM 14.31)

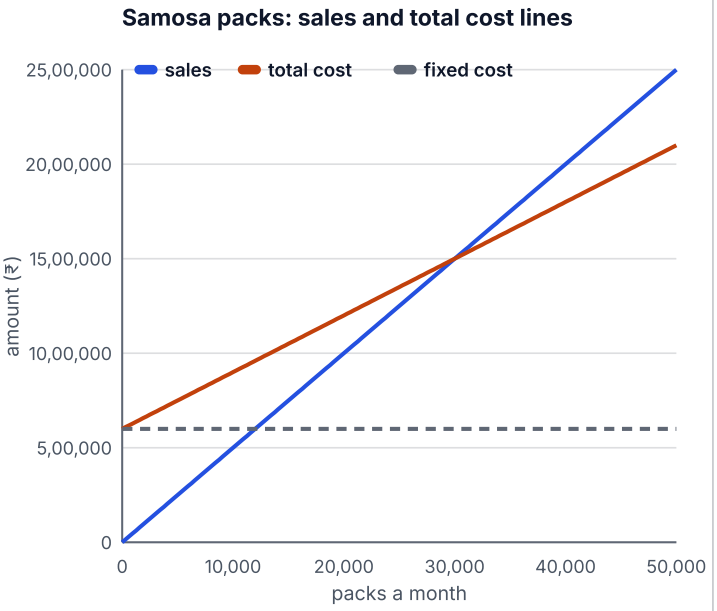
Worked examples

Worked example

Samosa packs: read the chart, then cut the price

Tapri's Indore factory can make 50,000 frozen samosa packs a month, and its fixed costs are ₹6,00,000 a month. At full capacity, sales would be ₹25,00,000 and

total cost ₹21,00,000. Ritu has drawn the sales and total cost lines on the chart below:



Blue: sales; orange: total cost; grey dashed: fixed cost.

Required:

(i) Calculate the selling price, variable cost and contribution per pack, and the P/V ratio.

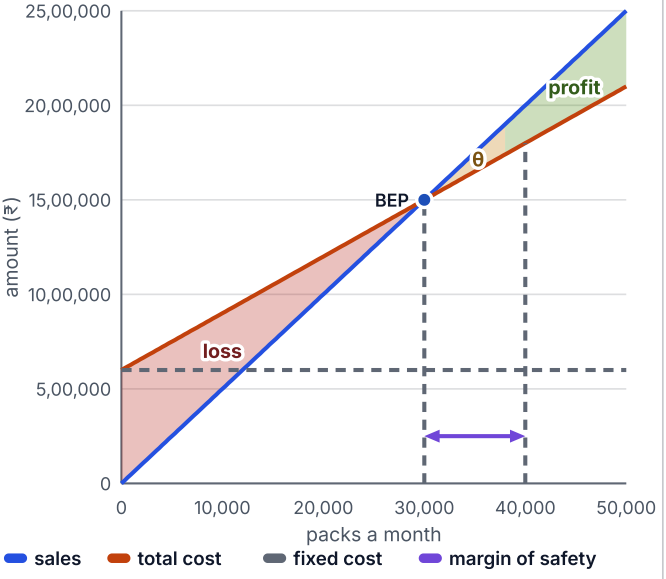
(ii) Calculate the break-even point in packs and in rupees, and show it on the chart.

(iii) For a budget of 40,000 packs, calculate the margin of safety in packs, in rupees and as a percentage, and the profit.

(iv) Kesar wants to cut the price to ₹45 a pack and expects to sell all 50,000 packs. Calculate the new break-even point, margin of safety and profit, and comment on the angle of incidence.

THE WORKING

Samosa packs: the chart, marked



The BEP is where the blue sales line crosses the orange total cost line: 30,000 packs and ₹15,00,000. The violet arrow is the margin of safety at the budget of 40,000 packs; θ (amber) is the angle of incidence.

(i) Working notes

1. Selling price per pack = ₹25,00,000 ÷ 50,000 = **₹50**.

2. Variable cost per pack = (₹21,00,000 – ₹6,00,000) ÷ 50,000 = **₹30**.
3. Contribution per pack = ₹50 – ₹30 = **₹20**; P/V ratio = ₹20 ÷ ₹50 × 100 = **40%**.

(ii) Break-even point

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{Contribution per pack}} = \frac{6{,}00{,}000}{20} = 30{,}000 \text{ packs}$$

Break-even sales = 30,000 × ₹50 = **₹15,00,000** (check: ₹6,00,000 ÷ 40%), 60% of capacity: on the chart, where the sales line crosses the total cost line.

(iii) At the budget of 40,000 packs

Particulars	Packs	Amount (₹)
Budgeted sales	40,000	20,00,000
Less: Break-even sales	30,000	15,00,000
Margin of safety	**10,000**	**5,00,000**

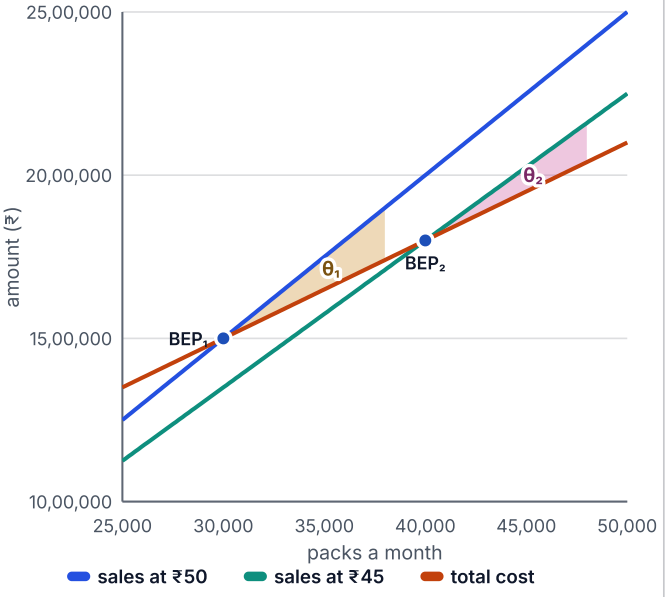
MOS ratio = ₹5,00,000 ÷ ₹20,00,000 × 100 = **25%**. Profit = ₹5,00,000 × 40% = **₹2,00,000**. Check: contribution 40,000 × ₹20 = ₹8,00,000, less fixed cost ₹6,00,000.

(iv) Price cut to ₹45, 50,000 packs

Contribution = ₹45 – ₹30 = ₹15 a pack; P/V ratio = 15 ÷ 45 = 33.33%.

Particulars	₹50, 40,000 packs	₹45, 50,000 packs
BEP (packs)	30,000	40,000
BEP as % of capacity	60%	80%
Break-even sales (₹)	15,00,000	18,00,000
Margin of safety (packs)	10,000	10,000
Margin of safety (₹)	5,00,000	4,50,000
MOS ratio	25%	20%
Profit (₹)	**2,00,000**	**1,50,000**

Price ₹50 against ₹45: the angle of incidence, close up



Close-up of the right half of the chart. Blue: sales at ₹50; teal: sales at ₹45; orange: total cost. BEP_1 and the amber wedge θ_1 belong to the ₹50 price, BEP_2 and the pink wedge θ_2 to ₹45: θ_2 opens later and narrower.

Comment. The cut takes a quarter off the contribution per pack, so the sales line is flatter and the angle of incidence narrower: profit grows more slowly after break-even. The BEP moves from 60% to 80% of capacity, and even with every pack sold, profit falls by ₹50,000. On these figures the price cut is not advisable.

Worked example

Chai premix: two years, one of them a loss

Tapri's chai premix division gives you the following results. Selling price, variable cost per unit and fixed cost were the same in both years.

Year	Sales (₹)	Profit / (Loss) (₹)
2024-25	1,50,00,000	(9,00,000)
2025-26	2,50,00,000	21,00,000

Required:

(i) Calculate the P/V ratio and the fixed cost.

(ii) Calculate the break-even sales.

(iii) Calculate the margin of safety for 2025-26 in rupees and as a percentage of sales.

(iv) Calculate the profit or loss if sales are ₹3,00,00,000.

(v) To fight BrewBucks' new instant coffee, Dev proposes a 10% price cut for 2026-27, with variable cost per unit and fixed cost unchanged. Calculate the sales needed to earn the 2025-26 profit, and the new break-even sales.

THE WORKING

(i) P/V ratio and fixed cost

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100 = \frac{21{,}00{,}000 - (-9{,}00{,}000)}{2{,}50{,}00{,}000 - 1{,}50{,}00{,}000} \times 100$$
$$= \frac{30{,}00{,}000}{1{,}00{,}00{,}000} \times 100 = 30\%$$

Fixed cost = contribution − profit of 2025-26 = ₹2,50,00,000 × 30% − ₹21,00,000 = **₹54,00,000**. Check with 2024-25: ₹45,00,000 − ₹54,00,000 = loss of ₹9,00,000.

(ii) Break-even sales = ₹54,00,000 ÷ 30% = **₹1,80,00,000**.

(iii) Margin of safety 2025-26 = ₹2,50,00,000 − ₹1,80,00,000 = **₹70,00,000**, which is **28%** of sales. Check: ₹70,00,000 × 30% = ₹21,00,000, the year's profit.

(iv) Profit at sales of ₹3,00,00,000 = ₹90,00,000 − ₹54,00,000 = **₹36,00,000**.

(v) After a 10% price cut

Working note: take the old price as ₹100. With a P/V ratio of 30%, the variable cost is ₹70. The new price is ₹90, the contribution ₹20, and the new P/V ratio 20 ÷ 90 = 2/9 (22.22%).

Contribution needed = ₹54,00,000 + ₹21,00,000 = ₹75,00,000.

$$\text{Sales needed} = \frac{75{,}00{,}000}{2/9} = ₹3{,}37{,}50{,}000$$

$$\text{New BES} = \frac{54{,}00{,}000}{2/9} = ₹2{,}43{,}00{,}000$$

Divide by the fraction 2/9, not a rounded 22.22%. In volume, ₹3,37,50,000 at ₹90 against ₹2,50,00,000 at ₹100 is 50% more units, just to stand still on profit.

Worked example

Diwali gift boxes: capacity, a semi-variable cost and the cash break-even

Kesar launches a new gift box every Diwali. This year's line can make 50,000 boxes and is selling 37,500 (75% of capacity) at ₹400 a box. The fixed part of packing and dispatch is paid in cash.

Particulars	Amount
Direct material	₹150 per box
Direct labour	₹60 per box
Variable overheads	₹30 per box
Packing and dispatch, semi-variable (variable part ₹40 per box)	₹21,00,000 at the current level
Fixed costs up to 80% capacity, including depreciation of ₹6,00,000	₹30,00,000
Additional fixed cost beyond 80% capacity	₹6,00,000

Required:

(i) Calculate the break-even point in boxes, in rupees and as a percentage of capacity.

(ii) Calculate the profit and the margin of safety at the current level.

(iii) Calculate the cash break-even point in boxes.

(iv) Kesar wants a profit of ₹15,00,000 with price and variable cost per box unchanged. Calculate the capacity utilisation needed and the margin of safety at that level.

THE WORKING

Working notes

1. Variable cost per box = ₹150 + ₹60 + ₹30 + ₹40 = ₹280; contribution = ₹400 − ₹280 = **₹120** a box; P/V ratio = 30%.
2. Fixed part of packing and dispatch = ₹21,00,000 − (37,500 × ₹40) = **₹6,00,000**.
3. Total fixed cost up to 80% capacity = ₹30,00,000 + ₹6,00,000 = **₹36,00,000**.

(i) BEP = ₹36,00,000 ÷ ₹120 = **30,000 boxes** = **₹1,20,00,000**, which is **60%** of capacity.

(ii) At 37,500 boxes

Particulars	Amount (₹)
Sales (37,500 × ₹400)	1,50,00,000
Less: Variable cost (37,500 × ₹280)	1,05,00,000
Contribution	**45,00,000**
Less: Fixed cost (WN 3)	36,00,000
Profit	**9,00,000**

Margin of safety = 37,500 − 30,000 = **7,500 boxes** = **₹30,00,000**, which is **20%** of sales. Check: ₹30,00,000 × 30% = ₹9,00,000.

(iii) Cash fixed costs = ₹36,00,000 − ₹6,00,000 depreciation = ₹30,00,000; cash BEP = ₹30,00,000 ÷ ₹120 = **25,000 boxes**.

(iv) Profit of ₹15,00,000

Step 1. Within 80% capacity: (₹36,00,000 + ₹15,00,000) ÷ ₹120 = 42,500 boxes, more than the 40,000 boxes that 80% allows. So the additional fixed cost is incurred, and fixed cost becomes ₹42,00,000.

Step 2.

$$\text{Boxes needed} = \frac{42,00,000 + 15,00,000}{120} = 47,500$$

$$\text{Capacity utilisation} = \frac{47,500}{50,000} = 95\%$$

New BEP = ₹42,00,000 ÷ ₹120 = 35,000 boxes. Margin of safety = 47,500 − 35,000 = **12,500 boxes** = **₹50,00,000**, 26.32% of sales of ₹1,90,00,000.

Convention: the new BEP uses the higher fixed cost, which Kesar commits to when he plans for 95%; ICAI's suggested answer to May 2026 Q2(b) does the same.

Exam angle

What the examiner keeps asking

In the exam

Where break-even questions appeared, May 2024 to September 2026

In the ICAI documents read for this lesson, break-even and margin of safety came up 15 times in 13 documents.

- **Descriptive, 3 to 7 marks.** Two years' sales and profit, then a changed cost: May 2024 Q5(b); MTP Series I Q1(c) for September 2025 and for September 2026 (4 marks each). Profit, P/V ratio, BES and fixed cost from a MOS in units and rupees: May 2026 Q5(b) (4 marks). Capacity, a semi-variable cost, cash BEP and a step in fixed cost: May 2026 Q2(b) (7 marks). BES of two machines: January 2025 Q1(a)(i) (3 marks). BES, profit and MOS%: RTP May 2024 Q11.
- **Multiple choice, 2 marks each**, often five on one case scenario: May 2026 MCQ 15, September 2025 MCQ 9, January 2026 MCQ 4 (multi-product); RTP January 2025 MCQ 7 (BEP, cash BEP, P/V ratio), RTP September 2025 MCQ 6; MTP Series I for September 2024 MCQs 6–10, Series II for January 2026 MCQ 3 and for May 2026 MCQ 11.
- **Angle of incidence**: in none of them, only in the Study Material's own exercises.

In the exam

The wording traps ICAI used

- "Sales could decrease by 50% before the company starts incurring losses" is a MOS ratio of 50% (RTP September 2025 MCQ 6).
- A loss in one of the two years (May 2024 Q5(b)): change in profit = profit − (−loss).
- "Variable cost will go up by 5%" (May 2024 Q5(b)): the suggested answer solved it both ways, 75% × 1.05 = 78.75% and 75% + 5% = 80%. State the reading you use.
- May 2026 Q2(b): split the semi-variable cost, take depreciation out of the cash BEP, and check whether the target crosses the step in fixed cost.
- The paper's own instruction: "Working notes should form part of the answer."

Trick

The three-line identity

BES × P/V ratio = fixed cost; MOS × P/V ratio = profit; so sales × P/V ratio = fixed cost + profit. Any two of sales, BES and MOS give the third. Conditions: one product (or a constant mix), with price, variable cost per unit and fixed cost constant over the range.

Trick

BES% + MOS% = 100%

On the same total sales, break-even sales and margin of safety are its two parts: if the MOS is 28% of sales, BES = sales × 72%.

Answer frame

The theory parts

The chapter's own theory questions (SM 14.52) include these three. Write the points in this order.

Short note on margin of safety

1. Definition: expected (actual) sales minus break-even sales, in units or rupees.
2. Formulas: sales − BES; profit ÷ P/V ratio; MOS ratio = (sales − BES) ÷ sales.
3. Meaning: the larger the margin of safety, the higher the chances of making profits.
4. A large angle of incidence with a high margin of safety: an extremely favourable position.
5. A two-line numerical example.

Short note on angle of incidence

1. The angle formed by the intersection of the sales line and the total cost line at the BEP.
2. It shows the rate at which profit is earned once the BEP is reached (below the BEP, the rate of loss).

3. The wider the angle, the greater the rate of earning profit; it widens with the P/V ratio.
4. A large angle with a high margin of safety: an extremely favourable position.
5. A labelled sketch.

Explain and illustrate the BEP with a break-even chart

1. Definition: no profit, no loss; total cost equals total sales.
2. Formula in units and in rupees.
3. The chart: costs and revenues up the side, activity along the bottom; fixed cost, total cost and sales lines.
4. The BEP where sales cut total cost; margin of safety and profit read off.
5. A labelled sketch with figures.

The numerical parts: three answers to the same question

Loses marks

"BEP = 30,000 boxes."

A bare figure: if it is wrong, nothing earns method marks.

Passes

"BEP = Fixed cost ÷ Contribution per unit = ₹36,00,000 ÷ ₹120 = 30,000 boxes."

Formula and figures, but ₹36,00,000 comes from nowhere: the semi-variable split is invisible.

Full marks

Working notes first (contribution per box, fixed part of the semi-variable cost, total fixed cost); then formula, figures and result with its unit; any reading of ambiguous wording stated as an assumption.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the loss that lost its minus sign

A loss of ₹9,00,000, then a profit of ₹21,00,000. The hurried answer takes the change in profit as ₹12,00,000, a P/V ratio of 12%, and every later part falls with it. **Fix:** a loss is a negative profit, so the change is ₹21,00,000 − (−₹9,00,000) = ₹30,00,000 and the P/V ratio 30%.

⚠ **Common mistake**

The margin of safety measured against the wrong sales

9,000 ÷ 15,000 = 60% divides the café's margin of safety by its break-even sales. **Fix:** divide by the total (actual or budgeted) sales: 9,000 ÷ 24,000 = 37.5%. Test: BES% and MOS% must add up to 100%.

⚠ **Common mistake**

Depreciation smuggled into the cash break-even point

Using the full fixed cost for the cash BEP, or taking out rent and salaries along with depreciation. **Fix:** leave out depreciation and other non-cash fixed costs, and nothing else.

⚠ **Common mistake**

The semi-variable cost that stayed in one piece

Treating all ₹21,00,000 of packing and dispatch as variable gives ₹56 a box, a contribution of ₹104 and a BEP of about 28,846 boxes instead of 30,000. **Fix:** split it first; the variable part joins the variable cost per unit, the fixed part the fixed cost.

Practice

Check yourself

1 A Tapri kiosk earned a profit of ₹72,000 last month at a P/V ratio of 24%. What was its margin of safety in rupees?

Answer: _____

2 Case scenario. Presenting next year's plan for the cold-chain division, Dev tells the board: "Our sales could fall by 40% before we make a loss. Every rupee of sales leaves 25 paise after variable costs. Fixed costs are ₹6,00,000." Ritu must put the projected sales on the slide. Which figure is correct?

- **A** ₹24,00,000
- **B** ₹40,00,000
- **C** ₹60,00,000
- **D** ₹15,00,000

3 A division's sales were ₹40,00,000 with a loss of ₹1,50,000 in one year, and ₹60,00,000 with a profit of ₹4,50,000 in the next, with no change in prices, unit variable cost or fixed cost. What are its break-even sales?

- **A** ₹30,00,000
- **B** ₹50,00,000
- **C** ₹45,00,000
- **D** ₹13,50,000

4 Which of these, on its own, widens the angle of incidence on a break-even chart?

- **A** An increase in fixed costs
- **B** A decrease in the variable cost per unit
- **C** An increase in the number of units sold
- **D** A decrease in fixed costs

5 Which of these fixed costs of the gift-box line are left out of its cash break-even point? Select all that apply.

- **A** Depreciation on the packing machine
- **B** Salary of the warehouse supervisor, paid monthly
- **C** Amortisation of a design licence paid for two years ago
- **D** Rent of the godown, paid monthly

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Contribution per unit = price − variable cost per unit; P/V ratio = contribution ÷ sales × 100.
- BEP (units) = fixed cost ÷ contribution per unit; BES (₹) = fixed cost ÷ P/V ratio.
- Cash BEP = cash fixed costs ÷ contribution per unit (depreciation and other non-cash costs out).
- Target profit: (fixed cost + desired profit) ÷ contribution per unit, or ÷ P/V ratio for rupees.
- MOS = total sales − BES = profit ÷ P/V ratio; MOS ratio = MOS ÷ total sales; profit = MOS × P/V ratio.

- Two periods: P/V ratio = change in profit ÷ change in sales, and a loss is a negative profit.
- Angle of incidence: where the sales line cuts the total cost line at the BEP; the wider, the faster profit is earned. Large angle plus high MOS: extremely favourable.
- Fixed cost up: BEP rises, angle unchanged. (Bucket first, then the glass.)

Answer key · Check yourself

1 **300000**

Margin of safety = profit ÷ P/V ratio = ₹72,000 ÷ 24% = ₹3,00,000.

2 **B** **₹40,00,000**

40% is the MOS ratio and 25 paise a rupee a P/V ratio of 25%. BES = ₹6,00,000 ÷ 25% = ₹24,00,000, the other 60% of sales, so sales = ₹24,00,000 ÷ 60% = ₹40,00,000. ₹24,00,000 is the BES itself; ₹60,00,000 divides by 40%.

3 **C** **₹45,00,000**

P/V ratio = ₹6,00,000 ÷ ₹20,00,000 = 30%; fixed cost = ₹18,00,000 − ₹4,50,000 = ₹13,50,000; BES = ₹13,50,000 ÷ 30% = ₹45,00,000. Dropping the loss's minus sign gives P/V 15% and ₹30,00,000; ₹13,50,000 is the fixed cost.

4 **B A decrease in the variable cost per unit**

A lower variable cost flattens the total cost line, so the lines open faster after the BEP (the P/V ratio rises). Fixed cost moves the total cost line without changing its slope; more units move along the same lines.

5 **A Depreciation on the packing machine**; **C Amortisation of a design licence paid for two years ago**

The cash BEP keeps only fixed costs payable in cash. Depreciation and amortisation are non-cash; salary and rent are paid in cash and stay in.

14.3 Break-even and profit-volume charts

How to draw and read the three CVP pictures ICAI teaches: the break-even chart, the contribution break-even chart and the profit-volume chart, and what each shows that the others hide.

In plain words

Kesar does not read tables

The board meets at nine. Ritu has a page of break-even figures for Tapri's new cookie line; Kesar has his reading glasses in another city.

> **Kesar:** Ritu, no tables. Draw it. If I can see it, I can decide it.
>
> **Ritu:** Three lines, then. One for sales, one for total cost, one for the fixed cost we pay even if we sell nothing. Where sales cross total cost, we break even.
>
> **Dev:** And where do I see contribution on that? It's the number I actually manage.
>
> **Ritu:** Not on that one. Swap the fixed cost line for a variable cost line and it appears. And for you, Kesar, a single line: profit, starting at minus the fixed cost.

That is the whole lesson. A **break-even chart** puts rupees up the side and activity (units, or sales) along the bottom, and draws a sales line, a total cost line and a fixed cost

line; the break-even point is where sales meet total cost. A **contribution break-even chart** draws the variable cost line instead of the fixed cost line, so contribution shows as the gap between sales and variable cost. A **profit-volume chart** throws away costs and revenues altogether and draws profit against activity: one line that starts at a loss equal to the fixed cost and crosses zero at the break-even point.

All three come from the same three numbers (price, variable cost per unit, fixed cost) and show the same break-even point. What differs is what you can read off them.

Think of it like

Three cameras on one cricket chase

The wide-angle camera shows both teams' totals growing (the break-even chart: sales and total cost). The stump camera shows what each ball actually added (the contribution chart). The scoreboard shows one number, runs still needed, falling to zero and then the winning margin (the profit-volume chart: a loss that shrinks to break-even and becomes profit). Same match, three views, one moment of victory.

Key idea

Every CVP chart is straight lines through two easy points. The break-even point is where sales cross total cost (or the profit line crosses zero), and each chart is chosen for what it lets you read: the break-even chart shows costs, the contribution chart adds contribution, the profit-volume chart shows profit directly.

How it works

ICAI lists two methods of break-even analysis (SM 14.14): **(A) algebraic computations**, the formulas of break-even point and margin of safety, and **(B) graphic presentations**, the three charts of this lesson.

Two points per line: drawing the break-even chart

Tapri's cookie line: price ₹250 a jar, variable cost ₹150, fixed cost ₹20,00,000 a month, capacity 40,000 jars, budget 30,000. ICAI's steps (SM 14.20):

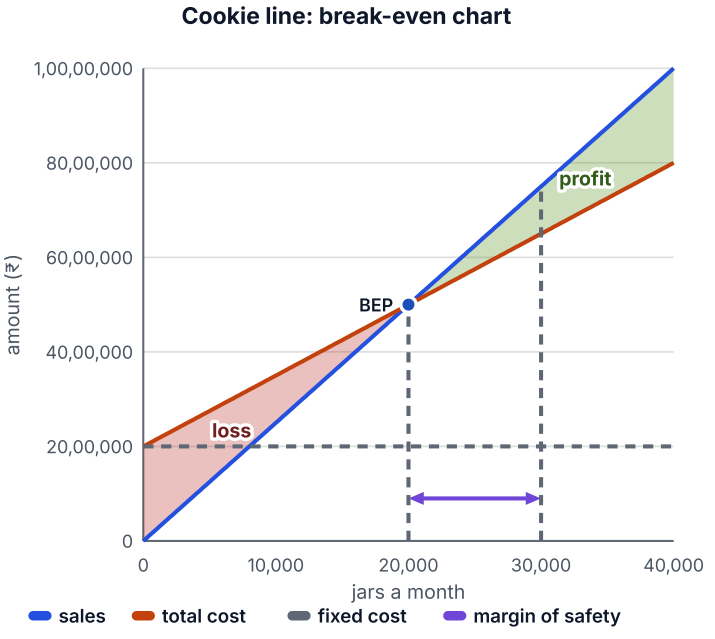
1. **Axes.** Costs and revenues on the vertical axis; the level of activity (here jars) on the horizontal axis. Choose a scale that reaches maximum activity and the highest sales figure.

2. **Fixed cost line**: horizontal at ₹20,00,000.

3 **Total cost line**: from the fixed cost on the vertical axis (₹20,00,000 at zero jars) to total cost at maximum activity (₹20,00,000 + 40,000 × ₹150 = ₹80,00,000).

4 **Sales line**: from the origin to sales at maximum activity (40,000 × ₹250 = ₹1,00,00,000).

5 **Read it.** The break-even point is where the sales line intersects the total cost line: 20,000 jars, ₹50,00,000. At the budget, the margin of safety (20,000 to 30,000 jars) and the profit (the gap between the lines, ₹75,00,000 − ₹65,00,000) can be measured off the chart.



Blue: sales; orange: total cost; grey dashed: fixed cost. They cross at the BEP, 20,000 jars and ₹50,00,000. Rose: loss; green: profit. The violet arrow is the margin of safety, 10,000 jars, at the budget of 30,000.

Interactive version: PassBook app, lesson Break-even and profit-volume charts

The contribution break-even chart: where contribution becomes visible

ICAI's complaint about the chart above: it cannot show contribution, "literally the backbone of marginal costing" (SM 14.20). The **contribution break-even chart** keeps the sales and total cost lines and replaces the fixed cost line by a **variable cost line from the origin** (zero at zero

activity; ₹60,00,000 at 40,000 jars). The total cost line runs parallel above it, ₹20,00,000 higher all the way. Contribution at any activity is the gap between the sales line and the variable cost line.

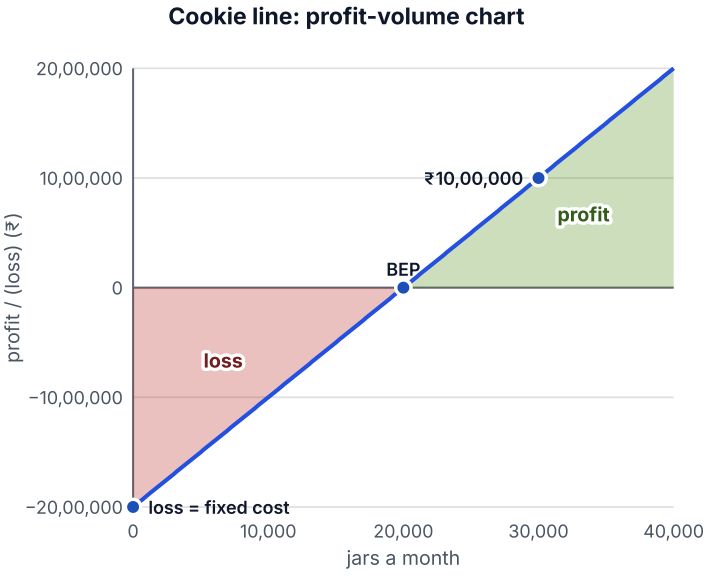


Blue: sales; orange: total cost; teal: variable cost. The amber area between sales and variable cost is contribution; the violet arrow measures it at 30,000 jars (₹30,00,000). The pink arrow is the fixed cost: the constant ₹20,00,000 gap between total cost and variable cost.

Interactive version: PassBook app, lesson Break-even and profit-volume charts

The profit-volume chart: one line, no costs

In the **profit-volume (P/V) chart** the vertical axis is profit (above) and loss (below), and the horizontal axis is drawn at zero profit. The line starts at a loss equal to the fixed cost at nil activity (−₹20,00,000); the second point is the break-even point or the profit at any level (₹10,00,000 at 30,000 jars). The break-even point is where the line cuts the horizontal axis (SM 14.21–14.22).



The blue profit line starts at −₹20,00,000 (the fixed cost) and rises ₹100 for every jar, the contribution per jar. It cuts the horizontal axis at the BEP, 20,000 jars; rose is the loss area, green the profit area.

The slope of the profit line is the contribution per unit (₹100 a jar); with sales in rupees along the bottom it is the P/V ratio (40 paise of extra profit for every extra rupee

of sales, with fixed cost unchanged): the chart draws the very relationship the ratio measures.

Interactive version: PassBook app, lesson Break-even and profit-volume charts

Where to read what

To read
Break-even point

Break-even chart
sales line cuts total cost line

Contribution chart
sales line cuts total cost line

Profit-volume chart
profit line cuts the horizontal axis

To read
Fixed cost

Break-even chart
height of the fixed cost line

Contribution chart
gap: total cost − variable cost

Profit-volume chart
the loss at zero activity

To read
Variable cost

Break-even chart
gap: total cost − fixed cost

Contribution chart
height of the variable cost line

Profit-volume chart
not shown

To read

Contribution

Break-even chart

not shown

Contribution chart

gap: sales − variable cost

Profit-volume chart

height of the line above −F

To read

Profit or loss

Break-even chart

gap: sales − total cost

Contribution chart

gap: sales − total cost

Profit-volume chart

height of the line

To read

Margin of safety

Break-even chart

expected activity − BEP, along the bottom

Contribution chart

the same

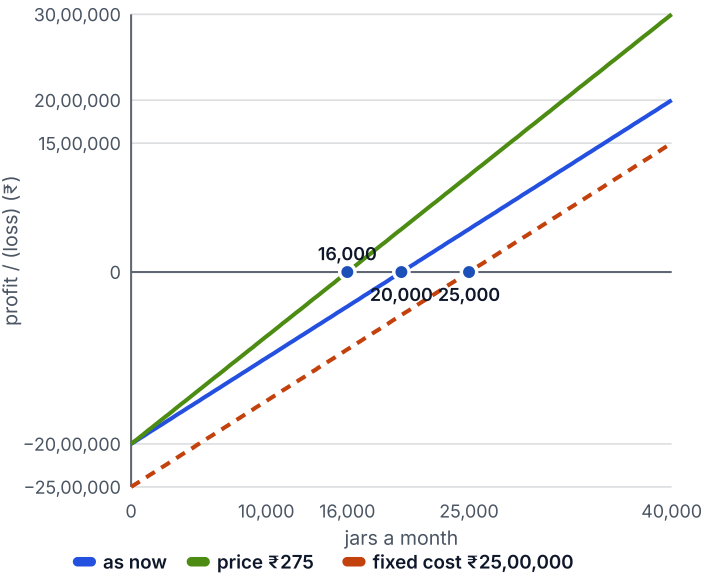
Profit-volume chart

the same

What a change does to the profit line

ICAI calls this the P/V chart's biggest advantage: it shows clearly the effect of any change in the variables on profit and the break-even point (SM 14.22).

Cookie line: a price rise and a fixed-cost rise on the P/V chart



Blue: the present line. Green: price up to ₹275, so the line pivots on the same −₹20,00,000 and climbs faster; the BEP moves left to 16,000 jars. Orange dashed: fixed cost up to ₹25,00,000, so the line drops parallel; the BEP moves right to 25,000 jars.

Change, all else equal	Profit line	BEP
Selling price up (or variable cost down)	steeper, same starting point	moves left
Selling price down (or variable cost up)	flatter, same starting point	moves right
Fixed cost up	parallel, lower	moves right
Fixed cost down	parallel, higher	moves left
Volume changes	a different point on the same line	no change

Formats and formulas

Each chart is plotted from a small table of points; write it as a working note above the graph. *Reproduce the line list for each chart from memory.*

Chart	Lines to draw	Point at zero activity	Point at maximum activity
Break-even	fixed cost	F	F
	total cost	F	F + V × units
	sales	0	price × units
Contribution break-even	variable cost	0	V × units
	total cost	F	F + V × units
	sales	0	price × units
Profit-volume	profit	−F	C × units − F

∑ **Formula**

The lines as equations (x = units)

$$\text{Sales} = p\,x \qquad \text{Total cost} = F + v\,x$$

$$\text{Variable cost} = v\,x \qquad \text{Profit} = (p - v)\,x - F$$

p = selling price per unit, v = variable cost per unit, F = fixed cost. With sales value S along the horizontal axis: profit = S × P/V ratio − F.

Σ **Formula**

Where two profit lines cross

$$\text{Units at which profits are equal} = \frac{F_2 - F_1}{C_2 - C_1}$$

C_1, C_2 = contribution per unit, F_1, F_2 = fixed costs of the two options. When the option with the higher contribution per unit also has the higher fixed cost (the usual trade-off), it earns more above this volume and the other earns more below it (the crossing in SM Example 5, 14.23; solved this way in January 2025 Q1(a)(ii)). If one option has both the higher contribution and the lower fixed cost, the lines never cross and it is better at every volume.

Definition

Break-even chart

A breakeven chart records costs and revenues on the vertical axis and the level of activity on the horizontal axis. The breakeven point is that point where the sales revenue line intersects the total cost line. (SM 14.20)

Definition

Contribution breakeven chart

Based on the same principles as a conventional breakeven chart except that it shows the variable cost line instead of the fixed cost line; contribution can be read as the difference between the sales revenue line and the variable cost line. (SM 14.20–14.21)

Definition

Profit-volume chart

A chart in which the vertical axis represents profits and losses and the horizontal axis is drawn at zero profit or loss; the breakeven point is where the profit line intersects the horizontal axis. (SM 14.21)

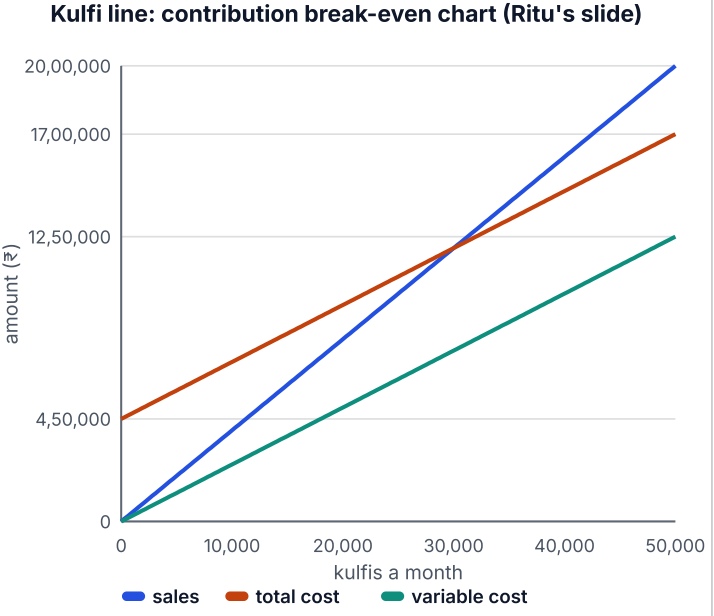
Worked examples

Worked example

The chart without the numbers: reading kulfi costs off a break-even chart

Ritu presented the kulfi line's contribution break-even chart to the board; the cost sheet behind it has gone

missing. Capacity is 50,000 kulfis a month and expected sales are 40,000. Here is her chart:



Blue: sales; orange: total cost; teal: variable cost. Each line is straight over the whole range.

Required:

(i) Read from the chart the selling price, the variable cost per kulfi and the fixed cost a month.

(ii) Calculate the break-even point and, at the expected sales, the margin of safety, the contribution and the profit; show them on the chart.

(iii) Kesar wants a profit of ₹2,40,000 a month at the same price. How many kulfis must be sold, and is it within capacity?

THE WORKING

(i) Readings from the chart

Line
Sales

At 0 kulfis (₹)
0

At 50,000 kulfis (₹)
20,00,000

Reading
Price = ₹20,00,000 ÷ 50,000 = **₹40**

Line
Variable cost

At 0 kulfis (₹)
0

At 50,000 kulfis (₹)
12,50,000

Reading
Variable cost = ₹12,50,000 ÷ 50,000 = **₹25** a kulfi

Line

Total cost

At 0 kulfis (₹)

4,50,000

At 50,000 kulfis (₹)

17,00,000

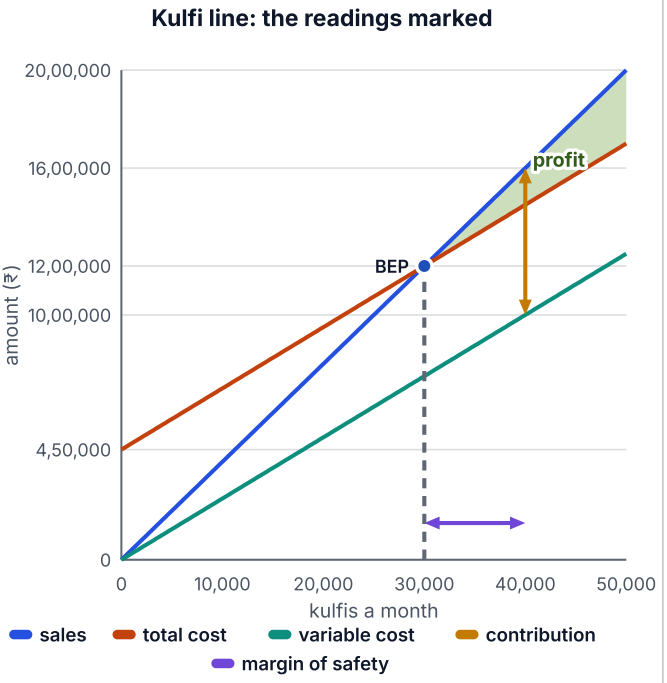
Reading

Fixed cost = total cost at nil activity = **₹4,50,000**; also the constant gap between total cost and variable cost (₹17,00,000 − ₹12,50,000)

(ii) Contribution = ₹40 − ₹25 = ₹15 a kulfi; P/V ratio = 37.5%.

Reading	Working	Amount
Break-even point	₹4,50,000 ÷ ₹15	30,000 kulfis (₹12,00,000)
Margin of safety	40,000 − 30,000	10,000 kulfis (₹4,00,000), 25%
Contribution	40,000 × ₹15 = ₹16,00,000 − ₹10,00,000	₹6,00,000
Profit	₹6,00,000 − ₹4,50,000	₹1,50,000

Check: margin of safety × P/V ratio = ₹4,00,000 × 37.5% = ₹1,50,000.



BEP at 30,000 kulfis (₹12,00,000), with a dotted line to the activity axis. At 40,000 kulfis the amber arrow is the contribution (₹16,00,000 − ₹10,00,000) and the height of the green wedge the profit (₹16,00,000 − ₹14,50,000). The violet arrow is the margin of safety.

(iii) Target profit of ₹2,40,000

$$\text{Kulfis} = \frac{\text{Fixed cost} + \text{Target profit}}{\text{Contribution per kulfi}} = \frac{4{,}50{,}000 + 2{,}40{,}000}{15} = 46{,}000$$

46,000 kulfis (sales of ₹18,40,000) is within the capacity of 50,000, but 6,000 above the expected sales: the target needs a 15% rise in volume at the present price.

Worked example

Cold coffee in summer and winter: two profit lines and a shut-down point

Tapri's cold coffee counters sell at ₹80 a cup from April to September and, to keep winter custom, at ₹70 from October to March. Variable cost is ₹50 a cup all year, and fixed costs are ₹6,00,000 for each half-year. Expected sales are 36,000 cups in summer and 28,000 cups in winter. If the counters were closed for the winter half-year, ₹2,50,000 of its fixed costs (lease and equipment) would still be incurred; the rest would be saved.

Required:

(i) Draw on one profit-volume graph a line for each half-year, and read each break-even point.

(ii) Read the profit or loss at the expected sales of each half-year.

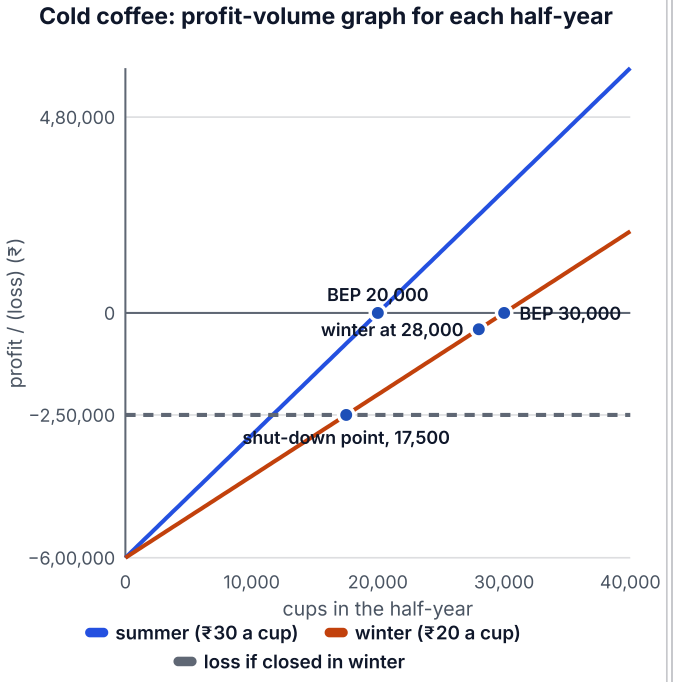
(iii) Show on the graph the loss if the counters close for winter, and advise whether to close. Below what winter volume would closing be better?

THE WORKING

Working notes

Particulars	Summer	Winter
Contribution per cup (₹)	30	20
P/V ratio	37.5%	28.57%
Loss at nil activity (₹)	(6,00,000)	(6,00,000)
Break-even point (cups)	20,000	30,000
Profit / (loss) at expected sales (₹)	4,80,000	(40,000)

Expected-sales points: summer 36,000 × ₹30 − ₹6,00,000 = ₹4,80,000; winter 28,000 × ₹20 − ₹6,00,000 = (₹40,000).



Blue: summer; orange: winter. Both start at the same loss of ₹6,00,000; the winter line is flatter because each cup contributes ₹10 less, so it breaks even later. Grey dashed: the ₹2,50,000 loss that remains if the counters close; the winter line crosses it at 17,500 cups.

(i) Summer break-even point: 20,000 cups (₹16,00,000). Winter: 30,000 cups (₹21,00,000).

(ii) Summer: profit ₹4,80,000. Winter: loss ₹40,000, because the expected 28,000 cups are 2,000 short of the winter break-even point.

(iii) Close for winter? Closing saves the variable cost and ₹3,50,000 of fixed cost (₹6,00,000 –

₹2,50,000), but loses the winter contribution.

Particulars	Stay open (₹)	Close (₹)
Contribution: 28,000 × ₹20	5,60,000	—
Fixed costs	(6,00,000)	(2,50,000)
Profit / (loss) for winter	**(40,000)**	**(2,50,000)**

Stay open: the winter loss is ₹2,10,000 smaller than the loss on closing. Closing is better only if winter sales fall below the volume whose contribution just covers the avoidable fixed cost: ₹3,50,000 ÷ ₹20 = **17,500 cups**, the point where the winter line meets the grey line.

Worked example

Four lines, one profit graph: the path and the line

Tapri's Bhopal outlets sell four lines. Fixed costs are ₹6,00,000 for the period.

Line	Sales (₹)	Variable cost (₹)
Samosa	8,00,000	6,00,000
Kulhad chai	12,00,000	4,80,000
Cold coffee	5,00,000	4,20,000
Premix (retail)	10,00,000	6,00,000

Required:

(i) Prepare a profit graph taking the lines in descending order of P/V ratio, and find the break-even point.

(ii) Kesar reads the break-even point as ₹10,00,000, where the path first crosses the axis. Explain why that is not the break-even point of the business.

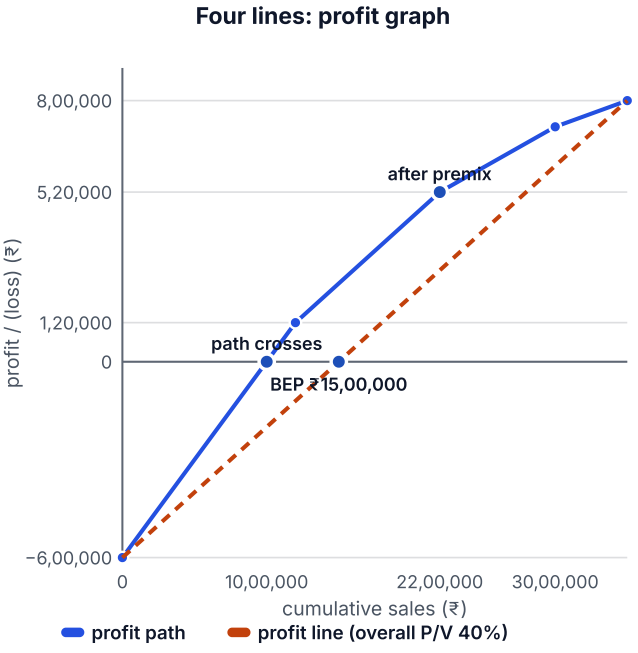
(iii) Read from the graph the profit if only the two lines with the highest P/V ratios were sold, fixed costs unchanged.

THE WORKING

Statement showing cumulative sales and profit (lines in descending order of P/V ratio)

Line	Contribution (₹)	P/V ratio	Cumulative sales (₹)	Cumulative profit (₹)
Kulhad chai	7,20,000	60%	12,00,000	1,20,000
Premix (retail)	4,00,000	40%	22,00,000	5,20,000
Samosa	2,00,000	25%	30,00,000	7,20,000
Cold coffee	80,000	16%	35,00,000	8,00,000

Cumulative profit = cumulative contribution − ₹6,00,000; at nil sales the loss is the fixed cost, ₹6,00,000.



Blue: the profit path, one segment per line, its slope that line's P/V ratio, steepest first. Orange dashed: the profit line from −₹6,00,000 to the final ₹8,00,000; it cuts the axis at the break-even point.

(i) Total contribution ₹14,00,000 on sales of ₹35,00,000 gives an overall P/V ratio of 40%; break-even point = ₹6,00,000 ÷ 40% = **₹15,00,000** of sales, where the profit line cuts the horizontal axis.

(ii) The path crosses zero at ₹6,00,000 ÷ 60% = ₹10,00,000 only because it assumes that kulhad chai, the line with the highest P/V ratio, is sold first and alone. The business sells all four lines together

in the present mix; the CVP assumption of a constant sales mix puts its break-even point on the straight profit line, whose slope is the overall P/V ratio. Read the break-even point off the line, not the path.

(iii) The point after the second segment: kulhad chai and premix alone earn a profit of **₹5,20,000** (contribution ₹11,20,000 − ₹6,00,000), ₹2,80,000 less than the ₹8,00,000 from all four lines. Every line has a positive contribution, so dropping any of them lowers profit unless fixed costs fall too; see short-term decisions.

Exam angle

What ICAI has asked

In the exam

Charts in the papers, May 2024 to September 2026

A search of the text of all 55 new-scheme Paper 4 documents (past papers with suggested answers, RTPs, MTPs) and all 56 old-scheme Paper 3 documents found no question that asks you to draw a chart. The chart questions are the Study Material's own: theory question 1, "EXPLAIN and ILLUSTRATE break-even point with the help of break-even chart"; practical problem 13, "DRAW a break-even chart showing the break-even point" and the new BEP after a price cut; Illustration 5, "DRAW a break-even chart showing

contribution and profit"; Illustration 6, "PREPARE a profit graph"; Example 5, two profit-volume lines on one graph.

The numbers behind the charts are asked all the time: January 2025 Q1(a)(ii), the sales level at which two machines are equally profitable (where two profit lines cross, 2 marks); old-scheme RTP November 2023 Q12(b), which of two processes gives more profit at the expected volume; MTP Series I for September 2025 Q1(c), the P/V ratio, BEP and margin of safety of a first half-year, then the sales of a loss-making second half (4 marks).

In the exam

If a chart is asked: what ICAI's own solved charts show

With no chart in a recent paper there is no suggested answer to copy; the Study Material's solved charts (Example 5, Illustrations 5 and 6, practical problem 13) carry these elements:

- The workings for each line (the profit at the chosen volume, the BEP), then the graph.
- Both axes labelled with what they measure and the unit ("Cost and Revenue (₹ '00000)", "Output ('000 units)"), on an even scale; a title where there are two situations.
- Every line named on the chart ("Sales line", "Total cost line", "Fixed cost line"); the BEP labelled, with a dotted line dropped to the activity axis.

- The readings asked for written out below the chart, with their arithmetic.
- A comment when the question compares two positions: ICAI's Example 5 comment names the BEP of each, the volume above which one is better, and which option gives the larger profit at the budgeted volume.

Trick

Two points, never more

Every line on these charts is straight: compute it at zero and at maximum activity and join them. On the P/V chart the two points can be (0, −F) and the BEP, or (0, −F) and the profit at the expected volume (SM 14.22). Condition: the CVP assumptions (linear costs and revenue in the relevant range).

Answer frame

"Explain the profit-volume chart and its advantage"

1. **What it plots:** profit and loss on the vertical axis, the horizontal axis drawn at zero profit, activity (units or sales) along it.
2. **How it is drawn:** a line from the loss at nil activity (equal to the fixed cost) to the profit at a given level or the BEP.

3. **Reading:** the BEP where the line cuts the horizontal axis; profit or loss at any activity as the height of the line.
4. **Advantage (ICAI's "biggest"):** it depicts clearly the effect on profit and the break-even point of any change in the variables (price, variable cost, fixed cost), by drawing a second line on the same graph.
5. **A labelled sketch with figures.**

"Distinguish the break-even chart from the contribution break-even chart"

Point	Break-even chart	Contribution break-even chart
Lines drawn	fixed cost, total cost, sales	variable cost, total cost, sales
Bottom line	horizontal at fixed cost	from the origin (variable cost)
Contribution	cannot be read (its major limitation)	gap between sales and variable cost lines
Fixed cost	height of the fixed cost line	constant gap between total cost and variable cost lines
BEP and profit	read in the same way on both	read in the same way on both

For "Explain and illustrate the break-even point with a break-even chart", use the frame in the break-even

lesson.

Three answers to "Draw a break-even chart"

Loses marks

A freehand sketch, unlabelled axes, the BEP circled but no figures.

The examiner cannot tell which line is which or check any reading.

Passes

Lines plotted to scale and the BEP figure written in.

No working notes for the points, axes without units, no margin of safety or profit shown.

Full marks

Working-note table of points; titled chart, labelled axes with units and an even scale; every line named; BEP labelled, with a dotted line to the activity axis; the readings asked for, with an arithmetic check; a comment if a change is compared.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the total cost line that left from the origin

Drawing total cost from zero, so it never sits above the sales line and the "break-even point" lands at the origin. **Fix:** total cost starts at the fixed cost on the vertical axis; only the sales line (and the variable cost line) leave the origin.

⚠ **Common mistake**

Contribution read off the wrong chart

Measuring "contribution" on a conventional break-even chart as the gap between sales and fixed cost. That gap is sales minus fixed cost, which means nothing. **Fix:** contribution needs a variable cost line: draw a contribution break-even chart and read sales minus variable cost.

⚠ **Common mistake**

The P/V line that started at zero

Starting the profit line at the origin, as if there were no fixed cost. **Fix:** at nil activity the loss equals the fixed cost; start at −F.

⚠ **Common mistake**

The profit path's first crossing called the break-even point

Reading ₹10,00,000 off the four-line graph. **Fix:** the break-even point of the mix is where the straight profit line from −F to the final profit cuts the axis (₹15,00,000).

Practice

Check yourself

1 On a profit-volume chart, Tapri's delivery kitchen shows a loss of ₹3,60,000 at zero activity and a profit of ₹2,40,000 at 40,000 meals. At how many meals does it break even?

Answer: _____

2 On a conventional break-even chart, what does the vertical gap between the total cost line and the fixed cost line measure at a given activity?

A Contribution

B Total variable cost

C Profit

D Margin of safety

3 Case scenario. Ritu draws two profit-volume lines for the cold-chain division: the present fleet, and a proposal with refrigerated trucks that has higher fixed costs and a lower variable cost per crate. The lines cross at 1,20,000 crates; next year's expected volume is 90,000 crates. Which of these is correct?

A The proposal is better, because its line is steeper

B The present fleet earns more profit at 90,000 crates

C Both earn the same at 90,000 crates, because the BEP does not depend on volume

D The proposal's break-even point must be lower than the present one

4 Tapri's fixed costs rise by ₹2,00,000 with price and variable cost unchanged. What happens to the line on its profit-volume chart?

- **A** It becomes steeper
- **B** It becomes flatter, starting from the same point
- **C** It moves down parallel to itself, so the break-even point moves right
- **D** It does not move, because fixed costs are not on a P/V chart

5 Which lines start at the origin on a contribution break-even chart? Select all that apply.

- **A** Sales
- **B** Variable cost
- **C** Total cost
- **D** Fixed cost

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Break-even chart: rupees up the side, activity along the bottom; fixed cost (horizontal), total cost (from F),

sales (from 0). BEP where sales cuts total cost.

- Contribution break-even chart: variable cost line from the origin instead of the fixed cost line; contribution = gap between sales and variable cost; fixed cost = constant gap between total cost and variable cost.
- Profit-volume chart: profit/loss up the side, horizontal axis at zero profit; the line starts at −F and cuts the axis at the BEP. Slope = contribution per unit (or the P/V ratio with sales value along the bottom).
- On the P/V chart, a price rise steepens the line (BEP left); a fixed cost rise drops it parallel (BEP right); volume moves along it.
- Two options cross where $(F_2 - F_1) \div (C_2 - C_1)$ units: above it the higher-contribution (higher-fixed-cost) option wins, below it the lower-fixed-cost one.
- Several products: the profit path segments slope by each P/V ratio; the BEP is on the straight line from −F to the final profit.
- Every line needs two points; label axes, lines and the BEP, and write the readings.

Answer key · Check yourself

1 **24000**

The line rises ₹6,00,000 over 40,000 meals: ₹15 a meal, the contribution per meal. BEP = ₹3,60,000 ÷ ₹15 = 24,000 meals.

2 **B Total variable cost**

Total cost = fixed cost + variable cost, so the gap above the fixed cost line is the variable cost. Contribution needs the contribution break-even chart (sales minus variable cost).

3 **B The present fleet earns more profit at 90,000 crates**

Below the crossing point the line with the lower fixed cost (the present fleet) is higher. The proposal's steeper line overtakes it only above 1,20,000 crates. A higher fixed cost with a higher contribution per unit can give a higher or lower BEP.

4 **C It moves down parallel to itself, so the break-even point moves right**

The slope is the contribution per unit, which has not changed; the starting loss is ₹2,00,000 larger, so the whole line drops by that amount and crosses zero later.

5 **A Sales**; **B Variable cost**

Sales and variable cost are both zero at zero activity. Total cost starts at the fixed cost, and a contribution chart has no separate fixed cost line.

14.4 Marginal versus absorption costing and profit reconciliation

Why the two methods report different profits when stock changes, ICAI's income statement for each, and how to reconcile them through the fixed overhead in opening and closing stock.

In plain words

The rent that hid in the kulhads

Tapri's kulhad factory made 10,000 cups this month and sold 8,000 at ₹15. Each cup uses ₹6 of clay, labour and firing; the factory's fixed overhead (rent, kiln lease, supervisor) is ₹40,000 a month, which the finance team absorbs at ₹4 a cup over the normal 10,000 cups.

> **Kesar:** Ritu's report says the kulhad unit made ₹32,000 this month. The finance team says ₹40,000. Same cups, same month. Whom do I believe?
>
> **Ritu:** Both, once you count the shelf. Finance put ₹4 of this month's rent into each of the 2,000 cups still in the godown, so ₹8,000 of rent is waiting there for next month. I charged the whole ₹40,000 to this month.
>
> **Kesar:** So the rent is hiding in the kulhads.
>
> **Ritu:** Until they are sold. Then it comes out again.

The finance team uses **absorption costing**: every production cost, fixed and variable, is charged to the

products, so a cup in stock carries ₹6 + ₹4 = ₹10. Ritu uses **marginal costing**: a cup carries only its ₹6 of variable cost, and the fixed overhead is a **period cost**, charged in full to the month in which it is incurred.

Sales are the same, the variable costs of the 8,000 cups sold are the same; the only difference is where ₹8,000 of fixed overhead sits this month. When the factory makes exactly what it sells, both methods show the same profit. When stock builds up, absorption costing shows more; when stock runs down, marginal costing shows more.

Think of it like

Tonight's gas, tomorrow's tiffin

You cook tonight and pack tomorrow's tiffin. Was the gas you burnt tonight's expense or part of tomorrow's lunch? Put a share of it into the tiffin and tonight's kitchen looks cheaper; charge it all tonight and tomorrow's lunch looks cheaper. The gas bill is the same either way; only the day that carries it changes.

Key idea

The two methods differ only in fixed production overhead. Absorption costing carries part of it forward in closing stock (and brings last period's part back with opening stock); marginal costing charges all of it to the period. So the difference in profit is exactly the fixed overhead in closing stock minus the fixed overhead in opening stock.

How it works

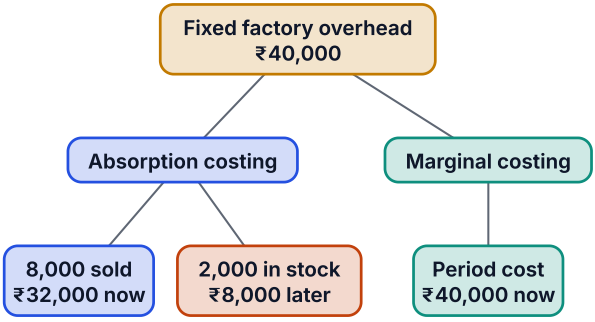
Where each cost goes

ICAI draws the two approaches as two routes (SM 14.43):

Cost	Absorption costing	Marginal costing
Direct materials, direct labour, variable factory overhead	cost of goods produced; expense when the goods are sold	the same
Fixed factory overhead	cost of goods produced; expense when the goods are sold	expense when incurred (period cost)
All selling and administration overhead	expense when incurred	expense when incurred

So under absorption costing the product (inventoriable) cost is total manufacturing cost, variable and fixed; under marginal costing it is variable manufacturing cost only (SM 1.35).

The kulhad month: where ₹40,000 of fixed overhead goes



Absorption costing charges this month only the ₹4 a cup on the cups sold and carries ₹8,000 forward in closing stock; marginal costing charges all ₹40,000 now. Absorption profit ₹40,000; marginal profit ₹32,000; difference ₹8,000 = 2,000 cups × ₹4.

Six steps through an absorption-versus-marginal question

1. **Fixed production overhead absorption rate** = budgeted fixed production overhead ÷ normal (budgeted) output. ICAI's Illustration 14: ₹1,60,000 ÷ 800 units = ₹200 a unit (SM 14.47).

2. **Stock movement in units**: closing stock = opening stock + production − sales, period by period.

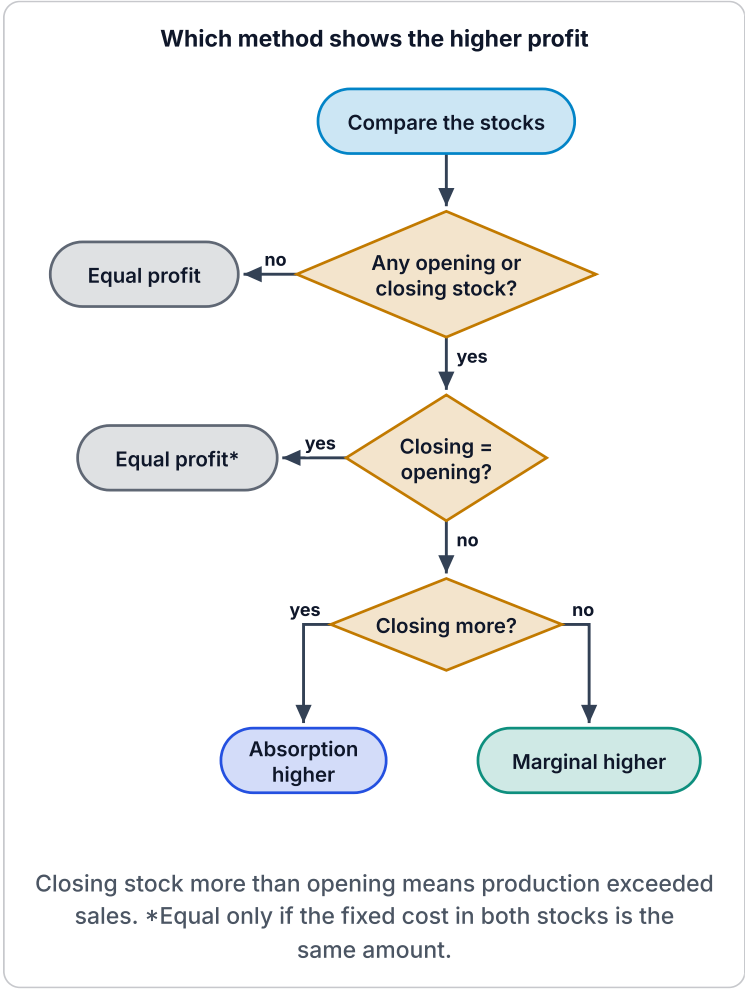
3 **Unit costs**: absorption = variable production cost + the absorption rate; marginal = variable production cost only. Opening stock is valued at the previous period's cost, closing stock at the current period's (SM 14.45).

4 **Under- or over-absorption** (absorption costing only) = actual fixed production overhead − overhead absorbed (units *produced* × rate). It is added to (under) or deducted from (over) cost of goods sold. Marginal costing charges the actual fixed overhead, so none arises (see under- and over-absorption).

5 **Selling and administration costs** are charged to the period under both; under marginal costing the variable ones are deducted before contribution.

6 **Reconcile**: absorption profit + fixed overhead in opening stock − fixed overhead in closing stock = marginal profit.

Which profit is higher? ICAI's four cases (SM 14.44)



Closing stock more than opening means production exceeded sales. *Equal only if the fixed cost in both stocks is the same amount.

1. **No opening and closing stock**: profit is equal under both.
2. **Opening stock equal to closing stock**: profit is equal, provided the fixed cost element in both stocks is the

same amount.

3. **Closing stock more than opening stock** (production more than sales): absorption profit is higher, because part of the fixed overhead in closing stock is carried forward to the next period.
4. **Opening stock more than closing stock** (production less than sales): marginal profit is higher, because fixed cost of the preceding period comes into this period's cost of goods sold through the opening stock.

Interactive version: PassBook app, lesson Marginal versus absorption costing and profit reconciliation

Formats and formulas

ICAI's two income statements (SM 14.45–14.46); the absorption one runs down the same lines as the cost sheet. *Reproduce both layouts from memory; the order of the lines is what earns the format marks.*

Income Statement (Absorption costing)

Particulars	Amount (₹)	Amount (₹)
Sales		xxx
Production costs:		
Direct material consumed		xxx
Direct labour cost		xxx

Particulars	Amount (₹)	Amount (₹)
Variable manufacturing overhead		xxx
Fixed manufacturing overhead		xxx
Cost of production		**xxx**
Add: Opening stock of finished goods (at cost of previous period's production)		xxx
Less: Closing stock of finished goods (at production cost of current period)		xxx
Cost of goods sold		**xxx**
Add (or less): Under (or over) absorption of fixed manufacturing overhead		xxx
Add: Administration costs	xxx	
Add: Selling and distribution costs	xxx	xxx
Total cost		**xxx**
Profit (Sales – Total cost)		**xxx**

Income Statement (Marginal costing)

Particulars	Amount (₹)
Sales	xxx
Variable manufacturing costs:	
– Direct material consumed	xxx
– Direct labour	xxx
– Variable manufacturing overhead	xxx
Cost of goods produced	**xxx**
Add: Opening stock of finished goods (at cost of previous period)	xxx
Less: Closing stock of finished goods (at current variable cost)	xxx
Cost of goods sold	**xxx**
Add: Variable administration, selling and distribution overhead	xxx
Total variable cost	**xxx**
Contribution (Sales – Total variable cost)	**xxx**
Less: Fixed costs (production, administration, selling and distribution)	xxx
Net profit	**xxx**

Under marginal costing the contributions of the various products are pooled, and the fixed overheads are met out

of this total contribution, which is also known as **gross margin** (SM 14.46). ICAI's printed statement also has a line "Add: Selling and distribution costs" with no amount between total variable cost and contribution; variable selling costs are already in the line above and fixed ones below, so nothing is entered there.

Reconciliation of profits (the layout of ICAI's suggested answers: SM practical problem 17, MTP Series I for May 2024 Q4(a)(iii))

Particulars	Amount (₹)
Profit as per absorption costing	xxx
Add: Fixed overhead in opening stock (opening stock under-valued in marginal costing)	xxx
Less: Fixed overhead in closing stock (closing stock under-valued in marginal costing)	xxx
Profit as per marginal costing	**xxx**

Interactive version: PassBook app, lesson Marginal versus absorption costing and profit reconciliation

∑ **Formula**

Fixed production overhead absorption rate

$$\text{Rate per unit} = \frac{\text{Budgeted fixed production overhead}}{\text{Normal (budgeted) output}}$$

Formula

Under- or over-absorption

$$\begin{aligned}&\text{Under (+) / over (−) absorption}\\&=\text{Actual fixed production overhead}\\&-\text{Units produced}\times\text{Rate}\end{aligned}$$

Formula

Difference in profit

$$\begin{aligned}&\text{Absorption profit}-\text{Marginal profit}\\&=\text{Fixed overhead in closing stock}\\&-\text{Fixed overhead in opening stock}\end{aligned}$$

Each stock's fixed overhead at the rate at which it was valued: closing stock at this period's rate, opening stock at the previous period's.

Definition

Absorption costing

Absorption costing is the practice of charging all costs, both variable and fixed to operations, processes or product. (SM 14.9) In it, fixed expenses are distributed over products on the basis of a pre-determined level of output, which leads to over- or under-recovery as actual output differs from the estimate. (SM 14.9)

Definition

Product (inventoriable) costs

Under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs. (SM 1.35)

Worked examples

Worked example **The bonus that lived in the godown**

Tapri's frozen paratha unit sells 4,00,000 packs a year at ₹50. Variable production cost is ₹28 a pack and variable selling cost ₹2 a pack sold. Fixed production overheads are ₹24,00,000 a year (actual as budgeted), absorbed on a normal output of 4,00,000 packs; fixed selling and administration overheads are ₹8,00,000. There is no opening stock.

The unit manager's bonus is 5% of the profit in the absorption costing report. The plan is to produce 4,00,000 packs. Near the year-end the manager proposes to produce 4,60,000 instead, "to use the spare freezer space"; the extra 60,000 packs will be in stock on 31 March, and sales will not change.

Required:

(i) Prepare the profit statement under absorption costing for the plan and for the manager's proposal.

(ii) Prepare the profit statement under marginal costing for both.

(iii) Reconcile the profits, calculate the manager's bonus in each case, and comment.

THE WORKING

Working notes

1. Absorption rate = ₹24,00,000 ÷ 4,00,000 = **₹6** a pack; full production cost = ₹28 + ₹6 = **₹34** a pack.
2. Under the proposal, overhead absorbed = 4,60,000 × ₹6 = ₹27,60,000 against actual ₹24,00,000: **over-absorbed ₹3,60,000**.
3. Closing stock under the proposal = 60,000 packs: ₹20,40,000 at ₹34 (absorption) or ₹16,80,000 at ₹28 (marginal).

(i) Statement of profit under absorption costing

Particulars	Plan (₹)	Proposal (₹)
Sales (4,00,000 × ₹50) (A)	2,00,00,000	2,00,00,000
Cost of production @ ₹34 (4,00,000 / 4,60,000 packs)	1,36,00,000	1,56,40,000
Less: Closing stock @ ₹34	—	(20,40,000)

Particulars	Plan (₹)	Proposal (₹)
Cost of goods sold	**1,36,00,000**	**1,36,00,000**
Less: Over-absorption (WN 2)	—	(3,60,000)
Add: Variable selling cost @ ₹2	8,00,000	8,00,000
Add: Fixed selling and administration	8,00,000	8,00,000
Cost of sales (B)	**1,52,00,000**	**1,48,40,000**
Profit (A – B)	**48,00,000**	**51,60,000**

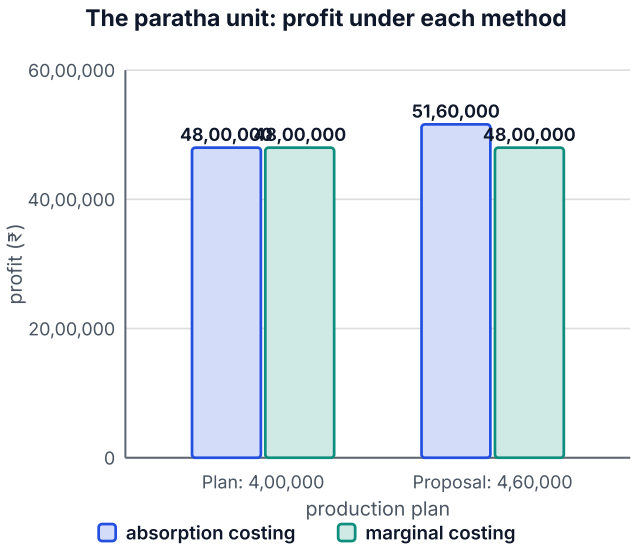
(ii) Statement of profit under marginal costing

Particulars	Plan (₹)	Proposal (₹)
Sales	2,00,00,000	2,00,00,000
Variable cost of production @ ₹28	1,12,00,000	1,28,80,000
Less: Closing stock @ ₹28	—	(16,80,000)
Variable cost of goods sold	**1,12,00,000**	**1,12,00,000**
Add: Variable selling cost	8,00,000	8,00,000
Contribution	**80,00,000**	**80,00,000**

Particulars	Plan (₹)	Proposal (₹)
Less: Fixed production overheads	24,00,000	24,00,000
Less: Fixed selling and administration	8,00,000	8,00,000
Profit	**48,00,000**	**48,00,000**

(iii) Reconciliation and bonus

Particulars	Plan (₹)	Proposal (₹)
Profit under absorption costing	48,00,000	51,60,000
Less: Fixed overhead in closing stock (60,000 × ₹6)	—	(3,60,000)
Profit under marginal costing	**48,00,000**	**48,00,000**
Manager's bonus at 5% of absorption profit	2,40,000	2,58,000



Blue: absorption costing; teal: marginal costing. Sales are the same under both plans; only the absorption profit moves with the extra production.

Comment. Producing 60,000 packs for stock raises the absorption profit by ₹3,60,000, the fixed overhead carried forward in closing stock, and the manager's bonus by ₹18,000, though not one extra pack is sold. The company spends ₹16,80,000 more on variable production for packs that sit in the freezer, and next year's profit will bear the ₹3,60,000 when they are sold. Marginal costing profit moves with sales, not production (SM 14.10): a bonus based on marginal costing profit, or a cap on closing stock, removes the incentive to produce for stock.

Worked example

Same stock, different profits: the chai premix year

Tapri's chai premix factory, 2025-26: selling price ₹150 a box; variable production cost ₹80 a box (the same as last year); variable selling cost ₹6 a box; fixed selling overheads ₹10,00,000. Fixed production overheads rose to ₹24,00,000 (₹19,20,000 last year) after a new plant lease; both years they are absorbed on a normal output of 1,60,000 boxes, and actual equalled budget. The opening stock of 12,000 boxes was made last year. During the year 1,50,000 boxes were produced and 1,50,000 sold, so 12,000 boxes, all from this year's production, remain at the year-end (FIFO).

Dev, reading the two reports: "The stock did not change, so the profits must be equal. One of you has made an error."

Required:

(i) Calculate the profit for the year by absorption costing and by marginal costing.

(ii) Reconcile the two profits, and comment on Dev's remark.

THE WORKING

Working notes

1. Absorption rates: last year ₹19,20,000 ÷ 1,60,000 = **₹12** a box; this year ₹24,00,000 ÷

1,60,000 = **₹15** a box.

2. Stock values: opening 12,000 boxes at ₹80 + ₹12 = ₹92 (absorption ₹11,04,000; marginal ₹9,60,000); closing 12,000 boxes at ₹80 + ₹15 = ₹95 (absorption ₹11,40,000; marginal ₹9,60,000).
3. Overhead absorbed = 1,50,000 × ₹15 = ₹22,50,000 against actual ₹24,00,000: **under-absorbed ₹1,50,000**.

(i) Income statement (absorption costing)

Particulars	Amount (₹)	Amount (₹)
Sales (1,50,000 × ₹150)		2,25,00,000
Cost of production (1,50,000 × ₹95)	1,42,50,000	
Add: Opening stock (12,000 × ₹92)	11,04,000	
Less: Closing stock (12,000 × ₹95)	(11,40,000)	
Cost of goods sold	**1,42,14,000**	
Add: Under-absorbed fixed production overhead (WN 3)	1,50,000	
Add: Variable selling cost (1,50,000 × ₹6)	9,00,000	

Particulars	Amount (₹)	Amount (₹)
Add: Fixed selling overheads	10,00,000	1,62,64,000
Profit		**62,36,000**

Income statement (marginal costing)

Particulars	Amount (₹)
Sales	2,25,00,000
Variable cost of production (1,50,000 × ₹80)	1,20,00,000
Add: Opening stock (12,000 × ₹80)	9,60,000
Less: Closing stock (12,000 × ₹80)	(9,60,000)
Add: Variable selling cost	9,00,000
Total variable cost	**1,29,00,000**
Contribution	**96,00,000**
Less: Fixed production overheads	24,00,000
Less: Fixed selling overheads	10,00,000
Profit	**62,00,000**

(ii) Reconciliation

Particulars	Amount (₹)
Profit as per absorption costing	62,36,000
Add: Fixed overhead in opening stock (12,000 × ₹12)	1,44,000
Less: Fixed overhead in closing stock (12,000 × ₹15)	(1,80,000)
Profit as per marginal costing	**62,00,000**

Comment. Dev is not right, and neither report is wrong. When opening and closing stocks are equal, the two profits are equal only if the fixed cost element in both stocks is the same amount (SM 14.44). Here the stock quantity is unchanged, but the opening boxes carry last year's ₹12 of fixed overhead and the closing boxes this year's ₹15, because the lease raised the fixed overheads. Absorption costing carries ₹36,000 more fixed overhead out of the year than it brought in, so its profit is ₹36,000 higher. The variable cost per box did not change, so marginal costing's stocks cancel exactly.

Exam angle

What ICAI has asked

In the exam

Where the topic appeared, May 2024 to September 2026

In the documents read for this lesson, marginal versus absorption costing came up in 10 places in 9 documents.

- **Profit statements under both methods, 6–10 marks.** MTP Series I for May 2024 Q4(a): three months of production and sales, profit statements by absorption and by marginal costing and a reconciliation (10 marks). MTP Series I for May 2025 Q5(a): profit for a quarter by both methods, from annual data at a 100% activity level (3 + 3 marks), the pattern of the Study Material's Illustration 14.
- **Multiple choice, 2 marks.** September 2025 MCQ 6: "Cost of production" and "Total Cost" as per absorption costing (cost of production: all manufacturing costs, fixed included; total cost: every cost, administration and selling included). MTP Series II for May 2025, MCQ case: total variable cost and fixed cost as per the marginal costing income statement.
- **Theory.** "Differentiate between Marginal and Absorption Costing" (old-scheme November 2020 Q6(e), 5 marks), answered with the Study Material's five points. Product costs "under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs": RTP May 2025 Q20(a) ("DEFINE Product costs"), and as an answer point in MTP Series I for May 2025 Q6(a), RTP

January 2026 Q19(a), MTP Series II for September 2026 Q6(b) and RTP May 2026 Q17(a).

In the exam

The traps ICAI builds in

- The absorption rate uses **normal** output, not actual (SM practical problem 17: 90% of capacity; MTP Series I for May 2025 Q5(a): normal level 4,400 units).
- Under- or over-absorption is based on units **produced**, not sold, and appears only in the absorption statement.
- Quarterly questions with annual data: fixed costs are a quarter of the year's (Illustration 14; MTP Series I for May 2025 Q5(a)).
- Opening stock at the **previous** period's cost, closing stock at the **current** period's (SM practical problem 17: opening stock at ₹13, the standard ₹11 + ₹2).
- The SM's own MCQs 4–6: sales = production gives equal profits (ICAI's key answer; it assumes the same fixed cost per unit in both stocks, which the chai premix example breaks); sales more than production gives higher marginal profit; the main difference is the treatment of fixed overheads.

Trick

The one-line reconciliation

Profit difference = (closing stock units × fixed rate in closing stock) − (opening stock units × fixed rate in opening stock). Compute it first: it checks both statements before you have finished either. Condition: under- or over-absorption written off to the period, as ICAI's format does.

Answer frame

"Discuss the points of difference between absorption costing and marginal costing" (ICAI's five, in order)

No.
1

Marginal costing
Only variable costs are considered for product costing and inventory valuation.

Absorption costing
Both fixed and variable costs are considered for product costing and inventory valuation.

No.

2

Marginal costing

Fixed costs are regarded as period costs; the profitability of different products is judged by their P/V ratio.

Absorption costing

Fixed costs are charged to the cost of production; each product bears a reasonable share of fixed cost, so its profitability is influenced by the apportionment of fixed costs.

No.

3

Marginal costing

Cost data presented highlight the total contribution of each product.

Absorption costing

Cost data are presented in the conventional pattern; net profit of each product is determined after subtracting fixed cost along with variable costs.

No.

4

Marginal costing

The difference in the magnitude of opening and closing stock does not affect the unit cost of production.

Absorption costing

The difference in the magnitude of opening and closing stock affects the unit cost of production due to the impact of related fixed cost.

No.

5

Marginal costing

Cost per unit remains the same irrespective of production, as it is valued at variable cost.

Absorption costing

Cost per unit reduces as production increases, because fixed cost per unit falls while variable cost per unit stays the same.

(SM 14.43–14.44) Add the difference in profit (the four cases) when the question asks about profits.

"Explain the difference in profit": the frame

1. **The cause:** the difference in stock valuation; under absorption costing stock carries fixed production overhead.
2. **The case that applies:** closing stock more than opening (absorption higher), less (marginal higher), equal (equal profits if the fixed cost in both stocks is the same).
3. **The amount:** fixed overhead in closing stock – fixed overhead in opening stock, with figures.
4. **The reconciliation statement**, from absorption profit to marginal profit.

Three answers to the paratha-bonus question

Loses marks

Profits for both plans with no working notes, closing stock valued at ₹34 under both methods.

Wrong stock value under marginal costing, and no way to award method marks.

Passes

Both statements correct, but no reconciliation and no explanation.

Part (iii) carries marks of its own.

Full marks

Working notes (unit costs, stock in units, under/over absorption); both statements in ICAI's layout; the reconciliation; the bonus; a comment naming the case (closing stock above opening) and its ₹ effect.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the commission that went into the godown

Valuing closing stock with the ₹2 selling cost included (₹36 or ₹30 a pack). **Fix:** stock is valued at production cost only, under both methods; selling costs are charged to the period.

⚠ **Common mistake**

Under-absorption counted on the packs sold

Working the proposal's absorbed overhead as 4,00,000 packs sold × ₹6 = ₹24,00,000, "nil" under- or over-absorption, instead of an over-absorption of ₹3,60,000. **Fix:** overhead is absorbed into production, so use units produced (4,60,000).

⚠ **Common mistake**

Under-absorption put into the marginal costing statement

Adding the chai premix factory's ₹1,50,000 under-absorption to the marginal statement as well. **Fix:** marginal costing charges actual fixed overhead (₹24,00,000) in full; no absorption, so nothing is under- or over-absorbed.

⚠ **Common mistake**

"Equal stocks, so equal profits"

Writing, as Dev did, that the chai premix profits must be equal because opening and closing stocks are both 12,000 boxes. **Fix:** equal only if the fixed cost in both stocks is the same amount; here ₹12 against ₹15 a box makes a ₹36,000 difference.

Practice

Check yourself

1 Tapri's cookie unit had an opening stock of 3,000 jars and a closing stock of 7,000 jars, both carrying fixed production overhead of ₹20 a jar. Its profit under marginal costing was ₹4,20,000. What was its profit under absorption costing (₹)?

Answer: _____

2 Case scenario. In March, Tapri's samosa factory sold more packs than it produced, with fixed production overhead absorbed at the same rate as in February. Dev's absorption costing report shows a profit of ₹3,10,000; Ritu's marginal costing report shows ₹3,40,000. Kesar asks which statement is correct.

A Ritu has made an error, because absorption costing always shows the higher profit

B Both can be right: stock fell, so ₹30,000 more fixed overhead came into March's cost with the opening stock than went out with the closing stock

C The ₹30,000 is March's under-absorbed overhead, which marginal costing ignores

D The difference shows that the closing stock was valued at selling price

3 Under absorption costing, which costs are included in the value of closing stock of finished goods? Select all that apply.

- **A** Direct materials
- **B** Fixed manufacturing overhead absorbed
- **C** Fixed selling and distribution overhead
- **D** Variable selling commission

More than one answer may be right.

4 Tapri's kulhad unit started April with no stock, made 12,000 cups and sold all of them. Its fixed factory overhead was under-absorbed by ₹6,000. How do its April profits under the two methods compare?

- **A** Absorption profit is ₹6,000 lower, because of the under-absorption
- **B** Absorption profit is ₹6,000 higher, because the overhead was under-absorbed
- **C** The two profits are equal
- **D** Marginal profit is higher by the fixed overhead absorbed in the cups sold

5 Fixed production overhead budgeted at ₹24,00,000 for a normal output of 1,20,000 units; actual overhead ₹24,00,000; actual production 1,10,000 units. What is the under-absorption (₹)?

Answer: _____

Answers: the answer key at the end of this lesson.

Recap

- Absorption costing charges all costs, fixed and variable, to products; marginal costing values products and stock at variable cost and charges fixed costs to the period.
- Product (inventoriable) cost: total manufacturing cost under absorption, variable manufacturing cost under marginal. Selling and administration costs are period costs under both.
- Fixed production overhead rate = budgeted fixed production overhead ÷ normal output; under/(over) absorption = actual − units produced × rate, adjusted in the absorption statement only.
- No stock: equal profits. Equal stocks: equal profits if their fixed cost is the same amount. Closing more than opening: absorption higher. Opening more than closing: marginal higher.

- Absorption profit − marginal profit = fixed overhead in closing stock − fixed overhead in opening stock.
- Reconciliation: absorption profit + fixed overhead in opening stock − fixed overhead in closing stock = marginal profit.
- The five ICAI differences: inventory valuation, fixed cost as period cost (P/V ratio), contribution highlighted, stock levels and unit cost, unit cost and volume.

Answer key · Check yourself

1 **500000**

Absorption profit = marginal profit + fixed overhead in closing stock − fixed overhead in opening stock = ₹4,20,000 + ₹1,40,000 − ₹60,000 = ₹5,00,000. Stock rose, so absorption profit is higher.

2 **B** **Both can be right: stock fell, so ₹30,000 more fixed overhead came into March's cost with the opening stock than went out with the closing stock**

When sales exceed production, opening stock is more than closing stock and marginal profit is higher (SM 14.44, case 4). Under-absorption is charged to the period under absorption costing and actual overhead under marginal costing, so it is not the difference.

3 **A Direct materials; B Fixed manufacturing overhead absorbed**

Absorption costing values stock at total manufacturing cost, variable and fixed. Selling and distribution costs, fixed or variable, are period costs under both methods.

4 **C The two profits are equal**

No opening and no closing stock, so no fixed overhead is carried in or out (SM 14.44, case 1). The ₹6,000 under-absorption is added to cost in the absorption statement, so absorption costing too charges April with the actual fixed overhead, exactly as marginal costing does.

5 **200000**

Rate = ₹24,00,000 ÷ 1,20,000 = ₹20 a unit; absorbed = 1,10,000 × ₹20 = ₹22,00,000; under-absorbed = ₹24,00,000 − ₹22,00,000 = ₹2,00,000. It is added to cost of goods sold in the absorption statement only.

14.5 Limiting factor and product mix

When one resource runs short, rank products by contribution per unit of that resource, fill demand in rank order, and earn the most from what you have.

In plain words

One fryer, two favourites

Tapri's Indore factory has one big fryer, and it is booked solid: 2,000 hours a month. Paneer rolls leave ₹30 of contribution a pack, aloo samosas ₹20.

> **Kesar:** Rolls earn ₹30, samosas ₹20. Fry rolls. Simple.
>
> **Ritu:** Per unit, please, and today the unit is the fryer hour. A pack of rolls needs half an hour of fryer, a pack of samosas a quarter of an hour.
>
> **Kesar:** So in one hour I get two packs of rolls or four packs of samosas... the samosa was the topper all along?

Two packs of rolls earn ₹60 in that hour; four packs of samosas earn ₹80. Fill the fryer with rolls and the month's contribution is 4,000 packs × ₹30 = ₹1,20,000; fill it with samosas and it is 8,000 packs × ₹20 = ₹1,60,000.

The fryer is the **limiting factor**, also called the **key factor**: the thing that stops Tapri making and selling more. Once something is short, the question is no longer "which product earns most per pack?" but "which product

earns most from one unit of the scarce thing?" That figure is the **contribution per unit of the limiting factor**. Rank the products on it, give the scarce resource to the top one until its demand is met, then to the next, and so on.

Think of it like

The exam-hall clock

In a three-hour paper, time is your limiting factor. A 10-mark question you can finish in 20 minutes earns half a mark a minute; a 16-mark question that eats an hour earns about a quarter. When the clock is short, you rank questions by marks per minute, not by marks, and you do the half-a-mark-a-minute ones first.

Key idea

When a resource is scarce, rank products by contribution per unit of that resource, not by contribution (or profit) per unit of product. Fixed costs that stay the same whatever the mix never change the ranking.

How it works

ICAI defines it in one line: limiting factor is anything which limits the activity of an entity; because it decides the level of sales and production, it is also known as the **key factor** (SM 14.36). The Study Material lists product mix under resource constraints among the short-term

decisions "related with excess demand": demand is there, the resources are not (SM 14.36).

Where limiting factors come from

Limiting (key) factor

Supply side

Men (employees)
Materials (raw material or supplies)
Machine (capacity)
Money (availability of fund or budget)

Demand side

Demand for the product
Nature of product
Regulatory and environmental requirement

ICAI's list, in ICAI's order (SM 14.36). Management's objective is to optimise the key resource to the maximum possible extent.

Five steps from scarce hours to the best mix

1 **Contribution per unit** of each product = selling price − variable cost. Leave fixed costs out: they are the same whatever the mix.

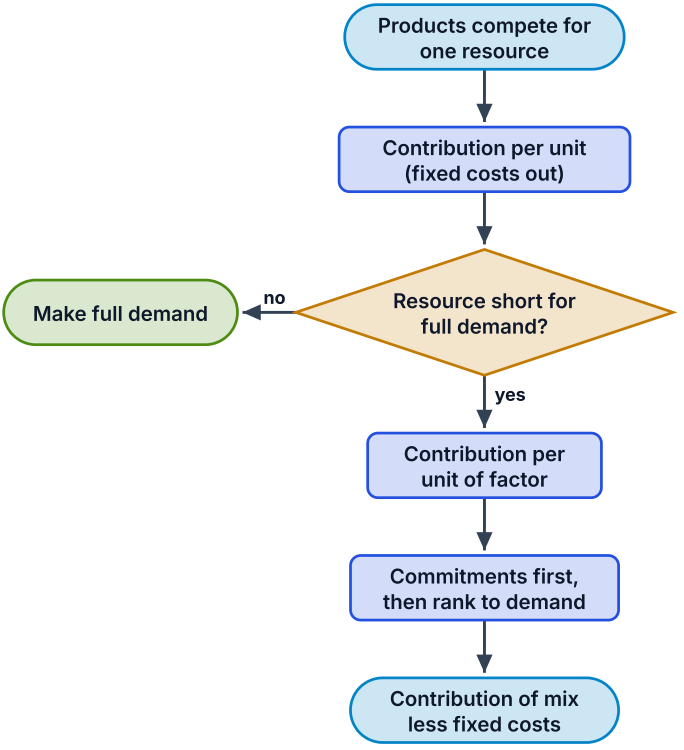
2 **Check that the factor really limits.** Work out how much of the resource the full demand needs. If there is enough, make every product with a positive contribution up to its demand; no ranking is needed.

3 **Contribution per unit of the limiting factor** = contribution per unit ÷ quantity of the factor one unit uses. If the question hides the quantity, derive it: hours = variable overhead per unit ÷ variable overhead rate per hour; kg = material cost per unit ÷ price per kg; labour hours = labour cost per unit ÷ wage rate per hour.

4 **Rank and allocate.** Highest contribution per unit of the factor first. Give the factor first to any minimum commitment (a contract, a product that must be made), then in rank order up to each product's maximum demand; the last product gets the balance.

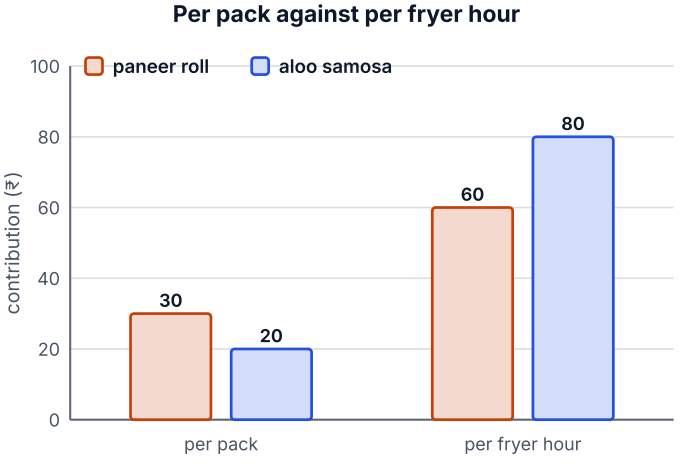
5 **Statement of profit.** Contribution of the chosen mix, less fixed costs.

The ranking method



Green: no constraint, so every product with a positive contribution is made up to its demand. Otherwise the answer ends with the profit statement of the chosen mix.

In the fryer story the ranking flips the moment you divide by the scarce hours:



Orange: paneer roll; blue: aloo samosa. Per pack the roll leads, ₹30 to ₹20; per fryer hour the samosa leads, ₹80 to ₹60. With the fryer full, the per-hour bars decide.

Interactive version: PassBook app, lesson Limiting factor and product mix

Which measure ranks the products

ICAI's formula is **profitability = contribution ÷ key factor** (SM 14.28). Put the scarce thing under the line:

What is short	Rank by
Nothing (enough of every resource)	No ranking: make every product with a positive contribution, up to demand
Machine or labour hours	Contribution per hour
Total units of capacity, shared (each unit of any product takes one)	Contribution per unit
Raw material	Contribution per kg (or per unit of material)
Total sales value	P/V ratio (contribution per rupee of sales)
Choosing to make or buy with scarce hours	Saving per hour = (buy price − variable cost of making) ÷ hours

This is why ICAI counts **dependence on key factors** among the limitations of marginal costing: the contribution of a product by itself is not a guide to optimum profitability unless it is linked with the key factor (SM 14.10).

Formats and formulas

Reproduce these three statements from memory; ICAI's suggested answers are built from them.

∑ **Formula**

Contribution per unit of the limiting factor

$$\text{Profitability} = \frac{\text{Contribution per unit}}{\text{Key factor per unit}}$$

ICAI writes it as Profitability = Contribution ÷ Key factor (SM 14.28). Key factor per unit: hours, kg, metres or any other unit of the scarce resource one unit of product uses.

∑ **Formula**

Recovering the quantity of the factor

$$\text{Hours per unit} = \frac{\text{VOH per unit}}{\text{VOH rate per hour}}$$

$$\text{Kg per unit} = \frac{\text{Material cost per unit}}{\text{Price per kg}}$$

VOH = variable overhead. Labour hours per unit = labour cost per unit ÷ wage rate per hour, in the same way.

Statement 1: ranking

Particulars	Product A	Product B	Product C
Selling price per unit	xxx	xxx	xxx
Less: Variable cost per unit	xxx	xxx	xxx
Contribution per unit	**xxx**	**xxx**	**xxx**
Key factor per unit (hours, kg)	xx	xx	xx
Contribution per unit of key factor	**xx**	**xx**	**xx**
Rank	II	I	III

Statement 2: optimum product mix

Product (in rank order)	Units	Key factor used	Key factor left
Minimum commitment, if any	xxx	xxx	xxx
Rank I (up to demand)	xxx	xxx	xxx
Rank II (up to demand)	xxx	xxx	xxx
Rank III (balance)	xxx	xxx	nil

Statement 3: profit

Product	Units	Contribution per unit (₹)	Total contribution (₹)
A	xxx	xx	xxx
B	xxx	xx	xxx
Total contribution			**xxx**
Less: Fixed costs			xxx
Profit			**xxx**

Definition

Limiting factor (key factor)

Limiting factor is anything which limits the activity of an entity. The factor is a key to determine the level of sale and production, thus it is also known as Key factor. (SM 14.36, 14.49)

Worked examples

Worked example **Three snacks, a fryer and a packing line**

Tapri's Indore factory fries and packs three snacks.

Data per pack:

Particulars	Aloo samosa	Paneer roll	Kachori
Selling price (₹)	120	200	150
Variable cost (₹)	80	140	105
Fryer hours	0.5	1.0	0.6
Packing-line hours	0.3	0.5	0.4
Maximum demand a month (packs)	10,000	6,000	5,000

The fryer can run 12,000 hours and the packing line 9,000 hours a month. Fixed overheads are ₹5,00,000 a month.

Required:

(i) Identify the key factor, and rank the products.

(ii) Determine the optimum product mix and calculate the maximum profit.

(iii) Kesar wants to fry in the order of contribution per pack. Calculate the profit under his plan and the profit it gives up.

THE WORKING

(i) Key factor and ranking

Resource

Fryer: 10,000 × 0.5 + 6,000 × 1.0 + 5,000 × 0.6

Hours for the full demand

14,000

Hours available

12,000

Short?

Yes

Resource

Packing line: 10,000 × 0.3 + 6,000 × 0.5 + 5,000 × 0.4

Hours for the full demand

8,000

Hours available

9,000

Short?

No

Only the fryer falls short of the full demand: **fryer hours are the key factor**. The packing line has spare hours whatever mix is chosen, so it does not affect the ranking.

Particulars	Aloo samosa	Paneer roll	Kachori
Contribution per pack (₹)	**40**	**60**	**45**
Fryer hours per pack	0.5	1.0	0.6
Contribution per fryer hour (₹)	**80**	**60**	**75**
Rank	I	III	II

(ii) Optimum mix and profit

Product (rank order)

Aloo samosa (full demand)

Packs

10,000

Fryer hours

5,000

Hours left

7,000

Product (rank order)

Kachori (full demand)

Packs

5,000

Fryer hours

3,000

Hours left

4,000

Product (rank order)

Paneer roll (balance, 4,000 ÷ 1.0)

Packs

4,000

Fryer hours

4,000

Hours left

nil

Check on the packing line: 10,000 × 0.3 + 5,000 × 0.4 + 4,000 × 0.5 = 7,000 hours, within the 9,000 available.

Product	Packs	Contribution per pack (₹)	Contribution (₹)
Aloo samosa	10,000	40	4,00,000
Kachori	5,000	45	2,25,000
Paneer roll	4,000	60	2,40,000
Total contribution			**8,65,000**
Less: Fixed overheads			5,00,000
Maximum profit			**3,65,000**

(iii) Kesar's order: paneer roll, kachori, aloo samosa

Paneer roll 6,000 packs use 6,000 hours; kachori 5,000 packs use 3,000; the last 3,000 hours make

6,000 packs of aloo samosa. Contribution = ₹3,60,000 + ₹2,25,000 + ₹2,40,000 = ₹8,25,000; profit = **₹3,25,000**, which is **₹40,000** less. Check: 2,000 fryer hours moved from samosas (₹80 an hour) to rolls (₹60 an hour) lose 2,000 × ₹20.



Blue: hours given in rank order of contribution per fryer hour (samosa, kachori, roll); orange: Kesar's order of contribution per pack (roll, kachori, samosa). Each segment's slope is that product's contribution per hour. Both paths use all 12,000 hours; the blue one ends ₹40,000 higher.

Worked example

Tea dust, a hotel contract and a broker

Tapri's premix division makes three chai premixes. Data per carton for the coming quarter:

Particulars (₹ per carton)	Masala	Elaichi	Ginger
Selling price	900	1,200	800
Tea dust @ ₹500 per kg	250	400	200
Other variable costs	230	320	300
Fixed overheads absorbed	120	150	100
Profit	300	330	200
Maximum demand (cartons)	30,000	20,000	25,000

The fixed overheads, absorbed above on budgeted demand, total ₹91,00,000 for the quarter and do not change with the mix. A poor Assam crop limits tea dust to 28,000 kg for the quarter. Tapri has a contract to supply 5,000 cartons of Elaichi to a hotel chain.

Required:

(i) Rank the products for the use of tea dust.

(ii) Determine the optimum mix and the profit for the quarter.

(iii) Calculate the contribution the hotel contract costs Tapri this quarter.

(iv) A broker offers 4,000 kg more tea dust at ₹650 per kg, all or nothing. Advise Tapri.

THE WORKING

(i) Ranking. The absorbed fixed overhead is added back: contribution = price − tea dust − other variable costs.

Particulars	Masala	Elaichi	Ginger
Contribution per carton (₹)	420	480	300
Tea dust per carton (kg)	0.5	0.8	0.4
Contribution per kg (₹)	**840**	**600**	**750**
Rank	I	III	II

By profit per carton Elaichi looks best (₹330); per kg of the scarce tea dust it is last.

(ii) Optimum mix. Tea dust for the full demand = 15,000 + 16,000 + 10,000 = 41,000 kg, against 28,000 kg available.

Product
Elaichi (contract)
Cartons
5,000
Tea dust (kg)
4,000
Kg left
24,000

Product
Masala (rank I, full demand)
Cartons
30,000
Tea dust (kg)
15,000
Kg left
9,000

Product
Ginger (rank II, balance 9,000 ÷ 0.4)
Cartons
22,500
Tea dust (kg)
9,000
Kg left
nil

Product	Cartons	Contribution (₹)
Masala	30,000	1,26,00,000
Ginger	22,500	67,50,000
Elaichi	5,000	24,00,000
Total contribution		**2,17,50,000**
Less: Fixed overheads		91,00,000
Profit		**1,26,50,000**

(iii) Cost of the contract. Without it, Ginger would get 10,000 kg (all 25,000 cartons) and Elaichi the last 3,000 kg (3,750 cartons): contribution ₹2,19,00,000. The contract costs **₹1,50,000**: 1,000 kg moved from Ginger (₹750 a kg) to Elaichi (₹600 a kg).

(iv) The broker's offer. The first 1,000 kg complete Ginger's demand (2,500 cartons × ₹300 = ₹7,50,000); the other 3,000 kg make 3,750 more cartons of Elaichi (₹18,00,000). Extra contribution = ₹25,50,000, already counting tea dust at ₹500 a kg. The premium is 4,000 × ₹150 = ₹6,00,000, so profit rises by **₹19,50,000**: accept the offer. Before any tea dust cost the 4,000 kg earn ₹25,50,000 + 4,000 × ₹500 = ₹45,50,000, so the lot is worth buying at

any price up to ₹45,50,000 ÷ 4,000 = ₹1,137.50 a kg.

Interactive version: PassBook app, lesson Limiting factor and product mix

Worked example

A ₹8,50,000 bank limit for the Diwali counters

Tapri's festive counters resell four bought-in gift items. The bank has sanctioned a seasonal limit of ₹8,50,000, which can be used only to pay suppliers for the goods; packing and other variable costs are paid out of daily collections. Every other resource is plentiful.

Item (per unit)	Selling price (₹)	Purchase cost (₹)	Other variable cost (₹)	Maximum demand (units)
Dry-fruit box	1,200	800	40	600
Kulhad gift set	600	250	50	1,000
Honey jar	450	300	30	800
Masala tin	300	120	30	1,500

Fixed costs of the counters for the season are ₹4,00,000.

Required:

(i) Identify the key factor and rank the items.

(ii) Determine the quantities to buy and the profit for the season.

(iii) The dry-fruit supplier offers to take half the price of each box on 60 days' credit, paid out of collections after the season. Revise the plan and state the gain.

THE WORKING

(i) Key factor. Purchases for the full demand = 600 × ₹800 + 1,000 × ₹250 + 800 × ₹300 + 1,500 × ₹120 = ₹11,50,000, against a limit of ₹8,50,000. The scarce resource is money (availability of funds, one of the Study Material's supply-side limiting factors, SM 14.36): rank by contribution per rupee of the bank limit used.

Particulars	Dry-fruit box	Kulhad gift set	Honey jar	Masala tin
Contribution per unit (₹)	360	300	120	150
Bank limit used per unit (₹)	800	250	300	120
Contribution per ₹ of limit	**0.45**	**1.20**	**0.40**	**1.25**
Rank	III	II	IV	I

The dry-fruit box earns the most per unit but ties up ₹800 of scarce funds for every box.

(ii) Purchase plan

Item (rank order)	Units	Limit used (₹)	Limit left (₹)	Contribution (₹)
Masala tin (full demand)	1,500	1,80,000	6,70,000	2,25,000
Kulhad gift set (full demand)	1,000	2,50,000	4,20,000	3,00,000
Dry-fruit box (balance, ₹4,20,000 ÷ ₹800)	525	4,20,000	nil	1,89,000
Honey jar	nil	—	—	—
Total contribution				**7,14,000**
Less: Fixed costs				4,00,000
Profit				**3,14,000**

(iii) With credit on half of each dry-fruit box. Each box now uses only ₹400 of the limit: contribution per rupee = ₹360 ÷ ₹400 = 0.90. The ranking becomes masala tin (1.25), kulhad gift set (1.20), dry-fruit box (0.90), honey jar (0.40).

Item	Units	Limit used (₹)	Contribution (₹)
Masala tin	1,500	1,80,000	2,25,000
Kulhad gift set	1,000	2,50,000	3,00,000
Dry-fruit box (full demand)	600	2,40,000	2,16,000
Honey jar (balance, ₹1,80,000 ÷ ₹300)	600	1,80,000	72,000
Total		**8,50,000**	**8,13,000**

Profit = ₹8,13,000 − ₹4,00,000 = ₹4,13,000: the credit offer adds **₹99,000** to profit, because it releases ₹2,10,000 of the bank limit (75 more boxes need ₹30,000; the other ₹1,80,000 buys 600 honey jars). Accept the offer, provided the supplier charges nothing extra for the credit.

Exam angle

In the exam

Where the limiting factor appeared

In the ICAI documents read for this lesson it drove the

answer in eight:

- **September 2026 Q1(a), 7 marks**: two products share a capacity of 770 units "put together", each has its own fixed expenses "incurred only if" it is made, and other fixed expenses continue anyway: "Calculate the optimum profit for the year in rupees giving the product mix (in units)." Each unit uses one unit of capacity, so contribution per unit ranks; then check that the product given the leftover capacity still covers its own fixed expenses.
- **September 2025 Q3(b), 6 marks**: "Determine the optimum product mix, considering material as the limiting factor, to generate maximum profit and calculate the maximum profit." Kg per unit came from the material cost at ₹20 per kg.
- **RTP September 2025 Q16**: machine hours fully used; the "total costs" per unit included fixed overheads absorbed per machine hour, to be added back; then a new product priced to replace the weaker one.
- **MTP Series I for September 2025 Q4(a), 6 marks**: make in-house or outsource three components with labour limited to 4,000 hours; rank by saving per labour hour, and buy outright the one cheaper outside.
- **MTP Series I for May 2026 Q4(a)(ii)**: capacity cut to 10,000 units, so a special order displaces regular sales (see short-term decisions).
- **Old scheme**: May 2023 Q4(c), 5 marks (machine hours hidden in "variable overheads ₹4 per machine

hour"); May 2022 Q3(a), 10 marks (labour hours, and "production of either garment must be at least 25% of the production of the other"); MTP Series II for November 2023 Q5(b), 10 marks (best product mix with bought-in prices; one product not available outside).

In the exam

The wording traps

- Quantities hidden in rupees: "variable overheads ₹40", absorbed at ₹4 per machine hour, is 10 hours, not 40.
- "Total cost" or "profit" per unit that includes absorbed fixed overheads: add them back before ranking.
- "Maximum sales potential" or "demand": a cap, so rank I rarely takes all the hours.
- A fixed ratio ("at least 25% of the other") or a contract: allocate it first, then rank.
- Fixed expenses "incurred only if" a product is made: compare the profit with and without that product before settling the mix.
- Two resources given: test each against the full demand; only the one that falls short is the key factor.

Trick

Divide by what is short

Rank on contribution ÷ (quantity of the scarce input per unit); for make-or-buy, on saving ÷ (scarce input per unit). Conditions: one binding resource, constant price and variable cost per unit, and known demand caps.

Answer frame

"What is a limiting factor?"

1. Definition in ICAI's words: anything which limits the activity of an entity; it determines the level of sale and production, hence key factor.
2. Supply side: men, materials, machine, money; demand side: demand, nature of product, regulatory and environmental requirements.
3. Management's objective: optimise the key resource to the maximum possible extent.
4. Decision rule: rank products by contribution per unit of the key factor (profitability = contribution ÷ key factor).
5. A two-line example with figures.

The numerical: three answers to "Determine the optimum mix"

Loses marks

"Rank by contribution per pack: rolls, kachoris, samosas; profit ₹3,25,000."

Ranked per pack, not per fryer hour: the arithmetic is right, the mix is wrong, and only the contribution line earns marks.

Passes

Ranking per fryer hour, then "10,000 samosas, 5,000 kachoris, 4,000 rolls; profit ₹3,65,000."

Right answer, but the hours per pack appear from nowhere and there is no allocation table: a slip anywhere would leave no method marks.

Full marks

Working note on hours per pack; ranking statement per fryer hour; the check that the hours fall short; allocation in rank order within demand; profit statement with fixed costs deducted.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the paneer roll that hogged the fryer

Ranking by contribution per pack (or by P/V ratio) when a resource is scarce. Kesar's order lost ₹40,000 a month. **Fix:** divide each contribution by the scarce quantity it uses, and rank on that.

⚠ **Common mistake**

Profit per unit dressed up as contribution

RTP-style data give "total cost" or "profit" per unit after fixed overhead absorption. Ranking on ₹330 profit put Elaichi first; on contribution per kg it was last. **Fix:** strip out absorbed fixed overheads; the total fixed cost is deducted once, at the end.

⚠ **Common mistake**

The rank I product that ate the whole factory

Giving all the hours to the top product when it can sell only 10,000 packs. **Fix:** allocate up to each product's maximum demand, then pass the balance down the ranking.

⚠ **Common mistake**

Rupees read as hours

"Variable overheads ₹12 per pack, absorbed at ₹24 per fryer hour" is half an hour, not 12 hours. **Fix:** divide the rupees by the rate per hour (or per kg) before dividing the contribution.

Practice

Check yourself

1 Product A has a contribution of ₹36 a unit and needs 3 machine hours; product B has ₹30 and needs 2 hours. Demand is 2,000 units of A and 3,000 of B, and 11,400 machine hours are available. What is the maximum total contribution (₹)?

Answer: _____

2 Case scenario. Ritu's distributor will take Tapri's products only up to a total invoice value of ₹50 lakh a month; every resource inside the factory is plentiful. Kesar asks which measure should rank the products. Which is correct?

- **A** Contribution per unit
- **B** P/V ratio
- **C** Contribution per machine hour
- **D** Profit per unit after absorbed fixed overheads

3 A kachori pack has variable overheads of ₹18, absorbed at ₹30 per fryer hour, and a contribution of ₹48. What is its contribution per fryer hour (₹)?

Answer: _____

4 Which of these does the Study Material list as supply-side limiting factors? Select all that apply.

- **A** Availability of skilled employees
- **B** Machine capacity
- **C** Demand for the product
- **D** Availability of funds
- **E** Regulatory and environmental requirements

More than one answer may be right.

5 Two products both have spare demand. X earns ₹50 a unit using 2 kg of a scarce chemical; Y earns ₹40 a unit using 1 kg. Kesar says "X first, it earns more." What should Ritu reply?

- **A** Agree: X has the higher contribution per unit
- **B** Make equal quantities of both
- **C** Y first: it earns ₹40 a kg against X's ₹25 a kg
- **D** Decide on the products' fixed overheads

Answers: the answer key at the end of this lesson.

Recap

- Limiting (key) factor: anything which limits the activity of an entity; it decides the level of sale and production (SM 14.36).
- Supply side: men, materials, machine, money; demand side: demand, nature of product, regulatory and environmental requirements.
- Profitability = contribution ÷ key factor: rank by contribution per unit of the scarce resource.
- First check the resource really falls short of the full demand; if not, make everything with a positive contribution.
- Derive hidden quantities: hours = VOH ÷ rate per hour; kg = material cost ÷ price per kg.
- Allocate commitments first, then rank order up to each product's demand; the last product takes the balance.
- Fixed costs: ignored in ranking, deducted once in the profit statement; absorbed fixed overhead per unit is added back. A product's own fixed cost is saved only if it drops out altogether.
- Make or buy with a scarce resource: rank by saving per unit of the resource; buy outright anything cheaper outside. (Marks per minute, not marks.)

Answer key · Check yourself

1 **154800**

Per hour: A ₹12, B ₹15, so B first: 3,000 units use 6,000 hours (₹90,000). The other 5,400 hours make 1,800 units of A (₹64,800). Total ₹1,54,800.

2 **B** **P/V ratio**

The scarce thing is sales value, so profitability = contribution ÷ key factor becomes contribution per rupee of sales: the P/V ratio. Machine hours are not short, and absorbed profit is never a ranking measure.

3 **80**

Fryer hours per pack = ₹18 ÷ ₹30 = 0.6 hours; contribution per hour = ₹48 ÷ 0.6 = ₹80. Dividing ₹48 by 18 "hours" is the classic slip.

4 **A** **Availability of skilled employees**; **B** **Machine capacity**; **D** **Availability of funds**

Supply side: men, materials, machine, money. Demand for the product and regulatory or environmental requirements are on the demand side (SM 14.36).

5 **C** **Y first: it earns ₹40 a kg against X's ₹25 a kg**

With the chemical short, the ranking is per kg: Y ₹40, X ₹25. Fixed overheads do not change with the mix.

14.6 Make or buy, shut down and other short-term decisions

Compare only the future costs and revenues that differ between the options: make or buy, shut down or continue, drop a product, accept a special order.

In plain words

The bun maska that "lost" money

Tapri's Lucknow café sends Kesar a report: the bun maska line lost ₹40,000 last month, after its share of the café's rent and head-office costs.

> **Kesar:** Bun maska is losing ₹40,000 a month. Band karo.
>
> **Ritu:** Per unit, please. Every month bun maska brings in ₹60,000 more than its buns, butter and gas cost. The ₹1,00,000 of rent and head-office cost we charge it will still be there tomorrow.
>
> **Kesar:** So the loss belongs to the rent, not the bun?
>
> **Ritu:** Drop the bun and the rent simply moves to the chai. And the café earns ₹60,000 less.

Ritu has used the one rule behind every short-term decision: compare only what **changes** between the options. The rent and head-office charge stay whatever Kesar decides, so they cannot decide anything. The ₹60,000 of **contribution** (sales minus variable costs) disappears if the line goes, so it decides everything. Costs

and revenues that are in the future and differ between the options are **relevant**; a cost already incurred (a **sunk cost**) never is. A benefit you give up by choosing one option (an **opportunity cost**) is relevant even though no cash is paid.

The same rule answers the other classic questions: make a part or buy it, close a plant for a slow season, drop a product, accept a cheap one-off order.

Think of it like

Cook or order in?

Tonight's dinner: cook at home or order in? The gas cylinder bought last month, the fridge and the rent are the same either way, so they are not part of the decision. Compare the groceries you will buy, plus whatever else you could have done with that evening, against the delivery bill.

Key idea

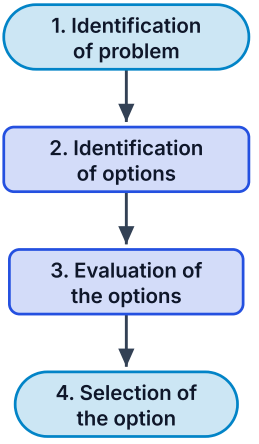
Decide on relevant costs: future costs and revenues that differ between the options, including benefits given up. Costs already incurred, and fixed costs that continue whatever you choose, do not decide anything.

How it works

Generally, short-term decisions are related with temporary gaps between demand and supply for

available resources (SM 14.36). ICAI frames every decision in four steps (SM 14.32–14.33):

The framework for decision making



Step 3 has two tasks: identify the costs and benefits of each option, then estimate their amount. Non-financial factors such as customer satisfaction, impact on existing market or customers and the ethics of the decision are evaluated too.

Which costs count

To be **identified** for measurement, a cost or benefit must pass two principles (SM 14.33):

- **Controllability**: it arises due to the choice of an option.
- **Relevance**: a controllable cost is relevant only if (a) it is a future cost and (b) it differs under the two options under consideration. Raw material for 1,000 extra

units is controllable, but irrelevant to "make in-house or send to job workers" if it costs the same either way.

To be **estimated**, two more (SM 14.35): **variability** (by how much the cost or benefit rises or falls because of the choice) and **traceability** (direct costs are assigned to the option; indirect costs are apportioned). Options are measured at directly traceable and variable costs.

ICAI's own relevance list, in ICAI's order (SM 14.34–14.35):

Cost
Historical cost

Relevant?
Irrelevant

ICAI's reason
Already incurred; does not affect the decision (book value of machinery)

Cost
Sunk cost

Relevant?
Irrelevant

ICAI's reason
Already paid (material in store with no replacement cost; cost of drawings, blueprints)

Cost
Committed cost

Relevant?
Irrelevant

ICAI's reason
Pre-agreed, cannot be revoked in normal circumstances (rate agreements, salaries); also a sunk cost

Cost

Opportunity cost

Relevant?

Relevant

ICAI's reason

The forgone potential benefit from the best rejected course of action

Cost

Notional or imputed cost

Relevant?

Relevant only if a benefit is actually forgone

ICAI's reason

Notional interest counts only if the company could earn that interest

Cost

Shut-down cost

Relevant?

Relevant

ICAI's reason

On suspending operations some fixed costs are avoided and extra ones may be incurred

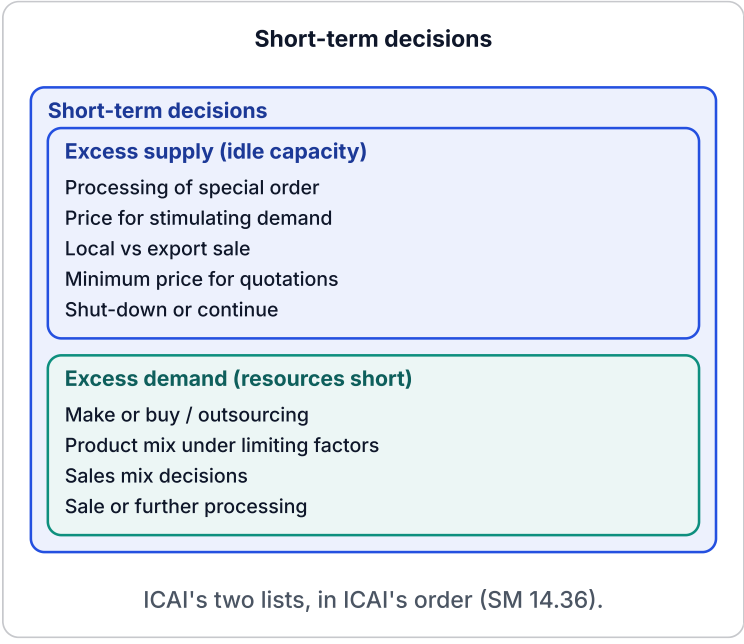
Note

Shut-down cost, two angles

Chapter 1 defines **shut down costs** as the fixed costs that continue to be incurred even when a plant is temporarily shut down, such as rent, rates and depreciation (SM 1.35). They are the cost of the "shut down" option, so they enter the decision to close. Once paid, they are history: ICAI's May 2025 suggested answer classifies rent already paid for a shed closed for two months as committed / shut-down cost, **non-relevant**.

The decisions, one rule each

The Study Material sorts short-term decisions into two families (SM 14.36):



ICAI's two lists, in ICAI's order (SM 14.36).

Decision

Make or buy

Compare

Buy price against the relevant cost of making: variable cost + avoidable fixed cost + contribution forgone if the capacity has another use

Choose

The cheaper

Decision

Shut down or continue (temporary)

Compare

Loss if continued against the shut-down cost (unavoidable fixed costs + extra costs of closing)

Choose

Shut down only if continuing loses more

Decision

Discontinue a product

Compare

Its contribution against its avoidable (specific) fixed costs, plus any rent or contribution from the freed capacity

Choose

Drop only if the saving exceeds the contribution lost

Decision

Special or export order

Compare

Price against variable cost + extra costs of the order + contribution lost on any regular sales it displaces

Choose

Accept if the net gain is positive

Two neighbours use the same rule: product mix and make-or-buy with a scarce resource rank by contribution (or saving) per unit of the key factor (limiting factor), and sell-or-process-further compares incremental revenue with incremental cost (sell at split-off or process further). A quotation built up from full cost is in estimated cost sheets, tenders and quotations, and Paper 6A applies the same relevant-cost rule to project cash flows in capital budgeting cash flows.

Formats and formulas

Reproduce these statements from memory: ICAI's suggested answers are built on this two-column comparison (the shut-down one is laid out as a "Statement showing operating loss").

Statement showing make or buy (per unit or in total)

Particulars	Make (₹)	Buy (₹)
Purchase price		xxx
Direct materials, labour, variable overheads	xxx	
Avoidable fixed costs	xxx	
Contribution forgone on the alternative use	xxx	
Relevant cost	**xxx**	**xxx**

Statement showing operating profit or loss: continue or shut down

Particulars	If continued (₹)	If shut down (₹)
Sales	xxx	
Less: Variable cost	xxx	
Contribution	**xxx**	
Less: Fixed cost	xxx	xxx (unavoidable)
Less: Additional shut-down costs		xxx
Profit / (loss)	**xxx**	**xxx**

∑ **Formula**

Shut-down point

$$\text{SDP} = \frac{\text{Fixed cost} - \text{Shut-down cost}}{\text{Contribution per unit}}$$

SDP = shut-down point in units; fixed cost = total fixed cost if the plant runs; shut-down cost = fixed costs that continue during the closure + additional costs of closing. Below this volume, closing loses less. In rupees, divide by the P/V ratio instead (SM practical problem 9: lock-up sales = avoidable fixed cost ÷ P/V ratio).

∑ **Formula**

Minimum price for a special order

$$\text{Minimum price} = \frac{\text{Relevant cost}}{\text{Units ordered}}$$

Relevant cost of the order = variable cost + costs specific to the order + contribution lost on any regular sales it displaces. With enough idle capacity the last part is nil.

Cost terms ICAI asks you to classify (Chapter 1, SM 1.34–1.35)

Term	ICAI's meaning
Differential cost	The change (increase or decrease) in total cost, variable and fixed, due to a change in activity level, technology, process or method of production
Imputed cost	Notional cost involving no cash outlay, such as interest on capital not actually paid
Opportunity cost	The value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action
Out-of-pocket cost	The portion of total cost that involves cash outflow; used in pricing in recession and make or buy; avoided if the proposal is not accepted

Term	ICAI's meaning
Shut down costs	Fixed costs which continue to be incurred, and cannot be avoided, during the temporary closure of a plant (rent, rates, depreciation)
Sunk costs	Historical costs incurred in the past; they play no role in decision making in the current period

Definition

Relevant cost

A cost is treated as relevant only if (a) it is a future cost and (b) it differs under two options under consideration. (SM 14.33)

Worked examples

Worked example **Desserts in the dock**

Tapri's Lucknow delivery kitchen sells three lines.

Figures for the quarter:

Particulars (₹)	Chai	Snacks	Desserts
Sales	30,00,000	25,00,000	15,00,000
Variable costs	12,00,000	15,00,000	10,50,000
Specific fixed costs	4,00,000	3,00,000	2,50,000

Each line's specific fixed costs (its own chef and equipment lease) would be saved if the line were dropped. Common fixed costs of ₹14,00,000 (kitchen rent, app platform fee, head office) are apportioned in the ratio of sales and would not change.

Required:

(i) Prepare a statement of profit by line, as the management report shows it.

(ii) Examine Kesar's proposal to discontinue Desserts.

(iii) A cloud kitchen offers ₹3,00,000 a quarter to rent the Desserts space. Does it change your advice?

THE WORKING

(i) Statement of profit by line

Particulars (₹)	Chai	Snacks	Desserts	Total
Sales	30,00,000	25,00,000	15,00,000	70,00,000
Less: Variable costs	12,00,000	15,00,000	10,50,000	37,50,000
Contribution	**18,00,000**	**10,00,000**	**4,50,000**	**32,50,000**
Less: Specific fixed costs	4,00,000	3,00,000	2,50,000	9,50,000
Contribution after specific fixed costs	14,00,000	7,00,000	2,00,000	23,00,000
Less: Common fixed costs (6 : 5 : 3)	6,00,000	5,00,000	3,00,000	14,00,000
Profit / (loss)	**8,00,000**	**2,00,000**	**(1,00,000)**	**9,00,000**

(ii) Discontinue Desserts? Dropping it loses its contribution of ₹4,50,000 and saves only its specific fixed costs of ₹2,50,000. The ₹3,00,000 of common cost does not disappear; it is spread over Chai and Snacks. Profit falls from ₹9,00,000 to **₹7,00,000**. Desserts contributes ₹2,00,000 towards common costs: the proposal should be **rejected**.

(iii) With the rent offer. Dropping Desserts and letting the space gives ₹7,00,000 + ₹3,00,000 = **₹10,00,000**, ₹1,00,000 more than keeping it. On the figures, discontinue and let the space, after checking that Chai and Snacks will not lose customers who order dessert with them.

Worked example

Khurja kulhads: make or buy

Tapri fires 2,00,000 kulhads a month in its own kiln at Khurja. Cost per kulhad:

Particulars	₹ per kulhad
Clay	1.20
Labour	1.00
Fuel and other variable overheads	0.60
Fixed overheads (kiln depreciation, shed rent, supervisor)	0.80
Total	**3.60**

A potters' co-operative offers kulhads at ₹3.10 delivered. If Tapri buys, the kiln supervisor (₹30,000 a month) would leave; the other fixed overheads would continue.

> **Kesar:** ₹3.10 against ₹3.60. Fifty paise a cup, two lakh cups. Sab adjust ho jayega: buy!

Required:

(i) Advise Tapri if the kiln has no other use.

(ii) In October the kiln could fire Diwali diyas instead, earning a contribution of ₹1,20,000 after all their own

costs, the supervisor's salary included. Advise for October.

(iii) State the highest price Tapri should pay the co-operative in an ordinary month.

THE WORKING

Working note. Fixed overheads = 2,00,000 × ₹0.80 = ₹1,60,000 a month, of which only the supervisor's ₹30,000 is avoidable; ₹1,30,000 continues whether Tapri makes or buys.

Statement showing make or buy (per month)

Particulars	Make (₹)	Buy (₹)
Purchase price (2,00,000 × ₹3.10)		6,20,000
Variable cost (2,00,000 × ₹2.80)	5,60,000	
Avoidable fixed cost: supervisor	30,000	
(i) Relevant cost, ordinary month	**5,90,000**	**6,20,000**
Contribution forgone on diyas	1,20,000	
(ii) Relevant cost, October	**7,10,000**	**6,20,000**

(i) Make: ₹30,000 a month cheaper. Kesar's ₹3.60 includes ₹0.65 a kulhad of fixed overheads (₹1,30,000) that would be paid anyway.

(ii) Buy in October and fire diyas: ₹90,000 better than making.

(iii) ₹5,90,000 ÷ 2,00,000 = **₹2.95** per kulhad.

Before switching, Tapri should also weigh the co-operative's quality and reliability in the Diwali rush, the non-financial side of step 3.

Interactive version: PassBook app, lesson Make or buy, shut down and other short-term decisions

Worked example

Manali in the snow: open daily, open on weekends, or close?

Tapri's Manali café sells chai at ₹60 a cup; variable cost is ₹36 a cup. For the snowbound winter quarter Ritu has

three options:

Option	Expected cups	Fixed costs for the quarter (₹)
Open every day	18,000	5,40,000
Open only Friday to Sunday (part-time staff, lower power load)	11,000	3,60,000
Close for the quarter	—	2,10,000 (lease, insurance, depreciation)

Closing also means a one-time cost of ₹60,000 for staff settlement, winterising the plumbing and the spring relaunch.

Required:

(i) Prepare a statement comparing the three options, and advise.

(ii) Calculate the shut-down point of each operating option.

(iii) Above what number of cups would opening every day be better than opening on weekends only?

THE WORKING

Contribution per cup = ₹60 − ₹36 = **₹24** under every option.

(i) Statement of operating result for the quarter

Particulars	Open daily (₹)	Weekends only (₹)	Close (₹)
Contribution (18,000 / 11,000 × ₹24)	4,32,000	2,64,000	—
Less: Fixed costs	5,40,000	3,60,000	2,10,000
Less: Closing and relaunch costs	—	—	60,000
Loss	**(1,08,000)**	**(96,000)**	**(2,70,000)**

Open on weekends only: it loses ₹12,000 less than opening daily and ₹1,74,000 less than closing. Closing is the worst option, because the café's contribution under either way of opening more than covers the fixed costs that closing would save.

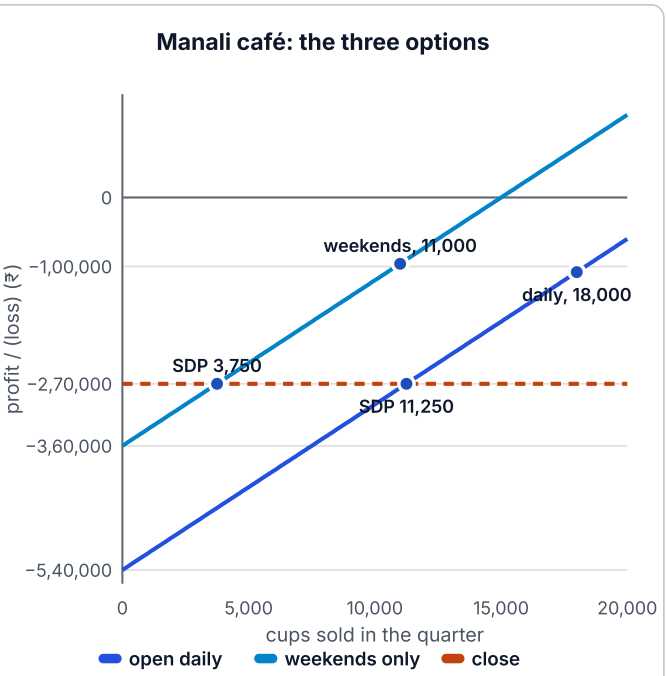
(ii) Shut-down points. Shut-down cost = fixed costs that continue + closing costs = ₹2,10,000 + ₹60,000 = ₹2,70,000.

$$\text{SDP} = \frac{\text{Total fixed cost} - \text{Shut-down cost}}{\text{Contribution per cup}}$$

Option	Working	Shut-down point
Open daily	(₹5,40,000 − ₹2,70,000) ÷ ₹24	11,250 cups
Weekends only	(₹3,60,000 − ₹2,70,000) ÷ ₹24	3,750 cups

Both expected volumes (18,000 and 11,000) are well above their shut-down points, which confirms that the café should not close.

(iii) Daily against weekends. Opening daily must lose no more than ₹96,000: ₹24 × cups − ₹5,40,000 ≥ −₹96,000, so cups ≥ ₹4,44,000 ÷ ₹24 = **18,500**. The expected 18,000 falls 500 short. If Ritu expects more than 18,500 cups from daily opening (a heavy-snow season brings more tourists), open daily; otherwise weekends only.



Blue: open daily; sky blue: weekends only; orange dashed: the ₹2,70,000 loss if the café closes. Each operating line meets the closing line at its shut-down point. The two operating lines are parallel (₹24 a cup) but start at different fixed costs, so they are read at their own expected volumes.

Interactive version: PassBook app, lesson Make or buy, shut down and other short-term decisions

Worked example

The airline order that could cost more than it pays

Tapri's premix plant can pack 10,00,000 sachets a month and sells 7,00,000 a month at ₹12; variable cost is ₹7 a sachet and fixed costs are ₹25,00,000 a month. Udaan Airways offers to buy 2,50,000 sachets for its November flights at ₹9 each, packed in the airline's own foil (₹0.40 a sachet extra), with a one-time design and plate cost of ₹30,000.

> **Kesar:** Idle machines, extra cash. What is there to think about?
>
> **Ritu:** The hotel chain that buys 1,20,000 of our sachets a month. Its buyer flies Udaan.

Ritu expects that if the order is accepted, the hotel chain (part of the regular 7,00,000) will learn the airline's price and demand ₹2 off a sachet for the next three months.

Required:

(i) Advise whether Tapri should accept the order if the regular price is not affected.

(ii) Advise again, taking the hotel chain's reaction into account.

(iii) Calculate the minimum price per sachet for the airline order in each case.

THE WORKING

Working note. Idle capacity = 10,00,000 − 7,00,000 = 3,00,000 sachets, so the whole order of 2,50,000 fits without displacing regular sales. Variable cost of an airline sachet = ₹7 + ₹0.40 foil = ₹7.40. Fixed costs of ₹25,00,000 are incurred anyway and are not relevant.

Statement of the effect of accepting the order

Particulars	Price unaffected (₹)	Hotel chain reacts (₹)
Sales (2,50,000 × ₹9)	22,50,000	22,50,000
Less: Variable cost (2,50,000 × ₹7.40)	18,50,000	18,50,000
Less: Design and plates	30,000	30,000
Contribution from the order	3,70,000	3,70,000
Less: Contribution lost on regular sales (1,20,000 × ₹2 × 3 months)	—	7,20,000
Net gain / (loss) from accepting	**3,70,000**	**(3,50,000)**

(i) Accept: profit for November rises by ₹3,70,000, from ₹10,00,000 (7,00,000 × ₹5 − ₹25,00,000) to ₹13,70,000.

(ii) Reject: the price concession to the hotel chain is an opportunity cost of the order, and it turns the gain into a loss of ₹3,50,000 over the three months.

(iii) Minimum price

Case	Relevant cost (₹)	Minimum price per sachet
Price unaffected	18,50,000 + 30,000 = 18,80,000	₹7.52
Hotel chain reacts	18,80,000 + 7,20,000 = 26,00,000	₹10.40

Before deciding, Tapri should weigh the non-financial factors the Study Material lists (SM 14.33): the impact on existing customers (can the airline's price be kept confidential, or the airline pack carry a different product description?), customer satisfaction, and the brand exposure on every November flight.

Exam angle

In the exam

Where these decisions appeared

In the ICAI documents read for this lesson:

- **Classifying costs**: May 2025 Q6(a), 5 marks: "identify the type of cost and state whether it is relevant/non-relevant in managerial decision making" for five situations (owned office with similar rent available nearby: opportunity or imputed cost, relevant; rent for a shed temporarily closed: committed or shut-down cost, non-relevant; market research already paid: sunk, non-relevant). Also RTP September 2026 Q16(b) and RTP May 2025 MCQ 1(ii).
- **Special and export orders**: MTP Series I for May 2026 Q4(a), 7 marks (accept with idle capacity; reject when a capacity cut forces lost regular sales); MTP Series II for January 2026 MCQs 4–5; MTP Series II for September 2025 MCQ 9; MTP Series I for January 2025 MCQs 6–10 (minimum price including the profit lost on Indian sales); MTP Series II for May 2026 Q2(a), 7 marks: a one-off order quoted at full cost plus 1/9 by company policy, a customer offering less, "State and explain the arguments for and against accepting", with a recommendation. The suggested answer re-costs the order at prime cost plus variable overheads only (no commission on a direct order, fixed overheads already recovered by budgeted sales) and accepts it.

- **Discontinuing a product**: RTP January 2025 Q10 (examine a recommendation to discontinue a product); RTP September 2025 Q16(iii).
- **Shut down**: old-scheme November 2023 Q4(b), 5 marks ("Whether the Noida plant be shut down? Find the shut-down point in units"); MTP Series II for November 2023 Q1(b) (shut-down point in units and as a percentage of capacity).
- **Make or buy**: RTP January 2026 Q17 (make or buy empty cans at three volumes, with an extra machine needed beyond present capacity, and the volume at which installing it pays); inside a key-factor question in MTP Series I for September 2025 Q4(a) (see limiting factor).

In the exam

The traps

- A product "making a loss" after apportioned common costs: test its contribution against its avoidable fixed costs.
- Full cost per unit against a buy price: strip out the fixed costs that continue.
- A quotation built from a full-cost pricing policy: re-cost the order with only the costs it adds (variable overheads at a variable rate, no commission if none is payable).
- "Must be accepted in full" with too little idle capacity: count the contribution lost on displaced sales.
- Shut-down cost in ICAI's answers = unavoidable fixed cost + additional cost of closing; the shut-down point divides the rest of the fixed cost by contribution per unit.

Trick

The "what changes?" column

Draw two columns, one per option, and write in only what differs between them. Anything that would sit in both columns at the same figure can be left out of both: the answer does not move. Condition: both columns must cover the same period and the same volume of the other products.

Answer frame

"Identify the type of cost and state whether it is relevant"

Write a three-column table, as ICAI's suggested answer does: **Situation | Type of cost | Relevant / Non-relevant**. One line per situation; give the ICAI term (opportunity, imputed or notional, sunk, committed, shut-down, out-of-pocket, differential), and the relevance follows from the two tests: future, and different between the options.

"Explain the framework for decision making"

1. **Identification of problem**: the goal set to reach the objectives, such as low-cost production; a cost leader must achieve the 3Es: economy in inputs, efficiency in process and operations, effectiveness in output (SM 14.32).
2. **Identification of options**: all possible options are explored. ICAI's list for low-cost production (SM 14.32–14.33): (a) purchase of inputs from a specialised market: local vs import; (b) make the input in its own factory: make or buy; (c) bulk purchase to avail a discount: how much to purchase; (d) make in-house: make vs outsource; (e) bulk processing: how much to produce; (f) an efficient machine: old machine vs new machine; (g) optimisation of key resources: product mix decisions.
3. **Evaluation of the options** against the objective: financial measures (impact on profit, market share, return on investment) and non-financial factors

(customer satisfaction, impact on existing market or customers, ethics); two tasks: identification of the costs and benefits of each option (controllability, relevance) and estimation of their amount (variability, traceability).

4. **Selection of the option**: the best option is selected and implemented.

"Should the special order be accepted?"

1. **Relevant cost of the order**: variable cost, costs specific to the order, and the contribution lost on any regular sales it displaces; fixed costs that continue (or are already recovered by budgeted sales) are excluded.
2. **Price against relevant cost**: the margin the order adds to profit.
3. **Capacity**: whether the order fits the idle capacity or displaces regular sales.
4. **Non-financial factors** (SM 14.33): impact on the existing market or customers (regular buyers may ask for the same price), customer satisfaction, ethics of the decision.
5. **Recommendation**: accept, reject, or accept with conditions, with the reason.

The numerical: shut down or continue

Loses marks

"Loss ₹1,08,000 if open daily, so close the café."

Compares a loss with nothing: closing has a cost too.

Passes

Both columns, the right decision, no shut-down point.

The second half of the question carries marks of its own.

Full marks

Contribution per unit; a column for each option (continued / shut down) with unavoidable fixed and closing costs; decision in words; shut-down point by formula, in units (and as a % of capacity where asked).

Common mistakes

⚠ **Common mistake**

Hall of Shame: the apportioned rent that killed the bun maska

Dropping a line because it shows a loss after apportioned common costs. Desserts' ₹1,00,000 "loss" hid a ₹2,00,000 contribution to common costs. **Fix:** compare its contribution with its avoidable fixed costs only.

⚠ **Common mistake**

Full cost against the buy price

₹3.60 a kulhad against ₹3.10 says buy; the relevant ₹2.95 says make. **Fix:** count only the variable cost, the fixed costs that would really disappear, and any contribution the capacity could earn elsewhere.

⚠ **Common mistake**

The special order with no room for it

Treating an order that must be accepted in full at its own contribution when only part of it fits the idle capacity, or ignoring a price concession it forces on regular customers. **Fix:** displaced regular sales and price concessions are opportunity costs: add the contribution lost to the cost of the order.

⚠ **Common mistake**

The feasibility report that keeps billing you

Counting money already spent, such as last year's market survey or the book value of old stock, as a cost of going ahead. **Fix:** sunk and committed costs are the same whichever option is chosen: leave them out.

Practice

Check yourself

1. A unit expects to sell 18,000 units in a slack quarter. Its fixed costs for the quarter are ₹4,00,000; if it closes, ₹1,50,000 of them continue and closing costs ₹30,000 more. What is the lowest contribution per unit (₹, two decimals) at which staying open is no worse than closing?

Answer: _____

2 Case scenario. Tapri has a one-year rate contract with a dairy for 2,000 litres of milk a day at ₹52 a litre, payable whether or not the milk is taken. About 300 litres a day are now left over and thrown away. Ananya proposes an iced-latte line that would use the surplus milk. How should the milk be costed in her proposal?

A At ₹52 a litre, the contract rate, as an out-of-pocket cost

B At the market price of milk, as an opportunity cost

C At ₹52 a litre plus a share of the dairy's delivery charges

D As a committed cost: irrelevant, because it is payable whether or not the line is started

3 A component costs ₹45 a unit in variable cost to make, plus ₹15 of absorbed fixed overheads that would continue if it were bought. An outside supplier quotes ₹52; the capacity has no other use. What should the company do?

- **A** Buy: ₹52 is less than the ₹60 cost of making
- **B** Make: it saves ₹7 a unit
- **C** Buy: it saves ₹8 a unit
- **D** Indifferent: the costs are equal

4 With ample idle capacity, a customer offers ₹80 a unit for 5,000 units of a product whose variable cost is ₹70; a special set-up costing ₹20,000 is needed. By how much would profit rise (₹)?

Answer: _____

5 Kesar is deciding whether to close the Manali café for the winter quarter. Which of these are relevant to the decision? Select all that apply.

- **A** Staff salaries that would be saved
- **B** Depreciation on the café's fittings, which continues
- **C** The cost of relaunching in spring
- **D** The contribution the winter sales would earn
- **E** Last year's renovation cost

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Relevant cost: a future cost that differs between the options (SM 14.33); sunk, historical and committed costs are irrelevant; opportunity cost is relevant; notional cost only if a benefit is really forgone.
- Framework: identify the problem, identify options, evaluate (controllability, relevance; variability, traceability; non-financial factors), select.
- Make or buy: buy price against variable cost + avoidable fixed cost + contribution forgone on the alternative use.

- Discontinue: drop only if the avoidable fixed cost (plus rent from freed space) exceeds the contribution lost; apportioned common costs stay.
- Shut down: compare the loss if continued with the shut-down cost (unavoidable fixed + closing costs).
- Shut-down point = (total fixed cost − shut-down cost) ÷ contribution per unit; in rupees ÷ P/V ratio.
- Special order: accept if price exceeds variable + specific costs; with no idle capacity add the contribution of displaced sales. Minimum price = relevant cost ÷ units.
- If it would sit in both columns at the same figure, it cannot decide anything. (The rent will be there tomorrow.)

Answer key · Check yourself

1 **12.2 to 12.23**

Shut-down cost = ₹1,50,000 + ₹30,000 = ₹1,80,000, so closing saves ₹4,00,000 − ₹1,80,000 = ₹2,20,000 of fixed cost. Staying open must earn at least that in contribution: ₹2,20,000 ÷ 18,000 = ₹12.22 a unit. Dividing the full ₹4,00,000 (₹22.22) is the break-even contribution, not the shut-down one.

2 **D** **As a committed cost: irrelevant, because it is payable whether or not the line is started**

The milk is paid for under the contract either way, and the surplus has no other use, so the cost does not differ between the options: a committed cost, irrelevant (SM 14.34). An opportunity cost would arise only if the surplus could be sold or used elsewhere.

3 **B** **Make: it saves ₹7 a unit**

The relevant cost of making is the ₹45 variable cost; the ₹15 continues either way. Making saves ₹52 − ₹45 = ₹7 a unit. Comparing with ₹60 is the full-cost trap.

4 **30000**

5,000 × (₹80 − ₹70) − ₹20,000 = ₹30,000. Fixed costs already being incurred are irrelevant.

5 **A** **Staff salaries that would be saved**; **C** **The cost of relaunching in spring**; **D** **The contribution the winter sales would earn**

Relevant: what differs between staying open and closing (salaries saved, relaunch cost, contribution lost). Depreciation continues either way, and last year's renovation is sunk.

14.7 Multi-product break-even analysis

With several products sharing fixed costs, break-even needs the sales mix: a weighted contribution per unit for a mix in units, a weighted P/V ratio for a mix in values.

In plain words

Chai, samosa and a question with no single answer

Tapri's Pune café broke even at 15,000 cups of chai a month. Now it sells samosas too, and a new fryer has lifted fixed costs to ₹2,07,000.

> **Kesar:** Ab samosa bhi! So what is the break-even now?
>
> **Ritu:** Per unit, please. But which unit? 17,250 cups and no samosas, or 20,700 samosas and no chai?
>
> **Kesar:** Whatever our customers buy, obviously.
>
> **Ritu:** Exactly. They buy three chais with every samosa. Give me that and I will give you one answer.

A cup of chai leaves ₹12 of contribution, a samosa ₹10. Every time the café sells its usual "batch" of three chais and one samosa, it collects ₹36 + ₹10 = ₹46. Fixed costs of ₹2,07,000 need 4,500 such batches: **13,500 cups and 4,500 samosas**. That is the break-even point for this **sales mix**, the proportion in which the products sell. If customers

start buying two samosas with every three chais, the answer changes.

So a business with several products has no break-even point of its own, only a break-even point **for a given mix**. You find the contribution of the mix (per batch, per average unit, or per rupee of sales), divide the shared fixed cost by it, and split the answer back in the mix.

Think of it like

The thali

A restaurant does not break even on rotis or on dal; it breaks even on thalis. Each thali holds a fixed mix of items, so it has one combined contribution, and fixed cost ÷ contribution per thali gives the thalis needed. Change what goes into the thali, and the number of thalis changes too.

Key idea

With products sharing fixed costs, the break-even point exists only for a stated sales mix. Weight the contributions by the mix, divide the fixed cost by the weighted figure, then split the result in the same mix.

How it works

ICAI's whole rule fits in three sentences: in a multi-product environment, where more than one product is made using a common fixed cost, the break-even formula

needs some adjustments; the contribution is calculated by taking weights for the products; the weights may be of sales mix quantity or sales mix values (SM 14.16). It rests on the fifth CVP assumption: the analysis covers a single product, or assumes that the proportion of the products stays constant as total units sold change (SM 14.12). Everything after the weighting (margin of safety, target profit) is the single-product method of break-even point and margin of safety applied to the mix.

One batch of the café's mix

One batch = 3 chai + 1 samosa ₹46

3 cups of chai		1 samosa	
contribution:	₹12 each	contribution:	₹10
total:	₹36	total:	₹10

The batch is the mix in miniature: its contribution, ₹46, plays the part a single product's contribution plays in an ordinary break-even.

Two routes, chosen by how the mix is given

1. **Mix in units** ("4 units of J for every 3 of K", "5 : 3 per batch"): composite contribution per unit = each product's contribution per unit × its share of units, added up. Or work per batch: contribution of the units in one batch.

2 **Composite break-even point** = common fixed cost ÷ composite contribution per unit (or per batch).

3 **Split it in the mix**: each product's break-even units = composite BEP × its share (or batches × its units per batch); × price for rupees.

4 **Mix in values** ("sales mix 40% : 35% : 25%", or each product's sales in rupees): composite P/V ratio = each product's P/V ratio × its share of sales value, added up. ICAI's tables call each term the **contribution per rupee of sales**. With actual figures, it is simply total contribution ÷ total sales.

5 **Composite break-even sales** = fixed cost ÷ composite P/V ratio; each product's share = composite BES × its sales mix %; ÷ price for units.

Which route? Read how the question gives the mix, then follow one column:

Step	Mix given in units	Mix given in values
Weight each product by	its share of units	its share of sales value
Weighted figure	contribution per unit (or per batch)	P/V ratio
Fixed cost ÷ weighted figure	BEP in units (or batches)	break-even sales in ₹
Split the answer by	the units ratio	the sales mix %

Either route gives the same answer when the mix is converted correctly: a units mix becomes a value mix by multiplying each product's units by its price.

Interactive version: PassBook app, lesson Multi-product break-even analysis

Change the mix, change the break-even

Total sales and fixed costs unchanged, a mix that shifts towards the products with the higher P/V ratio raises the composite P/V ratio, so it **lowers** break-even sales and **raises** profit; a shift towards the low-P/V products does the opposite. A product with a negative contribution pulls the composite down. That is why ICAI's questions so often replace one product with another and ask whether

the new mix is better (SM practical problem 20; January 2026 MCQs 1–5; RTP September 2026 Q15). The profit path that shows each product's slope on a P/V chart is drawn in break-even and profit-volume charts.

Interactive version: PassBook app, lesson Multi-product break-even analysis

Formats and formulas

Reproduce ICAI's statement from memory: it is the backbone of every value-mix answer (SM 14.77–14.78).

Particulars	A	B	C	Total
Selling price (₹)	xxx	xxx	xxx	
Less: Variable cost (₹)	xxx	xxx	xxx	
Contribution per unit (₹)	xx	xx	xx	
P/V ratio	x%	x%	x%	
Sales mix	x%	x%	x%	100%
Contribution per rupee of sales (P/V ratio × sales mix)	**x%**	**x%**	**x%**	**x%**

Then: total contribution = total sales × composite P/V ratio; less fixed costs = profit; break-even sales = fixed costs ÷ composite P/V ratio.

∑ **Formula**

Composite contribution per unit (mix in units)

$$\text{Composite contribution} = \sum (C \times w)$$

C = a product's contribution per unit; w = its units weight, its units in the mix ÷ total units in the mix. The result is a contribution per (average) unit. ICAI's Example 4: 4 units of J (₹40) to 3 of K (₹20) gives ₹40 × 4/7 + ₹20 × 3/7 = ₹31.43 (SM 14.16).

∑ **Formula**

Composite break-even point

$$\text{BEP (units)} = \frac{\text{Common fixed cost}}{\text{Composite contribution}}$$

$$\text{Product BEP} = \text{BEP} \times w$$

∑ **Formula**

Composite P/V ratio (mix in values)

$$\text{Composite P/V} = \sum (\text{P/V} \times \text{mix \%})$$

$$= \frac{\text{Total contribution}}{\text{Total sales}}$$

Mix % = each product's share of total sales value.

∑ Formula

Composite break-even sales

$$\text{Composite BES} = \frac{\text{Fixed cost}}{\text{Composite P/V}}$$

$$\text{Product BES} = \text{BES} \times \text{mix } \%$$

ICAI's Illustration 11 also shows a second, equivalent method: share the fixed cost among the products in the ratio of their weighted contributions, then each product's BEP = its share of fixed cost ÷ its own contribution per unit (SM 14.39).

Worked examples

Worked example **Namkeen and mathri, five to three**

Tapri's snacks division sells two packaged snacks. Namkeen sells at ₹60 a pack with a variable cost of ₹36; mathri sells at ₹80 with a variable cost of ₹52. They sell in the ratio of 5 packs of namkeen to 3 packs of mathri. Fixed costs are ₹10,20,000 a month, and the budget is 64,000 packs a month in the same mix.

Required:

(i) Calculate the composite break-even point, and the break-even packs and sales of each snack.

(ii) Calculate the margin of safety at the budget, in rupees and as a percentage of sales, and the budgeted profit.

(iii) Calculate the packs of each snack needed for a profit of ₹8,16,000 a month.

THE WORKING

Working note: composite contribution

Particulars	Namkeen	Mathri	Batch of 8
Contribution per pack (₹)	24	28	
Packs in the mix	5	3	8
Contribution in the mix (₹)	120	84	**204**
Sales in the mix (₹)	300	240	540

Composite contribution per pack = ₹204 ÷ 8 = **₹25.50**; composite P/V ratio = 204 ÷ 540 = 37.78%.

(i) Break-even point

$$\text{BEP} = \frac{10{,}20{,}000}{25.50} = 40{,}000 \text{ packs}$$

That is 5,000 batches of 8 (₹10,20,000 ÷ ₹204).

Snack	Break-even packs	Break-even sales (₹)
Namkeen (40,000 × 5/8)	25,000	15,00,000
Mathri (40,000 × 3/8)	15,000	12,00,000
Total	**40,000**	**27,00,000**

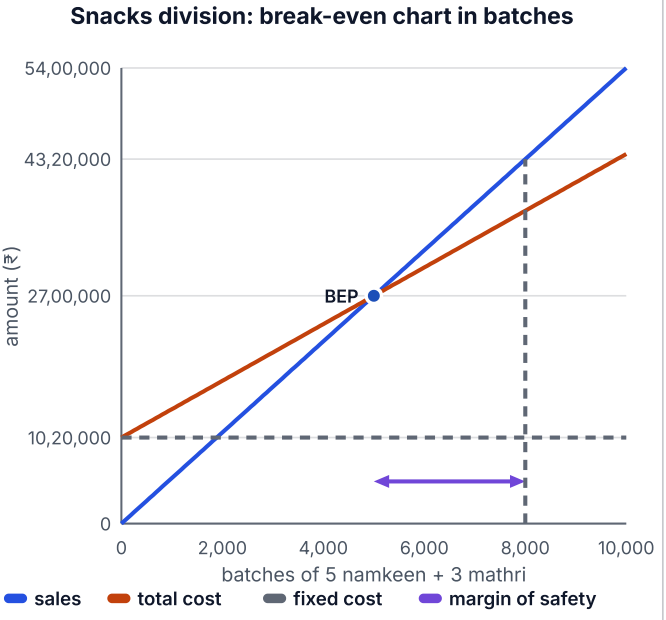
(ii) At the budget of 64,000 packs (40,000 namkeen, 24,000 mathri)

Budgeted sales = ₹24,00,000 + ₹19,20,000 = ₹43,20,000. Margin of safety = ₹43,20,000 − ₹27,00,000 = **₹16,20,000**, which is **37.5%** of sales. Profit = 64,000 × ₹25.50 − ₹10,20,000 = **₹6,12,000**; check: ₹16,20,000 × 204/540 = ₹6,12,000.

(iii) Profit of ₹8,16,000

$$\text{Packs} = \frac{10{,}20{,}000 + 8{,}16{,}000}{25.50} = 72{,}000$$

That is **45,000 packs of namkeen and 27,000 of mathri** (9,000 batches).



Blue: sales at ₹540 a batch; orange: total cost, ₹10,20,000 plus ₹336 a batch. They cross at 5,000 batches (₹27,00,000). The violet arrow is the margin of safety up to the budget of 8,000 batches (₹43,20,000).

Worked example

The chai-samosa combo: a third "product" in the mix

Tapri's Bhopal station kiosk sells chai at ₹30 (variable cost ₹12) and samosas at ₹20 (variable cost ₹8), in the ratio of 3 cups of chai to 2 samosas. Fixed costs are ₹6,24,000 a month, and the kiosk now sells 36,000 cups and 24,000 samosas a month.

Ananya proposes a combo of one chai and one samosa for ₹42. She expects monthly sales of 27,000 cups of chai alone, 9,000 samosas alone and 18,000 combos, keeping the ratio of 3 chai : 1 samosa : 2 combos. The combo's boxes and advertising add ₹36,000 a month to fixed costs.

Required:

(i) Calculate the present break-even point in units of each product and in sales, and the present profit.

(ii) Calculate the break-even point and the expected profit under the combo plan.

(iii) Advise Ananya, commenting on the composite P/V ratio.

THE WORKING

(i) Present mix: a batch of 3 chai + 2 samosas

Particulars	Chai	Samosa	Batch
Contribution per unit (₹)	18	12	
Units in the batch	3	2	5
Contribution of the batch (₹)	54	24	**78**
Sales of the batch (₹)	90	40	**130**

Composite P/V ratio = 78 ÷ 130 = 60%. Break-even batches = ₹6,24,000 ÷ ₹78 = 8,000: **24,000 cups of chai and 16,000 samosas**, break-even sales **₹10,40,000** (8,000 × ₹130).

Present volume = 12,000 batches: profit = 12,000 × ₹78 − ₹6,24,000 = **₹3,12,000** on sales of ₹15,60,000.

(ii) Combo plan: a batch of 3 chai + 1 samosa + 2 combos

The combo is treated as a product of its own: contribution = ₹42 − (₹12 + ₹8) = ₹22.

Particulars	Chai alone	Samosa alone	Combo	Batch
Contribution per unit (₹)	18	12	22	
Units in the batch	3	1	2	6
Contribution of the batch (₹)	54	12	44	**110**
Sales of the batch (₹)	90	20	84	**194**

Composite P/V ratio = 110 ÷ 194 = 56.70%. Fixed costs = ₹6,24,000 + ₹36,000 = ₹6,60,000. Break-even batches = ₹6,60,000 ÷ ₹110 = 6,000: **18,000 chai alone, 6,000 samosas alone and 12,000**

combos, break-even sales **₹11,64,000** (6,000 × ₹194).

Expected volume = 9,000 batches: profit = 9,000 × ₹110 − ₹6,60,000 = **₹3,30,000** on sales of ₹17,46,000.

Particulars	Present	Combo plan
Composite P/V ratio	60.00%	56.70%
Break-even sales (₹)	10,40,000	11,64,000
Expected sales (₹)	15,60,000	17,46,000
Margin of safety (% of sales)	33.33%	33.33%
Profit (₹)	**3,12,000**	**3,30,000**

(iii) Advice. The combo gives away ₹8 on every pair, so the composite P/V ratio falls from 60% to 56.70% and, with ₹36,000 more fixed cost, break-even sales rise by ₹1,24,000. Profit still rises by ₹18,000 because Ananya expects 9,000 more cups and 3,000 more samosas a month (45,000 cups and 27,000 samosas in all). The plan is worth trying only if that extra volume is reasonably certain: if total sales stayed at ₹15,60,000, the new mix would earn only ₹2,24,536 (₹15,60,000 × 110/194 − ₹6,60,000).

Worked example

The airport outlet: revenue share or fixed rent?

Tapri's airport outlet sells chai, coffee and snacks. Before rent, their P/V ratios are 65%, 50% and 30%, and they sell in the ratio of 40% : 35% : 25% of sales value. Fixed costs other than rent are ₹16,40,000 a quarter. The airport operator charges rent as a revenue share of 10% of sales. Expected sales are ₹70,00,000 a quarter.

Required:

(i) Calculate the composite P/V ratio and the break-even sales of the outlet and of each product.

(ii) Calculate the sales needed for a profit of ₹6,15,000 a quarter.

(iii) The operator offers instead a fixed rent of ₹6,55,000 a quarter. Calculate the sales at which the two rent options give the same profit, and advise at the expected sales.

THE WORKING

(i) Composite P/V ratio with the revenue share. A rent of 10% of sales is a variable cost: it takes 10 points off every product's P/V ratio.

Particulars	P/V ratio before rent
Chai	65%
Coffee	50%
Snacks	30%
Total	

Particulars	P/V ratio after the 10% revenue share
Chai	55%
Coffee	40%
Snacks	20%
Total	

Particulars	Sales mix
Chai	40%
Coffee	35%
Snacks	25%
Total	100%

Particulars	Contribution per rupee of sales
Chai	22%
Coffee	14%
Snacks	5%
Total	41%

Break-even sales = ₹16,40,000 ÷ 41% = **₹40,00,000**: chai ₹16,00,000, coffee ₹14,00,000, snacks ₹10,00,000 (40 : 35 : 25).

(ii) Target profit of ₹6,15,000

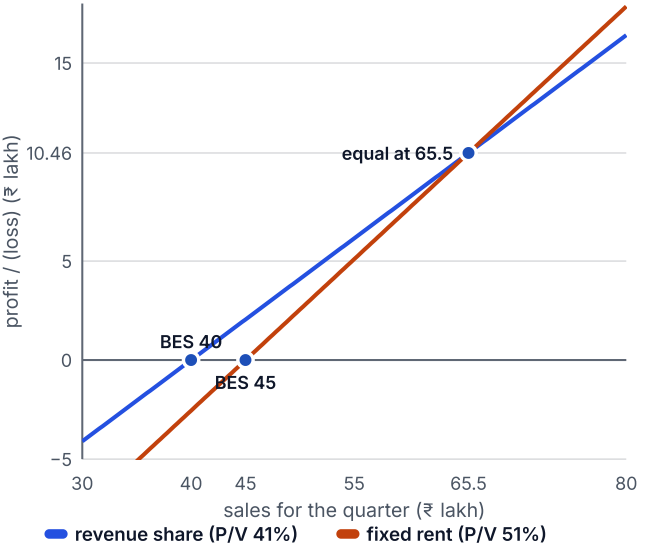
$$\text{Sales} = \frac{16{,}40{,}000 + 6{,}15{,}000}{41\%} = ₹55{,}00{,}000$$

(iii) Fixed rent of ₹6,55,000. The P/V ratios return to 65%, 50% and 30%: composite = 26% + 17.5% + 7.5% = **51%**. Fixed costs = ₹16,40,000 + ₹6,55,000 = ₹22,95,000; break-even sales = ₹22,95,000 ÷ 51% = ₹45,00,000.

Equal profit where 41% × S − ₹16,40,000 = 51% × S − ₹22,95,000, so 10% × S = ₹6,55,000 and **S = ₹65,50,000** (profit ₹10,45,500 either way).

Particulars	Revenue share	Fixed rent
Composite P/V ratio	41%	51%
Fixed costs (₹)	16,40,000	22,95,000
Break-even sales (₹)	40,00,000	45,00,000
Profit at expected sales of ₹70,00,000 (₹)	**12,30,000**	**12,75,000**

Airport outlet: profit under the two rent options



Blue: revenue share; orange: fixed rent, which starts from a bigger fixed cost but climbs faster. Below sales of ₹65.5 lakh the revenue share leaves more profit; above it the fixed rent does.

Advice. At the expected ₹70,00,000 the fixed rent earns ₹45,000 more a quarter. But the expected sales are only ₹4,50,000 above the indifference point, and the fixed rent raises break-even sales by ₹5,00,000: if a slow quarter brings sales below ₹65,50,000, the revenue share is better, and it shares the risk with the operator. Choose the fixed rent only if Ritu is confident that sales will stay above ₹65,50,000; otherwise keep the revenue share.

Exam angle

In the exam

Where multi-product break-even appeared

In the ICAI documents read for this lesson:

- **January 2026 MCQs 1–5, 2 marks each**: a three-product case: combined contribution-to-sales ratio at the existing level (from each product's sales in rupees), total profit, the new ratio when Product W replaces Z in a given mix, break-even sales and profit at the proposed mix.
- **RTP September 2026 Q15**: sales mix 35% : 35% : 30%, a proposal to replace C with P: "Will you advise the company to change over to production of P? Give reasons for your answer."
- **RTP May 2025 MCQ 7**: "Composite Break-even Batch" for ball pens and gel pens, the mix taken from expected sales of 6,000 and 3,600 units (5 : 3).

- **RTP September 2024 Q17** and **MTP Series II for September 2025 Q1(a), 5 marks**: a single product at a 60% margin of safety (net return on sales after tax), then a second product with "sales mix of X : Y would be 5 : 3": break-even units of each.
- **RTP January 2025 MCQ 1(ii)**: overall break-even units for three articles demanded in the ratio 2 : 3 : 5.
- The Study Material: Example 4, Illustration 11, practical problems 8 and 20.

In the exam

The traps

- A mix "per batch" or "for every": compute one batch's contribution, then the number of batches.
- Sales of each product given in rupees, not as percentages: composite P/V = total contribution ÷ total sales.
- A replacement product: recompute the whole composite, with the new mix, the new total sales and the new fixed cost.
- Rounding: ICAI's Illustration 11 gives 23,145.80 composite units and rounds each product to whole units (16,202 and 6,944).

Trick

Batch it

Turn any units mix into one "batch" (5 namkeen + 3 mathri). Fixed cost ÷ contribution per batch = break-even batches; multiply by each product's units in the batch. No fractions until the end. Condition: the mix stays constant at every volume.

Answer frame

"Advise whether the company should change over to the new mix"

1. Statement of contribution to sales ratio at the existing mix (P/V ratio × sales mix for each product, total).
2. Existing total contribution, profit and break-even sales.
3. The same statement and figures for the proposed mix, with its own sales and fixed costs.
4. Comparison: composite P/V ratio, break-even sales, profit (and margin of safety).
5. Conclusion in words, with the reason (higher composite P/V ratio and profit), and the assumption that the mix will be achieved.

Three answers to "Find the break-even point of each product"

Loses marks

Namkeen: ₹10,20,000 ÷ ₹24 = 42,500 packs; mathri: ₹10,20,000 ÷ ₹28 = 36,429 packs.

Each product is made to carry all the fixed cost, and the mix is ignored.

Passes

Composite contribution ₹25.50, composite BEP 40,000 packs.

Correct, but stops before splitting it between the products, which the question asked for.

Full marks

Working note on the mix (contribution of a batch), composite BEP by formula, split by the ratio in units and rupees, total checked against fixed cost ÷ composite P/V.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the P/V ratios that were simply averaged

(55% + 40% + 20%) ÷ 3 = 38.33% ignores that chai is 40% of the airport outlet's sales and snacks only 25%. It gives break-even sales of ₹42,78,261 instead of ₹40,00,000. **Fix:** weight each P/V ratio by its share of sales value, or divide total contribution by total sales.

⚠ **Common mistake**

The right weights on the wrong figures

A units ratio applied to P/V ratios, or a value mix applied to contributions per unit, mixes two different weightings. **Fix:** units weights go with contribution per unit; value weights go with P/V ratios. Convert one into the other with the prices if needed.

⚠ **Common mistake**

Splitting the answer in the wrong ratio

Splitting ₹27,00,000 of break-even sales 5 : 3 gives ₹16,87,500 of namkeen, but 5 : 3 is a ratio of packs. **Fix:** split composite units by the units ratio (25,000 and 15,000 packs), then multiply by the prices.

Practice

Check yourself

1 Products A and B sell in the ratio of 2 units of A to 3 units of B. A's contribution is ₹30 a unit and B's ₹15. Common fixed costs are ₹2,10,000. What is the composite break-even point in total units (A and B together)?

Answer: _____

2 Two products have P/V ratios of 60% and 30% and make up 30% and 70% of sales value. Fixed costs are ₹7,80,000. What are the composite break-even sales (₹)?

Answer: _____

3 Case scenario. Tapri's Indore kiosk sold chai worth ₹6,00,000 (P/V ratio 60%) and snacks worth ₹4,00,000 (P/V ratio 35%) last month. Fixed costs are ₹3,60,000. Ritu must report the break-even sales at this mix. Which figure is correct?

- **A** ₹7,20,000
- **B** ₹7,57,895
- **C** ₹6,00,000
- **D** ₹10,28,571

4 Total sales and fixed costs stay the same, but the mix shifts towards the product with the lowest P/V ratio. What happens?

- **A** Break-even sales fall and profit rises
- **B** Break-even sales rise and profit falls
- **C** Break-even sales and profit are unchanged, as total sales are unchanged
- **D** Only the margin of safety changes

5 X earns ₹40 a unit and Y ₹20; they sell one for one; fixed costs are ₹6,00,000. Kesar says X breaks even at 15,000 units and Y at 30,000. What is the correct break-even point?

- **A** 45,000 units: 15,000 X and 30,000 Y
- **B** 20,000 units: 10,000 X and 10,000 Y
- **C** 15,000 units of X only
- **D** 22,500 units: 11,250 X and 11,250 Y

Answers: the answer key at the end of this lesson.

Recap

- Several products sharing a fixed cost break even only for a stated sales mix (CVP assumption: the mix stays constant).

- Mix in units: composite contribution = Σ contribution per unit × units weight; composite BEP = common fixed cost ÷ composite contribution; split by the units ratio.
- Batch method: contribution of one batch of the mix; fixed cost ÷ that = batches.
- Mix in values: composite P/V = Σ P/V ratio × sales mix % (contribution per rupee of sales) = total contribution ÷ total sales; BES = fixed cost ÷ composite P/V; split by the sales mix %.
- A shift towards high-P/V products lowers break-even sales and raises profit; a replacement product means a new composite.
- Margin of safety = actual (or budgeted) sales − composite break-even sales; profit = MOS × composite P/V ratio.
- "Sales mix 5 : 3" with no unit stated: ICAI's answers treat it as units; state the assumption. (Break even on thalis, not on rotis.)

Answer key · Check yourself

1 **10000**

One batch (2 A + 3 B) contributes ₹60 + ₹45 = ₹105. Break-even batches = ₹2,10,000 ÷ ₹105 = 2,000, i.e. 4,000 A + 6,000 B = 10,000 units.

2 **2000000**

Composite P/V = 60% × 30% + 30% × 70% = 18% + 21% = 39%. BES = ₹7,80,000 ÷ 39% = ₹20,00,000.

3 **A ₹7,20,000**

Composite P/V = (₹3,60,000 + ₹1,40,000) ÷ ₹10,00,000 = 50%; BES = ₹3,60,000 ÷ 50% = ₹7,20,000. ₹7,57,895 uses the simple average 47.5%; ₹6,00,000 and ₹10,28,571 treat the kiosk as all chai or all snacks.

4 **B Break-even sales rise and profit falls**

The composite P/V ratio falls, so fixed cost ÷ composite P/V rises and each rupee of sales earns less contribution.

5 **B 20,000 units: 10,000 X and 10,000 Y**

Composite contribution = (₹40 + ₹20) ÷ 2 = ₹30 a unit; BEP = ₹6,00,000 ÷ ₹30 = 20,000 units, split 1 : 1. Kesar made each product carry all the fixed cost.

CHAPTER 15

Budgets and Budgetary Control

15.1 Budgets, budgetary control, the budget manual and key factor

What a budget is, how budgetary control compares actual with it, who runs the system, what the budget manual holds, and why the key factor is identified before the budgets are drawn up.

In plain words

Two crore cups, said on a Monday

At the Monday meeting Kesar announces next year's plan.

> **Kesar:** Next year we sell two crore cups, open fifteen cafés and launch a Holi thandai. Sab adjust ho jayega.
>
> **Ritu:** Per unit, please. How many cups in each city, each quarter? How much milk, how many staff, how many ovens?
>
> **Dev:** And who pays for fifteen cafés before the first cup is sold?
>
> **Kesar:** Details. You people do the details.

Ritu's "details" are the budget. A **budget** is a plan turned into numbers (cups, litres, hours, rupees) for a definite future period, written down and approved before that

period starts. Kesar's sentence is a wish; Ritu's spreadsheet, once the board approves it, is a commitment.

Then the year starts. Every month Ritu puts the actual figures next to the budget, finds where they differ and asks the person responsible what will be done about it: speed up, cut back, or, if the world has changed, revise the plan. That continuous comparison, with somebody answerable for each line, is **budgetary control**.

A system that many people feed needs a rulebook: who prepares which budget, on which form, by which date, and who approves it. That rulebook is the **budget manual**, and the people who run the system are the **budget committee** (the heads of the functions) acting through a **budget officer**.

One more thing comes before any of the numbers. Ritu checks the dairy contract: Tapri can get milk for only 1.6 crore cups. However loudly Kesar says "two crore", milk is the factor that limits everything else: sales, staff, cups, ovens. A factor that sets a limit to the total activity is the **key factor**, and the budget has to be built around it.

Think of it like

The family wedding

A wedding has a budget: guests, plates, decorations and rupees, fixed before the date. It has budgetary control: the aunt with a notebook who checks every bill against the plan and asks who ordered the extra band. It has a budget manual: the file that says who books the caterer, who approves payments and by when the guest list is closed. And it has a key factor: the hall seats 400. However many relatives Kesar wants to invite, the plates, the sweets and the return gifts are all set by 400.

Key idea

A budget is a plan in numbers for a definite future period, approved in advance; budgetary control is the continuous comparison of actual results with it, with responsibility fixed for the differences. The key factor sets the ceiling for the whole budget, so it is identified before the budgets are built.

How it works

Budget, forecast and budgeting

CIMA defines a **budget** as "A financial and/or quantitative statement, prepared and approved prior to a defined period of time of the policy to be pursued during that period for the purpose of attaining a given objective. It may include income, expenditure and employment of

capital" (SM 15.2). It is a blue-print of the projected plan of action expressed in quantitative terms for a specified period of time.

A **forecast** is an assessment of probable future events. A budget is built on forecasts, but it is a commitment or a target which the management seeks to attain: a forecast denotes some degree of flexibility while a budget denotes a definite target (SM 15.3). **Budgeting** is the process of designing, implementing and operating budgets.

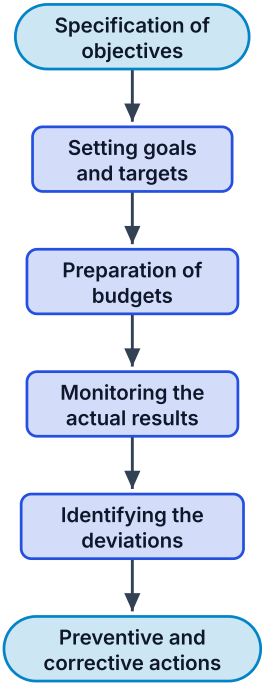
The **essential characteristics of a budget**, in ICAI's order (SM 15.3–15.4):

1. A budget is concerned for a definite future period.
2. A budget is a written document.
3. A budget is a detailed plan of all the economic activities of a business.
4. All the departments of a business unit should co-operate for the preparation of a business budget.
5. Budget is a means to achieve business objectives and it is not an end in itself.
6. Budget needs to be updated, corrected and controlled every time circumstances change; therefore, it is a continuous process.
7. Budget helps in planning, coordination and control.
8. Different types of budgets are prepared by industries according to business requirements.
9. A budget acts as a business barometer.

10. Budget is usually prepared in the light of past experiences.
11. Budget is a constant endeavour of the Management.

The budget cycle: from objectives to corrective action

The budget cycle (SM 15.5)



The first three steps plan. Monitoring the targets against actual results, narrowing the deviations and taking preventive and corrective action is what the Study Material calls budgetary control (SM 15.2).

The **objectives of budgeting** are three (SM 15.5–15.7): **planning** (targets in line with the organisation's objectives; planning precedes the budget), **direction and co-ordination** (of the budgetary units, the **responsibility centres**, each led by a manager answerable for its performance) and **controlling** (actual compared with targets, the feedback used to fine-tune). The main objective is to help achieve the overall objective of the organisation.

The **essential steps for preparing a budget** (SM 15.4):

1. Organisational structure must be clearly defined and responsibility assigned to identifiable units within the organisation.
2. Setting of clear objectives and reasonable targets, in consonance with the long-term plan of the organisation.
3. Objectives and responsibility should be clearly stated and communicated to the management or person responsible.
4. Budgets are prepared for the future periods based on the expected course of actions.
5. Budgets are updated for events not kept in mind while establishing them; hence budgets should be flexible enough for mid-term revision.
6. The entire organisation must be committed to the preparation and implementing of budgeting.
7. Budgets should be quantifiable, and the master budget should be broken down into various functional

budgets.

8. Budgets should be monitored periodically; variances of the actual outcomes analysed and responsibility fixed.
9. Budgetary performance needs to be linked effectively to the reward system.

Budgetary control: the comparison that never stops

Definition

Budgetary control

CIMA: "the establishment of budgets relating to the responsibilities of executives to the requirements of a policy and the continuous comparison of actual with budgeted results, either to secure by individual action, the objective of that policy or to provide a basis for its revision." It is the system of management control and accounting in which all the operations are forecasted and planned in advance to the extent possible and the actual results compared with the forecasted and planned results. (SM 15.7)

Budgetary control **involves** (SM 15.7): (1) establishment of budgets; (2) continuous comparison of actuals with budgets for achievement of targets; (3) revision of budgets after considering the changes in the circumstances; (4) fixation of the responsibility for failure to achieve the budget targets.

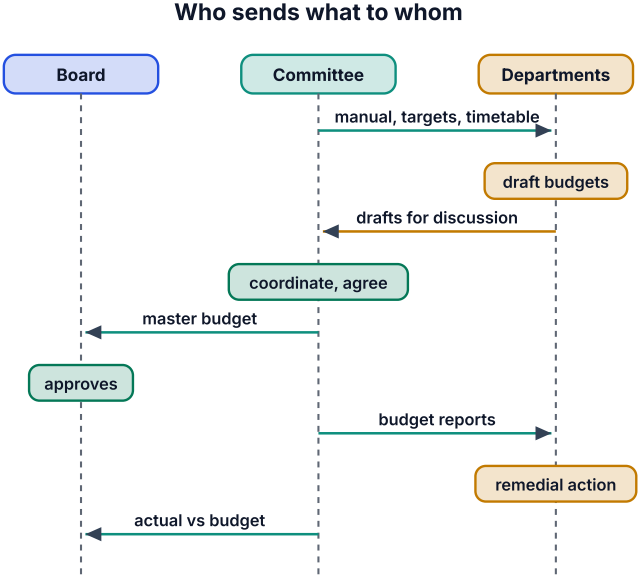
For it to work, the firm needs a proper basis of measurement, that is, a system of **standard costing** (standards and standard costing), and an organisation in which every line of authority and responsibility is defined, "from the Managing Director down to the Shop Foreman" (SM 15.9). Its objectives, the steps for establishing it, its advantages and its limitations are asked as 4- and 5-mark lists (Exam angle): the objectives, advantages and limitations are set out point by point in the Answer frame, the five steps in the second worked example.

Whether the comparison happens after the period (feedback control, ex-post corrective) or continuously during it (feedforward control, ex-ante preventive), and how budgets motivate the people who must meet them, are in control ratios, motivation, feedback and feed-forward.

Who runs it: the budget committee and the budget officer

The **budget committee** is a group of representatives of the various functions. Because a change in one function's target changes the others', the targets are discussed until a mutually agreed programme is decided: this is **coordination in budget-making** (SM 15.10). The **Chief Executive** is ultimately responsible for the budget programme, but a large part of the supervisory responsibility is better delegated to a **Budget Officer**, who should know the technical side of the business and report to the president or CEO. The responsibility for

introducing and implementing the system rests with the Budget Committee acting through the Budget Officer.



Committee: the budget committee acting through the budget officer; Departments: the functional heads. The arrows follow the Budget Officer's duties (SM 15.11, 15.16). Once the board approves the master budget, it is the standard every department works to (SM 15.55).

The **responsibilities of the Budget Committee and Budget Officer** (SM 15.11):

1. Assist in the preparation of the separate budget for various departments, coordinating the work of the accounts department (which normally compiles the budgets) with the functional departments like Sales, Production and Plant maintenance.

2. Forward the budget to the individual department heads who are responsible to implement it, and guide them in overcoming practical difficulties in its working.
3. Prepare the periodical budget reports for circulation to the individuals concerned.
4. Follow-up action to be taken on the budget reports.
5. Prepare an overall budget working report for discussion at the Budget Committee meetings, and ensure follow-up on the lines of action suggested by the Committee.
6. Prepare periodical reports for the Board meeting, comparing the budgeted Profit and Loss Account and Balance Sheet with the actual results attained.

The efficiency of the Budget Officer, and through him of the committee, is judged by the smooth working of the system and the agreement between the actual and the budgeted figures.

Note

Two compositions, both from the Study Material

One passage has all functional heads with "a member from the Board to preside over and guide the deliberations" (SM 15.10); another has "all the heads of departments along with the Managing Director as the Chairman" (SM 15.16). Use the one the question's context fits. In the chapter's own MCQ, the budget organisation is headed by the **budget director or budget controller** (SM 15.72).

Six steps to prepare the budget, key factor second

ICAI's section on the preparation of budgets (SM 15.16–15.18):

1. **Defining business or organisational objectives.** Written down, with the areas of control demarcated and the items of revenue and expenditure to be covered stated clearly.

2 **Identification of the key budget factor.** There are usually one or two key budget factors (sometimes more than two) which set a limit to the total activity. In India, non-availability of power sometimes does not allow production to increase in spite of heavy demand; similarly, lack of demand may limit production. Such a factor is known as the **key factor**; it must be identified and its influence on production or sales estimated properly while preparing the budget.

3 **Appointment of controller or officer.** Formulating a budget usually needs a whole-time senior executive, assisted by the Budget Committee. The Budget Controller or Officer coordinates and develops the budget programmes and prepares the manual of instructions, the **budget manual**.

4 **Budget manual.** A booklet specifying the objectives of the organisation in relation to its strategy; its typical contents are the model answer to the third worked example below.

5 **Budget period.** There is no general rule: the Budget Committee decides the length. Normally a calendar year or a period co-terminus with the financial year, divided into months, quarters or periods that coincide with the trading activity of the business.

6 **Standard of activity or output.** Past statistics cannot be completely relied upon, because the past mixes good and bad factors; apply them only when similar conditions are likely to repeat. In a progressive business a year's achievement should normally exceed the earlier years': what was good in the past is only fair for the current year.

Where the quantity that can be sold is the limit, sales are the **principal budget factor**, the case "in many business undertakings" (SM 15.30). Which product to favour when a factor is short is a marginal costing decision: limiting factor and product mix.

Definition

Budget manual

CIMA: "A document which sets out the responsibilities of the persons engaged in, the routine of, and the forms and records required for, budgetary control." A budget manual is a collection of documents that contains key information for those involved in the planning process. (SM 15.16–15.17)

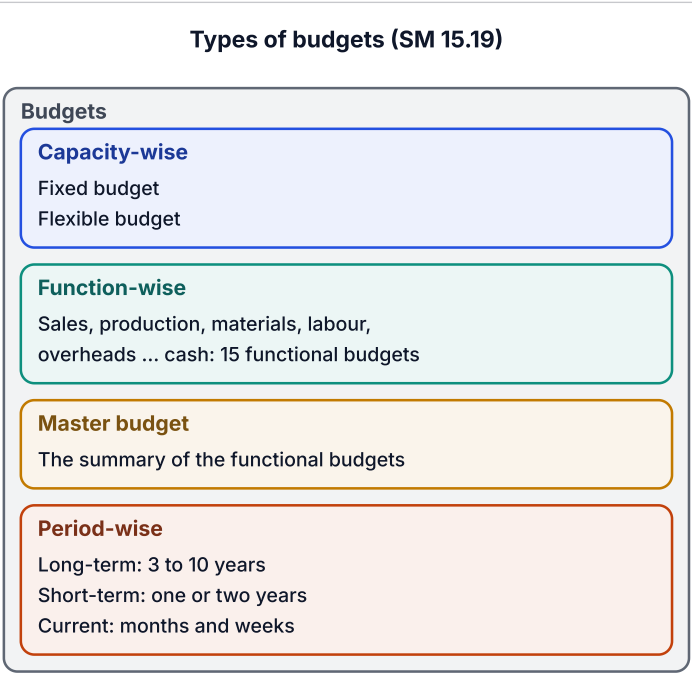
"Define Budget Manual. What are the salient features of Budget Manual?" was a 5-mark question in November 2022 (old scheme); the fourteen contents (SM 15.17–15.18) are in the third worked example.

The four families of budgets inside the master budget

The policy of a business for a defined period is represented by the **master budget**, whose detailed components are the **functional budgets**. These are broadly grouped under four heads, the **components of a budgetary control system** (SM 15.14):

Group	What it contains	Examples
Physical budgets	Information in quantitative terms	Quantity of sales, quantity of production, inventories, manpower
Cost budgets	Cost information	Manufacturing, selling, administration, research and development cost
Profit budgets	The ascertainment of profit	Sales budget, profit and loss budget
Financial budgets	The financial position	Cash budget, capital expenditure budget, budgeted balance sheet

Every kind of budget on one page



The four ways ICAI classifies budgets. Fixed and flexible budgets, the functional budgets and the master budget each have their own lesson.

The capacity-wise budgets are in flexible budgets, the functional budgets and the master budget in functional budgets and the master budget, the cash budget in the cash budget. The period-wise classes (SM 15.58–15.59):

- **Long-term budget**: covers more than a year, usually three to ten years; useful where the gestation period is long, as for makers of machinery or electricity.

- **Short-term budget**: generally one or two years, in monetary terms; used by consumer goods industries like sugar, cotton and textile.
- **Current budget**: months and weeks, relating to current activities. CIMA: a budget "established for use over a short period of time and is related to current conditions".

Formats and formulas

Reproduce these definitions from memory; ICAI asks for them in 1-mark slices of a 5-mark question.

Definition

Budget centre

A section of an organisation developed for the purpose of budgetary control, and intended to facilitate formulation of various budgets with the help of the head of the department. (SM 15.70)

Definition

Budget period

The period covered by a budget. There is no general rule governing its selection; the Budget Committee determines the length suitable for the business, normally a calendar year or a period co-terminus with the financial year, divided into months, quarters or periods coinciding with the trading activity of the business. (SM 15.18)

Definition

Key factor

A factor which sets a limit to the total activity: usually one or two (sometimes more than two), such as non-availability of power or lack of demand. It must be identified and its influence on production or sales estimated properly while preparing the budget. (SM 15.16)

Definition

Master budget

CIMA: "the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed." (SM 15.55)

∑ **Formula**

Budgeted activity under a key factor

$$\text{Limit} = \frac{\text{Quantity available}}{\text{Quantity per unit}}$$

$$\text{Budgeted output} = \text{the lowest of the limits}$$

Work out the limit of demand and of each resource (in units of output); the factor giving the lowest limit is the key factor. Conditions: one product, or a mix that stays fixed. With several products competing for one scarce resource, rank them by contribution per unit of that resource instead (limiting factor and product mix).

Worked examples

Worked example **Power cuts at the Dewas snack plant**

Tapri is budgeting its new frozen-snack plant at Dewas

for 2026-27. Ritu has collected the following:

Factor	Data for 2026-27
Demand (sales team's estimate)	9,60,000 packs
Fryers	2 fryers × 20 hours a day × 300 days, at 75 packs an hour per fryer
Skilled cooks	45 cooks × 2,400 hours a year; 6 minutes of cook time per pack
Potato filling (supply contract)	2,09,000 kg; 0.22 kg per pack
Power (sanctioned by the state discom)	12,60,000 kWh; 1.5 kWh per pack

Contribution is ₹18 a pack. Kesar wants the sales budget set at the sales team's 9,60,000 packs.

Required:

(i) Identify the key factor and the maximum output the budget can plan for.

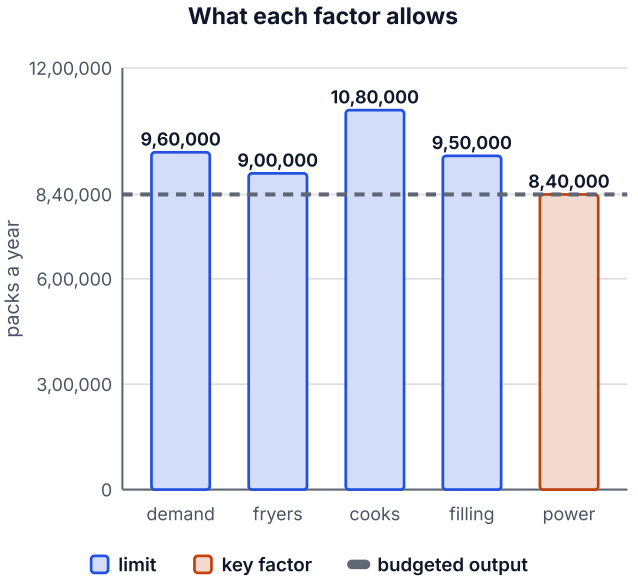
(ii) State the contribution the sales budget would overstate if it were set at 9,60,000 packs.

(iii) A rented diesel generator would add 1,80,000 kWh. Identify the key factor after renting it, and suggest how the remaining gap to demand could be met.

THE WORKING

(i) Limit set by each factor

Factor	Working	Limit (packs)
Demand	Given	9,60,000
Fryers	2 × 20 × 300 × 75	9,00,000
Skilled cooks	1,08,000 hours × 10 packs an hour	10,80,000
Potato filling	2,09,000 kg ÷ 0.22 kg	9,50,000
Power	**12,60,000 kWh ÷ 1.5 kWh**	**8,40,000**



Blue bars: limits that do not bind; orange bar: power, the lowest limit and so the key factor. The grey dashed line is the most the budget can plan for: every other budget is built to it.

Power sets the lowest limit: it is the **key factor**, and the budget can plan for at most **8,40,000 packs**. The sales, production, purchase and labour budgets are all built to this figure.

(ii) Setting sales at 9,60,000 packs plans 1,20,000 packs the plant cannot make: contribution overstated by 1,20,000 × ₹18 = **₹21,60,000**. Advertising should also be planned to the 8,40,000 packs: money should be spent on advertisement only when sufficient quantities of the product are ready for sale (SM 15.41).

(iii) With the generator: (12,60,000 + 1,80,000) ÷ 1.5 = 9,60,000 packs, equal to demand. The lowest limit is now the fryers at **9,00,000 packs**: removing one key factor reveals the next. The gap of 60,000 packs to demand needs two things:

1. **Fryers** (up to 9,50,000 packs): 50,000 ÷ 75 = about 667 more fryer-hours, from the Study Material's remedies for insufficient production facilities: working overtime, introducing shift working, sub-contracting or buying additional plant (SM 15.32).
2. **Potato filling** (beyond 9,50,000 packs): the contract covers only 9,50,000 packs, so filling becomes the next key factor. The last 10,000 packs need 10,000 × 0.22 = 2,200 kg more filling, from a second supplier, together with another 10,000 ÷ 75 = about 133 fryer-hours (800 in all).

Worked example

The chilli that nobody budgeted at Masala Mills

Masala Mills Ltd, a spice supplier to Tapri, works without budgets. Last year its purchase manager bought a full year's red chilli at the season's peak price, while its sales team, chasing volume, ran a 20% discount scheme. Each department thought it was doing well; nobody knew the combined result until the

annual accounts showed a loss. Now the bank wants to see the company's budget plans before it renews the cash credit limit. As a consultant:

(a) Identify and explain four objectives of a budgetary control system that would have helped Masala Mills, relating each to the facts.

(b) State the steps you would follow to establish the system, and what else the company needs for it to work.

MODEL ANSWER

(a) Objectives of a budgetary control system (four of the nine, chosen for these facts):

1. **Laying down the responsibilities** of each executive and other personnel, so that everyone knows what is expected of him and how he will be judged. The purchase manager and the sales head would each have had targets for which they were answerable.
2. **Providing a basis for comparison** of actual performance with the predetermined targets and investigation of deviations of actual performance and expenses from the budgeted figures, which helps in adopting corrective measures. The excess chilli purchases and the discounts would have shown up as deviations in the monthly budget reports, not in the annual accounts.
3. **Ensuring optimum use of available resources** to maximise profit or production, subject to the

limiting factors. Funds locked in a year's chilli stock would have been planned against the purchase and cash budgets.

4. **Co-ordinating various activities of the business**, centralising control and yet enabling management to decentralise responsibility and delegate authority in the overall interest of the business. Purchase and sales worked at cross purposes; budgets prepared together would have had to agree.

(b) Steps for establishing budgetary control:

1. Determining the objectives to be achieved over the budget period, and the policies that might be adopted to achieve them.
2. Determining the activities that should be undertaken to achieve the objectives.
3. Drawing up a plan or scheme of operation for each class of activity, in quantitative as well as monetary terms, for the budget period.
4. Laying out a system of comparison of actual performance by each person or department with the relevant budget, and determining the causes of variations.
5. Ensuring that corrective action is taken where the plan has not been achieved and, if that is not possible, revising the plan.

For the system to work, Masala Mills also needs a proper basis of measurement or standards with which to evaluate the efficiency of operations (a

system of standard costing), and an organisation in which all lines of authority and responsibility are properly defined. The bank's request is also met: management which has developed well-ordered budget plans, and operates accordingly, receives greater favour from credit agencies (SM 15.12).

Worked example **Kesar's one-page budget manual**

Kesar's draft "budget manual" for Tapri is one page long. It reads, in full:

1. The names of the fifteen functional budgets.
2. "Every department head prepares his own budget."
3. The budgeted profit and loss account for 2026-27, showing a profit of ₹42 crore.
4. "Budgets once approved are final. No revisions."
5. "Rest, sab adjust ho jayega."

Ritu has to tell the budget committee whether this is a budget manual.

Required: Examine each item of the draft against the meaning and contents of a budget manual, and state the contents Ritu should add.

MODEL ANSWER

Meaning. The budget manual is a booklet specifying the objectives of an organisation in

relation to its strategy. CIMA defines it as "a document which sets out the responsibilities of the persons engaged in, the routine of, and the forms and records required for, budgetary control." It is a collection of documents containing key information for those involved in the planning process.

The draft, item by item

Item	Examination
1. Names of the budgets	Not enough on its own: the manual shows, in a form of organisation chart, who is responsible for preparing each functional budget and how the budgets are interrelated
2. "Every head prepares his own"	Too vague: the manual states the functions and responsibilities of each executive for both preparation and execution, and the procedure and authority for approval
3. Budgeted profit and loss account	Does not belong: it is part of the master budget, an output of budgeting; the manual may state the key assumptions behind it, such as the rate of inflation
4. "No revisions"	Contrary to the manual: it sets out the manner in which budgets, after acceptance and issue, may be revised or amended, with the approval of top management

Item	Examination
5. "Rest, sab adjust ho jayega"	Not a content: the manual lays down the reports, forms and records, the accounts classification and the reporting of remedial action, so that nothing is adjusted informally

The draft is therefore not a budget manual.

Contents Ritu should add. A typical budget manual may include:

1. A statement regarding the objectives of the organisation and how they can be achieved through budgetary control.
2. A statement about the functions and responsibilities of each executive, both regarding preparation and execution of budgets.
3. Procedures to be followed for obtaining the necessary approval of budgets, the authority for granting approval stated in explicit terms, including whether one, two or more signatures are required on each document.
4. A form of organisation chart to show who is responsible for the preparation of each functional budget and the way in which the budgets are interrelated.
5. A timetable for the preparation of each budget.
6. The manner of scrutiny and the personnel to carry it out.

7. Reports, statements, forms and other records to be maintained.
8. The accounts classification to be employed, identical in the accounts and the budget department.
9. The reporting of the remedial action.
10. The manner in which budgets, after acceptance and issue, are to be revised or amended, on which action can be taken only with the approval of top management.
11. Preventing a "bottleneck", the late preparation of one budget holding up the preparation of all others.
12. Copies of all forms to be completed by those responsible for preparing budgets, with explanations concerning their completion.
13. A list of the organisation's account codes, with full explanations of how to use them.
14. Information concerning the key assumptions to be made by managers in their budgets, for example the rate of inflation and key exchange rates.

Exam angle

In the exam

Where these topics appeared, May 2024 to September 2026

In the new-scheme ICAI documents read for this lesson, the theory of budgets and budgetary control was asked 10 times in 10 documents, each time as a descriptive part (4 or 5 marks where marks are printed).

- **Limitations of budgetary control** (4 marks in the MTPs): MTP January 2025 Series I Q6(c); MTP May 2025 Series I Q6(c) ("any four"); RTP January 2026 Q19(c) ("any four").
- **Objectives**, as a case ("ABC Ltd has frequent cost overruns ... identify and explain four objectives"): MTP May 2025 Series II Q6(c) and MTP September 2026 Series II Q6(c), 4 marks each (each the "OR" choice of its part); the answers give all nine objectives, and the September 2026 one is marked "Any Four Points".
- **Steps**: "sequential steps to design, implement and monitor" a budgetary control system, MTP January 2025 Series II Q6(b) (5 marks), answered with the five steps for establishing budgetary control; "steps involved in the budgetary control technique", RTP September 2026 Q16(c), answered with the six steps for preparing budgets (objectives, key factor, controller, manual, period, standard of activity).
- **Definitions**: budget manual and budget period, two of five terms in September 2024 Q6(b) (5 marks).
- **Components**: "Discuss the four components of Budgetary Control System", January 2026 Q6(c) (4 marks).

- **Feedback and feedforward control**, May 2024 Q5(c) (4 marks): see feedback and feed-forward.
- Old scheme: budget manual and its salient features, November 2022 Q6(d) (5 marks); components, December 2021.

In the exam

The wording traps

- "Steps involved in the budgetary control technique" (RTP September 2026) and "steps for establishing budgetary control" (MTP January 2025) were answered from different lists. The Study Material's own answer key sends "Describe the steps involved in the budgetary control technique" to the five steps of section 5.3 (SM 15.84). Write the five steps, then add the key factor, budget manual and budget period from the preparation sequence: both answers are covered.
- "Components of budgetary control system" means the four groups (physical, cost, profit, financial), not the budget committee or the manual.
- "Any four" means four points with their one-line explanations, not nine one-word headings.

Trick

Limitations: "ET Can Eat No Rice"

Estimates, **T**ime factor, **C**o-operation required, **E**xpensive, **N**ot a substitute for management, **R**igid document: ICAI's six, in ICAI's order (SM 15.13–15.14).

Answer frame

"Discuss the objectives of a budgetary control system"

The nine points ICAI's answer gives, in order (SM 15.7–15.8); a 4-mark question takes any four with their explanation:

1. Portraying with precision the overall aims of the business and determining targets of performance for each section or department.
2. Laying down the responsibilities of each executive, so that everyone knows what is expected and how he will be judged.
3. Providing a basis for comparing actual performance with the predetermined targets and investigating deviations, which helps in adopting corrective measures.
4. Ensuring optimum use of available resources to maximise profit or production, subject to the limiting factors.
5. Co-ordinating the activities of the business, centralising control and yet enabling management to

decentralise responsibility and delegate authority.

6. Engendering a spirit of careful forethought, assessment of what is possible and an attempt at it: dynamism without being reckless.
7. Providing a basis for revision of current and future policies.
8. Drawing up long range plans with a fair measure of accuracy.
9. Providing a yardstick against which actual results can be compared.

"Advantages" and "limitations"

Advantages (SM 15.12)	One-line explanation
1. Efficiency	Business activities are conducted in an efficient manner
2. Control on expenditure	A yardstick for measuring the performance of individuals and departments
3. Finding deviations	Actual is compared with budget and deviations are communicated to management
4. Effective utilisation of resources	Men, material, machinery and money are planned for
5. Revision of plans	Helps review current trends and frame future policies

Advantages (SM 15.12)	One-line explanation
6. Implementation of standard costing	Creates suitable conditions for a standard costing system
7. Cost consciousness	Encourages cost consciousness and maximum utilisation of resources
8. Credit rating	Well-ordered budget plans receive greater favour from credit agencies

Limitations (SM 15.13–15.14)	One-line explanation
1. Based on estimates	Made on conditions at the time; revision is needed if they change
2. Time factor	Not executed automatically; management must not expect too much in the initial period
3. Co-operation required	Staff co-operation is usually not available initially; success depends on willing co-operation and teamwork
4. Expensive	Needs a proper organisation structure and consumes qualified manpower and equipment
5. Not a substitute for management	Only a managerial tool, to be intelligently applied
6. Rigid document	Should be flexible enough to take in changes in internal and external factors

"Distinguish between a forecast and a budget"

Point	Forecast	Budget
Nature	An assessment of probable future events	A financial or quantitative plan to be pursued over a period
Commitment	Denotes some degree of flexibility	A commitment, a definite target management seeks to attain
Order	Made first, on sales, production cost and finance	Prepared on the basis of the forecasts
Sales	Projects the available customer demand: reflects the environment	Shows how management intends to react to it (SM 15.31)
Common ground	Both relate to a defined period of time	Both relate to a defined period of time

Three answers to "What is a key factor?"

Loses marks

"The most important factor in the budget."

Vague: no limit, no example, no consequence.

Passes

"A factor which sets a limit to the total activity, such as lack of demand or shortage of power."

The definition with an example, but not what the budget does with it.

Full marks

The definition; "usually one or two, sometimes more"; examples (power, demand); it must be identified and its influence on production or sales estimated while preparing the budget; where sales are the limit they are the principal budget factor.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the 9,60,000-pack sales budget

Building the sales budget on demand when power allows 8,40,000 packs. Every budget downstream (production, materials, labour, cash) then plans for packs that cannot be made. **Fix:** find the limit set by each factor first; the lowest is the key factor and caps every budget.

⚠ **Common mistake**

Calling the forecast a budget

"Sales budget = what we expect to sell." **Fix:** a forecast is an assessment of probable events and stays flexible; a budget is the target management commits to and acts to achieve, built on the forecast.

⚠ **Common mistake**

The budget manual that is only a list of budgets

Fix: a budget manual sets out responsibilities, routines, forms and records for budgetary control (CIMA): approvals, timetable, organisation chart, account codes, revision procedure, key assumptions. The budgets themselves are not the manual.

⚠ **Common mistake**

Mixing up the three lists of steps

"Essential steps for preparing a budget" (nine, SM 15.4), "steps for establishing budgetary control" (five, SM 15.8–15.9) and "preparation of budgets" (six, SM 15.16–15.18) are different lists. **Fix:** read the question's verb and noun, then write the matching list; for "budgetary control technique", the five steps plus key factor, manual and period.

Practice

Check yourself

1 Case scenario. Tapri's cold-chain division is budgeting next year's ice-cream deliveries. Demand is estimated at 4,50,000 tubs. The freezer plant can make 4,20,000 tubs; refrigerated trucks can deliver 3,60,000 tubs; milk contracts cover 4,80,000 tubs. Which of the following is correct?

- **A** The key factor is demand; the sales budget is 4,50,000 tubs
- **B** The key factor is the freezer plant; the sales budget is 4,20,000 tubs
- **C** The key factor is delivery capacity; the sales budget is 3,60,000 tubs
- **D** There is no key factor, because milk covers demand

2 A plant has 6,000 machine hours and each unit needs 1.5 machine hours. Raw material of 18,000 kg is available at 4 kg per unit, and demand is 4,200 units. What is the maximum output the budget can plan for, in units?

Answer: _____

3 Ritu is drafting the contents page of Tapri's budget manual. Which of these belong in it? Select all that apply.

- **A** A timetable for the preparation of each budget
- **B** The manner in which a budget, once accepted and issued, may be revised
- **C** The budgeted profit and loss account and balance sheet approved by the board
- **D** The key assumptions managers must use, such as the rate of inflation
- **E** The standard cost card of each product

More than one answer may be right.

4 Which of these are limitations of a budgetary control system as listed by ICAI? Select all that apply.

- **A** Budgets are based on estimates
- **B** It is expensive to implement
- **C** It does not encourage cost consciousness
- **D** It is not a substitute for management
- **E** It harms the company's credit rating

More than one answer may be right.

5 Bunty, visiting Tapri, says: "A sales forecast and a sales budget are the same thing; both estimate next year's sales." Which reply is correct?

- **A** He is right; the two words are interchangeable
- **B** A forecast estimates the available demand; the budget is the target management commits to and acts to achieve
- **C** A forecast is prepared after the budget, to check it
- **D** A budget never uses a forecast

Answers: the answer key at the end of this lesson.

Recap

- Budget (CIMA): a financial and/or quantitative statement, prepared and approved before a defined period, of the policy to be pursued to attain a given objective.
- Forecast = assessment of probable events (flexible); budget = commitment, a definite target.
- Budgetary control: establish budgets, compare actual with budget continuously, revise when circumstances change, fix responsibility.

- Budget committee (functional heads) acts through the budget officer; the Chief Executive is ultimately responsible; the board approves the master budget.
- Preparation: objectives, key factor, controller, budget manual, budget period, standard of activity.
- Key factor: limits the total activity (power, demand); identify it first; the lowest limit binds and can move once removed.
- Budget manual: responsibilities, routine, forms and records for budgetary control: 14 typical contents.
- Components: physical, cost, profit, financial budgets; period-wise: long-term (3 to 10 years), short-term (one or two years), current (months, weeks). (The hall seats 400.)

Answer key · Check yourself

1 **C The key factor is delivery capacity; the sales budget is 3,60,000 tubs**

The lowest limit binds: trucks allow only 3,60,000 tubs, so delivery capacity is the key factor and every budget is built to 3,60,000. Demand and the plant are not the binding limits here.

2 **4000**

Machines allow 6,000 ÷ 1.5 = 4,000 units; material allows 18,000 ÷ 4 = 4,500 units; demand is 4,200. Machine hours are the key factor: 4,000 units.

3 **A A timetable for the preparation of each budget; B The manner in which a budget, once accepted and issued, may be revised; D The key assumptions managers must use, such as the rate of inflation**

The manual sets out the responsibilities, the routine, and the forms and records required for budgetary control (CIMA): the timetable, the revision procedure and the key assumptions are among its typical contents. The approved budgeted profit and loss account and balance sheet are the master budget itself, and standard cost cards belong to the standard costing system.

4 **A Budgets are based on estimates; B It is expensive to implement; D It is not a substitute for management**

ICAI's six limitations: based on estimates, time factor, co-operation required, expensive, not a substitute for management, rigid document. Cost consciousness and credit rating are among its advantages.

5 **B A forecast estimates the available demand; the budget is the target management commits to and acts to achieve**

The budget is prepared on the basis of forecasts but is a commitment: a forecast denotes flexibility, a budget a definite target. The sales budget is an active document, not merely a sales forecast.

15.2 Functional budgets and the master budget

How the sales budget drives the production, material, labour and overhead budgets, how each is laid out in ICAI's format, and how they roll up into the master budget.

In plain words

The budget that starts in the sales office

Budget week at Tapri. Ritu refuses to touch the factory's numbers until the sales team hands in theirs.

> **Kesar:** The factory is sitting idle while the sales people argue. Start with the samosas!
>
> **Ritu:** I cannot plan to fry samosas nobody will buy, or buy potatoes for samosas I will not fry.
>
> **Dev:** And I cannot arrange cash for potatoes nobody will fry.
>
> **Kesar:** So the whole company waits for one spreadsheet?
>
> **Ritu:** Per unit, please: yes. Sales first.

That is how a budget is built. The **sales budget** comes first: how many packs, at what price, in which channel. The **production budget** follows: sales, plus the stock Tapri wants left at the end, minus the stock it already has. Production decides how much material is used (the **material usage budget**), and usage, adjusted for the

material stocks wanted at the start and end, decides what must be bought (the **purchase budget**). The same production decides the hours of labour, the machine hours and the overheads. Costed and added up, they give the cost of production and the cost of goods sold.

Each of these is a **functional budget**: a budget for one function of the business (selling, producing, buying, administering). When all of them are ready, the budget officer summarises them into a budgeted profit and loss account and a budgeted balance sheet: the **master budget**, which the board approves.

Think of it like

The halwai's order book

Before Diwali a sweet shop works backwards. Orders for 500 kg of barfi, plus 50 kg to keep on the shelf, minus 30 kg left from yesterday: make 520 kg. Each kilo needs 600 g of khoya, so 312 kg of khoya will be used; keep 20 kg in the fridge for tomorrow, and 12 kg is already there, so buy 320 kg. The halwais needed, the gas cylinders burnt, the boxes ordered all follow from the 520 kg. The owner's final sheet of rupees in and rupees out is the master budget.

Key idea

Every functional budget is built from the one before it: sales, then production (sales + closing stock − opening stock), then materials (usage + closing stock − opening stock = purchases), labour and overheads. The master budget is their summary, approved and adopted as the standard to work to.

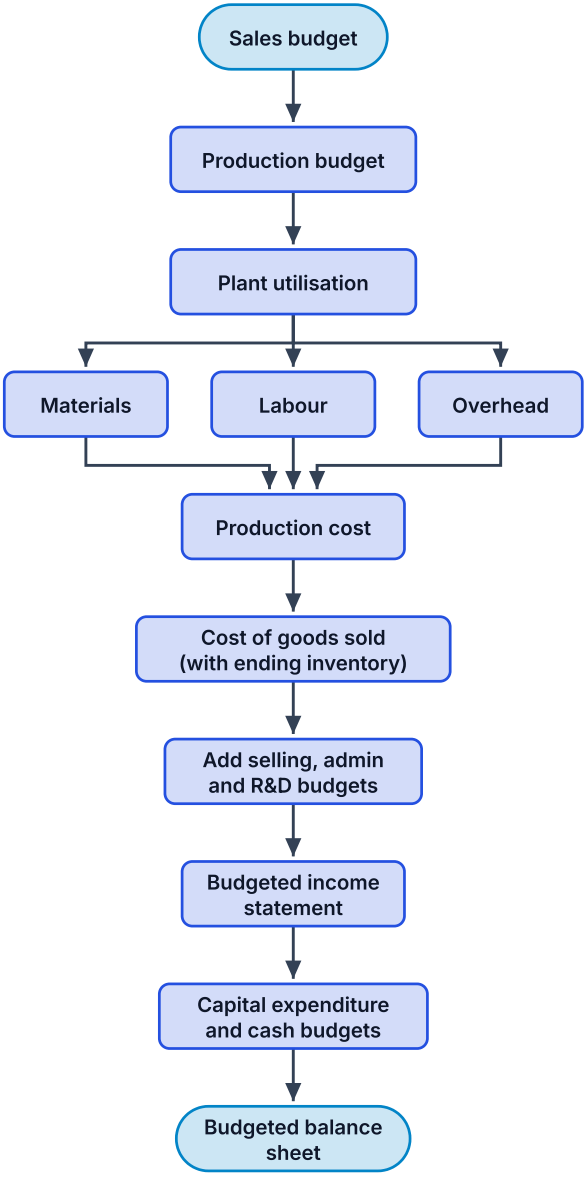
How it works

Fifteen functional budgets, one chain

A **functional budget** is one related to a function of the business, for example the production budget for the manufacturing function. Functional budgets are prepared for each function and are subsidiary to the master budget; which ones are prepared varies with the size and nature of the business (SM 15.29). ICAI's list:

(i) Sales; (ii) Production; (iii) Plant utilisation; (iv) Direct-material usage; (v) Direct-material purchase; (vi) Direct labour (personnel); (vii) Production or factory overhead; (viii) Production cost; (ix) Ending inventory; (x) Cost of goods sold; (xi) Selling and distribution cost; (xii) Administration expenses; (xiii) Research and development cost; (xiv) Capital expenditure; (xv) Cash budget.

How the functional budgets feed each other



How the Study Material's functional budgets feed one another, in the order it presents them (SM 15.29–15.56). Materials: the usage and purchase budgets; labour: direct labour; overhead: factory overhead. The budgeted income statement and balance sheet are the master budget.

Sales budget: the first domino

Sales forecast is the commencement of budgeting, so the sales budget assumes primary importance: the quantity which can be sold may be the **principal budget factor** in many business undertakings (SM 15.30). It is prepared for each product, with (1) the quantity of estimated sales and (2) the expected unit selling price, often by region or sales representative. Past sales volumes are the starting point, adjusted for (SM 15.30):

(i) backlog of unfulfilled sales orders; (ii) planned advertising and promotion; (iii) expected industry and general economic conditions; (iv) productive capacity; (v) projected pricing; (vi) findings of market research studies; (vii) relative product profitability; (viii) competition.

Kesar's favourite is (ii), planned advertising. Ritu's is (iv), productive capacity, which is why she keeps being left out of the ad shoots: a campaign that sells more than the fryers can make only sells disappointment.

The sales budget is not merely a sales forecast: a forecast projects the available customer demand, while the budget shows how management intends to react to it. It is an active rather than passive document (SM 15.31). It

may be classified by (1) products or groups of products; (2) areas, towns, salesmen and agents; (3) types of customers, for example Government, export, home sales and retail depots; (4) period: months, weeks (SM 15.31).

Production budget: sales, adjusted for stock

The production budget is prepared in two parts: a **production volume budget** in physical units and a **cost of production (manufacturing) budget** under material, labour and factory overhead (SM 15.32). The volume is based on (1) the sales budget, (2) the production capacity of the factory, (3) the planned increase or decrease in finished stocks and (4) the policy governing outside purchase.

When the sales budget asks for more than the facilities can make, ICAI's remedies are overtime, shift working, sub-contracting or buying more plant; when there is surplus capacity, more advertising, lower prices to increase sales, or sub-contracting the surplus (SM 15.32–15.33).

The **level of stocks** depends on three factors (SM 15.33): (1) seasonal industries, where stocks are built up in the off season for the peak; (2) a steady and uniform level of production, to use the plant fully and avoid retrenchment or lay-off of workers; (3) producing so that minimum

stocks are held, to avoid locking up funds. So the budget can show:

Policy	Capacity	Inventory carrying cost
Stabilised production (the same, often the maximum, every month)	Fully utilised	Varies with the stocks held
Stabilised stocks (minimum stock every month)	May be under-utilised	Lowest

Kesar prefers stabilised production (the fryers never rest); Dev prefers stabilised stocks (no cash asleep in the cold store). The production budget is where the two are traded off.

The **plant utilisation budget** states, in working hours, weight or other convenient units, the plant facilities needed to carry out the production budget. Its purposes (SM 15.34): (1) to determine the load on each process, cost centre or group of machines; (2) to show the overloaded ones, so that corrective action (overtime, sub-contracting, expansion) can be taken; (3) to dovetail the sales and production budgets where an overloaded process cannot be expanded; (4) where capacity is surplus, to boost sales to use it.

Materials: what is used, then what is bought

The **direct material usage budget** is compiled in three steps (SM 15.34–15.35): (1) specify the quality standards for each item of material; (2) set the standard quantity of each item, with an allowance for normal loss in process from past performance, test runs or technical estimates; (3) set standard prices, considering stock and contracts entered into. Usage cost = quantity required × standard price.

The **direct material purchase budget** starts from the production budget: quantities to be bought × expected unit purchase price. Two considerations govern it: (i) the **economic order quantity** and (ii) the **re-order point with safety stocks** to cover fluctuations in demand (stock levels, EOQ). Its timing is coordinated between the purchasing and production departments (SM 15.35). A purchases budget can be wider than the materials budget: it may include purchases of items other than materials (SM 15.73, MCQ 7).

Interactive version: PassBook app, lesson Functional budgets and the master budget

Labour and overheads

The **direct labour (personnel) budget** turns the production and plant utilisation budgets into standard hours for each grade of labour, divided into direct and indirect, and multiplies them by standard wage rates (SM 15.36). Its merits: (1) it defines the direct and indirect labour force required; (2) it lets the personnel

department plan recruitment and training ahead, so that labour turnover is kept to the minimum; (3) it reveals the labour cost for the manufacturing cost budget and the cash budget for financing the wage bill.

When workers are not fully efficient and some time is unproductive, the hours to be **paid** exceed the standard hours: standard hours ÷ efficiency (productivity) ratio gives the productive hours, and non-productive down-time is added on top: the hours paid while the fryer is being cleaned or the gas cylinder is late, real wages for zero samosas. Hours beyond the normal week are overtime, paid at a premium.

The **factory overhead budget** gives the total estimated cost of each item of indirect material, indirect labour and indirect expenses (power, fuel, fringe benefits, depreciation). It is useful for working out the **pre-determined overhead recovery rates**, and its departmental schedules split each cost into fixed and variable elements (SM 15.37). Fixed expenses are policy costs; indirect labour is estimated by work study (for example one supervisor for every twenty direct workers); indirect materials from the age and condition of plant (SM 15.37–15.38).

The **production cost budget** brings direct material, direct labour and factory overhead together. The **ending inventory budget** values the closing stocks of raw material and finished goods. The **cost of goods sold budget** adds the opening finished goods and deducts the

closing ones; without those stock adjustments the same statement is called the production cost budget (SM 15.38–15.40).

The remaining budgets, in one line each (SM 15.40–15.46): **selling and distribution cost** (selling cost is "the cost of seeking to create and stimulate demand and of securing orders"; split into fixed and variable); **administration expenses** (mostly policy costs, fixed in nature, estimated from past experience adjusted for anticipated changes); **research and development** (a limit on spending is set); **capital expenditure** (the planned outlay on fixed assets, under strict management control, often over several years); **cash** (receipts and payments by month: the cash budget). Their detailed lists are in Under the hood.

The master budget: the summary the board signs

Definition

Master budget

CIMA, London: "the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed." When all the necessary functional budgets have been prepared, the budget officer prepares the master budget, which may consist of a budgeted profit and loss account and a budgeted balance sheet: these are the budget summaries. Once approved by the board of directors, it represents a standard for the achievement of which all the departments will work. (SM 15.55)

Formats and formulas

Reproduce every format from memory: ICAI's suggested answers lay the statements out exactly like this, with a working note for every figure brought in.

∑ **Formula**

Production budget (units)

$$P = S + C - O$$

P = units to be produced; S = budgeted sales; C = desired closing stock; O = opening stock, all in units of finished goods. If good output is less than output (post-production rejects), divide the good units needed by (1 − rejection rate).

∑ **Formula**

Material usage and purchase budgets

$$U = P \times q \qquad B = U + C - O$$

U = usage; P = units to be produced; q = material per unit; B = quantity to be bought; C and O = closing and opening stocks of the material. Purchase cost = units to be purchased × unit price. A closing stock given as "N days' consumption" is usage × N ÷ days in the period.

∑ **Formula**

Hours to be paid in a labour budget

$$\text{Productive} = \frac{\text{Standard hours}}{\text{Efficiency ratio}}$$

$$\text{Paid} = \text{Productive} + \text{Down-time}$$

All in hours. Standard hours = units to be produced × standard hours per unit. Hours paid beyond the normal hours (workers × normal hours each) are overtime.

Production budget	Product A (units)	Product B (units)
Budgeted sales	xxx	xxx
Add: Desired closing stock	xxx	xxx
Total quantity required	**xxx**	**xxx**
Less: Opening stock	(xxx)	(xxx)
Units to be produced	**xxx**	**xxx**

Direct material purchase budget	Material X	Material Y
Units required for production	xxx	xxx
Add: Desired closing stock	xxx	xxx
Total requirement	**xxx**	**xxx**
Less: Opening stock	(xxx)	(xxx)
Units to be purchased	**xxx**	**xxx**
Unit price (₹)	xxx	xxx
Purchase cost (₹)	**xxx**	**xxx**

Cost of goods sold budget	Amount (₹)
Direct materials used	xxx
Direct labour	xxx
Factory overhead	xxx
Total manufacturing costs	**xxx**
Add: Finished goods (opening)	xxx
Less: Finished goods (closing)	(xxx)
Cost of goods sold	**xxx**

Budgeted income statement	Amount (₹)
Sales	xxx
Less: Cost of goods sold	xxx
Gross margin	**xxx**
Less: Selling and distribution expenses	xxx
Less: Administrative expenses	xxx
Profit before interest and taxes	**xxx**
Less: Interest	xxx
Profit before tax	**xxx**
Less: Income tax	xxx
Net profit	**xxx**

That is ICAI's budgeted income statement (SM 15.56). The budgeted balance sheet beside it shows share capital and retained income, represented by fixed assets less depreciation and the current assets (raw materials, finished goods, debtors, cash) less creditors. When a question gives only sales and costs, ICAI's "Master Budget" is the shorter statement of Illustration 6: sales, less direct materials and wages (prime cost), fixed and variable factory overhead (works cost), gross profit, less administration, selling and distribution expenses, net profit (SM 15.57–15.58).

Worked examples

Worked example **Pune cafés: the sales budget built café by café**

Tapri is preparing its 2026-27 sales budget for Pune café by café, not from last year's total:

1. 12 cafés trade all year; 4 new cafés open on 1 October 2026.
2. A café sells 600 cups a day from April to September and 750 cups a day from October to March. A new café sells only half that in its first month. Take 30 days in every month.
3. A cup is priced at ₹25 until 30 September and ₹27 from 1 October.
4. 40% of the cups sold are bought with a snack at ₹40; snack prices do not change.

Required:

(i) Prepare the sales budget for 2026-27 by half-year, in quantity and value, for cups and snacks.

(ii) Kesar says the October price rise adds more to sales than the four new cafés. Examine his claim.

THE WORKING

Working note: cups to be sold

Particulars	April–September	October–March
12 existing cafés (600 or 750 × 30 × 6 months)	12,96,000	16,20,000
4 new cafés, October (375 × 30)	–	45,000
4 new cafés, November–March (750 × 30 × 5)	–	4,50,000
Cups	**12,96,000**	**21,15,000**
Snacks (40% of cups)	5,18,400	8,46,000

(i) Sales budget for 2026-27

Particulars	April–September	October–March	Year
Cups (number)	12,96,000	21,15,000	34,11,000
Cups (₹25; ₹27)	3,24,00,000	5,71,05,000	8,95,05,000
Snacks (number)	5,18,400	8,46,000	13,64,400
Snacks (₹40)	2,07,36,000	3,38,40,000	5,45,76,000
Total sales (₹)	**5,31,36,000**	**9,09,45,000**	**14,40,81,000**

The budget is classified by product (cups and snacks) and by period (half-years), two of the classifications of a sales budget the Study Material lists (SM 15.31).

(ii) Kesar's claim

Source of sales in October–March	Amount (₹)
Price rise: 21,15,000 cups × ₹2	42,30,000
New cafés: 4,95,000 cups × ₹27	1,33,65,000
New cafés: 1,98,000 snacks × ₹40	79,20,000
New cafés in total	**2,12,85,000**

The claim is wrong: the four new cafés add ₹2,12,85,000 of sales, five times the ₹42,30,000 the price rise adds. (Profit is another matter: the new cafés also bring rent, salaries and materials, which the cost budgets will show.)

Worked example

Diwali at the premix plant: production within capacity

Tapri's premix plant expects to sell these cartons in the

festival season:

Month	Sales (cartons)
August	30,000
September	40,000
October	65,000
November	60,000
December	30,000
January	25,000

1. The plant can make at most 50,000 cartons a month. The opening stock on 1 August is 6,000 cartons, and each month's closing stock must be at least 20% of the next month's sales.
2. Each carton needs 3 kg of milk-tea blend. The blend in stock at each month end is to be 40% of the next month's usage, and 30,000 kg on 31 December; the opening stock on 1 August is 54,000 kg.
3. Blend costs ₹80 a kg up to October and ₹86 a kg from 1 November.

Required:

(i) Prepare the monthly production budget for August to December, making each month's requirement as late as the plant's capacity allows.

(ii) Prepare the blend purchase budget in kg and in rupees.

(iii) Comment on the stock this plan builds up, and state what else the plant could do when sales exceed its capacity.

THE WORKING

Working note 1: production needed if capacity were unlimited (sales + closing stock − opening stock)

Month	Sales	Add: Closing (20% of next)	Less: Opening	Production needed
August	30,000	8,000	6,000	32,000
September	40,000	13,000	8,000	45,000
October	65,000	12,000	13,000	64,000
November	60,000	6,000	12,000	54,000
December	30,000	5,000	6,000	29,000
Total	**2,25,000**			**2,24,000**

October and November need more than 50,000 cartons, so their excess must be made earlier.

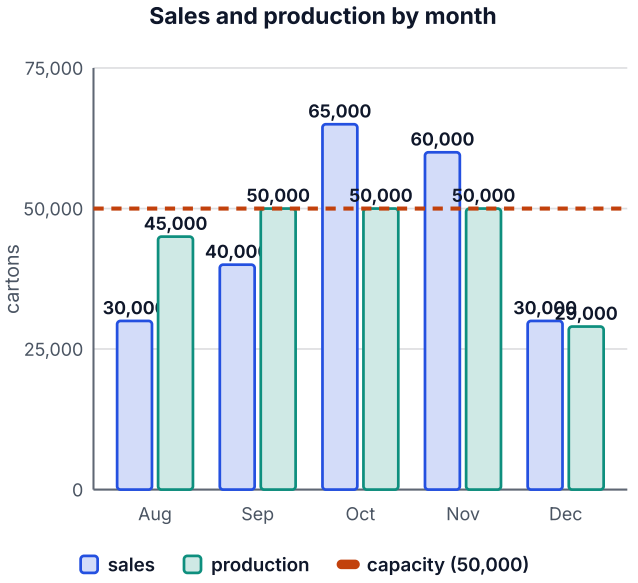
(i) Production budget within capacity (worked back from December)

Month	Needed	Add: made early for later months	Production	Excess moved to the month before
December	29,000	–	29,000	–
November	54,000	–	50,000	4,000
October	64,000	4,000	50,000	18,000
September	45,000	18,000	50,000	13,000
August	32,000	13,000	45,000	–
Total	**2,24,000**		**2,24,000**	

Check of finished stock (cartons):

Month	Opening	Add: Production	Less: Sales	Closing
August	6,000	45,000	30,000	21,000
September	21,000	50,000	40,000	31,000
October	31,000	50,000	65,000	16,000
November	16,000	50,000	60,000	6,000
December	6,000	29,000	30,000	5,000

Every closing stock is at least the 20% minimum (8,000; 13,000; 12,000; 6,000; 5,000).



Blue: sales; teal: production; orange dashed: the plant's capacity. Production runs at capacity from September to November and ahead of sales in August and September, building the stock that October and November draw down.

(ii) Blend purchase budget

Month	Usage (3 kg a carton)	Add: Closing	Less: Opening	Purchases (kg)
August	1,35,000	60,000	54,000	1,41,000
September	1,50,000	60,000	60,000	1,50,000
October	1,50,000	60,000	60,000	1,50,000
November	1,50,000	34,800	60,000	1,24,800
December	87,000	30,000	34,800	82,200
Total	**6,72,000**			**6,48,000**

Closing stock = 40% of the next month's usage (for example 40% × 1,50,000 = 60,000 kg at the end of August; 40% × 87,000 = 34,800 kg at the end of November).

Month	Purchases (kg)	Rate (₹)	Amount (₹)
August	1,41,000	80	1,12,80,000
September	1,50,000	80	1,20,00,000
October	1,50,000	80	1,20,00,000
November	1,24,800	86	1,07,32,800
December	82,200	86	70,69,200
Total	**6,48,000**		**5,30,82,000**

(iii) Comment. Because October and November ask for more than the plant can make, 18,000 cartons of their needs (14,000 for October, 4,000 for November) are made in August and September. The stock above the minimum is 13,000 cartons at the end of August, 18,000 at the end of September and 4,000 at the end of October: the finished stock rises to 31,000 cartons against a September minimum of 13,000. The plan uses the plant fully in the peak months, but the extra stock ties up funds and cold-store space for one to three months. When sales exceed the capacity of the facilities, ICAI's alternatives are overtime, shift working, sub-contracting or buying more plant (SM 15.32–15.33); a night shift in October and November would let the plant make each month's needs in that month and hold only the minimum stock.

Interactive version: PassBook app, lesson Functional budgets and the master budget

Worked example

Diwali packing: overtime or agency workers?

Tapri's packing line must pack 45,000 Diwali gift boxes and 7,200 corporate hampers in November. A gift box takes 0.30 standard hours and a hamper 0.75 standard hours.

1. The line has 100 permanent packers, each paid for 200 normal hours in November at ₹120 an hour. Their efficiency (productivity) ratio is 90%, and their non-productive down-time is 25% of productive hours.
2. Hours beyond the normal hours are paid as overtime at a premium of 50%.
3. A labour agency can instead supply casual packers at ₹150 an hour. They work at 75% efficiency and are paid only for the hours they pack (no down-time).

Required:

(i) Prepare the wages budget for November if the permanent packers do all the work, showing the hours to be paid.

(ii) Prepare the wages budget if the work beyond the permanent packers' normal hours is given to agency packers instead, and advise which plan Ritu should adopt.

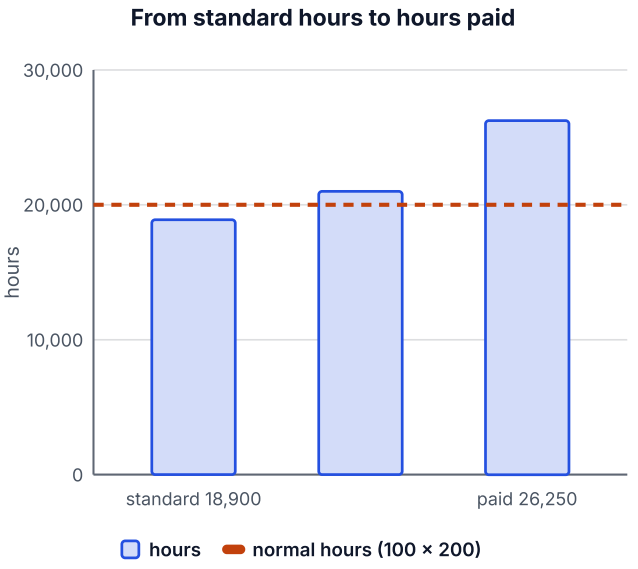
THE WORKING

Working note: standard hours = 45,000 × 0.30 + 7,200 × 0.75 = 13,500 + 5,400 = **18,900 hours**.

(i) Permanent packers only

Particulars	Hours
Standard hours	18,900
Productive hours (÷ 90%)	21,000
Add: Down-time (25% of productive)	5,250
Hours to be paid	**26,250**
Normal hours (100 × 200)	20,000
Overtime hours	**6,250**

Wages	Hours	Rate (₹)	Amount (₹)
Normal hours	20,000	120	24,00,000
Overtime (₹120 + 50%)	6,250	180	11,25,000
Total	**26,250**		**35,25,000**



Standard hours ÷ 90% give the productive hours; adding 25% down-time gives the hours paid. Everything above the orange dashed line (6,250 hours) is overtime at ₹180.

(ii) Permanent packers' normal hours plus agency packers

Particulars	Hours
Normal hours paid to permanent packers	20,000
Productive hours (20,000 ÷ 1.25)	16,000
Standard hours produced (16,000 × 90%)	14,400
Standard hours left for agency packers (18,900 − 14,400)	4,500
Agency hours to be paid (4,500 ÷ 75%)	**6,000**

Wages	Hours	Rate (₹)	Amount (₹)
Permanent packers, normal hours	20,000	120	24,00,000
Agency packers	6,000	150	9,00,000
Total	**26,000**		**33,00,000**

Advice. Compare the cost of one standard hour. An overtime hour paid gives 0.90 ÷ 1.25 = 0.72 standard hours, so a standard hour on overtime costs ₹180 ÷ 0.72 = ₹250. An agency hour gives 0.75 standard hours, so a standard hour costs ₹150 ÷ 0.75 = ₹200. The agency plan is cheaper by ₹35,25,000 − ₹33,00,000 = **₹2,25,000**, and Ritu should adopt it, provided the agency packers meet

> Tapri's quality and hygiene standards and the agency can supply 6,000 hours in November.

Worked example

Gift boxes for the festival quarter: the master budget

Tapri's gift-box unit gives these estimates for the October–December quarter. It will make more boxes than it sells, to hold stock for January's corporate orders.

Item	Estimate
Sales	1,20,000 boxes at ₹360
Production	1,30,000 boxes
Opening stock	5,000 boxes, valued at ₹230 each
Direct materials	₹150 a box
Direct wages	₹50 a box
Variable production overheads	₹20 a box
Fixed production overheads	₹26,00,000 for the quarter
Sales commission	4% of sales
Administration, selling and distribution (fixed)	₹30,00,000 for the quarter

Closing stock is valued at the quarter's production cost per box, and the opening stock is sold first.

Required: Prepare the production cost budget and the cost of goods sold budget as working notes, and the master budget (budgeted income statement) for the quarter. Comment on the effect of the stock build-up on the budgeted profit.

THE WORKING

Working note 1: production cost budget (1,30,000 boxes)

Particulars	Per box (₹)	Amount (₹)
Direct materials	150	1,95,00,000
Direct wages	50	65,00,000
Variable production overheads	20	26,00,000
Fixed production overheads (₹26,00,000 ÷ 1,30,000)	20	26,00,000
Production cost	**240**	**3,12,00,000**

Working note 2: cost of goods sold budget

Particulars	Amount (₹)
Production cost (WN 1)	3,12,00,000
Add: Opening stock (5,000 × ₹230)	11,50,000
Less: Closing stock (15,000 × ₹240)	(36,00,000)
Cost of goods sold	**2,87,50,000**

Closing stock = 5,000 + 1,30,000 − 1,20,000 = 15,000 boxes, all from this quarter's production because the opening stock is sold first.

Master budget for the quarter ending 31 December (budgeted income statement)

Particulars	(₹)	(₹)
Sales (1,20,000 × ₹360)		4,32,00,000
Less: Cost of goods sold (WN 2)		2,87,50,000
Gross margin		**1,44,50,000**
Less: Sales commission (4%)	17,28,000	
Less: Administration, selling and distribution	30,00,000	47,28,000
Net profit		**97,22,000**

Comment. The budgeted net profit is 22.5% of sales. Because 10,000 more boxes are made than are sold, the closing stock carries ₹20 a box of the quarter's fixed production overheads into January: 15,000 × ₹20 = ₹3,00,000 of fixed overheads sits in the closing stock instead of in this quarter's cost of goods sold. The budgeted profit therefore depends on the production plan as well as on the sales budget, which is why the master budget is prepared only after the production, cost and inventory budgets.

Exam angle

In the exam

Where functional budgets appeared, May 2024 to September 2026

In the new-scheme ICAI documents read for this lesson, functional budgets were asked 20 times in 16 documents: as 6 to 10-mark descriptive questions and as case-scenario MCQs.

- **Purchase and wages budgets with an efficiency ratio, down-time and overtime** (the chapter's Practical Problem 8 pattern), the most repeated: September 2026 Q2(a) (7 marks: wages budget only, 75% productivity, down-time 15% of productive hours, overtime above 45 hours a week), MTP September 2024 Series I Q5(a) (7 marks), MTP January 2025 Series II Q5(b) (6 marks), MTP January 2026 Series II Q3(a) (9 marks), MTP September 2026 Series I Q5(a) (10 marks), RTP January 2026 Q18.
- **Quarterly production, consumption and purchase budgets** (the Illustration 4 pattern): September 2025 Q4(a) (9 marks, with machine hours including maintenance), RTP September 2025 Q17, MTP May 2025 Series I Q4(b) (8 marks), RTP September 2026 Q14 (monthly, followed by variances).
- **Sales budget with price changes and extra campaign sales**: May 2026 Q4(a) (9 marks), with statements of budgeted and actual sales for the year.

- **Production budget with rejects and purchases with wastage**: MTP May 2024 Series I Q1 (5 + 5 marks with EOQ, then 4 marks on a supplier's limit). Units from sales revenue and a profit margin: RTP September 2024 Q18.
- **MCQs, 2 marks each**: RTP September 2024 MCQ 7 and MTP September 2026 Series II MCQ 13 (the same question: stock to be reduced by 40%); MTP September 2026 Series I MCQ 12 (raw material to buy); MTP September 2025 Series I MCQ 15 (a monthly production policy); and five-MCQ case scenarios: September 2026 MCQs 6–10 (production, consumption and purchase of two products' raw materials), MTP September 2025 Series I MCQs 6–10, MTP January 2026 Series I MCQs 6–10.
- **Master budget**: not in the new-scheme documents read; the Illustration 6 layout appeared in old-scheme MTPs (November 2020, May 2022).

In the exam

The traps ICAI sets

- "Increase over budgeted sales" (May 2026): apply the percentage to the previous year's **budget**, not its actual sales.
- Closing stock "N days' sales" or "N days' consumption": divide by the days in the budget period (four 5-day weeks = 20 days).
- Down-time is a percentage of **productive** hours, which are standard hours ÷ efficiency ratio; overtime is what is paid beyond workers × normal hours, at rate × (1 + premium).
- Quarter IV's "20% of next quarter" is the given year-end closing stock.
- The question's instruction: "Working notes should form part of the answer."

Trick

One line for every stock budget

Need = use + closing − opening. It gives production (use = sales of finished goods) and purchases (use = consumption of material). Condition: all three figures in the same unit and for the same period.

Answer frame

Theory parts ICAI asks

"List the functional budgets" (the chapter's own question asks for eight): sales, production, plant utilisation, direct material usage, direct material purchase, direct labour, factory overhead, production cost, ending inventory, cost of goods sold, selling and distribution cost, administration expenses, research and development cost, capital expenditure, cash. Write at least the number asked, in ICAI's order.

"Explain the production budget": definition (forecast of production, in two parts: volume and cost); based on the four (sales budget, capacity, planned change in finished stocks, outside purchase policy); the stock-level factors; stabilised production versus stabilised stocks; a format.

"State the considerations for the capital expenditure budget": the five considerations and, if asked, the six merits (Under the hood).

The numerical parts: three answers to the same wages budget

Loses marks

"Wages = 18,900 hours × ₹120 = ₹22,68,000."

Standard hours paid at the normal rate: no efficiency ratio, no down-time, no overtime.

Passes

Productive hours 21,000, down-time 5,250, hours paid 26,250 × ₹120 = ₹31,50,000.

The hours are right, but the 6,250 overtime hours are paid without their premium.

Full marks

Standard hours by product as a working note; ÷ 90%; + 25% down-time; normal hours (workers × normal hours each) at ₹120 and the excess at ₹180; total ₹35,25,000, with each step labelled.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the opening stock that was added

"Production = sales + closing stock + opening stock." The opening stock is already made: it reduces production. **Fix:** need = use + closing − opening, for finished goods and for materials alike.

⚠ **Common mistake**

Material bought for the sales instead of for production

Multiplying **sales** units by the material per unit. **Fix:** material usage comes from the **production** budget, which already includes the finished-goods stock change; then adjust for the material stocks.

⚠ **Common mistake**

The closing stock taken on the wrong base

"Closing stock of blend = 40% of next month's purchases" or "40% of this month's usage". **Fix:** read the base the question gives (next month's sales for products, next month's usage or N days' consumption for materials); for "N days", divide by the working days in the period.

⚠ **Common mistake**

Overtime without the premium

Fix: hours beyond the normal week are paid at the basic rate plus the premium (₹120 × 150% = ₹180 in the packing example). Down-time is added to productive hours before the overtime is found.

Practice

Check yourself

1 Tapri expects to sell 1,20,000 cartons of premix evenly through next year and wants to end the year with one month's sales in stock. It has 12,400 good cartons in stock now. 2% of the cartons produced fail the seal test after production and are destroyed. How many cartons must it produce?

Answer: _____

---

2 Case scenario. Ritu is budgeting Tapri's Diwali gift boxes. Sales are expected to be 30,000 boxes in October, 45,000 in November and 20,000 in December. Each month's closing stock is to be 40% of the next month's sales, and 10,000 boxes are in stock on 1 October. What is the production budget for November?

- **A** 35,000 boxes
- **B** 43,000 boxes
- **C** 27,000 boxes
- **D** 71,000 boxes

3 Production next quarter is 30,000 units, each needing 2 kg of material. The material stock is 8,000 kg and is to rise to 10,000 kg by the end of the quarter. How many kg must be purchased?

Answer: _____

4 A seasonal business keeps its production the same every month, at the maximum possible, to use the plant fully and avoid laying off workers. Which statement about this policy is correct?

- **A** Inventory carrying cost is the lowest possible
- **B** Capacity is fully utilised, but inventory carrying cost varies with the stocks held
- **C** Stocks stay at the minimum level throughout the year
- **D** It needs no production budget, only a sales budget

5 Which of these are factors for which past sales volumes are adjusted when preparing the sales budget? Select all that apply.

- **A** Backlog of unfulfilled sales orders
- **B** Planned advertising and promotion
- **C** The depreciation method of the plant
- **D** Productive capacity
- **E** Competition

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Functional budgets: one for each function (15 in ICAI's list), subsidiary to the master budget; the sales budget comes first.
- Production = budgeted sales + desired closing stock − opening stock (units); based on sales, capacity, stock change, outside purchase policy.
- Stabilised production: plant fully used, carrying cost varies; stabilised stocks: lowest carrying cost, capacity may be idle.
- Material usage = production × standard quantity; purchases = usage + closing stock − opening stock;

governed by EOQ and the re-order point with safety stock.

- Labour: standard hours ÷ efficiency ratio = productive hours; + down-time = hours paid; beyond normal hours is overtime at the premium rate.
- Cost of goods sold = materials + labour + factory overhead + opening finished goods − closing finished goods (without the stock adjustments: production cost budget).
- Master budget (CIMA): the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed: budgeted P&L and balance sheet.
- Need = use + closing − opening, every time. (Barfi first, khoya second.)

Answer key · Check yourself

1 **120000**

Good cartons needed = 1,20,000 + 10,000 (one month's sales) − 12,400 = 1,17,600. Since only 98% of output is good, production = 1,17,600 ÷ 0.98 = 1,20,000. Adding 2% to 1,17,600 instead (1,19,952) is not enough: the rejects are 2% of production, not of good output.

2 **A 35,000 boxes**

November: sales 45,000 + closing stock (40% of 20,000) 8,000 − opening stock (October's closing, 40% of 45,000) 18,000 = 35,000. 43,000 uses 1 October's stock as November's opening stock; 27,000 forgets the closing stock; 71,000 adds the opening stock.

3 **62000**

Usage = 30,000 × 2 = 60,000 kg; purchases = 60,000 + 10,000 − 8,000 = 62,000 kg.

4 **B Capacity is fully utilised, but inventory carrying cost varies with the stocks held**

Stabilised production uses the plant fully, so stocks build up in the off season and carrying cost varies with them. The lowest carrying cost belongs to stabilised stocks, which may under-utilise capacity.

5 **A Backlog of unfulfilled sales orders**; **B Planned advertising and promotion**; **D Productive capacity**; **E Competition**

ICAI's eight: backlog, planned advertising and promotion, industry and economic conditions, productive capacity, projected pricing, market research, relative product profitability, competition. The depreciation method does not change how much can be sold.

15.3 Flexible budgets

Why a budget fixed at one level of activity misleads, how a flexible budget recasts every cost to the level actually reached, and how to prepare one for several capacity levels.

In plain words

The café manager who "saved" ₹18,000

Tapri's Pune café budgeted ₹1,70,000 of costs for March, for 20,000 cups. Then a water pipe burst, the café shut for ten days, and it sold only 15,000 cups. March's costs came to ₹1,52,000.

> **Kesar:** ₹18,000 under budget! Give the manager a bonus.
>
> **Ritu:** Per unit, please. She sold 5,000 fewer cups. The milk, tea and kulhads for those cups were never bought.
>
> **Kesar:** So did she save money or not?
>
> **Ritu:** Ask the budget she should have had for 15,000 cups.

Ritu splits the café's costs by behaviour. Rent and salaries are **fixed**: ₹60,000 whatever the cups. Milk, tea and kulhads are **variable**: ₹4 a cup. Electricity is **semi-variable**: ₹10,000 for the lights and fridges plus ₹1 a cup for the boilers. For 15,000 cups the café should have spent

₹60,000 + ₹10,000 + 15,000 × ₹5 = ₹1,45,000. It spent ₹1,52,000: ₹7,000 **more** than it should have.

The ₹1,70,000 was a **fixed budget**: prepared for one level of activity and left unchanged whatever level is actually reached. The ₹1,45,000 is a **flexible budget**: the same plan recast for the activity actually attained, because it knows which costs move with volume and which do not. Compared with the fixed budget the manager looks thrifty; compared with the flexible budget she overspent.

Think of it like

The road-trip fuel budget

You budget ₹5,000 for fuel and ₹500 for tolls for a 500-km trip, then turn back after 300 km having spent ₹3,500 on fuel. "₹1,500 saved"? Fuel at ₹10 a km allowed ₹3,000 for 300 km: you overspent ₹500, perhaps on the hills. The toll was the same either way. A flexible budget is the fuel allowance for the distance you actually drove.

Key idea

A fixed budget is for one level of activity and does not change; a flexible budget recognises fixed, semi-variable and variable costs and changes with the level of activity attained. Judge actual costs only against the flexible budget for the actual level.

How it works

Fixed budget: one level, no change

A **fixed budget** is prepared on the basis of a standard or fixed level of activity and does not change with a change in the level of activities; it shows the expected results of a responsibility centre for only one activity level, and is also known as a **static budget** (SM 15.19–15.20). Fixed budgeting is used by many service companies and for some administrative functions of manufacturing companies, such as purchasing, engineering and accounting. It is suitable for fixed expenses; when the level attained differs from the budgeted level, it becomes ineffective as a tool of cost control.

Essential conditions for a fixed budget (SM 15.20): (1) the nature of business is not seasonal; (2) there is no impact of external factors on business activities; (3) demand for the product is certain and stable; (4) supply orders are received and issued regularly; (5) the market is normally domestic (or regular service exports); (6) no special labour or material is needed in production; (7) the supply of production inputs is regular; (8) there is a trend of price stability. These conditions are generally not all found in practice, hence a fixed budget is not suitable in business concerns.

Fixed budget: merits	Fixed budget: demerits
1. Very simple to understand	1. Does not suit a dynamic organisation and may give misleading results: a poor or good performance may remain unnoticed
2. Less time consuming	2. Not suitable for a long period
	3. Unsuitable when business conditions are changing constantly
	4. Accurate estimates are not possible

Flexible budget: a series of fixed budgets

Definition

Flexible budget

A budget which, by recognising the difference in behaviour between fixed and variable costs in relation to fluctuations in output, turnover, or other variable factors, is designed to change appropriately with such fluctuations. CIMA: "a budget which, by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained." (SM 15.21)

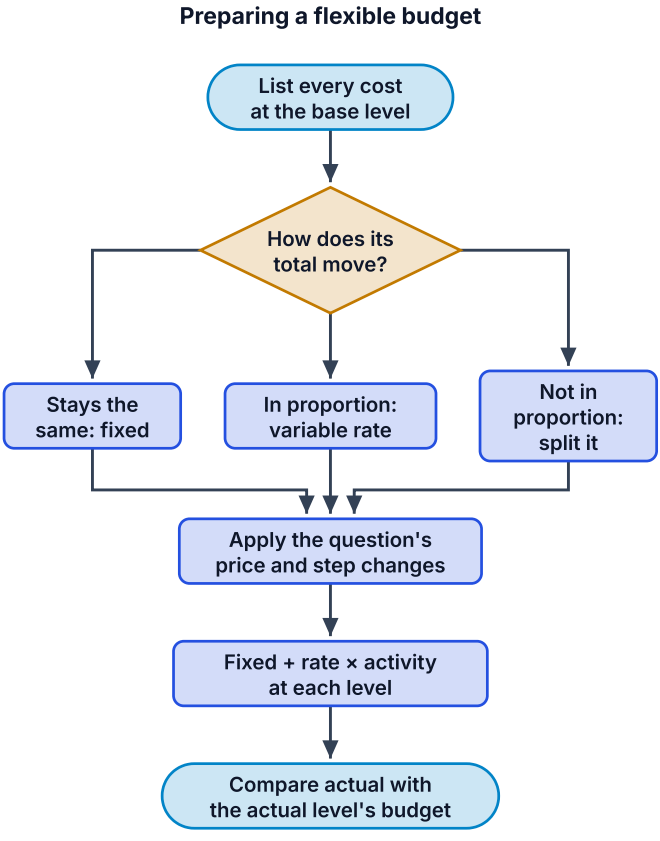
Unlike a static budget, a flexible budget shows the expected results of a responsibility centre for different activity levels: it can be viewed as a series of static

budgets for different levels of activity. It is especially useful in estimating and controlling factory costs and operating expenses, and more realistic because it considers how revenue and cost behave at different levels (SM 15.21). To prepare one, expenses are classified as (i) fixed, (ii) variable and (iii) semi-variable, and the semi-variable expenses are further segregated into fixed and variable parts (cost behaviour).

Flexible budgeting may be resorted to (SM 15.21): (i) in a new business venture, where demand is hard to forecast accurately; (ii) where the business depends on the fluctuations of nature (a wool trader sells plenty below freezing point and much less in warm weather); (iii) in a labour-intensive industry where production depends on the availability of labour.

Flexible budget: merits (SM 15.22)	Flexible budget: demerits (SM 15.22)
1. Sales, costs and profit can be calculated easily at various levels of capacity	1. Possible only with a proper accounting system and perfect knowledge of the factors of production and business circumstances
2. Adjustment to changed business conditions is very simple	2. Requires a system of standard costing
3. Helps decide the production level, showing budgeted costs and sales at each level, so management can pick the level that earns the profit predetermined by the owners	3. Very expensive and labour oriented
4. Shows the quantity to be produced to earn a determined profit	

Preparing one, step by step



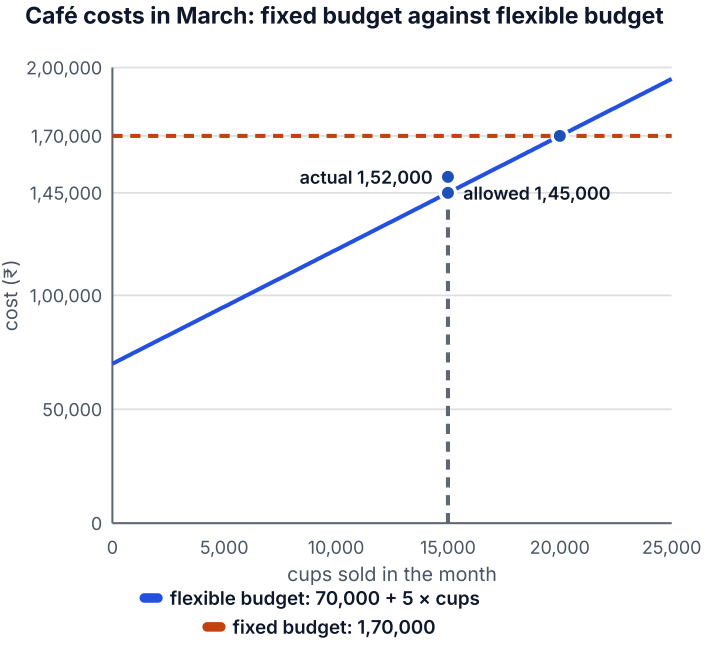
Classify, split, adjust, compute, compare. A cost whose total stays the same is fixed; one that rises in proportion is variable; one that rises but not in proportion is semi-variable, split into its fixed and variable parts by the high and low points method.

The split of a semi-variable cost uses the **high and low points method** (SM 15.28): the variable cost per unit is the change in the expense divided by the change in

output, and the fixed part is what remains of the total at either level. Then watch the question for three things ICAI likes to add: a price or cost rate that changes at certain levels, a fixed cost that steps up beyond a capacity level (an extra supervisor, more advertising), and a semi-variable cost that rises by a percentage within a band of activity (SM 15.23).

Interactive version: PassBook app, lesson Flexible budgets

The café on one chart



Blue: the cost the café should incur at each level; orange dashed: the fixed budget, the same whatever the cups. The lines meet at the budgeted 20,000 cups. At 15,000 cups the actual cost sits ₹18,000 below the orange line but ₹7,000 above the blue one.

The same chart explains a classic trap. At 20,000 cups the budget works out to ₹1,70,000 ÷ 20,000 = ₹8.50 a cup. Applying ₹8.50 to 15,000 cups gives ₹1,27,500, far below the correct ₹1,45,000, because the fixed ₹70,000 does not shrink with the cups: the cost per unit falls as activity rises. ICAI's Illustration 1 makes the point with a

recovery rate per hour of ₹0.61 at 70% and ₹0.50 at 100% (SM 15.23–15.24).

Formats and formulas

Reproduce the format from memory; ICAI's suggested answers show every level side by side, the fixed portions worked out in notes.

∑ **Formula**

Flexible budget allowance

$$\text{Allowance} = F + v \times A$$

F = fixed cost (including the fixed part of semi-variable costs); v = variable cost per unit of activity; A = activity attained, in units, hours, or sales value (for costs given as a percentage of sales).

∑ **Formula**

High and low points method

$$v = \frac{\text{Change in expense}}{\text{Change in output}}$$

$$F = \text{Total expense} - v \times \text{Output}$$

v = variable cost per unit; F = fixed cost. Use two levels at which the fixed cost is the same; the fixed part comes out equal at both.

Definition

Fixed budget

CIMA: "a budget designed to remain unchanged irrespective of the level of activity actually attained." (SM 15.19)

Flexible budget for ...	60%	80%	100%
Output (units)	xxx	xxx	xxx
Sales (A)	**xxx**	**xxx**	**xxx**
Variable costs, item by item	xxx	xxx	xxx
Variable part of semi-variable costs	xxx	xxx	xxx
Total variable cost (B)	**xxx**	**xxx**	**xxx**
Contribution (A – B)	**xxx**	**xxx**	**xxx**
Fixed costs, item by item	xxx	xxx	xxx
Fixed part of semi-variable costs	xxx	xxx	xxx
Step costs at the level	xxx	xxx	xxx
Total fixed cost (C)	**xxx**	**xxx**	**xxx**
Profit (A – B – C)	**xxx**	**xxx**	**xxx**

When only costs are asked ("flexible budget showing the total cost"), stop at total cost (B + C); ICAI often adds the

cost per unit or per hour at each level.

Worked examples

Worked example **Cold brew: the night shift and the export order**

Tapri's cold-brew plant can bottle 1,00,000 bottles a year; it now works at 60%. Ritu is preparing the budget for 60%, 80% and 100% of capacity:

1. The domestic market takes at most 70,000 bottles at ₹120 each. Bottles beyond that can only be exported, at ₹105 each, with export packing and freight of ₹7 a bottle.
2. Direct material costs ₹40 a bottle at every level.
3. The day shift can bottle 70,000 bottles at a direct labour cost of ₹15 a bottle. Bottles beyond that come from a night shift, whose labour is paid 20% more, and which needs a night supervisor at ₹1,50,000 a year.
4. Factory overheads were ₹14,40,000 in a year when the plant made 50,000 bottles and ₹16,80,000 in a year when it made 70,000; the fixed part is unchanged.
5. Administration and selling overheads are ₹6,00,000 a year (fixed), plus a commission of 5% on domestic sales.

Required:

(i) Prepare a flexible budget showing the contribution and profit at 60%, 80% and 100% capacity.

(ii) Comment on the level at which the plant should work, and state the smallest night-shift output that is worth running.

THE WORKING

Working notes

1. Factory overheads (high and low points): variable = (₹16,80,000 − ₹14,40,000) ÷ (70,000 − 50,000) = **₹12 a bottle**; fixed = ₹14,40,000 − 50,000 × ₹12 = **₹8,40,000**.
2. Bottles: 60% = 60,000, all domestic; 80% = 70,000 domestic + 10,000 export; 100% = 70,000 domestic + 30,000 export.
3. Labour: day shift ₹15; night shift ₹15 × 120% = **₹18 a bottle**, on the bottles beyond 70,000.
4. Commission: 5% × ₹120 = ₹6 a domestic bottle; none on exports.

(i) Flexible budget

Particulars	60% (₹)	80% (₹)	100% (₹)
Bottles	60,000	80,000	1,00,000
Domestic sales @ ₹120	72,00,000	84,00,000	84,00,000

Particulars	60% (₹)	80% (₹)	100% (₹)
Export sales @ ₹105	–	10,50,000	31,50,000
Sales (A)	**72,00,000**	**94,50,000**	**1,15,50,000**
Direct material @ ₹40	24,00,000	32,00,000	40,00,000
Direct labour, day shift @ ₹15	9,00,000	10,50,000	10,50,000
Direct labour, night shift @ ₹18	–	1,80,000	5,40,000
Variable factory overheads @ ₹12	7,20,000	9,60,000	12,00,000
Commission (5% of domestic sales)	3,60,000	4,20,000	4,20,000
Export packing and freight @ ₹7	–	70,000	2,10,000
Total variable cost (B)	**43,80,000**	**58,80,000**	**74,20,000**
Contribution (A − B)	**28,20,000**	**35,70,000**	**41,30,000**
Fixed factory overheads (WN 1)	8,40,000	8,40,000	8,40,000

Particulars	60% (₹)	80% (₹)	100% (₹)
Administration and selling (fixed)	6,00,000	6,00,000	6,00,000
Night supervisor	–	1,50,000	1,50,000
Fixed cost (C)	**14,40,000**	**15,90,000**	**15,90,000**
Profit (A − B − C)	**13,80,000**	**19,80,000**	**25,40,000**
Total cost per bottle (B + C)	97.00	93.38	90.10

(ii) Comment. Profit is highest at **100%** (₹25,40,000). A domestic bottle contributes ₹120 − ₹40 − ₹15 − ₹12 − ₹6 = ₹47; an export bottle from the night shift only ₹105 − ₹40 − ₹18 − ₹12 − ₹7 = ₹28. The lower price applies only to the extra bottles, not to the whole output, so every export bottle still adds ₹28, and the move from 80% to 100% adds 20,000 × ₹28 = ₹5,60,000. The night shift must first cover its supervisor: ₹1,50,000 ÷ ₹28 = 5,357.1, so it is worth running only if it makes at least **5,358 bottles**. The plant should work at 100%, provided the export buyer takes 30,000 bottles at ₹105.

Interactive version: PassBook app, lesson Flexible budgets

Worked example **Three cafés, one league table**

Kesar ranks the managers of Tapri's three Indore cafés every month by "actual cost against budget". Each café's monthly cost budget is built from the same cost behaviour, which Ritu has established:

- Milk, tea, sugar and kulhads: ₹6.00 a cup (variable).
- Electricity and gas: ₹8,000 a month plus ₹0.50 a cup (semi-variable).
- Rent and salaries: fixed, different for each café.

June	Rajwada	Vijay Nagar	Palasia
Budgeted cups	30,000	24,000	18,000
Cups actually sold	36,000	20,000	18,500
Rent and salaries (₹)	1,20,000	1,00,000	90,000
Actual total cost (₹)	3,54,000	2,46,000	2,20,000

Palasia's actual cost includes ₹6,000 for a boiler overhaul ordered and approved by head office. Kesar's June table puts Vijay Nagar first, Palasia second and Rajwada last, and he plans to move the Rajwada manager to a smaller café.

Required:

(i) Prepare each café's fixed budget for June and its flexible budget for the cups actually sold.

(ii) Compare actual costs with both budgets and rank the managers on the comparison that measures their performance.

(iii) Comment on Kesar's league table and his plan.

THE WORKING

(i) Budgets (fixed part of electricity ₹8,000; variable cost ₹6.00 + ₹0.50 = ₹6.50 a cup)

Particulars	Rajwada (₹)	Vijay Nagar (₹)	Palasia (₹)
Rent and salaries	1,20,000	1,00,000	90,000
Electricity and gas, fixed part	8,000	8,000	8,000
Fixed costs	**1,28,000**	**1,08,000**	**98,000**
Fixed budget: + ₹6.50 × budgeted cups	1,95,000	1,56,000	1,17,000
Fixed budget	**3,23,000**	**2,64,000**	**2,15,000**

Particulars	Rajwada (₹)	Vijay Nagar (₹)	Palasia (₹)
Flexible budget: + ₹6.50 × actual cups	2,34,000	1,30,000	1,20,250
Flexible budget	**3,62,000**	**2,38,000**	**2,18,250**

(ii) Comparison

Particulars	Actual cost
Rajwada (₹)	3,54,000
Vijay Nagar (₹)	2,46,000
Palasia (₹)	2,20,000

Particulars	Over / (under) the fixed budget
Rajwada (₹)	31,000
Vijay Nagar (₹)	(18,000)
Palasia (₹)	5,000

Particulars

Over / (under) the flexible budget

Rajwada (₹)

(8,000)

Vijay Nagar (₹)

8,000

Palasia (₹)

1,750

Particulars

Less: head office's boiler overhaul

Rajwada (₹)

–

Vijay Nagar (₹)

–

Palasia (₹)

(6,000)

Particulars

Controllable over / (under) the flexible budget

Rajwada (₹)

(8,000)

Vijay Nagar (₹)

8,000

Palasia (₹)

(4,250)

Particulars

As % of the flexible budget

Rajwada (₹)

(2.21%)

Vijay Nagar (₹)

3.36%

Palasia (₹)

(1.95%)

Particulars	Rank
Rajwada (₹)	I
Vijay Nagar (₹)	III
Palasia (₹)	II

(iii) Comment. Kesar's table compares each café's actual cost with a budget for a different number of cups, so it rewards lower sales: Vijay Nagar "saves" ₹18,000 only because it sold 4,000 fewer cups, and Rajwada "overspends" ₹31,000 only because it sold 6,000 more. Flexed to the cups actually sold, Rajwada spent ₹8,000 less than its allowance and is the best performer, while Vijay Nagar overspent by ₹8,000 and is the worst. Palasia's boiler overhaul was a head-office decision, so it is excluded from the manager's performance. The plan to move the Rajwada manager should be dropped; Vijay Nagar's overspending, on lower sales, is what needs explaining.

Interactive version: PassBook app, lesson Flexible budgets

Worked example

Rusks: a second oven line in the past data

Tapri's rusk unit sells its rusks at ₹90 a carton. Its production overheads in the last three years were:

Year	Output (cartons)	Production overheads (₹)
2023-24	40,000	9,20,000
2024-25	50,000	10,40,000
2025-26	65,000	13,70,000

A second oven line was commissioned at the start of 2025-26, adding ₹1,50,000 a year to fixed production overheads; with it the unit can bake 1,00,000 cartons a year. Other costs in 2025-26: direct materials ₹30 and direct labour ₹15 a carton; selling overheads ₹2,00,000 a year plus a commission of 5% on sales; administrative overheads ₹2,40,000 a year.

For 2026-27 the selling price is unchanged and these increases are expected: direct materials 10%, direct labour 8%, variable production overheads 5%, all fixed overheads 10%.

Required:

(i) Segregate the production overheads into fixed and variable elements.

(ii) Prepare a flexible budget for 2026-27 at 60,000, 70,000 and 80,000 cartons, showing the contribution and profit.

(iii) Kesar wants a profit of ₹10,00,000 in 2026-27. Calculate the output needed, and state whether the unit can produce it.

THE WORKING

(i) Segregation. Use two years with the same fixed overheads: 2023-24 and 2024-25 (2025-26 includes the second line).

Variable = (₹10,40,000 − ₹9,20,000) ÷ (50,000 − 40,000) = **₹12 a carton**; fixed = ₹9,20,000 − 40,000 × ₹12 = **₹4,40,000**.

Check 2025-26: ₹4,40,000 + ₹1,50,000 + 65,000 × ₹12 = ₹13,70,000. Taking 2023-24 and 2025-26 instead would give (₹13,70,000 − ₹9,20,000) ÷ 25,000 = ₹18 a carton, wrongly treating the new line's fixed cost as variable.

Working note: rates for 2026-27

Item	Working	2026-27
Direct materials	₹30 × 110%	₹33.00 a carton
Direct labour	₹15 × 108%	₹16.20 a carton
Variable production OH	₹12 × 105%	₹12.60 a carton
Commission	5% × ₹90	₹4.50 a carton
Fixed production OH	(₹4,40,000 + ₹1,50,000) × 110%	₹6,49,000
Fixed selling OH	₹2,00,000 × 110%	₹2,20,000
Administrative OH	₹2,40,000 × 110%	₹2,64,000

OH: overheads. Variable cost ₹66.30 a carton; contribution ₹90 – ₹66.30 = ₹23.70 a carton; fixed cost ₹11,33,000.

(ii) Flexible budget for 2026-27

Particulars	60,000 (₹)	70,000 (₹)	80,000 (₹)
Sales @ ₹90 (A)	**54,00,000**	**63,00,000**	**72,00,000**

Particulars	60,000 (₹)	70,000 (₹)	80,000 (₹)
Direct materials @ ₹33	19,80,000	23,10,000	26,40,000
Direct labour @ ₹16.20	9,72,000	11,34,000	12,96,000
Variable production OH @ ₹12.60	7,56,000	8,82,000	10,08,000
Commission @ ₹4.50	2,70,000	3,15,000	3,60,000
Variable cost (B)	**39,78,000**	**46,41,000**	**53,04,000**
Contribution (A – B)	**14,22,000**	**16,59,000**	**18,96,000**
Fixed production OH	6,49,000	6,49,000	6,49,000
Fixed selling OH	2,20,000	2,20,000	2,20,000
Administrative OH	2,64,000	2,64,000	2,64,000
Fixed cost (C)	**11,33,000**	**11,33,000**	**11,33,000**
Profit (A – B – C)	**2,89,000**	**5,26,000**	**7,63,000**

(iii) Output for a profit of ₹10,00,000 = (fixed cost + required profit) ÷ contribution per carton = (₹11,33,000 + ₹10,00,000) ÷ ₹23.70 = **90,000 cartons**. That is within the capacity of 1,00,000

cartons, so the unit can produce it, provided the market takes 90,000 cartons at ₹90. This is the use of a flexible budget that ICAI lists last: it shows the quantity to be produced to earn a determined profit.

Exam angle

In the exam

Where flexible budgets appeared, May 2024 to September 2026

In the new-scheme ICAI documents read for this lesson, flexible budgets were asked 17 times in 15 documents.

- **Profit or cost at several capacity levels** (the core numerical): May 2024 Q3(a) (7 marks: 60%, 70%, 90%, price falls, material rises, extra advertising above 80%); September 2024 Q3(b) (6 marks: total cost at 60%, 80%, 100%, transport per truck); MTP May 2026 Series I Q2(b) (6 marks: expense budget at two outputs); MTP September 2025 Series II Q3(b) (6 marks: before and after marketing efforts); RTP January 2025 Q18 (administration, selling and distribution costs at four levels).
- **From past years' costs to a new level with cost increases**: MTP September 2024 Series II Q5(b) (7 marks), the chapter's Practical Problem 4 pattern.
- **Step costs in a service setting**: May 2025 Q5(c) (7 marks, a college trip); MTP May 2024 Series II Q3(a) (10 marks, a school picnic with a break-even fee);

MTP May 2026 Series II MCQs 6–10 (five MCQs on a training workshop).

- **Harder variants**: RTP May 2025 Q19 (overtime premium above a volume and a learning effect); RTP May 2024 Q12 (the revenue budget case).
- **MCQs, 2 marks**: MTP May 2024 Series I MCQ 4 and MTP September 2026 Series I MCQ 15 (the same: overheads at 60% from the figures at 50% and 75%).
- **Theory, 4 or 5 marks**: demerits of fixed and flexible budgets, MTP September 2024 Series II Q6(c); suitability for a flexible budget, MTP January 2026 Series II Q4(c); "which budgeting system would you recommend", MTP September 2025 Series II Q6(b); differences between fixed and flexible budgets, RTP September 2025 Q18(c).

In the exam

The traps ICAI sets

- A transport, bus or batch cost per vehicle is a step cost: round the number of vehicles **up** (unless the loads divide exactly, as in September 2024, where it became a variable ₹1 a unit).
- A fixed cost that starts "above 80%" or "beyond 90%" is added only at the levels above it.
- The fixed portion of a semi-variable cost is worked out at the base level and held constant; percentage increases are applied to the variable rate and to the fixed total separately.
- "Profit 20% on sales" is 25% on cost.
- Compare actual cost with the flexible budget for the **actual** level; a fixed budget shows "savings" that are only lower volume.

Trick

Fixed cost survives, variable cost travels

Whatever the level: fixed cost stays as a total, variable cost moves as a rate × level. So total cost per unit always falls as the level rises (until a step cost or a price change intervenes). Condition: costs linear within the range, with step and price changes added separately.

Answer frame

"Distinguish between fixed and flexible budgets"

ICAI's five points, in order (SM 15.22–15.23):

Point
1. Change

Fixed budget
Does not change with the actual volume achieved: rigid or inflexible

Flexible budget
Can be re-cast for the activity level achieved: not rigid

Point
2. Levels

Fixed budget
One level of activity, one set of conditions; assumes no change, which is unrealistic

Flexible budget
Various budgets for different levels of activity

Point
3. Variance analysis

Fixed budget
All costs related to one level, so it does not give useful information

Flexible budget
Each cost analysed by behaviour, so it gives useful information

Point	
4. Cost and price	
Fixed budget	Cost ascertainment and price fixation mislead when budgeted and actual levels differ significantly
Flexible budget	Facilitates cost ascertainment, price fixation and quotations

Point	
5. Comparison	
Fixed budget	Comparing actual with budget is meaningless when activity levels differ
Flexible budget	A meaningful basis of comparison of actual with budget

"Explain briefly the concept of flexible budget"

1. The definition (the behaviour of fixed and variable costs; CIMA's wording with semi-variable costs).
2. Unlike a static budget: results for different activity levels; a series of static budgets.
3. Costs classified as fixed, variable and semi-variable, the last split into fixed and variable.
4. Where it is used: new ventures, businesses dependent on nature, labour-intensive industries.
5. A three-line numerical illustration (the café).

"Which budgeting system would you recommend?" (actuals far above a fixed budget)

ICAI's suggested answer (MTP September 2025 Series II Q6(b)): recommend a **flexible budget**; define it (both

definitions); state that unlike static budgets it shows expected results for different activity levels; then list its advantages: sales, costs and profit calculated easily at various levels; simple adjustment to changes in business conditions; helps select the production level that earns the predetermined profit; shows the quantity to be produced to earn a determined profit.

"Suitability for flexible budget" (SM 15.21–15.22)

1. Seasonal fluctuations in sales and/or production, for example the soft drinks industry.
2. A company which keeps introducing new products or changes the design of its products frequently.
3. Industries engaged in make-to-order business, like ship building.
4. An industry influenced by changes in fashion.
5. General changes in sales.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the league table that punished the busiest café

Comparing Rajwada's actual ₹3,54,000 with the ₹3,23,000 budgeted for 30,000 cups and calling the ₹31,000 difference an overspend. **Fix:** flex the budget to the 36,000 cups actually sold (₹3,62,000): the café stayed ₹8,000 within it. The same error works the other way: Vijay Nagar's "saving" of ₹18,000 against its fixed budget is only lower volume.

⚠ **Common mistake**

The budget rate per unit carried to another level

₹8.50 a cup at 20,000 cups applied to 15,000 cups gives ₹1,27,500. **Fix:** fixed costs do not shrink: allowance = fixed ₹70,000 + ₹5 × 15,000 = ₹1,45,000.

⚠ **Common mistake**

Semi-variable costs treated as wholly variable or wholly fixed

Treating the cold-brew plant's ₹16,80,000 of factory overheads at 70,000 bottles as all variable (₹24 a bottle) or all fixed. **Fix:** split it first, by the percentage the question gives or by the high and low points method (here ₹8,40,000 fixed and ₹12 a bottle); flex only the variable part. And pick two levels with the same fixed cost: the rusk unit's 2025-26 figure includes a second oven line.

⚠ **Common mistake**

8.33 vans

A fraction of a van, a bus or a batch: 1,000 crates ÷ 120 = 8.33 vans. **Fix:** vehicles and batches are hired whole: round up (9 vans), and cost the step.

Practice

Check yourself

1 A bakery's budgeted overheads are ₹2,40,000 at 40% capacity and ₹3,30,000 at 70%, with the same fixed cost at both levels. Its manager has been given an overhead allowance of ₹3,00,000 for next month. At what capacity level (in %) would that allowance be exactly right?

Answer: _____

2 Case scenario. Tapri's samosa factory sends crates to its cafés in vans that carry 120 crates each, at ₹1,800 a van trip. Loading labour costs ₹5 a crate, and the dispatch clerk's salary is ₹12,000 a month. The monthly flexible budget is ₹33,200 for 1,000 crates. By how much does the budget rise if the factory dispatches 1,100 crates?

- **A** ₹3,320
- **B** ₹500
- **C** ₹2,300
- **D** ₹1,800

3 A unit's budget at 8,000 machine hours is ₹96,000: ₹48,000 fixed and ₹6 an hour variable. It actually works 6,000 hours. What is the correct budget allowance?

- **A** ₹72,000
- **B** ₹84,000
- **C** ₹96,000
- **D** ₹36,000

4 For which of these businesses is a flexible budget most suitable?

- **A** A soft-drinks maker whose sales peak every summer
- **B** A firm with certain and stable demand, regular supplies and stable prices
- **C** A department all of whose costs are fixed
- **D** A firm that never changes its products and sells only in a steady domestic market

5 Which of these are ICAI's essential conditions for a fixed budget? Select all that apply.

- **A** The nature of the business is not seasonal
- **B** The demand for the product is certain and stable
- **C** The company keeps introducing new products
- **D** There is a trend of price stability
- **E** The business is make-to-order, like ship building

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Fixed (static) budget: one level of activity, unchanged whatever is attained; for stable, non-seasonal businesses and fixed expenses.
- Flexible budget (CIMA): recognises fixed, semi-variable and variable costs and changes with the level of activity attained; a series of static budgets.
- Budgeted cost = fixed cost + variable cost per unit × activity; split semi-variables first (high and low points: change in expense ÷ change in output).

- Add the question's steps: price changes at a level, extra fixed costs beyond a capacity, vehicles or batches rounded up.
- Judge performance against the flexible budget for the actual level: a fixed budget's "saving" may be only lower volume.
- Never carry a cost per unit from one level to another: fixed costs do not shrink.
- Fixed vs flexible: change, levels, variance analysis, cost and price, comparison.
- Suitable where sales are seasonal, products or designs keep changing, work is made to order, fashion rules, or sales change generally. (Fuel for the distance you actually drove.)

Answer key · Check yourself

1 **60**

Variable cost per 1% of capacity = ₹90,000 ÷ 30 = ₹3,000; fixed = ₹2,40,000 − 40 × ₹3,000 = ₹1,20,000. ₹3,00,000 = ₹1,20,000 + ₹3,000 × level, so level = ₹1,80,000 ÷ ₹3,000 = 60%. Dividing ₹3,00,000 by the ₹6,000 per 1% at 40% (₹2,40,000 ÷ 40) gives 50%, as if the fixed cost moved with activity.

2 **C** ₹2,300

Vans are hired whole: 1,000 crates need 9 trips (8.33 rounded up) and 1,100 crates need 10 (9.17 rounded up). The rise is one more trip, ₹1,800, plus loading 100 × ₹5 = ₹500: ₹2,300. ₹3,320 treats the whole ₹33,200 as variable (10% more); ₹500 forgets the extra van; ₹1,800 forgets the loading.

3 **B** ₹84,000

Fixed ₹48,000 + 6,000 × ₹6 = ₹84,000. ₹72,000 applies the ₹12 rate per hour at 8,000 hours (₹96,000 ÷ 8,000) to 6,000 hours, as if the fixed cost shrank; ₹96,000 is the fixed budget; ₹36,000 forgets the fixed cost.

4 **A** **A soft-drinks maker whose sales peak every summer**

Seasonal fluctuation in sales or production is the first case in ICAI's list of suitability. The other three describe the conditions under which a fixed budget works.

5 **A** **The nature of the business is not seasonal**; **B** **The demand for the product is certain and stable**; **D** **There is a trend of price stability**

The fixed budget needs a non-seasonal business, stable demand, regular supply orders and inputs, no special labour or material, a mainly domestic market, no external impact and stable prices. New products and make-to-order work are reasons for a flexible budget.

15.4 Cash budget

The month-by-month plan of cash receipts and payments that shows in advance when cash will run short or pile up, laid out the way ICAI's answers lay it out.

In plain words

Twenty lakh of profit and a call from the bank

Kesar launched Diwali gift boxes this year, as he launches something every festival. They sold out.

> **Kesar:** ₹20 lakh of profit by the end of November, Dev. So why is the bank manager calling me?
>
> **Dev:** Because profit is a promise and cash is a fact. The distributors pay us sixty days after the sale. The sweet-makers wanted their money in the month we sold.
>
> **Ritu:** Per month, please. In October and November we paid out ₹22 lakh and ₹34 lakh. Not one rupee came in until December.

The figures: boxes worth ₹30 lakh were sold in October, ₹50 lakh in November and ₹10 lakh in December. The boxes cost 60% of their price and the suppliers were paid in the month of sale; rent and staff cost ₹4 lakh a month; distributors pay two months after the sale. The line started September with ₹5 lakh in the bank.

The profit statement records a sale when the box is sold. The bank account records it when the distributor pays. So at the end of November the gift-box line has earned ₹20 lakh of profit and has a bank balance of minus ₹55 lakh: an overdraft. The cash arrives in December and January, two months too late for the bank manager's mood.

A **cash budget** would have shown this in August. For each month it lists the cash that will come in (the **receipts**), the cash that will go out (the **payments**) and the balance left at the end. It counts only cash, and it counts it in the month the cash actually moves.

That is its whole value: a shortage seen three months early can be arranged (a bank limit, a delayed purchase, a push on collections); a shortage discovered on salary day cannot.

Think of it like

The fuel gauge on a road trip

The route map says Indore to Goa is a fine trip: that is the profit budget. The fuel gauge says on which stretch of highway the tank runs dry: that is the cash budget. A careful driver reads the gauge before leaving and plans the petrol pumps. A trip can be worth taking and still end on the shoulder of the highway with an empty tank.

Key idea

A cash budget is a month-by-month plan of cash receipts and cash payments: only cash, and only in the month it moves. It shows in advance when cash will be short (so borrowing is arranged in time) and when it will be in excess (so it can be invested).

How it works

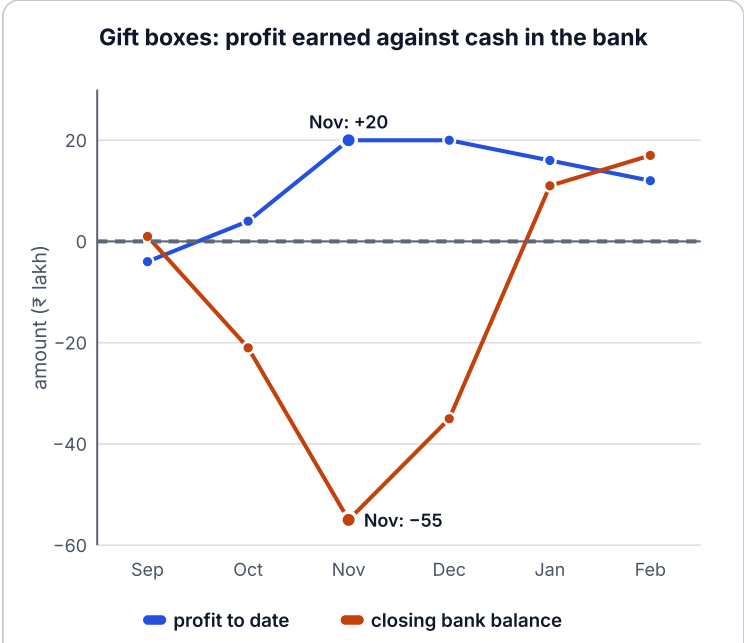
What ICAI means by a cash budget

The **cash budget** is a detailed budget of cash receipts and cash payments incorporating both revenue and capital items for the budget period (SM 15.54). It is usually in two parts:

- **Cash receipts**, estimated monthly: estimated cash sales, collections from debtors and anticipated receipts from other sources such as sale of assets and borrowings.
- **Cash disbursements**: estimated cash purchases, payments to creditors, employees' remuneration, bonus, advances to suppliers, budgeted capital expenditure for expansion, and so on.

It is the plan of receipts and payments of cash for the budget period, analysed to show the monthly flow of cash, drawn up in such a way that the balance can be forecasted at regular intervals.

The cash budget is the last of ICAI's fifteen functional budgets (SM 15.19) and one of the **financial budgets** of a budgetary control system, beside the capital expenditure budget and the budgeted balance sheet (SM 15.14). In the chapter's own MCQ it is the budget that is usually short-term (SM 15.74, key at 15.83). The gift boxes show why it is needed even when the profit budget looks healthy:

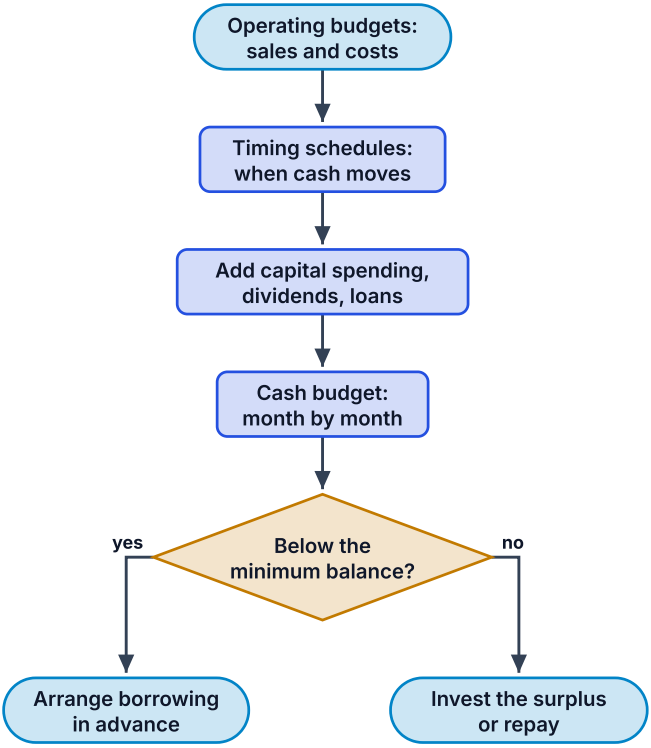


Blue: profit earned to date; orange: the bank balance at each month end. In November the line has earned ₹20 lakh and owes the bank ₹55 lakh, because sales are collected two months later while suppliers are paid in the month of sale. Only in January does the cash catch up.

Its numbers come from the other budgets: the sales budget, the direct materials purchases budget and the

selling and administrative expenses budget feed it, and so do the capital expenditure budget, dividend policies and plans for equity or long-term debt financing (SM 15.54–15.55; the capital expenditure budget "enables the cash budget to be completed", SM 15.46). The functional budgets are built in functional budgets and the master budget; this lesson re-times them into cash.

Where the cash budget's figures come from



The timing schedules are the working notes of every ICAI answer: collections from debtors, payments to creditors, wages and overheads, each moved to the month the cash moves. The closing balance of each month then goes into the budgeted balance sheet.

Seven steps from a sales forecast to a closing balance

1. **Set the columns**: one per month of the budget period (a quarter or six months in most questions).

2. **Receipts from sales.** Split sales into cash and credit. Cash sales are received in the month of sale. Credit sales are collected according to the collection pattern ("60% in the month after sale, 38% in the second month"): a working note spreads each month's credit sales over the months it is collected in. Any part written off as bad debts is never received.

3. **Other receipts**: sale of assets (the proceeds), loans raised, issue of shares or debentures, dividends from a subsidiary, and similar items, in the month they arrive.

4. **Payments for purchases.** Work out the purchases (questions often tie them to sales or production: "bought one month before sale", "50% of next month's sales"), split cash and credit, and apply the creditors' payment pattern in a second working note.

5 **Other payments**: wages (often part this month, part next), overheads (only the cash part, with their own lag), payments that fall only in some months (debenture interest, advance tax, GST), and special payments (dividends, loan repayments, capital expenditure, instalments on an asset).

6 **Leave out what is not cash**: depreciation, bad debts, provisions, and the profit or loss on selling an asset (only the proceeds are cash).

7 **Balance each month**: opening balance + receipts − payments = closing balance, which becomes next month's opening balance. Compare it with the minimum cash balance the business wants to keep: a shortfall is met by borrowing arranged in advance, a surplus can be invested or used to repay the borrowing.

Interactive version: PassBook app, lesson Cash budget

Cash, not profit: what goes in and what stays out

Item	In the cash budget?	When or how much
Credit sales	Yes	In the months collected

Item	In the cash budget?	When or how much
Bad debts	No	Never received
Cash discount allowed	Net amount	The cash actually received
Credit purchases	Yes	In the months paid
Depreciation, amortisation	No	Not a cash payment
Sale of a fixed asset	Proceeds only	Month of receipt
Asset bought on instalments	Each instalment	Month it is paid
Advance tax, interest, dividend	Yes	Month it is paid
Loan raised or repaid	Yes	Month it moves

Formats and formulas

ICAI's cash budget format

The Paper 6A Study Material prints this format, headed "... Co. Ltd., Cash Budget, Period ..." (FM Study Material 9.43–9.44); the suggested answer to RTP May 2026 Case Scenario 1 uses the same layout in Paper 4. *Reproduce it from memory.*

Particulars	Month 1 (₹)	Month 2 (₹)	Month 3 (₹)
Receipts:			
1. Opening balance	xxx	xxx	xxx
2. Collection from debtors	xxx	xxx	xxx
3. Cash sales	xxx	xxx	xxx
4. Loans from banks	xxx	xxx	xxx
5. Proceeds from issue of shares, debentures etc.	xxx	xxx	xxx
6. Miscellaneous receipts	xxx	xxx	xxx
7. Other items	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**
Payments:			
1. Payments to creditors	xxx	xxx	xxx
2. Wages	xxx	xxx	xxx
3. Overheads (factory rent, administration, selling and distribution)	xxx	xxx	xxx
4. Interest	xxx	xxx	xxx
5. Dividend	xxx	xxx	xxx
6. Corporate tax	xxx	xxx	xxx

Particulars	Month 1 (₹)	Month 2 (₹)	Month 3 (₹)
7. Capital expenditure	xxx	xxx	xxx
8. Other items	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**
Closing balance [Surplus (+) / Shortfall (–)]	**xxx**	**xxx**	**xxx**

The second layout: net cash flow first

Several Paper 4 suggested answers (MTP Series I for January 2025 Q4(b), MTP Series II for May 2025 Q5(b)) keep the opening balance out of the receipts:

Particulars	Month 1 (₹)	Month 2 (₹)
Total receipts (A)	xxx	xxx
Total payments (B)	xxx	xxx
Net cash flow (A – B)	**xxx**	**xxx**
Add: Opening balance	xxx	xxx
Closing balance	**xxx**	**xxx**

Both earn full marks. Pick one and never mix them: an opening balance counted inside the receipts *and* added again below is counted twice.

The working notes

Every cash budget answer stands on two schedules, shown as working notes:

Credit sales of	Amount (₹)	Collected in month 1	Collected in month 2
Month −1	xxx	xxx (2nd month %)	—
Month 0	xxx	xxx (1st month %)	xxx (2nd month %)
Month 1	xxx	—	xxx (1st month %)
Collection from debtors		**xxx**	**xxx**

The payments to creditors follow the same pattern, with purchases in place of sales.

∑ **Formula**

Closing balance

$$\text{Closing balance} = \text{Opening balance} + \text{Total receipts} - \text{Total payments}$$

Each month's closing balance is the next month's opening balance.

∑ **Formula**

Collection from debtors in a month

$$\text{Collection}_t = \sum_{k \geq 0} \text{Credit sales}_{t-k} \times p_k$$

p_k = the share of a month's credit sales collected k months after the sale ($k = 0$ is the month of sale). The shares add up to 100% less the bad debts.

∑ **Formula**

Collections and payments from opening and closing balances

$$\text{Receipts from debtors} = \text{Opening debtors}$$
$$+\text{Credit sales} - \text{Closing debtors}$$

$$\text{Purchases} = \text{Cost of goods sold}$$
$$+\text{Closing stock} - \text{Opening stock}$$

$$\text{Payment to creditors} = \text{Opening creditors}$$
$$+\text{Credit purchases} - \text{Closing creditors}$$

Used when a question gives budgeted balances instead of collection patterns (FM Study Material 9.48). Bad debts written off are also deducted from the receipts.

Formula

Surplus or deficit against a minimum balance

$$\text{Surplus or (deficit)} = \text{Opening balance} + \text{Receipts} - \text{Payments} - \text{Minimum cash balance}$$

A deficit is borrowed (or investments liquidated); a surplus is invested or repays the borrowing. The closing balance is then the minimum balance, or above it (FM Study Material 9.46).

Definition

Cash budget

A detailed budget of cash receipts and cash payments incorporating both revenue and capital items for the budget period. It represents the cash requirements of the business during the budget period: the plan of receipts and payments of cash, analysed to show the monthly flow of cash, drawn up in such a way that the balance can be forecasted at regular intervals. (SM 15.54)

Worked examples

Worked example

Chai premix, April to June: the full budget

Tapri's chai premix division gives you the following for

2026:

Month	Sales (₹)
February	40,00,000
March	45,00,000
April	50,00,000
May	60,00,000
June	55,00,000

1. 20% of sales are for cash. Of the credit sales, 60% are collected in the month after the sale and 38% in the second month after the sale; the rest is written off as bad debts.
2. Materials are bought in the month before the sale, at 50% of the sales value, all on credit. Creditors are paid in the second month after the purchase.
3. Wages are ₹6,00,000 a month; a wage revision raises them by 10% from 1 May. 80% of each month's wages are paid in that month and 20% in the next month.
4. Overheads are ₹8,00,000 a month, including depreciation of ₹1,50,000. They are paid in the month after they are incurred.
5. A packing machine costing ₹45,00,000 will be bought in April, payable in three equal instalments starting in the month after purchase.

6. An old delivery van with a book value of ₹3,00,000 will be sold for ₹2,50,000 in May.
7. Advance tax of ₹5,00,000 is payable in June.
8. The cash balance on 1 April 2026 is expected to be ₹3,00,000.

Required:

(i) Prepare a cash budget for April, May and June 2026.

(ii) Tapri keeps a minimum cash balance of ₹5,00,000. Identify any month in which the division needs finance, and how much.

THE WORKING

Which month's credit sales are collected when

Credit sales of	Amount (₹)	Apr (₹)	May (₹)	Jun (₹)
Feb	32,00,000	**12,16,000**	—	—
Mar	36,00,000	**21,60,000**	**13,68,000**	—
Apr	40,00,000	—	**24,00,000**	**15,20,000**
May	48,00,000	—	—	**28,80,000**
Collected		**33,76,000**	**37,68,000**	**44,00,000**

Working note 1. Credit sales are 80% of sales. Blue cells: 60% collected in the month after the sale; amber cells: 38% collected in the second month. The 2% bad debts (₹64,000 of February's, ₹72,000 of March's) never reach a column.

Working note 2: payments to creditors. Purchases of a month = 50% of the next month's sales, paid two months later. So April pays February's purchases (50% of March sales of ₹45,00,000); May pays March's (50% of ₹50,00,000); June pays April's (50% of ₹60,00,000).

Paid in	Purchases of	Amount (₹)
April	February	22,50,000
May	March	25,00,000
June	April	30,00,000

Working note 3: wages paid. Wages are ₹6,00,000 for March and April and ₹6,60,000 from May.

Month	80% now (₹)	20% of last (₹)	Paid (₹)
April	4,80,000	1,20,000	6,00,000
May	5,28,000	1,20,000	6,48,000
June	5,28,000	1,32,000	6,60,000

Working note 4: overheads. Cash overheads = ₹8,00,000 − ₹1,50,000 depreciation = ₹6,50,000 a month, each paid the following month. The machine instalments are ₹45,00,000 ÷ 3 = ₹15,00,000 in May, June and July.

(i) Cash budget for April to June 2026

Particulars	Apr (₹)	May (₹)	Jun (₹)
Receipts:			
Cash sales	10,00,000	12,00,000	11,00,000
Debtors (WN 1)	33,76,000	37,68,000	44,00,000
Sale of van	—	2,50,000	—
Total (A)	**43,76,000**	**52,18,000**	**55,00,000**
Payments:			
Creditors (WN 2)	22,50,000	25,00,000	30,00,000
Wages (WN 3)	6,00,000	6,48,000	6,60,000
Overheads (WN 4)	6,50,000	6,50,000	6,50,000
Machine	—	15,00,000	15,00,000
Advance tax	—	—	5,00,000
Total (B)	**35,00,000**	**52,98,000**	**63,10,000**
Net (A − B)	**8,76,000**	**(80,000)**	**(8,10,000)**
Opening balance	3,00,000	11,76,000	10,96,000
Closing balance	**11,76,000**	**10,96,000**	**2,86,000**

(ii) Finance needed. The closing balance stays above ₹5,00,000 in April and May. In June it falls to ₹2,86,000, so the division needs **₹2,14,000** (₹5,00,000 − ₹2,86,000) of short-term finance, arranged in advance for June. July brings the last machine instalment of ₹15,00,000, so Dev should look at July before choosing a one-month loan.

Notes: depreciation is left out (not cash); the van brings ₹2,50,000 in cash, and its ₹50,000 loss on sale does not appear; no July sales figure is needed, because the purchases made in June (for July's sales) are paid only in August.

Interactive version: PassBook app, lesson Cash budget

Worked example

Kulhad cafés: a minimum balance and an overdraft

Tapri's kulhad cafés in Pune expect these sales:

Month	Sales (₹)
June	20,00,000
July	18,00,000
August	16,00,000
September	22,00,000
October	30,00,000

1. 75% of sales are received in cash and UPI on the day; 25% are corporate catering orders, collected in the following month.
2. Purchases of milk, tea and snacks are 40% of each month's sales, half paid in the month of purchase and half in the following month.
3. Staff salaries are ₹5,00,000 a month, paid in the same month; a Diwali bonus of ₹3,00,000 is paid in October.
4. Rent and other expenses are ₹3,50,000 a month, including depreciation of ₹50,000, paid in the same month.
5. The fit-out of a new café, ₹8,00,000, is paid in August.

6. The cash balance on 1 July is ₹2,50,000. The cafés keep a minimum balance of ₹2,00,000. Any shortfall is met by a bank overdraft taken at the end of the month, repaid out of the first surplus above the minimum. Ignore interest.

Required: Prepare a cash budget for July to October, showing the overdraft taken and repaid.

THE WORKING

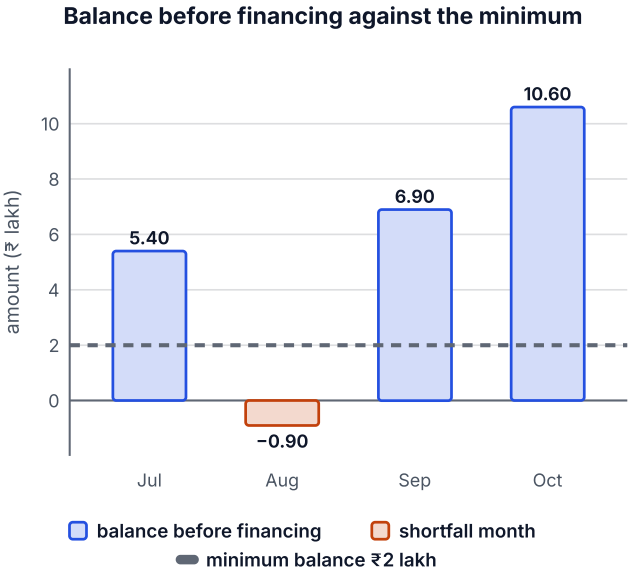
Working notes

1. Corporate collections (25% of last month's sales): July ₹5,00,000; August ₹4,50,000; September ₹4,00,000; October ₹5,50,000.
2. Purchases (40% of sales): June ₹8,00,000; July ₹7,20,000; August ₹6,40,000; September ₹8,80,000; October ₹12,00,000. Paid: July 3,60,000 + 4,00,000 = ₹7,60,000; August 3,20,000 + 3,60,000 = ₹6,80,000; September 4,40,000 + 3,20,000 = ₹7,60,000; October 6,00,000 + 4,40,000 = ₹10,40,000.
3. Cash part of rent and other expenses: ₹3,50,000 − ₹50,000 = ₹3,00,000 a month.

Cash budget, July to October

Particulars	Jul (₹)	Aug (₹)	Sep (₹)	Oct (₹)
Opening balance	2,50,000	5,40,000	2,00,000	4,00,000
Cash, UPI sales	13,50,000	12,00,000	16,50,000	22,50,000

Particulars	Jul (₹)	Aug (₹)	Sep (₹)	Oct (₹)
Corporate (WN 1)	5,00,000	4,50,000	4,00,000	5,50,000
Available (A)	**21,00,000**	**21,90,000**	**22,50,000**	**32,00,000**
Purchases (WN 2)	7,60,000	6,80,000	7,60,000	10,40,000
Salaries, bonus	5,00,000	5,00,000	5,00,000	8,00,000
Rent etc. (WN 3)	3,00,000	3,00,000	3,00,000	3,00,000
Café fit-out	—	8,00,000	—	—
Payments (B)	**15,60,000**	**22,80,000**	**15,60,000**	**21,40,000**
Before financing (A − B)	**5,40,000**	**(90,000)**	**6,90,000**	**10,60,000**
Overdraft taken (C)	—	2,90,000	—	—
Overdraft repaid (D)	—	—	2,90,000	—
Closing (A − B + C − D)	**5,40,000**	**2,00,000**	**4,00,000**	**10,60,000**



Blue bars: the balance before any financing; orange bar: August, the one month below the grey dashed minimum. August needs 2.90 lakh (from −0.90 up to 2.00); September's surplus repays it.

August's fit-out pushes the balance to minus ₹90,000, so the overdraft is ₹2,00,000 − (−₹90,000) = **₹2,90,000**: enough to restore the minimum, not merely to reach zero. September's ₹6,90,000 is ₹4,90,000 above the minimum, enough to repay all ₹2,90,000. Depreciation is left out throughout.

Exam angle

In the exam

Where the cash budget appeared, May 2024 to September 2026

In the Paper 4 documents read for this lesson, cash budgets came up in 4 documents, always as a numerical:

- **Descriptive.** A six-month cash budget for a toy maker, MTP Series I for January 2025 Q4(b) (8 marks): sales invoiced on the last day of the month and collected 50% one month and 50% two months later; toys produced two months before sale and materials paid two months after production; wages 75% in the month and 25% in the next, with a 25% rise from 1 March; variable overheads paid in the month after production and up 25% from April; a freehold property sold for ₹20,00,000 and a machine bought for ₹5,00,000 in June; depreciation given but not used; corporation tax in March. A three-month budget, MTP Series II for May 2025 Q5(b) (6 marks): collections of 80%, 10% and 8% in the three months after sale, the balance written off as bad debts; creditors 75% after one month and the balance after two; pallet jacks paid in four equal instalments "following the month of purchase".
- **Multiple choice, 2 marks each.** Sales receipts for one month from a cash and credit split, MTP Series II for September 2025 MCQ 15. A five-MCQ case scenario, RTP May 2026 Case Scenario 1: credit

sales twice cash sales, a 3% discount on cash sales, goods bought a month before sale at a 20% margin on selling price, 40% cash purchases, an advance tax instalment, and the closing cash balance.

- **Paper 6A** examines the same layout with a minimum balance, a short-term loan and investment of the excess: January 2025 Q1(a) (5 marks).
- **Theory.** No document read asked the objectives or advantages of the cash budget; they are in the Study Material (SM 15.55), set out in the Answer frame.

In the exam

The wording traps ICAI used

- "Invoiced on the last day of the month" and "paid one month after the sale": nothing is received in the month of sale.
- "Produced two months before they are sold, creditors paid two months after production" (MTP January 2025): the two lags cancel, and materials are paid in the month of sale. Draw the time line before writing a figure.
- A wage rise "from 1st March" applies to March's wages, so the 25% of March wages paid in April is at the new rate too.
- "Instalments following the month of purchase" (MTP May 2025): an order in July is first paid in August.
- Bad debts are the part of credit sales that never arrives: 80% + 10% + 8% leaves 2% that appears in

no month.

- RTP May 2026: a 3% discount on cash sales means the cash received is 97% of them; "20% profit margin on the selling price" makes purchases 80% of sales, not 100 ÷ 120. Its answer key prints ₹3,17,750 beside option (B) for the closing balance; option (B) and the working note both give ₹3,71,250.
- The paper's own instruction: "Working notes should form part of the answer."

Trick

The time-line row

Before any figure, write the months across the page and draw an arrow from each sale to the month it is collected, and from each purchase to the month it is paid. Every lag error is visible on the arrows. It works for any pattern of lags, as long as each month's collections are added from the arrows that land in it.

Answer frame

The theory parts

Objectives of preparing a cash budget (SM 15.55), in ICAI's order:

1. The probable cash position resulting from planned operations is assessed, so the excess or shortage of cash becomes clear; this helps in arranging short-term

borrowings in advance for a shortage, or investments when cash is in excess.

2. Cash can be coordinated in relation to total working capital, sales investment and debt.
3. A sound basis for credit and for current control of the cash position is established.
4. The effect of sudden and seasonal requirements, large stocks, delay in collection of receipts, etc. on the cash position of the organisation is revealed to the management.

Advantages of a cash budget (SM 15.55):

1. It aids in securing the optimum working capital needed for smooth running of the operations, and in planning payments to shareholders (the Study Material prints "option working capital").
2. It eases the strains of a cash shortage.
3. It facilitates temporary investment of cash wherever, and to whatever extent, it is found in excess.
4. It provides for normal growth.

"Define cash budget": the definition above, then its two parts (receipts and disbursements, with three examples of each) and that it is a financial budget prepared month by month.

The numerical: three answers to the same question

Loses marks

A single table of figures: "Collection from debtors, June: 44,00,000."

If one lag is wrong, nothing shows the method, and nothing earns method marks.

Passes

The cash budget with every line, but no working notes; depreciation correctly left out without a word.

The examiner cannot see whether 44,00,000 came from the right months.

Full marks

Working notes first (collections, creditors, wages, overheads), each showing which month's figure is paid when; then the budget in one consistent layout; a closing line on any shortfall against the minimum balance, and one sentence on items deliberately left out (depreciation, loss on sale).

Common mistakes

⚠ **Common mistake**

Hall of Shame: depreciation paid out of the bank

Treating ₹8,00,000 of overheads as cash in the premix example takes ₹1,50,000 a month out of the budget that never leaves the bank: June closes ₹4,50,000 too low. **Fix:** deduct depreciation (and any amortisation or provision) from the expense before it enters the payments.

⚠ **Common mistake**

The lag that slipped a month

"Collected in the month after sale" put into the month of sale, or "paid two months after purchase" put one month after. Every later month shifts with it. **Fix:** write the time line first, label each credit sale with the month it is collected, and check one figure by hand: April collects February's second instalment and March's first.

⚠ **Common mistake**

Collecting the bad debts anyway

Spreading 60% and 40% when the question says 60% and 38% and "the rest is written off". **Fix:** the collection percentages add up to less than 100%; the bad debts never become receipts.

⚠ **Common mistake**

The opening balance counted twice

Listing the opening balance as the first receipt and then adding it again below the net cash flow. **Fix:** choose one layout (opening balance inside the receipts, or net cash flow plus opening balance) and keep it for every month.

⚠ **Common mistake**

The asset's profit in place of its price

Entering the ₹50,000 loss on the van, or its ₹3,00,000 book value, as the receipt. **Fix:** a cash budget records the ₹2,50,000 actually received; profit or loss on sale is not cash.

Practice

Check yourself

1. A Tapri distributor's sales are ₹6,00,000 in January, ₹8,00,000 in February and ₹10,00,000 in March. 30% of sales are for cash, on which a 2% cash discount is allowed. Credit sales are collected 60% in the month after the sale and 37% in the second month; the rest is bad. What are the total cash receipts from sales in March, in rupees?

Answer: _____

2 Case scenario. Tapri Retail prices its packaged snacks at cost plus 25%. It keeps closing stock each month equal to half of the next month's cost of sales. All purchases are on credit and paid in the month after purchase. Sales are ₹8,00,000 in March, ₹10,00,000 in April, ₹12,00,000 in May and ₹9,00,000 in June. What is the payment to creditors in May?

- A ₹8,25,000
- B ₹8,80,000
- C ₹8,00,000
- D ₹8,40,000

3 Which of the following does not appear in a cash budget?

- A Proceeds from the sale of an old machine
- B An instalment paid on a new machine
- C Interest paid on a term loan
- D Depreciation on the machines

4 Debtors are ₹12,00,000 at the start of the quarter and are budgeted at ₹14,50,000 at its end. Credit sales for the quarter are ₹30,00,000, and ₹50,000 of debts will be written off as bad. What are the budgeted receipts from debtors for the quarter, in rupees?

Answer: _____

5 Which of these are objectives of preparing a cash budget as listed by ICAI? Select all that apply.

- **A** To assess the probable cash position, so that short-term borrowing or investment can be arranged in advance
- **B** To reveal the effect of seasonal requirements, large stocks and delayed collections on the cash position
- **C** To compute the profit of each month on an accrual basis
- **D** To coordinate cash with total working capital, sales, investment and debt

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Cash budget (SM 15.54): a detailed budget of cash receipts and cash payments, revenue and capital, for the budget period; two parts, receipts and disbursements; a financial budget, usually short-term and monthly.
- Only cash, only when it moves: credit sales in the months collected, purchases in the months paid; depreciation, bad debts and profit or loss on sale stay out.
- Working notes first: collections from debtors, payments to creditors, wages and overheads, each re-timed to the month of payment.
- Closing balance = opening balance + receipts − payments; it becomes next month's opening balance.
- Two layouts: opening balance inside the receipts (FM Study Material 9.43), or net cash flow plus opening balance; never both.
- Against a minimum balance: borrow the shortfall so the minimum is restored, repay or invest out of later surpluses.
- Objectives (SM 15.55): probable cash position (borrow or invest in time), coordination with working capital, basis for credit and cash control, effect of seasonal needs and delayed collections revealed.
- Profit is a promise, cash is a fact: read the fuel gauge before the trip.

Answer key · Check yourself

1 **785400**

Cash sales of March: 30% × ₹10,00,000 = ₹3,00,000, less the 2% discount: ₹2,94,000. February's credit sales, ₹5,60,000, are 60% collected: ₹3,36,000. January's credit sales, ₹4,20,000, are 37% collected: ₹1,55,400. Total ₹7,85,400; the 3% bad debts never arrive.

2 **B** **₹8,80,000**

At cost plus 25%, cost of sales is 100 ÷ 125 = 80% of sales: ₹6,40,000, ₹8,00,000, ₹9,60,000 and ₹7,20,000. May pays April's purchases = April's cost of sales + April's closing stock − April's opening stock = ₹8,00,000 + 50% × ₹9,60,000 − 50% × ₹8,00,000 = ₹8,80,000. ₹8,25,000 reads 25% as a margin on the selling price; ₹8,00,000 ignores the stock policy; ₹8,40,000 pays May's own purchases in May.

3 **D** **Depreciation on the machines**

Depreciation is a non-cash expense. The sale proceeds, the instalment and the interest are all cash moving in or out in a particular month.

4 **2700000**

Receipts = opening debtors + credit sales − bad debts written off − closing debtors = ₹12,00,000 + ₹30,00,000 − ₹50,000 − ₹14,50,000 = ₹27,00,000. Forgetting the bad debts gives ₹27,50,000: money that never arrives.

5 **A To assess the probable cash position, so that short-term borrowing or investment can be arranged in advance; B To reveal the effect of seasonal requirements, large stocks and delayed collections on the cash position; D To coordinate cash with total working capital, sales, investment and debt**

ICAI's four objectives (SM 15.55) are the three ticked and establishing a sound basis for credit and current control of the cash position. Profit on an accrual basis is the budgeted income statement's job; the cash budget ignores accruals.

15.5 Zero based budgeting and performance budgeting

How zero-based budgeting makes every activity justify its cost from the first rupee through ranked decision packages, and how performance budgeting ties money to functions, programmes, activities and results.

In plain words

The jingle nobody can hum

Budget season at Tapri. Kesar has a proposal for next year's marketing and training budget.

> **Kesar:** Last year's figure plus 10%. Simple. Sab adjust ho jayega.
>
> **Ritu:** Last year we paid ₹40 lakh for an FM radio jingle in 40 cities. Can anyone in this room hum it?
>
> **Dev:** *(after a silence)* The present value of that jingle is zero.
>
> **Ritu:** Then this year it starts at zero too. It gets money when it shows what the money buys.

Kesar's method is **traditional (incremental) budgeting**: start from last year's spending and argue about the increase. Ritu's is **zero-based budgeting (ZBB)**: every activity starts at zero, even one Tapri has run for years. Each activity is written up as a **decision package** (what it is, what it costs, what it delivers), the packages are judged

and ranked, and the money goes down the ranking until it runs out. An activity that cannot justify itself gets nothing.

Across town, the Malwa Nagar Nigam (the municipal corporation that licenses Tapri's cafés) has a different problem. Its street-lighting budget says "electricity ₹3.60 crore, salaries ₹1.20 crore, materials ₹0.80 crore, contractors ₹1.40 crore", and no councillor can tell what the city gets for ₹7 crore. A **performance budget** presents the same money by function, programme and activity, with the output each must deliver: how many lights kept burning, how many new lights put up, and what each light costs. The money is tied to results, and somebody is answerable for each result.

Think of it like

Three ways to pack for a trip

The incremental packer opens last trip's suitcase, leaves everything in and adds 10% more. The zero-based packer starts with an empty suitcase: every item must argue its way in, most important first, until the bag is full, and the snow boots stay home for a trip to Goa. The performance packer labels each item with the job it will do: "umbrella: dry on three rainy days".

Key idea

ZBB: every activity, old or new, must justify its cost from the first rupee; without justification the budget allowance is zero, and justified decision packages are ranked and funded in order of priority. Performance budgeting: the budget is presented in terms of functions, programmes and activities, linking the money (inputs) to the results expected (outputs), with responsibility fixed for those results.

How it works

Zero-based budgeting: the definition ICAI marks

Definition

Zero-based budgeting (ZBB)

A method of budgeting which requires each cost element to be specifically justified, though the activities to which the budget relates are not being undertaken for the first time. The cost of each activity has to be justified and without justification, the budget allowance is zero. (SM 15.59)

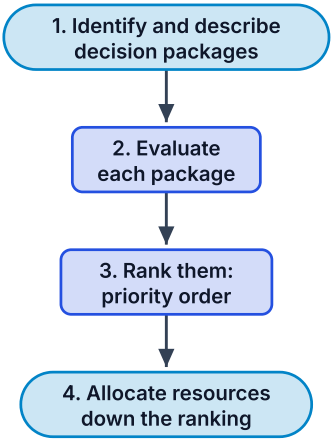
ZBB differs from the conventional system because it starts from scratch, not from trends or historical levels of expenditure. In the customary system, last year's figures are accepted as they are, or cut back, or increases are granted. ZBB starts from the premise that next period's budget is zero so long as the demand for a function,

process, project or activity is not justified for each rupee, from the first rupee spent (SM 15.59). Its other features, in the Study Material's order:

- It is an **emergent form of budgeting** that arises to overcome the limitations of incremental (traditional) budgeting.
- It is an **activity-based budgeting system**: budgets are prepared for each activity rather than for each functional department. Justification in the form of cost-benefit for the activity is required; the activities are then evaluated and prioritised by management on factors like **synchronisation with organisational objectives, availability of funds and regulatory requirement**.
- It suits **corporate and non-corporate entities** alike. Government departments, local bodies and not-for-profit organisations need to justify the benefits of spending on social programmes such as the mid-day meal, street lights or drinking water.
- In a company it is **best suited to discretionary costs**: research and development, training programmes, advertisement.

Four stages from a wish-list to a budget

The stages in zero-based budgeting (SM 15.60)



Stage 1 writes each programme or activity up with its specification, cost-benefit and other issues; stage 2 tests it against objectives, funds and regulatory requirements; stage 3 ranks it; stage 4 funds the ranked packages, as if budgeting for the first time.

1. **Identification and description of decision packages.** Decision packages are the programmes or activities on which a decision is required. Each is described for its technical specifications, its financial impact in the form of a cost-benefit analysis, and other issues such as environmental, regulatory and social ones.
2. **Evaluation of decision packages.** Each package is evaluated against factors like synchronisation with

organisational objectives, availability of funds and regulatory requirement.

3. **Ranking (prioritisation) of decision packages.** The evaluated packages are ranked by the priority of the activities. Because of this feature ZBB is also known as **priority-based budgeting**.
4. **Allocation of resources.** Resources are allocated to the ranked packages. Budgets are prepared as if for the first time, without reference to previous budgets.

Why bother: the advantages

The advantages of ZBB, in ICAI's order (SM 15.60–15.61):

1. It provides a **systematic approach** for evaluating different activities and ranking them in order of preference for the allocation of scarce resources.
2. It ensures that the functions undertaken by the organisation are **critical for the achievement of its objectives** and are being performed in the best possible way.
3. It gives management the opportunity to allocate resources only after a **thorough cost-benefit analysis**, so arbitrary cuts and enhancements are avoided.
4. Areas of **wasteful expenditure** can be easily identified and eliminated.
5. Departmental budgets are **closely linked with corporate objectives**.
6. It can be used to introduce and implement **management by objectives**, so it serves the objectives

of traditional budgeting and a variety of other purposes too.

The Study Material's separate list of why "zero-based budgeting is superior to traditional budgeting" (SM 15.61) repeats these points in short form: systematic evaluation, critical functions, cost-benefit allocation, wasteful expenditure eliminated with departmental budgets linked to corporate objectives, and management by objectives.

ZBB against traditional budgeting

Point
Orientation

Traditional budgeting
Accounting oriented; stress on the previous level of expenditure

Zero-based budgeting
Decision oriented and rational; all programmes, old and new, compete for scarce resources

Point
Starting point

Traditional budgeting
Past spending first, then demands for inflation and new programmes

Zero-based budgeting
Only the decision packages that enjoy priority over others

Point

Padding

Traditional budgeting

Some managers inflate requests so that after the cuts they still get what they want

Zero-based budgeting

Rational analysis of proposals; inflated requests are likely to be caught and exposed; approval only for result-oriented packages

Point

Clarity

Traditional budgeting

Not as clear and responsive

Zero-based budgeting

Clearer and more responsive

Point

Who justifies

Traditional budgeting

Top management decides why an amount is spent on a decision unit

Zero-based budgeting

The responsibility shifts to the manager of the decision unit

Point

Approach

Traditional budgeting

Routine

Zero-based budgeting

Straightforward; spotlights the packages with priority

(SM 15.61–15.62)

The catch: limitations

ZBB has two limitations in the Study Material (SM 15.62):

1. Creating the decision packages and ranking them has to be done on the basis of new data, a process that is **very tedious** for management.
2. The activities selected for ZBB are based on the traditional functional departments, so the scheme **may not be implemented properly**.

The first is why ZBB is never "the least time-consuming and least costly approach to budgeting".

Performance budgeting: money tied to results

Definition

Performance budgeting (PB)

Performance budgeting involves evaluation of the performance of an organisation in the context of both specific as well as overall objectives of the organisation. A performance budget is one which presents the purposes and objectives for which funds are required, the costs of the programmes proposed for achieving those objectives, and quantitative data measuring the accomplishments and work performed under each programme. (SM 15.62–15.63)

What follows from it (SM 15.63, 15.71):

- It needs complete clarity about both the short-term and the long-term objectives of the organisation.
- The responsibility of the various levels of management is **predetermined in terms of results** expected from them and the authority vested in them. PB fixes the responsibility of each executive and appraises performance continuously, so it is considered **synonymous with responsibility accounting**.
- It provides a meaningful relationship between estimated inputs and expected outputs as an integral part of the budgeting system.
- It is a technique of presenting budgets for costs and revenues **in terms of functions, programmes and activities**, correlating the physical and the financial aspects of the items in the budget.

Traditional budgeting against performance budgeting (SM 15.63):

Point	Traditional budgeting	Performance budgeting
Emphasis	The financial aspect, more than the physical aspect or performance	A relationship between inputs and outputs

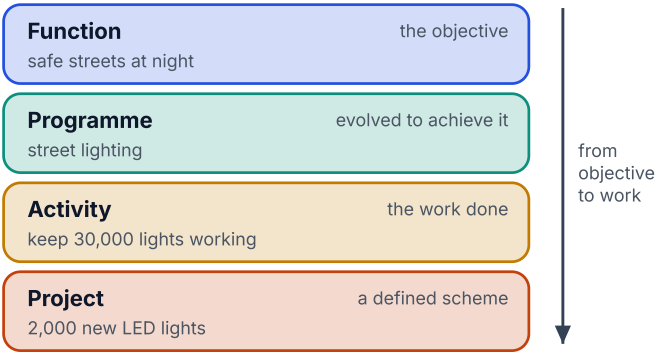
Point	Built around
Traditional budgeting	Objects or items of expenditure: salaries, stores and materials, rates, rents and taxes
Performance budgeting	Functions, the programmes that discharge them, and the activities those programmes involve

The ARC's sequence: function, programme, activity, project

According to the Administrative Reforms Commission (ARC), the basic steps in PB are (SM 15.63–15.64):

1. Establishing a meaningful functional, programme and activity classification of government operations.
2. Bringing the system of accounting and financial management in line with this classification.
3. Evolving suitable norms, yardsticks, work units of performance and unit costs, wherever possible, under each programme and activity, for their reporting and evaluation.

The ARC's terms, in their integrated sequence (SM 15.64)



The ARC uses the four terms in this order; "function" is used in the sense of "objective", and programmes are evolved to achieve it (SM 15.64). The small text under each term is the Malwa Nagar Nigam's street lighting; the right-hand words for activity and project are this lesson's gloss, not ICAI's wording.

In time horizon, PB is essentially a replacement of the traditional annual fiscal budget by a more output-oriented, but still annual, exercise (SM 15.64).

Worked examples

Worked example **Kesar's plus-ten-per-cent budget**

Tapri's marketing, training and product-development spending was ₹1.36 crore in 2025-26, spread over a radio jingle, digital ads, staff training and two new-product trials. For 2026-27 Kesar proposes "last year's figure plus 10% for every head". Ritu proposes zero-based budgeting instead. The board asks her for a note.

(a) Explain zero-based budgeting, and how each of its stages would deal with these costs.

(b) State whether ZBB suits these costs, giving reasons.

(c) State one limitation the board should expect.

MODEL ANSWER

(a) Zero-based budgeting. Zero-based budgeting is a method of budgeting which requires each cost element to be specifically justified, though the activities to which the budget relates are not being undertaken for the first time. The cost of each activity has to be justified; without justification, the budget allowance is zero. Unlike the traditional (incremental) system, in which last year's figures are accepted as they are, cut back or increased, ZBB starts from scratch, not from trends or historical levels of expenditure. It is an activity-based budgeting system: budgets are prepared for each activity rather than for each functional department, with justification in the form of cost-benefit, and the activities are evaluated and prioritised on factors like synchronisation with organisational objectives, availability of funds and regulatory requirement.

ZBB involves the following stages, applied here to Tapri's costs:

1. **Identification and description of decision packages**: the programmes or activities on which a decision is required, described for

technical specifications, financial impact (cost-benefit analysis) and other issues such as environmental, regulatory and social ones. The jingle, the digital ads, each training programme and each product trial becomes a separate package stating what it is, what it costs and what it is expected to deliver.

2. **Evaluation of decision packages** against synchronisation with organisational objectives, availability of funds, regulatory requirement, etc. A training programme that the food licences require would pass on regulatory grounds; the jingle must show a benefit for its cost.
3. **Ranking (prioritisation) of decision packages** by the priority of the activities; hence ZBB is also called priority-based budgeting.
4. **Allocation of resources** to the ranked packages, the budget being prepared as if for the first time, without reference to previous budgets: the ₹1.36 crore of 2025-26 is not the starting point, and packages below the cut-off receive nothing.

(b) Suitability. Marketing, training and product development are discretionary costs: their level is decided by management, not by the volume of output. In a corporate entity ZBB is best suited to discretionary costs like research and development, training programmes and advertisement. Kesar's method would carry every past activity forward, including ones that no longer serve Tapri's objectives (such as the radio jingle), and add 10% to

each. ZBB would require each activity to justify its cost and would fund only the ranked packages, so wasteful expenditure is identified and eliminated and the budget is linked to corporate objectives. ZBB is therefore suitable.

(c) Limitation. Creating the decision packages and ranking them has to be done on new data each time, which is very tedious for management. (Alternatively: the activities selected for ZBB are based on the traditional functional departments, so the scheme may not be implemented properly.)

Worked example **Ranking the decision packages**

Ritu's committee has ₹150 lakh for Tapri's 2026-27 growth budget. Six decision packages were submitted, evaluated and ranked as follows:

Rank	Package	Cost (₹ lakh)	Evaluation
1	A: Hygiene training for café staff	18	Required by the food licences

Rank
2
Package
B: Digital ads for premix, 10 new cities
Cost (₹ lakh)
45
Evaluation
Extra contribution ₹90 lakh

Rank
3
Package
F: Barista upskilling
Cost (₹ lakh)
15
Evaluation
Saves ₹25 lakh of wastage

Rank
4
Package
D: R&D on a sugar-free premix
Cost (₹ lakh)
35
Evaluation
Fits the 2027 health-range objective

Rank
5
Package
C: Printed wall calendars for distributors (run since 2015)
Cost (₹ lakh)
40
Evaluation
No measured sales effect

Rank
6

Package
E: Celebrity endorsement

Cost (₹ lakh)
60

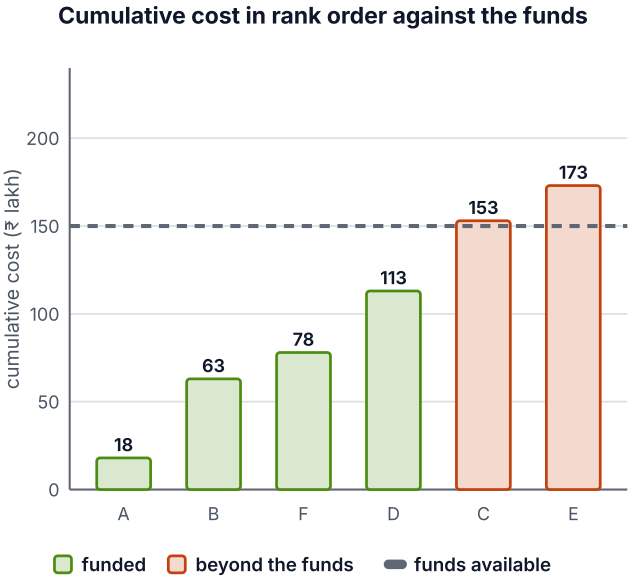
Evaluation
Extra contribution ₹45 lakh

Required:

(i) Allocate the ₹150 lakh under ZBB, and state the amount left unallocated.

(ii) Kesar asks for the calendars to receive the amount left over, "since they were there last year". Advise.

THE WORKING



Each bar is the total cost of the packages funded so far plus that package. Green: packages that fit under the grey dashed ₹150 lakh line; orange: C and E, each of which would take the total past it. The cut-off falls after rank 4 (D).

(i) Allocation in rank order

Rank	Package	Cost (₹ lakh)	Cumulative (₹ lakh)	Decision
1	A	18	18	Funded
2	B	45	63	Funded
3	F	15	78	Funded
4	D	35	113	Funded
5	C	40	153	Not funded
6	E	60	173	Not funded

Funds are allocated down the ranking. After D, ₹150 – ₹113 = **₹37 lakh** remains; C needs ₹40 lakh and E ₹60 lakh, so neither can be funded (each cumulative figure adds the package to the ₹113 lakh actually allocated). **₹113 lakh is allocated to A, B, F and D; ₹37 lakh is left unallocated.**

(ii) Advice. Under ZBB the budget allowance of an activity is zero unless it is justified, though the activity is not being undertaken for the first time. "They were there last year" is the traditional approach of referring first to past spending. Package C ranks fifth with no measured sales effect, and its cost exceeds the funds left. The ₹37 lakh should not be given to C on the strength of history. If the sales team submits a revised package

within ₹37 lakh with a measurable benefit, it can be evaluated and ranked like any other; otherwise the amount stays unallocated.

Worked example **Malwa Nagar Nigam's street lights**

The Malwa Nagar Nigam's traditional budget for street lighting in 2026-27 is:

Item of expenditure	Amount (₹ lakh)
Electricity charges	360
Salaries and wages	120
Materials and spares	80
Contract charges	140
Total	**700**

The city has 30,000 street lights. The salaries and ₹30 lakh of the materials maintain them; the electricity lights them; ₹50 lakh of materials and all the contract charges are for 2,000 new LED lights in new colonies.

Required:

(a) Recast the budget as a performance budget.

(b) State how it differs from the traditional budget.

(c) State what the Nigam must ensure to adopt performance budgeting.

MODEL ANSWER

(a) Performance budget for street lighting, 2026-27

Function: safe movement on the city's streets at night. Programme: street lighting.

Activity or project	Physical target	Cost (₹ lakh)	Unit cost (₹)
Energy for existing lights	30,000 lights lit	360	1,200 a light
Maintenance of existing lights	30,000 lights kept working	150	500 a light
New LED lights (project)	2,000 lights installed	190	9,500 a light
Total		**700**	

Working: maintenance = salaries ₹120 lakh + materials ₹30 lakh = ₹150 lakh; new lights = materials ₹50 lakh + contract charges ₹140 lakh = ₹190 lakh. Unit costs: ₹360 lakh ÷ 30,000; ₹150 lakh ÷ 30,000; ₹190 lakh ÷ 2,000. Performance indicators could include the share of lights found

working at night inspections and the time taken to repair a reported fault.

(b) Differences. The traditional budget gives more emphasis to the financial aspect than to the physical aspect or performance; the performance budget establishes a relationship between inputs and outputs, showing what each lakh is to achieve. The traditional budget is built around objects or items of expenditure (electricity, salaries, materials, contract charges); the performance budget around functions, the programmes that discharge them and the activities those programmes involve.

(c) For an enterprise to adopt performance budgeting, it is imperative that:

1. the objectives of the enterprise are spelt out in concrete terms;
2. the objectives are translated into specific functions, programmes, activities and tasks for the different levels of management, within the realities of fiscal constraints;
3. realistic and acceptable norms, yardsticks or standards and performance indicators are evolved and expressed in quantifiable physical units;
4. a style of management based upon a decentralised responsibility structure is adopted; and
5. an accounting and reporting system is developed to facilitate monitoring, analysis and

review of actual performance in relation to budgets.

Exam angle

In the exam

Where ZBB and PB appeared

ZBB or performance budgeting came up in 3 of the new-scheme documents read for this lesson (May 2024 to September 2026) and in 6 old-scheme documents (2019 to 2023):

- **Explain ZBB with its stages, 5 marks**: MTP Series I for May 2026 Q5(b), answered with the definition and the four stages. Old scheme: November 2019 Q6(e) and MTP Series I for December 2021 Q6(c) ("define ZBB and mention its various stages"); MTP Series I for May 2020 Q6(b) (stages only).
- **Advantages of ZBB**: MTP Series I for May 2023 Q6(a) (5 marks) and RTP November 2020 Q15(c), answered with the six advantages in ICAI's order.
- **Multiple choice, 2 marks**: MTP Series I for May 2024 MCQ 5 asked for the false statement about ZBB; the false one called it the least time-consuming and least costly approach.
- **Performance budgeting**: a definition worth one mark of a five-term question in September 2024 Q6(b) (5 marks), answered with the Study Material's paragraph; old scheme November 2020 Q6(b) (5

marks), "what are the important points an organisation should consider if it wants to adopt performance budgeting?", answered with the five imperatives.

In the exam

The wording traps

- ICAI's definition says "**though the activities to which the budget relates are not being undertaken for the first time**": ZBB re-justifies continuing activities. The old-scheme suggested answers to November 2019 Q6(e) and MTP Series I for December 2021 Q6(c) print the clause without "not"; write the Study Material's wording (SM 15.59, 15.71).
- Read the verb. For "define ZBB and mention its stages" or "briefly explain ZBB along with its steps" (MTP May 2026), ICAI's answers give the definition, its description and the four stage names in order; for "explain the stages" (MTP May 2020) the answer explains each stage in a line or two.
- "Points to consider to adopt performance budgeting" wants the five imperatives, not the ARC's three steps.

Trick

Stages: "I E R A"

Identify, **E**valuate, **R**ank, **A**llocate: identification and description of decision packages, evaluation, ranking, allocation of resources, in ICAI's order. For the ARC's sequence: **F**unction, **P**rogramme, **A**ctivity, **P**roject, from the objective down to the work.

Answer frame

"Explain zero-based budgeting and its stages" (5 marks)

1. The definition, word for word, including "though the activities ... are not being undertaken for the first time" and "without justification, the budget allowance is zero".
2. How it differs from traditional budgeting: from scratch, not from trends or historical levels; last year's figures are not accepted, cut or increased.
3. Activity-based; cost-benefit justification; evaluated and prioritised on objectives, funds and regulatory requirement.
4. Where it suits: corporate and non-corporate entities; social programmes; discretionary costs.
5. The four stages in ICAI's order: names only for "mention", a line on each for "explain".

"Distinguish between traditional budgeting and ZBB"

The six points of the table above (SM 15.61–15.62), written as a table, each row a pair of statements: traditional on one side, ZBB on the other.

"What should an enterprise consider to adopt performance budgeting?"

The five imperatives of the third worked example (objectives in concrete terms; translated into functions, programmes, activities and tasks; norms and indicators in physical units; decentralised responsibility; an accounting and reporting system for review).

Three answers to "Explain ZBB"

Loses marks

"In ZBB the budget starts from zero every year."

One line; no definition, no justification, no stages.

Passes

"ZBB requires each cost element to be justified for activities undertaken for the first time. Stages: identify, evaluate, rank, allocate."

The stages are there, but the definition has lost its "not", which reverses its meaning, and nothing is explained.

Full marks

Definition word for word, with "not" and "without justification, the budget allowance is zero"; the contrast with incremental budgeting; activity-based with cost-benefit justification; where it suits; the four stages in ICAI's order, each with its line when the question says "explain"; "priority-based budgeting" named at the ranking stage.

Common mistakes

⚠ **Common mistake**

Hall of Shame: "zero-based means a zero budget"

ZBB does not cut spending to zero. It starts each activity's allowance at zero and funds every package that is justified and ranked within the funds available. **Fix:** "zero" is the starting point of the argument, not the result.

⚠ **Common mistake**

ZBB "for new activities only"

Writing that ZBB applies to activities being undertaken for the first time. **Fix:** the definition says the opposite: each cost element is justified **though** the activities are **not** being undertaken for the first time. Continuing activities are the whole point.

⚠ **Common mistake**

Calling ZBB quick and cheap

Fix: creating decision packages and ranking them on new data is very tedious for management: that is ICAI's first limitation, and the false option in ICAI's own MCQ.

⚠ **Common mistake**

Performance budget, performance report, flexible budget

Answering "performance budgeting" with a report comparing actual and budget, or with a budget at several levels of activity. **Fix:** a performance budget presents funds by functions, programmes and activities with the physical results each must achieve, and fixes responsibility for those results.

Practice

Check yourself

1 Kesar agrees to zero-based budgeting "for the new cold-brew kiosks only. The 40 old cafés have run for years, so they keep last year's budget plus 10%." Which response is correct?

- **A** He is right: ZBB is meant for activities being undertaken for the first time
- **B** He is wrong: under ZBB each cost element is justified though the activities are not being undertaken for the first time, so the old cafés' costs must be justified too
- **C** He is right: ZBB needs a past level of expenditure to start from, and the kiosks have none
- **D** He is wrong: ZBB requires every café's budget to be cut to zero

2 A hospital trust has ₹100 lakh for new programmes. Its ranked decision packages cost, in rank order, ₹25 lakh, ₹40 lakh, ₹30 lakh, ₹50 lakh and ₹20 lakh. Funds are allocated strictly down the ranking. How much is left unallocated, in ₹ lakh?

Answer: _____

3 Case scenario. A state irrigation department has always presented its budget as "salaries ₹X, diesel ₹Y, cement ₹Z". Its new secretary wants a budget that shows the hectares to be irrigated under each programme and the cost per hectare, with each engineer answerable for the targets. Which technique is she asking for?

- **A** Performance budgeting
- **B** Zero-based budgeting
- **C** Flexible budgeting
- **D** Fixed budgeting

4 In what order does the Administrative Reforms Commission use its terms for performance budgeting?

- **A** Programme, function, project, activity
- **B** Activity, programme, function, project
- **C** Project, activity, programme, function
- **D** Function, programme, activity, project

5 Which of these are advantages of ZBB in ICAI's list? Select all that apply.

- **A** Areas of wasteful expenditure can be identified and eliminated
- **B** Resources are allocated only after a thorough cost-benefit analysis
- **C** Budgets can be prepared quickly from last year's figures
- **D** It can help introduce management by objectives

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- ZBB (SM 15.59): each cost element specifically justified, though the activities are not being undertaken for the first time; without justification the budget allowance is zero.
- Starts from scratch, not from trends; activity-based; cost-benefit justification; prioritised on objectives, funds and regulatory requirement.
- Suits corporate and non-corporate entities (mid-day meal, street lights, drinking water); in companies, discretionary costs (R&D, training, advertisement).

- Stages: identify and describe decision packages, evaluate, rank (hence priority-based budgeting), allocate resources.
- Six advantages: systematic evaluation, critical functions, cost-benefit allocation, waste eliminated, departmental budgets linked to corporate objectives, management by objectives. Two limitations: tedious, may not be implemented properly.
- Performance budgeting: funds presented by functions, programmes and activities with quantitative data on accomplishments; links inputs to outputs; synonymous with responsibility accounting.
- ARC: function, programme, activity, project; PB is still annual but output-oriented. To adopt it: concrete objectives, translated into functions to tasks, physical norms and indicators, decentralised responsibility, a reporting system.
- The jingle gets money when it shows what the money buys.

Answer key · Check yourself

1 **B** **He is wrong: under ZBB each cost element is justified though the activities are not being undertaken for the first time, so the old cafés' costs must be justified too**

ICAI's definition says each cost element is justified though the activities are not being undertaken for the first time (SM 15.59): continuing activities must justify themselves too. ZBB starts from zero, not from past spending, and "zero" is where the justification starts, not where the budget ends.

2 **5**

25 + 40 + 30 = 95. The fourth package needs 50 and only 5 remains; the fifth needs 20, which does not fit either. ₹5 lakh is left unallocated.

3 **A** **Performance budgeting**

Presenting funds by programme with physical output and unit costs, and fixing responsibility for the results, is performance budgeting. ZBB would ask each activity to justify its cost from zero; a flexible budget changes with the level of activity.

4 **D** **Function, programme, activity, project**

Function (in the sense of objective), programme (evolved to achieve it), activity, project (SM 15.64).

5 **A** Areas of wasteful expenditure can be identified and eliminated; **B** Resources are allocated only after a thorough cost-benefit analysis; **D** It can help introduce management by objectives

ICAI's six advantages include the three ticked (SM 15.60–15.61). Building on last year's figures is traditional budgeting, the very thing ZBB replaces.

15.6 Control ratios, budget variances, motivation, feedback and feed-forward

The six budget ratios that split a shortfall into speed, hours and days; how to read a budget variance; and how feedback, feed-forward and well-set targets make budgetary control work.

In plain words

Fast cooks, short month

Ritu's monthly report on the Dewas samosa line lands on Kesar's desk.

> **Kesar:** Ten per cent fewer samosas than the budget. Who was sleeping?
>
> **Ritu:** Nobody. Power cuts took 2,000 of the 10,000 hours we planned to work. In the hours we did work, the cooks were faster than standard.
>
> **Kesar:** Faster, and still short?
>
> **Ritu:** Faster by 12.5%, short of hours by 20%. Per hour, please, Kesar-ji.

The line had budgeted 10,000 hours. It actually worked 8,000. In those 8,000 hours it made samosas that should take 9,000 hours at the standard speed: its output was

worth 9,000 **standard hours**. Three comparisons tell the whole story:

- speed: 9,000 standard hours earned in 8,000 actual hours is 112.5%, the **efficiency ratio**;
- time: 8,000 of the 10,000 budgeted hours worked is 80%, the **capacity ratio**;
- output: 9,000 standard hours against the 10,000 budgeted is 90%, the **activity ratio**.

And 112.5% × 80% = 90%. These **control ratios** (ICAI's **budget ratios**) show how far actual performance is from the budget and why. A ratio of 100% or more is favourable; below 100%, unfavourable.

Next month the report shows oven maintenance ₹25,000 under budget, marked "favourable". Ritu is not pleased: the ovens were never serviced. A **budget variance** (actual against budget) means something only against the right yardstick and with its cause.

Two more questions decide whether all this helps. When is the comparison made: after the year, as a post-mortem (**feedback control**), or all through it, so the course can be corrected in time (**feed-forward control**)? And will the managers chase the targets at all? That depends on how the targets were set: **budgets and motivation**.

Think of it like

A rain-hit cricket chase

The plan: 50 overs. Rain allows only 40: the capacity ratio is 80%. The batters score 270, a total a par side needs 45 overs for: in 40 overs that is 112.5% efficiency. Against the 50-over plan the side delivered 45 overs' worth: the activity ratio is 90%. Feedback is the post-match review; feed-forward is checking the required run rate every over and changing the batting order before the match is lost. And a target the team believes it can chase is the one it plays hardest for.

Key idea

Activity ratio = efficiency ratio × capacity ratio: output against budget is speed times time. A budget variance is read against a budget flexed to the actual activity, with its cause and its controllability. Control works best when actuals are compared during the period (feed-forward) against targets the managers accept as achievable.

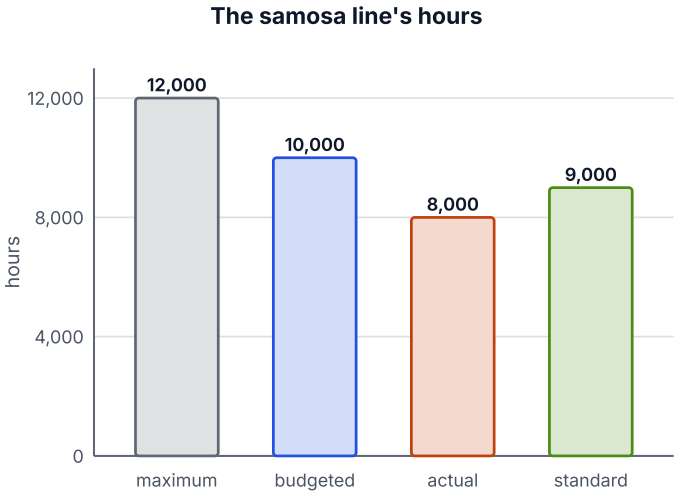
How it works

Four kinds of hours, one kind of day

Every control ratio compares two of these (the samosa line's figures are in the chart below; its full capacity

could work 12,000 hours):

- **Maximum possible hours**: the hours the full capacity could work in the period (maximum employees × hours a day × days).
- **Budgeted hours**: the hours planned to be worked in the budget period. In ICAI's illustration they are the employees "actual working" (budgeted to work) × hours a day × days (SM 15.69).
- **Actual hours** worked.
- **Standard hours for actual output**: the output achieved × the standard time per unit ("standard hours earned").
- **Budgeted working days** against the **available (actual) working days**.



Maximum possible 12,000 (grey), budgeted 10,000 (blue), actual 8,000 (orange), standard hours for the actual output 9,000 (green). Efficiency = green ÷ orange = 112.5%; activity = green ÷ blue = 90%; capacity = orange ÷ blue = 80%; standard capacity usage = blue ÷ grey = 83.33%; actual capacity usage = orange ÷ grey = 66.67%.

The six ratios ICAI lists

Budget ratios provide information about the performance level: the extent of deviation of actual performance from the budgeted performance, and whether it is favourable or unfavourable. If the ratio is 100% or more, the performance is considered favourable; below 100%, unfavourable (SM 15.67).

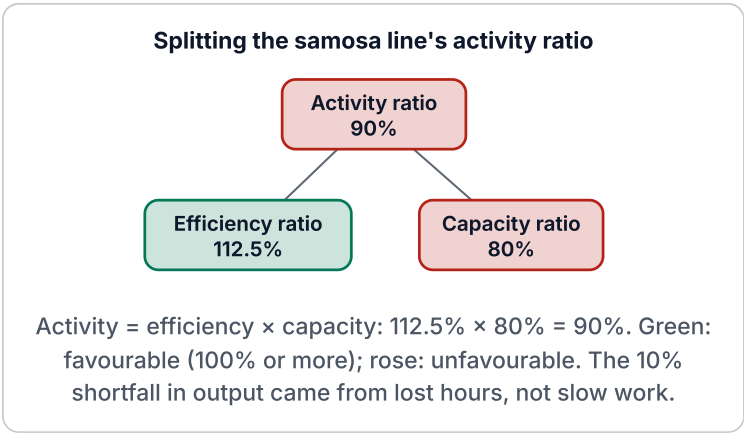
Ratio	Compares	It answers
Efficiency	Standard hours ÷ actual hours	Were the hours worked productive?
Activity	Standard hours ÷ budgeted hours	Did output reach the budget?
Calendar	Available days ÷ budgeted days	Were the budgeted days available?
Standard capacity usage	Budgeted hours ÷ maximum hours	How much capacity did the budget plan to use?
Actual capacity usage	Actual hours ÷ maximum hours	How much capacity was used?
Actual usage of budgeted capacity	Actual hours ÷ budgeted hours	Were the budgeted hours worked?

The last one is the ratio ICAI's MTPs call the **capacity ratio** or **capacity utilisation ratio** (actual hours ÷ budgeted hours). The formulas, each × 100, are in Formats and formulas.

Speed times time

Because standard ÷ budgeted = (standard ÷ actual) × (actual ÷ budgeted), the activity ratio is always the product of the other two. The chapter's own MCQ puts it the other way round: the efficiency ratio is the activity

ratio divided by the capacity ratio (SM 15.73, key at 15.83).



Activity = efficiency × capacity: 112.5% × 80% = 90%. Green: favourable (100% or more); rose: unfavourable. The 10% shortfall in output came from lost hours, not slow work.

Budget variances: reading the gap

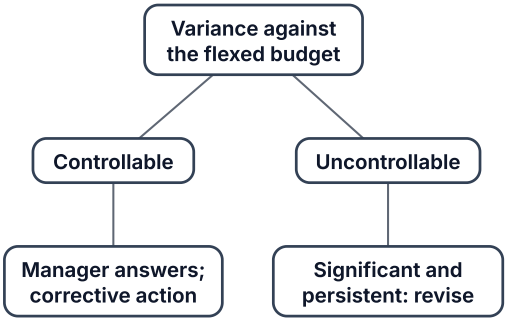
A **budget variance** is the difference between the actual and the budgeted figure of an item. As in standard costing, a variance is **favourable** when it is profitable for the company (actual cost below budget, actual sales above budget) and **adverse** when it causes loss (SM 13.12). Four rules make it useful:

1. **Compare like with like.** A fixed budget operates on one level of activity; when actual activity differs from it, comparing actual performance with it is meaningless, and a poor or good performance may remain unnoticed (SM 15.20, 15.23). Flex the budget to the actual activity first: flexible budgets.
2. **Favourable is not the same as efficient.** A favourable variance does not necessarily indicate efficient

performance, because it might have been arrived at by not carrying out the expenses mentioned in the budget (SM 15.72–15.73, key at 15.83).

3. **Report by exception, both ways.** A budget report prepared on the principle of exception shows both favourable and unfavourable variances (SM 15.73, key at 15.83).
4. **Ask who could have controlled it.** Controllable variances can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently; uncontrollable variances arise from conditions beyond its control. Responsibility centres are answerable for all adverse variances that could have been controlled; significant and persistent uncontrollable variances may call for a revision of the standard (SM 13.11–13.12).

What happens to a budget variance



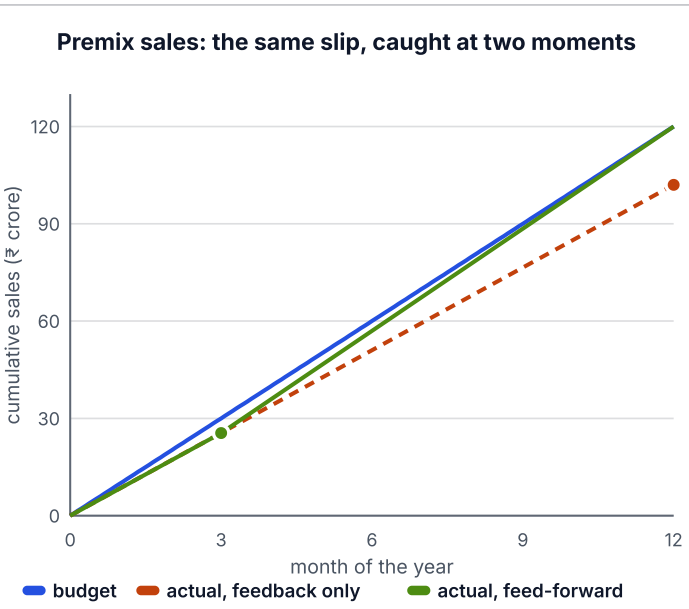
Budgetary control fixes responsibility for failure to achieve the budget and revises the budget when circumstances change (SM 15.7); the controllable or uncontrollable split decides which applies (SM 13.11–13.12).

Feedback and feed-forward: when the comparison happens

There are two types of budgetary control system based on the timing of action (SM 15.9–15.10).

Feedback control. The actual results for the budgeted period are collected and compared with the budgeted figures. Variance identification is done **after the completion of the budget period**; the variances are reported, and on the report corrective actions are taken, responsibility is fixed and, based on experience, future targets are modified. It is an ex-post corrective control system. It is common where the management information system (MIS) is not robust and data is obtained only after the books of account are finalised; it is less expensive to maintain but has limitations, and leaves the organisation on the looser side "in today's age of data warfare".

Feed-forward control is the opposite: an **ex-ante preventive** control mechanism. The budgets are set at the inception of the budgeted period and the actual results are **continuously monitored and compared**; the targets are kept as realistic as possible and are reviewed and reset if necessary. It requires a robust MIS supported by an integrated ERP system, so data is available as and when desired. It is very expensive, and beneficial where the business environment is dynamic and information gives an edge in competition.



An illustration. Sales run 15% behind budget (blue) from the start. Under feedback control (orange, dashed) the gap is found only in the year-end report (orange dot), ₹18 crore short. Under feed-forward control (green) the month-3 comparison (green dot) catches it, and corrective action restores the budget by the year end.

Budgets and motivation: targets people chase

Motivation is the driving force that converts efforts into results. A budget quantifies the desired results into targets, communicates them to executives at different levels and asks them to strive to achieve them; the behavioural side decides whether they will. A budgeting

process should therefore have these considerations to make it motivating (SM 15.14–15.15):

1. **Performance measurement.** The budget is first communicated to all executives, so that everybody knows the performance expected of him or her; achievement of targets is then considered in measuring and evaluating the performance of executives individually and departmentally. Rewards such as promotion, increment, performance related pay and bonus may be appropriate motivation factors.
2. **Achievable targets.** Targets consider practical aspects such as the availability of resources and the realism of the figures: neither very easy nor too tough. An unrealistic target has a reverse impact and may demotivate the executives.
3. **Optimum utilisation of resources.** An easily achievable target may under-utilise resources such as the potential skills of executives, while some pressure forces innovative ways of getting things done; so a balanced approach takes resources up to their effort zone, beyond the comfort zone.
4. **Involvement in the budgeting process.** Budgets that involve executives from all departments capture the requirements of all users; participative budgeting motivates the executives and gives them a sense of ownership, and involvement at the planning stage makes them accept the targets. Involvement at every stage may distort the objective of the budget, so a balance is followed.

Where a target sits (SM 15.15)

Unrealistic target — demotivates
beyond the effort zone

Balanced target — optimum use
past comfort, up to effort

Easy target — under-utilises
inside the comfort zone

Top to bottom, from the hardest target to the easiest. ICAI's balance: neither very easy nor too tough, with resources used up to the effort zone, though beyond the comfort zone. Placing the unrealistic target beyond the effort zone is this lesson's reading.

Formats and formulas

Reproduce the six formulas and the report layout from memory. (SM 15.68)

∑ **Formula**

Efficiency ratio

$$\text{Efficiency ratio} = \frac{\text{Standard hours}}{\text{Actual hours}} \times 100$$

Standard hours = standard hours for the actual output (standard hours earned).

∑ **Formula**

Activity ratio

$$\text{Activity ratio} = \frac{\text{Standard hours}}{\text{Budgeted hours}} \times 100$$

∑ **Formula**

Calendar ratio

$$\text{Calendar ratio} = \frac{\text{Available working days}}{\text{Budgeted working days}} \times 100$$

∑ **Formula**

Standard capacity usage ratio

$$\frac{\text{Budgeted hours}}{\text{Maximum possible hours}} \times 100$$

Maximum possible hours in the budgeted period.

∑ **Formula**

Actual capacity usage ratio

$$\frac{\text{Actual hours worked}}{\text{Maximum possible hours}} \times 100$$

Maximum possible working hours in the period.

∑ **Formula**

Actual usage of budgeted capacity ratio (the capacity ratio)

$$\frac{\text{Actual working hours}}{\text{Budgeted hours}} \times 100$$

ICAI's MTPs call it the capacity ratio or capacity utilisation ratio.

∑ **Formula**

How three of them fit together

$$\text{Activity ratio} = \text{Efficiency ratio} \times \text{Capacity ratio}$$

$$\text{Efficiency ratio} = \frac{\text{Activity ratio}}{\text{Capacity ratio}}$$

True whenever the capacity ratio is actual hours ÷ budgeted hours, because the actual hours cancel.

Which ratio from which data

You are given	Find first	Then
Employees (or machines), hours a day, days	Maximum and budgeted hours	Both capacity usage ratios
Output and standard time (or rate) a unit	Standard hours	Efficiency and activity ratios
A capacity usage ratio	Hours = ratio × maximum hours (budgeted from the standard ratio, actual from the actual)	The other ratios
Two of efficiency, capacity and activity	The third: activity = efficiency × capacity	The missing hours from the ratio's definition

Definition

Budget ratios

A ratio is a mathematical relationship between two or more related figures. Budget ratios provide information about the performance level, i.e., the extent of deviation of actual performance from the budgeted performance and whether the actual performance is favourable or unfavourable. (SM 15.67, 15.71)

Definition

Controllable and uncontrollable variances

Controllable variances are those which can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently. Uncontrollable variances are those which occur due to conditions beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place. (SM 13.11–13.12)

The budget report

The layout of ICAI's own answer comparing a master budget, a flexible budget and actual results (SM 15.92):

Particulars	Master budget (₹)	Flexible budget (₹)	Actual (₹)	Variance (₹)
Units	xxx	actual units	actual units	
Sales	xxx	xxx	xxx	xxx (F/A)
Variable costs, item by item	xxx	xxx	xxx	xxx (F/A)
Contribution	xxx	xxx	xxx	
Fixed overhead	xxx	xxx	xxx	xxx (F/A)
Net profit	**xxx**	**xxx**	**xxx**	**xxx (F/A)**

The variance column compares actual with the flexible budget, not with the master budget.

Worked examples

Worked example **Diwali at the Ratlam namkeen lines: all six ratios**

Tapri's namkeen factory at Ratlam measures its capacity in **line-hours** (one frying line running for one hour). For October 2026, the Diwali month, it gives you the following:

Particulars	Data
Frying lines installed	24
Frying lines in the budget	16
Working hours	Two shifts of 8 hours a day
Budgeted working days	25
Working days available	26 (one Sunday worked for Diwali orders)
Line-hours actually worked	7,040
Output	2,00,640 trays; standard output 30 trays a line-hour

Required:

(i) Calculate the efficiency, activity, calendar, standard capacity usage, actual capacity usage and actual usage of budgeted capacity ratios.

(ii) Kesar says output beat the budget "because the new temporary crews were fast". Use the ratios to say whether he is right, and comment on the month.

THE WORKING

Working notes

1. Maximum possible line-hours = 24 lines × 16 hours × 25 days = **9,600**.
2. Budgeted line-hours = 16 lines × 16 hours × 25 days = **6,400**.
3. Actual line-hours = **7,040** (given).
4. Standard line-hours for actual output = 2,00,640 trays ÷ 30 = **6,688**.
5. Budgeted working days = 25; available working days = 26.

(i) Ratios

Ratio	Working	Result
Efficiency	6,688 ÷ 7,040 × 100	95%
Activity	6,688 ÷ 6,400 × 100	104.5%
Calendar	26 ÷ 25 × 100	104%
Standard capacity usage	6,400 ÷ 9,600 × 100	66.67%
Actual capacity usage	7,040 ÷ 9,600 × 100	73.33%
Actual usage of budgeted capacity	7,040 ÷ 6,400 × 100	110%

(ii) Comment. Activity = efficiency × capacity: 95% × 110% = 104.5%. Output was 4.5% above the budget (favourable), but Kesar is wrong about the reason. The efficiency ratio is 95% (unfavourable): each line-hour produced only 95% of the standard output, which points at the new temporary crews and should be investigated. The extra output came from time: 10% more line-hours than budgeted (favourable), helped by the extra working day (calendar ratio 104%). The budget planned to use only two-thirds of the installed capacity (66.67%); 73.33% was used.

Worked example

Hosur packing unit: where did March's output go?

Tapri's cold-chain packing unit at Hosur packs chilled crates (0.40 standard hours each) and frozen crates (0.60 standard hours each). For March 2026:

Particulars	Data
Packers in the budget	40, for 8 hours a day on 25 working days
Maximum capacity	50 packers
Holi	The unit was closed on one budgeted working day
Hours actually worked	7,200
Crates packed	9,000 chilled and 5,400 frozen

Kesar blames the 14.5% shortfall in output against the budget on "lazy packers".

Required:

(i) Calculate the efficiency, activity, calendar, standard capacity usage, actual capacity usage and actual usage of budgeted capacity ratios.

(ii) Analyse the shortfall in standard hours between the hours lost to the holiday, the other hours not worked

and slower working, and comment on Kesar's view.

THE WORKING

Working notes

1. Budgeted hours = 40 × 8 × 25 = **8,000 hours**; maximum possible hours = 50 × 8 × 25 = **10,000 hours**.
2. Standard hours for actual output = 9,000 × 0.40 + 5,400 × 0.60 = 3,600 + 3,240 = **6,840 hours**.
3. Available working days = 25 − 1 = **24 days**.

(i) Ratios

Ratio	Working	Result
Efficiency	Standard hours ÷ actual hours = 6,840 ÷ 7,200	95.00%
Activity	Standard hours ÷ budgeted hours = 6,840 ÷ 8,000	85.50%
Calendar	Available working days ÷ budgeted working days = 24 ÷ 25	96.00%
Standard capacity usage	Budgeted hours ÷ maximum possible hours = 8,000 ÷ 10,000	80.00%

Ratio	Working	Result
Actual capacity usage	Actual hours ÷ maximum possible hours = 7,200 ÷ 10,000	72.00%
Actual usage of budgeted capacity	Actual hours ÷ budgeted hours = 7,200 ÷ 8,000	90.00%

Check: efficiency × actual usage of budgeted capacity = 95% × 90% = 85.5%, the activity ratio.

(ii) Where the 1,160 standard hours went (8,000 budgeted − 6,840 earned)

Cause	Hours	Share
Holiday: 40 packers × 8 hours × 1 day	320	27.6%
Other hours not worked: 8,000 − 320 − 7,200	480	41.4%
Slower working: 7,200 actual − 6,840 standard	360	31.0%
Shortfall	**1,160**	**100.0%**

Comment. Output was 85.5% of the budget (the activity ratio), a shortfall of 14.5%. Less than a third of it is slower working (efficiency 95%), which is the only part that "laziness" could explain. The holiday (calendar ratio 96%) was outside the

packers' control, and the 480 other hours not worked (absence or late starts) call for an attendance review before anyone is blamed. Kesar's view is not supported by the ratios.

Worked example

The pilot premix line: every cost over budget?

Tapri's pilot line for a new masala premix had this master budget and these actual results for June 2026. A late order from a hotel chain added 500 kg to the month.

Particulars	Master budget	Actual
Output and sales (kg)	5,000	5,500
Sales (₹)	10,00,000	10,78,000
Direct material (₹)	4,00,000	4,51,000
Direct labour (₹)	1,50,000	1,59,500
Variable overheads (₹)	1,00,000	1,15,500
Fixed overheads (₹)	1,20,000	1,26,000
Marketing campaign, fixed (₹)	50,000	10,000

The budget assumed ₹200 a kg for sales and ₹80, ₹30 and ₹20 a kg for material, labour and variable overheads. The launch campaign budgeted for June was postponed to August; the ₹10,000 spent in June was for its design. Kesar reads the comparison with the master budget, praises the sales team for ₹78,000 of extra sales and asks the production manager why every variable cost is over budget.

Required:

(i) Prepare a statement comparing the actual results with a flexible budget, and compute the variances.

(ii) Comment on Kesar's reading, on the marketing variance and on which variances an exception report should show.

THE WORKING

(i) Flexible budget compared with actual, June 2026

Particulars	Master (5,000 kg)	Flexible (5,500 kg)	Actual (5,500 kg)	Variance
Sales	10,00,000	11,00,000	10,78,000	22,000 A
Direct material	4,00,000	4,40,000	4,51,000	11,000 A
Direct labour	1,50,000	1,65,000	1,59,500	5,500 F
Variable overheads	1,00,000	1,10,000	1,15,500	5,500 A
Contribution	**3,50,000**	**3,85,000**	**3,52,000**	**33,000 A**
Fixed overheads	1,20,000	1,20,000	1,26,000	6,000 A
Marketing campaign	50,000	50,000	10,000	40,000 F
Net profit	**1,80,000**	**2,15,000**	**2,16,000**	**1,000 F**

The flexible budget applies the budgeted rates to the actual 5,500 kg: sales 5,500 × ₹200; material 5,500 × ₹80; labour 5,500 × ₹30; variable overheads 5,500 × ₹20. Fixed costs stay at the budget. (Flexing is simple here because the rates are given; separating fixed and semi-variable costs first is the method of flexible budgets. This example is about reading the variances.)

(ii) Comments.

1. *Kesar's reading.* Against the master budget every variable cost looks adverse (material ₹51,000, labour ₹9,500, variable overheads ₹15,500) only because 500 kg more were made:

comparing actual performance with a fixed budget at a different level of activity is meaningless. At the actual level, material is ₹11,000 adverse (₹82 a kg against ₹80) and variable overheads ₹5,500 adverse (₹21 against ₹20), but labour is ₹5,500 **favourable** (₹29 against ₹30). Sales are ₹78,000 above the master budget but ₹22,000 below the flexible budget: the 5,500 kg sold at an average of ₹196 a kg against ₹200. The sales team earned the volume; it owes an explanation for the price.

2. *Marketing.* The ₹40,000 favourable variance arises because the campaign was not carried out in June, not from efficiency: the spending has moved to August, and the delay may cost sales. A favourable variance does not necessarily indicate efficient performance. Without it, the month's operations are ₹1,000 – ₹40,000 = **₹39,000 adverse**, although profit is ₹36,000 above the master budget.
3. *Exception report.* On the principle of exception, the report should show both the favourable and the unfavourable significant variances: the sales price shortfall (₹22,000 A), material (₹11,000 A) and the marketing deferral (₹40,000 F), each with its reason and the manager responsible.

Exam angle

In the exam

Where these topics appeared, May 2024 to September 2026

In the new-scheme documents read for this lesson, control ratios were asked in 10 documents, feedback and feed-forward control in 2, controllable and uncontrollable variances in 2, and budgets and motivation in none.

- **Ratios, descriptive.** Working backwards from two ratios: May 2024 Q1(c) (4 marks; actual usage of budgeted capacity 86% and efficiency 110%) and January 2026 Q5(b) (5 marks; actual capacity usage 65% and activity 125%). Capacity, activity and efficiency ratios for two products in a month of 22 working days: MTP Series II for September 2026 Q1(c) (4 marks).
- **Ratios, multiple choice, 2 marks.** Activity ratio from standard, budgeted and actual hours: MTP Series I for September 2024 MCQ 15 and MTP Series I for May 2026 MCQ 12. Efficiency from capacity utilisation and activity: MTP Series II for September 2024 MCQ 14. Standard hours from budgeted hours, capacity utilisation and efficiency: MTP Series II for January 2026 MCQ 11.
- **The efficiency (productivity) ratio inside a wages budget**: MTP Series I for September 2024 Q5(a), MTP Series II for January 2025 Q5(b), RTP January 2026 Q18, MTP Series I for September 2026 Q5(a)

(an 85% target efficiency plus downtime of 15% of productive hours).

- **Feedback and feed-forward control**, "Discuss", 4 marks: May 2024 Q5(c); MTP Series II for May 2026 Q6(c) (an OR choice).
- **Controllable and uncontrollable variances**: two terms of a five-term definition question, September 2024 Q6(b); "Discuss", RTP September 2024 Q19(c).
- Old scheme: January 2021 Q5(b) (10 marks) asked capacity, efficiency and activity ratios with the fixed overhead variances; its budgeted hours were net of a 15% normal downtime.

In the exam

The traps ICAI set

- **Which employees make the budgeted hours?** In ICAI's illustration and in May 2024 Q1(c), budgeted hours are the "actual working" employees × hours × days; the maximum capacity employees give the maximum possible hours. Mixing them up swaps the two capacity ratios.
- **"Capacity ratio"** in the MTPs is actual hours ÷ budgeted hours (the Study Material's actual usage of budgeted capacity ratio), not budgeted ÷ maximum.
- **Normal downtime** reduces the budgeted hours before any ratio is taken (January 2021).
- **Two products**: standard hours are the sum over both products of output × standard hours a unit, and an

annual budget of hours is divided by 12 for the month (MTP Series II for September 2026).

- **Feedback or feed-forward**: ICAI's suggested answer (May 2024) reproduces the Study Material's two paragraphs. Get its key phrases in: "after the completion of the budget period", "ex-post corrective", "continuously monitored", "ex-ante preventive", "robust MIS" and "ERP".

Trick

E × C = A

Efficiency × **C**apacity = **A**ctivity: speed times time is output. Given any two, the third follows. Condition: the capacity ratio must be actual hours ÷ budgeted hours.

Answer frame

"Discuss feedback and feed-forward control" (4 marks)

Point	Feedback control	Feed-forward control
Timing	Variances identified after the completion of the budget period	Actual results continuously monitored and compared during the period
Nature	Ex-post corrective	Ex-ante preventive

Point	Feedback control	Feed-forward control
Action	Corrective action, responsibility fixed, future targets modified	Targets kept realistic, reviewed and reset if necessary
Information needed	Data available only after finalisation of books; MIS not robust	Robust MIS supported by an integrated ERP system
Cost and fit	Less expensive, but leaves the organisation on the looser side	Very expensive; suits a dynamic business environment where information gives a competitive edge

"How can budgets motivate executives?"

The four considerations in ICAI's order, each with a line: performance measurement (communicated targets, achievement used in evaluation, rewards); achievable targets (neither very easy nor too tough; unrealistic targets demotivate); optimum utilisation of resources (beyond the comfort zone, up to the effort zone); involvement in the budgeting process (participative budgeting gives ownership; a balance, not involvement at every stage).

The ratio numericals: three answers

Loses marks

"Efficiency 95%, activity 104.5%, capacity 110%."

Bare figures; one wrong hours figure and nothing earns method marks.

Passes

Each ratio with its formula and figures, but the budgeted hours appear without the lines × hours × days working.

Correct, with the base of every ratio unexplained.

Full marks

Working notes for maximum, budgeted, actual and standard hours and for the days; each ratio with formula, figures and result (ICAI prints 109.375% in full, SM 15.69); a one-line check (efficiency × capacity = activity) and a comment marking each ratio favourable or unfavourable.

Common mistakes

⚠ **Common mistake**

Hall of Shame: the activity ratio on actual hours

Dividing standard hours by actual hours and calling it activity. That is efficiency. **Fix:** activity compares output with the **budget**: standard hours ÷ budgeted hours. Check with E × C = A.

⚠ **Common mistake**

Maximum capacity used as the budget

Computing the Ratlam budgeted hours with all 24 installed lines: the standard capacity usage ratio becomes 100% and every other ratio shifts. **Fix:** the maximum capacity (installed lines, maximum employees) gives the maximum possible hours; the lines or employees budgeted to work ("actual working" in ICAI's data) give the budgeted hours.

⚠ **Common mistake**

The efficiency ratio upside down

With 9,750 actual hours and 10,000 standard hours, actual ÷ standard gives 97.5% where the answer is 102.56%, turning a good month bad. **Fix:** standard hours are always the numerator of the efficiency and activity ratios.

⚠ **Common mistake**

"Favourable, so well done"

A cost under budget praised without asking why. **Fix:** compare with the flexed budget, then ask the cause: an unspent campaign, skipped maintenance or a lower activity level is not efficiency.

Practice

Check yourself

1. A kulhad workshop budgeted 12,000 hours for a month. Its activity ratio was 99% and its efficiency ratio 110%. How many hours did it actually work?

Answer: _____

2 Case scenario. Tapri's app-delivery division changes its menu, prices and delivery zones every few weeks to fight BrewBucks. It has an integrated ERP system that gives daily data. Ritu must recommend a budgetary control system. Which recommendation is correct?

A Feedback control, because it is the less expensive system

B Feedback control, because variances are best identified after the budget period ends

C Feed-forward control, because the environment is dynamic and a robust MIS is available to monitor actuals continuously

D Neither; budgets cannot be used in a dynamic business

3 Ritu's June exception report for the premix pilot line lists the ₹11,000 adverse material variance but leaves out the ₹40,000 favourable marketing variance, "because it is good news". Is that right?

A Yes: a report on the principle of exception shows only the adverse variances

B Yes: a favourable variance always reflects efficient performance

C No: an exception report must list every variance, however small

D No: significant favourable variances belong in the report too, and this one comes from a campaign that was not run

4 Ritu reports for the samosa line: actual usage of budgeted capacity ratio 88%, efficiency ratio 110%. Kesar concludes that output was 10% above budget. Which statement is correct?

- **A** Kesar is right: an efficiency ratio of 110% means output was 10% above budget
- **B** Output was 96.8% of budget: 12% fewer hours than budgeted were worked, at 10% above standard speed
- **C** Output was 98% of budget: 110% − 12%
- **D** Output was 80% of budget: 88% ÷ 110%

5 Which of these are considerations ICAI lists for making a budgeting process motivating? Select all that apply.

- **A** Targets that are achievable, neither very easy nor too tough
- **B** Involvement of executives in the budgeting process
- **C** Keeping the targets secret until the year end, so managers do not relax
- **D** Using the achievement of targets in measuring and evaluating performance

More than one answer may be right.

Answers: the answer key at the end of this lesson.

Recap

- Efficiency = standard hours ÷ actual hours; activity = standard ÷ budgeted; calendar = available days ÷ budgeted days; all × 100.
- Standard capacity usage = budgeted ÷ maximum hours; actual capacity usage = actual ÷ maximum; actual usage of budgeted capacity (the capacity ratio) = actual ÷ budgeted.
- 100% or more is favourable; below 100%, unfavourable. Activity = efficiency × capacity.
- Budgeted hours come from the employees budgeted to work; maximum hours from maximum capacity; deduct normal downtime if given.
- Budget variances: compare with the flexed budget; favourable is not necessarily efficient; exception reports show both directions; controllable variances are the manager's to answer.
- Feedback: after the period, ex-post corrective, cheaper. Feed-forward: continuous, ex-ante preventive, needs robust MIS and ERP, suits dynamic businesses.
- Motivation: performance measurement, achievable targets, optimum utilisation of resources, involvement. (Rain cut the overs; the batters were fine.)

Answer key · Check yourself

1 **10800**

Standard hours = 99% × 12,000 = 11,880. Efficiency = standard ÷ actual, so actual = 11,880 ÷ 110% = 10,800 hours (a capacity ratio of 90%).

2 **C** **Feed-forward control, because the environment is dynamic and a robust MIS is available to monitor actuals continuously**

Feed-forward control is ex-ante preventive: actuals are monitored continuously and targets reset where necessary. It needs a robust MIS with an integrated ERP and suits a dynamic environment where information gives a competitive edge. Feedback is cheaper but acts only after the period.

3 **D** **No: significant favourable variances belong in the report too, and this one comes from a campaign that was not run**

A budget report on the principle of exception shows both favourable and unfavourable variances (SM 15.73, key at 15.83), and a favourable variance may come from expenses not carried out. "Exception" means the significant deviations, not every rupee.

4 **B Output was 96.8% of budget: 12% fewer hours than budgeted were worked, at 10% above standard speed**

The activity ratio, which compares output with the budget, is efficiency × actual usage of budgeted capacity = 110% × 88% = 96.8%. Efficiency compares output with the hours actually worked, not with the budget, so the faster work did not make up for the 12% of budgeted hours not worked. The ratios multiply; they do not add or divide.

5 **A Targets that are achievable, neither very easy nor too tough; B Involvement of executives in the budgeting process; D Using the achievement of targets in measuring and evaluating performance**

ICAI's four: performance measurement, achievable targets, optimum utilisation of resources, involvement in the budgeting process (SM 15.15). The budget must first be communicated to all executives, so secret targets fail the first consideration.